PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 6A

Financial Management

ICAI Question Bank

Every ICAI past-exam, RTP and MTP question for this paper, Nov 2019 to Sep 2026, sorted by chapter and sub-topic, each with ICAI's own answer, word for word.

976 questions (238 MCQs, 738 descriptive) from 65 ICAI documents · A4 edition · made 2026-10-21

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How this book works

- **Sorted by the syllabus.** Chapters and sub-topics follow the January 2027 syllabus (PassBook's lessons). Each question sits under the sub-topic it mainly tests. That placement is PassBook's judgement, most of all for old-scheme papers, which used different chapters.
- **Order inside a sub-topic:** exam papers first, then RTPs, then MTPs, newest attempt first.
- **Every card** shows where it comes from (EXAM RTP MTP), the attempt, ICAI's question number and marks. RTPs print no marks. old scheme marks papers from before May 2024.
- **Questions ICAI repeated** word for word, with the same answer, are printed once. The other documents are listed under *Also asked in* (29 repeats). Where ICAI answered the same question differently in two documents, both are printed, and each card points to the other: *Same question, different ICAI answer*.
- **A case scenario** stays together: the case, all its questions, then ICAI's answers in ICAI's order.
- **ICAI's words, unchanged.** Questions, options and answers are copied from ICAI's PDFs. Each card ends with the PDF pages. Lines marked *PassBook's note* are ours: why a question is out of syllabus, or where the Study Material now teaches a newer law (CARO 2020 for CARO 2016, say). ICAI's own slips are kept as printed.
- **Missing answers.** ICAI has not yet published Suggested Answers for the September 2026 exam (marked * below), and one old MTP's answers are no longer online. Those questions appear without an answer.
- **Out of syllabus** questions are in an appendix at the end (40), so they never get in the way.

Most asked sub-topics

How many separate ICAI documents asked about each sub-topic (repeats counted in every document that printed them). Revise the top rows first.

#	Sub-topic	Documents	Exams
1	Operating, financial and combined leverage Ch 6	64	17
2	Dividend theories: traditional, Walter, Gordon and Modigliani-Miller Ch 8	60	15
3	Building financial statements from ratios; limitations of ratios Ch 2	47	12
4	WACC and the marginal cost of capital Ch 4	44	10
5	NPV, profitability index and equivalent annual cost Ch 7	43	13
6	EBIT-EPS analysis, indifference point and financial break-even Ch 5	39	9
7	Capital structure theories: NI, NOI, traditional and Modigliani-Miller Ch 5	36	7
8	Management of receivables, credit policy, factoring and forfaiting Ch 9	32	7
9	Estimating working capital requirement Ch 9	28	6
10	Venture capital, lease financing, sale and lease back and other sources Ch 3	27	4
11	Objectives: profit maximisation, wealth maximisation and value creation Ch 1	24	7
12	Contemporary and international sources of finance Ch 3	22	8
13	Cost of equity and of retained earnings Ch 4	22	5
14	Long-term sources: equity, preference, debt and retained earnings Ch 3	20	6
15	The finance function, the financial manager and the controller Ch 1	19	6

Documents in this book

32 new-scheme and 33 old-scheme ICAI documents. A dash means ICAI published none for that attempt, or it is not in the bank.

Attempt	Exam	RTP	MTP
September 2026	✓*	✓	I, II
May 2026	✓	✓	I, II
January 2026	✓	✓	I, II
September 2025	✓	✓	I, II
May 2025	✓	✓	I, II
January 2025	✓	✓	I, II
September 2024	✓	✓	I, II
May 2024	✓	✓	I, II
November 2023 Old P8	✓	✓	I, II
May 2023 Old P8	✓	✓	I, II
November 2022 Old P8	✓	✓	I, II
May 2022 Old P8	✓	✓	I, II
December 2021 Old P8	✓	✓	I, II
July 2021 Old P8	✓	✓	I, II
January 2021 Old P8	✓	–	–
November 2020 Old P8	✓	✓	I
May 2020 Old P8	–	✓	I
November 2019 Old P8	✓	✓	I

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CHAPTER 1

Scope and Objectives of Financial Management

1.1 Objectives: profit maximisation, wealth maximisation and value creation

22 questions · asked in 24 ICAI documents · 7 exam papers

1 EXAM **May 2026** Q4(c) 2 marks

How do the concepts of profit maximization and wealth maximization differ in terms of objective, risk, time horizon and sustainability?

ICAI'S ANSWER

Particular	Profit Maximization	Wealth Maximisation
Objective	Large amount of profits.	Highest market value of shares.
Risk	Ignores risk or uncertainty	Recognises risk or certainty
Time Horizon	Emphasizes the short term gains.	Emphasizes the long term gains.
Sustainability	Not sustainable in long term. Sustainability is low.	Critical for the very existence of the business enterprise. Sustainability is high.

ICAI · P6 · May 2026 exam · question p. 21 · answer p. 22-23

2 EXAM **Sep 2025** Q4(c) 2 marks

Explain any two limitations of profit maximization objective of Financial Management.

ICAI'S ANSWER

Limitations of profit maximisation objective:

(i) The term profit is vague. It does not clarify what exactly it means.

(ii) Profit maximisation has to be attempted with a realisation of risks involved.

(iii) Profit maximisation as an objective does not take into account the time pattern of returns.

(iv) Profit maximisation as an objective is too narrow.

ICAI · P6 · Sep 2025 exam · question p. 18 · answer p. 19

3 EXAM **Jan 2025** Q4(b) 4 marks

Discuss the objectives and advantages of wealth maximization goal of Financial Management.

ICAI'S ANSWER

Objective of Wealth Maximisation:

Highest market value of shares.

Advantages of Wealth Maximisation:

(i) Emphasizes the long term gains

(ii) Recognises risk or uncertainty

(iii) Recognises the timing of returns

(iv) Considers shareholders' return.

ICAI · P6 · Jan 2025 exam · question p. 21 · answer p. 22

4 EXAM **May 2024** Q4(a) 4 marks

State with brief reasons whether the following statements are true or false:

(i) Maximising Market Price Per Share (MPS) as the financial objective which maximises the wealth of shareholders.

(ii) A combination of lower risk and higher return is known as risk return trade off and at this level of risk-return, profit is maximum.

(iii) Financial distress is a position when accounting profits of a firm are sufficient to meet its long-term obligations.

(iv) Angel investor is one who provides funds for start-up m exchange for an ownership/equity.

ICAI'S ANSWER

Statement	True or False	Reason
Maximising Market Price Per Share (MPS) as the financial objective which maximises the wealth of shareholders.	True	Maximizing MPS or Market value as the financial objective will ensure the maximizing shareholder's wealth.
A combination of lower risk and higher return is known as risk-return trade off and at this level of risk-return, profit is maximum.	False	There is a direct relationship between risk and profit. Higher the risk, higher is the possibility of profits. Stockholders expect greater returns from investments of higher risk and vice-versa.
Financial distress is a position when accounting profits of a firm are sufficient to meet its long-term obligations.	False	Financial distress is a position where Cash inflows of a firm are inadequate to meet all its current obligations.
Angel investor is one who provides funds for start-up in exchange for an ownership/equity.	True	Angel Financing is a form of an equity-financing where an angel investor provides capital for start-up or expansion, in exchange for an ownership/equity in the company.

ICAI · P6 · May 2024 exam · question p. 19 · answer p. 20

5 EXAM **Old P8 Nov 2023** Q6(c) 2 marks old scheme

Also asked in RTP Jan 2026 Q11(d)

What are disadvantages of Profit Maximization?

ICAI'S ANSWER

Disadvantages of Profit Maximisation objective of financial management.

(i) Emphasizes the short-term gains

(ii) Ignores risk or uncertainty

(iii) Ignores the timing of returns

(iv) Requires immediate resources.

ICAI · Old P8 · Nov 2023 exam · question p. 17 · answer p. 18

6 EXAM **Old P8 May 2022** Q6(c) 2 marks old scheme

State advantages of "Wealth Maximization" goals in Financial Management

ICAI'S ANSWER

Advantages of "Wealth Maximization" goals in Financial Management

(i) Emphasizes the long-term gains.

(ii) Recognises risk or uncertainty.

(iii) Recognises the timing of returns.

(iv) Considers shareholders' return.

ICAI · Old P8 · May 2022 exam · question p. 16 · answer p. 17

7 EXAM **Old P8 Jul 2021** Q6(c) 2 marks old scheme

List out the steps to be followed by the manager to measure and maximize the Shareholder's Wealth

ICAI'S ANSWER

For measuring and maximising shareholders' wealth, manager should follow:

Cash Flow approach not Accounting Profit

Cost benefit analysis

Application of time value of money.

ICAI · Old P8 · Jul 2021 exam · question p. 83 · answer p. 84

8 RTP **May 2025** Q10(a)

Same question, different ICAI answer: #10 RTP May 2024

Explain as to how the wealth maximisation objective is superior to the profit maximisation objective.

ICAI'S ANSWER

A firm's financial management may often have the following as their objectives:

(i) The maximisation of firm's profit.

(ii) The maximisation of firm's value / wealth.

The maximisation of profit is often considered as an implied objective of a firm. To achieve the aforesaid objective various type of financing decisions may be taken. Options resulting into maximisation of profit may be selected by the firm's decision makers. They even sometime may adopt policies yielding exorbitant profits in short run which may prove to be unhealthy for the growth, survival and overall interests of the firm. The profit of the firm in this case is measured in terms of its total accounting profit available to its shareholders.

The value/wealth of a firm is defined as the market price of the firm's stock. The market price of a firm's stock represents the focal judgment of all market participants as to what the value of the particular firm is. It takes into account present and prospective future earnings per share, the timing and risk of these earnings, the dividend policy of the firm and many other factors that bear upon the market price of the stock.

The value maximisation objective of a firm is superior to its profit maximisation objective due to following reasons.

1. The value maximisation objective of a firm considers all future cash flows, dividends, earning per share, risk of a decision etc. whereas profit maximisation objective does not consider the effect of EPS, dividend paid or any other returns to shareholders or the wealth of the shareholder.
2. A firm that wishes to maximise the shareholders wealth may pay regular dividends whereas a firm with the objective of profit maximisation may refrain from dividend payment to its shareholders.
3. Shareholders would prefer an increase in the firm's wealth against its generation of increasing flow of profits.
4. The market price of a share reflects the shareholders expected return, considering the long-term prospects of the firm, reflects the differences in timings of the returns, considers risk and recognizes the importance of distribution of returns.

The maximisation of a firm's value as reflected in the market price of a share is viewed as a proper goal of a firm. The profit maximisation can be considered as a part of the wealth maximisation strategy.

ICAI · P6 · RTP May 2025 · question p. 10 · answer p. 29-30

9 RTP **Sep 2024** Q12(a)

EXPLAIN the advantages and disadvantages of both profit maximization and wealth maximization goals

ICAI'S ANSWER

The table below highlights some of the advantages and disadvantages of both profit maximization and wealth maximization goals:

Goal	Objective	Advantages	Disadvantages
Profit Maximization	Large amount of profits	(i) Easy to calculate profits (ii) Easy to determine the link between financial decisions and profits.	(i) Emphasizes the short term gains (ii) Ignores risk or uncertainty (iii) Ignores the timing of returns (iv) Requires immediate resources.
Shareholders Wealth Maximisation	Highest market value of shares.	(i) Emphasizes the long term gains (ii) Recognises risk or uncertainty (iii) Recognises the timing of returns (iv) Considers shareholders' return.	(i) Offers no clear relationship between financial decisions and share price. (ii) Can lead to management anxiety and frustration.

ICAI · P6 · RTP Sep 2024 · question p. 10 · answer p. 29-30

10 RTP **May 2024** Q8(i)

Same question, different ICAI answer: #8 RTP May 2025

EXPLAIN as to how the wealth maximisation objective is superior to the profit maximisation objective

ICAI'S ANSWER

A firm's financial management may often have the following as their objectives:

(a) The maximisation of firm's profit.

(b) The maximisation of firm's value / wealth.

The maximisation of profit is often considered as an implied objective of a firm. To achieve the aforesaid objective various type of financing decisions may be taken. Options resulting into maximisation of profit may be selected by the firm's decision makers. They even sometime may adopt policies yielding exorbitant profits in short run which may prove to be unhealthy for the growth, survival and overall interests of the firm. The profit of the firm in this case is measured in terms of its total accounting profit available to its shareholders.

The value/wealth of a firm is defined as the market price of the firm's stock. The market price of a firm's stock represents the focal judgment of all market participants as to what the value of the particular firm is. It takes into account present and prospective future earnings per share, the timing and risk of these earnings, the dividend policy of the firm and many other factors that bear upon the market price of the stock.

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3. Shareholders would prefer an increase in the firm's wealth against its generation of increasing flow of profits.
4. The market price of a share reflects the shareholders expected return, considering the long-term prospects of the firm, reflects the differences in timings of the returns, considers risk and recognizes the importance of distribution of returns.

The maximisation of a firm's value as reflected in the market price of a share is viewed as a proper goal of a firm. The profit maximisation can be considered as a part of the wealth maximisation strategy.

ICAI · P6 · RTP May 2024 · question p. 8 · answer p. 27-28

11 RTP **Old P8 May 2022** Q10(a) old scheme

'Profit maximisation is not the sole objective of a company. It is at best a limited objective. If profit is given undue importance, a number of problems can arise.' DISCUSS four of such problems.

ICAI'S ANSWER

Profit maximisation cannot be the sole objective of a company. It is at best a limited objective. If profit is given undue importance, a number of problems can arise. Some of these have been discussed below:

(i) The term profit is vague. It does not clarify what exactly it means. It conveys a different meaning to different people. For example, profit may be in short term or long term period; it may be total profit or rate of profit etc.

(ii) Profit maximisation has to be attempted with a realisation of risks involved. There is a direct relationship between risk and profit. Many risky propositions yield high profit. Higher the risk, higher is the possibility of profits. If profit maximisation is the only goal, then risk factor is altogether ignored. This implies that finance manager will accept highly risky proposals also, if they give high profits. In practice, however, risk is very important consideration and has to be balanced with the profit objective.

(iii) Profit maximisation as an objective does not take into account the time pattern of returns. Proposal A may give a higher amount of profits as compared to proposal B, yet if the returns of proposal A begin to flow say 10 years later, proposal B may be preferred which may have lower overall profit but the returns flow is more early and quick.

(iv) Profit maximisation as an objective is too narrow. It fails to take into account the social considerations as also the obligations to various interests of workers, consumers, society, as well as ethical trade practices. If these factors are ignored, a company cannot survive for long. Profit maximization at the cost of social and moral obligations is a short sighted policy.

ICAI · Old P8 · RTP May 2022 · question p. 6 · answer p. 20

12 RTP **Old P8 Jul 2021** Q10(i) old scheme

"Profit Maximization cannot be the sole objective of a company". COMMENT.

ICAI'S ANSWER

Following are the reasons due to which Profit Maximization cannot be the sole objective of a company:

(a) The term profit is vague. It does not clarify what exactly it means. It conveys a different meaning to different people. For example, profit may be in short term or long-term period; it may be total profit or rate of profit etc.

(b) Profit maximisation has to be attempted with a realisation of risks involved. There is a direct relationship between risk and profit. Many risky propositions yield high profit. Higher the risk, higher is the possibility of profits. If profit maximisation is the only goal, then risk factor is altogether ignored. This implies that finance manager will accept highly risky proposals also, if they give high profits. In practice, however, risk is very important consideration and has to be balanced with the profit objective.

(c) Profit maximisation as an objective does not take into account the time pattern of returns. Proposal A may give a higher amount of profits as compared to proposal B, yet if the returns of proposal A begin to flow say 10 years later, proposal B may be preferred which may have lower overall profit but the returns flow is more early and quick.

(d) Profit maximisation as an objective is too narrow. It fails to take into account the social considerations as also the obligations to various interests of workers, consumers, society, as well as ethical trade practices. If these factors are ignored, a company cannot survive for long. Profit maximization at the cost of social and moral obligations is a short sighted policy.

ICAI · Old P8 · RTP Jul 2021 · question p. 7 · answer p. 21

13 RTP **Old P8 May 2020** Q10(i) old scheme

"The profit maximization is not an operationally feasible criterion." IDENTIFY.

ICAI'S ANSWER

The profit maximisation is not an operationally feasible criterion." This statement is true because profit maximisation can be a short-term objective for any organisation and cannot be its sole objective. Profit maximization fails to serve as an operational criterion for maximizing the owner's economic welfare. It fails to provide an operationally feasible measure for ranking alternative courses of action in terms of their economic efficiency. It suffers from the following limitations:

(a) Vague term: The definition of the term profit is ambiguous. Does it mean short term or long term profit? Does it refer to profit before or after tax? Total profit or profit per share?

(b) Timing of Return: The profit maximization objective does not make distinction between returns received in different time periods. It gives no consideration to the time value of money, and values benefits received today and benefits received after a period as the same.

(c) It ignores the risk factor.

(d) The term maximization is also vague.

ICAI · Old P8 · RTP May 2020 · question p. 6 · answer p. 14-15

14 MTP **Sep 2026 S-I** Q4(c) 2 marks

Also asked in MTP May 2026 S-I Q4(a)

The profit maximization is not an operationally feasible criterion." Comment on it.

ICAI'S ANSWER

The profit maximisation is not an operationally feasible criterion." This statement is true because profit maximisation can be a short-term objective for any organisation and cannot be its sole objective. Profit maximization fails to serve as an operational criterion for maximizing the owner's economic welfare. It fails to provide an operationally feasible measure for ranking alternative courses of action in terms of their economic efficiency. It suffers from the following limitations:

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(ii) Timing of Return: The profit maximization objective does not make distinction between returns received in different time periods. It gives no consideration to the time value of money, and values benefits received today and benefits received after a period as the same.

(iii) It ignores the risk factor.

(iv) The term maximization is also vague.

ICAI · P6 · MTP Sep 2026 · Series I · question p. 7 · answer p. 13

15 MTP **Jan 2026 S-II** Q4(c) 2 marks

HOW do we measure the value/wealth of a firm?

ICAI'S ANSWER

According to Van Horne, "Value of a firm is represented by the market price of the company's common stock. The market price of a firm's stock represents the focal judgment of all market participants as to what the value of the particular firm is. It takes into account present and prospective future earnings per share, the timing and risk of these earnings, the dividend policy of the firm and many other factors that bear upon the market price of the stock. The market price serves

as a performance index or report card of the firm's progress. It indicates how well management is doing on behalf of stockholders."

ICAI · P6 · MTP Jan 2026 · Series II · question p. 8 · answer p. 14

16 MTP **Sep 2025 S-II** Q4(a) 4 marks

ABC Ltd. is evaluating two investment opportunities. Project X offers high short-term accounting profits with aggressive cost-cutting measures, while Project Y provides lower immediate profits but improves customer satisfaction, brand loyalty, and long-term cash flows.

As a financial manager, WHICH project should the company choose to ensure value creation? JUSTIFY your answer by comparing the concepts of profit maximization, wealth maximization, and value creation.

ICAI'S ANSWER

Project Y should be chosen by the financial manager to ensure long-term value creation.

Profit maximization focuses on increasing short-term profits, often ignoring risk, sustainability, and timing of returns. Project X aligns with this approach by maximizing current profits, but may harm the firm's reputation, employee morale, and customer satisfaction.

Wealth maximization, on the other hand, is concerned with increasing the net present value of shareholders' wealth over the long run. It considers cash flows, risk, and time value of money. Project Y, despite lower immediate profits, enhances long-term cash flows and strengthens brand equity, which contributes to shareholder wealth.

Value creation refers to making strategic decisions that generate sustainable returns exceeding the cost of capital. Project Y builds intangible assets like customer loyalty and goodwill, which leads to long-term competitive advantage and value creation.

Therefore, based on wealth maximization and value creation principles, Project Y is the strategically sound choice.

ICAI · P6 · MTP Sep 2025 · Series II · question p. 9 · answer p. 13-14

17 MTP **May 2025 S-II** Q4(c) 2 marks

DISCUSS the conflicts in Profit versus Wealth maximization principle of the firm.

ICAI'S ANSWER

Conflict in Profit versus Wealth Maximization Principle of the Firm: Profit maximisation is a short-term objective and cannot be the sole objective of a company. It is at best a limited objective. If profit is given undue importance, a number of problems can arise like the term profit is vague, profit maximisation has to be attempted with a realisation of risks involved, it does not take into account the time pattern of returns and as an objective it is too narrow.

Whereas, on the other hand, wealth maximisation, as an objective, means that the company is using its resources in a good manner. If the share value is to stay high, the company has to reduce its costs and use the resources properly. If the company follows the goal of wealth maximisation, it means that the company will promote only those policies that will lead to an efficient allocation of resources.

ICAI · P6 · MTP May 2025 · Series II · question p. 9 · answer p. 17

18 MTP **Jan 2025 S-I** Q4(c) 2 marks

"XYZ Corp. has adopted a strategy to maximize short-term profits by increasing product prices significantly. ANALYZE why this might not be a feasible operational criterion for sustainable growth."

ICAI'S ANSWER

"The profit maximisation is not an operationally feasible criterion." This statement is true because Profit maximisation can be a short-term objective for any organisation and cannot be its sole objective. Profit maximization fails to serve as an operational criterion for maximizing the owner's economic welfare. It fails to provide an operationally feasible measure for ranking alternative courses of action in terms of their economic efficiency. It suffers from the following limitations:

(i) Vague term: The definition of the term profit is ambiguous. Does it mean short term or long term profit? Does it refer to profit before or after tax? Total profit or profit per share?

(ii) Timing of Return: The profit maximization objective does not make distinction between returns received in different time periods. It gives no consideration to the time value of money, and values benefits received today and benefits received after a period as the same.

(iii) It ignores the risk factor.

(iv) The term maximization is also vague.

ICAI · P6 · MTP Jan 2025 · Series I · question p. 7 · answer p. 12

19 MTP **May 2024 S-I** Q4(a) 4 marks

EXPLAIN the limitations of profit maximization objective of Financial Management.

ICAI'S ANSWER

Limitations of Profit Maximisation objective of financial management.

(i) **The term profit is vague. It does not clarify what exactly it means.** It conveys a different meaning to different people. For example, profit may be in short term or long term period; it may be total profit or rate of profit etc.

(ii) **Profit maximisation has to be attempted with a realisation of risks involved.** There is a direct relationship between risk and profit. Many risky propositions yield high profit. Higher the risk, higher is the possibility of profits. If profit maximisation is the only goal, then risk factor is altogether ignored. This implies that finance manager will accept highly risky proposals also, if they give high profits. In practice, however, risk is very important consideration and has to be balanced with the profit objective.

(iii) **Profit maximisation as an objective does not take into account the time pattern of returns.** Proposal A may give a higher amount of profits as compared to proposal B, yet if the returns of proposal A begin to flow say 10 years later, proposal B may be preferred which may have lower overall profit but the returns flow is more early and quick.

(iv) **Profit maximisation as an objective is too narrow.** It fails to take into account the social considerations as also the obligations to various interests of workers, consumers, society, as well as ethical trade practices. If these factors are ignored, a company cannot survive for long. Profit maximization at the cost of social and moral obligations is a short sighted policy.

ICAI · P6 · MTP May 2024 · Series I · question p. 5 · answer p. 6-7

20 MTP **Old P8 May 2023 S-II** Q6 10 marks old scheme

Attempt any 2

(i) DISTINGUISH between Profit maximisation vis-a-vis wealth maximization.

(ii) WRITE a note on agency problem and agency cost.

(iii) EXPLAIN the determinants of dividend decisions.

ICAI'S ANSWER

(i) It has traditionally been argued that the primary objective of a company is to earn profit; hence the objective of financial management is also profit maximisation. This implies that the finance manager has to make his decisions in a manner so that the profits of the concern are maximised. Each alternative, therefore, is to be seen as to whether or not it gives maximum profit.

However, profit maximisation cannot be the sole objective of a company. It is at best a limited objective. If profit is given undue importance, a number of problems can arise. Some of these have been discussed below:

(i) The term profit is vague. **It does not clarify what exactly it means.** It conveys a different meaning to different people. For example, profit may be in short term or long term period; it may be total profit or rate of profit etc.

(ii) Profit maximisation has to be attempted with a realisation of risks involved. There is a direct relationship between risk and profit. Many risky propositions yield high profit. Higher the risk, higher is the possibility of profits. If profit maximisation is the only goal, then risk factor is altogether ignored. This implies that finance manager will accept highly risky proposals also, if they give high profits. In practice, however, risk is very important consideration and has to be balanced with the profit objective.

(iii) Profit maximisation as an objective does not take into account the time pattern of returns. Proposal A may give a higher amount of profits as compared to proposal B, yet if the returns of proposal A begin to flow say 10 years later, proposal B may be preferred which may have lower overall profit but the returns flow is more early and quick.

(iv) Profit maximisation as an objective is too narrow. It fails to take into account the social considerations as also the obligations to various interests of workers, consumers, society, as well as ethical trade practices. If these factors are ignored, a company cannot survive for long. Profit maximization at the cost of social and moral obligations is a short sighted policy.

Wealth / Value Maximisation

We will first like to define what is Wealth / Value Maximization Model. Shareholders wealth are the result of cost benefit analysis adjusted with their timing and risk i.e. time value of money.

So, It is important that benefits measured by the finance manager are in terms of cash flow. Finance manager should emphasis on Cash flow for investment or financing decisions not on Accounting profit. The shareholder value maximization model holds that the primary goal of the firm is to maximize its market value and implies that business decisions should seek to increase the net present value of the economic profits of the firm. So for measuring and maximising shareholders wealth finance manager should follow:

A) Cash Flow approach not Accounting Profit

B) Cost benefit analysis

C) Application of time value of **money.**

How do we measure the value/wealth of a firm?

According to Van Horne, “Value of a firm is represented by the market price of the company's common stock. The market price of a firm's stock represents the focal judgment of all market participants as to what the value of the particular firm is. It takes into account present and prospective future earnings per share, the timing and risk of these earnings, the dividend policy of the firm and many other factors that bear upon the market price of the stock. The market price serves as a performance index or report card of the firm's progress. It indicates how well management is doing on behalf of stockholder’s”.

Why Wealth Maximization Works? Before we answer this question it is important to first understand and know what other goals a business enterprise may have. Some of the other goals a business enterprise may follow are:-

A) Achieving a higher growth rate

B) Attaining a larger market share

C) Gaining leadership in the market in terms of products and technology

D) Promoting employee welfare

E) Increasing customer satisfaction

F) Improving community life, supporting education and research, solving societal problems, etc.

Though, the above goals are important but the primary goal remains to be wealth maximization, as it is critical for the very existence of the business enterprise. If this goal is not met, public/institutions would lose confidence in the enterprise and will not invest further in the growth of the organization. If the growth of the organization is restricted than the other goals like community welfare will not get fulfilled.

Conflicts in Profit vs. Value maximisation principle

In any company, the management is the decision taking authority. As a normal tendency the management may pursue its own personal goals (profit maximization). But in an organization where there is a significant outside participation (shareholding, lenders etc.), the management may not be able to exclusively pursue its personal goals due to the constant supervision of the various stakeholders of the company-employees, creditors, customers, government, etc.

Every entity associated with the company will evaluate the performance of the management from the fulfilment of its own objective. The survival of the management will be threatened if the objective of any of the entities remains unfulfilled.

The wealth maximization objective is generally in accord with the interests of the various groups such as owners, employees, creditors and society, and thus, it may be consistent with the management objective of survival.

Owing to limitation (timing, social consideration etc.) in profit maximization, in today’s real world situations which is uncertain and multi-period in nature, wealth maximization is a better objective. Where the time period is short and degree of uncertainty is not great, wealth maximization and profit maximization amount to essentially the same.

The table below highlights some of the advantages and disadvantages of both profit maximization and wealth maximization goals:-

Goal	Objective	Advantages	Disadvantages
Profit Maximization	Large amount of profits	(i) Easy to calculate profits (ii) Easy to determine the link between financial decisions and profits.	(i) Emphasizes the short term gains (ii) Ignores risk or uncertainty (iii) Ignores the timing of returns (iv) Requires immediate resources.
Shareholders Wealth Maximisation	Highest market value of shares.	(i) Emphasizes the long term gains (ii) Recognises risk or uncertainty (iii) Recognises the timing of returns (iv) Considers shareholders' return.	(i) Offers no clear relationship between financial decisions and share price. (ii) Can lead to management anxiety and frustration.

Example: Profit maximization can be achieved in the short term at the expense of the long term goal, that is, wealth maximization. For example, a costly investment may experience losses in the short term but yield substantial profits in the long term. Also, a firm that wants to show a short term profit may, for example, postpone major repairs or replacement, although such postponement is likely to hurt its long term profitability.

(ii) Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders however in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a **principal agent relationship between managers and owners, which is known as Agency Problem**. In a nutshell, Agency Problem is the chances that managers may place personal goals ahead of the goal of owners Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

Addressing the agency problem

The agency problem arises if manager's interests are not aligned to the interests of the debt lender and equity investors. The agency problem of debt lender would be addressed by imposing negative covenants i.e. the managers cannot borrow beyond a point. This is one of the most important concepts of modern day finance and the application of this would be applied in the Credit Risk Management of Bank, Fund Raising, Valuing distressed companies.

Agency problem between the managers and shareholders can be addressed if the interests of the managers are aligned to the interests of the share- holders. It is easier said than done.

However, following efforts have been made to address these issues:

(A) Managerial compensation is linked to profit of the company to some extent and also with the long term objectives of the company.

(B) Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investo₹

(C) Effecting monitoring can be done.

(iii) The dividend policy is affected by the following factors:

1. Availability of funds: If the business is in requirement of funds, then retained earnings could be a good source. The reason being the saving of floatation cost and prevention of dilution of control which happens in case of new issue of equity shares to public.

2. Cost of capital: If the financing requirements are to be executed through debt (relatively cheaper source of finance), then it would be preferable to distribute more dividend. On the other hand, if the financing is to be done through fresh issue of equity shares, then it is better to use retained earnings as much as possible.

3. Capital structure: An optimum Debt Equity ratio should also be considered for the dividend decision.

4. Stock price: Stock price here means market price of the shares. Generally, higher dividends increase market value of shares and low dividends decrease the value.

5. Investment opportunities in hand: The dividend decision is also affected if there are investment opportunities in hand. In that situation, the company may prefer to retain more earnings.

6. Internal rate of return (IRR): If the internal rate of return (IRR) is more than the cost of retained earnings (K_r), it is better to distribute the earnings as much as possible.

7. Trend of industry: The investors depend on some industries for their regular dividend income. Therefore, in such cases, the firms have to pay dividend in order to survive in the market.

8. Expectation of shareholders: The shareholders can be categorised into two categories: (i) those who invests for regular income, & (ii) those who invests for growth. Generally, the investor prefers current dividend over the future growth.

9. Legal constraints: Section 123 of the Companies Act, 2013 which provides for declaration of dividend sates that Dividend shall be declared or paid by a company for any financial year only:

(a) out of the profits of the company for that year arrived at after providing for depreciation in accordance with the relevant provisions, or

(b) out of the profits of the company for any previous financial year or years arrived at after providing for depreciation in accordance with the relevant provisions and remaining undistributed, or

(c) out of both, or

(d) out of money provided by the Central Government or a State Government for the payment of dividend by the company in pursuance of a guarantee given by that Government.

It may be noted that, while computing the profits for payment of dividends any amount representing unrealised gains, notional gains or revaluation of assets and any change in carrying amount of an asset or of a liability on measurement of the asset or the liability at fair value shall be excluded.

10. Taxation: Before 1st April 2020, as per Section 115-O of Income Tax Act, 1961, dividend was subject to dividend distribution tax (DDT) in the hands of the company. Dividend on which DDT was paid, was to be exempted in the hands of the shareholder u/s 10(34). However, as per amendment made by the Finance Act 2020, the exemption u/s 10(34) shall not apply to dividend received on or after 1st April 2020 and the dividend income from shares held as investment shall be taxable under the head of 'Other income' at the applicable slab rate.

ICAI · Old P8 · MTP May 2023 · Series II · question p. 4 · answer p. 11-15

21 MTP **Old P8 Dec 2021 S-I** Q6(c) 2 marks old scheme

WRITE two main objectives of Financial Management.

ICAI'S ANSWER

Two main objectives of Financial Management

Profit Maximisation

It has traditionally been argued that the primary objective of a company is to earn profit; hence the objective of financial management is also profit maximisation. This implies that the finance manager has to make his decisions in a manner so that the profits of the concern are maximised. Each alternative, therefore, is to be seen as to whether or not it gives maximum profit.

Wealth / Value Maximisation

We will first like to define what is Wealth / Value Maximization Model. Shareholders wealth are the result of cost benefit analysis adjusted with their timing and risk i.e. time value of money.

So, Wealth = Present Value of benefits – Present Value of Costs

It is important that benefits measured by the finance manager are in terms of cash flow. Finance manager should emphasis on Cash flow for investment or financing decisions not on Accounting profit. The shareholder value maximization model holds that the primary goal of the firm is to maximize its market value and implies that business decisions should seek to increase the net present value of the economic profits of the firm.

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 4 · answer p. 11

22 MTP **Old P8 Jul 2021 S-I** Q6(a) 4 marks old scheme

DISCUSS the advantages and disadvantages of Wealth maximization principle.

ICAI'S ANSWER

Advantages and disadvantages of Wealth maximization principle.

Advantages:

(i) Emphasizes the long term gains

(ii) Recognises risk or uncertainty

(iii) Recognises the timing of returns

(iv) Considers shareholders' return.

Disadvantages:

(i) Offers no clear relationship between financial decisions and share price.

(ii) Can lead to management anxiety and frustration.

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 5 · answer p. 8-9

1.2 The finance function, the financial manager and the controller

18 questions · asked in 19 ICAI documents · 6 exam papers

23 EXAM **Sep 2026** Q4(c) 2 marks

Explain the key responsibilities of a Finance Executive in modern business world.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 19

24 EXAM Sep 2025 Q4(c) 2 marks

Explain two basic functions of Financial Management.

ICAI'S ANSWER

Basic function of financial management:

(a) **Investment decisions (I)**: These decisions relate to the selection of assets in which funds will be invested by a firm.

(b) **Financing decisions (F):** These decisions relate to acquiring the optimum finance to meet financial objectives and seeing that fixed and working capital are effectively managed.

(c) **Dividend decisions (D):** These decisions relate to the determination as to how much and how frequently cash can be paid out of the profits of an organisation as income for its owners/shareholders.

(d) **Working Capital Management (WCM):** Generally short term decision are reduced to management of current asset and current liability

ICAI · P6 · Sep 2025 exam · question p. 18 · answer p. 19

25 EXAM Old P8 Dec 2021 Q6(c) 2 marks old scheme

Explain in brief the phases of the evolution of financial management.

ICAI'S ANSWER

Evolution of Financial Management: Financial management evolved gradually over the past 50 years. The evolution of financial management is divided into three phases. Financial Management evolved as a separate field of study at the beginning of the century.

The three stages of its evolution are:

The Traditional Phase: During this phase, financial management was considered necessary only during occasional events such as takeovers, mergers, expansion, liquidation, etc. Also, when taking financial decisions in the organisation, the needs of outsiders (investment bankers, people who lend money to the business and other such people) to the business was kept in mind.

The Transitional Phase: During this phase, the day-to-day problems that financial managers faced were given importance. The general problems related to funds analysis, planning and control were given more attention in this phase.

The Modern Phase: Modern phase is still going on. The scope of financial management has greatly increased now. It is important to carry out financial analysis for a company. This analysis helps in decision making. During this phase, many theories have been developed regarding efficient markets, capital budgeting, option pricing, valuation models and also in several other important fields in financial management. Here, financial management is viewed as a supportive and facilitative function, not only for top management but for all levels of management.

ICAI · Old P8 · Dec 2021 exam · question p. 14 · answer p. 15-16

26 EXAM **Old P8 Jan 2021** Q6(a) 4 marks old scheme

State four tasks involved to demonstrate the importance of good Financial Management.

ICAI'S ANSWER

The best way to demonstrate the importance of good financial management is to describe some of the tasks that it involves:

- **Taking care** not to over-invest in fixed assets
- **Balancing** cash-outflow with cash-inflows
- **Ensuring** that there is a sufficient level of short-term working capital
- **Setting** sales revenue targets that will deliver growth
- **Increasing** gross profit by setting the correct pricing for products or services
- **Controlling** the level of general and administrative expenses by finding more cost-efficient ways of running the day-to-day business operations, and
- **Tax planning** that will minimize the taxes a business has to pay.

ICAI · Old P8 · Jan 2021 exam · question p. 13 · answer p. 14

27 EXAM **Old P8 Nov 2020** Q6(a) 4 marks old scheme

List out the role of Chief Financial Officer in today's World.

ICAI'S ANSWER

Role of Chief Financial Officer (CFO) in Today's World: Today, the role of chief financial officer, or CFO, is no longer confined to accounting, financial reporting and risk management. It's about being a strategic business partner of the chief executive officer, or CEO. Some of the role of a CFO in today's world are as follows-

- Budgeting
- Forecasting
- Managing M&As
- Profitability analysis (for example, by customer or product)
- Pricing analysis
- Decisions about outsourcing
- Overseeing the IT function.
- Overseeing the HR function.
- Strategic planning (sometimes overseeing this function).
- Regulatory compliance.
- Risk management

ICAI · Old P8 · Nov 2020 exam · question p. 13 · answer p. 13

28 EXAM **Old P8 Nov 2019** Q6(a) 3 marks old scheme

Same question, different ICAI answer: #39 Old P8 MTP Dec 2021 S-I

Briefly explain the three finance function decisions.

ICAI'S ANSWER

The finance functions are divided into long term and short term functions/ decisions:

Long term Finance Function Decisions

(i) Investment decisions (I): These decisions relate to the selection of assets in which funds will be invested by a firm. Funds procured from different sources have to be invested in various kinds of assets. Long term funds are used in a project for various fixed assets and also for current assets.

(ii) Financing decisions (F): These decisions relate to acquiring the optimum finance to meet financial objectives and seeing that fixed and working capital are effectively managed. The financial manager needs to possess a good knowledge of the sources of available funds and their respective costs and needs to ensure that the company has a sound capital structure, i.e. a proper balance between equity capital and debt.

(iii) Dividend decisions (D): These decisions relate to the determination as to how much and how frequently cash can be paid out of the profits of an organisation as income for its owners/shareholders. The owner of any profit-making organization looks for reward for his investment in two ways, the growth of the capital invested and the cash paid out as income; for a sole trader this income would be termed as drawings and for a limited liability company the term is *dividends*.

Short- term Finance Decisions/Function

Working capital Management (WCM): Generally short term decision is reduced to management of current asset and current liability (i.e., working capital Management).

ICAI · Old P8 · Nov 2019 exam · question p. 16 · answer p. 16-17

29 RTP **Sep 2026** Q10(a)

PQR Ltd. expanded its business by purchasing new machinery and increasing production. However, due to poor financial planning, the company is facing cash shortages, rising expenses, and difficulty managing its working capital. The management is now focusing on better budgeting, cost control, and cash flow management.

In the light of the above case, explain the importance of financial management and how it helps a business achieve growth, profitability, and financial stability.

ICAI'S ANSWER

Importance of Financial Management cannot be over-emphasized. It is, indeed, the key to successful business operations. Without proper administration of finance, no business enterprise can reach at its full potentials for growth and success. Money is to an enterprise, what oil is to an engine.

Financial Management is all about planning investment, funding the investment, monitoring expenses against budget and managing gains from the investments. Financial management means management of all matters related to an organization's finances.

The best way to demonstrate the importance of good financial management is to describe some of the tasks that it involves:-

(a) **Taking care** not to over-invest in fixed assets

(b) **Balancing** cash-outflow with cash-inflows

(c) **Ensuring** that there is a sufficient level of short-term working capital

(d) **Setting** sales revenue targets that will deliver growth

(e) **Increasing** gross profit by setting the correct pricing for products or services

(f) **Controlling** the level of general and administrative expenses by finding more cost-efficient ways of running the day-to-day business operations, and

(g) **Tax planning** that will minimize the taxes a business has to pay.

ICAI · P6 · RTP Sep 2026 · question p. 9 · answer p. 22-23

30 RTP **Jan 2025** Q11(a)

LIST the emerging issues (any four) affecting the future role of CFO.

ICAI'S ANSWER

Emerging Issues/Priorities Affecting the Future Role of Chief Financial Officer (CFO)

(i) **Regulation**: Regulation requirements are increasing and CFOs have an increasingly personal stake in regulatory adherence.

(ii) **Globalisation:** The challenges of globalisation are creating a need for finance leaders to develop a finance function that works effectively on the global stage and that embraces diversity.

(iii) **Technology:** Technology is evolving very quickly, providing the potential for CFOs to reconfigure finance processes and drive business insight through 'big data' and analytics.

(iv) **Risk:** The nature of the risks that organisations face are changing, requiring more effective risk management approaches and increasingly CFOs have a role to play in ensuring an appropriate corporate ethos.

(v) **Transformation:** There will be more pressure on CFOs to transform their finance functions to drive a better service to the business at zero cost impact.

(vi) **Stakeholder Management:** Stakeholder management and relationships will become important as increasingly CFOs become the face of the corporate brand.

(vii) **Strategy:** There will be a greater role to play in strategy validation and execution, because the environment is more complex and quick changing, calling on the analytical skills CFOs can bring.

(viii) **Reporting:** Reporting requirements will broaden and continue to be burdensome for CFOs.

(ix) **Talent and Capability:** A brighter spotlight will shine on talent, capability and behaviours in the top finance role.

ICAI · P6 · RTP Jan 2025 · question p. 10 · answer p. 28-29

31 RTP **Old P8 Nov 2023** Q10(b) old scheme

DESCRIBE the inter relationship between investing, financing, and dividend decisions.

ICAI'S ANSWER

Inter-relationship between Investment, Financing and Dividend Decisions

The finance functions are divided into three major decisions, viz., investment, financing, and dividend decisions. It is correct to say that these decisions are inter-related because the underlying objective of these three decisions is the same, i.e., maximisation of shareholders' wealth. Since investment, financing and dividend decisions are all interrelated, one must consider the joint impact of these decisions on the market price of the company's shares and these decisions should also be

solved jointly. The decision to invest in a new project needs the finance for the investment. The financing decision, in turn, is influenced by and influences dividend decision because retained earnings used in internal financing deprive shareholders of their dividends. An efficient financial management can ensure optimal joint decisions. This is possible by evaluating each decision in relation to its effect on the shareholders' wealth.

The above three decisions are briefly examined below in the light of their inter-relationship and to see how they can help in maximising the shareholders' wealth i.e., market price of the company's shares.

Investment decision: The investment of long-term funds is made after a careful assessment of the various projects through capital budgeting and uncertainty analysis. However, only that investment proposal is to be accepted which is expected to yield at least so much return as is adequate to meet its cost of financing. This has an influence on the profitability of the company and ultimately on its wealth.

Financing decision: Funds can be raised from various sources. Each source of funds involves different issues. The finance manager must maintain a proper balance between long-term and short-term funds. With the total volume of long-term funds, he must ensure a proper mix of loan funds and owner's funds. The optimum financing mix will increase return to equity shareholders and thus maximise their wealth.

Dividend decision: The finance manager is also concerned with the decision to pay or declare dividend. He assists the top management in deciding as to what portion of the profit should be paid to the shareholders by way of dividends and what portion should be retained in the business. An optimal dividend pay-out ratio maximises shareholders' wealth.

The above discussion makes it clear that investment, financing, and dividend decisions are interrelated and are to be taken jointly keeping in view their joint effect on the shareholders' wealth.

ICAI · Old P8 · RTP Nov 2023 · question p. 5 · answer p. 16

32 RTP **Old P8 Dec 2021** Q10(a) old scheme

DISCUSS the points that demonstrates the Importance of good financial management.

ICAI'S ANSWER

Points that demonstrate the "Importance of good financial management":

- **Taking care** not to over-invest in fixed assets
- **Balancing** cash-outflow with cash-inflows
- **Ensuring** that there is a sufficient level of short-term working capital
- **Setting** sales revenue targets that will deliver growth
- **Increasing** gross profit by setting the correct pricing for products or services
- **Controlling** the level of general and administrative expenses by finding more cost-efficient ways of running the day-to-day business operations, and
- **Tax planning** that will minimize the taxes a business has to pay.

ICAI · Old P8 · RTP Dec 2021 · question p. 5 · answer p. 17

33 MTP **Sep 2026 S-II** Q4(a) 4 marks

Also asked in MTP May 2024 S-II Q4(a)

WRITE short notes on Inter relationship between investment, financing and dividend decisions.

ICAI'S ANSWER

Inter-relationship between Investment, Financing and Dividend Decisions: The finance functions are divided into three major decisions, viz., investment, financing and dividend decisions. It is correct to say that these decisions are inter-related because the underlying objective of these three decisions is the same, i.e. maximisation of shareholders' wealth. Since investment, financing and dividend decisions are all interrelated, one has to consider the joint impact of these decisions on the market price of the company's shares and these decisions should also be solved jointly. The decision to invest in a new project needs the finance for the investment. The financing decision, in turn, is influenced by and influences dividend decision because retained earnings used in internal financing deprive shareholders of their dividends. An efficient financial management can ensure optimal joint decisions. This is possible by evaluating each decision in relation to its effect on the shareholders' wealth.

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Dividend decision: The finance manager is also concerned with the decision to pay or declare dividend. He assists the top management in deciding as to what portion of the profit should be paid to the shareholders by way of dividends and what portion should be retained in the business. An optimal dividend pay-out ratio maximises shareholders' wealth.

The above discussion makes it clear that investment, financing and dividend decisions are interrelated and are to be taken jointly keeping in view their joint effect on the shareholders' wealth.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 8 · answer p. 10-11

34 MTP **Jan 2026 S-I** Q4(a) 4 marks

DIFFERENTIATE between Financial Management and Financial Accounting.

ICAI'S ANSWER

Differentiation between Financial Management and Financial Accounting: Though financial management and financial accounting are closely related, still they differ in the treatment of funds and also with regards to decision - making.

Treatment of Funds: In accounting, the measurement of funds is based on the accrual principle. The accrual based accounting data do not reflect fully the financial conditions of the organisation. An organisation which has earned profit (sales less expenses) may said to be profitable in the accounting sense but it may not be able to meet its current obligations due to shortage of liquidity as a result of

say, uncollectible receivables. Whereas, the treatment of funds, in financial management is based on cash flows. The revenues are recognised only when cash is actually received (i.e. cash inflow) and expenses are recognised on actual payment (i.e. cash outflow). Thus, cash flow based returns help financial managers to avoid insolvency and achieve desired financial goals.

Decision-making: The chief focus of an accountant is to collect data and present the data while the financial manager's primary responsibility relates to financial planning, controlling and decision-making. Thus, in a way it can be stated that financial management begins where financial accounting ends.

ICAI · P6 · MTP Jan 2026 · Series I · question p. 7 · answer p. 14

35 MTP **Jan 2025 S-II** Q4(b) 4 marks

DESCRIBE some of the tasks that demonstrate the importance of good financial management

ICAI'S ANSWER

Some of the tasks that demonstrate the importance of good financial management

- Taking care not to over invest in fixed assets
- Balancing cash-outflows with cash-inflows
- Ensuring that there is a sufficient level of working capital
- Setting sales revenue targets that will deliver growth
- Increasing the Gross profit by setting the correct pricing for products or services
- Controlling the level of general and administration expenses by finding more cost-efficient ways of running the day-to-day business operations
- Tax Planning that will minimize the taxes a business has to pay

ICAI · P6 · MTP Jan 2025 · Series II · question p. 7 · answer p. 10

36 MTP **Sep 2024 S-I** Q4(c) 2 marks

DISCUSS the role of a chief financial officer.

ICAI'S ANSWER

The finance executive of an organisation plays an important role in the company's goals, policies, and financial success. His responsibilities include:

(a) Financial analysis and planning: Determining the proper amount of funds to employ in the firm, i.e. designating the size of the firm and its rate of growth.

(b) Investment decisions: The efficient allocation of funds to specific assets.

(c) Financing and capital structure decisions: Raising funds on favourable terms as possible i.e. determining the composition of liabilities.

(d) Management of financial resources (such as working capital).

(e) Risk management: Protecting assets.

ICAI · P6 · MTP Sep 2024 · Series I · question p. 6 · answer p. 10

37 MTP **Old P8 Nov 2023 S-II** Q6(c) 2 marks old scheme

Write a short note on role of finance controller.

ICAI'S ANSWER

Role of Financial Controller: The role of financial controller has undergone changes over the years. Until the middle of this century, its scope was limited to procurement of funds under major events in the life of the enterprise such as promotion, expansion, merger, etc. In the modern times, the role of financial controller includes besides procurement of funds, the three different kinds of decisions as well, namely, investment, financing and dividend.

The financial controller, in a bid to maximize shareholders' wealth, should strive to maximize returns in relation to the given risk; he should seek courses of actions that avoid unnecessary risks. To ensure maximum return, funds flowing in and out of the firm should be constantly monitored to assure that they are safeguarded and properly utilized.

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 4 · answer p. 9

38 MTP **Old P8 Nov 2022 S-I** Q6(a) 4 marks old scheme

A finance executive of an organisation plays an important role in the company's goals, policies, and financial success. WHAT his responsibilities include?

ICAI'S ANSWER

A finance executive of an organisation plays an important role in the company's goals, policies, and financial success. His responsibilities include:

(i) Financial analysis and planning: Determining the proper amount of funds to employ in the firm, i.e. designating the size of the firm and its rate of growth.

(ii) Investment decisions: The efficient allocation of funds to specific assets.

(iii) Financing and capital structure decisions: Raising funds on favourable terms as possible i.e. determining the composition of liabilities.

(iv) Management of financial resources (such as working capital).

(v) Risk management: Protecting assets.

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 4 · answer p. 6-7

39 MTP **Old P8 Dec 2021 S-I** Q6(a) 4 marks old scheme

Same question, different ICAI answer: #28 Old P8 Exam Nov 2019

BRIEFLY explain the three finance function decisions.

ICAI'S ANSWER

The three finance function decisions are as follows:

Investment decisions: These decisions relate to the **selection of assets in which funds will be invested by a firm.** Funds procured from different sources have to be invested in various kinds of assets. Long term funds are used in a project for various fixed assets and also for current assets. The investment of funds in a project has to be made after careful assessment of the various projects through capital budgeting. A part of long-term funds is also to be kept for financing the working capital requirements.

Financing decisions: These decisions relate to **acquiring the optimum finance** to meet financial objectives and seeing that fixed and working capital are effectively managed. The financial manager needs to possess a good knowledge of the sources of available funds and their respective costs and needs to ensure that the company has a sound capital structure, i.e. a proper balance between equity

capital and debt. Such managers also need to have a very clear understanding as to the difference between profit and cash flow, bearing in mind that profit is of little use unless the organisation is adequately supported by cash to pay for assets and sustain the working capital cycle.

Dividend decisions: These decisions relate to the **determination as to how much and how frequently cash can be paid out of the profits** of an organization as income for its owners/shareholders. The dividend decision thus has two elements – the amount to be paid out and the amount to be retained to support the growth of the organization, the latter being also a financing decision. The level and regular growth of dividends represent a significant factor in determining a profit-making company's market value, i.e. the value placed on its shares by the stock market.

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 4 · answer p. 10

40 MTP **Old P8 Nov 2019 S-I** Q6(a) 4 marks old scheme

DISCUSS the Inter relationship between investment, financing and dividend decisions.

ICAI'S ANSWER

Inter-relationship between Investment, Financing and Dividend Decisions: The finance functions are divided into three major decisions, viz., investment, financing and dividend decisions. It is correct to say that these decisions are inter-related because the underlying objective of these three decisions is the same, i.e. maximisation of shareholders' wealth. Since investment, financing and dividend decisions are all interrelated, one has to consider the joint impact of these decisions on the market price of the company's shares and these decisions should also be solved jointly. The decision to invest in a new project needs the finance for the investment. The financing decision, in turn, is influenced by and influences dividend decision because retained earnings used in internal financing deprive shareholders of their dividends. An efficient financial management can ensure optimal joint decisions. This is possible by evaluating each decision in relation to its effect on the shareholders' wealth.

The above three decisions are briefly examined below in the light of their inter-relationship and to see how they can help in maximising the shareholders' wealth i.e. market price of the company's shares.

Investment decision: The investment of long term funds is made after a careful assessment of the various projects through capital budgeting and uncertainty analysis. However, only that investment proposal is to be accepted which is expected to yield at least so much return as is adequate to meet its cost of financing. This have an influence on the profitability of the company and ultimately on its wealth.

Financing decision: Funds can be raised from various sources. Each source of funds involves different issues. The finance manager has to maintain a proper balance between long-term and short-term funds. With the total volume of long-term funds, he has to ensure a proper mix of loan funds and owner's funds. The optimum financing mix will increase return to equity shareholders and thus maximise their wealth.

Dividend decision: The finance manager is also concerned with the decision to pay or declare dividend. He assists the top management in deciding as to what portion of the profit should be paid to the shareholders by way of dividends and what portion should be retained in the business. An optimal dividend pay-out ratio maximises shareholders' wealth.

The above discussion makes it clear that investment, financing and dividend decisions are interrelated and are to be taken jointly keeping in view their joint effect on the shareholders' wealth.

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 4 · answer p. 7

1.3 Agency problem, financial environment, distress and insolvency

11 questions · asked in 12 ICAI documents · 2 exam papers

41 EXAM **Jan 2026** Q4(c) 2 marks

Define financial distress. Briefly explain the consequences of a prolonged financial distress.

ICAI'S ANSWER

Financial distress is a position where Cash inflows of a firm are inadequate to meet all its current obligations.

If distress continues for a long period of time, firm may have to sell its asset, even many times at a lower price. Further when revenue is inadequate to revive the situation, firm will not be able to meet its obligations and become insolvent.

ICAI · P6 · Jan 2026 exam · question p. 23 · answer p. 24

42 EXAM **May 2025** Q4(a) 4 marks

What is agency cost and what are its types? How can a company minimize agency cost and align the interest of manager and shareholder?

ICAI'S ANSWER

Meaning of Agency Cost: Agency Problem is the chances that managers may place personal goals ahead of the goal of owners. Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximize shareholders wealth.

Types of Agency Cost: Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

Following efforts have been made to address the issues of minimizing agency cost and align the interest of manager and shareholders:

- Managerial compensation is linked to the profit of the company to some extent and also with the long-term objectives of the company.
- Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors.
- Effecting monitoring can be done.

ICAI · P6 · May 2025 exam · question p. 23 · answer p. 24

43 RTP **Old P8 Nov 2023** Q10(a) old scheme

Same question, different ICAI answer: #45 Old P8 RTP Nov 2020

EXPLAIN agency problem and agency cost. How to address the issues of the same.

ICAI'S ANSWER

Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders, however, in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a **principal agent relationship between managers and owners, which is known as Agency**

Problem. In a nutshell, Agency Problem means that there is a chance that managers may place personal goals ahead of the goal of owners. Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

Addressing the agency problem

The agency problem arises if manager's interests are not aligned to the interests of the debt lender and equity investors. The agency problem of debt lender would be addressed by imposing negative covenants i.e. the managers cannot borrow beyond a point. This is one of the most important concepts of modern day finance and the application of this would be applied in the Credit Risk Management of Bank, Fund Raising, Valuing distressed companies.

Agency problem between the managers and shareholders can be addressed if the interests of the managers are aligned to the interests of the shareholders. It is easier said than done.

However, following efforts have been made to address these issues:

- Managerial compensation is linked to profit of the company to some extent and also with the long term objectives of the company.
- Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors.
- Effecting monitoring can be done.

ICAI · Old P8 · RTP Nov 2023 · question p. 5 · answer p. 15

44 RTP **Old P8 May 2022** Q10(b) old scheme

DISCUSS Agency problem and its cost. HOW it arises and HOW it can be addressed?

ICAI'S ANSWER

Agency Problem: Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders however in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a principal agent relationship between managers and owners, which is known as Agency Problem. In a nutshell, **Agency Problem is the chances that managers may place personal goals ahead of the goal of owners.**

Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

The agency problem arises if manager's interests are not aligned to the interests of the debt lender and equity investors. The agency problem of debt lender would be addressed by imposing negative covenants i.e. the managers cannot borrow beyond a point. This is one of the most important concepts of modern-day finance and the application of this would be applied in the Credit Risk Management of Bank, Fund Raising, Valuing distressed companies.

Agency problem between the managers and shareholders can be addressed if the interests of the managers are aligned to the interests of the shareholders. It is easier said than done.

However, following **efforts** have been made to **address these issues**:

- Managerial compensation is linked to profit of the company to some extent and also with the long-term objectives of the company.
- Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors.
- Effecting monitoring can be done.

ICAI · Old P8 · RTP May 2022 · question p. 6 · answer p. 20-21

45 RTP **Old P8 Nov 2020** Q10(a) old scheme

Same question, different ICAI answer: #43 Old P8 RTP Nov 2023

EXPLAIN agency problem and agency cost. How to address the issues of the same.

ICAI'S ANSWER

Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders, however, in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a **principal agent relationship between managers and owners, which is known as Agency Problem**. In a nutshell, Agency Problem is the chances that managers may place personal goals ahead of the goal of owners. Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring

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- Effecting monitoring can be done.

ICAI · Old P8 · RTP Nov 2020 · question p. 5 · answer p. 18

46 MTP **Sep 2025 S-I** Q4(b) 4 marks

Write a short note on financial distress and insolvency.

ICAI'S ANSWER

There are various factors like price of the product/ service, demand, price of inputs e.g. raw material, labour etc., which is to be managed by an organisation on a continuous basis. Proportion of debt also need to be managed by an organisation very delicately. Higher debt requires higher interest and if the cash inflow is not sufficient then it will put lot of pressure to the organisation. Both short term and long-term creditors will put stress to the firm. If all the above factors are not well managed by the firm, it can create situation known as distress, so financial distress is a position where Cash inflows of a firm are inadequate to meet all its current obligations.

Now if distress continues for a long period of time, firm may have to sell its asset, even many times at a lower price. Further when revenue is inadequate to revive the situation, firm will not be able to meet its obligations and become insolvent. So, insolvency basically means inability of a firm to repay various debts and is a result of continuous financial distress.

ICAI · P6 · MTP Sep 2025 · Series I · question p. 9 · answer p. 16-17

47 MTP **May 2025 S-I** Q4(a) 4 marks

EXPLAIN Agency Problem. Also EXPLAIN, how it can be addressed.

ICAI'S ANSWER

Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders however in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a **principal agent relationship between managers and owners, which is known as Agency Problem**.

Agency Problem can be addressed through following efforts:

- Managerial compensation is linked to profit of the company to some extent and also with the long term objectives of the company.
- Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors.
- Effecting monitoring can be done.

ICAI · P6 · MTP May 2025 · Series I · question p. 7 · answer p. 15

48 MTP **Sep 2024 S-II** Q4(a) 4 marks

The agency problem is one of the key concepts in corporate governance and financial management. On the light of this statement, EXPLAIN agency problem, consequences of agency problem and how to overcome the issue.

ICAI'S ANSWER

Though in a sole proprietorship firm, partnership etc., owners participate in management but in corporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory managers should act in the best interest of shareholders however in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a **principal agent relationship between managers and owners, which is known as Agency Problem**. In a nutshell, Agency Problem is the chances that managers may place personal goals ahead of the goal of owners. Agency Problem leads to Agency Cost. Agency cost is the additional cost

borne by the shareholders to monitor the manager and control their behaviour so as to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

Addressing the agency problem

The agency problem arises if manager's interests are not aligned to the interests of the debt lender and equity investors. The agency problem of debt lender would be addressed by imposing negative covenants i.e. the managers cannot borrow beyond a point. This is one of the most important concepts of modern day finance and the application of this would be applied in the Credit Risk Management of Bank, Fund Raising, Valuing distressed companies.

Agency problem between the managers and shareholders can be addressed if the interests of the managers are aligned to the interests of the share- holders. It is easier said than done.

However, following efforts have been made to address these issues:

- Managerial compensation is linked to profit of the company to some extent and also with the long term objectives of the company.
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- Effecting monitoring can be done.

ICAI · P6 · MTP Sep 2024 · Series II · question p. 6 · answer p. 8-9

49 MTP **Old P8 May 2023 S-I** Q6(b) 4 marks old scheme

Also asked in Old P8 MTP Nov 2020 S-I Q6(a)

DISCUSS Agency Problem and Agency Cost.

ICAI'S ANSWER

Agency Problem and Agency Cost: Though in a sole proprietorship firm, partnership etc., owners participate in management but incorporates, owners are not active in management so, there is a separation between owner/ shareholders and managers. In theory, managers should act in the best interest of shareholders however in reality, managers may try to maximise their individual goal like salary, perks etc., so there is a principal agent relationship between managers and owners, which is known as Agency Problem. In a nutshell, Agency Problem is the chances that managers may place personal goals ahead of the goal of owners. Agency Problem leads to Agency Cost. Agency cost is the additional cost borne by the shareholders to monitor the manager and control their behaviour to maximise shareholders wealth. Generally, Agency Costs are of four types (i) monitoring (ii) bonding (iii) opportunity (iv) structuring.

ICAI · Old P8 · MTP May 2023 · Series I · question p. 4 · answer p. 8

50 MTP **Old P8 May 2022 S-I** Q6(a) 4 marks old scheme

'Financial distress is a position where Cash inflows of a firm are inadequate to meet all its current obligations.'

Based on above mentioned context, EXPLAIN Financial Distress along with Insolvency.

ICAI'S ANSWER

Financial Distress and Insolvency: There are various factors like price of the product/ service, demand, price of inputs e.g., raw material, labour etc., which is to be managed by an organisation on

a continuous basis. Proportion of debt also need to be managed by an organisation very delicately. Higher debt requires higher interest and if the cash inflow is not sufficient then it will put lot of pressure to the organisation. Both short term and long-term creditors will put stress to the firm. If all the above factors are not well managed by the firm, it can create situation known as distress, so financial distress is a position where Cash inflows of a firm are inadequate to meet all its current obligations.

Now if distress continues for a long period of time, firm may have to sell its asset, even many times at a lower price. Further when revenue is inadequate to revive the situation, firm will not be able to meet its obligations and become insolvent. So, insolvency basically means inability of a firm to repay various debts and is a result of continuous financial distress.

ICAI · Old P8 · MTP May 2022 · Series I · question p. 4 · answer p. 7

51 MTP **Old P8 Jul 2021 S-II** Q6(b) 4 marks old scheme

DESCRIBE how agency problem can be addressed.

ICAI'S ANSWER

Agency problem between the managers and shareholders can be addressed if the interests of the managers are aligned to the interests of the share- holders. It is easier said than done.

However, following efforts have been made to address these issues:

- Managerial compensation is linked to profit of the company to some extent and also with the long term objectives of the company.
- Employee is also designed to address the issue with the underlying assumption that maximisation of the stock price is the objective of the investors.
- Effecting monitoring can be done.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 4 · answer p. 7

CHAPTER 2

Financial Analysis and Planning: Ratio Analysis

2.1 Liquidity and capital structure ratios

20 questions · asked in 10 ICAI documents · 3 exam papers

52 EXAM **Sep 2026** MCQ 6 2 marks MCQ

XYZ company Ltd had a current ratio of 2:1 in the year 2025 and working capital of ₹ 7,75,000. In the year 2026, it is expected that its current assets will be increased by 15 per cent. The management of the company desires to change its current ratio from 2:1 in the year 2025 to 2.5:1 in the year 2026.

What would be the percentage change in current liabilities and working capital from year 2025 to year 2026 ?

A 8% Decrease in current liabilities and 36% Increase in working capital.

B 8% Decrease in current liabilities and 38% Increase in working capital.

C 8% Increase in current liabilities and 38% Increase in working capital.

D 8% Increase in current liabilities and 36% Increase in working capital.

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P6 · Sep 2026 exam · question p. 5

53 EXAM **Jan 2026** Q1(a) 5 marks

From the books of AIL Limited following information is taken for the year 2024-25:

Gross profit margin (25% of Sales)	₹ 30,00,000
Inventory holding period	28 Days
Opening Inventory	₹ 8,00,000
Current Ratio	1.20
Quick Ratio	0.80
Selling & Distribution Expenses	₹ 12,00,000
Depreciation and other Non-cash Expenses	₹ 4,80,000

Assume 360 days in a year.

You are required to calculate:

(i) Inventory as on 31.03.2025

(ii) Working Capital

(iii) Basic Defense Interval

ICAI'S ANSWER

(i) Sales = $\frac{30,00,000}{0.25}$ = ₹ 1,20,00,000

Gross Profit = ₹ 30,00,000

Cost of goods sold (COGS) = Sales - Gross Profit = ₹ 90,00,000

$$\text{Average Inventory} = \frac{\text{Cost of goods sold (COGS)}}{\text{Inventory Turnover}}$$

$$= \frac{₹\ 90,00,000 \times 28}{360} = 7,00,000$$

$$\frac{\text{Opening Inventory} + \text{Closing Inventory}}{2} = 7,00,000$$

8,00,000+Closing Inventory = 7,00,000 x 2

Closing Inventory = ₹ 6,00,000

(ii) $$\text{Current Ratio} = \frac{\text{Current Assets (CA)}}{\text{Current Liabilities (CL)}}$$

Or, $\frac{CA}{CL} = \frac{1.2}{1}$

$\therefore$ CA = 1.2 CL .....(1)

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

$$= \frac{\text{Current Assets - 6,00,000}}{\text{Current Liabilities}}$$

Or, 0.8CL = CA - 6,00,000

CA = 0.8CL + 600000 .....(2)

Equating 1 and 2

1.2 CL = 0.8CL + 6,00,000

CL = ₹ 15,00,000

CA = 1.2 CL = ₹ 18,00,000

Working Capital = CA – CL = 18,00,000 – 15,00,000 = ₹ 3,00,000

(iii) $$\text{Basic Defense Interval Ratio} = \frac{(\text{Cash and Bank balances} + \text{Marketable Securities})}{\text{Operating Expenses(Cash)} \div \text{No. of days}}$$

Cash Operating Expenses = COGS + Selling and Dist. Exp – Depreciation and Non-cash expenditures

= 90,00,000 + 12,00,000 - 4,80,000

= ₹ 97,20,000

Quick Assets = 18,00,000 – 6,00,000 = ₹ 12,00,000

$$\text{Basic Defense Interval Ratio} = \frac{12,00,000}{97,20,000 \div 360} = \frac{12,00,000}{27,000} = 44.44 \text{ days}$$

ICAI · P6 · Jan 2026 exam · question p. 10 · answer p. 12-13

54 **EXAM** **Jan 2026** Q4(c) **2 marks**

BEE Ltd. has a current ratio of 2:1, showing enough short-term liquidity in the company. You are required to comment whether the current ratio will improve, decline, or remain unchanged.

Sr. No	Transaction	Impact
(i)	A debtor owing ₹ 60,000 becomes insolvent and only 75% is recovered in cash in full and final settlement.	
(ii)	Issued equity shares for ₹ 2,00,000, proceeds kept in current account with bank.	
(iii)	₹ 25,000 collected from customers in cash.	
(iv)	Goods costing 8,000 sold for ₹ 12,500 on credit.	

ICAI'S ANSWER

Sr. No.	Transaction	Impact on Current Ratio
(i)	A debtor owing ₹ 60,000 becomes insolvent and only 75% is recovered in cash	**Declines**
(ii)	Issued equity shares for ₹ 2,00,000, proceeds kept in current account with bank	**Improves**
(iii)	₹ 25,000 collected from customers in cash	**No change**
(iv)	Goods costing ₹ 8,000 sold for ₹ 12,500 on credit	**Improves**

ICAI · P6 · Jan 2026 exam · question p. 23 · answer p. 24

55 EXAM **Sep 2025** MCQ 8 1 mark MCQ

ABC Limited gives you the following information as on 31st March, 2025:

Current Liability : ₹ 1,60,000

Current ratio : 2.5

Liquid ratio (Quick ratio): 2

What will be the value of Stock?

- A ₹ 80,000
- B ₹ 1,20,000
- C ₹ 1,60,000
- D ₹ 64,000

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P6 · Sep 2025 exam · question p. 3 · answer p. 7

56 RTP **May 2026** Q2 MCQ

A firm's current liabilities comprise bank credit and other liabilities in the ratio of 2 : 1. The current liabilities of the firm are 3,00,000 and the current ratio is also 2 : 1. The quick ratio is 1 : 1. Thus, the amount of bank credit and inventory of the firm would be :

- A ₹ 3,00,000 and ₹ 1,00,000

B ₹ 2,00,000 and ₹ 3,00,000

C ₹ 3,00,000 and ₹ 2,00,000

D ₹ 1,00,000 and ₹ 3,00,000

ICAI'S ANSWER

(B) ₹ 2,00,000 and ₹ 3,00,000

Calculation of Current Assets

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Current Assets = Current Ratio × Current Liabilities

= 2 × ₹ 3,00,000 = ₹ 6,00,000

Calculation of Quick Assets

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Quick Assets = Quick Ratio × Current Liabilities

= 1 × ₹ 3,00,000 = ₹ 3,00,000

Inventory

Inventory = Current Assets - Quick Assets

= ₹ 6,00,000 - ₹ 3,00,000 = ₹ 3,00,000

Calculation of Bank Credit

Bank Credit: Other Liabilities = 2:1

Total Current Liabilities = ₹ 3,00,000

Bank Credit = ₹ 3,00,000 × 2/3 = ₹ 2,00,000

ICAI · P6 · RTP May 2026 · question p. 3 · answer p. 13-14

57 RTP **May 2026** Q3 MCQ

Same question, different ICAI answer: #60 MTP May 2026 S-I

The capital structure of JKL Limited consists of 12% debentures of ₹ 50 lakhs, 10% preference shares of ₹ 25 lakhs, 14% term loan of ₹ 15 lakhs and equity share capital of ₹ 48 lakhs. Its reserves & surpluses are ₹ 24 lakhs. What would be the capital-gearing ratio?

A 1.04

B 1.25

C 1.33

D 1.56

ICAI'S ANSWER

(B) 1.25

Capital Gearing Ratio

$$= \frac{\text{Pref. Share Capital} + \text{Debentures} + \text{Other Borrowed Funds}}{\text{Equity Share Capital} + \text{Reserves and Surplus} - \text{Losses}}$$

$$= \frac{₹25 \text{ lakhs} + ₹50 \text{ lakhs} + ₹15 \text{ lakhs}}{₹48 \text{ lakhs} + ₹24 \text{ lakhs}}$$

$$= \frac{₹90 \text{ lakhs}}{₹72 \text{ lakhs}} = 1.25$$

ICAI · P6 · RTP May 2026 · question p. 3 · answer p. 14

58 RTP **Old P8 Nov 2023** Q1 old scheme

From the following table of financial ratios of Prabhu Chemicals Limited, comment on various ratios given at the end:

Ratios	2021	2022	Average of Chemical Industry
Liquidity Ratios			
Current ratio	2.1	2.3	2.4
Quick ratio	1.4	1.8	1.4
Receivable turnover ratio	8	9	8
Inventory turnover	8	9	5
Receivables collection period	46 days	41 days	46 days
Operating profitability			
Operating income –ROI	24%	21%	18%
Operating profit margin	18%	18%	12%
Financing decisions			
Debt ratio	45%	44%	60%
Return			
Return on equity	26%	28%	18%

COMMENT on the following aspect of Prabhu Chemicals Limited

(i) Liquidity

(ii) Operating profits

(iii) Financing

(iv) Return to the shareholders

ICAI'S ANSWER

Ratios	Comment
Liquidity	Current ratio has improved from last year and matching the industry average. Quick ratio also improved than last year and above the industry average. The reduced inventory levels (evidenced by higher inventory turnover ratio) have led to better quick ratio in FY 2022 compared to FY 2021. Further the decrease in current liabilities is greater than the collective decrease in inventory and debtors as the current ratio have increase from FY2021 to FY 2022.
Operating Profits	Operating Income-ROI reduced from last year, but Operating Profit Margin has been maintained. This may happen due to decrease in operating cost. However, both the ratios are still higher than the industry average.
Financing	The company has reduced its debt capital by 1% and saved earnings for equity shareholders. It also signifies that dependency on debt compared to other industry players (60%) is low.
Return to the shareholders	Prabhu's ROE is 26 per cent in 2021 and 28 per cent in 2022 compared to an industry average of 18 per cent. The ROE is stable and improved over the last year.

ICAI · Old P8 · RTP Nov 2023 · question p. 1 · answer p. 6

59 MTP **Sep 2026 S-II** Case Scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Solstice Biotech Ltd. is engaged in the manufacture of advanced diagnostic testing kits, including a rapid screening kit that saw a sharp rise in demand following a nationwide public-health screening initiative. To meet this surge, production was scaled to a record high, and the company approached Continental Trade Bank for a loan to fund its working capital and investment requirements. Along with its loan proposal and project report, the company submitted its financial statements for the preceding three years, reproduced below.

Statement of Profit and Loss (₹ '000)

Particulars	2023–24	2024–25	2025–26
Sales — Cash	400	960	1,600
Sales — Credit	3,600	8,640	14,400
Total Sales	4,000	9,600	16,000
Cost of Goods Sold	2,480	5,664	9,600
Gross Profit	1,520	3,936	6,400
General, Administration & Selling Expenses	160	900	2,000
Depreciation	200	800	1,320
Interest Expense (on borrowings)	120	316	680
Profit Before Tax	1,040	1,920	2,400

Particulars	2023–24	2024–25	2025–26
Tax @ 30%	312	576	720
Profit After Tax	728	1,344	1,680

Balance Sheet (₹ '000)

Particulars	2023–24	2024–25	2025–26
Fixed Assets (net of depreciation)	3,800	5,000	9,400
Cash and Cash Equivalents	80	200	212
Accounts Receivable	600	3,000	4,200
Inventories	640	3,000	4,500
Total Assets	5,120	11,200	18,312
Equity Share Capital (₹ 10 each)	2,400	3,200	4,000
Other Equity	728	2,072	3,752
Non-Current Borrowings	1,472	2,472	5,000
Current Liabilities	520	3,456	5,560
Total Equity & Liabilities	5,120	11,200	18,312

Industry Average of Key Ratios

As on 31st March 2026, the company's closing balances of accounts receivable and inventories used for the year's turnover computations were ₹ 6,00,000 and ₹ 6,40,000 respectively.

Ratio	2023–24	2024–25	2025–26	Industry Avg.
Current ratio	2.54	1.80	1.60	2.30 : 1
Acid test (quick) ratio	1.31	0.93	0.79	1.20 : 1
Receivable turnover ratio	6.00	4.80	4.00	7 times
Inventory turnover ratio	3.88	3.11	2.56	4.85 times
Long-term debt to total debt	73.90%	41.70%	47.35%	24%
Debt-to-equity ratio	47.06%	46.89%	64.50%	35%
Net profit ratio	18.2%	14.0%	10.5%	18%
Return on total assets	14.22%	12.0%	9.17%	10%
Interest coverage ratio	9.67	7.08	4.53	10 times

Based on the above Case Scenario, answer the following questions:

MCQ 1 2 marks

Based on the FY 2025–26 figures, what are the Current Ratio and the Acid-test (Quick) Ratio of Solstice Biotech Ltd.?

- A 1.60 : 1 and 0.79 : 1
- B 2.30 : 1 and 1.20 : 1
- C 1.60 : 1 and 1.02 : 1
- D 1.80 : 1 and 0.93 : 1

MCQ 2 2 marks

What are the Receivable Turnover Ratio and Inventory Turnover Ratio of Solstice Biotech Ltd. for FY 2025–26, computed on average balances?

- A 3.43 times and 2.13 times
- B 6.00 times and 3.88 times
- C 4.00 times and 2.56 times
- D 4.80 times and 3.11 times

MCQ 3 2 marks

What are the Long-term Debt to Total Debt Ratio and the Debt-to-Equity Ratio of Solstice Biotech Ltd. for FY 2025–26?

- A 24.00% and 35.00%
- B 27.31% and 64.50%
- C 41.70% and 46.89%
- D 47.35% and 64.50%

MCQ 4 2 marks

What are the Net Profit Ratio and the Return on Total Assets of Solstice Biotech Ltd. for FY 2025–26?

- A 15.00% and 9.17%
- B 18.00% and 10.00%
- C 14.00% and 12.00%
- D 10.50% and 9.17%

MCQ 5 2 marks

The Interest Coverage Ratio of Solstice Biotech Ltd. fell to 4.53 times in FY 2025 –26 (industry average 10 times) while its Debt-to-Equity Ratio rose to 64.50% (industry average 35%). As the credit appraisal officer, what should be your primary concern in appraising this loan proposal?

- A Both figures are favourable, so the loan can be sanctioned without further scrutiny.
- B The falling interest coverage combined with rising leverage signals a reduced cushion to service debt and heightened repayment risk, warranting closer scrutiny before further lending.
- C The interest coverage ratio is irrelevant once the debt-to-equity ratio is known.
- D Since sales are growing, both ratios can be ignored in the loan decision.

ICAI'S ANSWERS

MCQ 1

(A) Current Assets (2025–26) = Cash & Cash Equivalents + Accounts Receivable + Inventories

= 212 + 4,200 + 4,500 = ₹ 8,912 thousand

Current Ratio = Current Assets ÷ Current Liabilities = 8,912 ÷ 5,560 = 1.60: 1

Quick Assets = Current Assets − Inventories = 8,912 − 4,500 = ₹ 4,412 thousand

Quick (Acid-test) Ratio = Quick Assets ÷ Current Liabilities = 4,412 ÷ 5,560

= 0.79: 1

MCQ 2

(C) Receivable Turnover Ratio = Credit Sales ÷ Average Accounts Receivable

Average Accounts Receivable = (Opening + Closing) ÷ 2 = (3,000 + 4,200) ÷ 2 = ₹ 3,600 thousand

= 14,400 ÷ 3,600 = 4.00 times

Inventory Turnover Ratio = COGS ÷ Average Inventory

Average Inventory = (3,000 + 4,500) ÷ 2 = ₹ 3,750 thousand

= 9,600 ÷ 3,750 = 2.56 times

MCQ 3

(D) Long-term Debt = Non-Current Borrowings = ₹ 5,000 thousand

Total Debt = Non-Current Borrowings + Current Liabilities = 5,000 + 5,560 = ₹ 10,560 thousand

Long-term Debt to Total Debt = (5,000 ÷ 10,560) × 100 = 47.35%

Shareholders' Equity = Equity Share Capital + Other Equity = 4,000 + 3,752 = ₹ 7,752 thousand

Debt-to-Equity Ratio = (Long-term Debt ÷ Shareholders' Equity) × 100 = (5,000 ÷ 7,752) × 100 = 64.50%

MCQ 4

(D) Net Profit Ratio = Profit After Tax ÷ Sales × 100 = 1,680 ÷ 16,000 × 100 = 10.5%

Return on Total Assets = Profit After Tax ÷ Total Assets × 100 = 1,680 ÷ 18,312 × 100 = 9.17%

MCQ 5

(B) Interest Coverage Ratio = EBIT ÷ Interest = (PBT + Interest) ÷ Interest = (2,400 + 680) ÷ 680 = 3,080 ÷ 680 = 4.53 times. A ratio of 4.53 times, well below the industry norm of 10 times, means earnings cover interest obligations with a much thinner margin than peers. Combined with a Debt-to-Equity Ratio of 64.50% (industry average 35%), the company is both more leveraged than its peers and less able to comfortably service that debt from operating earnings. For a loan officer, this combination rising leverage and weakening debt-servicing capacity is the key red flag in the proposal, not a reason for automatic approval.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 1-3 · answer p. 1, 1-2, 2

60 MTP **May 2026 S-I** MCQ 6 2 marks MCQ

Same question, different ICAI answer: #57 RTP May 2026

The capital structure of JKL Limited consists of 12% debentures of ₹ 50 lakhs, 10% preference shares of ₹ 25 lakhs, 14% term loan of ₹ 15 lakhs and equity share capital of ₹ 48 lakhs. Its reserves & surpluses are ₹ 24 lakhs. What would be the capital-gearing ratio?

A 1.04

B 1.25

C 1.33

D 1.56

ICAI'S ANSWER

(B) 1.25

$$\text{Capital gearing ratio} = \frac{(\text{Pref share capital} + \text{Debentures} + \text{Other borrowed funds})}{\text{Equity Share Capital} + \text{Reserves and Surplus - Losses}}$$

$$= \frac{₹25 \text{ lakhs} + ₹50 \text{ lakhs} + ₹15 \text{ lakhs}}{₹48 \text{ lakhs} + ₹24 \text{ lakhs}}$$

= ₹ 90 lakhs / ₹ 72 lakhs = 1.25

ICAI · P6 · MTP May 2026 · Series I · question p. 4 · answer p. 3

61 MTP **Sep 2025 S-I** MCQ (vi) 2 marks MCQ

A company has a debt-to-equity ratio of 2:1. It plans to raise additional capital such that the overall debt-to-equity ratio becomes 1:1. Assuming the company raises only equity for this purpose, by what percentage must the equity increase?

A 50%

B 100%

C 200%

D 300%

ICAI'S ANSWER

Answer key: **(B)**

(b) 100%

Let original equity = ₹ 100

Then, original debt = ₹ 200 (since D/E = 2:1)

Let new equity raised = ₹x

So, revised equity = ₹ 100 + ₹x

Debt remains ₹ 200 (as only equity is raised)

New D/E = 1:1

So, $\frac{200}{100 + x}$ =1

Or 100 + x = 200

Or x = 200 – 100 = 100

Percentage increase in equity

= $\frac{100}{100}$ x 100 = 100%

ICAI · P6 · MTP Sep 2025 · Series I · question p. 3-4 · answer p. 3

62 MTP **May 2025 S-I** Case scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Reema Industries is into trading business. Since its establishment it has seen a phenomenal growth in both its market share and profitability. The company enjoys the confidence of its shareholders who have been rewarded with growing dividends year after year. The company has never defaulted on its

loan payments and enjoys a favourable face with its lenders, which include financial institutions, commercial banks and other private debenture holders. Now the Reema Industries is looking to expand their business and for which they need further funds. Mr. Rishi, the CEO of the company wants their senior management to prepare the report including ratio analysis to be presented before investors. The Balance Sheet and other financial information are shown below.

Liabilities	₹	Assets		₹
Share Capital	1,00,000	Fixed Assets Less: Depreciation	1,70,000 14,000	1,56,000
Reserve and Surplus	80,000	Current Assets:		
9% Preference Share Capital	20,000	Cash		22,000
8% Debentures	50,000	Investments		25,000
Current Liabilities:		Sundry Debtors		30,000
Creditors	10,000	Stock		50,000
Bills Payable	15,000			
Outstanding Expenses	5,000			
Provision for Tax	3,000			
	2,83,000			2,83,000

Other information:

1. Net sales ₹ 1,00,000
2. Cost of goods sold ₹ 66,500
3. Net income before tax ₹ 20,000
4. Average creditor days is 60 days. Assume 360 days in a year.
5. Tax rate 30%

From the above financial information, Senior management asked you to calculate the following ratio for their analysis:

MCQ 1 2 marks

What is the Liquid ratio of the company?

A 2 times

B 2.33 times

C 3.22 times

D 3.84 times

MCQ 2 2 marks

What is the Sales to Proprietary ratio and Interest coverage ratio of the company?

A 0.5 : 1 and 6 times

B 0.8 : 1 and 4 times

C 1.5 : 1 and 3.5 times

D 1.2 : 1 and 2.5 times

MCQ 3 2 marks

What is the Debtor and Creditor turnover ratio of the company?

A 2 times and 6 times

B 3 times and 3.33 times

C 3.33 times and 6 times

D 2 times and 3 times

MCQ 4 2 marks

What is the Working Capital Turnover ratio (based on sales) of the company?

A 1.06 times

B 1.25 times

C 0.9 times

D 2.2 times

MCQ 5 2 marks

What is the Return on Investment of the company?

A 3.5%

B 14%

C 10%

D 7%

ICAI'S ANSWERS

MCQ 1

(B) 2.33 times

$$\text{Liquid Ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

Liquid Assets = Current Assets − Inventories

$$= \frac{77,000}{33,000} = 2.33 \text{ times}$$

MCQ 2

(A) 0.5 : 1 and 6 times

$$\text{Sales to Proprietary Ratio} = \frac{\text{Sales}}{\text{Proprietary fund}}$$

$$= \frac{1,00,000}{2,00,000} = 0.5 : 1$$

Proprietor's fund or Shareholder's fund

= Equity share capital + Preference share capital + Reserve and surplus

= 1,00,000 + 20,000 + 80,000 = 2,00,000

$$\text{Interest Coverage Ratio} = \frac{\text{Earnings before interest and taxes(EBIT)}}{\text{Interest}}$$

= 24,000/4000 = 6 times

EBIT = 20,000 + 4,000 = 24,000

Fixed interest charges = 8% on debentures of ₹ 50,000 = ₹ 4,000

MCQ 3

(C) 3.33 times and 6 times

$$\text{Debtors Turnover Ratio} = \frac{\text{Credit Sales}}{\text{Average Accounts Receivable}}$$

= 1,00,000/30,000 = 3.33 times

In the absence of credit sales and opening debtors, total sales is considered as credit sales and closing debtors as average debtors.

$$\text{Creditors Turnover Ratio} = \frac{\text{Annual Net Credit Purchases}}{\text{Average Accounts Payables}}$$

= 60,000/10,000 = 6 times

In the absence of opening creditors, closing creditors are treated as average creditors.

Credit Purchase = Creditors x 360 / 60

= 10,000 x 360/60 = 60,000

MCQ 4

(A) 1.06 times

$$\text{Working Capital Turnover Ratio} = \frac{\text{Sales}}{\text{Working Capital}}$$

= 1,00,000/94,000 = 1.06 times

Net Working Capital = Current assets – Current liabilities

= 1,27,000 – 33,000 = 94,000

MCQ 5

(D) 7%

$$\text{Return on Investment} = \frac{\text{Net Profit after taxes}}{\text{Shareholders' Funds}} \times 100$$

= 14,000/ 2,00,000 x 100

= 7%

Proprietor's fund or Shareholder's fund

= Equity share capital + Preference share capital + Reserve and surplus

= 1,00,000 + 20,000 + 80,000 = 2,00,000

ICAI · P6 · MTP May 2025 · Series I · question p. 1-2 · answer p. 1, 2

63 MTP **Old P8 Dec 2021 S-II** Q2 10 marks old scheme

Jensen and spencer pharmaceutical is in the business of manufacturing pharmaceutical drugs including the newly invented Covid vaccine. Due to increase in demand of Covid vaccines, the production had increased at all time high level and the company urgently needs a loan to meet the cash and investment requirements. It had already submitted a detailed loan proposal and project report to Expo-Impo bank, along with the financial statements of previous three years as follows:

Statement of Profit and Loss (In ₹ '000)

	2018–19	2019–20	2020–21
Sales			
Cash	400	960	1,600
Credit	3,600	8,640	14,400
Total sales	4,000	9,600	16,000
Cost of goods sold	2,480	5,664	9,600
Gross profit	1,520	3,936	6,400
Operating expenses:			
General, administration, and selling expenses	160	900	2,000
Depreciation	200	800	1,320
Interest expenses (on borrowings)	120	316	680
Profit before tax (PBT)	1,040	1,920	2,400
Tax @ 30%	312	576	720
Profit after tax (PAT)	728	1,344	1,680

BALANCE SHEET (In ₹ '000)

	2018–19	2019–20	2020–21
Assets			
Non-Current Assets			
Fixed assets (net of depreciation)	3,800	5,000	9,400
Current Assets			
Cash and cash equivalents	80	200	212
Accounts receivable	600	3,000	4,200
Inventories	640	3,000	4,500
Total	**5,120**	**11,200**	**18,312**
Equity & Liabilities			
Equity share capital (shares of ₹10 each)	2,400	3,200	4,000
Other Equity	728	2,072	3,752
Non-Current borrowings	1,472	2,472	5,000

	2018–19	2019–20	2020–21
Current liabilities	520	3,456	5,560
Total	**5,120**	**11,200**	**18,312**

INDUSTRY AVERAGE OF KEY RATIOS

Ratio	Sector Average
Current ratio	2.30:1
Acid test ratio (quick ratio)	1.20:1
Receivable turnover ratio	7 times
Inventory turnover ratio	4.85 times
Long-term debt to total debt	24%
Debt-to-equity ratio	35%
Net profit ratio	18%
Return on total assets	10%
Interest coverage ratio (times interest earned)	10

As a loan officer of Expo-Impo Bank, you are REQUIRED to apprise the loan proposal on the basis of comparison with industry average of key ratios considering closing balance for accounts receivable of ₹ 6,00,000 and inventories of ₹ 6,40,000 respectively as on 31st March, 2018.

ICAI'S ANSWER

(In ₹ '000)

Ratio	Formula	2018–19	2019–20	2020–21	Industry Average
Current ratio	Current Assets/Current Liabilities	1,320/520 = 2.54	6,200/3,456 = 1.80	8,912/5,560 = 1.60	2.30:1
Acid test ratio (quick ratio)	Quick Assets/Current Liabilities	680/520 = 1.31	3,200/3,456 = 0.93	4,412/5,560 = 0.79	1.20:1
Receivable turnover ratio	Credit Sales/Average Accounts Receivable	3,600/(600+600)/2 = 6	8,640/(600+3,000)/2 = 4.80	14,400/(3,000+4,200)/2 = 4	7 times
Inventory turnover ratio	COGS/Average Inventory	2,480/(640+640)/2 = 3.88	5,664/(640+3,000)/2 = 3.11	9,600/(3,000+4,500)/2 = 2.56	4.85 times

Ratio	Formula	2018–19	2019–20	2020–21	Industry Average
Long-term debt to total debt	Long term Debt/Total Debt × 100	1,472/1,992 × 100 = 73.90%	2,472/5,928 × 100 = 41.70%	5,000/10,560 × 100 = 47.35%	24%
Debt-to-equity ratio	Long term Debt/Shareholders' Equity × 100	1,472/3,128 × 100 = 47.06%	2,472/5,272 × 100 = 46.89%	5,000/7,752 × 100 = 64.50%	35%
Net profit ratio	Net Profit/Sales ×100	728/4,000 × 100 = 18.2%	1,344/9,600 × 100 = 14%	1,680/16,000 × 100 = 10.5%	18%
Return on total assets	Net Profit after taxes/Total assets ×100	728/5,120 × 100 = 14.22%	1,344/11,200 × 100 = 12%	1,680/18,312 × 100 = 9.17%	10%
Interest coverage ratio (times interest earned)	EBIT/Interest	1,160/120 = 9.67	2,236/316 = 7.08	3,080/680 = 4.53	10

Conclusion:

In the last two years, the current ratio and quick ratio are less than the ideal ratio (2:1 and 1:1 respectively) indicating that the company is not having enough resources to meet its current obligations. Receivables are growing slower. Inventory turnover is slowing down as well, indicating a relative build-up in inventories or increased investment in stock. High Long-term debt to total debt ratio and Debt to equity ratio compared to that of industry average indicates high dependency on long term debt by the company. The net profit ratio is declining substantially and is much lower than the industry norm. Additionally, though the Return on Total Asset(ROTA) is near to industry average, it is declining as well. The interest coverage ratio measures how many times a company can cover its current interest payment with its available earnings. A high interest coverage ratio means that an enterprise can easily meet its interest obligations, however, it is declining in the case of Jensen & Spencer and is also below the industry average indicating excessive use of debt or inefficient operations.

On overall comparison of the industry average of key ratios than that of Jensen & Spencer, the company is in deterioration position. The company's profitability has declined steadily over the period. However, before jumping to the conclusion relying only on the key ratios, it is pertinent to keep in mind the industry, the company dealing in with i.e. manufacturing of pharmaceutical drugs. The pharmaceutical industry is one of the major contributors to the economy and is expected to grow further. After the covid situation, people are more cautious towards their health and are going to spend relatively more on health medicines. Thus, while analysing the loan proposal, both the factors, financial and non-financial, needs to be kept in mind.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 2-3 · answer p. 3-5

2.2 Activity (turnover) ratios

11 questions · asked in 11 ICAI documents · 1 exam paper

64 EXAM **Sep 2026** MCQ 8 1 mark MCQ

The following information is available from the books of RC Ltd. :

Net profit margin : 10%

Total Assets : ₹ 100,00,000

Return on Assets : 15%

What would be the total asset turnover ratio ?

A 1.3

B 1.5

C 1.7

D 1.9

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P6 · Sep 2026 exam · question p. 6

65 RTP **Sep 2026** Q3 MCQ

MT Ltd.'s opening stock was ₹ 3,20,000 and the closing stock was ₹ 4,80,000. Sales during the year were ₹ 18,00,000 and the gross profit ratio was 30% on sales. Average accounts payable are ₹ 1,20,000.

Creditors Turnover Ratio = ?

A 11.83

B 13.75

C 14.00

D 15.50

ICAI'S ANSWER

(A) 11.83

Creditors Turnover Ratio = Purchases / Average Accounts Payable

Cost of Goods Sold = Opening Stock+ Purchases - Closing Stock

Purchases = ₹ 12,60,000 + ₹ 4,80,000 – ₹ 3,20,000

Purchases = ₹ 14,20,000

Average Accounts Payable = ₹ 1,20,000

Creditors Turnover Ratio = Purchases/Average Accounts Payable

Creditors Turnover Ratio = ₹ 14,20,000 / ₹ 1,20,000

Creditors Turnover Ratio = 11.833

Therefore, the Creditors Turnover Ratio is 11.833.

ICAI · P6 · RTP Sep 2026 · question p. 4 · answer p. 12

66 RTP Sep 2024 Q2 MCQ

KT Ltd.'s opening stock was ₹ 2,50,000 and the closing stock was ₹ 3,75,000. Sales during the year were ₹ 13,00,000 and the gross profit ratio was 25% on sales. Average accounts payable are ₹ 80,000.

Creditors Turnover Ratio =?

A 13.33

B 14.33

C 14.44

D 13.75

ICAI'S ANSWER

(D) 13.75

Creditors Turnover Ratio = Purchases / Average Accounts Payable

Cost of Goods Sold = Opening Stock+ Purchases - Closing Stock

Purchases = Cost of Goods Sold + Closing Stock - Opening Stock

Purchases = ₹ 9,75,000 + ₹ 3,75,000 - ₹ 2,50,000

Purchases = ₹ 11,00,000

Average Accounts Payable = ₹ 80,000

Creditors Turnover Ratio = Purchases/Average Accounts Payable

Creditors Turnover Ratio = ₹ 11,00,000 / ₹ 80,000

Creditors Turnover Ratio = 13.75

Therefore, the Creditors Turnover Ratio is 13.75.

ICAI · P6 · RTP Sep 2024 · question p. 5 · answer p. 15

67 RTP Old P8 May 2020 Q1 old scheme

MT Limited has the following Balance Sheet as on March 31, 2019 and March 31, 2020:

Balance Sheet

	₹ in lakhs	
	March 31, 2019	**March 31, 2020**
Sources of Funds:		
Shareholders' Funds	2,500	2,500
Loan Funds	3,500	3,000
	6,000	5,500
Applications of Funds:		
Fixed Assets	3,500	3,000
Cash and bank	450	400

	₹ in lakhs	
Receivables	1,400	1,100
Inventories	2,500	2,000
Other Current Assets	1,500	1,000
Less: Current Liabilities	(1,850)	(2,000)
	6,000	5,500

The Income Statement of the MT Ltd. for the year ended is as follows:

	₹ in lakhs	
	March 31, 2019	**March 31, 2020**
Sales	22,500	23,800
Less: Cost of Goods sold	(20,860)	(21,100)
Gross Profit	1,640	2,700
Less: Selling, General and Administrative expenses	(1,100)	(1,750)
Earnings before Interest and Tax (EBIT)	540	950
Less: Interest Expense	(350)	(300)
Earnings before Tax (EBT)	190	650
Less: Tax	(57)	(195)
Profits after Tax (PAT)	133	455

Required:

CALCULATE for the year 2019-20-

(a) Inventory turnover ratio

(b) Financial Leverage

(c) Return on Capital Employed (ROCE)

(d) Return on Equity (ROE)

(e) Average Collection period.

[Take 1 year = 365 days]

ICAI'S ANSWER

Ratios for the year 2019-2020

(a) Inventory turnover ratio

$$= \frac{\text{COGS}}{\text{Average Inventory}} = \frac{₹\ 21{,}100}{\frac{₹\ (2{,}500 + 2{,}000)}{2}} = 9.4$$

(b) Financial leverage

$$= \frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 950}{₹\ 650} = 1.46$$

(c) ROCE

$$= \frac{\text{EBIT (1- t)}}{\text{Average Capital Employed}} = \frac{₹\ 950\ (1-\ 0.3)}{₹\left(\frac{6{,}000 + 5{,}500}{2}\right)} = \frac{₹\ 665}{₹\ 5{,}750} \times 100 = 11.56\ \%$$

[Here Return on Capital Employed (ROCE) is calculated after Tax]

(d) ROE

$$= \frac{\text{Profits after tax}}{\text{Average shareholders' funds}} = \frac{₹\ 455}{₹\ 2{,}500} \times 100 = 18.2\%$$

(e) Average Collection Period

$$\text{Average Sales per day} = \frac{₹\ 23{,}800}{365} = ₹\ 65.20 \text{ lakhs}$$

$$\text{Average collection period} = \frac{\text{Average Receivables}}{\text{Average sales per day}}$$

$$= \frac{₹(1{,}400 + 1{,}100)}{2} = \frac{₹\ 1{,}250}{₹\ 65.2} = 19.17 \text{ days}$$

ICAI · Old P8 · RTP May 2020 · question p. 1-2 · answer p. 6-7

68 MTP May 2026 S-II Q1(c) 5 marks

PQR Limited has the following Balance Sheets as on March 31, 2026 and March 31, 2025:

Balance Sheet

	₹ in lakhs	
	March 31, 2026	**March 31, 2025**
Sources of Funds:		
Shareholders Funds	2,377	1,472
Loan Funds	3,570	3,083
	5,947	4,555
Applications of Funds:		
Fixed Assets	3,466	2,900
Cash and bank	489	470
Debtors	1,495	1,168
Stock	2,867	2,407
Other Current Assets	1,567	1,404
Less: Current Liabilities	(3,937)	(3,794)

₹ in lakhs	
5,947	4,555

The Income Statement of the PQR Ltd. for the year ended is as follows:

₹ in lakhs		
	March 31, 2026	**March 31, 2025**
Sales	22,165	13,882
Less: Cost of Goods sold	20,860	12,544
Gross Profit	1,305	1,338
Less: Selling, General and Administrative expenses	1,135	752
Earnings before Interest and Tax (EBIT)	170	586
Interest Expense	113	105
Profits before Tax	57	481
Tax	23	192
Profits after Tax (PAT)	34	289

CALCULATE for the year 2025-26:

(i) Inventory turnover ratio

(ii) Financial Leverage

(iii) Return on Capital Employed (ROCE)

(iv) Return on Equity (ROE)

(v) Average Collection period.

Take 365 days in a year

ICAI'S ANSWER

Ratios for the year 2025-2026

(i) Inventory turnover ratio

$$= \frac{\text{COGS}}{\text{Average Inventory}} = \frac{20{,}860}{\frac{(2{,}867 + 2{,}407)}{2}} = 7.91$$

(ii) Financial leverage

	2025-26	2024-25
$= \frac{\text{EBIT}}{\text{EBIT - I}}$	$= \frac{170}{57} = 2.98$	$= \frac{586}{481} = 1.22$

(iii) ROCE

$$= \frac{\text{EBIT (1-t)}}{\text{Average Capital Employed}} = \frac{57\,(1-0.4)}{\left(\frac{5{,}947+4{,}555}{2}\right)} = \frac{34.2}{5251} \text{ x } 100 = 0.651\%$$

[Here Return on Capital Employed (ROCE) is calculated after Tax]

(iv) ROE

$$= \frac{\text{Profits after tax}}{\text{Average shareholders' funds}} = \frac{34}{\frac{(2{,}377+1{,}472)}{2}} = \frac{34}{1{,}924.5} = 1.77\%$$

(v) Average Collection Period*

$$\text{Average Sales per day} = \frac{22{,}165}{365} = ₹\ 60.73 \text{ lakhs}$$

$$\text{Average collection period} = \frac{\text{Average Debtors}}{\text{Average sales per day}} = \frac{\frac{(1{,}495+1{,}168)}{2}}{60.73} = \frac{1331.5}{60.73}$$

= 22 days

ICAI · P6 · MTP May 2026 · Series II · question p. 5-6 · answer p. 5-6

69 MTP **Jan 2026 S-II Q1(a)** 5 marks

Following information relate to a concern:

Debtors Velocity	3 months
Credits Velocity	2 months
Stock Turnover Ratio	1.5
Gross Profit Ratio	25%
Bills Receivables	₹ 25,000
Bills Payables	₹ 10,000
Gross Profit	₹ 4,00,000
Fixed Assets to turnover Ratio	4

Closing stock of the period is ₹ 10,000 above the opening stock.

CALCULATE

(i) Sales and cost of goods sold

(ii) Sundry Debtors

(iii) Sundry Creditors

(iv) Closing Stock

(v) Fixed Assets

ICAI'S ANSWER

(i) **Determination of Sales and Cost of goods sold:**

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Sales}} \times 100$$

Or, $\frac{25}{100} = \frac{₹4,00,000}{\text{Sales}}$

Or, Sales = $\frac{4,00,000}{25}$ = ₹ 16,00,000

Cost of Goods Sold = Sales – Gross Profit = ₹ 16,00,000 - ₹ 4,00,000 = ₹ 12,00,000

(ii) **Determination of Sundry Debtors:**

Debtors velocity is 3 months or Debtors' collection period is 3 months,

So, Debtors' turnover ratio = $\frac{12 \text{ months}}{3 \text{ months}} = 4$

Debtors' turnover ratio = $\frac{\text{Credit Sales}}{\text{Average Accounts Receivable}} = \frac{₹16,00,000}{\text{Bills Receivable} + \text{Sundry Debtors}} = 4$

Or, Sundry Debtors + Bills receivable = ₹ 4,00,000

Sundry Debtors = ₹ 4,00,000 – ₹ 25,000 = ₹ 3,75,000

(iii) **Determination of Sundry Creditors:**

Creditors velocity of 2 months or credit payment period is 2 months.

So, Creditors' turnover ratio = $\frac{12 \text{ months}}{2 \text{ months}} = 6$

Creditors turnover ratio = $\frac{\text{Credit Purchases *}}{\text{Average Accounts Payables}} = \frac{₹12,10,000}{\text{Sundry Creditors} + \text{Bills Payables}} = 6$

So, Sundry Creditors + Bills Payable = ₹ 2,01,667

Or, Sundry Creditors + ₹ 10,000 = ₹ 2,01,667

Or, Sundry Creditors = ₹ 2,01,667 – ₹ 10,000 = ₹ 1,91,667

(iv) **Closing Stock**

Stock Turnover Ratio = $\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{₹12,00,000}{\text{Average Stock}}$ =1.5

So, Average Stock = ₹ 8,00,000

Now Average Stock = $\frac{\text{Opening Stock} + \text{Closing Stock}}{2}$

Or $\frac{\text{Opening Stock} + (\text{Opening Stock} + ₹10,000)}{2}$ = ₹ 8,00,000

Or, Opening Stock = ₹ 7,95,000

So, Closing Stock = ₹ 7,95,000 + ₹ 10,000 = ₹ 8,05,000

(v) **Calculation of Fixed Assets**

Fixed Assets Turnover Ratio = $\frac{\text{Cost of Goods Sold}}{\text{Fixed Assets}} = 4$

Or, $\frac{₹12,00,000}{\text{Fixed Assets}} = 4$

Or, Fixed Asset = ₹ 3,00,000

Workings:

***Calculation of Credit purchases:**

Cost of goods sold = Opening stock + Purchases – Closing stock

₹ 12,00,000 = ₹ 7,95,000 + Purchases – ₹ 8,05,000

₹ 12,00,000 + ₹ 10,000 = Purchases

₹ 12,10,000 = Purchases (credit).

Assumption:

(i) All sales are credit sales

(ii) All purchases are credit purchase

(iii) Stock Turnover Ratio and Fixed Asset Turnover Ratio may be calculated either on Sales or on Cost of Goods Sold.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 4-5 · answer p. 4-6

70 MTP **Jan 2025 S-II** MCQ 7 2 marks MCQ

If velocity of stock is 3 months, annual sales amount to Rs.6 lakh at 20% gross profit margin and opening stock is Rs.90,000; what is the closing stock value?

A Rs. 90,000

B Rs. 70,000

C Rs. 1,50,000

D Rs. 1,00,000

ICAI'S ANSWER

(C) COGS = Sales x (1−Gross Profit Margin)

COGS = 6,00,000 x (1−0.20) = 6,00,000 x 0.80 = 4,80,000

The velocity of stock is 3 months.

stock turnovers per year (12/3) = 4

Stock Turnover Ratio = COGS / Average Stock

Average Stock = 4,80,000/4 = 1,20,000

Average Stock = (Opening Stock + Closing Stock)/2

Closing Stock = 1,50,000

ICAI · P6 · MTP Jan 2025 · Series II · question p. 3 · answer p. 2

71 MTP **May 2024 S-II** Q3(a) 8 marks

EOC Ltd is a listed company and has presented the below abridged financial statements below.

Statement of Profit and Loss	₹	₹
Sales		1,25,00,000
Cost of goods sold		(76,40,000)
Gross Profit		48,60,000
Less: Operating Expenses		
Administrative Expenses	13,20,000	
Selling and Distribution Expenses	15,90,000	(29,10,000)

Statement of Profit and Loss	₹	₹
Operating Profit		19,50,000
Add: Non Operating Income		3,28,000
Less: Non Operating Expenses		(1,27,000)
Profit before Interest and taxes		21,51,000
Less: Interest		(4,39,000)
Profit before tax		17,12,000
Less: Taxes		(4,28,000)
Profit after Tax		12,84,000

Balance Sheet

Sources of Funds	₹	₹
Owned Funds		
Equity Share Capital	30,00,000	
Reserves and Surplus	18,00,000	48,00,000
Borrowed Funds		
Secured Loan	10,00,000	
Unsecured Loan	4,30,000	14,30,000
Total Funds Raised		**62,30,000**
Application of Funds		
Non-Current Assets		
Building	7,50,000	
Machinery	2,30,000	
Furniture	7,60,000	
Intangible Assets	50,000	17,90,000
Current Assets		
Inventory	38,60,000	
Receivables	39,97,000	
ST investments	3,00,000	
Cash and Bank	2,30,000	83,87,000

Sources of Funds	₹	₹
Less: Current Liabilities		
Creditors	25,67,000	
ST loans	13,80,000	(39,47,000)
Total Funds Employed		**62,30,000**

The company has set certain standards for the upcoming year financial status.

All the ratios are based on closing figures in financial statements.

Equity SC to Reserves=	1	
Net Profit Ratio=	15%	
Gross Profit Ratio=	50%	
Long Term Debt to Equity=	0.5	
Debtor Turnover=	100	Days
Creditor Turnover (based on COGS)=	100	Days
Inventory=	70%	of Opening inventory

Cash Balance is assumed to remain same for next year

You are required to -

(1) CALCULATE inventory turnover ratio in days for current year

(2) CALCULATE receivables turnover ratio in days for current year

(3) CALCULATE the projected receivables, inventory, payables and long term debt

ICAI'S ANSWER

$$\text{Inventory Turnover} = \frac{\text{Inventory}}{\text{COGS}} \times 365 = \frac{38,60,000 \times 365}{76,40,000} = 184.41 \text{ days} = 185 \text{ days (apx)}$$

$$\text{Receivables Turnover} = \frac{\text{Receivables}}{\text{Sales}} \times 365 = \frac{39,97,000 \times 365}{1,25,00,000} = 116.71 = 117 \text{ days (apx)}$$

Equity to Reserves = 1

Reserves = 1 x 30,00,000 = 30,00,000

Projected profit = 30,00,000 - 18,00,000 = 12,00,000

Net Profit Margin = 15%

12,00,000/ Sales = 0.15

Sales = 80,00,000

Gross Profit = 80,00,000 x 50% = 40,00,000

COGS = 80,00,000 - 40,00,000 = 40,00,000

$$\text{Projected Debtors Turnover} = 100 \text{ days} = \frac{\text{Closing Receivables}}{\text{Sales}} \text{ x } 365$$

$$100 = \frac{\text{Closing Receivables}}{80,00,000} \text{ x } 365$$

$$\text{Closing Receivables} = \frac{80,00,000 \text{ x } 100}{365} = 21,91,781$$

Projected Closing Inventory = 70% of opening inventory = 70% of 38,60,000 = 27,02,000

$$\text{Projected Creditor Turnover= 100 days} = \frac{\text{Closing Creditors}}{\text{COGS}} \text{x} 365$$

$$\text{Closing Creditors} = \frac{\text{COGS}}{365} \text{x} 100$$

$$\text{Closing Creditor} = \frac{40,00,000}{365} \text{x} 100 = 10,95,890$$

Equity Share Capital + Reserves = 30,00,000 + 30,00,000 = 60,00,000

Long Term Debt to Equity = 0.5

$$\frac{\text{LTD}}{60,00,000} = 0.5$$

Long Term Debt = 0.5 x 60,00,000

Long Term Debt = 30,00,000

ICAI · P6 · MTP May 2024 · Series II · question p. 5-7 · answer p. 9

72 MTP **Old P8 Nov 2023 S-II** Q1(d) 5 marks old scheme

ABC Ltd. has total sales of 12,00,000 all of which are credit sales. It has a gross profit ratio of 20% on sales and a current ratio of 2. The company's current liabilities are ₹ 3,00,000. Further, it has inventories of ₹ 1,00,000, marketable securities of ₹ 70,000 and cash of ₹ 50,000. From the above information:

(i) CALCULATE the average inventory if the expected inventory turnover ratio is three times?

(ii) Also CALCULATE the average collection period if the opening balance of debtors is expected to be ₹ 1,20,000.

Assume 360 days a year.

ICAI'S ANSWER

(i) Calculation of Average Inventory

Since gross profit is 20% of sales, the cost of goods sold should be 80% of the sales.

$$\text{Cost of goods sold} = 12,00,000 \text{ x } \frac{80}{100} = 9,60,000$$

$$\text{Inventory Turnover} = \frac{\text{Cost of goods sold}}{\text{Average Inventory}}$$

$$3 = \frac{9,60,000}{\text{Average Inventory}}$$

$$\text{Average Inventory} = \frac{9,60,000}{3} = 3,20,000$$

(ii) Calculation of Average Collection Period

$$\text{Average Collection Period} = \frac{\text{Average Receivables}}{\text{Credit Sales}} \text{ x } 360$$

$$\text{Where, Average Receivables} = \frac{\text{Opening Receivables} + \text{Closing Receivables}}{2}$$

Calculation of Closing balance of Receivables

	₹	₹
Current Assets (2 × 3,00,000)		6,00,000
Less: Inventories	1,00,000	
Less: Marketable Securities	70,000	
Less: Cash	50,000	2,20,000
Receivables (Closing Balance)		**3,80,000**

Now, Average Receivables = $\frac{1,20,000 + 3,80,000}{2} = 2,50,000$

So, Average Collection Period = $\frac{2,50,000}{12,00,000}$ x 360 = 75 days

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 2 · answer p. 2-3

73 MTP **Old P8 Nov 2022 S-I** Q1(c) 5 marks old scheme

PI Limited has the following Balance Sheet as on March 31, 2020 and March 31, 2021:

Balance Sheet

Particulars	March 31, 2020	March 31, 2021
Sources of Funds:		
Shareholders' Funds	87,500	87,500
Loan Funds	1,22,500	1,05,000
	2,10,000	1,92,500
Applications of Funds:		
Fixed Assets	87,500	1,05,000
Cash and bank	15,750	14,000
Receivables	49,000	38,500
Inventories	87,500	70,000
Other Current Assets	35,000	35,000
Less: Current Liabilities	(64,750)	(70,000)
	2,10,000	1,92,500

The Income Statement of the PI Ltd. for the year ended is as follows:

Particulars	March 31, 2020	March 31, 2021
Sales	7,87,500	8,33,000

Particulars	March 31, 2020	March 31, 2021
Less: Cost of Goods sold	(7,30,100)	(7,38,500)
Gross Profit	57,400	94,500
Less: Selling, General and Administrative expenses	(38,500)	(61,250)
Earnings before Interest and Tax (EBIT)	18,900	33,250
Less: Interest Expense	(12,250)	(10,500)
Earnings before Tax (EBT)	6,650	22,750
Less: Tax	(1,995)	(6,825)
Profits after Tax (PAT)	4,655	15,925

You are required to CALCULATE for the year 2020-21:

(i) Inventory turnover ratio

(ii) Financial Leverage

(iii) Return on Capital Employed (after tax)

ICAI'S ANSWER

Ratios for the year 2020-21

(i) Inventory turnover ratio

$$= \frac{\text{COGS}}{\text{Average Inventory}} = \frac{₹\ 7{,}38{,}500}{\frac{₹\ (87{,}500 + 70{,}000)}{2}} = 9.4$$

(ii) Financial leverage

$$= \frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 33{,}250}{₹\ 22{,}750} = 1.46$$

(iii) ROCE

$$= \frac{\text{EBIT (1- t)}}{\text{Average Capital Employed}} = \frac{₹\ 33{,}250\ (1\text{-}0.3)}{₹\left(\frac{2{,}10{,}000 + 1{,}92{,}500}{2}\right)} = \frac{₹\ 23{,}275}{₹\ 2{,}01{,}250} \times 100 = 11.56\ \%$$

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 1-2 · answer p. 2

74 MTP **Old P8 Dec 2021 S-I** Q1(c) 5 marks old scheme

ABC Ltd. has total sales of 10,00,000 all of which are credit sales. It has a gross profit ratio of 25% and a current ratio of 2. The company's current liabilities are ₹ 2,00,000. Further, it has inventories of ₹ 80,000, marketable securities of ₹ 50,000 and cash of ₹ 30,000. From the above information:

(i) CALCULATE the average inventory, if the expected inventory turnover ratio is three times?

(ii) Also CALCULATE the average collection period if the opening balance of debtors is expected to be ₹ 1,50,000.

Assume 360 days a year.

ICAI'S ANSWER

(i) Calculation of Average Inventory

Since gross profit is 25% of sales, the cost of goods sold should be 75% of the sales.

Cost of goods sold = 10,00,000 x $\frac{75}{100}$ = 7,50,000

Inventory Turnover = $\frac{\text{Cost of goods sold}}{\text{Average Inventory}}$

3 = $\frac{7,50,000}{\text{Average Inventory}}$

Average Inventory = $\frac{7,50,000}{3}$ = 2,50,000

(ii) Calculation of Average Collection Period

Average Collection Period = $\frac{\text{Average Debtors}}{\text{Credit Sales}}$ x 360

Where, Average Debtors = $\frac{\text{Opening Debtors} + \text{Closing Debtors}}{2}$

Calculation of Closing balance of debtors

	₹	₹
Current Assets (2 × 2,00,000)		4,00,000
Less: Inventories	80,000	
Marketable Securities	50,000	
Cash	30,000	1,60,000
Debtors Closing Balance		**2,40,000**

Now, Average Debtors = $\frac{1,50,000 + 2,40,000}{2}$ = 1,95,000

So, Average Collection Period = $\frac{1,95,000}{10,00,000}$ x 360 = 70.2 or 70 days

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 2 · answer p. 2-3

2.3 Profitability and market ratios, and the DuPont analysis

19 questions · asked in 12 ICAI documents · 6 exam papers

75 EXAM **May 2026** Case scenario · Q1(c)(i) to Q1(c)(iv) 5 marks

CASE SCENARIO

Following information is related to Exe Ltd. for the year ended 31st March, 2026:

Total dividend coverage ratio	2.863 times
Dividend yield on the equity shares	4%
Market price per equity share	₹ 40.00
Equity share capital of 10 each	12,00,000
Preference share capital of 10 each	₹ 3,00,000
Price-earnings ratio	8 times

You are required to calculate:

Q1(c)(i) 1 mark

Total dividend paid to equity shareholders

Q1(c)(ii) 1 mark

Earnings per share

Q1(c)(iii) 2 marks

Rate of preference dividend

Q1(c)(iv) 1 mark

Total profit after tax

ICAI'S ANSWERS

ICAI's working for this case

Dividend per Equity Share (DPS) = 4 % of ₹40.00 = ₹1.60

No. of Equity Shares = ₹12,00,000 / ₹10 = 1,20,000 shares

Q1(c)(i)

Total Dividend Paid to Equity Shareholders

1,20,000 × ₹1.60 = ₹1,92,000

Q1(c)(ii)

Calculation of Earnings Per Share (EPS)

$$\text{P/E ratio} = \frac{\text{MPS}}{\text{EPS}}$$

8 = 40 / EPS

EPS = ₹5

Q1(c)(iii)

Rate of Preference Dividend

Profit Available to Equity Shareholders

EPS × No. of Equity Shares = ₹5 × 1,20,000 = ₹6,00,000

Let preference dividend = P

Then, PAT = 6,00,000 + P

Total dividend =Equity Dividend+ Preference Dividend

=1,92,000 + P

Dividend coverage ratio = 2.863

$$2.863 = \frac{(6,00,000 + P)}{(1,92,000 + P)}$$

6,00,000+P = 2.863(1,92,000+P)

6,00,000+P = 5,49,696+2.863P

50,304 =1.863P

P = ₹ 27,000

Rate of Preference Dividend= 27,000/3,00,000 x 100=9%

Q1(c)(iv)

Total Profit after Tax

PAT = ₹6,00,000 + ₹27000 = ₹6,27,000

ICAI · P6 · May 2026 exam · question p. 10 · answer p. 12, 13

76 EXAM **Sep 2025** Q1(b) 5 marks

Diwan Limited has outlined its financial projections for the fiscal year 2025-26. The company plans to utilize total assets amounting to ₹ 50,00,000, with 35% of assets financed through debt at interest rate of 10.50% per annum. Projected sales revenue is ₹ 55,00,000. Direct costs are estimated at ₹ 30,00,000 and other operating expenses are estimated at ₹ 4,80,000. The applicable tax rate is 33%

You are required to calculate:

(i) Profit After Tax (PAT)

(ii) Net profit margin (After tax)

(iii) Return on Assets (After tax)

(iv) Asset turnover ratio

(v) Return on Equity

ICAI'S ANSWER

(i) **Income Statement**

Particulars	(₹)
Sales	55,00,000
Less: Direct Cost	30,00,000
Gross Profit	25,00,000
Less: Operating Expenses	4,80,000
EBIT	20,20,000

Particulars	(₹)
Less: Interest (50,00,000 × 35% x 10.5%)	1,83,750
Profit Before Tax	18,36,250
Less: Tax @ 33%	6,05,963
Profit After Tax	12,30,287

(ii) Net Profit Margin after tax $= \dfrac{\text{Net Profit after taxes}}{\text{Sales}}$

$= \dfrac{12,30,287}{55,00,000} = 22.37\%$

OR

$= \dfrac{\text{EBIT}(1-t)}{\text{Sales}}$

$= \dfrac{20,20,000\ (1-0.33)}{55,00,000} = 24.61\%$

(iii) Return on Assets (ROA) $= \dfrac{\text{EBIT}(1-t)}{\text{Average total assets}}$

$= \dfrac{20,20,000\ (1-0.33)}{50,00,000} = 27.07\%$

(iv) Asset turnover $= \dfrac{\text{Sales}}{\text{Assets}} = \dfrac{55,00,000}{50,00,000} = 1.1 \text{ times}$

(v) Return on equity $= \dfrac{\text{Net Profit after taxes}}{\text{Owners' equity}}$

$= \dfrac{₹\ 12,30,287}{65\% \text{ of } ₹\ 50,00,000}$

$= \dfrac{12,30,287}{32,50,000} = 37.85\%$

ICAI · P6 · Sep 2025 exam · question p. 8-9 · answer p. 11-12

77 EXAM **Sep 2024** MCQ 8 MCQ

ZX Limited has total assets of ₹ 7,20,000 and its Shareholders' equity is ₹ 4,50,000. The net profit margin of ZX Limited is 12.5% and asset turnover ratio is 1.5. Using the DuPont model, the return on equity of ZX Limited is calculated as :

A 7.03%

B 50%

C 11.72%

D 30%

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P6 · Sep 2024 exam · question p. 4 · answer p. 4

78 EXAM **Old P8 Nov 2022** Q1(b) 5 marks old scheme

The following figures are related to the trading activities of M Ltd.

Total assets	₹ 10,00,000
Debt to total assets	50%
Interest cost	10% per year
Direct Cost	10 times of the interest cost
Operating Exp.	₹ 1,00,000

The goods are sold to customers at a margin of 50% on the direct cost

Tax Rate is 30%

You are required to calculate

(i) Net profit margin

(ii) Net operating profit margin

(iii) Return on assets

(iv) Return on owner's equity

ICAI'S ANSWER

(i) Computation of Net Profit Margin

Debt = (10,00,000 x 50%) = ₹ 5,00,000

Interest cost = 5,00,000 x $\left(\frac{10}{100}\right)$ = ₹ 50,000

Direct cost = 50,000 x 10 = ₹ 5,00,000

Sales = 5,00,000 x 150% = ₹ 7,50,000

	(₹)
Gross profit = 7,50,000 – 5,00,000	= 2,50,000
Less: Operating expenses	= 1,00,000
∴ EBIT	= 1,50,000
Less: Interest	= 50,000
∴ EBT	= 1,00,000
Less: Tax @ 30%	= 30,000
∴ PAT	= 70,000

Net profit margin = $\left(\frac{70,000}{7,50,000}\right) \times 100 = 9.33\%$

(ii) Net Operating Profit margin

Net operating profit margin = $\left(\frac{\text{EBIT}}{\text{Sales}}\right) \times 100$

= $\left(\frac{1,50,000}{7,50,000}\right) \times 100 = 20\%$

(iii) Return on Assets

$$\text{Return on Assets} = \left[\left(\frac{\text{PAT} + \text{Interest}}{\text{Total Assets}}\right)\right] \text{ x } 100$$

$$= \left[\left(\frac{1,20,000}{10,00,000}\right)\right] \text{ x } 100] = 12\%$$

(OR)

$$\text{Return on Assets} = \frac{\text{EBIT}}{\text{Assets}} \times 100$$

$$= \frac{1,50,000}{10,00,000} \times 100 = 15\%$$

(OR)

$$= \frac{70,000}{10,00,000} \times 100 = 7\%$$

(OR)

$$= \left[\frac{1,50,000(1-0.3)}{10,00,000}\right] \text{ x } 100 = 10.5\%$$

(iv) Return on owner's equity

$$\text{Return} = \left(\frac{\text{PAT}}{\text{Owner's equity}}\right) \text{ x } 100$$

$$= \left(\frac{70,000}{5,00,000}\right) \text{ x } 100 = 14\%$$

ICAI · Old P8 · Nov 2022 exam · question p. 1-2 · answer p. 3-5

79 EXAM **Old P8 Nov 2020** Q1(a) 5 marks old scheme

Following information relates to RM Co. Ltd.

	(₹)
Total Assets employed	10,00,000
Direct Cost	5,50,000
Other Operating Cost	90,000

Goods are sold to the customers at 150% of direct costs.

50% of the assets being financed by borrowed capital at an interest cost of 8% per annum.

Tax rate is 30%.

You are required to calculate :

(i) Net profit margin

(ii) Return on Assets

(iii) Asset turnover

(iv) Return on owners' equity

ICAI'S ANSWER

Computation of net profit:

Particulars	(₹)
Sales (150% of ₹ 5,50,000)	8,25,000
Direct Costs	5,50,000
Gross profit	2,75,000
Other Operating Costs	90,000
Operating profit (EBIT)	1,85,000
Interest changes (8% of ₹ 5,00,000)	40,000
Profit before taxes (EBT)	1,45,000
Taxes (@ 30%)	43,500
Net profit after taxes (EAT)	**1,01,500**

(i) Net profit margin (After tax) = $\frac{\text{Profit after taxes}}{\text{Sales}} = \frac{₹\ 1,01,500}{₹\ 8,25,000}$ = **0.12303 or 12.303%**

Net profit margin (Before tax)= $\frac{\text{Profit before taxes}}{\text{Sales}} = \frac{₹\ 1,45,000}{₹\ 8,25,000}$ = **0.17576 or 17.576%**

(ii) Return on assets = $\frac{\text{EBIT (1 -T)}}{\text{Total Assets}} = \frac{₹\ 1,85,000\ (1 - 0.3)}{₹\ 10,00,000}$ = **0.1295 or 12.95%**

(iii) Asset turnover = $\frac{\text{Sales}}{\text{Assets}} = \frac{₹\ 8,25,000}{₹\ 10,00,000}$ = **0.825 times**

(iv) Return on owner's equity= $\frac{\text{Profit after taxes}}{\text{Owners equity}} = \frac{₹\ 1,01,500}{50\% \times ₹\ 10,00,000}$ = **0.203 or 20.3%**

ICAI · Old P8 · Nov 2020 exam · question p. 1 · answer p. 3

80 EXAM **Old P8 Nov 2019** Q1(a) 5 marks old scheme

Following information has been gathered from the books of Tram Ltd. the equity shares of which is trading in the stock market at ₹ 14.

Particulars	Amount (₹)
Equity Share Capital (face value ₹ 10)	10,00,000
10% Preference Shares	2,00,000
Reserves	8,00,000
10% Debentures	6,00,000
Profit before Interest and Tax for the year	4,00,000
Interest	60,000
Profit after Tax for the year	2,40,000

Calculate the following:

(i) Return on Capital Employed

(ii) Earnings per share

(iii) PE ratio

ICAI'S ANSWER

(i) Calculation of Return on capital employed (ROCE)

Capital employed = Equity Shareholders' funds + Debenture + Preference shares

= ₹ (10,00,000 + 8,00,000 + 6,00,000 + 2,00,000)

= ₹ 26,00,000

$$\text{Return on capital employed [ROCE-(Pre-tax)]} = \frac{\text{PBIT}}{\text{Capital Employed}} \times 100$$

$$= \frac{₹4,00,000}{₹26,00,000} \times 100$$

= 15.38% (approx.)

$$\text{Return on capital employed [ROCE-(Post-tax)]} = \frac{\text{Profit after Tax}}{\text{Capital Employed}} \times 100$$

$$= \frac{₹2,40,000}{₹26,00,000} \times 100$$

= 9.23% (approx.)

(ii) Calculation of Earnings per share

$$\text{Earnings per share} = \frac{\text{Earnings available to equity shareholders}}{\text{No of equity shares}}$$

$$= \frac{\text{Profit after tax-preference Dividend}}{\text{No of equity shares}}$$

$$= \frac{₹(2,40,000 - 20,000)}{₹1,00,000}$$

= ₹ 2.20

(iii) Calculation of PE ratio

$$\text{PE} = \frac{\text{Market Price per Share (MPS)}}{\text{Earning per Shares (EPS)}}$$

$$= \frac{₹14}{₹2.20} = 6.364 \text{ (approx.)}$$

ICAI · Old P8 · Nov 2019 exam · question p. 1 · answer p. 3

81 RTP Jan 2026 Q3 MCQ

As per extended Du Pont analysis:

ROE = NPM × TATR × x/y

Where, ROE = Return on Equity

NPM = Net Profit Margin

TATR = Total Asset Turnover Ratio

Here, x/y represents:

A Debt / Equity

B Total Assets / Sales

C Total Assets / Equity

D Equity / Debt

ICAI'S ANSWER

(C) Total Assets / Equity

ICAI · P6 · RTP Jan 2026 · question p. 3-4 · answer p. 14

82 RTP **Old P8 Dec 2021 Q1** old scheme

Following information has been gathered from the books of Cram Ltd. for the year ended 31st March 2021, the equity shares of which is trading in the stock market at ₹ 28:

Particulars	Amount (₹)
Equity Share Capital (Face value @ ₹ 20)	20,00,000
10% Preference Share capital	4,00,000
Reserves & Surplus	16,00,000
12.5% Debentures	12,00,000
Profit before Interest and Tax for the year	8,00,000

CALCULATE the following when company falls within 25% tax bracket:

(i) Return on Capital Employed

(ii) Earnings Per share

(iii) P/E Ratio

ICAI'S ANSWER

(i) Return on Capital Employed (ROCE)

$$\text{ROCE (Pre-tax)} = \frac{\text{Profit before interest and taxes (PBIT)}}{\text{Capital Employed}} \times 100$$

$$= \frac{₹\ 8,00,000}{₹\ 52,00,000} \times 100$$

= 15.38% (approx.)

$$\text{ROCE (Post-tax)} = \frac{\text{PBIT}(1 - t)}{\text{Capital Employed}} \times 100$$

$$= \frac{₹\ 8,00,000\ (1 - 0.25)}{₹\ 52,00,000} \times 100$$

= 11.54% (approx.)

(ii) Earnings Per share (EPS)

$$= \frac{\text{Profit available to equity share holders}}{\text{Number of equity shares outstanding}}$$

$$= \frac{₹\ 4,47,500}{1,00,000}$$

= ₹ 4.475

(iii) P/E Ratio

$$= \frac{\text{Market Price per Share (MPS)}}{\text{Earning per Share (EPS)}}$$

$$= \frac{₹\ 28}{₹\ 4.475}$$

= 6.26 times (approx.)

Workings:

(a) Income Statement

Particulars	Amount (₹)
Profit before Interest and Tax (PBIT)	8,00,000
Interest on Debentures (12.5% of ₹ 12,00,000)	(1,50,000)
Profit before Tax (PBT)	6,50,000
Tax @ 25%	(1,62,500)
Profit after Tax (PAT)	4,87,500
Preference Dividend (10% of ₹ 4,00,000)	(40,000)
Profit available to Equity shareholders	4,47,500

(b) Calculation of Capital Employed

= Equity Shareholder's Fund + Preference share Capital + Debentures

= (₹ 20,00,000 + ₹ 16,00,000) + ₹ 4,00,000 + ₹ 12,00,000 = ₹ 52,00,000

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83 RTP **Old P8 Jul 2021** Q1 old scheme

Given below are the estimations for the next year by Niti Ltd.:

Particulars	(₹ in crores)
Fixed Assets	5.20
Current Liabilities	4.68
Current Assets	7.80
Sales	23.00
EBIT	2.30

The company will issue equity funds of ₹ 5 crores in the next year. It is also considering the debt alternatives of ₹ 3.32 crores for financing the assets. The company wants to adopt one of the policies given below:

(₹ in crores)

Financing Policy	Short term debt @ 12%	Long term debt @ 16%	Total
Conservative	1.08	2.24	3.32
Moderate	2.00	1.32	3.32
Aggressive	3.00	0.32	3.32

Assuming corporate tax rate at 30%, CALCULATE the following for each of the financing policy:

(i) Return on total assets

(ii) Return on owner's equity

(iii) Net Working capital

(iv) Current Ratio

Also advise which Financing policy should be adopted if the company wants high returns.

ICAI'S ANSWER

(i) Return on total assets

$$\text{Return on total assets} = \frac{\text{EBIT } (1 - \text{T})}{\text{Total assets (FA + CA)}}$$

$$= \frac{₹\ 2.30 \text{ crores } (1- 0.3)}{₹\ 5.20 \text{ crores} + ₹\ 7.80 \text{ crores}}$$

$$= \frac{₹\ 1.61 \text{crores}}{₹\ 13 \text{ crores}} = 0.1238 \text{ or } 12.38\%$$

(ii) Return on owner's equity

(Amount in ₹)

	Financing policy (₹)		
	Conservative	**Moderate**	**Aggressive**
Expected EBIT	2,30,00,000	2,30,00,000	2,30,00,000
Less: Interest			
Short term Debt @ 12%	12,96,000	24,00,000	36,00,000
Long term Debt @ 16%	35,84,000	21,12,000	5,12,000
Earnings before tax (EBT)	1,81,20,000	1,84,88,000	1,88,88,000
Less: Tax @ 30%	54,36,000	55,46,400	56,66,400
Earnings after Tax (EAT)	1,26,84,000	1,29,41,600	1,32,21,600
Owner's Equity	5,00,00,000	5,00,00,000	5,00,00,000

	Financing policy (₹)		
Return on owner's equity = Net Profit after taxes (EAT)/Owners'equity	= 1,26,84,000/5,00,00,000 = 0.2537 or 25.37%	= 1,29,41,600/5,00,00,000 = 0.2588 or 25.88%	= 1,32,21,600/5,00,00,000 = 0.2644 or 26.44%

(iii) Net Working capital

(₹ in crores)

	Financing policy		
	Conservative	**Moderate**	**Aggressive**
Current Liabilities (Excluding Short Term Debt)	4.68	4.68	4.68
Short term Debt	1.08	2.00	3.00
Total Current Liabilities	5.76	6.68	7.68
Current Assets	7.80	7.80	7.80
Net Working capital = Current Assets - Current Liabilities	7.80 - 5.76 = 2.04	7.80 - 6.68 = 1.12	7.80 - 7.68 = 0.12

(iv) Current ratio

(₹ in crores)

	Financing policy		
	Conservative	**Moderate**	**Aggressive**
Current Ratio = Current Assets/Current Liabilities	= 7.80/5.76 = 1.35	= 7.80/6.68 = 1.17	= 7.80/7.68 = 1.02

Advise: It is advisable to adopt aggressive financial policy, if the company wants high return as the return on owner's equity is maximum in this policy i.e. 26.44%.

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84 RTP **Old P8 Nov 2019 Q1** old scheme

The following is the Profit and loss account and Balance sheet of KLM LLP.

Trading and Profit & Loss Account

Particulars	Amount (₹)	Particulars	Amount (₹)
To Opening stock	12,46,000	By Sales	1,96,56,000
To Purchases	1,56,20,000	By Closing stock	14,28,000
To Gross profit c/d	42,18,000		
	2,10,84,000		2,10,84,000

Particulars	Amount (₹)	Particulars	Amount (₹)
		By Gross profit b/d	42,18,000
To Administrative expenses	18,40,000	By Interest on investment	24,600
To Selling & distribution expenses	7,56,000	By Dividend received	22,000
To Interest on loan	2,60,000		
To Net profit	14,08,600		
	42,64,600		42,64,600

Balance Sheet as on.....

Capital & Liabilities	Amount (₹)	Assets	Amount (₹)
Capital	20,00,000	Plant & machinery	24,00,000
Retained earnings	42,00,000	Building	42,00,000
General reserve	12,00,000	Furniture	12,00,000
Term loan from bank	26,00,000	Sundry receivables	13,50,000
Sundry Payables	7,20,000	Inventory	14,28,000
Other liabilities	2,80,000	Cash & Bank balance	4,22,000
	1,10,00,000		1,10,00,000

You are required to COMPUTE:

(i) Gross profit ratio (ii) Net profit ratio (iii) Operating cost ratio

(iv) Operating profit ratio (v) Inventory turnover ratio (vi) Current ratio

(vii) Quick ratio (viii) Interest coverage ratio (ix) Return on capital employed

(x) Debt to assets ratio.

ICAI'S ANSWER

(i) Gross profit ratio = $\frac{\text{Gross profit}}{\text{Sales}} \times 100 = \frac{₹42,18,000}{₹1,96,56,000} \times 100 = 21.46\%$

(ii) Net profit ratio = $\frac{\text{Net profit}}{\text{Sales}} \times 100 = \frac{₹14,08,600}{₹1,96,56,000} \times 100 = 7.17\%$

(iii) Operating ratio = $\frac{\text{Operating cost}}{\text{Sales}} \times 100$

Operating cost = Cost of goods sold + Operating expenses

Cost of goods sold = Sales – Gross profit

= 1,96,56,000 - 42,18,000 = 1,54,38,000

Operating expenses = Administrative expenses + Selling & distribution expenses

= 18,40,000 + 7,56,000 = 25,96,000

Therefore, Operating ratio = $\frac{1,54,38,000 + 25,96,000}{1,96,56,000} \times 100$

= $\frac{1,80,34,000}{1,96,56,000} \times 100 = 91.75\%$

(iv) Operating profit ratio = 100 – Operating cost ratio

= 100 – 91.75% = 8.25%

(v) Inventory turnover ratio = $\frac{\text{Cost of goods sold}}{\text{Average stock}}$

= $\frac{1,54,38,000}{(14,28,000 + 12,46,000)/2}$

= $\frac{1,54,38,000}{13,37,000} = 11.55$ times

(vi) Current ratio = $\frac{\text{Current assets}}{\text{Current liablities}}$

Current assets = Sundry receivables + Inventory + Cash & Bank balance

= 13,50,000 + 14,28,000 + 4,22,000 = 32,00,000

Current liabilities = Sundry Payables + Other liabilities

= 7,20,000 + 2,80,000 = 10,00,000

Current ratio = $\frac{32,00,000}{10,00,000} = 3.2$ times

(vii) Quick Ratio = $\frac{\text{Current assets} - \text{Inventories}}{\text{Current liablities}}$

= $\frac{32,00,000 - 14,28,000}{10,00,000}$ =1.77 times

(viii) Interest coverage ratio = $\frac{\text{EBIDT}}{\text{Interest}} = \frac{\text{Net profit} + \text{Interest}}{\text{Interest}}$

= $\frac{14,08,600 + 2,60,000}{2,60,000} = 6.42$ times

(ix) Return on capital employed (ROCE) = $\frac{\text{EBIT}}{\text{Capital employed}} \times 100$

Capital employed = Capital + Retained earnings + General reserve + Term loan

= 20,00,000 + 42,00,000 + 12,00,000 + 26,00,000

= 1,00,00,000

Therefore, ROCE = $\frac{16,68,600}{1,00,00,000} \times 100 = 16.69\%$

(x) Debt to assets ratio = $\frac{\text{Debts}}{\text{Total assets}} \times 100 = \frac{26,00,000}{1,10,00,000} \times 100$ =23.64%

ICAI · Old P8 · RTP Nov 2019 · question p. 1-2 · answer p. 6-7

85 MTP **Jan 2026 S-II** Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

Virat Ltd., a steadily growing company in Mumbai, has been performing well in its current line of business. The management believes the timing is right to expand, and they have approached you—the company's financial analyst—to evaluate whether their planned new venture is financially sound.

At present, the company earns ₹ 20,00,000 in annual revenue. To generate this income, Virat Ltd. incurs several operating costs. About 40% of the revenue, or ₹ 8,00,000, goes toward the cost of goods

sold. Additionally, the company spends ₹ 2,00,000 on employee salaries, ₹ 50,000 on electricity and other overheads, and another ₹ 50,000 on advertising and promotional activities. After all these expenses, the company pays 35% tax on its profits.

To run the existing business, Virat Ltd. uses a total capital of ₹ 1.5 crore. This capital is funded partly through equity of ₹ 80,00,000, represented by 8,00,000 shares with a face value of ₹ 10 each, and partly through debt of ₹ 70,00,000, on which the company pays 9.95% interest per year.

Since the company is listed on the NSE, its performance is compared with the market index, which has produced a return of 14.95%. The company carries a beta of 1.10, indicating slightly higher risk than the market. For risk-free benchmarks, the latest RBI T-Bill rate is 5.45%.

Virat Ltd. is evaluating expansion into a new product line. Management has already conducted preliminary research and provided expected revenue and cost estimates for the next five years.

Expected Performance of the New Project

Year	2026	2027	2028	2029	2030
Revenue	9,50,000	10,45,000	11,49,500	12,64,450	13,90,895
COGS %	40%	42%	45%	47%	50%
COGS (₹)	3,80,000	4,38,900	5,17,275	5,94,291.5	6,95,447.5
Gross Profit	5,70,000	6,06,100	6,32,225	6,70,158.5	6,95,447.5
Employee Costs	45,000	47,250	49,612.5	52,093.13	54,697.78
Electricity/Other	5,000	5,100	5,202	5,306.04	5,412.16
Advertising	15,000	15,150	15,301.5	15,454.52	15,609.06

To begin this project, Virat Ltd. will need to invest ₹ 15,00,000 in fixed assets, with a useful life of 10 years and terminal (sale) value at the end of 5th as ₹ 7,50,000. This investment will be financed through a mix of ₹ 5,00,000 in equity (50,000 new shares at ₹ 10 each) and ₹ 10,00,000 in debt. For this company issued 10,000, 10% debentures of ₹ 100 each at a premium of 10% on 1.4.2026 to be matured after 5 years The debentures will be redeemed at par on maturity.

You are now required to analyse both the company's current performance and the financial feasibility of the new project using the information provided and answer the following questions (MCQ 1 to 5):

MCQ 1 2 marks

Calculate the profit or loss for the current year

A ₹ 1,52,500

B ₹ 1,42,500

C ₹ 1,38,500

D ₹ 1,32,275

MCQ 2 2 marks

Calculate the Return on Equity (ke) for the firm.

A 15.99%

B 15.90%

C 15.00%

D 14.95%

MCQ 3 2 marks

What us the Weighted Average Cost of Capital (WACC) to be considered for valuing the firm in current year.

A 12.73%

B 11.50%

C 9.97%

D 10.97%

MCQ 4 2 marks

Compute the cost of new debentures assuming 35% as tax rate.

A 4.28%

B 6.50%

C 10.00%

D 6.97%

MCQ 5 2 marks

What is NPV of the New Project using the discount rate of 12%?

A ₹ 3,50,000

B (₹ 3,118)

C ₹ 4,22,133

D ₹ 3,118

ICAI'S ANSWERS

MCQ 1

(D) ₹ 1,32,275

Particulars	Amount (₹)
Revenues	₹ 20,00,000
Less: COGS @ 40%	₹ 8,00,000
Gross Profit	₹ 12,00,000
Employee cost	₹ 2,00,000
Electricity and other cost	₹ 50,000
Other advt	₹ 50,000
Finance cost	₹ 6,96,500
Total Expenses	₹ 9,96,500
PBT	₹ 2,03,500

Particulars	Amount (₹)
Tax @35%	₹ 71,225
PAT	₹ 1,32,275

MCQ 2

(B) 15.90%

Cost of Equity using CAPM

Ke = Rf + (Rm – Rf) Beta

= 5.45% + (14.95%-5.45%)1.1

= 15.90%

MCQ 3

(B) 11.50%

Calculation of WACC

Source	Capital	Weight (W)	Cost before tax	After tax Cost (K)	WACC (W x C)
Equity	80,00,000	0.533	15.90%	15.90%	8.48%
Debt	70,00,000	0.467	9.95%	6.47%	3.02%
	1,50,00,000				11.50%

MCQ 4

(A) 4.28%

$$\text{Cost of debenture } (K_d) = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

I = Interest on debenture = 10% of ₹100 = ₹10

NP = Net Proceeds = 110% of ₹100 = ₹110

RV = Redemption value = ₹100

n = Period of debenture = 5 years

t = Tax rate = 35% or 0.35

$$K_d = \frac{₹10(1-0.35) + \frac{(₹100 - ₹110)}{5 \text{ years}}}{\frac{(₹100 + ₹110)}{2}}$$

$$\text{Or, } K_d = \frac{(₹10 \times 0.65) - ₹2}{₹105} = \frac{₹4.5}{₹105} = 0.0428 \text{ or } 4.28\%$$

MCQ 5

(C) ₹ 4,22,133

Calculation of Cash Inflows

Year	2026	2027	2028	2029	2030
Revenue	9,50,000	10,45,000	11,49,500	12,64,450	13,90,895
COGS %	40%	42%	45%	47%	50%
COGS (₹)	3,80,000	4,38,900	5,17,275	5,94,292	6,95,447
Gross Profit	5,70,000	6,06,100	6,32,225	6,70,159	6,95,448
Employee Costs	45,000	47,250	49,613	52,093	54,698
Electricity/Other	5,000	5,100	5,202	5,306	5,412
Advertising	15,000	15,150	15,302	15,455	15,609
Depreciation	1,50,000	1,50,000	1,50,000	1,50,000	1,50,000
Profit before tax	3,55,000	3,88,600	4,12,109	4,47,305	4,69,729
Less: Tax	1,24,250	1,36,010	1,44,238	1,56,557	1,64,405
Profit After Tax	2,30,750	2,52,590	2,67,871	2,90,748	3,05,324
Add: Depreciation	1,50,000	1,50,000	1,50,000	1,50,000	1,50,000
Net Cashflow	3,80,750	4,02,590	4,17,871	4,40,748	4,55,324 + 7,50,000 (TV of Asset)
PV Factor @12%	0.893	0.797	0.712	0.636	0.567
PV	3,40,010	3,20,864	2,97,524	2,80,316	6,83,419

NPV = 19,22,133 – 15,00,000 = ₹ 4,22,133

ICAI · P6 · MTP Jan 2026 · Series II · question p. 1-2 · answer p. 1, 2, 2-3

86 MTP **Old P8 Nov 2019 S-I** Q1(c) 5 marks old scheme

MNP Limited has made plans for the year 2019 -20. It is estimated that the company will employ total assets of Rs.50,00,000; 30% of assets being financed by debt at an interest cost of 9% p.a. The direct costs for the year are estimated at Rs. 30,00,000 and all other operating expenses are estimated at Rs. 4,80,000. The sales revenue are estimated at Rs. 45,00,000. Tax rate is assumed to be 40%. CALCULATE:

(i) Net profit margin (After tax);

(ii) Return on Assets (After tax);

(iii) Asset turnover; and

(iv) Return on Equity.

ICAI'S ANSWER

The net profit is calculated as follows:

	Rs.
Sales Revenue	45,00,000
Less: Direct Costs	30,00,000
Gross Profits	15,00,000
Less: Operating Expense	4,80,000
Earnings before Interest and tax (EBIT)	10,20,000
Less: Interest on debt (9% × 15,00,000)	1,35,000
Earnings before Tax) (EBT)	8,85,000
Less: Taxes (@ 40%)	3,54,000
Profit after Tax (PAT)	5,31,000

(i) Net Profit Margin (After Tax)

$$\text{Net Profit Margin} = \frac{\text{EBIT } (1 - t)}{\text{Sales}} \times 100 = \frac{\text{Rs.10,20,000} \times (1 - 0.4)}{\text{Rs.45,00,000}} = 13.6\%$$

(ii) Return on Assets (ROA) (After tax)

$$\text{ROA} = \frac{\text{EBIT } (1-t)}{\text{Total Assets}}$$

$$= \frac{\text{Rs.10,20,000 } (1 - 0.4)}{\text{Rs.50,00,000}} = \frac{\text{Rs.6,12,000}}{\text{Rs.50,00,000}}$$

= 0.1224 = 12.24 %

(iii) Asset Turnover

$$\text{Asset Turnover} = \frac{\text{Sales}}{\text{Assets}} = \frac{\text{Rs. 45,00,000}}{\text{Rs.50,00,000}} = 0.9$$

Asset Turnover = 0.9 times

(iv) Return on Equity (ROE)

$$\text{ROE} = \frac{\text{PAT}}{\text{Equity}} = \frac{\text{Rs.5,31,000}}{\text{Rs.35,00,000}} = 15.17\%$$

ROE = 15.17%

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 1 · answer p. 2-3

2.4 Building financial statements from ratios; limitations of ratios

51 questions · asked in 47 ICAI documents · 12 exam papers

87 EXAM Sep 2026 Q1(b) 5 marks

GPL Company Limited has furnished the following ratios and information for the year ended 31st March, 2026.

Inventory Turnover Ratio (Based on cost of goods sold)	8 times
Trade Debtors Velocity	36.5 days
Gross Profit Ratio	25%
Debt Equity Ratio	1:3
Capital Gearing Ratio	40%
Preference Share Capital	₹ 7,50,000

Additional information :

All sales were on credit. Average Trade Debtors exceeded Average Inventory by ₹ 1,81,250. Assume 365 days in a year.

You are required to calculate :

(i) Credit sales

(ii) Cost of goods sold

(iii) Average inventory

(iv) Average Debtors

(v) Proprietary Fund

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 14-15

88 EXAM Jan 2026 Q3(a) 8 marks

The Balance Sheet extract of ST Ltd. as on March 31 (current year) is as follows:

Liabilities	Amount (₹ in lakh)	Assets	Amount (₹ in lakh)
Share capital	260	Fixed Assets	600
Reserves	168	Inventories	360
Long-term loans	420	Receivables	270
Short-term loans	220	Cash & bank	78
Payables	144		
Provisions	96		
	1,308		1,308

Sales for the current year amounted to ₹ 900 lakh. For the next financial year ending March 31 (next year), sales are expected to increase by 20%.

The net profit margin (after tax) is expected to be 5% of sales, and the dividend payout ratio is anticipated to be 50%.

Total assets including cash & bank and current liabilities (payables and provisions) vary directly with sales.

You are required to:

(i) Estimate the amount of external funds required (EFR) for the next financial year.

(ii) Determine the sources and composition of funds to meet the requirement, given the following parameters:

(1) The current ratio must be maintained at 1.40.

(2) The ratio of fixed assets to long-term loans should be 1.50.

(3) The remaining fund is raised by issue of Equity shares.

(4) The additional funds are to be raised in the following order of preference:

(i) Short-term bank borrowings

(ii) Long-term loans

(iii) Equity

ICAI'S ANSWER

(i) External Funds Requirement (EFR)

Particulars	₹ in lakhs
Expected sales (₹ 900 + 20% of ₹ 900)	1,080.00
Net profit margin @ 5%	54.00
Dividend payout ratio @ 50%	27.00
Balance to be ploughed back (A)	**27.00**
Additional assets required [(₹1,308 – (144+96)*] × 20% (B)	213.60
Balance to be met from external sources (B – A)	**186.60**

*As total assets and current liabilities (payables & provisions) increase proportionately with sales.

Calculation of External Funds Requirement (EFR) can be presented in the following way-

Particulars	₹ in lakhs
A. Projected Level of Assets (1308 × 1.20)	1,569.60
B. Projected Level of Payables & Provisions (144+96) x 1.20	288.00
C. Retained Earnings (900 × 1.20 × 0.05 × 0.50	27.00
D. Existing (Share Cap. + Reserves + LTL + STL) (260+168+420+220)	1068.00
External Funds Requirement (EFR): A - (B+C+D)	186.60

(ii) Amount to be raised from different sources subject to given conditions:

(1) Amount to be raised from Short-Term Bank Borrowings

Particulars	₹ in lakhs
A. New Current Assets (708 lakhs + 20% of 708 lakhs)	849.60
B. New current liabilities as per given condition	606.86
C. *Less:* Existing current liabilities (240 lakhs + 20% of 240 lakhs+220 lakhs)	508
D. Additional funds to be raised from Short-term bank borrowings (B-C)	98.86

To maintain Current Ratio = 1.40

Required Current Liabilities = $\frac{849.60}{1.40}$ = 606.86 lakhs

(2) Amount to be raised from Long-Term Loans

Particulars	₹ in lakhs
Existing fixed assets	600.00
Increase @ 20%	120.00
New fixed assets	**720.00**

To maintain fixed assets to long-term loans = 1.50:

Long-term loans required = $\frac{720}{1.5}$ = 480.00 lakhs

Particulars	₹ in lakhs
Required long-term loans	480.00
Less: Existing long-term loans	420.00
Amount to be raised from long-term funds	**60.00**

(3) Amount to be raised from Equity Shares

Particulars	₹ in lakhs
External funds required	186.60
Less: From short-term funds	98.86
Less: From long-term funds	60.00
Balance to be raised by issue of Equity Shares	**27.74**

ICAI · P6 · Jan 2026 exam · question p. 19-20 · answer p. 20-22

89 EXAM May 2025 Q1(a) 5 marks

The following information is available for S Ltd. for the year ended 31st March, 2025:

Raw Material consumed	20% of COGS
Raw Material Inventory turnover ratio	4.00
Finished Goods Inventory holding period	0.75 month
Gross profit (based on COGS)	12.50%
Debtor collection period (all sales are credit sales)	3 months
Proprietary ratio	0.3125
Fixed Assets turnover ratio (based on sales)	3.00
Fixed Assets to Total Assets	40%

You are required to prepare a Balance Sheet as on 31st March, 2025 in the following format:

Liabilities	₹	Assets	₹
Shareholders' Fund (assume no Preference Shares)		Fixed Assets	12,00,000
Long-term Debt	15,00,000	Stock of Raw Material	
Current Liabilities		Stock of Finished Goods	
		Debtors	
		Cash	
Total		Total	

ICAI'S ANSWER

Workings:

(i) Calculation of Sales, Cost of Goods Sold and Gross Profit

$$\text{Fixed Assets Turnover} = \frac{\text{Sales}}{\text{Fixed Assets}}$$

$$3 = \frac{\text{Sales}}{12,00,000}$$

So, Sales = 12,00,000 x 3 = ₹ 36,00,000

$$\text{And Cost of Goods Sold} = 36,00,000 \times \frac{100}{(100 + 12.5)} = ₹\ 32,00,000$$

Therefore, Gross Profit = 32,00,000 x 12.5% = ₹ 4,00,000

(ii) Calculation of Stock of Raw Materials

Raw Material Consumed = COGS x 20%

= 32,00,000 x 20%

= ₹ 6,40,000

So, stock of raw material = $\frac{6,40,000}{4}$ = ₹ 1,60,000

(iii) Calculation of Debtors

Debtor Collection Period = $\frac{12}{\text{Receivable Turnover Ratio}}$ = 3 months

Or Debtors = 36,00,000 X $\frac{3}{12}$

= ₹ 9,00,000

(iv) Calculation of Total Assets

Total Asset = $\frac{\text{Fixed Assets}}{40\%}$

= 12,00,000 x $\frac{100}{40}$ = ₹ 30,00,000

So Total Liabilities including shareholder's fund = ₹ 30,00,000

(v) Calculation of Stock of Finished Goods

Stock of Finished Goods = 32,00,000 x $\frac{0.75}{12}$ = ₹ 2,00,000

(vi) Calculation of Current Assets and Fixed Assets

Current Assets = Total Assets – Fixed Assets

= 30,00,000 – 12,00,000 = ₹ 18,00,000

And Cash Balance = Current Assets – (Raw Material + Finished Goods + Debtors)

= 18,00,000 – (1,60,000 + 2,00,000 + 9,00,000)

= ₹ 5,40,000

(vii) Calculation of Shareholder's Fund

Proprietary Ratio = $\frac{\text{Shareholders Fund}}{\text{Total Assets}}$

0.3125 = $\frac{\text{Shareholders Fund}}{30,00,000}$

Shareholders Fund = ₹ 9,37,500

Balance Sheet as on 31st March 2025

Liabilities	(₹)	Assets	(₹)
Shareholders' Fund	9,37,500	Fixed Assets	12,00,000
Long-term Debt	15,00,000	Stock of Raw Material	1,60,000
Current Liabilities (Balancing Figure)	5,62,500	Stock of Finished Goods	2,00,000
		Debtors	9,00,000
		Cash	5,40,000
	30,00,000		**30,00,000**

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90 EXAM **Jan 2025** Case Scenario - II · MCQ 6 to MCQ 7

CASE SCENARIO - II

VP Ltd. provides the following financial information:

Current Ratio	1.5:1
Sales (80% Credit Sales)	₹ 150 Lakh
Inventory Turnover Ratio	6 Times
Average Collection Period	2 months
Gross Profit Ratio	20%
Quick Ratio	1:1

From the information given above, choose the correct answer to the following Q. No. 6 and 7:

MCQ 6

Inventory and Receivables would be:

A ₹ 20 Lakh and ₹ 20 Lakh

B ₹ 10 Lakh and ₹ 10 Lakh

C ₹ 10 Lakh and ₹ 20 Lakh

D ₹ 20 Lakh and ₹ 10 Lakh

MCQ 7

Current Assets and Current Liabilities would be:

A 30 Lakh and 20 Lakh

B 45 Lakh and 30 Lakh

C 60 Lakh and 40 Lakh

D 75 Lakh and 50 Lakh

ICAI'S ANSWERS

MCQ 6

Answer key: **(A)**

MCQ 7

Answer key: **(C)**

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91 EXAM **Jan 2025** Q3(b) 4 marks

The equity share capital of Sky Pack Ltd. as on 31st March, 2024 was ₹ 2,00,000. The relevant ratios of the

company are as follows:

Current debt to Total debt	0.35
Total debt to Owner's equity	0.65
Fixed assets to Owner's equity	0.55
Total assets turnover	2.5 times
Inventory turnover	10 times

You are required to prepare the Balance Sheet of Sky Pack Ltd. as on 31st March, 2024.

ICAI'S ANSWER

Sky Pack Ltd

Balance Sheet

Liabilities	₹	Assets	₹
Equity share capital	2,00,000	Fixed assets	1,10,000
Current debt	45,500	Cash (balancing figure)	1,37,500
Long term debt	84,500	Inventory	82,500
	3,30,000		3,30,000

Working Notes

1. Total debt = 0.65 x Equity share capital

= 0.65 x ₹ 2,00,000 = ₹ 1,30,000

Current debt to total debt = 0.35.

Current debt = 0.35 × ₹ 1,30,000 = ₹45,500

Long term debt = ₹1,30,000 - ₹45,500= ₹ 84,500

2. Fixed assets = 0.55 × Equity share Capital

= 0.55 × ₹ 2,00,000 = ₹ 1,10,000

3. Total assets to turnover = 2.5 Times : Inventory turnover = 10 Times

Hence, Inventory /Total assets = 2.5/10=1/4, Total assets = ₹ 3,30,000

Therefore Inventory = ₹ 3,30,000/4 = ₹ 82,500

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92 EXAM Sep 2024 Q1(c) 5 marks

Following information relates to MNP Limited for the year ended on 31s March, 2024:

Inventory turnover ratio (based on cost of goods sold)	7.5 times
Total assets turnover ratio	2.5 times
Long term debt to Shareholders' fund	0.6:1
Debtors collection period	30 days
Gross profit ratio	25% on sales
Current Ratio	2.9:1

Balance Sheet as on 31st March,2024

Liabilities	₹	Assets	₹
Equity share capital	6,00,000	Fixed Assets	?
Reserves & Surplus	3,00,000	Inventories	?
Long term debt	?	Debtors	?
Creditors	3,00,000	Cash	?
Total		Total	

You are required to complete the Balance Sheet of MNP Limited as on 31st March,2024. Assume a 360 days year and all sales are credit sales.

ICAI'S ANSWER

Working Notes:

Long term debt to Shareholder's fund = 0.6:1

Long term debt = 0.6 x ₹ 9,00,000 = ₹ 5,40,000

Total Assets = ₹ 17,40,000

Total Asset turnover ratio = $\frac{\text{Sales}}{\text{TotalAssets}}$ = 2.5 times

Sales = 2.5 x ₹ 17,40,000 = ₹ 43,50,000

Current ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$ = 2.9:1

Current Assets = 2.9 x ₹ 3,00,000 = ₹ 8,70,000

Fixed Assets = Total Assets - Current Assets

= ₹ 17,40,000 – ₹ 8,70,000 = ₹ 8,70,000

Gross profit ratio = 25% on sales

Gross Profit (GP) = ₹ 43,50,000 x 0.25 = ₹ 10,87,500

Cost of Good Sold (COGS) = Sales – GP

= ₹ 43,50,000 - ₹ 10,87,500

= ₹ 32,62,500

$$\text{Inventory Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}} = 7.5 \text{ times}$$

Inventory = ₹ 32,62,500/7.5 = ₹ 4,35,000

$$\text{Debtor Collection Period} = \frac{\text{Average Accounts Receivables}}{\text{Average Daily Credit Sales}} = 30 \text{ days}$$

Receivables = 30 days x ₹ 43,50,000/360 days = ₹ 3,62,500

Balance Sheet as on 31st March 2024

Liabilities	(₹)	Assets	(₹)
Share Capital	6,00,000	Fixed Assets	8,70,000
Reserve and Surplus	3,00,000	Inventories	4,35,000
Long-term loan	5,40,000	Debtors	3,62,500
Creditors	3,00,000	Cash (Balancing Figure)	**72,500**
	17,40,000		17,40,000

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93 EXAM **May 2024** Q1(a) 5 marks

Theme Ltd provides you the following information:

12.5 % Debt	₹ 45,00,000
Debt to Equity ratio	1.5 : 1
Return on Shareholder's fund	54%
Operating Ratio	85%
Ratio of operating expenses to Cost of Goods sold	2:6
Tax rate	25%
Fixed Assets	₹ 39,00,000
Current Ratio	1.8 : 1

You are required to calculate:

(i) Interest Coverage Ratio

(ii) Gross Profit Ratio

(iii) Current Assets

ICAI'S ANSWER

Working Notes:

Debt = ₹ 45,00,000

Interest = ₹ 45,00,000 x 12.5%= 5,62,500

Debt to Equity = 1.5:1 = $\frac{\text{Total Debt}}{\text{Shareholders' Equity}}$

Equity = ₹ 30,00,000

Return of Shareholder's funds = 54% = $\frac{\text{Net Profit after taxes}}{\text{Equity shareholders' fund}} \times 100$

Profit after tax (PAT) = 54% x Equity = ₹16,20,000

Profit before tax (PBT)(1-25%) = Profit after tax

= ₹16,20,000/75% = ₹21,60,000

Earning before interest and tax (EBIT) = PBT + Interest

= ₹21,60,000 + ₹ 5,62,500

= ₹27,22,500

(i) Interest Coverage Ratio = EBIT/Interest

= ₹27,22,500/₹5,62,500

= 4.84 Times

(ii) Operating Profit Ratio = 1 - Operating Ratio

= 1 – 0.85 = 0.15 or 15%

0.15 = $\frac{\text{Operating Profit}}{\text{Sales}} \times 100$

Sales = EBIT or Operating Profit /0.15

= ₹ 27,22,500 / 0.15

= ₹ 1,81,50,000

Operating ratio = $\frac{\text{Operating expenses}}{\text{Cost of goods sold (COGS)}}$ = 2 : 6 = 1 : 3

Operating expenses = 1/3COGS

Operating cost = Sales – Operating profit

= ₹ 1,81,50,000 - ₹ 27,22,500

= ₹ 1,54,27,500

₹ 1,54,27,500 = COGS + Operating expenses

₹ 1,54,27,500 = COGS + 1/3COGS

COGS = ₹ 1,15,70,625

Gross profit = Sales – COGS

= 1,81,50,000 – 1,15,70,625

= ₹ 65,79,375

Gross Profit ratio = $\frac{\text{Gross Profit}}{\text{Sales}} \times 100$

= 65,79,375/1,81,50,000

= 0.3625 or 36.25%

Gross profit and sales can be calculated in alternative way also. However, there will be no change in GP ratio i.e 36.25%

(iii) Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$

= 1.8

Current Assets = 1.8 Current Liabilities

Total of Balance sheet liability = Equity + Debt + Current Liabilities

=30,00,000+45,00,000+CL .....(2)

Total Balance sheet asset = Fixed Assets + Current Assets

= 39 lakhs + CA= 39 + 1.8CL.....(3)

Equating 2 and 3,

75,00,000 + CL = 39,00,000 + 1.8CL

0.8CL = 36,00,000

CL =₹ 45,00,000

Current Assets = 1.8 CL = 1.8 x 45 lakhs= ₹ **81,00,000**

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94 EXAM **Old P8 Nov 2023** Q1(a) 5 marks old scheme

You are available with following information of Brave Ltd:

Debtor's velocity	3 months
Stock velocity	6 months
Creditor's velocity	2 months
Gross profit ratio	20%

The gross profit for the year ended 31st March,2023 was ₹ 10,00,000. Stock for the same period was ₹ 40,000 more than what it was at the beginning of the year. Bills receivable were ₹ 1,20,000.

Form the above information you are required to calculate:

(i) Sales

(ii) Sundry debtors

(iii) Closing stock

ICAI'S ANSWER

(i) Determination of Sales:

Gross Profit Ratio = $\frac{\text{Gross Profit}}{\text{Sales}} \times 100$

Or, $\frac{20}{100} = \frac{₹\ 10,00,000}{\text{Sales}}$

Or, Sales = $\frac{10,00,00,000}{20}$ = ₹ 50,00,000

Cost of Goods Sold = Sales – Gross Profit

= ₹ 50,00,000 - ₹ 10,00,000 = ₹ 40,00,000

(ii) Determination of Sundry Debtors:

Debtors' velocity is 3 months or Debtors' collection period is 3 months,

So, Debtors' turnover ratio = $\frac{12\text{months}}{3\text{months}} = 4$

Debtors' turnover ratio = $\frac{\text{Credit Sales}}{\text{Average Accounts Receivable}}$

= $\frac{₹\ 50,00,000}{\text{Bills Receivable} + \text{Sundry Debtors}} = 4$

Or, Sundry Debtors + Bills receivable = ₹ 12,50,000

Sundry Debtors = ₹ 12,50,000 – ₹ 1,20,000 = ₹ 11,30,000

(iii) Determination of Closing Stock

Stock velocity is 6 months so Stock Turnover Ratio=2

Stock Turnover Ratio = $\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{₹\ 40,00,000}{\text{Average Stock}}$ =2

So, Average Stock = ₹ 20,00,000

Now Average Stock = $\frac{\text{Opening Stock} + \text{Closing Stock}}{2}$

Or $\frac{\text{Opening Stock} + (\text{Opening Stock} + ₹\ 40,000)}{2}$ = ₹ 20,00,000

Or, Opening Stock + ₹ 20,000 = ₹ 20,00,000

Or, Opening Stock = ₹ 19,80,000

So, Closing Stock = ₹ 19,80,000 + ₹ 40,000 = ₹ 20,20,000

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95 EXAM **Old P8 May 2023** Q2 10 marks old scheme

Following information and ratios are given in respect of AQUA Ltd. for the year ended 31st March, 2023:

Current ratio	4.0
Acid test ratio	2.5
Inventory turnover ratio (based on sales)	6
Average collection period (days)	70
Earnings per share	₹ 3.5
Current liabilities	₹ 3,10,000
Total assets turnover ratio (based on sales)	0.96
Cash ratio	0.43
Proprietary ratio	0.48
Total equity dividend	₹ 1,75,000

Equity dividend coverage ratio	1.60

Assume 360 days in a year.

You are required to complete Balance Sheet as on 31st March, 2023.

Balance Sheet as on 31st March, 2023.

Liabilities	₹	Assets	₹
Equity share capital (₹10 per share)	XXX	Fixed assets	XXX
Reserves & surplus	XXX	Inventory	XXX
Long-term debt	XXX	Debtors	XXX
Current liabilities	3,10,000	Loans & advances	XXX
		Cash & bank	XXX
Total	XXX	Total	XXX

ICAI'S ANSWER

(i) Current Ratio = 4

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = 4$$

$$\therefore \frac{\text{Current Assets}}{3,10,000} = 4$$

∴ Current Assets = ₹ 12,40,000

(ii) Acid Test Ratio = 2.5

$$\frac{\text{Current Assets - Inventory}}{\text{Current Liabilities}} = 2.5$$

$$\therefore \frac{12,40,000 - \text{Inventory}}{3,10,000} = 2.5$$

∴ 12,40,000 – Inventory = ₹ 7,75,000

Inventory = ₹ 4,65,000

(iii) Inventory Turnover Ratio (on Sales) = 6

$$\frac{\text{Sales}}{\text{Inventory}} = 6$$

$$\frac{\text{Sales}}{4,65,000} = 6$$

∴ Sales = ₹ 27,90,000

(iv) Debtors Collection Period = 70 days

∴ (Debtors / sales) x 360 = 70

∴ (Debtors / 27,90,000) x 360 = 70

Debtors = ₹ 5,42,500

(v) Total Assets Turnover Ratio (on Sales) = 0.96

$$\therefore \frac{\text{Sales}}{\text{Total Assets}} = 0.96$$

$\therefore \frac{27,90,000}{\text{Total Assets}} = 0.96$

Total Assets = ₹ 29,06,250

(vi) Fixed Assets (FA) = Total Assets – Current Assets

= 29,06,250 – 12,40,000

Fixed Assets = ₹ 16,66,250

(vii) Cash Ratio = $\frac{\text{Cash}}{\text{Current Liabilities}} = 0.43$

$\therefore \frac{\text{Cash}}{3,10,000} = 0.43$

$\therefore$ Cash = ₹ 1,33,300

(viii) Proprietary Ratio = $\frac{\text{Proprietary Fund}}{\text{Total Assets}} = 0.48$

$\therefore \frac{\text{Proprietary Fund}}{29,06,250} = 0.48$

$\therefore$ Proprietary Fund = ₹ 13,95,000

(ix) Equity Dividend Coverage Ratio = 1.6

or $\frac{\text{EPS}}{\text{DPS}} = \frac{3.5}{\text{DPS}}$

$\therefore$ DPS = ₹ 2.1875

DPS = $\frac{\text{Total Dividend}}{\text{Number of Equity Shares}}$

$\therefore 2.1875 = \frac{1,75,000}{\text{Number of Equity Shares}}$

$\therefore$ Number of Equity Shares = 80,000

$\therefore$ Equity Share Capital = 80,000 x 10 = ₹ 8,00,000

$\therefore$ Reserves &Surplus = 13,95,000 - 8,00,000 = ₹ 5,95,000

(x) Loans and Advances = Current Assets - (Inventory + Receivables + Cash & Bank)

= ₹ 12,40,000 - (₹ 4,65,000 + 5,42,500 + 1,33,300) = ₹ 99,200

Balance Sheet as on 31st March 2023

Liabilities	₹	Assets	₹
Equity Share Capital (₹ 10 per share)	8,00,000	Fixed Assets	16,66,250
Reserves & Surplus	5,95,000	Inventory	4,65,000
Long-term debt *(B/F)	12,01,250	Receivables	5,42,500
Current Liabilities	3,10,000	Loans & Advances	99,200
		Cash & Bank	1,33,300
Total	**29,06,250**	**Total**	**29,06,250**

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96 EXAM Old P8 May 2022 Q1(a) 5 marks old scheme

Following information and ratios are given for W Limited for the year ended 31st March, 2022:

Equity Share Capital of ₹ 10 each	₹ 10 lakhs
Reserves & Surplus to Shareholders' Fund	0.50
Sales / Shareholders' Fund	1.50
Current Ratio	2.50
Debtors Turnover Ratio	6.00
Stock Velocity	2 Months
Gross Profit Ratio	20%
Net Working Capital Turnover Ratio	2.50

You are required to calculate:

(i) Shareholders' Fund

(ii) Stock

(iii) Debtors

(iv) Current liabilities

(v) Cash Balance.

ICAI'S ANSWER

(i) Calculation of Shareholders' Fund:

$$\frac{\text{Reserve \& Surplus}}{\text{Shareholders' Funds}} = 0.5$$

$$\frac{\text{Reserve \& Surplus}}{\text{Equity Share Capital} + \text{Reserve \& Surplus}} = 0.5$$

$$\frac{\text{Reserve \& Surplus}}{10{,}00{,}000 + \text{Reserve \& Surplus}} = 0.5$$

Reserve & Surplus = 5,00,000 + 0.5 Reserve & Surplus

0.5 Reserve & Surplus = 5,00,000

Reserve & Surplus = 10,00,000

Shareholders' funds = 10,00,000 +10,00,000

Shareholders' funds = ₹ 20,00,000

(ii) Calculation of Value of Stock:

$$\frac{\text{Sales}}{\text{Shareholders' Funds}} = 1.5$$

Sales = 1.5 × 20,00,000

Sales = 30,00,000

Gross Profit = 30,00,000 × 20% = 6,00,000

Cost of Goods Sold = 30,00,000 – 6,00,000

= ₹ 24,00,000

Stock velocity = 2 months

$$\frac{\text{Average Stock}}{\text{Cost of Goods Sold}} \times 12 = 2$$

$$\frac{\text{Average Stock}}{24,00,000} \times 12 = 2$$

$$\text{Average Stock} = 24,00,000 \times \frac{2}{12}$$

Average stock = ₹ 4,00,000

(iii) Calculation of Debtors:

Debtors Turnover Ratio = 6

$$\therefore \frac{\text{Sales}}{\text{Average Debtor}} = 6$$

$$\therefore \frac{30,00,000}{\text{Average Debtor}} = 6$$

Average Debtors = ₹ 5,00,000

(iv) Calculation of Current Liabilities:

Net Working Capital Turnover ratio = 2.5

$$\frac{\text{Sales}}{\text{Current Assets} - \text{Current Liabilites}} = 2.5$$

$$\frac{30,00,000}{\text{Current Assets} - \text{Current Liabilites}} = 2.5$$

Current Assets – Current Liabilities = 12,00,000 ..... (1)

Current Ratio = 2.5

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = 2.5$$

Current Assets = 2.5 Current Liabilities .....(2)

From (1) & (2),

2.5 Current Liabilities – Current Liabilities = 12,00,000

1.5 Current Liabilities = 12,00,000

Current Liabilities = ₹ 8,00,000

(v) Calculation of Cash Balance:

Current Assets = 2.5 Current Liabilities

Current Assets = 2.5 (8,00,000)	= 20,00,000
(-) Debtors	(5,00,000)
(-) Stock	(4,00,000)
Cash Balance	**₹ 11,00,000**

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97 EXAM Old P8 Dec 2021 Q2 10 marks old scheme

Following are the data in respect of ABC Industries for the year ended 31st March, 2021:

Debt to Total assets ratio	:	0.40
Long-term debts to equity ratio	:	30%
Gross profit margin on sales	:	20%
Accounts receivables period	:	36 days
Quick ratio	:	0.9
Inventory holding period	:	55 days
Cost of goods sold	:	₹ 64,00,000

Liabilities	₹	Assets	₹
Equity Share Capital	20,00,000	Fixed assets	
Reserves & surplus		Inventories	
Long-term debts		Accounts receivable	
Accounts payable		Cash	
Total	50,00,000	Total	

Required:

Complete the Balance Sheet of ABC Industries as on 31st March, 2021. All calculations should be in nearest Rupee. Assume 360 days in a year.

ICAI'S ANSWER

Working Notes:

(1) Total liability = Total Assets = ₹ 50,00,000

Debt to Total Asset Ratio = 0.40

$$\frac{\text{Debt}}{\text{Total Assets}} = 0.40$$

Or, $\frac{\text{Debt}}{50,00,000} = 0.40$

So, **Debt = 20,00,000**

(2) Total Liabilities = ₹ 50,00,000

Equity share Capital + Reserves + Debt = ₹ 50,00,000

So, Reserves =₹ 50,00,000 - ₹ 20,00,000 - ₹ 20,00,000

So, Reserves & Surplus = ₹ 10,00,000

(3) $\frac{\text{Long term Debt}}{\text{Equity Shareholders' Fund}} = 30\%^*$

$\frac{\text{Long term Debt}}{(20,00,000+10,00,000)} = 30\%$

Long Term Debt = ₹ 9,00,000

(4) So, Accounts Payable = ₹ 20,00,000 – ₹ 9,00,000

Accounts Payable = ₹ 11,00,000

(5) Gross Profit to sales = 20%

Cost of Goods Sold = 80% of Sales = ₹ 64,00,000

Sales = $\frac{100}{80} \times 64{,}00{,}000$ **= 80,00,000**

(6) Inventory Turnover = $\frac{360}{55}$

$$\frac{\text{COGS}}{\text{Closing inventory}} = \frac{360}{55}$$

$$\frac{64{,}00{,}000}{\text{Closing inventory}} = \frac{360}{55}$$

Closing inventory = 9,77,778

(7) Accounts Receivable period = 36 days

$$\frac{\text{Accounts Receivable}}{\text{Credit sales}} \times 360 = 36$$

Accounts Receivable = $\frac{36}{360}$ × credit sales

= $\frac{36}{360}$ × 80,00,000 (assumed all sales are on credit)

Accounts Receivable = ₹ 8,00,000

(8) Quick Ratio = 0.9

$$\frac{\text{Quick Assets}}{\text{Current liabilities}} = 0.9$$

$$\frac{\text{Cash} + \text{Debtors}}{11{,}00{,}000} = 0.9$$

Cash + 8,00,000 = ₹ 9,90,000

Cash = ₹ 1,90,000

(9) Fixed Assets = Total Assets- Current Assets = 50,00,000 – (9,77,778+8,00,000+1,90,000) = **30,32,222**

Balance Sheet of ABC Industries as on 31st March 2021

Liabilities	(₹)	Assets	(₹)
Share Capital	20,00,000	Fixed Assets	**30,32,222**
Reserved surplus	**10,00,000**	Current Assets:	
Long Term Debt	**9,00,000**	Inventory	**9,77,778**
Accounts Payable	**11,00,000**	Accounts Receivables	**8,00,000**
		Cash	**1,90,000**
Total	50,00,000	**Total**	**50,00,000**

(***Note:** Equity shareholders' fund represent equity in 'Long term debts to equity ratio'. The question can be solved assuming only share capital as 'equity')

98 EXAM **Old P8 Jul 2021** Q3 10 marks old scheme

Masco Limited has furnished the following ratios and information relating to the year ended 31st March 2021:

Sales	₹ 75,00,000
Return on net worth	25%
Rate of income tax	50%
Share capital to reserves	6:4
Current ratio	2.5
Net profit to sales (After Income Tax)	6.50%
Inventory turnover (based on cost of goods sold)	12
Cost of goods sold	₹ 22,50,000
Interest on debentures	₹ 75,000
Receivables (includes debtors ₹ 1,25,000)	₹ 2,00,000
Payables	₹ 2,50,000
Bank Overdraft	₹ 1,50,000

You are required to:

(a) Calculate the operating expenses for the year ended 31st March, 2021.

(b) Prepare a balance sheet as on 31st March in the following format:

Liabilities	₹	Assets	₹
Share Capital		Fixed Assets	
Reserves and Surplus		Current Assets	
15% Debentures		Stock	
Payables		Receivables	
Bank Term Loan		Cash	

ICAI'S ANSWER

(a) Calculation of Operating Expenses for the year ended 31st March, 2021

Particulars	(₹)
Net Profit [@ 6.5% of Sales]	4,87,500
Add: Income Tax (@ 50%)	4,87,500

Particulars		(₹)
Profit Before Tax (PBT)		9,75,000
Add: Debenture Interest		75,000
Profit before interest and tax (PBIT)		10,50,000
Sales		75,00,000
Less: Cost of goods sold	22,50,000	
PBIT	10,50,000	33,00,000
Operating Expenses		42,00,000

(b) Balance Sheet as on 31st March, 2021

Liabilities	₹	Assets	₹
Share Capital	11,70,000	Fixed Assets	18,50,000
Reserve and Surplus	7,80,000	Current Assets	
15% Debentures	5,00,000	Stock	1,87,500
Payables	2,50,000	Receivables	2,00,000
Bank Overdraft(or Bank Term Loan)	1,50,000	Cash	6,12,500
	28,50,000		28,50,000

Working Notes:

(i) **Calculation of Share Capital and Reserves**

The return on net worth is 25%. Therefore, the profit after tax of ₹ 4,87,500 should be equivalent to 25% of the net worth.

$$\text{Net worth} \times \frac{25}{100} = ₹\ 4{,}87{,}500$$

$$\therefore \text{Net worth} = \frac{₹\ 4{,}87{,}500 \times 100}{25} = ₹\ 19{,}50{,}000$$

The ratio of share capital to reserves is 6:4

Share Capital = 19,50,000 x 6/10 = ₹ 11,70,000

Reserves = 19,50,000 x 4/10 = ₹ 7,80,000

(ii) **Calculation of Debentures**

Interest on Debentures @ 15% (as given in the balance sheet format) = ₹ 75,000

$$\therefore \text{Debentures} = \frac{75{,}000 \times 100}{15} = ₹\ 5{,}00{,}000$$

(iii) **Calculation of Current Assets**

Current Ratio = 2.5

Payables = ₹ 2,50,000

Bank overdraft = ₹ 1,50,000

Total Current Liabilities = ₹ 2,50,000 + ₹ 1,50,000 = ₹ 4,00,000

∴ Current Assets = 2.5 x Current Liabilities = 2.5 × 4,00,000 = ₹ 10,00,000

(iv) **Calculation of Fixed Assets**

Particulars	₹
Share capital	11,70,000
Reserves	7,80,000
Debentures	5,00,000
Payables	2,50,000
Bank Overdraft	1,50,000
Total Liabilities	28,50,000
Less: Current Assets	10,00,000
Fixed Assets	18,50,000

(v) **Calculation of Composition of Current Assets**

Inventory Turnover = 12

$$\frac{\text{Cost of goods sold}}{\text{Closing stock}} = 12$$

$$\text{Closing stock} = \frac{₹\ 22,50,000}{12} = \text{Closing stock} = ₹\ 1,87,500$$

Particulars	₹
Stock	1,87,500
Receivables	2,00,000
Cash (balancing figure)	6,12,500
Total Current Assets	10,00,000

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99 EXAM **Old P8 Jan 2021** Q1(a) 5 marks old scheme

From the following information, complete the Balance Sheet given below:

(i) Equity Share Capital : ₹ 2,00,000

(ii) Total debt to owner's equity : 0.75

(iii) Total Assets turnover : 2 times

(iv) Inventory turnover : 8 times

(v) Fixed Assets to owner's equity : 0.60

(vi) Current debt to total debt : 0.40

Balance Sheet of XYZ Co. as on March 31, 2020

Liabilities	Amount (₹)	Assets	Amount (₹)
Equity Shares Capital	2,00,000	Fixed Assets	?
Long term Debt	?	Current Assets:	
Current Debt	?	Inventory	?
		Cash	?

ICAI'S ANSWER

Balance Sheet of XYZ Co. as on March 31, 2020

Liabilities	Amount (₹)	Assets	Amount (₹)
Equity Share Capital	2,00,000	Fixed Assets	1,20,000
Long-term Debt	90,000	**Current Assets:**	
Current Debt	60,000	Inventory	87,500
		Cash (balancing figure)	1,42,500
	3,50,000		**3,50,000**

Working Notes

1. Total Debt = 0.75 x Equity Share Capital = 0.75 x ₹ 2,00,000 = ₹ 1,50,000

Further, Current Debt to Total Debt = 0.40.

So, Current Debt = 0.40 x ₹ 1,50,000 = ₹ 60,000

Long term Debt = ₹ 1,50,000 - ₹ 60,000 = ₹ 90,000

2. Fixed Assets = 0.60 x Equity Share Capital = 0.60 x ₹ 2,00,000 = ₹ 1,20,000

3. Total Assets to Turnover = 2 times; Inventory Turnover = 8 times

Hence, Inventory /Total Assets = 2/8 =1/4

Further, Total Assets = ₹ 2,00,000 + ₹ 1,50,000 = ₹ 3,50,000

Therefore, Inventory = ₹ 3,50,000/4 = ₹ 87,500

Cash in Hand = Total Assets – Fixed Assets – Inventory

= ₹ 3,50,000 - ₹ 1,20,000 - ₹ 87,500 = ₹ 1,42,500

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100 **RTP** **Sep 2026** **Q4**

The following information of ASD Ltd. relate to the year ended 31st March, 2026:

Net profit	8% of sales

Raw materials consumed	20% of Cost of Goods Sold
Direct wages	10% of Cost of Goods Sold
Stock of raw materials	3 months' usage
Stock of finished goods	6% of Cost of Goods Sold
Gross Profit	15% of Sales
Debt collection period	2 Months
(All sales are on credit)	
Current ratio	2 : 1
Fixed assets to Current assets	13 : 11
Fixed assets to sales	1 : 3
Long-term loans to Current liabilities	2 : 1
Capital to Reserves and Surplus	1 : 4

You are required to PREPARE-

(a) Profit & Loss Statement of ASD Limited for the year ended 31st March, 2026 in the following format.

Particulars	(₹)	Particulars	(₹)
To Direct Materials consumed	?	By Sales	?
To Direct Wages	?		
To Works (Overhead)	?		
To Gross Profit c/d	?		
	?		?
To Selling and Distribution Expenses	?	By Gross Profit b/d	?
To Net Profit	?		
	?		?

(b) Balance Sheet as on 31st March, 2026 in the following format.

Liabilities	(₹)	Assets	(₹)
Share Capital	?	Fixed Assets	1,30,00,000
Reserves and Surplus	?	Current Assets:	
Long term loans	?	Stock of Raw Material	?
Current liabilities	?	Stock of Finished Goods	?
		Debtors	?
		Cash	?
	?		?

ICAI'S ANSWER

Working Notes:

(i) Calculation of Sales

$$\frac{\text{Fixed Assets}}{\text{Sales}} = \frac{1}{3}$$

$$\therefore \frac{1,30,00,000}{\text{Sales}} = \frac{1}{3} \Rightarrow \text{Sales} = ₹\ 3,90,00,000$$

(ii) Calculation of Current Assets

$$\frac{\text{Fixed Assets}}{\text{Current Assets}} = \frac{13}{11}$$

$$\therefore \frac{1,30,00,000}{\text{Current Assets}} = \frac{13}{11} \Rightarrow \text{Current Assets} = ₹\ 1,10,00,000$$

(iii) Calculation of Raw Material Consumption and Direct Wages

	₹
Sales	3,90,00,000
Less: Gross Profit (15 % of Sales)	58,50,000
Cost of Goods sold	3,31,50,000

Raw Material Consumption (20% of Cost of Goods Sold) ₹ 66,30,000

Direct Wages (10% of Cost of Goods Sold) ₹ 33,15,000

(iv) Calculation of Stock of Raw Materials (= 3 months usage)

$$= 66,30,000 \times \frac{3}{12} = ₹\ 16,57,500$$

(v) Calculation of Stock of Finished Goods (= 6% of Cost of Goods Sold)

$$= 3,31,50,000 \times \frac{6}{100} = ₹\ 19,89,000$$

(vi) Calculation of Current Liabilities

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = 2$$

$$\frac{1,10,00,000}{\text{Current Liabilities}} = 2 \Rightarrow \text{Current Liabilities} = ₹\ 55,00,000$$

(vii) Calculation of Debtors

$$\text{Average collection period} = \frac{\text{Debtors}}{\text{Credit Sales}} \times 12 \text{ months}$$

$$\frac{\text{Debtors}}{3,90,00,000} \times 12 = 2 \Rightarrow \text{Debtors} = ₹\ 65,00,000$$

(viii) Calculation of Long-term Loan

$$\frac{\text{Long term Loan}}{\text{Current Liabilities}} = \frac{2}{1}$$

$$\frac{\text{Long term loan}}{55,00,000} = \frac{2}{1} \Rightarrow \text{Long term loan} = ₹\ 1,10,00,000$$

(ix) Calculation of Cash Balance

		₹
Current assets		1,10,00,000
Less: Debtors	65,00,000	
Raw materials stock	16,57,500	
Finished goods stock	19,89,000	1,01,46,500
Cash balance		8,53,500

(x) Calculation of Net worth

Fixed Assets		1,30,00,000
Current Assets		1,10,00,000
Total Assets		2,40,00,000
Less: Long term Loan	1,10,00,000	
Current Liabilities	55,00,000	1,65,00,000
Net worth		75,00,000

Net worth = Share capital + Reserves = ₹ 75,00,000

$$\frac{\text{Capital}}{\text{Reserves and Surplus}} = \frac{1}{4} \Rightarrow \text{Share Capital} = ₹\ 75,00,000 \times \frac{1}{5}$$

= ₹ 15,00,000

$$\text{Reserves and Surplus} = ₹\ 75,00,000 \times \frac{4}{5} = ₹\ 60,00,000$$

Profit and Loss Statement of ASD Ltd.

for the year ended 31st March, 2026

Particulars	(₹)	Particulars	(₹)
To Direct Materials consumed	66,30,000	By Sales	3,90,00,000
To Direct Wages	33,15,000		
To Works (Overhead) (Bal. fig.)	2,32,05,000		
To Gross Profit c/d (15% of Sales)	58,50,000		
	3,90,00,000		3,90,00,000
To Selling and Distribution Expenses (Bal. fig.)	27,30,000	By Gross Profit b/d	58,50,000
To Net Profit (8% of Sales)	31,20,000		
	58,50,000		**58,50,000**

Balance Sheet of ASD Ltd. as at 31st March, 2026

Liabilities	(₹)	Assets	(₹)
Share Capital	15,00,000	Fixed Assets	1,30,00,000
Reserves and Surplus	60,00,000	Current Assets:	
Long term loans	1,10,00,000	Stock of Raw Material	16,57,500
Current liabilities	55,00,000	Stock of Finished Goods	19,89,000
		Debtors	65,00,000
		Cash	8,53,500
	2,40,00,000		2,40,00,000

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101 RTP **May 2026** Q4

The following information pertains to P Limited for the year ended 31st March, 2025 :

Particulars	Amount / Ratio
Sales	₹ 3,60,00,000
Rate of Income Tax	40%
Return on Net Worth	30%
Share Capital to Reserves Ratio	6 : 4
Current Ratio	2 : 1
Percentage of Net Profit to Sales	8%

Particulars	Amount / Ratio
Inventory Turnover (based on cost of goods sold)	12
Cost of goods sold	₹ 1,44,00,000
Sundry Debtors	₹ 12,00,000
Sundry Creditors	₹ 16,00,000
Interest on 14% debentures	₹ 3,36,000

You are required to CALCULATE:

(i) Operating Expenses

(ii) Share Capital and Reserves

(iii) Closing Stock

(iv) Fixed Assets

ICAI'S ANSWER

(i) Operating Expenses

Net Profit = 8% on Sales = 8% x ₹ 3,60,00,000	₹ 28,80,000
Add: Income Tax = 40% x ₹ 28,80,000/60%	₹ 19,20,000
Profit Before tax	₹ 48,00,000
Add: Debenture Interest	₹ 3,36,000
Profit before interest and tax	₹ 51,36,000
Sales	₹ 3,60,00,000
Less:	
Cost of goods sold ₹ 1,44,00,000 Profit before interest & tax ₹ 51,36,000	₹ 1,95,36,000
Operating Expenses	₹ 1,64,64,000

(ii) Share Capital and Reserves

Return on Net worth is 30%, hence profit after tax of ₹ 28,80,000 is equivalent to 30% of Net Worth.

Net worth x 30/100 = ₹ 28,80,000

Net Worth = 28,80,000 x 100/30 = ₹ 96,00,000

Ratio of Share Capital to Reserves = 6:4

Share Capital = ₹ 96,00,000 x 6/10 = ₹ 57,60,000

Reserves =₹ 96,00,000 x 4/10 = ₹ 38,40,000

(iii) Closing Stock

Inventory Turnover (based on cost of goods sold) = 12

Cost of goods sold = ₹ 1,44,00,000

Inventory Turnover Ratio = Cost of goods sold/Closing Stock = 12

12 = ₹ 1,44,00,000/Closing Stock

Therefore, Closing Stock = ₹ 1,44,00,000/12

= ₹ 12,00,000

(iv) Fixed Assets

Calculation of Total liabilities

Interest on 14 % Debentures = ₹ 3,36,000

Therefore debentures = $\frac{₹3,36,000 \text{ x } 100}{14\%}$

= ₹ 24,00,000

Total of Liabilities Side

Share Capital	₹ 57,60,000
Reserves	₹ 38,40,000
Debentures	₹ 24,00,000
Sundry Creditors	₹ 16,00,000
Total Liabilities	₹ 1,36,00,000

Current Ratio = 2:1

Sundry Creditors = ₹ 16,00,000

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = 2$$

$$\frac{\text{Current Assets}}{\text{Sundry Creditors}} = 2$$

$$\frac{\text{Current Assets}}{₹16,00,000} = 2$$

Current Assets = 2 x ₹ 16,00,000

= ₹ 32,00,000

Fixed Assets = Total Liabilities – Current Assets

= ₹ 1,36,00,000 - ₹ 32,00,000

= ₹ 1,04,00,000

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102 RTP **Jan 2026** Q5

ABC Industries Ltd, a listed company, is a renowned player in the pharmaceutical sector. The company has availed various credit facilities from X bank, namely, Cash Credit Facility, Term Loan(s), Working Capital Demand Loan, Bank Guarantee(s) / Letter of Credit. This strategic mix of various credit facilities showcases the company's balanced approach to leveraging and risk management. The company has totally 1 Lakh Equity Shares of ₹ 10 each outstanding. The company's shares are traded at a premium of

20%. The company's capital yields a healthy return rate, indicative of its operational efficiency and market competitiveness. However, it operates in a high-tax environment with an income-tax rate of 50%, which significantly impacts its net earnings and available reinvestment capital.

At the beginning of the year 2024-25, the company went in for re-organisation of capital structure, with share capital remaining the same as follows:

(a) Share capital - 40%,

(b) 12% Preference Share Capital – 10%

(c) Other Shareholders' funds - 15%

(d) 15% Term Loan - 10%

(e) Current Liabilities - 25%

Term Loan is repayable in 5 instalments starting from 31.03.2025. Since the first instalment is not yet paid, as part of periodic review by the Bank and in order to ensure compliance with their internal and regulatory requirements, the Bank requested the company to furnish a Financial Soundness Report duly certified by your Chartered Accountant/Auditor. The report should inter alia cover various Ratios that:

1. estimates whether the business can pay short term debts.
2. measures the ability to meet current debt immediately.
3. measures the ability to meet the commitment of various debt services like interest, instalment etc.
4. measures the ability of business to meet the interest obligations.
5. measures the ability of business to pay the preference dividend.
6. shows how many times the cash flow before interest and taxes covers all fixed financing charges.

The Chief Executive officer asked the Statutory auditor to prepare a reply letter quoting the proper ratio that covers the issues raised by the bank in this format –

Issue	Ratio	Numerator	Denominator	Ideal Ratio/ Range	Ratio of the Company	Favourable/ Adverse

Closing fixed assets after providing 20% Depreciation constituted 60% of the total assets. Current assets comprises Inventories, Advance Tax, Prepaid Insurance and Receivables. The receivables (one-fifth of the current assets) to sales ratio revealed a credit period of 2 months. There were no cash sales. Operating profit margin being 15%. Also find out the ratios that

1. indicates composition of capital structure in terms of debt and equity
2. measures overall profits generated for each share in existence.
3. indicates market response of the equity shareholders' investment.
4. Indicates expectations of the equity investors about the earnings of the firm.

ICAI'S ANSWER

Equity Share Capital	40% of Total = Given	10,00,000
Total Liabilities	10,00,000 / 40%	25,00,000
Preference Share Capital	25,00,000 × 10%	2,50,000
Other Shareholders' funds	25,00,000 × 15%	3,75,000

15% Term Loan	25,00,000 × 10%	2,50,000
Current Liabilities	25,00,000 × 25%	6,25,000
Total Assets	Total Liabilities	25,00,000
Fixed Assets	60% of Total	15,00,000
Depreciation	$\frac{15,00,000}{80\%}$ x 20%	3,75,000
Current Assets	40% of Total	10,00,000
Receivables	1/5th of Current Assets	2,00,000
Stock, Advance Tax & Prepaid Insurance	Balance	8,00,000
Quick Assets	Only Receivables	2,00,000
Total Sales	$\frac{\text{Receivables}}{2}$ x 12	12,00,000
EBIT	12,00,000 × 15%	1,80,000
Interest	2,50,000 × 15%	37,500
EBT		1,42,500
PAT	1,42,500 × 50%	71,250
Preference dividend	2,50,000 × 12%	30,000
PAES		41,250
EPS (ratio measures overall profits generated for each share in existence)	41,250 / 1,00,000	0.41
Debt Equity ratio (indicates composition of capital structure in terms of debt and equity)	2,50,000 / 16,25,000	0.15
MPS	FV 10 + 20% Premium	12
Market value to Book value (indicates market response of the equity shareholders' investment)	12 Lakhs / 13.75 Lakhs	0.87
PE Ratio (indicates expectations of the equity investors about the earnings of the firm)	12/0.41	29.27 times

Issue	Ratio	Numerator	Denominator	Ideal Ratio/ Range	Ratio of the Company	Favourable/ Adverse
1	Current Ratio	Current Assets	Current Liabilities	2	1.6	Adverse
2	Quick Ratio	Quick Assets	Current Liabilities	1	0.32	Adverse

Issue	Ratio	Numerator	Denominator	Ideal Ratio/ Range	Ratio of the Company	Favourable/ Adverse
3	DSCR	Earnings available for Debt services	Interest & Instalments	2	6.1	Favourable
4	Interest coverage ratio	EBIT	Interest	> 1	4.8	Favourable
5	Preference Dividend coverage ratio	PAT	Preference dividend	> 1	2.375	Favourable
6	Fixed charges coverage ratio	EBIT + Depn	Interest + Repayment of loan	> 1	6.34	Favourable
7	Debt equity ratio	Debt	Equity	Not applicable	0.15	
8	EPS	Profit available for equity holders	No. of Equity Shares	Not applicable	0.41	
9	MV/BV	MV of Equity Shares	BV of Equity Shares	Not applicable	0.87	

Earnings available for Debt services = PAT + Interest + Depreciation + Instalments

= 71,250 + 37,500 + 3,75,000 + 50,000 = 5,33,750

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103 RTP **Sep 2025** Q4

Jamunapati Limited has furnished the following ratios and information relating to the year ended 31st March, 2024:

Sales	₹ 70,00,000
Return on net worth	25%
Rate of income tax	30%
Share capital to reserves	7:3
Current ratio	2
Net profit to sales	7%
Inventory turnover	10

Cost of goods sold	₹ 24,00,000
Interest on 15% debentures	₹ 1,05,000
Receivables	₹ 3,00,000
Payables	₹ 3,00,000

You are required to:

(a) CALCULATE the operating expenses for the year ended 31st March, 2024.

(b) PREPARE a Balance Sheet as on 31st March, 2024 in the following format:

Balance Sheet as on 31st March, 2024

Liabilities	₹	Assets	₹
Share Capital		Fixed Assets	
Reserve and Surplus		Current Assets	
15% Debentures		Stock	
Payables		Receivables	
		Cash	

ICAI'S ANSWER

(a) **Calculation of Operating Expenses for the year ended 31st March, 2024**

	(₹)
Net Profit [@ 7% of Sales]	4,90,000
Add: Income Tax (@ 30% of PBT)	2,10,000
Profit Before Tax (PBT)	7,00,000
Add: Debenture Interest	1,05,000
Profit before interest and tax (PBIT)	8,05,000
Now, Sales (given)	70,00,000
Less: Cost of goods sold	24,00,000
Gross Profit	46,00,000
Less: PBIT	8,05,000
Operating Expenses	**37,95,000**

(b) **Balance Sheet as on 31st March, 2024**

Liabilities	₹	Assets	₹
Share Capital	13,72,000	Fixed Assets	23,60,000
Reserve and Surplus	5,88,000	Current Assets:	
15% Debentures	70,00,000	Stock	2,40,000
Payables	3,00,000	Receivables	3,00,000
		Cash	60,000
	29,60,000		**29,60,000**

Working Notes:

(i) **Share Capital and Reserves and Surplus**

The return on net worth is 25%. Therefore, the profit after tax of ₹ 4,90,000 should be equivalent to 25% of the net worth.

$$\text{Net worth} \times \frac{25}{100} = ₹\ 4{,}90{,}000$$

$$\therefore \text{Net worth} = \frac{₹\ 4{,}90{,}000 \times 100}{25} = ₹\ 19{,}60{,}000$$

The ratio of share capital to reserves is 7:3

$$\text{Share Capital} = 19{,}60{,}000 \times \frac{7}{10} = ₹\ 13{,}72{,}000$$

$$\text{Reserves and Surplus} = 19{,}60{,}000 \times \frac{3}{10} = ₹\ 5{,}88{,}000$$

(ii) **Debentures**

Interest on Debentures @ 15% = ₹ 1,05,000

$$\therefore \text{Debentures} = \frac{1{,}05{,}000 \times 100}{15} = ₹\ 7{,}00{,}000$$

(iii) **Current Assets**

Current Ratio = 2

Payables = ₹ 3,00,000

∴ Current Assets = 2 Current Liabilities

= 2 × 3,00,000

= ₹ 6,00,000

It is assumed that current liabilities include payables only.

(iv) **Fixed Assets**

	₹
Share capital	13,72,000
Reserves and Surplus	5,88,000
Debentures	7,00,000
Payables	3,00,000
Total Liabilities	29,60,000
Less: Current Assets	6,00,000
Fixed Assets	23,60,000

(v) **Composition of Current Assets**

Inventory Turnover = 10

$$\frac{\text{Cost of goods sold}}{\text{Closing stock}} = 10$$

$$\text{Closing stock} = \frac{₹\ 24,00,000}{10} = ₹\ 2,40,000$$

Composition	₹
Stock	2,40,000
Receivables	3,00,000
Cash (balancing figure)	60,000
Total Current Assets	6,00,000

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104 RTP **May 2025** Q4

Using the information given below, PREPARE the Balance Sheet of Nevy Private Limited –

Particulars	Details
Stock turnover Ratio	15 times
Cash and Bank balance	10% of Current Assets (net off prepaid exp)
GP Ratio	20%
Creditors turnover (cost of goods sold) ratio	10 times
Debtors turnover ratio	12 times
Net Fixed Assets	25% of Total Liabilities
Depreciation	15% on Opening WDV

Particulars	Details
Current Ratio	1.6 : 1
Capital Gearing Ratio	0.6 : 1

All Purchases and Sales are assumed to be on credit basis.

Balance Sheet of Nevy Private Limited as of 31.03.2025

Particulars	Amount (₹)	Amount (₹)
A] Equities and Long Term Liabilities		
Share Capital	**36,00,000**	
Reserves and Surplus	**??**	
14% Bonds	**??**	**???**
B] Current Liabilities		
Trade Payables	**??**	
Outstanding expenses and provisions	**??**	
(*Net of Prepaid expenses of ₹7,50,000)		**45,00,000**
TOTAL		**?????**
C] Fixed Assets		
Opening WDV	**??**	
(-) Depreciation	**??**	**???**
D] Current Assets		
Inventory	**??**	
Trade Receivables	**??**	
Cash and Bank Balance	**??**	
		???
TOTAL		**?????**

(All the working notes should form part of your answer)

ICAI'S ANSWER

Balance Sheet of Nevy Private Limited as of 31.03.2025

Particulars	Notes	Amount (₹)	Amount (₹)
A] Equities and Long Term Liabilities			

Particulars	Notes	Amount (₹)	Amount (₹)
Share Capital		36,00,000	
Reserves and Surplus	WN-7	1,18,750	
14% Bonds	Bal. Fig.	22,31,250	59,50,000
B] Current Liabilities			
Trade Payables	WN-6	40,30,244	
Outstanding expenses and provisions	Bal. Fig.	4,69,756	45,00,000
(*Net of Prepaid expenses of ₹ 7,50,000)			
TOTAL			**1,04,50,000**
C] Fixed Assets			
Opening WDV	WN-3	32,94,118	
(-) Depreciation	WN-3	(4,94,118)	28,00,000 (WN-2)
D] Current Assets			
Inventory	WN-5	26,86,829	
Trade Receivables	WN-5	41,98,170	
Cash and Bank Balance	WN-4	7,65,000	76,50,000 (WN 1)
TOTAL			**1,04,50,000**

WN 1 – Calculation of Current Assets using Current Ratio

$$\text{Current Ratio} = \frac{\text{CA}}{\text{CL}}$$

CL = 45,00,000 + 7,50,000

= 52,50,000

1.6 = CA / 52,50,000

Therefore CA = 84,00,000

CA = Inventory + Trade Receivables + Cash Bank Balance + Prepaid Exp

84,00,000 = Inventory + Trade Receivables + Cash Bank Balance + 7,50,000

Therefore, total of Inventory + Trade Receivables + Cash Bank Balance = 76,50,000

WN 2 – Calculation of Fixed Assets & Total Assets

Fixed Assets = 25% of Total liabilities or 25% of Total Assets which means

Current Assets = 75% of Total liabilities or 75% of Total Assets

Fixed Assets + Current Assets = Total Assets/Total Liabilities

25 + 75 = 100

Fixed Assets = 84,00,000 x 25/75 = 28,00,000

Total Assets = 1,12,00,000

WN 3 – Calculation of Depreciation and Opening WDV of Fixed Assets

Opening WDV - Depreciation = Closing WDV

100 - 15 = 85

Therefore, Depreciation = 28,00,000 x 15/85 = 4,94,118

Opening WDV = 32,94,118

WN 4 – Calculation of Cash & Bank Balance

Cash & Bank Balance = 10% of (CA – Prepaid Exp)

Cash & Bank balance = 10% of (84,00,000 – 7,50,000)

Therefore, cash and bank balance = 7,65,000

WN 5 – Calculation of Inventory & Trade receivables

Total CA = Inventory + Trade Receivables + Cash Bank Balance + Prepaid Exp

84,00,000 = Inventory + Trade Receivables + 7,65,000 + 7,50,000

Inventory + Trade Receivables = 68,85,000

Now, Let Sales be X

GP Ratio = 20% = 0.2X

COGS = 80% = 0.8X

$$\text{Debtors T/O Ratio} = \frac{\text{Net Credit Sales}}{\text{Debtors}}$$

$$12 = \frac{X}{\text{Debtors}}$$

Debtors = X / 12

Inventory T/O Ratio = COGS/Inventory

15 = 0.8X/Inventory

Inventory = 0.8X/15

Inventory + Trade Receivables = 68,85,000

0.8X / 15 + X / 12 = 68,85,000

Therefore X = Sales = 5,03,78,050

COGS = 4,03,02,440

Trade Receivables = 41,98,170

Inventory = 26,86,829

WN 6 – Calculation of Trade Payables using Creditors Turnover Ratio

$$\text{Creditors T/O Ratio} = \frac{\text{COGS}}{\text{Trade Payables}}$$

$$10 = \frac{4,03,02,440}{\text{Trade Payables}}$$

Trade Payables = 40,30,244

WN 7 – Calculation of Reserves and Surplus & 14% Bonds

Total Capital Employed = Share Cap + R&S + Bonds

Capital Gearing Ratio = Capital bearing Fixed % /Capital not bearing fixed %

0.6 = Bonds/Share Capital + R&S

Therefore, Bonds = 21,60,000 + 0.6 R&S

Substituting the value of bonds in the above equation,

Total Capital Employed = Share Cap + R&S + (21,60,000 + 0.6 R&S)

59,50,000 = 36,00,000 + 1.6 R&S + 21,60,000

Therefore R&S = 1,18,750

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105 RTP **Jan 2025** Q4

Vardhaman Limited gives you the following information related for the year ending 31st March, 2024:

Particulars	Amount (₹)
Current Ratio	3:1
Loan funds to Owned Funds Ratio	1:3
Gross Profit Ratio	25%
Stock Turnover Ratio	10
Net Working Capital	₹ 5,00,000
Return on Total Assets (pre-tax)	15%
MPS	₹ 20
Total Assets Turnover Ratio	2.5
Opening stock	₹ 6,50,500
Fixed Assets	₹ 15,00,000
75,000 equity shares of	₹ 10 each
25,000, 12% Pref. Shares of	₹ 10 each
Depreciation	₹ 50,000
Interest on Debt	9%
Future Instalments	₹ 2,00,000

Tax rate applicable to the company is 25%

You are required to CALCULATE:

(i) Quick Ratio

(ii) Fixed Assets Turnover Ratio

(iii) Debt Service Coverage

(iv) Earnings per Share

(v) Price Earnings Ratio

ICAI'S ANSWER

WN 1: Calculation of Current Assets & Current Liabilities

Current Ratio = CA / CL = 3:1

Therefore, CA = 3CL

Net Working Capital = CA - CL = 5,00,000

= 3CL (-) CL = 5,00,000

Therefore, CL = 2,50,000,

CA = 7,50,000

WN 2: Calculation of Average Stock Value & Closing Stock

Total Assets = Fixed Assets + Current Assets

= 15 L + 7.5 L = 22.50 lakhs

Total Assets Turnover Ratio = Sales / Total Assets = 2.5 (given)

Therefore Sales = 22.5 lakhs X 2.5

Sales = 56,25,000

GP Margin = 25%, therefore COGS = 75% of Sales

COGS = 56.25 x 75% = 42,18,750

Stock Turnover Ratio = COGS / Average Stock = 10 (given)

Average Stock = 42,18,750 / 10 = 4,21,875

Average Stock = Op. Stock + Cl. Stock / 2

4,21,875 = 6,50,500 + Cl. Stock / 2

Cl Stock = 1,93,250

WN 3: Calculation of Cash Profit before Interest & Tax

Return on Total Assets (pre-tax) = (EBIT / Total Assets)

0.15 = EBIT / 22.50 lakhs

Therefore, EBIT = 3,37,500

Cash Profit before Int & Tax = EBIT + Depreciation

= 337500 + 50000

Cash Profit before Int & Tax = 3,87,500

WN 4 : Calculation of Loan Funds (Debt) & Owned Funds (Equity)

Debt to Equity = 1 : 3, which means 3 times Debt = Equity (Owned Funds)

As per the Accounting equation,

Equity + Debt + Current Liab. = Fixed Assets + Current Assets

3 Debt + Debt + 2,50,000 = 15,00,000 + 7,50,000

4 Debt = 20,00,000

Therefor Debt (Loan Funds) = **5,00,000**

Equity (Owned Funds) = **15,00,000**

WN 5: Calculation of Earnings Available to Eq. Share holders

Particulars	Amount (₹)
EBIT	3,37,500
(-) Int (5 lakhs x 9%)	(45,000)
EBT	2,92,500
(-) Tax @ 0.25	(73,125)
EAT	2,19,375
(-) Pref Div. (250000 × 12%)	(30,000)
Earnings For Eq. Sh Holders	1,89,375

1. Quick Ratio = {CA - Cl Stock}/CL

= 7,50,000 – 1,93,250 / 2,50,000

Quick Ratio = 2.23 : 1

2. Fixed Assets Turnover Ratio = Sales / Total Fixed Assets

= 56,25,000 / 15,00,000

Fixed Assets Turnover Ratio = 3.75 times

3. Debt Service Coverage Ratio = Cash profit before Int & Tax / Int + Instalments

= 3,87,500 / (45,000+2,00,000)

Debt Service Coverage Ratio = 1.58 times.

4. **EPS = Earnings for Eq. Shareholders / No of Eq. Shareholders**

= 1,89,375/75,000

EPS = ₹ 2.53

5. Price to Earnings Ratio = MPS / EPS

= 20 / 2.53

Price to Earnings Ratio = 7.91 times

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106 RTP Sep 2024 Q4

Following are the data in respect of LP enterprises for the year ended 31st March, 2024:

Debt to Total assets ratio	:	0.40
Long-term debts to equity ratio	:	30%
Gross profit margin on sales	:	20%
Accounts receivables period	:	36 days
Quick ratio	:	0.9
Inventory holding period	:	60 days
Cost of goods sold	:	₹ 64,00,000

Liabilities	₹	Assets	₹
Equity Share Capital	20,00,000	Fixed assets	
Reserves & surplus		Inventories	
Long-term debts		Accounts receivable	
Accounts payable		Cash	
Total	50,00,000	Total	

Required:

COMPLETE the Balance Sheet of LP enterprises as on 31st March, 2024. All calculations should be in nearest Rupee. Assume 360 days in a year.

ICAI'S ANSWER

Working Notes:

(1) Total liability = Total Assets = ₹ 50,00,000

Debt to Total Asset Ratio = 0.40

$$\frac{\text{Debt}}{\text{Total Assets}} = 0.40$$

$$\frac{\text{Debt}}{50,00,000} = 0.40$$

So, **Debt = 20,00,000**

(2) Total Liabilities = ₹ 50,00,000

Equity share Capital + Reserves + Debt = ₹ 50,00,000

So, Reserves = ₹ 50,00,000 - ₹ 20,00,000 - ₹ 20,00,000

So, Reserves & Surplus = ₹ 10,00,000

(3) $\frac{\text{Long term Debt}}{\text{Equity Shareholders' Fund}} = 30\%$

$$\frac{\text{Long term Debt}}{(20,00,000 + 10,00,000)} = 30\%$$

Long Term Debt = ₹ 9,00,000

(4) So, Accounts Payable = ₹ 20,00,000 – ₹ 9,00,000

Accounts Payable = ₹ 11,00,000

(5) Gross Profit to sales = 20%

Cost of Goods Sold = 80% of Sales = ₹ 64,00,000

Sales = $\frac{100}{80} \times 64,00,000$ **= 80,00,000**

(6) Inventory Turnover = $\frac{360}{60}$

$$\frac{\text{COGS}}{\text{Closing inventory}} = \frac{360}{60}$$

$$\frac{64,00,000}{\text{Closing inventory}} = \frac{360}{60}$$

Closing inventory = 10,66,667

(7) Accounts Receivable period = 36 days

$$\frac{\text{Accounts Receivable}}{\text{Credit sales}} \times 360 = 36$$

Accounts Receivable = $\frac{36}{360}$ × credit sales

= $\frac{36}{360}$ × 80,00,000 (assumed all sales are on credit)

Accounts Receivable = ₹ 8,00,000

(8) Quick Ratio = 0.9

$$\frac{\text{Quick Assets}}{\text{Current liabilities}} = 0.9$$

$$\frac{\text{Cash} + \text{Debtors}}{11,00,000} = 0.9$$

Cash + 8,00,000 = ₹ 9,90,000

Cash = ₹ 1,90,000

(9) Fixed Assets = Total Assets - Current Assets = 50,00,000 – (10,66,667 + 8,00,000 + 1,90,000)

(10) = **29,43,333**

Balance Sheet of LP enterprises as on 31st March 2024

Liabilities	(₹)	Assets	(₹)
Share Capital	20,00,000	Fixed Assets	29,43,333
Reserved surplus	10,00,000	Current Assets:	
Long Term Debt	9,00,000	Inventory	10,66,667
Accounts Payable	11,00,000	Accounts Receivables	8,00,000
		Cash	1,90,000
Total	50,00,000	Total	50,00,000

(*Note: Equity shareholders' fund represent equity in 'Long term debts to equity ratio'. The question can be solved assuming only share capital as 'equity')

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107 RTP **May 2024** Q1

From the following information and ratios, PREPARE the Balance Sheet as on 31st March 2023 and Income Statement for the year ended on that date for Limelite & Co.

Gross Profit	₹ 1,20,000
Shareholders' Funds	₹ 5,00,000
Gross Profit margin	40%
Net Profit Margin	10%
PBIT to PBT	2:1
Credit sales to Total sales	80%
Total Assets turnover	0.4 times
Inventory turnover (Use sales as turnover)	5 times
Average collection period (a 360 days year)	30 days
Current ratio	2
Operating expenses (excluding interest)	₹ 60,000
Long-term Debt to Equity	40%
Tax	Nil

ICAI'S ANSWER

Gross Profit = ₹ 1,20,000

Gross Profit Margin = 40%

$$\therefore \text{Sales} = \frac{\text{Gross Profit}}{\text{Gross Profit Margin}} = ₹\ 1,20,000 / 0.40 = ₹\ 3,00,000$$

Net profit (PBT) = 3,00,000 x 10% = ₹ 30,000

PBIT/PBT = 2

PBIT = 2 x 30,000

PBIT = 60,000

Interest = 60,000 – 30,000 = ₹ 30,000

Credit Sales to Total Sales = 80%

∴Credit Sales = ₹ 3,00,000×0.80 = ₹ 2,40,000

Total Assets Turnover = 0.4 times

$\therefore$Total Assets = $\frac{\text{Sales}}{\text{Total Assets Turnover}}$

= $\frac{₹\ 3{,}00{,}000}{0.4}$ = ₹ 7,50,000

Inventory turnover = 5 times

Inventory = $\frac{\text{Sales}}{\text{Inventory turnover}} = \frac{3{,}00{,}000}{5}$ = ₹ 60,000

Average Collection Period = 30 days

$\therefore$Debtors turnover = $\frac{360}{\text{Average Collection Period}}$ = 360/30 = 12

$\therefore$Debtors = $\frac{\text{Credit Sales}}{\text{Debtors turnover}} = \frac{₹\ 2{,}40{,}000}{12}$ = ₹ 20,000

Current ratio = 2

$2 = \frac{\text{Debtors} + \text{Inventory} + \text{Cash (Current Assets)}}{\text{Creditors (Current Liabilities)}}$

2 Creditors = (₹ 20,000 + ₹ 60,000 + Cash)

2 Creditors = ₹ 80,000 + Cash ---------- (i)

Long-term Debt to Equity = 40%

Shareholders' Funds (Equity) = ₹ 5,00,000

$\therefore$ Long-term Debt = ₹ 5,00,000 × 40% = ₹ 2,00,000

Creditors = Total Assets – (Shareholder's fund + Long term debt)

= ₹ 7,50,000 – (5,00,000 + 2,00,000) = ₹ 50,000

$\therefore$ Cash = (₹ 50,000×2) – ₹ 80,000 = ₹ 20,000 [From equation (i)]

Income Statement

	(₹)
Sales	3,00,000
Less: Cost of Goods Sold	1,80,000
Gross Profit	1,20,000
Less: Operating Expenses	60,000
PBIT	60,000
Less: Interest	30,000
Net Profit	30,000

Balance Sheet

Liabilities	₹	Assets		₹
Equity share capital	5,00,000	Fixed asset (bal. fig.)		6,50,000
Long term debt	2,00,000	Current assets:		
Current liability	50,000	Stock	60,000	
		Receivables	20,000	
		Cash	20,000	1,00,000
	7,50,000			7,50,000

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108 RTP **Old P8 Nov 2023** Q9 old scheme

Consider the following figures and ratios:

(i)	Sales for the year (all credit)	₹ 1,05,00,000
(ii)	Gross Profit ratio	35 percent
(iii)	Fixed assets turnover (based on cost of goods sold)	1.5
(iv)	Stock turnover (based on cost of goods sold)	6
(v)	Liquid ratio	1.5:1
(vi)	Current ratio	2.5:1
(vii)	Receivables (Debtors) collection period	1 month
(viii)	Reserves and surplus to Share capital	1:1.5
(ix)	Capital gearing ratio	0.7875
(x)	Fixed assets to net worth	1.3 : 1

You are required to PREPARE:

(a) Balance Sheet as on 31/3/2022 based on above details.

(b) The statement showing working capital requirement if the company wants to make a provision for contingencies @ 14 percent of net working capital.

ICAI'S ANSWER

Working Notes:

(i) Cost of Goods Sold = Sales – Gross Profit (35% of Sales)

= ₹ 1,05,00,000 – ₹ 36,75,000

= ₹ 68,25,000

(ii) Closing Stock = Cost of Goods Sold / Stock Turnover

$= \frac{₹\ 68,25,000}{6}$ ₹ = ₹ 11,37,500

(iii) Fixed Assets = Cost of Goods Sold / Fixed Assets Turnover

$= \frac{₹\ 68,25,000}{1.5}$ ₹

= ₹ 45,50,000

(iv) Current Assets:

Current Ratio = 2.5 and Liquid Ratio = 1.5

Inventories (Stock) = 2.5 – 1.5 = 1

Current Assets = Amount of Inventories (Stock) x $\frac{2.5}{1}$

= ₹11,37,500 x $\frac{2.5}{1}$ = ₹ 28,43,750

(v) Liquid Assets (Receivables and Cash)

= Current Assets – Inventories (Stock)

= ₹ 28,43,750 – ₹ 11,37,500

= ₹ 17,06,250

(vi) Receivables (Debtors) = Sales x $\frac{\text{Debtors Collection period}}{12}$

= ₹1,05,00,000 × $\frac{1}{12}$

= ₹ 8,75,000

(vii) Cash = Liquid Assets – Receivables (Debtors)

= ₹ 17,06,250 – ₹ 8,75,000 = ₹ 8,31,250

(viii) Net worth = $\frac{\text{Fixed Assets}}{1.3}$

= $\frac{₹45,50,000}{1.3}$ = ₹ 35,00,000

(ix) Reserves and Surplus

Reserves and Share Capital = Net worth

Net worth = 1 + 1.5 = 2.5

Reserves and Surplus = ₹35,00,000 x $\frac{1}{2.5}$

= ₹ 14,00,000

(x) Share Capital = Net worth – Reserves and Surplus

= ₹ 35,00,000 – ₹ 14,00,000

= ₹ 21,00,000

(xi) Current Liabilities = Current Assets/ Current Ratio

= $\frac{₹28,43,750}{2.5}$ = ₹ 11,37,500

(xii) Long-term Debts

Capital Gearing Ratio = Long-term Debts / Equity Shareholders' Fund

Long-term Debts = ₹ 35,00,000 × 0.7875 = ₹ 27,56,250

(a) Balance Sheet

Particulars	Figures as at 31-03-2022 (₹)	Figures as at 31-03-2021 (₹)
I. EQUITY AND LIABILITIES		
Shareholders' funds		
(a) Share capital	21,00,000	-
(b) Reserves and surplus	14,00,000	-
Non-current liabilities		
(a) Long-term borrowings	27,56,250	-
Current liabilities	11,37,500	-
TOTAL	**73,93,750**	**-**
II. ASSETS		
Non-current assets		
Fixed assets	45,50,000	-
Current assets		
Inventories	11,37,500	-
Trade receivables	8,75,000	-
Cash and cash equivalents	8,31,250	-
TOTAL	**73,93,750**	**-**

(b) Statement Showing Working Capital Requirement

Particulars	(₹)	(₹)
A. Current Assets		
(i) Inventories (Stocks)		11,37,500
(ii) Receivables (Debtors)		8,75,000
(iii) Cash in hand & at bank		8,31,250
Total Current Assets		28,43,750
B. Current Liabilities:		
Total Current Liabilities		11,37,500
Net Working Capital (A – B)		17,06,250

Particulars	(₹)	(₹)
Add: Provision for contingencies (14% of Net Working Capital)		2,38,875
Working capital requirement		19,45,125

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109 RTP **Old P8 May 2023** Q1 old scheme

From the following information, find out missing figures and REWRITE the balance sheet of Mukesh Enterprise.

Current Ratio = 2:1

Acid Test ratio = 3:2

Reserves and surplus = 20% of equity share capital

Long term debt = 45% of net worth

Stock turnover velocity = 1.5 months

Receivables turnover velocity = 2 months

You may assume closing Receivables as average Receivables.

Gross profit ratio = 20%

Sales is ₹ 21,00,000 (25% sales are on cash basis and balance on credit basis)

Closing stock is ₹ 40,000 more than opening stock.

Accumulated depreciation is 1/6 of original cost of fixed assets.

Balance sheet of the company is as follows:

Liabilities	(₹)	Assets	(₹)
Equity Share Capital	?	Fixed Assets (Cost)	?
Reserves & Surplus	?	Less: Accumulated. Depreciation	?
Long Term Loans	6,75,000	Fixed Assets (WDV)	?
Bank Overdraft	60,000	Stock	?
Creditors	?	Debtors	?
		Cash	?
Total	?	Total	?

ICAI'S ANSWER

Liabilities	(₹)	Assets	(₹)
Equity Share Capital	12,50,000	Fixed Assets (cost)	20,58,000
Reserves & Surplus	2,50,000	Less: Acc. Depreciation	(3,43,000)
Long Term Loans	6,75,000	Fixed Assets (WDV)	17,15,000
Bank Overdraft	60,000	Stock	2,30,000
Payables	4,00,000	Receivables	2,62,500
		Cash	4,27,500
Total	**26,35,000**	**Total**	**26,35,000**

Working Notes:

(i)

Sales	₹ 21,00,000
Less: Gross Profit (20%)	₹ 4,20,000
Cost of Goods Sold (COGS)	₹ 16,80,000

(ii) Receivables Turnover Velocity = $\frac{\text{Average Receivables}}{\text{Credit Sales}} \times 12$

$2 = \frac{\text{Average Receivables}}{₹\ 21{,}00{,}000 \times 75\%} \times 12$

Average Receivables = $\frac{₹\ 21{,}00{,}000 \times 75\% \times 2}{12}$

Average Receivables = ₹ 2,62,500

Closing Receivables = ₹ 2,62,500

(iii) Stock Turnover Velocity = $\frac{\text{Average Stock}}{\text{COGS}} \times 12$

Or $1.5 = \frac{\text{Average Stock}}{₹\ 16{,}80{,}000} \times 12$

Or Average Stock = $\frac{₹\ 16{,}80{,}000 \times 1.5}{12}$

Or Average Stock = ₹ 2,10,000

$\frac{\text{Opening Stock} + \text{Closing Stock}}{2}$ =₹ 2,10,000

Opening Stock + Closing Stock = ₹ 4,20,000.....(1)

Also, Closing Stock-Opening Stock = ₹ 40,000.....(2)

Solving (1) and (2), we get **closing stock = ₹ 2,30,000**

(iv) Current Ratio= $\frac{\text{Current Assets}}{\text{Current Liabilities}} = \frac{\text{Stock} + \text{Receivables} + \text{Cash}}{\text{Bank Overdraft} + \text{Creditors}}$

Or $2 = \frac{₹\ 2{,}30{,}000 + ₹\ 2{,}62{,}500 + \text{Cash}}{₹\ 60{,}000 + \text{Creditors}}$

Or ₹ 1,20,000 + 2 Payables = ₹ 4,92,500 + Cash

Or 2 Payables – Cash.= ₹ 3,72,500

Or Cash = 2 Payables – ₹ 3,72,500.....(3)

$$\text{Acid Test Ratio} = \frac{\text{Current Assests - Stock}}{\text{Current Liabilities}} = \frac{\text{Debtor} + \text{Cash}}{\text{Current Liabilities}}$$

$$\text{Or } \frac{3}{2} = \frac{₹\ 2{,}62{,}500 + \text{Cash}}{60{,}000 + \text{Creditors}}$$

Or ₹ 1,80,000 + 3 Payables = ₹ 5,25,000 + 2 Cash

Or 3 Payables – 2 Cash = ₹ 3,45,000 ..... (4)

Substitute (3) in (4)

Or 3 Payables – 2(2 Payables – ₹ 3,72,500) = ₹ 3,45,000

Or 3 Payables – 4 Payables + ₹ 7,45,000= ₹ 3,45,000

(Payables) = ₹ 3,45,000 - ₹ 7,45,000

Payables = ₹ 4,00,000

So, Cash = 2 x ₹ 4,00,000 – ₹ 3,72,5000

Cash = ₹ 4,27,500

(v) Long term Debt = 45% of Net Worth

Or ₹ 6,75,000 = 45% of Net Worth

Net Worth = ₹ 15,00,000

(vi) Equity Share Capital (ESC) + Reserves = ₹ 15,00,000

Or ESC + 0.2ESC = ₹ 15,00,000

Or 1.2 ESC = ₹ 15,00,000

Equity Share Capital (ESC) = ₹ 12,50,000

(vii) Reserves = 0.2 x ₹ 12,50,000

Reserves = ₹ 2,50,000

(viii) Total of Liabilities=Total of Assets

Or ₹ 12,50,000 + ₹ 2,50,000 + ₹ 6,75,000 +₹ 60,000 + ₹ 4,00,000 + Fixes Assets(FA) (WDV) + ₹ 2,30,000 + ₹ 2,62,000 +₹ 4,27,500

Or ₹ 26,35,000 = ₹ 9,20,000 + FA(WDV)

FA (WDV) =₹ 17,15,000

Now FA(Cost) – Depreciation = FA(WDV)

Or FA(Cost) – FA(Cost)/6 = ₹ 17,15,000

Or 5 FA(Cost)/6 = ₹ 17,15,000

Or FA(Cost) = ₹ 17,15,000x 6/5

So, FA(Cost) = ₹ 20,58,000

Depreciation = ₹ 20,58,000/6 = ₹ 3,43,000

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110 RTP **Old P8 Nov 2022** Q1 old scheme

The following information of ASD Ltd. relate to the year ended 31st March, 2022:

Net profit	8% of sales
Raw materials consumed	20% of Cost of Goods Sold
Direct wages	10% of Cost of Goods Sold
Stock of raw materials	3 months' usage
Stock of finished goods	6% of Cost of Goods Sold
Gross Profit	15% of Sales
Debt collection period	2 Months
(All sales are on credit)	
Current ratio	2 : 1
Fixed assets to Current assets	13 : 11
Fixed assets to sales	1 : 3
Long-term loans to Current liabilities	2 : 1
Capital to Reserves and Surplus	1 : 4

You are required to PREPARE-

(a) Profit & Loss Statement of ASD Limited for the year ended 31st March, 2022 in the following format.

Particulars	(₹)	Particulars	(₹)
To Direct Materials consumed	?	By Sales	?
To Direct Wages	?		
To Works (Overhead)	?		
To Gross Profit c/d	?		
	?		?
To Selling and Distribution Expenses	?	By Gross Profit b/d	?
To Net Profit	?		
	?		?

(b) Balance Sheet as on 31st March, 2022 in the following format.

Liabilities	(₹)	Assets	(₹)
Share Capital	?	Fixed Assets	1,30,00,000
Reserves and Surplus	?	Current Assets:	
Long term loans	?	Stock of Raw Material	?
Current liabilities	?	Stock of Finished Goods	?
		Debtors	?
		Cash	?
	?		?

ICAI'S ANSWER

Working Notes:

(i) Calculation of Sales

$$\frac{\text{Fixed Assets}}{\text{Sales}} = \frac{1}{3}$$

$$\therefore \frac{1,30,00,000}{\text{Sales}} = \frac{1}{3} \Rightarrow \text{Sales} = ₹\ 3,90,00,000$$

(ii) Calculation of Current Assets

$$\frac{\text{Fixed Assets}}{\text{Current Assets}} = \frac{13}{11}$$

$$\therefore \frac{1,30,00,000}{\text{Current Assets}} = \frac{13}{11} \Rightarrow \text{Current Assets} = ₹\ 1,10,00,000$$

(iii) Calculation of Raw Material Consumption and Direct Wages

	₹
Sales	3,90,00,000
Less: Gross Profit (15 % of Sales)	58,50,000
Cost of Goods sold	3,31,50,000

Raw Material Consumption (20% of Cost of Goods Sold) ₹ 66,30,000

Direct Wages (10% of Cost of Goods Sold) ₹ 33,15,000

(iv) Calculation of Stock of Raw Materials (= 3 months usage)

$$= 66,30,000 \times \frac{3}{12} = ₹\ 16,57,500$$

(v) Calculation of Stock of Finished Goods (= 6% of Cost of Goods Sold)

$$= 3,31,50,000 \times \frac{6}{100} = ₹\ 19,89,000$$

(vi) Calculation of Current Liabilities

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = 2$$

$$\frac{1,10,00,000}{\text{Current Liabilities}} = 2 \Rightarrow \text{Current Liabilities} = ₹\ 55,00,000$$

(vii) Calculation of Debtors

$$\text{Average collection period} = \frac{\text{Debtors}}{\text{Credit Sales}} \times 12 \text{ months}$$

$$\frac{\text{Debtors}}{3,90,00,000} \times 12 = 2 \Rightarrow \text{Debtors} = ₹\ 65,00,000$$

(viii) Calculation of Long-term Loan

$$\frac{\text{Long term Loan}}{\text{Current Liabilities}} = \frac{2}{1}$$

$$\frac{\text{Long term loan}}{55,00,000} = \frac{2}{1} \Rightarrow \text{Long term loan} = ₹\ 1,10,00,000$$

(ix) Calculation of Cash Balance

		₹
Current assets		1,10,00,000
Less: Debtors	65,00,000	
Raw materials stock	16,57,500	
Finished goods stock	19,89,000	1,01,46,500
Cash balance		8,53,500

(x) Calculation of Net worth

Fixed Assets		1,30,00,000
Current Assets		1,10,00,000
Total Assets		2,40,00,000
Less: Long term Loan	1,10,00,000	
Current Liabilities	55,00,000	1,65,00,000
Net worth		75,00,000

Net worth = Share capital + Reserves = ₹ 75,00,000

$$\frac{\text{Capital}}{\text{Reserves and Surplus}} = \frac{1}{4} \Rightarrow \text{Share Capital} = ₹\ 75,00,000 \times \frac{1}{5} = ₹\ 15,00,000$$

$$\text{Reserves and Surplus} = ₹\ 75,00,000 \times \frac{4}{5} = ₹\ 60,00,000$$

Profit and Loss Statement of ASD Ltd.

for the year ended 31st March, 2022

Particulars	(₹)	Particulars	(₹)
To Direct Materials consumed	66,30,000	By Sales	3,90,00,000
To Direct Wages	33,15,000		
To Works (Overhead) (Bal. fig.)	2,32,05,000		
To Gross Profit c/d (15% of Sales)	58,50,000		
	3,90,00,000		3,90,00,000
To Selling and Distribution Expenses (Bal. fig.)	27,30,000	By Gross Profit b/d	58,50,000
To Net Profit (8% of Sales)	31,20,000		
	58,50,000		**58,50,000**

Balance Sheet of ASD Ltd.

as at 31st March, 2022

Liabilities	(₹)	Assets	(₹)
Share Capital	15,00,000	Fixed Assets	1,30,00,000
Reserves and Surplus	60,00,000	Current Assets:	
Long term loans	1,10,00,000	Stock of Raw Material	16,57,500
Current liabilities	55,00,000	Stock of Finished Goods	19,89,000
		Debtors	65,00,000
		Cash	8,53,500
	2,40,00,000		**2,40,00,000**

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111 RTP **Old P8 May 2022** Q1 old scheme

FM Ltd. is in a competitive market where every company offers credit. To maintain the competition, FM Ltd. sold all its goods on credit and simultaneously received the goods on credit. The company provides

the following information relating to current financial year:

Debtors Velocity	3 months
Creditors Velocity	2 months
Stock Turnover Ratio (on Cost of Goods Sold)	1.5
Fixed Assets turnover Ratio (on Cost of Goods Sold)	4
Gross Profit Ratio	25%
Bills Receivables	₹ 75,000
Bills Payables	₹ 30,000
Gross Profit	₹ 12,00,000

FM Ltd. has the tendency of maintaining extra stock of ₹ 30,000 at the end of the period than that at the beginning.

DETERMINE:

(i) Sales and cost of goods sold

(ii) Sundry Debtors

(iii) Closing Stock

(iv) Sundry Creditors

(v) Fixed Assets

ICAI'S ANSWER

(i) Determination of Sales and Cost of goods sold:

$$\text{Gross Profit Ratio} = \frac{\text{Gross Profit}}{\text{Sales}} \times 100$$

$$\text{Or, } \frac{25}{100} = \frac{₹\ 12,00,000}{\text{Sales}}$$

$$\text{Or, Sales} = \frac{12,00,00,000}{25} = ₹\ 48,00,000$$

Cost of Goods Sold = Sales – Gross Profit

= ₹ 48,00,000 - ₹ 12,00,000 = ₹ 36,00,000

(ii) Determination of Sundry Debtors:

Debtors' velocity is 3 months or Debtors' collection period is 3 months,

$$\text{So, Debtors' turnover ratio} = \frac{12\text{months}}{3\text{months}} = 4$$

$$\text{Debtors' turnover ratio} = \frac{\text{Credit Sales}}{\text{Average Accounts Receivable}}$$

$$= \frac{₹\ 48,00,000}{\text{Bills Receivable} + \text{Sundry Debtors}} = 4$$

Or, Sundry Debtors + Bills receivable = ₹ 12,00,000

Sundry Debtors = ₹ 12,00,000 – ₹ 75,000 = ₹ 11,25,000

(iii) Determination of Closing Stock

Stock Turnover Ratio = $\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{₹\ 36,00,000}{\text{Average Stock}}$ =1.5

So, Average Stock = ₹ 24,00,000

Now Average Stock = $\frac{\text{Opening Stock} + \text{Closing Stock}}{2}$

Or $\frac{\text{Opening Stock} + (\text{Opening Stock} + ₹\ 30,000)}{2}$ = ₹ 24,00,000

Or 2 Opening Stock + ₹ 30,000= ₹48,00,000

Or 2 Opening Stock = ₹47,70,000

Or, Opening Stock = ₹ 23,85,000

So, Closing Stock = ₹ 23,85,000 + ₹ 30,000 = ₹ 24,15,000

(iv) Determination of Sundry Creditors:

Creditors' velocity of 2 months or credit payment period is 2 months.

So, Creditors' turnover ratio = $\frac{12 \text{ months}}{2 \text{ months}}$ = 6

Creditors turnover ratio = $\frac{\text{Credit Purchases *}}{\text{Average AccountsPayables}}$

= $\frac{₹\ 36,30,000}{\text{Sundry Creditors} + \text{Bills Payables}}$ = 6

So, Sundry Creditors + Bills Payable = ₹ 6,05,000

Or, Sundry Creditors + ₹ 30,000 = ₹ 6,05,000

Or, Sundry Creditors = ₹ 5,75,000

(v) Determination of Fixed Assets

Fixed Assets Turnover Ratio = $\frac{\text{Cost of GoodsSold}}{\text{Fixed Assets}}$ = 4

Or, $\frac{₹\ 36,00,000}{\text{Fixed Assets}}$ = 4

Or, Fixed Asset = ₹ 9,00,000

Workings:

***Calculation of Credit purchases:**

Cost of goods sold = Opening stock + Purchases – Closing stock

₹ 36,00,000 = ₹ 23,85,000 + Purchases – ₹ 24,15,000

Purchases (credit) = ₹ 36,30,000

Calculation of credit purchase also can be done as below:

Or Credit Purchases =Cost of goods sold +Difference in Opening Stock

Or Credit Purchases =36,00,000 +30,000=₹ 36,30,000

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112 RTP **Old P8 Nov 2020** Q1 old scheme

Following information has been provided from the books of M/s Laxmi & Co. for the year ending on 31st

March, 2020:

Net Working Capital	₹ 4,80,000
Bank overdraft	₹ 80,000
Fixed Assets to Proprietary ratio	0.75
Reserves and Surplus	₹ 3,20,000
Current ratio	2.5
Liquid ratio (Quick Ratio)	1.5

You are required to PREPARE a summarised Balance Sheet as at 31st March, 2020.

ICAI'S ANSWER

Working notes:

(i) Current Assets and Current Liabilities computation:

$$\frac{\text{Current assets}}{\text{Current liabilities}} = \frac{2.5}{1}$$

Or Current assets = 2.5 Current liabilities

Now, Working capital = Current assets − Current liabilities

Or ₹ 4,80,000 = 2.5 Current liability − Current liability

Or 1.5 Current liability = ₹ 4,80,000

∴ Current Liabilities = ₹ 3,20,000

So, Current Assets = ₹ 3,20,000 × 2.5 = ₹ 8,00,000

(ii) Computation of stock

$$\text{Liquid ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

$$\text{Or } 1.5 = \frac{\text{Current assets - Inventories}}{₹\ 3{,}20{,}000}$$

Or 1.5 × ₹ 3,20,000 = ₹ 8,00,000 − Inventories

Or Inventories = ₹ 8,00,000 – ₹ 4,80,000

Or Stock = ₹ 3,20,000

(iii) Computation of Proprietary fund; Fixed assets; Capital and Sundry creditors

$$\text{Fixed Asset to Proprietary ratio} = \frac{\text{Fixed assets}}{\text{Proprietary fund}} = 0.75$$

∴ Fixed Assets = 0.75 Proprietary fund (PF)[FA+NWC = PF]

or NWC = PF- FA [(i.e. .75 PF)]

and Net Working Capital (NWC) = 0.25 Proprietary fund

Or ₹ 4,80,000/0.25 = Proprietary fund

Or Proprietary fund = ₹ 19,20,000

and Fixed Assets = 0.75 proprietary fund

= 0.75 × ₹ 19,20,000 = ₹ 14,40,000

Capital = Proprietary fund – Reserves & Surplus

= ₹ 19,20,000 – ₹ 3,20,000 = ₹ 16,00,000

Sundry Creditors = (Current liabilities – Bank overdraft)

= (₹ 3,20,000 – ₹ 80,000) = ₹ 2,40,000

Balance Sheet as at 31st March, 2020

Liabilities	₹	Assets	₹
Capital	16,00,000	Fixed Assets	14,40,000
Reserves & Surplus	3,20,000	Stock	3,20,000
Bank overdraft	80,000	Other Current Assets	4,80,000
Sundry creditors	2,40,000		
	22,40,000		22,40,000

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113 RTP **Old P8 Nov 2020** Q9 old scheme

The following figures and ratios are related to a company:

(i) Sales for the year (all credit)	₹ 90,00,000
(ii) Gross Profit ratio	35 percent
(iii) Fixed assets turnover (based on cost of goods sold)	1.5
(iv) Stock turnover (based on cost of goods sold)	6
(v) Liquid ratio	1.5:1
(vi) Current ratio	2.5:1
(vii) Receivables (Debtors) collection period	1 month
(viii) Reserves and surplus to Share capital	1:1.5
(ix) Capital gearing ratio	0.7875
(x) Fixed assets to net worth	1.3 : 1

You are required to PREPARE:

(a) Balance Sheet of the company on the basis of above details.

(b) The statement showing working capital requirement, if the company wants to make a provision for contingencies @ 15 percent of net working capital.

ICAI'S ANSWER

Working Notes:

(i) Cost of Goods Sold = Sales – Gross Profit (35% of Sales)

= ₹ 90,00,000 – ₹ 31,50,000

= ₹ 58,50,000

(ii) Closing Stock = Cost of Goods Sold / Stock Turnover

= ₹ 58,50,000/6 = ₹ 9,75,000

(iii) Fixed Assets = Cost of Goods Sold / Fixed Assets Turnover

= ₹ 58,50,000/1.5

= ₹ 39,00,000

(iv) Current Assets:

Current Ratio = 2.5 and Liquid Ratio = 1.5

Inventories (Stock) = 2.5 – 1.5 = 1

Current Assets = Amount of Inventories (Stock) × 2.5/1

= ₹ 9,75,000 × 2.5/1 = ₹ 24,37,500

(v) Liquid Assets (Receivables and Cash)

= Current Assets – Inventories (Stock)

= ₹ 24,37,500 – ₹ 9,75,000

= ₹14,62,500

(vi) Receivables (Debtors) = Sales × Debtors Collection period /12

= ₹ 90,00,000 × 1/12

= ₹ 7,50,000

(vii) Cash = Liquid Assets – Receivables (Debtors)

= ₹14,62,500 – ₹ 7,50,000 = ₹ 7,12,500

(viii) Net worth = Fixed Assets /1.3

= ₹ 39,00,000/1.3 = ₹ 30,00,000

(ix) Reserves and Surplus

Reserves and Share Capital = Net worth

Net worth = 1 + 1.5 = 2.5

Reserves and Surplus = ₹ 30,00,000 × 1/2.5

= ₹ 12,00,000

(x) Share Capital = Net worth – Reserves and Surplus

= ₹ 30,00,000 – ₹ 12,00,000

= ₹ 18,00,000

(xi) Current Liabilities = Current Assets/ Current Ratio

= ₹ 24,37,500/2.5 = ₹ 9,75,000

(xii) Long-term Debts

Capital Gearing Ratio = Long-term Debts / Equity Shareholders' Fund

Long-term Debts = ₹30,00,000 × 0.7875 = ₹23,62,500

(a) Balance Sheet of the Company

Particulars	Figures as at 31-03-2020 (₹)	Figures as at 31-03-2019 (₹)
I. EQUITY AND LIABILITIES		
Shareholders' funds		
(a) Share capital	18,00,000	-
(b) Reserves and surplus	12,00,000	-
Non-current liabilities		
(a) Long-term borrowings	23,62,500	-
Current liabilities	9,75,000	-
TOTAL	**63,37,500**	**-**
II. ASSETS		
Non-current assets		
Fixed assets	39,00,000	-
Current assets		
Inventories	9,75,000	-
Trade receivables	7,50,000	-
Cash and cash equivalents	7,12,500	-
TOTAL	**63,37,500**	**-**

(b) Statement Showing Working Capital Requirement

Particulars	(₹)	(₹)
A. Current Assets		
(i) Inventories (Stocks)		9,75,000
(ii) Receivables (Debtors)		7,50,000
(iii) Cash in hand & at bank		7,12,500
Total Current Assets		24,37,500

Particulars	(₹)	(₹)
B. Current Liabilities:		
Total Current Liabilities		9,75,000
Net Working Capital (A – B)		14,62,500
Add: Provision for contingencies (15% of Net Working Capital)		2,19,375
Working capital requirement		16,81,875

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114 MTP **Sep 2026 S-I** Q3(b) 3 marks

The following financial information has been extracted from the records of M/s Durga & Co. as at 31st March, 2026:

Particulars	Amount / Ratio
Net Working Capital	₹ 4,80,000
Bank Overdraft	₹ 80,000
Fixed Assets to Proprietors' Fund Ratio	0.75 : 1
Reserves and Surplus	₹ 3,20,000
Current Ratio	2.5 : 1
Liquid (Quick) Ratio	1.5 : 1

Required:

Using the information provided above, prepare a Summarised Balance Sheet of M/s Durga & Co. as at 31st March, 2026, showing the major heads of assets and liabilities.

ICAI'S ANSWER

Working notes:

(i) **Current Assets and Current Liabilities computation:**

$$\frac{\text{Current assests}}{\text{Currect liabilities}} = \frac{2.5}{1}$$

Or Current assets = 2.5 Current liabilities

Now, Working capital = Current assets – Current liabilities

Or ₹ 4,80,000 = 2.5 Current liability – Current liability

Or 1.5 Current liability = ₹ 4,80,000

∴ Current Liabilities = ₹ 3,20,000

So, Current Assets = ₹ 3,20,000 × 2.5 = ₹ 8,00,000

(ii) **Computation of stock**

$$\text{Liquid ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

$$\text{Or } 1.5 = \frac{\text{Current assets - Inventories}}{₹3{,}20{,}000}$$

Or 1.5 × ₹ 3, 20,000 = ₹ 8,00,000 – Inventories

Or Inventories = ₹ 8,00,000 – ₹ 4, 80,000

Or Stock = ₹ 3,20,000

(iii) **Computation of Proprietary fund; Fixed assets; Capital and Sundry creditors**

$$\text{Fixed Asset to Proprietary ratio} = \frac{\text{Fixed assets}}{\text{Proprietary fund}} = 0.75$$

∴ Fixed Assets = 0.75 Proprietary fund (PF)[FA + NWC = PF]

or NWC = PF- FA [(i.e. .75 PF)]

and Net Working Capital (NWC)= 0.25 Proprietary fund

Or ₹ 4,80,000/0.25 = Proprietary fund

Or Proprietary fund = ₹ 19,20,000

and Fixed Assets = 0.75 proprietary fund = 0.75 × ₹ 19,20,000 = ₹ 14,40,000

Capital = Proprietary fund – Reserves & Surplus = ₹ 19,20,000 – ₹ 3,20,000 = ₹ 16,00,000

Sundry Creditors = (Current liabilities – Bank overdraft) = (₹ 3,20,000 – ₹ 80,000) = ₹ 2,40,000

Balance Sheet as at 31st March, 2020

Liabilities	₹	Assets	₹
Capital	16,00,000	Fixed Assets	14,40,000
Reserves & Surplus	3,20,000	Stock	3,20,000
Bank overdraft	80,000	Other Current Assets	4,80,000
Sundry creditors	2,40,000		
	22,40,000		22,40,000

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115 MTP **May 2026 S-I** Q1(a) 5 marks

The following information pertains to P Limited for the year ended 31st March, 2025:

Particulars	Amount
Sales	₹ 3,60,00,000
Rate of Income Tax	40%
Return on Net Worth	30%
Share Capital to Reserves Ratio	6 : 4

Particulars	Amount
Current Ratio	2 : 1
Percentage of Net Profit to Sales	8%
Inventory Turnover (based on cost of goods sold)	12
Cost of goods sold	₹ 1,44,00,000
Sundry Debtors	₹ 12,00,000
Sundry Creditors	₹ 16,00,000
Interest on 14% debentures	₹ 3,36,000

You are required to CALCULATE:

(i) Operating Expenses

(ii) Share Capital and Reserves

(iii) Closing Stock

(iv) Fixed Assets

ICAI'S ANSWER

(i) Operating Expenses

Net Profit = 8% on Sales = 8% x ₹ 3,60,00,000	₹ 28,80,000
Add: Income Tax = 40% x ₹ 28,80,000/60%	₹ 19,20,000
Profit Before tax	₹ 48,00,000
Add: Debenture Interest	₹ 3,36,000
Profit before interest and tax	₹ 51,36,000
Sales	₹ 3,60,00,000
Less:	
Cost of goods sold ₹ 1,44,00,000	
Profit before interest & tax ₹ 51,36,000	₹ 1,95,36,000
Operating Expenses	₹ 1,64,64,000

(ii) Share Capital and Reserves

Return on Net worth is 30%, hence profit after tax of ₹28,80,000 is equivalent to 30% of Net Worth.

Net worth x 30/100 = ₹ 28,80,000

Net Worth = 28,80,000 x 100/30

= ₹ 96,00,000

Ratio of Share Capital to Reserves = 6:4

Share Capital = ₹ 96,00,000 x 6/10 = ₹ 57,60,000

Reserves = ₹ 96,00,000 x 4/10 = ₹ 38,40,000

(iii) Closing Stock

Inventory Turnover (based on cost of goods sold) = 12

Cost of goods sold = ₹ 1,44,00,000

Inventory Turnover Ratio = Cost of goods sold / Closing Stock = 12

12 = ₹ 1,44,00,000/Closing Stock

Therefore, Closing Stock = ₹ 1,44,00,000/12

= ₹ 12,00,000

(iv) Fixed Assets

Calculation of Total liabilities

Interest on 14 % Debentures = ₹ 3,36,000

Therefore debentures = $\frac{₹3,36,000 \text{ x } 100}{14\%}$

= ₹ 24,00,000

Total of Liabilities Side

Share Capital	₹ 57,60,000
Reserves	₹ 38,40,000
Debentures	₹ 24,00,000
Sundry Creditors	₹ 16,00,000
Total Liabilities	₹ 1,36,00,000

Current Ratio = 2:1

Sundry Creditors = ₹ 16,00,000

Current Assets / Current Liabilities =2

Current Assets / Sundry Creditors =2

Current Assets /₹16,00,000 =2

Current Assets = 2 x ₹ 16,00,000

= ₹ 32,00,000

Fixed Assets = Total Liabilities – Current Assets

= ₹ 1,36,00,000 - ₹ 32,00,000

= ₹ 1,04,00,000

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116 MTP Jan 2026 S-I Q1(b) 5 marks

Following information is given for Prashuk Ltd.:

Fixed assets turnover ratio (Based on cost of sales)	8 times
Capital turnover ratio (Based on cost of sales)	2 times
Inventory Turnover	8 times
Receivable turnover	4 times
Payable turnover	6 times
GP Ratio	25%

Gross profit during the year amounts to ₹ 8,00,000. There is no long-term loan or overdraft. Reserve and surplus amount to ₹ 2,00,000. Ending inventory of the year is ₹ 20,000 above the beginning inventory. Based on the above information, DETERMINE the various assets and liabilities of the company.

ICAI'S ANSWER

(a) G.P. ratio = $\frac{\text{Gross Profit}}{\text{Sales}} = 25\%$

Sales = $\frac{\text{Gross Profit}}{25} \times 100 = \frac{₹\ 8,00,000}{25} \times 100$ = ₹ 32,00,000

(b) Cost of Sales = Sales – Gross profit

= ₹ 32,00,000 - ₹ 8,00,000

= ₹ 24,00,000

(c) Receivable turnover = $\frac{\text{Sales}}{\text{Debtors}} = 4$

= Debtors = $\frac{\text{Sales}}{4} = \frac{₹\ 32,00,000}{4}$ = ₹ 8,00,000

(d) Fixed assets turnover = $\frac{\text{Cost of Sales}}{\text{Fixed Assets}} = 8$

Fixed assets = $\frac{\text{Cost of Sales}}{8} = \frac{₹\ 24,00,000}{8}$ = ₹ 3,00,000

(e) Inventory turnover = $\frac{\text{Cost of Sales}}{\text{Average Stock}} = 8$

Average Stock = $\frac{\text{Cost of Sales}}{8} = \frac{₹24,00,000}{8}$ = ₹ 3,00,000

Average Stock = $\frac{\text{Opening Stock} + \text{Closing Stock}}{2}$

Average Stock = $\frac{\text{Opening Stock} + \text{Opening Stock} + 20,000}{2}$

Average Stock = Opening Stock + ₹ 10,000

Opening Stock = Average Stock - ₹ 10,000

= ₹ 3,00,000 - ₹10,000

= ₹ 2,90,000

Closing Stock = Opening Stock + ₹ 20,000

= ₹ 2,90,000 + ₹ 20,000 = ₹ 3,10,000

(f) Payable turnover = $\frac{\text{Purchase}}{\text{Creditors}} = 6$

Purchases = Cost of Sales + Increase in Stock

= ₹ 24,00,000 + ₹ 20,000 = ₹ 24,20,000

Creditors = $\frac{\text{Purchase}}{6} = \frac{₹24,20,000}{6}$ = ₹ 4,03,333

(g) Capital turnover = $\frac{\text{Cost of Sales}}{\text{Capital Employed}} = 2$

Capital Employed = $\frac{\text{Cost of Sales}}{2} = \frac{₹\ 24,00,000}{2}$ = ₹ 12,00,000

(h) Capital = Capital Employed – Reserves & Surplus

= ₹ 12,00,000 – ₹ 2,00,000 = ₹ 10,00,000

Balance Sheet of T Ltd as on.....

Liabilities	Amount (₹)	Assets	Amount (₹)
Capital	10,00,000	Fixed Assets	3,00,000
Reserve & Surplus	2,00,000	Stock	3,10,000
Creditors	4,03,333	Debtors	8,00,000
		Other Current Assets	1,93,333
	16,03,333		16,03,333

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117 MTP **Sep 2025 S-I** Q1(b) 5 marks

Gagan Pvt. Ltd. gives you the following information relating to the year ending 31st March, 2025:

(1)	Current Ratio	2.5 : 1
(2)	Debt-Equity Ratio	1:2
(3)	Return on Total Assets (After Tax)	15%
(4)	Total Assets Turnover Ratio	2
(5)	Gross Profit Ratio	25%
(6)	Stock Turnover Ratio	5
(7)	Net Working Capital	₹ 15,00,000
(8)	Fixed Assets	₹ 30,00,000
(9)	1,80,000 Equity Shares of	₹ 10 each
(10)	60,000, 10% Preference Shares of	₹ 10 each
(11)	Opening Stock	₹ 16,00,000

You are required to CALCULATE:

(a) Quick Ratio

(b) Fixed Assets Turnover Ratio

(c) Proprietary Ratio

(d) Earnings per Share

ICAI'S ANSWER

Workings Notes:

(i) Computation of Current Assets & Current Liabilities & Total Assets

Net Working Capital = Current Assets – Current Liabilities = 2.5 – 1 = 1.5

Thus, Current Assets = $\frac{\text{Net Working Capital} \times 2.5}{1.5}$

= $\frac{15,00,000 \times 2.5}{1.5}$ = ₹ 25,00,000

Current Liabilities (CL) = ₹ 25,00,000 – ₹ 15,00,000 = ₹ 10,00,000

Total Assets = Current Assets + Fixed Assets

= ₹ 25,00,000 + ₹ 30,00,000 = ₹ 55,00,000

(ii) Computation of Sales & Cost of Goods Sold

Sales = Total Assets Turnover × Total Assets

= 2 x (55,00,000)

= ₹ 1,10,00,000

Cost of Goods Sold = (100% – 25%) of Sales = 75% of Sales

= 75% × ₹ 1,10,00,000 = ₹ 82,50,000

(iii) Computation of Stock & Quick Assets

Average Stock = $\frac{\text{Cost of Good Sold}}{\text{Stock Turnover Ratio}} = \frac{82,50,000}{5}$

= ₹ 16,50,000

Closing Stock = (Average Stock × 2) – Opening Stock

= (₹ 16,50,000 × 2) – ₹ 16,00,000 = ₹ 17,00,000

Quick Assets = Current Assets – Closing Stock

= ₹ 25,00,000 – ₹ 17,00,000 = ₹ 8,00,000

(iv) Computation of Proprietary Fund

Debt-Equity Ratio = $\frac{\text{Debt}}{\text{Equity}} = \frac{1}{2}$

Or Equity = 2 Debt

Total Assets = Equity + Preference capital + Debt + CL

₹ 55,00,000 = 2 Debt + ₹ 6,00,000 + Debt + ₹ 10,00,000

3 Debt = 55,00,000 – 6,00,000 - 10,00,000

= 39,00,000

Debt = 13,00,000

Equity = ₹ 13,00,000 × 2 = ₹ 26,00,000

So, Proprietary Fund = Equity + Preference Capital = ₹ 26,00,000 + ₹ 6,00,000 = ₹ 32,00,000

(v) Computation of Profit after tax (PAT) = Total Assets × Return on Total Assets

= ₹ 55,00,000 × 15% = ₹ 8,25,000

Calculation of Ratios:

(a) Quick Ratio

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}} = \frac{8,00,000}{10,00,000} = 0.8$$

(b) Fixed Assets Turnover Ratio

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{Sales}}{\text{Fixed Assets}} = \frac{1,10,00,000}{30,00,000}$$

= 3.67

(c) Proprietary Ratio

$$\text{Proprietary Ratio} = \frac{\text{Proprietary fund}}{\text{Total Assets}} = \frac{32,00,000}{55,00,000} = 0.58$$

(d) Earnings per Equity Share (EPS)

$$\text{Earnings per Equity Share} = \frac{\text{PAT - Preference Share Dividend}}{\text{Number of Equity Shares}}$$

$$= \frac{8,25,000 - 60,000\ (10\%\ \text{of}\ 6,00,000)}{1,80,000} = ₹\ 4.25\ \text{per share}$$

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118 MTP Sep 2025 S-II Q1(b) 5 marks

Fortune Ltd. has furnished the following information relating to the year ended 31st March, 2024 and 31st March, 2025:

	31st March, 2024 (₹)	31st March, 2025 (₹)
Share Capital	60,00,000	60,00,000
Reserve and Surplus	30,00,000	40,00,000
Long term loan	40,00,000	40,00,000

- Net profit ratio: 8%
- Gross profit ratio: 20%
- Long-term loan has been used to finance 40% of the fixed assets.
- Stock turnover with respect to cost of goods sold is 4.
- Debtors represent 90 days of credit sales.
- The company holds cash equivalent to 1½ months cost of goods sold.
- Ignore taxation and assume 360 days in a year.
- All sales are credit sales.

You are required to PREPARE Balance Sheet as on 31st March, 2025 in the following format:

Liabilities	(₹)	Assets	(₹)
Share Capital	-	Fixed Assets	-
Reserve and Surplus	-	Sundry Debtors	-
Long-term loan	-	Closing Stock	-
Sundry Creditors	-	Cash in hand	-

ICAI'S ANSWER

(i) Change in Reserve & Surplus = ₹ 40,00,000 – ₹ 30,00,000 = ₹ 10,00,000

So, Net profit = ₹ 10,00,000

Net Profit Ratio = 8%

$\therefore$ Sales = $\frac{10,00,000}{8\%}$ = ₹ 1,25,00,000

(ii) Cost of Goods sold

= Sales – Gross profit Margin

= ₹ 1,25,000,000 – 20% of ₹ 1,25,00,000

= ₹ 1,00,00,000

(iii) Fixed Assets = $\frac{₹40,00,000}{40\%}$ = ₹ 1,00,00,000

(iv) Stock = $\frac{\text{Cost of Goods Sold}}{\text{Stock Turnover ratio}} = \frac{1,00,00,000}{4}$ = ₹ 25,00,000

(v) Debtors = $\frac{1,25,00,000}{360}$ x 90 = ₹ 31,25,000

(vi) Cash Equivalent = $\frac{1,00,00,000}{12}$ ×1.5= ₹ 12,50,000

Balance Sheet as on 31st March 2025

Liabilities	(₹)	Assets	(₹)
Share Capital	60,00,000	Fixed Assets	1,00,00,000
Reserve and Surplus	40,00,000	Sundry Debtors	31,25,000
Long-term loan	40,00,000	Closing Stock	25,00,000
Sundry Creditors (Balancing Figure)	28,75,000	Cash in hand	12,50,000
	1,68,75,000		1,68,75,000

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119 MTP **May 2025 S-II** Q3(a) 8 marks

Following information and ratios are given in respect of Simandhar Limited for the year ended

31.03.2025:

Particulars	Ratio
Liquid Ratio	2.0
Cash Asset Ratio	0.36
Current Ratio	3.50
Receivables Collection Period	30 days
Proprietary Ratio	0.72
Equity Dividend	₹ 2,50,000
Equity Dividend Coverage Ratio	2.10
Non - Current Assets turnover Ratio	0.80
EPS	₹ 3.0 per share
Stock Turnover Ratio	6.0 times
GP Ratio	1/5th of Sales

Assume 360 days in a year. Closing inventory is 20% more than opening inventory.

You are required to COMPLETE the Balance Sheet as of 31st March, 2025.

BALANCE SHEET

As on 31st March, 2025

Liabilities	Amount ₹	Assets	Amount ₹
Equity Share Capital (Face Value ₹ 10)	???	Fixed Assets	???
Reserves	???	Receivables	???
Long term Debentures	???	Inventory	???
Current Liabilities	2,80,000	Cash & Bank balance	???
		Short term advances	???
Total	**????**	**Total**	**????**

ICAI'S ANSWER

WN-1: Calculation of Total Current Assets using Current Ratio

$$\text{Current Ratio} = \frac{CA}{CL}$$

$$3.5 = \frac{CA}{2{,}80{,}000}$$

$\therefore$ CA = ₹ 9,80,000

WN-2: Calculation of Inventory using Liquid Ratio

$$\text{Liquid Ratio} = \frac{\text{CA- Inventory}}{CL}$$

$$2 = \frac{₹\ 9{,}80{,}000 - \text{Inventory}}{₹\ 2{,}80{,}000}$$

∴ Inventory = ₹ 4,20,000

WN-3: Calculation of Cash & Bank balance using Cash Asset Ratio

Cash asset ratio is nothing but just the cash ratio of the company

$$\text{Cash ratio} = \frac{\text{Cash \& Cash Equivalents} + \text{Marketable securities}}{CL}$$

$$0.36 = \frac{\text{Cash \& Cash Equivalents} + 0}{₹\ 2{,}80{,}000}$$

∴ Cash & Cash Equivalents = ₹ 1,00,800

WN-4: Calculation of Sales using Inventory turnover ratio

$$\text{Inventory turnover ratio} = \frac{COGS}{₹\ \text{Avg Inventory}}$$

$$\text{Avg Inventory} = \frac{\text{Opening} + \text{Closing}}{2}$$

Now, let the opening inventory be 'x' and as mentioned closing inventory would be 20% more than the opening inventory i.e. 1.2x

$$\therefore \text{Opening inventory} = \frac{₹\ 4{,}20{,}000 \times x}{(1.2x)}$$

∴ Opening inventory = ₹ 3,50,000

$$\therefore \text{Avg Inventory} = \frac{₹\ 4{,}20{,}000 + ₹\ 3{,}50{,}000}{2}$$

∴ Avg Inventory = ₹ 3,85,000

Now, using inventory turnover ratio

$$6 = \frac{COGS}{₹\ 3{,}85{,}000}$$

∴ COGS = ₹ 23,10,000

Sales = COGS + GP

x = ₹ 23,10,000 + 0.20x

∴ Sales = ₹ 28,87,500

WN-5: Calculation of Receivables using Receivables collection period

$$\text{Receivables Collection Period} = \frac{\text{Receivables}}{\text{Credit Sales}} \text{ x } 360$$

$$30 = \frac{\text{Receivables}}{₹\ 28{,}87{,}500} \text{ x } 360$$

∴ Receivables = ₹ 2,40,625

WN-6: Calculation of Short-term Advances

Current Assets = Inventory + Receivables + Cash & Cash equivalents + Short term advances

₹ 9,80,000 = ₹ 4,20,000 + ₹ 2,40,625 + ₹ 1,00,800 + x

∴ Short term advances = ₹ 2,18,575

WN-7: Calculation of Non-current Assets (Fixed asset) using non-current assets turnover ratio

Non-current asset turnover ratio = Sales / Non-current assets

$$0.80 = \frac{₹\ 28,87,500}{x}$$

∴ Non-current asset (Fixed assets) = ₹ 36,09,375

WN-8: Calculation of Net profit available for Equity shareholders using Equity dividend coverage ratio

$$\text{Equity dividend coverage ratio} = \frac{\text{Net Profit for Equity shareholders}}{\text{Equity Dividend}}$$

$$2.10 = \frac{x}{₹\ 2,50,000}$$

∴ Net Profit for Equity shareholders = ₹ 5,25,000

WN-9: Calculation of No of equity shares and amount of equity share capital

$$\text{EPS} = \frac{\text{Net Profit for Equity shareholders}}{\text{No of equity shares}}$$

$$3 = \frac{₹\ 5,25,000}{x}$$

∴ No of equity shares = ₹ 1,75,000

∴ Amount of Equity Share capital = 1,75,000 x ₹ 10 = ₹ 17,50,000

WN-10: Calculation of Reserves using Proprietary ratio

$$\text{Proprietary Ratio} = \frac{\text{Proprietor's funds}}{\text{Total Assets}}$$

Total assets = Non-current assets + Current Assets

= ₹ 36,09,375 + ₹ 9,80,000

= ₹ 45,89,375

$$0.72 = \frac{\text{Proprietor's funds}}{₹\ 45,89,375}$$

∴ Proprietor's funds = ₹ 33,04,350

∴ Reserves = ₹ 33,04,350 (-) ₹ 17,50,000

∴ Reserves = ₹ 15,54,350

Balance Sheet as of 31/03/2025

Liabilities	Amount (₹)	Assets	Amount (₹)
Equity Share Capital (Face Value ₹ 10) **(WN-9)**	17,50,000	Fixed Assets **(WN-7)**	36,09,375
Reserves **(WN-10)**	15,54,350	Receivables **(WN-5)**	2,40,625
Long term Debentures **(Bal. figure)**	10,05,025	Inventory **(WN-2)**	4,20,000
Current Liabilities	2,80,000	Cash & Bank balance **(WN-3)**	1,00,800
		Short term advances **(WN-6)**	2,18,575
TOTAL	**45,89,375**	**TOTAL**	**45,89,375**

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120 MTP **Jan 2025 S-I** Q2(a) 6 marks

From the following information pertaining to M/s Anya Co. Ltd., PREPARE its trading, Profit & Loss Account for the year ended on 31 March, 2024 and a summarized Balance Sheet as at that date:

	Amt in ₹
Current Ratio	2.5
Quick Ratio	1.3
Proprietary Ratio (Fixed Assets/ Proprietary Fund)	0.6
Gross Profit to Sale Ratio	10%
Debtors Velocity	40 days
Sales	730,000
Working Capital	120,000
Bank Overdraft	15,000
Share Capital	2,50,000

Closing Stock is 10% more than opening Stock.

Net Profit is 10% of Proprietary Funds.

ICAI'S ANSWER

Working Note:

1. Current Liabilities and Current Assets:

Let Current Liabilities be x

Given Current ratio = 2.5

Current Assets = 2.5x

Working Capital = 2.5x- x =1.5x

or x = 1,20,000/1.5 = 80,000

So Current Liabilities = 80,000

And Current Assets = 80,000 x 2.5 = 2,00,000

2. Closing Stock

Given, Quick Ratio = 1.3

$$\frac{\text{Current Assets - Closing Stock}}{\text{Current Liabilities - Bank Overdraft}} = 1.3$$

$$\frac{2,00,000 - \text{Closing Stock}}{80,000 - 15,000} = 1.3$$

or Closing Stock = 2,00,000-84,500 =1,15,500

Opening Stock = 1,15,000 x 100/110 =1,05,000

3. Debtors

Given Debtors Velocity = 40 days

$$\frac{\text{Debtors}}{\text{Sales}} \text{ x } 365 = 40$$

$$\text{Debtors} = \frac{7,30,000 \text{x} 40}{365} = 80,000$$

4. Gross Profit = 7,30,000 x 10/100 = 73,000

5. Proprietary Fund:

Proprietary Ratio = 0.6

$$\frac{\text{Fixed Assets}}{\text{Proprietary Fund}} = 0.6$$

$$\frac{\text{Working Capital}}{\text{Proprietary Fund}} \; 0.4$$

$$\text{Proprietary Fund} = \frac{1,20,000}{0.4} = 3,00,000$$

Fixed Assets = 3,00,000 x 0.6 = 1,80,000

Net Profit = 10% of Proprietary Fund = 30,000

M/s Anya Co Ltd.

Trading and Profit and loss Account for the year ended 31 March 2024

Particulars	Amount in ₹	Particulars	Amount in ₹
To Opening Stock	1,05,000	By Sales	7,30,000
To Purchase (Balancing Fig.)	6,67,500	By Closing Stock	1,15,500
To Gross Profit	73,000		
	8,45,500		**8,45,500**
To Operating Expenses (Balancing Figure)	43,000	By Gross Profit	73,000
To Net Profit	30,000		
	73,000		**73,000**

Balance Sheet as on 31 March 2024

Liabilities	Amount in ₹	Assets	Amount in ₹
Share Capital	2,50,000	Fixed Assets	1,80,000
Reserves & Surplus (Opening bal. + current profit)	50,000		
Current Liabilities		*Current Assets*	
Bank Overdraft	15,000	Stock	1,15,500
Other Current Liabilities	65,000	Debtors	80,000
		Other Current Assets	4,500
	3,80,000		**3,80,000**

ICAI · P6 · MTP Jan 2025 · Series I · question p. 5 · answer p. 6-7

121 MTP **Jan 2025 S-II Q1(a)** 5 marks

The financial statement and operating results of Alpha Limited revealed the following position as on 31st March, 2023:

—	Equity share capital (Rs. 10 fully paid share)	Rs. 20,00,000
—	Working capital	Rs. 6,00,000
—	Bank overdraft	Rs. 1,00,000
—	Current ratio	2.5 : 1
—	Liquidity ratio	1.5 : 1
—	Proprietary ratio (Net fixed assets/Proprietary fund)	.75 : 1
—	Cost of sales	Rs. 14,40,000
—	Debtors velocity	2 months

—	Stock turnover based on cost of sales	4 times
—	Gross profit ratio	20% of sales
—	Net profit ratio	15% of sales

Closing stock was 25% higher than the opening stock. There were also free reserves brought forward from earlier years. Current assets include stock, debtors and cash only. The current liabilities expect bank overdraft treated as creditors.

Expenses include depreciation of Rs. 90,000.

The following information was collected from the records for the year ended 31st March, 2024:

- Total sales for the year were 20% higher as compared to previous year.
- Balances as on 31st March, 2024 were : Stock Rs. 5,20,000, Creditors Rs. 4,15,000, Debtors Rs. 4,95,000 and Cash balance Rs. 3,10,000.
- Percentage of Gross profit on turnover has gone up from 20% to 25% and ratio of net profit to sales from 15% to 16%.
- A portion of Fixed assets was very old (book values Rs. 1,80,000) disposed for Rs. 90,000. (No depreciations to be provided on this item).
- Long-term investments were purchased for Rs. 2,96,600.
- Bank overdraft fully discharged.
- Percentage of depreciation to Fixed assets to be provided at the rate in the previous year.

PREPARE Balance Sheet as on 31st March, 2023 and 31st March, 2024.

ICAI'S ANSWER

Balance Sheets of Alpha Limited

Liabilities	₹		Assets	₹	
	31 March 2023	**31 March 2024**		**31 March 2023**	**31 March 2024**
Equity share capital (₹ 10 each fully paid)	20,00,000	20,00,000	Fixed Assets (₹18,90,000–₹90,000)	18,00,000	15,39,000
Reserve and Surplus (balancing)	1,30,000	1,30,000	Long term investment	–	2,96,600
Profit & Loss A/c (15% of sales)	2,70,000	6,15,600	**Current Assets** (₹ 10,00,000)		
Current Liabilities			Stock	4,00,000	5,20,000
Bank Overdraft	1,00,000	–	Sundry Debtors	3,00,000	4,95,000
Creditors	3,00,000	4,15,000	Cash at Bank (Balancing)	3,00,000	3,10,000

Liabilities	₹		Assets	₹	
Total	**28,00,000**	**31,60,600**	**Total**	**28,00,000**	**31,60,600**

Calculation for 31st March, 2023

(i) Calculation of Current Liabilities

Suppose that Current Liabilities = x, then current assets will be 2.5 x

Working capital = Current Assets – Current Liabilities

6,00,000 = 2.5x – x

x = 6,00,000 / 1.5 = ₹ 4,00,000 (C.L.)

Other Current Liabilities = Current Liabilities – Bank Overdraft

(Creditors) = 4,00,000 – 1,00,000 = ₹ 3,00,000

Current Assets = 2.5 x 4,00,000 = ₹ 10,00,000

(ii) Liquid Ratio = $\frac{\text{Liquid Assets}}{\text{Current Liabilities}}$

$1.5 = \frac{\text{Liquid Assets}}{4,00,000}$

Liquid assets = ₹6,00,000

Liquid assets = Current Assets – Stock

6,00,000 = 10,00,000 – Stock

So, Stock = ₹ 4,00,000

(iii) Calculation of fixed assets: Fixed assets to proprietary fund is 0.75, working capital is therefore 0.25 of proprietary fund. So,

Fixed Assets = 6,00,000 / 0.25 x 0.75 = ₹ 18,00,000

(iv) Sales = (14,40,000 / 80) × 100 = ₹ 18,00,000

(v) Debtors = $\frac{2}{12}$ x Sales

2 / 12 × 18,00,000 = ₹ 3,00,000

(vi) Net profit = 15% of ₹18,00,000 = ₹ 2,70,000

Calculation for the year 31st March, 2024

(vii) Sales = 18,00,000 + (18,00,000 × 0.2) = 21,60,000

(viii) Calculation of fixed assets

	₹		₹
To Opening balance	18,00,000	By Banks (Sale)	90,000
		By Loss on sales of Fixed asset	90,000
		By P & L (Dep.) (5% as in previous year)	81,000
		By Balance b/d	15,39,000
Total	18,00,000		18,00,000

(ix) Net profit for the year 2011, 16% × 21,60,000 = ₹ 3,45,600

Total Profit = 2,70,000 + 3,45,600 = ₹ 6,15,600

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122 MTP **Sep 2024 S-I** Q1(a) 5 marks

You are required to CALCULATE the Total Current Assets of Ananya Limited from the given information:

Stock Turnover	= 5 times
Sales (All credit)	= ₹ 7,20,000
Gross Profit Ratio	= 25%
Current Liabilities	= 2,40,000
Liquidity Ratio	= 1.25

Stock at the end is ₹ 30,000 more than stock in the beginning.

ICAI'S ANSWER

1. Cost of Goods Sold = Sales – Gross Profit

= ₹ 7,20,000 – 25% x ₹ 7,20,000 = ₹ 5,40,000

2. Stock Turnover = $\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{₹\ 5,40,000}{\text{Average Stock}}$ = 5 times.

Average Stock = $\frac{₹\ 5,40,000}{5}$ = ₹ 1,08,000

3. Let Opening Stock be x.

Closing Stock is ₹ 30,000 more than Opening Stock.

Closing Stock = (x + 30,000)

Average Stock = $\frac{x + x + 30,000}{2}$ = 1,08,000.

2x = 2,16,000 – 30,000

x = $\frac{1,86,000}{2}$ = 93,000 = Opening Stock.

Closing Stock = x + 30,000

= 93,000 + 30,000 = ₹ 1,23,000

4. Liquid Ratio = $\frac{\text{Liquid Assets}}{\text{Current Liabilities}} = \frac{\text{Liquid Assets}}{2,40,000}$ = 1.25.

Liquid Assets = ₹ 3,00,000

5. Current Assets = Liquid Assets + Closing Stock

= ₹ 3,00,000 + ₹ 1,23,000 = ₹ 4,23,000

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123 **MTP** **Sep 2024 S-II** Q3(b) **4 marks**

EPL Ltd. has furnished the following information relating to the year ended 31st March 2023 and 31st March, 2024:

	31st March, 2023	31st March, 2024
Share Capital	50,00,000	50,00,000
Reserve and Surplus	20,00,000	25,00,000
Long term loan	30,00,000	30,00,000

- Net profit ratio: 8%
- Gross profit ratio: 20%
- Long-term loan has been used to finance 40% of the fixed assets.
- Stock turnover with respect to cost of goods sold is 4.
- Debtors represent 90 days sales.
- The company holds cash equivalent to 1½ months cost of goods sold.
- Ignore taxation and assume 360 days in a year.

You are required to PREPARE Balance Sheet as on 31st March 2024 in following format:

Liabilities	(₹)	Assets	(₹)
Share Capital	-	Fixed Assets	-
Reserve and Surplus	-	Sundry Debtors	-
Long-term loan	-	Closing Stock	-
Sundry Creditors	-	Cash in hand	-

ICAI'S ANSWER

Change in Reserve & Surplus = ₹ 25, 00,000 – ₹ 20,00,000 = ₹ 5,00,000

So, Net profit = ₹ 5, 00,000

(i) Net Profit Ratio = 8%

∴ Sales = $\frac{5,00,000}{8\%}$ =₹ 62,50,000

(ii) Cost of Goods sold

= Sales – Gross profit Margin

= ₹ 62, 50,000 – 20% of ₹ 62, 50,000

= ₹ 50, 00,000

(iii) Fixed Assets = $\frac{₹\ 30,00,000}{40\%}$ =₹ 75,00,000

(iv) Stock = $\frac{\text{Cost of Goods Sold}}{\text{STR}} = \frac{50,00,000}{4}$ = ₹ 12,50,000

(v) Debtors = $\frac{62,50,000}{360} \times 90$ = ₹ 15,62,500

(vi) Cash Equivalent = $\frac{50,00,000}{12} \times 1.5$ = ₹ 6,25,000

Balance Sheet as on 31st March 2024

Liabilities	(₹)	Assets	(₹)
Share Capital	50,00,000	Fixed Assets	75,00,000
Reserve and Surplus	25,00,000	Sundry Debtors	15,62,500
Long-term loan	30,00,000	Closing Stock	12,50,000
Sundry Creditors (Balancing Figure)	4,37,500	Cash in hand	6,25,000
	1,09,37,500		1,09,37,500

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124 MTP **May 2024 S-I** Q1(c) 5 marks

ANVY Ltd. has furnished the following ratios and information for the year end 31st March, 2023:

Equity share capital	₹ 2,00,000
The relevant ratios of the company are as follows:	
Current debt to total debt	0.50
Total debt to Equity share capital	0.60
Fixed assets to Equity share capital	0.70
Total assets turnover	2.5 Times
Inventory turnover	10 Times

You are required to PREPARE the Balance Sheet of ANVY Ltd. as on 31st March, 2023.

ICAI'S ANSWER

ANVY Ltd

Balance Sheet as on 31st March, 2023

Liabilities	₹	Assets	₹
Equity share capital	2,00,000	Fixed assets	1,40,000
Current debt	60,000	Cash (balancing figure)	1,00,000
Long term debt	60,000	Inventory	80,000
	3,20,000		3,20,000

Working Notes

1. Total debt = 0.60 x Equity share capital = 0.60 × ₹ 2,00,000 = ₹ 1,20,000

Further, Current debt to total debt = 0.50. So, current debt = 0.50 x ₹1,20,000 = ₹ 60,000,

Long term debt = ₹1,20,000 - ₹60,000= ₹ 60,000

2. Fixed assets = 0.70 × Equity share Capital = 0.70 × ₹ 2,00,000 = ₹ 1,40,000

3. Total assets to turnover = 2.5 Times: Inventory turnover = 10 Times

Hence, Inventory /Total assets = 2.5/10=1/4, Total assets = ₹ 3,20,000

Therefore Inventory = ₹ 3,20,000/4 = ₹ 80,000

ICAI · P6 · MTP May 2024 · Series I · question p. 4 · answer p. 3-4

125 MTP Old P8 Nov 2023 S-I Q1(b) 5 marks old scheme

Following information has been provided from the books of Laxmi Pvt. Ltd. for the year ending on 31st March 2022:

Net Working Capital	₹ 5,40,000
Bank overdraft	₹ 1,00,000
Fixed Assets to Proprietary ratio	0.75
Reserves and Surplus	₹ 4,80,000
Current ratio	2.5
Liquid ratio (Quick Ratio)	1.5

You are required to PREPARE a summarised Balance Sheet as of 31st March 2022 assuming that there is no long-term debt.

ICAI'S ANSWER

Working notes:

(i) Computation of Current Assets and Current Liabilities

$$\frac{\text{Current assets}}{\text{Current liabilities}} = 2.5$$

Current assets = 2.5 Current liabilities

Now, Working capital = Current assets – Current liabilities

₹ 5,40,000 = 2.5 Current liability – Current liability

Or 1.5 Current liability = ₹ 5,40,000

∴ Current Liabilities = ₹ 3,60,000

So, Current Assets = ₹ 3,60,000 × 2.5 = ₹ 9,00,000

(ii) Computation of Inventories

$$\text{Liquid ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

$$1.5 = \frac{\text{Current assets - Inventories}}{₹\ 3{,}60{,}000}$$

1.5 × ₹ 3, 60,000 = ₹ 9,00,000 – Inventories

Inventories =₹ 9,00,000 – ₹ 5,40,000 = ₹ 3,60,000

(iii) Computation of Proprietary fund; Fixed assets; Capital and Sundry creditors

$$\text{Fixed Asset to Proprietary ratio} = \frac{\text{Fixed assets}}{\text{Proprietary fund}} = 0.75$$

∴ Fixed Assets = 0.75 Proprietary fund

Proprietary fund = Fixed Assets + Net Working Capital - Long Term Debt

= 0.75 Proprietary fund + ₹ 5,40,000 - 0

∴ Proprietary fund = ₹ 21,60,000

And Fixed Assets = 0.75 proprietary fund

= 0.75 × ₹ 21,60,000 = ₹ 16,20,000

Capital = Proprietary fund – Reserves & Surplus

= ₹ 21,60,000 – ₹ 4,80,000 = ₹ 16,80,000

Sundry Creditors = Current liabilities – Bank overdraft

= ₹ 3,60,000 – ₹ 1,00,000 = ₹ 2,60,000

Balance Sheet as of 31st March 2022

Liabilities	₹	Assets	₹
Capital	16,80,000	Fixed Assets	16,20,000
Reserves & Surplus	4,80,000	Inventories	3,60,000
Bank overdraft	1,00,000	Other Current Assets	5,40,000
Sundry creditors	2,60,000	(Balancing figure)	
	25,20,000		**25,20,000**

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 1 · answer p. 1-2

126 MTP **Old P8 May 2023 S-I** Q1(c) 5 marks old scheme

Based on the following particulars SHOW various assets and liabilities of Raina Ltd.

Fixed assets turnover ratio (Based on Cost of sales)	10 times
Capital turnover ratio (Based on Cost of sales)	3 times
Inventory Turnover	10 times
Receivable turnover	5 times
Payable turnover	5 times
GP Ratio	40%

Gross profit during the year amounts to Rs.15,00,000. There is no long-term loan or overdraft. Reserve and surplus amount to Rs.5,00,000. Ending inventory of the year is Rs. 40,000 above the beginning inventory.

ICAI'S ANSWER

G. P. ratio = $\frac{\text{Gross Profit}}{\text{Sales}} = 40$

(a) Sales = $\frac{\text{Gross Profit}}{40}$ x 100 = $\frac{15,00,000}{40}$ x100

= 37,50,000

(b) Cost of Sales = Sales Gross Profit = = ₹ 37,50,000 - ₹ 15,00,000

= ₹ 22,50,000

(c) Receivable turnover = $\frac{\text{Sales}}{\text{Receivables}} = 5$

= Receivables = $\frac{\text{Sales}}{5} = \frac{37,50,000}{5}$

= ₹7,50,000

(d) Fixed assets turnover = $\frac{\text{Cost of Sales}}{\text{Fixed Assets}}$ =10

Or Fixed assets = $\frac{\text{Cost of Sales}}{10} = \frac{₹\ 22,50,000}{10}$

= ₹ 2,25,000

(e) Inventory turnover = $\frac{\text{Cost of Sales}}{\text{Average Stock}} = 10$

Average Stock = $\frac{\text{Cost of Sales}}{10} = \frac{22,50,000}{10}$ = ₹ 2,25,000

Average Stock = $\frac{\text{Opening Stock} + \text{Closing stock}}{2} = \frac{\text{Opening stock} + \text{Opening stock} + 40,000}{2}$

Average Stock = Opening+ ₹ 20,000

Opening Stock= Average Stock- ₹ 20,000

Average Stock = ₹ 2,25,000 – ₹ 20,000

Opening Stock = ₹ 2,05,000

Closing Stock = Opening Stock + ₹ 40,000

Closing Stock = ₹ 2,05,000 +₹ 40,000 =₹ 2,45,000

(f) Payable turnover = $\frac{\text{Purchase}}{\text{Payables}} = 5$

Purchases = Cost of Sales + Increase in Stock

Purchases = ₹22,50,000 + ₹40,000 = ₹22,90,000

$$\text{Payables} = \frac{\text{Purchase}}{5} = \frac{₹\ 22,90,000}{5} = ₹4,58,000$$

$$\text{(h) Capital Employed} = \frac{\text{Cost of Sales}}{3} = \frac{₹\ 22,50,000}{3}$$

= ₹7,50,000

Equity share Capital = Capital Employed – Reserves & Surplus

= ₹7,50,000 – ₹5,00,000 = ₹2,50,000

Balance Sheet of T Ltd as on.....

Liabilities	₹	Assets	₹
Capital	2,50,000	Fixed Assets	2,25,000
Reserve & Surplus	5,00,000	Stock	2,45,000
Payables	4,58,000	Receivables	7,50,000
		Other Current Assets (balancing figure)	2,38,000
	14,58,000		14,58,000

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127 MTP **Old P8 May 2023 S-II** Q1(d) 5 marks old scheme

Using the following information, PREPARE the balance sheet:

Long-term debt to net worth	0.25
Total asset turnover	3
Average collection period	9 days
Inventory turnover	13
Gross profit margin	20%
Acid-test ratio	1.5

*Assume a 360-day year and all sales on credit.

Liabilities	₹	Assets	₹
Notes and payables	2,50,000	Cash	?
Long-term debt	?	Accounts receivable	?
Common stock	8,00,000	Inventory	?
Retained earnings	16,00,000	Plant and equipment	?
Total liabilities and equity	?	Total assets	?

ICAI'S ANSWER

Working Notes:

(i) Long term Debt

Long Term Debt/ Net worth = 0.25

Long Term Debt/ (8,00,000+16,00,000) = 0.25

Long term debt = 6,00,000

(ii) Total assets

Total liabilities and Equity = Notes and payables + Long-term debt + Common stock + Retained earnings

= 2,50,000+6,00,000+8,00,000+16,00,000

Total assets = Total liabilities and Equity = 32,50,000

(iii) Sales and Cost of Goods sold

Total asset turnover = 3 = Sales/ Total Assets = Sales/32,50,000

Sales = 97,50,000

Cost of goods sold = (100% - Gross Profit margin) x Sales

= (100% - 20%) x 97,50,000 =78,00,000.

(iv) Current Assets

Inventory turnover = 13 = COGS/ Inventory = 78,00,000/Inventory

Inventory = ₹ 6,00,000

Average collection period = 9 = Receivables/Sales x 360 = Receivables/ 97,50,000 x 360

Accounts receivables = 2,43,750

Acid-test ratio = 1.5 = (Cash+ Accounts Receivables) /Notes and Payables

= (Cash +2,43,750)/2,50,000 = 1.5

Cash = 1,31,250

(v) Plant and equipment

= Total Assets - Current Assets

= 32,50,000 - (1,31,250+2,43,750+6,00,000) = 22,75,000

Balance Sheet

Liabilities	₹	Assets	₹
Notes and payables	2,50,000	Cash	1,31,250
Long-term debt	6,00,000	Accounts receivable	2,43,750
Common stock	8,00,000	Inventory	6,00,000
Retained earnings	16,00,000	Plant and equipment	22,75,000
Total liabilities and equity	32,50,000	Total assets	32,50,000

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128 MTP **Old P8 Nov 2022 S-II** Q3 10 marks old scheme

From the following information and ratios, PREPARE the Balance sheet as at 31st March 2022 and Income statement for the year ended on that date for M/s Ganguly & Co -

Average Stock	₹10 lakh
Current Ratio	3:1
Acid Test Ratio	1:1
PBIT to PBT	2.2:1
Average Collection period (Assume 360 days in a year)	30 days
Stock Turnover Ratio (Use sales as turnover)	5 times
Fixed assets turnover ratio	0.8 times
Working Capital	₹10 lakh
Net profit Ratio	10%
Gross profit Ratio	40%
Operating expenses (excluding interest)	₹ 9 lakh
Long term loan interest	12%
Tax	Nil

ICAI'S ANSWER

1. Current Ratio = 3:1

   Current Assets (CA)/Current Liability (CL) = 3:1

   CA = 3CL

   WC = 10,00,000

   CA – CL = 10,00,000

   3CL – CL = 10,00,000

2CL = 10,00,000

$CL = \frac{10,00,000}{2}$

CL = ₹5,00,000

CA = 3 x 5,00,000

CA = ₹15,00,000

2. Acid Test Ratio = CA – Stock / CL = 1:1

$= \frac{15,00,000 - Stock}{5,00,000} = 1$

15,00,000 – stock = 5,00,000

Stock = ₹10,00,000

3. Stock Turnover ratio (on sales) = 5

$\frac{Sales}{Avg\ stock} = 5$

$\frac{Sales}{10,00,000} = 5$

Sales = ₹50,00,000

4. **Gross Profit** = 50,00,000 x 40% = **₹20,00,000**
   Net profit (PBT) = 50,00,000 x 10% = **₹5,00,000**

5. PBIT/PBT = 2.2

   PBIT = 2.2 x 5,00,000

   PBIT= 11,00,000

   Interest = 11,00,000 – 5,00,000 = **₹6,00,000**

   Long term loan = $\frac{6,00,000}{0.12}$ = **₹50,00,000**

6. Average collection period = 30 days

   Receivables = $\frac{30}{360}$ x 50.00.000 = 4,16,667

7. Fixed Assets Turnover Ratio = 0.8

   50,00,000/ Fixed Assets = 0.8

   Fixed Assets = **₹62,50,000**

Income Statement

	Amount (₹)
Sales	50,00,000
Less: Cost of Goods Sold	30,00,000
Gross Profit	20,00,000
Less: Operating Expenses	9,00,000
Less: Interest.	6,00,000
Net Profit	5,00,000

Balance sheet

Liabilities	Amount (₹)	Assets		Amount (₹)
Equity share capital	22,50,000	Fixed asset		62,50,000
Long term debt	50,00,000	Current assets:		
Current liability	5,00,000	Stock	10,00,000	
		Receivables	4,16,667	
		Other	83,333	15,00,000
	77,50,000			77,50,000

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129 MTP **Old P8 May 2022 S-I** Q1(b) 5 marks old scheme

Owner's equity of Yay Ltd. is ₹ 6,00,000. The financial ratios of the company are given below:

Current debt to total debt	0.4
Total debt to Owner's equity	0.6
Fixed assets to Owner's equity	0.6
Total assets turnover	2 times
Inventory turnover	8 times

COMPLETE the following Balance Sheet from the information given above:

Liabilities	(₹)	Assets	(₹)
Current Debt	-	Cash	-
Long-term Debt	-	Inventory	-
Total Debt	-	Total Current Assets	-
Owner's Equity	-	Fixed Assets	-
	9,60,000		-

ICAI'S ANSWER

Balance Sheet

Liabilities	(₹)	Assets	(₹)
Current debt	1,44,000	Cash (balancing figure)	3,60,000
Long term debt	2,16,000	Inventory	2,40,000
Total Debt	3,60,000	Total Current Assets	6,00,000
Owner's Equity	6,00,000	Fixed Assets	3,60,000
Total liabilities	9,60,000	Total Assets	9,60,000

Working Notes:

1. Total debt = 0.60 x Owner's Equity = 0.60 x ₹ 6,00,000 = ₹ 3,60,000

Further, Current debt to Total debt = 0.40.

So, Current debt = 0.40 × ₹ 3,60,000 = ₹ 1,44,000

Long term debt = ₹ 3,60,000 - ₹ 1,44,000 = ₹ 2,16,000

2. Fixed assets = 0.60 × Owner's Equity = 0.60 × ₹ 6,00,000 = ₹ 3,60,000
3. Total Assets = Total Liabilities = ₹ 9,60,000

Total assets to turnover = 2 Times; Inventory turnover = 8 Times

Hence, Inventory /Total assets = 2/8=1/4,

Therefore, Inventory = ₹ 9,60,000/4 = ₹ 2,40,000

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130 MTP **Old P8 May 2022 S-II** Q1(d) 5 marks old scheme

From the following information, you are required to PREPARE a summarised Balance Sheet for Rudra Ltd. for the year ended 31st March, 2022

Debt Equity Ratio	1:1
Current Ratio	3:1
Acid Test Ratio	8:3
Fixed Asset Turnover (on the basis of sales)	4
Stock Turnover (on the basis of sales)	6
Cash in hand	5,00,000
Stock to Debtor	1:1
Sales to Net Worth	4
Capital to Reserve	1:2
Gross Profit	20% of Cost

COGS to Creditor	10:1

Interest for entire year is yet to be paid on Long Term loan @ 10%.

ICAI'S ANSWER

Balance Sheet of Rudra Ltd.

Liabilities	Amount (₹)	Assets	Amount (₹)
Capital	10,00,000	Fixed Assets	30,00,000
Reserves	20,00,000	Current Assets:	
Long Term Loan @ 10%	30,00,000	Stock in Trade	20,00,000
Current Liabilities:		Debtors	20,00,000
Creditors	10,00,000	Cash	5,00,000
Other Short-term Current Liability (Other STCL)	2,00,000		
Outstanding Interest	3,00,000		
	75,00,000		**75,00,000**

Working Notes:

Let sales be ₹ x

Balance Sheet of Rudra Ltd.

Liabilities	Amount (₹)	Assets	Amount (₹)
Capital		Fixed Assets	x/4
Reserves		Current Assets:	
Net Worth	x/4	Stock in Trade	x/6
Long Term Loan @ 10%	x/4	Debtors	x/6
		Cash	5,00,000
Current liabilities:			
Creditors	x/12		
Other Short-term Current Liability			
Outstanding Interest			
Total Current Liabilities	x/9+5,00,000/3		
Total		**Total**	

1. Fixed Asset Turnover = 4 = $\frac{x}{\text{Fixed Assets}}$

Fixed Assets = $\frac{x}{4}$

2. Stock Turnover = 6 = $\frac{x}{\text{stock}}$

   Stock = $\frac{x}{6}$

3. Sales to net worth = 4 = $\frac{x}{\text{net worth}}$

   Net worth = $\frac{x}{4}$

4. Debt: Equity =1 : 1

   $\frac{\text{Long Term Loan}}{\text{Net worth}} = \frac{1}{1}$

   Long term loan = Net worth = $\frac{x}{4}$

5. Gross Profit to Cost = 20%

   $\frac{\text{G P}}{\text{Sales} - \text{G P}} = 20\%$

   $\frac{\text{G P}}{x - \text{G P}} = 20\%$

   GP = 0.2 x – 0.2 GP

   1.2 GP = 0.2 x

   G P = $\frac{0.2x}{1.2}$

   G P = x/6

   Cost of Goods Sold = x – x/6 = 5/6 x

6. COGS to creditors = 10:1

   $\frac{\text{COGS}}{\text{Creditors}} = \frac{10}{1}$

   $\frac{\frac{5}{6}x}{\text{creditors}} = \frac{10}{1}$

   Creditors = $\frac{5x}{60} = \frac{x}{12}$

7. $\frac{\text{Stock}}{\text{Debtor}} = 1$

   Debtor = Stock = $\frac{x}{6}$

8. Current Ratio = 3 : 1

   $\frac{\text{Stock+Debtors+Cash}}{\text{Current Liabilities}} = \frac{3}{1}$

   $\frac{\frac{x}{6} + \frac{x}{6} + 5{,}00{,}000}{\text{Current Liabilities}} = 3$

   $\frac{\frac{x}{3} + 5{,}00{,}000}{3} = \text{CL}$

   CL = $\frac{x}{9} + \frac{5{,}00{,}000}{3}$

9. CA = 3CL

   = $3\left(\frac{x}{9} + \frac{5{,}00{,}000}{3}\right)$

   CA = $\frac{x}{3}$ + 5,00,000

10. Net worth + Long Term Loan + Current Liability = Fixed Asset + Current Assets

    $\frac{x}{4} + \frac{x}{4} + \frac{x}{9} + \frac{5{,}00{,}000}{3} = \frac{x}{4} + \frac{x}{3}$ + 5,00,000

$$\frac{x}{4}+\frac{x}{9}-\frac{x}{3}=5{,}00{,}000-\frac{5{,}00{,}000}{3}$$

$$\frac{9x+4x-12x}{36}=\frac{15{,}00{,}000-5{,}00{,}000}{3}$$

$$\frac{x}{36}=\frac{10{,}00{,}000}{3}$$

x = 1,20,00,000

11. Now, from above calculations, we get,

-> Fixed Asset = $\frac{x}{4}=\frac{1{,}20{,}00{,}000}{4}$ = 30,00,000

-> Stock = $\frac{x}{6}=\frac{1{,}20{,}00{,}000}{6}$ = 20,00,000

-> Debtor = $\frac{x}{6}=\frac{1{,}20{,}00{,}000}{6}$ = 20,00,000

-> Net Worth = x / 4 = 30,00,000

Now, Capital to Reserve is 1 : 2

∴ Capital = ₹ 10,00,000

and, Reserve = ₹ 20,00,000

-> Long Term Loan = $\frac{x}{4}$ = 30,00,000

-> Outstanding Interest = 30,00,000×10% = 3,00,000

-> Creditors = $\frac{x}{12}=\frac{1{,}20{,}00{,}000}{12}$ = 10,00,000

-> Current Liabilities = Creditors + Other STCL + Outstanding Interest

$\frac{x}{9}+\frac{5{,}00{,}000}{3}$ = 10,00,000+ Other STCL + 3,00,000

$\frac{1{,}20{,}00{,}000}{9}+\frac{5{,}00{,}000}{3}$ = 13,00,000+ Other STCL

15,00,000 = Other STCL + 13,00,000

Other STCL = 2,00,000

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131 MTP **Old P8 May 2022 S-II** Q6(b) 4 marks old scheme

DISCUSS the limitations of financial ratios

ICAI'S ANSWER

The limitations of financial ratios are listed below:

(i) Diversified product lines: Many businesses operate a large number of divisions in quite different industries. In such cases ratios calculated on the basis of aggregate data cannot be used for inter-firm comparisons.

(ii) Financial data are badly distorted by inflation: Historical cost values may be substantially different from true values. Such distortions of financial data are also carried in the financial ratios.

(iii) Seasonal factors may also influence financial data.

(iv) To give a good shape to the popularly used financial ratios (like current ratio, debt- equity ratios, etc.): The business may make some year-end adjustments. Such window dressing can change the character of financial ratios which would be different had there been no such change.

(v) Differences in accounting policies and accounting period: It can make the accounting data of two firms non-comparable as also the accounting ratios.

(vi) There is no standard set of ratios against which a firm's ratios can be compared: Sometimes a firm's ratios are compared with the industry average. But if a firm desires to be above the average, then industry average becomes a low standard. On the other hand, for a below average firm, industry averages become too high a standard to achieve.

(vii) Financial ratios are inter-related, not independent: Viewed in isolation one ratio may highlight efficiency. But when considered as a set of ratios they may speak differently. Such interdependence among the ratios can be taken care of through multivariate analysis.

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132 MTP **Old P8 Dec 2021 S-I** Q4 10 marks old scheme

The following figures and ratios are related to a company:

(i)	Sales for the year (all credit)	₹ 30,00,000
(ii)	Gross Profit ratio	25 percent
(iii)	Fixed assets turnover (based on cost of goods sold)	1.5
(iv)	Stock turnover (based on cost of goods sold)	6
(v)	Liquid ratio	1 : 1
(vi)	Current ratio	1.5 : 1
(vii)	Receivables (Debtors) collection period	2 months
(viii)	Reserves and surplus to Share capital	0.6 : 1
(ix)	Capital gearing ratio	0.5
(x)	Fixed assets to net worth	1.20 : 1

You are REQUIRED to prepare:

(a) Balance Sheet of the company on the basis of above details.

(b) The statement showing working capital requirement, if the company wants to make a provision for contingencies @ 10 percent of net working capital including such provision.

ICAI'S ANSWER

Working Notes:

(i) Cost of Goods Sold = Sales – Gross Profit (25% of Sales)

= ₹ 30,00,000 – ₹ 7,50,000

= ₹ 22,50,000

(ii) Closing Stock = Cost of Goods Sold / Stock Turnover

= ₹ 22,50,000/6 = ₹ 3,75,000

(iii) Fixed Assets = Cost of Goods Sold / Fixed Assets Turnover

= ₹ 22,50,000/1.5

= ₹ 15,00,000

(iv) Current Assets:

Current Ratio = 1.5 and Liquid Ratio = 1

Stock = 1.5 – 1 = 0.5

Current Assets = Amount of Stock × 1.5/0.5

= ₹ 3,75,000 × 1.5/0.5 = ₹ 11,25,000

(v) Liquid Assets (Debtors and Cash)

= Current Assets – Stock

= ₹ 11,25,000 – ₹ 3,75,000

= ₹ 7,50,000

(vi) Debtors = Sales × Debtors Collection period /12

= ₹ 30,00,000 × 2 /12

= ₹ 5,00,000

(vii) Cash = Liquid Assets – Debtors

= ₹ 7,50,000 – ₹ 5,00,000 = ₹ 2,50,000

(viii) Net worth = Fixed Assets /1.2

= ₹ 15,00,000/1.2 = ₹ 12,50,000

(ix) Reserves and Surplus

Reserves and Share Capital = 0.6 + 1 = 1.6

Reserves and Surplus = ₹ 12,50,000 × 0.6/1.6

= ₹ 4,68,750

(x) Share Capital = Net worth – Reserves and Surplus

= ₹ 12,50,000 – ₹ 4,68,750

= ₹ 7,81,250

(xi) Current Liabilities = Current Assets/Current Ratio

= ₹ 11,25,000/1.5 = ₹ 7,50,000

(xii) Long-term Debts

Capital Gearing Ratio = Long-term Debts / Equity Shareholders' Fund

Long-term Debts = ₹ 12,50,000 × 0.5 = ₹ 6,25,000

(a) Preparation of Balance Sheet of a Company

Balance Sheet

Liabilities	Amount (₹)	Assets	Amount (₹)
Equity Share Capital	7,81,250	Fixed Assets	15,00,000
Reserves and Surplus	4,68,750	Current Assets	
Long-term Debts	6,25,000	Stock	3,75,000
Current Liabilities	7,50,000	Debtors	5,00,000
		Cash	2,50,000
	26,25,000		**26,25,000**

(b) Statement Showing Working Capital Requirement

	(₹)	(₹)
Current Assets		
(i) Stocks		3,75,000
(ii) Receivables (Debtors)		5,00,000
(iii) Cash in hand & at bank		2,50,000
A. Current Assets: Total		11,25,000
Current Liabilities		
B. Current Liabilities: Total		7,50,000
Net Working Capital (A – B)		3,75,000
Add: Provision for contingencies (1/9th of Net Working Capital)		41,667
Working capital requirement		**4,16,667**

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133 MTP Old P8 Jul 2021 S-I Q1(b) 5 marks old scheme

SN Ltd. has furnished the following ratios and information relating to the year ended 31st March 2021:

Share Capital	Rs. 6,25,000
Working Capital	Rs. 2,00,000
Gross Margin	25%
Inventory Turnover	5 times
Average Collection Period	1.5 months
Current Ratio	1.5:1
Quick Ratio	0.7:1
Reserves & Surplus to Bank & Cash	3 times

Further, the assets of the company consist of fixed assets and current assets, while its current liabilities comprise bank credit and others in the ratio of 3:1. Assume 360 days in a year.

You are required to PREPARE the Balance Sheet as on 31st March 2021.

(Note- Balance sheet may be prepared in traditional T Format.)

ICAI'S ANSWER

Workings:

1. Current Ratio = $\frac{\text{Current Assets(CA)}}{\text{Current Liabilities(CL)}} = \frac{1.5}{1}$

$\therefore$ CA = 1.5 CL

Also, CA - CL = Rs. 2,00,000

1.5 CL- CL = Rs. 2,00,000

CL = $\frac{\text{Rs. 2,00,000}}{0.5}$ = Rs. 4,00,000

CA = 1.5 × Rs. 4,00,000 = Rs. 6,00,000

2. Bank Credit (BC) to Other Current Liabilities (OCL) ratio = 3:1

$\frac{\text{Bank Credit (BC)}}{\text{Other Current Liabilities (OCL)}} = \frac{3}{1}$

BC = 3 OCL

Also, BC + OCL = CL

3 OCL + OCL = Rs. 4,00,000

OCL = $\frac{\text{Rs. 4,00,000}}{4}$ = Rs. 1,00,000

Bank Credit = 3 × Rs. 1,00,000 = Rs. 3,00,000

3. Quick Ratio = $\frac{\text{Current Assets - Inventories}}{\text{Current Liabilities}}$

0.7 = $\frac{\text{Rs. 6,00,000-Inventories}}{\text{Rs. 4,00,000}}$

Inventories = Rs. 6,00,000 – Rs. 2,80,000 = Rs. 3,20,000

4. Inventory Turnover = 5 times

$$\text{Inventory Turnover} = \frac{\text{Cost of goods sold (COGS)}}{\text{Average Inventory}}$$

$$\text{Average Inventory} = \frac{\text{Cost of goods sold (COGS)}}{\text{Inventory Turnover}}$$

COGS = Rs. 3,20,000 × 5 = Rs. 16,00,000

5. $\text{Gross Margin} = \frac{\text{Sales - COGS}}{\text{Sales}} \times 100 = 25\%$

$$\text{Sales} = \frac{16,00,000}{0.75} = \text{Rs. } 21,33,333.33$$

6. Average Collection Period (ACP) = 1.5 months = 45 days

$$\text{Debtors Turnover} = \frac{360}{\text{ACP}} = \frac{360}{45} = 8 \text{ times}$$

$$\text{Also, Debtors Turnover} = \frac{\text{Sales}}{\text{Average Debtors}}$$

$$\text{Hence, Debtors} = \frac{\text{Rs.}21,33,333.33}{8} = \text{Rs.}2,66,667$$

7. Bank & Cash = CA - (Debtors + Inventory)

= Rs. 6,00,000 – (Rs. 2,66,667 + 3,20,000) = Rs. 13,333

8. $\frac{\text{Reserves \& Surplus}}{\text{Bank \& Cash}} = 3$

Reserves & Surplus = 3 × Rs. 13,333 = Rs. 40,000

Balance Sheet of SN Ltd. as on 31st March 2021

Liabilities	(Rs.)	Assets	(Rs.)
Share Capital	6,25,000	Fixed Assets (Balancing Figure)	4,65,000
Reserves & Surplus	40,000		
Current Liabilities:		Current Assets:	
Bank Credit	3,00,000	Inventories	3,20,000
Other Current Liabilities	1,00,000	Debtors	2,66,667
		Bank & Cash	13,333
	10,65,000		10,65,000

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134 MTP Old P8 Jul 2021 S-II Q1(c) 5 marks old scheme

XYZ Ltd. has Owner's equity of Rs. 2,00,000 and the ratios of the company are as follows:

Current debt to total debt	0.3
Total debt to Owner's equity	0.5
Fixed assets to Owner's equity	0.6
Total assets turnover	2 times
Inventory turnover	10 times

COMPLETE the following Balance Sheet from the information given above:

Liabilities	(Rs.)	Assets	(Rs.)
Current Debt	-	Cash	-
Long-term Debt	-	Inventory	-
Total Debt	-	Total Current Assets	-
Owner's Equity	-	Fixed Assets	-

ICAI'S ANSWER

Balance Sheet

Liabilities	(Rs.)	Assets	(Rs.)
Current debt	30,000	Cash (balancing figure)	1,20,000
Long term debt	70,000	Inventory	60,000
Total Debt	1,00,000	Total Current Assets	1,80,000
Owner's Equity	2,00,000	Fixed Assets	1,20,000
Total liabilities	3,00,000	Total Assets	3,00,000

Workings:

1. Total debt = 0.50 x Owner's Equity = 0.50 x Rs. 2,00,000 = Rs. 1,00,000
   Further, Current debt to Total debt = 0.30

   So, Current debt = 0.30 × Rs. 1,00,000 = Rs. 30,000

   Long term debt = Rs. 1,00,000 - Rs. 30,000 = Rs. 70,000
2. Fixed assets = 0.60 × Owner's Equity = 0.60 × Rs. 2,00,000 = Rs. 1,20,000
3. Total Liabilities = Total Debt + Owner's Equity
   = Rs. 1,00,000 + Rs. 2,00,000 = Rs. 3,00,000

   Total Assets = Total Liabilities = Rs. 3,00,000

   Total assets to turnover = 2 Times; Inventory turnover = 10 Times

Hence, Inventory /Total assets = 2/10=1/5,

Therefore Inventory = Rs. 3,00,000/5 = Rs. 60,000

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135 MTP **Old P8 Nov 2020 S-I** Q1(d) 5 marks old scheme

Using the information given below, PREPARE the Balance Sheet of SKY Private Limited:

(i)	Current ratio	1.6 :1
(ii)	Cash and Bank balance	15% of total current assets
(iii)	Debtors turnover ratio	12 times
(iv)	Stock turnover (cost of goods sold) ratio	16 times
(v)	Creditors turnover (cost of goods sold) ratio	10 times
(vi)	Gross profit ratio	20%
(vii)	Capital gearing ratio	0.6
(viii)	Depreciation rate	15% on W.D.V.
(ix)	Net fixed Assets	20% of total assets

(Assume all purchase and sales are on credit)

Balance Sheet of SKY Private Limited as at 31.03.2020

Liabilities		Amount in ₹	Assets		Amount in ₹
Share Capital		25,00,000	**Fixed assets**		
Reserve & surplus		?	Opening WDV	?	
12% Long term debt		?	Less: Depreciation	?	?
Current liabilities					
Creditors	?		**Current Assets**		
Provisions & outstanding expenses	?	68,50,000	Stock	?	
			Debtors	?	
			Cash and bank balance	?	?
Total		**?**	**Total**		**?**

(Detailed working notes are not required to be shown)

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 2

136 MTP Old P8 May 2020 S-I Q4 10 marks old scheme

The following accounting information and financial ratios of A&R Limited relate to the year ended 31st March, 2020:

Inventory Turnover Ratio	6 Times
Creditors Turnover Ratio	10 Times
Debtors Turnover Ratio	8 Times
Current Ratio	2.4
Gross Profit Ratio	25%

Total sales Rs.6,00,00,000; cash sales 25% of credit sales; cash purchases Rs.46,00,000; working capital Rs.56,00,000; closing inventory is Rs.16,00,000 more than opening inventory.

You are required to CALCULATE:

(i) Average Inventory

(ii) Purchases

(iii) Average Debtors

(iv) Average Creditors

(v) Average Payment Period

(vi) Average Collection Period

(vii) Current Assets

(viii) Current Liabilities.

Take 365 days a year

ICAI'S ANSWER

(i) Computation of Average Inventory

Gross Profit = 25% of Rs.6,00,00,000 = Rs.1,50,00,000

Cost of goods sold (COGS) = Sales - Gross Profit

= Rs.6,00,00,000 – Rs.1,50,00,000

= Rs.4,50,00,000

$$\text{Inventory Turnover Ratio} = \frac{\text{COGS}}{\text{Average Inventory}}$$

$$6 = \frac{\text{Rs.4,50,00,000}}{\text{Average inventory}}$$

Average inventory = Rs.75,00,000

(ii) Computation of Purchases

Purchases = COGS + (Closing Stock – Opening Stock)

= Rs.4,50,00,000 + 16,00,000*

Purchases = Rs.4,66,00,000

* Increase in Stock = Closing Stock – Opening Stock = Rs.16,00,000

(iii) Computation of Average Debtors

Let Credit Sales be Rs.100, Cash sales = $\frac{25}{100} \times 100$ = Rs.25

Total Sales = 100 + 25= Rs.125

Total sales is Rs.125 credit sales is Rs.100

If total sales is Rs.6,00,00,000, then credit sales is = $\frac{\text{Rs. } 6,00,00,000 \times 100}{125}$

Credit Sales = Rs.4,80,00,000

Cash Sales = (Rs.6,00,00,000 – Rs.4,80,00,000) = Rs.1,20,00,000

Debtors Turnover Ratio = $\frac{\text{Net Credit Sales}}{\text{Average debtors}} = 8$

$= \frac{\text{Rs.4,80,00,000}}{\text{Average debtors}} = 8$

Average Debtors = $\frac{\text{Rs.4,80,00,000}}{8}$

Average Debtors = Rs.60,00,000

(iv) Computation of Average Creditors

Credit Purchases = Purchases – Cash Purchases

= Rs.4,66,00,000 – Rs.46,00,000 = Rs.4,20,00,000

Creditors Turnover Ratio = $\frac{\text{Credit Purchases}}{\text{Average Creditors}}$

$10 = \frac{\text{Rs.4,20,00,000}}{\text{Average Creditors}}$

Average Creditors = Rs.42,00,000

(v) Computation of Average Payment Period

Average Payment Period = $\frac{\text{Average Creditors}}{\text{Average Daily Credit Purchases}}$

$= \frac{\text{Rs. 42,00,000}}{\left(\frac{\text{Credit Purchases}}{365}\right)} = \frac{\text{Rs. 42,00,000}}{\left(\frac{\text{Rs. 4,20,00,000}}{365}\right)}$

$= \frac{\text{Rs.42,00,000}}{\text{Rs.4,20,00,000}} \times 365 = 36.5$ days

Alternatively

Average Payment Period = 365/Creditors Turnover Ratio

$= \frac{365}{10} = 36.5$ days

(vi) Computation of Average Collection Period

Average Collection Period = $\frac{\text{Average Debtors}}{\text{Net Credit Sales}} \times 365$

$= \frac{\text{Rs.60,00,000}}{\text{Rs.4,80,00,000}} \times 365 = 45.625$ days

Alternatively

Average collection period = $\frac{365}{\text{Debtors Turnover Ratio}}$

$= \frac{365}{8} = 45.625$ days

(vii) Computation of Current Assets

$$\text{Current Ratio} = \frac{\text{Current Assets (CA)}}{\text{Current Liabilities (CL)}} = 2.4$$

2.4 Current Liabilities = Current Assets

$$\text{or CL} = \frac{\text{CA}}{2.4}$$

Further, Working capital = Current Assets – Current liabilities

$$\text{So, Rs.56,00,000} = \text{CA} - \frac{\text{CA}}{2.4}$$

$$\text{Rs.56,00,000} = \frac{1.4\text{CA}}{2.4} \text{ Or, } 1.4 \text{ CA} = \text{Rs.1,34,40,000}$$

CA = Rs.96,00,000

(viii) Computation of Current Liabilities

$$\text{Current liabilities} = \frac{\text{Rs.96,00,000}}{2.4} = \text{Rs.40,00,000}$$

ICAI · Old P8 · MTP May 2020 · Series I · question p. 3 · answer p. 8-10

CHAPTER 3

Types of Financing

3.1 Long-term sources: equity, preference, debt and retained earnings

20 questions · asked in 20 ICAI documents · 6 exam papers

137 EXAM **Sep 2026** Q4(a) 4 marks

Under which circumstances raising funds through the issue of debentures is advantageous for a company ? Discuss.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 19

138 EXAM **May 2026** Q4(c) 2 marks

What are Junk Bonds and why are they called 'High Yield Bonds'?

ICAI'S ANSWER

High Yield Bonds (or Junk Bonds): Junk Bond is a bond that is rated below investment grade by credit rating agencies. It has a low credit rating and a high risk of default.

Because of the higher risk, investors are compensated with higher interest rates, which is why junk bonds are also called high-yield bonds.

ICAI · P6 · May 2026 exam · question p. 21 · answer p. 23

139 EXAM **May 2025** Q4(b) 4 marks

What are key features of bridge financing?

ICAI'S ANSWER

Key features of Bridge Financing

(i) Bridge finance refers to loans taken by a company normally from commercial banks for a short period because of pending disbursement of loans sanctioned by financial institutions.

(ii) The bridge loans are repaid/ adjusted out of the term loans as and when disbursed by the concerned institutions.

(iii) Bridge loans are normally secured by hypothecating movable assets, personal guarantees and demand promissory notes.

(iv) Generally, the rate of interest on bridge finance is higher as compared with that on term loans.

ICAI · P6 · May 2025 exam · question p. 23 · answer p. 25

140 EXAM **Sep 2024** Q4(a) 4 marks

Discuss any 2 advantages and 2 disadvantages of raising finance by issue of debentures.

ICAI'S ANSWER

Advantages of raising finance by issue of debentures are:

(i) The cost of debentures is much lower than the cost of preference or equity capital as the interest is tax-deductible. Also, investors consider debenture investment safer than equity or preferred investment and, hence, may require a lower return on debenture investment.

(ii) Debenture financing does not result in dilution of control.

(iii) In a period of rising prices, debenture issue is advantageous. The fixed monetary outgo decreases in real terms as the price level increases. In other words, the company has to pay a fixed rate of interest.

Disadvantages of debenture financing are:

(i) Debenture interest and the repayment of its principal amount is an obligatory payment.

(ii) The protective covenants associated with a debenture issue may be restrictive.

(iii) Debenture financing enhances the financial risk associated with the firm because of the reasons given in point (i).

(i) Since debentures need to be paid at the time of maturity, a large amount of cash outflow is needed at that time.

ICAI · P6 · Sep 2024 exam · question p. 22 · answer p. 22-23

141 EXAM **Old P8 Dec 2021** Q6(a) 4 marks old scheme

Write short notes on Bridge Finance and Clean Packing Credit.

ICAI'S ANSWER

Bridge Finance: Bridge finance refers to loans taken by a company normally from commercial banks for a short period because of pending disbursement of loans sanctioned by financial institutions. Though it is of short-term nature but since it is an important step in the facilitation of long-term loan, therefore it is being discussed along with the long term sources of funds. Normally, it takes time for financial institutions to disburse loans to companies. However, once the loans are approved by the term lending institutions, companies, in order not to lose further time in starting their projects, arrange short term loans from commercial banks. The bridge loans are repaid/ adjusted out of the term loans as and when disbursed by the concerned institutions. Bridge loans are normally secured by hypothecating movable assets, personal guarantees and demand promissory notes. Generally, the rate of interest on bridge finance is higher as compared with that on term loans.

Clean packing credit: This is an advance made available to an exporter only on production of a firm export order or a letter of credit without exercising any charge or control over raw material or finished goods. It is a clean type of export advance. Each proposal is weighed according to particular requirements of the trade and credit worthiness of the exporter. A suitable margin has to be maintained. Also, Export Credit Guarantee Corporation (ECGC) cover should be obtained by the bank .

ICAI · Old P8 · Dec 2021 exam · question p. 14 · answer p. 14-15

142 EXAM **Old P8 Jan 2021** Q6(c) old scheme

Explain in brief the following bonds:

(i) Callable Bonds

(ii) Puttable Bonds

ICAI'S ANSWER

(i) Callable bonds: A callable bond has a call option which gives the issuer the **right to redeem the bond before maturity** at a predetermined price known as the call price (Generally at a premium).

(ii) Puttable bonds: Puttable bonds give the investor a put option (i.e. the **right to sell** the bond) back to the company before maturity.

ICAI · Old P8 · Jan 2021 exam · question p. 14 · answer p. 14

143 RTP **Sep 2025** Q10(a)

Contemplate the list of features given below and IDENTIFY each one of them with the most relevant sources of funds:

(i) It is the most expensive source of funds.

(ii) It entails a high degree of risk since they have to be repaid as per the terms of agreement.

(iii) It supports businesses in their routine activities.

(iv) Business enterprise has options to raise capital from International markets also.

(v) This source of finance sometimes is the last option for startups which doesn't qualify for bank funding

ICAI'S ANSWER

The most relevant sources of funds:

Feature	Sources of Fund
(i) It is the most expensive source of funds.	**Equity**
(ii) It entails a high degree of risk since they have to be repaid as per the terms of agreement.	**Debentures**
(iii) It supports businesses in their routine activities.	**Funding from Banks**
(iv) Business enterprise has options to raise capital from International markets also.	**International Funding**
(v) This source of finance sometimes is the last option for startups which doesn't qualify for bank funding	**Angel Financing**

ICAI · P6 · RTP Sep 2025 · question p. 8-9 · answer p. 25

144 RTP **Sep 2024** Q12(b)

EXPLAIN in brief the features of various types of preference shares.

ICAI'S ANSWER

Various types of Preference shares can be as below:

Sl. No.	Type of Preference Shares	Salient Features
1.	Cumulative	Arrear Dividend will accumulate.
2.	Non-cumulative	No right to arrear dividend.
3.	Redeemable	Redemption should be done.
4.	Participating	Can participate in the surplus which remains after payment to equity shareholders.
5.	Non-Participating	Cannot participate in the surplus after payment of fixed rate of dividend.
6.	Convertible	Option of converting into equity shares.

ICAI · P6 · RTP Sep 2024 · question p. 10 · answer p. 30

145 RTP **Old P8 Jul 2021** Q10(ii) old scheme

DISCUSS the advantages and disadvantages of raising funds by issue of preference shares.

ICAI'S ANSWER

Advantages and disadvantages of raising funds by issue of preference shares

Advantages

(i) No dilution in EPS on enlarged capital base – On the other hand if equity shares are issued it reduces EPS, thus affecting the market perception about the company.

(ii) There is also the advantage of leverage as it bears a fixed charge (because companies are required to pay a fixed rate of dividend in case of issue of preference shares). Non-payment of preference dividends does not force a company into liquidity.

(iii) There is no risk of takeover as the preference shareholders do not have voting rights except where dividend payment are in arrears.

(iv) The preference dividends are fixed and pre-decided. Hence preference shareholders cannot participate in surplus profits as the ordinary shareholders can except in case of participating preference shareholders.

(v) Preference capital can be redeemed after a specified period.

Disadvantages

(i) One of the major disadvantages of preference shares is that preference dividend is not tax deductible and so does not provide a tax shield to the company. Hence, preference shares are costlier to the company than debt e.g. debenture.

(ii) Preference dividends are cumulative in nature. This means that if in a particular year preference dividends are not paid they shall be accumulated and paid later. Also, if these dividends are not paid, no dividend can be paid to ordinary shareholders. The non-payment of dividend to ordinary shareholders could seriously impair the reputation of the concerned company.

ICAI · Old P8 · RTP Jul 2021 · question p. 7 · answer p. 21-22

146 MTP **Sep 2026 S-II** Q4(c) 2 marks

Also asked in Old P8 MTP Dec 2021 S-II Q6(c)

EXPLAIN: Callable bonds and Puttable bonds.

ICAI'S ANSWER

(i) Callable bonds: A callable bond has a call option which gives the issuer the right to redeem the bond before maturity at a predetermined price known as the call price (Generally at a premium).

(ii) Puttable bonds: Puttable bonds give the investor a put option (i.e. the right to sell the bond) back to the company before maturity.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 8 · answer p. 12

147 MTP **Sep 2026 S-II** Q4(c) 2 marks

Also asked in Old P8 MTP Dec 2021 S-II Q6(c)

Briefly DESCRIBE bridge finance

ICAI'S ANSWER

Bridge Finance: Bridge finance refers to loans taken by a company normally from commercial banks for a short period because of pending disbursement of loans sanctioned by financial institutions. Though it is of short-term nature but since it is an important step in the facilitation of long-term loan, therefore it is being discussed along with the long term sources of funds. Normally, it takes time for financial institutions to disburse loans to companies. However, once the loans are approved by the term lending institutions, companies, in order not to lose further time in starting their projects, arrange short term loans from commercial banks. The bridge loans are repaid/ adjusted out of the term loans as and when disbursed by the concerned institutions. Bridge loans are normally secured by hypothecating movable assets, personal guarantees and demand promissory notes. Generally, the rate of interest on bridge finance is higher as compared with that on term loans.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 8 · answer p. 12

148 MTP **May 2026 S-I** Q4(b) 4 marks

EXPLAIN any four types of Preference Shares along with its Salient Features.

ICAI'S ANSWER

Various types of Preference shares can be as below:

Sl. No.	Type of Preference Shares	Salient Features
1	Cumulative	Arrear Dividend will accumulate.
2	Non-cumulative	No right to arrear dividend.
3	Redeemable	Redemption should be done.
4	Participating	Can participate in the surplus which remains after payment to equity shareholders.
5	Non-Participating	Cannot participate in the surplus after payment of fixed rate of Dividend.
6	Convertible	Option of converting into equity Shares.

ICAI · P6 · MTP May 2026 · Series I · question p. 8 · answer p. 11

149 MTP **Sep 2025 S-I** Q4(c) 2 marks

A company is looking to raise funds through an instrument that provides fixed periodic returns to the investor but does not grant voting rights. Interestingly, this instrument is considered to have characteristics of both equity and debt. It ranks above equity shareholders but below debenture holders in case of liquidation.

Which type of financing instrument should the company use and why?

ICAI'S ANSWER

Preference shares are a hybrid financial instrument. They offer a fixed dividend (like debt), but the capital raised is considered part of the equity of the company. Holders of preference shares:

(i) Do not have voting rights in most cases (unlike equity shareholders).

(ii) Rank above equity shareholders in case of company liquidation, but below debt holders like debenture holders.

(iii) Are repaid after creditors but before equity holders, making them less risky than equity but riskier than debt.

ICAI · P6 · MTP Sep 2025 · Series I · question p. 9 · answer p. 17

150 MTP **May 2025 S-II** Q4(b) 4 marks

EXPLAIN any four Features of Debentures or Bonds.

ICAI'S ANSWER

Features of Debentures or Bonds:

(i) Face Value: Debentures or bonds are denominated with some value, this denominated value is called face value of the debenture. Interest is calculated on the face value of the debenture. E.g. if a company issue 9% non- convertible debentures of ₹100 each, this means the face value is ₹ 100 and the interest @ 9% will be calculated on this face value.

(ii) Interest (Coupon) Rate: Each debenture bears a fixed interest (coupon) rate (except Zero coupon bond and Deep discount bond). Interest (coupon) rate is applied to face value of debenture to calculate interest, which is payable to the holders of debentures periodically (annually, semi-annually, etc.).

(iii) Maturity period: Debentures or Bonds has a fixed maturity period for redemption. However, in case of irredeemable debentures maturity period is not defined and it is taken as infinite.

(iv) Redemption Value: Redeemable debentures or bonds are redeemed on its specified maturity date. Based on the debt covenants, the redemption value is determined. Redemption value may vary from the face value of the debenture.

(v) Benefit of tax shield: The payment of interest to the debenture holders are allowed as expenses for the purpose of corporate tax determination. Hence, interest paid to the debenture holders save the tax liability of the company. Saving in the tax liability is also known as tax shield.

ICAI · P6 · MTP May 2025 · Series II · question p. 9 · answer p. 17

151 MTP **Old P8 Nov 2023 S-I** Q6(c) old scheme

Also asked in Old P8 MTP May 2020 S-I Q6(c)

"Financing a business through borrowing is cheaper than using equity." Briefly EXPLAIN.

ICAI'S ANSWER

"Financing a business through borrowing is cheaper than using equity".

(i) Debt capital is cheaper than equity capital from the point of its cost and interest being deductible for income tax purpose, whereas no such deduction is allowed for dividends.

(ii) Issue of new equity dilutes existing control pattern while borrowing does not result in dilution of control.

(iii) In a period of rising prices, borrowing is advantageous. The fixed monetary outgo decreases in real terms as the price level increases.

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 3 · answer p. 10

152 MTP **Old P8 May 2022 S-I** Q6(c) 2 marks old scheme

EXPLAIN in brief Callable bonds and Puttable bonds.

ICAI'S ANSWER

(i) Callable bonds: A callable bond has a call option which gives the issuer the right to redeem the bond before maturity at a predetermined price known as the call price (Generally at a premium).

(ii) Puttable bonds: Puttable bonds give the investor a put option (i.e. the right to sell the bond) back to the company before maturity.

ICAI · Old P8 · MTP May 2022 · Series I · question p. 4 · answer p. 8

153 MTP **Old P8 May 2022 S-II** Q6(a) 4 marks old scheme

DISCUSS in briefly any two long term sources of finance for a partnership firm.

ICAI'S ANSWER

The two sources of long-term finance for a partnership firm are as follows:

Loans from Commercial Banks: Commercial banks provide long term loans for the purpose of expansion or setting up of new units. Their repayment is usually scheduled over a long period of time. The liquidity of such loans is said to depend on the anticipated income of the borrowers.

As part of the long term funding for a partnership firm, the banks also fund the long term working capital requirement (it is also called WCTL i.e. working capital term loan).

Lease financing: Leasing is a general contract between the owner and user of the asset over a specified period of time. The asset is purchased initially by the lessor (leasing company) and thereafter leased to the user (lessee firm) which pays a specified rent at periodical intervals. Thus, leasing is an alternative to the purchase of an asset out of own or borrowed funds. Moreover, lease finance can be arranged much faster as compared to term loans from financial institutions.

ICAI · Old P8 · MTP May 2022 · Series II · question p. 4 · answer p. 10

154 MTP **Old P8 Dec 2021 S-I** Q6(c) 2 marks old scheme

BRIEFLY describe the financial needs of a business.

ICAI'S ANSWER

Financial Needs of a Business: Business enterprises need funds to meet their different types of requirements. All the financial needs of a business may be grouped into the following three

categories-

Long-term financial needs: Such needs generally refer to those requirements of funds which are for a period exceeding **5-10 years**. All investments in plant, machinery, land, buildings, etc., are considered as long-term financial needs.

Medium- term financial needs: Such requirements refer to those funds which are required for a period exceeding one year but not exceeding 5 years.

Short- term financial needs: Such type of financial needs arises to finance current assets such as stock, debtors, cash, etc. Investment in these assets is known as meeting of working capital requirements of the concern for a period not exceeding one year.

PassBook's note: ICAI prints this as the OR alternative to Q6(c), followed by the pair's [2 Marks]. The topic, the financial needs of a business, opens the January 2027 Study Material's Types of Financing chapter ("Financial Needs and Sources of Finance of a Business").

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 4 · answer p. 11

155 MTP **Old P8 Dec 2021 S-II** Q6(a) 4 marks old scheme

BRIEF out any four types of Preference shares along with its feature.

ICAI'S ANSWER

Sl. No.	Type of Preference Shares	Salient Features
1	Cumulative	Arrear Dividend will accumulate.
2	Non-cumulative	No right to arrear dividend.
3	Redeemable	Redemption should be done.
4	Participating	Can participate in the surplus which remains after payment to equity shareholders.
5	Non- Participating	Cannot participate in the surplus after payment of fixed rate of Dividend.
6	Convertible	Option of converting into equity Shares.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 5 · answer p. 9

156 MTP **Old P8 Jul 2021 S-I** Q6(b) 4 marks old scheme

DISCUSS in brief the characteristics of Debentures.

ICAI'S ANSWER

Characteristics of Debentures are as follows:

- Normally, debentures are issued on the basis of a debenture trust deed which lists the terms and conditions on which the debentures are floated.
- Debentures are either secured or unsecured.
- May or may not be listed on the stock exchange.
- The cost of capital raised through debentures is quite low since the interest payable on debentures can be charged as an expense before tax.

- From the investors' point of view, debentures offer a more attractive prospect than the preference shares since interest on debentures is payable whether or not the company makes profits.
- Debentures are thus instruments for raising long-term debt capital.
- The period of maturity normally varies from 3 to 10 years and may also increase for projects having high gestation period.

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 5 · answer p. 9

3.2 Venture capital, lease financing, sale and lease back and other sources

24 questions · asked in 27 ICAI documents · 4 exam papers

157 EXAM **Jan 2026** Q4(a) 4 marks

Discuss briefly the key features of seed capital assistance as a source of finance.

ICAI'S ANSWER

Seed Capital Assistance: The Seed Capital Assistance scheme is designed by IDBI for professionally or technically qualified entrepreneurs and/or persons possessing relevant experience, skills and entrepreneurial traits but lack adequate financial resources. All the projects eligible for financial assistance from IDBI, directly or indirectly through refinance are eligible under the scheme.

The Seed Capital Assistance is interest free but carries a service charge of one per cent per annum for the first five years and at increasing rate thereafter. However, IDBI will have the option to charge interest at such rate as may be determined by IDBI on the loan if the financial position and profitability of the company so permits during the currency of the loan. The repayment schedule is fixed depending upon the repaying capacity of the unit with an initial moratorium up-to five years.

ICAI · P6 · Jan 2026 exam · question p. 22 · answer p. 23

158 EXAM **Sep 2024** Q4(c) 2 marks

What is Leveraged Lease? Explain.

ICAI'S ANSWER

Leveraged Lease: Under this lease, a third party is involved besides lessor and the lessee. The lessor borrows a part of the purchase cost (say 80%) of the asset from the third party i.e., lender and asset so purchased is held as security against the loan. The lender is paid off from the lease rentals directly by the lessee and the surplus after meeting the claims of the lender goes to the lessor. The lessor is entitled to claim depreciation allowance.

ICAI · P6 · Sep 2024 exam · question p. 22 · answer p. 23

159 EXAM **Old P8 May 2023** Q6(c) 2 marks old scheme

Discuss features of Secured Premium Notes.

ICAI'S ANSWER

Features of Secured Premium Notes:

- SPN instruments are issued with a detachable warrant.
- These instruments are redeemable after a notified period of say 4 to 7 years.
- No interest is paid during the lock in period.

- The conversion of detachable warrant into equity shares will have to be done within time period notified by the company.

ICAI · Old P8 · May 2023 exam · question p. 20 · answer p. 21

160 EXAM **Old P8 Nov 2020** Q6(b) 4 marks old scheme

Explain in brief the methods of Venture Capital Financing.

ICAI'S ANSWER

Methods of Venture Capital Financing: Some common methods of venture capital financing are as follows-

(i) Equity financing: The venture capital undertakings generally require funds for a longer period but may not be able to provide returns to the investors during the initial stages. Therefore, the venture capital finance is generally provided by way of equity share capital. The equity contribution of venture capital firm does not exceed 49% of the total equity capital of venture capital undertakings so that the effective control and ownership remains with the entrepreneur.

(ii) Conditional loan: A conditional loan is repayable in the form of a royalty after the venture is able to generate sales. No interest is paid on such loans. In India venture capital financiers charge royalty ranging between 2 and 15 per cent; actual rate depends on other factors of the venture such as gestation period, cash flow patterns, risk and other factors of the enterprise. Some Venture capital financiers give a choice to the enterprise of paying a high rate of interest (which could be well above 20 per cent) instead of royalty on sales once it becomes commercially sound.

(iii) Income note: It is a hybrid security which combines the features of both conventional loan and conditional loan. The entrepreneur has to pay both interest and royalty on sales but at substantially low rates. IDBI's VCF provides funding equal to 80 – 87.50% of the projects cost for commercial application of indigenous technology.

(iv) Participating debenture: Such security carries charges in three phases — in the start-up phase no interest is charged, next stage a low rate of interest is charged up to a particular level of operation, after that, a high rate of interest is required to be paid.

ICAI · Old P8 · Nov 2020 exam · question p. 13 · answer p. 14

161 RTP **May 2026** Q8

EquipLease Financial Services Ltd. is a specialized equipment leasing company that has been operating in the industrial equipment financing space for over 15 years. The company's business model involves purchasing high-value machinery and leasing it to manufacturing companies, thereby earning returns through lease rentals while retaining ownership of the assets. The company has built a strong reputation for providing flexible financing solutions tailored to clients' cash flow patterns and business cycles. Their leasing arrangements help clients conserve capital while accessing state-of-the-art equipment.

EquipLease has recently purchased a specialized CNC precision machine with the following informations:

- **Purchase Cost:** ₹ 10,00,000 (paid upfront)
- **Expected Useful Life:** 4 years
- **Residual/Salvage Value:** Nil (technology becomes obsolete)
- **Depreciation for Tax Purposes:** Can be claimed by EquipLease under Straight line method (SLM) or Units of Production method(UOP)

- **Lease Rentals** earned by company is subject to tax.
- **EquipLease's Cost of Capital:** 10% per annum
- **Corporate Tax Rate:** 25%

TechManufacture Industries Ltd., a growing precision engineering company, has approached EquipLease expressing interest in leasing this CNC machine. TechManufacture has provided detailed information about their expected utilization pattern:

The machine's usage will follow a graduated ramp-up pattern based on TechManufacture's order book projections:

Year	Production Capacity Utilization	Relative Units
1	Low (Initial phase)	1 unit
2	Medium	2 units
3	High	3 units
4	Peak	4 units

This pattern is typical for businesses scaling up production with new equipment - starting slow as operators are trained and processes are optimized, then ramping up to full capacity. The Client wants the company to charge lease rentals based on the client's utilization pattern 1:2:3:4.

Mr. Vikram Malhotra, the Chief Credit Officer at EquipLease, is tasked with structuring an appropriate lease arrangement. He faces several strategic decisions:

i. HOW MUCH portion of the cost is recovered from tax savings on depreciation under both the methods viz. SLM & units of production method.

ii. FIND the portion of investment to be recovered from lease rentals under both the assumptions i.e. Depreciation under both the methods viz. SLM & units of production method.

iii. DETERMINE minimum lease rentals needed to recover the balance cost of investment after adjusting the Tax shields on the depreciation under both the depreciation methods if -

- EquipLease charge uniform (equal) rentals over 4 years
- Lease rentals follow the client's utilization pattern (1:2:3:4)

ICAI'S ANSWER

Calculation of PV of Tax savings under Straight line method (SLM)

Particulars	Result
Annual Depreciation	10 Lakhs / 4 Years = 2,50,000
PV of Tax savings on above	2,50,000 Lakhs x 25% x PVAF @ 10% i.e. 3.170 = 1,98,125

Calculation of PV of Tax savings under Units of Production (UOP)

Year	Depreciation	Tax savings	PVF @ 10%	PV of Tax Savings
1	1,00,000	25,000	0.909	22,725
2	2,00,000	50,000	0.826	41,300
3	3,00,000	75,000	0.751	56,325
4	4,00,000	1,00,000	0.683	68,300
				1,88,650

Let lease rental of 1st year be X

PV of after tax Lease rentals charged as per client's utilization pattern

Year	Lease rentals	After Tax Lease Rentals	PVF @ 10%	PV of Lease Rentals
1	X	0.75X	0.909	0.682X
2	2X	1.5X	0.826	1.239X
3	3X	2.25X	0.751	1.690X
4	4X	3X	0.683	2.049X
				5.660X

PV of after tax Lease rentals charged equally

Let annual lease rental be X

X x (1 – 25%) x 3.17(PVAF @ 10%) = 2.378X

Investment to be recovered from lease rentals

	In UOP Ratio		Equally	
Particulars	**SLM**	**UOP**	**SLM**	**UOP**
Investment	10,00,000	10,00,000	10,00,000	10,00,000
PV of Tax savings on Depreciation	1,98,125	1,88,650	1,98,125	1,88,650
Investment to be recovered	8,01,875	8,11,350	8,01,875	8,11,350
PV of after-tax Lease rentals	5.660X	5.660X	2.378X	2.378X
Lease rentals = X	1,41,674	1,43,348	3,37,206	3,41,190

Statement Showing PV of yearly lease rentals when charged in UOP Ratio

Year	Particulars	SLM	UOP
	Lease rentals = X	**1,41,674**	**1,43,348**
1	0.682X	96,622	97,763
2	1.239X	1,75,534	1,77,608
3	1.690X	2,39,429	2,42,258
4	2.049X	2,90,290	2,93,720
	PV of Total after tax lease rental	**8,01,875**	**8,11,350**

Statement Showing PV of yearly lease rentals when charged Equally

Year	Particulars	SLM	UOP
	Lease rentals = X	3,37,206	3,41,190
1	0.682X (0.909 × 0.75)	2,29,974	2,32,692
2	0.620X (0.826 × 0.75)	2,09,068	2,11,538
3	0.563X (0.751 × 0.75)	1,89,847	1,92,090
4	0.512X (0.683 × 0.75)	1,72,649	1,74,689
	PV of Total after tax lease rental	**8,01,539***	**8,11,009***

*** Difference is due to rounding off in PVAF**

ICAI · P6 · RTP May 2026 · question p. 8-9 · answer p. 23-25

162 RTP **May 2025** Q10(b)

Also asked in Old P8 RTP Dec 2021 Q10(b)

Explain some common methods of venture capital financing.

ICAI'S ANSWER

Some common methods of venture capital financing are as follows:

(i) ***Equity financing:*** The venture capital undertakings generally require funds for a longer period but may not be able to provide returns to the investors during the initial stages. Therefore, the venture capital finance is generally provided by way of equity share capital. The equity contribution of venture capital firm does not exceed 49% of the total equity capital of venture capital undertakings so that the effective control and ownership remains with the entrepreneur.

(ii) ***Conditional loan:*** A conditional loan is repayable in the form of a royalty after the venture is able to generate sales. No interest is paid on such loans. In India venture capital financiers charge royalty ranging between 2 and 15 per cent; actual rate depends on other factors of the venture such as gestation period, cash flow patterns, risk and other factors of the enterprise. Some Venture capital financiers give a choice to the enterprise of paying a high rate of interest (which could be well above 20 per cent) instead of royalty on sales once it becomes commercially sound.

(iii) ***Income note:*** It is a hybrid security which combines the features of both conventional loan and conditional loan. The entrepreneur has to pay both interest and royalty on sales but at substantially low rates. IDBI's VCF provides funding equal to 80 – 87.50% of the projects cost for commercial application of indigenous technology.

(iv) ***Participating debenture:*** Such security carries charges in three phases — in the start-up phase no interest is charged, next stage a low rate of interest is charged up to a particular level of operation, after that, a high rate of interest is required to be paid.

ICAI · P6 · RTP May 2025 · question p. 10 · answer p. 30-31

163 RTP **Old P8 Nov 2023** Q10(c) old scheme

STATE the meaning of debt securitization.

ICAI'S ANSWER

Debt Securitisation: It is a method of recycling of funds. It is especially beneficial to financial intermediaries to support the lending volumes. Assets generating steady cash flows are packaged together and against this asset pool, market securities can be issued, e.g., housing finance, auto loans, and credit card receivables.

Process of Debt Securitisation

(i) *The origination function* – A borrower seeks a loan from a finance company, bank. The credit worthiness of borrower is evaluated, and contract is entered into with repayment schedule structured over the life of the loan.

(ii) *The pooling function* – Similar loans on receivables are clubbed together to create an underlying pool of assets. The pool is transferred in favour of Special purpose Vehicle (SPV), which acts as a trustee for investors.

(iii) *The securitisation function* – SPV will structure, and issue securities based on asset pool. The securities carry a coupon and expected maturity which can be asset-based/mortgage based. These are generally sold to investors through merchant bankers. Investors are – pension funds, mutual funds, insurance funds.

PassBook's note: Debt securitisation is in the January 2027 Study Material (Types of Financing, section 5).

ICAI · Old P8 · RTP Nov 2023 · question p. 5 · answer p. 17

164 RTP **Old P8 May 2023** Q10(b) old scheme

EXPLAIN the process of Debt Securitisation.

ICAI'S ANSWER

Securitisation is a **process in which illiquid assets are pooled** into marketable securities that can be sold to investors. The process leads to the creation of financial instruments that represent ownership interest in, or are secured by a segregated income producing asset or pool of assets. These assets are generally secured by personal or real property such as automobiles, real estate, or equipment loans but in some cases are unsecured.

Example of Debt Securitisation:

A finance company has given a large number of car loans. It needs more money so that it is in a position to give more loans. One way to achieve this is to sell all the existing loans. But, in the absence of a liquid secondary market for individual car loans, this is not feasible.

However, a practical option is debt securitisation, in which the finance company sells its existing car loans already given to borrowers to the Special Purpose Vehicle (SPV). The SPV, in return pays to the company, which in turn continue to lend with this money. On the other hand, the SPV pools these loans and convert these into marketable securities. It means that now these converted securities can be issued to investors.

So, this process of debt securitization helps the finance company to raise funds and get the loans off its Balance Sheet. These funds also help the company disburse further loans. Similarly, the process is beneficial to the investors also as it creates a liquid investment in a diversified pool of car loans, which may be an attractive option to other fixed income instruments. The whole process is carried out in such a way that the original Receivables i.e. the car loan borrowers may not be aware of the transaction. They might have continued making payments the way they are already doing. However, these payments shall now be made to the new investors who have emerged out of this securitization process.

ICAI · Old P8 · RTP May 2023 · question p. 5 · answer p. 20

165 RTP **Old P8 Nov 2022** Q10(a) old scheme

Under financial lease, lessee bears the risk of obsolescence; while under operating lease, lessor bears the risk of obsolescence. In view of this, you are required to COMPARE the financial lease and operating lease.

ICAI'S ANSWER

Comparison between Financial Lease and Operating Lease

	Financial Lease	Operating Lease
1.	The risk and reward incident to ownership are passed on to the lessee. The lessor only remains the legal owner of the asset.	The lessee is only provided the use of the asset for a certain time. Risk incident to ownership belong wholly to the lessor.
2.	The lessee bears the risk of obsolescence.	The lessor bears the risk of obsolescence.
3.	The lessor is interested in his rentals and not in the asset. He must get his principal back along with interest. Therefore, the lease is non-cancellable by either party.	As the lessor does not have difficulty in leasing the same asset to other willing lessee, the lease is kept cancelable by the lessor.
4.	The lessor enters into the transaction only as financier. He does not bear the cost of repairs, maintenance or operations.	Usually, the lessor bears cost of repairs, maintenance or operations.
5.	The lease is usually full payout, that is, the single lease repays the cost of the asset together with the interest.	The lease is usually non-payout, since the lessor expects to lease the same asset over and over again to several users.

ICAI · Old P8 · RTP Nov 2022 · question p. 7 · answer p. 23

166 RTP **Old P8 Nov 2020** Q10(b) old scheme

COMPARE between Financial Lease and Operating Lease.

ICAI'S ANSWER

	Finance Lease	Operating Lease
1.	The risk and reward incident to ownership are passed on to the lessee. The lessor only remains the legal owner of the asset.	The lessee is only provided the use of the asset for a certain time. Risk incident to ownership belong wholly to the lessor.
2.	The lessee bears the risk of obsolescence.	The lessor bears the risk of obsolescence.
3.	The lessor is interested in his rentals and not in the asset. He must get his principal back along with interest. Therefore, the lease is non-cancellable by either party.	As the lessor does not have difficulty in leasing the same asset to other willing lessor, the lease is kept cancelable by the lessor.
4.	The lessor enters into the transaction only as financier. He does not bear the cost of repairs, maintenance or operations.	Usually, the lessor bears cost of repairs, maintenance or operations.
5.	The lease is usually full payout, that is, the single lease repays the cost of the asset together with the interest.	The lease is usually non-payout, since the lessor expects to lease the same asset over and over again to several users.

ICAI · Old P8 · RTP Nov 2020 · question p. 5 · answer p. 19

167 RTP **Old P8 May 2020** Q10(ii) old scheme

EXPLAIN the basics of debt securitisation process.

ICAI'S ANSWER

Process of Debt Securitisation:

(a) *The origination function* – A borrower seeks a loan from a finance company or a bank. The credit worthiness of borrower is evaluated and contract is entered into with repayment schedule structured over the life of the loan.

(b) *The pooling function* – Similar loans on receivables are clubbed together to create an underlying pool of assets. The pool is transferred in favour of Special purpose Vehicle (SPV), which acts as a trustee for investors.

(c) *The securitisation function* – SPV will structure and issue securities on the basis of asset pool. The securities carry a coupon and expected maturity which can be asset-based/mortgage based. These are generally sold to investors through merchant bankers. Investors are – pension funds, mutual funds, insurance funds.

The process of securitization is generally without recourse i.e. investors bear the credit risk and issuer is under an obligation to pay to investors only if the cash flows are received by him from the collateral. The benefits to the originator are that assets are shifted off the balance sheet, thus giving the originator recourse to off-balance sheet funding.

ICAI · Old P8 · RTP May 2020 · question p. 6 · answer p. 15

168 MTP **Sep 2026 S-I** Q4(a) 4 marks

Same question, different ICAI answer: #174 MTP May 2025 S-I

What is debt securitisation? Explain the basics of debt securitisation process.

ICAI'S ANSWER

Debt Securitisation: It is a method of recycling of funds. It is especially beneficial to financial intermediaries to support the lending volumes. Assets generating steady cash flows are packaged together and against this asset pool, market securities can be issued, e.g. housing finance, auto loans, and credit card receivables.

Process of Debt Securitisation

(i) *The origination function* – A borrower seeks a loan from a finance company or bank. The credit worthiness of borrower is evaluated and a contract is entered into with repayment schedule structure over the life of the loan.

(ii) *The pooling function* – Similar loans on receivables are clubbed together to create an underlying pool of assets. The pool is transferred in favour of Special Purpose Vehicle (SPV), which acts as a trustee for investors.

(iii) *The securitisation function* – The SPV structure and issue securities on the basis of these assets pool. The securities carry a coupon and expected maturity which can be asset-based/mortgage based. These are generally sold to investors through merchant bankers. Investors may be a pension funds, mutual funds, insurance funds.

The process of securitization is generally without recourse i.e. investors bear the credit risk and issuer is under an obligation to pay to investors only if the cash flows are received by him from the collateral. The benefits to the originator are that assets are shifted off the balance sheet, thus giving the originator recourse to off-balance sheet funding.

ICAI · P6 · MTP Sep 2026 · Series I · question p. 7 · answer p. 12

169 MTP **Sep 2026 S-I** Q4(b) 4 marks

Also asked in MTP Sep 2024 S-II Q4(b)

Operating leases and financial leases are traditionally the most important types of leases in financial management. However, in recent years, other types of leases have also gained significance due to their unique benefits and applications. IDENTIFY AND EXPLAIN at least four other types of leases that have become increasingly important in modern business practices.

ICAI'S ANSWER

(i) **Sales and Lease Back:** Under this type of lease, the owner of an asset sells the asset to a party (the buyer), who in turn leases back the same asset to the owner in consideration of a lease rentals. Under this arrangement, the asset is not physically exchanged but it all happen in records only. The main advantage of this method is that the lessee can satisfy himself completely regarding the quality of an asset and after possession of the asset convert the sale into a lease agreement. Under this transaction, the seller assumes the role of lessee (as the same asset which he has sold came back to him in the form of lease) and the buyer assumes the role of a lessor (as asset purchased by him was leased back to the seller). So, the seller gets the agreed selling price and the buyer gets the lease rentals.

(ii) **Leveraged Lease:** Under this lease, a third party is involved besides lessor and the lessee. The lessor borrows a part of the purchase cost (say 80%) of the asset from the third party i.e., lender and asset so purchased is held as security against the loan. The lender is paid off from the lease rentals directly by the lessee and the surplus after meeting the claims of the lender goes to the lessor. The lessor is entitled to claim depreciation allowance.

(iii) **Sales-aid Lease:** Under this lease contract, the lessor enters into a tie up with a manufacturer for marketing the latter's product through his own leasing operations, it is called a sales-aid lease. In consideration of the aid in sales, the manufacturer may grant either credit or a commission to the lessor. Thus, the lessor earns from both sources i.e. From lessee as well as the manufacturer.

(iv) **Close-ended and Open-ended Leases:** In the close-ended lease, the assets get transferred to the lessor at the end of lease, the risk of obsolescence, residual value etc., remain with the lessor being the legal owner of the asset. In the open-ended lease, the lessee has the option of purchasing the asset at the end of the lease period.

ICAI · P6 · MTP Sep 2026 · Series I · question p. 7 · answer p. 12-13

170 MTP **Jan 2026 S-I** Q4(b) 4 marks

EXPLAIN the methods of venture capital financing.

ICAI'S ANSWER

Some common methods of venture capital financing are as follows:

(i) ***Equity financing:*** The venture capital undertakings generally require funds for a longer period but may not be able to provide returns to the investors during the initial stages. Therefore, the venture capital finance is generally provided by way of equity share capital. The equity contribution of venture capital firm does not exceed 49% of the total equity capital of venture capital undertakings so that the effective control and ownership remains with the entrepreneur.

(ii) ***Conditional loan:*** A conditional loan is repayable in the form of a royalty after the venture is able to generate sales. No interest is paid on such loans. In India venture capital financiers charge royalty ranging between 2 and 15 per cent; actual rate depends on other factors of the venture such as gestation period, cash flow patterns, risk and other factors of the enterprise. Some Venture capital financiers give a choice to the enterprise of paying a high rate of interest (which could be well above 20 per cent) instead of royalty on sales once it becomes commercially sound.

(iii) ***Income note:*** It is a hybrid security which combines the features of both conventional loan and conditional loan. The entrepreneur has to pay both interest and royalty on sales but at substantially low rates. IDBI's VCF provides funding equal to 80 – 87.50% of the projects cost for commercial application of indigenous technology.

(iv) ***Participating debenture:*** Such security carries charges in three phases — in the start-up phase no interest is charged, next stage a low rate of interest is charged up to a particular level of operation, after that, a high rate of interest is required to be paid.

ICAI · P6 · MTP Jan 2026 · Series I · question p. 7 · answer p. 14-15

171 MTP **Jan 2026 S-II** Q4(a) 4 marks

Oceanic Retail Pvt. Ltd., a fast-growing chain of supermarkets, is planning to expand its presence across multiple cities. The company's CFO, Mr. Arvind Mehta, is evaluating different lease contracts for acquiring stores and equipment. The legal team has informed him that besides the common Finance Lease and Operating Lease, there are several other types of lease arrangements. EXPLAIN four other types of lease arrangements.

ICAI'S ANSWER

Other Types of Leases are explained below:

(i) **Sales and Lease Back:** Under this type of lease, the owner of an asset sells the asset to a party (the buyer), who in turn leases back the same asset to the owner in consideration of a lease rentals. Under this arrangement, the asset is not physically exchanged but it all happen in records only. The main advantage of this method is that the lessee can satisfy himself completely regarding the quality of an asset and after possession of the asset convert the sale into a lease agreement.

Under this transaction, the seller assumes the role of lessee (as the same asset which he has sold came back to him in the form of lease) and the buyer assumes the role of a lessor (as asset purchased by him was leased back to the seller). So, the seller gets the agreed selling price and the buyer gets the lease rentals.

(ii) **Leveraged Lease:** Under this lease, a third party is involved besides lessor and the lessee. The lessor borrows a part of the purchase cost (say 80%) of the asset from the third party i.e., lender and asset so purchased is held as security against the loan. The lender is paid off from the lease rentals directly by the lessee and the surplus after meeting the claims of the lender goes to the lessor. The lessor is entitled to claim depreciation allowance.

(iii) **Sales-aid Lease:** Under this lease contract, the lessor enters into a tie up with a manufacturer for marketing the latter's product through his own leasing operations, it is called a sales-aid lease. In consideration of the aid in sales, the manufacturer may grant either credit or a commission to the lessor. Thus, the lessor earns from both sources i.e. from lessee as well as the manufacturer.

(iv) **Close-ended and Open-ended Leases:** In the close-ended lease, the assets get transferred to the lessor at the end of lease, the risk of obsolescence, residual value etc., remain with the lessor being the legal owner of the asset. In the open-ended lease, the lessee has the option of purchasing the asset at the end of the lease period.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 8 · answer p. 13

172 MTP **Sep 2025 S-I** Q2(b) 2 marks

What is venture capital financing? Briefly describe its characteristics.

ICAI'S ANSWER

The venture capital financing refers to financing of new high risky venture promoted by qualified entrepreneurs who lack experience and funds to give shape to their ideas. In broad sense, under venture capital financing, venture capitalist make investment to purchase equity or debt securities from inexperienced entrepreneurs who undertake highly risky ventures with potential to succeed in future.

Some of the characteristics of Venture Capital financing are:

(i) It is basically an equity finance in new companies.

(ii) It can be viewed as a long-term investment in growth-oriented small/medium firms.

(iii) Apart from providing funds, the investor also provides support in the form of sales strategy, business networking and management expertise, enabling the growth of the entrepreneur.

ICAI · P6 · MTP Sep 2025 · Series I · question p. 8 · answer p. 11-12

173 MTP **Sep 2025 S-II** Q4(c) 2 marks

A company owns a large asset (a commercial building) and is facing a cash crunch. The CFO suggests a sale and leaseback arrangement. As a finance student, WHAT practical benefits and risks would you consider before supporting this decision?

ICAI'S ANSWER

Benefits of sale and leaseback arrangement:

(i) Immediate cash inflow to ease the cash crunch.

(ii) Continued use of the asset without ownership, avoiding disruption.

(iii) Off-balance sheet financing improves capital structure ratios.

Risks of sale and leaseback arrangement:

(i) Loss of ownership may mean higher long-term costs.

(ii) Lease obligations add to fixed costs and reduce flexibility.

Hence, sale and leaseback decision can be applied only if the lease terms are favorable and the company lacks better financing options.

ICAI · P6 · MTP Sep 2025 · Series II · question p. 10 · answer p. 15

174 MTP **May 2025 S-I** Q4(b) 4 marks

Also asked in Old P8 MTP May 2023 S-I Q6(a) · Old P8 MTP Nov 2019 S-I Q6(b)

Same question, different ICAI answer: #168 MTP Sep 2026 S-I

WHAT is debt securitisation? EXPLAIN the basics of debt securitisation process.

ICAI'S ANSWER

Debt Securitisation: It is a method of recycling of funds. It is especially beneficial to financial intermediaries to support the lending volumes. Assets generating steady cash flows are packaged together and against this asset pool, market securities can be issued, e.g. housing finance, auto loans, and credit card receivables.

Process of Debt Securitisation

(i) *The origination function* – A borrower seeks a loan from a finance company, bank, HDFC. The credit worthiness of borrower is evaluated and contract is entered into with repayment schedule structured over the life of the loan.

(ii) *The pooling function* – Similar loans on receivables are clubbed together to create an underlying pool of assets. The pool is transferred in favour of Special purpose Vehicle (SPV), which acts as a trustee for investors.

(iii) *The securitisation function* – SPV will structure and issue securities on the basis of asset pool. The securities carry a coupon and expected maturity which can be asset-based/mortgage based. These are generally sold to investors through merchant bankers. Investors are – pension funds, mutual funds, insurance funds.

The process of securitization is generally without recourse i.e. investors bear the credit risk and issuer is under an obligation to pay to investors only if the cash flows are received by him from the collateral. The benefits to the originator are that assets are shifted off the balance sheet, thus giving the originator recourse to off-balance sheet funding.

ICAI · P6 · MTP May 2025 · Series I · question p. 7 · answer p. 15-16

175 MTP **May 2025 S-II** Q4(a) 4 marks

EXPLAIN following types of leases:

(i) Sales and Lease Back

(ii) Leveraged Lease

(iii) Sales-Aid Lease

(iv) Close-Ended and Open-Ended Leases

ICAI'S ANSWER

(i) Sales and Lease Back: Under this type of lease, the owner of an asset sells the asset to a party (the buyer), who in turn leases back the same asset to the owner in consideration of a lease rentals. Under this arrangement, the asset is not physically exchanged but it all happen in records only. The main advantage of this method is that the lessee can satisfy himself completely regarding the quality of an asset and after possession of the asset convert the sale into a lease agreement.

Under this transaction, the seller assumes the role of lessee (as the same asset which he has sold came back to him in the form of lease) and the buyer assumes the role of a lessor (as asset purchased by him was leased back to the seller). So, the seller gets the agreed selling price and the buyer gets the lease rentals.

(ii) Leveraged Lease: Under this lease, a third party is involved besides lessor and the lessee. The lessor borrows a part of the purchase cost (say 80%) of the asset from the third party i.e., lender and asset so purchased is held as security against the loan. The lender is paid off from the lease rentals directly by the lessee and the surplus after meeting the claims of the lender goes to the lessor. The lessor is entitled to claim depreciation allowance.

(iii) Sales-Aid Lease: Under this lease contract, the lessor enters into a tie up with a manufacturer for marketing the latter's product through his own leasing operations, it is called a sales-aid lease. In consideration of the aid in sales, the manufacturer may grant either credit or a commission to the lessor. Thus, the lessor earns from both sources i.e. from lessee as well as the manufacturer.

(iv) Close-Ended and Open-Ended Leases: In the close-ended lease, the assets get transferred to the lessor at the end of lease, the risk of obsolescence, residual value etc., remain with the lessor being the legal owner of the asset. In the open-ended lease, the lessee has the option of purchasing the asset at the end of the lease period.

ICAI · P6 · MTP May 2025 · Series II · question p. 9 · answer p. 16

176 MTP **May 2024 S-I** Q4(b) 4 marks

WHAT are the methods of venture capital financing?

ICAI'S ANSWER

Some common methods of venture capital financing are as follows:

(i) Equity financing: The venture capital undertakings generally require funds for a longer period but may not be able to provide returns to the investors during the initial stages. Therefore, the venture capital finance is generally provided by way of equity share capital. The equity contribution of venture capital firm does not exceed 49% of the total equity capital of venture capital undertakings so that the effective control and ownership remains with the entrepreneur.

(ii) Conditional loan: A conditional loan is repayable in the form of a royalty after the venture is able to generate sales. No interest is paid on such loans. In India venture capital financiers charge royalty

ranging between 2 and 15 per cent; actual rate depends on other factors of the venture such as gestation period, cash flow patterns, risk and other factors of the enterprise. Some Venture capital financiers give a choice to the enterprise of paying a high rate of interest (which could be well above 20 per cent) instead of royalty on sales once it becomes commercially sound.

(iii) Income note: It is a hybrid security which combines the features of both conventional loan and conditional loan. The entrepreneur has to pay both interest and royalty on sales but at substantially low rates. IDBI's VCF provides funding equal to 80 – 87.50% of the projects cost for commercial application of indigenous technology.

(iv) Participating debenture: Such security carries charges in three phases — in the start-up phase no interest is charged, next stage a low rate of interest is charged up to a particular level of operation, after that, a high rate of interest is required to be paid.

ICAI · P6 · MTP May 2024 · Series I · question p. 5 · answer p. 7

177 MTP **Old P8 May 2023 S-I** Q6(c) 2 marks old scheme

Also asked in Old P8 MTP Jul 2021 S-I Q6(c)

DEFINE Security Premium Notes.

ICAI'S ANSWER

Secured Premium Notes: Secured Premium Notes is issued along with a detachable warrant and is redeemable after a notified period of say 4 to 7 years. The conversion of detachable warrant into equity shares will have to be done within time period notified by the company.

ICAI · Old P8 · MTP May 2023 · Series I · question p. 4 · answer p. 8

178 MTP **Old P8 Nov 2022 S-I** Q6(b) 4 marks old scheme

WHAT is the meaning of Venture Capital Financing. STATE some characteristics of it.

ICAI'S ANSWER

Venture Capital Financing: The venture capital financing refers to financing of new high risky venture promoted by qualified entrepreneurs who lack experience and funds to give shape to their ideas. In broad sense, under venture capital financing, venture capitalist make investment to purchase equity or debt securities from inexperienced entrepreneurs who undertake highly risky ventures with potential to succeed in future.

Some of the characteristics of Venture Capital financing are:

- It is basically an equity finance in new companies.
- It can be viewed as a long-term investment in growth-oriented small/medium firms.
- Apart from providing funds, the investor also provides support in form of sales strategy, business networking and management expertise, enabling the growth of the entrepreneur.

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 4 · answer p. 7

179 MTP **Old P8 Nov 2022 S-II** Q5(b) 2 marks old scheme

Write a short note on seed capital assistance.

ICAI'S ANSWER

Seed Capital Assistance: The seed capital assistance has been designed by IDBI for professionally or technically qualified entrepreneurs. All the projects eligible for financial assistance from IDBI,

directly or indirectly through refinance are eligible under the scheme. The project cost should not exceed ₹ 2 crores and the maximum assistance under the project will be restricted to 50% of the required promoter's contribution or ₹ 15 lacs whichever is lower.

The seed capital assistance is interest free but carries a security charge of one percent per annum for the first five years and an increasing rate thereafter

ICAI · Old P8 · MTP Nov 2022 · Series II · question p. 4 · answer p. 9

180 MTP **Old P8 Jul 2021 S-II** Q6(c) 2 marks old scheme

DEFINE Debt Securitisation.

ICAI'S ANSWER

Debt Securitisation is a process in which illiquid assets are pooled into marketable securities that can be sold to investors. The process leads to the creation of financial instruments that represent ownership interest in, or are secured by a segregated income producing asset or pool of assets. These assets are generally secured by personal or real property such as automobiles, real estate, or equipment loans but in some cases are unsecured.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 4 · answer p. 7

3.3 Short-term sources of finance

13 questions · asked in 15 ICAI documents · 3 exam papers

181 EXAM **May 2026** Q4(b) 4 marks

An Indian exporter has shipped goods to a foreign buyer, but the export proceeds are yet to be realised. He approaches the bank for immediate finance assistance.

(i) Identify the type of finance involved.

(ii) Explain any three forms of such type of finance.

ICAI'S ANSWER

Post-shipment Finance is the type of finance involved:

It takes the following forms:

(a) ***Purchase/discounting of documentary export bills***: Finance is provided to exporters by purchasing export bills drawn payable at sight or by discounting usance export bills covering confirmed sales and backed by documents including documents of the title of goods such as bill of lading, post parcel receipts, or air consignment notes.

(b) ***E.C.G.C. Guarantee:*** Post-shipment finance, given to an exporter by a bank through purchase, negotiation or discount of an export bill against an order, qualifies for post-shipment export credit guarantee. It is necessary, however, that exporters should obtain a shipment or contracts risk policy of E.C.G.C. Banks insist on the exporters to take a contracts shipments (comprehensive risks) policy covering both political and commercial risks. The Corporation, on acceptance of the policy, will fix credit limits for individual exporters and the Corporation's liability will be limited to the extent of the limit so fixed for the exporter concerned irrespective of the amount of the policy.

(c) ***Advance against export bills sent for collection:*** Finance is provided by banks to exporters by way of advance against export bills forwarded through them for collection, taking into account the creditworthiness of the party, nature of goods exported, usance, standing of drawee etc.

(d) ***Advance against duty draw backs, cash subsidy, etc.:*** To finance export losses sustained by exporters, bank advance against duty draw-back, cash subsidy etc., receivable by them against export performance. Such advances are of clean nature; hence necessary precaution should be exercised.

ICAI · P6 · May 2026 exam · question p. 21 · answer p. 22

182 EXAM **Old P8 Jul 2021** Q6(a) 4 marks old scheme

Explain in brief the forms of Post Shipment Finance.

ICAI'S ANSWER

Post-shipment Finance: It takes the following forms:

(a) **Purchase/discounting of documentary export bills**: Finance is provided to exporters by purchasing export bills drawn payable at sight or by discounting usance export bills covering confirmed sales and backed by documents including documents of the title of goods such as bill of lading, post parcel receipts, or air consignment notes.

(b) **E.C.G.C. Guarantee:** Post-shipment finance, given to an exporter by a bank through purchase, negotiation or discount of an export bill against an order, qualifies for post-shipment export credit guarantee. It is necessary, however, that exporters should obtain a shipment or contracts risk policy of E.C.G.C. Banks insist on the exporters to take a contracts shipments (comprehensive risks) policy covering both political and commercial risks. The Corporation, on acceptance of the policy, will fix credit limits for individual exporters and the Corporation's liability will be limited to the extent of the limit so fixed for the exporter concerned irrespective of the amount of the policy.

(c) **Advance against export bills sent for collection:** Finance is provided by banks to exporters by way of advance against export bills forwarded through them for collection, taking into account the creditworthiness of the party, nature of goods exported, usance, standing of drawee, etc.

(d) **Advance against duty draw backs, cash subsidy, etc.:** To finance export losses sustained by exporters, bank advance against duty draw-back, cash subsidy, etc., receivable by them against export performance. Such advances are of clean nature; hence necessary precaution should be exercised.

ICAI · Old P8 · Jul 2021 exam · question p. 83 · answer p. 83

183 EXAM **Old P8 Nov 2019** Q6(c) 4 marks old scheme

Briefly describe any four sources of short-term finance.

ICAI'S ANSWER

Sources of Short Term Finance: There are various sources available to meet short-term needs of finance. The different sources are discussed below-

(i) Trade Credit: It represents credit granted by suppliers of goods, etc., as an incident of sale. The usual duration of such credit is 15 to 90 days. It generates automatically in the course of business and is common to almost all business operations. It can be in the form of an 'open account' or 'bills payable'.

(ii) Accrued Expenses and Deferred Income: Accrued expenses represent liabilities which a company has to pay for the services which it has already received like wages, taxes, interest and dividends. Such expenses arise out of the day-to-day activities of the company and hence represent a spontaneous source of finance.

Deferred Income: These are the amounts received by a company in lieu of goods and services to be provided in the future. Since these receipts increases a company's liquidity, they are also considered to be an important sources of short-term finance.

(iii) Advances from Customers: Manufacturers and contractors engaged in producing or constructing costly goods involving considerable length of manufacturing or construction time usually demand advance money from their customers at the time of accepting their orders for executing their contracts or supplying the goods. This is a cost free source of finance and really useful.

(iv) Commercial Paper: A Commercial Paper is an unsecured money market instrument issued in the form of a promissory note. The Reserve Bank of India introduced the commercial paper scheme in the year 1989 with a view to enabling highly rated corporate borrowers to diversify their sources of short-term borrowings and to provide an additional instrument to investors.

(v) Treasury Bills: Treasury bills are a class of Central Government Securities. Treasury bills, commonly referred to as T-Bills are issued by Government of India to meet short term borrowing requirements with maturities ranging between 14 to 364 days.

(vi) Certificates of Deposit (CD): A certificate of deposit (CD) is basically a savings certificate with a fixed maturity date of not less than 15 days up to a maximum of one year.

(vii) Bank Advances: Banks receive deposits from public for different periods at varying rates of interest. These funds are invested and lent in such a manner that when required, they may be called back. Lending results in gross revenues out of which costs, such as interest on deposits, administrative costs, etc., are met and a reasonable profit is made. A bank's lending policy is not merely profit motivated but has to also keep in mind the socio- economic development of the country. Some of the facilities provided by banks are Short Term Loans, Overdraft, Cash Credits, Advances against goods, Bills Purchased/Discounted.

(viii) Financing of Export Trade by Banks: Exports play an important role in accelerating the economic growth of developing countries like India. Of the several factors influencing export growth, credit is a very important factor which enables exporters in efficiently executing their export orders. The commercial banks provide short-term export finance mainly by way of pre and post-shipment credit. Export finance is granted in Rupees as well as in foreign currency.

(ix) Inter Corporate Deposits: The companies can borrow funds for a short period say 6 months from other companies which have surplus liquidity. The rate of interest on inter corporate deposits varies depending upon the amount involved and time period.

(x) Certificate of Deposit (CD): The certificate of deposit is a document of title similar to a time deposit receipt issued by a bank except that there is no prescribed interest rate on such funds.

The main advantage of CD is that banker is not required to encash the deposit before maturity period and the investor is assured of liquidity because he can sell the CD in secondary market.

(xi) Public Deposits: Public deposits are very important source of short-term and medium term finances particularly due to credit squeeze by the Reserve Bank of India. A company can accept public deposits subject to the stipulations of Reserve Bank of India from time to time maximum up to 35 per cent of its paid up capital and reserves, from the public and shareholders. These deposits may be accepted for a period of six months to three years. Public deposits are unsecured loans; they should not be used for acquiring fixed assets since they are to be repaid within a period of 3 years. These are mainly used to finance working capital requirements.

ICAI · Old P8 · Nov 2019 exam · question p. 16 · answer p. 17-19

184 RTP **Sep 2025** Q10(b)

Trade credit and bank overdraft are two commonly used short-term financing sources but each has distinct features. On the light of the statement SHOW the difference between trade credit and bank overdraft.

ICAI'S ANSWER

Trade credit and bank overdraft are two commonly used short-term financing sources but each has distinct features which are shown below:

S. No.	Trade credit	Bank overdraft
1.	Trade credit is the credit extended by suppliers, allowing the firm to delay payment for goods or services.	Under this facility, customers are allowed to withdraw in excess of credit balance standing in their Current Account.
2.	The usual duration of such credit is 15 to 90 days.	Though overdrafts are repayable on demand, they generally continue for long periods by annual renewals of the limits.
3.	Trade credit is a type of interest free loan, therefore failure to avail this facility has an interest cost.	Interest is charged on daily balances
4.	It generates automatically in the course of business and is common to almost all business operations.	Borrower need to apply to bank for Bank overdraft facility and are operated in the same way as cash credit and current accounts.
5.	It enhances automatically with the increase in the volume of business and fast payment.	A fixed limit is, therefore, granted to the borrower within which the borrower is allowed to overdraw his account.

ICAI · P6 · RTP Sep 2025 · question p. 9 · answer p. 25-26

185 RTP **May 2024** Q8(ii)

EXPLAIN the importance of trade credit and accruals as source of working capital. What is the cost of these sources?

ICAI'S ANSWER

Trade credit and accruals as source of working capital refers to credit facility given by suppliers of goods during the normal course of trade. It is a short-term source of finance. SSI firms in particular are heavily dependent on this source for financing their working capital needs. The major advantages of trade credit are – easy availability, flexibility and informality.

There can be an argument that trade credit is a cost-free source of finance. But it is not. It involves implicit cost. The supplier extending trade credit incurs cost in the form of opportunity cost of funds invested in trade receivables. Generally, the supplier passes on these costs to the buyer by increasing the price of the goods or alternatively by not extending cash discount facility.

ICAI · P6 · RTP May 2024 · question p. 8 · answer p. 29

186 MTP **Sep 2025 S-I** Q4(c) 2 marks

A firm is managing its finances and aims to ensure that it has access to funds for the long term. It evaluates various sources like equity, debt, retained earnings, and also credit offered by suppliers.

Which of the above mentioned sources should be classified as a short-term source of finance and why by mentioning its characteristics in brief?

ICAI'S ANSWER

Trade credit is a facility offered by suppliers that allows the buyer to purchase goods and services now and pay later, typically within 30 to 90 days. While it may appear essential for operations and cash flow, it is actually a form of short-term financing. It is:

(i) A part of working capital management.

(ii) Used to fund day-to-day operational needs.

(iii) Thus, not suitable or classified as a long-term source of finance.

ICAI · P6 · MTP Sep 2025 · Series I · question p. 9 · answer p. 17

187 MTP **May 2024 S-I** MCQ 4 1 mark MCQ

What is the maximum period for which company can accept Public Deposits?

A 1 year

B 6 months

C 3 years

D 5 years

ICAI'S ANSWER

(C) 3 years

These deposits may be accepted for a period of six months to three years.

ICAI · P6 · MTP May 2024 · Series I · question p. 3 · answer p. 2

188 MTP **Old P8 Nov 2023 S-I** Q6(b) 4 marks old scheme

Also asked in Old P8 MTP Nov 2022 S-II Q6(b) · Old P8 MTP Nov 2020 S-I Q6(b)

EXPLAIN in brief the features of Commercial Papers.

ICAI'S ANSWER

Commercial Paper: A Commercial Paper is an unsecured money market instrument issued in the form of a promissory note. The Reserve Bank of India introduced the commercial paper scheme in the year 1989 with a view to enabling highly rated corporate borrowers to diversify their sources of short- term borrowings and to provide an additional instrument to investors. Subsequently, in addition to the Corporate, Primary Dealers and All India Financial Institutions have also been allowed to issue Commercial Papers. Commercial papers are issued in denominations of ₹ 5 lakhs or multiples thereof and the interest rate is generally linked to the yield on the one-year government bond.

All eligible issuers are required to get the credit rating from Credit Rating Information Services of India Ltd, (CRISIL), or the Investment Information and Credit Rating Agency of India Ltd (ICRA) or the Credit Analysis and Research Ltd (CARE) or the FITCH Ratings India Pvt. Ltd or any such other credit rating agency as is specified by the Reserve Bank of India.

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 3 · answer p. 9-10

189 MTP **Old P8 Nov 2023 S-II** Q6(a) 4 marks old scheme

DESCRIBE the different types of packing credit.

ICAI'S ANSWER

Different types of packing credit are-

(i) Clean packing credit: This is an advance made available to an exporter only on production of a firm export order or a letter of credit without exercising any charge or control over raw material or finished goods. It is a clean type of export advance. Each proposal is weighed according to particular requirements of the trade and credit worthiness of the exporter. A suitable margin has to be maintained. Also, Export Credit Guarantee Corporation (ECGC) cover should be obtained by the bank.

(ii) Packing credit against hypothecation of goods: Export finance is made available on certain terms and conditions where the exporter has pledgeable interest and the goods are hypothecated to the bank as security with stipulated margin. At the time of utilising the advance, the exporter is required to submit, along with the firm export order or letter of credit relative stock statements and thereafter continue submitting them every fortnight and/or whenever there is any movement in stocks.

(iii) Packing credit against pledge of goods: Export finance is made available on certain terms and conditions where the exportable finished goods are pledged to the banks with approved clearing agents who will ship the same from time to time as required by the exporter. The possession of the goods so pledged lies with the bank and is kept under its lock and key.

(iv) E.C.G.C. guarantee: Any loan given to an exporter for the manufacture, processing, purchasing, or packing of goods meant for export against a firm order qualifies for the packing credit guarantee issued by Export Credit Guarantee Corporation.

(v) Forward exchange contract: Another requirement of packing credit facility is that if the export bill is to be drawn in a foreign currency, the exporter should enter into a forward exchange contact with the bank, thereby avoiding risk involved in a possible change in the rate of exchange.

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 4 · answer p. 8-9

190 MTP **Old P8 Nov 2022 S-I** Q6(c) 2 marks old scheme

BRIEF OUT certain sources of finance- Inter Corporate Deposits and Certificate of Deposit.

ICAI'S ANSWER

Inter Corporate Deposits: The companies can borrow funds for a short period, say 6 months, from other companies which have surplus liquidity. The rate of interest on inter corporate deposits varies depending upon the amount involved and the time period.

Certificate of Deposit (CD): The certificate of deposit is a document of title similar to a time deposit receipt issued by a bank except that there is no prescribed interest rate on such funds.

The main advantage of CD is that banker is not required to encash the deposit before maturity period and the investor is assured of liquidity because he can sell the CD in secondary market.

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 4 · answer p. 7

191 MTP **Old P8 May 2022 S-I** Q6(b) 4 marks old scheme

Same question, different ICAI answer: #192 Old P8 MTP Dec 2021 S-II

EXPLAIN any four types of Packing Credit.

ICAI'S ANSWER

Types of Packing Credit-

(i) **Clean packing credit:** This is an advance made available to an exporter only on production of a firm export order or a letter of credit without exercising any charge or control over raw material or finished goods. It is a clean type of export advance. Each proposal is weighed according to particular requirements of the trade and credit worthiness of the exporter. A suitable margin has to be maintained. Also, Export Credit Guarantee Corporation (ECGC) cover should be obtained by the bank.

(ii) **Packing credit against hypothecation of goods:** Export finance is made available on certain terms and conditions where the exporter has pledge able interest and the goods are hypothecated to the bank as security with stipulated margin. At the time of utilising the advance, the exporter is required to submit, along with the firm export order or letter of credit relative stock statements and thereafter continue submitting them every fortnight and/or whenever there is any movement in stocks.

(iii) **Packing credit against pledge of goods:** Export finance is made available on certain terms and conditions where the exportable finished goods are pledged to the banks with approved clearing agents who will ship the same from time to time as required by the exporter. The possession of the goods so pledged lies with the bank and is kept under its lock and key.

(iv) **E.C.G.C. guarantee:** Any loan given to an exporter for the manufacture, processing, purchasing, or packing of goods meant for export against a firm order qualifies for the packing credit guarantee issued by Export Credit Guarantee Corporation.

(v) **Forward exchange contract:** Another requirement of packing credit facility is that if the export bill is to be drawn in a foreign currency, the exporter should enter into a forward exchange contact with the bank, thereby avoiding risk involved in a possible change in the rate of exchange.

ICAI · Old P8 · MTP May 2022 · Series I · question p. 4 · answer p. 7-8

192 MTP **Old P8 Dec 2021 S-II** Q6(b) 4 marks old scheme

Same question, different ICAI answer: #191 Old P8 MTP May 2022 S-I

EXPLAIN any four types of Packing Credit.

ICAI'S ANSWER

(i) **Clean packing credit:** This is an advance made available to an exporter only on production of a firm export order or a letter of credit without exercising any charge or control over raw material or finished goods. It is a clean type of export advance. Each proposal is weighed according to particular requirements of the trade and credit worthiness of the exporter. A suitable margin has to be maintained. Also, Export Credit Guarantee Corporation (ECGC) cover should be obtained by the bank.

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thereafter continue submitting them every fortnight and/or whenever there is any movement in stocks.

(iii) **Packing credit against pledge of goods:** Export finance is made available on certain terms and conditions where the exportable finished goods are pledged to the banks with approved clearing agents who will ship the same from time to time as required by the exporter. The possession of the goods so pledged lies with the bank and is kept under its lock and key.

(iv) **E.C.G.C. guarantee:** Any loan given to an exporter for the manufacture, processing, purchasing, or packing of goods meant for export against a firm order qualifies for the packing credit guarantee issued by Export Credit Guarantee Corporation.

(v) **Forward exchange contract:** Another requirement of packing credit facility is that if the export bill is to be drawn in a foreign currency, the exporter should enter into a forward exchange contact with the bank, thereby avoiding risk involved in a possible change in the rate of exchange.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 5 · answer p. 9

193 MTP **Old P8 Jul 2021 S-II** Q6(a) 4 marks old scheme

EXPLAIN in brief the features of Commercial Paper.

ICAI'S ANSWER

A **Commercial Paper** is an unsecured money market instrument issued in the form of a promissory note. The Reserve Bank of India introduced the commercial paper scheme in the year 1989 with a view to enabling highly rated corporate borrowers to diversify their sources of short-term borrowings and to provide an additional instrument to investors. Subsequently, in addition to the Corporate, Primary Dealers and All India Financial Institutions have also been allowed to issue Commercial Papers. Commercial papers are issued in denominations of Rs. 5 lakhs or multiples thereof and the interest rate is generally linked to the yield on the one-year government bond.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 4 · answer p. 7

3.4 Contemporary and international sources of finance

20 questions · asked in 22 ICAI documents · 8 exam papers

194 EXAM **Sep 2026** Q4(c) 2 marks

Describe any two features of green bond.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 19

195 EXAM **Sep 2025** Q4(b) 4 marks

Explain in brief the following types of bonds:

(i) Callable Bonds

(ii) Puttable Bonds

(iii) Masala Bonds

(iv) Drop Lock Bonds

ICAI'S ANSWER

(i) **Callable bonds:** A callable bond has a call option which gives the issuer the right to redeem the bond before maturity at a predetermined price known as the call price (Generally at a premium).

(ii) **Puttable bonds:** Puttable bonds give the investor a put option (i.e. the right to sell the bond) back to the company before maturity.

(iii) **Masala Bonds:** Masala (means spice) bond is an Indian name used for Rupee denominated bond that Indian corporate borrowers can sell to investors in overseas markets.

(iv) **Drop Lock Bond**: It is a Floating Rate Note with a normal floating rate. The floating rate bond would be automatically converted into fixed rate bond if interest rate falls below a predetermined level. The new fixed rate stays till the drop lock bond reaches its maturity.

ICAI · P6 · Sep 2025 exam · question p. 18 · answer p. 18-19

196 EXAM **Jan 2025** Q4(a) 4 marks

Explain the Environmental, Social and Governance linked Bonds.

ICAI'S ANSWER

Environmental, Social and Governance-linked bonds (ESG): These bonds carry a responsibility of the issuer company to prioritize optimal environmental, social and governance (ESG) factors. Investing in ESG bonds is considered as **socially responsible investing**. ESG bonds can be project-based - green bonds and social bonds; and target-based - sustainability-linked bonds (SLBs).

- ***Green bonds:*** These are the most popular ESG bonds that are issued by a financial, non-financial or public institution, where the bond proceeds are used to finance "green projects". Green projects are aimed at positive environmental and/or climate impact including the cultivation of eco-friendly technology.
- ***Social bonds:*** These bonds finance the socially impactful projects. The projects here are related to the social concerns such as Human rights, Equality, animal welfare etc.
- ***Sustainability-linked bonds (SLBs):*** These bonds are combination of green bonds and social bonds. Proceeds of SLBs are not meant for a specific project but for general corporate purpose to achieve Key Performance Indicator (KPIs).

ICAI · P6 · Jan 2025 exam · question p. 21 · answer p. 21-22

197 EXAM **Sep 2024** Q2(b) 2 marks

Explain Angel Financing.

ICAI'S ANSWER

Angel Financing is a form of an equity-financing in which individual or a group of individuals provides capital to entrepreneurs and early-stage businesses, or start-ups, in exchange for an ownership/equity in the company.

They may provide a one-time investment or an ongoing capital injection via a series of investments, Angel investors are looking for a higher rate of return than what is given by traditional investment

ICAI · P6 · Sep 2024 exam · question p. 15 · answer p. 16-17

198 EXAM **May 2024** Q4(c) 2 marks

Explain the features of crowd funding.

ICAI'S ANSWER

Crowd funding: crowdfunding means raising money for an individual or organisation from a group of people to fund a project, typically via internet (social media and crowdfunding websites). It generally involves collecting funds from family, friends, strangers, corporates and many more in exchange of equity (known as Equity funding), loans (known as P2P lending) or nothing at all (i.e. donation). This source of funding also helps start-up to substantiate demand for their product before entering into production.

In the crowdfunding process, three parties are involved i.e. fund raiser, mediator and fund investor. The platforms (mediator) may also charge certain fees in the form of processing fee, transaction fee, etc. either as a fixed amount or a percentage or in combination of both.

ICAI · P6 · May 2024 exam · question p. 20 · answer p. 23

199 EXAM Old P8 Nov 2023 Q6(a) 4 marks old scheme

Also asked in RTP Jan 2026 Q11(a)

Write the main features of Bulldog Bond.

ICAI'S ANSWER

Features of Bulldog Bond

- It is denominated in Bulldog Pound Sterling/Great Britain Pound (GBP)
- Issued in London
- Issuer Non- UK Company
- Regulations: Great Britain
- Purpose: Access of capital available in UK market
- Issue proceeds can be used to fund UK operation
- Issue proceeds can be used to fund a company's local opportunities

ICAI · Old P8 · Nov 2023 exam · question p. 17 · answer p. 17

200 EXAM Old P8 Nov 2022 Q6(a) 4 marks old scheme

These bonds are issued by non-US Banks and non-US corporations in US. What this bond is called and what are the other features of this Bond?

ICAI'S ANSWER

The Bond is called as Yankee Bond.

Features of the bond:

- These bonds are denominated in Dollars
- Bonds are to be registered in SEC (Securities and Exchange Commission)
- Bonds are issued in tranches
- Time taken can be up to 14 weeks

ICAI · Old P8 · Nov 2022 exam · question p. 15 · answer p. 15

201 EXAM Old P8 May 2022 Q6(c) 2 marks old scheme

Distinguish between American Depository Receipts and Global Depository Receipts.

ICAI'S ANSWER

Distinguish Between American Depository Receipts and Global Depository Receipts:

	American Depository Receipts	Global Depository Receipts
Meaning	It is a negotiable instrument which is issued by US bank, which represent the nazon-US Company stock that is being traded in US stock Exchange	It is a negotiable instrument which is issued by the international depository bank that represent the foreign company's stock trading world-wide.
Issued where	In the US domestic capital market.	European capital market.
Listed in	In the American Stock Exchange	In the Non-US Stock Exchange
Relevance	Foreign companies are able to trade in the US Stock Market.	Foreign companies can trade in any country's stock market other than that of the US.

Alternatively:

American Depository Receipts (ADRs): These are securities offered by **non-US companies who want to list on any of the US exchange**. Each ADR represents a certain number of a company's regular shares. ADRs allow US investors to buy shares of these companies without the costs of investing directly in a foreign stock exchange.

Global Depository Receipts (GDRs): These are negotiable certificates held in the bank of one country representing a **specific number of shares of a stock traded on the exchange of another country.** These financial instruments are used by companies to raise capital in either dollars or Euros. These are mainly traded in European countries and particularly in London.

ICAI · Old P8 · May 2022 exam · question p. 16 · answer p. 17

202 RTP **Sep 2026** Q10(b)

Global Infra Ltd., an Indian infrastructure company, plans to raise funds from international markets for its expansion project. The Finance Director is evaluating various foreign bond options.

Explain various foreign bonds for Global Infra Ltd. and their key features.

ICAI'S ANSWER

Foreign Bonds

Sl. No.	Name of Bond	Salient Features
1.	Foreign Currency Convertible Bond (FCCB)	• This bond comes at a very low rate of interest. • The advantage to the issuer is that the issuer can get foreign currency at a very low cost. • The risk is that in case the bond has to be redeemed on the date of maturity, the issuer has to make the payment and at that time the issuer may not have the money.
2.	Plain Vanilla Bond	• The issuer would pay the principal amount along with the interest rate. • This type of bond would not have any options. • This bond can be issued in the form of discounted bond or can be issued in the form of coupon bearing bond.

Sl. No.	Name of Bond	Salient Features
3.	Convertible Floating Rate Notes (FRN)	• A convertible FRN is issued by giving its holder an option to convert it into a longer term debt security with a specified coupon • It protects an investor against falling interest rate • The long- term debt security can be sold in the market and the investor can earn profit • Capital gain is not applicable to FRN
4.	Drop Lock Bond	• It is a Floating Rate Note with a normal floating rate • The floating rate bond would be automatically converted into fixed rate bond if interest rate falls below a predetermined level • The new fixed rate stays till the drop lock bond reaches its maturity • The difference between the convertible floating rate note and drop lock bond is that the former is a long option structure and the later one is a short option structure
5.	Variable Rate Demand Obligations	• A normal floating rate note with a nominal maturity • The holder of the floating rate note can sell the obligation back to the trustee at par plus accrued interest • It gives the investor an option to exit, so it is more liquid than the normal FRN
6.	Yield Curve Note (YCN)	• It is a structured debt security • Yield increases when prevailing interest rate declines • Yield decreases when prevailing interest rate increases • This is used to hedge the interest rate • This works like inverse floater
7.	Euro Bond	• Euro bonds are issued or traded in a country using a currency other than the one in which the bond is denominated. This means that the bond uses a certain currency, but operates outside the jurisdiction of the Central Bank that issues that currency. • Eurobonds are issued by multinational corporations, for example, a British company may issue a Eurobond in Germany, denominating it in U.S. dollars. • It is important to note that the term has nothing to do with the euro, and the prefix "euro-" is used more generally to refer to deposit outside the jurisdiction of the domestic central bank

ICAI · P6 · RTP Sep 2026 · question p. 9 · answer p. 23-26

203 **RTP** **May 2026** Q10(a)

Also asked in MTP May 2026 S-II Q4(a)

DISCUSS the features of international funding and angel financing in the context of corporate financing.

ICAI'S ANSWER

International Funding: With liberalization and globalization a business enterprise has options to raise capital from International markets also. Foreign Direct Investment (FDI) and Foreign Institutional Investors (FII) are two major routes for raising funds from foreign sources besides ADR's (American depository receipts) and GDR's (Global depository receipts). Obviously, the mechanism of procurement of funds has to be modified in the light of the requirements of foreign investors.

Angel Financing: Angel Financing is a form of an equity-financing where an angel investor is a wealthy individual who provides capital for start-up or expansion, in exchange for an ownership/equity in the company. Typically, angels, as they are known as, will invest around 25 to 60 per cent to help a company get started. This source of finance sometimes is the last option for startups which doesn't qualify for bank funding and are too small for venture capital financing.

ICAI · P6 · RTP May 2026 · question p. 10 · answer p. 27-28

204 RTP **Old P8 May 2023** Q10(a) old scheme

HIGHLIGHT the similarities and differences between Samurai Bond and Bull Dog Bond.

ICAI'S ANSWER

Samurai Bond	• Samurai bonds are denominated in Japanese Yen JPY • Issued in Tokyo • Issuer Non- Japanese Company • Regulations: Japanese • Purpose: Access of capital available in Japanese market • Issue proceeds can be used to fund Japanese operation • Issue proceeds can be used to fund a company's local opportunities. • It can also be used to hedge foreign exchange risk
Bulldog Bond	• It is denominated in Bulldog Pound Sterling/Great Britain Pound (GBP) • Issued in London • Issuer Non- UK Company • Regulations: Great Britain • Purpose: Access of capital available in UK market • Issue proceeds can be used to fund UK operation • Issue proceeds can be used to fund a company's local opportunities

ICAI · Old P8 · RTP May 2023 · question p. 5 · answer p. 19-20

205 RTP **Old P8 Nov 2022** Q10(b) old scheme

BRIEF OUT salient features of Samurai Bond.

ICAI'S ANSWER

Salient features of Samurai Bonds

- Samurai bonds are denominated in Japanese Yen JPY
- Issued in Tokyo
- Issuer Non- Japanese Company
- Regulations: Japanese
- Purpose: Access of capital available in Japanese market
- Issue proceeds can be used to fund Japanese operation
- Issue proceeds can be used to fund a company's local opportunities.
- It can also be used to hedge foreign exchange risk.

PassBook's note: The January 2027 Study Material names Samurai Bonds (Tokyo) among foreign bonds but does not list these features.

ICAI · Old P8 · RTP Nov 2022 · question p. 7 · answer p. 23

206 MTP **Sep 2026 S-II** Q3(b) 2 marks

Also asked in MTP May 2024 S-II Q3(b)

Name the various financial instruments dealt with in the international market.

ICAI'S ANSWER

Financial Instruments in the International Market

Some of the various financial instruments dealt with in the international market are:

(a) Euro Bonds

(b) Foreign Bonds

(c) Fully Hedged Bonds

(d) Medium Term Notes

(e) Floating Rate Notes

(f) External Commercial Borrowings

(g) Foreign Currency Futures

(h) Foreign Currency Option

(i) Euro Commercial Papers.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 8 · answer p. 10

207 MTP **Sep 2025 S-II** Q4(b) 4 marks

Is there any difference between crowd funding and peer to peer lending? EXPLAIN briefly.

ICAI'S ANSWER

Difference between Crowd funding and Peer-to-Peer (P2P) lending

Crowd funding: In simple terms, crowdfunding means raising money for an individual or organisation from a group of people to fund a project, typically via internet (social media and crowdfunding websites). It generally involves collecting funds from family, friends, strangers, corporates and many more in exchange of equity (known as Equity funding), loans (known as P2P lending) or nothing at all (i.e. donation). This source of funding also helps start-up to substantiate demand for their product before entering into production.

Peer-to-Peer (P2P) lending: It is that category of crowdfunding where lenders match with the borrowers in order to provide unsecured loans through online platform. The fund raised are paid back by the borrowers with interest, though this kind of lending involves certain risk of defaults (just as the banks bear in the case of conventional method of lending). Anyone interested in investing money under P2P lending can visit the P2P lending platforms and choose amongst borrowers considering risk & returns. Some of the platforms offering P2P lending are i2iFunding, Lendbox, Faircent, Rupee Circle, etc.

Hence, we can say that the crowd funding is a broader concept and P2P lending is basically a part of crowd funding.

ICAI · P6 · MTP Sep 2025 · Series II · question p. 10 · answer p. 14

208 MTP **Jan 2025 S-II** Q4(c) 2 marks

EXPLAIN the concept of Drop – Lock Bond (DL Bonds)

ICAI'S ANSWER

A drop lock is an arrangement whereby the interest rate on a floating-rate note becomes fixed if it falls to a specified level. Above that level the rate floats based on a benchmark market rate, typically with a semi-annual reset. In other words, drop lock bonds marry the attributes of both floating-rate securities and fixed-rate securities. The drop lock effectively sets a floor on the rate and a guaranteed minimum return to

ICAI · P6 · MTP Jan 2025 · Series II · question p. 7 · answer p. 10

209 MTP **Sep 2024 S-I** Q4(b) 4 marks

Briefly EXPLAIN the following –

i. Fully Hedged Bonds

ii. Medium Term Notes

iii. Floating Rate Notes

iv. Euro Commercial Papers

ICAI'S ANSWER

(i) **Fully Hedged Bonds:** In foreign bonds, the risk of currency fluctuations exists. Fully hedged bonds eliminate the risk by selling in forward markets the entire stream of principal and interest payments.

(ii) **Medium Term Notes (MTN):** Certain issuers need frequent financing through the Bond route including that of the Euro bond. However, it may be costly and ineffective to go in for frequent issues. Instead, investors can follow the MTN programme. Under this programme, several lots of bonds can be issued, all having different features e.g. different coupon rates, different currencies etc. The timing of each lot can be decided keeping in mind the future market opportunities. The entire documentation and various regulatory approvals can be taken at one point of time.

(iii) **Floating Rate Notes (FRN):** These are issued up to seven years maturity. Interest rates are adjusted to reflect the prevailing exchange rates. They provide cheaper money than foreign loans.

(iv) **Euro Commercial Papers (ECP):** ECPs are short term money market instruments. They have maturity period of less than one year. They are usually designated in US Dollars.

ICAI · P6 · MTP Sep 2024 · Series I · question p. 6 · answer p. 10

210 MTP **May 2024 S-II** Q4(c) 2 marks

EXPLAIN the concept of Indian depository receipts.

ICAI'S ANSWER

Concept of Indian Depository Receipts: The concept of the depository receipt mechanism which is used to raise funds in foreign currency has been applied in the Indian capital market through the issue of Indian Depository Receipts (IDRs). Foreign companies can issue IDRs to raise funds from Indian market on the same lines as an Indian company uses ADRs/GDRs to raise foreign capital. The IDRs are listed and traded in India in the same way as other Indian securities are traded.

ICAI · P6 · MTP May 2024 · Series II · question p. 7 · answer p. 12

211 MTP **Old P8 Nov 2022 S-I** Q6(c) 2 marks old scheme

STATE in brief four features of Plain Vanilla Bond.

ICAI'S ANSWER

Features of Plain Vanilla Bond:

- The issuer would pay the principal amount along with the interest rate.
- This type of bond would not have any options.
- This bond can be issued in the form of discounted bond or can be issued in the form of coupon bearing bond.

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 4 · answer p. 7

212 MTP **Old P8 May 2022 S-I** Q6(c) 2 marks old scheme

STATE in brief four features of Samurai Bond.

ICAI'S ANSWER

Features of Samurai Bond:

- Samurai bonds are denominated in Japanese Yen JPY
- Issued in Tokyo
- Issuer Non- Japanese Company
- Regulations: Japanese
- Purpose: Access of capital available in Japanese market
- Issue proceeds can be used to fund Japanese operation
- Issue proceeds can be used to fund a company's local opportunities.
- It can also be used to hedge foreign exchange risk

PassBook's note: Kept as printed. Ours: the January 2027 Study Material names Samurai Bonds (Tokyo) among foreign bonds but does not list these features.

ICAI · Old P8 · MTP May 2022 · Series I · question p. 4 · answer p. 8

213 MTP **Old P8 Jul 2021 S-I** Q6(c) 2 marks old scheme

DEFINE Masala bond.

ICAI'S ANSWER

Masala bond: Masala (means spice) bond is an Indian name used for Rupee denominated bond that Indian corporate borrowers can sell to investors in overseas markets. These bonds are issued outside India but denominated in Indian Rupees. NTPC raised Rs. 2,000 crore via masala bonds for its capital expenditure in the year 2016.

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 5 · answer p. 9

CHAPTER 4

Cost of Capital

4.1 Cost of debt and of preference capital

33 questions · asked in 16 ICAI documents · 7 exam papers

214 EXAM **May 2026** Q1(b) 5 marks

Z Ltd. has issued 8% convertible debentures of ₹ 100 each with a remaining maturity period of five years. The current market price per debenture is ₹ 85. At the time of maturity, debenture holders have an option to either redeem it at a premium of 5% or to convert each debenture into 5 equity shares. Current market price per equity share is ₹16. The company has just paid dividend of ₹ 8.03 per equity share. Seven years ago, it paid dividend of ₹ 5.00 per equity share.

Applicable tax rate is 30%.

Compute the post-tax cost of debentures using present value method.

Year	1	2	3	4	5	6	7
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683	0.621	0.564	0.513
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636	0.567	0.507	0.452

Interest Rate	1	2	3	4	5	6	7	8	9
$FVIF_{i,5}$	1.051	1.104	1.159	1.217	1.276	1.338	1.403	1.469	1.539
$FVIF_{i6}$	1.062	1.126	1.194	1.265	1.340	1.419	1.501	1.587	1.677
$FVIF_{i7}$	1.072	1.149	1.230	1.316	1.407	1.504	1.606	1.714	1.828

ICAI'S ANSWER

Calculation of growth rate in dividend

$8.03 = 5\ (1+g)^7$

$1.606 = (1+g)^7$

g = 7%

Price of share after 5 years

P_5 = 16 x FVIF 7%,5

= 16 x 1.403 = 22.45

Conversion value of shares = 22.45 × 5 = ₹ 112.25 (RV)

Redemption Value of Debenture (RV) = 100+5% premium = ₹ 105

Since, conversion value is higher, debenture holders will prefer conversion.

After-tax Interest = 8(1−0.30) = ₹ 5.60

Year	Cash Inflow (₹)	PVIF @ 10%	PV (₹)	PVIF @12%	PV (₹)
0	85 (Current Market Price)	1	(85)	1	(85)
1-5	5.60 (Interest)	3.79	21.224	3.605	20.188
5	112.25 (Conversion Value)	0.621	69.707	0.567	63.646
Total PV			5.931		-1.166

$$\text{IRR} = L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$$

$$= 10\% + \frac{5.931}{5.931 - (-1.166)}(12\% - 10\%) = 11.67\%$$

ICAI · P6 · May 2026 exam · question p. 9-10 · answer p. 12

215 **EXAM** **Jan 2026** Case Scenario - I · MCQ 1 to MCQ 5

CASE SCENARIO - I

AB Infra Projects Limited operates in the construction and infrastructure sector, executing large-scale public and private projects. The company has shown consistent profitability and strong project execution capabilities, which is supported by a balanced mix of equity and long-term debt financing. With several new infrastructure projects under consideration, the management now wants to review the cost of capital to ensure that the company's financing decisions are efficient and aligned with the company's strategic growth objectives. The finance department is evaluating the cost of equity, cost of preference shares and cost of debentures to determine the company's overall Weighted Average Cost of Capital (WACC) and maintain financial stability. The capital structure of AB Infra Projects Limited is as follows:

Particulars	Amount (₹)
Equity Shares (face value 10 per share)	60,00,000
12% Preference Share Capital	20,00,000
10% Convertible Debentures	40,00,000
Total Capital of the Company	1,20,00,000

The following are some of the additional information provided by the company relating to the above-mentioned capital structure:

(i) The risk-free rate of return (R_f) is 6%, equity beta (β) is 1.25 and equity risk premium (Rp) is 8%.

(ii) The face value of 12% preference shares is ₹ 100 each. These shares are irredeemable, having a flotation cost of 4%. The current market price of preference share is ₹ 100 per share.

(iii) 10% convertible debentures of ₹ 100 each are issued with a maturity period of 5 Years. At maturity, the debenture holders will have an option to convert the debenture into equity shares of the company in the ratio of 1: 4 (4 shares for each debenture). The current market price of the equity share is ₹ 120 each and company expects share prices to grow at an average rate of 5% per annum in the future. The current market price per debenture is ₹ 150.

(iv) The rate of corporate income tax is 30%.

Based on the above information, you are required to answer the following questions 1 to 5:

MCQ 1

What is the cost of equity share capital using Capital Asset Pricing Model (CAPM) Approach and cost of irredeemable preference share capital?

- A 14.5% and 13%
- B 15% and 11.5%
- C 16% and 12.5%
- D 17.5% and 12%

MCQ 2

What is the Redemption Value (RV) of debenture?

- A ₹ 612.62
- B ₹ 650.60
- C ₹ 480.00
- D ₹ 515.50

MCQ 3

What is the cost of the convertible debentures using approximation method?

- A 29.33%
- B 30.74%
- C 32.50%
- D 28.25%

MCQ 4

What is the weighted average cost of capital using market value weights?

- A 16.05%
- B 18.55%
- C 19.20%
- D 17.02%

MCQ 5

If the capital structure remains unchanged, determine the market price per equity share (MPS) when the net profit available to equity shareholders (after tax) is ₹ 48,00,000 and the company's P/E ratio is 21.

- A ₹ 168
- B ₹ 175
- C ₹ 150
- D ₹ 180

ICAI'S ANSWERS

MCQ 1

Answer key: **(C)**

MCQ 2

Answer key: **(A)**

MCQ 3

Answer key: **(B)**

MCQ 4

Answer key: **(D)**

MCQ 5

Answer key: **(A)**

ICAI · P6 · Jan 2026 exam · question p. 2-3 · answer p. 9

216 **EXAM** **Sep 2025** Case scenario · Q3(b)(i) to Q3(b)(ii) 3 marks

CASE SCENARIO

A company issues 20,000, 18% Debentures of ₹100 each. The debentures are redeemable after a period of 8 years. The cost of debentures using approximation method is 14.58%. The corporate tax rate is 30%. You are required to calculate:

Q3(b)(i) 2 marks

Issue price of the Debentures, if redeemable at par.

Q3(b)(ii) 1 mark

Revised Cost of Debentures using approximation method, if redeemable at 10% Premium after 8 years.

ICAI'S ANSWERS

Q3(b)(i)

$$\text{Cost of Redeemable Debenture } (K_d) = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$0.1458 = \frac{18(1-0.3) + \frac{(100 - NP)}{8}}{\frac{(100 + NP)}{2}}$$

$$0.1458 = \frac{(100.8 + 100 - NP) \times 2}{800 + 8NP}$$

116.64 + 1.166NP = 401.6 – 2NP

3.166 NP = 284.96

NP = 90

Q3(b)(ii)

$$\text{Revised Cost of Redeemable Debenture } (K_d) = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$= \frac{18(1-0.3) + \frac{(110 - 90)}{8}}{\frac{(110 + 90)}{2}}$$

= 15.1%

ICAI · P6 · Sep 2025 exam · question p. 16 · answer p. 17

217 EXAM May 2025 Case scenario · Q2(a)(i) to Q2(a)(iv) 8 marks

CASE SCENARIO

Capital structure of B Ltd. for the year ended 31st March, 2025 are as follows:

Particulars	Amount (₹)
Equity share capital @₹10 each	14,00,000
10% Preference share capital @₹1,000 each	10,00,000
Debenture @ ₹100 each	9,60,000
Bank Loan	6,40,000

- Risk-free rate of return is 14%, Market rate of return is 19% and beta of company is 1.20.
- 10% Preference shares are redeemable at ₹ 1,065.40 after 3 years.
- Interest on bank loan is 1.30 times of interest on debentures.
- Debentures are redeemable at par after 5 years. Floatation cost is ₹ 4 per debenture.
- Tax rate is 30%.
- Cost of capital is 14%.

You are required to calculate the following:

PV factors @ 10% and 14%

Year	1	2	3	4
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683
$PVIF_{0.14,t}$	0.877	0.769	0.675	0.592

Q2(a)(i) 1 mark

Cost of Equity.

Q2(a)(ii) 2 marks

Cost of preference share using YTM method.

Q2(a)(iii) 3 marks

Post-tax cost of debenture using approximation method.

Q2(a)(iv) 2 marks

Interest rate of bank loan.

ICAI'S ANSWERS

Q2(a)(i)

Calculation of Cost of Equity

Ke = Rf + β (Rm – Rf) = 14% + 1.2 (19% - 14%) = 20%

Q2(a)(ii)

Calculation of Cost of Preference Shares:

Current Market Price P0

= Preference Dividend (PD) X PVAF (r,14) + Redeemable Value (RV) X PVIF (r,14)

Or 1000 = 100 x PVAF (r,14) + 1065.4 x (r,14)

Calculation of NPV at discount rate of 10% and 14%

Year	Cash Flows (₹)	PVF @ 10%	PV (₹)	PVF @ 14%	PV (₹)
0	(1000)	1	(1000)	1	(1000)
1-3	100	2.487	248.70	2.322	232.20
3	1065.4	0.751	800.12	0.675	719.14
			+48.82		**-48.66**

IRR = $10\% + \frac{48.82}{48.82 - (-48.66)}$ x (14% - 10%)

= $10\% + \frac{48.82}{97.48}$ x 4%

= 10% + 2% = 12%

Q2(a)(iii)

Calculation of Post-tax Cost of Convertible Debentures:

Working Note:

Calculation of Interest rate on Bank term Loan & Debentures

Let the rate of interest on debentures be 'X'

Therefore, Rate of Interest on bank loan = 1.3X x (0.7) = 0.91X

Kd (Debentures) = $\frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$

= $\frac{100X(1-0.3) + \frac{(100-96)}{5}}{\frac{(100+96)}{2}}$

= $\frac{70X + 0.8}{98}$

Sources of capital	Amount of capital	Weights (W)	Cost (K)	W x K
Equity share capital	14,00,000	0.35	0.20	0.07
Preference share capital	10,00,000	0.25	0.12	0.03
Debentures	9,60,000	0.24	$\frac{70X + 0.8}{98}$	$\frac{16.8X + 0.192}{98}$
Bank Loan	6,40,000	0.16	0.91X	0.16 × 0.91X
	40,00,000	1.00		0.14

0.14 = 0.07 + 0.03 + $\frac{16.8X + 0.192}{98}$ + 0.16x(0.91X)

0.04 x 98 = 16.8 X +0.192 + 14.2688 X

3.728 = 31.0688X

X = 0.1199 or 12 %

So, interest on debentures = 12 %

$$\text{Kd (Debentures)} = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$= \frac{12(1-0.3) + \frac{(100-96)}{5}}{\frac{(100+96)}{2}}$$

$$= \frac{8.4 + 0.8}{98} = 9.387\% \text{ or } 9.39\%$$

Q2(a)(iv)

Interest Rate of bank loan

= 12 x 1.3 = 15.60%

ICAI · P6 · May 2025 exam · question p. 14-15 · answer p. 15, 15-16, 16-17, 17

218 EXAM **May 2024** Q2(a) 6 marks

The capital structure of Shine Ltd. as on 31.03.2024 is as under:

Particulars	Amount (₹)
Equity share capital off 10 each	45,00,000
15% Preference share capital of f 100 each	36,00,000
Retained earnings	32,00,000
13% Convertible Debenture off 100 each	67,00,000
11 % Term Loan	20,00,000
Total	2,00,00,000

Additional information:

(A) Company issued 13% Convertible Debentures of ₹ 100 each on 01.04.2023 with a maturity period of 6 years. At maturity, the debenture holders will have an option to convert the debentures into equity shares of the company in the ratio of 1 : 4 (4 shares for each debenture). The market price of the equity share is ₹ 25 each as on 31.03.2024 and the growth rate of the share is 6% per annum.

(B) Preference stock, redeemable after eight years, is currently selling at ₹ 150 per share.

(C) The prevailing default-risk free interest rate on 10-year GOI treasury bonds is 6%. The average market risk premium is 8% and the Beta (β) of the company is 1.54.

Corporate tax rate is 25% and rate of personal income tax is 20%.

You are required to calculate the cost of:

(i) Equity Share Capital

(ii) Preference Share Capital

(iii) Convertible Debenture

(iv) Retained Earnings

(v) Term Loan

ICAI'S ANSWER

(i) Cost of Equity Share capital

As per CAPM Model $K_e = R_f + ß\ (R_m - R_f)$

Rf = 6%

B = 1.54

Rm-Rf = 8%

Ke = 6% + 1.54(8%)

Ke = **18.32%**

(ii) Cost of Preference Share capital

n	= 8
Net Proceeds (NP)	= 150
Redemption Value (RV)	= 100
Preference Dividend (PD)	= 15

$$K_p = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$K_p = \frac{15 + \left(\frac{100 - 150}{8}\right)}{\left(\frac{100 + 150}{2}\right)}$$

K_p = **7%**

Alternatively, if we take NP as 100 and RV as 100, then solution can be done in the following way:

Cost of Preference Share capital

n	= 8
Net Proceeds (NP)	= 100
Redemption Value (RV)	= 150
Preference Dividend (PD)	= 15

$$K_p = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$K_p = \frac{15 + \left(\frac{150 - 100}{8}\right)}{\left(\frac{150 + 100}{2}\right)}$$

K_p = **17%**

(iii) Cost of convertible debenture

Cash Redemption Value (RV)	= 100
Share Redemption Value (RV):	
Value of share after 5 years	= 25 x $(1.06)^5$ = 33.46
Share Redemption Value (RV)	= 33.46 X 4=133.82

Therefore, investor will choose share redemption.

Redemption Value (RV)	= 133.82
Net Proceeds (NP)	= 100
n	= 5
Interest (I)	= 13
Tax (t)	= 25%

$$K_d = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$= \frac{13(1-0.25) + \frac{(133.82 - 100)}{5}}{\frac{(133.82 + 100)}{2}}$$

K_d = **14.13%**

(iv) Cost of Retained Earnings

$K_r = K_e$ (1-t_p)= 18.32 % x (1-0.20) = **14.66%**

We can also take cost of equity as cost of retained earnings,

Accordingly, $K_r = K_e$ = **18.32%**

(v) Cost of Term Loan

= 11% x (1-0.25) = **8.25%**

ICAI · P6 · May 2024 exam · question p. 8-9 · answer p. 9-11

219 EXAM **Old P8 May 2022** Q5 10 marks old scheme

A company issues:

- 15% convertible debentures of ₹ 100 each at par with a maturity period of 6 years. On maturity, each debenture will be converted into 2 equity shares of the company. The risk-free rate of return is 10%, market risk premium is 18% and beta of the company is 1.25. The company has paid dividend of ₹ 12.76 per share. Five year ago, it paid dividend of ₹ 10 per share. Flotation cost is 5% of issue amount.
- 5% preference shares of ₹ 100 each at premium of 10%. These shares are redeemable after 10 years at par. Flotation cost is 6% of issue amount.

Assuming corporate tax rate is 40%.

(i) Calculate the cost of convertible debentures using the approximation method.

(ii) Use YTM method to calculate cost of preference shares.

Year	1	2	3	4	5	6	7	8	9	10
$PVIF_{0.03,t}$	0.971	0.943	0.915	0.888	0.863	0.837	0.813	0.789	0.766	0.744
$PVIF_{0.05,t}$	0.952	0.907	0.864	0.823	0.784	0.746	0.711	0.677	0.645	0.614
$PVIFA_{0.03,t}$	0.971	1.913	2.829	3.717	4.580	5.417	6.230	7.020	7.786	8.530
$PVIFA_{0.05,t}$	0.952	1.859	2.723	3.546	4.329	5.076	5.786	6.463	7.108	7.722

Interest rate	1%	2%	3%	4%	5%	6%	7%	8%	9%
$FVIF_{i,5}$	1.051	1.104	1.159	1.217	1.276	1.338	1.403	1.469	1.539
$FVIF_{i,6}$	1.062	1.126	1.194	1.265	1.340	1.419	1.501	1.587	1.677
$FVIF_{i,7}$	1.072	1.149	1.230	1.316	1.407	1.504	1.606	1.714	1.828

ICAI'S ANSWER

(i) Calculation of Cost of Convertible Debentures:

Given that,

R_F = 10%

$R_m - R_f$ = 18%

B = 1.25

D_0 = 12.76

D_{-5} = 10

Flotation Cost = 5%

Using CAPM,

$K_e = R_f + \beta\,(R_m - R_f)$

= 10%+1.25 (18%)

= **32.50%**

Calculation of growth rate in dividend

$12.76 = 10\,(1+g)^5$

$1.276 = (1+g)^5$

$(1+5\%)^5 = 1.276$ ..... from FV Table

g = **5%**

Price of share after 6 years = $\frac{D_7}{k_e - g} = \frac{12.76(1.05)^7}{0.325 - 0.05}$

$P_6 = \frac{12.76 \times 1.407}{0.275}$

P_6 = 65.28

Redemption Value of Debenture (RV) = 65.28 × 2 = **130.56** (RV)

NP = 95

n = 6

$$K_d = \frac{\text{INT (1-t)} + \frac{(RV - NP)}{n}}{\frac{(RV - NP)}{2}} \times 100$$

$$= \frac{15\,(1 - 0.4) + \frac{(130.56 - 95)}{6}}{\frac{(130.56 + 95)}{2}} \times 100$$

$$= \frac{9 + 5.93}{112.78} \times 100$$

K_d **= 13.24%**

(ii) Calculation of Cost of Preference Shares:

Net Proceeds = 100 (1.1) - 6% of 100 (1.1)

= 110 - 6.60

= **103.40**

Redemption Value = 100

Year	Cash Flows (₹)	PVF @ 3%	PV (₹)	PVF @ 5%	PV (₹)
0	103.40	1	103.40	1	103.40
1-10	-5	8.530	-42.65	7.722	-38.61
10	-100	0.744	-74.40	0.614	-61.40
			-13.65		**3.39**

$$K_p = 3\% + \frac{5\% - 3\%}{[3.39 - (-13.65)]} \times 13.65$$

$$= 3\% + \frac{2\%}{17.04} \times 13.65$$

K_p **= 4.6021%**

ICAI · Old P8 · May 2022 exam · question p. 13-14 · answer p. 14-16

220 EXAM **Old P8 Nov 2020** Q1(d) 5 marks old scheme

TT Ltd. issued 20,000, 10% convertible debenture of ₹ 100 each with a maturity period of 5 years. At maturity the debenture holders will have the option to convert debentures into equity shares of the company in ratio of 1:5 (5 shares for each debenture). The current market price of the equity share is ₹ 20 each and historically the growth rate of the share is 4% per annum. Assuming tax rate is 25%. Compute the cost of 10% convertible debenture using Approximation Method and Internal Rate of Return Method.

PV Factor are as under:

Year	1	2	3	4	5
PV Factor @ 10%	0.909	0.826	0.751	0.683	0.621
PV Factor @ 15%	0.870	0.756	0.658	0.572	0.497

ICAI'S ANSWER

Determination of Redemption value:

Higher of-

(i) The cash value of debentures = ₹100

(ii) Value of equity shares = 5 shares × ₹ 20 $(1 + 0.04)^5$

= 5 shares × ₹ 24.333

= **₹121.665 rounded to ₹121.67**

₹121.67 will be taken as redemption value as it is higher than the cash option and attractive to the investors.

Calculation of Cost of 10% Convertible debenture

(i) Using Approximation Method:

$$K_d = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}} = \frac{10(1-0.25) + \frac{(121.67 - 100)}{5}}{\frac{(121.67 + 100)}{2}} = \frac{7.5 + 4.334}{110.835}$$

= **10.676%**

(ii) Using Internal Rate of Return Method

Year	Cash flows (₹)	Discount factor @ 10%	Present Value	Discount factor @ 15%	Present Value (₹)
0	100	1.000	(100.00)	1.000	(100.00)
1 to 5	7.5	3.790	28.425	3.353	25.148
5	121.67	0.621	75.557	0.497	60.470
NPV			**+3.982**		**-14.382**

$$\text{IRR} = L + \frac{NPV_L}{NPV_L - NPV_H}(H - L) = 10\% + \frac{3.982}{3.982 - (-14.382)}(15\% - 10\%)$$

= **0.11084 or 11.084% (approx.)**

ICAI · Old P8 · Nov 2020 exam · question p. 2 · answer p. 5

221 RTP **Sep 2025** Q3 MCQ

Consider the below mentioned statements:

1. A debt-equity ratio of 2:1 indicates that for every 1 unit of equity, the company can raise 2 units of debt.
2. The cost of floating a debt is greater than the cost of floating an equity issue.

State True or False:

A 1-True, 2-True

B 1-False, 2-True

C 1-False, 2-False

D 1-True, 2-False

ICAI'S ANSWER

Answer key: **(D)**

(d) 1-True, 2-False

ICAI · P6 · RTP Sep 2025 · question p. 4 · answer p. 13

222 MTP **Sep 2026 S-I** MCQ 7 2 marks MCQ

Kiran Industries Ltd. has issued bonds with a face value of ₹ 1,00,000 at an annual coupon rate of 7%. The bonds are currently trading at 93% of their face value. What is the approximate cost of debt for the company before taxes?

A 7.00%

B 7.53%

C 8.00%

D 6.65%

ICAI'S ANSWER

(B) Cost of Debt = Interest Payment / Market Price of Bond = (7,000 / 93,000) × 100 = 7.53%

ICAI · P6 · MTP Sep 2026 · Series I · question p. 3 · answer p. 2

223 MTP **Sep 2025 S-I** Q3(a) 5 marks

A company issues:

- 15% convertible debentures of ₹ 100 each at par with a maturity period of 6 years. On maturity, each debenture will be converted into 2 equity shares of the company. The risk-free rate of return is 8%, market risk premium is 16% and beta of the company is 1.20. The company has paid dividend of ₹ 13.38 per share. Five years ago, it paid dividend of ₹ 10 per share. Flotation cost is 6% of issue amount.
- 5% preference shares of ₹ 100 each at premium of 10%. These shares are redeemable after 10 years at par. Flotation cost is 5% of issue amount.

Assuming corporate tax rate is 40%.

(i) CALCULATE the cost of convertible debentures using the approximation method.

(ii) Use YTM method to CALCULATE cost of preference shares.

ICAI'S ANSWER

(i) Calculation of Cost of Convertible Debentures:

Given that,

R_f = 8%

$R_m - R_f$ = 16%

β = 1.20

D_0 = 13.38

Dividend 5 years ago = ₹ 10

Flotation Cost = 6%

Using CAPM,

$K_e = R_f + \beta (R_m - R_f)$

= 8%+1.20 (16%)

= 27.20%

Calculation of growth rate in dividend

$13.38 = 10 (1+g)^5$

$1.338 = (1+g)^5$

g = 6%

Price of share after 6 years = $\frac{D_7}{k_e - g} = \frac{13.38\,(1.06)^7}{0.272 - 0.06}$

$P_6 = \frac{13.38 \text{ x } 1.504}{0.212}$ = 94.92

Redemption Value of Debenture (RV) = 94.92 × 2 = 189.84 (RV)

NP = 94

n = 6

$K_d = \frac{\text{Int}(1-t) + \left(\frac{RV - NP}{n}\right)}{\frac{(RV - NP)}{2}} \times 100$

$= \frac{15(1-0.4) + \left(\frac{189.84 - 94}{6}\right)}{\frac{(189.84 + 94)}{2}} \times 100$

$= \frac{9 + 15.97}{141.92}$ x100

K_d = 17.59%

(ii) Calculation of Cost of Preference Shares:

Net Proceeds = 100 (1.1) - 5% of 100 (1.1)

= 110 – 5.50

= 104.50

Redemption Value = 100

Year	Cash Flows (₹)	PVF @ 3%	PV (₹)	PVF @ 5%	PV (₹)
0	104.50	1	104.50	1	104.50
1-10	-5	8.530	-42.65	7.722	-38.61
10	-100	0.744	-74.40	0.614	-61.40
			-12.55		**4.49**

$$K_p = 3\% + \frac{5\% - 3\%}{[(-12.55) - 4.49]} \text{ x } - 12.55$$

$$= 3\% + \frac{2\%}{17.04} \text{ x} 12.55$$

K_p = 4.4730%

ICAI · P6 · MTP Sep 2025 · Series I · question p. 8-9 · answer p. 12-13

224 MTP **May 2025 S-I** MCQ 8 1 mark MCQ

A Company issues ₹ 10,00,000, 12% debentures of ₹ 100 each. The debentures are redeemable after the expiry of fixed period of 7 years. The Company is in 35% tax bracket. Calculate the cost of debt after tax, if debentures are issued at 10% Premium.

A 9.00%

B 7.80%

C 9.71%

D 6.07%

ICAI'S ANSWER

(D) 6.07%

Calculation of Cost of Debt after tax:

$$\text{Cost of Debt (Kd)} = \frac{I(1-t) + \frac{RV - NP}{n}}{\frac{RV + NP}{2}}$$

Cost of 12% Debentures, if issued at 10% Premium:

$$\text{Kd} = \frac{₹1{,}20{,}000(1 - 0.35) + \frac{₹10{,}00{,}000 - ₹11{,}00{,}000}{7 \text{ years}}}{\frac{₹10{,}00{,}000 + ₹11{,}00{,}000}{2}} = \frac{₹78{,}000 - ₹14{,}286}{₹10{,}50{,}000}$$

= 0.0607 or 6.07%

ICAI · P6 · MTP May 2025 · Series I · question p. 3 · answer p. 4

225 MTP **May 2025 S-II** Q1(b) 5 marks

Bhanu Limited has issued 50,000; 10% convertible bonds of ₹ 100 each with a maturity period of 5 years. At the maturity, the bondholders will have an option to convert the bonds into the equity shares in the ratio of 1:4 (i.e 4 equity shares for each debenture). The equity shares of Bhanu Limited are listed and are currently traded at ₹ 30 each and historically the growth of the company's share has been 5% per annum over the years.

Of the total number of convertible bonds issued, 75% opt in for converting their bonds to the common stock at an agreed price which shall be at 5% premium to the intrinsic price at that point in time. However, the investors need to hold the shares for at least 2 years from the date of conversion.

The balance bonds would be redeemed by the company at a premium of 25% with a redemption cost of 2% incurred additionally at the time of redemption. Tax rate applicable to the company is at 25%. COMPUTE the cost of the convertible bonds using Approximation Method as well as the YTM method.

ICAI'S ANSWER

(A) Approximation Method:

(i) Calculation of Cost of Bonds - 75% Opting for the conversion

Value of equity shares at the end of 5th year will be calculated using the concept of time value of money.

FV = PV $(1 + r)^n$

= 30 $(1 + 0.05)^5$

= 38.29

Consideration Value (Agreed Price) = 38.29 + 5% premium

= ₹ 40.20 for each equity share

Each bond holder will receive 4 equity shares for each bond converted.

∴ ₹ that each bond holder will receive on conversion would be,

₹ 40.20 x 4 = ₹ 160.80

So, RV = ₹ 160.80, NP = ₹ 100, Int = ₹ 10, n = 5 years

$$\text{Cost of Bonds (Kd)} = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$\therefore \text{Kd} = \frac{10(1 - 0.25) + \frac{(160.8 - 100)}{5}}{\frac{(160.8 + 100)}{2}}$$

∴ Kd = 15.08%

(ii) Calculation of Cost of Bonds - Not Opting for the conversion

RV = 100 + 25% (Premium) = 125

Add: Redemption cost = (125 x 2%) = 2.5

Gross RV = 127.50

$$\text{Cost of Bonds (Kd)} = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$\therefore \text{Kd} = \frac{10(1 - 0.25) + \frac{(127.5 - 100)}{5}}{\frac{(127.5 + 100)}{2}}$$

∴ Kd = 11.43%

(B) Yield To Maturity (YTM) Method:

(i) Calculation of Cost of Bonds - 75% Opting for the conversion

Year	Cash Flows (₹)	DF @ 10%	Present Value (₹)	DF @ 18%	Present Value (₹)
0	(100)	1.000	(100)	1.000	(100)
1	7.5	0.909	6.818	0.847	6.353
2	7.5	0.826	6.195	0.718	5.385
3	7.5	0.751	5.633	0.609	4.568
4	7.5	0.683	5.123	0.515	3.863
5	7.5 + 160.80	0.621	104.514	0.437	73.547
	NPV		28.283		- 6.284

YTM (IRR) = $L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$

H = Higher rate = 18%

L = Lower rate = 10%

$\therefore$ IRR = $10 + \frac{28.283}{28.283 - (-6.284)}(18 - 10)$

Kd = 16.54%

(ii) Calculation of Cost of Bonds - Not Opting for the conversion

Year	Cash Flows (₹)	DF @ 10%	Present Value (₹)	DF @ 15%	Present Value (₹)
0	(100)	1.000	(100)	1.000	(100)
1	7.5	0.909	6.818	0.87	6.525
2	7.5	0.826	6.195	0.756	5.67
3	7.5	0.751	5.633	0.658	4.935
4	7.5	0.683	5.123	0.572	4.29
5	7.5 + 127.50	0.621	83.835	0.497	67.095
	NPV		7.604		-11.485

YTM (IRR) = $L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$

H = Higher rate = 15%

L = Lower rate = 10%

$\therefore$ IRR = $10 + \frac{7.604}{7.604 - (-11.485)}(15 - 10)$

Kd = 12%

ICAI · P6 · MTP May 2025 · Series II · question p. 6-7 · answer p. 6-9

226 MTP Jan 2025 S-II Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

MNP Ltd. is a multinational company having its operations spread mostly in India and neighbouring countries of India. The promotors of the company believed that capital structure of a company must be kept flexible and balanced, where proper mix should always be maintained between debt and equity. Such mix of debt and equity should be reviewed from time to time keeping in mind the changing situation of India and the global scenario.

The capital structure of MNP Ltd. is as under:

9% Debentures	Rs. 2,75,000
11% Preference shares	Rs. 2,25,000
Equity shares (face value: Rs. 10 per share)	Rs. 5,00,000
Total capital of the company	Rs. 10,00,000

The following are some of the additional information provided by MNP Ltd. relating to the above mentioned capital structure.

(i) Rs. 100 per debenture redeemable at par has 2% floatation cost and 10 years of maturity. The market price per debenture is Rs. 105.

(ii) Rs. 100 per preference share redeemable at par has 3% floatation cost and 10 years of maturity. The market price per preference share is Rs. 106.

(iii) Equity share has Rs. 4 floatation cost and market price per share of Rs. 24. The next year expected dividend is Rs. 2 per share with an annual growth of 5%. The firm has a practice of paying all earnings in the form of dividends.

(iv) Corporate Income-tax rate is 35%.

Since the company is a multinational company market value weights are preferred over book value weights when calculating the Weighted Average Cost of Capital (WACC) for several reasons. The company believes that market values reflect the current market perception of a company's financial health and future prospects. This is more relevant for calculating the cost of capital today, as investors base their decisions on current market conditions. Book values, based on historical accounting principles, may not accurately represent the true economic value of the company's capital components. Market values capture the actual cost that a company would incur if it were to raise new capital in the current market. Book values might not reflect the true cost of debt due to factors like changes in interest rates or creditworthiness. Similarly, book value of equity might not reflect the current investor expectations for future dividends and growth. Market values are readily available through stock prices and market interest rates. Obtaining accurate book values, especially for intangible assets, can be a complex and time-consuming process.

On the basis of this information provided above you are required to answer the following MCQs (1 to 5):

MCQ 1 2 marks

Calculate the cost of equity and choose the correct answer from the following?

A 14%

B 15%

C 16%

D 17%

MCQ 2 2 marks

Calculate the cost of debt and choose the correct answer from the following?

A 6.11%

B 5.48%

C 9%

D 10.55%

MCQ 3 2 marks

Calculate the cost of preference shares and choose the correct answer from the following?

A 10.57 %

B 5.11%

C 9%

D 10%

MCQ 4 2 marks

Calculate the WACC using market value weights and choose the correct answer from the following?

A 12.80 %

B 5.11%

C 9%

D 10.55%

MCQ 5 2 marks

What will be the current market price of MNP Ltd.'s equity shares if Ke = 10%, expected dividend is Rs. 2 per share and annual growth rate is 5% from the following options:

A 40 per share

B 20 per share

C 30 per share

D 45 per share

ICAI'S ANSWERS

MCQ 1

(B) $\text{Ke} = \frac{D1}{P0} + \text{g}$

$= \frac{2}{20} + 0.05 = 15\%$

MCQ 2

(B) $\text{K}_d = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}} = \frac{9(1-0.35) + \frac{(100 - 102.90)}{10}}{\frac{(100 + 102.90)}{2}} = 5.48\%$

MCQ 3

(A) $K_p = \dfrac{PD + \dfrac{(RV - NP)}{n}}{\dfrac{(RV + NP)}{2}}$

$K_p = \dfrac{11 + \left(\dfrac{100 - 102.82}{10}\right)}{\left(\dfrac{100 + 102.82}{2}\right)} = 10.57\%$

MCQ 4

(A) Calculation of WACC using market value weights

Source of capital	Market Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	**(a)**	**(b)**	**(c) = (a)×(b)**
Debentures (₹ 105 per debenture)	2,88,750	0.1672	0.0548	0.0092
Preference shares (₹ 106 per preference share)	2,38,500	0.1381	0.1057	0.0146
Equity shares (₹ 24)	12,00,000	0.6947	0.1500	0.1042
	17,27,250	1.00		0.1280

WACC (K_o) = 12.80%

MCQ 5

(A) Current Market Price = $\dfrac{D_1}{Ke - g}$

$= \dfrac{2}{0.10 - 0.05}$ = ₹ 40 per share

ICAI · P6 · MTP Jan 2025 · Series II · question p. 1-2 · answer p. 1, 2

227 MTP **Sep 2024 S-II** MCQ 7 2 marks MCQ

A company has issued bonds with a face value of ₹ 100,000 at an annual coupon rate of 8%. The bonds are currently trading at 95% of their face value. What is the approximate cost of debt for the company before taxes.

A 9.00%

B 7.65%

C 8.00%

D 8.42%

ICAI'S ANSWER

Answer key: **(D)**

(d) Calculation: Cost of Debt = (Interest Payment/ Market Price of Bond)

= (8,000 / 95,000) = 8.42%

ICAI · P6 · MTP Sep 2024 · Series II · question p. 3 · answer p. 2

228 MTP May 2024 S-II Case scenario · MCQ 1(1) to MCQ 1(5) 10 marks

CASE SCENARIO

Tiago Ltd is an all-equity company engaged in manufacturing of batteries for electric vehicles. There has been a surge in demand for their products due to rising oil prices. The company was established 5 years ago with an initial capital of ₹ 10,00,000 and since then it has raised funds by IPO taking the total paid up capital to ₹ 1 crore comprising of fully paid-up equity shares of face value ₹ 10 each. The company currently has undistributed reserves of ₹ 60,00,000. The company has been following constant dividend payout policy of 40% of earnings. The retained earnings by company are going to provide a return on equity of 20%. The current EPS is estimated as Rs 20 and prevailing PE ratio on the share of company is 15x. The company wants to expand its capital base by raising additional funds by way of debt, preference and equity mix. The company requires an additional fund of ₹ 1,20,00,000. The target ratio of owned to borrowed funds is 4:1 post the fund-raising activity. Capital gearing is to be kept at 0.4x.

The existing debt markets are under pressure due to ongoing RBI action on NPAs of the commercial bank. Due to challenges in raising the debt funds, the company will have to offer ₹ 100 face value debentures at an attractive yield of 9.5% and a coupon rate of 8% to the investors. Issue expenses will amount to 4% of the proceeds.

The preference shares will have a face value of ₹ 1000 each offering a dividend rate of 10%. The preference shares will be issued at a premium of 5% and redeemed at a premium of 10% after 10 years at the same time at which debentures will be redeemed.

The CFO of the company is evaluating a new battery technology to invest the above raised money. The technology is expected to have a life of 7 years. It will generate a after tax marginal operating cash flow of ₹ 25,00,000 p.a. Assume marginal tax rate to be 27%.

MCQ 1(1) 2 marks

Which of the following is best estimate of cost of equity for Tiago Ltd?

A 12.99%

B 11.99%

C 13.99%

D 14.99%

MCQ 1(2) 2 marks

Which of the following is the most accurate measure of issue price of debentures?

A 100

B 96

C 90.58

D 95.88

MCQ 1(3) 2 marks

Which of the following is the best estimate of cost of debentures to be issued by the company? (Using approximation method)

A 7.64%

B 6.74%

C 4.64%

D 5.78%

MCQ 1(4) 2 marks

Calculate the cost of preference shares using approximation method

A 10.23%

B 9.77%

C 12.12%

D 12.22%

MCQ 1(5) 2 marks

Which of the following best represents the overall cost of marginal capital to be raised?

A 10.52%

B 17.16%

C 16.17%

D 16.71%

ICAI'S ANSWERS

MCQ 1(1)

Answer key: **(D)**

(d) 14.99%

B = retention ratio=0.6, r=return on equity=20%, DPS=D0=20 x 0.4= 8,

MPS = P0 = EPS x PE = 20 x 15=300

G = b.r =0.6 x 20% = 12%

D1 = D0(1+g) = 8 (1.12) = 8.96

Ke = D1/P0 + g = 8.96/300 + 0.12 = 14.99%

MCQ 1(2)

Answer key: **(C)**

(c) 90.58

Price of debentures= PV of future cash flows for investor discounted at their yield

= 8 x PVAF(9.5%,10 years)+ 100 x PVF(9.5%, 10 years)

= 8 x 6.2788 + 100 x 0.4035

=50.2304 + 40.35

=90.58

MCQ 1(3)

Answer key: **(A)**

(a) 7.64%

NP = 90.58 x 96% = 86.96, RV = 100, Interest = 8, t = 0.27, n = 10

$$\text{Kd} = \frac{Int(1-t)+(RV-NP)/n}{(RV+NP)/2}$$

$$= \frac{8(1-0.27)+(100-86.96)/10}{(100+86.96)/2}$$

= 7.64%

MCQ 1(4)

Answer key: **(B)**

(b) 9.77%

$$\text{Kp} = \frac{PD + (RV - NP)/n}{(RV + NP)/2}$$

$$= \frac{100 + (1100 - 1050)/10}{(1100 + 1050)/2}$$

= 9.77%

MCQ 1(5)

Answer key: **(A)**

(a) 10.52%

	Existing	Total	Additional	
Equity Funds	1,60,00,000	2,00,00,000	40,00,000	
Preference Shares		24,00,000	24,00,000	
Debt		56,00,000	56,00,000	
	1,60,00,000	2,80,00,000	1,20,00,000	
Capital gearing =	0.4			
(PSC + Debt)/Equity =	0.4			
(Total Funds -Equity)/ Equity = 0.4				
(2.8 crores-Equity)/ equity = 0.4				
Equity =	2 crores			
Weighted avg cost of marginal capital		**Weights**	**Cost**	**W.C**
Equity Funds	40,00,000	0.333333333	14.99%	5.00%
Preference Shares	24,00,000	0.2	9.77%	1.952%
Debt	56,00,000	0.466666667	7.64%	3.565%
Total	**1,20,00,000**			**10.52%**

ICAI · P6 · MTP May 2024 · Series II · question p. 1-2 · answer p. 1, 1-2, 2

229 MTP **May 2024 S-II** MCQ 2 2 marks MCQ

Ranu & Co. has issued 10% debenture of face value 100 for ₹ 10 lakh. The debenture is expected to be sold at 5% discount. It will also involve floatation costs of ₹ 10 per debenture. The debentures are redeemable at a premium of 10% after 10 years. Calculate the cost of debenture if the tax rate is 30%.

A 9.74%

B 9.56%

C 8.25%

D 10.12%

ICAI'S ANSWER

Answer key: **(A)**

(a) 9.74%

$$K_d = \frac{I(1-t) + \frac{RV - NP}{n}}{\frac{RV + NP}{2}}$$

$$K_d = \left[\frac{10(0.7) + \frac{110 - 85}{10}}{97.5}\right]$$

= 9.50/97.5 = 9.74%

ICAI · P6 · MTP May 2024 · Series II · question p. 3 · answer p. 2

230 MTP **Old P8 Jul 2021 S-II** Q2(b) 3 marks old scheme

Development Finance Corporation issued zero interest deep discount bonds of face value of Rs. 1,50,000 each issued at Rs. 3,750 & repayable after 25 years. COMPUTE the cost of debt if there is no corporate tax.

ICAI'S ANSWER

Here,

Redemption Value (RV)= Rs.1,50,000

Net Proceeds (NP) = Rs. 3,750

Interest = 0

Life of bond = 25 years

There is huge difference between RV and NP therefore in place of approximation method we should use trial & error method.

FV = PV x $(1+r)^n$

1,50,000 = 3,750 x $(1+r)^{25}$

40 = $(1+r)^{25}$

Trial 1: r = 15%, $(1.15)^{25}$ = 32.919

Trial 2: r = 16%, $(1.16)^{25}$ = 40.874

Here:

L = 15%; H = 16%

NPV_L = 32.919 - 40 = - 7.081

NPV_H = 40.874 - 40 = + 0.874

$$\text{IRR} = L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$$

$$= 15\% + \frac{-7.081}{-7.081 - (0.874)} \times (16\% - 15\%) = 15.89\%$$

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 2 · answer p. 4

4.2 Cost of equity and of retained earnings

33 questions · asked in 22 ICAI documents · 5 exam papers

231 EXAM **Sep 2026** MCQ 7 2 marks MCQ

Alpha Ltd. has equity shares of ₹ 10 each, whose current market price is ₹ 20 per share. Its paid up capital also consists of 300, 8% Preference shares of ₹ 100 each. Recent audited accounts show a profit (after 30% tax) of ₹ 2,94,000. Depreciation and amortization amounted to ₹ 51,000. Cost of equity share (K_e) ascertained to be 15%.

What would be the number of outstanding equity shares of the company ?

- A 88,000 shares
- B 89,000 shares
- C 90,000 shares
- D 98,000 shares

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P6 · Sep 2026 exam · question p. 6

232 EXAM **Sep 2026** Q2(b) 3 marks

The shares of PK Company Limited and BK Company Limited are traded in the market. Analysts believe that both securities are correctly priced according to the CAPM. The following information have been provided :

Securities	Beta (β)	CAPM expected return (R_e)
PK Company Limited	0.8	12%
BK Company Limited	1.4	15%

You are required to calculate :

(i) Risk Free Rate (R_f)

(ii) Market Risk Premium (MRP)

(iii) Expected Return on Market Portfolio (R_m)

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 17

233 EXAM **May 2026** MCQ 7 2 marks MCQ

FG Ltd. is appraising a one-year investment plan, which requires an initial outlay of ₹ 1,00,000. At the end of term, it will be matured at ₹ 1,25,000. The plan has a beta of 1.3, risk free rate of return 10% and market rate of return 18%. Calculate the required rate of return and decide whether the plan is worthwhile.

- A 25%, The plan is not worthwhile.
- B 20.4%, The plan is not worthwhile.

C 20.4%, The plan is worthwhile.

D 25%, The plan is worthwhile.

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P6 · May 2026 exam · question p. 3 · answer p. 8

234 EXAM **May 2025** MCQ 8 1 mark MCQ

Q Ltd. is planning to pay dividend of ₹ 2 per share in next year. Growth rate of company is 8% p.a. Current market price per share is ₹ 51. Flotation cost is ₹ 1 per share.

What will be cost of equity?

A 12.23%

B 11.92%

C 12.00%

D 12.32%

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P6 · May 2025 exam · question p. 3-4 · answer p. 7

235 EXAM **Sep 2024** MCQ 7 MCQ

The capital structure of KPS Limited includes 5,00,000 equity shares of ₹ 10 each. The market price of equity share (cum-dividend) is ₹ 75 per share. The company has declared to pay dividend on equity shares @ ₹ 6 per share which will be paid within next three days. The company has a history of consistent growth in its dividends. It has been predicted that in the next year KPS Limited will pay dividend on its equity shares@ ₹ 7.59 per share. The rate of dividend growth will be maintained in foreseeable future. The cost of equity is calculated as:

A 36.5%

B 34.5%

C 37.5%

D 38.5%

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P6 · Sep 2024 exam · question p. 3-4 · answer p. 4

236 EXAM **Old P8 Nov 2020** Q6(c) 2 marks old scheme

Distinguish between Unsystematic Risk & Systematic Risk.

ICAI'S ANSWER

(i) Unsystematic Risk: This is also called company specific risk as the risk is related with the company's performance. This type of risk can be reduced or eliminated by diversification of the securities portfolio. This is also known as diversifiable risk.

(ii) Systematic Risk: It is the macro-economic or market specific risk under which a company operates. This type of risk cannot be eliminated by the diversification hence, it is non-diversifiable. The examples are inflation, Government policy, interest rate etc.

ICAI · Old P8 · Nov 2020 exam · question p. 13 · answer p. 14

237 RTP **Sep 2026** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

AURO Engineering Pvt. Ltd. is engaged in the manufacturing of industrial machinery and heavy engineering equipment used in infrastructure, automobile, and renewable-energy projects. Over the last few years, the company has achieved stable growth in revenues and profitability due to increasing demand from domestic and export markets. The management is presently evaluating its long-term financing strategy in view of its proposed expansion into high-capacity precision engineering equipment.

The General Manager - Finance of the company CA Jenisha K informed the Board of Directors that the company has been maintaining a balanced capital structure consisting of equity shares, preference shares, and long-term debentures. However, with the proposed expansion plans, the company is considering additional debt financing to take advantage of financial leverage and tax benefits associated with debt capital.

As on 31st March, 2026, the company had the following capital structure at book values:

Particulars	Amount (₹)
Equity Share Capital (1,00,00,000 shares)	30,00,00,000
11.5% Preference Shares	6,00,00,000
10% Debentures	10,00,00,000
Total Capital Employed	46,00,00,000

The equity shares of the company are currently quoted at ₹ 300 per share in the stock market. The company expects to pay a dividend of ₹ 15 per equity share next year, and the dividend is expected to grow at a constant rate of 5% per annum indefinitely. The applicable corporate tax rate for the company is 35%.

The management is now considering raising additional debt of ₹ 5,00,00,000 by issuing 12% debentures. The Finance Director expects that the proposed increase in financial leverage may improve shareholders' returns in the short run. As a result of the revised financing plan, the expected dividend per share is estimated to increase to ₹ 20 per share, while the dividend growth rate is expected to remain unchanged at 5% per annum. However, due to increased financial risk arising from higher leverage, the market price of the company's equity shares is expected to decline to ₹ 250 per share.

The Board of Directors has requested the finance team to evaluate the impact of the proposed financing decision on the company's overall cost of capital and advise whether the revised capital structure would be financially beneficial for the company.

You are required to answer the following the questions:

Q1(i)

What will be the Cost of Equity (Ke) under the existing capital structure?

- A 9%
- B 10%
- C 10.25%
- D 11%

Q1(ii)

What will be the existing Weighted Average Cost of Capital (WACC) of the company?

- A 9.12%
- B 9.43%
- C 9.85%
- D 10.20%

Q1(iii)

What will be the revised Cost of Equity (Ke) after raising additional debt?

- A 11%
- B 12%
- C 13%
- D 14%

Q1(iv)

What will be the after-tax cost of the newly issued debentures?

- A 6.80%
- B 7.20%
- C 7.80%
- D 8.40%

Q1(v)

What will be the revised Weighted Average Cost of Capital (WACC) after the proposed debt financing?

- A 10.12%
- B 10.48%
- C 11.04%
- D 11.62%

ICAI'S ANSWERS

Q1(i)

(B) 10%

Using Dividend Growth Model:

$$\text{Ke} = \frac{D1}{P0} + \text{g}$$

Where:

- D_1 = ₹ 15

- P_0 = ₹ 300
- g = 5%

$$Ke = \frac{15}{300} + 0.05$$

= 0.05 + 0.05

= 10%

Q1(ii)

(B) 9.43%

Cost of Preference Shares

Kp = 11.5/100 = 11.5%

After-tax Cost of Debentures

Kd = 10% (1 - 0.35) = 6.5%

WACC Computation

Source	Amount (₹)	Weight (W)	Cost (C)	Weighted Cost (W x C)
Equity Shares	30,00,00,000	0.6522	10%	6.52%
Preference Shares	6,00,00,000	0.1304	11.5%	1.50%
Debentures	10,00,00,000	0.2174	6.5%	1.41%
Total	46,00,000	1.0000		9.43%

Q1(iii)

(C) 13% Solution:

Using Dividend Growth Model:

$$Ke = \frac{20}{250} + 0.05$$

= 0.08 + 0.05

= 13%

Q1(iv)

(C) 7.80%

Kd = 12% (1 − 0.35)

= 7.8%

Q1(v)

(C) 11.04%

Revised Capital Structure

Particulars	Amount (₹)
Equity Share Capital	30,00,00,000
Preference Shares	6,00,00,000
Existing Debentures	10,00,00,000
New Debentures	5,00,00,000
Total Capital	51,00,00,000

Revised WACC Computation

Source	Weight (W)	Cost (C)	Weighted Cost (W x C)
Equity Shares	0.5882	13%	7.65%
Preference Shares	0.1176	11.5%	1.35%
Existing Debentures	0.1961	6.5%	1.27%
New Debentures	0.0981	7.8%	0.77%
Total	1.0000		11.04%

ICAI · P6 · RTP Sep 2026 · question p. 1-2 · answer p. 10, 10-11, 11, 11-12

238 RTP **Jan 2026** Q6

The capital structure of Shine Ltd. as on 31.03.2024 is as under:

Particulars	Amount (₹)
Equity share capital of ₹ 10 each	45,00,000
15% Preference share capital of ₹ 100 each	36,00,000
Retained earnings	32,00,000
13% Convertible Debenture of ₹ 100 each	67,00,000
11 % Term Loan	20,00,000
Total	2,00,00,000

Additional information:

(A) Company issued 13% Convertible Debentures of ₹ 100 each on 01.04.2023 with a maturity period of 6 years. At maturity, the debenture holders will have an option to convert the debentures into equity shares of the company in the ratio of 1 : 4 (4 shares for each debenture). The market price of the equity share is ₹ 25 each as on 31.03.2024 and the growth rate of the share is 6% per annum.

(B) Preference stock, redeemable after eight years at par, is currently selling at ₹ 150 per share.

(C) The prevailing default-risk free interest rate on 10-year GOI treasury bonds is 6%. The average market risk premium is 8% and the Beta (β) of the company is 1.54.

Corporate tax rate is 25%.

You are required to calculate the cost of:

(i) Equity Share Capital

(ii) Preference Share Capital

(iii) Convertible Debenture

(iv) Retained Earnings

(v) Term Loan

ICAI'S ANSWER

(i) **Cost of Equity Share capital**

As per CAPM Model $K_e = R_f + ß\ (R_m - R_f)$

Rf = 6%

B = 1.54

Rm-Rf = 8%

Ke = 6% + 1.54(8%)

Ke = **18.32%**

(ii) **Cost of Preference Share capital**

n = 8

Net Proceeds (NP) = 150

Redemption Value (RV) = 100

Preference Dividend (PD) = 15

$$K_p = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$K_p = \frac{15 + \left(\frac{100 - 150}{8}\right)}{\left(\frac{100 + 150}{2}\right)}$$

K_p = **7%**

(iii) **Cost of convertible debenture**

Cash Redemption Value (RV) = 100

Share Redemption Value (RV):

Value of share after 5 years = 25 x $(1.06)^5$ = 33.455

Share Redemption Value (RV) = 33.455 x 4 = 133.82

Therefore, investor will choose share redemption.

Redemption Value (RV) = 133.82

Net Proceeds (NP) = 100

n = 5

Interest (I) = 13

Tax (t) = 25%

$$K_d = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$= \frac{13(1-0.25) + \frac{(133.82 - 100)}{5}}{\frac{(133.82 + 100)}{2}}$$

K_d = **14.13%**

(iv) **Cost of Retained Earnings**

$K_r = K_e$ = **18.32%**

(v) **Cost of Term Loan**

= 11% x (1-0.25) = **8.25%**

ICAI · P6 · RTP Jan 2026 · question p. 6-7 · answer p. 17-19

239 RTP **May 2025** Q2 MCQ

The shares of ACB Ltd. are presently traded at ₹ 51 and the company is expected to pay dividends of ₹ 5 per share with a growth rate expected at 10% per annum. It plans to raise fresh equity share capital. The merchant banker has suggested that an underpricing of ₹ 2 is necessary in pricing the new issue besides involving a cost of ₹ 50 paise per share on miscellaneous expenses. The cost of new equity shares (assuming no change in dividend rate and growth rate) will be:

A 18%

B 17.5%

C 18.25%

D 20.31%

ICAI'S ANSWER

(D) 20.31%

$$\text{Ke} = \frac{D1}{NP} + \text{g}$$

Net proceeds per share = P0 - underpricing – Floatation cost

= 51−2−0.50=48.50

$$\text{Ke} = \frac{5}{48.50} + 0.10$$

= 20.31%

ICAI · P6 · RTP May 2025 · question p. 3 · answer p. 12

240 RTP **Jan 2025** Q11(b)

EXPLAIN any four Methods for Computation of Cost of Equity Capital.

ICAI'S ANSWER

Cost of equity capital is the rate of return which equates the present value of expected dividends with the market share price.

Methods for Computation of Cost of Equity Capital

- **Dividend Price Approach (:** Here, cost of equity capital is computed by dividing the expected dividend by market price per share.
  $K_e = \frac{D_1}{P_o}$
- **Earning/ Price Approach:** The advocates of this approach co-relate the earnings of the company with the market price of its share.
  $K_e = \frac{E}{P}$
- **Realized Yield Approach:** According to this approach, the average rate of return realized in the past few years is historically regarded as 'expected return' in the future. The yield of equity for the year is:
  $Y_t = \frac{D_t + P_t}{P_{t-1}}$
- **Capital Asset Pricing Model Approach (CAPM):** CAPM model describes the risk-return trade-off for securities. It describes the linear relationship between risk and return for securities.
  $K_e = R_f + ß\ (R_m - R_f)$

ICAI · P6 · RTP Jan 2025 · question p. 10 · answer p. 29

241 MTP **Sep 2026 S-II** MCQ 8 1 mark MCQ

The cost of equity capital is all of the following except:

A the minimum rate that a firm should earn on the equity-financed part of an investment.

B a return on the equity-financed portion of an investment that, at worst, leaves the market price of the stock unchanged.

C by far the most difficult component cost to estimate.

D generally lower than the before-tax cost of debt.

ICAI'S ANSWER

(D) Generally lower than the before-tax cost of debt.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 4 · answer p. 2

242 MTP **Sep 2026 S-II** Q2(a) 4 marks

Vertex Industries Ltd. is evaluating the cost of equity for its upcoming expansion project. The company's finance manager has collected the following market data:

- Risk-free rate of return (Rf) = 5%
- Expected return on the market portfolio (Rm) = 10%
- Beta (β) of Vertex Industries' equity = 2.0

Required

(a) Calculate the cost of equity (Ke) for Vertex Industries Ltd. using CAPM.

(b) Calculate and compare the cost of equity for two competitor firms: Competitor A (β = 1.2) and Competitor B (β = 0.8).

ICAI'S ANSWER

Cost of equity for Vertex Industries Ltd.

Given: Rf = 5%, Rm = 10%, β = 2.0

Ke = Rf + β (Rm − Rf)

Ke = 5% + 2.0 × (10% − 5%)

Ke = 5% + 2.0 × 5%

Ke = 5% + 10% = 15%

(b) Cost of equity for Competitor A and Competitor B

Competitor A (β = 1.2): Ke = 5% + 1.2 × (10% − 5%) = 5% + 6%

Ke = 11%

Competitor B (β = 0.8): Ke = 5% + 0.8 × (10% − 5%) = 5% + 4%

Ke = 9%

Company	Beta (β)	Cost of Equity (Ke)
Vertex Industries Ltd.	2.0	15%
Competitor A	1.2	11%
Competitor B	0.8	9%

Vertex Industries (β = 2.0) has the highest cost of equity at 15%, indicating the highest systematic (market-related) risk among the three firms.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 6 · answer p. 6-7

243 MTP **May 2026 S-I** Q1(b) 5 marks

Mr. Raman, a portfolio manager, is analyzing a set of securities using the **Capital Asset Pricing Model (CAPM)**. He has gathered the following information:

Security	Beta	CAPM Return
A Ltd	1.2	15.20%
B Ltd	0.8	12.80%
C Ltd	1.1	?
Cash in hand	?	Nil
Treasury Bill	?	?
Stock Market i.e. Nifty	?	?

(i) Complete the above table with all missing values.

(ii) CAPM explains the relationship between systematic risk and expected return on a security. DISCUSS the nature of the relationship between Beta and expected return under CAPM.

ICAI'S ANSWER

(i) According to CAPM

Rs = Rf + Beta x RP

15.20 = Rf + 1.2Rp .....(1)

12.80 = Rf + 0.8Rp .....(2)

By solving, 0.4Rp = 2.4.

So Risk Premium = 6%

Substituting in Eqn. 1

15.20 = Rf + 1.2*6%

Rf = 15.20 – 7.20 = 8%

Hence, Risk Free Return = 8%

Rp = Rm – Rf

6 = Rm – 8%

Rm = 14%

Rs (C Ltd.) = 8% + 1.1 x 6% = 14.6%

Security	Beta	CAPM Return
A Ltd	1.2	15.20%
B Ltd	0.8	12.80%
C Ltd	1.1	14.60%
Cash in hand	0	Nil
Treasury Bill	0	8.00%
Stock Market (Nifty)	1	14.00%

(ii) Risk-averse investors may prefer securities with low Beta (<1) i.e. B Ltd. Aggressive investors may choose high Beta (>1) securities for higher potential returns i.e. A or C Ltd.

ICAI · P6 · MTP May 2026 · Series I · question p. 5-6 · answer p. 6

244 MTP **May 2026 S-II** MCQ 6 2 marks MCQ

The shares of Ritik Ltd. are presently traded at ₹ 64 and the company is expected to pay dividends of ₹ 6 per share with a growth rate expected at 12% per annum. It plans to raise fresh equity share capital. The merchant banker has suggested that an underpricing of ₹ 3 is necessary in pricing the new issue besides involving a cost of ₹ 1 per share on miscellaneous expenses. The cost of new equity shares (assuming no change in dividend rate and growth rate) will be:

A 18%

B 17.5%

C 20.25%

D 22%

ICAI'S ANSWER

(D) 22%

$$\text{Ke} = \frac{D1}{NP} + g$$

Net proceeds per share = P0 - underpricing – Floatation cost

= 64 - 3 – 1 = 60

$$\text{Ke} = \frac{6}{60} + 0.12$$

= 22%

ICAI · P6 · MTP May 2026 · Series II · question p. 3 · answer p. 3

245 MTP **May 2026 S-II** Q4(c) 2 marks

DISCUSS the dividend-price approach, and earnings price approach to estimate cost of equity capital.

ICAI'S ANSWER

In dividend price approach, cost of equity capital is computed by dividing the current dividend by average market price per share. This ratio expresses the cost of equity capital in relation to what yield the company should pay to attract investors. It is computed as:

$$K_e = \frac{D_1}{P_0}$$

Where,

D_1 = Dividend per share in period 1

P_0 = Market price per share today

Whereas, on the other hand, the advocates of earnings price approach co-relate the earnings of the company with the market price of its share. Accordingly, the cost of ordinary share capital would be based upon the expected rate of earnings of a company. This approach is similar to dividend price approach, only it seeks to nullify the effect of changes in dividend policy.

ICAI · P6 · MTP May 2026 · Series II · question p. 9 · answer p. 11

246 MTP **Jan 2026 S-II** Q3(b) 3 marks

In October, 2025 AB Co.'s share was sold for ₹ 146 per share. A long term earnings growth rate of 7.5% is anticipated. AB Co. is expected to pay dividend of ₹ 3.36 per share.

(i) DETERMINE rate of return an investor can expect to earn assuming that dividends are expected to grow along with earnings at 7.5% per year in perpetuity?

(ii) It is expected that AB Co. will earn about 10% on book Equity and shall retain 60% of earnings. In this case, whether, there would be any change in growth rate and cost of Equity? ANALYSE.

ICAI'S ANSWER

(i) According to Dividend Discount Model approach the firm's expected or required return on equity is computed as follows:

$$= \frac{D_1}{P_0} + g$$

Where,

K_e = Cost of equity share capital

D_1 = Expected dividend at the end of year 1

P_0 = Current market price of the share.

g = Expected growth rate of dividend.

Therefore, $K_e = \frac{3.36}{146} + 7.5\%$

= 0.0230 + 0.075 = 0.098

Or, K_e = 9.80%

(ii) With rate of return on retained earnings (r) 10% and retention ratio (b) 60%, new growth rate will be as follows:

g = br

= 0.10 X 0.60 = 0.06

Accordingly, dividend will also get changed and to calculate this, first we shall calculate previous retention ratio (b_1) and then EPS assuming that rate of return on retained earnings (r) is same.

With previous Growth Rate of 7.5% and r =10% the retention ratio comes out to be:

0.075 = b_1 x 0.10

b_1 = 0.75 and payout ratio = 0.25

With 0.25 payout ratio the EPS will be as follows:

$\frac{3.36}{0.25}$ = 13.44

With new 0.40 (1 – 0.60) payout ratio the new dividend will be

D_1 = 13.44 X 0.40 = 5.376

Accordingly new K_e will be

$K_e = \frac{5.376}{146} + 6.0\%$

K_e = 9.68%

ICAI · P6 · MTP Jan 2026 · Series II · question p. 8 · answer p. 11-12

247 MTP **Sep 2025 S-I** MCQ (viii) 1 mark MCQ

A company's current market price per share is ₹ 200. It just paid a dividend of ₹ 10, and dividends are expected to grow at a constant rate of 5% per annum. What is the cost of equity as per the Dividend Discount Model (DDM)?

A 10.5%

B 5.0%

C 11.0%

D 10.25%

ICAI'S ANSWER

Answer key: **(D)**

(d) 10.25%

As per the Dividend Discount Model (DDM):

$$Ke = \frac{D_1}{P_0} + g$$

Where:

D0 = ₹ 10 (just paid dividend)

g = 5%

So, D1 = D0 × (1+g) = 10 × 1.05 = ₹ 10.50

P0 = ₹ 200

Now,

$$Ke = \frac{10.50}{200} + g$$

= 0.0525 + 0.05 = 0.1025 or 10.25%

ICAI · P6 · MTP Sep 2025 · Series I · question p. 4 · answer p. 3-4

248 MTP **May 2025 S-II** MCQ 8 1 mark MCQ

Out of the following information, calculate Risk free rate:

Rm = 10%

Ke = 15%

β=2

A 5%

B 7.5%

C 10%

D 15%

ICAI'S ANSWER

(A) 5%

Ke = Rf + β (Rm – Rf)

15 = Rf + 2(10 – Rf)

15 = Rf + 20 - 2Rf

Rf = 5

ICAI · P6 · MTP May 2025 · Series II · question p. 5 · answer p. 5

249 MTP **Sep 2024 S-I** Case scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Kaivalyabodhi Limited (KbL) has completed 35 years of operations in India. It has many subsidiary & associate companies in more than 100 countries. KbL's businesses include home and personal care, foods and beverages, and industrial, agricultural and other products. It is one of the largest producers of soaps and detergents in India. The company has grown organically as well as through acquisitions.

Over the years, the company has built a diverse portfolio of powerful brands, some being household names.

It is planning to acquire one of its competitors named Prestige Limited, which would enhance the growth of 'KbL'. The consideration amount will be 1.5X of its average Market Capitalization. Prestige limited has 1,30,000 outstanding equity shares and its shares were traded at an average market price of ₹ 45 as on the valuation date. The consideration amount will be paid equally in 5 years where the first installment is to be paid immediately. Prestige Limited has Ko of 15%

KbL will raise the funds required through debt and equity in the ratio of 30:70. The company requires the cost of capital estimates for evaluating its acquisitions, investment decisions and the performance of its businesses.

KbL's share price has grown from ₹ 150 to ₹ 301 in the last 5 years and it will continue to grow at the same rate. KbL pays dividends regularly. The company has recently paid a dividend of ₹ 8. For the calculation of equity, an average of 52 weeks high market price in the last 5 years is to be considered, which is as follows:

Yr 1	Yr 2	Yr 3	Yr 4	Yr 5
MPS 195	MPS 210	MPS 252	MPS 325	MPS 280

Ke calculated as per growth model holds a weight of 0.6.

The company also wishes to calculate the equity's expectation using CAPM which holds a weight of 0.4. The risk-free rate is assumed as the yield on long-term government bonds that the company regards as about 8%. KbL regards the market-risk premium to be equal to 11 per cent. Its estimation on the Beta is 0.78.

KbL will issue debentures with FV of ₹ 10,500 which is to be amortised equally over the life of 7 years. The company considers the effective rate of interest applicable to an 'AAA' rated company with a markup of 200 basis points as its coupon rate. It thinks that considering the trends over the years, 'AAA' rate is 7.5%.

Ignore taxation. Based on the above details, answer the question 1 to 5:

MCQ 1 2 marks

Calculate the cost of equity under both the methods

A 11%, 16%

B 18.65%, 10.34%

C 18.65%, 16.58%

D 16.5%, 9%

MCQ 2 2 marks

Calculate the overall cost of equity

A 17.82%

B 17.63%

C 15.37%

D 35.25%

MCQ 3 2 marks

Calculate the cost of debt, if the intrinsic value of debenture today is close to ₹ 9,740

A 15%

B 12%

C 9.5%

D 7.5%

MCQ 4 2 marks

Calculate the WACC & the amount of purchase consideration

A 18%, ₹ 90,00,000

B 15.21%, ₹ 87,75,000

C 16.07%, ₹ 87,75,000

D 15.94%, ₹ 58,50,000

MCQ 5 2 marks

Present Value of Purchase consideration is close to ₹

A 58,83,032

B 67,65,487

C 57,35,680

D 66,58,997

ICAI'S ANSWERS

MCQ 1

Answer key: **(C)**

(c) 18.65%, 16.58%

Ke under two approaches

Calculation of Ke (Using Gordon's Model)

Ke = $\frac{D1}{Po} + g$

Share Price has grown from 150 to 301 in 5 years,

$150\ (1+g)^5 = 301.$

$(1+g)^5 = 2.01$

Therefore, g = 15%, (From Annuity table – Re 1 after 5 years becomes ₹ 2.01 at rate of 15%)

D1 = 8 + 15% of 8 = 9.2

Po = Average of 52 weeks High price in last 5 years

Po = (195 + 210 +252 +325 +280) / 5

= 252.40

Ke = 9.2 / 252.40 + 0.15

= 18.65%

Calculation of Ke (Using CAPM)

Ke = Rf + (Rm – Rf) X Beta

= 8 + (11 x 0.78)

= 16.58%

MCQ 2

Answer key: **(A)**

(a) 17.82%

Overall Ke for the company

Approach	Cost of Equity (k)	Weight (w)	K x w
Gordon's	18.65%	0.6	11.19%
CAPM	16.58%	0.4	6.63%
			Total Ke = 17.82%

MCQ 3

Answer key: **(B)**

(b) 12%

Intrinsic Value of Debentures today is ₹ 9,740

WN 1 – Calculation of the Pattern of Future Cash flows

YR	PRINCIPAL (I)	INTEREST (II) = Coupon Rate = 9.5% (7.5% + 2%)	PV OF (I + II) @ 10%	PV OF (I + II) @ 15%
1	1,500	997.50	2270.45	2171.74
2	1,500	855	1946.28	1780.72
3	1,500	712.5	1662.28	1454.75
4	1,500	570	1413.84	1183.53
5	1,500	427.50	1196.83	958.31
6	1,500	285	1007.59	771.70
7	1,500	142.50	842.86	617.48
			10340.13	8938.23

$= 10\% + \frac{(10{,}340.13 - 9{,}740)}{(10{,}340.13 - 8{,}938.23)}$ x5% = 12.14% = 12% (approx.)

MCQ 4

Answer key: **(C)**

(c) 16.07%, ₹ 87,75,000

Ko = Wd x Kd + We x Ke

= 0.3 X 12 + 0.7 X 17.82

= 16.07%

Purchase Consideration using M-Cap method

= 1,30,000 eq shares x 45 MPS x 1.5X

= ₹ 87,75,000

MCQ 5

Answer key: **(D)**

(d) ₹ 66,58,997

It is to be paid equally over 5 years and first instalment is to be paid immediately at Yr 0

Discount rate will be the Ko calculated as above of the company and not 15% which is Ko of Prestige Limited

Year	Amount each year	PV @ 16.07%	PV (₹)
0	17,55,000	1.0000	17,55,000
1	17,55,000	0.8615	15,11,933
2	17,55,000	0.7423	13,02,737
3	17,55,000	0.6395	11,22,323
4	17,55,000	0.5510	9,67,005
	TOTAL PV		**66,58,997**

ICAI · P6 · MTP Sep 2024 · Series I · question p. 1-2 · answer p. 1, 2

250 MTP **May 2024 S-I** Q3(b) 2 marks

Same question, different ICAI answer: #252 Old P8 MTP Nov 2023 S-II

DISCUSS the dividend-price approach to estimate cost of equity capital.

ICAI'S ANSWER

In dividend price approach, cost of equity capital is computed by dividing the expected dividend by market price per share. This ratio expresses the cost of equity capital in relation to what yield the company should pay to attract investors. It is computed as:

$$K_e = \frac{D_1}{P_0}$$

Where,

Ke= Cost of equity

D = Expected dividend (also written as D1)

P0 = Market price of equity (ex- dividend)

ICAI · P6 · MTP May 2024 · Series I · question p. 5 · answer p. 6

251 MTP **Old P8 Nov 2023 S-I** Q1(d) 5 marks old scheme

ABC Company's equity share is quoted in the market at ₹ 30 per share currently. The company pays a dividend of ₹ 3 per share and the investor's market expects a growth rate of 7% per year.

You are required to:

(i) CALCULATE the company's cost of equity capital.

(ii) If the company issues 10% debentures of face value of ₹ 100 each and realises ₹ 95 per debenture while the debentures are redeemable after 10 years at a premium of 10%, CALCULATE cost of debenture using YTM?

Assume Tax Rate to be 50%.

ICAI'S ANSWER

(i) Cost of Equity Capital (K_e):

$$K_e = \frac{\text{Expected dividend per share}(D_1)}{\text{Market price per share}(P_0)} + \text{Growth rate(g)}$$

$$= \frac{₹\ 3 \times 1.07}{₹\ 30} + 0.07 = 0.177 \text{ or } 17.7\%$$

(ii) Cost of Debenture (K_d):

Using Present Value method (YTM)

Identification of relevant cash flows

Year	Cash flows
0	Current market price (P_0) = ₹ 95
1 to 10	Interest net of tax [I(1-t)] = 10% of ₹ 100 (1 – 0.5) = ₹ 5
10	Redemption value (RV) = ₹ 100 (1.10) = ₹ 110

Calculation of Net Present Values (NPV) at two discount rates

Year	Cash flows (₹)	Discount factor @ 5% (L)	Present Value (₹)	Discount factor @ 10% (H)	Present Value (₹)
0	(95)	1.000	(95.00)	1.000	(95.00)
1 to 10	5	7.722	38.61	6.145	30.725
10	110	0.614	67.54	0.386	42.46
NPV			+11.15		-21.815

Calculation of IRR

$$\text{IRR} = L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$$

$$= 5\% + \frac{₹\ 11.15}{₹\ 11.15\text{-}(₹\ \text{-}21.815)}(10\% - 5\%) = 5\% + \frac{₹\ 55.75}{₹\ 32.965} = 6.69\%$$

Therefore, K_d = 6.69%

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 2 · answer p. 2-3

252 MTP **Old P8 Nov 2023 S-II** Q6(c) 2 marks old scheme

Same question, different ICAI answer: #250 MTP May 2024 S-I

DISCUSS the dividend-price approach to estimate cost of equity capital.

ICAI'S ANSWER

In dividend price approach, cost of equity capital is computed by dividing the expected dividend by market price per share. This ratio expresses the cost of equity capital in relation to what yield the

company should pay to attract investors. It is computed as:

$$K_e = \frac{D_1}{P_0}$$

Where,

D_1 = Dividend per share in period 1

P_0 = Market price per share today.

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 4 · answer p. 9

253 MTP **Old P8 May 2022 S-I** Q1(d) 5 marks old scheme

(i) A company's equity share is quoted in the market at ₹ 125 per share currently. The company pays a dividend of ₹ 10 per share and the investor's market expects a growth rate of 6% per year. CALCULATE the company's cost of equity capital.

(ii) If the company issues 10% debentures of face value of ₹ 100 each and realises ₹ 98 per debenture while the debentures are redeemable after 12 years at a premium of 10%, CALCULATE cost of debenture using YTM?

Assume Tax Rate to be 50%.

ICAI'S ANSWER

(i) Cost of Equity Capital (Ke):

$$K_e = \frac{\text{Expected dividend per share (D1)}}{\text{Market price per share (P0)}} + \text{Growth rate(g)}$$

$$= \frac{₹\ 10 \times 1.06}{₹\ 125} + 0.06 = 0.1448 \text{ or } 14.48\%$$

(ii) Cost of Debenture (K_d):

Using Present Value method (YTM)

Identification of relevant cash flows

Year	Cash flows
0	Current market price (P_0) = ₹ 98
1 to 12	Interest net of tax [I(1-t)] = 10% of ₹ 100 (1 – 0.5) = ₹ 5
12	Redemption value (RV) = ₹ 100 (1.10) = ₹ 110

Calculation of Net Present Values (NPV) at two discount rates

Year	Cash flows (₹)	Discount factor @ 5% (L)	Present Value (₹)	Discount factor @ 10% (H)	Present Value (₹)
0	(98)	1.000	(98.00)	1.000	(98.00)
1 to 12	5	8.863	44.32	6.814	34.07
12	110	0.557	61.27	0.319	35.09
NPV			+7.59		-28.84

Calculation of IRR

$$\text{IRR} = \text{L} + \frac{\text{NPVL}}{\text{NPVL - NPVH}} \text{(H-L)}$$

$$= 5\% + \frac{7.59}{7.59 - (-28.84)} (10\%-5\%) = 6.04\%$$

Therefore, $K_d = 6.04\%$

ICAI · Old P8 · MTP May 2022 · Series I · question p. 2 · answer p. 2-3

254 MTP **Old P8 Dec 2021 S-II** Q1(a) 5 marks old scheme

XYZ Company's equity share is quoted in the market at ₹ 25 per share currently. The company pays a dividend of ₹ 5 per share and the investor's market expects a growth rate of 5% per year.

You are required to:

(i) CALCULATE the company's cost of equity capital.

(ii) If the company issues 12% debentures of face value of ₹ 100 each and realises ₹ 95 per debenture while the debentures are redeemable after 10 years at a premium of 12%, CALCULATE cost of debenture using YTM?

Assume tax rate to be 30%.

ICAI'S ANSWER

(i) Cost of Equity Capital (K_e):

$$K_e = \frac{\text{Expected dividend per share}(D_1)}{\text{Market price per share}(P_0)} + \text{Growth rate}(g)$$

$$= \frac{₹\ 5 \times 1.05}{₹\ 25} + 0.05 = 26\%$$

(ii) Cost of Debenture (K_d):

Using Present Value method (or YTM)

Identification of relevant cash flows

Year	Cash flows
0	Current market price (P_0) = ₹ 95
1 to 10	Interest net of tax [I(1-t)] = 12% of ₹ 100 (1 – 0.30) = ₹ 8.40
10	Redemption value (RV) = ₹ 100 (1.12) = ₹ 112

Calculation of Net Present Values (NPV) at two discount rates

Year	Cash flows	Discount factor @ 9% (L)	Present Value	Discount factor @ 10% (H)	Present Value
0	(95)	1.0000	(95.00)	1.0000	(95.00)
1 to 10	8.40	6.4176	53.91	6.1445	51.61
10	112	0.4224	47.31	0.3855	43.18
NPV			+6.22		-0.21

Calculation of IRR

IRR = $L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$

$= 9\% + \frac{6.22}{6.22 - (-0.21)}(10\% - 9\%) = 9\% + \frac{6.22}{6.43} = 9.97\%$

Therefore, K_d = 9.97%

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 1 · answer p. 1

255 MTP **Old P8 Jul 2021 S-II** Q1(d) 5 marks old scheme

In March, 2021 Tiruv Ltd.'s share was sold for Rs. 219 per share. A long term earnings growth rate of 11.25% is anticipated. Tiruv Ltd. is expected to pay dividend of Rs. 5.04 per share.

(i) DETERMINE the rate of return an investor can expect to earn assuming that dividends are expected to grow along with earnings at 11.25% per year in perpetuity?

(ii) It is expected that Tiruv Ltd. will earn about 15% on book equity and shall retain 60% of earnings. In this case, whether, there would be any change in growth rate and cost of equity? ANALYSE.

ICAI'S ANSWER

(i) According to Dividend Discount Model approach the firm's expected or required return on equity is computed as follows:

$K_e = \frac{D_1}{P_0} + g$

Where,

K_e = Cost of equity share capital

D_1 = Expected dividend at the end of year 1

P_0 = Current market price of the share.

g = Expected growth rate of dividend.

Therefore, $K_e = \frac{5.04}{219} + 0.1125 = 13.55\%$

(ii) With rate of return on retained earnings (r) of 15% and retention ratio (b) of 60%, new growth rate will be as follows:

g = br = 0.60 x 0.15 = 0.09 or 9%

Accordingly, dividend will also get changed and to calculate this, first we shall calculate previous retention ratio (b_1) and then EPS assuming that rate of return on retained earning (r) is same.

With previous Growth Rate of 11.25% and r =15%, the retention ratio comes out to be:

0.1125 = b_1 x 0.15

b_1 = 0.75 and payout ratio = 0.25

With 0.25 payout ratio, the EPS will be as follows:

EPS = $\frac{5.04}{0.25}$ = Rs. 20.16

With new payout ratio of 40% (1 – 0.60) the new dividend will be:

D_1 = Rs. 20.16 x 0.40 = Rs. 8.064

Accordingly new K_e will be:

$$K_e = \frac{8.064}{219} + 0.09 = 12.68\%$$

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 2 · answer p. 3

4.3 WACC and the marginal cost of capital

49 questions · asked in 44 ICAI documents · 10 exam papers

256 **EXAM** **Sep 2025** Q1(a) **5 marks**

The capital structure of RSA Limited is as under:

	(₹)
Equity Shares (₹ 10 per share)	1,00,00,000
8% Irredeemable Preference Shares (₹100 per share)	5,00,00,000

Additional Information:

(i) Equity shares are quoted at ₹ 60 each and a new issue priced at ₹ 60 will be fully subscribed; flotation cost will be ₹ 4 per share.

(ii) Issue price of the 8% Irredeemable preference shares was ₹ 45.

(iii) Current market price of the 8% irredeemable preference shares is ₹ 55.

RSA Limited has been paying dividend to its equity shareholders at a constant growth rate of 5% per year and the dividend paid this year was ₹ 2 per equity share.

You are required to calculate:

(i) The cost of equity using Gordon's model

(ii) The cost of the irredeemable preference share

(iii) The weighted average cost of capital using market value weights.

ICAI'S ANSWER

(i) $K_e = \frac{D_1}{P_0 - F} + g$

$= \frac{2(1+0.05)}{60-4} + 5$

= 8.75%

(ii) $K_P = \frac{PD}{P_0}$

$= \frac{8}{45}$

= 17.78%

(iii) **Calculation of WACC using market value weights**

Source of capital	Market Value	Weights	Cost of capital	WACC (K_o)
	(₹)	(a)	(b)	(c) = (a)×(b)
Equity shares	6,00,00,000	0.6857	0.0875	0.0600
Preference shares	2,75,00,000	0.3143	0.1778	0.0559
	8,75,00,000	1		0.1159

WACC (K_o) = 0.1159% or 11.59%

If P_0 is taken as current market price for calculation of preference share

(ii) $K_P = \frac{PD}{P_0}$

$= \frac{8}{55}$

= 14.55%

(iii) **Calculation of WACC using market value weights**

Source of capital	Market Value	Weights	Cost of capital	WACC (K_o)
	(₹)	(a)	(b)	(c) = (a)×(b)
Equity shares	6,00,00,000	0.6857	0.0875	0.06
Preference shares	2,75,00,000	0.3143	0.1455	0.0457
	8,75,00,000	1		0.1057

WACC (K_o) = 0.1057% or 10.57%

ICAI · P6 · Sep 2025 exam · question p. 8 · answer p. 9-10

257 EXAM **Jan 2025** Q2(a) 7 marks

The following information pertain to CMC Limited:

Number of Equity Shares	20,00,000
Book Value of 10% Convertible Debentures	₹ 1,00,00,000
Book Value of 12% Bank Term Loan	₹ 25,00,000
Market Price of Equity Share	₹ 55
Market Value of 10% Convertible Debenture	₹ 108
Face Value of Equity Share	₹ 10
Face Value of 10% Convertible Debenture	₹ 100
Beta coefficient of Equity shares of CMC Ltd.	1.5
Risk free rate of return	4.5%

Equity risk Premium	9%
Rate of taxation	30%

The company expects that the share prices will rise in future at an average rate of 6% per annum. The 10% convertible debentures of 100 each will be converted in six years' time into equity shares of the company in the ratio of 1: 4 (4 equity shares for each debenture).

The market value of 12% bank term loan is at par.

You are required to calculate:

(i) Cost of Equity Share Capital by applying Capital Asset Pricing Model (CAPM) Approach

(ii) Cost of Convertible Debenture by using approximation method,

(iii) Cost of Bank Term Loan

(iv) Weighted Average Cost of Capital using Market Value weights

ICAI'S ANSWER

(i) Cost of equity share

R_F = 4.5%

$R_m - R_f$ = 9%

B = 1.5

Using CAPM,

$K_e = R_f + \beta (R_m - R_f)$

= 4.5% + 1.5 (9%)

= 18%

(ii) Cost of convertible debenture (based on Market Value)

Price of share after 6 years = $55\ (1+0.06)^6$

= ₹ 78.018

Redemption Value of Debenture (RV) = 78.018 × 4 = **312.07** (RV)

NP = 108 **(MV)**

n = 6

$$K_d = \frac{INT(1-t) + \left(\frac{RV - NP}{n}\right)}{\frac{(RV + NP)}{2}} \times 100$$

$$= \frac{10\,(1-0.3) + \frac{(312.07 - 108)}{6}}{\frac{(312.07 + 108)}{2}} \times 100$$

$$= \frac{7 + 34.012}{210.035} \times 100$$

K_d = 19.52%

(iii) Cost of bank term loan

$K_d = I\ (1 - t)$

= 12% (1-0.3) = 8.4%

(iv) Calculation of Weighted Average Cost of Capital (WACC) using Market Value Weights

Source	(₹) in lakhs	Weight	Cost of Capital after tax	WACC
Equity Share Capital	11,00,00,000	0.892	0.180	0.1606
10% Convertible Debenture	1,08,00,000	0.088	0.195	0.0171
12% Bank Term Loan	25,00,000	0.020	0.084	0.0017
Total	12,33,00,000	1		0.1794

WACC = 17.94%

Alternative Solution to Part (ii) and (iv)

(ii) Cost of convertible debenture

Price of share after 6 years = $55\ (1+0.06)^6$

= ₹ 78.018

Redemption Value of Debenture (RV) = 78.018 × 4 = **312.07** (RV)

NP = 100

n = 6

$$K_d = \frac{INT(1-t) + \left(\frac{RV - NP}{n}\right)}{\frac{(RV - NP)}{2}} \times 100$$

$$= \frac{10\ (1-0.3) + \frac{(312.07 - 100)}{6}}{\frac{(312.07 + 100)}{2}} \times 100$$

$$= \frac{7 + 35.345}{206.035} \times 100$$

K_d = 20.55%

(iv) Calculation of Weighted Average Cost of Capital (WACC) using Market Value Weights

Source	(₹) in lakhs	Weight	Cost of Capital after tax	WACC
Equity Share Capital	11,00,00,000	0.892	0.180	0.1606
10% Convertible Debenture	1,08,00,000	0.088	0.205	0.0180
12% Bank Term Loan	25,00,000	0.020	0.084	0.0017
Total	12,33,00,000	1		0.1803

WACC = 18.03%

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258 EXAM **Sep 2024** Q2(a) 8 marks

Capital structure of T Limited as on 1st April,2024 is as under:

	₹
Equity Share Capital (₹ 10 per share)	50,00,000
10% Debentures (₹ 100 per Debenture)	40,00,000
12% Preference Share Capital (10,000 shares of ₹ 100 each)	10,00,000

Additional Information:

(1) The risk free rate of return is 10%. The Beta of T Ltd. is 1.75 and the return on market portfolio is 12%. The Equity shares have a current market price of ₹ 70 per share.

(2) The debentures are trading at a market price of ₹ 80 per debenture. The Debentures are to be redeemed after 5 years at par.

(3) Preference shares are redeemable after 5 years at a premium of 5%, presently selling at ₹ 104 per share.

(4) The Company pays tax at a rate of 30%.

(5) The Cost of Debentures are to be calculated on Yield to Maturity approach.

(6) The present value factors at 10% and 14% are:

Year	1	2	3	4	5
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683	0.621
$PVIF_{0.14,t}$	0.877	0.769	0.675	0.592	0.519

You are required to calculate Weighted Average Cost of Capital (after tax) of T Limited using Market value weights.

ICAI'S ANSWER

Cost of Equity Share Capital using Capital Asset Pricing Model (CAPM) Approach

$K_e = R_f + ß\ (R_m - R_f)$

$K_e = 0.10 + 1.75\ (0.12 - 0.10)$

$= 0.10 + 1.75\ (0.02) = 0.135$ or **13.5%**

Cost of Redeemable Debentures using Yield to Maturity (YTM) Approach

Step-1: Identification of relevant cash flows

Year	Cash flows
0	Current market price (P_0) = ₹ 80
1 to 5	Interest net of tax [I(1-t)] = 10% of ₹ 100 (1-0.30) = ₹ 7
5	Redemption value (RV) = Face value i.e. ₹ 100

Step- 2: Calculation of NPVs at two discount rates

Year	Cash flows (₹)	Discount factor @ 10% (L)	Present Value (₹)	Discount factor @ 14% (H)	Present Value (₹)
0	80	1.000	(80.000)	1.000	(80.000)
1 to 5	7	3.790	26.530	3.432	24.024
5	100	0.621	62.100	0.519	51.900
NPV			**+8.630**		**-4.076**

Step- 3: Calculation of Cost of Debentures (Kd)

$$Kd = L + \frac{NPVL}{NPVL\text{-}NPVH}(H\text{-}L) = 10\% + \frac{₹8.630}{₹8.630 - (₹ - 4.076)}(14\%\text{-}10\%) = \mathbf{12.72\%}$$

Cost of Redeemable Preference Share Capital using approximation method

$$K_p = \frac{PD + \frac{(RV\text{-}NP)}{n}}{\frac{(RV+NP)}{2}}$$

$$K_p = \frac{12 + \left(\frac{105 - 104}{5}\right)}{\left(\frac{105 + 104}{2}\right)}$$

= **11.67%**

Calculation of WACC using market value weights

Source of Capital	Market Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	**(a)**	**(b)**	**(c) = (a)×(b)**
Equity Share Capital (₹ 70 × 5,00,000 equity shares)	3,50,00,000	0.8919	0.1350	**0.1204**
10% Debentures (₹ 80 × 40,000)	32,00,000	0.0816	0.1272	**0.0104**
12% Preference Share Capital (₹ 104 × 10,000 shares)	10,40,000	0.0265	0.1167	**0.0031**
	3,92,40,000	1.000		**0.1339**

WACC (K_o) = 0.1339 or 13.39%

ICAI · P6 · Sep 2024 exam · question p. 14-15 · answer p. 15-16

259 EXAM **Old P8 Nov 2023** Q4 10 marks old scheme

Z Ltd. wishes to raise additional fund of ₹ 25,00,000 for meeting its investment plan. It has ₹ 5,25,000 in the form of retained earnings available for investment purposes. Further details are as following:

Combination of debt and equity	2:3
Cost of debt	
Upto ₹ 2,50,000	8% (before tax)

Above ₹ 2,50,000 and to upto ₹ 5,00,000	10% (before tax)
Beyond ₹ 5,00,000	12% (after tax)
Earning of company	₹ 50,00,000
Retention Ratio	40%
Expected growth of dividend	15%
Market price per share	₹ 500
Number of outstanding equity shares	1,00,000
Tax Rate	30%

You are required to calculate:

i. Cost of debt

ii. Cost of retained earnings and cost of equity

iii. Weighted average cost of capital

ICAI'S ANSWER

Equity	60% of ₹ 25,00,000	= ₹ 15,00,000
Debt	40% of ₹ 25,00,000	= ₹ 10,00,000

The capital structure after raising additional finance:

		(₹)
Shareholders' funds		
Equity Capital	(₹ 15,00,000 – ₹ 5,25,000)	9,75,000
Retained earnings		5,25,000
Debt (Interest at 8% p.a.)		2,50,000
(Interest at 10% p.a.)	(₹ 5,00,000 – ₹ 2,50,000)	2,50,000
(Interest at 12% p.a.)	(₹ 10,00,000 – ₹ 5,00,000)	5,00,000
Total Funds		25,00,000

(i) Determination of post-tax average cost of additional debt:

K_d = I (1 – t)

Where,

I = Interest Rate

t = tax-rate

On ₹ 2,50,000 = 8% (1 – 0.3) = 5.6% or 0.056

On ₹ 2,50,000 = 10% (1 – 0.3) = 7% or 0.07

On ₹ 5,00,000 = 12% or 0.12

Average Cost of Debt

$$= \frac{(₹\ 2{,}50{,}000 \times 0.056) + (₹\ 2{,}50{,}000 \times 0.07) + (₹\ 5{,}00{,}000 \times 0.12)}{₹\ 10{,}00{,}000} \times 100 = 9.15\%$$

(ii) Determination of cost of retained earnings and cost of equity by applying Dividend growth model:

$$K_e \text{ or } K_r = \frac{D_1}{P_0} + g = \frac{D_0(1+g)}{P_0} + g$$

Where,

D_0 = Dividend paid = 60% of EPS = 60% × ₹ 50 = ₹ 30

g = Growth rate = 15%

P_0 = Current market price per share = ₹ 500

So, K_e **or** $K_r = \frac{₹\ 30\ (1+0.15)}{₹\ 500} + 0.15 = 0.069 + 0.15 = 21.9\%$

(iii) Computation of overall weighted average after tax cost of additional finance:

Particulars	(₹)	Weights	Cost of funds	Weighted Cost (%)
Equity (including retained earnings)	15,00,000	0.60	21.9%	13.14
Debt	10,00,000	0.40	9.15%	3.66
WACC	25,00,000			**16.80**

Alternative Presentation

Particulars (1)	(₹) (2)	Cost of funds (3)	Product (2) x (3)
Equity (including retained earnings)	15,00,000	21.9%	3,28,500
Debt	10,00,000	9.15%	91,500
Total	25,00,000		4,20,000

WACC = (Product / Total book value) x 100 = (4,20,000 / 25,00,000) x 100 = **16.8%**

Alternative Solution for 4(ii) and 4(iii)

If we assume expected growth rate of Dividend as 5%.

(i) Determination of cost of retained earnings and cost of equity by applying Dividend growth model:

$$K_e \text{ or } K_r = \frac{D_1}{P_0} + g = \frac{D_0(1+g)}{P_0} + g$$

Where,

D_0 = Dividend paid = 60% of EPS = 60% × ₹ 50 = ₹ 30

g = Growth rate = 5%

P_0 = Current market price per share = ₹ 500

So, K_e **or** $K_r = \frac{₹\ 30\ (1 + 0.05)}{₹\ 500} + 0.05 = 0.063 + 0.05 = 11.3\%$

(ii) Computation of overall weighted average after tax cost of additional finance:

Particulars	(₹)	Weights	Cost of funds	Weighted Cost (%)
Equity (including retained earnings)	15,00,000	0.60	11.3%	6.78
Debt	10,00,000	0.40	9.15%	3.66
WACC	25,00,000			**10.44**

Alternative Presentation

Particulars (1)	(₹) (2)	Cost of funds (3)	(2) x (3)
Equity (including retained earnings)	15,00,000	11.3%	1,69,500
Debt	10,00,000	9.15%	91,500
Total	25,00,000		2,61,000

WACC = (Product / Total book value) x 100 = (2,61,000 / 25,00,000) x 100 = **10.44%**

ICAI · Old P8 · Nov 2023 exam · question p. 11 · answer p. 12-14

260 EXAM **Old P8 May 2023** Q4 10 marks old scheme

Capital structure of D Ltd. as on 31st March, 2023 is given below:

Particulars	₹
Equity share capital (₹ 10 each)	30,00,000
8% Preference share capital (₹ 100 each)	10,00,000
12% Debentures (₹ 100 each)	10,00,000

- Current market price of equity share is ₹ 80 per share. The company has paid dividend of ₹ 14.07 per share. Seven years ago, it paid dividend of ₹ 10 per share. Expected dividend is ₹ 16 per share.
- 8% Preference shares are redeemable at 6% premium after five years. Current market price per preference share is ₹ 104.
- 12% debentures are redeemable at 20% premium after 10 years. Flotation cost is ₹ 5 per debenture.
- The company is in 40% tax bracket.
- In order to finance an expansion plan, the company intends to borrow 15% Long-term loan of ₹ 30,00,000 from bank. This financial decision is expected to increase dividend on equity share from ₹ 16 per share to ₹ 18 per share. However, the market price of equity share is expected to decline from ₹ 80 to ₹ 72 per share, because investors' required rate of return is based on current market conditions.

Required:

(i) Determine the existing Weighted Average Cost of Capital (WACC) taking book value weights.

(ii) Compute Weighted Average Cost of Capital (WACC) after the expansion plan taking book value weights.

Interest Rate	1%	2%	3%	4%	5%	6%	7%
$FVIF_{i,5}$	1.051	1.104	1.159	1.217	1.276	1.338	1.403
$FVIF_{i,6}$	1.062	1.126	1.194	1.265	1.340	1.419	1.501
$FVIF_{i,7}$	1.072	1.149	1.230	1.316	1.407	1.504	1.606

ICAI'S ANSWER

(i) (a) Growth rate in Dividends

14.07 = 10 x FVIF (i,7 years)

FVIF (i,7 years) = 1.407

FVIF (5%, 7 years) = 1.407

i = 5%

Growth rate in dividend= 5%

(b) Cost of Equity

$$Ke = \frac{D_1}{P_0} + g$$

$$Ke = \frac{16}{80} + 0.05$$

Ke = 25%

(c) Cost of Preference Shares

$$Kp = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$Kp = \frac{8 + \frac{(106 - 104)}{5}}{\frac{(106 + 104)}{2}}$$

Kp = 8.4/105

Kp = 8%

(d) Cost of Debt

$$Kd = \frac{I(1 - t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$Kd = \frac{12(1 - 0.4) + \frac{(120 - 95)}{10}}{\frac{(120 + 95)}{2}}$$

Kd = (7.2+2.5)/107.5 = 9.02%

Kd = 9.02%

Calculation of existing Weighted Average Cost of Capital (WACC)

Capital	Amount (₹)	Weights	Cost	WACC
Equity Share Capital	30,00,000	0.6	25%	15.00%
Preference Share Capital	10,00,000	0.2	8%	1.60%
Debenture	10,00,000	0.2	9.02%	1.80%
	50,00,000	1		**18.40%**

Alternative presentation

(i) Computation of existing WACC on book value weights

Source (1)	Book value (₹) (2)	Weight (3)	Cost of capital (%) (4)	Product (2) x (4)
Equity share capital	30,00,000	0.60	25	7,50,000
Preference share capital	10,00,000	0.20	8	80,000
Debentures	10,00,000	0.20	9.02	90,200
Total	**50,00,000**	**1.00**		**9,20,200**

WACC = (Product / Total book value) x 100 = (9,20,200 /50,00,000) x 100 = **18.4%**

(ii) Cost of Long Term Debt = 15% (1-0.4) = 9%

Revised Ke = $\frac{18}{72}$ + 0.05 = 30%

Calculation of WACC after expansion taking book value weights

Capital	Amount	Weights	Cost	W.C
Equity Share Capital	30,00,000	0.3750	30%	11.25%
Preference Share Capital	10,00,000	0.1250	8%	1.00%
Debenture	10,00,000	0.1250	9.02%	1.13%
Long Term Debt	30,00,000	0.3750	9.00%	3.38%
	80,00,000	1.0000		**16.76%**

Alternative presentation

(i) Computation of WACC on book value weights after expansion

Source (1)	Book value (₹) (2)	Weight (3)	Cost of capital (%) (4)	Product (2) x (4)
Equity share capital	30,00,000	0.375	30	9,00,000
Preference share capital	10,00,000	0.125	8	80,000
Debentures	10,00,000	0.125	9.02	90,200
Long term loan	30,00,000	0.375	9	2,70,000
Total	**80,00,000**	**1.00**		**13,40,200**

WACC = (Product / Total book value) x 100 = (13,40,200 / 80,00,000) x 100 **= 16.76%**

ICAI · Old P8 · May 2023 exam · question p. 13-14 · answer p. 14-16

261 **EXAM** **Old P8 Nov 2022** Q1(c) **5 marks** old scheme

The following is the extract of the Balance Sheet of M/s KD Ltd.:

Particulars	Amount (₹)
Ordinary shares (Face Value ₹ 10/- per share)	5,00,000
Share Premium	1,00,000
Retained Profits	6,00,000
8% Preference Shares (Face Value ₹ 25/- per share)	4,00,000
12% Debentures (Face value ₹ 100/- each)	6,00,000
	22,00,000

The ordinary shares are currently priced at ₹ 39 ex-dividend and preference share is priced at ₹ 18 cum-dividend. The debentures are selling at 120 percent ex-interest. The applicable tax rate to KD Ltd. is 30 percent. KD Ltd.'s cost of equity has been estimated at 19 percent. Calculate the WACC (weighted average cost of capital) of KD Ltd. on the basis of market value.

ICAI'S ANSWER

Computation of WACC on the basis of market value

W.N. 1

Cum-dividend price of Preference shares = ₹ 18

Less: Dividend (8/100) x 25 = ₹ 2

∴ Market Price of Preference shares = ₹ 16

$K_p = \frac{2}{16}$ = 0.125 (or) 12.5%

No. of Preference shares = $\left(\frac{4,00,000}{25}\right)$ = 16,000

W.N. 2

Market price of Debentures = $\left(\frac{120}{100}\right)$ x 100 = ₹120

$K_d = \left[\frac{12\ (1-0.3)}{120}\right] = 0.07$ (or) 7%

No. of Debentures = $\left(\frac{6,00,000}{100}\right) = 6,000$

W.N.3

Market Price of Equity shares = ₹39

K_e (given) = 19% or 0.19

No. of Equity shares = $\frac{5,00,000}{10} = 50,000$

Sources	Market Value (₹)	Nos.	Total Market value (₹)	Weight	Cost of Capital	Product
Equity Shares	39	50,000	19,50,000	0.6664	0.19	0.1266
Preference Shares	16	16,000	2,56,000	0.0875	0.125	0.0109
Debentures	120	6,000	7,20,000	0.2461	0.07	0.0172
					WACC =	0.1547

WACC = 0.1547 or 15.47%

ICAI · Old P8 · Nov 2022 exam · question p. 2 · answer p. 5

262 EXAM **Old P8 Nov 2022** Q5(b) 6 marks old scheme

MR Ltd. is having the following capital structure, which is considered to be optimum as on 31.03.2022.

Equity share capital (50,000 shares)	₹ 8,00,000
12% Pref. share capital	₹ 50,000
15% Debentures	₹ 1,50,000
	₹ 10,00,000

The earnings per share (EPS) of the company were ₹ 2.50 in 2021 and the expected growth in equity dividend is 10% per year. The next year's dividend per share (DPS) is 50% of EPS of the year 202I. The current market price per share (MPS) is ₹ 25.00. The 15% new debentures can be issued by the company. The company's debentures are currently selling at ₹ 96 per debenture. The new 12% Pref. share can be sold at a net price of ₹ 91.50 (face value ₹ 100 each). The applicable tax rate is 30%.

You are required to calculate

(a) After tax cost of

(i) New debt,

(ii) New pref. share capital and

(iii) Equity shares assuming that new equity shares come from retained earnings.

(b) Marginal cost of capital,

How much can be spent for capital investment before sale of new equity shares assuming that retained earnings for next year investment is 50% of 2021?

ICAI'S ANSWER

(a) (i) After tax cost of new Debt:

$$K_d = I\frac{(1-t)}{P_1} = 15\frac{(1-0.3)}{96}$$

= 0.1094 (or) **10.94%**

(ii) After tax cost of New Preference share capital:

$$K_p = \frac{PD}{P_O} = \left(\frac{12}{91.5}\right) = 0.1311 \text{ (or) } \mathbf{13.11\%}$$

(iii) After tax cost of Equity shares:

$$K_e = \left(\frac{D_1}{P_O}\right) + g = \left[\frac{(2.50 \text{ x } 50\%)}{25}\right] + 0.10$$

= **0.15 (or) 15%**

(b) Marginal Cost of Capital

Type of capital	Proportions	Specific cost	Product
Equity Shares	0.80	0.15	0.12
Preference Shares	0.05	0.1311	0.0066
Debentures	0.15	0.1094	0.0164
∴ Marginal cost of capital			**0.1430**

(c) Amount that can be spend for capital investment

Retained earnings = 50% of EPS x No. of outstanding Equity shares

= 1.25 x 50,000

= ₹ 62,500

Proportion of equity (Retained earnings here) capital is 80% of total capital. Therefore,

₹ 62,500 is 80% of total capital.

∴ Amount of Capital Investment = $\frac{62,500}{0.80}$ = **₹ 78,125**

ICAI · Old P8 · Nov 2022 exam · question p. 12-13 · answer p. 14-15

263 EXAM **Old P8 Dec 2021** Q1(b) 5 marks old scheme

Book value of capital structure of B Ltd. is as follows:

Sources	Amount
12%, 6,000 Debentures @ ₹ 100 each	₹ 6,00,000
Retained earnings	₹ 4,50,000
4,500 Equity shares @ ₹ 100 each	₹ 4,50,000
	₹ 15,00,000

Currently, the market value of debenture is ₹ 110 per debenture and equity share is ₹ 180 per share. The expected rate of return to equity shareholder is 24% p.a. Company is paying tax @ 30%.

Calculate WACC on the basis of market value weights.

ICAI'S ANSWER

Calculation of Cost of Capital of debentures ignoring market value:

Cost of Debentures (K_d)= 12 (1 - .30) = 8.40%

Computation of Weighted Average Cost of Capital based on Market Value Weights

Source of Capital	Market Value (₹)	Weights to Total Capital	After tax Cost of capital (%)	WACC (%)
Debentures (6,000 nos. × ₹ 110)	6,60,000	0.45(approx.)	8.40	3.78
Equity Shares (4,500 nos. × ₹180)	8,10,000	0.55(approx.)	24.00	13.20
	14,70,000	1.00		16.98

Note: Cost of Debenture and Cost of equity considered as given without considering market value. Cost of sources of capital can be computed based on the Market price and accordingly Weighted Average Cost of Capital can be calculated as below:

Calculation of Cost of Capital for each source of capital considering market value of capital:

(1) Cost of Equity share capital:

$$K_e = \frac{\text{Earnings}}{\text{Market Price per share}} = \frac{24\% \times ₹\ 100}{₹\ 180} = 13.333\%$$

(2) Cost of Debentures (K_d)= $\frac{I(1-t)}{NP} = \frac{₹\ 12(1-0.3)}{₹\ 110} = 7.636\%$

Computation of Weighted Average Cost of Capital based on Market Value Weights

Source of Capital	Market Value (₹)	Weights to Total Capital	After tax Cost of capital (%)	WACC (%)
Debentures (6,000 nos. × ₹ 110)	6,60,000	0.45(approx.)	7.636	**3.44** (approx.)
Equity Shares (4,500 nos. × ₹ 180)	8,10,000	0.55(approx.)	13.333	**7.33** (approx.)
	14,70,000	1.00		**10.77** (approx.)

ICAI · Old P8 · Dec 2021 exam · question p. 1 · answer p. 3-4

264 EXAM **Old P8 Jul 2021** Q2 10 marks old scheme

Following are the information of TT Ltd.:

Particulars	
Earnings per share	₹ 10
Dividend per share	₹ 6

Particulars	
Expected growth rate in Dividend	6%
Current market price per share	₹ 120
Tax Rate	30%
Requirement of Additional Finance	₹ 30 lakhs
Debt Equity Ratio (For additional finance)	2:1
Cost of Debt	
0-5,00,000	10%
5,00,001 - 10,00,000	9%
Above 10,00,000	8%

Assuming that there is no Reserve and Surplus available in TT Ltd.

You are required to:

(a) Find the pattern of finance for additional requirement

(b) Calculate post tax average cost of additional debt

(c) Calculate cost of equity

(d) Calculate the overall weighted average after tax cost of additional finance.

ICAI'S ANSWER

(a) Pattern of raising additional finance

Equity	1/3 of ₹ 30,00,000	= ₹ 10,00,000
Debt	2/3 of ₹ 30,00,000	= ₹ 20,00,000

The capital structure after raising additional finance:

Particulars		(₹)
Shareholder's Funds		
Equity Capital		10,00,000
Debt (Interest at 10% p.a.)		5,00,000
(Interest at 9% p.a.)		5,00,000
(Interest at 8% p.a.)	(20,00,000–10,00,000)	10,00,000
Total Funds		**30,00,000**

(b) Determination of post-tax average cost of additional debt

K_d = I (1 – t)

Where,

I = Interest Rate

t = Corporate tax-rate

On First ₹ 5,00,000 = 10% (1 – 0.3) = 7% or 0.07

On Next ₹ 5,00,000 = 9% (1 – 0.3) = 6.3% or 0.063

On Next ₹ 10,00,000 = 8% (1 – 0.3) = 5.6% or 0.056

Average Cost of Debt

$= \frac{(₹\ 5{,}00{,}000 \times 0.07) + (₹\ 5{,}00{,}000 \times 0.063) + (₹\ 10{,}00{,}000 \times 0.056)}{₹\ 20{,}00{,}000}$ x 100 = **6.125%**

(c) Determination of cost of equity applying Dividend growth model:

$$K_e = \frac{D_1}{P_0} + g$$

Where,

K_e = Cost of equity

$D_1 = D_0$ (1+ g)

D_0 = Dividend paid

g = Growth rate = 6%

P_0 = Current market price per share = ₹ 120

$K_e = \frac{₹\ 6(1 + 0.06)}{₹\ 120} + 0.06 = \frac{₹\ 6.36}{₹\ 120} + 0.06$ = **0.113 or 11.3%**

(d) Computation of overall weighted average after tax cost of additional finance

Particulars	(₹)	Weights	Cost of funds	Weighted Cost (%)
Equity	10,00,000	1/3	11.3%	3.767
Debt	20,00,000	2/3	6.125 %	4.083
WACC	30,00,000			**7.85**

(Note: In the above solution different interest rate have been considered for different slab of Debt)

Alternative Solution

(a) Pattern of raising additional finance

Equity	1/3 of ₹ 30,00,000	= ₹ 10,00,000
Debt	2/3 of ₹ 30,00,000	= ₹ 20,00,000

The capital structure after raising additional finance:

Particulars	(₹)
Shareholders' Funds	
Equity Capital	10,00,000
Debt (Interest at 8% p.a.)	20,00,000
Total Funds	**30,00,000**

(b) Determination of post-tax average cost of additional debt

K_d = I (1 – t)

Where,

I = Interest Rate

t = Corporate tax-rate

K_d = 8% (1 – 0.3) = 5.6%

(c) Determination of cost of equity applying Dividend growth model:

$$K_e = \frac{D_1}{P_0} + g$$

Where,

K_e = Cost of equity

D_1 = D_0 (1+ g)

D_0 = Dividend paid

g = Growth rate =6%

P_0 = Current market price per share = ₹ 120

Then, $K_e = \frac{₹\ 6(1+0.06)}{₹\ 120} + 0.06 = \frac{₹\ 6.36}{₹\ 120} + 0.06 =$ **0.113 or 11.3%**

(d) Computation of overall weighted average after tax cost of additional finance

Particulars	(₹)	Weights	Cost of funds	Weighted Cost (%)
Equity	10,00,000	1/3	11.3%	3.767
Debt	20,00,000	2/3	5.6%	3.733
WACC	30,00,000			**7.50**

(Note: In the above solution single interest rate have been considered for Debt)

ICAI · Old P8 · Jul 2021 exam · question p. 69-70 · answer p. 70-72

265 EXAM Old P8 Jan 2021 Q4 10 marks old scheme

The Capital structure of PQR Ltd. is as follows:

	₹
10% Debenture	3,00,000
12% Preference Shares	2,50,000
Equity Share (face value ₹ 10 per share)	5,00,000
	10,50,000

Additional Information:

(i) ₹ 100 per debenture redeemable at par has 2% floatation cost & 10 years of maturity. The market price per debenture is ₹ 110.

(ii) ₹ 100 per preference share redeemable at par has 3% floatation cost & 10 years of maturity. The market price per preference share is ₹ 108.

(iii) Equity share has ₹ 4 floatation cost and market price per share of ₹ 25. The next year expected dividend is ₹ 2 per share with annual growth of 5%. The firm has a practice of paying all earnings in the form of dividends.

(iv) Corporate Income Tax rate is 30%.

Required:

Calculate Weighted Average Cost of Capital (WACC) using market value weights.

ICAI'S ANSWER

Workings:

1. Cost of Equity $(K_e) = \frac{D_1}{P_0 - F} + g = \frac{₹\ 2}{₹\ 25 - ₹\ 4} + 0.05 = 0.145$ (approx.)

2. Cost of Debt $(K_d) = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$

$= \frac{10(1-0.3) + \frac{(100-98)}{10}}{\frac{(100+98)}{2}} = \frac{7 + 0.2}{99} = 0.073$ (approx.)

3. Cost of Preference Shares $(K_p) = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$

$= \frac{12 + \frac{(100-97)}{10}}{\frac{(100+97)}{2}} = \frac{12 + 0.3}{98.5} = 0.125$ (approx.)

Calculation of WACC using market value weights

Source of capital	Market Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	**(a)**	**(b)**	**(c) = (a) × (b)**
10% Debentures (₹ 110 × 3,000)	3,30,000	0.178	0.073	0.013
12% Preference shares (₹ 108 × 2,500)	2,70,000	0.146	0.125	0.018
Equity shares (₹ 25 × 50,000)	12,50,000	0.676	0.145	0.098
	18,50,000	1.00		**0.129**

WACC (K_o) = **0.129 or 12.9%** (approx.)

ICAI · Old P8 · Jan 2021 exam · question p. 9 · answer p. 10

266 EXAM **Old P8 Nov 2019** Q5 10 marks old scheme

A Company wants to raise additional finance of ₹ 5 crore in the next year. The company expects to retain ₹ 1 crore earning next year. Further details are as follows:

(i) The amount will be raised by equity and debt in the ratio of 3: 1.

(ii) The additional issue of equity shares will result in price per share being fixed at ₹ 25.

(iii) The debt capital raised by way of term loan will cost 10% for the first ₹ 75 lakh and 12% for the next ₹ 50 lakh.

(iv) The net expected dividend on equity shares is ₹ 2.00 per share. The dividend is expected to grow at the rate of 5%.

(v) Income tax rate is 25%.

You are required:

(a) To determine the amount of equity and debt for raising additional finance.

(b) To determine the post-tax average cost of additional debt.

(c) To determine the cost of retained earnings and cost of equity.

(d) To compute the overall weighted average cost of additional finance after tax.

ICAI'S ANSWER

(a) Determination of the amount of equity and debt for raising additional finance:

Pattern of raising additional finance

Equity	3/4 of ₹ 5 Crore	**= ₹ 3.75 Crore**
Debt	1/4 of ₹ 5 Crore	**= ₹ 1.25 Crore**

The capital structure after raising additional finance:

Particulars		(₹ In crore)
Shareholders' Funds		
Equity Capital	(3.75 – 1.00)	2.75
Retained earnings		1.00
Debt (Interest at 10% p.a.)		0.75
(Interest at 12% p.a.)	(1.25-0.75)	0.50
Total Funds		5.00

(b) Determination of post-tax average cost of additional debt

K_d = I (1 – t)

Where,

I = Interest Rate

t = Corporate tax-rate

On ₹ 75,00,000 = 10% (1 – 0.25) = 7.5% or 0.075

On ₹ 50,00,000 = 12% (1 – 0.25) = 9% or 0.09

Average Cost of Debt

$$= \frac{(₹75,00,000 \times 0.075) + (₹50,00,000 \times 0.09)}{1,25,00,000} \text{ x } 100$$

$$= \frac{₹5,62,500 + ₹4,50,000}{1,25,00,000} \text{ x } 100 = 8.10\%$$

(c) Determination of cost of retained earnings and cost of equity (Applying Dividend growth model):

$$K_e = \frac{D_1}{P_0} + g$$

Where,

K_e = Cost of equity

$D_1 = D_O(1 + g)$

D_0 = Dividend paid (i.e = ₹ 2)

g = Growth rate

P_0 = Current market price per share

Then, $K_e = \frac{\text{Rs. } 2\ (1.05)}{\text{Rs. } 25} + 0.05 = \frac{\text{Rs. } 2.1}{\text{Rs. } 25}$ + 0.05 = 0.084 + 0.05 = 0.134 = 13.4%

Cost of retained earnings equals to cost of Equity i.e. 13.4%

(d) Computation of overall weighted average after tax cost of additional finance

Particular	(₹)	Weights	Cost of funds	Weighted Cost (%)
Equity (including retained earnings)	3,75,00,000	3/4	13.4%	10.05
Debt	1,25,00,000	1/4	8.1%	2.025
WACC	5,00,00,000			**12.075**

ICAI · Old P8 · Nov 2019 exam · question p. 14 · answer p. 14-16

267 EXAM **Old P8 Nov 2019** Q6(c) 4 marks old scheme

Explain the significance of Cost of Capital.

ICAI'S ANSWER

Significance of the Cost of Capital: The cost of capital is important to arrive at correct amount and helps the management or an investor to take an appropriate decision. The correct cost of capital helps in the following decision making:

(i) Evaluation of investment options: The estimated benefits (future cashflows) from available investment opportunities (business or project) are converted into the present value of benefits by discounting them with the relevant cost of capital. Here it is pertinent to mention that every investment option may have different cost of capital hence it is very important to use the cost of capital which is relevant to the options available. Here Internal Rate of Return (IRR) is treated as cost of capital for evaluation of two options (projects).

(ii) Performance Appraisal: Cost of capital is used to appraise the performance of a particulars project or business. The performance of a project or business in compared against the cost of capital which is known here as cut-off rate or hurdle rate.

(iii) Designing of optimum credit policy: While appraising the credit period to be allowed to the customers, the cost of allowing credit period is compared against the benefit/ profit earned by providing credit to customer of segment of customers. Here cost of capital is used to arrive at the present value of cost and benefits received.

ICAI · Old P8 · Nov 2019 exam · question p. 16 · answer p. 17

268 RTP **May 2026** Q5

Zebra Dynamics Ltd., a diversified engineering firm, is currently undergoing a strategic review of its capital financing. The Board of Directors has set strict hurdles for new project approvals. The Chief Financial Officer (CFO) has stated that the company's Cost of Capital based on Book Value weights is currently 11%, while the Cost of Capital based on Market Value weights is exactly 150 basis points higher.

Market Benchmarking Data

To determine the cost of equity, the finance team has gathered data on two comparable peers, A Ltd. and B Ltd, which operate in the same economic environment. Their returns are driven by the Capital Asset Pricing Model (CAPM):

Security	Beta	Expected CAPM Return
A Ltd.	1.2	15.20%
B Ltd.	0.8	12.80%

Capital Structure:

- Book Value Terms: The company maintains a traditional balance sheet with an equal ratio of Equity and Debt.
- Market Value Terms: Due to market optimism, the company's equity trades at a premium. The Market Value of Equity is 3 times its Book Value, while the debt trades at par (Market Value of Debt equals Book Value).

Debt Instrument Details

The company's debt capital is raised entirely through Debentures with the following characteristics:

- Coupon Rate: 7% per annum.
- Issue Details: The debentures were issued at a discount of ₹ 10 for ₹ 90.
- Redemption: They are redeemable at Par (₹ 100).
- Tenure: The debentures mature in 4 years.

You have been hired as a financial consultant to

(i) PROVIDE a detailed breakdown of the component costs.

(ii) CALCULATE the corporate tax rate applicable to the company assuming Discount on issue or premium on redemption are tax deductible.

(iii) CALCULATE the Beta of the company.

ICAI'S ANSWER

(i) Let the cost of Equity be 'X' and Cost of Debt be 'Y'

Calculation of WACC using Book value weights

Source of Capital	Cost of capital	Weights	WACC (K_o)
	(a)	**(b)**	**(c) = (a)×(b)**
Equity Share Capital	X	0.5	0.5X
Long term debts	Y	0.5	0.5Y
			11

Calculation of WACC using market value weights

Source of Capital	Cost of capital	MV Weights	WACC (K_o)
	(a)	**(b)**	**(c) = (a)×(b)**
Equity Share Capital	X	3/4 = 0.75	0.75X
Long term debts	Y	1/4 = 0.25	0.25Y
			11+1.5 = 12.5

0.5X + 0.5Y = 11 .....(1)

0.75X + 0.25Y = 12.5 .....(2)

Equating 1 & 2

Ke (X) = 14%

Substituting X in Eqn. 1

0.5x14 + 0.5Y = 11

7 + 0.5Y = 11

Post tax Kd (Y)= 8%

(ii) Pre Tax Cost of Redeemable Debenture (K_d) = $\frac{I + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$

Where,

I = Interest payment

NP = Net proceeds

RV = Redemption value of debentures

n = Remaining life of debentures

$$k_d = \frac{7 + \frac{(100 - 90)}{4}}{\frac{(100 + 90)}{2}}$$

$$k_d = \frac{7 + 2.5}{95}$$

Pre Tax Kd = 10%

Post tax Kd = Pre tax Kd (1–T)

8% = 10%(1–T)

Tax rate (T) = 20%

(iii) According to CAPM

Cost of Equity (K_e)= R_f + ß (R_m – R_f)

Where,

K_e = Cost of equity capital

R_f = Risk free rate of return

ß = Beta coefficient

R_m = Rate of return on market portfolio

(R_m – R_f) = Market risk premium

14% = 8% + ß x 6%

Beta coefficient = 1

Working Note:

A Ltd

15.20 = R_f + 1.2Rp .....(1)

B Ltd

12.80 = R_f + 0.8Rp .....(2)

By solving Eqn. 1 & 2,

Rp =6

So Risk Premium = 6%

Substituting Rp = 6 in Eqn 1

15.20 = R_f + 1.2x6%

R_f = 15.20 – 7.20 = 8%

Hence, Risk Free Return = 8%

Rp = R_m – R_f

6 = R_m – 8%

R_m = 14%

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269 RTP **Sep 2025** Q5

Lavanya Limited wishes to raise additional finance of ₹ 20 lakhs for meeting its investment plans. It has ₹ 4,20,000 in the form of retained earnings available for investment purposes. Further details are as following:

(1)	Debt / Equity mix	3:7
(2)	Cost of debt:	
	Upto ₹ 3,60,000	8% (before tax)
	Beyond ₹ 3,60,000	12% (before tax)
(3)	Earnings per share	₹ 5
(4)	Dividend pay out	40% of earnings
(5)	Retained Earnings	60%
(6)	Rate of Return on Retained Earnings	10%
(7)	Current market price per share	₹ 53
(8)	Tax rate	30%

You are required to:

(a) DETERMINE the pattern for raising the additional finance.

(b) DETERMINE the post-tax average cost of additional debt.

(c) DETERMINE the cost of retained earnings and cost of equity.

(d) COMPUTE the overall weighted average after tax cost of additional finance.

ICAI'S ANSWER

(a) **Pattern for raising the additional finance:**

Equity	70% of ₹ 20,00,000	= ₹14,00,000
Debt	30% of ₹ 20,00,000	= ₹ 6,00,000

The capital structure after raising additional finance:

		(₹)
Shareholders' funds		
Equity Capital	(₹ 14,00,000 – ₹ 4,20,000)	9,80,000
Retained earnings		4,20,000
Debt (Interest at 8% p.a.)		3,60,000
(Interest at 12% p.a.)	(₹ 6,00,000 – ₹ 3,60,000)	2,40,000
Total Funds		20,00,000

(b) **Determination of post-tax average cost of additional debt:**

K_d = I (1 – t)

Where,

I = Interest Rate

t = Corporate tax-rate

On ₹ 3,60,000 =8% (1 – 0.3) =5.6% or 0.056

On ₹ 2,40,000 =12% (1 – 0.3) = 8.4% or 0.084

Average Cost of Debt

$$= \frac{(₹\ 3{,}60{,}000 \times 0.056) + (₹\ 2{,}40{,}000 \times 0.084)}{₹\ 6{,}00{,}000} \times 100 = 6.72\%$$

(c) **Determination of cost of retained earnings and cost of equity by applying Dividend growth model:**

$$K_e \text{ or } K_r = \frac{D_1}{P_0} + g = \frac{D_0(1+g)}{P_0} + g$$

Where,

D_0 = Dividend paid = 40% of EPS = 40% × ₹ 5 = ₹ 2

g = Growth rate = Retained Earnings x Rate of Return

= 0.6 x 0.10 = 0.06 or 6%

P_0 = Current market price per share = ₹ 53

$$K_e \text{ or } K_r = \frac{2\ (1+0.6)}{53} = \frac{2.12}{53} + 0.06 = 0.04 + 0.06 = 0.10 \text{ or } 10\%$$

(d) **Computation of overall weighted average after tax cost of additional finance:**

Particulars	Amount (₹)	Weights	Cost of funds	Weighted Cost (%)
Equity (including retained earnings)	14,00,000	0.70	10%	7
Debt	6,00,000	0.30	6.72%	2.016
WACC	20,00,000			**9.016**

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270 **RTP** **May 2025** Q5

Paramhans Limited has a capital structure that consists of Equity share capital, Reserves & Surplus, Bank term loan, Debentures which are redeemable at a premium of 5% and Preference share capital redeemable at premium of 5%. The coupon rate on debentures is 1.5 times of that of bank term loan coupon rate whereas the preference dividend rate is 1.5 times of debentures' interest rate. Tenure for the bank term loan, debentures and preference share capital is 3 years, 5 years and 7 years respectively.

The current book value of the capital structure is as follows –

Particulars	Amount (₹)
Equity Share Capital (FV = ₹ 100)	25,00,000
Reserves And Surplus	10,00,000
Bank Term Loan	10,00,000
Debentures (FV = ₹ 100)	15,00,000
Preference Share Capital (FV = ₹ 100)	20,00,000
TOTAL	**80,00,000**

Tax rate applicable to the company is 25%.

Debentures are currently selling at a price of ₹ 96 whereas Preference shares are currently selling at ₹ 102. The equity shares of the company are quoted at ₹ 150 per share. The ongoing P/E ratio for the shares of Paramhans Limited is at 6.667 times. Paramhans Limited belongs to a risk class where the overall capitalization and discounting rate of the company is at 20%

CALCULATE –

(A) Rate of Interest on Bank term Loan & Debentures

(B) Rate of Preference dividend

(C) WACC using Market Value weights

ICAI'S ANSWER

(A) Calculation of Interest rate on Bank term Loan & Debentures

Let the rate of interest on bank term loan be 'X'

Therefore, Rate of Interest on debentures = 1.5X

Rate of Preference dividend = 1.5 x 1.5X = 2.25X (1.5 times of debentures interest rate)

Now, lets calculate Kd (term loan), Kd (debentures), Kp (Pref. shares) & Ke

Kd (Term loan) = Int (1 – t)

= X (1 – 0.25)

Kd (Term loan) = 0.75X

$$\text{Kd (Debentures)} = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

RV = 100 + 5% = ₹ 105

NP = Issue Price / Market price = ₹ 96

$$= \frac{150X(1 - 0.25) + \frac{(105 - 96)}{5}}{\frac{(105 + 96)}{2}}$$

Kd (Debentures) = $\frac{112.5X + 1.8}{100.5}$

$$\text{Kp} = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

RV = 100 + 5% = ₹ 105

NP = Issue price / Market price = ₹ 102

Therefore Kp = $\frac{225X + (105 - 102 \,/\, 7)}{(105 + 102) \,/\, 2}$

Kp = $\frac{225X + 0.4286}{103.5}$

$$\text{Ke} = \frac{1}{\text{PE Ratio}}$$

$$= \frac{1}{6.667}$$

Therefore, Ke = 15%

Kr = Ke = 15% (In absence of information on opportunity cost)

Overall Capitalization rate (Ko) = 20% (given)

Sources of capital	Amount of capital	Weights (W)	Cost (K)	W x K
Equity share capital	25,00,000	0.3125	15	4.6875
Reserves & surplus	10,00,000	0.1250	15	1.8750
Bank term loan	10,00,000	0.1250	0.75X	0.09375X
Debentures	15,00,000	0.1875	$\frac{112.5X + 1.8}{100.5}$	$\frac{21.094X + 0.3375}{100.5}$
Preference share capital	20,00,000	0.2500	$\frac{225X + 0.4286}{103.5}$	$\frac{56.25X + 0.1072}{103.5}$
	80,00,000	**1.0000**	**Ko /WACC**	**20%**

$$20 = 4.6875 + 1.8750 + 0.09375X + \frac{21.094X + 0.3375}{100.5} + \frac{56.25X + 0.1072}{103.5}$$

On solving the above equation,

$$13.4375 = 0.09375X + \frac{(2183.2\ X + 34.931 + 5653.125X + 10.774)}{10{,}401.75}$$

1,39,773.52 = 975.16X + 2,183.2 X + 5,653.125X + 45.705

1,39,727.815 = 8,811.485X

X = Rate of Interest on Bank term Loan = 15.86%

Rate of Interest on Debentures = 15.86 x 1.5 = 23.79%

Rate of Preference Dividend = 15.86 x 2.25 = 35.68%

(B) Calculation of WACC using Market Value weights

Sources	Market value	Weights (W)	Cost (K)	W x K
Equity share capital	37,50,000	0.456	15.000	6.83
Bank term loan	10,00,000	0.122	11.895	1.45
Debentures	14,40,000	0.175	17.843	3.12
Preference share capital	20,40,000	0.248	26.764	6.63
Total	**82,30,000**	**1**	**Ko /WACC**	**18.04**

WN – Calculation of Market Value of each capital sources

MV of Equity = 25,000 x 150 = 37,50,000

**MV of Reserves & Surplus = Already Included in the Equity Above

MV of Term Loan = 10,00,000

MV of Debentures = 15,000 x 96 = 14,40,000

MV of Pref. Shares = 20,000 x 102 = 20,40,000

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271 RTP **Jan 2025** Q5

The Capital Structure of Samyaktva Limited is as follows:

	Amount (in ₹)
12% Debentures	3,50,000
14% Pref. Shares	4,50,000
Equity shares (Face value of ₹ 10 each)	8,50,000
	16,50,000

Additional Information:

1. ₹ 100 per debentures redeemable at premium of 6% with floatation cost of 5% & 5 years of maturity. The current market price of the debenture is ₹ 115
2. ₹ 100 per preference shares redeemable at a premium of 10%, issued at discount of 2% with a floatation cost of 5% on the issue price. The current market price per preference share is ₹ 108. It has maturity of 10 years
3. An equity share has a floatation cost of ₹ 5 with a market price per share currently quoted at ₹ 30. Samyaktva Limited paid a last dividend of ₹ 4 and the company is expected to give an annual growth rate of 9% on the dividends. The company has a practice of paying all the earnings in the form of dividends.
4. Corporate Taxation rate is at 25%

CALCULATE WACC using market value weights

ICAI'S ANSWER

WN 1: Calculation of Cost of Debt

$$\text{Kd} = \frac{I(1-t) + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

RV = 100 + 6% = 106

n = term = 5 years

t = tax = 0.25

NP = Issue Price – Floatation cost

= 115 – 5% (Issue price will be at Market price and no Face Value)

= 109.25

$$\text{Kd} = \frac{12(1-0.25) + \frac{(106 - 109.25)}{5}}{\frac{(106 + 109.25)}{2}}$$

Therefore **Kd = 7.76%**

WN 2: Calculation of Cost of Preference Shares

$$\text{Kp} = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

RV = 100 + 10% = 110

n = term = 10 years

NP = Issue Price – Floatation cost

Issue Price = (108 – 2%) = 105.84

Net Proceeds = 105.84 (-) 5% = 100.55

$$\text{Kp} = \frac{14 + \frac{(110 - 100.55)}{10}}{\frac{(110 + 100.55)}{2}}$$

Therefore **Kp = 14.19%**

WN 3: Calculation of Cost of Equity

Since growth rate is given, Ke is to be calculated by using Gordon's formula

As per Gordon,

$$\text{Ke} = \frac{D_1}{P_0} + g$$

Where, D1 = Expected dividend at the end of Year 1

Po = Current Market Price (–) Floatation cost

G = growth rate in dividends

$$\text{Ke} = \frac{4 + 9\% \text{ x } 4}{30 - 5} + 0.09$$

Ke = 26.44%

Calculation of WACC using Market Value Weights

Sources	Amount of Capital (₹)	Weights (W)	Cost (K)	W X K
Debentures	4,02,500 (3,500 × 115)	0.1171	7.76 (WN 1)	0.9087
Preference shares	4,86,000 (4,500 × 108)	0.1413	14.19 (WN 2)	2.00
Equity shares	25,50,000 (85,000 × 30)	0.7416	26.44 (WN 3)	19.6079
	34,38,500			Ko = 22.52%

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272 RTP **Sep 2024** Q5

BS Ltd. has the following capital structure at book-value as on 31st March, 2024:

Particulars	(₹)
Equity share capital (10,00,000 shares)	3,00,00,000
11.5% Preference shares	60,00,000
10% Debentures	1,00,00,000
	4,60,00,000

The equity shares of the company are sold for ₹ 300. It is expected that the company will pay next year a dividend of ₹ 15 per equity share, which is expected to grow by 5% p.a. forever. Assume a 35% corporate tax rate.

Required:

(i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.

(ii) COMPUTE the new WACC, if the company raises an additional ₹ 50 lakhs debt by issuing 10 years 12% debentures but the yield on debentures of similar maturity and risk class is 13%; flotation cost is

2%. Face value of the debenture is ₹100. This would result in increasing the expected equity dividend to ₹ 20 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 250 per share.

ICAI'S ANSWER

(i) Computation of Weighted Average Cost of Capital based on existing capital structure

Source of Capital	Existing Capital structure	Weights	After tax cost of capital (%)	WACC (%)
	(₹)	(a)	(b)	(a) × (b)
Equity share capital (W.N.1)	3,00,00,000	0.652	10.00	6.52
11.5% Preference share capital	60,00,000	0.130	11.50	1.50
10% Debentures (W.N.2)	1,00,00,000	0.218	6.50	1.42
Total	4,60,00,000	1.000		9.44

Working Notes:

1. Cost of Equity Capital:

$$K_e = \frac{\text{Expected dividend}(D_1)}{\text{Current Market Price}(P_0)} + \text{Growth(g)}$$

$$= \frac{₹15}{₹300} + 0.05$$

$$= 10\%$$

2. Cost of 10% Debentures

$$K_d = \frac{\text{Interest}(1-t)}{\text{Net proceeds}}$$

$$= \frac{₹10,00,000(1-0.35)}{₹1,00,00,000}$$

$$= 0.065 \text{ or } 6.5\%$$

(ii) Computation of Weighted Average Cost of Capital based on new capital structure

Source of Capital	New Capital structure	Weights	After tax cost of capital (%)	WACC (%)
	(₹)	(a)	(b)	(a) x (b)
Equity share capital (W.N.3)	3,00,00,000	0.588	13.00	7.64
11.5% Preference share capital	60,00,000	0.118	11.50	1.36
10% Debentures (W.N.2)	1,00,00,000	0.196	6.50	1.27
12% Debentures (W.N.4)	50,00,000	0.098	9.21	0.90
Total	5,10,00,000	1.000		11.17

Working Notes:

3. Cost of Equity Capital:

$$K_e = \frac{₹20}{₹250} + 0.05$$

= 13%

4. Cost of 12% Debentures

$$K_d = \frac{I(1-t) + \left(\frac{RV - NP}{n}\right)}{\frac{RV + NP}{2}}$$

$$= \frac{₹12(1-0.35) + \left(\frac{₹100 - ₹90.31^*}{10 \text{ years}}\right)}{\frac{₹100 + ₹90.31^*}{2}}$$

$$= \frac{₹8.769}{₹95.155} = 0.0921$$

*Since yield on similar type of debentures is 13 per cent, the company would be required to offer debentures at discount.

Market price of debentures (approximation method)

= ₹ 12 ÷ 0.13 = ₹ 92.31

Sale proceeds from debentures = ₹ 92.31 – ₹ 2 (i.e., floatation cost) = ₹ 90.31

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273 RTP **May 2024** Q2

Totto Ltd. has following capital structure as on 31st December 2023, which is considered to be optimum:

	(₹)
12% Debenture	4,50,000
10% Preference share capital	1,50,000
Equity shares capital (2,00,000 shares)	24,00,000

The company's share has a current market price of ₹ 30.25 per share. The expected dividend per share in next year is 50 percent of the 2023 EPS. The EPS of last 10 years is as follows. The past trends are expected to continue:

Year	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
EPS (₹)	1.180	1.311	1.456	1.616	1.794	1.99	2.209	2.452	2.723	3.023

The company can issue 14 percent new debenture and 12 percent new preference share. The company's debenture is currently selling at ₹ 99. The new preference issue can be sold at a net price of ₹ 9.90, paying a dividend of ₹ 1.25 per share. The company's marginal tax rate is 50%.

(i) CALCULATE the after-tax cost (a) of new debts and new preference share capital, (b) of ordinary equity, assuming new equity comes from retained earnings.

(ii) CALCULATE the marginal cost of capital for the new funds raised.

(iii) How much can be spent for capital investment before new ordinary share must be sold? Marginal cost of capital remains to be constant. (Assuming that retained earnings available for next year's investment is 50% of 2023 earnings.)

(iv) What will be marginal cost of capital (cost of fund raised in excess of the amount calculated in part (iii) if the company can sell new ordinary shares of ₹ 22 per share? Assuming both the cost of debt and of preference share capital to be constant.

ICAI'S ANSWER

(i) **Calculation of after-tax cost of the followings:**

(a) New 14% Debentures $(K_d) = \frac{I(1-t)}{NP} = \frac{₹\ 14(1-0.5)}{₹\ 99}$

= 0.0707 or 7.07%

New 12% Preference Shares $(K_p) = \frac{PD}{NP} = \frac{₹\ 1.25}{₹\ 9.90}$

= 0.1263 or 12.63%

Where,

I = Interest

t = Tax rate

PD = Preference dividend

NP = Net proceeds

(b) Equity Shares (Retained Earnings) (K_e)

$= \frac{\text{Expected dividend}(D_1)}{\text{Current market price}(P_0)}$ +Growth rate(G)

$= \frac{50\%\ \text{of}\ ₹\ 3.023}{₹\ 30.25}$ +0.11* = 0.16 or 16%

* Growth rate (on the basis of EPS) is calculated as below :

$\frac{\text{EPS in current year} - \text{EPS in previous year}}{\text{EPS in previous year}} = \frac{₹\ 3.023 - ₹\ 2.723}{₹\ 2.723} = 0.11$

(Students may verify the growth trend by applying the above formula to last three or four years. Growth Rate is rounded off)

(ii) **Calculation of marginal cost of capital (on the basis of existing capital structure):**

Source of capital	Weight (a)	After tax Cost of capital (%) (b)	WACC (%) (a) × (b)
14% Debenture	0.15	7.07	1.0605
12% Preference shares	0.05	12.63	0.6315
Equity shares	0.80	16.00	12.800
Marginal cost of capital			14.492

(iii) The company can spend for capital investment before issuing new equity shares and without increasing its marginal cost of capital:

Retained earnings can be available for capital investment

= 50% of 2023 EPS × equity shares outstanding

= 50% of ₹ 3.023 × 2,00,000 shares = ₹3,02,300

Since, marginal cost of capital is to be maintained at the current level i.e. 14.492%, the retained earnings should be equal to 80% of total additional capital for investment.

Thus, investment before issuing equity $\left(\frac{₹\ 3{,}02{,}300}{80} \times 100\right)$

= ₹ 3,77,875

The remaining capital of ₹ 75,575 i.e. ₹ 3,77,875– ₹ 3,02,300 shall be financed by issuing 14% Debenture and 12% preference shares in the ratio of 3 : 1 respectively.

(iv) If the company spends more than ₹ 3,77,875 as calculated in part (iii) above, it will have to issue new shares at ₹ 22 per share.

The cost of new issue of equity shares will be:

$$K_e = \frac{\text{Expected dividend}(D_1)}{\text{Current market price}(P_0)} + \text{Growth rate}(g) = \frac{50\% \text{ of } ₹\ 3.023}{₹\ 22} + 0.11$$

= 0.1787 or 17.87%

Calculation of marginal cost of capital (assuming the existing capital structure will be maintained):

Source of capital	Weight (a)	Cost (%) (b)	WACC (%) (a) × (b)
14% Debenture	0.15	7.07	1.0605
12% Preference shares	0.05	12.63	0.6315
Equity shares	0.80	17.87	14.296
Marginal cost of capital			15.988

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274 RTP **Old P8 Nov 2023** Q2 old scheme

Jason Limited is planning to raise additional finance of ₹ 20 lakhs for meeting its new project plans. It has ₹ 4,20,000 in the form of retained earnings available for investment purposes. Further details are as following:

Debt / Equity Mix	30 / 70
Cost of Debt	
Upto ₹ 3,60,000	8 % (before tax)
Beyond ₹ 3,60,000	12 % (before tax)
Earnings per share	₹ 4
Dividend pay-out	50% of earnings
Current Market Price per share	₹ 44
Expected Growth rate in Dividend	10 %
Tax	40%

You are required:

(a) To determine the cost of retained earnings and cost of equity.

(b) To determine the post-tax average cost of additional debt.

(c) To determine the pattern for raising the additional finance, and

(d) Compute the overall weighted average after tax cost of additional finance.

ICAI'S ANSWER

(a) Cost of Equity / Retained Earnings (using dividend growth model)

$Ke = \frac{D_1}{P_0}$

where D1 = Do (1 + g) = 2 (1 + .10) = 2.2

$Ke = \frac{2.2}{44} + 0.10 = 0.15$ or 15 %

(b) Cost of Debt (Post Tax)

Kd = I (1-t)

Upto 3,60,000 Kd = .08 (1-0.4) = 0.048

Beyond 3,60,000 = .12 (1-0.4) = 0.072

Thus, post-tax cost of additional debt = 0.048 x 3,60,000 / 6,00,000 + 0.072 x 2,40,000/ 6,00,000 = 0.0288 + 0.0288 = 0.0576 or 5.76%

(c) Pattern for Raising Additional Finance

Debt = 20,00,000 x 30% = 6,00,000

Equity = 20,00,000 x 70 % = 14,00,000

Out of this total equity amount of ₹ 14,00,000 -

Equity Shares = 14,00,000 – 4,20,000

= 9,80,000

And Retained Earnings = 4,20,000

(d) Overall Weighted Average after tax cost of additional finance

WACC = Kd x Debt Mix + Ke x Equity Mix = 0.0576 x 30% + 0.15 x 70% = 0.01728 + 0.105 = 0.1223 or 12.23% (approx.)

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275 RTP **Old P8 May 2023** Q2 old scheme

Amrit Corporation has the following book value capital structure:

Equity Capital (50 lakh shares of ₹ 10 each).	₹ 5,00,00000
15% Preference share (50,000 shares ₹ 100 each)	₹ 50,00,000
Retained earnings	₹ 4,00,00,000
Debentures 14% (2,50,000 debentures ₹ 100 each)	₹ 2,50,00,000
Term loan 13%	₹ 4,00,00000

The companies last year earnings per share was ₹ 5, and it maintains a dividend pay-out ratio of 60% and returns on equity is 10%. The market price per share is ₹ 20.8. Preference share redeemable after 10

years is currently selling for ₹ 90 per share. Debentures redeemable after 6 years are currently selling for ₹ 75 per debenture. The income tax rate is 40%.

(a) CALCULATE the Weighted Average Cost of Capital (WACC) using market value proportions.

(b) DETERMINE the Marginal Cost of Capital (MACC) if it needs ₹ 5,00,00000 next year assuming the amount will be raised by 60% equity, 20% debt and 20% retained earnings. Equity issues will fetch a net price of ₹ 14 and cost of debt will be 13% before tax up to ₹ 40,00,000 and beyond ₹ 40,00,000 it will be 15% before tax.

ICAI'S ANSWER

(a) Calculation of Cost of Equity

(i) D0 = ₹ 5x 60%

D0 = ₹ 3

g = b x r

= (1-0.6) x 10% = 4%

D1 = D0 x (1 + g)

= 3 x (1 + 4%)

= 3 x 1.04 = 3.12

$$Ke = \frac{D_1}{P_0} + g$$

$$Ke = \frac{3.12}{20.8} + 0.04$$

Ke = 19%

(ii) Calculation of Cost of Preference Shares

N =10 years

NP = ₹ 90

PD = ₹ 15

RV = ₹ 100

$$Kp = \frac{PD + (RV - NP) / N}{(RV + NP)} \text{ x100}$$

$$Kp = \frac{15 + (100 - 90) / 10}{(100 + 90) / 2} \text{ x100}$$

Kp = 16/95 x 100

Kp= 16.84%

(iii) Calculation of Cost of Debentures

N = 6 years

NP = ₹ 75

Interest = ₹ 14

RV = ₹ 100

T = 40%

$$Kd = \frac{\text{int}\,(1 - t) + (RV - NP) / N}{(RV + NP) / 2} \times 100$$

$$Kd = \frac{14 \times (1 - 0.4) + (100 - 75) / 6}{(100 + 75) / 2} \times 100$$

$$Kd = \frac{8.4 - 4.17}{87.5} \times 100$$

Kd=14.37%

(iv) Cost of Term Loan

Kd = Interest rate (1-t)

Kd = 13% (1-40%)

Kd = 7.8%

Calculation of Weighted Average Cost of Capital (WACC) (using market weights)

Capital	Cost of Capital	Market Value		Market Value Weights	Product (Cost x weights)
Equity	19.00%	20.8 x 50,00,000	₹10,40,00,000	0.6218	11.81%
Preference Shares	16.84%	90 x 50,000	₹ 45,00,000	0.0269	0.45%
Debentures	14.37%	75 x 2,50,000	₹ 1,87,50,000	0.1121	1.61%
Term Loan	7.80%		₹ 4,00,00,000	0.2392	1.87%
Total			**₹16,72,50,000**	1	**15.74%**

WACC= 15.74%

(b) Calculation of Marginal Cost of Capital (MACC)

The required capital of ₹ 50,000,000 will be raised as follows:

Equity = 60% of ₹ 50,000,000 = ₹ 30,000,000

Deby = 20% of ₹ 50,000,000 = ₹10,000,000

Retained Earnings= 20% of ₹ 50,000,000 = ₹ 10,000,000

$$\text{Marginal Cost of Equity} = \frac{3.12}{1.4} + 0.04$$

= 26.28%

Marginal Cost of Debt

$$\text{Cost of Debt (before tax)} = \frac{13\% \text{ of } ₹\ 40,00,000 + 15\% \text{ of } ₹\ 60,00,000}{₹\ 1,00,00,000}$$

$$= \frac{₹\ 5,20,000 + ₹\ 9,00,000}{₹\ 1,00,00,000} = 14.2\%$$

Cost of Debt (after tax). = 14.2% (1-t)

= 14.2% (1-0.4)

= 8.52%

Calculation of marginal cost of capital

Capital	Cost of Capital	Value	Weights	Product (Cost x weights)
Equity	26.28%	₹ 3,00,00,000	0.6	15.77%
Reserves	26.28%	₹ 1,00,00,000	0.2	5.26%
Debt	8.52%	₹ 1,00,00,000	0.2	1.70%
Total		**₹ 5,00,00,000**	**1**	**22.73%**

Marginal Cost of Capital (MACC) = 22.73%

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276 RTP **Old P8 Nov 2022** Q2 old scheme

Bounce Ltd. evaluates all its capital projects using discounting rate of 15%. Its capital structure consists of equity share capital, retained earnings, bank term loan and debentures redeemable at par.

Rate of interest on bank term loan is 1.5 times that of debenture. Remaining tenure of debenture and bank loan is 3 years and 5 years respectively. Book value of equity share capital, retained earnings and bank loan is ₹ 10,00,000, ₹ 15,00,000 and ₹ 10,00,000 respectively. Debentures which are having book value of ₹ 15,00,000 are currently trading at ₹ 97 per debenture. The ongoing P/E multiple for the shares of the company stands at 5. You are required to CALCULATE the rate of interest on bank loan and debentures if tax rate applicable is 25%.

ICAI'S ANSWER

Let the rate of Interest on debenture be x

∴ Rate of Interest on loan = 1.5x

$$\therefore K_d \text{ on debentures} = \frac{\text{Int (1-t)} + \frac{\text{RV-NP}}{\text{n}}}{\frac{\text{RV+ NP}}{2}}$$

$$= \frac{100x(1-0.25) + \frac{100-97}{3}}{\frac{100+97}{2}}$$

$$= \frac{75x+1}{98.5}$$

∴ K_d on bank loan = 1.5x (1 – 0.25) = 1.125x

$$K_e = \frac{EPS}{MPS} = \frac{1}{MPS/EPS} = \frac{1}{P/E} = \frac{1}{5} = 0.2$$

$$K_Y = K_e = 0.2$$

Computation of WACC

Capital	Amount (₹)	Weights	Cost	Product
Equity	10,00,000	0.2	0.2	0.04
Reserves	15,00,000	0.3	0.2	0.06
Debentures	15,00,000	0.3	(75x+1)/98.5	(22.5x + 0.3)/98.5
Bank Loan	10,00,000	0.2	1.125x	0.225x
	50,00,000	1		0.1 + 0.225x + $\frac{22.5x + 0.3}{98.5}$

WACC = 15%

$\therefore 0.1 + 0.225x + \frac{22.5x}{98.5} + \frac{0.3}{98.5} = 0.15$

$\therefore 9.85+22.1625x+22.5x+0.3 = (0.15)\ (98.5)$

$\therefore 44.6625x = 14.775 - 9.85 - 0.3$

$\therefore 44.6625x = 4.625$

$\therefore x = \frac{4.625}{44.6625}$

$\therefore x = 10.36\ \%$

$\therefore$ Rate of interest on debenture = x = 10.36%

Rate of interest on Bank loan = 1.5x = (1.5) (10.36%) = 15.54%.

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277 RTP **Old P8 May 2022** Q2 old scheme

The information relating to book value (BV) and market value (MV) weights of Ex Limited is given below:

Sources	Book Value (₹)	Market Value (₹)
Equity shares	2,40,00,000	4,00,00,000
Retained earnings	60,00,000	-
Preference shares	72,00,000	67,50,000
Debentures	18,00,000	20,80,000

Additional information:

I. Equity shares are quoted at ₹ 130 per share and a new issue priced at ₹ 125 per share will be fully subscribed; flotation costs will be ₹ 5 per share on face value.

II. During the previous 5 years, dividends have steadily increased from ₹ 10 to ₹ 16.105 per share. Dividend at the end of the current year is expected to be ₹ 17.716 per share.

III. 15% Preference shares with face value of ₹ 100 would realise ₹ 105 per share.

IV. The company proposes to issue 11-year 15% debentures but the yield on debentures of similar maturity and risk class is 16%; flotation cost is 2% on face value.

V. Corporate tax rate is 30%.

You are required to DETERMINE the weighted average cost of capital of Ex Limited using both the weights.

ICAI'S ANSWER

(i) Cost of Equity $(K_e) = \frac{D_1}{P_0 - F} + g = \frac{₹\ 17.716}{₹\ 125 - ₹\ 5} + 0.10^*$

K_e = 0.2476

*Calculation of g:

₹ 10 $(1+g)^5$ = ₹ 16.105

Or, $(1+g)^5 = \frac{16.105}{10}$ = 1.6105

Table (FVIF) suggests that ₹ 1 compounds to ₹ 1.6105 in 5 years at the compound rate of 10 percent. Therefore, g is 10 per cent.

(ii) Cost of Retained Earnings $(K_r) = \frac{D_1}{P_0} + g = \frac{₹\ 17.716}{₹\ 130} + 0.10 = 0.2363$

(iii) Cost of Preference Shares $(K_p) = \frac{PD}{P_0} = \frac{₹\ 15}{₹\ 105} = 0.1429$

(iv) Cost of Debentures $(K_d) = \frac{I(1-t) + \left(\frac{RV - NP}{n}\right)}{\frac{RV + NP}{2}}$

$= \frac{₹\ 15\ (1\text{-}0.30) + \left(\frac{₹\ 100 - ₹\ 91.75^*}{11 \text{ years}}\right)}{\frac{₹\ 100 + ₹\ 91.75^*}{2}}$

$= \frac{₹\ 15 \times 0.70 + ₹\ 0.75}{₹\ 95.875} = \frac{₹\ 11.25}{₹\ 95.875} = 0.1173$

*Since yield on similar type of debentures is 16 per cent, the company would be required to offer debentures at discount.

Market price of debentures (approximation method)

= ₹ 15 ÷ 0.16 = ₹ 93.75

Sale proceeds from debentures = ₹ 93.75 – ₹ 2 (i.e., floatation cost) = ₹91.75

Market value (P_0) of debentures can also be found out using the present value method:

P_0 = Annual Interest × PVIFA (16%, 11 years) + Redemption value × PVIF (16%, 11 years)

P_0 = ₹ 15 × 5.0287 + ₹ 100 × 0.1954

P_0 = ₹ 75.4305 + ₹ 19.54 = ₹ 94.9705

Net Proceeds = ₹ 94.9705 – 2% of ₹ 100 = ₹ 92.9705

Accordingly, the cost of debt can be calculated

Total Cost of capital [BV weights and MV weights]

(Amount in (₹) lakh)

Source of capital	Weights		Specific Cost (K)	Total cost	
	BV	MV		(BV × K)	(MV × K)
Equity Shares	240	320**	0.2476	59.4240	79.2320
Retained Earnings	60	80**	0.2363	14.1780	18.9040
Preference Shares	72	67.50	0.1429	10.2888	9.6458
Debentures	18	20.80	0.1173	2.1114	2.4398
Total	390	488.30		86.0022	110.2216

**Market Value of equity has been apportioned in the ratio of Book Value of equity and retained earnings i.e., 240:60 or 4:1.

Weighted Average Cost of Capital (WACC):

Using Book Value = $\frac{₹\ 86.0022}{₹\ 390}$ = 0.2205 or 22.05%

Using Market Value = $\frac{₹\ 110.2216}{₹\ 488.30}$ = 0.2257 or 22.57%

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278 RTP **Old P8 Dec 2021** Q2 old scheme

Kalyanam Ltd. has an operating profit of ₹ 34,50,000 and has employed Debt which gives total Interest Charge of ₹ 7,50,000. The firm has an existing Cost of Equity and Cost of Debt as 16% and 8% respectively. The firm has a new proposal before it, which requires funds of ₹ 75 Lakhs and is expected to bring an additional profit of ₹ 14,25,000. To finance the proposal, the firm is expecting to issue an additional debt at 8% and will not be issuing any new equity shares in the market. Assume no tax culture.

You are required to CALCULATE the Weighted Average Cost of Capital (WACC) of Kalyanam Ltd.:

(i) Before the new Proposal

(ii) After the new Proposal

ICAI'S ANSWER

Workings:

(a) Value of Debt = $\frac{\text{Interest}}{\text{Cost of debt } (K_d)}$

= $\frac{₹\ 7,50,000}{0.08}$ = ₹ 93,75,000

(b) Value of equity capital = $\frac{\text{Operating profit - Interest}}{\text{Cost of equity } (K_e)}$

= $\frac{₹\ 34,50,000 - ₹\ 7,50,000}{0.16}$ = ₹ 1,68,75,000

(c) New Cost of equity (K_e) after proposal

= $\frac{\text{Increased Operating profit - Interest on Increased debt}}{\text{Equity capital}}$

= $\frac{(₹\ 34,50,000 + ₹\ 14,25,000) - (₹\ 7,50,000 + ₹\ 6,00,000)}{₹\ 1,68,75,000}$

$$= \frac{₹\ 48,75,000 - ₹\ 13,50,000}{₹\ 1,68,75,000} = \frac{₹\ 35,25,000}{₹\ 1,68,75,000} = 0.209 \text{ or } 20.9\%$$

(i) Calculation of Weighted Average Cost of Capital (WACC) before the new proposal

Sources	Amount (₹)	Weight	Cost of Capital	WACC
Equity	1,68,75,000	0.6429	0.160	0.1029
Debt	93,75,000	0.3571	0.080	0.0286
Total	**2,62,50,000**	**1**		**0.1315 or 13.15 %**

(ii) Calculation of Weighted Average Cost of Capital (WACC) after the new proposal

Sources	Amount (₹)	Weight	Cost of Capital	WACC
Equity	1,68,75,000	0.5000	0.209	0.1045
Debt	1,68,75,000	0.5000	0.080	0.0400
Total	**3,37,50,000**	**1**		**0.1445 or 14.45 %**

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279 RTP **Old P8 Jul 2021** Q2 old scheme

Indel Ltd. has the following capital structure, which is considered to be optimum as on 31st March, 2021:

Particulars	(₹)
14% Debentures	60,000
11% Preference shares	20,000
Equity Shares (10,000 shares)	3,20,000
	4,00,000

The company share has a market price of ₹ 47.20. Next year dividend per share is 50% of year 2020 EPS. The following is the uniform trend of EPS for the preceding 10 years which is expected to continue in future.

Year	EPS (₹)	Year	EPS (₹)
2011	2.00	2016	3.22
2012	2.20	2017	3.54
2013	2.42	2018	3.90
2014	2.66	2019	4.29
2015	2.93	2020	4.72

The company issued new debentures carrying 16% rate of interest and the current market price of debenture is ₹ 96.

Preference shares of ₹ 18.50 (with annual dividend of ₹ 2.22 per share) were also issued. The company is in 30% tax bracket.

(A) CALCULATE after tax:

(i) Cost of new debt

(ii) Cost of new preference shares

(iii) New equity share (assuming new equity from retained earnings)

(B) CALCULATE marginal cost of capital when no new shares are issued.

(C) DETERMINE the amount that can be spent for capital investment before new ordinary shares must be sold, assuming that the retained earnings for next year's investment is 50 percent of earnings of 2020.

(D) COMPUTE marginal cost of capital when the fund exceeds the amount calculated in (C), assuming new equity is issued at ₹ 40 per share?

ICAI'S ANSWER

(A) (i) Cost of new debt

$$K_d = \frac{I(1-t)}{P_0}$$

$$= \frac{₹\ 16(1-0.3)}{₹\ 96} = 0.11667$$

(ii) Cost of new preference shares

$$K_p = \frac{₹\ 2.22}{₹\ 18.5} = 0.12$$

(iii) Cost of new equity shares

$$K_e = \frac{D_1}{P_0} + g$$

$$= \frac{₹\ 2.36}{₹\ 47.20} + 0.10$$

= 0.05 + 0.10 = 0.15

Calculation of g when there is a uniform trend (on the basis of EPS)

$$= \frac{\text{EPS (2012) - EPS (2011)}}{\text{EPS (2011)}} = \frac{₹\ 2.20 - ₹\ 2.00}{₹\ 2.00} = 0.10 \text{ or } 10\%$$

Calculation of D_1

D_1 = 50% of 2020 EPS = 50% of ₹ 4.72 = ₹ 2.36

(B) Calculation of marginal cost of capital

Type of Capital	Proportion	Specific Cost	Product
(1)	**(2)**	**(3)**	**(2) × (3) = (4)**
Debentures	0.15	0.11667	0.0175
Preference Share	0.05	0.1200	0.0060
Equity Share	0.80	0.1500	0.1200
		Marginal cost of capital	0.1435

(C) The company can spend the following amount without increasing marginal cost of capital and without selling the new shares:

Retained earnings = 50% of EPS of 2020 × outstanding equity shares

= 50% of ₹ 4.72 × 10,000 shares = ₹ 23,600

The ordinary equity (Retained earnings in this case) is 80% of total capital.

So, ₹ 23,600 = 80% of Total Capital

∴ Capital investment before issuing equity shares = $\frac{₹\ 23,600}{0.80}$ = ₹ 29,500

(D) If the company spends in excess of ₹ 29,500, it will have to issue new equity shares at ₹ 40 per share.

∴ The cost of new issue of equity shares will be:

$$K_e = \frac{D_1}{P_0} + g = \frac{₹\ 2.36}{₹\ 40} + 0.10 = 0.159$$

The marginal cost of capital will be:

Type of Capital	Proportion	Specific Cost	Product
(1)	**(2)**	**(3)**	**(2) × (3) = (4)**
Debentures	0.15	0.11667	0.0175
Preference Shares	0.05	0.1200	0.0060
Equity Shares (New)	0.80	0.1590	0.1272
		Marginal cost of capital	0.1507

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280 RTP **Old P8 Nov 2020** Q2 old scheme

CALCULATE the WACC using the following data by using:

(a) Book value weights

(b) Market value weights

The capital structure of the company is as under:

Particulars	(₹)
Debentures (₹ 100 per debenture)	5,00,000
Preference shares (₹ 100 per share)	5,00,000
Equity shares (₹ 10 per share)	10,00,000
	20,00,000

The market prices of these securities are:

Debentures	₹ 105 per debenture
Preference shares	₹ 110 per preference share
Equity shares	₹ 24 each.

Additional information:

(i) ₹ 100 per debenture redeemable at par, 10% coupon rate, 4% floatation costs, 10-year maturity.

(ii) ₹ 100 per preference share redeemable at par, 5% coupon rate, 2% floatation cost and 10-year maturity.

(iii) Equity shares has ₹ 4 floatation cost and market price ₹ 24 per share.

The next year expected dividend is ₹ 1 with annual growth of 5%. The firm has practice of paying all earnings in the form of dividend.

Corporate tax rate is 30%. Use YTM method to calculate cost of debentures and preference shares.

ICAI'S ANSWER

(i) Cost of Equity (K_e)

$= \frac{D_1}{P_0 - F} + g = \frac{₹\ 1}{₹\ 24 - ₹\ 4} + 0.05 = 0.1$ or 10%

(ii) Cost of Debt (K_d)

Current market price (P_0) – floatation cost = I(1-t) × PVAF(r,10) + RV × PVIF(r,10)

₹ 105 – 4% of ₹ 105 = ₹ 10 (1-0.3) × PVAF (r,10) + ₹ 100 × PVIF (r,10)

Calculation of NPV at discount rate of 5% and 7%

Year	Cash flows (₹)	Discount factor @ 5%	Present Value	Discount factor @ 7%	Present Value (₹)
0	100.8	1.000	(100.8)	1.000	(100.8)
1 to 10	7	7.722	54.05	7.024	49.17
10	100	0.614	61.40	0.508	50.80
NPV			+14.65		-0.83

Calculation of IRR

$\text{IRR} = 5\% + \frac{14.65}{14.65 - (-0.83)}(7\% - 5\%) = 5\% + \frac{14.65}{15.48}(7\% - 5\%) = 6.89\%$

Cost of Debt (K_d) = 6.89%

(iii) Cost of Preference shares (K_p)

Current market price (P_0) – floatation cost = PD × PVAF(r,10) + RV × PVIF(r,10)

₹ 110 – 2% of ₹ 110 = ₹ 5 × PVAF (r,10) + ₹ 100 × PVIF (r,10)

Calculation of NPV at discount rate of 3% and 5%

Year	Cash flows (₹)	Discount factor @ 3%	Present Value	Discount factor @ 5%	Present Value (₹)
0	107.8	1.000	(107.8)	1.000	(107.8)
1 to 10	5	8.530	42.65	7.722	38.61
10	100	0.744	74.40	0.614	61.40
NPV			+9.25		-7.79

Calculation of IRR

$$\text{IRR} = 3\% + \frac{9.25}{9.25 - (-7.79)}(5\% - 3\%) = 3\% + \frac{9.25}{17.04}(5\% - 3\%) = 4.08\%$$

Cost of Preference Shares (K_p) = 4.08%

(a) Calculation of WACC using book value weights

Source of capital	Book Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	**(a)**	**(b)**	**(c) = (a)×(b)**
10% Debentures	5,00,000	0.25	0.0689	0.01723
5% Preference shares	5,00,000	0.25	0.0408	0.0102
Equity shares	10,00,000	0.50	0.10	0.05000
	20,00,000	1.00		0.07743

WACC (K_o) = 0.07743 or 7.74%

(b) Calculation of WACC using market value weights

Source of capital	Market Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	**(a)**	**(b)**	**(c) = (a)×(b)**
10% Debentures (₹ 105× 5,000)	5,25,000	0.151	0.0689	0.0104
5% Preference shares (₹ 110× 5,000)	5,50,000	0.158	0.0408	0.0064
Equity shares (₹ 24× 1,00,000)	24,00,000	0.691	0.10	0.0691
	34,75,000	1.000		0.0859

WACC (K_o) = 0.0859 or 8.59%

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281 RTP **Old P8 May 2020** Q2 old scheme

PK Ltd. has the following book-value capital structure as on March 31, 2020.

	(₹)
Equity share capital (10,00,000 shares)	2,00,00,000
11.5% Preference shares	60,00,000
10% Debentures	1,00,00,000
	3,60,00,000

The equity shares of the company are sold for ₹ 200. It is expected that the company will pay next year a dividend of ₹ 10 per equity share, which is expected to grow by 5% p.a. forever. Assume a 35% corporate tax rate.

Required:

(i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.

(ii) COMPUTE the new WACC, if the company raises an additional ₹50 lakhs debt by issuing 12% debentures. This would result in increasing the expected equity dividend to ₹12.40 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 160 per share.

ICAI'S ANSWER

(i) Computation of Weighted Average Cost of Capital based on existing capital structure

Source of Capital	Existing Capital structure (₹)	Weights	After tax cost of capital (%)	WACC (%)
		(a)	(b)	(a) × (b)
Equity share capital (W.N.1)	2,00,00,000	0.555	10.00	5.55
11.5% Preference share capital	60,00,000	0.167	11.50	1.92
10% Debentures (W.N.2)	1,00,00,000	0.278	6.50	1.81
	3,60,00,000	1.000		9.28

Working Notes (W.N.):

1. Cost of equity capital:

$$K_e = \frac{\text{Expected Dividend } (D_1)}{\text{Current Market Price per share } (P_0)} + \text{Growth } (g)$$

$$= \frac{₹10}{₹200} + 0.05 = 10\%$$

2. Cost of 10% Debentures:

$$= \frac{I(1-t)}{NP} = \frac{₹10,00,000(1-0.35)}{₹1,00,00,000} = 0.065 \text{ or } 6.5\%$$

(ii) Computation of Weighted Average Cost of Capital based on new capital structure

Source of Capital	New Capital structure (₹)	Weights	After tax cost of capital (%)	WACC (%)
		(b)	(a)	(a) × (b)
Equity share capital (W.N. 3)	2,00,00,000	0.488	12.75	6.10
Preference share	60,00,000	0.146	11.50	1.68
10% Debentures (W.N. 2)	1,00,00,000	0.244	6.50	1.59
12% Debentures (W.N.4)	50,00,000	0.122	7.80	0.95
	4,10,00,000	1.00		10.32

Working Notes (W.N.):

3. Cost of equity capital:

$$K_e = \frac{\text{Expected Dividend } (D_1)}{\text{Current Market Price per share } (P_0)} + \text{Growth } (g)$$

$$= \frac{₹12.4}{₹160} + 0.05 = 0.1275 \text{ or } 12.75\%$$

4. Cost of 12% Debentures

$$= \frac{₹6,00,000(1-0.35)}{₹50,00,000} = 0.078 \text{ or } 7.8\%$$

$$K_d = \frac{₹2,40,000(1-0.35)}{₹20,00,000} = 0.078 \text{ or } 7.8\%$$

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282 RTP **Old P8 Nov 2019** Q2 old scheme

KM Ltd. has the following capital structure on September 30, 2019:

Sources of capital	(₹)
Equity Share Capital (40,00,000 Shares of ₹ 10 each)	4,00,00,000
Reserves & Surplus	4,00,00,000
12% Preference Shares	2,00,00,000
9% Debentures	6,00,00,000
	16,00,00,000

The market price of equity share is ₹60. It is expected that the company will pay next year a dividend of ₹6 per share, which will grow at 10% forever. Assume 40% income tax rate.

You are required to COMPUTE weighted average cost of capital using market value weights.

ICAI'S ANSWER

Workings:

(i) Cost of Equity (Ke) = $\frac{D_1}{P_0} + g = \frac{₹6}{₹60} + 0.10 = 0.20 = 20\%$

(ii) Cost of Debentures (Kd) = I (1 - t) = 0.09 (1 - 0.4) = 0.054 or 5.4%

Computation of Weighted Average Cost of Capital (WACC using market value weights)

Source of capital	Market Value of capital (₹)	Weight	Cost of capital (%)	WACC (%)
9% Debentures	6,00,00,000	0.1875	5.40	1.01
12% Preference Shares	2,00,00,000	0.0625	12.00	0.75
Equity Share Capital (₹60 × 40,00,000 shares)	24,00,00,000	0.7500	20.00	15.00
Total	32,00,00,000	1.00		16.76

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283 MTP **Sep 2026 S-I** Q1(b) 5 marks

Using the financial information provided below, calculate the Weighted Average Cost of Capital (WACC). Determine the WACC by applying both the book value weights and the market value weights and present the results separately for each method. Calculate the WACC using the following data by using:

The capital structure of the company is as under:

Particulars	(₹)
Debentures (₹ 100 per debenture)	5,00,000
Preference shares (₹ 100 per share)	5,00,000
Equity shares (₹ 10 per share)	10,00,000
	20,00,000

The current market prices of the company's securities are as follows: each debenture is quoted at ₹ 105, each preference share is trading at ₹ 110, and each equity share has a market price of ₹ 24."

Additional information:

(i) ₹ 100 per debenture redeemable at par, 10% coupon rate, 4% floatation costs, 10-year maturity.

(ii) ₹ 100 per preference share redeemable at par, 5% coupon rate, 2% floatation cost and 10-year maturity.

(iii) Equity shares has ₹ 4 floatation cost and market price ₹ 24 per share.

The next year expected dividend is ₹ 1 with annual growth of 5%. The firm has practice of paying all earnings in the form of dividend. Corporate tax rate is 30%. Use YTM method to calculate cost of debentures and preference shares.

ICAI'S ANSWER

(i) Cost of Equity (K_e) = $\frac{D1}{P0 - F} + g = \frac{₹1}{₹24 - ₹4} + 0.05 = 0.1$ or 10%

(ii) Cost of Debt (K_d) Current market price (P_0) – floatation cost = I(1-t) × PVAF(r,10) + RV × PVIF(r,10) ₹ 105 – 4% of ₹ 105 = ₹ 10 (1-0.3) × PVAF (r,10) + ₹ 100 × PVIF (r,10) Calculation of NPV at discount rate of 5% and 7%

Year	Cash flows (₹)	Discount factor @ 5%	Present Value	Discount factor @ 7%	Present Value (₹)
0	100.8	1.000	(100.8)	1.000	(100.8)
1 to 10	7	7.722	54.05	7.024	49.17
10	100	0.614	61.40	0.508	50.80
NPV			+14.65		-0.83

Calculation of IRR

$$\text{IRR} = 5\% + \frac{14.65}{14.65 - (-0.83)}(7\% - 5\%) = 5\% + \frac{14.65}{15.48}(7\% - 5\%) = 6.89\%$$

Cost of Debt (K_d) = 6.89%

(iii) Cost of Preference shares (K_p) Current market price (P_0) – floatation cost = PD × PVAF(r,10) + RV × PVIF(r,10) ₹ 110 – 2% of ₹ 110 = ₹ 5 × PVAF (r,10) + ₹ 100 × PVIF (r,10)

Calculation of NPV at discount rate of 3% and 5%

Year	Cash flows (₹)	Discount factor @ 3%	Present Value	Discount factor @ 5%	Present Value (₹)
0	107.8	1.000	(107.8)	1.000	(107.8)
1 to 10	5	8.530	42.65	7.722	38.61
10	100	0.744	74.40	0.614	61.40
NPV			+9.25		-7.79

Calculation of IRR

$$\text{IRR} = 3\% + \frac{9.25}{9.25 - (-7.79)}(5\% - 3\%) = 3\% + \frac{9.25}{17.04}(5\% - 3\%) = 4.08\%$$

Cost of Preference Shares (K_p) = 4.08%

(a) Calculation of WACC using book value weights

Source of capital	Book Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	**(a)**	**(b)**	**(c) = (a)×(b)**
10% Debentures	5,00,000	0.25	0.0689	0.01723
5% Preference shares	5,00,000	0.25	0.0408	0.0102
Equity shares	10,00,000	0.50	0.10	0.05000
	20,00,000	1.00		0.07743

WACC (K_o) = 0.07743 or 7.74%

(b) Calculation of WACC using market value weights

Source of capital	Market Value	Weights	After tax cost of capital	WACC (K_o)
	(₹)	(a)	(b)	(c) = (a)×(b)
10% Debentures (₹ 105× 5,000)	5,25,000	0.151	0.0689	0.0104
5% Preference shares (₹ 110× 5,000)	5,50,000	0.158	0.0408	0.0064
Equity shares (₹ 24× 1,00,000)	24,00,000	0.691	0.10	0.0691
	34,75,000	1.000		0.0859

WACC (K_o) = 0.0859 or 8.59%

ICAI · P6 · MTP Sep 2026 · Series I · question p. 4-5 · answer p. 4-5

284 MTP **Sep 2026 S-II** MCQ 7 2 marks MCQ

If the weighting of equity in total capital is 1/3, that of debt is 2/3, the return on equity is 15% that of debt is 10% and the corporate tax rate is 32%, what is the Weighted Average Cost of Capital (WACC)?

A 10.533%

B 7.533%

C 9.533%

D 11.350%

ICAI'S ANSWER

(C) 9.533%

Weighted Average Cost of Capital (WACC)

After-tax Cost of Debt:

10% × (1 - 0.32) = 6.8%

$$\text{WACC} = \frac{1}{3}(15\%) + \frac{2}{3}(6.8\%)$$

= 5% + 4.533% = 9.533%

ICAI · P6 · MTP Sep 2026 · Series II · question p. 4 · answer p. 2

285 MTP **May 2026 S-II** Q3(a) 7 marks

The capital structure of RST Ltd., as at 31st March, 2025, is as under:

	₹
Equity Share Capital (₹ 100 each)	6,00,00,000
10% Cumulative Preference Shares (₹ 100 each)	2,00,00,000
12% Debentures	8,00,00,000
Retained Earnings	4,00,00,000

The equity shares are traded in the market and current market price of the share is ₹ 150 per share. The market value of preference share is ₹ 120 each. Market value of debentures is ₹ 8,60,00,000.

For the year ended on 31st March, 2025, the company has paid equity dividend @24% and anticipates that it will grow by 5% every year. The tax rate applicable to the company is 25%.

You are required to COMPUTE:

(i) Cost of capital of each source of finance for the company.

(ii) Weighted Average Cost of Capital (WACC) on the basis of book value weights.

(iii) Weighted Average Cost of Capital (WACC) based on market value weights.

ICAI'S ANSWER

(i) Cost of Equity – $K_e = \frac{D_1}{P_0} + g = \frac{₹\ 24(1+0.05)}{₹\ 150} + 0.05$

= 0.168 + 0.05 = 0.218 or 21.8%

Cost of Preference share = 10/100 = 0.10 or 10%

Alternatively - If cost of preference share is calculated on the basis of market price of preference share

Cost of Preference share = 10/120= 0.0833 or 8.33%

Cost of long term debt = I (1 - t) = 12 % (1 - 0.25) = 9%

Cost of retained earning i.e. K_e= 21.8%

(ii) **Weighted Average Cost of Capital (book value weights)**

Source	Amount (₹)	Weight	After tax cost of capital %	WACC %
Equity	6,00,00,000	0.3	21.8	6.54
Retained earnings	4,00,00,000	0.2	21.8	4.36
10% Preference shares	2,00,00,000	0.1	10	1
12% Long term debt	8,00,00,000	0.4	9	3.6
TOTAL	**20,00,00,000**	**1**		**15.5**

(iii) **Weighted Average Cost of Capital (market value weights)**

Source	Amount (₹)	Weight	After tax cost of capital %	WACC %
Equity	5,40,00,000	0.27	21.8	5.886
Retained Earning	3,60,00,000	0.18	21.8	3.924
10% Preference Shares	2,40,00,000	0.12	10	1.2
12% Long term debt	8,60,00,000	0.43	9	3.87
TOTAL	**20,00,00,000**	**1.0**		**14.88**

Alternatively using cost of preference share on the basis of market price

Source	Amount (₹)	Weight	After tax cost of capital %	WACC
Equity	5,40,00,000	0.27	21.8	5.886
Retained Earning	3,60,00,000	0.18	21.8	3.924
10% Preference Shares	2,40,00,000	0.12	8.33	1
12% Long term debt	8,60,00,000	0.43	9	3.87
TOTAL	**20,00,00,000**	**1.0**		**14.68**

ICAI · P6 · MTP May 2026 · Series II · question p. 8 · answer p. 8-9

286 MTP **Jan 2026 S-I** MCQ 8 1 mark MCQ

A company has a financial structure where equity is 60% of its total debt plus equity. Its cost of equity is 10% and gross loan interest is 9%. Corporation tax is paid at 30%. What is the company's weighted average cost of capital (WACC)?

A 7.55%

B 7.80%

C 8.52%

D 8.05%

ICAI'S ANSWER

(C) 8.52%

After-tax cost of debt

Kd = 9% (1-0.30) = 6.3%

WACC = (E/V) Ke + (D/V) Kd

= 0.6 (10%) + 0.4 (6.3%)

= 6% + 2.52% = 8.52%

ICAI · P6 · MTP Jan 2026 · Series I · question p. 4 · answer p. 4

287 MTP **Jan 2026 S-I** Q1(a) 5 marks

Divine Limited has the following capital structure, which it considers to be optimal:

Capital Structure	Weightage (in percentage)
Debt	25
Preference Shares	15
Equity Shares	60
	100

Divine Limited's expected net income this year is ₹34,285.72, its established dividend payout ratio is 30 per cent, its tax rate is 40 per cent, and investors expect earnings and dividends to grow at a constant

rate of 9 per cent in the future. It paid a dividend of ₹ 3.60 per share last year, and its shares currently sell at a price of ₹ 54 per share.

Divine Limited requires additional funds which it can obtain in the following ways:

- Preference Shares: New preference shares with a dividend of ₹ 11 can be sold to the public at a price of ₹95 per share.
- Debt: Debt can be sold at an interest rate of 12 per cent.

You are required to:

(i) DETERMINE the cost of each capital structure component; and

(ii) COMPUTE the weighted average cost of capital (WACC) of Divine Limited.

ICAI'S ANSWER

(i) Computation of Costs of Different Components of Capital Equity Shares

$$K_e = \frac{D_1}{P_0} + g = \frac{D_0(1+g)}{P_0} + g$$

$$= \frac{₹\ 3.60(1.09)}{₹\ 54} + 0.09 = 0.0727 + 0.09 = 16.27\%.$$

Preference Shares

$$K_p = \frac{\text{Preference Share Dividend}}{P_0} = \frac{₹11}{₹\ 95} = 11.58\%.$$

Debt at K_d = 12%

K_d (1 – T) = 12 (1 – 0.4) = 12% (0.6) = 7.20%.

(ii) Weighted Average Cost of Capital (WACC)

WACC = $W_d K_d$ (1 – T) + $W_p K_p$ + $W_e K_e$

WACC = 0.25 (7.2%) + 0.15 (11.58%) + 0.60 (16.27%)

= 1.8 + 1.737 + 9.762 = 13.30%.

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288 MTP **Jan 2026 S-II** Q4(b) 4 marks

TechWave Ltd. plans to buy new machinery worth ₹ 50 lakh to increase production. The Finance Manager says that before investing, the company must understand the cost of capital. In this regard you are required to EXPLAIN the meaning of cost of capital. Also, briefly STATE three reasons why cost of capital is significant for decision-making at TechWave Ltd.

ICAI'S ANSWER

Cost of capital is the return expected by the providers of capital (i.e. shareholders, lenders and the debt-holders) to the business as a compensation for their contribution to the total capital.

The cost of capital is important to arrive at correct amount and helps the management or an investor to take an appropriate decision. The correct cost of capital helps in the following decision making:

(i) **Evaluation of investment options:** The estimated benefits (future cash flows) from available investment opportunities (business or project) are converted into the present value of benefits by **discounting them with the relevant cost of capital**. Here it is pertinent to mention that every investment option may have different cost of capital hence it is very important to use the cost of capital which is relevant to the options available.

(ii) **Financing Decision:** When a finance manager has to choose one of the two sources of finance, he can simply compare their cost and choose the source which has lower cost. Besides cost, he also considers financial risk and control.

(iii) **Designing of optimum credit policy:** While appraising the credit period to be allowed to the customers, the cost of allowing credit period is compared against the benefit/ profit earned by providing credit to customer of segment of customers. Here cost of capital is used to arrive at the present value of cost and benefits received.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 8 · answer p. 13-14

289 MTP **Sep 2025 S-II** MCQ 6 2 marks MCQ

ABC Ltd. has the following capital structure based on market values:

Equity: ₹ 60 lakhs

Debt: ₹ 40 lakhs

Cost of Equity (Retained Earnings): 14%

Cost of New Equity (after accounting for 5% flotation cost): 15%

Cost of Debt: 10%

Corporate Tax Rate: 30%

The company plans to raise additional capital by issuing new equity instead of using retained earnings. The capital structure weights will remain unchanged. What will be the impact on the Weighted Average Cost of Capital (WACC)?

A WACC will increase to 11.8%

B WACC will remain unchanged at 11.2%

C WACC will decrease to 8.8%

D WACC will increase only if debt increases

ICAI'S ANSWER

(A) WACC will increase to 11.8%

Initial WACC (using retained earnings):

$$= \frac{60}{60+40} \times 14\% + \frac{40}{60+40} \times 10\% \times (1 - 0.30)$$

$= 0.6 \times 14\% + 0.4 \times 7\%$

$= 8.4\% + 2.8\% = 11.2\%$

WACC with new equity (15% cost of equity due to flotation cost):

$= 0.6 \times 15\% + 0.4 \times 7\%$

$= 9\% + 2.8\% = 11.8\%$

ICAI · P6 · MTP Sep 2025 · Series II · question p. 3 · answer p. 3

290 MTP **Sep 2025 S-II** Q1(c) 5 marks

ABC Ltd. has the following capital structure, which is considered to be optimum as on 31st March, 2025.

Particulars	Amount (₹)
14% Debentures	60,000
11% Preference shares	20,000
Equity Shares (10,000 shares)	3,20,000
	4,00,000

The company share has a market price of ₹ 19.67. Next year dividend per share is 50% of year 2024 EPS. Following is the uniform trend of EPS for the preceding 10 years which is expected to continue in future:

Year	EPS (₹)	Year	EPS (₹)
2015	1.00	2020	1.61
2016	1.10	2021	1.77
2017	1.21	2022	1.95
2018	1.33	2023	2.15
2019	1.46	2024	2.36

The company issued new debentures carrying 16% rate of interest and the current market price of debenture is ₹ 96.

Preference shares of ₹ 9.20 (with annual dividend of ₹ 1.1 per share) were also issued. The company is in 50% tax bracket.

(A) CALCULATE after tax:

(i) Cost of new debt

(ii) Cost of new preference shares

(iii) Cost of new equity share (assuming new equity from retained earnings)

(B) CALCULATE marginal cost of capital when no new shares are issued.

(C) DETERMINE the amount that can be spent for capital investment before new ordinary shares must be sold. Assuming that the retained earnings for next year's investment is 50 percent of 2024.

(D) COMPUTE marginal cost of capital when the fund exceeds the amount calculated in (C), assuming new equity is issued at ₹ 25 per share.

ICAI'S ANSWER

(A) (i) Cost of new debt

$$K_d = \frac{I(1-t)}{P_0}$$

$$= \frac{₹16\ (1-0.5)}{₹96} = 0.0833$$

(ii) Cost of new preference shares

$$K_p = \frac{PD}{P_0} = \frac{₹1.1}{₹9.2} = 0.12$$

(iii) Cost of new equity shares

$K_e = \frac{D_1}{P_0} + g$

$= \frac{₹1.18}{₹19.67} + 0.10 = 0.06 + 0.10 = 0.16$

Calculation of g when there is a uniform trend (on the basis of EPS)

$= \frac{\text{EPS (2016)} - \text{EPS (2015)}}{\text{EPS (2015)}}$

$= \frac{₹1.10 - ₹1.00}{₹1.00} = 0.10 \text{ or } 10\%$

Calculation of D_1

D_1 = 50% of 2024EPS = 50% of 2.36 = ₹ 1.18

(B) Calculation of marginal cost of capital

Type of Capital	Proportion	Specific Cost	Product
(1)	**(2)**	**(3)**	**(2) × (3) = (4)**
Debenture	0.15	0.0833	0.0125
Preference Share	0.05	0.1200	0.0060
Equity Share	0.80	0.1600	0.1280
Marginal cost of capital			**0.1465**

(C) The company can spend the following amount without increasing marginal cost of capital and without selling the new shares:

Retained earnings = 50% of EPS of 2024 × outstanding equity shares

= 0.50 × ₹2.36 × 10,000 shares = ₹ 11,800

The ordinary equity (Retained earnings in this case) is 80% of total capital

So, ₹11,800 = 80% of Total Capital

∴ Capital investment before issuing equity shares = $\frac{₹11,800}{0.80}$ = ₹ 14,750

(D) If the company spends in excess of ₹ 14,750, it will have to issue new equity shares at ₹25 per share.

∴ The cost of new issue of equity shares will be = $\frac{D_1}{P_0} + g$

$= \frac{₹1.18}{₹25} + 0.10 = 0.1472$

The marginal cost of capital will be:

Type of Capital	Proportion	Specific Cost	Product
(1)	**(2)**	**(3)**	**(2) × (3) = (4)**
Debentures	0.15	0.0833	0.0125
Preference Shares	0.05	0.1200	0.0060
Equity Shares (New)	0.80	0.1472	0.1178
			0.1363

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291 MTP **May 2025 S-I** Q2(a) 6 marks

Q Ltd. has the following capital structure at book-value as on 31st March 2024:

Particulars	(₹)
Equity share capital (10,00,000 shares)	4,00,00,000
12% Preference shares	80,00,000
11% Debentures	2,00,00,000
	6,80,00,000

The equity shares of the company are sold for ₹ 400. It is expected that next year the company will pay a dividend of ₹ 20 per equity share, which is expected to grow by 5% p.a. forever. Assume a 30% corporate tax rate.

Required:

(i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.

(ii) COMPUTE the new WACC, if the company raises an additional ₹ 50 lakhs debt by issuing 12% debentures. This would result in increasing the expected equity dividend to ₹ 25 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 300 per share.

ICAI'S ANSWER

(i) Computation of Weighted Average Cost of Capital based on existing capital structure

Source of Capital	Existing Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) × (b)
Equity share capital (W.N.1)	4,00,00,000	0.588	10.00	5.88
12% Preference share capital	80,00,000	0.118	12.00	1.42
11% Debentures (W.N.2)	2,00,00,000	0.294	7.70	2.26
Total	6,80,00,000	1.000		9.56

Working Notes:

1. Cost of Equity Capital:

$$\text{Ke} = \frac{\text{Expected dividend } (D_1)}{\text{Current Market Price } (P_0)} + \text{Growth}(g)$$

$$= \frac{20}{400} + 0.05$$

$$= 10\%$$

2. Cost of 10% Debentures

$$\text{Kd} = \frac{\text{Interest}(1-t)}{\text{Net proceeds}}$$

$$= \frac{22{,}00{,}000\ (1\text{- }0.30)}{2{,}00{,}00{,}000}$$

= 0.077 or 7.7%

(ii) Computation of Weighted Average Cost of Capital based on new capital structure

Source of Capital	New Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) x (b)
Equity share capital (W.N.3)	4,00,00,000	0.548	13.33	7.30
12% Preference share capital	80,00,000	0.110	12.00	1.32
11% Debentures (W.N.2)	2,00,00,000	0.274	7.70	2.11
12% Debentures (W.N.4)	50,00,000	0.068	8.40	0.57
Total	7,30,00,000	1.000		11.30

Working Notes:

3. Cost of Equity Capital:

$$\text{Ke} = \frac{25}{300} + 0.05$$

= 13.33%

4. Cost of 12% Debentures

$$Kd = \frac{6,00,000\ (1- 0.30)}{50,00,000}$$

= 0.084 or 8.4%

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292 MTP **Jan 2025 S-I** Q1(b) 5 marks

P Ltd. has the following capital structure at book-value as on 31st March, 2024:

Particulars	(₹)
Equity share capital (1,00,000 shares)	10,00,000
12% Preference shares	15,00,000
10% Debentures	15,00,000
	40,00,000

Additional Information:

1. The equity shares of P Ltd. are currently traded at ₹ 100 per share.
2. The company expects to pay a dividend of ₹ 5 per equity share next year, with dividends projected to grow perpetually at a rate of 5% p.a.
3. The corporate tax rate is 35%.

Requirements:

1. CALCULATE the Weighted Average Cost of Capital (WACC) based on the current capital structure.
2. RECALCULATE the WACC if the company raises an additional ₹ 5 lakhs of debt by issuing 12% debentures. This change will result in:
   - An increase in the expected equity dividend to ₹ 7 per share while the growth rate remains constant at 5%.
   - A decrease in the market price of equity shares to ₹90 per share

ICAI'S ANSWER

(i) Computation of Weighted Average Cost of Capital based on existing capital structure

Source of Capital	Existing Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) × (b)
Equity share capital (W.N.1)	10,00,000	0.250	10.000	2.500
12% Preference share capital	15,00,000	0.375	12.000	4.500
10% Debentures (W.N.2)	15,00,000	0.375	6.500	2.438
Total	40,00,000	1.000		9.438

Working Notes:

1. Cost of Equity Capital:

$$K_e = \frac{\text{Expected dividend}(D_1)}{\text{Current Market Price}(P_0)} + \text{Growth}(g)$$

$$= \frac{₹5}{₹100} + 0.05$$

= 10%

2. Cost of 10% Debentures

$$K_d = \frac{\text{Interest}(1\text{- } t)}{\text{Net proceeds}}$$

$$= \frac{₹1{,}50{,}000\ (1 - 0.35)}{₹15{,}00{,}000}$$

= 0.065 or 6.5%

(ii) Computation of Weighted Average Cost of Capital based on new capital structure

Source of Capital	New Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) x (b)
Equity share capital (W.N.3)	10,00,000	0.222	12.777	2.836
12% Preference share capital	15,00,000	0.334	12.000	4.000
10% Debentures (W.N.2)	15,00,000	0.333	6.500	2.165
12% Debentures (W.N.4)	5,00,000	0.111	7.800	0.866
Total	45,00,000	1.000		9.867

Working Notes:

3. Cost of Equity Capital:

$$K_e = \frac{₹7}{₹90} + 0.05$$

= 12.777%

4. Cost of 12% Debentures

$$K_d = \frac{₹60{,}000\ (1 - 0.35)}{₹5{,}00{,}000}$$

= 0.078 or 7.8%

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293 MTP **Sep 2024 S-I** MCQ 7 2 marks MCQ

Assuming Ke = 11%, Kd = 8% and Ko = 10%, Debt Equity ratio of the company

A 2:3

B 3:2

C 1:2

D 2:1

ICAI'S ANSWER

Answer key: **(C)**

(c) 1:2

Item	Cost	Weight	Product
Debt	8%	W	8W
Equity	11%	1 – W	11 – 11W
			WACC = 10

Wd = 1/3 and We = 2/3 Debt Equity Ratio = 1/2

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294 MTP **Sep 2024 S-I** Q1(b) 5 marks

Gitarth Limited has a current debt equity ratio of 3:7. The company is presently considering several alternative investment proposals costing less than ₹ 25 lakhs. The company will always raise the funds required without disturbing its current capital structure ratio.

The cost of raising debt and equity are as follows-

Cost of Project	Kd	Ke
Upto 5 lakhs	10%	12%
Above 5 lakhs & upto 10 lakhs	12%	13.5%
Above 10 lakhs & upto 20 lakhs	13%	15%
Above 20 lakhs	14%	16%

Corporate tax rate is 30%, CALCULATE:

i) Cut off rate for two Projects I & Project II whose fund requirements are 15 lakhs & ₹ 26 lakhs respectively.

ii) If a project is expected to give an after-tax return of 13%, determine under what conditions it would be acceptable.

ICAI'S ANSWER

Calculation of slab wise Overall Cost of Capital

(i)

Project Cost	Capital Source	Weights (w)	Cost (k)	w x k (%)
Upto 5 Lakhs	Debt	0.3	10	3
	Equity	0.7	12	8.4

Project Cost	Capital Source	Weights (w)	Cost (k)	w x k (%)
			Ko	**11.4**
Above 5 lakhs upto 10 lakhs	Debt	0.3	12	3.6
	Equity	0.7	13.5	9.45
			Ko	**13.05**
Above 10 lakhs upto 20 lakhs	Debt	0.3	13	3.9
	Equity	0.7	15	10.5
			Ko	**14.4**
Above 20 lakhs	Debt	0.3	14	4.2
	Equity	0.7	16	11.2
			Ko	**15.4**

Cost of Raising funds for Project I

Total Capital	Ko(%)	Total Cost (in ₹)
5,00,000	11.40	57,000
5,00,000	13.05	65,250
5,00,000	14.40	72,000
15,00,000		**1,94,250**

Overall COC (%) = Total Cost (in ₹) / Total Capital

= 1,94,250/15,00,000 * 100

= 12.95 %

Cost of Raising funds for Project II

Total Capital	Ko(%)	Total Cost (in ₹)
5,00,000	11.4	57,000
5,00,000	13.05	65,250
10,00,000	14.4	1,44,000
6,00,000	15.4	92,400
26,00,000		**3,58,650**

Overall COC (%) = 358650 / 2600000 * 100 = **13.79%**

(ii) If any project is expected to give an after-tax return of 13%, it can be accepted only if the maximum Overall COC (%) of that project equals 13% or less, as at 13%, project would be at break-even i.e earning 13% from the project and incurring 13% COC.

So, under that scenario, Project I can be taken as its COC is 12.95% whereas Project II can't be taken as its COC is 13.79%.

Maximum Value of the Project that can be taken at 13% is approx. (Using IRR technique Interpolation)

At 15 Lakhs Ko = 12.95%

At 26 Lakhs Ko = 13.79%

By interpolation, maximum value of Project at 13% will be

15 Lakhs + {(0.05 x 11)/0.84}

= 15.6548 lakhs

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295 MTP **Sep 2024 S-II** Q3(a) 6 marks

Calculate the WACC using the following data by using Market Value weights:

Particulars	₹
Equity Shares (₹ 10 per equity share)	15,00,000
Reserves & Surplus	5,00,000
Preference Shares (₹ 100 per preference share)	7,50,000
Debentures (₹ 100 per debenture)	5,50,000

The market prices of these securities are:

Debentures - ₹ 105 per debenture,

Preference shares - ₹115 per preference share

Equity shares - ₹ 27 per equity share

Additional information:

(1) ₹ 100 FV per debenture redeemable at premium of 10%, 10% coupon rate, 4% floatation costs, 10-year maturity.

(2) ₹ 100 FV per preference share redeemable at par, 12% coupon rate, 2% floatation cost and 10-year maturity.

(3) Equity shares have ₹ 4.5 floatation cost and market price of 27 per share.

The last dividend paid by the company was ₹ 2 which is expected to grow at an annual growth rate of 9%. The firm has the practice of paying all earnings as a dividend.

The corporate tax rate is 25%. To calculate the overall cost of debt & preference shares, take the average of their respective costs using YTM & approximation method.

ICAI'S ANSWER

WN-1 : Calculation of Cost of Debt (Kd)

Approximation Method = $\frac{\text{Int } (1\text{-t})+(\text{RV}-\text{NP})/\text{N}}{(\text{RV}+\text{NP})\ /\ 2}$

RV = 100 + 10% = 110, NP = 105 - 4% = 100.8

= $\frac{10\ (1\text{- }0.25)+(110-100.8)/10}{(110+100.8)\ /\ 2}$ = 7.99%

YTM Method:

CMP (Po) (-) Floatation Cost = {Int(1-t) × PVAF (r%,10years)} + {RV × PVIF (r%,10th Year)}

105 - 4% = {10 (1 – 0.25) × PVAF (r%,10 years)} + {110 × PVIF (r%,10th year)}

Using trial and error method, NPV at 5% & 10%

Year	Cash flows	Disc Factor @ 5%	PV (₹)	Disc Factor @ 10%	PV (₹)
0	-100.8	1	-100.8	1	-100.8
1 to 10	7.5	7.7217	57.91275	6.1446	46.0845
10	110	0.6139	67.529	0.3855	42.405
			24.64175		**-12.3105**

IRR = 5+$\frac{24.64175}{24.64175-(-12.3105)}$ X (10-5) = **8.33%**

Therefore overall cost of debt (Kd) = (7.99 + 8.33) / 2 = 8.16%

WN-2 : Calculation of Cost of Preference (Kp)

Approximation Method = $\frac{\text{Pref. Div.}+(\text{RV}-\text{NP})/\text{N}}{(\text{RV}+\text{NP})\ /\ 2}$

RV = 100 NP = 115 - 2% = 112.7

= $\frac{12+(100-112.7)/10}{(100+112.7)\ /\ 2}$ **= 10.09%**

YTM Method:

CMP (Po) (-) Floatation Cost = {Pref Div × PVAF (r%,10years)} + {RV × PVIF (r%,10th Year)}

115 - 2% = {12 × PVAF (r%,10 years)} + {100 × PVIF (r%,10th year)}

Using trial and error method, NPV at 5% & 10%

Year	Cash flows	Disc Factor @ 5%	PV (₹)	Disc Factor @ 10%	PV (₹)
0	-112.7	1	-112.7	1	-112.7
1 to 10	12	7.7217	92.6604	6.1446	73.7352
10	100	0.6139	61.39	0.3855	38.55
			41.3504		**-0.4148**

IRR = 5+$\frac{41.3504}{41.3504-(-0.4148)}$ X (10-5) = **9.95%**

Therefore, overall cost of debt (Kp) = (10.09 + 9.95) / 2 = 10.02%

WN-3 : Calculation of Cost of equity (Ke)

Ke = {D1 / (Po - Floatation)} + G

= {2+9% / 27 – 4.5} + 0.09

= 18.69%

Calculation of WACC using market value weights

Source of Capital	Working	Market Value	Weights	Cost (K)	WACC (Ko)
		(₹)	**(A)**	**(B)**	**(A x B)**
Equity	27 × 150000	40,50,000	0.7377	18.69	13.7877
Reserves	Included in equity	-	-	-	-
Preference	115 × 7500	8,62,500	0.1571	10.02	1.5741
Debentures	105 × 5500	5,77,500	0.1052	8.16	0.8584
		54,90,000	**1**		**16.22%**

WACC (Ko) = 16.22%

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296 MTP **May 2024 S-I** Q3(a) 8 marks

Ram Ltd evaluates all its capital projects using discounting rate of 16%. Its capital structure consists of equity share capital, retained earnings, bank term loan and debentures redeemable at par. Rate of interest on bank term loan is 1.4 times that of debenture. Remaining tenure of debenture and bank loan is 4 years and 6 years respectively. Book value of equity share capital, retained earnings and bank loan is ₹ 20,00,000, ₹ 30,00,000 and ₹ 20,00,000 respectively. Debentures which are having book value of ₹ 30,00,000 are currently trading at ₹ 98 per debenture. The ongoing PE multiple for the shares of the company stands at 4.

You are required to:

(i) CALCULATE the rate of interest on bank loan and

(ii) CALCULATE the rate of interest on debentures

Tax rate applicable is 30%.

ICAI'S ANSWER

Working Note:

Let the rate of Interest on debenture be x

∴ Rate of Interest on loan = 1.4x

$$\therefore \text{ kd on debentures} = \frac{Int(1-t) + \frac{RV - NP}{n}}{\frac{RV + NP}{2}}$$

$$= \frac{100x(1-0.30)+\frac{100-98}{4}}{\frac{100+98}{2}}$$

$$= \frac{70x+0.5}{99}$$

∴ Kd on bank loan = 1.4 x (1 – 0.30) = 0.98x

Ke = $\frac{EPS}{MPS} = \frac{1}{MPS/EPS} = \frac{1}{PE} = \frac{1}{4}$ = 0.25

Ke = 0.25

Computation of WACC

Capital	Amount	Weights	Cost	Product
Equity	20,00,000	0.2	0.25	0.05
Reserves	30,00,000	0.3	0.25	0.075
Debentures	30,00,000	0.3	(70x+0.5)/99	(21x+0.15)/99
Bank Loan	20,00,000	0.2	0.98x	0.196x
	1,00,00,000	1		0.125+0.196x + (21x + 0.15)/99

WACC = 16%

∴ 0.125+0.196x+ $\frac{21x+0.15}{99}$ = 0.16

∴ 12.375+19.404x+21x+0.15 = (0.16)(99)

∴ 40.404x = 15.84 – 12.525

∴ 40.404x = 3.315

∴ x = $\frac{3.315}{40.404}$

∴ x = 8.20%

(i) Rate of interest on debenture = x = 8.20%

(ii) Rate of interest on Bank loan = 1.4x = (1.4)(8.20%) = 11.48%.

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297 MTP **Old P8 Nov 2023 S-II** Q3 10 marks old scheme

Q Ltd. has the following capital structure at book-value as on 31st March 2022:

Particulars	(₹)
Equity share capital (10,00,000 shares)	4,00,00,000
12% Preference shares	80,00,000
11% Debentures	2,00,00,000
	6,80,00,000

The equity shares of the company are sold for ₹ 400. It is expected that the company will pay next year a dividend of ₹ 20 per equity share, which is expected to grow by 5% p.a. forever. Assume a 30% corporate tax rate.

Required:

(i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.

(ii) COMPUTE the new WACC, if the company raises an additional ₹ 50 lakhs debt by issuing 12% debentures. This would result in increasing the expected equity dividend to ₹ 25 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 300 per share.

ICAI'S ANSWER

(i) Computation of Weighted Average Cost of Capital based on existing capital structure

Source of Capital	Existing Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) × (b)
Equity share capital (W.N.1)	4,00,00,000	0.588	10.00	5.88
12% Preference share capital	80,00,000	0.118	12.00	1.42
11% Debentures (W.N.2)	2,00,00,000	0.294	7.70	2.26
Total	6,80,00,000	1.000		9.56

Working Notes:

1. Cost of Equity Capital:

$$K_e = \frac{\text{Expected dividend } (D_1)}{\text{Current Market Price } (P_0)} + \text{Growth(g)}$$

$$= \frac{20}{400} + 0.05$$

= 10%

2. Cost of 10% Debentures

$$K_d = \frac{\text{Interest}(1-t)}{\text{Net proceeds}}$$

$$= \frac{22,00,000\ (1\text{-}0.30)}{2,00,00,000}$$

= 0.077 or 7.7%

(ii) Computation of Weighted Average Cost of Capital based on new capital structure

Source of Capital	New Capital structure (₹)	Weights (a)	After tax cost of capital (%) (b)	WACC (%) (a) x (b)
Equity share capital (W.N.3)	4,00,00,000	0.548	13.33	7.30
12% Preference share capital	80,00,000	0.110	12.00	1.32
11% Debentures (W.N.2)	2,00,00,000	0.274	7.70	2.11
12% Debentures (W.N.4)	50,00,000	0.068	8.40	0.57
Total	7,30,00,000	1.000		11.30

Working Notes:

3. Cost of Equity Capital:

$$K_e = \frac{25}{300} + 0.05$$

$= 13.33\%$

4. Cost of 12% Debentures

$$K_d = \frac{6,00,000\ (1\text{-}0.30)}{50,00,000}$$

$= 0.084$ or 8.4%

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298 MTP **Old P8 Nov 2022 S-I** Q1(a) 5 marks old scheme

The capital structure of a Company is given below:

Source of capital	Book Value (₹)
Equity shares @ ₹ 100 each	24,00,000
9% Cumulative preference shares @ ₹ 100 each	4,00,000
11% Debentures	12,00,000
	40,00,000

The company had paid equity dividend @ 25% for the last year which is likely to grow @ 5% every year. The current market price of the company's equity share is ₹ 200.

Considering corporate tax @ 30%, you are required to CALCULATE:

(i) Cost of capital for each source of capital.

(ii) Weighted average cost of capital.

ICAI'S ANSWER

(i) Calculation of Cost of Capital for each source of capital:

(a) Cost of Equity share capital:

$$K_e = \frac{D_0(1+g)}{\text{Market Price per share}(P_0)} + g = \frac{25\% \times ₹100(1+0.05)}{₹200} + 0.05$$

$$= \frac{₹26.25}{₹200} + 0.05 = 0.18125 \text{ or } \mathbf{18.125\%}$$

(b) Cost of Preference share capital (K_p) = **9%**

(c) Cost of Debentures (K_d) = r (1 – t)

= 11% (1 – 0.3) = **7.7%**

(ii) Weighted Average Cost of Capital

Source	Amount (₹)	Weights	After tax Cost of Capital (%)	WACC (%)
		(a)	**(b)**	**(c) = (a) × (b)**
Equity share	24,00,000	0.60	18.125	10.875
9% Preference share	4,00,000	0.10	9.000	0.900
11% Debentures	12,00,000	0.30	7.700	2.310
	40,00,000	1.00		**14.085**

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299 MTP **Old P8 May 2022 S-II** Q5 10 marks old scheme

The capital structure of RV Limited as on 31st March, 2022 as per its Balance Sheet is as follows:

Particulars	₹
Equity shares of ₹ 10 each	25,00,000
10% Preference shares of ₹ 100 each	5,00,000
Retained earnings	5,00,000
13% debentures of ₹ 100 each	20,00,000

The market price of equity shares is ₹ 50 per share. Expected dividend on equity shares is ₹ 3 per share. The dividend per share is expected to grow at the rate of 8%.

Preference shares are redeemable after eight years and the current market price is ₹ 80 per share.

Debentures are redeemable after five years and are currently selling at ₹ 90 per debenture.

The tax rate applicable to the company is 35%.

CALCULATE weighted average cost of capital using:

(i) Book value proportions

(ii) Market value proportions

ICAI'S ANSWER

Working Notes:

(i) Cost of Equity (K_e)

$\frac{D_1}{P}$ +g = $\frac{\text{Rs. } 3}{\text{Rs. } 50}$ + 0.08 = 0.14 i.e. 14%

(ii) Cost of preference shares (K_p)

$$\frac{D + \frac{RV - NP}{n}}{\frac{RV + NP}{2}} = \frac{10 + \frac{(100 - 80)}{8}}{\frac{100 + 80}{2}} = \frac{12.5}{90} = 0.1389 = 13.89\%$$

(iii) Cost of debenture (K_d)

$$\frac{I(1-t) + \frac{RV - NP}{n}}{\frac{RV + NP}{2}} = \frac{13(1 - 0.35) + \frac{(100 - 90)}{5}}{\frac{100 + 90}{2}} = \frac{8.45 + 2}{95} = 0.11 \text{ i.e. } 11\%$$

Or,

$$\left[\frac{I + \frac{RV - NP}{n}}{\frac{RV + NP}{2}}\right](1 - t) = \left[\frac{13 + \frac{(100 - 90)}{5}}{\frac{100 + 90}{2}}\right](1 - 0.35) = 0.1026 \text{ i.e. } 10.26\%$$

Weighted Average cost of capital (Book Value)

	Amount (₹)	Weight (W)	Cost (K)	W x K
Equity shares	25,00,000	0.4546	0.14	0.0636
Preference shares	5,00,000	0.0909	0.1389	0.0126
Retained Earnings	5,00,000	0.0909	0.14	0.0127
Debentures	20,00,000	0.3636	0.1026	0.0373
	55,00,000			0.1262

Or (if K_d is 11%) the WACC = 0.1289

Thus, WACC (Book value based) = 12.62% or 12.89%

Weighted Average cost of capital (Market Value)

	Amount (₹)	Weight (W)	Cost (K)	W x K
Equity shares	1,25,00,000	0.85	0.14	0.119
Preference shares	4,00,000	0.028	0.1389	0.0039
Debentures	18,00,000	0.122	0.1026	0.0125
	1,47,00,000			0.1354

Or (if K_d is 11%) the WACC = 0.1363

Thus, WACC (Market value based) = 13.54% or 13.63%

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300 MTP Old P8 Dec 2021 S-I Q1(b) 5 marks old scheme

The following is the capital structure of Sharda Ltd. as on 31.12.2020:

	(₹)
Equity shares: 2,00,000 shares (of ₹ 100 each)	2,00,00,000
9% Preference Shares (of ₹ 100 each)	60,00,000
8% Debentures	90,00,000
	3,50,00,000

The market price of the company's share is ₹ 120 and it is expected that a dividend of ₹ 12 per share would be declared for the year 2021. The dividend growth rate is 5% and the company is in the 30% tax bracket.

(i) CALCULATE the company's weighted average cost of capital.

(ii) Further, in order to finance an expansion plan, the company intends to borrow a fund of ₹ 2 crores bearing 12% rate of interest. In this situation, WHAT will be the company's revised weighted average cost of capital? This financing decision is expected to increase dividend from ₹ 12 to ₹ 14 per share. However, the market price of equity share is expected to decline from ₹ 120 to ₹ 115 per share.

In case of both (i) and (ii) above, use market value weight while calculating weighted average cost of capital.

ICAI'S ANSWER

(i) Computation of the weighted average cost of capital

Source of finance (a)	Market Value of capital (₹)	Weight (b)	After tax Cost of capital (%) (c)	WACC (%) (d) = (b) × (c)
Equity share (Working note 1) [₹120 × 2,00,000 shares]	2,40,00,000	0.6154	15	9.231
9% Preference share	60,00,000	0.1538	9	1.3842
8% Debentures	90,00,000	0.2308	5.60	1.2925
	3,90,00,000	1.0000		11.9077

(ii) Computation of Revised Weighted Average Cost of Capital

Source of finance (a)	Market Value of capital (₹)	Weight (b)	After tax Cost of capital (%) (c)	WACC (%) (d) = (b) × (c)
Equity shares (Working note 2) [₹115 × 2,00,000 shares]	2,30,00,000	0.3966	17.17	6.8096
9% Preference shares	60,00,000	0.1034	9.00	0.9306
8% Debentures	90,00,000	0.1552	5.60	0.8691
12% Loan	2,00,00,000	0.3448	8.40	2.8963
	5,80,00,000	1.0000		11.5056

Working Notes:

(1) Cost of Equity Shares

K_e = {Dividend Per Share (D_1)/Market Price Share (P_0)} + Growth Rate

= 12/120 + 0.05

= 0.15 or 15%

(2) Revised cost of equity shares (K_e)

Revised K_e = 14/115 + 0.05

= 0.1717 or 17.17%

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301 MTP **Old P8 Jul 2021 S-I** Q2 10 marks old scheme

CALCULATE the WACC by using Market value weights.

The capital structure of the company is as under:

	(Rs.)
Debentures (Rs.100 per debenture)	10,00,000
Preference shares (Rs.100 per share)	10,00,000
Equity shares (Rs.10 per share)	20,00,000
	40,00,000

The market prices of these securities are:

Debentures	Rs. 115 per debenture
Preference shares	Rs. 120 per preference share
Equity shares	Rs. 265 each.

Additional information:

(1) Rs.100 per debenture redeemable at par, 10% coupon rate, 2% floatation cost, 10-year maturity.

(2) Rs.100 per preference share redeemable at par, 5% coupon rate, 2% floatation cost and 10-year maturity.

(3) Equity shares have a floatation cost of Rs. 1 per share.

The next year expected dividend is Rs. 5 with an annual growth of 15%. The firm has the practice of paying all earnings in the form of dividend.

Corporate tax rate is 30%. Use YTM method to calculate cost of debentures and preference shares.

ICAI'S ANSWER

(i) Cost of Equity (K_e)

$$= \frac{D_1}{P_0\text{-}F} + g = \frac{\text{Rs. } 5}{\text{Rs.265-Re.1}} + 0.15 = 0.1689 \text{ or } 16.89\%$$

(ii) Cost of Debt (K_d)

Calculation of NPV at discount rate of 5% and 7%

Year	Cash flows (Rs.)	Discount factor @ 5%	Present Value	Discount factor @ 7%	Present Value (Rs.)
0	112.7	1.000	(112.7)	1.000	(112.7)
1 to 10	7	7.722	54.05	7.024	49.17
10	100	0.614	61.40	0.508	50.80
NPV			+2.75		-12.73

Calculation of IRR

$$\text{IRR} = 5\% + \frac{2.75}{2.75\text{-}(-12.73)}(7\% - 5\%) = 5\% + \frac{2.75}{15.48}(7\% - 5\%) = 5.36\%$$

Cost of Debt (K_d) = 5.36%

(iii) Cost of Preference shares (K_p)

Calculation of NPV at discount rate of 2% and 5%

Year	Cash flows (Rs.)	Discount factor @ 2%	Present Value	Discount factor @ 5%	Present Value (Rs.)
0	117.6	1.000	(117.6)	1.000	(117.6)
1 to 10	5	8.983	44.92	7.722	38.61
10	100	0.820	82.00	0.614	61.40
NPV			+9.32		-17.59

Calculation of IRR

$$\text{IRR} = 2\% + \frac{9.32}{9.32\text{-}(-17.59)}(5\%\text{-}2\%) = 2\% + \frac{9.32}{26.91}(5\%\text{-}2\%) = 3.04\%$$

Cost of Preference Shares (K_p) = 3.04%

Calculation of WACC using market value weights

Source of capital	Market Value	Weights	After tax cost of capital	WACC (Ko)
	(Rs.)	**(a)**	**(b)**	**(c) = (a)×(b)**
10% Debentures (Rs.115× 10,000)	11,50,000	0.021	0.0536	0.00113
5% Preference shares (Rs.120× 10,000)	12,00,000	0.022	0.0304	0.00067
Equity shares (Rs.265 × 2,00,000)	5,30,00,000	0.957	0.1689	0.16164
	5,53,50,000	1.000		0.16344

WACC (K_o) = 0.16344 or 16.344%

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302 MTP **Old P8 Nov 2020 S-I** Q3 10 marks old scheme

P Ltd. has the following capital structure at book-value as on 31st March, 2020:

Particulars	(₹)
Equity share capital (10,00,000 shares)	3,00,00,000
11.5% Preference shares	60,00,000
10% Debentures	1,00,00,000
	4,60,00,000

The equity shares of the company are sold for ₹ 300. It is expected that the company will pay next year a dividend of ₹ 15 per equity share, which is expected to grow by 5% p.a. forever. Assume a 35% corporate tax rate.

Required:

(i) COMPUTE weighted average cost of capital (WACC) of the company based on the existing capital structure.

(ii) COMPUTE the new WACC, if the company raises an additional ₹ 50 lakhs debt by issuing 12% debentures. This would result in increasing the expected equity dividend to ₹ 20 and leave the growth rate unchanged, but the price of equity share will fall to ₹ 250 per share.

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

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303 MTP Old P8 May 2020 S-I Q1(b) 5 marks old scheme

ABC Limited has the following book value capital structure:

Equity Share Capital (1 crore shares @ Rs.10 each)	Rs.1,000 lakh
Reserves and Surplus	Rs.2,250 lakh
9% Preference Share Capital (5 lakh shares @ Rs.100 each)	Rs.500 lakh
8.5% Debentures (1.5 lakh debentures @ Rs.1,000 each)	Rs.1,500 lakh
12% Term Loans from Financial Institutions	Rs.500 lakh

- The debentures of ABC Limited are redeemable at par after five years and are quoting at Rs.985 per debenture.
- The current market price per equity share is Rs.60. The prevailing default-risk free interest rate on 10-year GOI Treasury Bonds is 5.5%. The average market risk premium is 7%. The beta of the company is 1.85
- The preference shares of the company are redeemable at 10% premium after 5 years is currently selling at Rs.102 per share.

The applicable income tax rate for the company is 35%.

Required:

CALCULATE weighted average cost of capital of the company using market value weights.

ICAI'S ANSWER

Working Notes:

(1) Computation of cost of debentures (K_d) :

$$K_d = \frac{\text{Rs.85}(1-0.35) + \frac{(1{,}000-985)}{5}}{\frac{(1{,}000+985)}{2}} = \frac{55.25+3}{992.5} = 0.0586 \text{ or } 5.86\%$$

(2) Computation of cost of term loans (K_T) :

= r (1– t)

= 0.12 (1– 0.35) = 0.078 or 7.8%

(3) Computation of cost of preference capital (K_P) :

$$K_p = \frac{\text{Preference Dividend} + (\text{RV - NP}) / \text{n}}{(\text{RV} + \text{NP}) / 2}$$

$$= \frac{\text{Rs.9} + \frac{(110-102)}{5}}{\frac{(110+102)}{2}} = \frac{9+1.6}{106} = 0.1 \text{ or } 10\%$$

(4) Computation of cost of equity (K_e) :

= R_f + ß(R_m – R_f)

Or, = Risk free rate + (Beta × Risk premium)

= 0.055 + (1.85 × 0.07) = 0.1845 or 18.45%

Calculation of Weighted Average cost of capital Using market value weights

Source of Capital	Market value of capital structure (Rs. in lakh)	Weights	After tax cost of capital (%)	WACC (%)
Equity share capital (1 crore shares × Rs.60)	6,000	0.71	18.45	13.09
9% Preference share capital (5 lakh shares × Rs.102)	510	0.06	10.00	0.60
8.5 % Debentures (1.5 lakh × Rs.985)	1,477.5	0.17	5.86	0.99
12% Term loans	500	0.06	7.80	0.47
	8,487.50	1.000		15.15

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304 MTP **Old P8 Nov 2019 S-I** Q4 10 marks old scheme

ABC Ltd. has the following capital structure which is considered to be optimum as on 31st March, 2019

	(Rs.)
14% Debentures	30,00,000
11% Preference shares	10,00,000
Equity Shares (10,000 shares)	1,60,00,000
	2,00,00,000

The company share has a market price of Rs. 236. Next year dividend per share is 50% of year 2019 EPS. The following is the trend of EPS for the preceding 10 years which is expected to continue in future.

Year	EPS (Rs.)	Year	EPS Rs.)
2010	10.00	2015	16.10
2011	11.00	2016	17.70
2012	12.10	2017	19.50
2013	13.30	2018	21.50
2014	14.60	2019	23.60

The company issued new debentures carrying 16% rate of interest and the current market price of debenture is Rs. 96.

Preference share Rs. 9.20 (with annual dividend of Rs. 1.1 per share) were also issued. The company is in 50% tax bracket.

(A) CALCULATE after tax:

(i) Cost of new debt

(ii) Cost of new preference shares

(iii) New equity share (consuming new equity from retained earnings)

(B) CALCULATE marginal cost of capital when no new shares are issued.

(C) COMPUTE the amount that can be spent for capital investment before new ordinary shares must be sold. Assuming that retained earnings for next year's investment are 50 percent of 2019.

(D) COMPUTE marginal cost of capital when the funds exceeds the amount calculated in (C), assuming new equity is issued at Rs. 200 per share?

ICAI'S ANSWER

(A) (i) Cost of new debt

$$K_d = \frac{I(1-t)}{P_0}$$

$$= \frac{16\ (1-0.5)}{96} = 0.0833$$

(ii) Cost of new preference shares

$$K_p = \frac{PD}{P_0} = \frac{1.1}{9.2} = 0.12$$

(iii) Cost of new equity shares

$$K_e = \frac{D_1}{P_0} + g$$

$$= \frac{11.80}{236} + 0.10 = 0.05 + 0.10 = 0.15$$

Calculation of D_1

D_1 = 50% of 2019 EPS = 50% of 23.60 = Rs. 11.80

(B) Calculation of marginal cost of capital

Type of Capital	Proportion	Specific Cost	Product
(1)	**(2)**	**(3)**	**(2) × (3) = (4)**
Debenture	0.15	0.0833	0.0125
Preference Share	0.05	0.12	0.0060
Equity Share	0.80	0.15	0.1200
Marginal cost of capital			0.1385

(C) The company can spend the following amount without increasing marginal cost of capital and without selling the new shares:

Retained earnings = (0.50) (236 × 10,000) = Rs. 11,80,000

The ordinary equity (Retained earnings in this case) is 80% of total capital

11,80,000 = 80% of Total Capital

Capital investment before issuing equity $= \frac{\text{Rs.}11,80,000}{0.80}$ = Rs.14,75,000

(D) If the company spends in excess of Rs.14,75,000 it will have to issue new shares.

The cost of new issue will be $= \frac{\text{Rs. } 11.80}{200} + 0.10 = 0.159$

The marginal cost of capital will be:

Type of Capital	Proportion	Specific Cost	Product
(1)	**(2)**	**(3)**	**(2) × (3) = (4)**
Debentures	0.15	0.0833	0.0125
Preference Shares	0.05	0.1200	0.0060
Equity Shares (New)	0.80	0.1590	0.1272
			0.1457

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 3-4 · answer p. 5-6

CHAPTER 5

Financing Decisions: Capital Structure

5.1 Capital structure theories: NI, NOI, traditional and Modigliani-Miller

40 questions · asked in 36 ICAI documents · 7 exam papers

305 EXAM **Sep 2026** Q2(a) 7 marks

A Ltd. and B Ltd. both the companies belong to the same risk-class and are identical in all respect except that A Ltd. has no debt capital, while B Ltd. has 60,000, 12.5% Debentures of ₹ 100 each. The earnings before interest and taxes of each company is ₹ 19,50,000. The equity capitalisation rate of A Ltd. is 15 per cent. Corporate tax rate is 30 per cent.

Considering M-M approach with tax, you are required to :

(i) Determine the total market value of A Ltd. and B Ltd.

(ii) Calculate equity capitalisation rate of B Ltd.

(iii) Calculate overall cost of capital of A Ltd. and B Ltd.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 17

306 EXAM **Sep 2025** MCQ 6 2 marks MCQ

PRAX Ltd. is an all-equity financed company, currently valued at ₹ 2,100 lakhs, with a cost of capital of 18%. The company is contemplating a strategic financial restructuring by:

(i) Repurchasing ₹ 325 lakhs of equity shares.

(ii) Replacing the repurchased equity with 16% debentures.

The Company maintains constant Earnings Before Interest and Taxes (EBIT) and following a policy of distributing entire earnings as dividend. Corporate tax rate is 35%.

What would be the market value of Equity after structure modification as per Modigliani and Miller (MM) Approach?

A ₹ 1,888.75 Lakh

B ₹ 2,213.75 Lakh

C ₹ 1,456.75 Lakh

D ₹ 2,538.75 Lakh

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P6 · Sep 2025 exam · question p. 2-3 · answer p. 7

307 EXAM **May 2025** Q3(b) 4 marks

Z Ltd. is an unlevered company. It wants to repurchase its equity shares of ₹ 300 lakhs by issue of 12% debentures of same amount. Current market value of Z Ltd. is ₹ 1,400 lakhs. Its cost of capital is 18%. The company will maintain same level of EBIT in future years. Dividend pay-out ratio is 100%. Company pays tax at a rate of 30%.

As per Modigliani and Miller approach, due to such change in capital structure, what will be impact on the following?

(i) Market Value of Z Ltd.

(ii) Overall cost of capital

(iii) Cost of equity

ICAI'S ANSWER

Market Value of Equity = $\frac{\text{Net income (NI) for equity holders}}{K_e}$

₹ 1,400 lakhs = $\frac{\text{Net income (NI) for equity holders}}{0.18}$

Net Income to equity holders/EAT = ₹ 252 lakhs

Therefore, EBIT = $\frac{\text{EAT}}{(1-t)} = \frac{252 \text{ lakhs}}{(1-0.3)}$ = ₹ 360 lakhs

Income Statement

	All Equity (₹ In lakhs)	Equity & Debt (₹ In lakhs)
EBIT (as calculated above)	360	360
Interest on ₹ 300 lakhs @ 12%	-	36
EBT	360	324
Tax @ 30%	108	97.2
Income available to equity holders	252	226.8

(i) Market value of Z Ltd

Market value of levered firm = Value of unlevered firm + Tax Advantage

= ₹ 1,400 lakhs + (₹ 300 lakhs x 0.3)

= ₹ 1,490 lakhs

Change in market value of the company = 1,490 lakhs – 1,400 lakhs

= ₹ 90 lakhs

The impact is that the market value of the company has increased by ₹ 90 lakhs due to replacement of equity with debt.

(ii) Overall Cost of Capital

Market Value of Equity = Market value of levered firm - Equity repurchased

= ₹ 1,490 lakhs – ₹ 300 lakhs = ₹ 1,190 lakhs

Cost of Equity (K_e) = $\frac{\text{Net Income to equity holders}}{\text{Market value of equity}} \times 100$

= $\frac{₹ 226.8 \text{ lakhs}}{₹ 1190 \text{ lakhs}} \times 100$ =19.06% (approx.)

Cost of debt (K_d) = I (1 - t) = 12 (1 - 0.3) = 8.4%

Components	Amount (₹ In lakhs)	Cost of Capital %	Weight	WACC (K_o) %
Equity	1,190	19.06	0.80(approx.)	15.25
Debt	300	8.4	0.20(approx.)	1.68
	1490		1	16.93

(**Note:** If we take the weight of Equity as 0.7986 and weight of Debt as 0.2013 then WACC will be 16.91%).

Impact on Market value and Overall cost of capital of Z Ltd.

Particulars	Before restructure	After restructure	Impact
Market value of company	₹ 1,400 lakhs	₹ 1,490 Lakhs	Increase due to import of debt
Cost of equity	18%	19.06%	Increase
Overall cost of capital	18%	16.93%	Decrease

So, the impact is that the Overall Cost of Capital or K_o has fallen by 1.07% (18% - 16.93%) due to the benefit of tax relief on debt interest payment.

(iii) Cost of Equity

The impact is that cost of equity has risen by 1.06% (19.06% - 18%) due to the presence of financial risk i.e. introduction of debt in capital structure.

Note: Cost of Capital and Cost of equity can also be calculated with the help of following formulas,

Cost of Capital (K_o) = K_{eu} [1 – (t x L)]

Where,

K_{eu} = Cost of equity in an unlevered company

t = Tax rate

$$L = \frac{\text{Debt}}{\text{Debt} + \text{Equity}}$$

$$\text{So, } K_o = 0.18 \left[1 - \left(0.3 \times \frac{₹\ 300 \text{ lakhs}}{₹\ 1490 \text{ lakhs}}\right)\right] = 0.1691 \text{ or } 16.91\% \text{ (approx.)}$$

$$\text{Cost of Equity } (K_e) = K_{eu} + (K_{eu} - K_d)\ \frac{\text{Debt } (1\text{-t})}{\text{Equity}}$$

Where,

K_{eu} = Cost of equity in an unlevered company

K_d = Cost of debt

t = Tax rate

$$\text{So, } K_e = 0.18 + \left((0.18 - 0.12) \times \frac{₹\ 300 \text{ lakhs } (1\text{-}0.3)}{₹\ 1190 \text{ lakhs}}\right)$$

= 0.1906 or 19.06%

ICAI · P6 · May 2025 exam · question p. 18-19 · answer p. 21-23

308 EXAM May 2025 Q4(c) 2 marks

What is hierarchy of financing under 'Pecking Order' theory, and why does it exist?

ICAI'S ANSWER

Hierarchy of financing Pecking order:

Pecking order theory suggests that managers may use various sources for raising of fund in the following order:

1. Managers first choice is to use internal finance.
2. In absence of internal finance, they can use secured debt, unsecured debt, hybrid debt etc.
3. Managers may issue new equity shares as a last option.

The hierarchy of financing Pecking order exist to Minimize the cost of capital, financial control and avoiding negative market signals.

ICAI · P6 · May 2025 exam · question p. 23 · answer p. 25

309 EXAM May 2024 Q2(b) 4 marks

Following data is available in respect of Levered and Unlevered companies having same business risk:

Capital employed = ₹ 2,00,000, EBIT = ₹ 25,000 and Ke = 12.5%

Sources	Levered Company (f)	Unlevered Company (₹)
Debt (@8%)	75,000	Nil
Equity	1,25,000	2,00,000

An investor is holding 12% shares in levered company. Calculate the increase in annual earnings of investor if he switches over his holding from Levered to Unlevered company.

ICAI'S ANSWER

1. Valuation of firms

Particulars	Levered Firm (₹)	Unlevered Firm (₹)
EBIT	25,000	25,000
Less: Interest on debt (8% × ₹ 75,000)	6,000	Nil
Earnings available to Equity shareholders	19,000	25,000
Ke	12.5%	12.5%
Value of Equity (S) (Earnings available to Equity shareholders/ Ke)	1,52,000	2,00,000
Debt (D)	75,000	Nil
Value of Firm (V) = S + D	2,27,000	2,00,000

Value of Levered company is more than that of unlevered company. Therefore, investor will sell his shares in levered company and buy shares in unlevered company. To maintain the level of risk he will borrow proportionate amount and invest that amount also in shares of unlevered company.

2. Investment & Borrowings

	₹
Sell shares in Levered company (₹ 1,52,000 × 12%)	18,240
Borrow money (₹ 75,000 × 12%)	9,000
Buy shares in Unlevered company	27,240

3. Change in Return

	₹
Income from shares in Unlevered company (₹ 27,240 × 12.5%)	3,405
Less: Interest on loan (₹ 9,000 × 8%)	720
Net Income from unlevered firm	2,685
Less: Income from Levered firm (₹ 18,240 × 12.5%)	2,280
Incremental Income due to arbitrage	405

Solution can also be done in the following way:

Valuation of firms

Particulars	Levered Firm (₹)	Unlevered Firm (₹)
EBIT	25,000	25,000
Less: Interest on debt (8% × ₹ 75,000)	6,000	Nil
Earnings available to Equity shareholders	19,000	25,000
Ke	12.5%	12.5%
Value of Equity (S) (Earnings available to Equity shareholders/ Ke)	1,52,000	2,00,000
Debt (D)	75,000	Nil
Value of Firm (V) = S + D	2,27,000	2,00,000

Value of Levered company is more than that of unlevered company. Therefore, investor will sell his shares in levered company and buy shares in unlevered company.

Arbitrage Process:

If investor have 12% shares of levered company, value of investment in equity shares is 12% of ₹ 1,52,000 i.e. ₹ 18,240 and return will be 12% of ₹19,000 = ₹ 2,280.

Alternate Strategy will be:

Sell 12% shares of levered firm for ₹ 18,240 and borrow 12% of levered firm's debt i.e. ₹ 9,000 (12% of ₹ 75,000) and invest the money i.e. 12% in unlevered firm's stock:

Total resources /Money investor have = ₹ 18,240 + ₹ 9,000 = ₹ 27,240 and investor invest 12% of ₹ 2,00,000 = ₹ 24,000

Surplus cash available with investor is = ₹ 27,240 – ₹ 24,000 = ₹ 3,240

Investor return = 12% EBIT of unlevered firm – Interest to be paid on borrowed funds

i.e. = 12% of ₹ 25,000 – 8% of ₹ 9,000 = ₹ 3,000 – ₹ 720 = ₹ 2,280

Now, return remains the same i.e. ₹ 2,280 which investor is getting from levered company before investing in unlevered company but still have ₹ 3,240 excess money available with investor. Hence, investor is better off by doing arbitrage.

ICAI · P6 · May 2024 exam · question p. 9 · answer p. 12-14

310 EXAM **Old P8 Nov 2022** Q5(a) 4 marks old scheme

The following are the costs and values for the firms A and B according to the traditional approach.

	Firm A	Firm B
Total value of firm, V (in ₹)	50,000	60,000
Market value of debt, D (in ₹)	0	30,000
Market value of equity, E (in ₹)	50,000	30,000
Expected net operating income (in ₹)	5,000	5,000
Cost of debt (in ₹)	0	1,800
Net Income (in ₹)	5,000	3,200
Cost of equity, Ke = NI/V	10.00%	10.70%

(i) Compute the Equilibrium value for Firm A and B in accordance with the M -M approach. Assume that (a) taxes do not exist and (b) the equilibrium value of Ke is 9.09%.

(ii) Compute Value of Equity and Cost of Equity for both the firms.

ICAI'S ANSWER

(i) Computation of Equilibrium value of Firms A & B under MM Approach:

As per MM approach K_O is equal to K_{eu}

$\therefore K_O = K_{eu}$ (1 – t) = 9.09 (1 – 0) = 9.09

Particulars	A	B
EBIT (NOI) (₹)	5000	5000
$K_{O\,(\%)}$	9.09	9.09
Equilibrium value (₹) (NOI/Ko) X 100	55005.5	55005.5
	$\frac{5,000}{9.09}$ x 100	$\frac{5,000}{9.09}$ x 100

(ii) Computation of value of equity and cost of equity of Firms A & B

Particulars	A	B
Equilibrium value (₹)	55,005.5	55,005.5
Less: Value of Debt	-	30,000
Value of Equity	55,005.5	25,005.5

Cost of Equity of Firm A (unlevered) = 9.09

Cost of Debt of Firm B (K_d) (levered) = (1800/30000) x 100 = 6%

Cost of Equity of Firm B (Levered) = K_O + (K_O - K_d) x (Debt / Equity)

= 9.09 + (9.09 – 6) x (30000/25005.5)

= 9.09 + 3.09 x 1.2 = 9.09 + 3.71 = **12.80%**

(OR)

Cost of Equity of Firm B (Levered) = $\left(\frac{\text{NI}}{\text{Value of Equity}}\right)$ x 100

= $\left(\frac{3200}{25005.5}\right)$ x 100 = **12.8%**

ICAI · Old P8 · Nov 2022 exam · question p. 12 · answer p. 13-14

311 EXAM **Old P8 Jul 2021** Q1(b) 5 marks old scheme

The details about two companies R Ltd. and S Ltd. having same operating risk are given below:

Particulars	R Ltd.	S Ltd.
Profit before interest and tax	₹ 10 lakhs	₹ 10 lakhs
Equity share capital ₹ 10 each	₹ 17 lakhs	₹ 50 lakhs
Long term borrowings @ 10%	₹ 33 lakhs	-
Cost of Equity (K_e)	18%	15%

You are required to:

(1) Calculate the value of equity of both the companies on the basis of M.M. Approach without tax.

(2) Calculate the Total Value of both the companies on the basis of M.M. Approach without tax.

ICAI'S ANSWER

(1) Computation of value of equity on the basis of MM approach without tax

Particulars	R Ltd. (₹ in lakhs)	S Ltd. (₹ in lakhs)
Profit before interest and taxes	10	10
Less: Interest on debt (10% × ₹ 33,00,000)	3.3	-
Earnings available to Equity shareholders	6.7	10
K_e	18%	15%
Value of Equity (Earnings available to Equity shareholders/K_e)	**37.222**	**66.667**

(2) Computation of total value on the basis of MM approach without tax

Particulars	R Ltd. (₹ in lakhs)	S Ltd. (₹ in lakhs)
Value of Equity (S) (as calculated above)	37.222	66.667
Debt (D)	33	-
Value of Firm (V) = S + D	**70.222**	**66.667**

ICAI · Old P8 · Jul 2021 exam · question p. 64 · answer p. 68

312 EXAM **Old P8 Jan 2021** Q3 10 marks old scheme

A Limited and B Limited are identical except for capital structures. A Ltd. has 60 per cent debt and 40 per cent equity, whereas B Ltd. has 20 per cent debt and 80 per cent equity. (All percentages are in market-value terms.) The borrowing rate for both companies is 8 per cent in a no-tax world, and capital markets are assumed to be perfect.

(a) (i) If X, owns 3 per cent of the equity shares of A Ltd., determine his return if the Company has net operating income of ₹ 4,50,000 and the overall capitalization rate of the company, (K_o) is 18 per cent.

(ii) Calculate the implied required rate of return on equity of A Ltd.

(b) B Ltd. has the same net operating income as A Ltd.

(i) Calculate the implied required equity return of B Ltd.

(ii) Analyse why does it differ from that of A Ltd.

ICAI'S ANSWER

(a) Value of A Ltd. = $\frac{\text{NOI}}{K_o} = \frac{₹\ 4{,}50{,}000}{18\%}$ = ₹ 25,00,000

(i) Return on Shares of X on A Ltd.

Particulars	Amount (₹)
Value of the company	25,00,000
Market value of debt (60% x ₹ 25,00,000)	15,00,000
Market value of shares (40% x ₹ 25,00,000)	**10,00,000**
Particulars	**Amount (₹)**
Net operating income	4,50,000
Interest on debt (8% × ₹ 15,00,000)	1,20,000
Earnings available to shareholders	**3,30,000**
Return on 3% shares (3% × ₹ 3,30,000)	**9,900**

(ii) Implied required rate of return on equity of A Ltd. $= \frac{₹\ 3,30,000}{₹\ 10,00,000} =$ **33%**

(b) (i) Calculation of Implied rate of return of B Ltd.

Particulars	Amount (₹)
Total value of company	25,00,000
Market value of debt (20% × ₹ 25,00,000)	5,00,000
Market value of equity (80% × ₹ 25,00,000)	**20,00,000**
Particulars	**Amount (₹)**
Net operating income	4,50,000
Interest on debt (8% × ₹ 5,00,000)	40,000
Earnings available to shareholders	**4,10,000**

Implied required rate of return on equity $= \frac{₹\ 4,10,000}{₹\ 20,00,000} =$ **20.5%**

(ii) Implied required rate of return on equity of B Ltd. is lower than that of A Ltd. because B Ltd. uses less debt in its capital structure. As the equity capitalisation is a linear function of the debt-to-equity ratio when we use the net operating income approach, the decline in required equity return offsets exactly the disadvantage of not employing so much in the way of "cheaper" debt funds.

ICAI · Old P8 · Jan 2021 exam · question p. 8 · answer p. 8-9

313 RTP **May 2026** Q6

Same question, different ICAI answer: #317 RTP Sep 2024

The following data relate to two companies belonging to the same risk class:

Particulars	A Ltd.	B Ltd.
Expected Net Operating Income	₹ 18,00,000	₹ 18,00,000
12% Debt	₹ 54,00,000	-
Equity Capitalization Rate	?	18

Required:

(a) DETERMINE the total market value, equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per M.M. Approach.

(b) DETERMINE the total market value, equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per M.M. Approach.

ICAI'S ANSWER

(a) Assuming no tax as per MM Approach.

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis

Market Value of 'B Ltd' [Unlevered(u)]

Total Value of Unlevered Firm (V_u) = [NOI/k_e]

= 18,00,000/0.18

= ₹ 1,00,00,000

K_e of Unlevered Firm (given) = 0.18

K_o of Unlevered Firm (Same as above) = k_e (as there is no debt)

= 0.18

Market Value of 'A Ltd' [Levered Firm (I)]

Total Value of Levered Firm (V_L) = V_u + (Debt× Nil)

= ₹ 1,00,00,000 + (54,00,000 × nil)

= ₹ 1,00,00,000

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC)

	Particulars	A Ltd.	B Ltd.
A	Net Operating Income (NOI)	18,00,000	18,00,000
B	*Less:* Interest on Debt (I)	6,48,000	-
C	Earnings of Equity Shareholders (NI)	11,52,000	18,00,000
D	Overall Capitalization Rate (k_o)	0.18	0.18
E	Total Value of Firm (V = NOI/k_o)	1,00,00,000	1,00,00,000
F	*Less:* Market Value of Debt	54,00,000	-
G	Market Value of Equity (S)	46,00,000	1,00,00,000

	Particulars	A Ltd.	B Ltd.
H	Equity Capitalization Rate [k_e = NI /S]	0.2504	0.18
I	Weighted Average Cost of Capital [WACC] (k_o)	0.18	0.18

Assuming 40% taxes as per MM Approach

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis Approach

Market Value of 'B Ltd' [Unlevered(u)]

Total Value of unlevered Firm (V_u) = [NOI (1 - t)/ke]

= 18,00,000(1 – 0.40)] / 0.18

= ₹ 60,00,000

K_e of unlevered Firm (given) = 0.18

K_o of unlevered Firm (Same as above = k_e as there is no debt)

= 0.18

Market Value of 'A Ltd' [Levered Firm (I)]

Total Value of Levered Firm (V_L) = V_u + (Debt× Tax)

= ₹ 60,00,000 + (54,00,000 × 0.40)

= ₹ 81,60,000

Computation of Weighted Average Cost of Capital (WACC) of 'B Ltd.'

= 18% (i.e. $K_e = K_o$)

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC) of A Ltd

Particulars	A Ltd.
Net Operating Income (NOI)	18,00,000
Less: Interest on Debt (I)	6,48,000
Earnings Before Tax (EBT)	11,52,000
Less: Tax @ 40%	4,60,800
Earnings for equity shareholders (NI)	6,91,200
Total Value of Firm (V) as calculated above	81,60,000
Less: Market Value of Debt	54,00,000
Market Value of Equity (S)	27,60,000
Equity Capitalization Rate [k_e = NI/S]	0.2504
Weighted Average Cost of Capital (k_o)*	13.23

*Computation of WACC A Ltd.

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	27,60,000	0.338	0.2504	0.0846
Debt	54,00,000	0.662	0.072*	0.0477
Total	81,60,000			0.1323

*K_d= 12% (1- 0.4) = 12% × 0.6 = 7.2%

WACC = 13.23%

ICAI · P6 · RTP May 2026 · question p. 6 · answer p. 19-21

314 RTP **Jan 2026** Q7

J Ltd. is an all equity financed company with a market value of ₹ 25,00,000 and cost of equity (Ke) 21%. The company wants to buyback equity shares worth ₹ 5,00,000 by issuing and raising 15% perpetual debt of the same amount. Rate of tax may be taken as 30%. After the capital restructuring and applying MM Model (with taxes), you are required to calculate:

(i) Market value of J Ltd.

(ii) Cost of Equity (Ke)

(iii) Weighted average cost of capital (using market weights) and comment on it.

ICAI'S ANSWER

Market Value of Equity = ₹ 25,00,000

K_e = 21%

$$\frac{\text{Net income (NI) for equity - holders}}{K_e} = \text{Market Value of Equity}$$

$$\frac{\text{Net income (NI) for equity holders}}{0.21} = 25,00,000$$

Net income for equity holders = 5,25,000

EBIT= 5,25,000/0.7 = 7,50,000

	All Equity	Debt and Equity
EBIT	7,50,000	7,50,000
Interest to debt-holders	-	(75,000)
EBT	7,50,000	6,75,000
Taxes (30%)	(2,25,000)	(2,02,500)
Income available to equity shareholders	5,25,000	4,72,500
Income to debt holders plus income available to shareholders	5,25,000	5,47,500

Present value of tax-shield benefits = ₹ 5,00,000 × 0.30 = ₹ 1,50,000

(i) **Value of Restructured firm**

= ₹ 25,00,000 + ₹ 1,50,000 = ₹ 26,50,000

(ii) **Cost of Equity (K_e)**

Total Value = ₹ 26,50,000

Less: Value of Debt = ₹ 5,00,000

Value of Equity = ₹ 21,50,000

$K_e = \frac{4,72,500}{21,50,000} = 0.219 = 21.98\%$

(iii) **WACC (on market value weight)**

Cost of Debt (after tax) = 15% (1- 0.3) = 0.15 (0.70) = 0.105 = 10.5%

Components of Costs	Amount	Cost of Capital (%)	Weight	WACC (%)
Equity	21,50,000	21.98	0.81	17.80
Debt	5,00,000	10.50	0.19	2.00
	26,50,000			19.80

Comment: At present the company is all equity financed. So, $K_e = K_o$ i.e. 21%. However, after restructuring, the K_o would be reduced to 19.80% and K_e would increase from 21% to 21.98%.

ICAI · P6 · RTP Jan 2026 · question p. 7 · answer p. 19-20

315 RTP Jan 2025 Q3 MCQ

Abhi Ltd is an all equity financed company. It is considering replacing ₹ 275 lakhs equity shares with 15% debentures of the same amount. Current Market value of the company is 1750 lakhs with cost of capital at 20%. Future EBITs are going to be constant and entire earnings are going to be distributed. Corporate Tax Rate can be assumed to be 30%. What will be the new cost of equity of the firm?

A 19.11%

B 17.53%

C 10.50%

D 20.62%

ICAI'S ANSWER

(D) 20.62%

Current PAT = 1750 x 20% = 350

Current PBT = Future EBIT = 350/0.7 = 500

Future PBT = 500 - 275x15% = 458.75

Future PAT = 458.75 x 70% = 321.125

Value (L) = Value (UL) + Debt x t = 1750+275 x 30% = 1832.5

Value of Equity= 1832.5-275 = 1557.5

Ke = 321.125/1557.5 = 20.62%

ICAI · P6 · RTP Jan 2025 · question p. 4 · answer p. 14

316 RTP Sep 2024 Q6

Company XYZ is unlevered and has a cost of equity of 20 percent and a total market value of ₹ 10,00,00,000. Company ABC is identical to XYZ in all respects except that it uses debt finance in its capital structure with a market value of ₹ 4,00,00,000 and a cost of 10 percent. FIND the market value of equity, weighted average cost of capital and cost of equity of ABC if the tax advantage of debt is 25 percent.

ICAI'S ANSWER

Computation of Market Value of Equity of Company ABC

Total market value of Company ABC

V_{ABC} =V $_{XYZ}$ + Bt.....(i)

Where,

V_{ABC} = Market value of leveraged company.

V_{XYZ} = Market value of unleveraged company.

B = Market value of debt.

t = Tax rate.

Now, given

V_{xyz} = ₹ 10,00,00,000

B = ₹ 4,00,00,000

t = 25%

By substituting values in equation (i) above, we have

V_{ABC} = ₹ 10,00,00,000 + ₹ 4,00,00,000 × 0.25%

= ₹ 11,00,00,000

The Market Value of Equity (s) of Company ABC,

= ₹ 11,00,00,000 – ₹ 4,00,00,000

= ₹ 7,00,00,000

Weighted Average Cost of Capital of Company ABC

$WACC_{ABC}$= $WACC_{XYZ}$[1- Bt/V_{ABC}]

$$= 20\% \left[1 - \frac{4,00,00,000}{11,00,00,000} \times 0.25\right]$$

= 18.18%

Where,

$WACC_{ABC}$ is the weighted average cost of capital of the levered company ABC

$WACC_{XYZ}$ is the weighted average cost of capital of the unlevered company XYZ.

Cost of Equity of company ABC

R_{Eabc} = R_{Exyz}+ [(1 - t)B/E(R_{Exyz}- R_B)]

20%+[(1-.25)4,00,00,000/7,00,00,000(.20-.10)]

24.28% approx.

Where,

R_{EABC} is the cost of equity in the levered Company ABC.

R_{Exyz} is the cost of equity in the unlevered Company XYZ.

E is the market value of equity.

B is the market value of debt.

R_B is the cost of debt

ICAI · P6 · RTP Sep 2024 · question p. 7 · answer p. 20-21

317 RTP **Sep 2024** Q7

Same question, different ICAI answer: #313 RTP May 2026

The following data relate to two companies belonging to the same risk class:

Particulars	A Ltd.	B Ltd.
Expected Net Operating Income	₹ 18,00,000	₹ 18,00,000
12% Debt	₹ 54,00,000	-
Equity Capitalization Rate	-	18

Required:

(a) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per M.M. Approach.

(b) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per M.M. Approach.

ICAI'S ANSWER

(a) Assuming no tax as per MM Approach.

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis

Market Value of 'B Ltd' [Unlevered(u)]

Total Value of Unlevered Firm (V_u) = [NOI/k_e] = 18,00,000/.18 = ₹ 1,00,00,000

K_e of Unlevered Firm (given) = 0.18

K_o of Unlevered Firm (Same as above = k_e as there is no debt) = 0.18

Market Value of 'A Ltd' [Levered Firm (I)]

Total Value of Levered Firm (V_L) = V_u + (Debt× Nil) = ₹ 1,00,00,000 + (54,00,000 × nil)

= ₹ 1,00,00,000

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC)

	Particulars	A Ltd.	B Ltd.
A	Net Operating Income (NOI)	18,00,000	18,00,000
B	Less: Interest on Debt (I)	6,48,000	-
C	Earnings of Equity Shareholders (NI)	11,52,000	18,00,000

	Particulars	A Ltd.	B Ltd.
D	Overall Capitalization Rate (k_o)	0.18	0.18
E	Total Value of Firm (V = NOI/k_o)	1,00,00,000	1,00,00,000
F	Less: Market Value of Debt	54,00,000	-
G	Market Value of Equity (S)	46,00,000	1,00,00,000
H	Equity Capitalization Rate [k_e = NI /S]	0.2504	0.18
I	Weighted Average Cost of Capital [WACC] (k_o)	0.18	0.18

Assuming 40% taxes as per MM Approach

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis Approach

Market Value of 'B Ltd' [Unlevered(u)]

Total Value of unlevered Firm (V_u) = [NOI(1 - t)/ke] = 18,00,000 (1 – 0.40)] / 0.18

= ₹ 60,00,000

K_e of unlevered Firm (given) = 0.18

K_o of unlevered Firm (Same as above = k_e as there is no debt) = 0.18

Market Value of 'A Ltd' [Levered Firm (I)]

Total Value of Levered Firm (V_L) = V_u + (Debt× Tax) = ₹ 60,00,000 + (54,00,000 × 0.40)

= ₹ 81,60,000

Computation of Weighted Average Cost of Capital (WACC) of 'B Ltd.'

= 18% (i.e. K_e = K_o)

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC) of A Ltd

Particulars	A Ltd.
Net Operating Income (NOI)	18,00,000
Less: Interest on Debt (I)	6,48,000
Earnings Before Tax (EBT)	11,52,000
Less: Tax @ 40%	4,60,800
Earnings for equity shareholders (NI)	6,91,200
Total Value of Firm (V) as calculated above	81,60,000
Less: Market Value of Debt	54,00,000
Market Value of Equity (S)	27,60,000
Equity Capitalization Rate [k_e = NI/S]	.2504

Particulars	A Ltd.
Weighted Average Cost of Capital (k_o)*	13.23

*Computation of WACC A Ltd

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	27,60,000	0.338	0.2504	0.0846
Debt	54,00,000	0.662	0.072*	0.0477
Total	81,60,000			0.1323

*K_d= 12% (1- 0.4) = 12% × 0.6 = 7.2%

WACC = 13.23%

ICAI · P6 · RTP Sep 2024 · question p. 7 · answer p. 21-23

318 RTP **May 2024** Q3

Also asked in MTP Sep 2026 S-I Q1(a)

Following data is available in respect of two companies having same business risk:

Capital employed = ₹ 3,00,000, EBIT = ₹ 45,000 and K_e = 12.5%

Sources	A Ltd	B Ltd
	Levered Company (₹)	**Unlevered Company (₹)**
Debt (@10%)	1,50,000	Nil
Equity	1,50,000	3,00,000

An investor is holding 20% shares in levered company. CALCULATE the increase in annual earnings of investor if he switches his holding from Levered to Unlevered company.

ICAI'S ANSWER

(i) **Valuation of firms**

Particulars	A Ltd	B Ltd
	Levered Firm (₹)	**Unlevered Firm (₹)**
EBIT	45,000	45,000
Less: Interest on debt (10% × ₹ 1,50,000)	15,000	Nil
Earnings available to Equity shareholders	30,000	45,000
Ke	12.5%	12.5%
Value of Equity (S) (Earnings available to Equity shareholders/Ke)	2,40,000	3,60,000

Particulars	A Ltd	B Ltd
Debt (D)	1,50,000	Nil
Value of Firm (V) = S + D	3,90,000	3,60,000

Value of Levered company is more than that of unlevered company. Therefore, investor will sell his shares in levered company and buy shares in unlevered company. To maintain the level of risk he will borrow proportionate amount and invest that amount also in shares of unlevered company.

(ii) **Investment & Borrowings**

	₹
Sell shares in Levered company (₹ 2,40,000 × 20%)	48,000
Borrow money (₹ 1,50,000 × 20%)	30,000
Buy shares in Unlevered company	78,000

(iii) **Change in Return**

	₹
Income from shares in Unlevered company (₹ 78,000 × 12.5%)	9,750
Less: Interest on loan (₹ 30,000 × 10%)	3,000
Net Income from unlevered firm	6,750
Less: Income from Levered firm (₹ 48,000 × 12.5%)	6,000
Incremental Income due to arbitrage	750

ICAI · P6 · RTP May 2024 · question p. 5 · answer p. 15-16

319 RTP **Old P8 May 2022** Q3 old scheme

The following data relates to two companies belonging to the same risk class:

Particulars	Bee Ltd.	Cee Ltd.
12% Debt	₹ 27,00,000	-
Equity Capitalization Rate	-	18
Expected Net Operating Income	₹ 9,00,000	₹ 9,00,000

You are required to:

(a) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per M.M. Approach.

(b) DETERMINE the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per M.M. Approach.

ICAI'S ANSWER

(a) Assuming no tax as per MM Approach.

Calculation of Value of Firms 'Bee Ltd.' and 'Cee Ltd' according to MM Hypothesis

Market Value of 'Cee Ltd' [Unlevered(u)]

Total Value of Unlevered Firm (V_u) = [NOI/k_e] = 9,00,000/0.18 = ₹ 50,00,000

K_e of Unlevered Firm (given) = 0.18

K_o of Unlevered Firm (Same as above = k_e as there is no debt) = 0.18

Market Value of 'Bee Ltd' [Levered Firm (I)]

Total Value of Levered Firm (V_L) = V_u + (Debt× Nil)

= ₹ 50,00,000 + (27,00,000 × nil)

= ₹ 50,00,000

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC)

Particulars	Bee Ltd.
Net Operating Income (NOI)	9,00,000
Less: Interest on Debt (I)	3,24,000
Earnings of Equity Shareholders (NI)	5,76,000
Overall Capitalization Rate (k_o)	0.18
Total Value of Firm (V = NOI/k_o)	50,00,000
Less: Market Value of Debt	27,00,000
Market Value of Equity (S)	23,00,000
Equity Capitalization Rate [k_e = NI /S]	0.2504
Weighted Average Cost of Capital (k_o)* k_o = (k_e×S/V) + (k_d×D/V)	0.18

*Computation of WACC Bee Ltd

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	23,00,000	0.46	0.2504	0.1152
Debt	27,00,000	0.54	0.12*	0.0648
Total	50,00,000			0.18

*K_d = 12% (since there is no tax)

WACC = 18%

(b) Assuming 40% taxes as per MM Approach

Calculation of Value of Firms 'Bee Ltd.' and 'Cee Ltd' according to MM Hypothesis

Market Value of 'Cee Ltd' [Unlevered(u)]

Total Value of unlevered Firm (V_u) = [NOI (1 - t)/ke] = 9,00,000 (1 – 0.40)] / 0.18

= ₹ 30,00,000

K_e of unlevered Firm (given) = 0.18

K_o of unlevered Firm (Same as above = k_e as there is no debt) = 0.18

Market Value of 'Bee Ltd' [Levered Firm (I)]

Total Value of Levered Firm (V_L) = V_u + (Debt× Tax)

= ₹ 30,00,000 + (27,00,000 × 0.4)

= ₹ 40,80,000

Computation of Weighted Average Cost of Capital (WACC) of 'Cee Ltd.'

= 18% (i.e. $K_e = K_o$)

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC) of Bee Ltd

Particulars	Bee Ltd. (₹)
Net Operating Income (NOI)	9,00,000
Less: Interest on Debt (I)	3,24,000
Earnings Before Tax (EBT)	5,76,000
Less: Tax @ 40%	2,30,400
Earnings for equity shareholders (NI)	3,45,600
Total Value of Firm (V) as calculated above	40,80,000
Less: Market Value of Debt	27,00,000
Market Value of Equity (S)	13,80,000
Equity Capitalization Rate [k_e = NI/S]	0.2504
Weighted Average Cost of Capital (k_o)* k_o = (k_e×S/V) + (k_d×D/V)	13.23

*Computation of WACC Bee Ltd.

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	13,80,000	0.338	0.2504	0.0846
Debt	27,00,000	0.662	0.072*	0.0477
Total	40,80,000			0.1323

*K_d= 12% (1- 0.4) = 12% × 0.6 = 7.2%

WACC = 13.23%

ICAI · Old P8 · RTP May 2022 · question p. 2 · answer p. 10-12

320 RTP Old P8 Dec 2021 Q3 old scheme

Blue Ltd., an all equity financed company is considering the repurchase of ₹ 275 lakhs equity shares and to replace it with 15% debentures of the same amount. Current market value of the company is ₹ 1,750 lakhs with its cost of capital of 20%. The company's Earnings before Interest and Taxes (EBIT) are expected to remain constant in future years. The company also has a policy of distributing its entire earnings as dividend.

Assuming the corporate tax rate as 30%, you are required to CALCULATE the impact on the following on account of the change in the capital structure as per Modigliani and Miller (MM) Approach:

(i) Market value of the company

(ii) Overall Cost of capital

(iii) Cost of equity

ICAI'S ANSWER

Workings:

Market Value of Equity = $\frac{\text{Net income (NI) for equity holders}}{K_e}$

₹ 1,750 lakhs = $\frac{\text{Net income (NI) for equity holders}}{0.20}$

Net Income to equity holders/EAT = ₹ 350 lakhs

Therefore, EBIT = $\frac{\text{EAT}}{(1-\text{t})} = \frac{₹\ 350\ \text{lakhs}}{(1-0.3)}$ = ₹ 500 lakhs

Income Statement

	All Equity (₹ In lakhs)	Equity & Debt (₹ In lakhs)
EBIT (as calculated above)	500	500
Interest on ₹ 275 lakhs @ 15%	-	41.25
EBT	-	458.75
Tax @ 30%	500	137.63
Income available to equity holders	150	321.12
	350	

(i) Market value of the company

Market value of levered firm = Value of unlevered firm + Tax Advantage

= ₹ 1,750 lakhs + (₹ 275 lakhs x 0.3)

= ₹ 1,832.5 lakhs

Change in market value of the company = ₹ 1,832.5 lakhs – ₹ 1,750 lakhs

= ₹ 82.50 lakhs

The impact is that the market value of the company has increased by ₹ 82.50 lakhs due to replacement of equity with debt.

(ii) Overall Cost of Capital

Market Value of Equity = Market value of levered firm - Equity repurchased

= ₹ 1,832.50 lakhs – ₹ 275 lakhs = ₹ 1,557.50 lakhs

Cost of Equity (K_e) = (Net Income to equity holders / Market value of equity) × 100

= (₹ 321.12 lakhs / ₹ 1,557.50 lakhs) × 100 = 20.62%

Cost of debt (K_d) = I (1 - t) = 15 (1 - 0.3) = 10.50%

Components	Amount (₹ In lakhs)	Cost of Capital %	Weight	WACC (K_o) %
Equity	1,557.50	20.62	0.85	17.53
Debt	275.00	10.50	0.15	1.58
	1,832.50		1	19.11

The impact is that the Overall Cost of Capital or K_o has fallen by 0.89% (20% - 19.11%) due to the benefit of tax relief on debt interest payment.

(iii) Cost of Equity

The impact is that cost of equity has risen by 0.62% (20.62% - 20%) due to the presence of financial risk i.e. introduction of debt in capital structure.

Note: Cost of Capital and Cost of equity can also be calculated with the help of following formulas, though there will be no change in the final answers.

Cost of Capital (K_o) = K_{eu} [1 – (t x L)]

Where,

K_{eu} = Cost of equity in an unlevered company

t = Tax rate

$$L = \frac{\text{Debt}}{\text{Debt} + \text{Equity}}$$

So, $K_o = 0.20 \left[1 - \left(0.3 \times \frac{₹\ 275\ \text{lakhs}}{₹\ 1{,}832.5\ \text{lakhs}}\right)\right]$ = 0.191 or 19.10% (approx.)

Cost of Equity (K_e) = K_{eu} + (K_{eu} – K_d) $\frac{\text{Debt (1- t)}}{\text{Equity}}$

Where,

K_{eu} = Cost of equity in an unlevered company

K_d = Cost of debt

t = Tax rate

So, $K_e = 0.20 + \left((0.20 - 0.15) \times \frac{₹\ 275\ \text{lakhs}\ (1 - 0.3)}{₹\ 1{,}557.5\ \text{lakhs}}\right)$ = 0.2062 or 20.62%

ICAI · Old P8 · RTP Dec 2021 · question p. 1-2 · answer p. 8-9

321 RTP **Old P8 Jul 2021** Q3 old scheme

Same question, different ICAI answer: #330 MTP May 2025 S-II

Zordon Ltd. has net operating income of ₹ 5,00,000 and total capitalization of ₹ 50,00,000 during the current year. The company is contemplating to introduce debt financing in capital structure and has

various options for the same. The following information is available at different levels of debt value:

Debt value (₹)	Interest rate (%)	Equity capitalization rate (%)
0	-	10.00
5,00,000	6.0	10.50
10,00,000	6.0	11.00
15,00,000	6.2	11.30
20,00,000	7.0	12.40
25,00,000	7.5	13.50
30,00,000	8.0	16.00

Assuming no tax and that the firm always maintains books at book values, you are REQUIRED to calculate:

(i) Amount of debt to be employed by firm as per traditional approach.

(ii) Equity capitalization rate, if MM approach is followed.

ICAI'S ANSWER

(a) Amount of debt to be employed by firm as per traditional approach

Calculation of Equity, W_d and W_e

Total Capital (₹)	Debt (₹)	W_d	Equity value (₹)	W_e
(a)	**(b)**	**(b)/(a)**	**(c) = (a) - (b)**	**(c)/(a)**
50,00,000	0	-	50,00,000	1.0
50,00,000	5,00,000	0.1	45,00,000	0.9
50,00,000	10,00,000	0.2	40,00,000	0.8
50,00,000	15,00,000	0.3	35,00,000	0.7
50,00,000	20,00,000	0.4	30,00,000	0.6
50,00,000	25,00,000	0.5	25,00,000	0.5
50,00,000	30,00,000	0.6	20,00,000	0.4

Statement of Weighted Average Cost of Capital (WACC)

K_e	W_e	K_d	W_d	K_eW_e	K_dW_d	K_o
(1)	**(2)**	**(3)**	**(4)**	**(5) = (1) x (2)**	**(6) = (3) x (4)**	**(7) = (5) + (6)**
0.100	1.0	-	-	0.100	-	0.100
0.105	0.9	0.060	0.1	0.095	0.006	0.101
0.110	0.8	0.060	0.2	0.088	0.012	0.100
0.113	0.7	0.062	0.3	0.079	0.019	0.098
0.124	0.6	0.070	0.4	0.074	0.028	0.102
0.135	0.5	0.075	0.5	0.068	0.038	0.106
0.160	0.4	0.080	0.6	0.064	0.048	0.112

So, amount of Debt to be employed = ₹ 15,00,000 as WACC is minimum at this level of debt i.e. 9.8%.

(b) As per MM approach, cost of the capital (Ko) remains constant and cost of equity increases linearly with debt.

$$\text{Value of a firm} = \frac{\text{Net Operating Income (NOI)}}{K_0}$$

$$₹\ 50,00,000 = \frac{₹\ 5,00,000}{K_o}$$

$$K_o = \frac{₹\ 5,00,000}{₹\ 50,00,000} = 10\%$$

Statement of Equity Capitalization rate (ke) under MM approach

Debt (₹)	Equity (₹)	Debt/Equity	K_o	K_d	K_o - K_d	$K_e = K_o + (K_o - K_d)$ Debt/Equity
(1)	**(2)**	**(3) = (1)/(2)**	**(4)**	**(5)**	**(6) = (4) -(5)**	**(7) = (4) + (6) x (3)**
0	50,00,000	0	0.10	-	0.100	0.100
5,00,000	45,00,000	0.11	0.10	0.060	0.040	0.104
10,00,000	40,00,000	0.25	0.10	0.060	0.040	0.110
15,00,000	35,00,000	0.43	0.10	0.062	0.038	0.116
20,00,000	30,00,000	0.67	0.10	0.070	0.030	0.120
25,00,000	25,00,000	1.00	0.10	0.075	0.025	0.125
30,00,000	20,00,000	1.50	0.10	0.080	0.020	0.130

ICAI · Old P8 · RTP Jul 2021 · question p. 2-3 · answer p. 10-11

322 MTP **Sep 2026 S-I** MCQ 8 1 mark MCQ

Meera Enterprises is considering changing its capital structure by increasing its debt ratio from 30% to 50%. What is the likely impact on the company's cost of equity, assuming all other factors remain constant?

A Cost of equity will be unaffected by debt ratio

B Cost of equity will remain unchanged

C Cost of equity will decrease

D Cost of equity will increase

ICAI'S ANSWER

(D) Cost of equity will increase. As Meera Enterprises raises its debt ratio, financial risk to equity shareholders rises (higher fixed interest obligations and greater earnings volatility for equity holders), so equity investors demand a higher return to compensate — increasing the cost of equity.

ICAI · P6 · MTP Sep 2026 · Series I · question p. 4 · answer p. 2

323 MTP **May 2026 S-I** Q2(a) 6 marks

Zanshu Ltd. is an all equity financed company with a market value of ₹ 25,00,000 and cost of equity (Ke) 21%. The company wants to buyback equity shares worth ₹5,00,000 by issuing and raising 15% perpetual debt of the same amount. Rate of tax may be taken as 30%. After the capital restructuring and applying MM Model (with taxes), you are required to CALCULATE:

(i) Market value of Zanshu Ltd.

(ii) Cost of Equity (Ke)

(iii) Weighted average cost of capital (using market weights) and comment on it.

ICAI'S ANSWER

Market Value of Equity = ₹ 25,00,000

Ke = 21%

$$\frac{\text{Net income (NI) for equity - holders}}{K_e} = \text{Market Value of Equity}$$

$$\frac{\text{Net income (NI) for equity holders}}{0.21} = 25,00,000$$

Net income for equity holders = 5,25,000

EBIT= 5,25,000/0.7 = 7,50,000

	All Equity	Debt and Equity
EBIT	7,50,000	7,50,000
Interest to debt-holders	-	75,000
EBT	7,50,000	6,75,000
Taxes (30%)	2,25,000	2,02,500
Income available to equity shareholders	5,25,000	4,72,500
Income to debt holders plus income available to shareholders	5,25,000	5,47,500

Present value of tax-shield benefits = ₹ 5,00,000 × 0.30 = ₹ 1,50,000

(i) Value of Restructured firm

= ₹ 25,00,000 + ₹ 1,50,000 = ₹ 26,50,000

(ii) Cost of Equity (Ke)

Total Value = ₹ 26,50,000

Less: Value of Debt = ₹ 5,00,000

Value of Equity = ₹ 21,50,000

$$Ke = \frac{4,72,500}{21,50,000} = 0.219 = 21.98\%$$

(iii) WACC (on market value weight)

Cost of Debt (after tax) = 15% (1- 0.3) = 0.15 (0.70) = 0.105 = 10.5%

Components of Costs	Amount	Cost of Capital (%)	Weight	WACC (%)
Equity	21,50,000	21.98	0.81	17.80
Debt	5,00,000	10.50	0.19	2.00
	26,50,000			19.80

Comment: At present the company is all equity financed. So, Ke = Ko i.e. 21%. However, after restructuring, the Ko would be reduced to 19.80% and Ke would increase from 21% to 21.98%.

ICAI · P6 · MTP May 2026 · Series I · question p. 6 · answer p. 7-8

324 MTP **May 2026 S-I** Q4(c) 2 marks

Pecking order theory suggests that managers may use various sources for raising of fund in an order. EXPLAIN the preferred order of financing sources suggested by this theory.

ICAI'S ANSWER

1. Managers first choice is to use internal finance.
2. In absence of internal finance, they can use secured debt, unsecured debt, hybrid debt etc.
3. Managers may issue new equity shares as a last option.

ICAI · P6 · MTP May 2026 · Series I · question p. 8 · answer p. 12

325 MTP **May 2026 S-II** Q1(a) 5 marks

DCF Limited has its Earnings Before Interest and Taxes (EBIT) of ₹ 60,00,000. The company is evaluating options to introduce debt financing in its capital structure. Following information are related to various options of debt financing in the capital structure:

Option	Debt in capital structure	Cost of debt (Kd)	Cost of equity (Ke)
1	₹ 60,00,000	10%	12%
2	₹ 75,00,000	13%	15%
3	₹ 90,00,000	14%	16%
4	₹ 1,05,00,000	14%	16%

Assuming that the company operates in zero tax regime.

WHICH option of debt financing in the capital structure would you recommend as per Traditional Approach of the capital structure theory?

ICAI'S ANSWER

Particulars	1	2	3	4
EBIT	60,00,000	60,00,000	60,00,000	60,00,000
Less:				
Interest	6,00,000	9,75,000	12,60,000	14,70,000
Earning available for equity holders [A]	54,00,000	50,25,000	47,40,000	45,30,000
Equity capitalization rate [B]	12%	15%	16%	16%
Market Value of equity [A]/[B]	4,50,00,000	3,35,00,000	2,96,25,000	2,83,12,500
Market value of debt (given)	60,00,000	75,00,000	90,00,000	105,00,000
Total Market Value	5,10,00,000	4,10,00,000	3,86,25,000	3,88,12,500
Overall cost of capital = $\frac{\text{EBIT} \times 100}{\text{Total Market Value}}$	11.76%	14.63%	15.53%	15.46%

The capital structure having debt of ₹ 60,00,000 is recommended as overall cost of capital is minimum at 11.76%.

ICAI · P6 · MTP May 2026 · Series II · question p. 4 · answer p. 4

326 MTP **Jan 2026 S-I** Q3(a) 6 marks

Vastupal Limited is currently a levered a company having 15% convertible debt of ₹ 250 lakhs in its capital structure. The current market value of Vastupal Limited stands at ₹ 1500 lakhs with the overall capitalization rate for the common stock holders is at 20% considering financial and non-financial risks.

Owing to the regulatory norms applicable to Vastupal Limited, the company is planning to become an all equity financed company gradually over a period of time. Currently, company plans to convert 70% of debt to equity and the balance after 3 years. It's Earnings before Interest & Taxes (EBIT) are expected to remain constant in future as well. The entire earnings are distributed to the common stock holders and the applicable corporate tax rate is at 20%.

You are required to calculate the impact on the following on account of the change in the capital structure as per Modigliani and Miller (MM) Hypothesis:

(i) The market value of the company and its impact

(ii) It's cost of equity, and its impact

(iii) It's cost of capital and its impact

Also CALCULATE the market value of the totally unlevered company & the cost of equity i.e. after 3 years when the balance 30% convertible debt is also fully converted to equity.

ICAI'S ANSWER

(i) Market Value of Vastupal Limited (After conversion)

Value of Levered Co (VL) = Value of Substantial Unlevered Co (VU) + Tax advantage

1,500 lakhs = VU + (250 x 70% x 20%)

∴ Market Value of Substantial Unlevered Co (VU) = 1,465 Lakhs

Alternatively, Market Value of Substantial Unlevered Co (VU) can be calculated as below:

MV of Substantial Unlevered Co (30% Debt)

= MV of Totally Unlevered Co + Tax advantage

= {1500–0.20(250)} + (250 x 30% x 20%)

= 1,450 + 15

Market Value of Substantial Unlevered Co (VU) = 1,465 Lakhs

The impact is that the market value of the company has decreased by 35 lakhs (1500 lakhs - 1465 lakhs)

(ii) Cost of Equity (Ke) – After conversion

WN-1 Preparation of Income Statement

Particulars	Before Conversion	After Conversion
EBIT	350.00	350.00
(-) Int	(37.50) (250 × 15%)	(11.25) (250 × 30% x 15%)
EBT	312.50	338.75
(-) Tax @ 20%	(62.50)	(67.75)
Dividend	250.00	271.00

MV of Equity (Before conversion) = $\frac{\text{Dividend}}{\text{Ke}}$

(1,500 – 250) = $\frac{\text{Dividend}}{0.2}$

∴ Dividend = 250 Lakhs

MV of Equity (Before Conversion) = MV of Co (Before conversion) - MV of Debt

= 1,500 – 250 = 1,250

MV of Equity (After Conversion) = MV of Co (After conversion) - MV of Remaining Debt

= 1,465 - {250 x 30%}

= 1,390 lakhs

∴ **Ke (after Conversion)** = $\frac{\text{Dividend}}{\text{MV of Equity}}$

= 271/ 1,390

= 19.49%

The impact is that cost of equity has fallen by 0.50% i.e. 20.00% - 19.49% due to reduction in the financial risk.

(iii) Cost of Capital (Ko) – After Conversion

Components	Amount (in lakhs)	Weights (W)	Cost of Capital (K)	WACC (W x K)
Equity	1390.00	0.9488	19.49	18.49
Debt	75.00	0.0512	12.00 15 (1 – 0.2)	0.61
	1465.00			19.10%

Cost of Capital (Ko) – Before Conversion

Components	Amount (in lakhs)	Weights (W)	Cost of Capital (K)	WACC (W x K)
Equity	1250.00	0.8333	20.00	16.67
Debt	250.00	0.1667	12.00 15 (1 – 0.2)	2.00
	1500.00			18.67%

The impact is that the overall cost of capital has increased by 0.43% i.e. 19.10% - 18.67% due to non-availability of tax benefit owing to the reduction in the debt.

WN-2: Calculation of Market Value of the company & Ke after the end of 3 years

MV of Substantial Unlevered Co = MV of Totally Unlevered Co + Tax advantage

1,465 lakhs = MV of Totally Unlevered Co + (250 x 30% x 20%)

∴ **MV of Totally Unlevered Co = 1,450 lakhs**

Calculation of Ke of Totally Unlevered Co

$$\text{Ke} = \frac{\text{Dividend}}{\text{MV of Equity}}$$

Particulars	Amount
EBIT	350.00
(-) Int	0.00
EBT	350.00
(-) Tax @20%	(70.00)
Dividend	280.00

∴ Ke = 280/1,450

∴ **Ke = 19.31%**

ICAI · P6 · MTP Jan 2026 · Series I · question p. 6-7 · answer p. 11-13

327 MTP **Jan 2026 S-II Q1(b)** 5 marks

Deepa Limited has Earnings Before Interest and Taxes (EBIT) amounting to ₹ 60,00,000. The company is considering the introduction of debt into its capital structure and is evaluating different levels of debt

financing. The details of the alternative debt options, along with the corresponding cost of debt and cost of equity, are given below:

Option	Debt in Capital Structure	Cost of Debt (Kd)	Cost of Equity (Ke)
1	₹ 60,00,000	10%	12%
2	₹ 75,00,000	13%	15%
3	₹ 90,00,000	14%	16%
4	₹ 1,05,00,000	14%	16%

Assuming that the company operates in zero tax regime.

WHICH option of debt financing in the capital structure would you recommend as per Traditional Approach of the capital structure theory?

ICAI'S ANSWER

Particulars	1	2	3	4
EBIT	60,00,000	60,00,000	60,00,000	60,00,000
Less:				
Interest	6,00,000	9,75,000	12,60,000	14,70,000
Earning available for equity holders [A]	54,00,000	50,25,000	47,40,000	45,30,000
Equity capitalization rate [B]	12%	15%	16%	16%
Market Value of equity [A]/[B]	4,50,00,000	3,35,00,000	2,96,25,000	2,83,12,500
Market value of debt (given)	60,00,000	75,00,000	90,00,000	105,00,000
Total Market Value	5,10,00,000	4,10,00,000	3,86,25,000	3,88,12,500
Overall cost of capital = EBIT x 100/Total market value	11.76%	14.63%	15.53%	15.46%

The capital structure having debt of ₹ 60,00,000 is recommended as overall cost of capital is minimum at 11.76%.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 5 · answer p. 6

328 MTP **Sep 2025 S-I** Q3(b) 5 marks

The following data relates to two companies belonging to the same risk class:

Particulars	A Ltd.	B Ltd.
Expected Net Operating Income	₹ 19,00,000	₹ 19,00,000
12% Debt	₹ 57,00,000	-
Equity Capitalization Rate	-	19%

REQUIRED:

(a) Determine the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming no taxes as per M.M. Approach.

(b) Determine the total market value, Equity capitalization rate and weighted average cost of capital for each company assuming 40% taxes as per M.M. Approach.

ICAI'S ANSWER

(i) Assuming no tax as per MM Approach

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis

Market Value of 'B Ltd' [Unlevered(u)]

Total Value of Unlevered Firm (Vu) = [NOI/ke] = 19,00,000/0.19 = ₹ 1,00,00,000

Ke of Unlevered Firm (given) = 0.19

Ko of Unlevered Firm (Same as above = Ke as there is no debt) = 0.19

Market Value of 'A Ltd' [Levered Firm (I)]

Total Value of Levered Firm (VL) = Vu + (Debt× Nil)

= ₹ 1,00,00,000 + (57,00,000 × nil)

= ₹1,00,00,000

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC)

	Particulars	A Ltd.	B Ltd.
A.	Net Operating Income (NOI)	₹ 19,00,000	₹ 19,00,000
B.	Less: Interest on Debt (I)	₹ 6,84,000	-
C.	Earnings of Equity Shareholders (NI)	₹ 12,16,000	₹ 19,00,000
D	Overall Capitalization Rate (ko)	0.19	0.19
E	Total Value of Firm (V = NOI/ko)	₹ 1,00,00,000	₹ 1,00,00,000
F	Less: Market Value of Debt (D)	₹ 57,00,000	-
G	Market Value of Equity (S)	₹ 43,00,000	₹ 1,00,00,000
H	Equity Capitalization Rate [ke = NI /S]	0.2828	0.19
I	Weighted Average Cost of Capital [WACC (ko)]*ko = (ke×S/V) + (kd×D/V)	0.19	0.19

*Computation of WACC A Ltd.

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	₹ 43,00,000	0.43	0.2828	0.1216
Debt	₹ 57,00,000	0.57	0.12*	0.0684
Total	₹ 1,00,00,000			0.19

*Kd = 12% (since there is no tax)

WACC = 19%

(ii) Assuming 40% taxes as per MM Approach

Calculation of Value of Firms 'A Ltd.' and 'B Ltd' according to MM Hypothesis

Market Value of 'B Ltd' [Unlevered(u)]

Total Value of unlevered Firm (Vu) = [NOI (1 - t)/ke]

= 19,00,000 (1 – 0.40)] / 0.19 = ₹ 60,00,000

Ke of unlevered Firm (given) = 0.19

Ko of unlevered Firm (Same as above = Ke as there is no debt) = 0.19

Market Value of 'A Ltd' [Levered Firm (I)]

Total Value of Levered Firm (VL) = Vu + (Debt× Tax)

= ₹ 60,00,000 + (₹ 57,00,000 × 0.4) = ₹ 82,80,000

Computation of Weighted Average Cost of Capital (WACC) of 'B Ltd.'

= 19% (i.e. Ke = Ko)

Computation of Equity Capitalization Rate and Weighted Average Cost of Capital (WACC) of A Ltd

Particulars	A Ltd. (₹)
Net Operating Income (NOI)	19,00,000
Less: Interest on Debt (I)	6,84,000
Earnings Before Tax (EBT)	12,16,000
Less: Tax @ 40%	4,86,400
Earnings for equity shareholders (NI)	7,29,600
Total Value of Firm (V) as calculated above	82,80,000
Less: Market Value of Debt	57,00,000
Market Value of Equity (S)	25,80,000
Equity Capitalization Rate [ke = NI/S]	0.2828
Weighted Average Cost of Capital (ko)* ko = (ke×S/V) + (kd×D/V)	13.77%

*Computation of WACC A Ltd.

Component of Capital	₹	Weight	Cost of Capital	WACC
Equity	25,80,000	0.312	0.2828	0.0882
Debt	57,00,000	0.688	0.072*	0.0495
Total	82,80,000			0.1377

*Kd = 12% (1- 0.4) = 12% × 0.6 = 7.2%

WACC = 13.77%

ICAI · P6 · MTP Sep 2025 · Series I · question p. 9 · answer p. 13-16

329 MTP **May 2025 S-I** MCQ 6 2 marks MCQ

Z Ltd.'s operating income (before interest and tax) is ₹ 9,00,000. The firm's cost of debt is 10 per cent and currently firm employs ₹ 30,00,000 of debt. The overall cost of capital of firm is 12 per cent. What is the cost of equity.

A 13.3%

B 15.2%

C 16.0%

D 12.5%

ICAI'S ANSWER

(A) 13.3%

Calculation of Cost of Equity

Calculation of value of firm (v) = $\frac{\text{EBIT}}{\text{Overall cost of capital } (K_o)}$

= $\frac{9,00,000}{0.12}$ = Rs.75,00,000

Market value of equity (S) = V – Debts

= 75,00,000 – 30,00,000 = ₹. 45,00,000

Market value of debts (D) = 30,00,000

K_e (Cost of equity) = $K_o\left(\frac{V}{S}\right) - K_d\left(\frac{D}{S}\right)$

= $0.12\left(\frac{75,00,000}{45,00,000}\right) - 0.10\left(\frac{30,00,000}{45,00,000}\right)$

= 0.20 – 0.067 = 0.133 × 100

K_e = 13.3%.

ICAI · P6 · MTP May 2025 · Series I · question p. 3 · answer p. 3

330 MTP **May 2025 S-II** Q2(b) 6 marks

Same question, different ICAI answer: #321 Old P8 RTP Jul 2021

Zordon Ltd. has net operating income of ₹ 5,00,000 and total capitalization of ₹ 50,00,000 during the current year. The company is contemplating to introduce debt financing in capital structure and has

various options for the same. The following information is available at different levels of debt value:

Debt value (₹)	Interest rate (%)	Equity capitalization rate (%)
0	-	10.00
5,00,000	6.0	10.50
10,00,000	6.0	11.00
15,00,000	6.2	11.30
20,00,000	7.0	12.40
25,00,000	7.5	13.50
30,00,000	8.0	16.00

Assuming no tax and that the firm always maintains books at book values, you are REQUIRED to calculate:

(i) Amount of debt to be employed by firm as per traditional approach.

(ii) Equity capitalization rate, if MM approach is followed.

ICAI'S ANSWER

(i) Amount of debt to be employed by firm as per traditional approach

Calculation of Equity, Wd and We

Total Capital (₹)	Debt (₹)	Wd	Equity value (₹)	We
(a)	**(b)**	**(b)/(a)**	**(c) = (a) - (b)**	**(c)/(a)**
50,00,000	0	-	50,00,000	1.0
50,00,000	5,00,000	0.1	45,00,000	0.9
50,00,000	10,00,000	0.2	40,00,000	0.8
50,00,000	15,00,000	0.3	35,00,000	0.7
50,00,000	20,00,000	0.4	30,00,000	0.6
50,00,000	25,00,000	0.5	25,00,000	0.5
50,00,000	30,00,000	0.6	20,00,000	0.4

Statement of Weighted Average Cost of Capital (WACC)

Ke	We	Kd	Wd	Ke We	KdWd	Ko
(1)	(2)	(3)	(4)	(5) = (1) x (2)	(6) = (3) x (4)	(7) = (5) + (6)
0.100	1.0	-	-	0.100	-	0.100
0.105	0.9	0.060	0.1	0.095	0.006	0.101
0.110	0.8	0.060	0.2	0.088	0.012	0.100
0.113	0.7	0.062	0.3	0.079	0.019	0.098
0.124	0.6	0.070	0.4	0.074	0.028	0.102
0.135	0.5	0.075	0.5	0.068	0.038	0.106
0.160	0.4	0.080	0.6	0.064	0.048	0.112

So, amount of Debt to be employed = ₹ 15,00,000 as WACC is minimum at this level of debt i.e. 9.8%.

(ii) As per MM approach, cost of the capital (Ko) remains constant and cost of equity increases linearly with debt.

$$\text{Value of a firm} = \frac{\text{Net Operating Income (NOI)}}{K_0}$$

$$₹\ 50,00,000 = \frac{₹\ 5,00,000}{K_o}$$

$$\text{Ko} = \frac{₹\ 5,00,000}{₹\ 50,00,000} = 10\%$$

Statement of Equity Capitalization rate (ke) under MM approach

Debt (₹)	Equity (₹)	Debt/Equity	Ko	Kd	Ko - Kd	Ke = Ko + (Ko - Kd) Debt/Equity
(1)	**(2)**	**(3) = (1)/(2)**	**(4)**	**(5)**	**(6) = (4) -(5)**	**(7) = (4) + (6) x (3)**
0	50,00,000	0	0.10	-	0.100	0.100
5,00,000	45,00,000	0.11	0.10	0.060	0.040	0.104
10,00,000	40,00,000	0.25	0.10	0.060	0.040	0.110
15,00,000	35,00,000	0.43	0.10	0.062	0.038	0.116
20,00,000	30,00,000	0.67	0.10	0.070	0.030	0.120
25,00,000	25,00,000	1.00	0.10	0.075	0.025	0.125
30,00,000	20,00,000	1.50	0.10	0.080	0.020	0.130

ICAI · P6 · MTP May 2025 · Series II · question p. 8 · answer p. 11-12

331 MTP **Jan 2025 S-II Q1(b)** 5 marks

Theta Limited is expecting an annual earning of Rs. 3 Lakhs before paying any interest and taxes. The company has Rs. 10 lakhs of 10% debentures in its capital structure. The capitalisation rate is 12.5%. You

are required to calculate the value of Theta Limited as per the NI approach. Also, COMPUTE the overall cost of capital.

ICAI'S ANSWER

EBIT = ₹ 3,00,000

Less: Interest = ₹ 10,00,000 × 10% = ₹ 1,00,000

Earnings available to equity shareholders = ₹ 2,00,000

Equity capitalization rate = 12.5%

Market value of equity = $\frac{₹\ 2,00,000}{12.5\%}$ = ₹ 16,00,000

Market value of debt = ₹ 10,00,000

Market value of the firm = ₹ 26,00,000

Overall cost of capital = $\frac{₹\ 3,00,000 \times 100}{₹\ 26,00,000}$ = 11.54%

ICAI · P6 · MTP Jan 2025 · Series II · question p. 5 · answer p. 4

332 MTP **Sep 2024 S-I** MCQ 8 1 mark MCQ

Given:

Earnings available to the equity shareholders ₹ 30 Lakhs,

Cost of equity is 15%,

Debt outstanding ₹ 150 Lakhs

Value of the firm will be –

A ₹ 200 Lakhs

B ₹ 250 Lakhs

C ₹ 350 Lakhs

D ₹ 300 Lakhs

ICAI'S ANSWER

Answer key: **(C)**

(c) ₹ 350 Lakhs

Value of Equity = 30 Lakhs ÷ 15% = ₹ 200 Lakhs

Value of Debt = ₹ 150 Lakhs

Value of Firm = 200 Lakhs + 150 Lakhs = ₹ 350 Lakhs

ICAI · P6 · MTP Sep 2024 · Series I · question p. 3 · answer p. 3

333 MTP **Sep 2024 S-II** MCQ 8 1 mark MCQ

A company is considering changing its capital structure by increasing its debt ratio from 40% to 55%. What is the likely impact on the company's cost of equity, assuming all other factors remain constant?

A Cost of equity will be unaffected by debt ratio

B Cost of equity will remain unchanged

C Cost of equity will decrease

D Cost of equity will increase

ICAI'S ANSWER

Answer key: **(D)**

(d) Cost of equity will increase. As the company increases its debt ratio, the financial risk increases, which typically leads to an increase in the cost of equity as equity investors demand a higher return for the additional risk.

ICAI · P6 · MTP Sep 2024 · Series II · question p. 3 · answer p. 2

334 MTP **May 2024 S-I** Q1(b) 5 marks

Capital structure (in market-value terms) of AN Ltd is given below:

Company	Debt	Equity
AN Ltd.	50%	50%

The borrowing rate for the company is 10% in a no-tax world and capital markets are assumed to be perfect.

Required:

(i) If Mr. R, owns 8% of the equity shares of AN Ltd., DETERMINE his return if the Company has net operating income of ₹ 10,00,000 and the overall capitalization rate of the company (Ko) is 20%.

(ii) CALCULATE the implied required rate of return on equity of AN Ltd.

ICAI'S ANSWER

Value of AN Ltd. $= \frac{NOI}{K_o} = \frac{₹\ 10{,}00{,}000}{20\%} = ₹\ 50{,}00{,}000$

(i) Return on Shares of Mr. R on AN Ltd.

Particulars	Amount (₹)
Value of the company	50,00,000
Market value of debt (50% x ₹ 50,00,000)	25,00,000
Market value of shares (50% x ₹ 50,00,000)	**25,00,000**
Particulars	**Amount (₹)**
Net operating income	10,00,000
Interest on debt (10% × ₹ 25,00,000)	2,50,000
Earnings available to shareholders	**7,50,000**
Return on 8% shares (8% × ₹ 7,50,000)	**60,000**

(ii) Implied required rate of return on equity of AN Ltd. $= \frac{₹\ 7{,}50{,}000}{₹\ 25{,}00{,}000}$

= **30%**

ICAI · P6 · MTP May 2024 · Series I · question p. 3-4 · answer p. 3

335 MTP Old P8 May 2023 S-II Q4 10 marks old scheme

Following data is available in respect of two companies having same business risk:

Capital employed = ₹ 12,00,000, EBIT = ₹ 2,40,000 and Ke = 15%

Sources	Dumbo Ltd (₹)	Jumbo Ltd (₹)
Debt (@12%)	4,00,000	Nil
Equity	8,00,000	12,00,000

An investor is holding 20% shares in the levered company. CALCULATE the increase in annual earnings of investor if arbitrage process is undertaken.

Also EXPLAIN the arbitrage process if Ke = 20% for Dumbo Ltd instead of 15%.

ICAI'S ANSWER

(I). Valuation of firms

Particulars	Dumbo Ltd (₹)	Jumbo Ltd (₹)
EBIT	2,40,000	2,40,000
Less: Interest on debt (12% × ₹ 4,00,000)	48,000	Nil
Earnings available to Equity shareholders	1,92,000	2,40,000
K_e	15%	15%
Value of Equity (S) (Earnings available to Equity shareholders/K_e)	12,80,000	16,00,000
Debt (D)	4,00,000	Nil
Value of Firm (V) = S + D	16,80,000	16,00,000

Value of Levered company is more than that of unlevered company. Therefore, investor will sell his shares in levered company and buy shares in unlevered company. To maintain the level of risk he will borrow proportionate amount and invest that amount also in shares of unlevered company

(II). Investment & Borrowings

	₹
Sell shares in Levered company (12,80,000 × 20%)	2,56,000
Borrow money (4,00,000 × 20%)	80,000
Buy shares in Unlevered company	3,36,000

(III). Change in Return

	₹
Income from shares in Unlevered company (2,40,000 × 3,36,000/16,00,000)	50,400
Less: Interest on loan (80,000 × 12%)	9,600
Net Income from unlevered firm 40,800	
Less: Income from Levered firm (1,92,000 × 20%)	38,400
Incremental Income due to arbitrage	2,400

Arbitrage process if Ke = 20%

(I). Valuation of firms

Particulars	Dumbo Ltd (₹)	Jumbo Ltd (₹)
EBIT	2,40,000	2,40,000
Less: Interest on debt (12% × ₹ 4,00,000)	48,000	Nil
Earnings available to Equity shareholders	1,92,000	2,40,000
K_e	20%	15%
Value of Equity (S) (Earnings available to Equity shareholders/K_e)	9,60,000	16,00,000
Debt (D)	4,00,000	Nil
Value of Firm (V) = S + D	13,80,000	16,00,000

Value of unlevered company is more than that of levered company. Therefore, investor will sell his shares in unlevered company and buy proportionate shares and debt in levered company i.e. 20% share.

(II). Investment & Borrowings

	₹
Sell shares in unlevered company (16,00,000 × 20%)	3,20,000
Buy shares in levered company (9,60,000 × 20%)	1,92,000
Buy Debt of levered company	1,28,000

(III). Change in Return

	₹
Income from shares in levered company (1,92,000 × 20%)	38,400
Add: Interest on debt of levered (1,28,000 × 12%)	15,360
Net Income from levered firm	53,760
Less: Income from unlevered firm (2,40,000 × 20%)	48,000
Incremental Income due to arbitrage	5,760

ICAI · Old P8 · MTP May 2023 · Series II · question p. 3 · answer p. 8-9

336 MTP **Old P8 Nov 2022 S-I** Q5(a) 8 marks old scheme

Leo Ltd. has a net operating income of ₹ 21,60,000 and the total capitalisation of ₹ 120 lakhs. The company is evaluating the options to introduce debt financing in the capital structure and the following information is available at various levels of debt value.

Debt value (₹)	Interest rate (%)	Equity Capitalisation rate (%)
0	N.A.	12.00
10,00,000	7.00	12.50
20,00,000	7.00	13.00
30,00,000	7.50	13.50
40,00,000	7.50	14.00
50,00,000	8.00	15.00
60,00,000	8.50	16.00
70,00,000	9.00	17.00
80,00,000	10.00	20.00

You are required to COMPUTE the equity capitalization rate if MM approach is followed. Assume that the firm operates in zero tax regime and calculations to be based on book values.

ICAI'S ANSWER

As per MM approach, cost of the capital (K_o) remains constant, and cost of equity increases linearly with debt.

Value of a Firm = $\frac{\text{NOI}}{K_0}$

$\therefore 1,20,00,000 = \frac{21,60,000}{k_0}$

$\therefore K_0 = \frac{21,60,000}{1,20,00,000} = 18\%$

Under MM approach, $k_e = k_0 + \frac{D}{E}(k_0 - k_d)$

Statement of equity capitalization under MM approach

Debt Value (₹)	Equity Value (₹)	Debt/ Equity	K_d (%)	K_o (%)	K_o-k_d (%)	$K_e = K_o+(K_o-K_d)$ (D/E) (%)
-	1,20,00,000	0.0000	NA	18.00	18.00	18.00
10,00,000	1,10,00,000	0.0909	7.00	18.00	11.00	19.00
20,00,000	1,00,00,000	0.2000	7.00	18.00	11.00	20.20
30,00,000	90,00,000	0.3333	7.50	18.00	10.50	21.50
40,00,000	80,00,000	0.5000	7.50	18.00	10.50	23.25
50,00,000	70,00,000	0.7143	8.00	18.00	10.00	25.14
60,00,000	60,00,000	1.0000	8.50	18.00	9.50	27.50
70,00,000	50,00,000	1.4000	9.00	18.00	9.00	30.60
80,00,000	40,00,000	2.0000	10.00	18.00	8.00	34.00

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 4 · answer p. 6

337 MTP **Old P8 May 2022 S-I** Q2 10 marks old scheme

PRI Ltd. and SHA Ltd. are identical, however, their capital structure (in market-value terms) differs as follows:

Company	Debt	Equity
PRI Ltd.	60%	40%
SHA Ltd.	20%	80%

The borrowing rate for both companies is 8% in a no-tax world and capital markets are assumed to be perfect.

(a) (i) If Mr. Rhi, owns 6% of the equity shares of PRI Ltd., DETERMINE his return if the Company has net operating income of ₹ 9,00,000 and the overall capitalization rate of the company (K_o) is 18%.

(ii) CALCULATE the implied required rate of return on equity of PRI Ltd.

(b) SHA Ltd. has the same net operating income as PRI Ltd.

(i) CALCULATE the implied required equity return of SHA Ltd.

(ii) ANALYSE why does it differ from that of PRI Ltd.

ICAI'S ANSWER

Value of PRI Ltd. = $\frac{NOI}{K_o} = \frac{₹\ 9,00,000}{18\%}$ = ₹ 50,00,000

(a) (i) Return on Shares of Mr. Rhi on PRI Ltd.

Particulars	Amount (₹)
Value of the company	50,00,000
Market value of debt (60% x ₹ 50,00,000)	30,00,000
Market value of shares (40% x ₹ 50,00,000)	**20,00,000**
Particulars	**Amount (₹)**
Net operating income	9,00,000
Interest on debt (8% × ₹ 30,00,000)	2,40,000
Earnings available to shareholders	**6,60,000**
Return on 6% shares (6% × ₹ 6,60,000)	**39,600**

(ii) Implied required rate of return on equity of PRI Ltd. = $\frac{₹\ 6,60,000}{₹\ 20,00,000}$ = **33%**

(b) (i) Calculation of Implied rate of return of SHA Ltd.

Particulars	Amount (₹)
Total value of company	50,00,000
Market value of debt (20% × ₹ 50,00,000)	10,00,000
Market value of equity (80% × ₹ 50,00,000)	**40,00,000**
Particulars	**Amount (₹)**
Net operating income	9,00,000
Interest on debt (8% × ₹ 10,00,000)	80,000
Earnings available to shareholders	**8,20,000**

Implied required rate of return on equity = $\frac{₹\ 8,20,000}{₹\ 40,00,000}$ = **20.5%**

(ii) Implied required rate of return on equity of SHA Ltd. is lower than that of PRI Ltd. because SHA Ltd. uses less debt in its capital structure. As the equity capitalisation is a linear function of the debt-to-equity ratio when we use the net operating income approach, the decline in required equity return offsets exactly the disadvantage of not employing so much in the way of "cheaper" debt funds.

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338 MTP **Old P8 May 2022 S-II** Q1(a) 5 marks old scheme

Following data is available in respect of two companies having same business risk:

Capital employed = ₹ 4,00,000, EBIT = ₹ 60,000 and K_e = 12.5%

Sources	Levered Company (₹)	Unlevered Company (₹)
Debt (@10%)	2,00,000	Nil
Equity	2,00,000	4,00,000

An investor is holding 15% shares in levered company. CALCULATE the increase in annual earnings of investor if he switches his holding from Levered to Unlevered company.

ICAI'S ANSWER

Valuation of firms

Particulars	Levered Firm (₹)	Unlevered Firm (₹)
EBIT	60,000	60,000
Less: Interest on debt (10% × ₹ 2,00,000)	20,000	Nil
Earnings available to Equity shareholders	40,000	60,000
K_e	12.5%	12.5%
Value of Equity (S) (Earnings available to Equity shareholders/K_e)	3,20,000	4,80,000
Debt (D)	2,00,000	Nil
Value of Firm (V) = S + D	5,20,000	4,80,000

Value of Levered company is more than that of unlevered company. Therefore, investor will sell his shares in levered company and buy shares in unlevered company. To maintain the level of risk he will borrow proportionate amount and invest that amount also in shares of unlevered company.

Investment & Borrowings	(₹)
Sell shares in Levered company (₹ 3,20,000 × 15%)	48,000
Borrow money (₹ 2,00,000 × 15%)	30,000
Buy shares in Unlevered company	78,000

Change in Return	(₹)
Income from shares in Unlevered company (₹ 78,000 × 12.5%)	9,750
Less: Interest on loan (₹ 30,000 × 10%)	3,000
Net Income from unlevered firm	6,750
Less: Income from Levered firm (₹ 48,000 × 12.5%)	6,000
Incremental Income due to arbitrage	750

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339 MTP Old P8 Dec 2021 S-I Q1(a) 5 marks old scheme

In respect of two companies having same business risk, following information is given:

Capital employed = ₹ 4,00,000; EBIT = ₹ 60,000; K_e = 12%

Sources	Levered Company (₹)	Unlevered Company (₹)
Debt (@10%)	1,50,000	Nil
Equity	1,50,000	3,00,000

Investor is holding 20% shares in levered company. CALCULATE increase in annual earnings of investor if he switches his holding from Levered to Unlevered company.

ICAI'S ANSWER

1. Valuation of firms

Particulars	Levered Firm (₹)	Unlevered Firm (₹)
EBIT	60,000	60,000
Less: interest (1,50,000 × 10%)	15,000	Nil
Earnings available to Equity Shareholder/K_e	45,000	60,000
	12%	12%
Value of Equity	3,75,000	5,00,000
Debt	1,50,000	Nil
Value of Firm	5,25,000	5,00,000

Value of Levered company is more than that of unlevered company. Therefore, investor will sell his shares in levered company and buy shares in unlevered company. To maintain the level of risk he will borrow proportionate amount and invest that amount also in shares of unlevered company.

2. Investment & Borrowings

Sell shares in Levered company (3,75,000×20%)	75,000
Borrow money (1,50,000 × 20%)	30,000
Buy shares in Unlevered company	1,05,000

3. Change in Return

Income from shares in Unlevered company (1,05,000 × 12%)	12,600
Less: interest on loan (30,000 × 10%)	3,000
Net Income from unlevered firm	9,600
Income from Levered firm (75000 × 12%)	9,000
Incremental Income due to arbitrage	600

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 1 · answer p. 1

340 MTP **Old P8 Dec 2021 S-II** Q5(b) 2 marks old scheme

'Pecking order theory' suggests manager to use various sources for raising of fund in certain order. BRIEF out that order.

ICAI'S ANSWER

Pecking order theory suggests that managers may use various sources for raising of fund in the following order:

1. Managers first choice is to use **internal finance.**
2. In absence of internal finance, they can use secured **debt**, unsecured debt, hybrid debt etc.
3. Managers may issue new **equity** shares as a last option.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 5 · answer p. 8-9

341 MTP **Old P8 Jul 2021 S-II** Q1(a) 5 marks old scheme

Kee Ltd. and Lee Ltd. are identical in every respect except for capital structure. Kee Ltd. does not employ debt in its capital structure, whereas Lee Ltd. employs 12% debentures amounting to Rs. 20 lakhs. Assuming that:

(i) All assumptions of MM model are met;

(ii) The income tax rate is 30%;

(iii) EBIT is Rs. 5,00,000 and

(iv) The equity capitalization rate of Kee Ltd. is 25%.

CALCULATE the average value of both the Companies.

ICAI'S ANSWER

Kee Ltd. (pure Equity) i.e. unlevered company:

EAT = EBT (1 – t)

= EBIT (1 - 0.3) = Rs. 5,00,000 × 0.7 = Rs. 3,50,000

(Here, EBIT = EBT as there is no debt)

Value of unlevered company Kee Ltd. = $\frac{\text{EAT}}{\text{Equity capitalization rate}}$

= $\frac{\text{Rs. } 3{,}50{,}000}{25\%}$ = Rs. 14,00,000

Lee Ltd. (Equity and Debt) i.e levered company:

Value of levered company = Value of Equity + Value of Debt

= Rs. 14,00,000 + (Rs. 20,00,000 × 0.3)

= Rs. 20,00,000

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342 MTP **Old P8 May 2020 S-I** Q1(d) 5 marks old scheme

A&R Ltd. is an all equity financed company with a market value of Rs.25,000 lakh and cost of equity (K_e) 18%. The company wants to buyback equity shares worth Rs.5,000 lakh by issuing and raising 10%

debentures redeemable at 10% premium after 5 years. Rate of tax may be taken as 35%. Applying Modigliani-Miller (MM) (with taxes), you are required to CALCULATE after restructuring:

(i) Market value of A&R Ltd.

(ii) Cost of Equity (K_e)

(iii) Weighted average cost of capital (using market weights).

ICAI'S ANSWER

Value of a company (V) = Value of equity (S) + Value of debt (D)

A&R Ltd. is all equity financed company, its value would equal to value of equity.

Market value of equity = = $\frac{\text{Net Income (NI)}}{K_e}$

In the question, market value of equity is Rs.25,000 lakh and cost of equity (K_e) is 18%. The Net Income (NI) is calculated as follows:

$\frac{\text{Net income (NI) for equity - holders}}{K_e}$ = Market Value of Equity

$\frac{\text{Net income (NI) for equity holders}}{0.18}$ = 25,000 lakh

Net income for equity holders = 4,500 lakh

Net Income (NI) is after tax income, the before tax income would be

EBT= $\frac{4{,}500\text{lakh}}{(1-0.35)}$ = 6,923.07 lakh.

Since, A&R Ltd. is an all equity financed and there is no interest expense, so here EBT is equal to EBIT.

After issuing 10% debentures, the A&R Ltd would become a levered company.

(i) The value of A&R Ltd. after issuing debentures would be calculated as follows:

Value of a levered company (V_g)

= Value of an unlevered company (V_u) + Tax benefit (TB)

= Rs.25,000 lakh + (Rs.5,000 lakh × 35%)

= Rs.25,000 + Rs.1,750 = Rs.26,750

(ii) Cost of Equity (K_e)

Total Value	= Rs.26,750 lakh
Less: Value of Debt	= Rs. 5,000 lakh
Value of Equity	= Rs. 21,750

K_e = $\frac{4{,}175\text{lakh}}{21{,}750\text{lakh}}$ =0.1919 = 19.19%

(iii) WACC (on market value weight)

Components of Costs	Amount (lakh)	Cost of Capital (%)	Weight	WACC (%)
Equity	21,750	19.19	0.81	15.54
Debt	5,000	8.10	0.19	1.54
	26,750			17.08

Workings Note:

1. (Rs. in lakh)

	All Equity	Debt and Equity
EBIT (as calculated above)	6,923.07	6,923.07
Interest to debt-holders	-	500.00
EBT	6,923.07	6,423.07
Taxes (35%)	2,423.07	2,248.07
Income available to equity shareholders	4,500.00	4,175.00
Income to debt holders plus income available to shareholders	4,500.00	4,675.00

2. Cost of Debenture (K_d) $= \dfrac{\text{Rs.}500(1-0.35)+\dfrac{(5{,}500-5{,}000)}{5}}{\dfrac{(5{,}500+5{,}000)}{2}}$

$= \dfrac{\text{Rs.}325+100}{5{,}250} = 0.081 \text{ or } 8.1\%$

ICAI · Old P8 · MTP May 2020 · Series I · question p. 2 · answer p. 3-4

343 MTP **Old P8 May 2020 S-I** Q6(a) 4 marks old scheme

EXPLAIN in brief the Pecking order theory.

ICAI'S ANSWER

This theory states that firms prefer to issue debt when they are positive about future earnings. Equity is issued when they are doubtful and internal finance is insufficient.

The pecking order theory argues that the capital structure decision is affected by manager's choice of a source of capital that gives higher priority to sources that reveal the least amount of information.

Pecking order theory suggests that managers may use various sources for raising of fund in the following order.

1. Managers first choice is to use internal finance
2. In absence of internal finance they can use secured debt, unsecured debt, hybrid debt etc.
3. Managers may issue new equity shares as a last option.

So briefly under this theory rules are

Rule 1: Use internal financing first.

Rule 2: Issue debt next

Rule 3: Issue of new equity shares at last

ICAI · Old P8 · MTP May 2020 · Series I · question p. 4 · answer p. 11

344 MTP **Old P8 Nov 2019 S-I** Q1(d) 5 marks old scheme

A Ltd. and B Ltd. are identical in every respect except capital structure. A Ltd. does not employ debts in its capital structure whereas B Ltd. employs 12% Debentures amounting to Rs.100 lakhs. Assuming that :

(i) All assumptions of M-M model are met;

(ii) Income-tax rate is 30%;

(iii) EBIT is Rs. 25,00,000 and

(iv) The Equity capitalization rate of 'A' Ltd. is 20%.

CALCULATE the value of both the companies and also find out the Weighted Average Cost of Capital for both the companies.

ICAI'S ANSWER

(i) Calculation of Value of 'A Ltd.' and 'B Ltd' according to MM Hypothesis

Market Value of 'A Ltd' (Unlevered)

$$V_u = \frac{\text{EBIT}\ (1-t)}{K_e} = \frac{\text{Rs.25,00,000}\ (1-0.30)}{20\%} = \frac{\text{Rs.17,50,000}}{20\%} = \text{Rs. 87,50,000}$$

Market Value of 'B Ltd.' (Levered)

$V_g = V_u + \text{TB}$

= Rs. 87,50,000 + (Rs.1,00,00,000 × 0.30)

= Rs. 87,50,000 + Rs.30,00,000 = Rs.1,17,50,000

(ii) Computation of Weighted Average Cost of Capital (WACC)

WACC of 'A Ltd.' = 20% (i.e. $K_e = K_o$)

WACC of 'B Ltd.'

	B Ltd. (Rs.)
EBIT	25,00,000
Interest to Debt holders	(12,00,000)
EBT	13,00,000
Taxes @ 30%	(3,90,000)
Income available to Equity Shareholders	9,10,000
Total Value of Firm	1,17,50,000
Less: Market Value of Debt	(1,00,00,000)
Market Value of Equity	17,50,000
Return on equity (K_e) = 9,10,000 / 17,50,000	0.52

Computation of WACC B. Ltd

Component of Capital	Amount	Weight	Cost of Capital	WACC
Equity	17,50,000	0.149	0.52	0.0775
Debt	1,00,00,000	0.851	0.084*	0.0715
Total	1,17,50,000			0.1490

*K_d= 12% (1- 0.3) = 12% × 0.7 = 8.4%

WACC = 14.90%

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 1-2 · answer p. 3

5.2 EBIT-EPS analysis, indifference point and financial break-even

55 questions · asked in 39 ICAI documents · 9 exam papers

345 EXAM **Sep 2026** Q3(b) 4 marks

The following information is available from RST Company Ltd :

No. of equity shares outstanding	2,50,000 shares
10% long term debt	₹ 10,00,000
10% Pref. shares of ₹ 10 each	₹ 2,00,000
Fixed Cost	₹ 50,000
Variable cost ratio	60% of sales
Tax rate	25%

You are required to determine the amount of sale assuming that :

(i) All earnings are wiped out by the payment of interest and preference dividend; and

(ii) EPS = ₹ 3

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 19

346 EXAM **Sep 2025** Q3(a) 7 marks

AVS Limited is planning to diversify its cotton manufacturing business. The Chief Financial Officer (CFO) intends to raise fund of ₹ 22,00,000 for this project. The current estimates of Earnings Before Interest and Taxes (EBIT) from the new project are ₹ 5,00,000 per annum. The company's share is currently selling at ₹ 120 and is expected to decline to ₹ 110, in case the funds are borrowed more than ₹ 10,00,000. It is stated that the cost of debt will be 14% up to ₹ 8,00,000, 16% for additional amount of ₹ 2,00,000 above ₹ 8,00,000 and 18% for additional amount above ₹ 10,00,000. The tax rate applicable to the company is 30%. the following options are in consideration of the company.

Option	Debt	Equity
I	60%	40%
II	50%	50%
III	40%	60%

Considering the objective of maximising Earning Per Share (EPS), which option of financing should the company choose?

ICAI'S ANSWER

Particulars	Options		
	Option-I: 60% Debt	**Option-II: 50% Debt**	**Option-III: 40% Debt**
	₹	₹	₹
Debt	13,20,000	11,00,000	8,80,000
Equity	8,80,000	11,00,000	13,20,000
EBIT	5,00,000	5,00,000	5,00,000
Interest on Debts:			
- on debt upto ₹ 8 lacs @14%	(1,12,000)	(1,12,000)	(1,12,000)
- on debt above 8 lacs and upto 10 lacs @ 16%	(32,000)	(32,000)	(12,800)
- on debt above 10 lacs @ 18%	(57,600)	(18,000)	---
Profit before taxes	2,98,400	3,38,000	3,75,200
Taxes @ 40%	(89,520)	(1,01,400)	(1,12,560)
Profit after taxes	2,08,880	2,36,600	2,62,640
Number of shares	8,000	10,000	11,000
Earnings per share	**26.11**	**23.66**	**23.88**

Since Option 1 i.e 60% debt provide the maximum EPS, company should choose Option 1 (60% debt and 40% equity) for financing.

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347 **EXAM** **Jan 2025** Case Scenario - I · MCQ 1 to MCQ 5

CASE SCENARIO - I

Coral Ltd. is an agri-business company that operates in two segments - animal feed and crop protection. The company's Research and Development Department has been instrumental in its growth and success.

The existing capital structure of Coral Ltd. is as follows:

Particulars	Amount (₹)
Equity Shares (10,00,000 shares of ₹ 10 each)	1,00,00,000
15% Debentures (30,000 Debentures of ₹ 100 each)	30,00,000

Coral Ltd. desires to expand its horizon in breeding high-yielding and disease-resistant seeds for increasing agricultural productivity. The company requires additional funds amounting ₹ 100 lakh to finance its business expansion plan. The expected earnings before interest and taxes after this additional investment will be ₹ 76 lakh. The applicable corporate income tax rate is 30%.

The company has two alternatives for raising this additional fund:

Particulars	Plan - I	Plan – II
Equity shares of ₹ 10 each to be issued at a premium of ₹ 15 per share	30%	10%
13% Debentures of ₹ 100 each to be issued at par	70%	50%
7.15% Preference Shares of ₹10 each to be issued at par	-	40%

You are required to answer the following questions 1 to 5:

MCQ 1

What would be the Earnings Per Share (EPS) of the company in Plan-I and Plan-II?

A ₹ 4.37 and ₹4.26

B ₹ 3.36 and ₹ 3.88

C ₹ 3.90 and ₹ 4.10

D ₹ 4.25 and ₹ 4.50

MCQ 2

What would be the Market Price per Share (MPS) of the company if Price Earnings Ratio (PE ratio) in Plan-1 is 12 times and Plan-II is 15 times?

A ₹ 46.80 and ₹ 61.50

B ₹ 40.32 and ₹ 58.20

C ₹ 51.00 and ₹ 67.50

D ₹ 52.44 and ₹ 63.90

MCQ 3

What would be the financial Break Even Point (BEP) in Plan-I and Plan-II?

A ₹ 13,75,000 and ₹ 15,10,000

B ₹ 13,70,000 and ₹ 15,00,000

C ₹ 13,65,000 and ₹ 15,15,000

D ₹ 13,60,000 and ₹ 15,08,571

MCQ 4

What would be the indifference point between Plan-1 and Plan-11?

A ₹ 34,33,333

B ₹ 34,40,000

C ₹ 35,15,000

D ₹ 35,22,222

MCQ 5

What would be the Earnings Per Share (EPS) in Plan-1 and Plan-II at the indifference point as calculated by you above?

A ₹ 1.30 and ₹ 1.30

B ₹ 1.65 and ₹ 1.75

C ₹ 1.50 and ₹ 1.50

D ₹ 1.80 and ₹ 1.90

ICAI'S ANSWERS

MCQ 1

Answer key: **(C)**

MCQ 2

Answer key: **(A)**

MCQ 3

Answer key: **(D)**

MCQ 4

Answer key: **(B)**

MCQ 5

Answer key: **(A)**

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348 **EXAM** **Sep 2024** Q3(b) 5 marks

ER Private Limited has a paid-up capital ₹ 2,50,000 consisting of 25,000 Equity shares of ₹ 10 each. The Market price per share is ₹ 24 with PE ratio of 8. The company is planning to purchase a plant which will cost ₹ 5,00,000. This plant is expected to yield earnings before interest and taxes of ₹ 2,00,000 per annum. It has two alternatives to finance the plant:

Alternatives	Equity	Debt
A	100%	-
B	50%	50%

Other information is as under:

(i) Cost of debt is 12%.

(ii) Equity shares of face value of ₹ 10 each will be issued at a premium of ₹ 10 per share.

(iii) PE ratio of Leveraged company will be 7.

(iv) Tax rate -40%.

Advise which alternative is the most suitable to raise the funds for additional capital, keeping in mind to maximize the benefit to its Shareholders.

ICAI'S ANSWER

Calculation of No. of Equity Shares and Existing Earnings before Interest and Taxes

Particulars	Existing	Alternative A: Issue Equity shares only	Alternative B: Issue Equity Shares and 12% Debentures of equal amount
Number of Equity Shares			
- Existing	25,000	25,000	25,000
- Newly issued		25,000 (₹ 5,00,000/₹ (10+10))	12,500 (₹ 2,50,000/₹(10+10))
Total no. of Equity Shares	25,000	50,000	37,500
Calculation of Existing Earnings before Interest and Taxes (EBIT)			
Market Price per share (MPS)	₹ 24		
Price-Earnings Ratio (PE Ratio)	8 times	8 times	7 times
Earning per share (EPS) = MPS/PE Ratio	3		
Earnings after Tax (EAT) = EPS x No. of Equity shares	**75,000**		
Earning before Tax (EBT) = EAT/0.6 (or EBIT as Interest nil)	1,25,000		

Calculation of EPS and MPS under two financial alternatives

Particulars	Existing	Alternative A	Alternative B
Earnings before Interest and Tax:			
- Existing EBIT	1,25,000	1,25,000	1,25,000
- From New Project		2,00,000	2,00,000
	1,25,000	3,25,000	3,25,000

Particulars	Existing	Alternative A	Alternative B
Less: Interest on 12% Debentures	-	-	30,000
Earnings before Tax (EBT)	1,25,000	3,25,000	2,95,000
Less: Tax @ 40%	50,000	1,30,000	1,18,000
Earnings after Tax (EAT)	**75,000**	**1,95,000**	**1,77,000**
EPS = EAT/ No. of Equity Shares	**3.00**	**3.90**	**4.72**
Market Price per share (MPS) = EPS x Price-Earning Ratio	**24.00**	**31.20**	**33.04**

Advise: Alternative B i.e., issue of 12% Debentures is most suitable to maximize the market price per share.

Alternatively, Solution can also be presented in following way:

Calculation of EPS and MPS under two financial alternatives

Particulars	Existing	Alternative A	Alternative B
Earnings before Interest and Tax:		2,00,000	2,00,000
- From New Project		2,00,000	2,00,000
Less: Interest on 12% Debentures	-	-	30,000
Earnings before Tax (EBT)		2,00,000	1,70,000
Less: Tax @ 40%		80,000	68,000
Earnings after Tax (EAT) from new project		1,20,000	1,02,000
Earnings from Existing (PAT)	75,000	75,000	75,000
Total Earnings After Tax (EAT)	75,000	1,95,000	1,77,000
Number of Shares	25,000	50,000	37,500
EPS = EAT/ No. of Equity Shares	**3.00**	**3.90**	**4.72**
Market Price per share (MPS) = EPS x Price-Earning Ratio	**24.00**	**31.20**	**33.04**

Advise: Alternative B i.e., issue of 12% Debentures is most suitable to maximize the market price per share.

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349 EXAM **Old P8 Nov 2023** Q2 10 marks old scheme

The data of K Textiles Lid, are given as follows:

Particulars	Amount (₹)
Profit Before Interest and Tax	50,00,000

Particulars	Amount (₹)
Less: Interest on debentures @ 10%	10,00,000
Profit before tax	40,00,000
Less: Income tax @ 50%	20,00,000
Profit after tax	20,00,000
No. of equity shares (₹ 10 each)	10,00,000
EPS	2
PE Ratio	10
Market price per share	20

The Company is planning to start a new project needs to be having a total capital outlay of ₹ 40,00,000. You are informed that a debt equity ratio [D/D+E] higher than 36% pushes the Ke (cost of equity) up to 12.5%, means reducing the PE ratio to 8 and rises the interest rate on additional amount borrowed to 12%. Retained earnings of the company is ₹ 1.4 crores.

Find out the probable price of share if:

- The additional funds are raised as a loan
- The amount is raised by issuing equity shares.

ICAI'S ANSWER

In this question, EBIT after proposed extension is not given. Therefore, we can assume that existing return on capital employed will be maintained.

Working notes:

1. Return on Capital Employed = $\frac{\text{EBIT}}{\text{Capital Employed}} = \frac{₹\ 50,00,000}{₹\ 3,40,00,000}$ =14.70%

Capital Employed = Debt + Equity

= ₹ 1,00,00,000 + (₹ 1,00,00,000 + ₹ 1,40,00,000)

= ₹ 3,40,00,000

2. Proposed EBIT = Proposed Capital Employed x Return on capital employed

= (₹ 3,40,00,000 + ₹ 40,00,000) x 14.70% = ₹ 55,86,000

3. Debt Equity Ratio = $\frac{\text{Debt}}{\text{Debt} + \text{Equity}}$

Option1: Loan option

Debt = ₹ 1,00,00,000 + ₹ 40,00,000 = ₹ 1,40,00,000

Equity = ₹ 2,40,00,000

Debt Equity ratio= $\frac{1.4 \text{ cr.}}{1.4 \text{ cr.} + 2.40 \text{ cr.}}$ = 36.84%

Debt equity ratio has crossed the limit of 36%, hence, PE ratio in this case will be 8 times and additional borrowing will be at the rate of 12%.

Option2: Equity option

Debt = ₹ 1,00,00,000

Equity = ₹ 2,40,00,000 + ₹ 40,00,000 = ₹ 2,80,00,000

Debt Equity ratio = $\frac{1 \text{ cr.}}{1 \text{ cr.} + 2.8 \text{ cr.}}$ =26.32%

Debt equity ratio has not crossed the limit of 36% hence PE ratio in this case will remain at 10 times.

4. Number of equity shares to be issued in case of equity option @ ₹ 20 per share

= ₹ 40,00,000 / ₹ 20 = 2,00,000

Calculation of EPS and MPS under two financial options

Particulars	Financial Options	
	Option I	**Option II**
	12% additional loan of 40,00,000	**10,00,000 equity shares @ ₹ 10 and 2,00,000 equity shares @ ₹ 20**
	(₹)	**(₹)**
Profit before interest and Tax (PBIT)	55,86,000	55,86,000
Less: Interest on old debentures @ 10%	10,00,000	10,00,000
Less: Interest on additional loan (new) @ 12% on ₹ 40,00,000	4,80,000	Nil
Profit before tax	41,06,000	45,86,000
Less: Taxes @ 50%	20,53,000	22,93,000
Earnings for equity shareholders (EAT/Profit after tax)	20,53,000	22,93,000
Number of Equity Shares	10,00,000	12,00,000
Earnings per Share (EPS)	2.05	1.91
Price/ Earnings ratio	8	10
Market price per share (MPS)	16.42	19.11

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350 EXAM Old P8 May 2023 Q3 10 marks old scheme

The following information pertains to CIZA Ltd.:

	₹
Capital Structure:	
Equity share capital (₹ 10 each)	8,00,000
Retained earnings	20,00,000
9% Preference share capital (₹ 100 each)	12,00,000
12% Long-term loan	10,00,000
Interest coverage ratio	8
Income tax rate	30%
Price – earnings ratio	25

The company is proposed to take up an expansion plan, which requires an additional investment of ₹ 34,50,000. Due to this proposed expansion, earnings before interest and taxes of the company will increase by ₹ 6,15,000 per annum. The additional fund can be raised in following manner:

- By issue of equity shares at present market price, or
- By borrowing 16% Long-term loans from bank.

You are informed that Debt-equity ratio (Debt/ Shareholders' fund) in the range of 50% to 80% will bring down the price-earnings ratio to 22 whereas; Debt-equity ratio over 80% will bring down the price-earnings ratio to 18.

Required:

Advise which option is most suitable to raise additional capital so that the Market Price per Share (MPS) is maximized.

ICAI'S ANSWER

Working notes:

(i) Interest Coverage ratio = 8

$$\frac{\text{EBIT}}{\text{Interest}} = 8$$

$$\frac{\text{EBIT}}{1{,}20{,}000} = 8$$

So, EBIT = ₹ 9,60,000

(ii) Proposed Earnings Before Interest & Tax = 9,60,000 + 6,15,000 = ₹ 15,75,000

Option 1: Equity option

Debt = ₹ 10,00,000

Shareholders Fund = 8,00,000+20,00,000+12,00,000+34,50,000 = ₹ 74,50,000

Debt Equity ratio(Debt/Shareholders fund) = $\frac{10{,}00{,}000}{74{,}50{,}000} = 13.42\%$

P/E ratio in this case will be 25 times

Option 2: Debt option

Debt = 10,00,000+34,50,000 = ₹ 44,50,000

Shareholders Fund = 8,00,000+20,00,000+12,00,000 = ₹ 40,00,000

Debt Equity ratio(Debt/Shareholders fund) = $\frac{44,50,000}{40,00,000}$ = 111.25%

Debt equity ratio has crossed the limit of 80% hence PE ratio in this case will remain at 18 times.

Number of Equity Shares to be issued = ₹ 34,50,000/ ₹ 150 = 23,000

(iii) Calculation of Earnings per Share and Market Price per share

Particulars	₹
Current Earnings Before Interest & Tax	9,60,000
Less: Interest	1,20,000
Earnings Before Tax	8,40,000
Less: Taxes	2,52,000
Earnings After Tax	5,88,000
Less: Preference Dividend (@9%)	1,08,000
Net earnings for Equity shareholders	4,80,000
Number of equity shares	80,000
Earnings Per Share	**6**
Price-earnings ratio	25
Market Price per share	**150**

Calculation of EPS and MPS under two financial options

Particulars	Financial Options	
	Option I	**Option II**
	Equity Shares Issued (₹)	**16% Long Term Debt Raised (₹)**
Earnings before interest and Tax (EBIT)	15,75,000	15,75,000
Less: Interest on old debentures @ 12%	1,20,000	1,20,000
Less: Interest on additional loan (new) @ 16% on ₹ 34,50,000	NIL	5,52,000
Earnings before tax	14,55,000	9,03,000

Particulars	Financial Options	
Less: Taxes @ 30%	4,36,500	2,70,900
(EAT/Profit after tax)	**10,18,500**	**6,32,100**
Less: Preference Dividend (@9%)	1,08,000	1,08,000
Net Earnings available to Equity shareholders	**9,10,500**	**5,24,100**
Number of Equity Shares	1,03,000	80,000
Earnings per Share (EPS)	**8.84**	**6.55**
Price/ Earnings ratio	25	18
Market price per share (MPS)	**221**	**117.9**

Advise: Equity option has higher Market Price per Share therefore company should raise additional fund through equity option.

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351 EXAM **Old P8 May 2022** Q4 10 marks old scheme

The particulars relating to Raj Ltd. for the year ended 31st March, 2022 are given as follows:

Output (units at normal capacity)	1,00,000
Selling price per unit	₹ 40
Variable cost per unit	₹ 20
Fixed cost	₹ 10,00,000

The capital structure of the company as on 31st March, 2022 is as follows:

Particulars	Amount in ₹
Equity share capital (1,00,000 shares of ₹ 10 each)	10,00,000
Reserves and surplus	5,00,000
Current liabilities	5,00,000
Total	**20,00,000**

Raj Ltd. has decided to undertake an expansion project to use the market potential that will involve ₹ 20 lakhs. The company expects an increase in output by 50%. Fixed cost will be increased by ₹ 5,00,000 and variable cost per unit will be decreased by 15%. The additional output can be sold at the existing selling price without any adverse impact on the market.

The following alternative schemes for financing the proposed expansion program are planned:

(Amount in ₹)

Alternative	Debt	Equity Shares
1	5,00,000	Balance
2	10,00,000	Balance
3	14,00,000	Balance

Current market price per share is ₹ 200.

Slab wise interest rate for fund borrowed is as follows:

Fund limit	Applicable interest rate
Up-to ₹ 5,00,000	10%
Over₹ 5,00,000 and up-to ₹ 10,00,000	15%
Over ₹ 10,00,000	20%

Find out which of the above-mentioned alternatives would you recommend for Raj Ltd. with reference to the EPS, assuming a corporate tax rate is 40%?

ICAI'S ANSWER

Alternative 1 = Raising Debt of ₹ 5 lakh + Equity of ₹ 15 lakh

Alternative 2 = Raising Debt of ₹ 10 lakh + Equity of ₹ 10 lakh

Alternative 3 = Raising Debt of ₹ 14 lakh + Equity of ₹ 6 lakh

Calculation of Earnings per share (EPS)

Particulars	FINANCIAL ALTERNATIVES		
	Alternative 1	**Alternative 2**	**Alternative 3**
	(₹)	**(₹)**	**(₹)**
Expected EBIT [W. N. (a)]	19,50,000	19,50,000	19,50,000
Less: Interest [W. N. (b)]	(50,000)	(1,25,000)	(2,05,000)
Earnings before taxes (EBT)	**19,00,000**	**18,25,000**	**17,45,000**
Less: Taxes @ 40%	7,60,000	7,30,000	6,98,000
Earnings after taxes (EAT)	**11,40,000**	**10,95,000**	**10,47,000**
Number of shares [W. N. (d)]	1,07,500	1,05,000	1,03,000
Earnings per share (EPS)	**10.60**	**10.43**	**10.17**

Conclusion: Alternative 1 (i.e. Raising Debt of ₹ 5 lakh and Equity of ₹ 15 lakh) is recommended which maximises the earnings per share.

Working Notes (W.N.):

(a) Calculation of Earnings before Interest and Tax (EBIT)

Particulars		
Output (1,00,000 + 50%)	(A)	1,50,000
Selling price per unit		₹ 40
Less: Variable cost per unit (₹ 20 – 15%)		₹ 17
Contribution per unit	(B)	₹ 23
Total contribution	**(A x B)**	**₹ 34,50,000**
Less: Fixed Cost (₹ 10,00,000 + ₹ 5,00,000)		₹ 15,00,000
EBIT		**₹ 19,50,000**

(b) Calculation of interest on Debt

Alternative		(₹)	Total (₹)
1	(₹ 5,00,000 × 10%)		50,000
2	(₹ 5,00,000 × 10%)	50,000	
	(₹ 5,00,000 × 15%)	75,000	1,25,000
3	(₹ 5,00,000 × 10%)	50,000	
	(₹ 5,00,000 × 15%)	75,000	
	(₹ 4,00,000 × 20%)	80,000	2,05,000

(c) Number of equity shares to be issued

$$\text{Alternative 1} = \frac{₹\ (20,00,000 - 5,00,000)}{₹\ 200\ (\text{Market price of share})} = \frac{₹\ 15,00,000}{₹\ 200} = 7,500 \text{ shares}$$

$$\text{Alternative 2} = \frac{₹\ (20,00,000 - 10,00,000)}{₹\ 200\ (\text{Market price of share})} = \frac{₹\ 10,00,000}{₹\ 200} = 5,000 \text{ shares}$$

$$\text{Alternative 3} = \frac{₹\ (20,00,000 - 14,00,000)}{₹\ 200\ (\text{Market price of share})} = \frac{₹\ 6,00,000}{₹\ 200} = 3,000 \text{ shares}$$

(d) Calculation of total equity shares after expansion program

	Alternative 1	Alternative 2	Alternative 3
Existing no. of shares	1,00,000	1,00,000	1,00,000
Add: issued under expansion program	7,500	5,000	3,000
Total no. of equity shares	**1,07,500**	**1,05,000**	**1,03,000**

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352 EXAM Old P8 Dec 2021 Q3 10 marks old scheme

Earnings before interest and tax of a company are ₹ 4,50,000. Currently the company has 80,000 Equity shares of ₹ 10 each, retained earnings of ₹ 12,00,000. It pays annual interest of ₹ 1,20,000 on 12% Debentures. The company proposes to take up an expansion scheme for which it needs additional fund of ₹ 6,00,000. It is anticipated that after expansion, the company will be able to achieve the same return on investment as at present.

It can raise fund either through debts at rate of 12% p.a. or by issuing Equity shares at par. Tax rate is 40%.

Required:

Compute the earning per share if:

(i) The additional funds were raised through debts.

(ii) The additional funds were raised by issue of Equity shares.

Advise whether the company should go for expansion plan and which sources of finance should be preferred.

ICAI'S ANSWER

Working Notes:

(1) **Capital employed before expansion plan:**

	(₹)
Equity shares (₹ 10 × 80,000 shares)	8,00,000
Debentures {(₹ 1,20,000/12) × 100}	10,00,000
Retained earnings	12,00,000
Total capital employed	**30,00,000**

(2) **Earnings before interest and tax (EBIT) =** 4,50,000

(3) **Return on Capital Employed (ROCE):**

$$\text{ROCE} = \frac{\text{EBIT}}{\text{Capital employed}} \times 100 = \frac{₹\ 4{,}50{,}000}{₹\ 30{,}00{,}000} \times 100 = \mathbf{15\%}$$

(4) **Earnings before interest and tax (EBIT) after expansion scheme:**

After expansion, capital employed = ₹ 30,00,000 + ₹ 6,00,000 = ₹ 36,00,000

Desired EBIT = 15% × ₹ 36,00,000 = **₹ 5,40,000**

(i) & (ii) Computation of Earnings Per Share (EPS) under the following options:

	Present situation	Expansion scheme Additional funds raised as	
		Debt (i)	**Equity (ii)**
	(₹)	(₹)	(₹)
Earnings before Interest and Tax (EBIT)	4,50,000	5,40,000	5,40,000

	Present situation	Expansion scheme Additional funds raised as	
Less: Interest - Old Debt	1,20,000	1,20,000	1,20,000
- New Debt	--	**72,000** (₹ 6,00,000 × 12%)	--
Earnings before Tax (EBT)	3,30,000	3,48,000	4,20,000
Less: Tax (40% of EBT)	1,32,000	1,39,200	1,68,000
PAT/EAT	1,98,000	**2,08,800**	**2,52,000**
No. of shares outstanding	80,000	80,000	1,40,000
Earnings per Share (EPS)	2.475 (₹ 1,98,000/80,000)	**2.610** (₹ 2,08,800/80,000)	**1.800** (₹ 2,52,000/1,40,000)

Advise to the Company: When the expansion scheme is financed by additional debt, the EPS is higher. Hence, the company **should finance the expansion** scheme by **raising debt**.

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353 EXAM Old P8 Nov 2020 Q3 10 marks old scheme

J Ltd. is considering three financing plans. The-key information is as follows:

(a) Total investment to be raised ₹ 4,00,000.

(b) Plans showing the Financing Proportion:

Plans	Equity	Debt	Preference Shares
X	100%	-	-
Y	50%	50%	-
Z	50%	-	50%

(c) Cost of Debt 10%

Cost of preference shares 10%

(d) Tax Rate 50%

(e) Equity shares of the face value of ₹10 each will be issued at a premium of ₹ 10 per share.

(f) Expected EBIT is ₹ 1,00,000.

You are required to compute the following for each plan :

(i) Earnings per share (EPS)

(ii) Financial break even point

(iii) Indifference Point between the plans and indicate if any of the plans dominate.

ICAI'S ANSWER

(i) Computation of Earnings per Share (EPS)

Plans	X (₹)	Y (₹)	Z (₹)
Earnings before interest & tax (EBIT)	1,00,000	1,00,000	1,00,000
Less: Interest charges (10% of ₹ 2,00,000)	--	(20,000)	--
Earnings before tax (EBT)	1,00,000	80,000	1,00,000
Less: Tax @ 50%	(50,000)	(40,000)	(50,000)
Earnings after tax (EAT)	50,000	40,000	50,000
Less: Preference share dividend (10% of ₹2,00,000)	--	--	(20,000)
Earnings available for equity shareholders (A)	50,000	40,000	30,000
No. of equity shares (B) Plan X = ₹ 4,00,000/ ₹ 20 Plan Y = ₹ 2,00,000 / ₹ 20 Plan Z = ₹ 2,00,000 / ₹ 20	20,000	10,000	10,000
E.P.S (A ÷ B)	**2.5**	**4**	**3**

(ii) Computation of Financial Break-even Points

Financial Break-even point = Interest + Preference dividend/(1 - tax rate)

Proposal 'X' = **0**

Proposal 'Y' = **₹ 20,000** (Interest charges)

Proposal 'Z' = Earnings required for payment of preference share dividend = ₹ 20,000 ÷ (1- 0.5 Tax Rate) = **₹ 40,000**

(iii) Computation of Indifference Point between the plans

Combination of Proposals

(a) Indifference point where EBIT of proposal "X" and proposal 'Y' is equal

$$\frac{(\text{EBIT})(1\text{-}0.5)}{20{,}000 \text{ shares}} = \frac{(\text{EBIT-}\ ₹\ 20{,}000)(1\text{-}0.5)}{10{,}000 \text{ shares}}$$

0.5 EBIT = EBIT – ₹ 20,000

EBIT = **₹ 40,000**

(b) Indifference point where EBIT of proposal 'X' and proposal 'Z' is equal:

$$\frac{(\text{EBIT})(1\text{-}0.5)}{20{,}000 \text{ shares}} = \frac{\text{EBIT}(1\text{-}0.5) - ₹\ 20{,}000}{10{,}000 \text{ shares}}$$

0.5 EBIT = EBIT- ₹ 40,000

0.5 EBIT = ₹ 40,000

$$\text{EBIT} = \frac{₹\ 40{,}000}{0.5} = ₹\ \mathbf{80{,}000}$$

(c) Indifference point where EBIT of proposal 'Y' and proposal 'Z' are equal

$$\frac{(\text{EBIT- ₹ 20,000})(1\text{-}0.5)}{10{,}000 \text{ shares}} = \frac{\text{EBIT}(1\text{-}0.5) \text{ - ₹ 20,000}}{10{,}000 \text{ shares}}$$

0.5 EBIT – ₹ 10,000 = 0.5 EBIT – ₹ 20,000

There is no indifference point between proposal 'Y' and proposal 'Z'

Analysis: It can be seen that financial proposal 'Y' dominates proposal 'Z', since the financial break-even-point of the former is only ₹ 20,000 but in case of latter, it is ₹ 40,000. EPS of plan 'Y' is also higher.

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354 RTP **Sep 2025** Q6

ABC Ltd. is planning to raise ₹50,00,000 for a major expansion project. The firm is evaluating three financing options:

(i) Equity Financing Only

Issuing new shares at ₹10 each (Issue price: ₹10, but flotation cost per share: ₹0.50).

Existing equity shares: 10,00,000.

Market price per share: ₹20.

(ii) Debt Financing Only

Borrow ₹50,00,000 at 12% interest.

Flotation cost on debt: 2% of the face value.

The lender imposes a debt covenant limiting the interest coverage ratio (EBIT/Interest) to a minimum of 2.5.

(iii) Mix of Equity and Debt Financing

Raise ₹25,00,000 through equity (floatation cost: ₹0.50/share at ₹ 10 issue price).

Raise ₹25,00,000 through debt at 12% interest (floatation cost: 2%).

Additional Information

(i) Project EBIT: ₹15,00,000 annually.

(ii) Corporate tax rate: 30%.

(iii) Existing shares: 10,00,000.

(iv) Risk-free rate: 6%, Market return: 14%, Equity beta: 1.2.

From the above information:

1. CALCULATE the net proceeds from each financing method (after floatation costs) and determine the number of new shares issued in Option (i) and Option (iii).
2. For debt options, CHECK if the interest coverage ratio complies with the covenant (minimum 2.5).
3. COMPUTE EPS under each option.
4. COMPUTE the WACC under each option, considering the impact of floatation costs on capital raised.

ADVISE which financing option is best from both an EPS and WACC perspective.

ICAI'S ANSWER

(i) **Calculation of Net Proceeds after Floatation Cost and Number of New Shares in Option (i) and (iii)**

Equity Option

Net proceeds per share = ₹ 10 – ₹ 0.50 = ₹9.50

Number of shares issued = ₹ 50,00,000 / ₹ 9.50 = **5,26,316 shares**

Debt Option

Flotation cost = 2% of ₹ 50,00,000 = ₹1,00,000

Net proceeds = ₹ 50,00,000 – ₹1,00,000 = **₹49,00,000**

But project requires ₹ 50,00,000.

So gross borrowing = ₹50,00,000 / (1 – 0.02) = **₹51,02,041**

Interest = 12% × ₹51,02,041 = **₹6,12,245**

Mix Option (₹25 lakh Equity, ₹25 lakh Debt)

Equity

Net proceeds/share = ₹ 9.50

Shares issued = ₹ 25,00,000 / ₹ 9.50 = **2,63,158 shares**

Debt

Net debt proceeds = ₹ 25,00,000.

Gross borrowing = ₹25,00,000 / (1 – 0.02) = ₹25,51,020

Interest = 12% × ₹25,51,020 = **₹3,06,122**

(ii) **Calculation of Interest Coverage Ratio (ICR)**

ICR = EBIT / Interest

Debt Option:

EBIT = ₹ 15,00,000

Interest = ₹ 6,12,245

ICR = ₹ 15,00,000 / ₹ 6,12,245 ≈ **2.45 (Fails covenant)**

Mix of Equity and Debt Option

Interest = ₹ 3,06,122

ICR = ₹ 15,00,000 / ₹ 3,06,122 ≈ **4.90 (Passes covenant)**

(iii) **Calculation of Earnings Per Share (EPS)**

$$\text{EPS} = \frac{(\text{EBIT-Interest})\ (1\text{-Tax})}{\text{Total Shares}}$$

Equity Option

Interest = ₹ 0

Net Income = ₹ 15,00,000 × (1 – 0.30) = ₹ 10,50,000

Total Shares = 10,00,000 + 5,26,316 = **15,26,316**

EPS = ₹ 10,50,000 / 15,26,316 = **₹ 0.688**

Debt Option (Fails covenant)

Interest = ₹ 6,12,245

Net Income = (₹ 15,00,000 – ₹ 6,12,245) × (1 – 0.30)

= ₹ 6,21,428.5

Total Shares = 10,00,000

EPS = ₹ 6,21,428.5 / 10,00,000

= **₹ 0.621** (But violates ICR)

Mix Option

Interest = ₹ 3,06,122

Net Income = (₹ 15,00,000 – ₹ 3,06,122) × 0.70 = ₹ 8,35,714.6

Total Shares = 10,00,000 + 2,63,158 = **12,63,158**

EPS = ₹ 8,35,714.6 / 12,63,158 ≈ **₹ 0.662**

(iv) **Calculation of WACC**

Cost of Equity (Using CAPM)

Cost of Equity = $Rf+\beta(Rm–Rf)$

= $6\%+1.2 \times (14\%–6\%) = 6\%+9.6\%=15.6\%$

Now, WACC = (EV×Ke) + (DV×Kd×(1−t))

Equity Option

E = ₹ 50,00,000

D = ₹ 0

WACC = 15.6%

Debt Option

E = ₹ 0

D = ₹ 51,02,041

Kd = 12%

After-tax cost of debt = 12% × (1 – 0.30) = 8.4%

WACC = 8.4% (Lowest WACC but fails covenant)

Mix Option

E = ₹ 25,00,000, D = ₹ 25,51,020

Total Value = ₹ 50,51,020

Ke = 15.6%, Kd (after tax) = 8.4%

WACC = (Weight of Ke x Cost of Equity) + (Weight of Kd x Cost of Debt)

= (0.495 x 0.156) + (0.505 x 0.084)

= 0.0772 + 0.0424 = 0.1196 or 11.96%

Conclusion & Recommendation

Option	EPS	WACC	Covenant Compliant
Equity	₹ 0.688	15.6%	Not Applicable
Debt	₹ 0.621	8.4%	**No** (ICR < 2.5)
Mix	₹ 0.662	11.96%	Yes

Hence, the recommended option is Mixed Financing as it complies with the debt covenant, has higher EPS than Debt Option, and lower WACC than Equity Option.

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355 RTP **May 2025** Q6

Namra Limited provides you with the following information –

Particulars	Amount (₹)
Operating Profit	6,20,000
Less: Interest on Debentures @ 10%	(80,000)
EBT	5,40,000
Less: Tax @ 20%	(1,08,000)
PAT	4,32,000
Less: 14% Preference Dividend	(1,12,000)
Earnings for Equity Share Holders	3,20,000
No of Equity Shares (₹ 10 Each)	16,000
EPS	20

The Reserves & Surplus of the company is at ₹ 9,00,000 and Namra Limited requires additional funds of ₹ 15,00,000 for modernization and expansion. The current capitalization rate for the equity is 20% and company has a policy to retain 40% of its earnings. The debentures and preference shares are trading at premium of 10% & 25% to its current book value respectively. The fair value of equity shares is calculated by dividing the number of equity shares to the Overall Value of the firm. New equity shares for expansion will be issued 15% discount to the current fair value price.

Return on Capital Employed (ROCE) which is based on the total value of the firm, will increase by 10% to its current rate after expansion and modernization. If the capital gearing ratio goes above 2.50 then interest rate on additional debt will increase by 200 basis points and dividend on preference shares would increase by half a percentage.

You are required to ADVISE on the below two financial plan to be selected based on earnings.

(i) Two-third amount is raised through debenture and remaining by preference share.

(ii) Issue of equity shares only.

ICAI'S ANSWER

Particulars	Additional Funds : Debentures = 15,00,000 x 2/3 Pref = 15,00,000 x 1/3	Additional Funds : 100% Equity
Operating Profit **Wn – 1**	9,55,776	9,55,776
Less: Interest Exp		
Current	(80,000)	(80,000)
Additional (10,00,000 X 12%) **Wn - 2**	(1,20,000)	-
EBT	7,55,776	8,75,776
Less: Tax @ 20%	(1,51,155)	(1,75,155)
EAT	6,04,621	7,00,621
Less: Preference Dividend		
Current	(1,12,000)	(1,12,000)
Additional (5,00,000 X 14.5%) **Wn - 2**	(72,500)	-
Earnings for Equity Shareholders	4,20,121	5,88,621
No of Equity Shares		
Current	16,000	16,000
Additional **Wn - 3**	-	7,550
EPS	**26.26**	**25.00**

WN 1 – Calculation of EBIT after expansion

As given in the question, ROCE is based on the total value of firm, so first step would be to calculate the total value of the firm

Value of Firm = Value of Debt + Value of Pref shares + Value of Equity + Value of R/S

$$\text{Value of Debt (Vd)} = \frac{\text{Interest (₹)}}{\text{Interest \%}} + 10\% \text{ premium}$$

$$= \frac{80{,}000}{0.10} + 10\% \text{ premium}$$

= ₹ 8,80,000

Value of Preference (Vp) = $\frac{\text{Pref Div (₹)}}{\text{Pref Div \%}}$ + 25% premium

= $\frac{1,12,000}{0.14}$ + 25% premium

= 10,00,000

Value of Equity share capital (Ve) = $\frac{\text{Dividend}}{\text{Ke}}$

= $\frac{3,20,000 \text{ x } 0.6}{0.20}$

= 9,60,000

Value of R/S = 9,00,000

Therefore, Value of Firm (Vf) = 8,80,000 + 10,00,000 + 9,60,000 + 9,00,000

= 37,40,000

ROCE (Before expansion) = $\frac{\text{EBIT}}{\text{Total value of the firm}}$

= $\frac{6,20,000}{37,40,000}$

= 16.58%

ROCE (After expansion) = 16.58 + 1.658 (i.e 16.58 + 10%)

= 18.24%

EBIT (After expansion) = (37,40,000 + 15,00,000) x 18.24%

= ₹ 9,55,776

WN 2 – Calculation of Interest on additional debt and Preference dividend on additional Preference share capital

Condition – If the capital gearing ratio goes above 2.50, then additional debt raised would be at higher rate of interest and additional Preference shares would also be raised at higher preference dividend rate

Capital gearing ratio when Additional funds are raised through additional debt and preference share capital.

Capital gearing ratio = $\frac{8,00,000 + 8,00,000 + 15,00,000}{1,60,000 + 9,00,000}$

Capital gearing ratio = 2.92

Since it is greater than 2.50, Interest on Debt = 10% + 2% (200 basis points) = 12%

Preference Dividend = 14% + 0.5% = 14.5%

WN 3 – Calculation of Additional No of Equity shares when funds are raised through equity

Fair value of equity shares before issuing new equity share

= $\frac{\text{Total Value of the firm}}{\text{No of existing equity shares}}$

= $\frac{37,40,000}{16,000}$

= ₹ 233.75

Issue Price = 233.75 - 15% Discount = ₹ 198.69

Therefore, No of New Equity shares to be issued

$$= \frac{\text{Additional Funds to be raised}}{\text{Fair value}}$$

$$= \frac{15,00,000}{198.69}$$

= 7549.45 shares approx. 7550 shares

Comment – It is advisable for Namra Limited to raise the additional funds through a mix of debentures and preference as EPS is maximum

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356 RTP **Jan 2025** Q6

Ritu Limited in the expansion stage and it provides you the following information:

	(₹)
Profit (EBIT)	5,00,000
Less: Interest on Debenture @ 10%	(1,00,000)
EBT	4,00,000
Less Income Tax @ 30%	(1,20,000)
	2,80,000
No. of Equity Shares (₹ 10 each)	50,000
Earnings per share (EPS)	5.6
Price /EPS (PE) Ratio	10

The company has reserves and surplus of ₹ 10,00,000 and required ₹ 5,00,000 further for modernisation. Return on Capital Employed (ROCE) is constant. Debt (Debt/ Equity) Ratio lesser than 2 will raise the P/E Ratio to 12. Interest rate on additional debts is 12%. You are required to ASCERTAIN the probable price of the share.

(i) If the additional capital are raised as debt; and

(ii) If the amount is raised by issuing equity shares at ruling market price.

ICAI'S ANSWER

Ascertainment of probable price of shares of Akash limited

Particulars	Plan-I	Plan-II
	If ₹ 5,00,000 is raised as debt (₹)	**If ₹ 5,00,000 is raised by issuing equity shares (₹)**
Earnings Before Interest and Tax (EBIT) {20% of new capital i.e. 20% of (₹ 25,00,000 + ₹ 5,00,000)} (Refer working note1)	6,00,000	6,00,000

Particulars	Plan-I	Plan-II
Less: Interest on old debentures (10% of ₹ 10,00,000)	(1,00,000)	(1,00,000)
Less: Interest on new debt (12% of ₹ 5,00,000)	(60,000)	--
Earnings Before Tax (EBT)	4,40,000	5,00,000
Less: Tax @ 30%	(1,32,000)	(1,50,000)
Earnings for equity shareholders (EAT)	3,08,000	3,50,000
No. of Equity Shares (refer working note 2)	50,000	58,929
Earnings per Share (EPS)	₹ 6.16	₹ 5.94
Price/ Earnings (P/E) Ratio (refer working note 3)	12	10
Probable Price Per Share (PE Ratio × EPS)	₹ 73.92	₹ 59.40

Working Notes:

1. Calculation of existing Return of Capital Employed (ROCE):

	(₹)
Equity Share capital (50,000 shares × ₹ 10)	5,00,000
10% Debentures $\left(₹1,00,000 \times \frac{100}{10}\right)$	10,00,000
Reserves and Surplus	10,00,000
Total Capital Employed	25,00,000
Earnings before interest and tax (EBIT) (given)	5,00,000
ROCE = $\frac{₹5,000,000}{₹25,00,000} \times 100$	20%

2. Number of Equity Shares to be issued in Plan-II:

$= \frac{₹5,00,000}{₹56} = 8{,}929 \text{shares}$

Thus, after the issue total number of shares = 50,000+ 8,929 = 58,929 shares

3. Debt/Equity Ratio if ₹ 5,00,000 is raised as debt:

$= \frac{₹15,00,000}{₹15,00,000} = 1$

As the debt equity ratio is less than 2 the P/E ratio will be increase to 12 in Plan-I

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357 RTP Old P8 Nov 2023 Q3 old scheme

Prakash Limited provides you the following information:

	(₹)
Profit (EBIT)	3,00,000
Less: Interest on Debenture @ 10%	(50,000)
EBT	2,50,000
Less Income Tax @ 50%	(1,25,000)
	1,25,000
No. of Equity Shares (₹ 10 each)	25,000
Earnings per share (EPS)	5
Price /EPS (PE) Ratio	10

The company has reserves and surplus of ₹ 7,50,000 and required ₹ 5,00,000 further for modernisation. Return on Capital Employed (ROCE) is constant. Debt (Debt/ Debt + Equity) Ratio higher than 40% will bring the P/E Ratio down to 8 and increase the interest rate on additional debts to 12%. You are required to ASCERTAIN the probable price of the share.

(i) If the additional capital is raised as debt; and

(ii) If the amount is raised by issuing equity shares at ruling market price

ICAI'S ANSWER

Ascertainment of probable price of shares of Prakash limited

Particulars	Plan-I	Plan-II
	If ₹ 5,00,000 is raised as debt (₹)	**If ₹ 5,00,000 is raised by issuing equity shares (₹)**
Earnings Before Interest and Tax (EBIT) {20% of new capital i.e., 20% of (₹15,00,000 + ₹ 5,00,000)} (Refer working note1)	4,00,000	4,00,000
Less: Interest on old debentures (10% of ₹5,00,000)	(50,000)	(50,000)
Less: Interest on new debt (12% of ₹5,00,000)	(60,000)	--
Earnings Before Tax (EBT)	2,90,000	3,50,000
Less: Tax @ 50%	(1,45,000)	(1,75,000)
Earnings for equity shareholders (EAT)	1,45,000	1,75,000
No. of Equity Shares (refer working note 2)	25,000	35,000
Earnings per Share (EPS)	₹ 5.80	₹ 5.00

Particulars	Plan-I	Plan-II
Price/ Earnings (P/E) Ratio (refer working note 3)	8	10
Probable Price Per Share (PE Ratio × EPS)	₹ 46.40	₹ 50

Working Notes:

1. Calculation of existing Return of Capital Employed (ROCE):

	(₹)
Equity Share capital (25,000 shares × ₹10)	2,50,000
10% Debentures (₹ 50,000 × 100/10)	5,00,000
Reserves and Surplus	7,50,000
Total Capital Employed	15,00,000
Earnings before interest and tax (EBIT) (given)	3,00,000
ROCE = ₹ 3,00,000/₹ 15,00,000 × 100	20%

2. Number of Equity Shares to be issued in Plan-II:

$$= \frac{₹\ 5{,}00{,}000}{₹\ 50} = 10{,}000 \text{ Shares}$$

Thus, after the issue total number of shares = 25,000+ 10,000 = 35,000 shares

3. Debt/Equity Ratio if ₹ 5,00,000 is raised as debt:

$$= \frac{₹\ 10{,}00{,}000}{₹\ 20{,}00{,}000} \times 100 = 50\%$$

As the debt equity ratio is more than 40% the P/E ratio will be brought down to 8 in Plan-I

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358 RTP **Old P8 May 2023** Q3 old scheme

Current Capital Structure of XYZ Ltd is as follows:

Equity Share Capital of 7 lakh shares of face value ₹ 20 each

Reserves of ₹ 10,00,000

9% bonds of ₹ 3,00,00,000

11% preference capital: 3,00,000 shares of face value ₹ 50 each

Additional Funds required for XYZ Ltd are ₹ 5,00,00,000.

XYZ Ltd is evaluating the following alternatives:

I. Proposed alternative I: Raise the funds via 25% equity capital and 75% debt at 10%. PE ratio in such scenario would be 12.

II. Proposed alternative II: Raise the funds via 50% equity capital and rest from 12% Preference capital .PE ratio in such scenario would be 11.

Any new equity capital would be issued at a face value of ₹ 20 each. Any new preferential capital would be issued at a face value of ₹ 20 each. Tax rate is 34%

DETERMINE the indifference point under both the alternatives.

ICAI'S ANSWER

Current Capital Structure

Equity Share Capital	₹ 20 × 7 lakhs	₹ 1,40,00,000
Reserves		₹ 10,00,000
9% Bonds		₹ 3,00,00,000
11% Preference Share Capital	₹ 50 × 3 lakhs	₹ 1,50,00,000
Total Capital Employed		**₹ 6,00,00,000**

Proposed Capital Structure

Capital	Working	Proposal I	Proposal II
Capital to be raised		**₹5,00,00,000**	**₹5,00,00,000**
Equity	50000000 × 25%	₹ 1,25,00,000	-
	50000000 × 50%	-	₹ 2,50,00,000
Debt @ 10%	50000000 × 75%	₹ 3,75,00,000	-
Preference Shares @ 12%	50000000 × 50%	-	₹ 2,50,00,000
Combined Capital		**Amount (proposal 1)**	**Amount (proposal 2)**
Equity		₹ 2,65,00,000	₹ 3,90,00,000
Reserves		₹ 10,00,000	₹ 10,00,000
9% Bond		₹ 3,00,00,000	₹ 3,00,00,000
10% Debt		₹ 3,75,00,000	-
11% Preference Shares		₹ 1,50,00,000	₹ 1,50,00,000
12% Preference Shares		-	₹ 2,50,00,000
		₹ 11,00,00,000	**₹ 11,00,00,000**

Interest for Proposal I = ₹ 3,00,00,000 x 9% + ₹ 3,75,00,000 x 10%

= ₹ 27,00,000 + ₹ 37,50,000

= ₹ 64,50,000

Preference Dividend for Proposal I = ₹ 1,50,00,000 x 11% = ₹ 16,50,000

Interest for Proposal II = ₹ 3,00,00,000 x 9% = ₹ 27,00,000

Preference Dividend for Proposal II = ₹ 1,50,00,000 x 11% + ₹ 2,50,00,000 x 12%

= ₹ 16,50,000 + ₹ 30,00,000 = ₹ 46,50,000

Let the indifference point be ₹ X

For Proposal I,

$$EPS = \frac{(X - ₹\ 64,50,000) \times 0.66 - ₹\ 16,50,000}{13,25,000} \quad (1)$$

For Proposal II,

$$EPS = \frac{(X - ₹\ 27,00,000) \times 0.66 - ₹\ 46,50,000}{19,50,000} \quad (2)$$

Equating (1) and (2),

$$EPS = \frac{(X - ₹\ 64,50,000) \times 0.66 - ₹\ 16,50,000}{13,25,000} = \frac{(X - ₹\ 27,00,000) \times 0.66 - ₹\ 46,50,000}{19,50,000}$$

$$\frac{0.66\ X\ ₹\ 42,57,000 - ₹\ 16,50,000}{1,325} = \frac{0.66X - ₹\ 17,82,000 - ₹\ 46,50,000}{1,950}$$

$$\frac{0.66X - ₹\ 59,07,000}{53} \text{ ..... } = \frac{0.66X - ₹\ 64,32,000}{78}$$

51.48X – ₹ 46,07,46,000 = 37.98X - ₹34,08,96,000

16.5X = ₹ 11,98,50,000

Indifference Point = X = ₹ 72,63,636.36

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359 RTP **Old P8 Nov 2022** Q3 old scheme

ABC Limited provides you the following information:

	(₹)
Profit (EBIT)	2,80,000
Less: Intt. on Debt @10%	40,000
EBT	2,40,000
Less: Income Tax @ 50%	1,20,000
	1,20,000
No. of Equity Shares (₹ 10 each)	30,000
Earnings per share (EPS)	4
Price / EPS (P/E) Ratio	10
Ruling Market price per share	40

The company has undistributed reserves of ₹ 7,00,000 and needs ₹ 4,00,000 further for expansion. This investment is expected to earn the same rate as funds already invested. You are informed that a debt equity (debt/ debt +equity) ratio higher than 32% will push the P/E ratio down to 8 and raise the interest rate on additional borrowings (debentures) to 12%. You are required to ASCERTAIN the probable price of the share.

(i) If the additional funds are raised as debt; and

(ii) If the amount is raised by issuing equity shares at ruling market price of ₹ 40 per share.

ICAI'S ANSWER

Ascertainment of probable price of shares

Particulars	Plan (i) (If ₹ 4,00,000 is raised as debt) (₹)	Plan (ii) (If ₹ 4,00,000 is raised by issuing equity shares) (₹)
Earnings Before Interest (EBIT) 20% on (14,00,000 + 4,00,000)	3,60,000	3,60,000
Less: Interest on old debentures @ 10% on 4,00,000	40,000	40,000
	3,20,000	3,20,000
Less: Interest on New debt @ 12% on ₹ 4,00,000	48,000	-
Earnings Before Tax (After interest)	2,72,000	3,20,000
Less: Tax @ 50%	1,36,000	1,60,000
Earnings for equity shareholders (EAIT)	1,36,000	1,60,000
Number of Equity Shares (in numbers)	30,000	40,000
Earnings per Share (EPS)	4.53	4.00
Price/ Earnings Ratio	8	10
Probable Price Per Share	36.24 (8 × 4.53)	40 (10 × 4)

Working Notes:

	(₹)
1. Calculation of Present Rate of Earnings	
Equity Share capital (30,000 x ₹ 10)	3,00,000
10% Debentures $\left(40{,}000 \times \frac{100}{10}\right)$	4,00,000
Reserves (given)	7,00,000
	14,00,000
Earnings before interest and tax (EBIT) given	2,80,000
Rate of Present Earnings = $\left(\frac{2{,}80{,}000}{14{,}00{,}000} \times 100\right)$	20%
2. Number of Equity Shares to be issued in Plan $\left(\frac{4{,}00{,}000}{40}\right)$	10,000
Thus, after the issue total number of shares	30,000 + 10,000 = 40,000
3. Debt/Equity Ratio if ₹ 4,00,000 is raised as debt:	$\left(\frac{8{,}00{,}000}{18{,}00{,}000} \times 100\right)$ = 44.44%

As the debt equity ratio is more than 32% the P/E ratio shall be 8 in plan (i)

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360 RTP **Old P8 Nov 2020** Q3 old scheme

Xylo Ltd. is considering two alternative financing plans as follows:

Particulars	Plan – A (₹)	Plan – B (₹)
Equity shares of ₹ 10 each	8,00,000	8,00,000
Preference Shares of ₹ 100 each	-	4,00,000
12% Debentures	4,00,000	-
	12,00,000	12,00,000

The indifference point between the plans is ₹ 4,80,000. Corporate tax rate is 30%. CALCULATE the rate of dividend on preference shares.

ICAI'S ANSWER

Computation of Rate of Preference Dividend

$$\frac{(\text{EBIT - Interest})\ (1 - t)}{\text{No. of Equity Shares } (N_1)} = \frac{(\text{EBIT } (1 - t) - \text{Preference Dividend}}{\text{No. of Equity Shares } (N_2)}$$

$$\frac{(₹\ 4{,}80{,}000 - ₹\ 48{,}000)\ \text{x}\ (1 - 0.30)}{80{,}00{,}000 \text{ shares}} = \frac{₹\ 4{,}80{,}000(1 - 0.30) - \text{Preference Dividend}}{80{,}00{,}000 \text{ shares}}$$

$$\frac{(₹\ 3{,}02{,}400}{80{,}00{,}000 \text{ shares}} = \frac{₹\ 3{,}36{,}000 - \text{Preference Dividend}}{80{,}00{,}000 \text{ shares}}$$

₹ 3,02,400 = ₹ 3,36,000 – Preference Dividend

Preference Dividend = ₹ 3,36,000 – ₹ 3,02,400 = ₹ 33,600

$$\text{Rate of Dividend} = \frac{\text{Preference Dividend}}{\text{Preference share capital}} \times 100$$

$$= \frac{₹\ 33,600}{4,00,000} \times 100 = 8.4\%$$

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361 RTP **Old P8 May 2020** Q3 old scheme

CALCULATE the level of earnings before interest and tax (EBIT) at which the EPS indifference point between the following financing alternatives will occur.

(i) Equity share capital of ₹60,00,000 and 12% debentures of ₹40,00,000.

Or

(ii) Equity share capital of ₹40,00,000, 14% preference share capital of ₹20,00,000 and 12% debentures of ₹40,00,000.

Assume the corporate tax rate is 35% and par value of equity share is ₹100 in each case.

ICAI'S ANSWER

Computation of level of earnings before interest and tax (EBIT)

In case, alternative (i) is accepted, then the EPS of the firm would be:

$$\text{EPS}_{\text{Alternative (i)}} = \frac{(\text{EBIT} - \text{Interest})\ (1 - \text{tax rate})}{\text{No. of equity shares}}$$

$$= \frac{(\text{EBIT} - 0.12 \times ₹40,00,000)\ (1 - 0.35)}{60,000 \text{ shares}}$$

In case, alternative (ii) is accepted, then the EPS of the firm would be:

$$\text{EPS}_{\text{Alternative (ii)}} = \frac{(\text{EBIT} - 0.12 \times ₹40,00,000)(1 - 0.35) - (0.14 \times ₹20,00,000)}{40,000 \text{ shares}}$$

In order to determine the indifference level of EBIT, the EPS under the two alternative plans should be equated as follows:

$$\frac{(\text{EBIT} - 0.12 \times ₹40,00,000)(1 - 0.35)}{60,000 \text{ shares}} = \frac{(\text{EBIT} - 0.12 \times ₹40,00,000)(1 - 0.35) - (0.14 \times ₹20,00,000)}{40,000 \text{ shares}}$$

Or $\frac{0.65\ \text{EBIT} - ₹3,12,000}{3} = \frac{0.65\ \text{EBIT} - ₹5,92,000}{2}$

Or 1.30 EBIT – ₹6,24,000 = 1.95 EBIT – ₹17,76,000

Or (1.95 – 1.30) EBIT = ₹17,76,000 – ₹6,24,000 = ₹11,52,000

Or $\text{EBIT} = \frac{₹11,52,000}{0.65}$

Or EBIT = ₹17,72,308

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362 RTP **Old P8 Nov 2019** Q3 old scheme

The management of RT Ltd. wants to raise its funds from market to meet out the financial demands of its long-term projects. The company has various combinations of proposals to raise its funds. You are given

the following proposals of the company:

Proposal	Equity shares (%)	Debts (%)	Preference shares (%)
P	100	-	-
Q	50	50	-
R	50	-	50

(i) Cost of debt and preference shares is 12% each.

(ii) Tax rate –40%

(iii) Equity shares of the face value of ₹10 each will be issued at a premium of ₹10 per share.

(iv) Total investment to be raised ₹8,00,00,000.

(v) Expected earnings before interest and tax ₹3,60,00,000.

From the above proposals the management wants to take advice from you for appropriate plan after computing the following:

- Earnings per share
- Financial break-even-point

COMPUTE the EBIT range among the plans for indifference.

ICAI'S ANSWER

(i) Computation of Earnings per Share (EPS)

Plans	P (₹)	Q (₹)	R (₹)
Earnings before interest & tax (EBIT)	3,60,00,000	3,60,00,000	3,60,00,000
Less: Interest charges	--	(48,00,000)	--
Earnings before tax (EBT)	3,60,00,000	3,12,00,000	3,60,00,000
Less : Tax @ 40%	(1,44,00,000)	(1,24,80,000)	(1,44,00,000)
Earnings after tax (EAT)	2,16,00,000	1,87,20,000	2,16,00,000
Less : Preference share dividend	--	--	(48,00,000)
Earnings available for equity shareholders	2,16,00,000	1,87,20,000	1,68,00,000
No. of equity shares	40,00,000	20,00,000	20,00,000
E.P.S	5.40	9.36	8.40

(ii) Computation of Financial Break-even Points

Proposal 'P' = 0

Proposal 'Q' = ₹48,00,000 (Interest charges)

Proposal 'R' = Earnings required for payment of preference share dividend i.e. ₹48,00,000 ÷ 0.6 = ₹80,00,000

(iii) Computation of Indifference Point between the Proposals

Combination of Proposals

(a) Indifference point where EBIT of proposal "P" and proposal 'Q' is equal

$$\frac{\text{EBIT}(1-0.4)}{40{,}00{,}000 \text{ shares}} = \frac{(\text{EBIT} - ₹48{,}00{,}000)(1-0.4)}{20{,}00{,}000 \text{ shares}}$$

0.6 EBIT = 1.2 EBIT – ₹57,60,000

EBIT = ₹96,00,000

(b) Indifference point where EBIT of proposal 'P' and proposal 'R' is equal:

$$\frac{\text{EBIT}(1-0.40)}{40{,}00{,}000 \text{ shares}} = \frac{\text{EBIT}(1-0.40) - ₹48{,}00{,}000}{20{,}00{,}000 \text{ shares}}$$

$$\frac{0.6\text{EBIT}}{40{,}00{,}000 \text{ shares}} = \frac{0.6\text{EBIT} - ₹48{,}00{,}000}{20{,}00{,}000 \text{ shares}}$$

0.30 EBIT = 0.6 EBIT – ₹48,00,000

$$\text{EBIT} = \frac{₹48{,}00{,}000}{0.30} = ₹1{,}60{,}00{,}000$$

(c) Indifference point where EBIT of proposal 'Q' and proposal 'R' are equal

$$\frac{(\text{EBIT} - ₹48{,}00{,}000)(1-0.4)}{20{,}00{,}000 \text{ shares}} = \frac{\text{EBIT}(1-0.4) - ₹48{,}00{,}000}{20{,}00{,}000 \text{ shares}}$$

There is no indifference point between proposal 'Q' and proposal 'R'

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363 MTP **Sep 2026 S-II** Q3(a) 8 marks

ABC Limited is a manufacturing company with an all-equity capital structure. The firm is experiencing growth opportunities and needs to make strategic decisions regarding its capital structure to finance expansion plans. The company currently has 1,00,000 ordinary shares @ ₹ 10 in issue. Its current operational metrics are:

- Return on Capital Employed (ROCE): 31.25% [Capital Employed = Face Value of Equity Share Capital]
- Operating Margin (EBIT / Revenue): 31.25%
- Tax Rate: 50%

ABC Limited has identified investment opportunities requiring additional capital equal to 25% of its existing equity base. Management is evaluating three financing alternatives:

Option 1 — Equity Financing: Issue ordinary shares at face value

Option 2 — Debt Financing: Borrow at 8% per annum

Option 3 — Preference Share Financing: Issue preference shares carrying an 8% per annum dividend

Preliminary analysis shows that the difference in EPS between the Debt option and the Preference Share option works out to ₹ 0.10 per share.

You are required to:

1. Identify which financing alternative, Debt or Preference Shares yields the higher EPS, and explain the reason for the difference.

2. Find the Existing Equity Base, Face Value of Equity Shares, EBIT and Revenue.

3. If the additional finance is raised entirely through equity shares issued at face value, how many new shares should be issued?

4. Assuming the same EBIT, determine the Earnings Per Share (EPS) under each of the three financing alternatives.

ICAI'S ANSWER

1. The Debt option yields the higher EPS. Although the interest rate on debt (8%) equals the dividend rate on preference shares (8%), and both raise the identical amount of additional capital, the two are treated differently for tax purposes:

- Interest on debt is a charge against profit, it is deducted before computing tax, so it earns a tax shield of (Interest × Tax Rate).
- Preference dividend is an appropriation of profit, it is paid out of profit after tax, so it earns no tax shield.

Because of this tax shield, the after-tax cost of debt is lower than the after-tax cost of preference capital, so more residual earning is left for equity shareholders under the Debt optio, producing a higher EPS.

2. Existing Equity Share Capital (= Capital Employed) is taken as ₹ 10,00,000, with 1,00,000 shares in issue.

The EPS gap between Debt and Preference arises purely from the tax shield on interest, and this relationship holds regardless of the size of the existing equity base:

- Incremental earnings from choosing Debt over Preference = Tax shield on interest = X × 8% × 50% = 0.04X
- Spread over the unchanged 1,00,000 shares: $\frac{0.04x}{1,00,000} = 0.10$

$\Rightarrow$ 0.04X = ₹ 10,000

$\Rightarrow$ X = ₹ $\frac{10,000}{0.04}$ = ₹ 2,50,000

- EBIT = ROCE × Capital Employed = 31.25% × ₹ 10,00,000 = ₹ 3,12,500
- Revenue = EBIT ÷ Operating Margin = ₹ 3,12,500 ÷ 31.25% = ₹ 10,00,000

3. New shares = Additional Funds ÷ Face Value = ₹ 2,50,000 ÷ ₹ 10 = 25,000 shares

Total shares after issue = 1,00,000 (existing) + 25,000 (new) = 1,25,000 shares

4. **Evaluation of alternative options**

Particulars	Option 1 (Equity)	Option 2 (Debt)	Option 3 (Pref. Shares)
Revenue (₹)	10,00,000	10,00,000	10,00,000
EBIT (31.25% of Capital Employed) (₹)	3,12,500	3,12,500	3,12,500
Less: Interest (8% of ₹ 2,50,000) (₹)	Nil	20,000	Nil
Earnings Before Tax (₹)	3,12,500	2,92,500	3,12,500
Less: Tax @ 50% (₹)	1,56,250	1,46,250	1,56,250

Particulars	Option 1 (Equity)	Option 2 (Debt)	Option 3 (Pref. Shares)
Earnings After Tax (₹)	1,56,250	1,46,250	1,56,250
Less: Preference Dividend (8% of ₹ 2,50,000) (₹)	—	—	20,000
Earnings available to equity shareholders (₹)	**1,56,250**	**1,46,250**	**1,36,250**
Number of Equity Shares	1,25,000	1,00,000	1,00,000
Earnings Per Share (EPS)	**₹ 1.25**	**₹ 1.46**	**₹ 1.36**

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364 MTP **May 2026 S-II** MCQ 7 2 marks MCQ

X Ltd. is considering the following two alternative financing plans:

	Plan – I (₹)	Plan – II (₹)
Equity shares of ₹ 10 each	4,00,000	4,00,000
12% Debentures	2,00,000	-
Preference Shares of ₹ 100 each	-	2,00,000
	6,00,000	6,00,000

The indifference point between the plans is ₹ 2,40,000. Corporate tax rate is 30%.

What is the dividend on preference shares:

A ₹ 16,800

B ₹ 25,000

C ₹ 36,600

D ₹ 10,200

ICAI'S ANSWER

(A) ₹ 16,800

Computation of Rate of Preference Dividend

$$\frac{(\text{EBIT - Interest})\ (1\text{- t})}{\text{No. of Equity Shares } (N_1)} = \frac{\text{EBIT } (1\text{- t}) \text{ - Preference Dividend}}{\text{No. of Equity Shares } (N_2)}$$

$$\frac{(₹2{,}40{,}000 - ₹24{,}000)\ (1 - 0.30)}{40{,}000 \text{ shares}} = \frac{₹2{,}40{,}000\ (1 - 0.30) - \text{Preference Dividend}}{40{,}000 \text{ shares}}$$

$$\frac{₹2{,}16{,}000\ (1 - 0.30)}{40{,}000 \text{ shares}} = \frac{₹1{,}68{,}000 - \text{Preference Dividend}}{40{,}000 \text{ shares}}$$

₹ 1,51,200 = ₹ 1,68,000 – Preference Dividend

Preference Dividend = ₹ 1,68,000 – ₹ 1,51,200 = ₹ 16,800

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365 MTP **Jan 2026 S-I** Q4(c) 2 marks

DISCUSS financial break-even and EBIT-EPS indifference analysis.

ICAI'S ANSWER

Financial Break-even and EBIT-EPS Indifference Analysis

Financial break-even point is the minimum level of EBIT needed to satisfy all the fixed financial charges i.e. interest and preference dividend. It denotes the level of EBIT for which firm's EPS equals zero. If the EBIT is less than the financial breakeven point, then the EPS will be negative but if the expected level of EBIT is more than the breakeven point, then more fixed costs financing instruments can be taken in the capital structure, otherwise, equity would be preferred.

EBIT-EPS analysis is a vital tool for designing the optimal capital structure of a firm. The objective of this analysis is to find the EBIT level that will equate EPS regardless of the financing plan chosen.

$$\frac{(\text{EBIT} - I_1)(1-T)}{E_1} = \frac{(\text{EBIT} - I_2)(1-T)}{E_2}$$

Where,

EBIT	=	Indifference point
E_1	=	Number of equity shares in Alternative 1
E_2	=	Number of equity shares in Alternative 2
I_1	=	Interest charges in Alternative 1
1_2	=	Interest charges in Alternative 2
T	=	Tax-rate

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366 MTP Jan 2026 S-II MCQ 6 2 marks MCQ

A company is considering two financing plans:

- Plan A (Equity only): 40,000 equity shares
- Plan B (Debt + Equity): Debt of ₹ 12,00,000 at 10% interest and 25,000 equity shares

Tax rate = 30%.

The EBIT indifference point between Plan A and Plan B is:

A ₹ 4,00,000

B ₹ 4,80,000

C ₹ 6,00,000

D ₹ 3,20,000

ICAI'S ANSWER

(D) ₹ 3,20,000

Plan A

$$\text{EPS}_A = \frac{\text{EBIT} \times (1-0.30)}{40{,}000} = \frac{0.70\ \text{EBIT}}{40{,}000}$$

Plan B

$$EPS_B = \frac{(EBIT - 1,20,000)(1 - 0.30)}{25,000} = \frac{0.70(EBIT - 1,20,000)}{25,000}$$

Equating Plan A and Plan B

$$\frac{0.70\ EBIT}{40,000} = \frac{0.70(EBIT - 1,20,000)}{25,000}$$

$$\frac{EBIT}{40,000} = \frac{EBIT - 1,20,000}{25,000}$$

EBIT = ₹ 3,20,000

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367 MTP **Sep 2025 S-I** Case scenario · Q2(a)(1) to Q2(a)(4) 8 marks

CASE SCENARIO

Majestic Ltd., a manufacturing company, is evaluating its financial performance and considering an expansion plan. The financial controller provides the following information for the financial year 2025:

Equity Share Capital	₹ 10,00,000 (₹ 10 per share)
Retained Earnings	₹ 10,00,000
12% Debentures	₹ 41,66,667
EBIT	₹ 20,00,000
Interest Expense	₹ 5,00,000
Contribution Margin Ratio	40%
Tax Rate	30%
Operating Leverage	2.5
Financial Leverage	1.333
Combined Leverage	3.333
Working Capital	20,00,000
Current Ratio	2
Fixed Asset Turnover Ratio	3

The company is now considering an expansion project costing ₹ 25,00,000. It anticipates the following changes post-expansion in 2026:

(i) Sales are expected to increase by ₹50,00,000.

(ii) Variable costs will constitute 55% of sales.

(iii) Additional fixed costs due to expansion are ₹ 10,00,000.

The company is evaluating two financing options for the expansion:

Option 1: Raise ₹ 25,00,000 by issuing new equity shares at ₹ 10 each.

Option 2: Raise ₹ 25,00,000 by issuing new debt at 12% interest.

From the above information, you are required to:

Q2(a)(1) 3 marks

Prepare the Income Statement and Balance Sheet of 2025 based on the data provided.

Q2(a)(2) 2 marks

Compute the EPS under each financing option.

Q2(a)(3) 2 marks

Compute the Operating Leverage, Financial Leverage and Combined Leverage of both the financing options.

Q2(a)(4) 1 mark

Recommend the best financing option based on the EPS and impact of leverage giving reasons.

ICAI'S ANSWERS

ICAI's working for this case

Working Notes:

Contribution = DOL × EBIT = 2.5 × 20,00,000 = ₹ 50,00,000

$$\text{Sales} = \frac{\text{Contribution}}{\text{ContributionMarginRatio}} = \frac{50,00,000}{0.40} = ₹\ 1,25,00,000$$

Calculation of Current Assets and Current Liabilities

Let Current Liabilities be X

Then, 2X - X = 20,00,000

So, Current Liabilities = ₹ 20,00,000

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Current Assets = 2 Current Liabilities

So, Current Assets = ₹ 20,00,000 x 2 = ₹ 40,00,000

Calculation of Fixed Assets

Sales = ₹ 1,25,00,000

$$\text{Fixed Assets Turnover Ratio} = \frac{\text{Sales}}{\text{Fixed Assets}}$$

So, Fixed Assets = ₹ 1,25,00,000/3 = ₹ 41,66,667

Q2(a)(1)

Income Statement for the Financial Year ended 2025

Particulars	Amount (₹)
Sales	1,25,00,000
Less: Variable Costs @ 60%	75,00,000
Contribution	50,00,000

Particulars	Amount (₹)
Less: Fixed Costs (b.f.)	30,00,000
EBIT	20,00,000
Less: Interest	5,00,000
EBT	15,00,000
Less: Tax (30%)	4,50,000
EAT	10,50,000
EPS (10,50,000/1,00,000)	₹ 10.50

Balance Sheet as at Year 2025

Liabilities	₹	Assets	₹
Equity Capital	10,00,000	Fixed Assets	41,66,667
Retained Earnings	10,00,000	Current Assets	40,00,000
12% Debentures	41,66,667		
Current Liabilities	20,00,000		
	81,66,667		**81,66,667**

Q2(a)(2)

Calculation of EPS in each option under expansion plan

Post-Expansion Financials (2026)

Sales = 1,25,00,000 + 50,00,000

= ₹ 1,75,00,000

Variable Cost (55%) = 1,75,00,000 x 55% = ₹ 96,25,000

Contribution = ₹ 78,75,000

Fixed Costs = ₹ 30,00,000 + ₹ 10,00,000

= ₹ 40,00,000

EBIT = ₹ 38,75,000

Particulars	Equity Option	Debt Option
EBIT	38,75,000	38,75,000
Interest (old)	5,00,000	5,00,000
Interest (new)	-	3,00,000
EBT	33,75,000	30,75,000
Tax@30%	10,12,500	9,22,500
EAT	23,62,500	21,52,500
No. of Shares	3,50,000	1,00,000
EPS	6.75	21.525

Q2(a)(3)

Calculation of Operating Leverage, Financial Leverage and Combined Leverage for each option under expansion plan

	2025	Equity Option (2026)	Debt Option (2026)
EPS	₹ 10.50	₹ 6.75	₹ 21.525
Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}}$	2.5	= $\frac{78,75,000}{38,75,000}$ = 2.03	= $\frac{78,75,000}{38,75,000}$ = 2.03
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	1.333	= $\frac{38,75,000}{33,75,000}$ = 1.15	= $\frac{38,75,000}{30,75,000}$ = 1.26
Combined Leverage = OL x FL	3.333	= 2.03 × 1.15 = 2.33	= 2.03 × 1.26 = 2.56

Q2(a)(4)

Recommendation of the best financing option

Debt financing results in a higher EPS of ₹ 21.525 and offer better returns to shareholders. Though financial risk increases slightly, the leverage ratios remain within manageable limits. Therefore, if the company is confident in its ability to generate stable cash flow, debt financing is the better option.

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368 MTP **Sep 2025 S-II** Q2(b) 5 marks

The following data are presented in respect of Excellent Automation Ltd.:

	(₹)
Profit before interest and tax	78,00,000
Less: Interest on debentures @ 12%	18,00,000
Profit before tax	60,00,000

	(₹)
Less: Income tax @ 50%	30,00,000
Profit After tax	30,00,000
No. of equity shares (of ₹ 10 each)	12,00,000
EPS	2.5
PE Ratio	10
Market price per share	25

The company is planning to start a new project requiring a total capital outlay of ₹ 60,00,000. You are informed that a debt equity ratio (D/D+E) higher than 35%, pushes the Ke up to 12.5%, means reducing the PE ratio to 8 and rises the interest rate on additional amount borrowed to 14%. FIND OUT the probable price of share if:

(i) the additional funds are raised as a loan.

(ii) the amount is raised by issuing equity shares.

(Note: Retained earnings of the company is ₹ 1.8 crore)

ICAI'S ANSWER

In this question, EBIT after proposed extension is not given. Therefore, we can assume that existing return on capital employed will be maintained.

Working notes:

1. Return on Capital Employed = $\frac{\text{EBIT}}{\text{Capital Employed}} = \frac{78,00,000}{4,50,00,000} = 17.33\%$

Capital Employed = Debt + Equity

= ₹ 1,50,00,000 + (₹ 1,20,00,000 + ₹ 1,80,00,000)

= ₹ 4,50,00,000

2. Proposed EBIT = Proposed Capital Employed x Return on capital employed

= (₹ 4,50,00,000 + ₹ 60,00,000) x 17.33%

= ₹ 88,38,300

3. Debt Equity Ratio = $\frac{\text{Debt}}{\text{Debt} + \text{Equity}}$

Option1: Loan option

Debt = ₹ 1,50,00,000 + ₹ 60,00,000 = ₹ 2,10,00,000

Equity = ₹ 3,00,00,000

Debt Equity ratio = $\frac{2.1 \text{ cr.}}{2.1 \text{ cr.} + 3 \text{ cr.}} = 41.18\%$

Debt equity ratio has crossed the limit of 35%, hence, PE ratio in this case will be 8 times and additional borrowing will be at the rate of 14%.

Option2: Equity option

Debt = ₹ 1,50,00,000

Equity = ₹ 3,00,00,000 + ₹ 60,00,000 = **₹ 3,60,00,000**

Debt Equity Ratio = $\frac{₹1.5 \text{ cr.}}{₹1.5 \text{ cr} . + ₹3.6 \text{ cr.}} = 29.41\%$

Debt equity ratio has not crossed the limit of 35% hence PE ratio in this case will remain at 10 times.

4. Number of equity shares to be issued in case of equity option @ ₹ 25 per share = ₹ 60,00,000 / ₹ 25 = 2,40,000

Calculation of EPS and MPS under two financial options

Particulars	Financial Options	
	Option I	**Option II**
	14% additional loan of 60,00,000 (₹)	**12,00,000 equity shares @ ₹ 10 and 2,40,000 equity shares @ ₹ 25 (₹)**
Profit before interest and Tax (PBIT)	88,38,300	88,38,300
Less: Interest on old debentures @ 12%	18,00,000	18,00,000
Less: Interest on additional loan (new) @ 14% on ₹ 60,00,000	8,40,000	Nil
Profit before tax	61,98,300	70,38,300
Less: Taxes @ 50%	30,99,150	35,19,150
Earnings for equity shareholders (EAT/Profit after tax)	30,99,150	35,19,150
Number of Equity Shares	12,00,000	14,40,000
Earnings per Share (EPS)	2.58	2.44
Price/ Earnings ratio	8	10
Market price per share (MPS)	20.64	24.40

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369 MTP **Sep 2025 S-II** Q3(b) 2 marks

Briefly EXPLAIN the process for analysing Optimal Capital Structure.

ICAI'S ANSWER

Objective of financial management is to **maximize wealth**. Therefore, one should choose a capital structure which maximizes wealth. For this purpose, following analysis should be done:

(1) EBIT-EPS-MPS analysis: Chose a capital structure which maximizes market price per share. For that, start with same EBIT for all capital structures and calculate EPS. Thereafter, either multiply EPS by price earning ratio or divide it by cost of equity to arrive at MPS.

(2) Indifference Point analysis: In above analysis, we have considered value at a given EBIT only. What will happen if EBIT changes? Will it change your decision also? To answer this question, you can do indifference point analysis.

(3) Financial Break-Even Point (BEP) analysis: With change in capital structure, financial risk also changes. Though this risk has already been considered in PE ratio or in cost of equity in point one above, but one may calculate and consider it separately also by calculating Financial BEP.

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370 MTP **May 2025 S-I** Q3(a) 3 marks

Perfact Limited is considering a total investment of ₹ 27 lakhs. You are required to CALCULATE the level of earnings before interest and tax (EBIT) at which the EPS indifference point between the following financing alternatives will occur:

(i) Equity share capital of ₹ 18,00,000 and 14% debentures of ₹ 9,00,000.

Or

(ii) Equity share capital of ₹ 15,00,000, 16% preference share capital of ₹ 5,00,000 and 14% debentures of ₹ 7,00,000.

Assume the corporate tax rate is at 25% and par value of equity share is ₹ 10 in each case. Also CALCULATE the Financial Break-Even Point (FBEP) for both the plans.

ICAI'S ANSWER

At Indifference Point, EPS of Plan (i) = EPS of Plan (ii)

Plan (i) Plan (ii)

$$\frac{(\text{EBIT} - I_1)(1-T)}{E_1} = \frac{(\text{EBIT} - I_2)(1-T) - \text{PD}}{E_2}$$

$$\frac{(\text{EBIT} - 1{,}26{,}000)(1-0.25)}{1{,}80{,}000} = \frac{(\text{EBIT} - 98{,}000)(1-0.25) - 80{,}000}{1{,}50{,}000}$$

0.75 EBIT = 4,48,500

EBIT = Indifference point = 5,98,000

A) Financial BEP = Int + (Pref Div / 1-t)

Plan (i) = 1,26,000 + 0

FBEP for Plan (i) = 1,26,000

Plan (ii) = 98,000 + {80,000/ (1 - 0.25)}

FBEP for Plan (ii) = 2,04,666

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371 MTP **May 2025 S-I** Q4(c) 2 marks

DISCUSS optimal capital structure and HOW to analyse it.

ICAI'S ANSWER

Objective of financial management is to **maximize wealth**. Therefore, one should choose a capital structure which maximizes wealth. For this purpose, following analysis should be done:

(1) EBIT-EPS-MPS analysis: Chose a capital structure which maximizes market price per share. For that, start with same EBIT for all capital structures and calculate EPS. Thereafter, either multiply EPS by price earning ratio or divide it by cost of equity to arrive at MPS.

(2) Indifference Point analysis: In above analysis, we have considered value at a given EBIT only. What will happen if EBIT changes? Will it change your decision also? To answer this question, you can do indifference point analysis.

(3) Financial Break-Even Point (BEP) analysis: With change in capital structure, financial risk also changes. Though this risk has already been considered in PE ratio or in cost of equity in point one above, but one may calculate and consider it separately also by calculating Financial BEP.

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372 MTP **Jan 2025 S-I** Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

Small bus Company is into manufacturing mini buses. Since its establishment it has seen a phenomenal growth in both its market share and profitability. The financial statements (Statement of P&L and Balance Sheet) are shown below. The company enjoys the confidence of its shareholders who have been rewarded with growing dividends year after year. Last year too, the company had announced 20 per cent dividend, which was the highest in the automobile sector. The company has never defaulted on its loan payments and enjoys a favourable face with its lenders, which include financial institutions, commercial banks and other private debenture holders. The competition in the bus industry has increased in the past few years and the company foresees further intensification of competition with the entry of several foreign bus manufacturers; many of whom are market leaders in their respective countries. The mini bus segment especially, will witness entry of foreign majors in the near future, with latest technology being offered to the Indian customer. Small bus company's management realises the need for large scale investment in upgradation of technology and improvement of manufacturing facilities to beat competition.

While on one hand, the competition in the industry has been intensifying, on the other hand, there has been a slowdown in the Indian economy, which has not only reduced the demand for buses, but also led to adoption of price cutting strategies by various bus manufacturers.

The Company needs ₹ 3,12,50,000 for the investment in technology and improvement of manufacturing facilities. Company has three options for the funds:

I The Company may issue 31,25,000 equity shares at ₹ 10 per share.

II The Company may issue 15,62,500 equity shares at ₹ 10 per share and 1,56,250 debentures of ₹ 100 denomination bearing an 9% rate of interest.

III The Company may issue 15,62,500 equity shares at ₹ 10 per share and 1,56,250 preference shares at ₹ 100 per share bearing an 10% rate of dividend.

The company's earnings before interest and taxes after investment is ₹ 37,50,000. Income tax rate applicable to the company is 40%.

Based on the above facts, the management of the company asked you to answer the following questions (MCQs 1 to 5):

MCQ 1 2 marks

What is the EPS under financial plan I?

A ₹ 0.50

B ₹ 0.62

C ₹ 0.72

D ₹ 0.44

MCQ 2 2 marks

What is the EPS under financial plan II?

A ₹ 0.70

B ₹ 0.90

C ₹ 0.42

D ₹ 1.10

MCQ 3 2 marks

What is the EPS under financial plan III?

A ₹ 0.44

B ₹ 0.70

C ₹ 0.85

D ₹ 1.20

MCQ 4 2 marks

What is the EBIT-EPS indifference points by formulae between Financing Plan I and Plan II?

A ₹ 28,12,500.00

B ₹ 29,00,000.00

C ₹ 32,50,666.66

D ₹ 45,15,253.56

MCQ 5 2 marks

What is the EBIT-EPS indifference points by formulae between Financing Plan I and Plan III?

A ₹ 36,36,666.66

B ₹ 45,25,000.00

C ₹ 28,56,256.25

D ₹ 52,08,333.33

ICAI'S ANSWERS

MCQ 1

(C) ₹ 0.72

Computation of EPS under financial plan I: Equity Financing

	(₹)
EBIT	37,50,000.00
Interest	-
EBT	37,50,000.00
Less: Taxes 40%	(15,00,000.00)
PAT	22,50,000.00
No. of equity shares	31,25,000.00
EPS	0.72

MCQ 2

(B) ₹ 0.90

Computation of EPS under financial plan II: Debt – Equity Mix

	(₹)
EBIT	37,50,000.00
Less: Interest	(14,06,250.00)
EBT	23,43,750.00
Less: Taxes 40%	(9,37,500.00)
PAT	14,06,250.00
No. of equity shares	15,62,500.00
EPS	0.90

MCQ 3

(A) ₹ 0.44

Computation of EPS under financial plan III: Preference Shares – Equity Mix

	(₹)
EBIT	37,50,000.00
Less: Interest	-
EBT	37,50,000.00
Less: Taxes (40%)	(15,00,000.00)
PAT	22,50,000.00
Less: Pref. dividend	(15,62,500.00)

	(₹)
PAT for equity shareholders	6,87,500.00
No. of Equity shares	15,62,500.00
EPS	0.44

MCQ 4

(A) ₹ 28,12,500

EBIT – EPS Indifference Point- Plan I and Plan II:

$$\frac{(\text{EBIT}) \times (1\text{- } T_C)}{N_1} = \frac{(\text{EBIT} - \text{Interest}) \times (1 - T_C)}{N_2}$$

$$\frac{\text{EBIT}(1 - 0.40)}{31,25,000} = \frac{(\text{EBIT} - 14,06,250) \times (1 - 0.40)}{15,62,500}$$

0.6EBIT = 1.2 EBIT – 16,87,500

= ₹ 28,12,500

MCQ 5

(D) ₹ 52,08,333.33

EBIT – EPS Indifference Point: Plan I and Plan III

$$\frac{\text{EBIT}(1 - T_c)}{N_1} = \frac{\text{EBIT}(1 - T_c) - \text{Pref. Div.}}{N_2}$$

$$\frac{\text{EBIT}(1 - 0.4)}{31,25,000} = \frac{\text{EBIT}(1 - 0.4) - 15,62,500}{15,62,500}$$

0.6EBIT = 1.2EBIT – 31,25,000

EBIT = ₹ 52,08,333.33

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373 MTP **Sep 2024 S-II** Q1(a) 5 marks

X Ltd is willing to raise funds for its New Project which requires an investment of ₹ 84 Lakhs. The Company has two options:

Option I: To issue Equity Shares (₹ 10 each) only

Option II: To avail Term Loan at an interest rate of 12%. But in this case, as insisted by the Financing Agencies, the Company will have to maintain a Debt–Equity proportion of 2:1.

The Corporate Tax Rate is 30%. FIND out the point of indifference for the project.

ICAI'S ANSWER

Let the EBIT at the Indifference Point level be E

Particulars	Alternative 1	Alternative 2
Description	**Fully Equity of 84 Lakhs**	**Debt = 56 Lakhs, Equity = 28 Lakhs**
EBIT	E	E

Particulars	Alternative 1	Alternative 2
Less: Interest at 12% of ₹ 56 Lakhs	Nil	6.72
EBT	E	E – 6.72
Less: Tax at 30%	0.3 E	0.3 E – 2.016
EAT	0.7 E	0.7 E – 4.704
Less: Preference Dividend	Nil	Nil
Residual Earnings	0.7 E	0.7 E – 4.704
No. of Equity Shares (Face Value ₹ 10)	8.4 Lakh Shares	2.8 Lakh Shares
EPS = $\frac{\text{Residual Earnings}}{\text{No. of Equity Shares}}$	$\frac{0.7\ E}{8.4\ \text{Lakh Shares}}$	$\frac{0.7\ E - 4.704}{2.8\ \text{Lakh Shares}}$

For indifference between the above alternatives, EPS should be equal.

So, $\frac{0.7\ E}{8.4\ \text{Lakh Shares}} = \frac{0.7\ E - 4.704}{2.8\ \text{Lakh Shares}}$

On cross multiplication and simplification, 2.1 E – 14.112 = 0.7 E. So, 1.4 E = 14.112

So, $E = \frac{14.112}{1.4} = 10.08$

So, for same EPS, required EBIT = ₹ 10.08 Lakhs. EPS at that level = ₹ 0.84

Note: Presentation of solution may differ.

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374 MTP **Sep 2024 S-II** Q4(c) 2 marks

EXPLAIN the Relationship between EBIT-EPS-MPS

ICAI'S ANSWER

The basic objective of financial management is to design an appropriate capital structure which can provide the highest wealth, i.e., highest MPS, which in turn depends on EPS.

Given a level of EBIT, EPS will be different under different financing mix depending upon the extent of debt financing. The effect of leverage on the EPS emerges because of the existence of fixed financial charge i.e., interest on debt, financial fixed dividend on preference share capital. The effect of fixed financial charge on the EPS depends upon the relationship between the rate of return on assets and the rate of fixed charge. If the rate of return on assets is higher than the cost of financing, then the increasing use of fixed charge financing (i.e., debt and preference share capital) will result in increase in the EPS. This situation is also known as favourable financial leverage or Trading on Equity. On the other hand, if the rate of return on assets is less than the cost of financing, then the effect may be negative and, therefore, the increasing use of debt and preference share capital may reduce the EPS of the firm.

The fixed financial charge financing may further be analysed with reference to the choice between the debt financing and the issue of preference shares. Theoretically, the choice is tilted in favour of debt financing for two reasons: (i) the explicit cost of debt financing i.e., the rate of interest payable

on debt instruments or loans is generally lower than the rate of fixed dividend payable on preference shares, and (ii) interest on debt financing is tax-deductible and therefore the real cost (after-tax) is lower than the cost of preference share capital.

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375 MTP **May 2024 S-I** MCQ 2 2 marks MCQ

Ramu Ltd. wants to implement a project for which ₹ 25 lakhs is required. Following financing options are at hand:

Option 1:

Equity Shares	25,000 @ ₹ 100

Option 2:

Equity Shares	10,000 @ ₹ 100
12% Preference Shares	5,000@ ₹ 100
10% Debentures	10,000@ ₹ 100

What is the indifference point & EPS at that level of EBIT assuming corporate tax to be 35%.

A ₹ 2,94,872; ₹ 11.80

B ₹ 3,20,513; ₹ 8.33

C ₹ 2,94,872; ₹ 7.67

D ₹ 3,20513; ₹ 12.82

ICAI'S ANSWER

(B) ₹ 3,20,513; ₹ 8.33

$$\frac{(EBIT - I)(1-t) - D_p}{N_1} = \frac{(EBIT - I)(1-t) - D_p}{N_2}$$

$$\frac{(x-0)(1-0.35)}{25{,}000} = \frac{(x-1{,}00{,}000)(1-0.35) - 60{,}000}{10{,}000}$$

x = EBIT = ₹ 3,20,513

At EBIT of ₹ 3,20,513, EPS under both options will be the same i.e., ₹ 8.33 per share

ICAI · P6 · MTP May 2024 · Series I · question p. 2-3 · answer p. 2

376 MTP **May 2024 S-II** Q2(b) 6 marks

The GT Limited is willing to expand its business for which it requires an additional finance of ₹ 50,00,000. At present, the capital structure of the company is as under:

- 7,00,000 Equity shares of ₹ 10 each
- 10% Debentures ₹ 63,00,000
- 12% Term loan ₹ 54,00,000
- Retained earnings ₹ 1,30,00,000

At present, the company's EBIT is ₹ 54,00,000. However, the company, after expansion, expects ROI 2% greater than the present ROI, Income Tax Rate is 30%.

Following two options, for getting additional finance, are available-

(a) To raise funds as term loan @ 12%

(b) To raise funds by issuing 1,00,000 equity shares at ₹ 20 per share and balance by issuing 11% debentures at par.

Required:

(i) FIND out the market price of shares, if the P/E ratio is 10.

(ii) RECOMMEND the suitable option of raising funds with reason.

ICAI'S ANSWER

Expected return on capital employed

Capital Employed = Debt + Equity

= (₹ 63,00,000 + ₹ 54,00,000) + (₹ 70,00,000 + ₹ 1,30,00,000)

= ₹ 3,17,00,000

Return on capital employed/ROI = $\left(\frac{\text{EBIT}}{\text{Capital employed}}\right)$ x 100

At present:

= $\left(\frac{54,00,000}{3,17,00,000}\right)$ x 100

= 17.03%

Now company expects 2% more as ROI

So, Expected ROI = 17.03% + 2%

= 19.03%

Proposed EBIT = Proposed Capital Employed x Return on capital employed

= (₹ 3,17,00,000 + ₹ 50,00,000) x 19.03% = ₹ 69,84,010

(i) Market Price per Share:

Particular	Financial Options	
	Option – I 12% term loan of ₹ 50,00,000	**Option II 1,00,000 equity shares @ ₹ 20 and 11% debentures of ₹ 30,00,000**
	(₹)	**(₹)**
EBIT	69,84,010	69,84,010
Less: Interest		
- 10% on old debentures	6,30,000	6,30,000
- 11% on new debentures	-	3,30,000
- 12% on old term loan	6,48,000	6,48,000
- 12% on new term loan	6,00,000	
Total Interest	18,78,000	16,08,000

Particular	Financial Options	
EBT	51,06,010	53,76,010
Less Tax @ 30%	15,31,803	16,12,803
EAT	35,74,207	37,63,207
No. of equity shares	7,00,000	8,00,000
Earnings per share	**5.11**	**4.70**
P/E ratio	10	10
Market Price per Share = EPS x P/E ratio	**51.06**	**47.04**

(ii) Recommendation:

The option I is better and may be opted as both EPS and MPS are higher.

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377 MTP **Old P8 Nov 2023 S-I** Q1(a) 5 marks old scheme

Bhaskar Manufactures Ltd. have Equity Share Capital of ₹ 5,00,000 (face value ₹100) to meet the expenditure of an expansion programme, the company wishes to raise ₹ 3,00,000 and is having following four alternative sources to raise the funds:

Plan A: To have full money from equity shares.

Plan B: To have ₹ 1 lakhs from equity and ₹ 2 lakhs from borrowing from the financial institution @ 10% p.a.

Plan C: Full money from borrowing @ 10% p.a.

Plan D: ₹1 lakh in equity and ₹ 2 lakhs from preference shares at 8% p.a.

The company is expected to have an earning of ₹ 1,50,000. The corporate tax is 50%. Suggest a suitable plan of the above four plans to raise the required funds.

ICAI'S ANSWER

Statement showing the EPS under the four plans

	Plan A	Plan B	Plan C	Plan D
Equity share capital	₹ 8,00,000	₹ 6,00,000	₹ 5,00,000	₹ 6,00,000
8% Pref. Share capital	-	-	-	₹ 2,00,000
Borrowing @ 10%	-	₹ 2,00,000	₹ 3,00,000	-
	₹ 8,00,000	₹ 8,00,000	₹ 8,00,000	₹ 8,00,000
E.B.I.T	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000
Less: Interest @ 10%		₹ 20,000	₹ 30,000	

	Plan A	Plan B	Plan C	Plan D
E.B.T	₹ 1,50,000	₹ 1,30,000	₹1,20,000	₹ 1,50,000
Less: Tax	₹ 75,000	₹65,000	₹60,000	₹ 75,000
Less: Pref Divided				₹ 16,000
Earnings available to equity share holders	₹ 75,000	₹ 65,000	₹ 60,000	₹ 59,000
No.of equity shares (₹100)	8,000	6,000	5,000	6,000
Earning per share	**₹ 9.38**	**₹ 10.83**	**₹ 12.00**	**₹ 9.83**

Plan C given the highest EPS and therefore to be accepted.

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 1 · answer p. 1

378 MTP **Old P8 Nov 2023 S-I** Q4 10 marks old scheme

A company needs ₹ 42,50,000 for the construction of a new plant. The following three plans are feasible:

I The company may issue 4,25,000 equity shares at ₹ 10 per share.

II The company may issue 2,12,500 equity shares at ₹ 10 per share and 21,250 debentures of ₹ 100 denominations bearing an 8% rate of interest.

III The company may issue 2,12,500 equity shares at ₹ 10 per share and 21,250 cumulative preference shares at ₹ 100 per share bearing an 8% rate of dividend.

(i) The company's earnings before interest and taxes are ₹ 75,000, ₹ 1,50,000, ₹ 3,00,000, ₹ 4,50,000 and ₹ 7,50,000. DETERMINE earnings per share under each of three financial plans? Assume a corporate income tax rate of 40%.

(ii) IDENTIFY which alternative would you recommend and why?

(iii) DETERMINE the EBIT-EPS indifference points by formulae between Financing Plan I and Plan II and Plan I and Plan III.

ICAI'S ANSWER

(i) Computation of EPS under three-financial plans.

Plan I: Equity Financing

	(₹)	(₹)	(₹)	(₹)	(₹)
EBIT	75,000	1,50,000	3,00,000	4,50,000	7,50,000
Interest	0	0	0	0	0
EBT	75,000	1,50,000	3,00,000	4,50,000	7,50,000
Less: Tax @ 40%	30,000	60,000	1,20,000	1,80,000	3,00,000
PAT	45,000	90,000	1,80,000	2,70,000	4,50,000
No. of equity shares	4,25,000	4,25,000	4,25,000	4,25,000	4,25,000
EPS	0.11	0.21	0.42	0.64	1.06

Plan II: Debt – Equity Mix

	(₹)	(₹)	(₹)	(₹)	(₹)
EBIT	75,000	1,50,000	3,00,000	4,50,000	7,50,000
Less: Interest	1,70,000	1,70,000	1,70,000	1,70,000	1,70,000
EBT	(95,000)	(20,000)	1,30,000	2,80,000	5,80,000
Less: Tax @ 40%	38,000*	8000*	52,000	1,12,000	2,32,000
PAT	(57,000)	(12,000)	78,000	1,68,000	3,48,000
No. of equity shares	2,12,500	2,12,500	2,12,500	2,12,500	2,12,500
EPS	(₹ 0.27)	(0.056)	0.37	0.79	1.64

* The Company can set off losses against the overall business profit or may carry forward it to next financial years.

Plan III: Preference Shares – Equity Mix

	(₹)	(₹)	(₹)	(₹)	(₹)
EBIT	75,000	1,50,000	3,00,000	4,50,000	7,50,000
Less: Interest	0	0	0	0	0
EBT	75,000	1,50,000	3,00,000	4,50,000	7,50,000
Less: Tax @ 40%	30,000	60,000	1,20,000	1,80,000	3,00,000
PAT	45,000	90,000	1,80,000	2,70,000	4,50,000
Less: Pref. dividend	1,70,000*	1,70,000*	1,70,000	1,70,000	1,70,000
PAT after Pref. dividend.	(1,25,000)	(80,000)	10,000	1,00,000	2,80,000
No. of Equity shares	2,12,500	2,12,500	2,12,500	2,12,500	2,12,500

	(₹)	(₹)	(₹)	(₹)	(₹)
EPS	(0.59)	(0.38)	0.05	0.47	1.32

* In case of cumulative preference shares, the company must pay cumulative dividend to preference shareholders, when company earns sufficient profits.

(ii) From the above EPS computations tables under the three financial plans we can see that when EBIT is ₹ 4,50,000 or more, Plan II: Debt-Equity mix is preferable over the Plan I and Plan III, as rate of EPS is more under this plan. On the other hand, an EBIT of less than ₹4,50,000, Plan I: Equity Financing has higher EPS than Plan II and Plan III. Plan III Preference Share-Equity mix is not acceptable at any level of EBIT, as EPS under this plan is lower.

The choice of the financing plan will depend on the performance of the company and other macro-economic conditions. If the company is expected to have higher operating profit Plan II: Debt – Equity Mix is preferable. Moreover, debt financing gives more benefit due to availability of tax shield.

(iii) EBIT – EPS Indifference point: Plan I and Plan II

$$\frac{\text{EBIT}_1 \times (1-t)}{\text{No.of equity shares}(N_1)} = \frac{(\text{EBIT}_2 - \text{Interest}) \times (1-t)}{\text{No.of equity shares}(N_2)}$$

$$\frac{\text{EBIT}(1-0.40)}{4{,}25{,}000 \text{ shares}} = \frac{(\text{EBIT} - ₹1{,}70{,}000) \times (1-0.40)}{2{,}12{,}500 \text{shares}}$$

0.6 EBIT = 1.2 EBIT – ₹2,04,000

$$\text{EBIT} = \frac{₹2{,}04{,}000}{0.6} = ₹\ 3{,}40{,}000$$

Indifference points between Plan I and Plan II is ₹ 3,40,000

EBIT – EPS Indifference Point: Plan I and Plan III

$$\frac{\text{EBIT}_1 \times (1-t)}{\text{No.of equity shares}(N_1)} = \frac{\text{EBIT}_3 \times (1-t) - \text{Pref.dividend}}{\text{No.of equity shares}(N_3)}$$

$$\frac{\text{EBIT}_1(1-0.40)}{4{,}25{,}000 \text{shares}} = \frac{\text{EBIT}_3(1-0.40) - \text{Rs.}1{,}70{,}000}{2{,}12{,}500 \text{shares}}$$

0.6 EBIT = 1.2 EBIT – ₹ 3,40,000

$$\text{EBIT} = \frac{₹3{,}40{,}000}{0.6} = ₹\ 5{,}66{,}667$$

Indifference points between Plan I and Plan III is ₹ 5,66,667.

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 2-3 · answer p. 6-8

379 MTP **Old P8 Nov 2023 S-II** Q2 10 marks old scheme

A Company earns a profit of ₹7,00,000 per annum after meeting its interest liability of ₹1,00,000 on 10% debentures. The Tax rate is 40%. The number of Equity Shares of ₹10 each are 1,00,000 and the retained earnings amount to ₹20,00,000. The company proposes to take up an expansion scheme for which a sum of ₹10,00,000 is required. It is anticipated that after expansion, the company will be able to achieve the same return on investment as at present. The funds required for expansion can be raised either through debt at the rate of 12% or by issuing equity shares at par.

Required:

(i) COMPUTE the Earnings per Share (EPS), if:

- The additional funds were raised as debt

- The additional funds were raised by issue of equity shares.

(ii) ADVISE the company as to which source of finance is preferable.

ICAI'S ANSWER

Working Notes:

1. Capital employed before expansion plan:

	(₹)
Equity shares (₹10 × 1,00,000 shares)	10,00,000
Debentures {(₹1,00,000/10) × 100}	10,00,000
Retained earnings	20,00,000
Total capital employed	40,00,000

2. Earnings before the payment of interest and tax (EBIT):

	(₹)
Profit (EBT)	7,00,000
Add: Interest	1,00,000
EBIT	8,00,000

3. Return on Capital Employed (ROCE):

$$\text{ROCE} = \frac{\text{EBIT}}{\text{Capital employed}} \times 100 = \frac{\text{Rs.8,00,000}}{\text{Rs.40,00,000}} \times 100 = 20\%$$

4. Earnings before interest and tax (EBIT) after expansion scheme:

After expansion, capital employed = ₹40,00,000 + ₹10,00,000

= ₹ 50,00,000

Desired EBIT = 20% × ₹50,00,000 = ₹10,00,000

(i) Computation of Earnings Per Share (EPS) under the following options:

	Present situation	Expansion scheme Additional funds raised as	
		Debt	**Equity**
	(₹)	**(₹)**	**(₹)**
Earnings before Interest and Tax (EBIT)	8,00,000	10,00,000	10,00,000
Less: Interest - Old capital	1,00,000	1,00,000	1,00,000
- New capital	--	1,00,000 (₹10,00,000 × 10%)	--
Earnings before Tax (EBT)	7,00,000	8,00,000	9,00,000

	Present situation	Expansion scheme Additional funds raised as	
Less: Tax (40% of EBT)	2,80,000	3,20,000	3,60,000
PAT	4,20,000	4,80,000	5,40,000
No. of shares outstanding	1,00,000	1,00,000	2,00,000
Earnings per Share (EPS)	4.20 $\left(\frac{₹4,20,000}{1,00,000}\right)$	4.80 $\left(\frac{₹4,80,000}{1,00,000}\right)$	2.70 $\left(\frac{₹5,40,000}{2,00,000}\right)$

(ii) Advise to the Company: When the expansion scheme is financed by additional debt, the EPS is higher. Hence, the company should finance the expansion scheme by raising debt.

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 2 · answer p. 3-4

380 MTP **Old P8 May 2023 S-I** Q1(d) 5 marks old scheme

Aeron We Ltd. is considering two alternative financing plans as follows:

Particulars	Plan – A (₹)	Plan – B (₹)
Equity shares of ₹ 100 each	90,00,000	90,00,000
Preference Shares of ₹ 100 each	-	20,00,000
9% Debentures	20,00,000	-
	1,10,00,000	1,10,00,000

The indifference point between the plans is ₹7,60,000. Corporate tax rate is 25%. CALCULATE the rate of dividend on preference shares.

ICAI'S ANSWER

Computation of Rate of Preference Dividend

$$\frac{(\text{EBIT- Interest})\ (1\text{ -t})}{\text{No. of Equity Shares } (N_1)} = \frac{(\text{EBIT}(1\text{-t}) - \text{Preference Dividend}}{\text{No. of Equity Shares } (N_2)}$$

$$\frac{(7,60,000 - 1,80,000)\ \text{x}\ (1\text{-}0.25)}{90,000\ \text{shares}} = \frac{7,60,000\ (1\text{-}0.25) - \text{Preference Dividend}}{90,000\ \text{shares}}$$

$$\frac{4,35,000}{90,000\ \text{shares}} = \frac{5,70,000 - \text{Preference Dividend}}{90,000\ \text{shares}}$$

₹ 4,35,000 = ₹ 5,70,000 – Preference Dividend

Preference Dividend = ₹ 5,70,000 – ₹ 4,35,000 = ₹ 1,35,000

$$\text{Rate of Dividend} = \frac{\text{Preference Dividend}}{\text{Preference share capital}} \times 100$$

$$= \frac{1,35,000}{20,00,000} \times 100 = 6.75\ \%$$

ICAI · Old P8 · MTP May 2023 · Series I · question p. 2 · answer p. 4

381 MTP Old P8 May 2023 S-I Q2 10 marks old scheme

RML Limited needs ₹6,50,00,000 for the Expansion purposes. The following three plans are feasible:

(I) The Company may issue 6,50,000 equity shares at ₹100 per share.

(II) The Company may issue 4,00,000 equity shares at ₹100 per share and 2,50,000 debentures of ₹100 denomination bearing a 9% rate of interest.

(III) The Company may issue 4,00,000 equity shares at ₹100 per share and 2,50,000 cumulative preference shares at ₹100 per share bearing a 9% rate of dividend.

(i) If the Company's earnings before interest and taxes are ₹15,62,500, ₹22,50,000, ₹62,50,000, ₹93,75,000 and ₹1,56,25,000, CALCULATE the earnings per share under each of three financial plans? Assume a Corporate Income tax rate of 25%.

(ii) WHICH alternative would you recommend and why?

ICAI'S ANSWER

Computation of EPS under three-financial plans.

Plan I: Equity Financing

	(₹)	(₹)	(₹)	(₹)	(₹)
EBIT	15,62,500	22,50,000	62,50,000	93,75,000	1,56,25,000
Interest	0	0	0	0	0
EBT	15,62,500	22,50,000	62,50,000	93,75,000	1,56,25,000
Less: Tax @ 25%	3,90,625	5,62,500	15,62,500	23,43,750	39,06,250
PAT	11,71,875	16,87,500	46,87,500	70,31,250	1,17,18,750
No. of equity shares	6,50,000	6,50,000	6,50,000	6,50,000	6,50,000
EPS	1.80	2.60	7.21	10.82	18.03

Plan II: Debt – Equity Mix

	(₹)	(₹)	(₹)	(₹)	(₹)
EBIT	15,62,500	22,50,000	62,50,000	93,75,000	1,56,25,000
Less: Interest	22,50,000	22,50,000	22,50,000	22,50,000	22,50,000
EBT	(6,87,500)	0	40,00,000	71,25,000	1,33,75,000
Less: Tax @ 25%	1,71,875*	0	10,00,000	17,81,250	33,43,750
PAT	(5,15,625)	0	30,00,000	53,43,750	1,00,31,250
No. of equity shares	4,00,000	4,00,000	4,00,000	4,00,000	4,00,000
EPS (₹)	(1.29)	0.00	7.50	13.36	25.08

* The Company can set off losses against the overall business profit or may carry forward it to next financial years.

Plan III: Preference Shares – Equity Mix

	(₹)	(₹)	(₹)	(₹)	(₹)
EBIT	15,62,500	22,50,000	62,50,000	93,75,000	1,56,25,000
Less: Interest	0	0	0	0	0
EBT	15,62,500	22,50,000	62,50,000	93,75,000	1,56,25,000
Less: Tax @ 25%	3,90,625	5,62,500	15,62,500	23,43,750	39,06,250
PAT	11,71,875	16,87,500	46,87,500	70,31,250	1,17,18,750
Less: Pref. dividend *	22,50,000	22,50,000	22,50,000	22,50,000	22,50,000
PAT after Pref. dividend.	(10,78,125)	(5,62,500)	24,37,500	47,81,250	94,68,750
No. of Equity shares	4,00,000	4,00,000	4,00,000	4,00,000	4,00,000
EPS	(2.70)	(1.41)	6.09	11.95	23.67

* In case of cumulative preference shares, the company has to pay cumulative dividend to preference shareholders.

(ii) In case of lower EBIT Plan I i.e Equity Financing is better however in case of higher EBIT Plan II i.e Debt=Equity Mix is best.

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382 MTP **Old P8 Nov 2022 S-II** Q2 10 marks old scheme

The financial advisor of Sun Ltd is confronted with following two alternative financing plans for raising ₹ 10 lakhs that is needed for plant expansion and modernization

Alternative I: Issue 80% of funds with 14% Debenture [Face value (FV) ₹ 100] at par and redeem at a premium of 10% after 10 years and balance by issuing equity shares at $33\frac{1}{3}$ % premium.

Alternative II: Raise 10% of funds required by issuing 8% Irredeemable Debentures [Face value (FV) ₹ 100] at par and the remaining by issuing equity shares at current market price of ₹125.

Currently, the firm has an Earnings per share (EPS) of ₹ 21

The modernization and expansion programme is expected to increase the firm's Earnings before Interest and Taxation (EBIT) by ₹ 200,000 annually.

The firm's condensed Balance Sheet for the current year is given below:

Balance Sheet as on 31.3.2022

Liabilities	Amount (₹)	Assets	Amount (₹)
Current Liabilities	5,00,000	Current Assets	16,00,000
10% Long Term Loan	15,00,000	Plant & Equipment (Net)	34,00,000
Reserves & Surplus	10,00,000		
Equity Share Capital (FV: ₹ 100 each)	20,00,000		
TOTAL	50,00,000	TOTAL	50,00,000

However, the finance advisor is concerned about the effect that issuing of debt might have on the firm. The average debt ratio for firms in industry is 35%.He believes if this ratio is exceeded, the P/E ratio of the company will be 7 because of the potentially greater risk.

If the firm increases its equity capital by more than 10 %, he expects the P/E ratio of the company will increase to 8.5 irrespective of the debt ratio.

Assume Tax Rate of 25%. Assume target dividend pay-out under each alternative to be 60% for the next year and growth rate to be 10% for the purpose of calculating Cost of Equity

SUGGEST with reason which alternative is better on the basis of each of the below given criteria:

I. Earnings per share (EPS) & Market Price per share (MPS)

II. Financial Leverage

III. Weighted Average Cost of Capital & Marginal Cost of Capital (using Book Value weights)

ICAI'S ANSWER

Calculation of Equity Share capital and Reserves and surplus:

Alternative 1:

$$\text{Equity Share capital} = ₹20,00,000 + \frac{₹2,00,000 \text{ x } 100}{133.3333} = ₹21,50,000$$

$$\text{Reserves} = ₹10,00,000 + \frac{₹2,00,000 \text{ x } 33.3333}{133.3333} = ₹10,50,000$$

Alternative 2:

$$\text{Equity Share capital} = ₹\ 20,00,000 + \frac{₹\ 9,00,000 \text{x} 100}{125} = ₹27,20,000$$

$$\text{Reserves} = ₹10,00,000 + \frac{₹\ 9,00,000 \text{ x} 25}{125} = ₹11,80,000$$

Capital Structure Plans

Amount in ₹

Capital	Alternative 1	Alternative 2
Equity Share capital	21,50,000	27,20,000
Reserves and surplus	10,50,000	11,80,000
10% long term debt	15,00,000	15,00,000
14% Debentures	8,00,000	-
8% Irredeemable Debentures	-	1,00,000
Total Capital Employed	55,00,000	55,00,000

Computation of Present Earnings before interest and tax (EBIT)

EPS (₹)	21
No. of equity shares	20,000
Earnings for equity shareholders (I x II) (₹)	4,20,000
Profit Before Tax (III/75%) (₹)	5,60,000
Interest on long term loan (1500000 × 10%) (₹)	1,50,000
EBIT (IV + V) (₹)	7,10,000

EBIT after expansion = ₹7,10,000 +₹ 2,00,000 = ₹9,10,000

Evaluation of Financial Plans on the basis of EPS, MPS and Financial Leverage

Amount in ₹

Particulars	Alternative I	Alternate II
EBIT	9,10,000	9,10,000
Less: Interest: 10% on long term loan	(1,50,000)	(1,50,000)
14% on Debentures	(1,12,000)	Nil
8% on Irredeemable Debentures	Nil.	(8000)
PBT	6,48,000	7,52,000
Less: Tax @25%	(1,62,000)	(1,88,000)
PAT	4,86,000	5,64,000
No. of equity shares	21,500	27,200
EPS	**22.60**	**20.74**
Applicable P/E ratio (Working Note 1)	7	8.5

Particulars	Alternative I	Alternate II
MPS (EPS X P/E ratio)	**158.2**	**176.29**
Financial Leverage EBIT/PBT	**1.40**	**1.21**

Working Note 1

	Alternative I	Alternative II
Debt:		
₹15,00,000 +₹8,00,000	23,00,000	-
₹15,00,000 +₹1,00,000	-	16,00,000
Total capital Employed (₹)	55,00,000	55,00,000
Debt Ratio (Debt/Capital employed)	=0.4182	=0.2909
	=41.82%	=29.09%
Change in Equity: ₹21,50,000-₹20,00,000	1,50,000	
₹27,20,000-₹20,00,000		7,20,000
Percentage change in equity	7.5%	**36%**
Applicable P/E ratio	**7**	**8.5**

Calculation of Cost of equity and various type of debt

	Alternative I	Alternative II
A) Cost of equity		
EPS	22.60	20.74
DPS (EPS X 60%)	13.56	12.44
Growth (g)	10%	10%
Po (MPS)	158.2	176.29
Ke= Do (1 + g)/ Po	13.56 (1.1)/158.2 =9.43%	12.44 (1.1)/176.29 =7.76%
B) Cost of Debt:		
10% long term debt	10% + (1-0.25) = 7.5%	10% +(1-0.25) = 7.5%
14% redeemable debentures	$\frac{14(1-0.25)+(110-100/10)}{110+100/2}$ = 10.5 + 1 / 10.5 = 10.95%	nil

	Alternative I	Alternative II
8% irredeemable debenture	NA	8000(1-0.25)/1,00,00 = 6%

Calculation of Weighted Average cost of capital (WACC)

	Alternative 1			Alternative 2		
Capital	**Weights**	**Cost (%)**	**WACC**	**Weights**	**Cost (%)**	**WACC**
Equity Share Capital	0.3909	9.43	3.69%	0.4945	7.76	3.84%
Reserves and Surplus	0.1909	9.43	1.80%	0.2145	7.76	1.66%
10% Long term Debt	0.2727	7.50	2.05%	0.2727	7.50	2.05%
14% Debenture	0.1455	10.95	1.59%			
8% Irredeemable Debentures	-			0.0182	6	0.11%
			9.12%			**7.66%**

Calculation Marginal Cost of Capital (MACC)

	Alternative 1			Alternative 2		
Capital	**Amount(weight)**	**Cost (%)**	**MACC**	**Amount (weight)**	**Cost (%)**	**MACC**
Equity Share Capital	₹ 1,50,000(0.15)	9.43	1.41%	₹7,20,000(0.72)	7.76	5.59%
Reserves and Surplus	₹ 50,000(0.05)	9.43	0.47%	₹1,80,000(0.18)	7.76	1.40%
14% Debenture	₹ 8,00,000(0.80)	10.95	8.76%	-		0.00%
8% Irredeemable Debentures	-			₹1,00,000(0.10)	6	0.60%
Total Capital Employed	₹10,00,000		**10.65%**	₹10,00,000		**7.58%**

Summary of solution:

	Alternate I	Alternate II
Earning per share (EPS)	22.60	20.74
Market price per share (MPS)	158.20	176.29
Financial leverage	1.4043	1.2101
Weighted Average cost of capital (WACC)	9.12%	7.66%
Marginal cost of capital (MACC)	10.65%	7.58%

Alternative 1 of financing will be preferred under the criteria of EPS, whereas Alternative II of financing will be preferred under the criteria of MPS, Financial leverage, WACC and marginal cost of capital.

ICAI · Old P8 · MTP Nov 2022 · Series II · question p. 2-3 · answer p. 3-6

383 MTP **Old P8 May 2022 S-II** Q2(a) 7 marks old scheme

The Modern Chemicals Ltd. requires ₹ 25,00,000 for a new plant. This plant is expected to yield earnings before interest and taxes of ₹ 5,00,000. While deciding about the financial plan, the company considers the objective of maximising earnings per share. It has three alternatives to finance the project- by raising debt of ₹ 2,50,000 or ₹ 10,00,000 or ₹ 15,00,000 and the balance, in each case, by issuing equity shares. The company's share is currently selling at ₹ 150, but is expected to decline to ₹ 125 in case the funds are borrowed in excess of ₹ 10,00,000. The funds can be borrowed at the rate of 10% upto ₹ 2,50,000, at 15% over ₹ 2,50,000 and upto ₹ 10,00,000 and at 20% over ₹ 10,00,000. The tax rate applicable to the company is 50%. ANALYSE, which form of financing should the company choose?

ICAI'S ANSWER

Calculation of Earnings per share for three alternatives to finance the project

Particulars	Alternatives		
	I	**II**	**III**
	To raise debt of ₹2,50,000 and equity of ₹ 22,50,000	**To raise debt of ₹ 10,00,000 and equity of ₹ 15,00,000**	**To raise debt of ₹ 15,00,000 and equity of ₹ 10,00,000**
	(₹)	**(₹)**	**(₹)**
Earnings before interest and tax	5,00,000	5,00,000	5,00,000
Less: Interest on debt at the rate of	25,000 (10% on ₹ 2,50,000)	1,37,500 (10% on ₹ 2,50,000) (15% on ₹ 7,50,000)	2,37,500 (10% on ₹ 2,50,000) (15% on ₹ 7,50,000) (20% on ₹ 5,00,000)
Earnings before tax	4,75,000	3,62,500	2,62,500
Less: Tax (@ 50%)	2,37,500	1,81,250	1,31,250

Particulars	Alternatives		
Earnings after tax: (A)	2,37,500	1,81,250	1,31,250
Number of shares : (B) (Refer to working note)	15,000	10,000	8,000
Earnings per share: (A)/(B)	15.833	18.125	16.406

So, the earning per share (EPS) is higher in alternative II i.e. if the company finance the project by raising debt of ₹ 10,00,000 and issue equity shares of ₹ 15,00,000. Therefore, the company should choose this alternative to finance the project.

Working Note:

	Alternatives		
	I	II	III
Equity financing : (A)	₹ 22,50,000	₹ 15,00,000	₹ 10,00,000
Market price per share : (B)	₹ 150	₹ 150	₹ 125
Number of equity share: (A)/(B)	15,000	10,000	8,000

ICAI · Old P8 · MTP May 2022 · Series II · question p. 2 · answer p. 6-7

384 MTP **Old P8 Dec 2021 S-I** Q2 10 marks old scheme

Sophisticated Limited is considering three financing plans. The key information is as follows:

(a) Total investment amount to be raised ₹ 4,00,000

(b) Plans of Financing Proportion:

Plans	Equity	Debt	Preference Shares
A	100%	-	-
B	50%	50%	-
C	50%	-	50%

(c) Cost of debt 10%

Cost of preference shares 10%

(d) Tax rate 30%

(e) Equity shares of the face value of ₹ 10 each will be issued at a premium of ₹ 10 per share.

(f) Expected EBIT is ₹ 10,00,000.

You are required to DETERMINE for each plan: -

(i) Earnings per share (EPS)

(ii) The financial break-even point.

(iii) Indicate if any of the plans dominate and compute the EBIT range among the plans for indifference.

ICAI'S ANSWER

(i) Computation of Earnings per share (EPS)

Plans	A	B	C
Earnings before interest and tax (EBIT)	10,00,000	10,00,000	10,00,000
Less: Interest charges	---	(20,000) (10% × ₹2 lakh)	---
Earnings before tax (EBT)	10,00,000	9,80,000	10,00,000
Less: Tax (@ 30%)	(3,00,000)	(2,94,000)	(3,00,000)
Earnings after tax (EAT)	7,00,000	6,86,000	7,00,000
Less: Preference Dividend	---	---	(20,000) (10% × ₹2 lakh)
Earnings available for Equity shareholders (A)	7,00,000	6,86,000	6,80,000
No. of Equity shares (B)	20,000 (₹ 4 lakh ÷ ₹ 20)	10,000 (₹ 2 lakh ÷ ₹ 20)	10,000 (₹ 2 lakh ÷ ₹ 20)
EPS ₹ [(A) ÷ (B)]	35	68.6	68

(ii) Calculation of Financial Break-even point

Financial break-even point is the earnings which are equal to the fixed finance charges and preference dividend.

Plan A: Under this, plan there is no interest or preference dividend payment. Hence, the Financial Break-even point will be zero.

Plan B: Under this plan, there is an interest payment of ₹ 20,000 and no preference dividend. Hence, the Financial Break-even point will be ₹ 20,000 (Interest charges).

Plan C: Under this plan, there is no interest payment but an after tax preference dividend of ₹ 20,000 is paid. Hence, the Financial Break- even point will be before tax earnings of ₹ 28,571 (i.e. ₹ 20,000 ÷ 0.7)

(iii) Computation of indifference point between the plans.

The indifference between two alternative methods of financing is calculated by applying the following formula.

$$\frac{(EBIT - I_1)(1 - T)}{E_1} = \frac{(EBIT - I_2)(1 - T)}{E_2}$$

Where,

EBIT = Earnings before interest and tax.

I_1 = Fixed charges (interest or pref. dividend) under Alternative 1

l_2 = Fixed charges (interest or pref. dividend) under Alternative 2

T = Tax rate

E_1 = No. of equity shares in Alternative 1

E_2 = No. of equity shares in Alternative 2

Now, we can calculate indifference point between different plans of financing.

(a) Indifference point where EBIT of Plan A and Plan B is equal.

$$\frac{(EBIT-0)(1-0.3)}{20000}=\frac{(EBIT-20{,}000)(1-0.3)}{10{,}000}$$

0.7 EBIT (10,000) = (0.7 EBIT – 14,000) (20,000)

7,000 EBIT = 14,000 EBIT – 28 crores

EBIT = 40,000

(b) Indifference point where EBIT of Plan A and Plan C is equal

$$\frac{(EBIT-0)(1-0.3)}{20000}=\frac{(EBIT-0)(1-0.3)-20{,}000}{10{,}000}$$

0.7 EBIT (10,000) = (0.7 EBIT – 20,000) (20,000)

7000 EBIT = 14,000 EBIT – 40 crores

EBIT = 57,142.86

(c) Indifference point where EBIT of Plan B and Plan C are equal

$$\frac{(EBIT-20{,}000)(1-0.3)}{10000}=\frac{(EBIT-0)(1-0.3)-20{,}000}{10{,}000}$$

(0.7 EBIT – 14,000) (10,000) = (0.7 EBIT – 20,000) (10,000)

7,000 EBIT – 14 crore = 7,000 EBIT - 20 crore

There is no indifference point between the financial plans B and C.

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 2 · answer p. 4-5

385 MTP **Old P8 Dec 2021 S-II** Q1(b) 5 marks old scheme

ABC Limited is setting up a project with a capital outlay of ₹ 90,00,000. It has two alternatives in financing the project cost.

Alternative-I: 100% equity finance by issuing equity shares of ₹ 10 each

Alternative-II: Debt-equity ratio 2:1 (issuing equity shares of ₹ 10 each)

The rate of interest payable on the debts is 18% p.a. The corporate tax rate is 30%. CALCULATE the indifference point between the two alternative methods of financing.

ICAI'S ANSWER

Calculation of Indifference point between the two alternatives of financing.

Alternative-I By issue of 9,00,000 equity shares of ₹10 each amounting to ₹ 90 lakhs. No financial charges are involved.

Alternative-II By raising the funds in the following way:

Debt = ₹ 60 lakhs

Equity = ₹ 30 lakhs (3,00,000 equity shares of ₹ 10 each)

Interest payable on debt = 60,00,000× $\frac{18}{100}$= ₹ 10,80,000

The difference point between the two alternatives is calculated by:

$$\frac{(EBIT - I_1)(1-T)}{E_1} = \frac{(EBIT - I_2)(1-T)}{E_2}$$

$$\frac{(EBIT - 0)(1-0.30)}{9{,}00{,}000} = \frac{(EBIT - 10{,}80{,}000)(1-0.30)}{3{,}00{,}000}$$

$$\frac{(EBIT)(0.70)}{9{,}00{,}000} = \frac{(EBIT - 10{,}80{,}000)(0.70)}{3{,}00{,}000}$$

$$\frac{EBIT(0.70)}{3} = \frac{0.70(EBIT - 10{,}80{,}000)}{1}$$

EBIT = 3EBIT−32,40,000

−2 EBIT = −32,40,000

EBIT = $\frac{32{,}40{,}000}{2}$

EBIT = ₹ 16,20,000

Therefore, at EBIT of ₹ 16,20,000, earnings per share for the two alternatives is equal.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 1 · answer p. 1-2

386 MTP **Old P8 Jul 2021 S-I** Q1(d) 5 marks old scheme

HN Limited is considering total investment of Rs. 20 lakhs. You are required to CALCULATE the level of earnings before interest and tax (EBIT) at which the EPS indifference point between the following financing alternatives will occur:

(i) Equity share capital of Rs. 12,00,000 and 14% debentures of Rs. 8,00,000.

Or

(ii) Equity share capital of Rs. 8,00,000, 16% preference share capital of Rs. 4,00,000 and 14% debentures of Rs. 8,00,000.

Assume the corporate tax rate is 30% and par value of equity share is Rs.10 in each case.

ICAI'S ANSWER

Computation of level of earnings before interest and tax (EBIT)

In case alternative (i) is accepted, then the EPS of the firm would be:

$$\text{EPS}_{\text{Alternative (i)}} = \frac{(\text{EBIT} - \text{Interest})\ (1 - \text{tax rate})}{\text{No. of equity shares}} = \frac{(\text{EBIT} - 0.14 \times 8{,}00{,}000)\ (1 - 0.3)}{1{,}20{,}000 \text{ shares}}$$

In case the alternative (ii) is accepted, then the EPS of the firm would be

$$\text{EPS}_{\text{Alternative (ii)}} = \frac{(\text{EBIT} - \text{Interest})\ (1 - \text{tax rate}) - \text{PD}}{\text{No. of equity shares}}$$

$$= \frac{(\text{EBIT} - 0.14 \times 8{,}00{,}000)\ (1 - 0.3) - 0.16 \times 4{,}00{,}000}{80{,}000 \text{ shares}}$$

In order to determine the indifference level of EBIT, the EPS under the two alternative plans should be equated as follows:

$$\frac{(\text{EBIT} - 0.14 \times 8{,}00{,}000)\ (1 - 0.3)}{1{,}20{,}000 \text{ shares}} = \frac{(\text{EBIT} - 0.14 \times 8{,}00{,}000)\ (1 - 0.3) - 0.16 \times 4{,}00{,}000}{80{,}000 \text{ shares}}$$

Or, $\frac{0.7\text{EBIT} - 78{,}400}{1{,}20{,}000} = \frac{0.7\text{EBIT} - 1{,}42{,}400}{80{,}000}$

Or 1.40 EBIT – Rs. 1,56,800 = 2.10 EBIT – Rs. 4,27,200

Or 0.70 EBIT = Rs. 2,70,400

Or EBIT = $\frac{2{,}70{,}400}{0.7}$

Or EBIT = Rs. 3,86,285.71 (approx.)

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 2 · answer p. 3-4

387 MTP **Old P8 Nov 2020 S-I** Q2 10 marks old scheme

Sinha Steel Ltd. requires ₹ 30,00,000 for a new plant which expects to yield earnings before interest and taxes of ₹ 5,00,000. While deciding about the financial plan, the company considers the objective of maximizing earnings per share. It has three alternatives to finance the project as follows -

Alternative	Debt	Equity Shares
1	₹ 2,50,000	balance
2	₹ 10,00,000	balance
3	₹ 15,00,000	balance

The company's share is currently selling at ₹ 200, but is expected to decline to ₹ 160 in case the funds are borrowed in excess of ₹ 10,00,000.

Slab wise interest rate for fund borrowed are as follows -

Fund Limit	Applicable Interest rate
up-to ₹ 2,50,000	10%
over ₹ 2,50,000 and up-to ₹ 10,00,000	15%
over ₹ 10,00,000	20%

The tax rate applicable to the company is 50 percent.

ANALYSE which form of financing should the company choose?

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 3

388 MTP **Old P8 Nov 2019 S-I** Q1(b) 5 marks old scheme

RPS Company presently has Rs. 36,00,000 in debt outstanding bearing an interest rate of 10 per cent. It wishes to finance a Rs. 40,00,000 expansion programme and is considering three alternatives: additional debt at 12 per cent interest, preferred stock with an 11 per cent dividend, and the sale of common stock at Rs. 16 per share. The company presently has 8,00,000 shares of common stock outstanding and is in a 40 per cent tax bracket.

(i) If earnings before interest and taxes are presently Rs. 15,00,000, CALCULATE earnings per share for the three alternatives, assuming no immediate increase in profitability?

(ii) CALCULATE indifference point between debt and common stock.

ICAI'S ANSWER

(i)

(Rs. in thousands)

	Debt	*Preferred Stock*	*Common Stock*
	Rs.	*Rs.*	*Rs.*
EBIT	1,500	1,500	1,500
Interest on existing debt	360	360	360
Interest on new debt	480	–	–
Profit before taxes	660	1,140	1,140
Taxes	264	456	456
Profit after taxes	396	684	684
Preferred stock dividend	–	440	–
Earnings available to common shareholders	396	244	684
Number of shares	800	800	1,050
Earnings per share	.495	.305	.651

(ii) Mathematically, the indifference point between debt and common stock is (Rs in thousands):

$$\frac{\text{EBIT}^* - \text{Rs. } 840}{800} = \frac{\text{EBIT}^* - \text{Rs. } 360}{1{,}050}$$

EBIT* (1,050) – Rs. 840(1,050) = EBIT* (800) – Rs. 360 (800)

250EBIT* = Rs. 5,94,000

EBIT* = Rs. 2,376

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 1 · answer p. 1-2

5.3 Capital structure planning, determinants and over- and under-capitalisation

15 questions · asked in 11 ICAI documents · 4 exam papers

389 EXAM **May 2026** Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

XYZ Ltd. is engaged in manufacturing of an electronic component, to be used by industries producing electronic gadgets for domestic use. The company is a leveraged company, currently paying annual interest of ₹ 33 lakhs on its 11% debt. The company is financially prudent. It has a balance of ₹ 50 lakhs as retained earnings. The company maintains Long-term Debt to Equity ratio of 2: 1. The face value per share of the company is ₹ 10.

Due to operating in a competitive market, the marketing manager of the company has submitted a report exhibiting significant potential for sales growth. The production manager has endorsed the

idea and has suggested adopting latest technology at the workplace. In case the company implements new technology, the production efficiency will increase which will bring down the overall cost of production. To procure and install this new technology, the company requires an additional fund of ₹ 3 crores. This additional fund will be arranged in such a manner that it will not impact the present Long-term Debt to Equity ratio. The additional debt will carry interest rate of 12% p.a. As a result of implementing the new technology, it is anticipated that ROI will increase by 3% from present ROI of 15%. Applicable tax rate is 30%.

Based on the facts mentioned above pertaining to XYZ Ltd., you are required to answer the following questions 1 to 5:

MCQ 1 2 marks

What is the equity share capital outstanding before the additional fund is introduced?

A ₹ 1.00 crores

B ₹ 1.50 crores

C ₹ 0.50 crores

D ₹ 2.00 crores

MCQ 2 2 marks

What is the total capital employed before and after additional financing?

A ₹ 4.00 crores and ₹7.00 crores

B ₹ 5.00 crores and ₹8.00 crores

C ₹ 4.50 crores and ₹ 7.50 crores

D ₹ 5.50 crores and ₹ 8.50 crores

MCQ 3 2 marks

Compute the EPS after additional financing. What will be the percentage increase in EPS of the company, after introducing the additional fund?

A ₹ 2.73, 13.04%

B ₹ 2.18, 35.65%

C ₹ 2.73, 11.54%

D ₹ 2.18, 26.28%

MCQ 4 2 marks

What will be the expected MPS after the change in funds, if company expects a P/E ratio of 20?

A ₹ 43.68

B ₹ 54.60

C ₹ 70.00

D ₹ 47.60

MCQ 5 2 marks

What will be the post-tax weighted average cost of debt after additional financing?

A 7.70%

B 8.05%

C 9.00%

D 7.98%

ICAI'S ANSWERS

MCQ 1

Answer key: **(A)**

MCQ 2

Answer key: **(C)**

MCQ 3

Answer key: **(A)**

MCQ 4

Answer key: **(B)**

MCQ 5

Answer key: **(D)**

ICAI · P6 · May 2026 exam · question p. 1 · answer p. 8

390 EXAM **Sep 2024** Q4(c) 2 marks

What are the remedies for over-capitalisation?

ICAI'S ANSWER

Following steps may be adopted to avoid the negative consequences of over-capitalisation:

(i) Company should go for thorough reorganization.

(ii) Buyback of shares.

(iii) Reduction in claims of debenture-holders and creditors.

(iv) Value of shares may also be reduced. This will result in sufficient funds for the company to carry out replacement of assets.

ICAI · P6 · Sep 2024 exam · question p. 22 · answer p. 24

391 EXAM **Old P8 Nov 2023** Q6(c) 2 marks old scheme

Also asked in RTP Jan 2026 Q11(c)

What are the causes of over-capitalization?

ICAI'S ANSWER

Over-capitalisation arises due to following reasons:

(i) Raising more money through issue of shares or debentures than company can employ profitably.

(ii) Borrowing huge amount at higher rate than rate at which company can earn.

(iii) Excessive payment for the acquisition of fictitious assets such as goodwill etc.

(iv) Improper provision for depreciation, replacement of assets and distribution of dividends at a higher rate.

(v) Wrong estimation of earnings and capitalization

ICAI · Old P8 · Nov 2023 exam · question p. 17 · answer p. 18

392 EXAM **Old P8 Nov 2022** Q6(c) 2 marks old scheme

The firm has more capital than its requirements. What is this situation called? Give two consequences of it.

ICAI'S ANSWER

The situation is called as Over Capitalization.

Consequences of Over Capitalization:

- Considerable reduction in the rate of dividend and interest payments.
- Reduction in the market price of shares
- Resorting to "Window dressing"
- Some companies may opt for reorganization. However, sometimes the matter gets worse and the company may go into liquidation.

ICAI · Old P8 · Nov 2022 exam · question p. 15 · answer p. 16

393 EXAM **Old P8 Nov 2022** Q6(c) 2 marks old scheme

What are the important factors considered for deciding the source and quantum of capital?

ICAI'S ANSWER

The source and quantum of capital is decided keeping in mind the following factors:

(i) Control: Capital structure should be designed in such a manner that existing shareholders continue to hold majority stake

(ii) Risk: Capital structure should be designed in such a manner that financial risk of a company does not increase beyond tolerable limit.

(iii) Cost: Overall cost of capital remains minimum.

ICAI · Old P8 · Nov 2022 exam · question p. 15 · answer p. 16

394 MTP **Sep 2025 S-II** Q4(c) 2 marks

A company is currently debt-free and is considering raising ₹10 crore. As a finance manager, what practical factors would you evaluate before deciding whether to raise funds through debt, equity, or a mix of both? JUSTIFY your approach briefly.

ICAI'S ANSWER

Before deciding the capital structure, one should evaluate:

(i) **Cost of capital** – Debt is cheaper due to tax shield, but excessive debt increases financial risk.

(ii) **Control considerations** – Equity dilutes ownership; debt doesn't.

(iii) **Current interest rate and market conditions** – Favorable debt terms or bullish equity market can guide the choice.

(iv) **Cash flow stability** – If the company has stable cash flows, it can safely take on debt.

If the company has stable earnings and wants to retain control, a mix with a reasonable portion of debt to optimize cost is favourable.

ICAI · P6 · MTP Sep 2025 · Series II · question p. 10 · answer p. 14

395 MTP **May 2025 S-II** Q4(c) 2 marks

EXPLAIN in brief the concept of Over Capitalisation.

ICAI'S ANSWER

It is a situation where a firm has more capital than it needs or in other words assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest. This situation mainly arises when the existing capital is not effectively utilized on account of fall in earning capacity of the company while company has raised funds more than its requirements. The chief sign of over-capitalisation is the fall in payment of dividend and interest leading to fall in value of the shares of the company.

ICAI · P6 · MTP May 2025 · Series II · question p. 9 · answer p. 18

396 MTP **Jan 2025 S-I** Q4(b) 4 marks

You are a financial consultant for a company that has a very high capital base but low earnings per share (EPS). EXPLAIN over-capitalization. What are the causes and consequences of over-capitalization?"

ICAI'S ANSWER

Over-capitalization and its Causes and Consequences

It is a situation where a firm has more capital than it needs or in other words assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest.

Causes of Over Capitalization

Over-capitalisation arises due to following reasons:

(i) Raising more money through issue of shares or debentures than company can employ profitably.

(ii) Borrowing huge amount at higher rate than rate at which company can earn.

(iii) Excessive payment for the acquisition of fictitious assets such as goodwill etc.

(iv) Improper provision for depreciation, replacement of assets and distribution of dividends at a higher rate.

(v) Wrong estimation of earnings and capitalization.

Consequences of Over-Capitalisation

Over-capitalisation results in the following consequences:

(i) Considerable reduction in the rate of dividend and interest payments.

(ii) Reduction in the market price of shares.

(iii) Resorting to "window dressing".

(iv) Some companies may opt for reorganization. However, sometimes the matter gets worse and the company may go into liquidation.

ICAI · P6 · MTP Jan 2025 · Series I · question p. 7 · answer p. 11-12

397 MTP **May 2024 S-I** Q4(c) 2 marks

WHAT is 'Optimum Capital Structure'?

ICAI'S ANSWER

Optimum Capital Structure: The capital structure is said to be optimum when the firm has selected such a combination of equity and debt so that the wealth of firm is maximum. At this capital

structure, the cost of capital is minimum and the market price per share i.e. value of the firm is maximum.

ICAI · P6 · MTP May 2024 · Series I · question p. 5 · answer p. 7

398 MTP **Old P8 Nov 2022 S-I** Q5(b) 2 marks old scheme

BRIEF OUT the remedies for Over-Capitalisation.

ICAI'S ANSWER

Remedies for Over-Capitalisation: Following steps may be adopted to avoid the negative consequences of over-capitalisation-

(i) Company should go for thorough reorganization.

(ii) Buyback of shares.

(iii) Reduction in claims of debenture-holders and creditors.

(iv) Value of shares may also be reduced. This will result in sufficient funds for the company to carry out replacement of assets.

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 4 · answer p. 6

399 MTP **Old P8 May 2020 S-I** Q6(b) 4 marks old scheme

EXPLAIN Over-capitalisation. STATE its causes and consequences.

ICAI'S ANSWER

Over-capitalization and its Causes and Consequences

It is a situation where a firm has more capital than it needs or in other words assets are worth less than its issued share capital, and earnings are insufficient to pay dividend and interest.

Causes of Over Capitalization

Over-capitalisation arises due to following reasons:

(i) Raising more money through issue of shares or debentures than company can employ profitably.

(ii) Borrowing huge amount at higher rate than rate at which company can earn.

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Consequences of Over-Capitalisation

Over-capitalisation results in the following consequences:

(i) Considerable reduction in the rate of dividend and interest payments.

(ii) Reduction in the market price of shares.

(iii) Resorting to "window dressing".

(iv) Some companies may opt for reorganization. However, sometimes the matter gets worse and the company may go into liquidation.

ICAI · Old P8 · MTP May 2020 · Series I · question p. 4 · answer p. 11-12

CHAPTER 6

Financing Decisions: Leverages

6.1 Operating, financial and combined leverage

110 questions · asked in 64 ICAI documents · 17 exam papers

400 EXAM **Sep 2026** Q1(a) 5 marks

ABC Company Ltd. has witnessed substantial growth during recent past. On a year to year basis comparison, it has been observed that in the year 2025-26, Sales and Earnings before Interest and Taxes (EBIT) of the company would be increased by 10% and 20% respectively coupled with an increase in Earnings Per Share (EPS) from ₹ 15 to ₹ 18.

In the Year 2026-27, the management of the company is planning for an expansion that will require a capital investment of ₹ 50,00,000. The management has decided to raise the amount entirely by issue of 14% Debentures of ₹ 100 each at par. The expected EBIT of the company after expansion will be ₹ 35,00,000.

Required :

(i) Calculate operating and financial leverage of the company on year to year basis comparison.

(ii) Determine the amount of debt capital in the capital structure of the company in the year 2025-26.

(iii) Calculate financial leverage of the company after expansion.

(iv) When the cost of debt capital is more than the return on investment, will it be beneficial for the company to use debt capital in its capital structure ?

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 13-14

401 EXAM **May 2026** Case scenario · Q1(a)(i) to Q1(a)(ii) 5 marks

CASE SCENARIO

Finance manager of A Ltd. noticed that a 5% increase in sales leads to increase of ₹ 9,000 in operating profit. The company operates with a contribution margin of 40%, fixed annual operating cost of ₹1,35,000 and interest obligation of ₹ 30,000.

You are required to calculate:

Q1(a)(i) 2 marks

Current level of sales of the company.

Q1(a)(ii) 3 marks

Operating, financial and combined leverage.

ICAI'S ANSWERS

Q1(a)(i)

Current level of Sales:

Let the present sales be = ₹X

or 5% of Sales×40% = 9,000

or 0.05X x 0.40 = 9,000

or 0.02X = 9,000

Current Sales (X) = ₹ 4,50,000

P/V ratio = Change in profit/ Change in sales

0.40 = 9,000/X

X = 9,000/0.40

= Rs 22,500

Hence, total sales;

= Change in sales/ % change in sales

= ₹ 22,500/0.05

= ₹ 4,50,000

Income Statement

Particulars	(₹)
Sales	4,50,000
Contribution (40%)	1,80,000
Less: Fixed annual operating cost	(1,35,000)
EBIT (Earnings before interest and tax)	45,000
Less: Interest	(30,000)
EBT (Earnings before tax)	15,000

Q1(a)(ii)

Leverage

Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ 1,80,000}{₹\ 45,000} = 4$

Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 45,000}{₹\ 15,000} = 3$

Combined Leverage = Operating Leverage × Financial Leverage

= 4 × 3 = 12

Or,

Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 1,80,000}{₹\ 15,000} = 12$

ICAI · P6 · May 2026 exam · question p. 9 · answer p. 10-11, 11

402 EXAM Jan 2026 Q2(a) 5 marks

RK Limited provided following data for the year ended on 31-03-2025:

Sales	₹ 50,00,000
EPS	₹ 2.50
Tax Rate	₹ 40%
Fixed cost excluding interest	₹ 10,80,000

The Capital structure is as under:

Equity Share capital of ₹ 10/- each	₹ 18,00,000
10% Preference share of ₹ 100/- each	₹ 9,00,000
12% Debentures	₹ 75,00,000

You are required to:

(i) Calculate Earnings before Interest and Tax (EBIT).

(ii) Calculate Combined Leverage.

(iii) Advise, the level of sales where Earning before Tax (EBT) will be zero.

ICAI'S ANSWER

(i) Calculation of Earnings before Interest and Tax (EBIT)

Particulars	Amount (₹)
Sales Revenue	50,00,000
Less: Variable Cost (Balancing figure)	21,20,000
Contribution	**28,80,000**
Less: Fixed Cost (excluding interest)	10,80,000
Earnings Before Interest & Tax (EBIT)	**18,00,000**
Less: Interest on Debentures (12% of ₹ 75,00,000)	9,00,000
Earnings Before Tax (EBT)	**9,00,000**
Less: Tax @ 40%	3,60,000
Profit After Tax (PAT)	**5,40,000**
Less: Preference Dividend (10% of ₹ 9,00,000)	90,000
Profit available to Equity Shareholders (1,80,000 × 2.50)	**4,50,000**

(ii) Combined Leverage = $\frac{\text{Contribution}}{\text{EBT - Pref. Dividend (Pre - tax)}}$

$$= \frac{₹\ 28,80,000}{₹\ 9,00,000 - 1,50,000} = 3.84$$

(iii) EBT zero means 100% reduction in EBT. Since combined leverage is 3.84, sales have to be dropped by 100/3.84 = 26.042%. Hence new sales will be ₹ 50,00,000 × (100 – 26.042) = ₹ 36,97,900.

Therefore, at ₹ 36,97,900. level of sales, the Earnings before Tax of the company will be equal to zero.

Combined Leverage and Sales can also be calculated without deducting preference dividend in following way:

(ii) Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 28,80,000}{₹\ 9,00,000} = 3.2$

(iii) EBT zero means 100% reduction in EBT. Since combined leverage is 3.2, sales have to be dropped by 100/3.2 = 31.25%. Hence new sales will be ₹ 50,00,000 × (100 – 31.25) = ₹ 34,37,500.

Therefore, at ₹ 34,37,500. level of sales, the Earnings before Tax of the company will be equal to zero.

ICAI · P6 · Jan 2026 exam · question p. 15 · answer p. 16-17

403 EXAM **Jan 2026** Q3(b) 2 marks

Explain in what ways financial leverage operates as a "double edged sword".

ICAI'S ANSWER

When the cost of 'fixed cost fund' is less than the return on investment, financial leverage will help to increase return on equity and EPS. The firm will also benefit from the saving of tax on interest on debts etc. However, when cost of debt will be more than the return it will affect return of equity and EPS unfavourably and as a result firm can be under financial distress. Therefore, financial leverage is also known as "double edged sword".

ICAI · P6 · Jan 2026 exam · question p. 20 · answer p. 22

404 EXAM **Sep 2025** Case scenario · Q2(a)(i) to Q2(a)(v) 7 marks

CASE SCENARIO

M/s KRY Limited is a mid-sized company engaged in manufacturing and sales of Industrial equipment. The capital structure of the company is as under:

Equity Share Capital (12,500 Shares of ₹100 each)	₹12,50,000
6% Debentures	₹ 50,00,000
8% Bank of Loan	XXXXX
Other information are as under:	
Total Sales	₹ 75,00,000
P/V Ratio	40%
Operating Leverage	2.4
Corporate Tax Rate	3.84
Corporate Tax Rate	30%

P/E Ratio	8

You are required to calculate following:

Q2(a)(i) 1 mark

Earnings Before Interest and Tax

Q2(a)(ii) 1 mark

Fixed Cost excluding interest

Q2(a)(iii) 2 marks

Amount of Bank Loan and Bank Interest

Q2(a)(iv) 2 marks

Earning Per Share

Q2(a)(v) 1 mark

Market Price Per Share

ICAI'S ANSWERS

Q2(a)(i)

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}}$$

$$2.4 = \frac{75{,}00{,}000 \text{ x } 0.40}{\text{EBIT}}$$

EBIT = 12,50,000

Q2(a)(ii)

Fixed Cost excluding interest= Contribution – EBIT

= 30,00,000 – 12,50,000 = 17,50,000

Q2(a)(iii)

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}}$$

$$3.84 = \frac{30{,}00{,}000}{\text{EBT}}$$

EBT = 7,81,250

Total Interest = EBIT – EBT = 4,68,750

Interest on bank loan = Total Interest – Interest on debentures

= 4,68,750 – 3,00,000 = 1,68,750

Amount of bank loan = 1,68,750/8%

= 21,09,375

Q2(a)(iv)

Particulars	(₹)
EBT (Earnings before tax)	7,81,250
Less: Tax @ 30%	2,34,375
PAT (Profit after tax)	5,46,875
No. of shares	12,500
Earnings per share (EPS)	43.75

Q2(a)(v)

Market price per share = EPS x PE Ratio

= 43.75 x 8 = 350

ICAI · P6 · Sep 2025 exam · question p. 13 · answer p. 14, 15

405 EXAM **May 2025** MCQ 6 2 marks MCQ

A company has net worth of ₹ 5,00,000. Its debt to equity ratio is 2. Interest on debt is 10%. The company earns an operating profit of ₹ 4,00,000. Tax rate is 30%.

What will be the financial leverage of the company?

A 1.11

B 1.33

C 1.43

D 1.90

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P6 · May 2025 exam · question p. 3 · answer p. 7

406 EXAM **May 2025** Case scenario · Q1(c)(i) to Q1(c)(iii) 5 marks

CASE SCENARIO

Following information relates to A Ltd. for the year ended 31st March, 2025:

Profit volume ratio	24%
Operating leverage	2.00
Financial leverage	1.50
Interest Expenses	₹ 12,000
Tax rate	30%
Number of Equity Shares	1,000

You are required to:

Q1(c)(i) 2 marks

Prepare Income Statement for the year ended 31st March, 2025.

Q1(c)(ii) 1 mark

Calculate EPS.

Q1(c)(iii) 2 marks

Calculate percentage change in earnings per share, if sales increase by 5%.

ICAI'S ANSWERS

ICAI's working for this case

Workings:

(a) Financial Leverage = $\frac{\text{EBIT}}{\text{EBT i.e EBIT} - \text{Interest}}$

So, $1.5 = \frac{\text{EBIT}}{\text{EBIT} - 12{,}000}$

Or, 1.5 (EBIT – 12,000) = EBIT

Or, 0.5 EBIT = 18,000

Or, EBIT = ₹ 36,000

(b) Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}}$ Or, $2 = \frac{\text{Contribution}}{₹\ 36{,}000}$

Or, Contribution = ₹ 72,000

Sales = $\frac{\text{Contribution}}{\text{P/V Ratio}}$

= $\frac{₹\ 72{,}000}{0.24}$ = ₹ 3,00,000

(c) Fixed Cost = Contribution – EBIT

= 72,000 – 36,000

Or, Fixed cost = ₹ 36,000

Q1(c)(i)

Income Statements for the year ended 31st March 2025

Particulars	A Ltd. (₹)
Sales	3,00,000
Less: Variable cost (76%)	2,28,000
Contribution	72,000
Less: Fixed Cost	36,000
Earnings before interest and tax (EBIT)	36,000
Less: Interest	12,000
Earnings before tax (EBT)	24,000
Less: Tax @ 30%	7,200
Earnings after tax (EAT)	16,800

Q1(c)(ii)

Calculation of EPS

$$\text{EPS} = \frac{\text{Profit after tax}}{\text{No. of Equity Shares}} = \frac{16{,}800}{1{,}000} = ₹16.8$$

Q1(c)(iii)

Calculation of % change in EPS if Sales increase by 5%

Degree of Combined Leverage (DCL) = Operating Leverage x Financial Leverage

= 2 x 1.5 = 3

Alternatively, solution can be presented in following way:

$$\text{DCL} = \frac{\text{Contribution}}{\text{EBT}} = \frac{72{,}000}{24{,}000} = 3$$

$$\text{So, } \frac{\%\text{ change in EPS}}{\%\text{ change in Sales}} = 3$$

Or % change in EPS = DCL x % change in Sales

= 3 x 0.05 = 0.15 or 15%

ICAI · P6 · May 2025 exam · question p. 9 · answer p. 13, 14

407 EXAM **May 2025** Q4(c) 2 marks

"The total risk of any business is the combination of degree of operating and financial risk". In the light of the above statement, you are required to consider the first two columns of the given table and give your comments in the 3rd column.

Your comments should depict the total risk profile by using the most appropriate amongst the following three words only:

Lower, Higher and Moderate

Also select the best combination of DOL and DFL from the given table.

DOL	DFL	Comments
Low	High	
High	Low	
High	High	

ICAI'S ANSWER

Combination of DOL and DFL

DOL	DFL	Comments
Low	High	Moderate total risk. Best combination. Higher financial risk is balanced by lower total business risk.
High	Low	Moderate total risk. Not a good combination. Lower EBIT due to higher DOL and lower advantage of trading on equity due to low DFL.
High	High	Higher total risk. Very risky combination.

408 EXAM **Jan 2025** Case scenario · Q1(b)(i) to Q1(b)(iv) 5 marks

CASE SCENARIO

Following is the Balance Sheet of EXIM Ltd. as on 31st March, 2024:

Liabilities	₹	Assets	₹
Equity Share Capital of ₹ 100 each	20,00,000	Fixed Assets	50,00,000
Retained Earnings	4,00,000	Current Assets	30,00,000
12.5% Debenture	40,00,000		
Current Liabilities	16,00,000		
	80,00,000		80,00,000

The additional information is given as under:

Fixed costs per annum (exclusive interest)	: ₹ 16,00,000
Variable operating cost ratio	: 70%
Total Assets turnover ratio	: 2.5
Income tax rate	: 30%

You are required to calculate:

Q1(b)(i) 2 marks

Earnings Per Share

Q1(b)(ii) 1 mark

Operating Leverage

Q1(b)(iii) 1 mark

Financial Leverage

Q1(b)(iv) 1 mark

Combined Leverage

ICAI'S ANSWERS

ICAI's working for this case

Workings:

Total Assets = ₹ 80 lakhs

Total Asset Turnover Ratio = 2.5

Hence, Total Sales = 80 × 2.5 = ₹ 200 lakhs

Computation of Profit after Tax (PAT)

	(₹) in lakhs
Sales	200.00

	(₹) in lakhs
Less: Variable Operating Cost @ 70%	140.00
Contribution	60.00
Less: Fixed Cost (other than Interest)	16.00
EBIT	44.00
Less: Interest on Debentures (12.5% × ₹ 40 lakhs)	5.00
PBT	39.00
Less: Tax @ 30%	11.70
PAT	27.30

Q1(b)(i)

Earnings per Share

$$\text{EPS} = \frac{\text{PAT}}{\text{Number of Equity Shares}} = \frac{₹27.30 \text{ lakhs}}{20{,}000} = ₹\ 136.5$$

Q1(b)(ii)

Operating Leverage

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹60 \text{ lakhs}}{₹44 \text{ lakhs}} = 1.363$$

Q1(b)(iii)

Financial Leverage

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{PBT}} = \frac{₹44 \text{ lakhs}}{₹39 \text{ lakhs}} = 1.128$$

Q1(b)(iv)

Combined Leverage

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{PBT}}$$

Or,

= Operating Leverage × Financial Leverage

= 1.363 x 1.128 = 1.538

ICAI · P6 · Jan 2025 exam · question p. 9 · answer p. 11, 12

409 EXAM **Sep 2024** MCQ 6 MCQ

A company has sales of ₹ 6,00,000, variable cost of ₹ 2,40,000, fixed operating cost of ₹ 2,70,000. The financial leverage is 2.5. The company wants to double its EBIT. The percentage change in ales required in order to double its EBIT will be :

A 50%

B 25%

C 40%

D 80%

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P6 · Sep 2024 exam · question p. 3 · answer p. 4

410 **EXAM** **Sep 2024** Q1(a) **5 marks**

Financial information for the year 2023-24 of two companies, N Limited and C Limited are as under:

Details	N Limited	C Limited
Equity share capital (₹ 100 each)	₹ 10,00,000	₹ 8,00,000
Debt	₹ 5,00,000@10%	₹ 7,00,000@8%
Fixed Cost	3,00,000	3,36,000
Combined Leverage	8	4.5
Financial Leverage	2	1.5

You are required to calculate:

(i) Contribution for N Ltd. and C Ltd.

(ii) Margin of safety in % for N Ltd. and C. Ltd.

(iii) Sales of C Ltd.

ICAI'S ANSWER

(i) Calculation of Contribution

N Limited	C Limited
Financial Leverage (FL) = $\frac{EBIT}{EBT}$ or $\frac{EBIT}{EBIT - Interest}$	
2 = $\frac{EBIT}{EBIT - 50,000}$	1.5 = $\frac{EBIT}{EBIT - 56,000}$
2 EBIT – 1,00,000 = EBIT	1.5 EBIT – 84,000 = EBIT
EBIT = ₹ 1,00,000	**EBIT = ₹ 1,68,000**
EBT = ₹ 50,000	EBT = ₹ 1,12,000
Combined Leverage (CL) = $\frac{Contribution}{EBT}$	
8 = Contribution / 50,000	4.5 = Contribution / 1,12,000
Contribution= ₹ 4,00,000	**Contribution= ₹ 5,04,000**

(ii) Calculation of Margin of safety (MOS) in %

$$MOS = \frac{Contribution - Fixed\ Cost}{Contribution} = \frac{EBIT}{Contribution}$$

N Limited	C Limited
MOS = 1,00,000/4,00,000 **= 25%**	MOS = 1,68,000/5,04,000 **= 33.33%**

Part (ii) can also be presented in following way:

Calculation of Margin of safety (MOS) in %

MOS = 1/operating leverage (OL)

OL = CL/FL

N Limited	C Limited
OL = 8/2 = 4	OL = 4.5/1.5 = 3
MOS = 1/4 **= 25%**	MOS = 1/3 **= 33.33%**

(iii) Sales of C Limited

Let assume that PV ratio is 40%

PV Ratio = $\frac{\text{Contribution}}{\text{Sales OR Sales}} = \frac{\text{Contribution}}{\text{PV Ratio}}$

Sales = $\frac{5,04,000}{0.40}$ **= ₹ 12,60,000**

Part (iii) of the solution can be solved by any alternative assumption.

ICAI · P6 · Sep 2024 exam · question p. 9 · answer p. 11-12

411 **EXAM** **May 2024** Q1(b) **5 marks**

Alpha Limited has provided following information:

Equity Share Capital	25,000 Shares @ ₹ 100 per Share
15% Debentures	10,000 Debentures@ ₹ 750/- per Debenture
Sales	50 Lakhs units@ ₹ 20 per unit
Variable Cost	₹ 12.50 per unit
Fixed Costs	₹ 175.00 Lakhs

Due to recent policy changes and entry of foreign competitors in the sector, Alpha Limited expects the sales may decline by 15-20%, However, selling price and other costs will remain the same. Corporate Taxes will continue@20%.

You are required to calculate the decrease in Earnings per share, Degree of Operating Leverage and Financial Leverage separately if sales are declined by (i) 15%; and (ii) 20%;

ICAI'S ANSWER

Income Statement with required calculations

Particulars	(₹)	(₹)	(₹)
	Existing	**Sales declined by 15%**	**Sales declined by 20%**

Particulars	(₹)	(₹)	(₹)
Sales in units	50,00,000	42,50,000	40,00,000
Sales price per unit	20	20	20
Variable Cost per unit	(12.50)	(12.50)	(12.50)
Contribution per unit	7.5	7.5	7.5
Contribution	3,75,00,000	3,18,75,000	3,00,00,000
Fixed expenses	(1,75,00,000)	(1,75,00,000)	(1,75,00,000)
EBIT	2,00,00,000	1,43,75,000	1,25,00,000
Debenture Interest	(11,25,000)	(11,25,000)	(11,25,000)
EBT	1,88,75,000	1,32,50,000	1,13,75,000
Tax @ 20%	(37,75,000)	(26,50,000)	(22,75,000)
Profit after tax (PAT)	1,51,00,000	1,06,00,000	91,00,000
No. of shares	25,000	25,000	25,000
Earnings per share (EPS) = PAT/No. of shares	₹ 1,51,00,000/25,000 = ₹ 604	₹ 1,06,00,000/25,000 = ₹424	₹ 91,00,000/25,000 = ₹364
(i) Decrease in EPS		= ₹180 **Or** % Decrease in EPS = 180/604 × 100 = **29.80%**	= ₹240 **Or** % Decrease in EPS = 240/604 × 100 = **39.73%**
(ii) Operating leverage = Contribution/EBIT Or Degree of Operating leverage = Percentage change in EBIT/Percentage change in sales		= ₹ 3,18,75,000/₹ 1,43,75,000 = **2.22** **Or** 28.125/15 = **1.875**	= ₹ 3,00,00,000/₹ 1,25,00,000 = **2.40** **Or** 37.50/20 **1.875**
(iii) Financial Leverage = EBIT/EBT Or Degree of Financial Leverage = Percentage change in EPS/Percentage change in EBIT		= ₹ 1,43,75,000/₹ 1,32,50,000 = **1.08** **Or** 29.80/28.125 = **1.06**	= ₹ 1,25,00,000/₹ 1,13,75,000 = **1.10** **Or** 39.735/37.50 = **1.06**

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412 EXAM Old P8 Nov 2023 Q1(b) 5 marks old scheme

The following details of Shiva Ltd. for the year ended 31st March,2023 are given below:

Operating Leverage	1.4
Combined Leverage	2.8
Fixed Cost (Excluding Interest)	₹ 2.04 lakhs
Sales	₹ 30 lakhs
12% Debentures of ₹ 10 each	₹ 21.25 lakhs
Equity Share Capital of ₹ 10 each	₹ 17.00 lakhs
Income Tax Rate	30%

Required:

(i) Calculate P/V ratio and Earning Per Share (EPS)

(ii) If the company belongs to an industry, whose assets turnover is 1.5, does it have a high or low assets turnover?

(iii) Financial Leverage

ICAI'S ANSWER

(i) P/V Ratio and Earning per share (EPS)

$$\text{Operating leverage} = \frac{\text{Contribution(C)}}{\text{Contribution - Fixed Cost (FC)}}$$

$$1.4 = \frac{C}{C - 2,04,000}$$

Or, C = 1.4 (C – 2,04,000)

Or, C = 1.4 C – 2,85,600

Or, Contribution = ₹ 7,14,000

$$\text{Now, P/V ratio} = \frac{\text{Contribution (C)}}{\text{Sales (S)}} \times 100 = \frac{7,14,000}{30,00,000} \times 100 = 23.8\%$$

Therefore, P/V Ratio = 23.80%

EBT = Contribution – Fixed Cost – Interest

= ₹ 7,14,000 – ₹ 2,04,000 – (12% × ₹ 21,25,000)

= ₹ 5,10,000 – ₹ 2,55,000

= ₹ 2,55,000

PAT = EBT(1-T) = ₹ 2,55,000(1-0.3) = ₹ 1,78,500

$$\text{EPS} = \frac{\text{Profit after tax}}{\text{No. of equity shares}}$$

$$\text{EPS} = \frac{₹\ 1,78,500}{1,70,000 \text{ shares}} = ₹\ 1.05$$

(ii) Assets turnover

$$\text{Assets turnover} = \frac{\text{Sales}}{\text{Total Assets*}} = \frac{₹\ 30,00,000}{₹\ 17,00,000 + ₹\ 21,25,000} = 0.7843$$

0.7843 < 1.5 means lower than industry turnover.

*Total Asset = Equity share capital + 12% Debentures

(iii) Financial leverage

Combined Leverage = Operating Leverage (OL) × Financial Leverage (FL)

2.8 = 1.4 × FL

Or, FL = 2

Financial Leverage = 2

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413 EXAM **Old P8 May 2023** Q1(d) 5 marks old scheme

Following information is given for X Ltd.:

Total contribution (₹)	4,25,000
Operating leverage	3.125
15% Preference shares (₹ 100 each)	1,000
Number of equity shares	2,500
Tax rate	50%

Calculate EPS of X Ltd., if 40% decrease in sales will result EPS to zero.

ICAI'S ANSWER

(i) Operating Leverage (OL) = $\frac{\text{Contribution}}{\text{EBIT}}$ Or, 3.125 = $\frac{₹\ 4,25,000}{\text{EBIT}}$ Or EBIT

= ₹ 1,36,000

(ii) Degree of Combined Leverage (CL) = $\frac{\%\ \text{Change in EPS}}{\%\ \text{Change in Sales}} = \frac{100}{40} = 2.5$

(iii) Combined Leverage = OL × FL = 3.125 × FL

So, Financial Leverage = 2.5 /3.125 = 0.8

(iv) Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{1,36,000}{\text{EBT}} = 0.8$

So, EBT = $\frac{1,36,000}{0.80}$ = ₹ 1,70,000

Calculation of EPS of X Ltd

Particulars	(₹)
EBT	1,70,000
Less: Tax (50%)	85,000
EAT	85,000
Preference Dividend	15,000
Net Earnings for Equity Shareholders	70,000
Number of equity shares	2,500
EPS	**28**

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414 EXAM **Old P8 May 2023** Q6(c) 2 marks old scheme

Briefly explain concept of "Trading on Equity" in financial leverage analysis.

ICAI'S ANSWER

Financial Leverage as 'Trading on Equity':

Financial leverage indicates the use of funds with fixed cost like long term debts and preference share capital along with equity share capital which is known as trading on equity. The basic aim of financial leverage is to increase the earnings available to equity shareholders using fixed cost fund. A firm is known to have a positive/favourable leverage when its earnings are more than the cost of debt. If earnings are equal to or less than cost of debt, it will be a negative/unfavourable leverage. When the quantity of fixed cost fund is relatively high in comparison to equity capital it is said that the firm is 'trading on equity".

ICAI · Old P8 · May 2023 exam · question p. 20 · answer p. 21

415 EXAM **Old P8 Nov 2022** Q2 10 marks old scheme

The following information is available for SS Ltd.

Profit volume (PV) ratio	30%
Operating leverage	2.00
Financial leverage	1.50
Loan	₹ 1,25,000
Post-tax interest rate	5.6%
Tax rate	30%
Market Price per share (MPS)	₹ 140
Price Earnings Ratio (PER)	10

You are required to:

(1) Prepare the Profit-Loss statement of SS Ltd. and

(2) Find out the number of equity shares.

ICAI'S ANSWER

(1) Preparation of Profit – Loss Statement

Working Notes:

1. Post tax interest	5.60%
Tax rate	30%
Pre tax interest rate = (5.6/70) x 100	8%
Loan amount	₹ 1,25,000
Interest amount = 1,25,000 × 8%	₹ 10,000

Financial Leverage (FL) = $\left(\frac{\text{EBIT}}{\text{EBT}}\right) = \left[\frac{\text{EBIT}}{(\text{EBIT} - \text{Interest})}\right] = \left[\frac{\text{EBIT}}{(\text{EBIT} - 10{,}000)}\right]$

$1.5 = \left[\frac{\text{EBIT}}{(\text{EBIT} - 10{,}000)}\right]$

1.5 EBIT -15000 = EBIT

1.5 EBIT – EBIT = 15,000

0.5 EBIT = 15,000

∴ **EBIT = ₹ 30,000**

EBT = EBIT – Interest = 30,000 – 10,000 = ₹ 20,000

2. Operating Leverage (OL) = $\frac{\text{Contribution}}{\text{EBIT}}$

$2 = \frac{\text{Contribution}}{30{,}000}$

Contribution = ₹ 60,000

3., **Fixed cost** = Contribution – Profit

= 60,000 – 30,000 = **₹ 30,000**

4., Sales = $\frac{\text{Contribution}}{\text{PV Ratio}}$

$= \frac{60{,}000}{30\%}$ = **₹ 2,00,000**

5. If PV ratio is 30%, then the variable cost is 70% on sales.

∴ **Variable cost** = 2,00,000 x 70% = **₹ 1,40,000**

Profit – Loss Statement

Particulars	₹
Sales	2,00,000
Less: Variable cost	1,40,000

Particulars	₹
Contribution	60000
Less: Fixed cost	30,000
EBIT	30,000
Less: Interest	10,000
EBT	20,000
Less: Tax @ 30%	6,000
EAT	**14,000**

(2) Calculation of no. of Equity shares

Market Price per Share (MPS) = ₹140

Price Earnings Ratio (PER) = 10

WKT,

$$\textbf{EPS} = \frac{\text{MPS}}{\text{PER}} = \frac{140}{10} = \textbf{₹ 14}$$

Total earnings (EAT) = ₹ 14,000

∴ No. of Equity Shares = 14,000 / 14 = **1000**

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416 EXAM **Old P8 May 2022** Q2 10 marks old scheme

Details of a company for the year ended 31st March, 2022 are given below:

Sales	₹ 86 lakhs
Profit Volume (P/V) Ratio	35%
Fixed Cost excluding interest expenses	₹ 10 lakhs
10% Debt	₹ 55 lakhs
Equity Share Capital of ₹ 10 each	₹ 75 lakhs
Income Tax Rate	40%

Required:

(i) Determine company's Return on Capital Employed (Pre-tax) and EPS.

(ii) Does the company have a favourable financial leverage?

(iii) Calculate operating and combined leverages of the company.

(iv) Calculate percentage change in EBIT, if sales increases by 10%.

(v) At what level of sales, the Earning before Tax (EBT) of the company will be equal to zero?

ICAI'S ANSWER

Income Statement

Particulars	Amount (₹)
Sales	86,00,000
Less: Variable cost (65% of 86,00,000)	55,90,000
Contribution (35% of 86,00,000)	30,10,000
Less: Fixed costs	10,00,000
Earnings before interest and tax (EBIT)	20,10,000
Less: Interest on debt (@ 10% on ₹ 55 lakhs)	5,50,000
Earnings before tax (EBT)	14,60,000
Tax (40%)	5,84,000
PAT	8,76,000

(i) ROCE (Pre-tax) = $\frac{\text{EBIT}}{\text{Capital employed}} \times 100 = \frac{\text{EBIT}}{\text{Equity} + \text{Debt}} \times 100$

$\frac{₹\ 20,10,000}{₹\ (75,00,000+55,00,000)} \times 100$ = **15.46%**

EPS (PAT/No. of equity shares) **1.168 or ₹ 1.17**

(ii) ROCE is 15.46% and Interest on debt is 10%. Hence, it has a **favourable financial leverage.**

(iii) Calculation of Operating, Financial and Combined leverages:

Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ 30,10,000}{₹\ 20,10,000}$ = **1.497 (approx.)**

Financial Leverage = $\frac{\text{EBIT}}{\text{EBIT}} = \frac{₹\ 20,10,000}{₹\ 14,60,000}$ = 1.377 (approx.)

Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 30,10,000}{₹\ 14,60,000}$ = 2.062 (approx.)

Or, = Operating Leverage × Financial Leverage = 1.497 × 1.377 **= 2.06 (approx.)**

(iv) Operating leverage is 1.497. So, if sales are increased by 10%.

EBIT will be increased by 1.497 × 10% i.e. 14.97% (approx.)

(v) Since the combined Leverage is 2.062, sales have to drop by 100/2.062 i.e. 48.50% to bring EBT to Zero.

Accordingly, New Sales = ₹ 86,00,000 × (1 - 0.4850)

= ₹ 86,00,000 × 0.515

= ₹ 44,29,000 (approx.)

Hence, at ₹ 44,29,000 sales level, EBT of the firm will be equal to Zero.

ICAI · Old P8 · May 2022 exam · question p. 7 · answer p. 7-8

417 EXAM Old P8 Dec 2021 Q5 10 marks old scheme

Information of A Ltd. is given below:

- Earnings after tax: 5% on sales
- Income tax rate: 50%
- Degree of Operating Leverage: 4 times
- 10% Debenture in capital structure: ₹ 3 lakhs
- Variable costs: ₹ 6 lakhs

Required:

(i) From the given data complete following statement:

Sales	XXXX
Less: Variable costs	₹ 6,00,000
Contribution	XXXX
Less: Fixed costs	XXXX
EBIT	XXXX
Less: Interest expenses	XXXX
EBT	XXXX
Less: Income tax	XXXX
EAT	XXXX

(ii) Calculate Financial Leverage and Combined Leverage.

(iii) Calculate the percentage change in earning per share, if sales increased by 5%.

ICAI'S ANSWER

(i) Working Notes

Earning after tax (EAT)is 5% of sales

Income tax is 50%

So, EBT is 10% of Sales

Since Interest Expenses is ₹ 30,000

EBIT = 10% of Sales + ₹30,000 ..... (Equation i)

Now Degree of operating leverage = 4

So, $\frac{\text{Contribution}}{\text{EBIT}} = 4$

Or, Contribution = 4 EBIT

Or, Sales – Variable Cost = 4 EBIT

Or, Sales – ₹ 6,00,000 = 4 EBIT ..... (Equation ii)

Replacing the value of EBIT of equation (i) in Equation (ii)

We get, Sales – ₹ 6,00,000 = 4 (10% of Sales + ₹ 30,000)

Or, Sales – ₹ 6,00,000 = 40% of Sales + ₹ 1,20,000

Or, 60% of Sales = ₹ 7,20,000

So, Sales = $\frac{₹\ 7{,}20{,}000}{60\%}$ =**₹ 12,00,000**

Contribution = Sales – Variable Cost = ₹ 12,00,000 – ₹ 6,00,000 =₹ 6,00,000

EBIT = $\frac{₹\ 6{,}00{,}000}{4}$ = **₹ 1,50,000**

Fixed Cost = Contribution – EBIT = ₹ 6,00,000 – ₹ 1,50,000 = ₹ 4,50,000

EBT = EBIT – Interest = ₹ 1,50,000 – ₹ 30,000 = **₹ 1,20,000**

EAT = 50% of ₹ 1,20,000 = ₹ 60,000

Income Statement

Particulars	(₹)
Sales	**12,00,000**
Less: Variable cost	6,00,000
Contribution	6,00,000
Less: Fixed cost	4,50,000
EBIT	**1,50,000**
Less: Interest	30,000
EBT	**1,20,000**
Less: Tax (50%)	60,000
EAT	60,000

(ii) Financial Leverage= $\frac{\text{EBIT}}{\text{EBT}} = \frac{1{,}50{,}000}{1{,}20{,}000}$ **= 1.25 times**

Combined Leverage = Operating Leverage × Financial Leverage

= 4×1.25 = **5 times**

Or,

Combined Leverage = $\frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{EBT}}$

Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 6{,}00{,}000}{₹\ 1{,}20{,}000}$ = **5 times**

(iii) Percentage Change in Earnings per share

Combined Leverage = $\frac{\%\ \text{Change in EPS}}{\%\ \text{change in Sales}} = 5 = \frac{\%\ \text{change in EPS}}{5\%}$

∴ % Change in EPS = **25%**

Hence, if sales increased by 5 %, EPS will be increased by 25 %.

ICAI · Old P8 · Dec 2021 exam · question p. 12 · answer p. 12-14

418 EXAM Old P8 Jul 2021 Q5 10 marks old scheme

A company had the following balance sheet as on 31st March, 2021:

Liabilities	₹ in Crores	Assets	₹ in Crores
Equity Share Capital (75 lakhs Shares of ₹ 10 each)	7.50	Building	12.50
Reserves and Surplus	1.50	Machinery	6.25
15% Debentures	15.00	Current Assets	
Current Liabilities	6.00	Stock	3.00
		Debtors	3.25
		Bank Balance	5.00
	30.00		30.00

The additional information given is as under:

Fixed cost per annum (excluding interest)	₹ 6 crores
Variable operating cost ratio	60%
Total assets turnover ratio	2.5
Income-tax rate	40%

Calculate the following and comment:

(i) Earnings per share

(ii) Operating Leverage

(iii) Financial Leverage

(iv) Combined Leverage

ICAI'S ANSWER

Total Assets	= ₹ 30 crores
Total Asset Turnover Ratio	= 2.5
Hence, Total Sales	= 30 × 2.5 = ₹ 75 crores

Computation of Profit after Tax (PAT)

Particulars	(₹ in crores)
Sales	75.00
Less: Variable Operating Cost @ 60%	45.00
Contribution	30.00
Less: Fixed Cost (other than Interest)	6.00

Particulars	(₹ in crores)
EBIT/PBIT	24.00
Less: Interest on Debentures (15% × 15)	2.25
EBT/PBT	21.75
Less: Tax @ 40%	8.70
EAT/ PAT	13.05

(i) Earnings per Share

$$\text{EPS} = \frac{\text{PAT}}{\text{Number of Equity Shares}} = \frac{13.05}{0.75} = ₹\ 17.40$$

It indicates the amount the company earns per share. Investors use this as a guide while valuing the share and making investment decisions. It is also an indicator used in comparing firms within an industry or industry segment.

(ii) Operating Leverage

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{30}{24} = 1.25$$

It indicates the choice of technology and fixed cost in cost structure. It is level specific. When firm operates beyond operating break-even level, then operating leverage is low. It indicates sensitivity of earnings before interest and tax (EBIT) to change in sales at a particular level.

(iii) Financial Leverage

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{PBT}} = \frac{24}{21.75} = 1.103$$

The financial leverage is very comfortable since the debt service obligation is small vis-à-vis EBIT.

(iv) Combined Leverage

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{PBT}} = \frac{30}{21.75} = 1.379$$

Or,

= Operating Leverage × Financial Leverage

= 1.25 × 1.103 = 1.379

The combined leverage studies the choice of fixed cost in cost structure and choice of debt in capital structure. It studies how sensitive the change in EPS is vis-à-vis change in sales. The leverages operating, financial and combined are used as measurement of risk.

ICAI · Old P8 · Jul 2021 exam · question p. 81 · answer p. 81-83

419 EXAM Old P8 Jan 2021 Q2 10 marks old scheme

The information related to XYZ Company Ltd. for the year ended 31st March, 2020 are as follows:

Equity Share Capital of ₹ 100 each	₹ 50 Lakhs
12% Bonds of ₹ 1000 each	₹ 30 Lakhs
Sales	₹ 84 Lakhs
Fixed Cost (Excluding Interest)	₹ 7.5 Lakhs
Financial Leverage	1.39
Profit-Volume Ratio	25%
Market Price per Equity Share	₹ 200
Income Tax Rate Applicable	30%

You are required to compute the following:

(i) Operating Leverage

(ii) Combined Leverage

(iii) Earning per share

(iv) Earning Yield

ICAI'S ANSWER

Workings:

1. Profit Volume Ratio = $\frac{\text{Contribution}}{\text{Sales}}$ x 100

So, 25 = $\frac{\text{Contribution}}{₹\ 84,00,000}$ x 100

Contribution = $\frac{₹\ 84,00,000 \text{ x } 25}{100}$ = ₹ 21,00,000

2. Financial leverage = $\frac{\text{EBIT}}{\text{EBT}}$

Or, 1.39 = $\frac{₹\ 13,50,000 \text{ (as calculated above)}}{\text{EBT}}$

EBT = ₹ 9,71,223

3. Income Statement

Particulars	(₹)
Sales	84,00,000
Less: Variable Cost (Sales - Contribution)	(63,00,000)
Contribution	21,00,000
Less: Fixed Cost	(7,50,000)
EBIT	13,50,000

Particulars	(₹)
Less: Interest (EBIT - EBT)	(3,78,777)
EBT	9,71,223
Less: Tax @ 30%	(2,91,367)
Profit after Tax (PAT)	**6,79,856**

(i) Operating Leverage = $\frac{\text{Contribution}}{\text{Earnings before interest and tax (EBIT)}}$

= $\frac{₹\ 21,00,000}{₹\ 13,50,000}$ = **1.556** (approx.)

(ii) Combined Leverage = Operating Leverage x Financial Leverage

= 1.556 x 1.39 = **2.163** (approx.)

Or, $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 21,00,000}{₹\ 9,71,223}$ = **2.162** (approx.)

(iii) Earnings per Share (EPS)

EPS = $\frac{\text{PAT}}{\text{No. of shares}} = \frac{₹\ 6,79,856}{50,000}$ = **₹ 13.597**

(iv) Earning Yield

= $\frac{\text{EPS}}{\text{Market Price}}$ x 100 = $\frac{₹\ 13.597}{₹\ 200}$ x 100 = **6.80%** (approx.)

Note: The question has been solved considering Financial Leverage given in the question as the base for calculating total interest expense including the interest of 12% Bonds of ₹ 30 Lakhs. The question can also be solved in other alternative ways.

ICAI · Old P8 · Jan 2021 exam · question p. 6 · answer p. 6-8

420 EXAM **Old P8 Nov 2020** Q5 10 marks old scheme

The following data is available for Stone Ltd. :

	(₹)
Sales	5,00,000
(-) Variable cost @ 40%	2,00,000
Contribution	3,00,000
(-) Fixed cost	2,00,000
EBIT	1,00,000
(-) Interest	25,000
Profit before tax	75,000

Using the concept of leverage, find out

(i) The percentage change in taxable income if EBIT increases by 10%.

(ii) The percentage change in EBIT if sales increases by 10%.

(iii) The percentage change in taxable income if sales increases by 10%.

Also verify the results in each of the above case.

ICAI'S ANSWER

(i) Degree of Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 1{,}00{,}000}{₹\ 75{,}000}$ = **1.333 times**

So, If EBIT increases by 10% then Taxable Income (EBT) will be increased by 1.333 × 10 = **13.33% (approx.)**

Verification

Particulars	Amount (₹)
New EBIT after 10% increase (₹ 1,00,000 + 10%)	1,10,000
Less: Interest	25,000
Earnings before Tax after change (EBT)	85,000

Increase in Earnings before Tax = ₹ 85,000 - ₹ 75,000 = ₹ 10,000

So, percentage change in Taxable Income (EBT) = $\frac{₹\ 10{,}000}{₹\ 75{,}000} \times 100$ = **13.333%, hence verified**

(ii) Degree of Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ 3{,}00{,}000}{₹\ 1{,}00{,}000}$ = **3 times**

So, if sale is increased by 10% then EBIT will be increased by 3 × 10 = **30%**

Verification

Particulars	Amount (₹)
New Sales after 10% increase (₹ 5,00,000 + 10%)	5,50,000
Less: Variable cost (40% of ₹ 5,50,000)	2,20,000
Contribution	3,30,000
Less: Fixed costs	2,00,000
Earnings before interest and tax after change (EBIT)	1,30,000

Increase in Earnings before interest and tax (EBIT) = ₹ 1,30,000 - ₹ 1,00,000 = ₹ 30,000

So, percentage change in EBIT = $\frac{₹\ 30{,}000}{₹\ 1{,}00{,}000} \times 100$ = **30%, hence verified.**

(iii) Degree of Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 3{,}00{,}000}{₹\ 75{,}000}$ = **4 times**

So, if sale is increased by 10% then Taxable Income (EBT) will be increased by 4 × 10 = **40%**

Verification

Particulars	Amount (₹)
New Sales after 10% increase (₹ 5,00,000 + 10%)	5,50,000
Less: Variable cost (40% of ₹ 5,50,000)	2,20,000
Contribution	3,30,000
Less: Fixed costs	2,00,000
Earnings before interest and tax (EBIT)	1,30,000
Less: Interest	25,000
Earnings before tax after change (EBT)	1,05,000

Increase in Earnings before tax (EBT) = ₹ 1,05,000 - ₹ 75,000 = ₹ 30,000

So, percentage change in Taxable Income (EBT) = $\frac{₹\ 30,000}{₹\ 75,000} \times 100$ = **40%, hence verified**

ICAI · Old P8 · Nov 2020 exam · question p. 11 · answer p. 11-13

421 **EXAM** **Old P8 Nov 2019** Q2 **10 marks** **old scheme**

The Balance Sheet of Gitashree Ltd. is given below:

Liabilities		(₹)
Shareholders' fund		
Equity share capital of ₹ 10 each	₹ 1,80,000	
Retained earnings	₹ 60,000	2,40,000
Non-current liabilities 10% debt		2,40,000
Current liabilities		1,20,000
		6,00,000
Assets		
Fixed Assets		4,50,000
Current Assets		1,50,000
		6,00,000

The company's total asset turnover ratio is 4. Its fixed operating cost is ₹ 2,00,000 and its variable operating cost ratio is 60%. The income tax rate is 30%.

Calculate:

(i) (a) Degree of Operating leverage.

(b) Degree of Financial leverage.

(c) Degree of Combined leverage.

(ii) Find out EBIT if EPS is (a) ₹ 1 (b) ₹ 2 and (c) ₹ 0.

ICAI'S ANSWER

Working Notes:

Total Assets = ₹ 6,00,000

Total Asset Turnover Ratio i.e. = $\frac{\text{Total Sales}}{\text{Total Assets}} = 4$

Hence, Total Sales = ₹ 6,00,000 × 4 = ₹ 24,00,000

Computation of Profits after Tax (PAT)

Particulars	(₹)
Sales	24,00,000
Less: Variable operating cost @ 60%	14,40,000
Contribution	9,60,000
Less: Fixed operating cost (other than Interest)	2,00,000
EBIT (Earning before interest and tax)	7,60,000
Less: Interest on debt (10% × 2,40,000)	24,000
EBT (Earning before tax)	7,36,000
Less: Tax 30%	2,20,800
EAT (Earning after tax)	5,15,200

(i) (a) Degree of Operating Leverage

Degree of Operating leverage = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹9,60,000}{₹7,60,000}$ = 1.263 (approx.)

(b) Degree of Financial Leverage

Degree of Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{₹9,60,000}{₹7,60,000}$ = 1.033 (approx.)

(c) Degree of Combined Leverage

Degree of Combined Leverage = $\frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{EBT}} = \frac{\text{Contribution}}{\text{EBT}}$

= $\frac{₹9,60,000}{₹7,60,000}$ = 1.304 (approx.)

Or

Degree of Combined Leverage = Degree of Operating Leverage × Degree of Financial Leverage

= 1.263 × 1.033 = 1.304 (approx.)

(ii) (a) If EPS is Re. 1

EPS = $\frac{(\text{EBIT} - \text{Interest})(1 - \text{tax})}{\text{No of equity shares}}$

Or, 1 = $\frac{(\text{EBIT} - ₹24,000)\ (1 - 0.30)}{18,000}$

Or, EBIT = ₹ 49,714 (approx.)

(b) If EPS is ₹ 2

$$2 = \frac{(\text{EBIT} - ₹24{,}000)\ (1 - 0.30)}{18{,}000}$$

Or, EBIT = ₹ 75,429 (approx.)

(c) If EPS is ₹ 0

$$0 = \frac{(\text{EBIT} - ₹24{,}000)\ (1 - 0.30)}{18{,}000}$$

Or, EBIT = ₹ 24,000

Alternatively, if EPS is 0 (zero), EBIT will be equal to interest on debt i.e. ₹ 24,000.

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422 RTP **Sep 2026** Q2 MCQ

Total Assets & Current liabilities of the Manju Limited are 50 lakhs & 10 lakhs respectively. ROCE is 15%, measure of business operating risk is at 3.5 & P/V ratio is 70%. Calculate Sales.

A 21 lakhs

B 30 lakhs

C 37.50 lakhs

D 40 lakhs

ICAI'S ANSWER

(B) ROCE = EBIT / Total Capital Employed

Total Capital Employed = Total Assets – Current Liabilities

= 50 lakhs – 10 lakhs

= 40 lakhs

EBIT = 40 lakhs x 15%

= 6 lakhs

Now, OL of 3.5 = Contribution / EBIT

Therefore Contribution = 6 Lakhs X 3.5 = 21 lakhs

Sales = Contribution/PV Ratio=21 lakhs/0.7=30 lakhs

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423 RTP **Sep 2026** Q5

Anirban Ltd. is a well-established manufacturing company engaged in producing industrial components for various engineering industries. The management intends to assess the company's financial performance and capital structure efficiency using key financial ratios and leverage measures. The following information has been extracted from the company's latest financial statements.

Capital Structure of Anirban Ltd.

Source of Finance	Amount (₹)
Equity Share Capital (₹10 per share)	8,00,000
10% Preference Share Capital (₹100 per share)	5,00,000
12% Debentures (₹100 each)	7,00,000
Total Capital Employed	20,00,000

Additional Information:

1. The company reported a Profit After Tax (PAT) of ₹ 2,80,000 during the year.
2. The applicable corporate tax rate is 30%.
3. Total operating expenses, including depreciation of ₹ 96,800, are equal to 1.5 times the Earnings Before Interest and Tax (EBIT).
4. Equity shareholders were paid a dividend at the rate of 15%.
5. The market price per equity share is ₹ 23.

Required:

Based on the information provided above, calculate the following:

(i) Operating Leverage and Financial Leverage.

(ii) Preference Dividend Cover and Equity Dividend Cover.

(iii) Earnings Yield Ratio.

(iv) Price–Earnings (P/E) Ratio.

Show all necessary workings and assumptions clearly.

ICAI'S ANSWER

Working Notes:

	(₹)
Net Profit after Tax	2,80,000
Tax @ 30%	1,20,000
EBT	4,00,000
Interest on Debentures	84,000
EBIT	4,84,000
Total Operating Expenses (1.5 times of EBIT)	7,26,000
Sales	12,10,000

(i) **Operating Leverage**

Variable operating expenses=₹ 7,26,000-96,800=₹6,29,200

$$= \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ (12,10,000 - 6,29,200)}{₹\ 4,84,000} = \frac{₹\ 5,80,800}{₹\ 4,84,000} = 1.2 \text{ times}$$

Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{4,84,000}{4,00,000}$ = 1.21 times

(ii) **Cover for Preference Dividend**

= $\frac{\text{PAT}}{\text{Preference Share Dividend}} = \frac{₹\ 2,80,000}{₹\ 50,000}$ = 5.6 times

Cover for Equity Dividend

= $\frac{(\text{PAT - Preference Dividend})}{\text{Equity Share Dividend}} = \frac{₹\ (2,80,000 - 50,000)}{₹\ 1,20,000}$

= $\frac{₹\ 2,30,000}{₹\ 1,20,000}$ = 1.92 times

(iii) **Earning Yield Ratio**

= $\frac{\text{EPS}}{\text{Market Price}} \times 100$

= $\left(\frac{\frac{2,30,000}{80,000}}{23} \times 100\right)$

= $\frac{2.875}{23} \times 100 = 12.5\%$

(iv) **Price – Earnings Ratio (PE Ratio)**

= $\frac{\text{Market Price}}{\text{EPS}} = \frac{23}{2.875}$

= 8 times

Note: All operating expenses, other than depreciation, are variable costs.

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424 RTP **May 2026** Q7

ABC Manufacturing Limited produces industrial components with a well-established market presence. The company operates under a special tax-free status granted by the government, which exempts it from corporate income tax obligations.

Financial Information

The company's financial parameters for the current fiscal year are as follows:

Capital Structure Details

- Total Capital Invested: ₹ 20,00,000 (₹ 20 lakhs)
- Cost of Equity (Ke): 12% per annum
- Cost of Debt (Kd): 6% per annum
- Weighted Average Cost of Capital (WACC): 9% per annum

Operational Information

- Annual Sales Revenue: ₹ 24,00,000 (₹ 24 lakhs)
- Variable Costs: 50% of sales revenue
- Fixed Costs: ₹ 10,00,000 (₹ 10 lakhs) per annum

General corporate tax is 25%

As a financial analyst, you have been tasked with conducting a comprehensive financial analysis of ABC Manufacturing Limited. Specifically, you are required to:

i. DETERMINE the optimal debt-equity composition that results in the given weighted average cost of capital of 9%.

ii. PREPARE a comprehensive Income Statement showing the company's profitability from operations through net income when the cost of debt is 6%, 10% & 12%.

iii. CALCULATE the Return on Investment (ROI) to assess how efficiently the company is utilizing its total capital invested.

iv. CALCULATE the Return on Equity (ROE) to evaluate the returns generated for equity shareholders.

v. FIND the different leverages in the following format

Leverages	Computation	Dependent on	Indicates effect of ____ on
Leverages showing level of business risk			
Leverages showing level of finance risk			
Leverages showing level of combined risk			

vi. EXPLAIN the concept of Leverage as a double edged sword by explaining the Leverage Effect when ROI > Kd, ROI = Kd, ROI < Kd.

vii. DERIVE ROE from ROI in the following format –

Cost of debt =	6%	10%	12%
Return on Investment % of Equity			
Surplus / Deficit from debt payments in %			
Surplus / Deficit from debt payments in INR			
Return on Equity in INR=ROI + Surplus – Deficit			
Return on Equity in % = Return on Equity in INR/Equity			

ICAI'S ANSWER

Optimal debt-equity composition

Let the weight of equity be W

Item	Cost	Weight	Product
Equity	12	W	12W
Debt	6	1–W	6 – 6W
		1	**9%**

9 = 12W + 6 – 6W

3 = 6W

W = 0.5.

So, Weight of Equity is 0.50

So, Weight of Debt is 1 – 0.50 = 0.50

Total Capital = 20 Lakhs. Hence, Debt and equity 10 Lakhs each.

Income Statement

Cost of debt =	6%	10%	12%
Sales	24,00,000	24,00,000	24,00,000
Less: Variable Costs (50%)	12,00,000	12,00,000	12,00,000
Contribution	12,00,000	12,00,000	12,00,000
Less: Fixed Costs (Given)	10,00,000	10,00,000	10,00,000
EBIT	2,00,000	2,00,000	2,00,000
Less: Interest	60,000	1,00,000	1,20,000
EBT / PAT	1,40,000	1,00,000	80,000
Return on Investment = EBIT/ Capital employed	10%	10%	10%
Return on equity = PAT / Equity Capital	14%	10%	8%

Leverages showing level of different risk

Leverages showing level of	Leverages	Computation	Dependent on	Indicates effect of ... on ...
Business risk	Operating Leverage (DOL)	Contribution ÷ EBIT = 12,00,000 ÷ 2,00,000 = 6	Fixed Operating Costs	Sales on EBIT
Financial Risk	Financial Leverage (DFL)	EBIT ÷ EBT = 2,00,000 ÷1,40,000 = 1.43	Fixed Financial Costs	EBIT on EPS
Combined Risk	Combined Leverage (DCL)	Contribution ÷ EBT or DOL × DFL = 6 × 1.43 = 8.58	Both Fixed Operating & Fixed Financial Costs	Sales on EPS

Return on Investment = EBIT/ Capital employed

= 2,00,000/20,00,000 =10%

When the cost of 'fixed cost fund' is less than the return on investment, financial leverage will help to increase return on equity and EPS. The firm will also benefit from the saving of tax on interest on debts etc. However, when cost of debt will be more than the return it will affect return of equity and EPS unfavourably and as a result firm can be under financial distress. Therefore, financial leverage is also known as "**double edged sword**".

Effect on EPS and ROE:

When, ROI (10%) > Interest (6%) – Favourable – Advantage

When, ROI (10%) < Interest (12%) – Unfavourable – Disadvantage

When, ROI (10%) = Interest (10%) – Neutral – Neither advantage nor disadvantage

Income Statement

Cost of debt =	6%	10%	12%
Return on Investment = 10% of Equity	1,00,000	1,00,000	1,00,000
Surplus / Deficit from debt payments in %	10 – 6 = 4%	10 – 10 = 0	10 – 12 =- 2%
Surplus / Deficit from debt payments in INR = % of Debt	40,000	Nil	(20,000)
Return on Equity in INR = ROI + Surplus – Deficit	1,40,000	1,00,000	80,000
Return on Equity in % = Return on Equity in INR / Equity	14%	10%	8%

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425 RTP **Jan 2026** Q4 MCQ

If Degree of Financial Leverage (DFL) = 5/4, which one of the following will be the interest coverage ratio?

A 5

B 4

C 3/4

D 1.33

ICAI'S ANSWER

(A) 5

DFL = EBIT/(EBIT-I) = 5/4

4EBIT = 5EBIT -5I

EBIT = 5I

Interest Coverage Ratio = EBIT/Interest

= 5I/I = 5

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426 RTP **Jan 2026** Q8

Following Balance Sheet and Income Statement have been obtained from the books of accounts of Pinaca Pvt. Ltd.

Balance Sheet as on March 31st 2025

Liabilities	Amount (₹)	Assets	Amount (₹)
Equity Capital (₹ 10 per share)	80,00,000	Net Fixed Assets	1,00,00,000
10% Debt	60,00,000	Current Assets	90,00,000
Retained Earnings	35,00,000		
Current Liabilities	15,00,000		
	1,90,00,000		1,90,00,000

Income Statement for the year ending March 31st 2025

Particulars	Amount (₹)
Sales	34,00,000
Less: Operating expenses (including ₹ 6,00,000 depreciation)	(12,00,000)
EBIT	22,00,000
Less: Interest	(6,00,000)
Earnings before tax	16,00,000

The tax rate applicable to the company is 35 percent.

(i) DETERMINE the degree of operating, financial and combined leverages at the current sales level, if all operating expenses, other than depreciation, are variable costs.

(ii) If total assets remain at the same level, but sales (i) increase by 20 percent and (ii) decrease by 20 percent, COMPUTE the earnings per share at the new sales level?

ICAI'S ANSWER

(i) **Degree of operating, financial and combined leverages at the current sales level-**

$$\text{DOL} = \frac{\text{Contribution}}{\text{EBIT}}$$

$$= \frac{₹34,00,000 - ₹6,00,000}{₹22,00,000} = 1.27$$

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBT}}$$

$$= \frac{₹22,00,000}{₹16,00,000} = 1.375$$

$$\text{DCL} = \text{DOL} \times \text{DFL} = 1.27 \times 1.38 = 1.75$$

(ii) **Earnings per share at the new sales level (Amount in ₹)**

Particulars	Increase by 20%	Decrease by 20%
Sales level	40,80,000	27,20,000
Less: Variable expenses	7,20,000	4,80,000
Less: Fixed cost	6,00,000	6,00,000

Particulars	Increase by 20%	Decrease by 20%
Earnings before interest and taxes	27,60,000	16,40,000
Less: Interest	6,00,000	6,00,000
Earnings before taxes	21,60,000	10,40,000
Less: Taxes @35%	7,56,000	3,64,000
Earnings after taxes (EAT)	14,04,000	6,76,000
Number of equity shares	8,00,000	8,00,000
EPS	1.76	0.85

Working Notes:

Variable Costs = ₹ 6,00,000 (total cost - depreciation)

Variable Costs at:

(i) Sales level, ₹ 40,80,000 = ₹ 7,20,000 (increase by 20%)

(ii) Sales level, ₹ 27,20,000 = ₹ 4,80,000 (decrease by 20%)

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427 RTP **Sep 2025** Q2 MCQ

A manufacturing company which is presently growing at a good rate has a Financial Leverage of 3. Its present borrowings are as follows:

12% Term Loan	₹ 60,00,000
Bank borrowings@10%	₹ 40,00,000
Public Deposits@8%	₹ 20,00,000

The Interest Coverage Ratio will be

A 1.0

B 1.5

C 2.0

D 2.5

ICAI'S ANSWER

Answer key: **(B)**

(b) Working Notes

Calculation of Interest

12% Term Loan	₹ 7,20,000
Bank borrowings@10%	₹ 4,00,000
Public Deposits@8%	₹ 1,60,000
Total Interest	₹ 12,80,000

Financial Leverage = EBIT/EBIT – Interest

3 = EBIT/EBIT – 12,80,000

Or EBIT = 19,20,000

So, Interest Coverage Ratio = EBIT/Interest

= 19,20,000/12,80,000

= 1.5

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428 RTP **Sep 2025** Q7

X Limited and Y Limited are two mid-sized companies operating in the same competitive industry. Both companies have recently undergone a financial performance review to assess their operational efficiency, cost structure, and overall financial risk. You, as a financial analyst, have been provided with selective financial indicators and are required to draw insights and comparisons based on leverage analysis and income statement reconstruction.

The management of X Limited has disclosed that the company is currently operating with a Margin of Safety (M/S) ratio of 0.1667. In contrast, Y Limited has a Margin of Safety that is twice as high as that of X Limited.

Both companies maintain a Financial Leverage of 3. Their variable cost ratios are 60% for X Limited and 50% for Y Limited.

In terms of financing costs, X Limited incurs an annual interest expense of ₹30,000. Y Limited, however, incurs an interest cost that is 300% higher than X Limited. Both companies are subject to a corporate tax rate of 30%, which affects their net profitability after interest and taxes.

You are required to PREPARE Income statement for both the companies and IDENTIFY the company which is better placed with reasons based on leverages.

ICAI'S ANSWER

Company X

(i) Financial Leverage = $\frac{\text{EBIT}}{\text{EBT i.e EBIT} - \text{Interest}}$

So, $3 = \frac{\text{EBIT}}{\text{EBIT} - ₹\ 30{,}000}$

Or, 3 (EBIT – 30,000) = EBIT

Or, 2 EBIT = 90,000

Or, EBIT = 45,000

(ii) Margin of safety = 0.1667

Operating Leverage = 1/Margin of safety

= 1/0.1667 = 6

$= \frac{\text{Contribution}}{\text{EBIT}}$

Or, $6 = \frac{\text{Contribution}}{₹\ 45{,}000}$

Or, Contribution = ₹ 2, 70,000

$\text{Sales} = \frac{\text{contribution}}{\text{P/V Ratio}(1 - \text{variable cost ratio})}$

$= \frac{₹\ 2{,}70{,}000}{40\%} = ₹\ 6{,}75{,}000$

(iii) Fixed Cost = Contribution – EBIT

= ₹ 2,70,000 – 45,000

Or, Fixed cost = ₹ 2,25,000

Company Y

(i) Financial Leverage $= \frac{\text{EBIT}}{\text{EBT i.e EBIT} - \text{Interest}}$

So, $3 = \frac{\text{EBIT}}{\text{EBIT} - ₹\ 1{,}20{,}000}$

Or, 3 (EBIT – ₹ 1,20,000) = EBIT

Or, 3 EBIT – ₹ 3,60,000 = EBIT

Or, EBIT = ₹ 1,80,000

(ii) Margin of safety = 0.1667 x 2 = 0.3333

Operating Leverage = 1/Margin of safety

= 1/0.3333 = 3

$= \frac{\text{Contribution}}{\text{EBIT}}$

Or, $3 = \frac{\text{Contribution}}{₹\ 1{,}80{,}000}$

Or, Contribution = ₹ 5,40,000

$\text{Sales} = \frac{\text{Contribution}}{\text{P/V Ratio } (1 - \text{variable cost ratio})}$

$= \frac{₹\ 5{,}40{,}000}{50\%} = ₹\ 10{,}80{,}000$

(iii) Fixed Cost = Contribution – EBIT

= ₹ 5,40,000 – ₹ 1,80,000

Or, Fixed cost = ₹ 3,60,000

(iv) Interest = ₹ 30,000+₹ 30,000x300% = ₹ 1,20,000

Income Statements of X Ltd and Y Ltd

	X Ltd (₹)	Y Ltd (₹)
Sales	6,75,000	10,80,000

	X Ltd (₹)	Y Ltd (₹)
Less: Variable cost	4,05,000	5,40,000
Contribution	2,70,000	5,40,000
Less: Fixed Cost	2,25,000	3,60,000
Earnings before interest and tax (EBIT)	45,000	1,80,000
Less: Interest	30,000	1,20,000
Earnings before tax (EBT)	15,000	60,000
Less: Tax @ 30%	4,500	18,000
Earnings after tax (EAT)	10,500	42,000

Comment based on Leverage

Comment based on leverage – Company Y is better than company X of the following reasons:

(i) Capacity of Company Y to meet interest liability is same that of companies X (from EBIT/Interest ratio)

$$[X = \frac{₹\ 45,000}{₹\ 30,000} = 1.5,\ Y = \frac{₹\ 1,80,000}{₹\ 1,20,000} = 1.5]$$

(ii) However, Company Y has lesser financial risk as the total risk (business and financial) of company Y is lower (combined leverage of Company X is 18 and Company Y is 9).

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429 RTP **May 2025** Q3 MCQ

RGT Infrastructure Company has a degree of operating leverage of 3 at a sales level of ₹ 7,00,000 and operating income of ₹2,20,000. Fall in sales of the company by 10% will result in decrease in operating income of:

A ₹ 80,000

B ₹ 55,000

C ₹ 66,000

D ₹ 40,000

ICAI'S ANSWER

(C) ₹ 66,000

% Change in Operating Income = Degree of Operating Leverage (DOL) × % Change in Sales

% Change in Operating Income = 3 × (−10%) =−30%

Decrease in Operating Income = 30% of 2,20,000

= ₹ 66,000

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430 RTP May 2025 Q7

Details of Kshitij Limited are given below for the year ended 31st March 2025 –

Particulars	Details
Sales	₹ 180 lakhs
Fixed Cost (Excl Interest)	₹ 45 lakhs
B.E.P Sales	₹ 120 lakhs
Equity Share Capital of ₹ 100 Each	₹ 150 lakhs
Income Tax Rate	25%
Cost Of Debt (Kd)	9%
Debt	₹ 90 lakhs

Required to CALCULATE -

(A) Operating, Financial, Combined Leverage & P/V Ratio

(B) Return on Capital Employed and EPS

(C) Does Kshitij Limited have favorable Financial Leverage?

(D) % Change in EPS, if EBIT increases or decreases by 15%

(E) At what level of Sales, the PAT will be equal to 3/4th of its current value.

ICAI'S ANSWER

Particular	Amount (₹)
Sales	180 lakhs
Contribution (180 × 37.5%) **WN - 1**	**67.50 lakhs**
(-) Fixed Cost	(45.00) lakhs
EBIT	**22.50 lakhs**
(-) Interest Exp **WN - 2**	(10.80) lakhs
EBT	**11.70 lakhs**
(-) Tax @ 25%	(2.925) lakhs
EAT	**8.775 lakhs**
No of Equity Shares	**1.50 lakhs**
EPS	**₹ 5.85**

(A) Calculation of OL, FL, CL & P/V Ratio

WN 1 – Calculation of P/V Ratio

B.E.P Sales = $\frac{\text{Fixed Cost}}{\text{PV Ratio}}$

PV Ratio = 45 / 120 = 37.5%

WN 2 – Calculation of Interest Exp

Kd = Interest (1 – t)

9 = Interest (1 – 0.25)

Interest = 12%

Interest Exp = 90 Lakhs x 12% = ₹ 10.8

OL = $\frac{1}{\text{Margin of Safety (MOS)}}$

MOS = $\frac{\text{Actual Sales} - \text{BEP Sales}}{\text{Actual Sales}}$

= 60/180

= 0.3333

OL = $\frac{\mathbf{1}}{\mathbf{0.33333}}$ = 3

OR

OL = $\frac{\text{Contribution}}{\text{EBIT}}$

= 67.50 / 22.50

OL = 3

FL = $\frac{\text{EBIT}}{\text{EBT}}$

= 22.50 / 11.70

FL = 1.9231

CL = OL x FL

= 3 x 1.9231

CL = 5.7693

(B) ROCE & EPS

EPS =₹ 5.85 (From Income statement above)

ROCE = $\frac{\text{EBIT}}{\text{Capital Employed}}$

= $\frac{22.50}{240}$

= 9.375%

Capital Employed = Equity + Debt

= 150 + 90 = 240 lakhs

(C) Since ROCE = 9.375% < Interest = 12%, Kshitij Limited doesn't have a favorable financial leverage.

(D) Financial leverage measures the relationship for % change EPS due to changes in EBIT

$$FL = \frac{\%\text{ change in EPS}}{\%\text{ change in EBIT}}$$

$$1.9231 = \frac{\%\text{ change in EPS}}{15}$$

= 28.8465%

Therefore, EPS will increase or decrease by 28.8465%, if EBIT increases or decreases by 15%

(E) Current PAT = 8.775 lakhs

3/4th of current PAT = 6.58125 lakhs

So, it means PAT decreases by 25%

Combined leverage measures the relationship for % change in PAT due to changes in sales

$$CL = \frac{\%\text{ change in PAT}}{\%\text{ change in Sales}}$$

$$5.7693 = \frac{25}{\%\text{ change in Sales}}$$

% Change in Sales = 4.333%

New Sales level = 180 lakhs - 4.33%

New Sales level = ₹172.20 Lakhs

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431 **RTP** **Jan 2025** **Q7**

From the following financial data of Company X and Company Y:

(i) PREPARE their Income Statements.

(ii) CALCULATE Margin of Safety for both the Companies

(iii) CALCULATE Percentage change in EPS for both the companies, if percentage change in sales is 25%

(in ₹)

	Company X	Company Y
Variable Cost	72,000	65% of Sales
Fixed Cost	35,000	-
Interest Expenses	12,000	6,000
Financial Leverage	4:1	-
Operating Leverage	-	5:1
Income Tax Rate	30%	30%
Sales	-	1,45,000

ICAI'S ANSWER

(i) **Income Statement**

Particulars	Co. X (₹)	Co. Y (₹)
Sales	1,23,000 **(WN 2)**	1,45,000
(-) Variable Cost	(72,000)	(94,250) (65% on sales)
Contribution	51,000 **(WN 2)**	50,750
(-) Fixed Cost	(35,000)	(40,600)
EBIT	16,000 **(WN 1)**	10,150 **(WN 3)**
(-) Interest	(12,000)	(6,000)
EBT	4,000	4,150
(-) Tax @ 30%	(1,200)	(1,245)
EAT	**2,800**	**2,905**

WN 1: Calculation of EBIT for Co. X using Financial Leverage

$$\text{FL} = \frac{\text{EBIT}}{\text{EBT}} \text{ or } \frac{\text{EBIT}}{\text{EBIT} - \text{Interest}}$$

$$4 = \frac{\text{EBIT}}{\text{EBIT} - 12{,}000}$$

EBIT = ₹ 16,000

EBT = ₹ 16,000 – ₹ 12,000 = ₹ 4,000

WN 2: Calculation of Contribution and Sales using reverse mechanism

Contribution = EBIT + Fixed Cost

= ₹ 16,000 + ₹ 35,000

Contribution = ₹ 51,000

Sales = Contribution + Variable Cost

Sales = ₹1,23,000

WN 3: Calculation of EBIT for Co. Y using Operating leverage

OL = Contribution / EBIT

$$5 = \frac{50{,}750}{\text{EBIT}}$$

EBIT = ₹ 10,150

(ii) Margin of Safety (MOS) is inversely proportionate to the Operating Leverage as higher the safety margin lower would be the business risk

$$\text{MOS} = \frac{1}{\text{OL}}$$

$$\text{Operating Leverage (Co. X)} = \frac{51{,}000}{16{,}000}$$

Operating Leverage (Co. X) = 3.1875 : 1

Therefore, MOS for Co. X = 1 / 3.1875

MOS for Co. X = 31.37%

Operating Leverage (Co. Y) = 5 : 1

Therefore, MOS for Co. X = $\frac{1}{5}$

MOS for Co. Y = 20%

(iii) Combined leverage measures the percentage change in EPS due to percentage change in sales

Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}}$

Combined Leverage (Co. X) = $\frac{51,000}{4,000}$

= 12.75

Combined Leverage = $\frac{\%\text{ change in EPS}}{\%\text{ change in sales}}$

$12.75 = \frac{\%\text{ change in EPS}}{25\%}$

% change in EPS (Co. X) = 318.75%

Combined Leverage (Co. Y) = $\frac{50,750}{4,150}$

= 12.23

$12.23 = \frac{\%\text{ change in EPS}}{25\%}$

% change in EPS (Co. Y) = 305.75%

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432 RTP **Sep 2024** Q3 MCQ

Also asked in MTP May 2026 S-I MCQ 8

A firm has sales of ₹ 75,00,000, variable cost of ₹ 42,00,000 and fixed cost of ₹ 6,00,000.

It has a debt of ₹ 45,00,000 at 9% and equity of ₹ 55,00,000. Does it have favourable financial leverage?

A. ROI is less than interest on loan funds and hence it has no favourable financial leverage.

B. ROI is equal to interest on loan funds and hence it has favourable financial leverage.

C. ROI is greater than interest on loan funds and hence it has favourable financial leverage.

D. ROI is greater than interest on loan funds and hence it has unfavourable financial leverage.

ICAI'S ANSWER

(C) ROI is greater than interest on loan funds and hence it has favourable financial leverage.

EBIT = 75,00,000 - 42,00,000 - 6,00,000 = 27,00,000,

ROI = 27,00,000/(45,00,000 + 55,00,000) = 27%,

Rate of Interest lower than Return on investment.

Therefore, there is favourable leverage.

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433 RTP Sep 2024 Q8

Following data of PC Ltd. under Situations 1, 2 and 3 and Financial Plan A and B is given:

Installed Capacity (units)		3,600
Actual Production and Sales (units)		2,400
Selling price per unit (₹)		30
Variable cost per unit (₹)		20
Fixed Costs (₹):	Situation 1	3,000
	Situation 2	6,000
	Situation 3	9,000

Capital Structure:

Particulars	Financial Plan	
	A	**B**
Equity	₹ 15,000	₹ 22,500
Debt	₹ 15,000	₹ 7,500
Cost of Debt	12%	12%

Required:

(i) CALCULATE the operating leverage and financial leverage.

(ii) FIND out the combinations of operating and financial leverage which give the highest value and the least value.

ICAI'S ANSWER

(i) Operating Leverage

	Situation 1	Situation 2	Situation 3
	(₹)	**(₹)**	**(₹)**
Sales (S)			
2,400 units @ ₹ 30 per unit	72,000	72,000	72,000
Less: Variable Cost (VC) @ ₹ 20 per unit	48,000	48,000	48,000
Contribution (C)	24,000	24,000	24,000
Less: Fixed Cost (FC)	3,000	6,000	9,000
EBIT	21,000	18,000	15,000
Operating Leverage = $\frac{C}{EBIT}$	$\frac{₹24,000}{₹21,000}$ = 1.14	$\frac{₹24,000}{₹18,000}$ = 1.33	$\frac{₹24,000}{₹15,000}$ = 1.60

Financial Leverage

	Financial Plan	
	A (₹)	**B (₹)**
Situation 1		
EBIT	21,000	21,000
Less: Interest on debt (₹ 15,000 × 12%); (₹ 7,500 × 12%)	1,800	900
EBT	19,200	20,100
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	$\frac{₹21,000}{₹19,200}$ = 1.09	$\frac{₹21,000}{₹20,100}$ = 1.04
Situation 2		
EBIT	18,000	18,000
Less: Interest on debt	1,800	900
EBT	16,200	17,100
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	$\frac{₹18,000}{₹16,200}$ = 1.11	$\frac{₹18,000}{₹17,100}$ = 1.05
Situation 3		
EBIT	15,000	15,000
Less: Interest on debt	1,800	900
EBT	13,200	14,100
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	$\frac{₹15,000}{₹13,200}$ = 1.14	$\frac{₹15,000}{₹14,100}$ = 1.06

(ii) Combined Leverages

CL = OL x FL

		Financial Plan	
		A (₹)	**B (₹)**
(a)	Situation 1	1.14 × 1.09 = 1.24	1.14 × 1.04 = 1.19
(b)	Situation 2	1.33 × 1.11 = 1.48	1.33 × 1.05 = 1.40
(c)	Situation 3	1.60 × 1.14 = 1.82	1.60 × 1.06 = 1.70

The above calculations suggest that the highest value is in Situation 3 financed by Financial Plan A and the lowest value is in the Situation 1 financed by Financial Plan B.

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434 RTP May 2024 Q4

Same question, different ICAI answer: #445 MTP Sep 2026 S-I

From the following financial data of Company A and Company B, PREPARE their Income Statements.

	Company A (₹)	Company B (₹)
Variable Cost	88,000	50% of sales
Fixed Cost	26,500	-
Interest Expenses	14,000	11,000
Financial Leverage	5 : 1	-
Margin of Safety	-	0.25
Income Tax Rate	30%	30%
EBIT	-	14,000

ICAI'S ANSWER

Income Statements of Company A and Company B

	Company A (₹)	Company B (₹)
Sales	1,32,000	1,12,000
Less: Variable cost	88,000	56,000
Contribution	44,000	56,000
Less: Fixed Cost	26,500	42,000
Earnings before interest and tax (EBIT)	17,500	14,000
Less: Interest	14,000	11,000
Earnings before tax (EBT)	3,500	3,000
Less: Tax @ 30%	1,050	900
Earnings after tax (EAT)	2,450	2,100

Working Notes:

Company A

(i) Financial Leverage = $\frac{\text{EBIT}}{\text{EBT i.e EBIT - Interest}}$

So, $5 = \frac{\text{EBIT}}{\text{EBIT - 14,000}}$

Or, 5 (EBIT – 14,000) = EBIT

Or, 4 EBIT = 70,000

Or, EBIT = ₹17,500

(ii) Contribution = EBIT + Fixed Cost

= ₹ 17,500 + ₹ 26,500 = ₹ 44,000

(iii) Sales = Contribution + Variable cost

= ₹ 44,000 + ₹ 88,000

= ₹ 1,32,000

Company B

(i) Operating Leverage =1/Margin of Safety = $\frac{\text{Contribution}}{\text{EBIT}}$

$1/0.25 = \frac{\text{Contribution}}{₹14,000}$

$4 = \frac{\text{Contribution}}{₹14,000}$

Contribution = ₹14,000 x 4 = ₹56,000

(ii) Fixed Cost = Contribution – EBIT = 56,000 – 14,000 = ₹ 42,000

(iii) Contribution = 50% of Sales (as Variable Cost is 50% of Sales)

Sales = 56,000 x 2 = ₹1,12,000

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435 RTP **Old P8 Nov 2023 Q4** old scheme

The capital structure of ABC Ltd. for the year ended 31st March 2022 consisted as follows:

Particulars	Amount in ₹
Equity share capital (face value ₹ 100 each)	20,00,000
10% debentures (₹ 100 each)	20,00,000

During the year 2021-22, sales decreased to 1,00,000 units as compared to 1,20,000 units in the previous year. However, the selling price stood at ₹ 15 per unit and variable cost at ₹ 10 per unit for both the years. The fixed expenses were at ₹ 2,00,000 p.a. and the income tax rate is 30%.

You are required to CALCULATE the following:

(a) The degree of financial leverage at 1,20,000 units and 1,00,000 units.

(b) The degree of operating leverage at 1,20,000 units and 1,00,000 units.

(c) The percentage change in EPS.

ICAI'S ANSWER

Sales in units	1,20,000 (₹)	1,00,000 (₹)
Sales Value	18,00,000	15,00,000
Variable Cost	(12,00,000)	(10,00,000)
Contribution	6,00,000	5,00,000
Fixed expenses	(2,00,000)	(2,00,000)

Sales in units	1,20,000 (₹)	1,00,000 (₹)
EBIT	4,00,000	3,00,000
Debenture Interest	(2,00,000)	(2,00,000)
EBT	2,00,000	1,00,000
Tax @ 30%	(60,000)	(30,000)
Profit after tax (PAT)	1,40,000	70,000
(i) Financial Leverage= EBIT/EBT	= 4,00,000/2,00,000 = 2	= 3,00,000/1,00,000 = 3
(ii) Operating leverage = Contribution/EBIT	6,00,000/4,00,000 = 1.50	= 5,00,000/3,00,000 = 1.67
(iii) Earnings per share (EPS)	1,40,000/20,000 = ₹ 7	70,000/20,000 = ₹ 3.5
Decrease in EPS	= ₹ 7 – ₹ 3.5 = ₹ 3.5	
% decrease in EPS	3.5/7 = x 100 = 50%	

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436 RTP **Old P8 May 2023** Q4 old scheme

The selected financial data for A, B and C companies for the current year ended 31st March are as follows:

Particulars	A	B	C
Variable Expenses as a % of sales	60	50	40
Interest	₹ 1,00,000	₹ 4,00,000	₹ 6,00,000
Degree of Operating Leverage	4:1	3:1	2.5:1
Degree of Financial Leverage	3:1	5:1	2.5:1
Income Tax Rate	30%	30%	30%

(a) PREPARE income statement for A, B and C companies

(b) COMMENT on the financial position and structure of these companies

ICAI'S ANSWER

Income Statement of companies A, B and C

Particulars	A	B	C
Sales	₹15,00,000	₹30,00,000	₹41,66,667
Less: Variable Expenses	₹9,00,000	₹15,00,000	₹16,66,667
Contribution	₹6,00,000	₹15,00,000	₹25,00,000

Particulars	A	B	C
Less: Fixed Cost	₹4,50,000	₹10,00,000	₹15,00,000
EBIT	₹1,50,000	₹5,00,000	₹10,00,000
Less: Interest	₹1,00,000	₹4,00,000	₹6,00,000
PBT	₹50,000	₹1,00,000	₹4,00,000
Less: Tax @ 30%	₹15,000	₹30,000	₹1,20,000
PAT	₹35,000	₹70,000	₹2,80,000

Working Notes:

(i) Degree of Financial Leverage = $\frac{\text{EBIT}}{\text{EBIT - Interest}}$

DFL x (EBIT – Int) = EBIT

DFL x EBIT – Int x DFL= EBIT

DFL x EBIT – EBIT =Int x DFL

EBIT(DFL – 1) = Int x DFL

$\text{EBIT} = \frac{\text{int x DFL}}{\text{DFL - 1}}$

For A,

$\text{EBIT}_A = \frac{₹\ 1{,}00{,}000 \text{ x } 3}{3 - 1}$

EBIT_A **= ₹150000**

For B

$\text{EBIT}_B = \frac{₹\ 4{,}00{,}000 \text{ x } 5}{5 - 1}$

EBIT_B **= ₹500000**

For C

$\text{EBIT}_c = \frac{₹\ 6{,}00{,}000 \text{ x } 2.5}{2.5 - 1}$

EBIT_c **=10,00,000**

(ii) DOL= $\frac{\text{Contribution}}{\text{EBIT}}$

Contribution = DOL x EBIT

Contribution_A = 4 x ₹1,50,000

Contribution_A **= ₹6,00,000**

Contribution_B = 3 x ₹5,00,000

Contribution_B **= ₹15,00,000**

Contribution_C = 2.5 x ₹10,00,000

Contribution_C **= ₹25,00,000**

(iii) Fixed Cost = Contribution – EBIT

Fixed Cost_A= ₹6,00,000 – ₹1,50,000 = ₹4,50,000

Fixed Cost_B =₹15,00,000 – ₹5,00,000 = ₹10,00,000

Fixed Cost_C = ₹25,00,000 – ₹10,00,000 = ₹15,00,000

(iv) Contribution= Sales – VC

VC= Sales – Contribution

Sales x VC Ratio= Sales – Contribution

Contribution= Sales – Sales x VC Ratio

Contribution=Sales(1-VCR)

$$\text{Sales} = \frac{\text{Contribution}}{\text{1- VCR}}$$

Sales_A = ₹6,00,000/(1-0.6) = ₹15,00,000

Sales_B = ₹15,00,000/(1-0.5) = ₹30,00,000

Sales_C = ₹25,00,000/(1-0.4) = ₹41,66,667

Of all the companies, A has the highest degree of Operating Leverage, B has highest degree of Financial Leverage and C is equally leveraged on both Operating and Financial fronts. If we consider combined leverage companies will have the leverages of 12, 15 and 6.25 (by multiplying both operating and financial leverages). This means A is undertaking a higher degree of operating risk while B is undertaking a higher degree of financial risk.

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437 RTP **Old P8 Nov 2022** Q4 old scheme

Debu Ltd. currently has an equity share capital of ₹ 1,30,00,000 consisting of 13,00,000 Equity shares. The company is going through a major expansion plan requiring to raise funds to the tune of ₹ 78,00,000. To finance the expansion the management has following plans:

Plan-I : Issue 7,80,000 Equity shares of ₹ 10 each.

Plan-II : Issue 5,20,000 Equity shares of ₹ 10 each and the balance through long-term borrowing at 12% interest p.a.

Plan-III : Issue 3,90,000 Equity shares of ₹ 10 each and 39,000, 9% Debentures of ₹ 100 each.

Plan-IV : Issue 3,90,000 Equity shares of ₹ 10 each and the balance through 6% preference shares.

EBIT of the company is expected to be ₹ 52,00,000 p.a.

Considering corporate tax rate @ 40%, you are required to-

(i) CALCULATE EPS in each of the above plans.

(ii) ASCERTAIN financial leverage in each plan and comment.

ICAI'S ANSWER

Sources of Capital	Plan I	Plan II	Plan III	Plan IV
Present Equity Shares	13,00,000	13,00,000	13,00,000	13,00,000
New Issue	7,80,000	5,20,000	3,90,000	3,90,000
Equity share capital (₹)	2,08,00,000	1,82,00,000	1,69,00,000	1,69,00,000
No. of Equity shares	20,80,000	18,20,000	16,90,000	16,90,000
12% Long term loan (₹)	–	26,00,000	–	–
9% Debentures (₹)	–	–	39,00,000	–
6% Preference Shares (₹)	–	–	–	39,00,000

Computation of EPS and Financial Leverage

Sources of Capital	Plan I	Plan II	Plan III	Plan IV
EBIT (₹)	52,00,000	52,00,000	52,00,000	52,00,000
Less: Interest on 12% Loan (₹)	–	3,12,000	–	–
Less: Interest on 9% debentures (₹)	–	–	3,51,000	–
EBT (₹)	52,00,000	48,88,000	48,49,000	52,00,000
Less: Tax@ 40%	20,80,000	19,55,200	19,39,600	20,80,000
EAT (₹)	31,20,000	29,32,800	29,09,400	31,20,000
Less: Preference Dividends (₹)	–	–	–	2,34,000
(a) Net Earnings available for equity shares (₹)	31,20,000	29,32,800	29,09,400	28,86,000
(b) No. of equity shares	20,80,000	18,20,000	16,90,000	16,90,000
(c) EPS (a ÷ b) (₹)	1.50	1.61	1.72	1.71
Financial leverage $\left(\frac{EBIT}{EBT}\right)$	1.00	1.06	1.07	1.08*

* Financial Leverage in the case of Preference dividend = $\left(\frac{\text{EBIT}}{(\text{EBIT - Interest}) - \left(\frac{\text{Dp}}{(1-t)}\right)}\right)$

$= \left(\frac{52,00,000}{(52,00,000 - 0) - \left(\frac{2,34,000}{(1-.40)}\right)}\right) = \left(\frac{52,00,000}{48,10,000}\right) = 1.08$

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438 RTP Old P8 May 2022 Q4 old scheme

Company P and Q are having same earnings before tax. However, the margin of safety of Company P is 0.20 and, for Company Q, is 1.25 times than that of Company P. The interest expense of Company P is ₹ 1,50,000 and, for Company Q, is 1/3rd less than that of Company P. Further, the financial leverage of Company P is 4 and, for Company Q, is 75% of Company P.

Other information is given as below:

Particulars	Company P	Company Q
Profit volume ratio	25%	33.33%
Tax rate	45%	45%

You are required to PREPARE Income Statement for both the companies.

ICAI'S ANSWER

Income Statement

Particulars	Company P (₹)	Company Q (₹)
Sales	40,00,000	18,00,000
Less: Variable Cost	30,00,000	12,00,000
Contribution	10,00,000	6,00,000
Less: Fixed Cost	8,00,000	4,50,000
EBIT	2,00,000	1,50,000
Less: Interest	1,50,000	1,00,000
EBT	50,000	50,000
Tax (45%)	22,500	22,500
EAT	27,500	27,500

Workings:

(i) Margin of Safety

For Company P = 0.20

For Company Q = 0.20 x 1.25 = 0.25

(ii) Interest Expenses

For Company P = ₹ 1,50,000

For Company Q = ₹ 1,50,000 (1-1/3) = ₹ 1,00,000

(iii) Financial Leverage

For Company P = 4

For Company Q = 4 x 75% = 3

(iv) EBIT

For Company A

Financial Leverage = EBIT/(EBIT- Interest)

4 = EBIT/(EBIT- ₹ 1,50,000)

4EBIT – ₹ 6,00,000 = EBIT

3EBIT = ₹ 6,00,000

EBIT = ₹ 2,00,000

For Company B

Financial Leverage = EBIT/(EBIT - Interest)

3 = EBIT/(EBIT – ₹ 1,00,000)

3EBIT – ₹ 3,00,000 = EBIT

2EBIT = ₹ 3,00,000

EBIT = ₹ 1,50,000

(v) Contribution

For Company A

Operating Leverage = 1/Margin of Safety

= 1/0.20 = 5

Operating Leverage = Contribution/EBIT

5 = Contribution/₹ 2,00,000

Contribution = ₹ 10,00,000

For Company B

Operating Leverage = 1/Margin of Safety

= 1/0.25 = 4

Operating Leverage = Contribution/EBIT

4 = Contribution/₹ 1,50,000

Contribution = ₹ 6,00,000

(vi) Sales

For Company A

Profit Volume Ratio = 25%

Profit Volume Ratio = Contribution/Sales × 100

25% = ₹ 10,00,000/Sales

Sales = ₹ 10,00,000/25%

Sales = ₹ 40,00,000

For Company B

Profit Volume Ratio = 33.33%

Therefore, Sales = ₹ 6,00,000/33.33%

Sales = ₹ 18,00,000

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439 RTP **Old P8 Dec 2021 Q4** old scheme

The following particulars relating to Navya Ltd. for the year ended 31st March 2021 is given:

Output	1,00,000 units at normal capacity
Selling price per unit	₹ 40
Variable cost per unit	₹ 20
Fixed cost	₹ 10,00,000

The capital structure of the company as on 31st March, 2021 is as follows:

Particulars	₹
Equity share capital (1,00,000 shares of ₹ 10 each)	10,00,000
Reserves and surplus	5,00,000
7% debentures	10,00,000
Current liabilities	5,00,000
Total	30,00,000

Navya Ltd. has decided to undertake an expansion project to use the market potential, that will involve ₹ 10 lakhs. The company expects an increase in output by 50%. Fixed cost will be increased by ₹ 5,00,000 and variable cost per unit will be decreased by 10%. The additional output can be sold at the existing selling price without any adverse impact on the market.

The following alternative schemes for financing the proposed expansion programme are planned:

(i) Entirely by equity shares of ₹ 10 each at par.

(ii) ₹ 5 lakh by issue of equity shares of ₹ 10 each and the balance by issue of 6% debentures of ₹ 100 each at par.

(iii) Entirely by 6% debentures of ₹ 100 each at par.

FIND out which of the above-mentioned alternatives would you recommend for Navya Ltd. with reference to the risk and return involved, assuming a corporate tax of 40%.

ICAI'S ANSWER

Statement showing Profitability of Alternative Schemes for Financing

(₹ in '00,000)

Particulars	Existing	Alternative Schemes		
		(i)	(ii)	(iii)
Equity Share capital (existing)	10	10	10	10

Particulars	Existing	Alternative Schemes		
New issues	-	10	5	-
	10	**20**	**15**	**10**
7% debentures	10	10	10	10
6% debentures	-	-	5	10
	20	**30**	**30**	**30**
Debenture interest (7%)	0.7	0.7	0.7	0.7
Debenture interest (6%)	-	-	0.3	0.6
	0.7	**0.7**	**1.0**	**1.3**
Output (units in lakh)	1	1.5	1.5	1.5
Contribution per. unit (₹) (Selling price - Variable Cost)	**20**	**22**	**22**	**22**
Contribution (₹ lakh)	**20**	**33**	**33**	**33**
Less: Fixed cost	10	15	15	15
EBIT	**10**	**18**	**18**	**18**
Less: Interest (as calculated above)	0.7	0.7	1.0	1.3
EBT	**9.3**	**17.3**	**17**	**16.7**
Less: Tax (40%)	3.72	6.92	6.8	6.68
EAT	**5.58**	**10.38**	**10.20**	**10.02**
Operating Leverage (Contribution /EBIT)	**2.00**	**1.83**	**1.83**	**1.83**
Financial Leverage (EBIT/EBT)	**1.08**	**1.04**	**1.06**	**1.08**
Combined Leverage (Contribution/EBT)	**2.15**	**1.91**	**1.94**	**1.98**
EPS (EAT/No. of shares) (₹)	**5.58**	**5.19**	**6.80**	**10.02**
Risk	**-**	**Lowest**	**Lower than option (3)**	**Highest**
Return	**-**	**Lowest**	**Lower than option (3)**	**Highest**

From the above figures, we can see that the Operating Leverage is same in all alternatives though Financial Leverage differs. Alternative (iii) uses the maximum amount of debt and result into the highest degree of financial leverage, followed by alternative (ii). Accordingly, risk of the company

will be maximum in these options. Corresponding to this scheme, however, maximum EPS (i.e., ₹ 10.02 per share) will be also in option (iii).

So, if Navya Ltd. is ready to take a high degree of risk, then alternative (iii) is strongly recommended. In case of opting for less risk, alternative (ii) is the next best option with a reduced EPS of ₹ 6.80 per share. In case of alternative (i), EPS is even lower than the existing option, hence not recommended.

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440 RTP **Old P8 Jul 2021 Q4** old scheme

Following information has been extracted from the accounts of newly incorporated Textyl Pvt. Ltd. for the Financial Year 2020-21:

Sales	₹ 15,00,000
P/V ratio	70%
Operating Leverage	1.4 times
Financial Leverage	1.25 times

Using the concept of leverage, find out and verify in each case:

(i) The percentage change in taxable income if sales increase by 15%.

(ii) The percentage change in EBIT if sales decrease by 10%.

(iii) The percentage change in taxable income if EBIT increase by 15%.

ICAI'S ANSWER

Workings:

1. Contribution = Sales x P/V ratio
   = ₹ 15,00,000 x 70% = ₹ 10,50,000
2. Operating Leverage = $\frac{\text{Contribution}}{\text{Earnings before interest and tax (EBIT)}}$
   Or, $1.4 = \frac{₹\ 10,50,000}{\text{EBIT}}$
   EBIT = ₹ 7,50,000
3. Financial leverage = $\frac{\text{EBIT}}{\text{EBT}}$
   Or, $1.25 = \frac{₹\ 7,50,000}{\text{EBT}}$
   EBT = ₹ 6,00,000
4. Fixed Cost = Contribution – EBIT
   = ₹ 10,50,000 – ₹ 7,50,000 = ₹ 3,00,000
5. Interest = EBIT – EBT
   = ₹ 7,50,000 – ₹ 6,00,000 = ₹ 1,50,000
6. Income Statement

Particulars	Amount (₹)
Sales	15,00,000
Less: Variable cost (30% of ₹ 15,00,000)	4,50,000
Contribution (70% of ₹ 15,00,000)	10,50,000
Less: Fixed costs	3,00,000
Earnings before interest and tax (EBIT)	7,50,000
Less: Interest	1,50,000
Earnings before tax (EBT)	6,00,000

(i) Combined Leverage = $\frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 10{,}50{,}000}{₹\ 6{,}00{,}000}$ = **1.75 times**

Or, Combined Leverage = Operating Leverage x Financial Leverage

= 1.4 x 1.25 = **1.75 times**

So, if sales is increased by 15% then taxable income (EBT) will be increased by 1.75 × 15% = **26.25%**

Verification

Particulars	Amount (₹)
New Sales after 15% increase (₹ 15,00,000 + 15% of ₹ 15,00,000)	17,25,000
Less: Variable cost (30% of ₹ 17,25,000)	5,17,500
Contribution (70% of ₹ 17,25,000)	12,07,500
Less: Fixed costs	3,00,000
Earnings before interest and tax (EBIT)	9,07,500
Less: Interest	1,50,000
Earnings before tax after change (EBT)	7,57,500

Increase in Earnings before tax (EBT) = ₹ 7,57,500 - ₹ 6,00,000 = ₹ 1,57,500

So, percentage change in Taxable Income (EBT) = $\frac{₹\ 1{,}57{,}500}{₹\ 6{,}00{,}000} \times 100$ = **26.25%, hence verified.**

(ii) Degree of Operating Leverage (Given) = **1.4 times**

So, if sales is decreased by 10% then EBIT will be decreased by 1.4 × 10% = **14%**

Verification

Particulars	Amount (₹)
New Sales after 10% decrease (₹ 15,00,000 - 10% of ₹ 15,00,000)	13,50,000
Less: Variable cost (30% of ₹ 13,50,000)	4,05,000
Contribution (70% of ₹ 13,50,000)	9,45,000
Less: Fixed costs	3,00,000
Earnings before interest and tax after change (EBIT)	6,45,000

Decrease in Earnings before interest and tax (EBIT) = ₹ 7,50,000 - ₹ 6,45,000 = ₹ 1,05,000

So, percentage change in EBIT = $\frac{₹\ 1,05,000}{₹\ 7,50,000} \times 100$ = **14%, hence verified.**

(iii) Degree of Financial Leverage (Given) = **1.25 times**

So, if EBIT increases by 15% then Taxable Income (EBT) will be increased by 1.25 × 15% = **18.75%**

Verification

Particulars	Amount (₹)
New EBIT after 15% increase (₹ 7,50,000 + 15% of ₹ 7,50,000)	8,62,500
Less: Interest	1,50,000
Earnings before Tax after change (EBT)	7,12,500

Increase in Earnings before Tax = ₹ 7,12,500 - ₹ 6,00,000 = ₹ 1,12,500

So, percentage change in Taxable Income (EBT) = $\frac{₹\ 1,12,500}{₹\ 6,00,000} \times 100$ = **18.75%, hence verified.**

ICAI · Old P8 · RTP Jul 2021 · question p. 3 · answer p. 11-14

441 RTP **Old P8 Nov 2020 Q4** old scheme

The capital structure of PS Ltd. for the year ended 31st March, 2020 consisted as follows:

Particulars	Amount in ₹
Equity share capital (face value ₹ 100 each)	10,00,000
10% debentures (₹ 100 each)	10,00,000

During the year 2019-20, sales decreased to 1,00,000 units as compared to 1,20,000 units in the previous year. However, the selling price stood at ₹ 12 per unit and variable cost at ₹ 8 per unit for both the years. The fixed expenses were at ₹ 2,00,000 p.a. and the income tax rate is 30%.

You are required to CALCULATE the following:

(a) The degree of financial leverage at 1,20,000 units and 1,00,000 units.

(b) The degree of operating leverage at 1,20,000 units and 1,00,000 units.

(c) The percentage change in EPS.

ICAI'S ANSWER

Sales in units	1,20,000	1,00,000
	(₹)	(₹)
Sales Value	14,40,000	12,00,000
Variable Cost	(9,60,000)	(8,00,000)
Contribution	4,80,000	4,00,000
Fixed expenses	(2,00,000)	(2,00,000)
EBIT	2,80,000	2,00,000
Debenture Interest	(1,00,000)	(1,00,000)
EBT	1,80,000	1,00,000
Tax @ 30%	(54,000)	(30,000)
Profit after tax (PAT)	1,26,000	70,000

(i) Financial Leverage= $\frac{\text{EBIT}}{\text{EBT}}$	$= \frac{₹\ 2,80,000}{₹\ 1,80,000}$ = 1.56	$= \frac{₹\ 2,00,000}{₹\ 1,00,000}$ = 2
(ii) Operating leverage = $\frac{\text{Contribution}}{\text{EBIT}}$	$\frac{₹\ 4,80,000}{₹\ 2,80,000}$ = 1.71	$= \frac{₹\ 4,00,000}{₹\ 2,00,000}$ = 2
(iii) Earnings per share (EPS)	$\frac{₹\ 1,26,000}{₹\ 10,000}$ = ₹ 12.6	$\frac{₹\ 70,000}{₹\ 10,000}$ = ₹ 7
Decrease in EPS	= ₹ 12.6 – ₹ 7 = ₹ 5.6	
% decrease in EPS	$\frac{5.6}{12.6}$ = x 100 = 44.44%	

ICAI · Old P8 · RTP Nov 2020 · question p. 2 · answer p. 9-10

442 RTP **Old P8 May 2020** Q4 old scheme

The following information is related to YZ Company Ltd. for the year ended 31st March, 2020:

Equity share capital (of ₹ 10 each)	₹ 50 lakhs
12% Bonds of ₹ 1,000 each	₹ 37 lakhs
Sales	₹ 84 lakhs
Fixed cost (excluding interest)	₹ 6.96 lakhs
Financial leverage	1.49
Profit-volume Ratio	27.55%
Income Tax Applicable	40%

You are required to CALCULATE:

(i) Operating Leverage;

(ii) Combined leverage; and

(iii) Earnings per share.

Show calculations up-to two decimal points.

ICAI'S ANSWER

Computation of Profits after Tax (PAT)

Particulars	Amount (₹)
Sales	84,00,000
Contribution (Sales × P/V ratio)	23,14,200
Less: Fixed cost (excluding Interest)	(6,96,000)
EBIT (Earnings before interest and tax)	16,18,200
Less: Interest on debentures (12% × ₹37 lakhs)	(4,44,000)
Less: Other fixed Interest (balancing figure)	(88,160)
EBT (Earnings before tax)	10,86,040*
Less: Tax @ 40%	4,34,416
PAT (Profit after tax)	6,51,624

(i) Operating Leverage:

$$= \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ 23,14,200}{₹\ 16,18,200} = 1.43$$

(ii) Combined Leverage:

= Operating Leverage × Financial Leverage

= 1.43 × 1.49 = 2.13

Or,

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{EBT}}$$

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 23,14,200}{₹10,86,040} = 2.13$$

$$\text{*Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 16,18,200}{\text{EBT}} = 1.49$$

$$\text{So, EBT} = \frac{₹\ 16,18,200}{1.49} = ₹10,86,040$$

Accordingly, other fixed interest

= ₹ 16,18,200 - ₹ 10,86,040 - ₹ 4,44,000 = ₹ 88,160

(iii) Earnings per share (EPS):

$$= \frac{\text{PAT}}{\text{No. of shares outstanding}} = \frac{₹\ 6,51,624}{5,00,000 \text{ equity shares}} = ₹\ 1.30$$

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443 RTP **Old P8 Nov 2019** Q4 old scheme

The following summarises the percentage changes in operating income, percentage changes in revenues, and betas for four listed firms.

Firm	Change in revenue	Change in operating income	Beta
A Ltd.	35%	22%	1.00
B Ltd.	24%	35%	1.65
C Ltd.	29%	26%	1.15
D Ltd.	32%	30%	1.20

Required:

(i) CALCULATE the degree of operating leverage for each of these firms. Comment also.

(ii) Use the operating leverage to EXPLAIN why these firms have different beta.

ICAI'S ANSWER

(i) Degree of operating leverage = $\frac{\%\ \text{Change in Operating income}}{\%\ \text{Change in Revenues}}$

A Ltd.	=	0.22 / 0.35	=	0.63
B Ltd.	=	0.35 / 0.24	=	1.46
C Ltd.	=	0.26 / 0.29	=	0.90
D Ltd.	=	0.30 / 0.32	=	0.94

It is level specific.

(ii) High operating leverage leads to high beta. So when operating leverage is lowest i.e. 0.63, Beta is minimum (1) and when operating leverage is maximum i.e. 1.46, beta is highest i.e. 1.65

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444 MTP **Sep 2026 S-I** MCQ 6 2 marks MCQ

Total Assets & Current Liabilities of Suyash Limited are ₹ 45 lakhs & ₹ 5 lakhs respectively. ROCE is 20%, measure of business operating risk (operating leverage) is 3, and P/V ratio is 80%. Calculate Sales.

A 18 lakhs

B 30 lakhs

C 24 lakhs

D 32 lakhs

ICAI'S ANSWER

(B) ROCE = EBIT / Total Capital Employed

Total Capital Employed = Total Assets – Current Liabilities = 45 lakhs – 5 lakhs = 40 lakhs

EBIT = 40 lakhs × 20% = 8 lakhs

Operating Leverage (3) = Contribution / EBIT

Therefore, Contribution = 8 lakhs × 3 = 24 lakhs

Sales = Contribution / P/V Ratio = 24 lakhs / 0.80 = 30 lakhs

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445 MTP **Sep 2026 S-I** Q1(c) 5 marks

Same question, different ICAI answer: #434 RTP May 2024

From the following financial data of Company A and Company B, PREPARE their Income Statements.

	Company A (₹)	Company B (₹)
Variable Cost	88,000	50% of sales
Fixed Cost	26,500	-
Interest Expenses	14,000	11,000
Financial Leverage	5 : 1	-
Margin of Safety	-	0.25
Income Tax Rate	30%	30%
EBIT	-	14,000

ICAI'S ANSWER

Income Statements of Company A and Company B

	Company A (₹)	Company B (₹)
Sales	1,32,000	1,12,000
Less: Variable cost	88,000	56,000
Contribution	44,000	56,000
Less: Fixed Cost	26,500	42,000
Earnings before interest and tax (EBIT)	17,500	14,000
Less: Interest	14,000	11,000
Earnings before tax (EBT)	3,500	3,000
Less: Tax @ 30%	1,050	900
Earnings after tax (EAT)	2,450	2,100

Working Notes:

Company A

(i) Financial Leverage = $\frac{\text{EBIT}}{\text{EBT i.e EBIT - Interest}}$

So, $5 = \frac{EBIT}{EBIT - 14,000}$

Or, 5 (EBIT – 14,000) = EBIT

Or, 4 EBIT = 70,000

Or, EBIT = ₹17,500

(ii) Contribution = EBIT + Fixed Cost = ₹ 17,500 + ₹ 26,500 = ₹ 44,000

(iii) Sales = Contribution + Variable cost = ₹ 44,000 + ₹ 88,000 = ₹ 1,32,000

Company B

(i) Operating Leverage =1/Margin of Safety = $\frac{Contribution}{EBIT}$ 1/0.25

$= \frac{Contribution}{₹14,000}$ $4 = \frac{Contribution}{₹14,000}$

Contribution = ₹14,000 x 4 = ₹56,000

(ii) Fixed Cost = Contribution – EBIT = 56,000 – 14,000 = ₹ 42,000

(iii) Contribution = 50% of Sales (as Variable Cost is 50% of Sales) Sales = 56,000 x 2 = ₹1,12,000

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446 MTP **Sep 2026 S-I** Q4(c) 2 marks

Also asked in MTP Sep 2024 S-II Q4(c)

EXPLAIN Financial Leverage as a 'Double edged Sword'

ICAI'S ANSWER

When the cost of 'fixed cost fund' is less than the return on investment, financial leverage will help to increase return on equity and EPS. The firm will also benefit from the saving of tax on interest on debts etc. However, when cost of debt will be more than the return it will affect return of equity and EPS unfavourably and as a result firm can be under financial distress. Therefore, financial leverage is also known as "**double edged sword".**

Effect on EPS and ROE:

When, ROI > Interest – Favourable – Advantage

When, ROI < Interest – Unfavourable – Disadvantage

When, ROI = Interest – Neutral – Neither advantage nor disadvantage

ICAI · P6 · MTP Sep 2026 · Series I · question p. 7 · answer p. 14

447 MTP **Sep 2026 S-II** Q2(b) 6 marks

Sunrise Foods Limited, a mid-sized packaged-foods company, is reviewing its cost structure and capital mix before approaching its board with next year's financial plan. Its Balance Sheet as on 31st March 2026

is given below:

Liabilities & Equity	₹ (in crores)	Assets	₹ (in crores)
Equity Share Capital (1.20 crore shares of ₹ 10 each)	12	Fixed Assets (Net)	32
Reserves and Surplus	3	Current Assets	18
12% Debentures	25		
Current Liabilities	10		
Total	**50**	**Total**	**50**

The company's finance team has also shared the following details:

- Fixed costs per annum (excluding interest): ₹ 10 crores
- Variable operating cost ratio: 60%
- Total Assets turnover ratio: 3
- Income-tax rate: 30%

As the company's financial analyst, you are required to CALCULATE the following and comment on what each result means for Sunrise Foods:

(i) Earnings per share

(ii) Operating Leverage

(iii) Financial Leverage

(iv) Combined Leverage

ICAI'S ANSWER

Total Assets = ₹ 50 crores

Total Assets Turnover Ratio = Sales ÷ Total Assets = 3

Hence, Total Sales = 50 × 3 = ₹ 150 crores

Computation of Profit after Tax (PAT)

Particulars	₹ in crores
Sales	150.00
Less: Variable operating cost @ 60%	90.00
Contribution	**60.00**
Less: Fixed cost (other than interest)	10.00
EBIT (Earnings before interest and tax)	**50.00**
Less: Interest on debentures (12% × 25)	3.00
EBT (Earnings before tax)	**47.00**
Less: Tax @ 30%	14.10

Particulars	₹ in crores
EAT (Earnings after tax)	**32.90**

(i) Earnings per Share

EPS = Earnings available to equity shareholders ÷ Number of equity shares

EPS = 32.90 crores ÷ 1.20 crore shares = ₹ 27.42

(ii) Operating Leverage

Operating Leverage = Contribution ÷ EBIT = 60 ÷ 50 = 1.20

An operating leverage of 1.20 shows that fixed costs form a modest part of Sunrise Foods' cost structure. A 1% change in sales would change EBIT by about 1.20%, so the company's operating profit is only mildly sensitive to swings in sales volume.

(iii) Financial Leverage

Financial Leverage = EBIT ÷ EBT = 50 ÷ 47 = 1.06

The financial leverage is quite low, indicating that Sunrise Foods' interest burden is small compared to its operating profit, the debenture obligation is well covered by EBIT, leaving the company financially comfortable.

(iv) Combined Leverage

Combined Leverage = Operating Leverage × Financial Leverage = 1.20 × 1.06 = 1.28

A combined leverage of 1.28 means a 1% increase in sales would raise EPS by roughly 1.28%, Sunrise Foods carries a moderate overall risk from the combination of its fixed operating costs and its use of debt.

Together, the operating, financial and combined leverages give the board a quick read on how sensitive the company's profits and shareholder earnings are to changes in sales, which is useful context before deciding on any further borrowing or capacity expansion.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 6-7 · answer p. 7-8

448 MTP **May 2026 S-I** Q2(b) 4 marks

The following particulars are related to PQR Ltd:

Sales ₹ 1,50,00,000

Variable Cost ₹ 84,00,000

Fixed Cost ₹ 12,00,000

The company has borrowed ₹ 90,00,000 at 10% and its equity capital is ₹ 1,10,00,000.

You are required to CALCULATE:

(i) Return on Investment (ROI).

(ii) Operating and Financial Leverages.

(iii) Combined Leverage.

(iv) Earnings Before Interest and Taxes (EBIT), if the sale is dropped to ₹ 1,00,00,000.

ICAI'S ANSWER

(i) ROI = EBIT/Investment

EBIT = Sales - VC – FC

= ₹ 1,50,00,000 – 84,00,000 – 12,00,000

= ₹ 54,00,000

ROI = ₹ 54,00,000/₹ 2,00,00,000 = 27%

(ii) Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹66,00,000}{₹54,00,000}$ = 1.222 times

Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{₹54,00,000}{₹45,00,000}$ = 1.2 times

(iii) Combined Leverage = OL × FL = 1.222 × 1.2 = 1.466 times.

(iv) Operating Leverage = 1.222

$1.222 = \frac{\Delta \text{ EBIT}}{1/3}$

Δ EBIT = 0.4074

New EBIT = ₹ 54,00,000 – 40.74% of 54,00,000

= ₹ 32,00,000

ICAI · P6 · MTP May 2026 · Series I · question p. 6-7 · answer p. 9

449 MTP **May 2026 S-I** Q4(c) 2 marks

A company is analyzing its risk profile using combined leverage. It observes that its Degree of Operating Leverage (DOL) is low while its Degree of Financial Leverage (DFL) is high. The management believes this combination helps balance overall risk. Based on the concept of combined leverage, EXPLAIN whether this is a suitable risk combination and JUSTIFY your answer briefly.

ICAI'S ANSWER

A combination of low DOL and high DFL is considered the best balanced combination of risk.

- Low DOL indicates lower operating (business) risk and higher EBIT
- High DFL indicates higher financial risk due to the use of debt.

This combination is suitable because the Higher financial risk is balanced by lower total business risk, resulting in a moderate and well-balanced total risk.

ICAI · P6 · MTP May 2026 · Series I · question p. 8 · answer p. 11-12

450 MTP **May 2026 S-II** Q1(b) 5 marks

Zenith Manufacturing Ltd., a mid-sized production company, is currently reviewing the impact of its capital structure on profitability and risk. The company has financed its operations through a mix of equity and debt.

The capital structure of Zenith Manufacturing Ltd. consists of:

- Equity share capital of ₹ 15,00,000 (shares of ₹ 100 each), and
- 10% debentures amounting to ₹ 15,00,000.

During the current financial year, the company experienced a 20% increase in sales volume, rising from 1,50,000 units to 1,80,000 units. The management is interested in understanding how this increase in activity level affects earnings and leverage. The selling price is ₹ 15 per unit, variable costs amount to ₹ 8 per unit and fixed expenses amount to ₹ 3,00,000. The income tax rate is assumed to be 30 per cent.

(I) You are required to CALCULATE the following:

(i) The percentage increase in earnings per share.

(ii) The degree of financial leverage at 1,50,000 units and 1,80,000 units.

(iii) The degree of operating leverage at 1,50,000 units and 1,80,000 units.

(II) COMMENT on the behaviour of operating and financial leverage in relation to increase of production from 1,50,000 to 1,80,000 units.

ICAI'S ANSWER

(I) **Determination of earnings per share (EPS)**

Sales Level	1,50,000 Units	1,80,000 Units
	(₹)	**(₹)**
Sales Revenue	22,50,000	27,00,000
Variable Cost	12,00,000	14,40,000
Contribution	10,50,000	12,60,000
Fixed Cost	3,00,000	3,00,000
EBIT	**7,50,000**	**9,60,000**
Interest	1,50,000	1,50,000
EBT	**6,00,000**	**8,10,000**
Tax (30%)	1,80,000	2,43,000
EAT	4,20,000	5,67,000
No. of Shares	15,000	15,000
EPS (₹)	**28.00**	**37.80**

(i) The percentage increase in earnings per share = $\frac{37.8 - 28}{28} \times 100$

= 35%

(ii) DFL at 1,50,000 units = $\frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 7,50,000}{₹\ 6,00,000} = 1.25$

DFL at 1,80,000 units = $\frac{\text{EBIT}}{\text{EBT}} = \frac{₹\ 9,60,000}{₹\ 8,10,000} = 1.19$

(iii) DOL at 1,50,000 units = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ 10,50,000}{₹\ 7,50,000} = 1.40$

DOL at 1,80,000 units = $\frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ 12,60,000}{₹\ 9,60,000} = 1.31$

(II) With the increase in production and sales from 1,50,000 units to 1,80,000 units, the EPS has increased by 35%, indicating improved profitability. Further, both operating leverage and financial leverage have decreased, which implies:

- Reduction in business risk (due to lower operating leverage), and
- Reduction in financial risk (due to lower financial leverage).

Hence, the overall risk profile of the company declines with an increase in of production from 1,50,000 to 1,80,000 units.

ICAI · P6 · MTP May 2026 · Series II · question p. 5 · answer p. 4-5

451 MTP **May 2026 S-II** Q4(c) 2 marks

Also asked in Old P8 MTP May 2022 S-II Q2(b)

"Operating risk is associated with cost structure, whereas financial risk is associated with capital structure of a business concern." Critically EXAMINE this statement.

ICAI'S ANSWER

"Operating risk is associated with cost structure whereas financial risk is associated with capital structure of a business concern".

Operating risk refers to the risk associated with the firm's operations. It is represented by the variability of earnings before interest and tax (EBIT). The variability in turn is influenced by revenues and expenses, which are affected by demand of firm's products, variations in prices and proportion of fixed cost in total cost. If there is no fixed cost, there would be no operating risk. Whereas financial risk refers to the additional risk placed on firm's shareholders as a result of debt and preference shares used in the capital structure of the concern. Companies that issue more debt instruments would have higher financial risk than companies financed mostly by equity.

ICAI · P6 · MTP May 2026 · Series II · question p. 9 · answer p. 11

452 MTP **Jan 2026 S-I** MCQ 6 2 marks MCQ

Consider the following information for Atharv Ltd.:

	₹ in lakhs
EBIT (Earnings before Interest and Tax)	15,750
Earnings before Tax (EBT):	7,000
Fixed Operating costs:	1,575

Calculate percentage change in earnings per share, if sales increase by 5%.

A 20%

B 19.65%

C 18.5%

D 12.375%

ICAI'S ANSWER

(D) 12.375%

Operating Leverage (OL)

$$= \frac{\text{Contribution}}{\text{EBIT}} = \frac{\text{EBIT} + \text{Fixed Cost}}{\text{EBIT}} = \frac{₹\ 15{,}750 + ₹\ 1{,}575}{15{,}750} = 1.1$$

Financial Leverage (FL)

$$= \frac{\text{EBIT}}{\text{EBT}} = \frac{15{,}750}{7{,}000} = 2.25$$

Combined Leverage (CL)

$= 1.1 \times 2.25 = 2.475$

Percentage Change in Earnings per share

$$\text{DCL} = \frac{\%\ \text{change in EPS}}{\%\ \text{change in Sales}}$$

$$2.475 = \frac{\%\ \text{change in EPS}}{5\%}$$

$\therefore$ % change in EPS = 12.375%.

Hence if sales is increased by 5%, EPS will be increased by 12.375%.

ICAI · P6 · MTP Jan 2026 · Series I · question p. 3 · answer p. 3-4

453 MTP **Jan 2026 S-I** Q2(b) 2 marks

DIFFERENTIATE between Business risk and Financial risk.

ICAI'S ANSWER

Business Risk and Financial Risk: Business risk refers to the risk associated with the firm's operations. It is an unavoidable risk because of the environment in which the firm has to operate and the business risk is represented by the variability of earnings before interest and tax (EBIT). The variability in turn is influenced by revenues and expenses. Revenues and expenses are affected by demand of firm's products, variations in prices and proportion of fixed cost in total cost.

Whereas, Financial risk refers to the additional risk placed on firm's shareholders as a result of debt use in financing. Companies that issue more debt instruments would have higher financial risk than companies financed mostly by equity. Financial risk can be measured by ratios such as firm's financial leverage multiplier, total debt to assets ratio etc.

ICAI · P6 · MTP Jan 2026 · Series I · question p. 6 · answer p. 10

454 MTP **Jan 2026 S-I** Q3(b) 4 marks

The Capital structure of Magnus Ltd. is as follows:

	(₹)
Equity Share of ₹ 10 each	8,00,000
10% Preference Share of ₹ 100 each	5,00,000
12% Debentures of ₹ 100 each	7,00,000
	20,00,000

Additional Information:

- Profit after tax (Tax Rate 30%) are ₹ 2,80,000

- Operating Expenses (including Depreciation ₹ 96,800) are 1.5 times of EBIT
- Equity Dividend paid is 15%
- Market price of Equity Share is ₹ 23

CALCULATE:

(i) Operating and Financial Leverage

(ii) Cover for preference and equity dividend

ICAI'S ANSWER

Working Notes:

	(₹)
Net Profit after Tax	2,80,000
Tax @ 30%	1,20,000
EBT	4,00,000
Interest on Debentures	84,000
EBIT	4,84,000
Operating Expenses (1.5 times of EBIT)	7,26,000
Sales	12,10,000

(i) **Operating Leverage**

$$= \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹\ (12,10,000 - 6,29,200)}{₹\ 4,84,000} = \frac{₹\ 5,80,800}{₹\ 4,84,000} = 1.2 \text{ times}$$

Financial Leverage $= \frac{\text{EBIT}}{\text{EBT}} = \frac{4,84,000}{4,00,000} = 1.21$ times

(ii) **Cover for Preference Dividend**

$$= \frac{\text{PAT}}{\text{Preference Share Dividend}}$$

$$= \frac{₹\ 2,80,000}{₹\ 50,000} = 5.6 \text{ times}$$

Cover for Equity Dividend

$$= \frac{(\text{PAT - Preference Dividend})}{\text{Equity Share Dividend}} = \frac{₹\ (2,80,000 - 50,000)}{₹\ 1,20,000}$$

$$= \frac{₹\ 2,30,000}{₹\ 1,20,000} = 1.92 \text{ times}$$

ICAI · P6 · MTP Jan 2026 · Series I · question p. 7 · answer p. 13-14

455 MTP Jan 2026 S-II MCQ 7 2 marks MCQ

A company operates at a production level of 5,000 units. The contribution is ₹ 60 per unit, operating leverage is 6, combined leverage is 24. If tax rate is 30%, what would be its earnings after tax?

A ₹ 8,750

B ₹ 30,000

C ₹ 9,500

D ₹ 10,000

ICAI'S ANSWER

(A) ₹ 8,750

Computation of Earnings after tax (EAT) or Profit after tax (PAT)

Total contribution = 5,000 units x ₹ 60/unit = ₹ 3,00,000

Operating leverage (OL) x Financial leverage (FL) = Combined leverage (CL)

$\therefore 6 \times FL = 24 \therefore FL = 4$

$\therefore OL = \frac{\text{Contribution}}{\text{EBIT}} \therefore 6 = \frac{₹3,00,000}{\text{EBIT}} \therefore \text{EBIT} = ₹\ 50,000$

$FL = \frac{\text{EBIT}}{\text{EBT}} \therefore 4 = \frac{₹50,000}{\text{EBT}} \therefore \text{EBT} = ₹\ 12,500$

Since tax rate is 30%, therefore, Earnings after tax = 12,500 x 0.70 = ₹ 8,750

Earnings after tax (EAT) = ₹ 8,750

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456 MTP **Jan 2026 S-II** Q3(a) 7 marks

ShirtCo is a garment manufacturing company producing and selling 10,000 shirts every year.

The company's cost and revenue structure is as follows:

Particulars	Amount (₹ / per unit)
Selling Price	500
Variable Cost	200
Fixed Cost (Total)	25,00,000

The company's Chief Accountant has calculated the Degree of Operating Leverage (DOL) as 6 times and the Degree of Combined Leverage (DCL) as 12 times, based on the current level of production and sales.

He claims that "If sales increase by 10%, the EBIT will increase by 60%."

As a financial analyst, you are asked to verify whether his contention is correct under the following independent situations:

1. Case 1: Production and sales quantity increase by 10%, while selling price and costs remain unchanged.
2. Case 2: The selling price per shirt increases by 10%, while the sales quantity and cost remains unchanged.
3. Case 3: The selling price per shirt increases by 10%, but the price elasticity of demand is 1.2, leading to a change in sales quantity accordingly.

Required:

(i) CALCULATE the new EBIT under each case.

(ii) DETERMINE the percentage change in EBIT compared to the original level under each case.

(iii) COMMENT on the accuracy of the Chief Accountant's claim in each situation. If not correct, give reasons.

(iv) Assuming the company is tax exempt and the company has 10% Debentures amounting to INR 25 Lakhs and 10% Preference Shares amounting to INR 10 Lakhs, Comment whether the financial leverage computed by the Chief Accountant is correct. If not, COMPUTE the correct DFL and DCL. [Consider the base case for this part. Ignore the changes in quantity and price.]

ICAI'S ANSWER

Calculation of Current EBIT

Particulars	Amount (₹)
Sales (10,000 × 500)	50,00,000
Variable cost (10,000 × 200)	20,00,000
Contribution	30,00,000
Fixed cost	25,00,000
EBIT	**5,00,000**

Case 1 — Quantity (production & sales) increases by 10% (price & costs unchanged)

New quantity = 10,000 × 1.10 = **11,000 units**

Particulars	Amount (₹)
Contribution (11,000 × 300)	33,00,000
Fixed cost	25,00,000
New EBIT	**8,00,000**

% Change in EBIT

$$\frac{8,00,000 - 5,00,000}{5,00,000} \times 100 = 60\%$$

Conclusion (Case 1): The accountant's statement is correct here. A 10% rise in quantity produces a 60% rise in EBIT.

Case 2 — Selling price per unit increases by 10% (quantity unchanged; variable cost unchanged)

New price = 500 × 1.10 = **550 pu**

Variable cost per unit = 200

Contribution per unit = 350

Quantity = 10,000

Particulars	Amount (₹)
Contribution (10,000 × 350)	35,00,000
Fixed cost	25,00,000
New EBIT	**10,00,000**

% Change in EBIT

$$\frac{10,00,000 - 5,00,000}{5,00,000} \times 100 = 100\%$$

Conclusion (Case 2): The accountant's 60% prediction is not correct. A 10% price increase (with volume fixed) doubles EBIT here — a 100% increase — because unit contribution rises from 300 to 350 (a 16.67% increase in contribution per unit) while fixed costs stay the same.

Case 3 — Price increases by 10% but demand responds with price elasticity = 1.2

Price elasticity of demand (absolute value) = 1.2 means a 10% price increase causes a 12% decrease in quantity (elastic demand). Price and Quantity demanded are inversely correlated.

New price = 550 pu (10% ↑)

Particulars	Amount (₹)
Contribution per unit	350
Total contribution (8,800 × 350)	30,80,000
Fixed cost	25,00,000
New EBIT	**5,80,000**

% Change in EBIT

$$\frac{5,80,000 - 5,00,000}{5,00,000} \times 100 = 16\%$$

Conclusion (Case 3): The accountant's 60% prediction is wrong. Because quantity falls (elastic demand) the net effect is only a 16% increase in EBIT.

Reason: Operating Leverage means percentage change in EBIT with respect to percentage change in sales quantity. OL is a function of Fixed cost, variable contribution margin and Sales Volume. Hence, OL will be justified if there is change in sales volume i.e. Quantity.

(iv) DCL = DOL x DFL

12 = 6 x DFL

DFL = 2

Earnings Before Tax (EBT)

EBT = 5,00,000 - 2,50,000 = 2,50,000

Correct Degree of Financial Leverage (DFL)

$$\textbf{DFL} = \frac{\text{EBIT}}{\text{EBT} - \text{Preference Dividend}}$$

$$= \frac{5,00,000}{2,50,000 - 1,00,000}$$

= 3.33

Hence, DCL = 6 x 3.33 = 20

Therefore, the financial leverage computed by the Chief Accountant is not correct

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457 MTP **Sep 2025 S-II** MCQ 7 2 marks MCQ

A company has the following details:

Sales	₹ 50,00,000
Variable Costs	₹ 30,00,000
Fixed Operating Costs	₹ 10,00,000
Interest Expense	₹ 5,00,000

If sales increase by 10%, what will be the percentage increase in Earnings Per Share (EPS)?

A 10%

B 20%

C 30%

D 40%

ICAI'S ANSWER

(D) 40%

Calculation of Degree of Operating Leverage (DOL)

Contribution = Sales - Variable Costs

= ₹ 50,00,000 - ₹30,00,000 = ₹ 20,00,000

EBIT = Contribution - Fixed Costs

= ₹ 20,00,000 - ₹ 10,00,000 = ₹ 10,00,000

Calculation of Degree of Combined Leverage (DCL)

$$\text{DCL} = \frac{\text{Contribution}}{\text{EBT}}$$

= ₹ 20,00,000 / ₹ 5,00,000 = 2

Applying DCL to % change in sales

If sales increase by 10%, EPS will increase by:

DCL × % change in sales = 4 × 10% = 40%

ICAI · P6 · MTP Sep 2025 · Series II · question p. 3-4 · answer p. 4

458 MTP **Sep 2025 S-II** Q2(a) 5 marks

ABC Engineering Ltd., a mid-sized capital-intensive manufacturing company, is evaluating the risk-return profile of its operations. The company currently operates at 75% of its production capacity. The following information relates to its current operations:

Income Statement at 75% Capacity

Particulars	Amount (₹)
Sales Revenue	7,50,00,000
Variable Cost	4,50,00,000
Fixed Operating Costs	1,20,00,000
EBIT (Earnings Before Interest & Tax)	?
Interest on Debt (12% Debentures)	?
EBT (Earnings Before Tax)	?
Tax Rate	30%

Additional Information:

1. The company has equity share capital of ₹2,00,00,000 (shares of ₹ 10 each, fully paid-up).

2. The current Debt-Equity ratio is 0.75:1, and the company is considering increasing its production to 90% capacity.

3. At 90% capacity:

- Sales and variable cost per unit will increase proportionately.
- Fixed operating costs will increase by ₹ 10,00,000 due to additional maintenance, supervisory staff, and overheads.
- To finance the additional working capital and fixed overheads, the company is considering issuing additional ₹ 50,00,000 in 13% debentures.

4. The management wants to analyze the impact of increased capacity on Operating Leverage, Financial Leverage, and Combined Leverage and the change in EPS (Earnings Per Share) under the new financial plan.

You are required to:

(i) At 75% Capacity (Current Scenario)

(a) CALCULATE EPS by filling the missing figures.

(b) CALCULATE Degree of Operating Leverage (DOL), Degree of Financial Leverage (DFL) and Degree of Combined Leverage (DCL).

(ii) At 90% Capacity with Revised Financial Plan

(a) CALCULATE new DOL, DFL, and DCL.

(b) Revised EPS.

(iii) ADVICE as to whether the company proceed with the capacity expansion and debt issue.

ICAI'S ANSWER

(i) Current Scenario at 75% Capacity

(a) Calculation of EBIT, EBT, Profit after tax and Earning Per Share

Particulars	Amount (₹)
Sales Revenue	7,50,00,000
Less: Variable Cost	4,50,00,000
Contribution	3,00,00,000
Less: Fixed Cost	1,20,00,000
EBIT	1,80,00,000
Less: Interest (1,50,00,000 × 12%)	18,00,000
EBT	1,62,00,000
Less: Tax@30%	48,60,000
Profit after tax	1,13,40,000
No. of Equity Shares	20,00,000
Earnings Per Share	5.67

(b) Calculation of Various Leverages

Degree of Operating Leverage (DOL):

$$\text{DOL} = \frac{\text{Contribution}}{\text{EBIT}} = 3,00,00,000 / 1,80,00,000 = 1.67$$

Degree of Financial Leverage (DFL):

$$\text{DFL} = \frac{\text{EBIT}}{\text{EBT}} = 1,80,00,000 / 1,62,00,000 = 1.11$$

Degree of Combined Leverage (DCL):

$$\text{DCL} = \text{DOL} \times \text{DFL} = 1.67 \times 1.11 = 1.85$$

(ii) New Scenario at 90% Capacity

(a) Revised calculation of EBIT, EBT, Profit after tax and Earning Per Share:

Particulars	Amount (₹)
Sales Revenue [7,50,00,000 × (90/75)]	9,00,00,000
Less: Variable Cost [4,50,00,000 × (90/75)]	5,40,00,000
Contribution	3,60,00,000
Less: Fixed Cost (1,20,00,000 + 10,00,000)	1,30,00,000
EBIT	2,30,00,000
Less: Interest (1,50,00,000 × 12%) + (50,00,000 × 13%)	24,50,000
EBT	2,05,50,000

Particulars	Amount (₹)
Less: Tax@30%	61,65,000
Profit after tax	1,43,85,000
No. of Equity Shares	20,00,000
Earning Per Share (1,43,85,000 / 20,00,000)	7.19

(b) Revised Calculation of Various Leverages

DOL = 3,60,00,000 / 2,30,00,000 = 1.57

DFL = 2,30,00,000 / 2,05,50,000 = 1.12

DCL = 1.57 × 1.12 = 1.76

(c) At higher capacity utilization, the company is able to generate higher EBIT and EPS. However, it takes on slightly more financial risk by issuing 13% debentures. The company should go ahead with the capacity expansion and new financing plan, as the overall risk has been slightly reduced as indicated in the combined leverage while shareholder return by way of EPS has significantly increased.

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459 MTP May 2025 S-I MCQ 7 2 marks MCQ

A company operates at a production level of 1,000 units. The contribution is ₹ 60 per unit, operating leverage is 6, and combined leverage is 24. If tax rate is 30%, what would be its earnings after tax?

A ₹ 1,600

B ₹ 1,750

C ₹ 1,500

D ₹ 1,230

ICAI'S ANSWER

(B) ₹1,750

Computation of Earnings after tax

Contribution = ₹ 60 × 1,000 = ₹ 60,000

Operating Leverage (OL) × Financial Leverage (FL) = Combined Leverage (CL)

6 × Financial Leverage = 24

∴ Financial Leverage = 4

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹60,000}{\text{EBIT}} = 6$$

$$\text{EBIT} = \frac{60,000}{6} = ₹10,000$$

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = 4$$

$$\therefore \text{EBT} = \frac{\text{EBIT}}{4} = \frac{10,000}{4} = ₹2,500$$

EBIT- Earnings before Interest and tax.

EBT- Earnings before tax.

Since tax rate = 30%

Earnings after Tax (EAT) = EBT (1 – 0.30) [30% is tax rate]

= ₹2,500 (0.70)

∴ Earnings after Tax (EAT) = ₹1,750

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460 MTP **May 2025 S-I** Q1(b) 5 marks

The following information is available for Punyakalash Limited

Margin of Safety	0.40
Financial Leverage	1.50
Debt	1,50,000
Tax Rate	25%
Earnings Yield	12%
Interest Rate (Post tax)	9%
MPS	125
PV Ratio	30%

PREPARE Income statement and find out the number of equity shares.

ICAI'S ANSWER

Income Statement Punyakalash Limited

Particulars	Amount (₹)
Sales **(WN 4)**	4,50,000
(-) VC	3,15,000
Contribution **(WN 4)**	1,35,000
(-) Fixed Cost	81,000
EBIT **(WN 3)**	54,000
(-) Interest **(WN 2)**	18,000
EBT	36,000
(-) Tax	9,000
EAT / Net profit available to equity shareholders	27,000

$$\text{No of Equity shares} = \frac{\text{Net profit available to equity shareholders}}{\text{EPS}}$$

= 27,000 / 15

= 1,800 shares

WN 1 Calculation of EPS using Earnings yield ratio

Earnings yield ratio = EPS / MPS X 100

Therefore, EPS = 15

WN 2 Calculation of Interest cost

$$\text{Interest rate (pre-tax)} = \frac{\text{Interest rate (post tax)}}{1-t}$$

Therefore, Int Rate (pre-tax) = 9 / 0.75 = 12%

Interest Cost = 1,50,000 X 12% = 18,000

WN 3 Calculation of EBIT using FL

$$\text{FL} = \frac{\text{EBIT}}{\text{EBIT} - \text{Int}}$$

$$1.5 = \frac{\text{EBIT}}{\text{EBIT} - 18{,}000}$$

Therefore EBIT = 54,000

WN 4 Calculation of Operating Leverage (OL), Contribution & Sales

$$\text{OL} = \frac{1}{\text{MOS}} = \frac{1}{0.4} = 2.5$$

$$\text{OL} = \frac{\text{Contribution}}{\text{EBIT}}$$

$$2.5 = \frac{\text{Contribution}}{54{,}000}$$

Therefore Contribution = 1,35,000

$$\text{Sales} = \frac{\text{Contribution}}{\text{PV Ratio}} = 1{,}35{,}000 / 0.3 = 4{,}50{,}000$$

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461 MTP **May 2025 S-I** Q1(c) 5 marks

From the following information for financial year 2023-24,

Financial Leverage (FL) = 4

P/V Ratio = 40%

Tax rate = 30%

Depreciation (part of manufacturing overheads) = ₹ 10,000

Preference dividend is 15% of Operating Profit

Cash Breakeven Sales = ₹ 2,25,000

Equity Share Capital = ₹ 1,00,000

Reserves & Surplus as on 31.03.2023 = ₹ 35637

Particulars	Amount (₹)
Sales	???

Particulars	Amount (₹)
(-) Variable cost	???
Contribution	???
(-) Fixed cost	???
EBIT	???
(-) Interest exp	57,400
EBT	???
(-) Tax	???
EAT	???
(-) Preference dividend	???
Earnings for equity shareholders	???

CALCULATE –

i. Complete the Income statement for FY 2023-24

ii. Operating Leverage & Combined Leverage

iii. Percentage change in EPS, if sales increase and decreases by 7%

iv. Calculate Return on Equity Shareholders funds on 31.03.24

v. Amount of Debt, if post tax interest rate is 6.65%

ICAI'S ANSWER

WN1 – Calculation of EBIT using Financial Leverage

$$FL = \frac{EBIT}{(EBIT\text{-}Int.) - \frac{Dp}{1\text{-}t}}$$

Let EBIT be 'X'

Preference Dividend (Dp) = 0.15X

$$4 = \frac{X}{(X\text{-}57{,}400) - \frac{0.15X}{1\text{-}0.3}}$$

3.16X – 2,29,600 = X

X = EBIT = ₹ 1,06,296

Preference Dividend = 1,06,296 x 0.15 = ₹ 15,944

WN2 – Calculation of Total Fixed Cost

Cash Fixed Cost = Cash Breakeven sales x PV Ratio

= 2,25,000 x 0.40

= ₹ 90,000

Total Fixed Cost = 90,000 + 10,000 (Depreciation)

= ₹ 1,00,000

Therefore Contribution = 1,06,296 + 1,00,000 = 2,06,296

i. Income Statement for FY 2023-24:

Particulars	Amount (₹)
Sales (2,06,296 / 0.40)	**5,15,740**
(-) Variable Cost	(3,09,444)
Contribution	**2,06,296**
(-) Fixed Cost **(See Wn 2)**	(1,00,000)
EBIT **(See Wn 1)**	**1,06,296**
(-) Interest Exp	(57,400)
EBT	**48,896**
(-) Tax	(14,669)
EAT	**34,227**
(-) Preference Dividend (1,06,296 X 15%)	**(15,944)**
Earnings For Equity Shareholders	**18,283**

ii. Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}}$

= 2,06,296 / 1,06,296

= 1.94 times

Combined Leverage = OL x FL

= 4 x 1.94

= 7.76 times

iii. Combined leverage is the study from Sales to Eps, where combined leverage measures the % change in EPS due to increase/decrease in % change in sales

CL = % Change in EPS / % Change in sales

7.76 = % Change in EPS / 7

Therefore, % Change in EPS = 54.32%

If sales increase by 7%, then EPS will rise by 54.32% and on the contrary if sales decrease by 7%, then EPS will fall by 54.32%.

iv. Return on Equity shareholders' funds

$= \frac{\text{Earnings for Equity holders}}{\text{Total Eq. Shareholders' funds}}$

= 18,283 / 1,53,920

= 11.88%

v. Calculation of Amount of Debt

$$\text{Pre-tax Interest \%} = \frac{\text{Interest rate (post tax)}}{1 - t}$$

= 6.65/0.70

= 9.5%

$$\text{Debt Amount} = \frac{\text{Interest (Rs.)}}{\text{Interest}(\%)}$$

= 57,400 / 0.095

= ₹ 6,04,210

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462 MTP **May 2025 S-I** Q4(c) 2 marks

EXPLAIN the concept of "Double edged sword" in Financial leverage analysis.

ICAI'S ANSWER

When the cost of 'fixed cost fund' is less than the return on investment, financial leverage will help to increase return on equity and EPS. The firm will also benefit from the saving of tax on interest on debts etc. However, when cost of debt will be more than the return it will affect return of equity and EPS unfavourably and as a result firm can be under financial distress. Therefore, financial leverage is also known as "**double edged sword**".

Effect on EPS and ROE:

When, ROI > Interest – Favourable – Advantage

When, ROI < Interest – Unfavourable – Disadvantage

When, ROI = Interest – Neutral – Neither advantage nor disadvantage

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463 MTP **May 2025 S-II** MCQ 7 2 marks MCQ

The details of XYZ Ltd. are as follows:

Variable Cost Ratio = 60%

Operating Leverage = 5

Combined Leverage = 15

EBIT = ₹ 30,000

Tax Rate = 30%

Calculate the amount of interest.

A ₹ 40,000

B ₹ 30,000

C ₹ 20,000

D ₹ 24,000

ICAI'S ANSWER

(C) ₹ 20,000

Combined Leverage = Operating Leverage x Financial Leverage

15 = 5 x Financial Leverage

Financial Leverage = 3

Now, Financial Leverage = $\frac{\text{EBIT}}{\text{EBIT - Interest}}$

Or 3 = $\frac{30{,}000}{30{,}000 - \text{Interest}}$

90,000 – 3 Interest = 30,000

3 Interest = 60,000

So, Interest = ₹ 20,000

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464 MTP **May 2025 S-II** Q1(c) 5 marks

CALCULATE the operating leverage, financial leverage and combined leverage from the following data under Situation I and II and Financial Plan A and B:

Installed Capacity	4,000 units
Actual Production and Sales	75% of the Capacity
Selling Price	₹ 30 per unit
Variable Cost	₹ 15 per unit

Fixed Cost:

Under Situation I	₹ 15,000
Under Situation-II	₹ 20,000

Capital Structure:

	Financial Plan	
	A (₹)	**B (₹)**
Equity	10,000	15,000
Debt (Rate of Interest at 20%)	10,000	5,000
	20,000	20,000

ICAI'S ANSWER

(i) Operating leverages:

Particulars	Situation-I (₹)	Situation-II (₹)
Sales (3,000 units @ ₹ 30/- per unit)	90,000	90,000
Less: Variable Cost (VC) @ ₹ 15 per unit	(45,000)	(45,000)
Contribution (C)	45,000	45,000
Less: Fixed Cost (FC)	15,000	20,000
EBIT	30,000	25,000
Operating Leverage (C/EBIT)	45,000/30,000	45,000/25,000
	= 1.5	= 1.8

(ii) Financial Leverages:

	A (₹)	B (₹)
Situation I:		
EBIT	30,000	30,000
Less: Interest on debt	(2,000)	(1,000)
EBT	28,000	29,000
Financial Leverage (EBIT/EBT)	30,000/28,000	30,000/29,000
	= 1.07	= 1.03
Situation-II:		
EBIT	25,000	25,000
Less: Interest on debt	(2,000)	(1,000)
EBT	23,000	24,000
Financial Leverage (EBIT/EBT)	25,000/23,000	25,000/24,000
	= 1.09	= 1.04

(iii) Combined Leverages:

		A (₹)	B (₹)
(a)	Situation I	1.5 × 1.07 = 1.61	1.5 × 1.03 = 1.55
(b)	Situation II	1.8 × 1.09 = 1.96	1.8 × 1.04 = 1.87

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465 MTP **Jan 2025 S-I** MCQ 6 2 marks MCQ

A company has a degree of operating leverage is 2 and degree of financial leverage is 3. If the sales of the company increase by 5% during the next quarter, the Earning Per Share (EPS) will increase by?

A 20%

B 30%

C 50%

D 60%

ICAI'S ANSWER

(B) 30%

The formula for Degree of Combined Leverage (DCL) is:

DCL=DOL×DFL

DCL=2×3=6

The percentage change in EPS is:

% Δ EPS = DCL × % Δ Sales

% Δ EPS = 6 × 5% = 30%

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466 MTP **Jan 2025 S-I** Q1(a) 5 marks

ABC Industries is a mid-sized company manufacturing consumer goods. Last quarter, the company reported sales of ₹ 2,00,000. The production process involves significant variable costs, which account for 50% of the sales value. Additionally, the company incurs ₹ 40,000 as fixed operating costs for rent, utilities, and management expenses. ABC Industries has also borrowed funds, leading to ₹ 10,000 as annual interest on long-term debt.

The company is currently planning to launch a new marketing campaign aimed at boosting sales by 10%. As a financial analyst at ABC Industries, you are required to:

1. CALCULATE the combined leverage.
2. ILLUSTRATE the impact of the 10% sales increase using the combined leverage.

ICAI'S ANSWER

Statement showing Computation of Combined leverage

	₹
Sales	2,00,000
Less: Variable costs (50%)	1,00,000
Contribution	1,00,000
Less: Fixed operating costs	40,000
EBIT	60,000
Less: Interest	10,000
Taxable Income (PBT)	50,000

Combined leverage = $\frac{C}{PBT} = \frac{1,00,000}{50,000} = 2$

The combined leverage of '2' indicates that with every increase of ₹ 1 in sales, the taxable income will increase by ₹ 2 (i.e. 1×2). This can be verified by the following computations when the sales increase by 10%

	₹
Sales	2,20,000
Less: variable costs (50%)	1,10,000
Contribution	1,10,000
Less: Fixed operating costs	40,000
EBIT	70,000
Less: Interest	10,000
Taxable Income (PBT)	60,000

It is clear from the above computation that on account of increase in sales by 10%, the profit before tax has increased by 20%.

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467 MTP **Jan 2025 S-II** MCQ 6 2 marks MCQ

EBIT = 4,00,000

EBT = 3,00,000

Sales = 16,00,000

Which of the following is / are correct?

1. DFL is 1.33
2. Interest coverage ratio is 3
3. Operating profit margin is 25%

Select the correct answer using the code given below:

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 3 only

ICAI'S ANSWER

(C) DFL = $\frac{\text{EBIT}}{\text{EBT}}$

DFL = 4,00,000/3,00,000 = 1.33

Interest Coverage Ratio = $\frac{\text{EBIT}}{\text{Interest Expense}}$

= 4,00,000/1,00,000 = 4

Operating Profit Margin = $\frac{\text{Sales}}{\text{EBIT}}$x100

Operating Profit Margin = (4,00,000/16,00,000) × 100 = 25%

ICAI · P6 · MTP Jan 2025 · Series II · question p. 3 · answer p. 2

468 MTP **Jan 2025 S-II** MCQ 8 1 mark MCQ

Margin of safety is affected if:

1. P/V ratio changes
2. Fixed cost changes
3. Volume of sales changes

A 1 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

ICAI'S ANSWER

(D) 1, 2 and 3

ICAI · P6 · MTP Jan 2025 · Series II · question p. 3 · answer p. 2

469 MTP Jan 2025 S-II Q1(c) 5 marks

The following data relates to Beta Limited:

	Rs.
Sales	2,00,000
Less: Variable Expenses (30%)	60,000
Contribution	1,40,000
Fixed operating expenses	1,00,000
EBIT	40,000
Less: Interest	5,000
EBT	35,000

(i) CALCULATE by what percentage will EBT increase if sales increases by 6 percent.

(ii) CALCULATE by what percentage will EBIT increase if there is 10 per cent increase in sales?

(iii) CALCULATE by what percentage EBT increase if EBIT increases by 6 per cent?

ICAI'S ANSWER

(i) Increase in taxable income if sales increase by 6%.

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 1,40,000}{₹\ 35,000} = 4$$

If the sales increases by 6%, EBT will increase by 24%. (4 × 6%)

(ii) Increase in EBIT if sales increase by 10%.

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{Earnings before interest and tax}} = \frac{₹1,40,000}{₹\ 40,000} = 3.5$$

If sales increases by 10%, EBIT will increase by (3.5× 10) 35%.

(iii) Increase in taxable income if EBIT increase by 6%.

$$\text{Financial Leverage} = \frac{\text{Earnings before interest and tax(EBIT)}}{\text{EBT}} = \frac{₹\ 40,000}{₹35,000}$$

=1.14

If EBIT increases by 6%, EBT will increase by 6.8%. (1.14 × 6%)

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470 MTP Sep 2024 S-I MCQ 6 2 marks MCQ

X ltd has actual Sales of ₹ 20 lakhs and its Break-even sales are at ₹ 15 lakhs. The degree of total risk involved in the company is 6.5. Calculate the % impact on EPS, if EBIT is affected by 12%.

A 40%

B 78%

C 312%

D 19.5%

ICAI'S ANSWER

Answer key: **(D)**

(d) 19.5%

Financial Leverage (FL) indicates % impact in EPS, if EBIT is affected by 12%

FL = Combined Leverage (CL) /Operating Leverage (OL)

CL = 6.5 (Measure of total risk)

OL = 1 / Margin of Safety

Margin of Safety (MOS) = $\frac{\text{Actual Sales} - \text{B.E Sales}}{\text{Actual Sales}}$

MOS = 20 lakhs – 15 lakhs / 20 lakhs = 0.25

Therefore, OL = 1 / 0.25 = 4

So, FL = 6.5 / 4 = 1.625

So % Change in EPS = 12 x 1.625 = 19.5%

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471 MTP **Sep 2024 S-I** Q1(c) 5 marks

From the following details of X Ltd, PREPARE the Income Statement for the year ended 31st December:

Financial Leverage	2
Interest	₹ 2,000
Operating Leverage	3
Variable Cost as a Percentage of Sales	75%
Income Tax Rate	30%

ICAI'S ANSWER

Income Statement

DFL = $\frac{\text{EBIT}}{\text{EBT}} = \frac{\text{EBT} + \text{Interest}}{\text{EBT}} = \frac{\text{EBT} + 2{,}000}{\text{EBT}} = \frac{2}{1}$.

EBT + ₹ 2000 = 2 EBT.

EBT = ₹ 2,000

EBIT = EBT + Interest = ₹ 2000 + ₹ 2000 = ₹ 4,000.

Contribution $\frac{\text{Contribution}}{\text{EBIT}} = \frac{\text{Contribution}}{4{,}000} = \frac{3}{1}$

Sales $\frac{\text{Contribution}}{\text{PVR}} = \frac{12{,}000}{25\%}$	48,000
Less: Variable Cost Given = 75%	(36,000)
Contribution	12,000
Less: Fixed Cost(Contribution - EBIT = ₹ 12,000 – ₹ 4,000)	(8,000)
EBIT	4,000
Less: Interest	(2,000)
EBT	2,000
Less: Tax at 30%	(600)
EAT	1,400

ICAI · P6 · MTP Sep 2024 · Series I · question p. 4 · answer p. 6

472 MTP **Sep 2024 S-I** Q4(c) 2 marks

WHAT is the range of DOL?

ICAI'S ANSWER

DOL can never be between zero and one. It can be zero or less or it can be one or more.

When Sales is much higher than BEP sales, DOL will be slightly more than one. With decrease in sales, DOL will increase. At BEP, DOL will be infinite. When sales is slightly less than BEP, DOL will be negative infinite. With further reduction in sale, DOL will move towards zero. At zero sales, DOL will also be zero.

ICAI · P6 · MTP Sep 2024 · Series I · question p. 6 · answer p. 10

473 MTP **Sep 2024 S-II** Case scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Mathangi Ltd. is a News broadcasting channel having its broadcasting Centre in Chennai. There are total 200 employees in the organisation including top management. As a part of employee benefit expenses, the company serves tea to its employees, which is outsourced from a third-party. The company offers tea three times a day to each of its employees. The third-party charges ₹ 10 for each cup of tea. The company works for 200 days in a year.

Looking at the substantial amount of expenditure on tea, the finance department has proposed to the management an installation of a master tea vending machine from Nirmal Ltd which will cost ₹ 5,00,000 with a useful life of five years. Upon purchasing the machine, the company will have to incur annual maintenance which will require a payment of ₹ 25,000 every year. The machine would require electricity consumption of 500 units p.m. and current incremental cost of electricity for the company is ₹ 24 per unit. Apart from these running costs, the company will have to incur ₹ 8,00,000 for consumables like milk, tea powder, paper cup, sugar etc. The company is in the 25% tax bracket. Straight line method of depreciation is allowed for the purpose of taxation.

Nirmal Ltd sells 100 master tea vending machines. Variable cost is ₹ 4,50,000 per machine and fixed operating cost is ₹ 25,00,000. Capital Structure of Mathangi Ltd and Nirmal Ltd consists of the following –

Particulars	Mathangi Ltd.	Nirmal Ltd.
Equity Share Capital (Face value ₹ 10 each)	40,00,000	40,00,000
Reserves & Surplus	25,00,000	50,00,000
12% Preference Share Capital	12,00,000	Nil
15% Debentures	20,00,000	40,00,000

Risk free rate of return = 5%, Market return = 10%, Beta of the Mathangi Ltd. = 1.9

You are required to answer the following five questions based on the above details:

MCQ 1 2 marks

If sales of Nirmal Ltd are up by 10%, impact on its EBIT is

A 30%

B 60%

C 5%

D 20%

MCQ 2 2 marks

Combined leverage of Nirmal Ltd is

A 1.63

B 2.63

C 1.315

D 2

MCQ 3 2 marks

Discount rate that can be applied for making investment decisions of Mathangi Ltd is

A 12%

B 13.52%

C 15%

D 20%

MCQ 4 2 marks

Incremental cash flow after tax per annum attributable to Mathangi Ltd due to investment in the machine is

A ₹ 2,39,438

B ₹ 1,98,250

C ₹ 98,250

D ₹ 1,31,000

MCQ 5 2 marks

Net present value of investment in the machine by Mathangi Ltd is

A ₹ 6,88,522

B ₹ 1,88,522

C ₹ 9,91,250

D ₹ 4,91,250

ICAI'S ANSWERS

MCQ 1

Answer key: **(D)**

(d)

MCQ 2

Answer key: **(B)**

(b)

Particulars	Computation	Result
Sales	100 × 5,00,000	5,00,00,000
Less Variable cost	100 × 4,50,000	4,50,00,000
Contribution		**50,00,000**
Less Fixed cost		25,00,000
EBIT		**25,00,000**
Less Interest	15% × 40,00,000	6,00,000
EBT		**19,00,000**

Operating leverage = Contribution ÷ EBIT = 50 Lakhs ÷ 25 Lakhs = 2 times

Operating leverage = % Change in EBIT ÷ % Change in Sales i.e. if sales increase by 10%, EBIT increase by 20%.

Financial leverage = EBIT ÷ EBT = 25 Lakhs ÷ 19 Lakhs = 1.315 times

Combined leverage = Operating leverage × Financial leverage = 2 × 1.315 = 2.63 times

MCQ 3

Answer key: **(B)**

(b)

Particulars	Weights	Cost in %	Weights × Cost
Share Capital	40,00,000	5 + 1.9 × (10 – 5) = 14.5	5,80,000
Reserves & Surplus	25,00,000	14.5	3,62,500
Preference Share Capital	12,00,000	12	1,44,000
15% Debentures	20,00,000	15 × (1 – 25%) = 11.25	2,25,000
Total	**97,00,000**	**Total Cost**	**13,11,500**

Discount rate = WACC = 13,11,500 ÷ 97,00,000 × 100 = 13.52%

MCQ 4

Answer key: **(B)**

(b)

Particulars	Computation	Result
Savings in Tea cost	200 Employees × 200 days × 3 times × ₹ 10	12,00,000
Less: Annual maintenance		(25,000)
Less: Cost of Electricity	500 units × ₹ 24 per unit × 12 months	(1,44,000)
Less: Consumables		(8,00,000)
Less: Depreciation	5,00,000 ÷ 5 years	(1,00,000)
Profit before tax		**1,31,000**
Less: Tax	1,31,000 × 25%	32,750
Profit after tax		**98,250**
Add: Depreciation		1,00,000
Cash flow after tax	98,250 + 1,00,000	**1,98,250**

MCQ 5

Answer key: **(B)**

(b)

Year	Particulars	Cash flow	PVF@13.52%	PV
0	Initial investment	5,00,000	1	(5,00,000)
1 to 5	Savings	1,98,250	3.473	6,88,522
	Net present value			**1,88,522**

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474 MTP **Sep 2024 S-II** MCQ 6 2 marks MCQ

Total Assets & Current liabilities of the Vitrag Limited are 50 lakhs & 10 lakhs respectively. ROCE is 15%, measure of business operating risk is at 3.5 & P/V ratio is 70%. Calculate Sales.

A 21 lakhs

B 30 lakhs

C 37.50 lakhs

D 40 lakhs

ICAI'S ANSWER

Answer key: **(B)**

(b) ROCE = EBIT / Total Capital Employed

Total Capital Employed = Total Assets – Current Liabilities

= 50 lakhs – 10 lakhs

= 40 lakhs

EBIT = 40 lakhs x 15%

= 6 lakhs

Now, OL of 3.5 = Contribution / EBIT

Therefore Contribution = 6 Lakhs X 3.5 = 21 lakhs

Sales = **Contribution / PV Ratio = 21 lakhs / 0.7 = 30 lakhs**

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475 MTP **May 2024 S-I** MCQ 3 2 marks MCQ

"If EBIT increases by 6%, net profit increases by 6.9%. If sales increase by 6%, net profit will increase by 24%.

Financial leverage must be -....."

A 1.19

B 1.13

C 1.12

D 1.15

ICAI'S ANSWER

(D) 1.15

FL= % change in NP/%change in EBIT=6.9/6=1.15

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476 MTP **May 2024 S-I** Q4(c) 2 marks

EXPLAIN the concept of Financial Leverage as 'Trading on Equity'.

ICAI'S ANSWER

Financial leverage indicates the use of funds with fixed cost like long term debts and preference share capital along with equity share capital which is known as trading on equity. The basic aim of

financial leverage is to increase the earnings available to equity shareholders using fixed cost fund.

A firm is known to have a positive/favourable leverage when its earnings are more than the cost of debt. If earnings are equal to or less than cost of debt, it will be an negative/unfavourable leverage. When the quantity of fixed cost fund is relatively high in comparison to equity capital it is said that the firm is "trading on equity".

ICAI · P6 · MTP May 2024 · Series I · question p. 5 · answer p. 8

477 MTP **May 2024 S-II** MCQ 3 2 marks MCQ

Given Data: Sales is ₹ 10,00,000, Break even sales is ₹ 6,00,000.

What is the Degree of operating leverage?

A 3

B 2

C 2.5

D 2.2

ICAI'S ANSWER

Answer key: **(C)**

(c) 2.5

Margin of safety = (sales – BEP sales)/sales x 100

= 40%

Degree of operating leverage = 1/MOS

= 1/40% = 2.5

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478 MTP **May 2024 S-II** Q2(a) 4 marks

From the following PREPARE Income statement of company P and Q.

	P	Q
No. of Equity Shares	1,00,000	70,000
Financial leverage	3 : 1	4 : 1
Operating Leverage	2 : 1	3 : 1
Variable cost to sales	67%	50%
Interest	₹ 5,50,000	₹ 6,00,000
Income tax rate	30%	30%

Also CALCULATE EPS of the company.

ICAI'S ANSWER

Income statement

	Particulars	P	Q
		(₹)	**(₹)**
	Sales	50,00,000	48,00,000
(–)	Variable Cost	33,50,000	24,00,000
	Contribution	16,50,000	24,00,000
	Fixed Cost	8,25,000	16,00,000
	EBIT	8,25,000	8,00,000
(–)	Interest	5,50,000	6,00,000
	EBT	2,75,000	2,00,000
(–)	Tax	82,500	60,000
	EAT	1,92,500	1,40,000
(÷)	No. of Shares	1,00,000	70,000
	EPS	**₹ 1.93**	**₹ 2.00**

Working Note :

1. Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}} = \frac{\text{EBIT}}{(\text{EBIT} - \text{Int.})}$

Let the EBIT be X

P	Q
3 = X/(X – 5,50,000) 3(X – 5,50,000) = X 3X – 16,50,000 = X 2X = 16,50,000	4 = X/(X – 6,00,000) 4(X – 6,00,000) = X 4X – 24,00,000 = X 3X = 24,00,000
X = 8,25,000	**X = 8,00,000**

2. Operating Leverage = Contribution/EBIT

Let the Contribution be X

P	Q
2 = X/8,25,000	3= X/8,00,000
X = 16,50,000	**X = 24,00,000**

3. Sales

Let the Sales be 100

Sales – Variable Cost = Contribution

	P		Q
Contribution =	100 – 67	=	100 – 50
=	33	=	50
Sales =			
	P		**Q**
For 33	= 16,50,000	For 50	= 24,00,000
For 100	**= 50,00,000**	For 100	**= 48,00,000**

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479 MTP **Old P8 Nov 2023 S-I** Q3 10 marks old scheme

Following are the selected financial information of A Ltd. and B Ltd. for the current Financial Year:

	A Ltd.	B Ltd.
Variable Cost Ratio	60%	50%
Interest	₹ 30,000	₹ 1,20,000
Operating Leverage	6	3
Financial Leverage	4	3
Tax Rate	30%	30%

You are required to FIND out:

(i) EBIT

(ii) Sales

(iii) Fixed Cost

(iv) Identify the company which is better placed with reasons based on leverages.

ICAI'S ANSWER

Company A

(i) Financial Leverage = $\frac{\text{EBIT}}{\text{EBT i.e EBIT} - \text{Interest}}$

So, $4 = \frac{\text{EBIT}}{\text{EBIT - ₹ 30,000}}$

Or, 4 (EBIT – 30,000) = EBIT

Or, 3 EBIT = 1,20,000

Or, EBIT = 40,000

(ii) Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}}$ Or, $6 = \frac{\text{Contribution}}{₹\ 40{,}000}$

Or Contribution = ₹ 2, 40,000

$$\text{Sales} = \frac{\text{Contribution}}{\text{P/V Ratio (1 - variable cost ratio)}} = \frac{₹\ 2{,}40{,}000}{40\%} = ₹\ 6{,}00{,}000$$

(iii) Fixed Cost = Contribution – EBIT

= ₹ 2, 40,000 – 40,000

Or Fixed cost = ₹ 2,00,000

Company B

$$\text{(i) Financial Leverage} = \frac{\text{EBIT}}{\text{EBT i.e EBIT} - \text{Interest}}$$

$$\text{So, } 3 = \frac{\text{EBIT}}{\text{EBIT -1,20,000}}$$

Or, 3 (EBIT – ₹1,20,000) = EBIT

Or, 3 EBIT -₹ 3,60,000 = EBIT

Or, 2 EBIT = ₹ 3,60,000

Or, EBIT = ₹ 1,80,000

$$\text{(ii) Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}}$$

$$\text{Or, } 3 = \frac{\text{Contribution}}{₹1{,}80{,}000}$$

Or, Contribution = ₹ 5,40,000

$$\text{Sales} = \frac{\text{Contribution}}{\text{P/V Ratio (1 - variable cost ratio)}} = \frac{₹\ 5{,}40{,}000}{50\%} = ₹\ 10{,}80{,}000$$

(iii) Fixed Cost = Contribution – EBIT

= ₹ 5,40,000 – ₹ 1,80,000

Or, Fixed cost = ₹ 3,60,000

Income Statements of Company A and Company B

	Company A (₹)	Company B (₹)
Sales	6,00,000	10,80,000
Less: Variable cost	3,60,000	5,40,000
Contribution	2,40,000	5,40,000
Less: Fixed Cost	2,00,000	3,60,000
Earnings before interest and tax (EBIT)	40,000	1,80,000
Less: Interest	30,000	1,20,000
Earnings before tax (EBT)	10,000	60,000
Less: Tax @ 30%	3,000	18,000
Earnings after tax (EAT)	7,000	42,000

Comment based on Leverage

Comment based on leverage – Company B is better than company A of the following reasons:

- Capacity of Company B to meet interest liability is better than that of companies A (from EBIT/Interest ratio)
  [A = $\frac{₹40,000}{₹30,000}$ = 1.33, B = $\frac{₹1,80,000}{₹1,20,000}$ = 1.50]

  Company B has the least financial risk as the total risk (business and financial) of company B is lower (combined leverage of Company A – 24 and Company B- 9)

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480 MTP **Old P8 Nov 2023 S-I** Q6(a) 4 marks old scheme

EXPLAIN the difference between Business risk and financial risk.

ICAI'S ANSWER

Business Risk and Financial Risk

Business risk refers to the risk associated with the firm's operations. It is an unavoidable risk because of the environment in which the firm must operate, and the business risk is represented by the variability of earnings before interest and tax (EBIT). The variability in turn is influenced by revenues and expenses. Revenues and expenses are affected by demand of firm's products, variations in prices and proportion of fixed cost in total cost.

On the other hand, financial risk refers to the additional risk placed on firm's shareholders because of debt use in financing. Companies that issue more debt instruments would have higher financial risk than companies financed mostly by equity. Financial risk can be measured by ratios such as firm's financial leverage multiplier, total debt to assets ratio etc.

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481 MTP **Old P8 Nov 2023 S-II** Q1(a) 5 marks old scheme

The capital structure of AB Ltd. for the year ended 31st March, 2023 consisted as follows:

Particulars	Amount in ₹
Equity share capital (face value ₹ 100 each)	20,00,000
10% debentures (₹ 100 each)	10,00,000

During the year 2022-23, sales decreased to 2,00,000 units as compared to 2,20,000 units in the previous year. However, the selling price stood at ₹ 10 per unit and variable cost at ₹ 6 per unit for both the years. The fixed expenses were at ₹ 4,00,000 p.a. and the income tax rate is 30%.

You are required to CALCULATE the following:

(i) The degree of financial leverage at 2,20,000 units and 2,00,000 units.

(ii) The degree of operating leverage at 2,20,000 units and 2,00,000 units.

(iii) The percentage change in EPS.

ICAI'S ANSWER

Income Statement with required calculations

Particulars	(₹)	(₹)
Sales in units	2,20,000	2,00,000
Sales Value	22,00,000	20,00,000
Variable Cost	(13,20,000)	(12,00,000)
Contribution	8,80,000	8,00,000
Fixed expenses	(4,00,000)	(4,00,000)
EBIT	4,80,000	4,00,000
Debenture Interest	(1,00,000)	(1,00,000)
EBT	3,80,000	3,00,000
Tax @ 30%	(1,14,000)	(90,000)
Profit after tax (PAT)	2,66,000	2,10,000
No. of shares	20,000	20,000
(i) Financial Leverage $= \frac{\text{EBIT}}{\text{EBT}}$	$= \frac{₹\ 4,80,000}{₹\ 3,80,000}$ = 1.26	$= \frac{₹\ 4,00,000}{₹\ 3,00,000}$ = 1.33
(i) Operating Leverage $= \frac{\text{Contribution}}{\text{EBIT}}$	$= \frac{₹\ 8,80,000}{₹\ 4,80,000}$ = 1.83	$= \frac{₹\ 8,00,000}{₹\ 4,00,000}$ = 2
(iii) Earnings per share (EPS) $= \frac{\text{PAT}}{\text{No. of shares}}$	$= \frac{₹\ 2,66,000}{20,000}$ = ₹ 13.3	$= \frac{₹\ 2,10,000}{20,000}$ = ₹ 10.5
Decrease in EPS	= ₹ 13.3 – ₹ 10.5 = ₹ 2.8	
% decrease in EPS	$= \frac{2.8}{13.3} \times 100$ = 21.05%	

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482 MTP **Old P8 May 2023 S-I** Q1(a) 5 marks old scheme

Following are the selected financial Information of Alt Car Limited for the year ended 31st March 2022:

Financial Leverage	3
Interest	₹ 85,000
Operating Leverage	2
Variable cost as a percentage of sales	85%
Income tax rate	25%

You are required to PREPARE the Income Statement.

ICAI'S ANSWER

(i) Financial Leverage = $\frac{\text{EBIT}}{\text{EBIT - Interest}}$

Or, 3 = $\frac{\text{EBIT}}{\text{EBIT - Interest}}$

Or, 3 = $\frac{\text{EBIT}}{\text{EBIT -₹ 85000}}$

Or, EBIT = ₹1,27,500

(ii) Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}}$

Or, = $\frac{\text{Contribution}}{1,27,500}$ =2

Or, Contribution = ₹ 2,55,000

(iii) Sales = $\frac{\text{Contribution}}{\text{P / V Ratio}} = \frac{2,55,000}{15\%}$ = ₹17,00,000

(iv) Now, Contribution – Fixed cost = EBIT

Or ₹ 2,55,000 – Fixed cost = ₹1,27,500

Or Fixed Cost =₹1,27,500

Income Statement for the year ended 31st March 2022

Particulars	₹
Sales	17,00,000
Less: Variable Cost (85% of Rs.17,00,000)	(14,45,000)
Contribution	2,55,000
Less: Fixed Cost (Contribution - EBIT)	(1,27,500)
Earnings Before Interest and Tax (EBIT)	1,27,500
Less: Interest	(85,000)
Earnings Before Tax (EBT)	42,500
Less: Income Tax @ 25%	(10,625)
Earnings After Tax (EAT or PAT)	31,875

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483 MTP **Old P8 May 2023 S-II** Q1(c) 5 marks old scheme

Manchow Limited and Noodles Limited are generating same level of Operating Income. The margin of safety for Manchow Ltd is 0.4 and for Noodles Limited it is 1.25 times of Manchow Ltd. The Interest expense of Manchow Limited is ₹ 22,50,000 and it is 40% lower for Noodles Limited. Financial Leverages of Manchow Limited and Noodles Limited are 3 and 2 respectively. Profit Volume Ratio for both companies stand as 40% and 50% respectively. Assuming a tax rate of 30%, PREPARE income statement for both companies

ICAI'S ANSWER

Particulars	Manchow Ltd (₹)	Noodle Ltd (₹)
Sales	2,10,93,750	1,08,00,000
Less: Variable Cost	1,26,56,250	54,00,000
Contribution	84,37,500	54,00,000
Less: Fixed Cost	50,62,500	27,00,000
EBIT	33,75,000	27,00,000
Less: Interest	22,50,000	13,50,000
EBT	11,25,000	13,50,000
Less: Tax	3,37,500	4,05,000
PAT	7,87,500	9,45,000

Workings:

(i) Margin of Safety

For Manchow Ltd= 0.4

For Noodles Ltd= 0.4 x 1.25 = 0.5

(ii) Interest Expense

For Manchow Ltd = ₹ 22,50,000

For Noodles Ltd = ₹ 22,50,000 x 60%= ₹ 13,50,000

(iii) For Manchow Ltd:

Financial Leverage = 3

$$\frac{\text{EBIT}}{\text{EBT}} = \frac{\text{EBIT}}{\text{EBIT-Interest}} = 3$$

$$\frac{\text{EBIT}}{\text{EBIT - 22,50,000}} = 3$$

EBIT = 3 EBIT- 67,50,000

67,50,000 = 2 EBIT

EBIT = 33,75,000

For Noodles Ltd:

Financial Leverage = 2

$$\frac{\text{EBIT}}{\text{EBT}} = \frac{\text{EBIT}}{\text{EBIT-Interest}} = 2$$

$$\frac{\text{EBIT}}{\text{EBIT-13,50,000}} = 2$$

EBIT = 2 EBIT-27,00,000

EBIT =27,00,000

(iv) Contribution:

For Manchow Ltd

Operating Leverage = 1/ Margin of Safety

= 1/0.4

= 2.5

Operating Leverage = Contribution/EBIT

2.5 = Contribution/33,75,000

Contribution = 84,37,500

For Noodles Ltd

Operating Leverage = 1/ Margin of Safety

= 1/0.5

=2

Operating Leverage = Contribution/EBIT

2 = Contribution/27,00,000

Contribution = 54,00,000

(v) Sales:

For Manchow Ltd

P/V Ratio = 40%

P/V Ratio = Contribution/Sales

0.4 = 84,37,500/Sales

Sales = 2,10,93,750

For Noodles Ltd

P/V Ratio = 50%

P/V Ratio = Contribution/Sales

0.5 = 54,00,000/Sales

Sales = 1,08,00,000

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484 MTP **Old P8 Nov 2022 S-I** Q1(b) 5 marks old scheme

Following information is provided relating to SVB Ltd.:

Sales price	₹ 21 per unit
Variable cost	₹ 13.50 per unit
Break-even point	30,000 units

You are required to CALCULATE operating leverage at sales volume 37,500 units and 45,000 units.

ICAI'S ANSWER

Computation of Operating Leverage (OL)

Selling Price = ₹ 21 per unit

Variable Cost = ₹ 13.50 per unit

Fixed Cost = BEP × (Selling price – Variable cost) = 30,000 × (21 – 13.50) = 30,000 × 7.5 = 2,25,000

Particulars	For 37,500 units (₹)	For 45,000 units (₹)
Sales (@ ₹ 21 /unit)	7,87,500	9,45,000
Less: Variable Cost (@ 13.50 /unit)	5,06,250	6,07,500
Contribution	2,81,250	3,37,500
Less: Fixed Cost	2,25,000	2,25,000
Earnings before Interest and tax (EBIT)	56,250	1,12,500
Operating Leverage $\left(\frac{\text{Contribution}}{\text{EBIT}}\right)$	$\left(\frac{2,81,250}{56,250}\right)$	$\left(\frac{3,37,500}{1,12,500}\right)$
Operating Leverage	5 times	3 times

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 1 · answer p. 1-2

485 MTP Old P8 Nov 2022 S-II Q5(a) 8 marks old scheme

The following information is related to Navya Company Ltd. for the year ended 31st March 2022:

Equity share capital (₹ 10 each)	₹ 65,50,000
12% Bonds of ₹ 1,00 each	₹ 60,91,400
Sales	₹ 111 lakhs
Fixed cost (excluding interest)	₹ 7,15,000
Financial leverage	1.55
Profit-volume Ratio	25%
Income Tax Applicable	30%

You are required to CALCULATE:

(i) Operating Leverage.

(ii) Combined leverage; and

(iii) Earnings per share.

Show calculations upto two decimal points.

ICAI'S ANSWER

Income Statement

Particulars	Amount (₹)
Sales	1,11,00,000
Contribution (Sales × P/V ratio)	27,75,000
Less: Fixed cost (excluding Interest)	(7,15,000)
EBIT (Earnings before interest and tax)	20,60,000
Less: Interest on debentures (12% × ₹ 60,91,400)	(7,30,968)
EBT (Earnings before tax)	13,29,032
Less: Tax @ 30%	3,98,710
PAT (Profit after tax)	9,30,322

(i) Operating Leverage:

$$= \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹27,75,000}{₹20,60,000} = 1.35$$

(ii) Combined Leverage:

= Operating Leverage × Financial Leverage

= 1.35 × 1.55 = 2.09 (Approx)

Or,

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{EBT}}$$

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}} = \frac{₹\ 20,60,000}{₹13,29,032} = 2.09 \text{ (Approx)}$$

(iii) Earnings per share (EPS):

$$= \frac{\text{PAT}}{\text{No.of shares outstanding}} = \frac{₹\ 9,30,322}{6,55,000 \text{ equity shares}} = ₹\ 1.42$$

ICAI · Old P8 · MTP Nov 2022 · Series II · question p. 3-4 · answer p. 9

486 MTP **Old P8 May 2022 S-I** Q1(c) 5 marks old scheme

The capital structure of Roshan Ltd. for the year ended 31st March, 2022 consisted as follows:

Particulars	Amount (₹' 000)
Equity share capital (face value ₹ 100 each)	1,50,000
10% debentures (₹ 100 each)	1,50,000

During the year 2021-22, sales of the company decreased to 15,00,000 units as compared to 18,00,000 units in the previous year. However, the selling price stood at ₹ 120 per unit and variable cost at ₹ 80 per unit for both the years. The fixed expenses were at ₹ 3 crore p.a. and the income tax rate is 30%.

You are required to CALCULATE the following:

(i) The degree of financial leverage at 18,00,000 units and 15,00,000 units.

(ii) The degree of operating leverage at 18,00,000 units and 15,00,000 units.

(iii) The percentage change in EPS.

ICAI'S ANSWER

Income Statement with required calculations

Particulars	Previous Year	Current Year
Sales (in units)	18,00,000	15,00,000
No. of shares	15,00,000	15,00,000
	(₹' 000)	**(₹' 000)**
Sales Value	2,16,000	1,80,000
Variable Cost	(1,44,000)	(1,20,000)
Contribution	72,000	60,000
Fixed expenses	(30,000)	(30,000)
EBIT	42,000	30,000
Debenture Interest	(15,000)	(15,000)
EBT	27,000	15,000
Tax @ 30%	(8,100)	(4,500)
Profit after tax (PAT)	18,900	10,500
(i) Financial Leverage $= \frac{\text{EBIT}}{\text{EBT}}$	$= \frac{₹\ 42,000}{₹\ 27,000}$ = 1.56	$= \frac{₹\ 30,000}{₹\ 15,000}$ = 2
(ii) Operating leverage $= \frac{\text{Contribution}}{\text{EBIT}}$	$= \frac{₹\ 72,000}{₹\ 42,000}$ = 1.71	$= \frac{₹\ 60,000}{₹\ 30,000}$ = 2
(iii) Earnings per share (EPS) $= \frac{\text{PAT}}{\text{No. of shares}}$	$= \frac{₹\ 18,900}{₹\ 1,500}$ = ₹ 12.6	$= \frac{₹\ 10,500}{₹\ 1,500}$ = ₹ 7
Decrease in EPS	= ₹ 12.6 – ₹ 7 = ₹ 5.6 % decrease in EPS = $\frac{5.6}{12.6} \times 100$ = 44.44%	

ICAI · Old P8 · MTP May 2022 · Series I · question p. 1-2 · answer p. 2

487 MTP **Old P8 May 2022 S-II** Q1(b) 5 marks old scheme

From the given details, PREPARE Income Statement for Alpha Ltd. and Beta Ltd.

Particulars	Alpha Ltd.	Beta Ltd.
Operating Leverage	1.875	1.800
Financial Leverage	1.600	1.250
PV Ratio	60%	50%
Profit after tax	₹ 3,00,000	₹ 2,40,000
Tax rate	40%	40%

ICAI'S ANSWER

Particulars	Alpha Ltd. (₹)	Beta Ltd. (₹)	
Sales	25,00,000	18,00,000	
Less: Variable Cost	10,00,000	9,00,000	(Bal. fig.)
Contribution	15,00,000	9,00,000	
Less: Fixed Cost	7,00,000	4,00,000	(Bal. fig.)
EBIT	8,00,000	5,00,000	
Less: Interest	3,00,000	1,00,000	(Bal. fig.)
PBT	5,00,000	4,00,000	
Less: Tax (40%)	2,00,000	1,60,000	
PAT	3,00,000	2,40,000	

Working Note:

Particulars	Alpha Ltd.	Beta Ltd.
PAT	₹ 3,00,000	₹ 2,40,000
Tax Rate (t)	40%	40%
∴ PBT = PAT/(I-t)	$\frac{3,00,000}{1-0.4}$ = 5,00,000	$\frac{2,40,000}{1-0.4}$ = 4,00,000
Finance Leverage	1.60	1.25
∴ EBIT = PBT × FL	5,00,000 × 1.6 = 8,00,000	4,00,000 × 1.25 = 5,00,000
Operating Leverage	1.875	1.800
∴ Contribution = EBIT × OL	8,00,000 × 1.875 = 15,00,000	5,00,000 × 1.8 = 9,00,000
PV ratio	60%	50%
∴ Sales = $\frac{\text{Contribution}}{\text{PV ratio}}$	$\frac{15,00,000}{.60}$ = 25,00,000	$\frac{9,00,000}{.50}$ = 18,00,000

488 MTP **Old P8 Dec 2021 S-I** Q5(a) 8 marks old scheme

The following details of PQR Limited for the year ended 31st March, 2021 are given below:

Operating leverage	1.4
Combined leverage	2.8
Fixed Cost (Excluding interest)	₹ 2.10 lakhs
Sales	₹ 40.00 lakhs
10% Debentures of ₹ 100 each	₹ 25.00 lakhs
Equity Share Capital of ₹ 10 each	₹ 20.00 lakhs
Income tax rate	30 per cent

REQUIRED:

(i) Calculate Financial leverage

(ii) Calculate P/V ratio and Earning per Share (EPS)

(iii) If the company belongs to an industry, whose assets turnover is 1.6, does it have a high or low assets turnover?

(iv) At what level of sales, the Earning before Tax (EBT) of the company will be equal to zero?

In the question, assume that 10% Debentures and Share Capital consists of total liabilities.

ICAI'S ANSWER

(i) Financial leverage

Combined Leverage = Operating Leverage x Financial Leverage

So, financial leverage = Combined Leverage/Operating Leverage

= 2.8/1.4 = 2

(ii) P/V Ratio and EPS

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{Contribution} - \text{Fixed Cost}}$$

$$1.4 = \frac{\text{Contribution}}{\text{Contribution} - 2{,}10{,}000}$$

1.4 Contribution – 2,94,000 = Contribution

0.4 Contribution = 2,94,000

Contribution = 7,35,000

$$\text{Now, P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \text{ x } 100 = \frac{7{,}35{,}000}{40{,}00{,}000} \text{ x } 100 = 18.375\%$$

$$\text{EPS} = \frac{\text{Profit after tax (PAT)}}{\text{No. of equity shares}}$$

Earning before tax (EBT) = Contribution – Fixed Cost – Interest

= 7,35,000 – 2,10,000 – 2,50,000

= 2,75,000

Profit after tax = EBT – Tax @ 30%

= 2,75,000 – 82,500

= 1,92,500

$$\text{EPS} = \frac{1,92,500}{2,00,000} = 0.9625$$

(iii) Asset Turnover

Total Assets = Equity Share Capital + Debentures = ₹ 20 lakhs + ₹ 25 lakhs = ₹ 45 lakhs

$$\text{Asset Turnover} = \frac{\text{Sales}}{\text{Total Assets}} = \frac{40,00,000}{45,00,000} = 0.89$$

0.89 < 1.6, means lower than industry turnover.

(iv) EBT zero means 100% reduction in EBT. Since combined leverage is 2.8, sales have to be dropped by 100/2.8 = 35.71%. Hence new sales will be

40,00,000 x (100% - 35.71%) = 25,71,600

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 3-4 · answer p. 9-10

489 MTP **Old P8 Dec 2021 S-II** Q1(c) 5 marks old scheme

The capital structure of PS Ltd. for the year ended 31st March, 2021 consisted as follows:

Particulars	Amount in ₹
Equity share capital (face value ₹ 10 each)	10,000
10% debentures (₹ 100 each)	1,00,000

During the year 2020-21, sales decreased to 10,000 units as compared to 12,000 units in the previous year. However, the selling price stood at ₹ 12 per unit and variable cost at ₹ 8 per unit for both the years. The fixed expenses were at ₹ 20,000 p.a. and the income tax rate is 30%.

You are required to CALCULATE the following:

(i) The degree of financial leverage at 12,000 units and 10,000 units.

(ii) The degree of operating leverage at 12,000 units and 10,000 units.

(iii) The percentage change in EPS due to change in units sold.

ICAI'S ANSWER

Sales in units	12,000	10,000
	(₹)	**(₹)**
Sales Value	1,44,000	1,20,000
Variable Cost	(96,000)	(80,000)
Contribution	48,000	40,000
Fixed expenses	(20,000)	(20,000)

Sales in units	12,000	10,000
EBIT	28,000	20,000
Debenture Interest	(10,000)	(10,000)
EBT	18,000	10,000
Tax @ 30%	(5,400)	(3,000)
Profit after tax (PAT)	12,600	7,000
(i) Financial Leverage= EBIT/EBT	= ₹ 28,000/₹ 18,000 = 1.56	= ₹ 20,000/₹ 10,000 = 2
(ii) Operating leverage = Contribution/EBIT	= ₹ 48,000/₹ 28,000 = 1.71	= ₹ 40,000/₹ 20,000 = 2
(iii) Earnings per share (EPS)	= ₹ 12,600/₹ 1,000 = ₹ 12.6	= ₹ 7,000/₹ 1,000 = ₹ 7
Decrease in EPS	= ₹ 12.6 – ₹ 7 = ₹ 5.6	
% decrease in EPS	= 5.6/12.6 X 100 = 44.44%	

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 1-2 · answer p. 2-3

490 MTP **Old P8 Jul 2021 S-I** Q1(c) 5 marks old scheme

Following information are related to four firms of the same industry:

Firm	Change in Revenue	Change in Operating Income	Change in Earning per Share
P	25%	23%	30%
Q	27%	30%	26%
R	24%	36%	20%
S	20%	30%	20%

For all the firms, FIND OUT:

(i) Degree of operating leverage, and

(ii) Degree of combined leverage.

ICAI'S ANSWER

Calculation of Degree of Operating leverage and Degree of Combined leverage

Firm	Degree of Operating Leverage (DOL) = % change in Operating Income/% change in Revenue	Degree of Combined Leverage (DCL) = % change in EPS/% change in Revenue
P	23%/25% = 0.92	30%/25% = 1.2
Q	30%/27% = 1.11	26%/27% = 0.96
R	36%/24% = 1.50	20%/24% = 0.83
S	30%/20% = 1.50	20%/20% = 1.00

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 1-2 · answer p. 3

491 MTP **Old P8 Jul 2021 S-II** Q5 10 marks old scheme

Following data of MT Ltd. under Situations 1, 2 and 3 and Financial Plan A and B is given:

Installed Capacity (units)		3,600
Actual Production and Sales (units)		2,400
Selling price per unit (Rs.)		30
Variable cost per unit (Rs.)		20
Fixed Costs (Rs.):	Situation 1	3,000
	Situation 2	6,000
	Situation 3	9,000

Capital Structure :

Particulars	Financial Plan	
	A	**B**
Equity	Rs. 15,000	Rs. 22,500
Debt	Rs. 15,000	Rs. 7,500
Cost of Debt	12%	12%

Required:

(i) CALCULATE the operating leverage and financial leverage.

(ii) FIND out the combinations of operating and financial leverage which give the highest value and the least value.

ICAI'S ANSWER

(i) Operating Leverage

	Situation 1	Situation 2	Situation 3
	(Rs.)	**(Rs.)**	**(Rs.)**
Sales (S)			
2,400 units @ Rs. 30 per unit	72,000	72,000	72,000
Less: Variable Cost (VC) @ Rs. 20 per unit	48,000	48,000	48,000
Contribution (C)	24,000	24,000	24,000
Less: Fixed Cost (FC)	3,000	6,000	9,000
EBIT	21,000	18,000	15,000
Operating Leverage = $\frac{C}{EBIT}$	$\frac{\text{Rs. } 24,000}{\text{Rs. } 21,000}$ = 1.14	$\frac{\text{Rs. } 24,000}{\text{Rs. } 18,000}$ = 1.33	$\frac{\text{Rs. } 24,000}{\text{Rs. } 15,000}$ = 1.60

Financial Leverage

	Financial Plan	
	A (Rs.)	**B (Rs.)**
Situation 1		
EBIT	21,000	21,000
Less: Interest on debt (Rs. 15,000 × 12%);(Rs. 7,500 × 12%)	1,800	900
EBT	19,200	20,100
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	$\frac{\text{Rs. } 21,000}{\text{Rs. } 19,200}$ = 1.09	$\frac{\text{Rs. } 21,000}{\text{Rs. } 20,100}$ = 1.04
Situation 2		
EBIT	18,000	18,000
Less: Interest on debt	1,800	900
EBT	16,200	17,100
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	$\frac{\text{Rs. } 18,000}{\text{Rs. } 16,200}$ = 1.11	$\frac{\text{Rs. } 18,000}{\text{Rs. } 17,100}$ = 1.05
Situation 3		
EBIT	15,000	15,000
Less: Interest on debt	1,800	900
EBT	13,200	14,100
Financial Leverage = $\frac{\text{EBIT}}{\text{EBT}}$	$\frac{\text{Rs. } 15,000}{\text{Rs. } 13,200}$ = 1.14	$\frac{\text{Rs. } 15,000}{\text{Rs. } 14,100}$ = 1.06

(ii) Combined Leverages

CL = OL x FL

		Financial Plan	
		A (Rs.)	**B (Rs.)**
(a)	Situation 1	1.14 × 1.09 = 1.24	1.14 × 1.04 = 1.19
(b)	Situation 2	1.33 × 1.11 = 1.48	1.33 × 1.05 = 1.40
(c)	Situation 3	1.60 × 1.14 = 1.82	1.60 × 1.06 = 1.70

The above calculations suggest that the highest value is in Situation 3 financed by Financial Plan A and the lowest value is in the Situation 1 financed by Financial Plan B.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 3-4 · answer p. 6-7

492 MTP **Old P8 Jul 2021 S-II** Q6(c) 2 marks old scheme

EXPLAIN the principles of "Trading on equity".

ICAI'S ANSWER

The use of long-term fixed interest-bearing debt and preference share capital along with equity share capital is called financial leverage or trading on equity. The use of long-term debt increases the earnings per share if the firm yields a return higher than the cost of debt. The earnings per share also increase with the use of preference share capital but due to the fact that interest is allowed to be deducted while computing tax, the leverage impact of debt is much more. However, leverage can operate adversely also if the rate of interest on long-term loan is more than the expected rate of earnings of the firm. Therefore, it needs caution to plan the capital structure of a firm.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 4 · answer p. 7

493 MTP **Old P8 Nov 2020 S-I** Q5(a) 8 marks old scheme

Following Balance Sheet and Income Statement have been obtained from the books of accounts of Benaca Pvt. Ltd.

Balance Sheet as on March 31st 2020

Liabilities	Amount (₹)	Assets	Amount (₹)
Equity Capital (₹10 per share)	80,00,000	Net Fixed Assets	1,00,00,000
10% Debt	60,00,000	Current Assets	90,00,000
Retained Earnings	35,00,000		
Current Liabilities	15,00,000		
	1,90,00,000		1,90,00,000

Income Statement for the year ending March 31st 2020

Particulars	Amount (₹)
Sales	34,00,000
Less: Operating expenses (including ₹ 6,00,000 depreciation)	(12,00,000)
EBIT	22,00,000
Less: Interest	(6,00,000)
Earnings before tax	16,00,000

The tax rate applicable to the company is 35 percent.

(i) DETERMINE the degree of operating, financial and combined leverages at the current sales level, if all operating expenses, other than depreciation, are variable costs.

(ii) If total assets remain at the same level, but sales (i) increase by 20 percent and (ii) decrease by 20 percent, COMPUTE the earnings per share at the new sales level?

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

494 MTP **Old P8 May 2020 S-I** Q1(a) 5 marks old scheme

The data relating to two companies are as given below:

	Company A	Company B
Equity Capital	Rs.6,00,00,000	Rs.3,50,00,000
15% Debentures	Rs.40,00,000	Rs.65,00,000
Output (units) per annum	6,00,000	1,50,000
Selling price/ unit	Rs.60	Rs.500
Fixed Costs per annum	Rs.70,00,000	Rs.1,40,00,000
Variable Cost per unit	Rs.30	Rs.275

You are required to CALCULATE the Operating leverage, Financial leverage and Combined leverage of the two Companies.

ICAI'S ANSWER

Computation of Operating leverage, Financial leverage and Combined leverage of two companies

	Company A	Company B
Output units per annum	6,00,000	1,50,000
	(Rs.)	**(Rs.)**
Selling price / unit	60	500
Sales revenue	3,60,00,000 (6,00,000 units × Rs.60)	7,50,00,000 (1,50,000 units × Rs.500)
Less: Variable costs	1,80,00,000 (6,00,000 units × Rs.30)	4,12,50,000 (1,50,000 units × Rs.275)
Contribution (C)	1,80,00,000	3,37,50,000
Less: Fixed costs	70,00,000	1,40,00,000
EBIT (Earnings before Interest and tax)	1,10,00,000	1,97,50,000
Less: Interest @ 15% on debentures	6,00,000	9,75,000
PBT	1,04,00,000	1,87,75,000

Operating Leverage = Contribution/EBIT	1.64 (Rs.1,80,00,000 ÷ 1,10,00,000)	1.71 (Rs.3,37,50,000 ÷ Rs. 1,97,50,000)
Financial Leverage = EBIT/PBT	1.06 (Rs.1,10,00,000 ÷ Rs.1,04,00,000)	1.05 (Rs.1,97,50,000 ÷ Rs. 1,87,75,000)
Combined Leverage = DOL × DFL	1.74 (1.64 × 1.06)	1.80 (1.71 × 1.05)

ICAI · Old P8 · MTP May 2020 · Series I · question p. 1 · answer p. 1

495 MTP **Old P8 Nov 2019 S-I** Q2(a) 3 marks old scheme

B LLP. has the following balance sheet and Income statement information:

Balance Sheet as on March 31st 2019

Liabilities	(Rs.)	Assets	(Rs.)
Partners' Capital	80,00,000	Net Fixed Assets	1,00,00,000
Term Loan	60,00,000	Inventories	45,00,000
Retained Earnings	35,00,000	Trade Receivables	40,50,000
Trade Payables	15,00,000	Cash & Bank	4,50,000
	1,90,00,000		1,90,00,000

Income Statement for the year ending March 31st 2019

	(Rs.)
Sales	34,00,000
Operating expenses (including Rs. 6,00,000 depreciation)	12,00,000
EBIT	22,00,000
Less: Interest	6,00,000
Earnings before tax	16,00,000
Less: Taxes	5,60,000
Net Earnings (EAT)	10,40,000

COMPUTE the degree of operating, financial and combined leverages at the current sales level, if all operating expenses, other than depreciation, are variable costs.

ICAI'S ANSWER

Computation of Degree of Operating (DOL), Financial (DFL) and Combined leverages (DCL).

$$\text{DOL} = \frac{\text{Rs. } 34,00,000 - \text{Rs. } 6,00,000}{\text{Rs. } 22,00,000} = 1.27$$

$$\text{DFL} = \frac{\text{Rs.}22,00,000}{\text{Rs. } 16,00,000} = 1.38$$

$$DCL = DOL \times DFL = 1.27 \times 1.38 = 1.75$$

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 2 · answer p. 3

Investment Decisions

7.1 Capital budgeting and estimating relevant cash flows

46 questions · asked in 18 ICAI documents · 5 exam papers

496 EXAM **Sep 2026** Case Scenario – I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO – I

NPK Ltd. is a Start-up incorporated 2 years ago with a paid up equity capital of ₹ 20 lakhs and recognised by the Department for Promotion of Industry and Internal Trade (DPIIT). Start-up India initiative of the Government of India envisages for building up a robust start-up ecosystem in the country for nurturing innovation and providing opportunities to budding entrepreneurs. NPK Ltd. has been engaged in creating innovative solutions for the defence sector. NPK Ltd., being an eligible start-up applied for income tax holiday and got 100% tax exemption on profits for the coming three consecutive years.

NPK Ltd. has proposed to invest ₹ 800 lakh as capital cost in a project having life of 8 years. It would generate Earnings before Depreciation, Interest and Taxes (EBDIT) of ₹ 220 lakh per annum throughout the life of the project. Depreciation will be charged under Straight-Line Method and the same is allowable under the income tax rules. Under Seed Financing Scheme, a financial institution has sanctioned a loan of ₹ 800 lakh under a subsidised interest rate of 10 per cent per annum. The entire loan amount is repayable at the end of the life of the project.

After the three consecutive years of tax holiday, the tax rate applicable for NPK Ltd. will be 30 per cent.

Year	1	2	3	4	5	6	7	8
$PVIF_{7\%,t}$	0.935	0.873	0.816	0.763	0.713	0.666	0.623	0.582
$PVIF_{10\%,t}$	0.909	0.826	0.751	0.683	0.621	0.564	0.513	0.467

Based on the facts mentioned above relating to NPK Ltd., you are required to answer the following questions 1 to 5 :

MCQ 1 2 marks

What will be the after tax cost of debt for the first 3 years and thereafter for the rest of the life of the project ?

A 7% for the first 3 years and thereafter 10%.

B 10% for the first 3 years and thereafter 7%.

C 10% for the first 3 years and thereafter 13%.

D 13% for the first 3 years and thereafter 10%.

MCQ 2 2 marks

What will be the net cash inflow after tax per year for the first three years of the life of the project ?

A ₹ 128 lakh

B ₹ 140 lakh

C ₹ 196 lakh

D ₹ 220 lakh

MCQ 3 2 marks

What will be the net cash inflow after tax per year for 4th year to 8th year of the life of the project ?

A ₹ 128 lakh

B ₹ 144 lakh

C ₹ 184 lakh

D ₹ 193 lakh

MCQ 4 2 marks

What will be the Pay-Back Period of the project ?

A 3.76 years

B 4.86 years

C 5.06 years

D 5.72 years

MCQ 5 2 marks

What will be the Net Present Value (NPV) of the project ?

A ₹ 163.888 lakh

B ₹ 164.488 lakh

C ₹ 362.768 lakh

D ₹ 396.884 lakh

ICAI'S ANSWERS

MCQ 1

Answer key: **ICAI has not published the key yet.**

MCQ 2

Answer key: **ICAI has not published the key yet.**

MCQ 3

Answer key: **ICAI has not published the key yet.**

MCQ 4

Answer key: **ICAI has not published the key yet.**

MCQ 5

Answer key: **ICAI has not published the key yet.**

ICAI · P6 · Sep 2026 exam · question p. 2-3

497 **EXAM** **Sep 2026** Q3(a) 6 marks

Auto Industries Limited manufactures auto components on lathe machine which has a book value of ₹ 1 lakh. The machine being depreciated at 20% has another five years of its useful life remaining but can be sold for ₹ 25,000 today. Auto Industries Limited is contemplating to replace this machine with a new one which delivers more output and is cost efficient too. The cost of the machine is ₹ 4 lakh with a life of five years and depreciation of 20%. The company estimates that with the new machine, it will be able to make extra sales of ₹ 0.50 lakh and save cost by ₹ 0.75 lakh for the next five years. With the increased sales, the requirement of working capital would also change and the company would need ₹ 30,000 more.

At the end of the five years the benefits of increased revenue and decreased cost will aggregate to ₹ 50,000 p.a. for indefinite time. The company has WACC of 15% and pays taxes at 25%.

Given present value of annuity of ₹ 1 for 5 years @15% is 3.3521.

Present value of ₹ 1 @15% for 5th year is 0.4971.

Advise the management of the company whether the machine should be replaced or not ?

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 18

498 EXAM **Jan 2025** Q4(c) 2 marks

State the concept of exclusion of Financing Cost Principle.

ICAI'S ANSWER

The **exclusion** of financing costs principle means that:

(i) The interest on long-term debt is ignored while computing profits and taxes.

(ii) The expected dividends are deemed irrelevant in cash flow analysis.

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499 EXAM **Sep 2024** Case Scenario 1 · MCQ 1 to MCQ 5

CASE SCENARIO 1

RS Limited is manufacturing selling soft drinks in India. The production process involves one important process which increases the shelf life of the soft drinks. Presently, the machine used for this purpose is an old one, in which wastage due to breakage of glass bottles is considerably high and due to its limited capacity, the company is not in a position to increase its production.

The production manager has approached the CEO of RS Limited for purchasing an automated machine, which will drastically reduce the wastage due to breakage during the process of increasing shelf life of soft drinks. The automated machine will support increase in production. The production manager is confident that acquisition of the automated machine will be beneficial for the company.

Other information is as under :

- With the introduction of automated machine, additional sales and related costs over the next five years would be as follows:

Year	Additional Sales Unit	Selling price per unit (₹)	Variable Manufacturing, Selling and Distribution cost per unit (₹)	Additional fixed Selling & Distribution Cost (₹)
1	20,000	30	20	25,000
2	25,000	30	20	30,000
3	30,000	35	20	30,000
4	32,000	35	22	35,000
5	28,000	35	22	35,000

- Cost of acquisition of automated machine is ₹ 5,00,000. Residual value of the automated machine at the end of its life of 5 years will be ₹ 50,000. Depreciation on automated machine will be under Straight line method.
  Depreciation is not included in the cost stated above.

- The Production Manager has estimated the cost savings (before tax) due to reduction in breakages as under:

	Year 1	Year 2	Year 3	Year 4	Year 5
Savings Cost due reduction in breakages	₹ 15,000	₹ 15,000	₹ 20,000	₹ 20,000	₹ 20,000

- The machine which is being used at present has zero written down value and if sold, would fetch an amount of ₹ 10,000 only.
- The cost of capital of the company is 10%. The tax rate applicable for the company is 30%. Ignore capital gain taxes.
  P.V Factors of ₹ 1 at year end at 10%:

	Year 1	Year 2	Year 3	Year 4	Year 5
P.V Factor of ₹ 1	0.909	0.826	0.751	0.683	0.621

You are required to answer the following Questions 1 to 5 :

MCQ 1

What is the Profit before Taxes for the Year 2, Year 3 and Year 4 of the investment proposal ?

A ₹ 2,35,000, ₹ 4,40,000, ₹ 4,01,000

B ₹ 1,45,000, ₹ 3,50,000, ₹ 3,11,000

C ₹ 2,05,000, ₹ 4,10,000, ₹ 3,66,000

D ₹ 1,40,000, ₹ 3,60,000, ₹ 3,31,000

MCQ 2

What is the Cash Inflow after Taxes for the Year 1, Year 2 and Year 3 of the investment proposal ?

A ₹ 1,50,000, ₹ 1,85,000, ₹ 3,45,000

B ₹1,65,000, ₹ 1,95,500, ₹ 3,55,000

C ₹ 1,60,000, ₹ 1,91,500, ₹ 3,35,000

D ₹ 1,70,000, ₹ 1,90,000, ₹ 3,40,000

MCQ 3

What is the Discounted Cash Inflow after Taxes for the Year 1, Year 2 and Year 3 of the investment proposal ?

A ₹ 1,49,985, ₹ 1,61,483, ₹ 2,66,605

B ₹ 1,36,350, ₹ 1,52,810, ₹ 2,59,095

C ₹ 1,54,530, ₹ 1,56,940, ₹ 2,55,340

D ₹ 1,45,440, ₹ 1,58,179, ₹ 2,51,585

MCQ 4

What is the Net Present Value of the investment proposal?

A ₹ 3,78,990.30

B ₹ 4,54,980.60

C ₹ 4,74,890.40

D ₹ 3,89,260.70

MCQ 5

What is the Discounted Payback period of the investment proposal ?

A 2.74 years

B 2.87 years

C 2.38 years

D 2.48 years

ICAI'S ANSWERS

MCQ 1

Answer key: **(B)**

MCQ 2

Answer key: **(C)**

MCQ 3

Answer key: **(D)**

MCQ 4

Answer key: **(C)**

MCQ 5

Answer key: **(A)**

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500 EXAM **Old P8 May 2023** Q5 10 marks old scheme

Four years ago, Z Ltd. had purchased a machine of ₹ 4,80,000 having estimated useful life of 8 years with zero salvage value. Depreciation is charged using SLM method over the useful life. The company want to replace this machine with a new machine. Details of new machine are as below:

- Cost of new machine is ₹ 12,00,000, Vendor of this machine is agreed to take old machine at a value of ₹ 2,40,000. Cost of dismantling and removal of old machine will be ₹ 40,000. 80% of net purchase price will be paid on spot and remaining will be paid at the end of one year.
- Depreciation will be charged @ 20% p.a. under WDV method.
- Estimated useful life of new machine is four years and it has salvage value of ₹ 1,00,000 at the end of year four.
- Incremental annual sales revenue is ₹ 12,25,000.
- Contribution margin is 50%.
- Incremental indirect cost (excluding depreciation) is ₹ 1,18,750 per year.
- Additional working capital of ₹ 2,50,000 is required at the beginning of year and ₹ 3,00,000 at the beginning of year three. Working capital at the end of year four will be nil.
- Tax rate is 30%.
- Ignore tax on capital gain.

Z Ltd. will not make any additional investment, if it yields less than 12%

Advice, whether existing machine should be replaced or not.

Year	1	2	3	4	5
$PVIF_{0.12, t}$	0.893	0.797	0.712	0.636	0.567

ICAI'S ANSWER

Working Notes:

(i) Calculation of Net Initial Cash Outflow

Particulars	₹
Cost of New Machine	12,00,000
Less: Sale proceeds of existing machine	2,00,000
Net Purchase Price	10,00,000
Paid in year 0	8,00,000
Paid in year 1	2,00,000

(ii) Calculation of Additional Depreciation

Year	1	2	3	4
	₹	₹	₹	₹
Opening WDV of machine	10,00,000	8,00,000	6,40,000	5,12,000
Depreciation on new machine @ 20%	2,00,000	1,60,000	1,28,000	1,02,400
Closing WDV	8,00,000	6,40,000	5,12,000	4,09,600
Depreciation on old machine (4,80,000/8)	60,000	60,000	60,000	60,000
Incremental depreciation	**1,40,000**	**1,00,000**	**68,000**	**42,400**

(iii) Calculation of Annual Profit before Depreciation and Tax (PBDT)

Particulars	Incremental Values (₹)
Sales	12,25,000
Contribution	6,12,500
Less: Indirect Cost	1,18,750
Profit before Depreciation and Tax (PBDT)	4,93,750

Calculation of Incremental NPV

Year	PVF @ 12%	PBTD (₹)	Incremental Depreciation (₹)	PBT (₹)	Tax @ 30% (₹)	Cash Inflows (₹)	PV of Cash Inflows (₹)
	(1)	(2)	(3)	(4)	(5) = (4) x 0.30	(6) = (4) – (5) + (3)	(7) = (6) x (1)
1	0.893	4,93,750	1,40,000	3,53,750	106,125	3,87,625	3,46,149.125

Year	PVF @ 12%	PBTD (₹)	Incremental Depreciation (₹)	PBT (₹)	Tax @ 30% (₹)	Cash Inflows (₹)	PV of Cash Inflows (₹)
2	0.797	4,93,750	1,00,000	3,93,750	1,18,125	3,75,625	2,99,373.125
3	0.712	4,93,750	68,000	4,25,750	1,27,725	3,66,025	2,60,609.800
4	0.636	4,93,750	42,400	4,51,350	1,35,405	3,58,345	2,27,907.420
			*			*	**11,34,039.470**
Add: PV of Salvage (₹ 1,00,000 x 0.636)							63,600
Less: **Initial Cash Outflow** - Year 0							**8,00,000**
Year 1 (₹ 2,00,000 × 0.893)							**1,78,600**
Less: Working Capital - Year 0							2,50,000
Year 2 (₹ 3,00,000 × 0.797)							2,39,100
Add: Working Capital released - Year 4 (₹ 5,50,000 × 0.636)							3,49,800
Incremental Net Present Value							**79,739.470**

Since the incremental NPV is positive, existing machine should be replaced.

Alternative Presentation

Computation of Outflow for new Machine:

	₹
Cost of new machine	12,00,000
Replaced cost of old machine	2,40,000
Cost of removal	40,000
Net Purchase price	**10,00,000**
Outflow at year 0	8,00,000
Outflow at year 1	2,00,000

Computation of additional deprecation

Year	1	2	3	4
	₹	₹	₹	₹
Opening WDV of machine	10,00,000	8,00,000	6,40,000	5,12,000
Depreciation on new machine @ 20%	2,00,000	1,60,000	1,28,000	1,02,400
Closing WDV	8,00,000	6,40,000	5,12,000	4,09,600
Depreciation on old machine (4,80,000/8)	60,000	60,000	60,000	60,000
Incremental depreciation	**1,40,000**	**1,00,000**	**68,000**	**42,400**

Computation of NPV

	Year	0	1	2	3	4
		₹	₹	₹	₹	₹
1.	Increase in sales revenue		12,25,000	12,25,000	12,25,000	12,25,000
2.	Contribution		6,12,500	6,12,500	6,12,500	6,12,500
3.	Increase in fixed cost		1,18,750	1,18,750	1,18,750	1,18,750
4.	Incremental Depreciation		1,40,000	1,00,000	68,000	42,400
5.	Net profit before tax [1-(2+3+4)]		3,53,750	3,93,750	4,25,750	4,51,350
6.	Net Profit after tax (5 x 70%)		2,47,625	2,75,625	2,98,025	3,15,945
7.	Add: Incremental depreciation		1,40,000	1,00,000	68,000	42,400
8.	Net Annual cash inflows (6 + 7)		3,87,625	3,75,625	3,66,025	3,58,345
9.	Release of salvage value					1,00,000
10.	(investment)/disinvestment in working capital	(2,50,000)		(3,00,000)		5,50,000
11.	Initial cost	(8,00,000)	(2,00,000)			
12.	Total net cash flows	(10,50,000)	1,87,625.0	75,625	3,66,025	10,08,345
13.	Discounting Factor	1	0.893	0.797	0.712	0.636
14.	Discounted cash flows (12 x 13)	(10,50,000)	1,67,549.125	60,273.125	2,60,609.800	641307.420

NPV = (1,67,549 + 60,273 + 2,60,610 + 6,41,307) - 10,50,000 = **₹ 79,739**

Since the NPV is positive, existing machine should be replaced.

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501 EXAM Old P8 Jul 2021 Q4 10 marks old scheme

An existing company has a machine which has been in operation for two years, its estimated remaining useful life is 4 years with no residual value in the end. Its current market value is ₹ 3 lakhs. The management is considering a proposal to purchase an improved model of a machine gives increase output. The details are as under:

Particulars	Existing Machine	New Machine
Purchase Price	₹ 6,00,000	₹ 10,00,000
Estimated Life	6 years	4 years
Residual Value	0	0
Annual Operating days	300	300
Operating hours per day	6	6
Selling price per unit	₹ 10	₹ 10
Material cost per unit	₹ 2	₹ 2
Output per hour in units	20	40
Labour cost per hour	₹ 20	₹ 30
Fixed overhead per annum excluding depreciation	₹ 1,00,000	₹ 60,000
Working Capital	₹ 1,00,000	₹ 2,00,000
Income-tax rate	30%	30%

Assuming that - cost of capital is 10% and the company uses written down value of depreciation @ 20% and it has several machines in 20% block.

Advice the management on the Replacement of Machine as per the NPV method.

The discounting factors table given below:

Discounting Factors	Year 1	Year 2	Year 3	Year 4
10%	0.909	0.826	0.751	0.683

ICAI'S ANSWER

(i) Calculation of Net Initial Cash Outflows:

Particulars	₹
Purchase Price of new machine	10,00,000
Add: Net Working Capital	1,00,000
Less: Sale proceeds of existing machine	3,00,000
Net initial cash outflows	8,00,000

(ii) Calculation of annual Profit Before Tax and depreciation

Particulars		Existing machine	New Machine	Differential
(1)		**(2)**	**(3)**	**(4) = (3) – (2)**
Annual output		36,000 units	72,000 units	36,000 units
		₹	₹	₹
(A)	Sales revenue @ ₹ 10 per unit	3,60,000	7,20,000	3,60,000
(B)	Cost of Operation			
	Material @ ₹ 2 per unit	72,000	1,44,000	72,000
	Labour			
	Old = 1,800 × ₹ 20	36,000		
	New = 1,800 × ₹ 30		54,000	18,000
	Fixed overhead excluding depreciation	1,00,000	60,000	(40,000)
	Total Cost (B)	2,08,000	2,58,000	50,000
Profit Before Tax and depreciation (PBTD) (A – B)		**1,52,000**	**4,62,000**	**3,10,000**

(iii) Calculation of Net Present value on replacement of machine

Year	PBTD	Depreciation @ 20% WDV	PBT	Tax @ 30%	PAT	Net cash flow	PVF @ 10%	PV
(1)	**(2)**	**(3)**	**(4 = 2-3)**	**(5)**	**(6 = 4-5)**	**(7 = 6 + 3)**	**(8)**	**(9 = 7 × 8)**
1	3,10,000	1,40,000	1,70,000	51,000	1,19,000	2,59,000	0.909	2,35,431.000
2	3,10,000	1,12,000	1,98,000	59,400	1,38,600	2,50,600	0.826	2,06,995.600
3	3,10,000	89,600	2,20,400	66,120	1,54,280	2,43,880	0.751	1,83,153.880
4	3,10,000	71,680	2,38,320	71,496	1,66,824	2,38,504	0.683	1,62,898.232
								7,88,478.712
Add: Release of net working capital at year end 4 (1,00,000 x 0.683)								68,300.000

Year	PBTD	Depreciation @ 20% WDV	PBT	Tax @ 30%	PAT	Net cash flow	PVF @ 10%	PV
Less: Initial Cash Outflow								8,00,000.000
NPV								**56,778.712**

Advice: Since the incremental NPV is positive, existing machine should be replaced.

Working Notes:

1. Calculation of Annual Output

Annual output = (Annual operating days x Operating hours per day) x output per hour

Existing machine = (300 x 6) x 20 = 1,800 x 20 = 36,000 units

New machine = (300 x 6) x 40 = 1,800 x 40 = 72,000 units

2. Base for incremental depreciation

Particulars		₹
WDV of Existing Machine		
Purchase price of existing machine		6,00,000
Less: Depreciation for year 1	1,20,000	
Depreciation for Year 2	96,000	2,16,000
WDV of Existing Machine (i)		**3,84,000**
Depreciation base of New Machine		
Purchase price of new machine		10,00,000
Add: WDV of existing machine		3,84,000
Less: Sales value of existing machine		3,00,000
Depreciation base of New Machine (ii)		**10,84,000**
Base for incremental depreciation [(ii) – (i)]		**7,00,000**

(Note: The above solution have been done based on incremental approach)

Alternatively, solution can be done based on Total Approach as below:

(i) Calculation of depreciation:

Existing Machine						
	Year 1	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**
Opening balance	6,00,000	4,80,000	3,84,000	3,07,200	2,45,760	1,96,608.00
Less: Depreciation @ 20%	1,20,000	96,000	76,800	61,440	49,152	39,321.60
WDV	4,80,000	3,84,000	3,07,200	2,45,760	1,96,608	1,57,286.40

New Machine				
	Year 1	**Year 2**	**Year 3**	**Year 4**
Opening balance	**10,84,000***	8,67,200	6,93,760	5,55,008.00
Less: Depreciation @ 20%	**2,16,800**	**1,73,440**	**1,38,752**	**1,11,001.60**
WDV	8,67,200	6,93,760	5,55,008	4,44,006.40

* As the company has several machines in 20% block, the value of Existing Machine from the block calculated as below shall be added to the new machine of ₹ 10,00,000:

WDV of existing machine at the beginning of the year	₹ 3,84,000
Less: Sale Value of Machine	₹ 3,00,000
WDV of existing machine in the block	₹ 84,000

Therefore, opening balance for depreciation of block = ₹ 10,00,000 + ₹ 84,000 = ₹ 10,84,000

(ii) Calculation of annual cash inflows from operation:

Particulars		EXISTING MACHINE			
		Year 3	**Year 4**	**Year 5**	**Year 6**
Annual output (300 operating days x 6 operating hours x 20 output per hour)		**36,000 units**	**36,000 units**	**36,000 units**	**36,000 units**
		₹	**₹**	**₹**	**₹**
(A)	**Sales revenue @ ₹ 10 per unit**	**3,60,000.00**	**3,60,000.00**	**3,60,000.00**	**3,60,000.00**

Particulars		EXISTING MACHINE			
(B)	Less: Cost of Operation				
	Material @ ₹ 2 per unit	72,000.00	72,000.00	72,000.00	72,000.00
	Labour @ ₹ 20 per hour for (300 × 6) hours	36,000.00	36,000.00	36,000.00	36,000.00
	Fixed overhead	1,00,000.00	1,00,000.00	1,00,000.00	1,00,000.00
	Depreciation	76,800.00	61,440.00	49,152.00	39,321.60
	Total Cost (B)	**2,84,800.00**	**2,69,440.00**	**2,57,152.00**	**2,47,321.60**
	Profit Before Tax (A – B)	75,200.00	90,560.00	1,02,848.00	1,12,678.40
	Less: Tax @ 30%	22,560.00	27,168.00	30,854.40	33,803.52
	Profit After Tax	**52,640.00**	**63,392.00**	**71,993.60**	**78,874.88**
	Add: Depreciation	76,800.00	61,440.00	49,152.00	39,321.60
	Add: Release of Working Capital				1,00,000.00
	Annual Cash Inflows	**1,29,440.00**	**1,24,832.00**	**1,21,145.60**	**2,18,196.48**

Particulars		NEW MACHINE			
		Year 1	**Year 2**	**Year 3**	**Year 4**
Annual output (300 operating days x 6 operating hours x 40 output per hour)		**72,000 units**	**72,000 units**	**72,000 units**	**72,000 units**
		₹	**₹**	**₹**	**₹**
(A)	**Sales revenue @ ₹ 10 per unit**	**7,20,000.00**	**7,20,000.00**	**7,20,000.00**	**7,20,000.00**

Particulars		NEW MACHINE			
(B)	Less: Cost of Operation				
	Material @ ₹ 2 per unit	1,44,000.00	1,44,000.00	1,44,000.00	1,44,000.00
	Labour @ ₹ 30 per hour for (300 × 6) hours	54,000.00	54,000.00	54,000.00	54,000.00
	Fixed overhead	60,000.00	60,000.00	60,000.00	60,000.00
	Depreciation	2,16,800.00	1,73,440.00	1,38,752.00	1,11,001.60
	Total Cost (B)	**4,74,800.00**	**4,31,440.00**	**3,96,752.00**	**3,69,001.60**
	Profit Before Tax (A – B)	2,45,200.00	2,88,560.00	3,23,248.00	3,50,998.40
	Less: Tax @ 30%	73,560.00	86,568.00	96,974.40	1,05,299.52
	Profit After Tax	1,71,640.00	2,01,992.00	2,26,273.60	2,45,698.88
	Add: Depreciation	2,16,800.00	1,73,440.00	1,38,752.00	1,11,001.60
	Add: Release of Working Capital				2,00,000.00
	Annual Cash Inflows	**3,88,440.00**	**3,75,432.00**	**3,65,025.60**	**5,56,700.48**

(iii) Calculation of Incremental Annual Cash Flow:

Particulars	Year 1 (₹)	Year 2 (₹)	Year 3 (₹)	Year 4 (₹)
Existing Machine (A)	1,29,440.00	1,24,832.00	1,21,145.60	2,18,196.48
New Machine (B)	3,88,440.00	3,75,432.00	3,65,025.60	5,56,700.48
Incremental Annual Cash Flow (B – A)	2,59,000.00	2,50,600.00	2,43,880.00	3,38,504.00

(iv) Calculation of Net Present Value on replacement of machine:

Year	Incremental Annual Cash Flow (₹) (A)	Discounting factor @ 10% (B)	Present Value of Incremental Annual Cash Flow (₹) (A x B)
1	2,59,000.00	0.909	2,35,431.000
2	2,50,600.00	0.826	2,06,995.600
3	2,43,880.00	0.751	1,83,153.880
4	3,38,504.00	0.683	2,31,198.232
Total Incremental Inflows			**8,56,778.712**
Less: Net Initial Cash Outflows (Working note)			8,00,000.000
Incremental NPV			**56,778.712**

Advice: Since the incremental NPV is positive, existing machine should be replaced.

Working Note:

Calculation of Net Initial Cash Outflows:

Particulars	₹
Cost of new machine	10,00,000
Less: Sale proceeds of existing machine	3,00,000
Add: incremental working capital required (₹ 2,00,000 – ₹ 1,00,000)	1,00,000
Net initial cash outflows	**8,00,000**

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502 RTP **Sep 2025** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

Mint Technologies, a startup in the IT infrastructure space, is planning to enter the computer networking market with a specialized offering—VPN connectivity solutions for enterprises with distributed office networks. The company recently won a significant tender from a major national bank to set up secure VPN connections across 100 branches. The leadership team is excited but cautious, as this move requires significant upfront investment and ongoing operational capability.

To evaluate the financial viability of this project, you have been hired as a consultant.

- They have forecasted their numbers for 5 years and they have assumed that they will grow at industry average (3%) afterwards.
- Mint will invest all the money for CAPEX in the infrastructure and equipment. Mint is planning to issue 50Mn shares in lieu of the amount.

- Mint will have to keep upgrading the infrastructure and equipment to cover different offices of various clients.
- Mint expects an initial investment of around ₹ 150 Mn in network access as an initial setup cost for connecting different offices. Mint needs to invest additional ₹10 Mn every year starting from beginning and is expected to grow 10% every year.
- The equipment investment varies with number of branches/ offices. This varies with time and currently it is ₹ 0.1 Mn per office.
- Initially, Mint has won a tender of 100 branches of a bank and expecting it to grow at 30% for next 5 years.
- Mint is expecting to have a maintenance cost of 10% for equipment and 15% for network.
- To handle all this, current manpower cost is ₹ 1 Mn
- Mint has entered into an agreement with local authority and has agreed to share 15% of the revenue with the same.
- Mint is expecting to generate revenue of ₹ 0.5 Mn per office and to remain competitive in the market, it is not going to increase with inflation.
- Capital expenditure and operating expenses will grow 10% every year.
- Tax slab in the country is 30% and depreciation of the equipment and network is assumed to be 10%. Discounting rate is 20%.
- Assume all the expenditure including initial setup cost from year 1.

Based on the above information, you are required to answer following 5 MCQs:

Q1(i)

What is the CFAT for year 1?

A -15,33,50,000

B 1,00,17,000

C 1,75,73,810

D 2,74,29,367

Q1(ii)

What is the CFAT for year 5?

A -15,33,50,000

B 1,00,17,000

C 4,02,25,268

D 2,74,29,367

Q1(iii)

Calculate the terminal value

A ₹ 24,37,17,801

B ₹ 9,79,74,556

C ₹ 6,38,54,064

D ₹ 21,32,20,955

Q1(iv)

What is the total present value of the mint technologies?

- A ₹ 1,89,77,337
- B ₹ 1,32,20,955
- C ₹ 1,67,06,548
- D ₹ 89,97,791

Q1(v)

Calculate the IRR for the company

- A 22.60%
- B 15.39%
- C 20.25%
- D 24.68%

ICAI'S ANSWERS

Q1(i)

Answer key: **(A)**

(a) -15,33,50,000

PassBook's note: ICAI's key (a) -15,33,50,000 is the "Net CFAT (I)" (after capex) of Year 1 in ICAI's working (q1v's answer).

Q1(ii)

Answer key: **(C)**

(c) 4,02,25,268

Q1(iii)

Answer key: **(A)**

(a) ₹ 24,37,17,801

Q1(iv)

Answer key: **(C)**

(c) ₹ 1,67,06,548

Q1(v)

(d) 24.68%

Calculation of CFAT

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5
Number of Offices (A)	100	130	169	219*	284*
Revenue Per office (B)	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000
Total Revenue (C) (A x B)	5,00,00,000	6,50,00,000	8,45,00,000	10,95,00,000	14,20,00,000
Operating Cost					
Share of Local Authority	75,00,000	97,50,000	1,26,75,000	1,64,25,000	2,13,00,000
Manpower Cost	10,00,000	14,30,000	20,44,900	29,14,890	41,58,044

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5
Maintenance Cost for Equipment	10,00,000	13,30,000	18,01,900	24,67,400	34,19,065
Maintenance Cost for Network	2,40,00,000	2,56,50,000	2,74,65,000	2,94,61,500	3,16,57,650
Total Operating Cost (D)	3,35,00,000	3,81,60,000	4,39,86,800	5,12,68,790	6,05,34,759
EBITDA (E) (C-D)	1,65,00,000	2,68,40,000	4,05,13,200	5,82,31,210	8,14,65,241
Less: Depreciation	1,70,00,000	1,84,30,000	2,01,11,900	2,21,08,400	2,45,24,165
EBIT (F)	-5,00,000	84,10,000	2,04,01,300	3,61,22,810	5,69,41,076
Less: Taxes	-1,50,000	25,23,000	61,20,390	1,08,36,843	1,70,82,323
PAT (G)	-3,50,000	58,87,000	1,42,80,910	2,52,85,967	3,98,58,753
Add: Depreciation	1,70,00,000	1,84,30,000	2,01,11,900	2,21,08,400	2,45,24,165
CFAT (H)	1,66,50,000	2,43,17,000	3,43,92,810	4,73,94,367	6,43,82,918
Less: Capex	17,00,00,000	1,43,00,000	1,68,19,000	1,99,65,000	2,41,57,650
Net CFAT (I)	-15,33,50,000	1,00,17,000	1,75,73,810	2,74,29,367	4,02,25,268

***Note:** Since offices cannot be represented in fractions, the number should be rounded down to the nearest whole number.

Working Note

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5
Cost					
Manpower Cost (With Inflation)	10,00,000	11,00,000	12,10,000	13,31,000	14,64,100
Manpower Cost per office (With Inflation)	10,000	11,000	12,100	13,310	14,641
CAPEX					
Additional Branches	100	30	39	50	65
Network CAPEX w/o inflation	16,00,00,000	1,00,00,000	1,00,00,000	1,00,00,000	1,00,00,000
Network CAPEX With inflation	16,00,00,000	1,10,00,000	1,21,00,000	1,33,10,000	1,46,41,000

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5
Additional Equipment CAPEX/Branch (with inflation)	1,00,000	1,10,000	1,21,000	1,33,100	1,46,410
CAPEX Schedule					
Network Capex	16,00,00,000	1,10,00,000	1,21,00,000	1,33,10,000	1,46,41,000
Cumulative Capex in Network	16,00,00,000	17,10,00,000	18,31,00,000	19,64,10,000	21,10,51,000
Additional Equipment CAPEX (in the Year)	1,00,00,000	33,00,000	47,19,000	66,55,000	95,16,650
Cumulative Additional Equipment CAPEX	1,00,00,000	1,33,00,000	1,80,19,000	2,46,74,000	3,41,90,650

Terminal Value (TV) using Gordon Growth Model (GGM)

$$\text{Terminal Value} = \frac{\text{Terminal Year Cash Flow x } (1+g)}{Ke - g}$$

$$= \frac{4,02,25,268\ (1+0.03)}{20\% - 3\%} = ₹\ 24,37,17,801$$

Calculation of Total PV

Year	CF	PVAF@20%	PV
1	-15,33,50,000	0.833	-12,77,86,555
2	1,00,17,000	0.694	69,51,798
3	1,75,73,810	0.579	1,01,75,236
4	2,74,29,367	0.482	1,32,20,955
5	4,02,25,268	0.402	1,61,70,558
5	24,37,17,801	0.402	9,79,74,556
Total PV			**1,67,06,548**

Calculation of IRR

Year	Cash Flows	PVAF@20%	PV	PVAF@25%	PV
1	-15,33,50,000	0.8333	-12,77,86,555	0.800	-12,26,80,000
2	1,00,17,000	0.694	69,51,798	0.640	64,10,880
3	1,75,73,810	0.579	1,01,75,236	0.512	89,97,791
4	2,74,29,367	0.482	1,32,20,955	0.410	1,12,46,040
5	4,02,25,268	0.402	1,61,70,558	0.328	1,31,93,888
5	24,37,17,801	0.402	9,79,74,556	0.262	6,38,54,064
			1,67,06,548		**-1,89,77,337**

$$\text{IRR} = L + \frac{NPV_L}{NPV_L - NPV_H}(H - L)$$

H = Higher rate = 25%

L = Lower rate = 20%

$$\text{IRR} = 20 + \frac{1,67,06,548}{1,67,06,548 - (-1,89,77,337)}(25 - 20)$$

= 20 + 4.68= 24.68%

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503 RTP **May 2025** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

Linty is a small-sized firm manufacturing company. Its manufacturing plant is situated in Chattisgarh. Currently, company is labour oriented due to which there is less production, delay in deliveries and more defects in production. The management of the company is considering the proposal to purchase a new automatic machine which will carry out some operations which are at present performed by manual labour. There are two alternative models of the machine that are available in the market. Machine TMT 1 and TMT 2. If machine is replaced, it would provide labor saving and reduce the defects as well. It is expected to have to have an economic life of 10 years for both the models. The following details are collected:

	Machine	
	TMT 1 (₹)	**TMT 2 (₹)**
Cost of Machine	45,00,000	50,00,000
Estimated saving in direct wages per annum	15,00,000	20,00,000
Estimated saving in scrap per annum	5,00,000	6,00,000
Estimated additional cost of indirect material per annum	2,00,000	2,00,000
Estimated additional cost of indirect labour per annum	1,50,000	1,80,000
Estimated additional cost of repairs and maintenance per annum	4,00,000	8,00,000

Depreciation is charged using straight line method over the useful life. Company is in 35 percent tax bracket and expected rate of return may be 15 percent.

Being a finance manager of the company, you are required to evaluate the alternatives by answering the followings:

Q1(i)

What is the annual saving from Machine TMT 1?

- A ₹ 5,20,000
- B ₹ 5,98,000
- C ₹ 9,70,000
- D ₹ 10,98,000

Q1(ii)

What is the annual saving from Machine TMT 2?

- A ₹ 5,20,000
- B ₹ 5,98,000
- C ₹ 9,70,000
- D ₹ 10,98,000

Q1(iii)

What is the payback period of Machine TMT 1 and TMT 2 Respectively?

- A 3.60 years and 4.60 years
- B 4.25 years and 4.42 years
- C 4.63 years and 4.55 years
- D 4.55 years and 4.42 years

Q1(iv)

What is the Accounting (Average) Rate of Return of Machine TMT 1 and TMT 2 Respectively?

- A 20% and 22%
- B 23.11% and 23.92%
- C 22.21% and 23.11%
- D 23.92% and 22.21%

Q1(v)

What is the Profitability Index (PI) of Machine TMT 1 and TMT 2 Respectively?

- A 1.10 and 1.05
- B 0.98 and 1.01
- C 1.19 and 1.08
- D 1.08 and 1.10

ICAI'S ANSWERS

Q1(i)

(A) ₹ 5,20,000

Q1(ii)

(B) ₹ 5,98,000

Working Notes:

$$\text{Depreciation on Machine TMT 1} = \frac{45,00,000}{10}$$

= ₹ 4,50,000

$$\text{Depreciation on Machine TMT 2} = \frac{50,00,000}{10}$$

= ₹ 5,00,000

Particulars	Machine TMT 1 (₹)	Machine TMT 2 (₹)
Annual Savings:		
Direct Wages	15,00,000	20,00,000
Scraps	5,00,000	6,00,000
Total Savings (A)	20,00,000	26,00,000
Annual Estimated Cash Cost :		
Indirect Material	2,00,000	2,00,000
Indirect Labour	1,50,000	1,80,000
Repairs and Maintenance	4,00,000	8,00,000
Total Cost (B)	7,50,000	11,80,000
Annual Cash Savings (A-B)	12,50,000	14,20,000
Less: Depreciation	4,50,000	5,00,000
Annual Savings before Tax	8,00,000	9,20,000
Less: Tax @ 35%	2,80,000	3,22,000
Annual Savings /Profits after tax	5,20,000	5,98,000
Add: Depreciation	4,50,000	5,00,000
Annual Cash Inflows	9,70,000	10,98,000

Q1(iii)

(C) 4.63 and 4.55 years

$$\text{Payback Period} = \frac{\text{Total Initial Capital Investment}}{\text{Annual expected after tax net cashflow}}$$

$$\text{Machine TMT 1} = \frac{45,00,000}{9,70,000}$$

= 4.63 years

$$\text{Machine TMT 2} = \frac{50,00,000}{10,98,000}$$

= 4.55 years

Q1(iv)

(B) 23.11% and 23.92%

Accounting (Average) Rate of Return (ARR)

$$= \frac{\text{Average Annual Net Savings}}{\text{Average investment}} \text{ x } 100$$

$$\text{Machine TMT 1} = \frac{5,20,000}{22,50,000} \text{ x } 100$$

= 23.11%

$$\text{Machine TMT 2} = \frac{5,98,000}{25,00,000} \text{ x } 100$$

= 23.92%

Q1(v)

(D) 1.08 and 1.10

Present Value Cash Inflow = Annual Cash Inflow x PV factor at 15% Machine TMT 1

= 9,70,000 x 5.019 = ₹ 48,68,430

Machine TMT 2 = 10,98,000 x 5.019 = ₹ 55,10,862

$$\text{Profitability Index or PV Index} = \frac{\text{Present Value of Cash Inflow}}{\text{Investment}}$$

$$\text{Machine TMT 1} = \frac{48,68,430}{45,00,000}$$

= 1.08

$$\text{Machine TMT 2} = \frac{55,10,862}{50,00,000}$$

= 1.10

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504 **RTP** **Sep 2024** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

Mr. Ronak, a doctor by profession, has his own private hospital at Goa having specialization in cardiac treatments. However, now-a-days, Goa not only being a place for the tourists, but is also a place for business delegates, cultural people, politicians, students and other classes of people. Gradually, Goa is opening new windows for businesses and getting recognition as an important tourist and leisure hub in South West India.

There are a number of hotels and resorts at Goa. However, the need still exists for more hotel services, in particular with the excellent service, and because of the large number of visitors from all over the country and all walks of life always favour Goa state for their recreation.

Mr. Ronak although being a doctor by profession is contemplating to establish a five-star hotel at Goa. The hotel will consist of 5 floors. The hotel will include 40 normal rooms and 8 deluxe suites, as well as a restaurant and couple of conference rooms with a small wedding hall on the ground floor. Following are the estimated occupancy rate including fare composition in the Table 1. Being a five-star hotel, breakfast would be complementary but lunch and dinner are on a-la-carte basis.

Table 1: Hotel accommodation, estimated occupancy rate and fare.

Types of Facility	Numbers	Occupancy Rate	Average Rent Per Room Per Day	Growth Rate in Rent
Normal Room	40	33% or 120 Days	₹ 8000	12%
Deluxe Suites	8	33% or 120 Days	₹ 25,000	9%
Conference with Wedding Hall	2	40 days	₹ 3,00,000	9%
Restaurant	1	All days	₹ 27,000 sales per day	8%

For the sake of simplicity in calculation, growth rate to be applied only once after completion of 10 years.

The estimated cost of land will be ₹ 250 million and the construction cost will be ₹ 100 million. The estimated salvage value at the end of 15th year will be 25% of the cost of construction. The cost of furniture will be of ₹ 1,50,000 for each normal room and ₹ 3,80,000 for each deluxe suite. The cost of the furniture for the conference room with wedding hall will be ₹ 7,00,000 each and for restaurant it will be 10,00,000. In addition, the hotel will require 4 elevators at different locations and will be costing around ₹ 35,00,000 each. The cost of buying and installing electronic appliances like TV sets, Air conditioners, Fridge etc. will be around ₹ 35 million. Elevators would be depreciated at a rate of 5% p.a. Electronic appliances will have a salvage value of 15% of its acquisition cost at the end of 15 years.

The hotel will be built by renowned builder named **'Harihar Infrastructure'**. The builder estimated that building will survive for 15 years. The required furniture will be supplied by the local reputed furniture company named **Veru Furnishings Ltd**. They ensured that furniture will go for 10 years very smoothly. At the end of tenth year, new furniture for normal rooms and deluxe suites will be bought and old furniture for the same will be sold by the hotel owner. The owner of the hotel estimates that he would be able to purchase the required furniture at 15% higher price than the previous purchase price. The salvage values of the furniture at the end of tenth year will be 5% of their purchase prices with no book value remaining. Furniture at restaurant, conference and wedding hall will not require any major changes as such except for minor renovation which will cost ₹ 20,00,000 in total at the end of 12th year. Any scrap generated on account of such renovation will be sold at ₹ 1,75,000.

In order to boost the tourism industry at Goa, the state govt will be granting subsidy of 15% on the initial capex incurred, it will be paid at the time of cost incurred and additional subsidy of 10% on annual revenue expenses for the first 3 years of operation, but will be credited directly in the bank account only at the end of 5th year and the same shall be non-taxable.

The total annual recurring expenses will be ₹ 1,80,00,000/-. It includes salaries to managers, staff and employees, utilities expenses, house keeping and security services' contract, AMC for electronic appliances, restaurant supplies and materials, other miscellaneous expenses, etc. After the end of 10 years, annual recurring expenses will increase at a rate of 10% which is to be applied once. Furthermore, the hotel authority is determined to provide the best and professional hotel services to the clients by offering training to the employees. They decided to spend ₹ 5,00,000 per year for the purpose of training of the employees.

The hotel project will be entitled to enjoy tax holiday for the first five years after which the corporate tax rate of 25% will also be applied for the hotel. The Cost of equity for the company is 12% and the estimated hurdle rate by considering the structure of capital of the proposed hotel is fixed at 15%.

(Depreciation to be taken on SLM basis and assume 360 days in a year. Ignore depreciation on furniture used in restaurant, conference and wedding hall)

Based on above, please answer to the following MCQs.

Q1(i)

The amount of net initial investment required is:

A ₹ 41.044 Crores

B ₹ 34.887 Crores

C ₹ 6.156 Crores

D ₹ 40.74 Crores

Q1(ii)

NPV of the project is:

A ₹ 7.0532 Cr

B ₹ 8.4029 Cr

C ₹ 8.4935 Cr

D ₹ 2.4700 Cr

Q1(iii)

Pay Back period of the project to recover the initial investment is:

A 5.12 years

B 12.02 years

C 11.80 years

D 4.46 years

Q1(iv)

Estimated Recurring accounting profit/(loss) for first three years are:

A ₹ 7.0928 Cr p.a

B ₹ 6.9078 Cr p.a

C ₹ 6.9937 Cr p.a

D ₹ 9.6120 Cr p.a

Q1(v)

IRR of the project is:

A 16.25%

B 19.39%

C 15%

D 12%

ICAI'S ANSWERS

Q1(i)

(B) ₹ 34.887 Crores

Amount Initial Investment required:

(A) Cost of Land & Construction Cost = 250 + 100 = 350 million i.e 35,00,00,000

(B) Furniture Cost

Normal Rooms = 40 x 1,50,000 = 60,00,000

Suite rooms = 8 x 3,80,000 = 30,40,000

Conference and wedding halls = 2 x 7,00,000

= 14,00,000

Restaurant = 10,00,000

(C) Elevators = 4 x 35,00,000 = 1,40,00,000

(D) Electronic Appliances = 3,50,00,000

Gross Investment Required = ₹ 41,04,40,000

Less: 15% Govt Subsidy on Capex **=** ₹ (6,15,66,000)

Net Initial Investment to be incurred by Hotel = ₹ 34,88,74,000

Q1(ii)

(A) ₹ 7.0532 Cr

PV of Cash Inflow = ₹ 42.2317 Cr

As per WN - 2

(-) PV of Cash Outflow = ₹ 35.1785 Cr

As per WN - 1

NPV = 7.0532 CR

Note: Discounting Rate would be the hurdle rate and not cost of equity as hurdle rate means the overall cost of capital

WN 1 - Calculation of PV of Cash Outflows

		(₹)	DF @ 15%	PV (₹)
Year 1	Initial Net Investment	4,88,74,000	1.0000	34,88,74,000
At the end of 10th Year				
Year 10	Purchase of new furniture (At 15% higher price)			
	Normal Rooms	69,00,000	0.2472	17,05,680
	Suite Rooms	34,96,000	0.2472	8,64,211
At the end of 12th Year				

		(₹)	DF @ 15%	PV (₹)
Year 12	Renovation at restaurant, conference and wedding halls (Net)	18,25,000	0.1869	3,41,093
				35,17,84,983

WN – 2: Calculation of PV of Cash Inflows

	Year 1 to 5 (₹)	Year 6 to 10 (₹)	Year 11 to 15 (₹)
Sales			**(Apply growth rate here)**
Normal Rooms	3,84,00,000	3,84,00,000	4,30,08,000
Suites	2,40,00,000	2,40,00,000	2,61,60,000
Conf & Hall	2,40,00,000	2,40,00,000	2,61,60,000
Restaurant	97,20,000	97,20,000	1,04,97,600
Total Sales (A)	**9,61,20,000**	**9,61,20,000**	**10,58,25,600**
Less:			
Annual Recurring Exp (Excl Depreciation)	1,80,00,000	1,80,00,000	1,98,00,000
Training Exp	5,00,000	5,00,000	5,00,000
Depreciation			
Building	50,00,000	50,00,000	50,00,000
Elevators	7,00,000	7,00,000	7,00,000
Electronic App	19,83,333	19,83,333	19,83,333
Furniture (Old)	8,58,800	8,58,800	
Furniture (New)			10,39,600
TOTAL EXP (B)	**2,70,42,133**	**2,70,42,133**	**2,90,22,933**
NPBT (A - B)	6,90,77,867	6,90,77,867	7,68,02,667
(-) Tax	Nil (Tax Holiday)	1,72,69,466	1,92,00,666
NPAT	6,90,77,867	5,18,08,400	5,76,02,000
(+) Depreciation	85,42,133	85,42,133	87,22,933
(+) Cash Inflows from Operation	7,76,20,000	6,03,50,533	6,63,24,933

	Year 1 to 5 (₹)	Year 6 to 10 (₹)	Year 11 to 15 (₹)
PVAF@ 15%	3.3522	1.6666	0.8285
PV of Cash Inflows from Operations	26,01,97,764	10,05,80,199	5,49,50,207

(+) PV of Other Cash Inflows

In year 5 - Govt Subsidy on first 3 years of **Annual Revenue Exp** = 55,50,000 x 0.4972 = ₹ 27,59,460

In year 10 - Salvage Value of Old Furniture = 4,52,000 x 0.2472 = ₹ 1,11,734

In year 15 - Salvage of building and electronic appliance = 3,02,50,000 x 0.1229 = ₹ 37,17,725

Therefore, **Total PV of Cash Inflows = ₹ 42,23,17,089**

Q1(iii)

(D) 4.46 years

Total Net Initial Investment Incurred = ₹ 34,88,74,000

Cumulative of Total Cash Inflows (not discounted cash inflows)

Year	Total Cash Inflows (₹)	Cumulative of Cash Inflows (₹)
1.	7,76,20,000	7,76,20,000
2.	7,76,20,000	15,52,40,000
3.	7,76,20,000	23,28,60,000
4.	7,76,20,000	31,04,80,000
5.	8,31,70,000 (Govt Subsidy of 55,50,000 added here)	39,36,50,000
6.	6,03,50,533	45,40,00,533

From the above table, it can be seen that the initial net investment incurred is getting recovered after 4th year but before the end of 5th year i.e. somewhere between 4th & 5th Year.

Payback period = $4 + \frac{3,83,94,000}{8,31,70,000}$

= 4 + 0.46 = 4.46 years

Q1(iv)

(A) ₹ 7.0928 Cr p.a.

Accounting profit = NPAT + Govt Subsidy on Revenue Expense

= 6,90,77,867 + (1,85,00,000 x 10%)

= 6,90,77,867 + 18,50,000

= 7,09,27,867 per annum

Q1(v)

(B) 19.39%

DF @ 15% NPV = ₹ 7.0532 Cr

DF @ 20% NPV = 34,10,68,926.17 - 35,07,57,719.00

= (₹ 0.9688 Cr)

$$\text{IRR} = \text{Lower Rate \%} + \frac{\{\text{NPV @Lower rate \% X (Higher Rate \% - Lower Rate\%)}\}}{\text{NPV @ Lower rate \% - NPV @ Higher Rate \%}}$$

IRR = 15 + (7.0532 *5 / 8.0220)

= 15 + 4.396

= 19.396%

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505 **RTP** **May 2024** Q5

HMR Ltd. is considering replacing a manually operated old machine with a fully automatic new machine. The old machine had been fully depreciated for tax purpose but has a book value of ₹ 2,50,000 on 31st March. The machine has begun causing problems with breakdowns and it cannot fetch more than ₹ 40,000 if sold in the market at present. It will have no realizable value after 10 years. The company has been offered ₹ 1,50,000 for the old machine as a trade in on the new machine which has a price (before allowance for trade in) of ₹ 6,00,000. The expected life of new machine is 10 years with salvage value of ₹ 35,000.

Further, the company follows written down value method depreciation @ 10% but for tax purpose, straight line method depreciation is used considering that this is the only machine in the block of assets. A working capital of ₹ 50,000 will be needed and it will be released at the end of tenth year.

Given below are the expected sales and costs from both old and new machine:

	Old machine	New machine
Annual output	60,000 units	80,000 units
Selling price per unit	₹ 18	₹ 18
Annual operating hours	2,800	2,800
Material cost per unit	₹ 5	₹ 5
Labour cost per hour	₹ 50	₹ 75
Indirect cash cost per annum	₹ 1,00,000	₹ 1,75,000

From the above information, ANALYSE whether the old machine should be replaced or not if the opportunity cost of capital of the Company is 10%?

The Income tax rate is 30%. Further assume that book profit is treated as ordinary income for tax purpose.

Also ESTIMATE the internal rate of return of the replacement decision.

All calculations to be calculated to 3 decimal places.

ICAI'S ANSWER

Workings:

(i) **Initial Cash Outflow:**

	Amount (₹)
Cost of new machine	6,00,000
Less: Sale Price of existing machine Net of Tax (₹ 1,50,000 × 0.70)	1,05,000
	4,95,000

(ii) **Terminal Cash Flows:**

New Machine

	Amount (₹)
Salvage value of Machine	35,000
Less: Depreciated WDV {₹ 6,00,000 - (₹ 56,500 × 10 years)}	35,000
Short Term Capital Gain (STCG)	Nil
Tax	Nil
Net Salvage Value (cash flows)	35,000

(iii) **Computation of additional cash flows (yearly)**

Particulars	Existing machine	New Machine	Incremental
(1)	(2)	(3)	(4)=(3)–(2)
Annual output	60,000 units	80,000 units	20,000 units
	₹	₹	₹
(A) Sales revenue @ ₹ 18 per unit	10,80,000	14,40,000	3,60,000
(B) Less: Cost of Operation			
Material @ ₹ 5 per unit	3,00,000	4,00,000	1,00,000
Labour			
Old = 2,800 x ₹ 50	1,40,000		70,000
New = 2,800 x ₹ 75		2,10,000	
Indirect cash cost	1,00,000	1,75,000	75,000
Total Cost (B)	5,40,000	7,85,000	2,45,000
Profit Before Tax and depreciation (PBTD) (A – B)	5,40,000	6,55,000	1,15,000

Particulars	Existing machine	New Machine	Incremental
Less: Depreciation $\left(\frac{6,00,000-35,000}{10}\right)$			56,500
Earning after depreciation before Tax			58,500
Less: Tax @30%			17,550
Earning after depreciation and Tax			40,950
Add: Depreciation			56,500
Net Cash inflow			97,450

Analysis: Since the Incremental Cash flow is positive, the old machine should be replaced.

Note: As mentioned in the question WDV of Machine is zero for tax purpose hence no depreciation shall be provided in existing machine.

(iv) **Calculation of IRR**

Computation of NPV @ 10%

	Period	Cash flow (₹)	PVF @ 10%	PV (₹)
Incremental cash flows	1-10	97,450	6.144	5,98,733
Add: Release of Working Capital	10	50,000	0.386	19,300
Add: Terminal year cash	10	35,000	0.386	13,510
				6,31,543
Less: Initial cash outflow	0	4,95,000	1	4,95,000
Less: Working capital	0	50,000	1	50,000
			NPV	86,543

Since NPV computed in Part (i) is positive. Let us discount cash flows at higher rate say at 20%

	Period	Cash flow (₹)	PVF @ 20%	PV (₹)
Incremental cash flows	1-10	97,450	4.192	4,08,510
Add: Release of Working Capital	10	50,000	0.162	8,100
Add: Terminal year cash	10	35,000	0.162	5,670
				4,22,280
Less: Initial cash outflow	0	4,95,000	1	4,95,000
Less: Working capital	0	50,000	1	50,000
			NPV	(1,22,720)

Now we use interpolation formula:

$$10\% + \frac{86,543}{86,543 - (-1,22,720)} \times 10\%$$

$$10\% + \frac{86,543}{2,09,263} \times 10\%$$

IRR = 10% + 4.14% = 14.14%

Summary of Results

		Decision
Incremental Cash Flow	₹ 97,450	Accept
IRR	14.14% > Cost of Capital (10%)	Accept

ICAI · P6 · RTP May 2024 · question p. 6-7 · answer p. 18-20

506 RTP **Old P8 Nov 2020** Q5 old scheme

A large profit making company is considering the installation of a machine to process the waste produced by one of its existing manufacturing process to be converted into a marketable product. At present, the waste is removed by a contractor for disposal on payment by the company of ₹ 150 lakh per annum for the next four years. The contract can be terminated upon installation of the aforesaid machine on payment of a compensation of ₹ 90 lakh before the processing operation starts. This compensation is not allowed as deduction for tax purposes.

The machine required for carrying out the processing will cost ₹ 600 lakh to be financed by a loan repayable in 4 equal instalments commencing from end of the year 1. The interest rate is 14% per annum. At the end of the 4th year, the machine can be sold for ₹ 60 lakh and the cost of dismantling and removal will be ₹ 45 lakh.

Sales and direct costs of the product emerging from waste processing for 4 years are estimated as under:

(₹ In lakh)

Year	1	2	3	4
Sales	966	966	1,254	1,254
Material consumption	90	120	255	255
Wages	225	225	255	300
Other expenses	120	135	162	210
Factory overheads	165	180	330	435
Depreciation (as per income tax rules)	150	114	84	63

Initial stock of materials required before commencement of the processing operations is ₹ 60 lakh at the start of year 1. The stock levels of materials to be maintained at the end of year 1, 2 and 3 will be ₹ 165 lakh and the stocks at the end of year 4 will be nil. The storage of materials will utilise space which would otherwise have been rented out for ₹ 30 lakh per annum. Labour costs include wages of 40 workers, whose transfer to this process will reduce idle time payments of ₹ 45 lakh in the year - 1 and ₹

30 lakh in the year - 2. Factory overheads include apportionment of general factory overheads except to the extent of insurance charges of ₹ 90 lakh per annum payable on this venture. The company's tax rate is 30%.

Present value factors for four years are as under:

Year	1	2	3	4
PV factors @14%	0.877	0.769	0.674	0.592

ADVISE the management on the desirability of installing the machine for processing the waste. All calculations should form part of the answer.

ICAI'S ANSWER

Statement of Operating Profit from processing of waste (₹ in lakh)

Year	1	2	3	4
Sales :(A)	966	966	1,254	1,254
Material consumption	90	120	255	255
Wages	180	195	255	300
Other expenses	120	135	162	210
Factory overheads (insurance only)	90	90	90	90
Loss of rent on storage space (opportunity cost)	30	30	30	30
Interest @14%	84	63	42	21
Depreciation (as per income tax rules)	150	114	84	63
Total cost: (B)	744	747	918	969
Profit (C)=(A)-(B)	222	219	336	285
Tax (30%)	66.6	65.7	100.8	85.5
Profit after Tax (PAT)	155.4	153.3	235.2	199.5

Statement of Incremental Cash Flows (₹ in lakh)

Year	0	1	2	3	4
Material stock	(60)	(105)	-	-	165
Compensation for contract	(90)	-	-	-	-
Contract payment saved	-	150	150	150	150
Tax on contract payment	-	(45)	(45)	(45)	(45)
Incremental profit	-	222	219	336	285

Year	0	1	2	3	4
Depreciation added back	-	150	114	84	63
Tax on profits	-	(66.6)	(65.7)	(100.8)	(85.5)
Loan repayment	-	(150)	(150)	(150)	(150)
Profit on sale of machinery (net)	-	-	-	-	15
Total incremental cash flows	(150)	155.4	222.3	274.2	397.5
Present value factor	1.00	0.877	0.769	0.674	0.592
Present value of cash flows	(150)	136.28	170.95	184.81	235.32
Net present value					577.36

Advice: Since the net present value of cash flows is ₹ 577.36 lakh which is positive the management should install the machine for processing the waste.

Notes:

(i) Material stock increases are taken in cash flows.

(ii) Idle time wages have also been considered.

(iii) Apportioned factory overheads are not relevant only insurance charges of this project are relevant.

(iv) Interest calculated at 14% based on 4 equal instalments of loan repayment.

(v) Sale of machinery- Net income after deducting removal expenses taken. Tax on Capital gains ignored.

(vi) Saving in contract payment and income tax thereon considered in the cash flows.

ICAI · Old P8 · RTP Nov 2020 · question p. 3 · answer p. 10-11

507 MTP **Sep 2026 S-I** Case Scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Jennie Components Ltd. is a leading manufacturer of industrial metal sheets and precision-engineered components supplied to automobile, infrastructure, and construction companies across India. Over the last few years, the company has expanded its production capacity significantly due to rising demand from electric vehicle manufacturers and large-scale infrastructure projects. However, the manufacturing operations of the company generate substantial quantities of metal scrap and chemical waste as a by-product of the production process.

At present, the company disposes of the waste through an external waste-management agency and incurs disposal charges of ₹ 0.50 per kilogram. Under normal operating conditions, the company generates approximately 20,00,000 kilograms of waste annually. Recently, increasing environmental compliance requirements and rising waste-management costs prompted the management to explore whether the waste material could be commercially utilized instead of being discarded.

After conducting detailed technical research and laboratory testing involving an expenditure of ₹ 12,00,000, the company's technical team discovered that the waste could be processed further and

converted into reusable industrial filler material used in cement, paint, and construction chemical industries. Market surveys conducted by the marketing department indicated strong demand for the processed material, and management expects that the entire quantity processed each year can be sold without any difficulty at a selling price of ₹ 5 per kilogram.

The proposed further-processing activity would require installation of a specialized processing machine costing ₹ 1,20,00,000. The machine is expected to have a useful life of 10 years with no salvage value at the end of its life. Depreciation on the machine would be charged under the straight-line method. The processing activity would involve additional variable processing cost of 2.50 per kilogram along with annual advertising expenditure of ₹ 4,00,000 and additional fixed operating expenses of ₹6,00,000 per annum, excluding depreciation.

The Finance Director CA K. Aananda informed the Board that the company's cost of capital is 15% and the applicable corporate tax rate is 50%. The present value annuity factor for ₹ 1 for 10 years at 15% is 5.019. The management further clarified that there would be no loss of quantity during processing and all annual cash flows are expected to occur at the end of each year.

The Board of Directors has requested the finance team to evaluate the financial viability of processing the waste further instead of disposing it externally and advise whether the proposal would enhance shareholders' wealth.

MCQ 1 2 marks

What will be the annual incremental contribution before depreciation and tax from processing the waste further?

A ₹ 40,00,000

B ₹ 45,00,000

C ₹ 50,00,000

D ₹ 50,50,000

MCQ 2 2 marks

What will be the annual after-tax cash inflow from the further-processing proposal?

A ₹ 24,00,000

B ₹ 27,00,000

C ₹ 31,00,000

D ₹ 38,00,000

MCQ 3 2 marks

What will be the Present Value of annual after-tax cash inflows from the proposal?

A ₹ 1,32,57,000

B ₹ 1,55,58,900

C ₹ 1,44,82,500

D ₹ 1,67,25,000

MCQ 4 2 marks

What will be the Net Present Value (NPV) of the further-processing proposal?

A ₹ 15,58,900

B ₹ 24,00,000

C ₹ 35,58,900

D ₹ 47,50,000

MCQ 5 2 marks

What will be the Profit Before Tax (PBT) generated annually from the further-processing proposal after considering depreciation?

A ₹ 25,00,000

B ₹ 32,00,000

C ₹ 38,00,000

D ₹ 44,00,000

ICAI'S ANSWERS

MCQ 1

(C) ₹ 50,00,000

Revenue from Sale of Processed Waste = 20,00,000 kg × ₹ 5 = ₹ 1,00,00,000

Add: Disposal Cost Saved = 20,00,000 kg × ₹ 0.50 = ₹ 10,00,000

Total Relevant Benefits = ₹ 1,10,00,000

Less: Variable Processing Cost = 20,00,000 kg × ₹ 2.50 = ₹ 50,00,000

Less: Additional Fixed Costs

Particulars	Amount (₹)
Advertising Expenses	4,00,000
Additional Fixed Cost	6,00,000
Total	10,00,000

Incremental Contribution Before Depreciation & Tax = 1,10,00,000 – 50,00,000 – ₹ 10,00,000 = ₹ 50,00,000

MCQ 2

(C) ₹ 31,00,000 Incremental Contribution Before Depreciation & Tax = ₹ 50,00,000

Less: Depreciation = ₹ 1,20,00,000 / 10 = ₹ 12,00,000

Profit Before Tax = 50,00,000 – 12,00,000 = ₹ 38,00,000

Less: Tax @ 50% = ₹ 19,00,000

Profit After Tax = ₹ 19,00,000

Add Back Depreciation = 19,00,000 + 12,00,000 = ₹ 31,00,000

MCQ 3

(B) ₹ 1,55,58,900

Annual After-tax Cash Inflow = ₹ 31,00,000

Present Value of Annual Cash Inflows = ₹ 31,00,000 × 5.019 = ₹ 1,55,58,900

MCQ 4

(C) ₹ 35,58,900

Particulars	Amount (₹)
PV of Annual Cash Inflows	1,55,58,900
Less: Initial Investment	1,20,00,000
Net Present Value	35,58,900

Since NPV is positive, the company should process the waste further instead of disposing it externally.

MCQ 5

(C) ₹ 38,00,000

Incremental Contribution Before Depreciation & Tax From earlier computation:

Particulars	Amount (₹)
Sales Revenue from Processed Waste	1,00,00,000
Add: Disposal Cost Saved	10,00,000
Total Benefits	1,10,00,000
Less: Variable Processing Cost	(50,00,000)
Less: Additional Fixed Costs	(10,00,000)
Contribution Before Depreciation & Tax	50,00,000

Less: Depreciation = Cost of Machine/Useful Life = ₹ 1,20,00,000/10 = ₹ 12,00,000

Profit Before Tax (PBT) = 50,00,000 – 12,00,000 = ₹ 38,00,000

ICAI · P6 · MTP Sep 2026 · Series I · question p. 1-2 · answer p. 1, 2

508 MTP **Sep 2026 S-II** MCQ 6 2 marks MCQ

Anand is considering automating his glass factory with the purchase of a machine costing ₹ 10,95,000. Shipping and installation costs are ₹ 25,000. He has calculated that the automation would result in savings of ₹ 1,25, 000. The machine has a useful life of 3 years. What will be the incremental cash outlay of this automation?

A ₹ 11,20,000

B ₹ 9,70,000

C ₹ 9,95,000

D ₹ 10,70,000

ICAI'S ANSWER

(A) ₹ 11,20,000

Incremental Initial Cash Outlay = Cost of machine + Shipping & Installation

= ₹ 10,95,000 + ₹ 25,000

= ₹ 11,20,000

509 MTP Jan 2026 S-I Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

ABC Ltd is a growing renewable energy manufacturing company specializing in Parabolic Solar Cookers, which are widely used for rural cooking programs, NGO sustainability projects, and government-sponsored green initiatives. Over the past few years, ABC Ltd has seen a steady rise in demand due to increasing awareness of clean energy solutions and state-level subsidies.

Currently, ABC Ltd fabricates the metal frame, stand, and support structures of the solar cooker at their facility. However, one critical component—the Concentrator Dish (the reflective parabola that focuses sunlight)—is being purchased from an external vendor, PQR Ltd, at a cost of ₹ 2,000 per unit.

Recently, ABC Ltd has faced several operational challenges with the outsourced dish supplier like Supply Delays, Quality Inconsistency, Rising Procurement Costs, large government order requiring strict timelines and quality standards.

Because of these concerns, management is considering backward integration—bringing dish manufacturing in-house to gain greater control over cost, quality, and delivery.

To produce the concentrator dishes internally, management is evaluating the cost structure:

- Raw Material: ₹ 700 per unit
- Labour Cost: ₹ 300 per unit
- Other Variable Costs: ₹ 100 per unit

Total Variable Cost per unit = ₹ 1,100

To manufacture the dish internally, ABC Ltd must purchase new machinery costing ₹ 25,00,000 with:

- Useful life: 5 years
- Depreciation rate: 20% per year (straight-line)
- Tax rate: 30%
- Discount rate: 10%
- Expected annual production: 1,000 units
- Present value annuity factor for 10% over 5 years: 3.7908

The leadership team wants to understand whether in-house manufacturing is financially viable compared to continuing purchases from PQR Ltd. Based on the above facts you are required to answer the following questions (MCQs 1 to 5):

MCQ 1 2 marks

What is present value of outflow incase of concentrator dish purchased from PQR Ltd for 5 years after considering taxes?

A 50,40,325

B 43,52,789

C 31,25.852

D 53,07,120

MCQ 2 2 marks

What is present value of outflow after considering taxes if ABC Ltd is planning to manufacture concentrator dish?

- A 45,80,296
- B 48,50,296
- C 46,85,029
- D 53,25,841

MCQ 3 2 marks

Suppose only 700 units are needed in a particular year. Which conclusion is correct?

- A The investment is still cash-positive even at 700 units
- B The investment becomes cash-negative at 700 units
- C The depreciation cancels out tax benefits at 700 units
- D ABC should outsource if output falls below 900 units

MCQ 4 2 marks

ABC Ltd has found 2 new vendors selling concentrator dish. XYZ Ltd for ₹ 2100/- per unit and UVW Ltd for ₹1900/- per unit. Which option should ABC Ltd choose to minimize the cost?

- A Manufacture the Concentrator dish inhouse
- B Purchase from PQR Ltd
- C Purchase from XYZ Ltd
- D Purchase from UVW Ltd

MCQ 5 2 marks

UVW Ltd is negotiating with ABC Ltd and asking for final price for which they would want to purchase it. What should be price per unit at which ABC Ltd can purchase it from UVW Ltd?

- A 2,000
- B 1,965
- C 1,828
- D 1,900

ICAI'S ANSWERS

MCQ 1

(D) 53,07,120

Total Cost per year = 2,000 per unit x 1,000 units

= 20,00,000

Total Cost after adjusting taxes = 20,00,000 x (1-0.3) = 14,00,000

Present Value of cost at discount rate of 10% p. a

= Cost per Anum x Annuity factor

= 14,00,000 x 3.7908

= 53,07,120/-

MCQ 2

(B) 48,50,296

1. Cost of Machinery = ₹ 25,00,000/- outflow in year 0.

2. Present value of cost of machinery = ₹ 25,00,000

3. Annual outflows

Raw Material	700
Labour	300
Other Variable Cost	100
Variable Cost	1,100
Total Cost (1,100 × 1,000 units)	11,00,000
Add: Depreciation	5,00,000
Total Cost	16,00,000
Total cost net of taxes (16,00,000x x70/100)	11,20,000
Less: Depreciation	5,00,000
Net Cash Outflow	6,20,000

Present Value of cash outflow for 5 years = 6,20,000 x annuity factor

= 6,20,000 x 3.7908

= 23,50,296

Total Outflows = 25,00,000+23,50,296

= 48,50,296

MCQ 3

(A) The investment is still cash-positive even at 700 units

Fixed investment cost (depreciation) = ₹ 5,00,000

Per-unit variable savings = ₹ 2,000 – ₹ 1,100 = ₹ 900

Actual annual savings = 700 × 900 = ₹ 6,30,000

Profit before tax = 6,30,000 − 5,00,000 = 1,30,000

Tax = 30% → 39,000

Profit after tax = 91,000

Add back depreciation = 5,00,000

Cash inflow = ₹ 5,91,000

MCQ 4

(A) Manufacture the Concentrator dish inhouse

1. Manufacture of Concentrator dish

Total Outflows after adjusting taxes = 48,50,296

2. Purchase from PQR Ltd.

Total Outflows after adjusting taxes = 53,07,120

3. Purchase from XYZ Ltd.

Cost per unit charged by XYZ Ltd. is ₹ 2100/- per unit which is greater than PQR Ltd. Cost of purchasing from PQR Ltd. is already greater than manufacturing it. So, this need not be calculated.

4. Purchase from UVW Ltd.

Cost per Unit = 1,900

Total Cost = 1,900 x 1,000 units = 19,00,000

Total cost after adjusting taxes = 19,00,000 x 70/100

= 13,30,000

Present Value of cost at discount rate of 10% p.a. = Cost per Anum x Annuity factor

= 13,30,000 x 3.7908

= 50,41,764

MCQ 5

(C) 1,828

So, the cost of Manufacturing is least considering all options and so ABC Ltd should be purchasing it from UVW Ltd only if total cost of purchasing it from UVW Ltd is equal to manufacturing it.

Total outflow of manufacturing concentrator dish = 48,50,296

Lets do a reverse calculation

PV of cost at discount rate of 10% p.a. = Cost per Annum x Annuity factor

48,50,296 = Cost per Annum x 3.7908

Cost per Annum = 48,50,296/3.7908 = 12,79,491

Cost including Taxes = $\frac{12,79,491}{70 \text{ x } 100}$ = 18,27,845

Cost per unit = $\frac{18,27,845}{1000}$ = 1,827.8

ICAI · P6 · MTP Jan 2026 · Series I · question p. 1-2 · answer p. 1, 1-2, 2, 2-3, 3

510 MTP **Jan 2026 S-II** Q1(c) 5 marks

Orion Aerospace Ltd. manufactures highly customized components for satellites launched by the France and Germany. The company operates through eight locations worldwide, including its central headquarters. To streamline the budgeting process and reduce escalating travel costs, the Finance Director, Ms. Ananya Rao, is considering the introduction of video conferencing as an alternative to physical meetings.

In previous years, the company sent two executives from each location to the central headquarters twice a year for budget discussions. The average travel cost per executive (including airfare, accommodation, and meals) is ₹ 27,000 per trip.

Under the proposed plan, Orion Aerospace Ltd. would install a video conferencing system at each location at a cost of ₹ 8,25,000 per location. In addition, the system would incur an average telephone

transmission cost of ₹ 300 per hour, and it is estimated that 48 hours of transmission time per year would be required to complete the budgeting process. The company depreciates such equipment using the straight-line method over a useful life of five years, with no residual value.

As an alternative, the company may opt to use locally rented video conferencing facilities, which are available at a cost of ₹ 1,500 per hour, plus ₹ 400 per hour towards average telephone charges.

Additional Information:

The company follows a policy of minimum annual cash savings of 10% of the existing travel cost before approving any new cost-saving technology.

You are a Senior Officer in the Finance Department. Based on the above information, you are required to EVALUATE the proposal and advise Ms. Ananya Rao on whether Orion Aerospace Ltd. should implement video conferencing, and if so, WHICH alternative would be financially preferable.

ICAI'S ANSWER

Option I : Cost of travel, in case Video Conferencing facility is not provided

Total Trip = No. of Locations × No. of Persons × No. of Trips per Person = 7 × 2 × 2 = 28 Trips

Total Travel Cost (including air fare, hotel accommodation and meals) (28 trips × ₹ 27,000 per trip)

= ₹ 7,56,000

Option II : Video Conferencing Facility is provided by Installation of Own Equipment at Different Locations

Cost of Equipment at each location (₹ 8,25,000 × 8 locations) = ₹ 66,00,000

Economic life of Machines (5 years). Annual depreciation (66,00,000/5) = ₹ 13,20,000

Annual transmission cost (48 hrs. transmission × 8 locations × ₹ 300 per hour) = ₹ 1,15,200

Annual cost of operation (13,20,000 + 1,15,200) **= ₹ 14,35,200**

Option III : Engaging Video Conferencing Facility on Rental Basis

Rental cost (48 hrs. × 8 location × ₹ 1,500 per hr) = ₹ 5,76,000

Telephone cost (48 hrs.× 8 locations × ₹ 400 per hr.) = ₹ 1,53,600

Total rental cost of equipment (5,76,000 + 1,53,600) **= ₹ 7,29,600**

Minimum required savings = 10% of existing travel cost

Required minimum savings = 10% of ₹ 7,56,000 = ₹ 75,600

Actual Savings Using Rented Facilities = ₹ 7,56,000 - ₹ 7,29,600 **= ₹ 26,400**

Analysis: The annual cost is minimum, if video conferencing facility is engaged on rental basis. However, saving is less than 10% of existing travel cost. Therefore, no change is suggested.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 5-6 · answer p. 6-7

511 MTP **May 2024 S-I** Q2(b) 2 marks

Also asked in Old P8 MTP Nov 2022 S-II Q6(c)

WRITE a short note on "Cut-off Rate".

ICAI'S ANSWER

Cut-off Rate: It is the minimum rate which the management wishes to have from any project. Usually this is based upon the cost of capital. The management gains only if a project gives return of more than the cut - off rate. Therefore, the cut - off rate can be used as the discount rate or the opportunity cost rate.

ICAI · P6 · MTP May 2024 · Series I · question p. 4 · answer p. 5

512 MTP Old P8 May 2022 S-II Q4 10 marks old scheme

Manoranjan Ltd is a News broadcasting channel having its broadcasting Centre in Mumbai. There are total 200 employees in the organisation including top management. As a part of employee benefit expenses, the company serves tea or coffee to its employees, which is outsourced from a third-party. The company offers tea or coffee three times a day to each of its employees. 120 employees prefer tea all three times, 40 employees prefer coffee all three times and remaining prefer tea only once in a day. The third-party charges ₹ 10 for each cup of tea and ₹ 15 for each cup of coffee. The company works for 200 days in a year.

Looking at the substantial amount of expenditure on tea and coffee, the finance department has proposed to the management an installation of a master tea and coffee vending machine which will cost ₹ 10,00,000 with a useful life of five years. Upon purchasing the machine, the company will have to enter into an annual maintenance contract with the vendor, which will require a payment of ₹ 75,000 every year. The machine would require electricity consumption of 500 units p.m. and current incremental cost of electricity for the company is ₹ 12 per unit. Apart from these running costs, the company will have to incur the following consumables expenditure also:

(1) Packets of Coffee beans at a cost of ₹ 90 per packet.

(2) Packet of tea powder at a cost of ₹ 70 per packet.

(3) Sugar at a cost of ₹ 50 per Kg.

(4) Milk at a cost of ₹ 50 per litre.

(5) Paper cup at a cost of 20 paise per cup.

Each packet of coffee beans would produce 200 cups of coffee and same goes for tea powder packet. Each cup of tea or coffee would consist of 10g of sugar on an average and 100 ml of milk.

The company anticipate that due to ready availability of tea and coffee through vending machines its employees would end up consuming more tea and coffee. It estimates that the consumption will increase by on an average 20% for all class of employees. Also, the paper cups consumption will be 10% more than the actual cups served due to leakages in them.

The company is in the 25% tax bracket and has a current cost of capital at 12% per annum. Straight line method of depreciation is allowed for the purpose of taxation. You as a financial consultant is required to ADVISE on the feasibility of acquiring the vending machine.

PV factors @ 12%:

Year	1	2	3	4	5
PVF	0.8929	0.7972	0.7118	0.6355	0.5674

ICAI'S ANSWER

A. Computation of CFAT (Year 1 to 5)

Particulars		Amount (₹)
(a) Savings in existing Tea & Coffee charges	(120 × 10 ×3) + (40 ×15 × 3) + (40 ×10 × 1) x 200 days	11,60,000
(b) AMC of machine		(75,000)
(c) Electricity charges	500 ×12 ×12	(72,000)
(d) Coffee Beans	(W.N.) 144 × 90	(12,960)
(e) Tea Powder	(W.N.) 480 × 70	(33,600)
(f) Sugar	(W.N.) 1248 × 50	(62,400)
(g) Milk	(W.N.) 12480 × 50	(6,24,000)
(h) Paper Cup	(W.N.) 1,37,280 × 0.2	(27,456)
(i) Depreciation	10,00,000/5	(2,00,000)
Profit before Tax		**52,584**
(-) Tax @ 25%		(13,146)
Profit after Tax		**39,438**
Depreciation		2,00,000
CFAT		**2,39,438**

B. Computation of NPV

Year	Particulars	CF	PVF @ 12%	PV
0	Cost of machine	(10,00,00)	1	(10,00,000)
1-5	CFAT	2,39,438	3.6048	8,63,126
Net Present Value				**(1,36,874)**

Since NPV of the machine is negative, it should not be purchased.

Working Note:

Computation of Qty of consumable

No. of Tea Cups = [(120 × 3 × 200 days) + (40 × 1 × 200 days) × 1.2 = 96,000

No. of Coffee cups = 40 × 3 × 200 days × 1.2 = 28,800

$$\text{No. of coffee beans packet} = \frac{28,800}{200} = 144$$

$$\text{No. of Tea Powder Packets} = \frac{96,000}{200} = 480$$

$$\text{Qty of Sugar} = \frac{(96,000 + 28,800) \times 10\text{ g}}{1,000\text{ g}} = 1248\text{ kgs}$$

Qty of Milk = $\frac{(96{,}000 + 28{,}800) \times 100 \text{ ml}}{1{,}000 \text{ ml}}$ = 12,480 litres

No. of paper cups = (96,000 + 28,800) × 1.1 = 1,37,280

ICAI · Old P8 · MTP May 2022 · Series II · question p. 3 · answer p. 8-9

513 MTP **Old P8 Dec 2021 S-II** Q3 10 marks old scheme

Superb Ltd. constructs customized parts for satellites to be launched by USA and Canada. The parts are constructed in eight locations (including the central headquarter) around the world. The Finance Director, Ms. Kuthrapali, chooses to implement video conferencing to speed up the budget process and save travel costs. She finds that, in earlier years, the company sent two officers from each location to the central headquarter to discuss the budget twice a year. The average travel cost per person, including air fare, hotels and meals, is ₹ 27,000 per trip. The cost of using video conferencing is ₹ 8,25,000 to set up a system at each location plus ₹ 300 per hour average cost of telephone time to transmit signals. A total 48 hours of transmission time will be needed to complete the budget each year. The company depreciates this type of equipment over five years by using straight line method. An alternative approach is to travel to local rented video conferencing facilities, which can be rented for ₹ 1,500 per hour plus ₹ 400 per hour averge cost for telephone charges. You are Senior Officer of Finance Department. You have been asked by Ms. Kuthrapali to EVALUATE the proposal and SUGGEST if it would be worthwhile for the company to implement video conferencing.

ICAI'S ANSWER

Option I : Cost of travel, in case Video Conferencing facility is not provided

Total Trip = No. of Locations × No. of Persons × No. of Trips per Person = 7×2×2 = 28 Trips

Total Travel Cost (including air fare, hotel accommodation and meals) (28 trips × ₹ 27,000 per trip) **= ₹ 7,56,000**

Option II : Video Conferencing Facility is provided by Installation of Own Equipment at Different Locations

Cost of Equipment at each location (₹ 8,25,000 × 8 locations) = ₹ 66,00,000

Economic life of Machines (5 years). Annual depreciation (66,00,000/5) = ₹ 13,20,000

Annual transmission cost (48 hrs. transmission × 8 locations × ₹ 300 per hour) = ₹ 1,15,200

Annual cost of operation (13,20,000 + 1,15,200) **= ₹ 14,35,200**

Option III : Engaging Video Conferencing Facility on Rental Basis

Rental cost (48 hrs. × 8 location × ₹ 1,500 per hr) = ₹ 5,76,000

Telephone cost (48 hrs.× 8 locations × ₹ 400 per hr.) = ₹ 1,53,600

Total rental cost of equipment (5,76,000 + 1,53,600) **= ₹ 7,29,600**

Analysis: The annual cash outflow is minimum, if video conferencing facility is engaged on rental basis. Therefore, Option III is suggested.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 3-4 · answer p. 5

7.2 Payback, discounted payback and ARR

18 questions · asked in 12 ICAI documents · 2 exam papers

514 EXAM **May 2024** Q4(c) 2 marks

Discuss the relevance of Payback reciprocal in capital budgeting decisions.

ICAI'S ANSWER

Reciprocal of the payback would be a close approximation of the Internal Rate of Return if the life of the project is at least twice the payback period and the project generates equal amount of the annual cash inflows.

The payback reciprocal is a helpful tool for quick estimation of rate of return of a project provided its life is at least twice the payback period.

It may be calculated as follows:

Payback Reciprocal = Average annual cash flows/initial Investment

Or

Payback Reciprocal = 1 / payback period

ICAI · P6 · May 2024 exam · question p. 19 · answer p. 22

515 EXAM **Old P8 Jul 2021** Q6(c) 2 marks old scheme

Explain the limitations of Average Rate of Return.

ICAI'S ANSWER

Limitations of Average Rate of Return

- The accounting rate of return technique, like the payback period technique, **ignores the time value of money** and considers the value of all cash flows to be equal.
- The technique uses accounting numbers that are dependent on the organization's **choice of accounting procedures,** and different accounting procedures, e.g., depreciation methods, can lead to substantially different amounts for an investment's net income and book values.
- The method **uses net income rather than cash flows**; while net income is a useful measure of profitability, the net cash flow is a better measure of an investment's performance.
- Furthermore, inclusion of only the book value of the invested asset **ignores** the fact that a project can require **commitments of working capital** and other outlays that are not included in the book value of the project.

ICAI · Old P8 · Jul 2021 exam · question p. 83 · answer p. 84

516 RTP **Sep 2026** Q7

The management of P Limited is evaluating two mutually exclusive investment alternatives, namely Machine I and Machine II, for acquisition. The company has a cost of capital of 12% and is subject to a corporate tax rate of 30%.

The relevant details of the two machines are provided below:

Particulars	Machine I	Machine II
Initial Cost of Machine	₹ 10,00,000	₹ 15,00,000
Useful Life	5 years	6 years
Annual Income Before Depreciation and Tax	₹ 3,45,000	₹ 4,55,000

Additional Information:

- Depreciation is to be charged using the Straight-Line Method (SLM) over the useful life of each machine.

Required:

(i) Compute the following capital budgeting indicators for each machine:

- Discounted Payback Period (DPP)
- Net Present Value (NPV)
- Internal Rate of Return (IRR)

(ii) Based on the results obtained, recommend which machine should be selected by the management of P Limited, providing appropriate justification.

The present value factors of Re. 1 are as follows:

Year	1	2	3	4	5	6
At 12%	.893	.797	.712	.636	.567	.507
At 13%	.885	.783	.693	.613	.543	.480
At 14%	.877	.769	.675	.592	.519	.456
At 15%	.870	.756	.658	.572	.497	.432
At 16%	.862	.743	.641	.552	.476	.410

ICAI'S ANSWER

(i) **Computation of Discounted Payback Period, Net Present Value (NPV) and Internal Rate of Return (IRR) for Two Machines**

Calculation of Cash Inflows

	Machine - I (₹)	Machine - II (₹)
Annual Income before Tax and Depreciation	3,45,000	4,55,000
Less: Depreciation		
Machine – I: 10,00,000 /5	2,00,000	-
Machine – II: 15,00,000 / 6	-	2,50,000
Income before Tax	1,45,000	2,05,000

	Machine - I (₹)	Machine - II (₹)
Less: Tax @ 30 %	43,500	61,500
Income after Tax	1,01,500	1,43,500
Add: Depreciation	2,00,000	2,50,000
Annual Cash Inflows	3,01,500	3,93,500

		Machine – I			Machine – II		
Year	**P.V. of Re.1 @12%**	**Cash flow**	**P.V.**	**Cumulative P.V**	**Cash flow**	**P.V.**	**Cumulative P.V.**
1	0.893	3,01,500	2,69,240	2,69,240	3,93,500	3,51,396	3,51,396
2	0.797	3,01,500	2,40,296	5,09,536	3,93,500	3,13,620	6,65,016
3	0.712	3,01,500	2,14,668	7,24,204	3,93,500	2,80,172	9,45,188
4	0.636	3,01,500	1,91,754	9,15,958	3,93,500	2,50,266	11,95,454
5	0.567	3,01,500	1,70,951	10,86,909	3,93,500	2,23,115	14,18,569
6	0.507	-	-	-	3,93,500	1,99,505	16,18,074

Discounted Payback Period for:

Machine – I

$$\text{Discounted Payback Period} = 4 + \frac{(10,00,000 - 9,15,958)}{1,70,951}$$

$$= 4 + \frac{84,042}{1,70,951} = 4 + 0.4916$$

= 4.49 years or 4 years and 5.9 months

Machine – II

$$\text{Discounted Payback Period} = 5 + \frac{(15,00,000 - 14,18,569)}{1,99,505} = 5 + \frac{81,431}{1,99,505}$$

= 5 + 0.4082

= 5.41 years or 5 years and 4.9 months

Net Present Value for:

Machine – I

NPV = ₹ 10,86,909 – 10,00,000 = ₹ 86,909

Machine – II

NPV = ₹ 16,18,074 – 15,00,000 = ₹ 1,18,074

Internal Rate of Return (IRR) for:

Machine – I

P.V. Factor = $\frac{\text{Initial Investment}}{\text{Annual Cash Inflow}} = \frac{10,00,000}{3,01,500}$ = 3.3167

PV factor falls between 15% and 16%

Present Value of Cash inflow at 15% and 16% will be:

Present Value at 15% = 3.353 x 3,01,500 = 10,10,930

Present Value at 16% = 3.274 x 3,01,500 = 9,87,111

IRR = $15 + \frac{10,10,930 - 10,00,0000}{10,10,930 - 9,87,111} \times (16 - 15) = 15 + \frac{10,930}{23,819} \times 1$

= 15.4588%

= 15.46%

Machine - II

P.V. Factor = $\frac{15,00,000}{3,93,500}$ = 3.8119

Present Value of Cash inflow at 14% and 15% will be:

Present Value at 14% = 3.888 x 3,93,500 = 15,29,928

Present Value at 15% = 3.785 x 3,93,500 = 14,89,398

IRR = $14 + \frac{15,29,928 - 15,00,000}{15,29,928 - 14,89,398} \times (15 - 14) = 14 + \frac{29,928}{40,530} \times 1$= 14.7384%

= 14.74%

(ii) **Advise to the Management**

Ranking of Machines in terms of the Three Methods

	Machine - I	Machine - II
Discounted Payback Period	I	II
Net Present Value	II	I
Internal Rate of Return	I	II

Advise: Since Machine - I has better ranking than Machine – II, therefore, Machine – I should be selected

ICAI · P6 · RTP Sep 2026 · question p. 7-8 · answer p. 17-20

517 RTP **Old P8 Nov 2019** Case scenario · Q10(a) to Q10(c) old scheme

CASE SCENARIO

Write short notes on the following:

Q10(a)

STATE the meaning of Payback Reciprocal.

Q10(b)

STATE the functions of treasury department.

Q10(c)

DESCRIBE the Inter relationship between investment, financing and dividend decisions.

ICAI'S ANSWERS

Q10(a)

As the name indicates it is the reciprocal of payback period. A major drawback of the payback period method of capital budgeting is that it does not indicate any cut off period for the purpose of investment decision. It is, however, argued that the reciprocal of the payback would be a close approximation of the Internal Rate of Return (later discussed in detail) if the life of the project is at least twice the payback period and the project generates equal amount of the annual cash inflows. In practice, the payback reciprocal is a helpful tool for quick estimation of rate of return of a project provided its life is at least twice the payback period.

The payback reciprocal can be calculated as follows:

$$\text{Payback Reciprocal} = \frac{\text{Average annual cash in flow}}{\text{Initial investment}}$$

Q10(b)

1. **Cash Management:** It involves efficient cash collection process and managing payment of cash both inside the organisation and to third parties.

There may be complete centralization within a group treasury or the treasury may simply advise subsidiaries and divisions on policy matter viz., collection/payment periods, discounts, etc.

Treasury will also manage surplus funds in an investment portfolio. Investment policy will consider future needs for liquid funds and acceptable levels of risk as determined by company policy.

2. **Currency Management:** The treasury department manages the foreign currency risk exposure of the company. In a large multinational company (MNC) the first step will usually be to set off intra-group indebtedness. The use of matching receipts and payments in the same currency will save transaction costs. Treasury might advise on the currency to be used when invoicing overseas sales.

The treasury will manage any net exchange exposures in accordance with company policy. If risks are to be minimized then forward contracts can be used either to buy or sell currency forward.

3. **Fund Management:** Treasury department is responsible for planning and sourcing the company's short, medium and long-term cash needs. Treasury department will also participate in the decision on capital structure and forecast future interest and foreign currency rates.

4. **Banking:** It is important that a company maintains a good relationship with its bankers. Treasury department carry out negotiations with bankers and act as the initial point of contact with them. Short-term finance can come in the form of bank loans or through the sale of commercial paper in the money market.

5. **Corporate Finance:** Treasury department is involved with both acquisition and divestment activities within the group. In addition, it will often have responsibility for investor relations. The latter activity has assumed increased importance in markets where share-price performance is regarded as crucial and may affect the company's ability to undertake acquisition activity or, if the price falls drastically, render it vulnerable to a hostile bid.

Q10(c)

Inter-relationship between Investment, Financing and Dividend Decisions: The finance functions are divided into three major decisions, viz., investment, financing and dividend decisions. It is correct to say that these decisions are inter-related because the underlying objective of these three decisions is the same, i.e. maximisation of shareholders' wealth. Since investment, financing and dividend decisions are all interrelated, one has to consider the joint impact of these decisions on the market price of the company's shares and these decisions should also be solved jointly. The decision

to invest in a new project needs the finance for the investment. The financing decision, in turn, is influenced by and influences dividend decision because retained earnings used in internal financing deprive shareholders of their dividends. An efficient financial management can ensure optimal joint decisions. This is possible by evaluating each decision in relation to its effect on the shareholders' wealth.

The above three decisions are briefly examined below in the light of their inter-relationship and to see how they can help in maximising the shareholders' wealth i.e. market price of the company's shares.

Investment decision: The investment of long term funds is made after a careful assessment of the various projects through capital budgeting and uncertainty analysis. However, only that investment proposal is to be accepted which is expected to yield at least so much return as is adequate to meet its cost of financing. This have an influence on the profitability of the company and ultimately on its wealth.

Financing decision: Funds can be raised from various sources. Each source of funds involves different issues. The finance manager has to maintain a proper balance between long-term and short-term funds. With the total volume of long-term funds, he has to ensure a proper mix of loan funds and owner's funds. The optimum financing mix will increase return to equity shareholders and thus maximise their wealth.

Dividend decision: The finance manager is also concerned with the decision to pay or declare dividend. He assists the top management in deciding as to what portion of the profit should be paid to the shareholders by way of dividends and what portion should be retained in the business. An optimal dividend pay-out ratio maximises shareholders' wealth.

The above discussion makes it clear that investment, financing and dividend decisions are interrelated and are to be taken jointly keeping in view their joint effect on the shareholders' wealth

ICAI · Old P8 · RTP Nov 2019 · question p. 5 · answer p. 14-15, 15, 15-16

518 MTP **May 2026 S-II** Case Scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Orion Manufacturing Ltd., a mid-sized engineering company, is facing increasing competition and rising labour costs. To improve efficiency and reduce dependency on manual labour, the management is considering automating a part of its production process.

After preliminary research, the company has shortlisted two advanced machines: Model X-Pro and Model Z-Elite. Both machines can perform the same operations currently handled by workers, but differ in cost and efficiency levels.

The finance team has gathered the following data to evaluate the investment decision.

Particulars		Model X-Pro	Model Z-Elite
Cost of Machine	(₹)	20,00,000	25,00,000
Estimated working life		5 years	5 years
Estimated saving in direct wages per annum	(₹)	7,00,000	9,00,000
Estimated saving in scrap per annum	(₹)	60,000	1,00,000
Estimated additional cost of indirect material per annum	(₹)	30,000	90,000
Estimated additional cost of indirect labour per annum	(₹)	40,000	50,000
Estimated additional cost of repairs and maintenance		45,000	85,000

Depreciation will be charged on a straight line method. Corporate tax rate is 30 percent and expected rate of return may be 12 percent.

You are being the finance manager of the company required to analyse both the machine in relation to financial details and answer the following questions (MCQ 1 to 5)

MCQ 1 2 marks

Pay-back Period of Machine Model X-Pro is:

A 2.55 years

B 3 years

C 3.50 years

D 4 years

MCQ 2 2 marks

Pay-back Period of Machine Model Z-Elite is:

A 2 years

B 2.52 years

C 3 years

D 3.61 Years

MCQ 3 2 marks

Accounting (Average) Rate of Return (ARR) on the basis of average investment is:

A X-Pro: 17.15%, Z-Elite: 15.40%

B X-Pro: 16%, Z-Elite: 18.2%

C X-Pro: 18.2%, Z-Elite: 20%

D X-Pro: 20%, Z-Elite: 22.16%

MCQ 4 2 marks

Profitability Index (P.V. Index) of Model X-Pro is:

A 1.05

B 1.10

C 1.03

D 1.20

MCQ 5 2 marks

Profitability Index (P.V. Index) of Model Z-Elite is:

A 1.02

B 1.08

C 1.12

D 1.00

ICAI'S ANSWERS

ICAI's working for this case

Working Notes:

Depreciation on Machine X-Pro = $\frac{20,00,000}{5}$

= 4,00,000

Depreciation on Machine Z-Elite = $\frac{25,00,000}{5}$ = 5,00,000

Particulars	Machine X-Pro (₹)	Machine Z-Elite (₹)
Annual Savings:		
Direct Wages	7,00,000	9,00,000
Scraps	60,000	1,00,000
Total Savings (A)	7,60,000	10,00,000
Annual Estimated Cash Cost :		
Indirect Material	30,000	90,000
Indirect Labour	40,000	50,000
Repairs and Maintenance	45,000	85,000
Total Cost (B)	1,15,000	2,25,000
Annual Cash Savings (A-B)	6,45,000	7,75,000
Less: Depreciation	4,00,000	5,00,000
Annual Savings before Tax	2,45,000	2,75,000
Less: Tax @ 30%	73,500	82,500
Annual Savings /Profits after tax	1,71,500	1,92,500
Add: Depreciation	4,00,000	5,00,000
Annual Cash Inflows	5,71,500	6,92,500

MCQ 1

(C) 3.50 years

Payback Period Machine X-Pro = $\frac{\text{Total Initial Capital Investment}}{\text{Annual expected after tax net cashflow}}$

= $\frac{20,00,000}{5,71,500}$ = 3.50 Years

MCQ 2

(D) 3.61 Years

Payback Period Machine Z-Elite = $\frac{25,00,000}{6,92,500}$ = 3.61 Years

MCQ 3

(A) X-Pro: 17.15%, Z-Elite: 15.40%

Accounting (Average) Rate of Return (ARR)

ARR = $\frac{\text{Average Annual Net Savings}}{\text{Average investment}} \times 100$

Machine X-Pro = $\frac{1,71,500}{10,00,000} \times 100 = 17.15\%$

Machine Z-Elite = $\frac{1,92,500}{12,50,000} \times 100 = 15.4\%$

MCQ 4

(C) 1.03

Present Value Cash Inflow = Annual Cash Inflow x PV factor at 12%

Machine X-PRO Present Value Cash Inflow = 5, 71,500 x 3.605 = ₹ 20, 60,258

PV Index = $\frac{\text{Present Value of Cash Inflow}}{\text{Investment}}$

PV Index Machine X-PRO = $\frac{20,60,258}{20,00,000}$ = 1.03

MCQ 5

(D) 1.00

Machine Z-Elite Present Value Cash Inflow = 6, 92,500 x 3.605 = ₹ 24, 96,463

PV Index Machine Z-Elite = $\frac{24,96,463}{25,00,000}$ = 0.998 = 1.0 approx.

ICAI · P6 · MTP May 2026 · Series II · question p. 1-2 · answer p. 2, 3

519 MTP **May 2025 S-II** Q1(a) 5 marks

A firm can make investment in either of the following two projects. The firm anticipates its cost of capital to be 10%. The pre-tax cash flows of the projects for five years are as follows:

Year	0	1	2	3	4	5
Project A (₹)	(2,00,000)	35,000	80,000	90,000	75,000	20,000
Project B (₹)	(2,00,000)	2,18,000	10,000	10,000	4,000	3,000

Ignore Taxation.

An amount of ₹ 35,000 will be spent on account of sales promotion in year 3 in case of Project A. This has not been taken into account in calculation of pre-tax cash flows.

The discount factors are as under:

Year	0	1	2	3	4	5
PVF (10%)	1	0.91	0.83	0.75	0.68	0.62

You are required to CALCULATE for each project:

(i) The Discounted Payback period

(ii) Desirability factor

(iii) Net Present Value

ICAI'S ANSWER

Calculation of Present Value of cash flows

Year	PV factor @ 10%	Project A		Project B	
		Cash flows (₹)	**Discounted Cash flows (₹)**	**Cash flows (₹)**	**Discounted Cash flows (₹)**
0	1.00	(2,00,000)	(2,00,000)	(2,00,000)	(2,00,000)
1	0.91	35,000	31,850	2,18,000	1,98,380
2	0.83	80,000	66,400	10,000	8,300
3	0.75	55,000(90,000-35,000)	41,250	10,000	7,500
4	0.68	75,000	51,000	4,000	2,720
5	0.62	20,000	12,400	3,000	1,860
Net Present Value			2,900		18,760

(i) Discounted Payback period for the projects:

Project-A: The cumulative discounted cash inflows upto year 4 is ₹1,90,500 and remaining amount required to equate the cash outflow is ₹ 9,500 i.e. (₹ 2,00,000 – ₹ 1,90,500) which will be recovered from year-5 cash inflow. Hence, Payback period will be calculated as below:

$4 \text{ years} + \frac{₹\ 9{,}500}{₹\ 12{,}400}$ = 4.766 years Or 4 years 9.19 months Or 4 years 9 months and 6 days.

Project-B: The cash inflow in year-1 is ₹1,98,380 and remaining amount required to equate the cash outflow is ₹ 1,620 i.e. (₹ 2,00,000 – ₹ 1,98,380) which will be recovered from year-2 cash inflow. Hence, Payback period will be calculated as below:

$1 \text{ year} + \frac{₹\ 1{,}620}{₹\ 8{,}300}$ = 1.195 years Or 1 Year 2.34 months Or 1 Year 2 months and 10 days.

(ii) Desirability factor of the projects

$$\text{Desirability Factor (Profitability Index)} = \frac{\text{Discounted value of Cash Inflows}}{\text{Discounted value of Cash Outflows}}$$

Project A = $\frac{₹\ 2,02,900}{₹\ 2,00,000} = 1.01$

Project B = $\frac{₹\ 2,18,760}{₹\ 2,00,000} = 1.09$

(iii) Net Present Value (NPV) of the projects:

Please refer the above table.

Project A- ₹ 2,900

Project B- ₹ 18,760

ICAI · P6 · MTP May 2025 · Series II · question p. 6 · answer p. 5-6

520 MTP Jan 2025 S-I MCQ 7 2 marks MCQ

Following are the data on a capital project being evaluated by the management of Aman Ltd.

Particulars	Project A
Annual cost saving	₹ 1,80,000
Useful life	5 years
Internal rate of return	10%
Salvage value	0
PVAF (15,4 years)	3.79

Based upon the information, the payback period of the project will be

A 2.652

B 2.850

C 3.790

D 3.855

ICAI'S ANSWER

(C) 3.79

Initial Investment = Annual Cost Savings × PVAF

Annual cost savings = ₹ 1,80,000

PVAF (10%, 5 years) = 3.79

Initial Investment = 1,80,000 × 3.79 = 6,82,200

Payback Period = Initial Investment/ Annual Cost Savings

= 6,82,200/1,80,000

= 3.79 years

ICAI · P6 · MTP Jan 2025 · Series I · question p. 3 · answer p. 2-3

521 MTP May 2024 S-II MCQ 4 1 mark MCQ

A project requires an initial investment of ₹ 20,000 and it would give annual cash inflow of ₹ 4,000. The useful life of the project is estimated to be 10 years. What is payback reciprocal/Approximated IRR?

A 20%

B 15%

C 25%

D 12%

ICAI'S ANSWER

Answer key: **(A)**

(a) 20%

$$\text{Payback Reciprocal} = \frac{\text{Average annual cash in flow}}{\text{Initial investment}}$$

$$= \frac{₹\ 4{,}000 \times 100}{₹\ 20{,}000} = 20\%$$

ICAI · P6 · MTP May 2024 · Series II · question p. 3 · answer p. 2-3

522 MTP **May 2024 S-II** Q4(c) 2 marks

Also asked in Old P8 MTP Nov 2019 S-I Q6(c)

EXPLAIN the concept of discounted payback period.

ICAI'S ANSWER

Concept of Discounted Payback Period

Payback period is time taken to recover the original investment from project cash flows. It is also termed as break even period. The focus of the analysis is on liquidity aspect and it suffers from the limitation of ignoring time value of money and profitability. Discounted payback period considers present value of cash flows, discounted at company's cost of capital to estimate breakeven period i.e. it is that period in which future discounted cash flows equal the initial outflow. The shorter the period, better it is. It also ignores post discounted payback period cash flows.

ICAI · P6 · MTP May 2024 · Series II · question p. 7 · answer p. 11-12

523 MTP **Old P8 Nov 2023 S-I** Q5 10 marks old scheme

A firm can make investment in either of the following two projects. The firm anticipates its cost of capital to be 10%. The pre-tax cash flows of the projects for five years are as follows:

Year	0	1	2	3	4	5
Project A (₹)	(3,00,000)	55,000	1,20,000	1,30,000	1,05,000	40,000
Project 8 (₹)	(3,00,000)	3,18,000	20,000	20,000	8,000	6,000

Ignore Taxation.

An amount of ₹ 45,000 will be spent on account of sales promotion in year 3 in case of Project A. This has not been considered in calculation of pre-tax cash flows.

The discount factors are as under:

Year	0	1	2	3	4	5
PVF (10%)	1	0.91	0.83	0.75	0.68	0.62

You are required to calculate for each project:

(i) The payback period

(ii) The discounted payback period

(iii) Desirability factor

(iv) Net Present Value

ICAI'S ANSWER

Calculation of Present Value of cash flows

Year	PV factor @ 10%	Project A		Project B	
		Cash flows (₹)	Discounted Cash flows	Cash flows (₹)	Discounted Cash flows
0	1.00	(3,00,000)	(3,00,000)	(3,00,000)	(3,00,000)
1	0.91	55,000	50,050	3,18,000	2,89,380
2	0.83	1,20,000	99,600	20,000	16,600
3	0.75	85,000(1,30,000-45,000)	63,750	20,000	15,000
4	0.68	1,05,000	71,400	8,000	5,440
5	0.62	40,000	24,800	6,000	3,720
Net Present Value			**9,600**		**30,140**

(i) The Payback period of the projects:

Project-A: The cumulative cash inflows up-to year 3 is ₹2,60,000 and remaining amount required to equate the cash outflow is ₹ 40,000 i.e. (₹ 3,00,000 – ₹ 2,60,000) which will be recovered from year-4 cash inflow. Hence, Payback period will be calculated as below:

$$3 \text{ years} + \frac{40,000}{1,05,000} = 3.381 \text{ years or 3 years, 4 months, 9 days (approx.)}$$

Project-B: The cash inflow in year-1 is ₹ 3,18,000 and the amount required to equate the cash outflow is ₹ 3,00,000, which can be recovered in a period less than a year. Hence, Payback period will be calculated as below:

$$\frac{3,00,000}{3,18,000} = 0.943 \text{ years or 11 months}$$

(ii) Discounted Payback period for the projects:

Project-A: The cumulative discounted cash inflows up-to year 4 is ₹ 2,84,800 and remaining amount required to equate the cash outflow is ₹ 15,200 i.e. (₹ 3,00,000 – ₹ 2,84,800) which will be recovered from year-5 cash inflow. Hence, Payback period will be calculated as below:

$$4 \text{ years} + \frac{15,200}{24,800} = 4.613 \text{ years or 4 years, 2 months, and 11 days}$$

Project-B: The cash inflow in year-1 is ₹2,89,380 and remaining amount required to equate the cash outflow is ₹ 10,620 i.e. (₹ 3,00,000 – ₹ 2,89,380) which will be recovered from year-2 cash inflow. Hence, Payback period will be calculated as below:

$1 \text{ year} + \frac{10,620}{16,600} = 1.640$ years or 1 Year, 7 months and 23 days.

(iii) Desirability factor of the projects

$$\text{Desirability Factor (Profitability Index)} = \frac{\text{Discounted value Cash Inflows}}{\text{Discounted value of Cash Outflows}}$$

Project A $= \frac{3,09,600}{3,00,000} = 1.032$

Project B $= \frac{3,30,140}{3,00,000} = 1.100$

(iv) Net Present Value (NPV) of the projects:

Please refer the above table.

Project A- ₹ 9,600

Project B- ₹ 30,140

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524 MTP **Old P8 May 2022 S-II** Q6(c) 2 marks old scheme

EXPLAIN the term 'Payback reciprocal'.

ICAI'S ANSWER

Financial ratios provide clues but not conclusions. These are tools only in the hands of experts because there is no standard ready-made interpretation of financial ratios

As the name indicates it is the **reciprocal of payback period**. A major drawback of the payback period method of capital budgeting is that it does not indicate any cut off period for the purpose of investment decision. It is, however, argued that the reciprocal of the payback would be a close approximation of the Internal Rate of Return (later discussed in detail) if the life of the project is at least twice the payback period and the project generates equal amount of the annual cash inflows. In practice, the payback reciprocal is a helpful tool for quickly estimating the rate of return of a project provided its life is at least twice the payback period.

The payback reciprocal can be calculated as follows:

$$\text{Payback Reciprocal} = \frac{\text{Average annual cash in flow}}{\text{Initial investment}}$$

ICAI · Old P8 · MTP May 2022 · Series II · question p. 4 · answer p. 11

525 MTP **Old P8 Nov 2020 S-I** Q4 10 marks old scheme

A firm can make investment in either of the following two projects. The firm anticipates its cost of capital to be 10%. The pre-tax cash flows of the projects for five years are as follows:

Year	0	1	2	3	4	5
Project A (₹)	(2,00,000)	35,000	80,000	90,000	75,000	20,000
Project 8 (₹)	(2,00,000)	2,18,000	10,000	10,000	4,000	3,000

Ignore Taxation.

An amount of ₹ 35,000 will be spent on account of sales promotion in year 3 in case of Project A. This has not been taken into account in calculation of pre-tax cash flows.

The discount factors are as under:

Year	0	1	2	3	4	5
PVF (10%)	1	0.91	0.83	0.75	0.68	0.62

You are required to calculate for each project:

(i) The payback period

(ii) The discounted payback period

(iii) Desirability factor

(iv) Net Present Value

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 3-4

7.3 NPV, profitability index and equivalent annual cost

60 questions · asked in 43 ICAI documents · 13 exam papers

526 EXAM May 2026 Q2(a) 7 marks

R Ltd. wants to reduce the dependency of manual workforce in the operation of its factory by adopting automatic machine. It has two options under consideration Machine M and Machine N. Both the machines serve the same job and are also able to satisfy the purpose of the company.

Details of both the machines are given below:

	Machine M	Machine N
Purchase cost (₹)	4,50,000	3,50,000
Estimated useful life	5 years	7 years
Estimated salvage value (% of cost)	10%	10%
Annual saving to wages (₹)	11,00,000	12,00,000
Runing cost per year (excl. depreciation) (₹)	9,00,000	11,00,000
Annual saving in scrap due to use of machine in place of manual work force (₹)	1,00,000	1,20,000

The Company's cost of capital is 9%. Corporate tax rate is 25%. Depreciation will be charged on SLM basis.

(i) Calculate NPV and Profitability index for each machine.

(ii) Which machine should be selected on the basis of Equivalent Annualized Criterion?

Year	1	2	3	4	5	6	7
$PVIF_{0.09,\,t}$	0.917	0.842	0.772	0.708	0.650	0.596	0.547
$PVIA_{0.09,\,t}$	0.917	1.759	2.531	3.239	3.889	4.485	5.032

ICAI'S ANSWER

Evaluation of Alternatives

Working Notes:

$$\text{Depreciation on Machine M} = \frac{(4{,}50{,}000 - 45{,}000)}{5} = ₹\ 81{,}000$$

$$\text{Depreciation on Machine N} = \frac{(3{,}50{,}000 - 35{,}000)}{7} = ₹\ 45{,}000$$

(i) NPV and PI for each machine

Particulars	Machine M (₹)	Machine N (₹)
Annual Savings:		
Direct Wages	11,00,000	12,00,000
Scraps	1,00,000	1,20,000
Total Savings (A)	12,00,000	13,20,000
Annual Running Cost (B)	9,00,000	11,00,000
Annual Cash Savings (A-B)	3,00,000	2,20,000
Less: Depreciation	81,000	45,000
Annual Savings before Tax	2,19,000	1,75,000
Less: Tax @ 25%	54,750	43,750
Annual Savings /Profits after tax	1,64,250	1,31,250
Add: Depreciation	81,000	45,000
Annual Cash Inflows	2,45,250	1,76,250
PVIFA	3.889	5.032
PV of Annual Cash Inflows	9,53,777.25	8,87,890
PV of Salvage Value	45,000 × 0.650 = ₹ 29,250	35,000 × 0.547 = ₹ 19,145
Total PV of Inflows	9,83,027.25	9,06,035
NPV	9,83,027.25- 4,50,000 = ₹ 5,33,027.25	9,06,035- 3,50,000 = ₹ 5,56,035
PV Index = Present Value of Cash Inflow/Investment	9,83,027.25/4,50,000 =2.18	9,06,035/3,50,000 =2.59

(ii) Equivalent Annualized NPV

Machine M = $\frac{5,33,027.25}{3.889}$ = ₹ 1,37,060/-

Machine N = $\frac{5,56,035}{5.032}$ = ₹ 1,10,500/-

So machine M should be selected on the basis of Equivalent Annualized Criterion.

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527 **EXAM** **Jan 2026** Q2(b) **5 marks**

Algo Steel Ltd. is a well-established company in the steel manufacturing sector, successfully operating for the past 20 years. The existing capital structure comprises of 80% equity and 20% debt. Company's Cost of Equity (K) is 13.5%, reflecting the return expected by shareholders. Its debt component comprises of a bank loan carrying an interest rate of 8% per annum. The company is subject to taxation and taxes are to be considered in project evaluation. After preliminary evaluation, the Board of Directors has shortlisted a project with the following details:

Particulars	
Initial Investment	₹ 24,00,000
Estimated Salvage Value	10% of Initial Investment
Expected Life of the Project	4 years
Tax Savings on Depreciation	₹ 1,35,000 p.a.
Method of Depreciation	Straight Line Method (SLM)

Particulars	Year 1	Year 2	Year 3	Year 4
Expected Cash Flows before depreciation & tax (in ₹)	12,00,000	20,00,000	18,00,000	16,00,000
PV Factor @ 10.00%	0.909	0.826	0.751	0.683
PV Factor @ 12.00%	0.893	0.797	0.712	0.636

Required:

(i) Calculate Average Rate of Return (ARR) based on Average Investment.

(ii) Calculate Net Present Value (NPV) by using Weighted Average Cost of Capital (WACC).

ICAI'S ANSWER

Annual Depreciation = $\frac{21,60,000\ (24,00,000 - 2,40,000)}{4}$

= ₹ 5,40,000

Tax Saving on Depreciation = Depreciation × Tax Rate

1,35,000 = 5,40,000 x Tax Rate

Tax Rate = 25%

(i) Average Rate of Return (ARR)

Statement of Profit after Tax

Particulars	Year 1 (₹)	Year 2 (₹)	Year 3 (₹)	Year 4 (₹)
Cash flow before dep. & tax	12,00,000	20,00,000	18,00,000	16,00,000
Less: Depreciation	5,40,000	5,40,000	5,40,000	5,40,000
Profit before tax	**6,60,000**	**14,60,000**	**12,60,000**	**10,60,000**
Less: Tax @ 25%	1,65,000	3,65,000	3,15,000	2,65,000
Profit after tax (PAT)	**4,95,000**	**10,95,000**	**9,45,000**	**7,95,000**

$$\text{Average Rate of Return (ARR)} = \frac{\text{Average annual net income (PAT)}}{\text{Average Investment}}$$

Total PAT = 4,95,000 + 10,95,000 + 9,45,000 + 7,95,000

= ₹ 33,30,000

$$\text{Average PAT} = \frac{33,30,000}{4} = ₹\ 8,32,500$$

Average Investment = (₹ 24,00,000 + ₹ 2,40,000)/2 = ₹ 13,20,000

Or, Average Investment = ½ (Initial Investment – Salvage Value) + Salvage Value

= ½ (₹ 24,00,000 - ₹ 2,40,000) + ₹ 2,40,000

= ₹ 13,20,000

$$\text{ARR} = \frac{8,32,500}{13,20,000} \times 100 = 63.068\% \text{ or } 63.07\%$$

(ii) Calculation of WACC

WACC = (0.80 x 13.5%) + {0.20 x 6% (8% x 0.75)}

=10.8% + 1.2%

= 12%

Statement of Cash Flows after Tax

Particulars	Year 1 (₹)	Year 2 (₹)	Year 3 (₹)	Year 4 (₹)	Year 4 (₹)
Profit after tax (PAT)	4,95,000	10,95,000	9,45,000	7,95,000	
Add: Depreciation	5,40,000	5,40,000	5,40,000	5,40,000	
Cash Flow after Tax (CFAT)	10,35,000	16,35,000	14,85,000	13,35,000	2,40,000 (Salvage Value
PV Factor @12%	0.893	0.797	0.712	0.636	0.636
PV	**9,24,255**	**13,03,095**	**10,57,320**	**8,49,060**	**1,52,640**

Total PV of Inflows = ₹ 42,86,370

NPV = PV of Inflows – Initial Investment

= 42,86,370 – 24,00,000

= ₹18,86,370

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528 EXAM **Sep 2025** MCQ 7 2 marks MCQ

KL Limited is replacing an existing machine. The replacement cost of the machine is ₹ 1,00,000 now and it requires maintenance of ₹ 20,000 at the end of every year for five years. At the end of Fifth year, it would have a salvage value of 10,000. The cost of capital is 14%.

$PVIFA_{0.14,5}$ is 3.433 and $PVIF_{0.14,5}$ is 0.519.

What will be the Equivalent Annual Cost (EAC) for the machine?

A ₹ 57,617

B ₹ 45,217

C ₹ 47,617

D ₹ 60,217

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P6 · Sep 2025 exam · question p. 3 · answer p. 7

529 EXAM **Sep 2025** Q1(c) 5 marks

Global Beverage Corporation is considering replacing one of its bottling machines with a newer and more efficient model. The existing machine was purchased five years ago for ₹ 16,00,000 and has a total useful life of ten years. Currently, the company can sell this machine for ₹ 9,60,000. The new machine costs ₹ 32,00,000 having an estimated salvage value of ₹ 3,20,000 and a useful life of five years. With the new machine, annual contribution margin is projected to increase from ₹ 16,00,000 to ₹ 18,40,000 and operating efficiencies are expected to yield further annual savings of ₹ 3,20,000. Depreciation is calculated on a straight-line method over the machine's five-year life. The company's cost of capital is 12% and corporate tax rate is 33%.

You are required to calculate the Net Present Value of new machine.

Given: $PVIFA_{0.12,5}$ is 3.605 and $PVIF_{0.12,5}$ is 0.567

ICAI'S ANSWER

Calculation of Net Initial Cash Outflows:

Particulars	₹
Cost of new machine	32,00,000
Less: Sale proceeds of existing machine	(9,60,000)
Add: Tax paid on sale of existing machine {(9,60,000 – 8,00,000) x 33%}	52,800
Net initial cash outflows	22,92,800

Calculation of Net Present Value of new machine

Particulars	₹
Incremental Contribution	2,40,000
Add: Cost saving	3,20,000
Savings	**5,60,000**
Less: Incremental Depreciation {5,76,000-1,60,000)	4,16,000
Incremental Profit Before Tax	1,44,000
Less: Tax @ 33%	47,520
Profit After Tax	96,480
Add: Depreciation	4,16,000
Cash Inflow	5,12,480
PVIFA	3.605
PV of Cash Inflow	**18,47,490**
Add: PV of Salvage Value of new machine (₹ 3,20,000 × 0.567)	1,81,440
Total PV of Cash Inflow	**20,28,930**
Less: Initial Cash Outflow	22,92,800
NPV	**(2,63,870)**

If Tax on sale of existing machine is not taken

Calculation of Net Initial Cash Outflows:

Particulars	₹
Cost of new machine	32,00,000
Less: Sale proceeds of existing machine	(9,60,000)
Net initial cash outflows	22,40,000

Accordingly, NPV Will be ₹ 20,28,930 - ₹ 22,40,000 = (₹ 2,11,070)

ICAI · P6 · Sep 2025 exam · question p. 9 · answer p. 12-13

530 EXAM May 2025 MCQ 7 2 marks MCQ

T Ltd. is looking for a capital project in order to replace its existing old machine. It got two proposals to

consider; details of which are given below:

	Proposal X	Proposal Y
Initial investment	₹ 6,50,000	₹ 7,80,000
Estimated useful life	5 Years	3 Years
Annal cash inflows	₹ 1,90,000	₹ 3,50,000
Cost of capital	10%	10%

Year	1	2	3	4	5
$PVIF_{0.10,t}$	0909	0.826	0.751	0.683	0.621
$PVIFA_{0.10,t}$	0.909	1.736	2.487	3.170	3.791

What will be Equivalent Annual NPV for Proposal X and Proposal Y?

A ₹ 70,290.00, ₹ 90,450.00

B ₹ 18,541.28, ₹ 36,369.12

C ₹ 1,90,000.00, ₹ 3,50,000.00

D ₹ 77,326.73, ₹ 99,504.95

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P6 · May 2025 exam · question p. 3 · answer p. 7

531 **EXAM** **Jan 2025** Q3(a) **6 marks**

SRT Limited manufactures steel rods and is now considering to purchase a new aluminium smelting and moulding plant. This plant will have the cost of ₹ 20,00,000 to purchase and install the plant. It has a useful life of 5 years with a residual value of ₹ 1,00,000. Production and sales from the new plant are expected to be 1,00,000 units per year. Other estimates are as follows:

Selling Price	₹150 per unit
Direct Cost	₹100 per unit

Fixed cost (including depreciation) is ₹ 8,00,000 per annum. Marketing and promotion cost not included in the above will be ₹ 1,00,000 and ₹ 1,60,000 for years 1 and 2, respectively. Additionally, investment in debtors and stocks will increase in year 1 by ₹ 1,50,000 and ₹ 2,00,000, respectively. Creditors will also increase by ₹ 1,00,000 in year 1. Thus, debtors, stocks, and creditors will be recouped at the end of the fifth year.

The cost of capital is 18%. Corporate tax is 30% and is paid in the year in which profits are made. Depreciation is tax deductible. The company follows straight line method of depreciation.

Required:

(i) Calculate the Net Present Value and Profitability Index of the project.

(ii) Advise SRT Limited whether the plant should be purchased.

The PV factors at 18% are:

Year	1	2	3	4	5
PV factor	0.847	0.718	0.609	0.516	0437

ICAI'S ANSWER

Calculation Net Present Value (NPV)

Years	1	2	3	4	5	Total
Units	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000	
(I) Sales (₹ 150 per unit)	1,50,00,000	1,50,00,000	1,50,00,000	1,50,00,000	1,50,00,000	
(II) Cost						
Direct cost (₹100 per unit)	1,00,00,000	1,00,00,000	1,00,00,000	1,00,00,000	1,00,00,000	
Marketing cost	1,00,000	1,60,000				
Depreciation	3,80,000	3,80,000	3,80,000	3,80,000	3,80,000	
Fixed cost	4,20,000	4,20,000	4,20,000	4,20,000	4,20,000	
Total cost	1,09,00,000	1,09,60,000	1,08,00,000	1,08,00,000	1,08,00,000	
Profit (I - II)	41,00,000	40,40,000	42,00,000	42,00,000	42,00,000	
Tax @30%	12,30,000	12,12,000	12,60,000	12,60,000	12,60,000	
Profit After tax	28,70,000	28,28,000	29,40,000	29,40,000	29,40,000	
Add back depreciation	3,80,000	3,80,000	3,80,000	3,80,000	3,80,000	
Cash inflow after tax	32,50,000	32,08,000	33,20,000	33,20,000	33,20,000	
Add: WC released					2,50,000	
Add: Residual value of machine					1,00,000	
Net cash inflow after tax	32,50,000	32,08,000	33,20,000	33,20,000	36,70,000	

Years	1	2	3	4	5	Total
Present value factor	0.847	0.718	0.609	0.516	0.437	
Present value cash inflow	27,52,750	23,03,344	20,21,880	17,13,120	16,03,790	1,03,94,884
Less: initial cash outflow (20,00,000 + 2,50,000)						22,50,000
NPV						81,44,884

Initial Investment =22,50,000 (assumed working capital invested at start of the year)

Profitability Index (PI)

$$= \frac{\text{Sum of discounted cash inflows}}{\text{Initial cash outlay or Total discounted cash outflow (as the case may)}}$$

= 1,03,94,884/22,50,000 = 4.62

Advise: Since the plant has a positive NPV and the Profitability Index is >1, therefore, it should be purchased.

Note: If we assume that working capital increased at end of the first year then cash outflow will be (20,00,000 + 2,11,750 (2,50,000 x 0.847) = 22,11,750, NPV 81,83,134 and Profitability Index will be = 1,03,94,884 / 22,11,750 = 4.70.

ICAI · P6 · Jan 2025 exam · question p. 18 · answer p. 19-20

532 EXAM **May 2024** Q3(a) 7 marks

HCP Ltd. is a leading manufacturer of railway parts for passenger coaches and freight wagons. Due to high wastage of material and quality issues in production, the General Manager of the company is considering the replacement of machine A with a new CNC machine B. Machine A has a book value of ₹ 4,80,000 and remaining economic life is 6 years. It could be sold now at ₹ 1,80,000 and zero salvage value at the end of sixth year. The purchase price of Machine B is ₹ 24,00,000 with economic life of 6 years. It will require ₹ 1,40,000 for installation and ₹ 60,000 for testing. Subsidy of 15% on the purchase price of the machine B will be received from Government at the end of 1st year. Salvage value at the end of sixth year will be ₹ 3,20,000.

The General manager estimates that the annual savings due to installation of machine B include a reduction of three skilled workers with annual salaries of ₹ 1,68,000 each, ₹ 4,80,000 from reduced wastage of materials and defectives and ₹ 3,50,000 from loss in sales due to delay in execution of purchase orders. Operation of Machine B will require the services of a trained technician with annual salary of t 3,90,000 and annual operation and maintenance cost will increase by ₹ 1,54,000. The company's tax rate is 30% and it's required rate of return is 14%. The company follows straight line method of depreciation. Ignore tax savings on loss due to sale of existing machine.

The present value factors at 14% are:

Years	0	1	2	3	4	5	6
PV Factor	1	0.877	0.769	0.675	0.592	0.519	0.456

Required:

(i) Calculate the Net Present Value and Profitability Index and advise the company for replacement decision.

(ii) Also calculate the discounted pay-back period.

ICAI'S ANSWER

Calculation of Net Initial Cash Outflows:

Particulars	₹
Cost of new machine	24,00,000
Less: Sale proceeds of existing machine	(1,80,000)
Add: Installation	1,40,000
Add: Testing	60,000
Less: Subsidy from government (15% of 24,00,000) x 0.877	(3,15,720)
Net initial cash outflows	21,04,280

Calculation of Incremental Depreciation

Particulars	₹
Depreciation on existing machine (4,80,000/6) (i)	80,000
Depreciation base of New Machine	
Cost of new machine	24,00,000
Add: Installation	1,40,000
Add: Testing	60,000
Less: Subsidy from government	(3,60,000)
Less: Salvage value at the end of 6 th year	(3,20,000)
Depreciation base of New Machine	19,20,000
Depreciation on New Machine (19,20,000/6) (ii)	3,20,000
Incremental depreciation [(ii) – (i)]	2,40,000

Computation of Annual Operating Cash flow after tax (CFAT)

Particulars	Amount (₹)	Amount (₹)
Savings in cost		
Cost of 3 skilled workers (₹1,68,000 × 3)	5,04,000	
Reduced wastage of material	4,80,000	
Saving in loss of sales	3,50,000	
Total		13,34,000
Less: Increase in cost		
Salary to trained technician	3,90,000	
Increase in annual operation and maintenance cost	1,54,000	
Total		(5,44,000)
Incremental Saving before tax and depreciation		7,90,000
Less: Incremental Depreciation		(2,40,000)
Incremental PBT		5,50,000
Less: Tax @30%		(1,65,000)
PAT		3,85,000
Add: Depreciation		2,40,000
Incremental CFAT		6,25,000

Calculation of NPV

Particulars	Year	Net Cashflow (₹)	PVF @ 14%	PV (₹)
Net initial cash outflows	0	(24,20,000)	1	(21,04,280)
Incremental CFAT	1 to 6	6,25,000	3.888	24,30,000
Salvage Value of New Machine	6	3,20,000	0.456	1,45,920
PV of inflows				25,75,920
Net Present Value				4,71,640

$$\text{Profitability Index} = \frac{\text{Sum of discounted cash inflows}}{\text{Initial cash outlay or Total discounted cash outflow (as the case may)}}$$

= 25,75,920/21,04,280= 1.224

Advise: Since the NPV is positive and PI is greater than 1, the company should replace the machine

Computation of Discounted Payback Period

Year	Cashflow	PVF @ 14%	PV of CFs (₹)	Cumulative PV (₹)
1	6,25,000	0.877	5,48,125	5,48,125
2	6,25,000	0.769	4,80,625	10,28,750
3	6,25,000	0.675	4,21,875	14,50,625
4	6,25,000	0.592	3,70,000	18,20,625
5	6,25,000	0.519	3,24,375	21,45,000
6	9,45,000	0.456	4,30,920	25,75,920

Discounted Payback Period

$$= 4 + \frac{21,04,280 - 18,20,625}{3,24,375}$$

= 4.87 years

If we take subsidy in cash inflow of 1st year, then solution can also be done in the following way:

Calculation of Net Initial Cash Outflows:

Particulars	₹
Cost of new machine	24,00,000
Less: Sale proceeds of existing machine	(1,80,000)
Add: Installation	1,40,000
Add: Testing	60,000
Net initial cash outflows	24,20,000

Note: However, Incremental Depreciation and CFAT will remain same.

Calculation of NPV

Particulars	Year	Net Cashflow (₹)	PVF @ 14%	PV (₹)
Net initial cash outflows	0	(24,20,000)	1	(24,20,000)
Subsidy	1	3,60,000	0.877	3,15,720
Incremental CFAT	1 to 6	6,25,000	3.888	24,30,000
Salvage Value of New Machine	6	3,20,000	0.456	1,45,920
PV of inflows				28,91,640
Net Present Value				4,71,640

$$\text{Profitability Index} = \frac{\text{Sum of discounted cash in flows}}{\text{Initial cash outlay or Total discounted cash outflow (as the case may)}}$$

= 28,91,640 /24,20,000 = 1.195

Advise: Since the NPV is positive and PI is greater than 1, the company should replace the machine

Computation of Discounted Payback Period

Year	Cashflow	PVF @ 14%	PV of CFs (₹)	Cumulative PV (₹)
1	9,85,000	0.877	8,63,845	8,63,845
2	6,25,000	0.769	4,80,625	13,44,470
3	6,25,000	0.675	4,21,875	17,66,345
4	6,25,000	0.592	3,70,000	21,36,345
5	6,25,000	0.519	3,24,375	24,60,720
6	9,45,000	0.456	4,30,920	28,91,640

Discounted Payback Period

$$= 4 + \frac{24,20,000 - 21,36,345}{3,24,375}$$

= 4.87 years

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533 EXAM **Old P8 Nov 2023** Q3 10 marks old scheme

ABC Ltd. is considering to purchase a machine which is priced at ₹ 5,00,000. The estimated life of machine is 5 years and has an expected salvage value of ₹ 45,000 at the end of 5 years. It is expected to generate revenues of ₹ 1,50,000 per annum for five years. The annual operating cost of the machine is ₹ 28,125, Corporate Tax Rate is 20% and the cost of capital is 10%.

You are required to analyse whether it would be profitable for the company to purchase the machine by using;

(i) Payback period Method

(ii) Net Present value method

(iii) Profitability Index Method

ICAI'S ANSWER

Computation of Annual Cash Flows

Particular	(₹)
Revenue	1,50,000
Less: Operating Cost	(28,125)
Less: Depreciation (5,00,000 - 45,000)/5	(91,000)
Profit before Tax	30,875
Less: Tax	(6,175)
Profit after Tax	24,700
Add: Depreciation	91,000
Annual Cash Inflows	1,15,700

(i) Computation of Payback Period

Year	Cash Flows	Cumulative Present Value
1	1,15,700	1,15,700
2	1,15,700	2,31,400
3	1,15,700	3,47,100
4	1,15,700	4,62,800
5 (Including Salvage)	1,60,700	6,23,500

Amount to be recovered in 5th year cash flow = ₹ 5,00,000 – ₹ 4,62,800 = ₹ 37,200

Payback period = 4 years + $\frac{37,200}{1,60,700}$ = 4.23 years

Since the payback period is less than the life of machinery, the company may purchase the machine.

(ii) Computation of Net Present Value

Year	Cash Flows	PVF @10%	Present Value
0	(5,00,000)	1.000	(5,00,000)
1 - 5	1,15,700	3.791	4,38,594
5	45,000	0.621	27,941
Net Present Value			(33,465)

Since the net present value (NPV) is negative, the company should not purchase the machine.

(iii) Computation of Profitability Index (PI)

$$\text{Profitability Index (PI)} = \frac{\text{Sum of present value of net cash inflow}}{\text{Initial cash outflow}}$$

$$= \frac{₹4,38,594 + ₹\ 27,941}{₹\ 5,00,000} = 0.93$$

Since the profitability index is less than 1, the company should not purchase the machine.

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534 EXAM **Old P8 Nov 2022** Q3 10 marks old scheme

A firm is in need of a small vehicle to make deliveries. It is intending to choose between two options. One option is to buy a new three wheeler that would cost ₹ 1,50,000 and will remain in service for 10 years.

The other alternative is to buy a second hand vehicle for ₹ 80,000 that could remain in service for 5 years. Thereafter the firm, can buy another second hand vehicle for ₹ 60,000 that will last for another 5 years.

The scrap value of the discarded vehicle will be equal to it written down value (WDV). The firm pays 30% tax and is allowed to claim depreciation on vehicles @ 25% on WDV basis.

The cost of capital of the firm is 12%.

You are required to advise the best option.

Given:

t	1	2	3	4	5	6	7	8	9	10
PVIF (t,12%)	0.892	0.797	0.711	0.635	0.567	0.506	0.452	0.403	0.360	0.322

ICAI'S ANSWER

Selection of Investment Decision

Tax shield on Purchase of New vehicle

Year	WDV	Dep. @ 25%	Tax shield @ 30%
1	1,50,000	37,500	11,250
2	1,12,500	28,125	8,437
3	84,375	21,094	6,328
4	63,281	15,820	4,746
5	47,461	11,865	3,560
6	35,596	8,899	2,670
7	26,697	6,674	2,002
8	20,023	5,006	1,502
9	15,017	3,754	1,126
10	11,263	2,816	845
11	8,447	Scrap value	

Tax shield on Purchase of Second hand vehicles

Year	WDV	Dep. @ 25%	Tax shield @ 30%	
1	80,000	20,000	6,000	
2	60,000	15,000	4,500	
3	45,000	11,250	3,375	
4	33,750	8,437	2,531	
5	25,313	6,328	1,898	Scrap value = ₹ 18,985
6	60,000	15,000	4,500	
7	45,000	11,250	3,375	
8	33,750	8,437	2,531	
9	25,313	6,328	1,898	
10	18,985	4,746	1,424	Scrap value = ₹ 14,239

Calculation of PV of Net outflow of New Vehicle

Year	Cash OF/IF	PV Factor	PV of OF/IF
0	1,50,000	1	1,50,000
1	(11,250)	0.892	(10,035)
2	(8,437)	0.797	(6,724)
3	(6,328)	0.711	(4,499)
4	(4,746)	0.635	(3,014)
5	(3,560)	0.567	(2,018)
6	(2,670)	0.506	(1,351)
7	(2,002)	0.452	(905)
8	(1,502)	0.403	(605)
9	(1,126)	0.360	(405)
10	(845 + 8447)	0.322	(2,992)
		PVNOF	**1,17,452**

Calculation of PV of Net outflow of Second hand Vehicles

Year	Cash OF/IF	PV Factor	PV of OF/IF
0	80,000	1	80,000

Year	Cash OF/IF	PV Factor	PV of OF/IF
1	(6,000)	0.892	(5,352)
2	(4,500)	0.797	(3,587)
3	(3,375)	0.711	(2,400)
4	(2,531)	0.635	(1,607)
5	(60000 – 18985 – 1898) = 39,117	0.567	22,179
6	(4,500)	0.506	(2,277)
7	(3,375)	0.452	(1,525)
8	(2,531)	0.403	(1,020)
9	(1,898)	0.360	(683)
10	(1424 + 14239) = (15,663)	0.322	(5,043)
		PVNOF	**78,686**

Advise: The PV of net outflow is low in case of buying the second hand vehicles. Therefore, it is advisable to buy second hand vehicles.

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535 EXAM **Old P8 Nov 2022** Q4 10 marks old scheme

A hospital is considering to purchase a diagnostic machine costing ₹ 80,000. The projected life of the machine is 8 years and has an expected salvage value of ₹ 6,000 at the end of 8 years. The annual operating cost of the machine is ₹ 7,500. It is expected to generate revenues of ₹ 40,000 per year for eight years. Presently, the hospital is outsourcing the diagnostic work and is earning commission income of ₹ 12,000 per annum.

Consider tax rate of 30% and Discounting Rate as 10%.

Advise:

Whether it would be profitable for the hospital to purchase the machine?

Give your recommendation as per Net Present Value method and Present Value Index method under below mentioned two situations:

(i) If Commission income of ₹ 12,000 p.a. is before taxes.

(ii) If Commission income of ₹ 12,000 p.a. is net of taxes.

Given:

t	1	2	3	4	5	6	7	8
PVIF (t, 10%)	0.909	0.826	0.751	0.683	0.621	0.564	0.513	0.467

ICAI'S ANSWER

Analysis of Investment Decisions

Determination of Cash inflows	Situation-(i) Commission Income before taxes	Situation-(ii) Commission Income after taxes
Cash flow up-to 7th year:		
Sales Revenue	40,000	40,000
Less: Operating Cost	(7,500)	(7,500)
	32,500	32,500
Less: Depreciation (80,000 – 6,000) ÷ 8	(9,250)	(9,250)
Net Income	23,250	23,250
Tax @ 30%	(6,975)	(6,975)
Earnings after Tax (EAT)	16,275	16,275
Add: Depreciation	9,250	9,250
Cash inflow after tax per annum	25,525	25,525
Less: Loss of Commission Income	(8,400)	(12,000)
Net Cash inflow after tax per annum	17,125	13,525
In 8th Year:		
Net Cash inflow after tax	17,125	13,525
Add: Salvage Value of Machine	6,000	6,000
Net Cash inflow in year 8	**23,125**	**19,525**

Calculation of Net Present Value (NPV) and Profitability Index (PI)

	Particulars	PV factor @10%	Situation-(i) [Commission Income before taxes]	Situation-(ii) [Commission Income after taxes]
A	Present value of cash inflows (1st to 7th year)	4.867	83,347.38 (17,125 × 4.867)	65,826.18 (13,525 × 4.867)
B	Present value of cash inflow at 8th year	0.467	10,799.38 (23,125 × 0.467)	9,118.18 (19,525 × 0.467)
C	PV of cash inflows		94,146.76	74,944.36

	Particulars	PV factor @10%	Situation-(i) [Commission Income before taxes]	Situation-(ii) [Commission Income after taxes]
D	Less: Cash Outflow	1.00	(80,000)	(80,000)
E	**Net Present Value (NPV)**		**14,146.76**	**(5,055.64)**
F	**PI = (C÷D)**		**1.18**	**0.94**

Recommendation: The hospital may consider purchasing of diagnostic machine in situation (i) where commission income is 12,000 before tax as NPV is positive and PI is also greater than 1. Contrary to situation (i), in situation (ii) where the commission income is net of tax, the recommendation is reversed to not purchase the machine as NPV is negative and PI is also less than 1.

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536 EXAM Old P8 May 2022 Q3 10 marks old scheme

Alpha Limited is a manufacturer of computers. It wants to introduce artificial intelligence while making computers. The estimated annual saving from introduction of the artificial intelligence (AI) is as follows:

- reduction of five employees with annual salaries of ₹ 3,00,000 each
- reduction of ₹ 3,00,000 in production delays caused by inventory problem
- reduction in lost sales ₹ 2,50,000 and
- Gain due to timely billing ₹ 2,00,000

The purchase price of the system for installation of artificial intelligence is ₹ 20,00,000 and installation cost is ₹ 1,00,000. 80% of the purchase price will be paid in the year of purchase and remaining will be paid in next year.

The estimated life of the system is 5 years and it will be depreciated on a straight-line basis.

However, the operation of the new system requires two computer specialists with annual salaries of ₹ 5,00,000 per person.

In addition to above, annual maintenance and operating cost for five years are as below:

(Amount in ₹)

Year	1	2	3	4	5
Maintenance & Operating Cost	2,00,000	1,80,000	1,60,000	1,40,000	1,20,000

Maintenance and operating cost are payable in advance.

The company's tax rate is 30% and its required rate of return is 15%.

Year	1	2	3	4	5
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683	0.621
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636	0.567
$PVIF_{0.15,t}$	0.870	0.756	0.658	0.572	0.497

Evaluate the project by using Net Present Value and Profitability Index.

ICAI'S ANSWER

Computation of Annual Cash Flow after Tax

Particulars	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Savings in Salaries		15,00,000	15,00,000	15,00,000	15,00,000	15,00,000
Reduction in Production Delays		3,00,000	3,00,000	3,00,000	3,00,000	3,00,000
Reduction in Lost Sales		2,50,000	2,50,000	2,50,000	2,50,000	2,50,000
Gain due to Timely Billing		2,00,000	2,00,000	2,00,000	2,00,000	2,00,000
Salary to Computer Specialist		(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)
Maintenance and Operating Cost (payable in advance)		(2,00,000)	(1,80,000)	(1,60,000)	(1,40,000)	(1,20,000)
Depreciation (21 lakhs/5)		(4,20,000)	(4,20,000)	(4,20,000)	(4,20,000)	(4,20,000)
Gain Before Tax		6,30,000	6,50,000	6,70,000	6,90,000	7,10,000
Less: Tax (30%)		1,89,000	1,95,000	2,01,000	2,07,000	2,13,000
Gain After Tax		4,41,000	4,55,000	4,69,000	4,83,000	4,97,000
Add: Depreciation		4,20,000	4,20,000	4,20,000	4,20,000	4,20,000
Add: Maintenance and Operating Cost (payable in advance)		2,00,000	1,80,000	1,60,000	1,40,000	1,20,000

Particulars	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Less: Maintenance and Operating Cost (payable in advance)	(2,00,000)	(1,80,000)	(1,60,000)	(1,40,000)	(1,20,000)	-
Net CFAT	(2,00,000)	8,81,000	8,95,000	9,09,000	9,23,000	10,37,000

Note: Annual cash flows can also be calculated Considering tax shield on depreciation & maintenance and operating cost. There will be no change in the final cash flows after tax.

Computation of NPV

Particulars	Year	Cash Flows (₹)	PVF	PV (₹)
Initial Investment (80% of 20 Lacs)	0	16,00,000	1	16,00,000
Installation Expenses	0	1,00,000	1	1,00,000
Instalment of Purchase Price	1	4,00,000	0.870	3,48,000
PV of Outflows (A)				**20,48,000**
CFAT	0	(2,00,000)	1	(2,00,000)
CFAT	1	8,81,000	0.870	7,66,470
CFAT	2	8,95,000	0.756	6,76,620
CFAT	3	9,09,000	0.658	5,98,122
CFAT	4	9,23,000	0.572	5,27,956
CFAT	5	10,37,000	0.497	5,15,389
PV of Inflows (B)				28,84,557
NPV (B-A)				**8,36,557**
Profitability Index (B/A)				**1.408 or 1.41**

Evaluation: Since the NPV is positive (i.e. ₹ 8,36,557) and Profitability Index is also greater than 1 (i.e. 1.41), Alpha Ltd. may introduce artificial intelligence (AI) while making computers.

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537 EXAM Old P8 Dec 2021 Q4 10 marks old scheme

Stand Ltd. is contemplating replacement of one of its machines which has become outdated and inefficient. Its financial manager has prepared a report outlining two possible replacement machines.

The details of each machine are as follows:

	Machine 1	Machine 2
Initial investment	₹ 12,00,000	₹ 16,00,000
Estimated useful life	3 years	5 years
Residual value	₹ 1,20,000	₹ 1,00,000
Contribution per annum	₹ 11,60,000	₹ 12,00,000
Fixed maintenance costs per annum	₹ 40,000	₹ 80,000
Other fixed operating costs per annum	₹ 7,20,000	₹ 6,10,000

The maintenance costs are payable annually in advance. All other cash flows apart from the initial investment assumed to occur at the end of each year. Depreciation has been calculated by straight line method and has been included in other fixed operating costs. The expected cost of capital for this project is assumed as 12% p.a.

Required:

(i) Which machine is more beneficial, using Annualized Equivalent Approach? Ignore tax.

(ii) Calculate the sensitivity of your recommendation in part (i) to changes in the contribution generated by machine 1.

Year	1	2	3	4	5	6
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636	0.567	0.507
$PVIFA_{0.12,t}$	0.893	1.690	2.402	3.038	3.605	4.112

ICAI'S ANSWER

(i) Calculation of Net Cash flows

Machine 1

Other fixed operating costs (excluding depreciation) = 7,20,000–[(12,00,000–1,20,000)/3] = ₹ 3,60,000

Year	Initial Investment	Contribution	Fixed maintenance costs	Other fixed operating costs (excluding depreciation)	Residual Value	Net cash flow
	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
0	(12,00,000)		(40,000)			(12,40,000)
1		11,60,000	(40,000)	(3,60,000)		7,60,000
2		11,60,000	(40,000)	(3,60,000)		7,60,000
3		11,60,000		(3,60,000)	1,20,000	9,20,000

Machine 2

Other fixed operating costs (excluding depreciation) = 6,10,000–[(16,00,000–1,00,000)/5] = ₹ 3,10,000

Year	Initial Investment	Contribution	Fixed maintenance costs	Other fixed operating costs (excluding depreciation)	Residual Value	Net cash flow
	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
0	(16,00,000)		(80,000)			(16,80,000)
1		12,00,000	(80,000)	(3,10,000)		8,10,000
2		12,00,000	(80,000)	(3,10,000)		8,10,000
3		12,00,000	(80,000)	(3,10,000)		8,10,000
4		12,00,000	(80,000)	(3,10,000)		8,10,000
5		12,00,000		(3,10,000)	1,00,000	9,90,000

Calculation of Net Present Value

Year	12% discount factor	Machine 1		Machine 2	
		Net cash flow (₹)	**Present value (₹)**	**Net cash flow (₹)**	**Present value (₹)**
0	1.000	(12,40,000)	(12,40,000)	(16,80,000)	(16,80,000)
1	0.893	7,60,000	6,78,680	8,10,000	7,23,330
2	0.797	7,60,000	6,05,720	8,10,000	6,45,570
3	0.712	9,20,000	6,55,040	8,10,000	5,76,720
4	0.636			8,10,000	5,15,160
5	0.567			9,90,000	5,61,330
NPV @ 12%			**6,99,440**		**13,42,110**
PVAF @ 12%			2.402		3.605
Equivalent Annualized Criterion			**2,91,190.674**		**3,72,291.262**

Recommendation: Machine 2 is more beneficial using Equivalent Annualized Criterion.

(ii) Calculation of sensitivity of recommendation in part (i) to changes in the contribution generated by machine 1

Difference in Equivalent Annualized Criterion of Machines required for changing the recommendation in part (i) = 3,72,291.262- 2,91,190.674 = **₹ 81,100.588**

$\therefore$ Sensitivity relating to contribution = $\frac{₹\ 81{,}100.588}{₹\ 11{,}60{,}000.00} \times 100 = 6.991$ or **7% yearly**

Alternatively,

The annualized equivalent cash flow for machine 1 is lower by ₹ (3,72,291.262– 2,91,190.674) = **₹81,100.588** than for machine 2. Therefore, it would need to increase contribution for **complete 3 years** before the decision would be to invest in this machine.

Sensitivity w.r.t contribution = 81,100.588 / (11,60,000 × 2.402) x100 = **2.911%**

PassBook's note: One item: requirements (i) and (ii) share one marks note (10) and one connected answer. Old-scheme mapping (ours): requirement (i), the annualised equivalent (equivalent annual) approach, is in the current Paper 6A syllabus (Investment Decisions: equivalent annual cost).

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538 EXAM Old P8 Jan 2021 Q5 10 marks old scheme

A company wants to buy a machine, and two different models namely A and B are available. Following further particulars are available:

Particulars	Machine-A	Machine-B
Original Cost (₹)	8,00,000	6,00,000
Estimated Life in years	4	4
Salvage Value (₹)	0	0

The company provides depreciation under Straight Line Method. Income tax rate applicable is 30%.

The present value of ₹ 1 at 12% discounting factor and net profit before depreciation and tax are as under:

Year	Net Profit Before Depreciation and tax		PV Factor
	Machine-A ₹	Machine-B ₹	
1.	2,30,000	1,75,000	0.893
2.	2,40,000	2,60,000	0.797
3.	2,20,000	3,20,000	0.712
4.	5,60,000	1,50,000	0.636

Calculate:

1. NPV (Net Present Value)
2. Discounted pay-back period
3. PI (Profitability Index)

Suggest: Purchase of which machine is more beneficial under Discounted pay-back period method, NPV method and PI method.

ICAI'S ANSWER

Workings:

(i) Calculation of Annual Depreciation

$$\text{Depreciation on Machine – A} = \frac{₹\ 8,00,000}{4} = ₹\ 2,00,000$$

$$\text{Depreciation on Machine – B} = \frac{₹\ 6,00,000}{4} = ₹\ 1,50,000$$

(ii) Calculation of Annual Cash Inflows

Particulars	Machine-A (₹)			
	1	2	3	4
Net Profit before Depreciation and Tax	2,30,000	2,40,000	2,20,000	5,60,000
Less: Depreciation	2,00,000	2,00,000	2,00,000	2,00,000
Profit before Tax	30,000	40,000	20,000	3,60,000
Less: Tax @ 30%	9,000	12,000	6,000	1,08,000
Profit after Tax	21,000	28,000	14,000	2,52,000
Add: Depreciation	2,00,000	2,00,000	2,00,000	2,00,000
Annual Cash Inflows	**2,21,000**	**2,28,000**	**2,14,000**	**4,52,000**

Particulars	Machine-B (₹)			
	1	2	3	4
Net Profit before Depreciation and Tax	1,75,000	2,60,000	3,20,000	1,50,000
Less: Depreciation	1,50,000	1,50,000	1,50,000	1,50,000
Profit before Tax	25,000	1,10,000	1,70,000	0
Less: Tax @ 30%	7,500	33,000	51,000	0
Profit after Tax	17,500	77,000	1,19,000	0
Add: Depreciation	1,50,000	1,50,000	1,50,000	1,50,000
Annual Cash Inflows	**1,67,500**	**2,27,000**	**2,69,000**	**1,50,000**

(iii) Calculation of PV of Cash Flows

	Machine – A				Machine - B		
Year	**PV of Re 1 @ 12%**	**Cash flow (₹)**	**PV (₹)**	**Cumulative PV (₹)**	**Cash flow (₹)**	**PV (₹)**	**Cumulative PV (₹)**
1	0.893	2,21,000	1,97,353	1,97,353	1,67,500	1,49,578	1,49,578
2	0.797	2,28,000	1,81,716	3,79,069	2,27,000	1,80,919	3,30,497
3	0.712	2,14,000	1,52,368	5,31,437	2,69,000	1,91,528	5,22,025
4	0.636	4,52,000	2,87,472	8,18,909	1,50,000	95,400	6,17,425

1. NPV (Net Present Value)

Machine – A

NPV = ₹ 8,18,909 - ₹ 8,00,000 = **₹ 18,909**

Machine – B

NPV = ₹ 6,17,425 – ₹ 6,00,000 = **₹ 17,425**

2. Discounted Payback Period

Machine – A

$$\text{Discounted Payback Period} = 3 + \frac{₹\ 8,00,000 - ₹\ 5,31,437}{₹\ 2,87,472}$$

= 3 + 0.934

= 3.934 years or 3 years 11.21 months

Machine – B

$$\text{Discounted Payback Period} = 3 + \frac{₹\ 6,00,000 - ₹\ 5,22,025}{₹\ 95,400}$$

= 3 + 0.817

= 3.817 years or 3 years 9.80 months

3. PI (Profitability Index)

Machine – A

$$\text{Profitability Index} = \frac{₹\ 8,18,909}{₹\ 8,00,000} = 1.024$$

Machine – B

$$\text{Profitability Index} = \frac{₹\ 6,17,425}{₹\ 6,00,000} = 1.029$$

Suggestion:

Method	Machine - A	Machine - B	Suggested Machine
Net Present Value	₹ 18,909	₹ 17,425	**Machine A**
Discounted Payback Period	3.934 years	3.817 years	**Machine B**
Profitability Index	1.024	1.029	**Machine B**

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539 EXAM **Old P8 Nov 2020** Q1(b) 5 marks old scheme

CK Ltd. is planning to buy a new machine. Details of which are as follows:

Cost of the Machine at the commencement	₹ 2,50,000
Economic Life of the Machine	8 year
Residual Value	Nil
Annual Production Capacity of the Machine	1,00,000 units
Estimated Selling Price per unit	₹ 6
Estimated Variable Cost per unit	₹ 3

Estimated Annual Fixed Cost (Excluding depreciation)	₹ 1,00,000
Advertisement Expenses in 1st year in addition of annual fixed cost	₹ 20,000
Maintenance Expenses in 5th year in addition of annual fixed cost	₹ 30,000
Cost of Capital	12%

Ignore Tax.

Analyse the above mentioned proposal using the Net Present Value Method and advice.

P.V. factor @ 12% are as under:

Year	1	2	3	4	5	6	7	8
PV Factor	0.893	0.797	0.712	0.636	0.567	0.507	0.452	0.404

ICAI'S ANSWER

Calculation of Net Cash flows

Contribution = (₹ 6 – ₹ 3) × 1,00,000 units = **₹ 3,00,000**

Fixed costs (excluding depreciation) = ₹ 1,00,000

Year	Capital (₹)	Contribution (₹)	Fixed costs (₹)	Advertisement/ Maintenance expenses (₹)	Net cash flow (₹)
0	(2,50,000)				(2,50,000)
1		3,00,000	(1,00,000)	(20,000)	1,80,000
2		3,00,000	(1,00,000)		2,00,000
3		3,00,000	(1,00,000)		2,00,000
4		3,00,000	(1,00,000)		2,00,000
5		3,00,000	(1,00,000)	(30,000)	1,70,000
6		3,00,000	(1,00,000)		2,00,000
7		3,00,000	(1,00,000)		2,00,000
8		3,00,000	(1,00,000)		2,00,000

Calculation of Net Present Value

Year	Net cash flow (₹)	12% discount factor	Present value (₹)
0	(2,50,000)	1.000	(2,50,000)
1	1,80,000	0.893	1,60,740
2	2,00,000	0.797	1,59,400
3	2,00,000	0.712	1,42,400
4	2,00,000	0.636	1,27,200

Year	Net cash flow (₹)	12% discount factor	Present value (₹)
5	1,70,000	0.567	96,390
6	2,00,000	0.507	1,01,400
7	2,00,000	0.452	90,400
8	2,00,000	0.404	80,800
			7,08,730

Advise: CK Ltd. should buy the new machine, as the net present value of the proposal is positive i.e ₹ 7,08,730.

ICAI · Old P8 · Nov 2020 exam · question p. 1-2 · answer p. 3-4

540 EXAM **Old P8 Nov 2019** Q6(b) 3 marks old scheme

Explain the steps while using the equivalent annualized criterion.

ICAI'S ANSWER

Equivalent Annualized Criterion: This method involves the following steps-

(i) Compute NPV using the WACC or discounting rate.

(ii) Compute Present Value Annuity Factor (PVAF) of discounting factor used above for the period of each project.

(iii) Divide NPV computed under step (i) by PVAF as computed under step (ii) and compare the values.

ICAI · Old P8 · Nov 2019 exam · question p. 16 · answer p. 17

541 RTP **Jan 2026** Q2 MCQ

Please consider the following data:

Project	NPV (₹ '000)	I (₹ '000)
A	15	60
B	20	100
C	25	150

Given that NPV = PV – I and PI = PV / I, where PV is the present value of future net cash inflows and I is the initial investment on a project. Based on the data furnished above, which one of the following statements is correct?

A Project A is recommended based on PI criterion

B Project B is recommended based on NPV criterion

C Project C is recommended based on PI criterion

D Project A is recommended based on NPV criterion

ICAI'S ANSWER

(A) Project A is recommended based on PI criterion

Project A: PV = 15 + 60 = 75

Project B: PV = 20 + 100 = 120

Project C: PV = 25 + 150 = 175

PI = PV / I

Project A: PI = 75/60 = 1.25

Project B: PI = 120/100 = 1.20

Project C: PI = 175/150 ≈ 1.167

ICAI · P6 · RTP Jan 2026 · question p. 3 · answer p. 14

542 **RTP** **Jan 2025** **Q9**

A company is considering the proposal to take up a new project which requires investment of ₹ 850 lakhs in plant & machinery and ₹ 150 lakhs in working capital. The project is expected to yield the following Cash flows before tax and depreciation over the next five years:

Year	Amount (₹ in Lakhs)
1	290
2	320
3	360
4	390
5	270

The desired rate of return from the project is 14% and assets must be depreciated at 20% on a written down value basis. The scrap value at the end of the five-year period may be taken as ₹ 140 lakhs. The income tax applicable to the company is 20%. This is the only asset in the entire block. Capital gains tax is at 15% (for capital loss as well)

You are required to CALCULATE the net present value of the project and advise the management to take appropriate decisions. Also calculate the Internal Rate of Return and Desirability factor of the Project.

Note: Present values of Re. 1 at different rates of interest are as follows:

Year	14%	16%	20%
1	0.88	0.86	0.83
2	0.77	0.74	0.69
3	0.67	0.64	0.58
4	0.59	0.55	0.48
5	0.52	0.48	0.40

ICAI'S ANSWER

(A) **Calculation of NPV**

WN 1 : Calculation of Present Value of Cash Outflow (PV CO)

(i) Initial Investment = ₹ 850 lakhs

(ii) Working capital outlay = ₹ 150 lakhs

Therefore, total PV CO = ₹ 1000 lakhs

WN 2 : Calculation of Present Value of Cash Inflows (PV CI)

Cash flows before tax are given i.e. nothing but NPBDT

Amount (₹ in lakhs)

Year	1	2	3	4	5
NPBDT	290.00	320.00	360.00	390.00	270.00
(-) Dep	170.00	136.00	108.80	87.04	69.63
NPBT	120.00	184.00	251.20	302.96	200.37
(-) Tax	24.00	36.80	50.24	60.59	40.07
NPAT	96.00	147.20	200.96	242.37	160.29
(+) Dep	170.00	136.00	108.80	87.04	69.63
CFAT	266.00	283.20	309.76	329.41	229.93
(+) Working Capital Release					150.00
(+) Scrap					140.00
PV Factor @ 14%	0.88	0.77	0.67	0.59	0.52
PV CI	**234.08**	**218.06**	**207.54**	**194.35**	**270.36**

(i) Total PV CI = ₹ 1124.40 Lakhs

WN 3 : Calculation of Present Value of tax savings on short term Capital loss

	₹ in Lakhs
WDV at end of 5th year	278.53
(-) Sale value	140.00
Loss on sale	138.53
Tax savings on above @ 15%	20.78

PV of tax savings on short term capital loss (STCL) = Tax saving x PV factor (14%, 5th year)

= 20.78 x 0.52

= ₹ 10.81 lakhs

NPV = PV CI + PV of tax savings on STCL - PV CO

= 1124.40 + 10.81 - 1000

NPV = ₹ 135.20 lakhs

Advise: Since the NPV of the project is positive, project should be accepted

(B) **Calculation of IRR**

IRR is that discounting rate where NPV = 0 (point where PV of all CI = PV Co)

We know that @ 14%, NPV is ₹ 135.20, so by trial-and-error method we need to calculate that rate where NPV equals 0.

When Discounting rate is 16%

	1	2	3	4	5
CFAT	266.00	283.20	309.76	329.41	229.93
(+) Working Capital Release					150.00
(+) Scrap					140.00
PV Factor @ 14%	0.86	0.74	0.64	0.55	0.48
PV CI	**228.76**	**209.57**	**198.25**	**181.17**	**249.56**

PV CI = 1067.31

(+) PV of tax savings on STCL = 9.97 {20.78 x 0.48}

(-) PV CO = (1000)

NPV = ₹ 77.29

Since NPV is positive at 16% as well, we need to go for Trial II at 20%

When Discounting rate is 20%

	1	2	3	4	5
CFAT	266.00	283.20	309.76	329.41	229.93
(+) Working Capital Release					150.00
(+) Scrap					140.00
PV Factor @ 14%	0.83	0.69	0.58	0.48	0.4
PV CI	220.78	195.41	179.66	158.12	207.97

PV CI = 961.94

(+) PV of tax savings on STCL = 8.31 {20.78 x 0.40}

(-) PV CO = (1000)

NPV = ₹ (29.75)

Since NPV is negative at 20%, IRR lies somewhere between 16% and 20%

IRR = $LR + \frac{\text{NPV at LR}}{\text{NPV at LR - NPV at HR}} \times (HR - LR)$

LR = Lower Rate (16% here) HR = Higher Rate (20% here)

IRR = $16 + \frac{77.29}{77.29 - (-29.75)} \times (20 - 16)$

IRR = 18.89%

(C) **Calculation of Desirability Factory (Profitability Index)**

PI = TOTAL PV CI / PV CO

PI = 1135.21 / 1000

PI = 1.13521

ICAI · P6 · RTP Jan 2025 · question p. 8-9 · answer p. 24-26

543 RTP **Jan 2025** Q11(c)

Do the profitability index and the NPV criterion of evaluating investment proposals lead to the same acceptance-rejection and ranking decisions? In what SITUATIONS will they give conflicting results?

ICAI'S ANSWER

In the most of the situations the Net Present Value Method (NPV) and Profitability Index (PI) yield same accept or reject decision. In general items, under PI method a project is acceptable if profitability index value is greater than 1 and rejected if it less than 1. Under NPV method a project is acceptable if Net present value of a project is positive and rejected if it is negative. Clearly a project offering a profitability index greater than 1 must also offer a net present value which is positive. But a conflict may arise between two methods if a choice between mutually exclusive projects has to be made. Consider the following example:

	Project A	Project B
PV of Cash inflows	3,00,000	1,60,000
Initial cash outflows	1,00,000	40,000
Net present value	2,00,000	1,20,000
P.I	$\frac{3,00,000}{1,00,000}$ = 3	$\frac{1,60,000}{40,000}$ = 4

According to NPV method, project A would be preferred, whereas according to profitability index method project B would be preferred.

This is because Net present value gives ranking on the basis of absolute value of rupees, whereas, profitability index gives ranking on the basis of ratio. Although PI method is based on NPV, it is a better evaluation technique than NPV in a situation of capital rationing.

ICAI · P6 · RTP Jan 2025 · question p. 10 · answer p. 30

544 RTP **Old P8 Nov 2023** Q5 old scheme

PQR Limited is considering buying a new machine which would have a useful economic life of five years, at a cost of ₹ 40,00,000 and a scrap value of ₹ 5,00,000, with 80 per cent of the cost being payable at the start of the project and 20 per cent at the end of the first year. The machine would produce 80,000 units per annum of a new product with an estimated selling price of ₹ 400 per unit. Direct costs would

be ₹ 375 per unit and annual fixed costs, including depreciation calculated on a straight- line basis, would be ₹ 10,40,000 per annum.

In the first year and the second year, special sales promotion expenditure, not included in the above costs, would be incurred, amounting to ₹ 1,25,000 and ₹ 1,75,000 respectively.

EVALUATE the project using the NPV method of investment appraisal, assuming the company's cost of capital to be 12 percent.

ICAI'S ANSWER

Calculation of Net Cash flows

Contribution = (400 – 375) × 80,000 = ₹ 20,00,000

Fixed costs = 10,40,000 – [(40,00,000 – 5,00,000)/5] = ₹ 3,40,000

Year	Capital (₹)	Contribution (₹)	Fixed costs (₹)	Promotion (₹)	Net cash flow (₹)
0	(32,00,000)				(32,00,000)
1	(8,00,000)	20,00,000	(3,40,000)	(1,25,000)	7,35,000
2		20,00,000	(3,40,000)	(1,75,000)	14,85,000
3		20,00,000	(3,40,000)		16,60,000
4		20,00,000	(3,40,000)		16,60,000
5	5,00,000	20,00,000	(3,40,000)		21,60,000

Calculation of Net Present Value

Year	Net cash flow (₹)	12% discount factor	Present value (₹)
0	(32,00,000)	1.000	(32,00,000)
1	7,35,000	0.893	6,56,355
2	14,85,000	0.797	11,83,545
3	16,60,000	0.712	11,81,920
4	16,60,000	0.636	10,55,760
5	21,60,000	0.567	12,24,720
			21,02,300

The net present value of the project is **₹21,02,300**.

ICAI · Old P8 · RTP Nov 2023 · question p. 3 · answer p. 9-10

545 RTP **Old P8 May 2023** Q5 old scheme

Dharma Ltd, an existing profit-making company, is planning to introduce a new product with a projected life of 8 years. Initial equipment cost will be ₹ 240 lakhs and additional equipment costing ₹ 26 lakhs will be needed at the beginning of third year. At the end of 8 years, the original equipment will have resale value equivalent to the cost of removal, but the additional equipment would be sold for ₹ 2

lakhs. Working Capital of ₹ 25 lakhs will be needed at the beginning of the operations. The 100% capacity of the plant is of 4,00,000 units per annum, but the production and sales volume expected are as under:

Year	Capacity (%)
1	20
2	30
3-5	75
6-8	50

A sale price of ₹ 100 per unit with a profit volume ratio (contribution/sales) of 60% is likely to be obtained. Fixed operating cash cost are likely to be ₹ 16 lakhs per annum. In addition to this the advertisement expenditure will have to be incurred as under:

Year	1	2	3-5	6-8
Expenditure (₹ Lakhs each year)	30	15	10	4

The company is subjected to 50% tax rate and consider 12% to be an appropriate cost of capital. Straight line method of depreciation is followed by the company. ADVISE the management on the desirability of the project.

ICAI'S ANSWER

Calculation of Cash Flow After tax

	Year	1	2	3 to 5	6 to 8
A	Capacity	20%	30%	75%	50%
B	Units	80000	120000	300000	200000
C	Contribution p.u.	₹60	₹60	₹60	₹60
D	Contribution	₹48,00,000	₹72,00,000	₹1,80,00,000	₹1,20,00,000
E	Fixed Cash Cost	₹16,00,000	₹16,00,000	₹16,00,000	₹16,00,000
	Depreciation				
F	Original Equipment (₹240Lakhs/8)	₹30,00,000	₹30,00,000	₹30,00,000	₹30,00,000
G	Additional Equipment (₹24Lakhs/6)	--	--	₹4,00,000	₹4,00,000
H	Advertisement Expenditure	₹30,00,000	₹15,00,000	₹10,00,000	₹4,00,000
I	Profit Before Tax (D-E-F-G-H)	₹ (28,00,000)	₹11,00,000	₹1,20,00,000	₹66,00,000

	Year	1	2	3 to 5	6 to 8
J	Tax savings/ (expenditure)	₹14,00,000	₹ (5,50,000)	₹ (60,00,000)	₹ (33,00,000)
K	Profit After Tax	₹ (14,00,000)	₹5,50,000	₹60,00,000	₹33,00,000
L	Add: Depreciation (F+G)	₹30,00,000	₹30,00,000	₹34,00,000	₹34,00,000
M	Cash Flow After Tax	₹16,00,000	₹35,50,000	₹94,00,000	₹67,00,000

Calculation of NPV

Year	Particulars	Cash Flows	PV factor	PV
0	Initial Investment	₹ (2,40,00,000)	1.000	₹ (2,40,00,000)
0	Working Capital Introduced	₹ (25,00,000)	1.000	₹ (25,00,000)
1	CFAT	₹16,00,000	0.893	₹ 14,28,800
2	CFAT	₹ 35,50,000	0.797	₹ 28,29,350
2	Additional Equipment	₹ (26,00,000)	0.797	₹ (20,72,200)
3	CFAT	₹ 94,00,000	0.712	₹ 66,92,800
4	CFAT	₹ 94,00,000	0.636	₹ 59,78,400
5	CFAT	₹ 94,00,000	0.567	₹ 53,29,800
6	CFAT	₹ 67,00,000	0.507	₹ 33,96,900
7	CFAT	₹ 67,00,000	0.452	₹ 30,28,400
8	CFAT	₹ 67,00,000	0.404	₹ 27,06,800
8	WC Released	₹ 25,00,000	0.404	₹ 10,10,000
8	Salvage Value	₹ 2,00,000	0.404	₹ 80,800
	Net Present Value			**₹39,09,850**

Since the NPV is positive, the proposed project should be implemented.

ICAI · Old P8 · RTP May 2023 · question p. 3 · answer p. 14-15

546 RTP **Old P8 Nov 2022** Q5 old scheme

K. K. M. M Hospital is considering purchasing an MRI machine. Presently, the hospital is outsourcing the work received relating to MRI machine and is earning commission of ₹ 6,60,000 per annum (net of tax).

The following details are given regarding the machine:

	(₹)
Cost of MRI machine	90,00,000
Operating cost per annum (excluding Depreciation)	14,00,000
Expected revenue per annum	45,00,000
Salvage value of the machine (after 5 years)	10,00,000
Expected life of the machine	5 years

Assuming tax rate @ 40%, whether it would be profitable for the hospital to purchase the machine?

Give your RECOMMENDATION under:

(i) Net Present Value Method, and

(ii) Profitability Index Method.

PV factors at 10% are given below:

Year	1	2	3	4	5
PV factor	0.909	0.826	0.751	0.683	0.620

ICAI'S ANSWER

Determination of Cash inflows

Elements	(₹)
Sales Revenue	45,00,000
Less: Operating Cost	14,00,000
	31,00,000
Less: Depreciation (90,00,000 – 10,00,000)/5	16,00,000
Net Income	15,00,000
Tax @ 40%	6,00,000
Earnings after Tax (EAT)	9,00,000
Add: Depreciation	16,00,000
Cash inflow after tax per annum	25,00,000
Less: Loss of Commission Income	6,60,000
Net Cash inflow after tax per annum	18,40,000
In 5th Year:	
New Cash inflow after tax	18,40,000

Elements	(₹)
Add: Salvage Value of Machine	10,00,000
Net Cash inflow in year 5	28,40,000

Calculation of Net Present Value (NPV)

Year	CFAT	PV Factor @10%	Present Value of Cash inflows
1 to 4	18,40,000	3.169	58,30,960
5	28,40,000	0.620	17,60,800
			75,91,760
Less: Cash Outflows			90,00,000
NPV			(14,08,240)

$$\text{Profitability Index} = \frac{\text{Sum of discounted cash inflows}}{\text{Present value of cash outflows}} = \frac{75,91,760}{90,00,000} = 0.844$$

Advise: Since the net present value is negative and profitability index is also less than 1, therefore, the hospital should not purchase the MRI machine.

ICAI · Old P8 · RTP Nov 2022 · question p. 3-4 · answer p. 13-14

547 RTP **Old P8 May 2022** Q5 old scheme

ABC & Co. is considering whether to replace an existing machine or to spend money on revamping it. ABC & Co. currently pays no taxes. The replacement machine costs ₹ 18,00,000 now and requires maintenance of ₹ 2,00,000 at the end of every year for eight years. At the end of eight years, it would have a salvage value of ₹ 4,00,000 and would be sold. The existing machine requires increasing amounts of maintenance each year and its salvage value fall each year as follows:

Year	Maintenance (₹)	Salvage (₹)
Present	0	8,00,000
1	2,00,000	5,00,000
2	4,00,000	3,00,000
3	6,00,000	2,00,000
4	8,00,000	0

The opportunity cost of capital for ABC & Co. is 15%.

REQUIRED:

When should the company replace the machine?

The following present value table is given for you:

Year	Present value of ₹ 1 at 15% discount rate
1	0.8696
2	0.7561
3	0.6575
4	0.5718
5	0.4972
6	0.4323
7	0.3759
8	0.3269

ICAI'S ANSWER

ABC & Co.

Equivalent Annual Cost (EAC) of new machine

		(₹)
(i)	Cost of new machine now	18,00,000
	Add: PV of annual repairs @ ₹ 2,00,000 per annum for 8 years (₹ 2,00,000 × 4.4873)	8,97,460
		26,97,460
	Less: PV of salvage value at the end of 8 years (₹ 4,00,000×0.3269)	1,30,760
		25,66,700
	Equivalent annual cost (EAC) (₹ 25,66,700/4.4873)	5,71,992

PV of cost of replacing the old machine in each of 4 years with new machine

Scenario	Year	Cash Flow	PV @ 15%	PV
		(₹)		**(₹)**
Replace Immediately	0	(5,71,992)	1.00	(5,71,992)
	0	8,00,000	1.00	8,00,000
				2,28,008
Replace in one year	1	(5,71,992)	0.8696	(4,97,404)
	1	(2,00,000)	0.8696	(1,73,920)
	1	5,00,000	0.8696	4,34,800

Scenario	Year	Cash Flow	PV @ 15%	PV
				(2,36,524)
Replace in two years	1	(2,00,000)	0.8696	(1,73,920)
	2	(5,71,992)	0.7561	(4,32,483)
	2	(4,00,000)	0.7561	(3,02,440)
	2	3,00,000	0.7561	2,26,830
				(6,82,013)
Replace in three years	1	(2,00,000)	0.8696	(1,73,920)
	2	(4,00,000)	0.7561	(3,02,440)
	3	(5,71,992)	0.6575	(3,76,085)
	3	(6,00,000)	0.6575	(3,94,500)
	3	2,00,000	0.6575	1,31,500
				(11,15,445)
Replace in four years	1	(2,00,000)	0.8696	(1,73,920)
	2	(4,00,000)	0.7561	(3,02,440)
	3	(6,00,000)	0.6575	(3,94,500)
	4	(5,71,992)	0.5718	(3,27,065)
	4	(8,00,000)	0.5718	(4,57,440)
				(16,55,365)

Advice: The company should replace the old machine immediately because the PV of cost of replacing the old machine with new machine is least.

ICAI · Old P8 · RTP May 2022 · question p. 3 · answer p. 14-15

548 RTP **Old P8 Dec 2021** Q5 old scheme

HMR Ltd. is considering replacing a manually operated old machine with a fully automatic new machine. The old machine had been fully depreciated for tax purpose but has a book value of ₹ 2,40,000 on 31st March 2021. The machine has begun causing problems with breakdowns and it cannot fetch more than ₹ 30,000 if sold in the market at present. It will have no realizable value after 10 years. The company has been offered ₹ 1,00,000 for the old machine as a trade in on the new machine which has a price (before allowance for trade in) of ₹ 4,50,000. The expected life of new machine is 10 years with salvage value of ₹ 35,000.

Further, the company follows straight line depreciation method but for tax purpose, written down value method depreciation @ 7.5% is allowed taking that this is the only machine in the block of assets.

Given below are the expected sales and costs from both old and new machine:

	Old machine (₹)	New machine (₹)
Sales	8,10,000	8,10,000
Material cost	1,80,000	1,26,250
Labour cost	1,35,000	1,10,000
Variable overhead	56,250	47,500
Fixed overhead	90,000	97,500
Depreciation	24,000	41,500
PBT	3,24,750	3,87,250
Tax @ 30%	97,425	1,16,175
PAT	2,27,325	2,71,075

From the above information, ANALYSE whether the old machine should be replaced or not if required rate of return is 10%? Ignore capital gain tax.

PV factors @ 10%:

Year	1	2	3	4	5	6	7	8	9	10
PVF	0.909	0.826	0.751	0.683	0.621	0.564	0.513	0.467	0.424	0.386

ICAI'S ANSWER

Workings:

1. **Calculation of Base for depreciation or Cost of New Machine**

Particulars	(₹)
Purchase price of new machine	4,50,000
Less: Sale price of old machine	1,00,000
	3,50,000

2. **Calculation of Profit before tax as per books**

Particulars	Old machine (₹)	New machine (₹)	Difference (₹)
PBT as per books	3,24,750	3,87,250	62,500
Add: Depreciation as per books	24,000	41,500	17,500
Profit before tax and depreciation (PBTD)	3,48,750	4,28,750	80,000

Calculation of Incremental NPV

Year	PVF @ 10%	PBTD (₹)	Dep. @ 7.5% (₹)	PBT (₹)	Tax @ 30% (₹)	Cash Inflows (₹)	PV of Cash Inflows (₹)
	(1)	(2)	(3)	(4)	(5) = (4) x 0.30	(6) = (4) – (5) + (3)	(7) = (6) x (1)
1	0.909	80,000.00	26,250.00	53,750.00	16,125.00	63,875.00	58,062.38
2	0.826	80,000.00	24,281.25	55,718.75	16,715.63	63,284.38	52,272.89
3	0.751	80,000.00	22,460.16	57,539.84	17,261.95	62,738.05	47,116.27
4	0.683	80,000.00	20,775.64	59,224.36	17,767.31	62,232.69	42,504.93
5	0.621	80,000.00	19,217.47	60,782.53	18,234.76	61,765.24	38,356.21
6	0.564	80,000.00	17,776.16	62,223.84	18,667.15	61,332.85	34,591.73
7	0.513	80,000.00	16,442.95	63,557.05	19,067.12	60,932.88	31,258.57
8	0.467	80,000.00	15,209.73	64,790.27	19,437.08	60,562.92	28,282.88
9	0.424	80,000.00	14,069.00	65,931.00	19,779.30	60,220.70	25,533.58
10	0.386	80,000.00	13,013.82	66,986.18	20,095.85	59,904.15	23,123.00
							3,81,102.44
Add: PV of Salvage value of new machine (₹ 35,000 × 0.386)							13,510.00
Total PV of incremental cash inflows							3,94,612.44
Less: Cost of new machine							3,50,000.00
Incremental Net Present Value							44,612.44

Analysis: Since the Incremental NPV is positive, the old machine should be replaced.

ICAI · Old P8 · RTP Dec 2021 · question p. 3 · answer p. 11-12

549 RTP **Old P8 Jul 2021** Q5 old scheme

The General Manager of Merry Ltd. is considering the replacement of five-year-old equipment. The company has to incur excessive maintenance cost of the equipment. The equipment has zero written down value. It can be modernized at a cost of ₹ 1,40,000 enhancing its economic life to 5 years. The equipment could be sold for ₹ 30,000 after 5 years. The modernization would help in material handling and in reducing labour , maintenance & repairs costs.

The company has another alternative to buy a new machine at a cost of ₹ 3,50,000 with an economic life of 5 years and salvage value of ₹ 60,000. The new machine is expected to be more efficient in reducing costs of material handling, labour, maintenance & repairs, etc.

The annual cost are as follows:

	Existing Equipment (₹)	Modernization (₹)	New Machine (₹)
Wages & Salaries	45,000	35,500	15,000
Supervision	20,000	10,000	7,000
Maintenance	25,000	5,000	2,500
Power	30,000	20,000	15,000
	1,20,000	70,500	39,500

Assuming tax rate of 50% and required rate of return of 10%, should the company modernize the equipment or buy a new machine?

PV factor at 10% are as follows:

Year	1	2	3	4	5
PV factor	0.909	0.826	0.751	0.683	0.621

ICAI'S ANSWER

Workings:

Calculation of Depreciation:

$$\text{On Modernized Equipment} = \frac{₹\ 1{,}40{,}000 - ₹\ 30{,}000}{5 \text{ years}} = ₹\ 22{,}000 \text{ p.a.}$$

$$\text{On New machine} = \frac{₹\ 3{,}50{,}000 - ₹\ 60{,}000}{5 \text{ years}} = ₹\ 58{,}000 \text{ p.a.}$$

(i) Calculation of Incremental annual cash inflows/ savings:

Particulars	Existing Equipment (₹)	Modernization		New Machine	
		Amount (₹)	Savings (₹)	Amount (₹)	Savings (₹)
	(1)	(2)	(3)=(1)-(2)	(4)	(5)=(1)-(4)
Wages & Salaries	45,000	35,500	9,500	15,000	30,000
Supervision	20,000	10,000	10,000	7,000	13,000
Maintenance	25,000	5,000	20,000	2,500	22,500
Power	30,000	20,000	10,000	15,000	15,000
Total	**1,20,000**	**70,500**	**49,500**	**39,500**	**80,500**
Less: Depreciation (Refer Workings)			22,000		58,000
Total Savings			**27,500**		**22,500**

Particulars	Existing Equipment (₹)	Modernization	New Machine
Less: Tax @ 50%		13,750	11,250
After Tax Savings		**13,750**	**11,250**
Add: Depreciation		22,000	58,000
Incremental Annual Cash Inflows		**35,750**	**69,250**

(ii) Calculation of Net Present Value (NPV)

Particulars	Year	Modernization (₹)	New Machine (₹)
Initial Cash outflow (A)	0	1,40,000.00	3,50,000.00
Incremental Cash Inflows	1-5	1,35,492.50 (₹ 35,750 × 3.790)	2,62,457.50 (₹ 69,250 × 3.790)
Salvage value	5	18,630.00 (₹ 30,000 × 0.621)	37,260.00 (₹ 60,000 × 0.621)
PV of Cash inflows (B)		1,54,122.50	2,99,717.50
Net Present Value (B - A)		14,122.50	(50,282.50)

Advise: The company should modernize its existing equipment and not buy a new machine because NPV is positive in modernization of equipment.

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550 RTP **Old P8 May 2020** Q5 old scheme

A company is considering the proposal of taking up a new project which requires an investment of ₹800 lakhs on machinery and other assets. The project is expected to yield the following earnings (before depreciation and taxes) over the next five years:

Year	Earnings (₹ in lakhs)
1	320
2	320
3	360
4	360
5	300

The cost of raising the additional capital is 12% and assets have to be depreciated at 20% on written down value basis. The scrap value at the end of the five year period may be taken as zero. Income-tax applicable to the company is 40%.

You are required to CALCULATE the net present value of the project and advise the management to take appropriate decision. Also CALCULATE the Internal Rate of Return of the Project.

Note: Present values of Re. 1 at different rates of interest are as follows:

Year	10%	12%	14%	16%	20%
1	0.91	0.89	0.88	0.86	0.83
2	0.83	0.80	0.77	0.74	0.69
3	0.75	0.71	0.67	0.64	0.58
4	0.68	0.64	0.59	0.55	0.48
5	0.62	0.57	0.52	0.48	0.40

ICAI'S ANSWER

(i) Calculation of Net Cash Flow

(₹ in lakhs)

Year	Profit before dep. and tax	Depreciation (20% on WDV)	PBT	PAT	Net cash flow
(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(3) + (5)**
1	320	800 × 20% = 160	160	96	256
2	320	(800 – 160)× 20% = 128	192	115.20	243.20
3	360	(640 – 128)× 20% = 102.4	257.6	154.56	256.96
4	360	(512 – 102.4)× 20% = 81.92	278.08	166.85	248.77
5	300	(409.6 – 81.92) = 327.68*	–27.68	–16.61	311.07

*this is treated as a short term capital loss.

(ii) Calculation of Net Present Value (NPV)

(₹ in lakhs)

Year	Net Cash Flow	12%		16%		20%	
		D.F	**P.V**	**D.F**	**P.V**	**D.F**	**P.V**
1	256	0.89	227.84	0.86	220.16	0.83	212.48
2	243.20	0.80	194.56	0.74	179.97	0.69	167.81
3	256.96	0.71	182.44	0.64	164.45	0.58	149.03
4	248.77	0.64	159.21	0.55	136.82	0.48	119.41
5	311.07	0.57	177.31	0.48	149.31	0.40	124.43
			941.36		850.71		773.16
	Less: Initial Investment		800.00		800.00		800.00

Year	Net Cash Flow	12%	16%	20%
	NPV	141.36	50.71	-26.84

(iii) Advise: Since Net Present Value of the project at 12% = 141.36 lakhs, therefore the project should be implemented.

(iv) Calculation of Internal Rate of Return (IRR)

$$\text{IRR} = 16\% + \frac{50.71 \times 4}{50.71 - (-26.84)}$$

$$= 16\% + \frac{2.03}{77.55} = 16\% + 2.62\% = 18.62\%.$$

ICAI · Old P8 · RTP May 2020 · question p. 3-4 · answer p. 10-11

551 RTP **Old P8 Nov 2019** Q5 old scheme

MTR Limited is considering buying a new machine which would have a useful economic life of five years, at a cost of ₹25,00,000 and a scrap value of ₹3,00,000, with 80 per cent of the cost being payable at the start of the project and 20 per cent at the end of the first year. The machine would produce 75,000 units per annum of a new product with an estimated selling price of ₹300 per unit. Direct costs would be ₹285 per unit and annual fixed costs, including depreciation calculated on a straight- line basis, would be ₹8,40,000 per annum.

In the first year and the second year, special sales promotion expenditure, not included in the above costs, would be incurred, amounting to ₹1,00,000 and ₹1,50,000 respectively.

EVALUATE the project using the NPV method of investment appraisal, assuming the company's cost of capital to be 15 percent.

ICAI'S ANSWER

Calculation of Net Cash flows

Contribution = (300 – 285) × 75,000 = ₹11,25,000

Fixed costs = 8,40,000 – [(25,00,000 – 3,00,000)/5] = ₹4,00,000

Year	Capital (₹)	Contribution (₹)	Fixed costs (₹)	Adverts (₹)	Net cash flow (₹)
0	(20,00,000)				(20,00,000)
1	(5,00,000)	11,25,000	(4,00,000)	(1,00,000)	1,25,000
2		11,25,000	(4,00,000)	(1,50,000)	5,75,000
3		11,25,000	(4,00,000)		7,25,000
4		11,25,000	(4,00,000)		7,25,000
5	3,00,000	11,25,000	(4,00,000)		10,25,000

Calculation of Net Present Value

Year	Net cash flow (₹)	12% discount factor	Present value (₹)
0	(20,00,000)	1.000	(20,00,000)
1	1,25,000	0.892	1,11,500
2	5,75,000	0.797	4,58,275
3	7,25,000	0.711	5,15,475
4	7,25,000	0.635	4,60,375
5	10,25,000	0.567	5,81,175
			1,26,800

The net present value of the project is ₹1,26,800.

ICAI · Old P8 · RTP Nov 2019 · question p. 3 · answer p. 9-10

552 **MTP** **Sep 2026 S-II** Q1(c) **5 marks**

Nova Green Mobility Ltd., a well-established and profitable manufacturer of electrical components, is evaluating a proposal to enter the electric -vehicle (EV) charging equipment business by setting up a new plant to manufacture smart EV charging stations. The project calls for a capital outlay of ₹ 1,000 lakhs and working capital of ₹ 150 lakhs. The plant is designed for an annual production capacity of 3 lakh units, and over the project's five-year life, capacity utilisation is projected as follows:

Year	1	2	3	4	5
Capacity Utilisation (%)	50	65	80	100	100

Each charging unit is expected to sell at an average price of ₹ 600, yielding a contribution margin of 60%. Annual fixed costs (excluding depreciation) are estimated at ₹ 500 lakhs from the third year onwards, and at ₹ 200 lakhs and ₹ 350 lakhs in the first and second years respectively.

The capital asset is expected to fetch a scrap value of ₹ 200 lakhs at the end of the fifth year, and depreciation for income-tax purposes is charged on the written-down-value method at 40% p.a. The applicable income-tax rate is 30%, and the cost of capital is 12%. A further ₹ 200 lakhs of working capital will need to be invested at the end of the third year, and no capital gains tax is applicable to the transaction.

You are required to compute the Net Present Value (NPV) of the project. Nova Green Mobility Ltd. is preparing to present the proposal to Secure Venture Capital Firm, which invests in a project only if it adds more than ₹ 100 lakhs to shareholder wealth. Assess whether Nova Green Mobility Ltd. would qualify for funding from Secure Venture Capital Firm on this basis.

ICAI'S ANSWER

Statement Showing Computation of Cash Flow After Tax (CFAT)

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5
Capacity Utilisation	50%	65%	80%	100%	100%

Particulars	Year 1	Year 2	Year 3	Year 4	Year 5
Units Produced	1,50,000	1,95,000	2,40,000	3,00,000	3,00,000
Contribution per Unit (₹ 600 × 60%)	360	360	360	360	360
Total Contribution (₹)	5,40,00,000	7,02,00,000	8,64,00,000	10,80,00,000	10,80,00,000
Less: Fixed Cost (₹)	2,00,00,000	3,50,00,000	5,00,00,000	5,00,00,000	5,00,00,000
Less: Depreciation (₹) (W.N.)	4,00,00,000	2,40,00,000	1,44,00,000	86,40,000	51,84,000
Profit Before Tax (₹)	(60,00,000)	1,12,00,000	2,20,00,000	4,93,60,000	5,28,16,000
Less: Tax @ 30% (₹)	(18,00,000)	33,60,000	66,00,000	1,48,08,000	1,58,44,800
Profit After Tax (₹)	(42,00,000)	78,40,000	1,54,00,000	3,45,52,000	3,69,71,200
Add: Depreciation (₹)	4,00,00,000	2,40,00,000	1,44,00,000	86,40,000	51,84,000
Cash Flow After Tax (₹)	3,58,00,000	3,18,40,000	2,98,00,000	4,31,92,000	4,21,55,200

Statement Showing Computation of Net Present Value

Year	Description	Cash Flow (₹)	PVF @12%	Present Value (₹)
0	Initial Investment	(10,00,00,000)	1.000	(10,00,00,000)
0	Working Capital Introduced	(1,50,00,000)	1.000	(1,50,00,000)
3	Additional Working Capital	(2,00,00,000)	0.7118	(1,42,36,000)
1	Cash Flow After Tax	3,58,00,000	0.8929	3,19,65,820
2	Cash Flow After Tax	3,18,40,000	0.7972	2,53,82,848
3	Cash Flow After Tax	2,98,00,000	0.7118	2,12,11,640
4	Cash Flow After Tax	4,31,92,000	0.6355	2,74,48,516
5	Cash Flow After Tax	4,21,55,200	0.5674	2,39,18,860
5	Working Capital Released	3,50,00,000	0.5674	1,98,59,000
5	Sale of Scrap	2,00,00,000	0.5674	1,13,48,000
	Net Present Value			**3,18,98,684**

Working Note: Computation of Depreciation (WDV Method @ 40% p.a.)

Year	Opening WDV (₹)	Depreciation @ 40% (₹)	Closing WDV (₹)
1	10,00,00,000	4,00,00,000	6,00,00,000
2	6,00,00,000	2,40,00,000	3,60,00,000
3	3,60,00,000	1,44,00,000	2,16,00,000
4	2,16,00,000	86,40,000	1,29,60,000
5	1,29,60,000	51,84,000	77,76,000

The Net Present Value of the project works out to ₹ 3,18,98,684 (approximately ₹ 319 lakhs), which is positive and exceeds the ₹ 100 lakhs threshold set by Secure Venture Capital Firm. Nova Green Mobility Ltd. is therefore in a position to secure funding from Secure Venture Capital Firm for this project.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 5-6 · answer p. 5-6

553 MTP **May 2026 S-I** Case Scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

XYZ Limited is a mid-sized manufacturing company operating in the industrial equipment sector. The company has been in operation for over 15 years and has established a strong market presence. As part of the annual financial review, the CFO has tasked you, a financial analyst, with analyzing the company's capital structure, determining the cost of capital components and evaluate a proposed investment project. The finance team has compiled the following information from the company's latest financial statements and market data:

Balance Sheet Data (Book Values)

- **Equity Share Capital:** ₹ 15 lakhs
- **Reserves & Surplus:** ₹ 5 Lakhs
- **Book Value of Debt:** ₹ 10 lakhs

Market Data

The market has assigned different valuations to the company's securities:

- **Market Value of Equity:** The current market valuation of equity is 3 times its book value
- **Market Value of Debt:** The current market valuation of debt is 2 times its book value

Weighted Average cost of capital(WACC)

The finance team has calculated the Weighted Average cost of capital of the firm under two different approaches:

1. **As per Book Value:** 10%
2. **As per Market Value:** 10.5%

The company is considering a new project requiring an initial investment of ₹ 20 lakhs. The project has the following additional features:

- **Working Capital Requirement:** ₹ 3 lakhs (fully recoverable at end of Year 3)
- **Salvage Value:** ₹ 2 lakhs (end of Year 3)

- **Depreciation:** Straight-line over 3 years.

Expected Cash Inflows (before depreciation & tax):

Year	Cash Inflows (₹ lakhs)
1	8
2	9
3	10

Assume the tax rate is 25% and the investment rate is 2% more than the Pre tax cost of debt. The project is evaluated based investment rate.

	Year 1	Year 2	Year 3
6%	0.943	0.890	0.840
8%	0.926	0.857	0.794
10%	0.909	0.826	0.751
12%	0.893	0.797	0.712

You are required to answer the following questions (MCQs 1 to 5):

MCQ 1 2 marks

Cost of equity is –

A Ke = 6%

B Ke = 8%

C Ke = 12%

D Ke = 10%

MCQ 2 2 marks

Cost of debt is –

A Kd = 12%

B Kd = 10%

C Kd = 12%

D Kd = 6%

MCQ 3 2 marks

The Net Present Value (NPV) of the project is –

A ₹ 1.35 lakhs

B -₹ 0.55 lakhs

C ₹ 1.14 lakhs

D -₹ 0.95 lakhs

MCQ 4 2 marks

The Profitability Index (PI) of the project is –

A 1.05

B 1.08

C 1.10

D 0.98

MCQ 5 2 marks

The Discounted Payback Period is approximately –

A 2.32 years

B 2.60 years

C 2.95 years

D 2.89 years

ICAI'S ANSWERS

MCQ 1

(C) Ke = 12%,

MCQ 2

(D) Kd = 6%

Working Note:

Let the cost of equity be 'X' and cost of debt be 'Y'

WACC under BV

Item	Cost	Weight	Product
Equity	X	20 Lakhs /30 Lakhs = 2/3	2X/3
Debt	Y	10 Lakhs /30 Lakhs = 1/3	Y/3
		1	**10%**

$$10 = \frac{2X}{3} + \frac{Y}{3}$$

30 = 2X + Y ....(1)

WACC under MV

Item	Cost	Weight	Product
Equity	X	60 Lakhs /80 Lakhs = 3/4	3X/4
Debt	Y	20 Lakhs /80 Lakhs = 1/4	Y/4
		1	**10.5%**

$$10.5 = \frac{3X}{4} + \frac{Y}{4}$$

42 = 3X + Y ....(2)

By Solving both equations

42 = 3X + Y

30 = 2X + Y

Cost of equity (X) = 12%

Substituting in Equation 1,

30 = 24 + Y

Cost of Debt (Y) = 6%

MCQ 3

(C) ₹ 1.14 lakhs

Statement Showing Calculation of Cash Inflows

Particulars	Year 1	Year 2	Year 3
Cash Inflows (before dep & tax)	8.00	9.00	10.00
Less: Depreciation	6.00	6.00	6.00
Profit Before Tax (PBT)	2.00	3.00	4.00
Less: Tax @ 25%	0.50	0.75	1.00
Profit After Tax (PAT)	1.50	2.25	3.00
Add: Depreciation	6.00	6.00	6.00
Operating Cash Flow	**7.50**	**8.25**	**9.00**
Add: Salvage Value	–	–	2.00
Add: Working Capital Recovery	–	–	3.00
Total Cash Inflow	**7.50**	**8.25**	**14.00**

Investment rate = (6% / 0.75) + 2% = 10%

Statement showing calculation of PV

Year	CF	PV Factor @ 10%	PV (₹ in lakhs)
0	23	1	23
1	7.50	0.909	6.82
2	8.25	0.826	6.81
3	14.00	0.751	10.51
NPV			**1.14**

MCQ 4

(A) 1.05

$$\text{Profitability Index (PI)} = \frac{\text{Sum of discounted cash in flows}}{\text{Initial cash outlay or Total discounted cash outflow (as the case may)}}$$

= 24.14 / 23

= 1.05 (approx)

MCQ 5

(D) 2.89 years

Year	Discounted cash inflows	Cumulative Discounted cash inflows
1	6.82	6.82
2	6.81	13.63
3	10.51	24.14

Discounted Payback Period = 2 years + $\left(\frac{23 - 13.63}{10.51}\right)$ = 2.89 years

ICAI · P6 · MTP May 2026 · Series I · question p. 1-3 · answer p. 1, 1-2, 2, 3

554 MTP **May 2026 S-I** Q1(c) 5 marks

A company wants to invest in a machinery that would cost ₹ 50,000 at the beginning of year 1. It is estimated that the net cash inflows from operations will be ₹ 18,000 per annum for 3 years, if the company opts to service a part of the machine at the end of year 1 at ₹ 10,000. In such a case, the scrap value at the end of year 3 will be ₹ 12,500. However, if the company decides not to service the part, then it will have to be replaced at the end of year 2 at ₹ 15,400. But in this case, the machine will work for the 4th year also and get operational cash inflow of ₹ 18,000 for the 4th year. It will have to be scrapped at the end of year 4 at ₹ 9,000. Assuming cost of capital at 10% and ignoring taxes, will you RECOMMEND the purchase of this machine based on the net present value of its cash flows?

If the supplier gives a discount of ₹ 5,000 for purchase, WHAT would be your decision? (The present value factors at the end of years 0, 1, 2, 3, 4, 5 and 6 are respectively 1, 0.9091, 0.8264, 0.7513, 0.6830, 0.6209 and 0.5644).

ICAI'S ANSWER

Option I : Purchase Machinery and Service Part at the end of Year 1.

Net Present value of cash flow @ 10% per annum discount rate.

$$NPV = -50{,}000 + \frac{18{,}000}{(1.1)} + \frac{18{,}000}{(1.1)^2} + \frac{18{,}000}{(1.1)^3} - \frac{10{,}000}{(1.1)} + \frac{12{,}500}{(1.1)^3}$$

= – 50,000 + 18,000 (0.9091 + 0.8264 + 0.7513) – (10,000 × 0.9091) + (12,500 × 0.7513)

= – 50,000 + (18,000 × 2.4868) – 9,091 + 9,391

= – 50,000 + 44,762 – 9,091 + 9,391

NPV = – 4,938

Since, Net Present Value is negative; therefore, this option is not to be considered.

If Supplier gives a discount of ₹ 5,000 then,

NPV = +5,000 – 4,938 = + 62

In this case, Net Present Value is positive but very small; therefore, this option may not be advisable.

Option II : Purchase Machinery and Replace Part at the end of Year 2.

$$NPV = -50{,}000 + \frac{18{,}000}{(1.1)} + \frac{18{,}000}{(1.1)^2} + \frac{18{,}000}{(1.1)^3} - \frac{15{,}400}{(1.1)^2} + \frac{27{,}000}{(1.1)^4}$$

= – 50,000 + 18,000 (0.9091 + 0.8264 + 0.7513) – (15,400 × 0.8264) + (27,000 × 0.6830)

= – 50,000 + 18,000 (2.4868) – (15,400 × 0.8264) + (27,000 × 0.6830)

= – 50,000 + 44,762 – (15,400 × 0.8264) + (27,000 × 0.6830)

= – 50,000 + 44,762 – 12,727+ 18,441

= – 62,727+ 63,203 = +476

Net Present Value is positive, but very low as compared to the investment.

If the Supplier gives a discount of ₹ 5,000, then

NPV = 5,000 + 476 = 5,476

Decision: Option II is worth investing as the net present value is positive and higher as compared to Option I.

ICAI · P6 · MTP May 2026 · Series I · question p. 6 · answer p. 6-7

555 MTP **Sep 2025 S-I** Case scenario · MCQ (i) to MCQ (v) 10 marks

CASE SCENARIO

ABC Ltd., a high-end furniture manufacturer specializes in classic, handcrafted designs that combine style and utility. The company, which was founded with a passion for quality craftsmanship and environmentally friendly methods, provides a wide variety of furniture for homes and workplaces, such as dining tables, living room sets, bedroom collections, and custom-built items.

However, lately the company has been facing delays in product delivery due to limited storage and inefficient distribution. To enhance its supply chain and improve customer satisfaction, the company has decided to establish two strategically located warehouses. This move aims to reduce delivery time, optimize inventory management, and streamline logistics operations.

ABC Ltd. previously relied on a single warehouse near its production facility, causing bottlenecks in order fulfillment and increasing transportation costs for distant markets. After analyzing demand trends and logistics inefficiencies, the company plans to set up one warehouse – Baton either in the northern region, closer to key suppliers, and or in the southern region- Katon, near major customer hubs.

The further details of either of the two warehouses to be introduced are as follows:

Warehouse Baton

The Initial cost of the warehouse is ₹ 40,00,000 and its useful life is 5 years. It's annual net savings before tax is projected as follows:

Year 1: ₹ 12,00,000

Year 2: ₹ 14,00,000

Year 3: ₹ 10,00,000

Year 4: ₹ 10,00,000

Year 5: ₹ 13,00,000

Further, the Mid-life major maintenance is expected to be ₹ 5,00,000 in Year 3, salvage value - ₹ 5,00,000 and Shutdown and disposal cost - ₹ 1,00,000 in Year 5.

Warehouse Katon:

The Initial cost of the warehouse is ₹ 60,00,000 and its useful life is 8 years. It's annual net savings before tax is projected as follows:

Year 1–2: ₹ 10,00,000

Year 3–6: ₹ 15,00,000

Year 7–8: ₹ 12,00,000

Maintenance cost details:

Year 4: ₹ 2,00,000

Year 7: ₹ 3,00,000

Further, the salvage value is expected to be ₹ 8,00,000, and shutdown and disposal cost to be ₹ 1,50,000 in Year 8. The company is subject to tax rate of 30% and considers 10% to be an appropriate after-tax cost of capital. The company follows straight-line method of depreciation.

You are being a finance manager, has been asked the following questions:

MCQ (i) 2 marks

The NPV of Baton, factoring in tax shields from depreciation and expenses is .....

A ₹ (79,590)

B ₹ (53,365)

C ₹ (45,365)

D ₹ (36,834)

MCQ (ii) 2 marks

The Equivalent Annual Cost (EAC) of Baton is .....

A ₹ (12,310)

B ₹ (14,071)

C ₹ (16,834)

D ₹ (18,535)

MCQ (iii) 2 marks

The NPV of Katon, factoring in tax shields from depreciation and expenses is .....

A ₹ (41,667)

B ₹ (6,91,666)

C ₹ (8,41,667)

D ₹ (9,62,789)

MCQ (iv) 2 marks

The Equivalent Annual Cost (EAC) of Katon is .....

A ₹ (7810)

B ₹ (1,29,646)

C ₹ (1,57,763)

D ₹ (7,235)

MCQ (v) 2 marks

Which among the following is correct?

A Warehouse Baton should be chosen as it has the lower Equivalent Annual Cost despite having higher upfront cost.

B Warehouse Katon should be chosen as it has the lower Equivalent Annual Cost despite having higher upfront cost.

C Warehouse Baton should be chosen as it has the lower Equivalent Annual Cost and upfront cost.

D Warehouse Katon should be chosen as it has the lower Equivalent Annual Cost and upfront cost.

ICAI'S ANSWERS

MCQ (i)

Answer key: **(B)**

(b) ₹ (53,365)

Warehouse Baton (5-year life)

Straight-Line Depreciation

$$\text{Depreciation} = \frac{40,00,000 - 4,00,000}{5} = ₹\ 7,20,000$$

After-Tax Cash Flows

Year (1)	Savings (2)	Maintenance (3)	Depreciation (4)	Taxable Income (5)	After-tax Profit (6)	Cash Inflow (7) (6+4)	Present Value@10% (8)
1	12,00,000	0	7,20,000	4,80,000	3,36,000	10,56,000	9,59,904
2	14,00,000	0	7,20,000	6,80,000	4,76,000	11,96,000	9,88,374
3	10,00,000	5,00,000	7,20,000	- 2,20,000	- 1,54,000	5,66,000	4,25,236
4	10,00,000	0	7,20,000	2,80,000	1,96,000	9,16,000	6,25,628
5	13,00,000	0	7,20,000	5,80,000	4,06,000	11,26,000	6,99,133
5	4,00,000						2,48,360
						Total	₹ 39,46,635

Net Present Value (NPV)

$$\text{NPV} = -40,00,000 + 39,46,635 = ₹\ (53,365)$$

MCQ (ii)

Answer key: **(B)**

(b) ₹ (14077)

Equivalent Annual Cost (EAC)

$$\frac{\text{NPV}}{\text{PVIFA}(10,5)} = \frac{(53,365)}{3.791} = ₹\ (14077)$$

MCQ (iii)

Answer key: **(A)**

(a) ₹ (41,667)

Warehouse Katon (8-year life)

Straight-Line Depreciation

$$\text{Depreciation} = \frac{60,00,000 - 6,50,000}{8} = ₹\ 6,68,750 \text{ per year}$$

After-Tax Cash Flows

Year (1)	Savings (2)	Maintenance (3)	Depreciation (4)	Taxable Income (5)	After-tax Profit (6)	Cash Inflow (7) (6+4)	Present Value@10% (8)
1	10,00,000	0	6,68,750	3,31,250	2,31,875	9,00,625	8,18,668
2	10,00,000	0	6,68,750	3,31,250	2,31,875	9,00,625	7,44,277
3	15,00,000	0	6,68,750	8,31,250	5,81,875	12,50,625	9,39,595
4	15,00,000	2,00,000	6,68,750	6,31,250	4,41,875	11,10,625	7,58,557
5	15,00,000	0	6,68,750	8,31,250	5,81,875	12,50,625	7,76,513
6	15,00,000	0	6,68,750	8,31,250	5,81,875	12,50,625	7,05,853
7	12,00,000	3,00,000	6,68,750	2,31,250	1,61,875	8,30,625	4,26,194
8	12,00,000		6,68,750	5,31,250	3,71,875	10,40,625	4,85,452
8	6,50,000						3,03,225
	Total						₹ 59,58,334

Net Present Value (NPV)

NPV=– 60,00,000 + 59,58,334

= ₹ (41,667)

MCQ (iv)

Answer key: **(A)**

(a) ₹ (7,810)

Equivalent Annual Cost (EAC)

$$\frac{\text{NPV}}{\text{PVIFA}(10,8)} = \frac{(41,667)}{5.335} = ₹\ (7,810)$$

MCQ (v)

Answer key: **(B)**

(b) Conclusion and Recommendation:

Warehouse	NPV (₹)	EAC (₹)
Baton	(53,365)	(14,071)
Katon	(41,667)	(7810)

From the above, it can be concluded that warehouse Katon should be chosen as it has the lower Equivalent Annual Cost, despite having higher upfront costs.

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556 **MTP** **Sep 2025 S-II** Case Scenario · MCQ 1 to MCQ 5

CASE SCENARIO

Medilab Diagnostics Pvt. Ltd., a fast-growing healthcare diagnostics chain in Maharashtra, operates several diagnostic centers, offering services such as pathology, radiology, and preventive health screenings.

One of their most profitable centers in Pune recently experienced downtime due to failure of its 10-year-old ultrasound scanner. The machine has reached the end of its useful life, causing disruptions

in patient appointments, especially in obstetrics and abdominal imaging.

To prevent further delays, the procurement and finance teams have evaluated three options:

Two new ultrasound scanner models with identical features are available for purchase from different brands:

Brand	Cost of Machine	Life of Machine	Maintenance Cost			Rate of Depreciation
			Year 1-5	Year 6-10	Year 11-15	
Sonex	₹ 7,20,000	15 years	₹ 24,000	₹34,000	₹45,000	4%
EchoMed	₹ 5,40,000	10 years	₹ 36,000	₹ 58,000	-	6%

The residual value of both machines will drop by 1/3 of the purchase price in the first year, and thereafter shall be depreciated annually at the specified rate.

Alternatively, EchoMed model can be taken on rent, which will be returned to the vendor after use, under the following terms and conditions:

- Annual Rent shall be paid in the beginning of each year and for first year it shall be ₹ 1,18,000
- Annual Rent for the subsequent 4 years shall be ₹ 1,18,500
- Annual Rent for the final 5 years shall be ₹ 1,26,750

The rental agreement can be terminated early by paying a penalty of ₹ 1,10,000. This penalty would be reduced by ₹ 11,000 each year of the period of rental agreement.

The cost of capital for Medilab Diagnostics is 12%.

Being a head of the finance department, you are required to answer the following question of the management (MCQs 1 to 5):

MCQ 1

What is the residual value of the Sonex and Echomed at the end of their life?

A Sonex: ₹ 1,34,690; EchoMed: ₹ 76,997

B Sonex: ₹ 64,000; EchoMed: ₹ 57,000

C Sonex: ₹ 31,043; EchoMed: ₹ 26,278

D Sonex: ₹ 76,800; EchoMed: ₹ 68,400

MCQ 2

If Sonex is used for 15 years, what is the Equivalent Annual Cost of the machine?

A ₹ 1,28,000

B ₹ 1,25,600

C ₹ 1,30,004

D ₹ 1,34,230

MCQ 3

If Echomed is used for 10 years, what is the Equivalent Annual Cost of the machine?

A ₹ 1,28,861

B ₹ 1,35,640

C ₹ 1,32,100

D ₹ 1,27,400

MCQ 4

If Echomed is taken on rent for 10 years, what is the Equivalent Annual Cost of the machine?

A ₹ 1,22,100

B ₹ 1,35,977

C ₹ 1,28,861

D ₹ 1,18,000

MCQ 5

Which is the most economical option if the machine is likely to be used for a period of 5 years?

A Purchase Sonex

B Purchase EchoMed

C Rent EchoMed

D Rent Sonex

ICAI'S ANSWERS

MCQ 1

(D) Sonex: ₹ 76,800; EchoMed: ₹ 68,400

(a) Scrap Value of Machine of Brand Sonex

= ₹ 7,20,000 – ₹ 2,40,000 – (₹ 7,20,000 × 0.04 × 14)

= ₹ 7,20,000 – ₹ 2,40,000 – ₹ 4,03,200 = **₹ 76,800**

(b) Scrap Value of Machine of Brand EchoMed

= ₹ 5,40,000 – ₹ 1,80,000 – (₹ 5,40,000 × 0.06 × 9)

= ₹ 5,40,000 – ₹ 1,80,000 – ₹ 2,91,600 = **₹ 68,400**

MCQ 2

(D) ₹ 1,34,230

Present Value (PV) of Cost if Machine of Brand Sonex is Purchased

Period	Cash Outflow (₹)	PVF @ 12%	Present Value (₹)
0	7,20,000	1	7,20,000
1 – 5	24,000	3.605	86,520
6 – 10	34,000	2.045	69,530
11 – 15	45,000	1.161	52,245
15	(76,800)	0.183	(14,054)
			9,14,241

PVAF for 1-15 years 6.811

Equivalent Annual Cost = $\frac{₹9,14,241}{6.811}$ = **₹ 1,34,230**

MCQ 3

(B) ₹ 1,35,640

Present Value (PV) of Cost if Machine of Brand EchoMed is Purchased

Period	Cash Outflow (₹)	PVF @ 12%	Present Value (₹)
0	5,40,000	1	5,40,000
1 – 5	36,000	3.605	1,29,780
6 – 10	58,000	2.045	1,18,610
10	(68,400)	0.322	(22,025)
			7,66,365

PVAF for 1-10 years 5.65

Equivalent Annual Cost = $\frac{₹7,66,365}{5.65}$ = **₹ 1,35,640**

MCQ 4

(B) ₹ 1,35,977

Present Value (PV) of Cost if Machine of Brand EchoMed is Taken on Rent

Period	Cash Outflow (₹)	PVF @ 12%	Present Value (₹)
0	1,18,000	1.000	1,18,000
1 – 4	1,18,500	3.037	3,59,880
5 – 9	1,26,750	2.291	2,90,389
			7,68,269

PVAF for 1-10 years 5.65

Equivalent Annual Cost = $\frac{₹7,68,269}{5.65}$ = **₹ 1,35,977**

MCQ 5

(C) Rent EchoMed

If Machine is Used for 5 Years

(a) Scrap Value of Machine of Brand Sonex

= ₹ 7,20,000 – ₹ 2,40,000 – (₹ 7,20,000 × 0.04 × 4)

= ₹ 7,20,000 – ₹ 2,40,000 – ₹ 1,15,200 = ₹ 3,64,800

(b) Scrap Value of Machine of Brand EchoMed

= ₹ 5,40,000 – ₹ 1,80,000 – (₹ 5,40,000 × 0.06 × 4)

= ₹ 5,40,000 – ₹ 1,80,000 – ₹ 1,29,600 = ₹ 2,30,400

Present Value (PV) of Cost if Machine of Brand Sonex is Purchased

Period	Cash Outflow (₹)	PVF @ 12%	Present Value (₹)
0	7,20,000	1	7,20,000
1 – 5	24,000	3.605	86,520
5 (Scrap)	(3,64,800)	0.567	(2,06,842)
			₹ 5,99,678

Present Value (PV) of Cost if Machine of Brand EchoMed is Purchased

Period	Cash Outflow (₹)	PVF @ 12%	Present Value (₹)
0	5,40,000	1	5,40,000
1 – 5	36,000	3.605	1,29,780
5 (Scrap)	(2,30,400)	0.567	(1,30,637)
			₹ 5,39,143

Present Value (PV) of Cost if Machine of Brand EchoMed is Taken on Rent

Period	Cash Outflow (₹)	PVF @ 12%	Present Value (₹)
0	1,18,000	1	1,18,000
1 – 4	1,18,500	3.037	3,59,885
5 (Penalty)	55,000	0.567	31,185
			₹ 5,09,070

Since the Present Value of total cost is least in case of taking EchoMed scanner on rent, Medilab should take EchoMed on rent if the machine is to be used for only 5 years.

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557 MTP **May 2025 S-I** Q3(b) 7 marks

Millenial Limited, a Quick commerce (Q-Comm) startup company engaged into the business of deliveries is in a need of a delivery vehicles. The company is considering two different options –

Option A - Buying a brand new 4 electric delivery vehicles that would cost ₹ 1,50,000 each with a GST of 5% not eligible for set off. Electric vehicles would be eligible for a government subsidy of ₹ 20,000 each but only to be received at the end of the year. The life of the delivery vehicle would be 10 years. Scrap value is to be considered at 10% on Gross cost.

Option B – The other alternative is to buy the vehicles needed on a secondhand basis for ₹ 1,00,000 each which will remain in service for a period of 5 years and after 5 years company done capital work on the vehicles for ₹ 70,000 each after which they can be used for another 5 years. The scrap value of the spare parts replaced at the end of 5 years would be ₹ 32,000 whereas at the end of the 10th year scrap value of vehicle would be ₹ 4,000 each.

Millenial Limited's applicable both corporate tax rate and capital gain tax rate is at 15% whereas the vehicles would be depreciated at 20% on WDV basis. The Q-Comm delivery industry's average required rate of return is 15%. You are required to evaluate both the options and advise on the best one.

The revenue and cash expenses for the Q-comm company is expected at ₹ 15,00,000 p.a. and ₹ 10,00,000 p.a. respectively.

ICAI'S ANSWER

Evaluation of option A – buying brand new vehicles

The cash inflows would be in the form of 'cash flows from operations', government subsidy, scrap value and tax savings on capital loss if any.

A] Calculation of PV Cash Inflows from Operations -

Year	1	2	3	4	5	6	7	8	9	10
Revenue	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000
Less: Cash Exp	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000
NPBDT	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000
Less: Depre	1,26,000	1,00,800	80,640	64,512	51,610	41,288	33,030	26,424	21,139	16,911
NPBT	3,74,000	3,99,200	4,19,360	4,35,488	4,48,390	4,58,712	4,66,970	4,73,576	4,78,861	4,83,089
Less: Tax @ 15%	56,100	59,880	62,904	65,323	67,259	68,807	70,046	71,036	71,829	72,463
NPAT	3,17,900	3,39,320	3,56,456	3,70,165	3,81,132	3,89,905	3,96,925	4,02,540	4,07,032	4,10,626
Add: Depre	1,26,000	1,00,800	80,640	64,512	51,610	41,288	33,030	26,424	21,139	16,911
Cash flow after tax from operation	4,43,900	4,40,120	4,37,096	4,34,677	4,32,742	4,31,193	4,29,955	4,28,964	4,28,171	4,27,537
PVAF @ 15%	0.870	0.757	0.658	0.572	0.497	0.432	0.376	0.327	0.284	0.247
PV of Cash flows	3,86,193	3,33,171	2,87,609	2,48,635	2,15,073	1,86,275	1,61,663	1,40,271	1,21,601	1,05,602

PV = ₹ 21,86,092

Since GST on purchase of electric vehicle is not eligible for set off, it would be added to the cost of vehicle

Initial Outflow = 1,50,000 x 4 = ₹ 6,00,000

(+) 5% GST = ₹ 30,000

Purchase price / Gross Cost = ₹ 6,30,000

B] Calculation of PV on Government subsidy –

Government subsidy received at the end of Year 1 = 20,000 x 4 = ₹ 80,000

PVIF at 15%, Year 1 = 0.870

Therefore, PV on Government subsidy = ₹ 69,600

C] Calculation of PV of Scrap value

Scrap value received at the end of Year 10 = 10% on Gross Cost

= 6,30,000 x 0.1

= 63,000

Therefore, PV of scrap = 63,000 x 0.247 = ₹ 15,561

D] Calculation of PV on Tax savings due to Capital Loss at the end of 10th year

WDV of assets at the end of 10th year = 6,30,000 - 5,62,354 = ₹ 67,646

Less : Sale Value = ₹ 63,000

Capital Loss = ₹ 4,646

Tax savings at 15% on above loss = ₹ 697

PV on Tax savings due to Capital Loss = 697 x 0.247 = ₹ 172

NPV of Net Outflow = PV on Cash Inflows from Operations + PV on govt subsidy + PV on Scrap + PV on Tax Savings due to Capital Loss - Initial Outflow

= 21,86,092 + 69,600 + 15,561 + 172 - 6,30,000

NPV = ₹ 16,41,425

Evaluation of option B – buying second hand vehicles

A] Calculation of PV of Cash Outflow

Purchase of 4 vehicles at the beginning = 1,00,000 x 4 = ₹ 4,00,000

Rework of 4 vehicles at the end of 5th year = (70,000 x 4) x 0.497

= ₹ 1,39,160

Total Gross Cash Outflow = 4,00,000 + 1,39,160 = ₹ 5,39,160

B] Calculation of PV of Cash Inflows from Operations –

Year	1	2	3	4	5	6	7	8	9	10
Revenue	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000	15,00,000
Less: Cash Exp	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000	10,00,000
NPBDT	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000	5,00,000
Less: Depre	80,000	64,000	51,200	40,960	32,769	75,814	60,652	48,521	38,817	31,054
NPBT	4,20,000	4,36,000	4,48,800	4,59,040	4,67,231	4,24,186	4,39,348	4,51,479	4,61,183	4,68,946
Less: Tax @ 15%	63,000	65,400	67,320	68,856	70,085	63,628	65,902	67,722	69,177	70,342
NPAT	3,57,000	3,70,600	3,81,480	3,90,184	3,97,146	3,60,558	3,73,446	3,83,757	3,92,006	3,98,604
Add: Depre	80,000	64,000	51,200	40,960	32,769	75,814	60,652	48,521	38,817	31,054
Cash flow after tax from operation	4,37,000	4,34,600	4,32,680	4,31,144	4,29,915	4,36,372	4,34,098	4,32,278	4,30,823	4,29,658
PVAF @ 15%	0.870	0.757	0.658	0.572	0.497	0.432	0.376	0.327	0.284	0.247
PV of Cash flows	3,80,190	3,28,992	2,84,703	2,46,614	2,13,668	1,88,513	1,63,221	1,41,355	1,22,354	1,06,126

PV = ₹ 21,75,736

C] Calculation of WDV at the end 5th year (for depreciation purpose)

WDV at the end of 5th Year = ₹ 1,31,072(4,00,000 – 2,68,928)

Add: Rework = ₹ 2,80,000 (70,000 x 4)

Less: Sale Value = (₹ 32,000)

WDV on which Depreciation will be calculated = ₹ 3,79,072

D] Calculation of PV of Scrap

PV of Scrap sale at the end of 5th Year = ₹ (8,000 x 4) x 0.497 = ₹ 15,904

PV of Scrap sale at the end of 10th year = ₹ (4,000 x 4) x 0.247 = ₹ 3,952

Total PV on Scrap Sale = 15,904 + 3,952 = ₹ 19,856

E] Calculation of PV on Tax savings due to Capital Loss (At Year 10)

WDV as per Income tax at the end of 10th Year = ₹ 1,24,214(3,79,072 – 2,54,858)

Less: Sale Value (4,000 x 4) = (₹ 16,000)

Capital Loss = ₹ 1,08,214

Tax @ 15% on Capital Loss = ₹ 16,232

PV on Tax Savings (16,232 x 0.247) = ₹ 4,009

NPV = PV on Cash Inflows from operations + PV on Scrap + PV on Tax Savings due to Capital Loss - Initial Outflow

= 21,75,736 + 19,856 + 4,009 - 5,39,160

NPV = ₹ 16,60,441

Comment - Since NPV for Option B is more, Millenial Limited should go for buying second hand e-delivery vehicles instead every 5 years instead of owning brand new vehicles.

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558 MTP Jan 2025 S-I Q1(c) 5 marks

Vyom Limited, an IT conglomerate, is planning to take over Aryayash Limited, a startup company incorporated 2 years ago but holding a lot of prospects. To determine the buyout consideration, Vyom Limited has approached you as a Finance controller to estimate the fair value of the startup company today based on future earnings estimates. Following details of the startup company are as below -

Expected Sales in the coming year are ₹ 25 lakhs with P/V ratio of 40%. The sales are expected to grow at a rate of 20% for the next 2 years, to 40% for another 2 years, 25% in the 6th year and thereafter cash flows will grow at a steady rate of 10%. Fixed cost for the upcoming year is expected to be 12 lakhs for the first two years, ₹ 10 lakhs thereafter. Loss in any year can be set-off only against the profits of the immediate next year.

Corporate taxes applicable are 25% & 20% to Vyom Limited & Aryayash Limited respectively. Vyom Limited's desired rate of return is 15% & Cost of Capital of Aryayash Limited is 17%.

As a finance controller, CALCULATE the Fair value of Aryayash Limited.

ICAI'S ANSWER

Fair Value of Company = Present Value all future cash flows discounted at the expected Rate of return of acquiring company.

WN 1 – Calculation of Cash flows ₹ in Lakhs

YEAR	1	2	3	4	5	6
Contribution (40% on sales)	10	12	14.4	20.16	28.22	35.28
(-) Fixed Cost	-12	-12	-10	-10	-10	-10
NPBT (A)	-2	0	4.4	10.16	18.22	25.28
(-) Losses Set Off	0	0	-2(Setoff)	0	0	0
Taxable Income	0	0	2.4	10.16	18.22	25.28
(-) Tax @ 25% **(B)**	0	0	0.6	2.54	4.55	6.32
Cash Flow (A – B)	**-2**	**0**	**3.8**	**7.62**	**13.66**	**18.96**
PV OF CASH FLOWS @ 15%	-1.740	0	2.50	4.35	6.79	8.19

Total PV of cash flows (yr 1 to 6) = 20.08 lakhs

(+) PV of cash flow at terminal value (end of Year 6) = $\frac{18.96 + 10\%}{0.15 - 0.10}$

= 417.12 Lakhs

Therefore, PV of above = 417.12 X PV factor (15%, 6th Year) = 180.20 lakhs

Total fair value of Aryayash limited = 20.08 + 180.20 = 200.28 Lakhs

Note – 1. Discounting rate should be the desired rate of acquiring company i.e. of Vyom Limited

2. Terminal value of cash flows means the cash flows at that point from where it would grow at constant rate. Here it assumed that from 7th year, Cash flows/NPAT will grow at a constant rate and not sales

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559 MTP **Jan 2025 S-II Q3** 10 marks

Hemsparsh Private Limited is globally recognized consultancy firm having its presence in various countries across the globe and is currently headquartered at Ahmedabad, India.

It plans to commence a new branch in the Australia owing to the untapped opportunities available there in the outsourcing business. The company hired a professional for the preparation of the Project report and the fee paid was Rs 2,00,000. The company also incurred Rs 5,00,000 in the form R&D costs. As per the project report, the Company will require an initial fund outlay of Rs 25 crores for buying property & setting up the other infrastructure. It will also require working capital amounting to Rs 5 crore. The company is planning to operate for a very long period of time, however for the sake of simplicity, calculations shall end at the end of the 10th year. The Earnings before tax but after deducting Interest Exp (EBT) estimated would be as follows –

YEAR	EBT (Amount in Rs)
1	2,00,00,000
2	2,50,00,000

YEAR	EBT (Amount in Rs)
3	4,00,00,000
4	4,75,00,000
5	6,00,00,000
6	6,40,00,000
7	6,15,00,000
8	5,25,00,000
9	3,80,00,000
10	2,90,00,000

The above amounts also include an allocated common cost of Rs 12,50,000. Company will distribute 10% dividends every year on post-tax earnings. Company intends to borrow funds of 3 crores at a post-tax Interest rate of 6.5% in India. As per the tax treaty between India & Australia (Tax Agreement between two nations), first 3 years are tax free and from 4th year 75% of corporate taxes are to be paid in the country where it is headquartered and balance in the other nation. Total Corporate tax rate applicable to the company is 30%. However, tax on capital gains is to be paid at 15%, only in the headquarters. Salvage value for depreciation purpose is estimated at Rs. 90,00,000. The assets would be disposed of in the market at Rs. 3,50,00,000 at the end. Hemsparsh Private Limited desires a premium of 3% to the current MCLR of 12% (Marginal Cost of Funds based Lending Rate). Assume no other assets in the block.

CALCULATE NPV for the project and advise only from Indian law perspective. If the company wishes to recoup its investment within 3.5 years, STATE any two measures that the company shall take.

ICAI'S ANSWER

Calculation of NPV (Amount in crores)

Year	1	2	3	4	5	6	7	8	9	10
EBT	2.000	2.500	4.000	4.750	6.000	6.400	6.150	5.250	3.800	2.900
Add: Interest	0.195	0.195	0.195	0.252	0.252	0.252	0.252	0.252	0.252	0.252
Add: Allocated Common Cost	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125
Project Profit Before Tax	2.320	2.820	4.320	5.127	6.377	6.777	6.527	5.627	4.177	3.277
Less: Tax	-	-	-	1.154	1.435	1.525	1.469	1.266	0.940	0.737
Profit After Tax	2.320	2.820	4.320	3.973	4.942	5.252	5.058	4.361	3.237	2.539
Add: Depreciation	2.410	2.410	2.410	2.410	2.410	2.410	2.410	2.410	2.410	-

Year	1	2	3	4	5	6	7	8	9	10
Cash Inflows	4.730	5.230	6.730	6.383	7.352	7.662	7.468	6.771	5.647	2.539
Add: Release Of Working Capital	-	-	-	-	-	-	-	-	-	5.000
Add: Net Cash Inflow from sale of asset (Net Of Tax) (WN-3)	-	-	-	-	-	-	-	-	-	3.471
Total Cash Inflows	4.730	5.230	6.730	6.383	7.352	7.662	7.468	6.771	5.647	11.010
DF @ 15%	0.870	0.756	0.658	0.572	0.497	0.432	0.376	0.327	0.284	0.247
PV Cash Inflow	4.113	3.955	4.425	3.650	3.655	3.312	2.808	2.213	1.605	2.722

TOTAL PV CI = 32.458 Crores

(-) TOTAL PVCO = 30.000 Crores (Initial Outlay + Working Capital)

NPV = 2.458 Crores

ADVISE - Since NPV is positive, company should go for the project.

Notes -

1. Allocated common costs are to be excluded from cash inflows
2. Dividend distribution are deemed irrelevant for cash flow analysis
3. Discounting rate = MCLR + premium = 12 + 3 = 15%
4. Interest exp is to be excluded from the cash inflows as it is already getting covered in the discounting rate above
5. Professional fees paid for project report and R&D costs being sunk costs are irrelevant for decision making

WN 1 – Calculation of applicable taxes each year

For the first 3 years, tax will be zero and for the next 7 years tax rate applicable would 22.5% (30 x 0.75) as balance tax will be paid in Australia, so it will have no relevance under India perspective calculations.

WN – 2 Calculation of interest expense each year

Since post tax interest rate is given in the question, firstly it needs to be converted to pre-tax rate. However, for the first 3 years of the project, post-tax and pre-tax rate would be same owing to zero taxes

Interest Expense (first 3 years) = 3,00,00,000 X 6.5% = 19,50,000 or 0.195 crores

Interest Expense (next 7 years) = 3,00,00,000 x 8.39% = 25,17,000 or 0.2517 crores

Pre-tax Interest Rate = $\frac{\text{Post tax Rate}}{1 - \text{India Tax Rate}}$

= 6.5 / (1 - 0.225)

= 8.39%

WN 3 – CALCULATION OF CAPITAL GAINS INCOME IN YEAR 10

Cost of Asset remaining in the block at the beginning of Year 10

= 3,31,00,000 (2,41,00,000 + 90,00,000)

(+) New Asset purchased during the year = 0

(-) Sale Value of the Asset = 3,50,00,000

Capital Gains Income before tax = 19,00,000

(-) Capital Gains tax = 19,00,000 x 15% = 2,85,000

Net Cash Inflow after tax = 3,50,00,000- 2,85,000

= 3,47,15,000

B) Current Payback Period = 4 + 1.927 /7.352

= 4.262 years

Target Payback Period = 3.5 years

Some key measures to reduce your Payback period are as follows (Only illustrative):

i. Emphasizing on reduction of operational costs

ii. Improving marketing thereby resulting into higher sales

iii. Incorporate product-led growth strategies

iv. Judicious efforts in bringing down the overall cost of capital thereby reducing the discounting rate and in turn better Payback period.

v. Leveraging out the presence of the fixed cost

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560 MTP **Sep 2024 S-I** Q3(a) 8 marks

Parmarth Limited is a manufacturer of computers. Owing to recent developments in Artificial Intelligence (AI), it is planning to introduce AI in its computer process. This would result into an estimated annual savings as follows:

(i) Savings of ₹ 3,50,000 in production delays caused by inventory problem.

(ii) Savings in Salaries of 5 employees with an annual pay of ₹ 4,20,000 per annum

(iii) Reduction in Lost sales of ₹ 1,75,000

(iv) Gain due to timely billing is ₹ 3,25,000

The project would result in annual maintenance and operating costs as follows, which are to be paid in advance (at the beginning)

YEAR	1	2	3	4	5
COST	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000

Furthermore, the new system would need 2 AI specialists' professional drawing salaries of ₹ 6,50,000 per annum per person. The purchase price of the new system for installing AI into computers would involve an outlay of ₹ 21,50,000 and installation cost of ₹ 1,50,000.

75% of the total value for depreciation would be paid in the year of purchase and the balance would be paid at the end of the 1st year. The new system will be sold for ₹ 1,90,000. This is the only asset in the block for Income tax purpose.

The life of the system would be 5 years with the hurdle rate of 12%. Depreciation will be charged at 40% on WDV basis, corporate tax rate is 25% and capital gains tax rate is at 20%.

CALCULATE NPV and advise the management on the acceptability of the proposal. Also calculate ARR & PI.

ICAI'S ANSWER

Calculation of Present value of cash inflows (PVCI)

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Savings in cost due to Production Delays	-	3,50,000	3,50,000	3,50,000	3,50,000	3,50,000
Savings in Salaries	-	21,00,000	21,00,000	21,00,000	21,00,000	21,00,000
Reduction in lost sales	-	1,75,000	1,75,000	1,75,000	1,75,000	1,75,000
Gain due to timely billing	-	3,25,000	3,25,000	3,25,000	3,25,000	3,25,000
	-	29,50,000	29,50,000	29,50,000	29,50,000	29,50,000
Less:						
Salary of AI specialists	-	13,00,000	13,00,000	13,00,000	13,00,000	13,00,000
Annual Maint. & Op Cost	-	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000
NPBDT	-	14,70,000	14,50,000	15,30,000	15,40,000	15,20,000
(-) Depreciation	-	9,20,000	5,52,000	3,31,200	1,98,720	1,19,232
NPBT	-	5,50,000	8,98,000	11,98,800	13,41,280	14,00,768
(-) Tax @ 25%	-	1,37,500	2,24,500	2,99,700	3,35,320	3,50,192
NPAT	-	4,12,500	6,73,500	8,99,100	10,05,960	10,50,576
(+) Depreciation	-	9,20,000	5,52,000	3,31,200	1,98,720	1,19,232
(+) Annual Maint. & Op Cost	-	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000
Gross Cash Inflows	-	15,12,500	14,25,500	13,50,300	13,14,680	12,99,808

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
(-) Annual Maint. & Op Cost actually paid	1,80,000	2,00,000	1,20,000	1,10,000	1,30,000	-
Net Cash Inflows	-1,80,000	13,12,500	13,05,500	12,40,300	11,84,680	12,99,808
(+) Sale Value at the end of life	-	-	-	-	-	1,90,000
	-1,80,000	13,12,500	13,05,500	12,40,300	11,84,680	14,89,808
PV Factor @ 12%	1	0.8929	0.7` 972	0.7118	0.6355	0.5674
PV of Cash Inflows	-1,80,000	11,71,875	10,40,737	8,82,821	7,52,886	8,45,357
Total PV of Cash Inflows	45,13,675					

Calculation of Present value of cash outflows (PVCO)

As mentioned in the question, 75% of the depreciable value will be paid at the beginning. Depreciable value means purchase price plus the installation cost.

	Year 0	Year 1
Purchase Price & Installation Cost	17,25,000	5,75,000
PV Factor @ 12%	1	0.8929
PVCO	17,25,000	5,13,418

(2) Total PVCO = 22,38,418

(3) PV of Tax on Capital Gains (Only asset in the block) - 5th Year end

Capital Gains = Sale Price (-) Closing WDV at 5th year

= 1,90,000 (-) 1,78,848

= 11,152

Tax @ 20% on above = 2230.40

PV = 2,230.40 x 0.5674 = 1,266

Net PVCI = PVCI - PV of Tax on Capital Gains

= 45,13,675 - 1,266 = 45,12,409

NPV = Net PVCI – PVCO

= 45,12,409 - 22,38,418

= 22,73,991

(II) PI = PVCI / PVCO = 45,12,409/ 22,38,418 = 2.0158

(III) ARR = Average NPAT / Initial Investment

= 8,08,327.2/23,00,000 x 100 = 35.145%

Note – ARR is calculated based on Initial Investment, similarly it can be calculated based on Average Investment

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561 MTP **Sep 2024 S-II** Q1(b) 5 marks

Mr. Anand is thinking of buying a Share at ₹ 500 whose Face Value per share is ₹ 100. He is expecting a bonus at the ratio 1 : 5 at the end of the fourth year. Annual expected dividend is 20% and the same rate is expected to be maintained on the expanded capital base. He intends to sell the Shares at the end of seventh year at an expected price of ₹ 900 each. Incidental Expenses for purchase and sale of Shares are estimated to be 5% of the Market Price. Assuming a Discount rate of 12% per annum, COMPUTE the Net Present Value from the acquisition of the shares.

ICAI'S ANSWER

Computation of PV of Future Cash Flows

Year	Nature	Cash Flow	DF @ 12%	DCF
1	Dividends (₹ 100 × 20%)	20	0.893	17.86
2	Dividends (₹ 100 × 20%)	20	0.797	15.94
3	Dividends (₹ 100 × 20%)	20	0.712	14.24
4	Dividends (₹ 100 × 20%)	20	0.636	12.72
5	Dividends (₹ 100 × 1.2 × 20%)	24	0.567	13.61
6	Dividends (₹ 100 × 1.2 × 20%)	24	0.507	12.17
7	Dividends (₹ 100 × 1.2 × 20%)	24	0.452	10.85
7	Net Sale Proceeds (₹ 900 × 1.2 – 5%)	1,026	0.452	463.75
	Present Value of Cash Inflows			561.14
0	Less: Initial Investment (₹ 500 + 5%)	525	1	525.00
	Net Present Value			36.14

Note: At the end of Year 4, Anand will have 1.2 Share i.e. 1 Bought Share + 1/5th Bonus Share.

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562 MTP **May 2024 S-I** Q2(a) 8 marks

NC Ltd. Is considering purchasing a new machine to increase its production facility. At present, it uses an old machine which can process 5,000 units of TVs per week. NC could replace it with new machine, which is product specific and can produce 15,000 units per week. New machine cost ₹ 100 crores and requires the working capital of ₹ 3 crores, which will be released at the end of 5th year. The new machine is expected to have a salvage value of ₹ 20 crores.

The company expects demand for TVs to be 10,000 units per week. Each TV sells for ₹ 30,000 and has Profit Volume Ratio (PV) of 0.10. The company works for the 56 weeks in the year. Additional fixed costs (excluding depreciation) are estimated to increase by ₹ 10 crores. The company is subject to a 40% tax

rate and its after-tax cost of capital is 20%. The relevant rate of depreciation is 25 % for both taxation and accounts. The company uses the WDV method of depreciation. The existing machine will have no scrap value.

You are required to:

ADVISE whether the company should replace the old machine.

(Decimal may be taken up to 2 units)

ICAI'S ANSWER

Cash inflows after tax (CFAT)

Particular	₹
Current production (units per week)	5,000 units
New capacity (units per week)	15,000 units
Demand (units per week)	10,000 units
Increase in sales (units per week) A.	5,000 units
Contribution per unit (₹ 30,000 × 0.10) B.	3,000
Increase in contribution A x B x 56	84 crores
Less: Additional fixed cost	10 crores
Increase in profit	74 crores
Less: Tax @ 40%	29.6 crores
Profit after tax	44.4 crores

Tax shield due to depreciation

Year	Depreciation (₹ in Crore)	Tax Shield (₹ in Crore)	PV Factor @ 20%	Total Present Value (₹ in Crore)
1	25.00	10	0.83	8.33
2	18.75	7.5	0.69	5.18
3	14.06	5.62	0.58	3.26
4	10.55	4.22	0.48	2.03
5	7.91	3.16	0.40	1.27
Total				20.07

Tax shield on capital loss = (23.73-20.00) x 30% = ₹ 1.12 crores

Net Present Value (NPV)

Particulars	Year	Cash Flow (₹ in Crores)	PVAF @ 20%	Present Value (₹ in Crores)
Initial Investment	0	(100)	1	(100)
Working capital	0	(3)	1	(3)
Profit after tax	1-5	44.4	2.99	132.76
Salvage value	5	20	0.40	8.00
Tax shield on Depreciation	1-5			20.07
Tax shield on capital loss	5	1.12	0.40	0.45
Release of Working Capital	5	3	0.40	1.20
NPV				59.47

The company is advised to replace the old machine since the NPV of the new machine is positive.

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563 MTP **May 2024 S-II** Q1(c) 5 marks

An existing profitable company, RMC World Ltd. is considering a new project for manufacture of home automation gadget involving a capital expenditure of ₹ 1000 Lakhs and working capital of ₹ 150 Lakhs. The capacity of the plants for an annual production of 3 lakh units and capacity utilization during 5 year life of the project is expected to be as indicated below:

Year	1	2	3	4	5
Capacity Utilization (%)	50	65	80	100	100

The average price per unit of product is expected to be ₹600 netting a contribution of 60 percent. The annual fixed costs, excluding depreciation, are estimated to be ₹500 Lakhs per annum from the third year onwards. For the first and second year, it would be ₹ 200 lakhs and ₹ 350 lakhs respectively.

Scrap value of the capital asset at the end of 5th year is ₹ 200 Lakhs. Depreciation on capital asset is provided on written down value basis @ 40% p.a. for income tax purpose. The rate of income tax may be taken at 30%. The cost of capital is 12%. At end of the third year an additional investment of ₹ 200 lakhs would be required for working capital. There is no capital gain tax applicable.

COMPUTE the NPV of the project. RMC World Ltd. is about to make a presentation to Secure Venture Capital Firm. Secure Venture Capital Firms will invest in any project if the net addition to shareholder wealth from the project is above ₹ 100 lakhs.

ICAI'S ANSWER

Calculation of Cash Flow after Tax

	Year 1	Year 2	Year 3	Year 4	Year 5
Capacity	50%	65%	80%	100%	100%
Units	1,50,000	1,95,000	2,40,000	3,00,000	3,00,000

	Year 1	Year 2	Year 3	Year 4	Year 5
Contribution p.u.	360	360	360	360	360
(600 × 60%)					
Total Contribution	5,40,00,000	7,02,00,000	8,64,00,000	10,80,00,000	10,80,00,000
Less: Fixed Asset	2,00,00,000	3,50,00,000	5,00,00,000	5,00,00,000	5,00,00,000
Less: Depreciation (W.N.)	4,00,00,000	2,40,00,000	1,44,00,000	86,40,000	51,84,000
PBT	(60,00,000)	1,12,00,000	2,20,00,000	4,93,60,000	5,28,16,000
Less: Tax	(18,00,000)	33,60,000	66,00,000	1,48,08,000	1,58,44,800
PAT	(42,00,000)	78,40,000	1,54,00,000	3,45,52,000	3,69,71,200
Add: Depreciation	4,00,00,000	2,40,00,000	1,44,00,000	86,40,000	51,84,000
CFAT	3,58,00,000	3,18,40,000	2,98,00,000	4,31,92,000	4,21,55,200

Calculation of NPV

Year	Description	Cash Flow	PVF @12%	PV
0	Initial Investment	(10,00,00,000)	1	(10,00,00,000)
0	WC introduced	(1,50,00,000)	1	(1,50,00,000)
3	WC introduced	(2,00,00,000)	0.7118	(1,42,36,000)
1	CFAT	3,58,00,000	0.8929	3,19,65,820
2	CFAT	3,18,40,000	0.7972	2,53,82,848
3	CFAT	2,98,00,000	0.7118	2,12,11,640
4	CFAT	4,31,92,000	0.6355	2,74,48,516
5	CFAT	4,21,55,200	0.5674	2,39,18,860
5	WC released	3,50,00,000	0.5674	1,98,59,000
5	Scrap Sale	2,00,00,000	0.5674	1,13,48,000
	Net Present Value			**3,18,98,684**

Working Notes (W.N.)

Calculation of Depreciation

Year	Opening WDV	Depreciation	Closing WDV
1	10,00,00,000	4,00,00,000	6,00,00,000
2	6,00,00,000	2,40,00,000	3,60,00,000
3	3,60,00,000	1,44,00,000	2,16,00,000
4	2,16,00,000	86,40,000	1,29,60,000
5	1,29,60,000	51,84,000	77,76,000

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564 MTP **Old P8 May 2023 S-I Q4** 10 marks old scheme

Yellow bells Ltd. wants to replace its old machine with new automatic machine. The old machine had been fully depreciated for tax purpose but has a book value of ₹3,50,000 on 31st March 2022. The machine cannot fetch more than ₹45,000 if sold in the market at present. It will have no realizable value after 10 years. The company has been offered ₹1,60,000 for the old machine as a trade in on the new machine which has a price (before allowance for trade in) of ₹6,50,000. The expected life of new machine is 10 years with salvage value of ₹63,000.

Further, the company follows straight line depreciation method but for tax purpose, written down value method depreciation @ 9% is allowed taking that this is the only machine in the block of assets.

Given below are the expected sales and costs from both old and new machine:

	Old machine (₹)	New machine (₹)
Sales	11,74,500	11,74,500
Material cost	2,61,000	1,83,063
Labour cost	1,95,750	1,59,500
Variable overhead	81,563	68,875
Fixed overhead	1,30,500	1,41,375
Depreciation	34,800	60,175
Profit Before Tax (PBT)	4,70,888	5,61,513
Tax @ 25%	1,17,722	1,40,378
Profit After Tax (PAT)	3,53,166	4,21,134

From the above information, ANALYSE whether the old machine should be replaced or not if required rate of return is 10%? Ignore capital gain tax.

PV factors @ 10%:

Year	1	2	3	4	5	6	7	8	9	10
PVF	0.909	0.826	0.751	0.683	0.621	0.564	0.513	0.467	0.424	0.386

ICAI'S ANSWER

(i) Calculation of Base for depreciation or Cost of New Machine

Particulars	(₹)
Purchase price of new machine	6,50,000
Less: Sale price of old machine	1,60,000
	4,90,000

(ii) Calculation of Profit before tax as per books

Particulars	Old machine	New machine	Difference
	(₹)	**(₹)**	**(₹)**
PBT as per books	4,70,888	5,61,513	90,625
Add: Depreciation as per books	34,800	60,175	25,375
Profit before tax and depreciation (PBTD)	5,05,688	6,21,688	1,16,000

Calculation of Incremental NPV

Year	PVF	PBTD	Dep. @ 9%	PBT	Tax @ 25%	Cash Inflows	PV of Cash Inflows
	@ 10%	**(₹)**	**(₹)**	**(₹)**	**(₹)**	**(₹)**	**(₹)**
	1	**2**	**3**	**4(2-3)**	**(5) = (4) x 0.25**	**(6) = (4) – (5) + (3)**	**(7) = (6) x (1)**
1	0.909	1,16,000.00	44,100.00	71,900.00	17,975.00	98,025.00	89,104.73
2	0.826	1,16,000.00	40,131.00	75,869.00	18,967.25	97,032.75	80,149.05
3	0.751	1,16,000.00	36,519.21	79,480.79	19,870.20	96,129.80	72,193.48
4	0.683	1,16,000.00	33,232.48	82,767.52	20,691.88	95,308.12	65,095.45
5	0.621	1,16,000.00	30,241.56	85,758.44	21,439.61	94,560.39	58,722.00
6	0.564	1,16,000.00	27,519.82	88,480.18	22,120.05	93,879.95	52,948.29
7	0.513	1,16,000.00	25,043.03	90,956.97	22,739.24	93,260.76	47,842.77
8	0.467	1,16,000.00	22,789.16	93,210.84	23,302.71	92,697.29	43,289.63
9	0.424	1,16,000.00	20,738.14	95,261.86	23,815.47	92,184.53	39,086.24
10	0.386	1,16,000.00	18,871.70	97,128.30	24,282.07	91,717.93	35,403.12

Year	PVF	PBTD	Dep. @ 9%	PBT	Tax @ 25%	Cash Inflows	PV of Cash Inflows
							5,83,834.77
Add: PV of Salvage value of new machine (₹ 63,000 × 0.386)							24,318.00
Total PV of incremental cash inflows							6,08,152.77
Less: Cost of new machine [as calculated in point(i)]							4,90,000.00
Incremental Net Present Value							1,18,152.77

Analysis: Since the Incremental NPV is positive, the old machine should be replaced.

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565 MTP **Old P8 May 2023 S-II** Q1(a) 5 marks old scheme

Rambow Ltd. is contemplating purchasing machinery that would cost ₹ 10,00,000 plus GST @ 18% at the beginning of year 1. Cash inflows after tax from operations have been estimated at ₹ 2,56,000 per annum for 5 years. The company has two options for the smooth functioning of the machinery - one is service, and another is replacement of parts. The company has the option to service a part of the machinery at the end of each of the years 2 and 4 at ₹ 1,00,000 plus GST @ 18% for each year. In such a case, the scrap value at the end of year 5 will be ₹ 76,000. However, if the company decides not to service the part, then it will have to be replaced at the end of year 3 at ₹ 3,00,000 plus GST@ 18% and in this case, the machinery will work for the 6th year also and get operational cash inflow of ₹ 1,86,000 for the 6th year. It will have to be scrapped at the end of year 6 at ₹ 1,36,000.

Assume cost of capital at 12% and GST paid on all inputs including capital goods are eligible for input tax credit in the same month as and when incurred.

(i) DECIDE whether the machinery should be purchased under option 1 or under option 2 or it shouldn't be purchased at all.

(ii) If the supplier gives a discount of ₹ 90,000 for purchase, WHAT would be your decision?

Note: The PV factors at 12% are:

Year	0	1	2	3	4	5	6
PV Factor	1	0.8928	0.7972	0.7118	0.6355	0.5674	0.5066

ICAI'S ANSWER

Option I: Purchase Machinery and Service Part at the end of Year 2 and 4.

Net Present value of cash flow @ 12% per annum discount rate.

NPV (in ₹) = - 10,00,000 + 2,56,000 x (0.8928+0.7972+0.7118+0.6355+0.5674) – (1,00,000 x 0.7972+1,00,000 x 0.6355) + (76,000 x 0.5674)

= - 10,00,000 + (2,56,000 x 3.6047) – 1,43,270+43,122.4

= - 10,00,000 + 9,22,803.2 – 1,43,270+ 43,122.4

NPV = - 1,77,344.4

Since Net Present Value is negative; therefore, this option is not to be considered.

If Supplier gives a discount of ₹ 90,000, then:

NPV (in ₹) = + 90,000 - 1,77,344.4 = -87,344.4

In this case, Net Present Value is still negative; therefore, this option may not be advisable

Option II: Purchase Machinery and Replace Part at the end of Year 2.

NPV (in ₹)= - 10,00,000 + 2,56,000 x (0.8928+0.7972+0.7118+0.6355+0.5674) – (3,00,000 x 0.7118) + (1,86,000 x 0.5066+1,36,000 x 0.5066)

= - 10,00,000 + (2,56,000 x 3.6047) – 2,13,540+1,63,125.2

= - 10,00,000 + 9,22,803.2 – 2,13,540+1,63,125.2

NPV = - 1,27,611.6

Net Present Value is negative, the machinery should not be purchased.

If the Supplier gives a discount of ₹ 90,000, then:

NPV (in ₹) = 90,000 - 1,27,611.6 = - 37,611.6

In this case, Net Present Value is still negative; therefore, this option may not be advisable.

Decision: The Machinery should not be purchased as it will earn a negative NPV in both options of repair and replacement.

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566 MTP **Old P8 May 2023 S-II** Q5 10 marks old scheme

Genzy Ltd. is planning to introduce a new product with a project life of 10 years. The initial equipment cost will be ₹ 2.5 crores. At the end of 10 years, the equipment will have a resale value of 50 lakhs. A working capital of ₹ 30,00,000 will be needed and it will be released at the end of the tenth year. The project will be financed with the following capital sources.

Particulars	Amount (₹)	Issue Price (Market price)
Equity Share Capital of Face value ₹ 10 each	1,50,00,000	₹30
Debentures of face value ₹ 100 each with a maturity of 10 years	90,00,000	₹90
Preference shares of ₹ 100 each with a maturity of 10 years	60,00,000	₹96

The existing yield on T-bills is averaging 8% p.a. The systematic risk measure for the proposed project is 1.6. NSE NIFTY is expected to yield 14% p.a. on average for the foreseeable future. Debenture holders have been promised a coupon of 12% and preference shareholders have been committed a dividend of 15%.

The sales volumes over 10 years have been estimated as follows:

Year	1	2	3-5	6-8	9-10
Units per year	70,000	98,000	2,10,000	2,50,000	1,20,000

A sales price of ₹ 300 per unit is expected and variable expenses will amount to 60% of sales revenue. Fixed cash operating costs will amount to ₹ 40,00,000 per year. The loss of any year will be set off from the profits of subsequent years.

The company is subject to a 30 per cent tax rate. The company follows straight line method of depreciation which is to be assumed to be admissible for tax purpose also .

CALCULATE the net present value of the project for the company and advise the management to take appropriate decision.

The PV factors are to be taken as rounded figures upto 2 decimals. Use market value weights to COMPUTE overall cost of capital.

ICAI'S ANSWER

Cost of Equity

Ke = Rf + Beta * (Rm – Rf)

Ke = 8% + 1.6 * (14% - 8%)

Ke = 8% + (1.6 * 6%)

Ke = 17.6%

1. Cost of Redeemable Debentures (Post-Tax)

$$Kd = \frac{\text{Int (1-t)} + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$Kd = \frac{12,00,000 * (1 - 30\%) + ((1,00,00,000 - 90,00,000) / 10)}{(1,00,00,000 + 90,00,000) / 2}$$

$$Kd = \frac{8,40,000 + 1,00,000}{95,00,000}$$

Kd = 9.89%

2. Cost of Redeemable Preference Shares

$$Kp = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

$$Kp = \frac{(62,50,000 * 15\%) + ((62,50,000 - 60,00,000) / 10)}{(62,50,000 + 60,00,000) / 2}$$

$$Kp = \frac{9,37,500 + 25,000}{61,25,000}$$

Kp = 15.71%

3. Weighted Average Cost of Capital (WACC) – Book Value Method

Source of Capital	Market Value	Weights	After Tax Cost of Capital	WACC
Equity Share Capital	1,50,00,000	0.5	17.6%	0.088
Debentures	90,00,000	0.3	9.89%	0.030
Preference Share Capital	60,00,000	0.2	15.71%	0.031
	3,00,00,000	**1.000**		**0.149**

WACC = 14.9%

4. Computation of CFAT

	(year 1 to year 4)					
Sr. No.	**Particulars / Year**	**1**	**2**	**3-5**	**6-8**	**9-10**
A	Sale Price p.u.	300	300	300	300	300
	Sale units	70,000	98,000	2,10,000	2,50,000	1,20,000
C	Sales (A x B)	2,10,00,000	2,94,00,000	6,30,00,000	7,50,00,000	3,60,00,000
D	Variable Cost p.u.	180	180	180	180	180
E	Variable Cost (B x D)	1,26,00,000	1,76,40,000	3,78,00,000	4,50,00,000	2,16,00,000
F	Contribution (C - E)	84,00,000	1,17,60,000	2,52,00,000	3,00,00,000	1,44,00,000
G	Less: Fixed Cost	40,00,000	40,00,000	40,00,000	40,00,000	40,00,000
H	PBDT (F-G)	44,00,000	77,60,000	2,12,00,000	2,60,00,000	1,04,00,000
I	Less: Depreciation (2,50,00,000-50,00,000) / 10	20,00,000	20,00,000	20,00,000	20,00,000	20,00,000
J	PBT	24,00,000	57,60,000	1,92,00,000	2,40,00,000	84,00,000
K	Less: Taxes @ 30%	7,20,000	17,28,000	57,60,000	72,00,000	25,20,000
L	PAT	16,80,000	40,32,000	1,34,40,000	1,68,00,000	58,80,000
M	Add: Depreciation	20,00,000	20,00,000	20,00,000	20,00,000	20,00,000
N	CFAT	36,80,000	60,32,000	1,54,40,000	1,88,00,000	78,80,000

5. Computation of NPV

Sr. No.	Particulars / Year	1	2	3-5	6-8	9-10
I	CFAT	36,80,000	60,32,000	1,54,40,000	1,88,00,000	78,80,000
II	PVAF @ 14.9%	0.87	0.76	(0.66+0.57+0.50) = 1.73	(0.43+0.38+0.33) = 1.14	(0.29+0.25) = 0.54
III	PV of CFATs (I x II)	32,01,600	45,84,320	2,67,11,200	2,14,32,000	42,55,200
IV	Salvage + Release of WC					80,00,000
V	PVF @ 14.9%					0.25
VI	PV of Salvage (IV x V)					20,00,000

PV of Inflows = 32,01,600 + 45,84,320 + 2,67,11,200 + 2,14,32,000 + 42,55,200 + 20,00,000

PV of Inflows = 6,21,84,320

PV of Outflows = Investment + Introduction of Working Capital

PV of Outflows = 2,50,00,000 + 30,00,000

PV of Outflows = 2,80,00,000

NPV = PV of Inflows – PV of Outflows

NPV = 6,21,84,320 - 2,80,00,000

NPV = 3,41,84,320

The management should consider taking up the project as the Net Present Value of the Project is Positive.

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567 MTP **Old P8 Nov 2022 S-I** Q2 10 marks old scheme

Embros Ltd. is planning to invest in a new product with a project life of 8 years. Initial equipment cost will be ₹ 35 crores. Additional equipment costing ₹ 2.50 crores will be purchased at the end of the third year from the cash inflow of this year. At the end of 8th year, the original equipment will have no resale value, but additional equipment can be sold at 10% of its original cost. A working capital of ₹ 4 crores will be needed, and it will be released at the end of 8th year. The project will be financed with sufficient amount of equity capital.

The sales volumes over eight years have been estimated as follows:

Year	1	2	3	4–5	6–8
Units	14,40,000	21,60,000	52,00,000	54,00,000	36,00,000

Sales price of ₹ 120 per unit is expected and variable expenses will amount to 60% of sales revenue. Fixed cash operating costs will amount ₹ 3.60 crores per year. The loss of any year will be set off from the profits of subsequent year. The company follows straight line method of depreciation and is subject to 30% tax rate. Considering 12% after-tax cost of capital for this project, you are required to CALCULATE the net present value (NPV) of the project and advise the management to take appropriate decision.

PV factors @ 12% are:

Year	1	2	3	4	5	6	7	8
	.893	.797	.712	.636	.567	.507	.452	.404

ICAI'S ANSWER

Calculation of year-wise Cash Inflow *(₹ in crores)*

Year	Sales	VC (60% of Sales Value)	FC	Dep.	Profit	Tax (@30%)	PAT	Dep.	Cash inflow
1	17.28	10.368	3.6	4.375	(1.063)	-	(1.0630)	4.375	3.312
2	25.92	15.552	3.6	4.375	2.393	0.3990*	1.9940	4.375	6.369
3	62.4	37.44	3.6	4.375	16.985	5.0955	11.8895	4.375	16.2645
4–5	64.8	38.88	3.6	4.825#	17.495	5.2485	12.2465	4.825	17.0715
6–8	43.2	25.92	3.6	4.825	8.855	2.6565	6.1985	4.825	11.0235

*(30% of 2.393 – 30% of 1.063) = 0.7179 – 0.3189 = 0.3990

#4.375 + (2.50 - .25)/5 = 4.825

Calculation of Cash Outflow at the beginning

Particulars	₹
Cost of New Equipment	35,00,00,000
Add: Working Capital	4,00,00,000
Outflow	39,00,00,000

Calculation of NPV

Year	Cash inflows	PV factor	NPV
	(₹)		(₹)
1	3,31,20,000	.893	2,95,76,160
2	6,36,90,000	.797	5,07,60,930
3	16,26,45,000 - 2,50,00,000 = 13,76,45,000	.712	9,80,03,240
4	17,07,15,000	.636	10,85,74,740
5	17,07,15,000	.567	9,67,95,405
6	11,02,35,000	.507	5,58,89,145
7	11,02,35,000	.452	4,98,26,220
8	11,02,35,000 + 4,00,00,000 + 25,00,000 = 15,27,35,000	.404	6,17,04,940
	Present Value of Inflow		55,11,30,780

Year	Cash inflows	PV factor	NPV
	Less: Out flow		39,00,00,000
	Net Present Value		**16,11,30,780**

Advise: Since the project has a positive NPV, it may be accepted.

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568 MTP **Old P8 Nov 2022 S-II** Q6(c) 2 marks old scheme

What do you UNDERSTAND by desirability factor/profitability index?

ICAI'S ANSWER

Desirability Factor/Profitability Index

In certain cases, we have to compare a number of proposals each involving different amount of cash inflows. One of the methods of comparing such proposals is to work out what is known as the 'Desirability factor' or 'Profitability index'. In general terms, a project is acceptable if its profitability index value is greater than 1.

Mathematically, the desirability factor is calculated as below:

$$\frac{\text{Sum of Discounted Cash inflows}}{\text{Initial Cash outlay or Total Discounted Cash outflow (as the case may be)}}$$

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569 MTP **Old P8 May 2022 S-I** Q4 10 marks old scheme

A manufacturing company is presently paying a garbage disposer company ₹ 0.50 per kilogram to dispose-off the waste resulting from its manufacturing operations. At normal operating capacity, the waste is about 2,00,000 kilograms per year.

After spending ₹ 1,20,000 on research, the company discovered that the waste could be sold for ₹ 5 per kilogram if it was processed further. Additional processing would, however, require an investment of ₹ 12,00,000 in new equipment, which would have an estimated life of 10 years with no salvage value. Depreciation would be calculated by straight line method.

No change in the present selling and administrative expenses is expected except for the costs incurred in advertising ₹ 40,000 per year, if the new product is sold. Additional processing costs would include variable cost of ₹ 2.50 per kilogram of waste put into process along with fixed cost of ₹ 60,000 per year (excluding Depreciation).

There will be no losses in processing, and it is assumed that the total waste processed in a given year will be sold in the same year. Estimates indicate that 2,00,000 kilograms of the product could be sold each year.

The management when confronted with the choice of disposing off the waste or processing it further and selling it, seeks your ADVICE. Which alternative would you RECOMMEND? Assume that the firm's cost of capital is 15% and it pays on an average 50% Tax on its income.

Consider Present value of Annuity of ₹ 1 per year @ 15% p.a. for 10 years as 5.019.

ICAI'S ANSWER

Evaluation of Alternatives:

Savings in disposing off the waste

Particulars	(₹)
Outflow (2,00,000 × ₹ 0.50)	1,00,000
Less: tax savings @ 50%	50,000
Net Outflow per year	**50,000**

Calculation of Annual Cash inflows in Processing of waste Material

Particulars	Amount (₹)	Amount (₹)
Sale value of waste (₹ 5 × 2,00,000 kilograms)		10,00,000
Less: Variable processing cost (₹ 2.50 × 2,00,000 kilograms)	5,00,000	
Less: Fixed processing cost	60,000	
Less: Advertisement cost	40,000	
Less: Depreciation	1,20,000	(7,20,000)
Earnings before tax (EBT)		**2,80,000**
Less: Tax @ 50%		(1,40,000)
Earnings after tax (EAT)		1,40,000
Add: Depreciation		1,20,000
Annual Cash inflows		**2,60,000**

Total Annual Benefits = Annual Cash inflows + Net savings (adjusting tax) in disposal cost

= ₹ 2,60,000 + ₹ 50,000 = ₹ 3,10,000

Calculation of Net Present Value

Year	Particulars	Amount (₹)
0	Investment in new equipment	(12,00,000)
1 to 10	Total Annual benefits × $PVAF_{(10\text{ years},\ 15\%)}$ ₹ 3,10,000 × 5.019	15,55,890
	Net Present Value	**3,55,890**

Recommendation: Processing of waste is a better option as it gives a positive Net Present Value.

Note- Research cost of ₹ 1,20,000 is not relevant for decision making as it is sunk cost.

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570 MTP Old P8 Dec 2021 S-I Q3 10 marks old scheme

Sadbhavna Limited is a manufacturer of computers. It wants to introduce artificial intelligence while making computers. It estimates that the annual savings from the artificial intelligence (AI) include a reduction of five employees with annual salaries of ₹ 3,00,000 each, ₹ 3,00,000 from reduction in production delays caused by inventory problem, reduction in lost sales ₹ 2,50,000 and ₹ 2,00,000 from billing issues.

The purchase price of the system for installation of artificial intelligence is ₹ 20,00,000 with installation cost of ₹ 1,00,000. The life of the system is 5 years and it will be depreciated on a straight-line basis. The salvage value is zero which will be its market value after the end of its life of five years.

However, the operation of the new system for AI requires two computer specialists with annual salaries of ₹ 5,00,000 per person. Also, the estimated maintenance and operating expenses of 1,50,000 is required.

The company's tax rate is 30% and its required rate of return is 12%.

From the above information:

(i) CALCULATE the initial cash outflow and annual operating cash flow over its life of 5 years.

(ii) Further, EVALUATE the project by using Payback Period, Net Present Value and Profitability Index.

(iii) You are also REQUIRED to obtain the cash flows and NPV on the assumption that book salvage value for depreciation purposes is ₹ 2,00,000 even though the machine is having no real worth in terms of its resale value. Also, the book salvage value of ₹ 2,00,000 is allowed for tax purposes.

Also COMMENT on the acceptability of the project in (ii) and (iii) above.

ICAI'S ANSWER

(i) Project's Initial Cash Outlay

Cost	20,00,000
Installation Expenses	1,00,000
Total Cash Outflow	21,00,000
Depreciation per year = 21,00,000/5 =	4,20,000

Project's Operating Cash Flows over its 5-year life

Savings (A)

Reduction in salaries (₹ 3,00,000 × 5)	15,00,000
Reduction in production delays	3,00,000
Reduction in lost sales	2,50,000
Gains due to timely billing	2,00,000
	22,50,000

Costs (B)

- Depreciation	4,20,000
- Additional Specialist Cost (₹ 5,00,000 × 2)	10,00,000
- Maintenance Cost	1,50,000
	15,70,000
Increase in Profit before tax **(A – B)**	6,80,000
Less: Tax @ 30%	2,04,000
Profit after tax	**4,76,000**

Cash Inflows = Profit after tax + Depreciation

= 4,76,000 + 4,20,000 = **8,96,000**

(ii) Evaluation of the project by using NPV Method

Year	Cash Inflows	PVAF (12%,5y)	Total PV
1-5	8,96,000	3.605	32,30,080
Less: Total Initial Cash Outflow			21,00,000
Net Present Value			**11,30,080**

Since NPV is positive, therefore, the project is acceptable.

Evaluation of the project by using Profitability Index Method

Profitability Index = Present Value of Cash Inflows/Present Value of Cash Outflows

= 32,30,080/21,00,000

= 1.538

Since, the profitability index is more than 1, the project is acceptable.

Calculation of the Project's Payback*

Year	Net Cash Flow	Cumulative Cash Flow
1	8,96,000	8,96,000
2	8,96,000	17,92,000
3	8,96,000	26,88,000
4	8,96,000	35,84,000
5	8,96,000	44,80,000

Here, the payback period is 2 years plus a fraction of the 3rd year

So, payback period = 2 years + 3,08,000/8,96,000

= 2.34 years

* Payback period may also be solved directly as follows: 21,00,000/8,96,000 = 2.34 years

(iii) Project's cash flows and NPV assuming that the book salvage for depreciation purpose is ₹ 2,00,000

Depreciation = (₹ 21,00,000 – 2,00,000)/5 = 3,80,000

Cash Inflows for the years 1 to 5 are:

Savings (calculated as earlier)		22,50,000
Less: Costs		
- Depreciation	3,80,000	
- Additional Specialists cost	10,00,000	
- Maintenance cost	1,50,000	15,30,000
Profit before tax		**7,20,000**
Less: Tax @ 30%		2,16,000
Profit after tax		5,04,000
Cash Inflow (5,04,000 + 3,80,000)		8,84,000

Calculation of NPV

It may be noted that at the end of year 5, the book value of the project would be ₹ 2,00,000 but its realizable value is nil. So, the capital loss of ₹ 2,00,000 will result in tax savings of ₹ 60,000 (i.e., ₹ 2,00,000 x 30%), as the capital loss is available for tax purposes in view of the information given. Therefore, at the end of year 5, there would be an additional inflow of ₹ 60,000. The NPV may now be calculated as follows:

Year	Cash Flow (₹)	PVAF (12%, n)	PV
1-5	8,84,000	3.605	31,86,820
5	60,000	0.567	34,020
PV of inflows			32,20,840
Outflows			21,00,000
NPV			11,20,840

As the NPV of the project is positive, the project is acceptable.

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571 MTP **Old P8 Jul 2021 S-I** Q5 10 marks old scheme

GG Pathology Lab Ltd. is using 2D sonography machine which has reached the end of its useful life. The lab is intending to upgrade along with the technology by investing in 3D sonography machine as per the

choices preferred by the patients. Following new 3D sonography machine of two different brands with same features is available in the market:

Brand	Cost of machine	Life of machine	Maintenance Cost (Rs.)			SLM Depreciation rate
	(Rs.)	(Rs.)	Year 1-5	Year 6-10	Year 11-15	(%)
X	15,00,000	15	50,000	70,000	98,000	6
Y	10,00,000	10	70,000	1,15,000	-	6

Residual Value of machines shall be dropped by 10% and 40% of Purchase price for Brand X and Y respectively in the first year and thereafter shall be depreciated at the rate mentioned above on the original cost.

Alternatively, the machine of Brand Y can also be taken on rent to be returned back to the owner after use on the following terms and conditions:

- Annual Rent shall be paid in the beginning of each year and for first year it shall be Rs. 2,24,000. Annual Rent for the subsequent 4 years shall be Rs. 2,25,000.
- Annual Rent for the final 5 years shall be Rs. 2,70,000.
- The Rent/Agreement can be terminated by GG Labs by making a payment of Rs. 2,20,000 as penalty. This penalty would be reduced by Rs. 22,000 each year of the period of rental agreement.

You are required to:

(i) ADVISE which brand of 3D sonography machine should be acquired assuming that the use of machine shall be continued for a period of 20 years.

(ii) STATE which of the option is most economical if machine is likely to be used for a period of 5 years?

The cost of capital of GG Labs is 12%.

The present value factor of Rs. 1 @ 12% for different years is given as under:

Year	PVF	Year	PVF
1	0.893	9	0.361
2	0.797	10	0.322
3	0.712	11	0.287
4	0.636	12	0.257
5	0.567	13	0.229
6	0.507	14	0.205
7	0.452	15	0.183
8	0.404	16	0.163

ICAI'S ANSWER

Since the life span of each machine is different and time span exceeds the useful lives of each modeI, we shall use Equivalent Annual Cost method to decide which brand should be chosen.

(i) If machine is used for 20 years

(a) Residual value of machine of brand X

= [Rs. 15,00,000 – (1 - 0.10)] - (Rs. 15,00,000 × 0.06 × 14) = Rs. 90,000

(b) Residual value of machine of brand Y

= [Rs. 10,00,000 – (1 - 0.40)] - (Rs. 10,00,000 × 0.06 × 9) = Rs. 60,000

Present Value (PV) of cost if machine of brand X is purchased

Period	Cash Outflow (Rs.)	PVF @ 12%	PV (Rs.)
0	15,00,000	1.000	15,00,000
1-5	50,000	3.605	1,80,250
6-10	70,000	2.046	1,43,220
11-15	98,000	1.161	1,13,778
15	(90,000)	0.183	(16,470)
			19,20,778

PVAF for 1-15 years = 6.812

Equivalent Annual Cost = $\frac{\text{Rs.}19,20,778}{6.812}$ = Rs. 2,81,969.76

Present Value (PV) of cost if machine of brand Y is purchased

Period	Cash Outflow (Rs.)	PVF @ 12%	PV (Rs.)
0	10,00,000	1.000	10,00,000
1-5	70,000	3.605	2,52,350
6-10	1,15,000	2.046	2,35,290
10	(60,000)	0.322	(19,320)
			14,68,320

PVAF for 1-10 years = 5.651

Equivalent Annual Cost = $\frac{\text{Rs.}14,68,320}{5.651}$ = Rs. 2,59,833.66

Present Value (PV) of cost if machine of brand Y is taken on rent

Period	Cash Outflow (Rs.)	PVF @ 12%	PV (Rs.)
0	2,24,000	1.000	2,24,000
1-4	2,25,000	3.038	6,83,550
5-9	2,70,000	2.291	6,18,570
			15,26,120

PVAF for 1-10 years = 5.651

Equivalent Annual Cost = $\frac{\text{Rs.}15,26,120}{5.651}$ = Rs. 2,70,061.94

Decision: Since Equivalent Annual Cash Outflow is least in case of purchase of Machine of brand Y the same should be purchased.

(ii) If machine is used for 5 years

(a) Scrap value of machine of brand X

= [Rs. 15,00,000 – (1 - 0.10)] - (Rs. 15,00,000 × 0.06 × 4) = Rs. 9,90,000

(b) Scrap value of machine of brand Y

= [Rs. 10,00,000 – (1 - 0.40)] - (Rs. 10,00,000 × 0.06 × 4) = Rs. 3,60,000

Present Value (PV) of cost if machine of brand X is purchased

Period	Cash Outflow (Rs.)	PVF @ 12%	PV (Rs.)
0	15,00,000	1.000	15,00,000
1-5	50,000	3.605	1,80,250
5	(9,90,000)	0.567	(5,61,330)
			11,18,920

Present Value (PV) of cost if machine of brand Y is purchased

Period	Cash Outflow (Rs.)	PVF @ 12%	PV (Rs.)
0	10,00,000	1.000	10,00,000
1-5	70,000	3.605	2,52,350
5	(3,60,000)	0.567	(2,04,120)
			10,48,230

Present Value (PV) of cost if machine of brand Y is taken on rent

Period	Cash Outflow (Rs.)	PVF @ 12%	PV (Rs.)
0	2,24,000	1.000	2,24,000
1-4	2,25,000	3.038	6,83,550
5	1,10,000*	0.567	62,370
			9,69,920

* [Rs. 2,20,000 - (Rs. 22,000 × 5) = Rs. 1,10,000]

Decision: Since Cash Outflow is least in case of rent of Machine of brand Y the same should be taken on rent.

PassBook's note: Its requirements (i) and (ii) are answered as one connected solution on the same data and stay one item. Ours: the question is the equivalent annual cost of machines with unequal lives, which the January 2027 Study Material teaches (Investment Decisions, Equivalent Annualized Criterion).

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 4-5 · answer p. 7-8

572 MTP **Old P8 Jul 2021 S-II Q4** 10 marks old scheme

City Clap Ltd. is in the business of providing housekeeping services. There is a proposal before the company to purchase a mechanized cleaning system for a sum of Rs. 40 lakhs. The present system of the company is to use manual labour for the cleaning job. You are provided with the following information:

Proposed Mechanized System:	
Cost of the machine	Rs. 40 lakhs
Life of the machine	7 years
Depreciation (on straight line basis)	15%
Operating cost of mechanized system	Rs. 20 lakhs per annum
Present system (Manual):	
Manual labour	350 persons
Cost of manual labour	Rs. 15,000 per person per annum

The company has an after-tax cost of fund at 10% per annum.

The applicable tax rate is 50%.

PV factor for 7 years at 10% are as follows:

Years	1	2	3	4	5	6	7
P.V. factor	0.909	0.826	0.751	0.683	0.621	0.564	0.513

You are required to DETERMINE whether it is advisable to purchase the mechanized cleaning system. Give your recommendations with workings.

ICAI'S ANSWER

Calculation of NPV

	(Rs.)	(Rs.)
Cost of Manual System (Rs. 15,000 × 350)		52,50,000
Less: Cost of Mechanised System:		
Operating Cost	20,00,000	
Depreciation (Rs. 40,00,000 × 0.15)	6,00,000	26,00,000
Saving per annum		26,50,000
Less: Tax (50%)		13,25,000
Saving after tax		13,25,000
Add: Depreciation		6,00,000
Cash flow per annum		19,25,000
Cumulative PV Factor for 7 years @ 10%		4.867
Present value of cash flow for 7 years		93,68,975
Less: Cost of the Machine		40,00,000
NPV		53,68,975

The mechanized cleaning system should be purchased since NPV is positive by Rs. 53,68,975.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 3 · answer p. 5

573 MTP **Old P8 Nov 2019 S-I** Q2(b) 7 marks old scheme

H Ltd. is considering a new product line to supplement its range of products. It is anticipated that the new product line will involve cash investments of Rs.70,00,000 at time 0 and Rs.1,00,00,000 in year 1. After-tax cash inflows of Rs. 25,00,000 are expected in year 2, Rs.30,00,000 in year 3, Rs.35,00,000 in year 4 and Rs.40,00,000 each year thereafter through year 10. Although the product line might be viable after year 10, the company prefers to be conservative and end all calculations at that time.

(i) If the required rate of return is 15 per cent, FIND OUT the net present value of the project? Is it acceptable?

(ii) COMPUTE NPV if the required rate of return were 10 per cent?

(iii) COMPUTE the internal rate of return?

ICAI'S ANSWER

(i)

Year	Cash flow	Discount Factor (15%)	Present value
	(Rs.)		**(Rs.)**
0	(70,00,000)	1.000	(70,00,000)
1	(1,00,00,000)	0.870	(87,00,000)
2	25,00,000	0.756	18,90,000
3	30,00,000	0.658	19,74,000
4	35,00,000	0.572	20,02,000
5–10	40,00,000	2.163	86,52,000
		Net Present Value	(11,82,000)

As the net present value is negative, the project is unacceptable.

(ii) Similarly, NPV at 10% discount rate can be computed as follows:

Year	Cash flow	Discount Factor (10%)	Present value
	(Rs.)		**(Rs.)**
0	(70,00,000)	1.000	(70,00,000)
1	(1,00,00,000)	0.909	(90,90,000)
2	25,00,000	0.826	20,65,000
3	30,00,000	0.751	22,53,000
4	35,00,000	0.683	23,90,500
5–10	40,00,000	2.974	1,18,96,000
		Net Present Value	25,14,500

Since NPV = Rs.25,14,500 is positive, hence the project would be acceptable.

(iii) $\text{IRR} = \text{LR} + \dfrac{\text{NPVat LR}}{\text{NPVat LR} - \text{NPV at HR}} \times (\text{HR} - \text{LR})$

$= 10\% + \dfrac{\text{Rs.25,14,500}}{\text{Rs.25,14,500} - (-)11,82,000} \times (15\% - 10\%)$

= 10% + 3.4012 or 13.40%

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 2 · answer p. 4

7.4 IRR, MIRR and the NPV-IRR conflict

15 questions · asked in 11 ICAI documents · 4 exam papers

574 EXAM **May 2026** MCQ 8 1 mark MCQ

Details of a project are given below:

Initial outlay	₹ 800 lakhs
NPV at 16% discount rate	₹ 50.71 lakhs
NPV at 20% discount rate	₹ (26.84) lakhs

What will be Internal rate of return (IRR) of the project? If cost of capital is 18.50%, should the project be accepted?

A 16.00%, Yes

B 18.62%, Yes

C 18.00%, Yes

D 18.62%, No

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P6 · May 2026 exam · question p. 3 · answer p. 8

575 EXAM **May 2025** Case scenario · Q3(a)(i) to Q3(a)(v) 6 marks

CASE SCENARIO

Following data are given for a capital project:

Annual interim cash inflows for first two years	₹ 1,00,000
Annual interim cash inflows for next two years	₹ 80,000
Useful life	4 Years
Salvage value at end of the project	₹ 50,000
Internal rate of return	12%
Cost of capital	10%

You are required to calculate the following:

Year	1	2	3	4
$PVIF_{0.09,t}$	0.917	0.842	0.772	0.708
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683
$PVIF_{0.11,t}$	0.901	0.812	0.731	0.659
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636
$PVIF_{0.10,t}$	1.100	1.210	1.331	1.464
$PVIF_{0.12,t}$	1.120	1.254	1.405	1.574

Q3(a)(i) 1 mark

Initial investment

Q3(a)(ii) 1 mark

Net present value

Q3(a)(iii) 1 mark

Profitability Index

Q3(a)(iv) 1 mark

Discounted payback period

Q3(a)(v) 2 marks

MIRR

ICAI'S ANSWERS

Q3(a)(i)

Initial Investment

Using IRR = 12%, the NPV is zero. So, the initial investment is equal to the present value of inflows at 12%.

Year	Cash Inflows (₹)	PVF@12%	Present Value (₹)
1	1,00,000	0.893	89,300
2	1,00,000	0.797	79,700
3	80,000	0.712	56,960
4	80,000	0.636	50,880
4	50,000 (salvage)	0.636	31,800
		Total	**3,08,640**

Total Present Value = ₹ 3,08,640. So, Initial Investment = ₹ 3,08,640.

Q3(a)(ii)

Net Present Value (NPV)

Year	Cash Flows (₹)	PVF@10%	Present Value (₹)
0	(3,08,640)	1	(3,08,640)
1	1,00,000	0.909	90,900
2	1,00,000	0.826	82,600
3	80,000	0.751	60,080
4	80,000	0.683	54,640
4	50,000 (salvage)	0.683	34,150
	Sum of PV of Cash Inflows		3,22,370
NPV	(3,22,370-3,08,640)		**13,730**

Q3(a)(iii)

Profitability Index (PI)

$$\text{PI} = \frac{\text{Sum of PV of Cash Inflows}}{\text{Initial Cash Outflow}} = \frac{₹\ 3{,}22{,}370}{₹\ 3{,}08{,}640} = 1.0445$$

Q3(a)(iv)

Discounted Payback Period:

Year	Cash Flows (₹)	PVF@10%	PV (₹)	Cumulative PV (₹)
1	1,00,000	0.909	90,900	90,900
2	1,00,000	0.826	82,600	1,73,500
3	80,000	0.751	60,080	2,33,580
4	1,30,000 (including salvage)	0.683	88,790	3,22,370

So, discounted payback = 3 years + $\frac{₹\ 3{,}08{,}640 - ₹\ 2{,}33{,}580}{₹\ 88{,}790}$ = 3.845 years OR 3.85 years OR 3 years 10 months 4 days approx.

Q3(a)(v)

Modified Internal Rate of Return (MIRR)

Terminal value of cash inflows compounded to year 4 using 10%:

Year	Cash Inflows (₹)	@ 10% reinvestment rate factor	Present Value (₹)
1	1,00,000	1.331	1,33,100
2	1,00,000	1.210	1,21,000
3	80,000	1.100	88,000
4	80,000	1.000	80,000
4	50,000 (salvage)	1.000	50,000
		Total	**4,72,100**

Total Terminal Value = ₹ 4,72,100

Now, 3,08,640/4,72,100 = 0.6537 or 0.654

If we see 0.654 in year of present value discount table, the rate which is coming is around 11.20%. So, MIRR is around 11.20%.

Alternatively, solution can be presented in following way for MIRR:

Total Return = 4,72,100/3,08,640 = 1.5296

$(1+r)^4 = 1.5296$

MIRR = $(1.5296)^{(1/4)} - 1$ = 11.20% (Approx)

ICAI · P6 · May 2025 exam · question p. 18 · answer p. 19, 20, 20-21

576 EXAM **Old P8 May 2022** Q6(a) 4 marks old scheme

Identify the limitations of Internal Rate of Return.

ICAI'S ANSWER

Limitations of Internal Rate of Return (IRR)

- The calculation **process is tedious** if there is more than one cash outflow interspersed between the cash inflows; there can be multiple IRR, the interpretation of which is difficult.
- The IRR approach **creates a peculiar situation** if we compare two projects with different inflow/outflow patterns.
- It is assumed that under this method all the future cash inflows of a proposal are reinvested at a rate equal to the IRR. **It ignores a firm's ability to re-invest** in portfolio of different rates.
- If mutually exclusive projects are considered as investment options which have considerably different cash outlays. A project with a larger fund commitment but lower IRR contributes more in terms of absolute NPV and increases the shareholders' wealth. In such situation **decisions based only on IRR criterion may not be correct**.

ICAI · Old P8 · May 2022 exam · question p. 16 · answer p. 16

577 EXAM **Old P8 Jan 2021** Q6(c) 2 marks old scheme

Define Internal Rate of Return (IRR)

ICAI'S ANSWER

Internal rate of return: Internal rate of return for an investment proposal is the discount rate that equates the present value of the expected cash inflows with the initial cash outflow.

ICAI · Old P8 · Jan 2021 exam · question p. 13 · answer p. 14

578 RTP **May 2026** Q10(c)

STATE the acceptance rule under Internal Rate of Return (IRR) method of capital budgeting.

ICAI'S ANSWER

Acceptance Rule of IRR method.

The use of IRR, as a criterion to accept capital investment decision involves a comparison of IRR with the required rate of return known as cut-off rate.

If IRR ≥ Cut-off Rate or WACC	Accept the Proposal
If IRR < Cut-off Rate or WACC	Reject the Proposal

ICAI · P6 · RTP May 2026 · question p. 11 · answer p. 29

579 RTP **Jan 2026** Q9

ABC Manufacturing Ltd is a mid–sized engineering and fabrication company that supplies precision components to the automobile and heavy machinery sectors. In order to maintain competitiveness and expand its market share, the company has proposed a new capital investment project involving the purchase and installation of advanced Computer Numerical Control (CNC) machines and automated assembly lines.

The estimated cost of the project is to be financed partly through internal accruals and largely through term loans from XYZ Bank. ABC Manufacturing has approached the bank requesting project financing of the initial outlay, citing strong growth potential in demand for precision-engineered products over the next decade.

Financial Projections Provided by ABC Manufacturing Ltd.

The company's feasibility report outlines the following:

- The simple payback period is estimated at 5 years, assuming cumulative cash inflows cover the initial investment within that timeframe without considering the time value of money.
- The discounted payback period is calculated at 7 years, after discounting projected cash inflows at the weighted average cost of capital (WACC).
- The project's Modified Internal Rate of Return (MIRR) is projected at 10.9%, which is much lower than the incremental borrowing rate.
- Terminal Value of the annual cash inflows reinvested at the rate of 10% is ₹ 11,43,589.
- Cash inflows are expected to be received equally and steadily at the end of every year over the life of the assets. However, the same Cash inflows are expected to be received even after the simple payback period due to efficiency gains, market expansion, and economies of scale.

(Note: Round off Initial investment to nearest Lakhs).

Management's Position

ABC Manufacturing argues that:

1. The project is strategically essential for long-term survival in a competitive industry.
2. The simple payback of 5 years indicates that the investment risk is acceptable.
3. Although the discounted payback of 7 years is relatively long, the project will continue to generate positive inflows 1 year beyond this period.
4. MIRR of 10.9% should be interpreted in light of industry cyclicality, expected market growth, and non-financial benefits such as improved brand reputation and quality standards.

Bank's Concerns

The credit appraisal team at XYZ Bank has raised several concerns:

- Reliance on payback methods alone ignores long-term profitability, and risk factors.
- Industry volatility, competition, and technological obsolescence may affect the reliability of projected cash flows.

The credit team must now prepare a recommendation on whether to finance the project and, if so, under what conditions. Assuming you were the credit analyst, prepare justification report with a reasoned argument based on other financial considerations namely, PI, NPV, IRR & ARR in the following format.

Appraisal technic	Acceptable range	Outcome for this asset	Recommendation
NPV			
MIRR			
Simple payback			
Discount payback			
ARR			
PI			

Note: For ARR, use the formula Profit after tax

Consider the following PVAF:

PVAF @ 9% for 7 years = 5 & PVAF @ 9% for 8 years = 5.535

ICAI'S ANSWER

Under MIRR, if the cash flows are equal over the life of the asset

Initial investment x $(1 + MIRR)^n$ = CFAT x FVAF

Terminal Value (CFAT x FVAF) = ₹ 11,43,589

Initial investment x $(1+10.9\%)^8$ = ₹ 11,43,589

Initial investment = 11,43,589 / $(1+10.9\%)^8$ = 5,00,000 (rounded off to nearest lakh)

Under simple payback period, if the cash flows are equal over the life of the asset

$$\text{Payback Period} = \frac{\text{Total initial capital investment}}{\text{Annual expected after - tax net cash flow}}$$

Initial investment = CFAT x Simple payback period in years

5,00,000 = 5 x CFAT

CFAT = 1,00,000

Under Discounted payback period, if the cash flows are equal over the life of the asset

$$\text{PVAF} = \frac{\text{Total initial capital investment}}{\text{Annual expected after - tax net cash flow}}$$

Initial investment = CFAT x PVAF @ WACC% for 7 years

5,00,000 = 1,00,000 x PVAF @ WACC% for 7 years

PVAF @ WACC% for 7 years = 5

So, WACC = 9%

NPV = Sum of Present Values of Cash inflows – Cost of the Project

= 1,00,000 x 5.535 – 5,00,000 = 53,500

$$\text{Profitability Index} = \frac{\text{Sum of present value of net cash inflow}}{\text{Initial cash outflow}}$$

$$= \frac{₹5,53,500}{₹5,00,000} = 1.107$$

PAT = CFAT – Depreciation

$$= 1,00,000 - \frac{5,00,000}{8} = 37,500$$

$$\text{ARR} = \frac{\text{Profit after Depreciation}}{\text{Investment in the begining of the year}} \text{ x } 100$$

$$= \frac{37,500}{5,00,000} = 7.5\%$$

Appraisal technic	Acceptable range	Outcome for this asset	Recommendation
NPV	> 0	53,500	Accept
MIRR	MIRR > WACC	10.9% > 9%	Accept
Simple payback	Payback < Life	5 < 8	Accept
Discount payback	Payback < Life	7 < 8	Accept
ARR	ARR > WACC	7.5% < 9%	Reject
PI	> 1	1.107	Accept

ICAI · P6 · RTP Jan 2026 · question p. 8-11 · answer p. 21-22

580 MTP **May 2025 S-II** Q3(b) 2 marks

WRITE a short note on internal rate of return.

ICAI'S ANSWER

Internal Rate of Return: It is that rate at which discounted cash inflows are equal to the discounted cash outflows. In other words, it is the rate which discounts the cash flows to zero. It can be stated in the form of a ratio as follows:

$$\frac{\text{Cash inflows}}{\text{Cash Outflows}} = 1$$

This rate is to be found by trial and error method. This rate is used in the evaluation of investment proposals. In this method, the discount rate is not known but the cash outflows and cash inflows are known.

In evaluating investment proposals, internal rate of return is compared with a required rate of return, known as cut-off rate. If it is more than cut-off rate the project is treated as acceptable; otherwise project is rejected.

ICAI · P6 · MTP May 2025 · Series II · question p. 9 · answer p. 15-16

581 MTP **Jan 2025 S-I** Q4(c) 2 marks

DEFINE Modified Internal Rate of Return method.

ICAI'S ANSWER

Modified Internal Rate of Return (MIRR): There are several limitations attached with the concept of the conventional Internal Rate of Return. The MIRR addresses some of these deficiencies. For example, it eliminates multiple IRR rates; it addresses the reinvestment rate issue and produces results, which are consistent with the Net Present Value method.

Under this method, all cash flows, apart from the initial investment, are brought to the terminal value using an appropriate discount rate (usually the cost of capital). This results in a single stream of cash inflow in the terminal year. The MIRR is obtained by assuming a single outflow in the zeroth year and the terminal cash inflow as mentioned above. The discount rate which equates the present value of the terminal cash inflow to the zeroth year outflow is called the MIRR.

ICAI · P6 · MTP Jan 2025 · Series I · question p. 7 · answer p. 12

582 MTP **Old P8 Nov 2023 S-II** Q1(c) 5 marks old scheme

A company proposes to install a machine involving a Capital Cost of ₹72,00,000. The life of the machine is 5 years and its salvage value at the end of the life is nil. The machine will produce the net operating income after depreciation of ₹13,60,000 per annum. The Company's tax rate is 35%.

The Net Present Value factors for 5 years are as under:

Discounting Rate	:	14	15	16	17	18	19
Cumulative factor	:	3.43	3.35	3.27	3.20	3.13	3.06

You are required to COMPUTE the internal rate of return (IRR) of the proposal.

ICAI'S ANSWER

Computation of cash inflow per annum	₹
Net operating income per annum	13,60,000
Less: Tax @ 35%	4,76,000
Profit after tax	8,84,000
Add: Depreciation (₹72,00,000 / 5 years)	14,40,000
Cash inflow	23,24,000

The IRR of the investment can be found as follows:

NPV = – ₹ 72,00,000 + ₹ 23,24,000 ($PVAF_5$, r) = 0

or PVA F_5 r (Cumulative factor) = $\frac{₹72,00,000}{₹23,24,000}$ = 3.09

Computation of Internal Rate of Return (IRR)

Discounting rate	15%	19%
Cumulative factor	3.35	3.06
Total NPV (₹)	77,85,400 (₹23,24,000 × 3.35)	71,11,440 (₹23,24,000 × 3.06)
Internal outlay (₹)	72,00,000	72,00,000
Surplus (Deficit) (₹)	5,85,400	(88,560)

$$IRR = LR + \frac{\text{NPV at LR}}{\text{NPV at LR - NPV at HR}} \times (HR - LR)$$

$$= 15\% + \frac{5,85,400}{5,85,400 - (-88,560)} \times (19\% - 15\%)$$

= 15% +3.47 =18.47%

Note: Lower rate can be 18% or less than 18%. However, there will be no change in the final answer.

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 1 · answer p. 2

583 MTP **Old P8 Nov 2022 S-II** Q6(a) 4 marks old scheme

DISTINGUISH between Net Present Value and Internal Rate of Return.

ICAI'S ANSWER

NPV versus IRR: NPV and IRR methods differ in the sense that the results regarding the choice of an asset under certain circumstances are mutually contradictory under two methods. In case of mutually exclusive investment projects, in certain situations, they may give contradictory results such that if the NPV method finds one proposal acceptable, IRR favours another. The different rankings given by the NPV and IRR methods could be due to size disparity problem, time disparity problem and unequal expected lives.

The net present value is expressed in financial values whereas internal rate of return (IRR) is expressed in percentage terms.

In the net present value cash flows are assumed to be re-invested at cost of capital rate. In IRR reinvestment is assumed to be made at IRR rates.

ICAI · Old P8 · MTP Nov 2022 · Series II · question p. 4 · answer p. 9-10

584 MTP **Old P8 May 2020 S-I** Q1(c) 5 marks old scheme

A company proposes to install a machine involving a Capital Cost of Rs.72,00,000. The life of the machine is 5 years and its salvage value at the end of the life is nil. The machine will produce the net operating income after depreciation of Rs.13,60,000 per annum. The Company's tax rate is 35%.

The Net Present Value factors for 5 years are as under:

Discounting Rate	:	14	15	16	17	18	19
Cumulative factor	:	3.43	3.35	3.27	3.20	3.13	3.06

You are required to COMPUTE the internal rate of return (IRR) of the proposal.

ICAI'S ANSWER

Computation of cash inflow per annum	Rs.
Net operating income per annum	13,60,000
Less: Tax @ 35%	4,76,000
Profit after tax	8,84,000
Add: Depreciation (Rs.72,00,000 / 5 years)	14,40,000
Cash inflow	23,24,000

The IRR of the investment can be found as follows:

$NPV = -$ Rs. 72,00,000 + Rs. 23,24,000 $(PVAF_5, r) = 0$

or PVA F_5 r (Cumulative factor) $= \frac{Rs.72,00,000}{Rs.23,24,000} = 3.09$

Computation of Internal Rate of Return (IRR)

Discounting rate	15%	19%
Cumulative factor	3.35	3.06
Total NPV (Rs.)	77,85,400 (Rs.23,24,000 × 3.35)	71,11,440 (Rs.23,24,000 × 3.06)
Internal outlay (Rs.)	72,00,000	72,00,000
Surplus (Deficit) (Rs.)	5,85,400	(88,560)

$$IRR = LR + \frac{NPV \text{ at LR}}{NPV \text{ at LR - NPV at HR}} \times (HR - LR)$$

$$= 15\% + \frac{5,85,400}{5,85,400 - (-88,560)} \times (19\% - 15\%)$$

$= 15\% + 3.47 = 18.47\%$

ICAI · Old P8 · MTP May 2020 · Series I · question p. 2 · answer p. 2-3

7.5 Capital rationing

1 question · asked in 1 ICAI document · 1 exam paper

585 EXAM **Old P8 Nov 2019** Q1(d) 5 marks old scheme

A company has ₹ 1,00,000 available for investment and has identified the following four investments in which to invest.

Project	Investment (₹)	NPV (₹)
C	40,000	20,000
D	1,00,000	35,000
E	50,000	24,000
F	60,000	18,000

You are required to optimize the returns from a package of projects within the capital spending limit if-

(i) The projects are independent of each other and are divisible.

(ii) The projects are not divisible.

ICAI'S ANSWER

(i) Optimizing returns when projects are independent and divisible.

Computation of NPVs per Re. 1 of Investment and Ranking of the Projects

Project	Investment (₹)	NPV (₹)	NPV per Re. 1 invested (₹)	Ranking
C	40,000	20,000	0.50	1
D	1,00,000	35,000	0.35	3
E	50,000	24,000	0.48	2
F	60,000	18,000	0.30	4

Building up of a Package of Projects based on their Rankings

Project	Investment (₹)	NPV (₹)
C	40,000	20,000
E	50,000	24,000
D (1/10th of Project)	10,000	3,500
Total	1,00,000	47,500

The company would be well advised to invest in Projects C, E and D (1/10th) and reject Project F to optimise return within the amount of ₹ 1,00,000 available for investment.

(ii) Optimizing returns when projects are indivisible.

Package of Project	Investment (₹)	Total NPV (₹)
C and E	90,000 (40,000 + 50,000)	44,000 (20,000 + 24,000)
C and F	1,00,000 (40,000 + 60,000)	38,000 (20,000 + 18,000)
Only D	1,00,000	35,000

The company would be well advised to invest in Projects C and E to optimise return within the amount of ₹ 1,00,000 available for investment.

ICAI · Old P8 · Nov 2019 exam · question p. 2 · answer p. 6

CHAPTER 8

Dividend Decision

8.1 Forms of dividend, buy-back and determinants of dividend policy

8 questions · asked in 7 ICAI documents · 3 exam papers

586 EXAM **Jan 2026** MCQ 6 MCQ

Mr. X, a senior executive of BPC Electronics Limited has holding of 42,000 shares of the company. The shares are currently trading in the stock exchange at ₹ 480 per share. Mr. X has a policy of reinvesting the amount of any dividends received back into the shares of the company. BPC Electronics Limited has declared and paid a dividend of ₹ 30 per share.

What is the total number of shares that Mr. X would hold after he reinvests the dividend received in shares of BPC Electronics Limited?

A 45,500 shares

B 44,800 shares

C 45,000 shares

D 44,625 shares

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P6 · Jan 2026 exam · question p. 4 · answer p. 9

587 EXAM **Jan 2026** MCQ 8 MCQ

ZED Ltd. is expecting to pay a dividend of ₹ 50 per share in the next financial year 2025-26. The company's speed of adjustment is 25%, the target pay-out ratio is 40%, and the earnings per share (EPS) for the year amount to ₹ 200.

What was the dividend paid for the financial year 2024-25 using Lintner's Model?

A ₹ 30 per share

B ₹ 40 per share

C ₹ 60 per share

D ₹ 50 per share

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P6 · Jan 2026 exam · question p. 5 · answer p. 9

588 EXAM **Sep 2025** Q4(a) 4 marks

Discuss any 2 advantages and limitations of Stock Splits.

ICAI'S ANSWER

Advantages of Stock Splits

1. It makes the share affordable to small investors.
2. Number of shares may increase the number of shareholders; hence the potential of investment may increase.

Limitations of Stock Splits

1. Additional expenditure needs to be incurred on the process of stock split.
2. Low share price may attract speculators or short term investors, which are generally not preferred by any company

ICAI · P6 · Sep 2025 exam · question p. 18 · answer p. 18

589 EXAM **Old P8 May 2023** Q6(a) 4 marks old scheme

List out the conditions, framed by SEBI, which a company needs to fulfil in order to issue of bonus shares.

ICAI'S ANSWER

To issue Bonus shares, a Company needs to fulfill all the conditions given by Securities Exchange Board of India (SEBI):

(i) As per SEBI, the bonus shares are issued not in lieu of cash dividends.

(ii) A bonus issue should be authorized by Article of Association (AOA) and not to be declared unless all partly paid-up shares have been converted into fully paid-up shares.

(iii) The Company should not have defaulted on re-payment of loan, interest, and any statutory dues.

(iv) Bonus shares are to be issued only from share premium and free reserves and not from capital reserve on account of fixed assets revaluation.

ICAI · Old P8 · May 2023 exam · question p. 20 · answer p. 20-21

590 MTP **Jan 2026 S-II** Q4(c) 2 marks

EXPLAIN any common form of dividend.

ICAI'S ANSWER

Generally, the dividend can be of the following forms (depending upon some factors that will be discussed later):

1. **Cash dividend:** It is the most **common form of dividend**. Cash here means cash, cheque, warrant, demand draft, pay order or directly through Electronic Clearing Service (ECS) but not in kind.
2. **Share repurchases**: A share repurchase is transaction in which company buys back its own shares using corporate cash. This is done by lot of corporates these days.
3. **Stock dividend (Bonus Shares):** It is a **distribution of shares in lieu of cash dividend**. When the company issues new shares to its existing shareholders without any consideration it is called bonus shares.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 8 · answer p. 14

591 MTP **May 2025 S-II** MCQ 6 2 marks MCQ

Mr. A is holding 2,00,000 shares of ABC Ltd. It is presently trading on NSE at cum dividend price of ₹ 100 per share. Mr. A has a policy to re-invest the amount of any dividend received into the shares of ABC Ltd. again. If ABC Ltd. has declared a dividend of ₹ 20 per share, determine the no. of shares that Mr. A would hold after he invests dividend in the shares of ABC Ltd.

A 1,00,000

B 2,50,000

C 2,80,000

D 50,000

ICAI'S ANSWER

(B) 2,50,000 shares

Ex-dividend price is ₹ 80 (100 – 20).

The total amount of dividend received is ₹ 40,00,000 (2,00,000 shares x ₹ 20 per share) which is re-invested at the rate of ₹ 80 per share.

Thus, additional shares purchased would be ₹ 40,00,000/₹ 80 = 50,000 shares

So, Mr. A would now hold 2,50,000 shares i.e. (2,00,000 + 50,000)

ICAI · P6 · MTP May 2025 · Series II · question p. 4-5 · answer p. 4

592 MTP Jan 2025 S-II Q4(c) 2 marks

MENTION any one advantage of stock dividend – to the company as well as to the investor

ICAI'S ANSWER

Advantage to the Company - Stock dividends are suitable in the situation of cash crunch and deficiency faced by the company and suitable when restrictions are imposed by lenders to pay the cash dividend

Advantage to the investor – Improves liquidity in the hands of the investors as bonus shares leads to breaking down of higher priced shares into lower priced shares and hence give a choice to shareholders to sell some of the lower priced shares and get some liquidity

ICAI · P6 · MTP Jan 2025 · Series II · question p. 7 · answer p. 10

593 MTP Sep 2024 S-II Q1(c) 5 marks

Paarath Limited had recently repurchased 20,000 equity shares at a premium of 10% to its prevailing market price. The book value per share (after repurchasing) is ₹ 193.20.

Other Details of the company are as follows:

Earnings of the company (before buyback) = ₹ 18,00,000

Current MPS is ₹ 270 with a P/E Ratio of 18.

CALCULATE the Book Value per share of the company before the re-purchase.

ICAI'S ANSWER

i. No of Eq. Shares (before buyback) = Total Earnings (before buyback)/EPS

= 18,00,000/(270/18)

= 1,20,000 shares

ii. Buyback price = 270 + 10% premium = 297

iii. No of Eq. shares (after buyback) = 1,20,000 (-) 20,000 = 1,00,000 shares

iv. Total Book Value of Equity (after buyback) = 1,00,000 X 193.20 = 1,93,20,000

Now,

Total BV of Eq. (after buyback) = Total BV of Eq.(before buyback) (-) Amt of buyback

1,93,20,000 = x (-) (20,000 X 297)

Therefore x = Total BV (before buyback)

= 2,52,60,000

BV per share (before buyback) = 2,52,60,000 / 1,20,000

= 210.50 per share

ICAI · P6 · MTP Sep 2024 · Series II · question p. 4 · answer p. 4

8.2 Dividend theories: traditional, Walter, Gordon and Modigliani-Miller

79 questions · asked in 60 ICAI documents · 15 exam papers

594 EXAM **May 2026** Case scenario · Q3(a)(i) to Q3(a)(iii) 4 marks

CASE SCENARIO

K Ltd. provides the following information:

Net profit	₹ 50,00,000
10% Preference share capital @ ₹100 each	₹ 2,00,00,000
Number of equity share outstanding	5,00,000
Rate of return on investment	25%
Cost of equity	15%

You are required to:

Q3(a)(i) 2 marks

Calculate EPS and also calculate market price per equity share as per Walter's Model with the retention ratio of 20%.

Q3(a)(ii) 1 mark

If the company aims to keep the market price of its equity share at ₹ 60 each, what retention ratio should be adopted?

Q3(a)(iii) 1 mark

What would be optimum dividend payout ratio as per Gordon's Model?

ICAI'S ANSWERS

ICAI's working for this case

	(₹) in lakhs
Net Profit	50
Less: Preference dividend	20
Earning for equity shareholders	30

Q3(a)(i)

Earning per share = 30,00,000/5,00,000 = ₹ 6.00

Market Price as per Walters model

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$P = \frac{4.8 + \frac{0.25}{0.15}(6 - 4.8)}{0.15}$$

P = ₹ 45.33

Q3(a)(ii)

Retention ratio when market price is ₹ 60 each

Let, the dividend per share be D to get share price of 6O

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$\text{or } 60 = \frac{D + \frac{0.25}{0.15}(6 - D)}{0.15}$$

$$\text{or } 9 = \frac{0.15D + 1.5 - 0.25D}{0.15}$$

or 1.35 = 0.15D +1.5 - 0.25D

or 0.10D = 0.15

D = ₹ 1.50

Retention ratio= 4.5/6 = 0.75 = 75%

Q3(a)(iii)

As per Gordon's model, when r > Ke, optimum dividend payout ratio is 'Zero'.

ICAI · P6 · May 2026 exam · question p. 17 · answer p. 18, 18-19, 19

595 **EXAM** **May 2026** Q4(a) **4 marks**

MM Hypothesis of dividend policy is based on certain assumptions. What are those assumptions without which the hypothesis cannot hold? Explain briefly.

ICAI'S ANSWER

MM hypothesis is based on the following assumptions:

- **Perfect capital markets:** The firm operates in a market in which all investors are rational and information is freely available to all.
- **No taxes:** There are no taxes or no tax discrimination between dividend income and capital appreciation (capital gain). It means there is no difference in taxation of dividend income or capital gain. This assumption is necessary for the universal applicability of the theory, since the tax rates may be different in different countries.
- **Fixed investment policy:** It is necessary to assume that all investment should be financed through equity only, since implication after using debt as a source of finance may be difficult to understand. Further, the impact will be different in different cases.
- **No floatation or transaction cost:** Similarly, these costs may differ from country to country or market to market.
- **Risk of uncertainty does not exist**. Investors are able to forecast future prices and dividend with certainty and one discount rate is appropriate for all securities and all time periods.

ICAI · P6 · May 2026 exam · question p. 21 · answer p. 21-22

596 EXAM Jan 2026 Q1(c) 5 marks

PQR Ltd. is a leading company operating in the coal sector, engaged in the extraction, processing, and distribution of coal for industrial and energy purposes.

The following information is provided by the management of PQR Ltd.

Particulars	
Issued Share Capital	₹ 24,00,000
Dividend Paid	10 per Share
Growth Rate	20% p.a. in perpetuity
Cost of Equity (K_e)	25%

You are required to calculate:

(i) Using Gordon Model:

(a) The current market price of the share.

(b) Number of issued shares, if market price per share (MPS) is equal to face value per share.

(ii) Using Modigliani and Miller (MM) Hypothesis:

(a) The market price of the share at the end of year if dividend is paid. Use the current market price calculated in Gordon model as P_o.

(b) What will be the amount of earnings required for the company so that the MM hypothesis holds true, if the company has made new investments of ₹ 5,00,000 during the period and issued 1,000 new equity shares?

ICAI'S ANSWER

(i) (a) $P_0 = \frac{E_1(1-b)}{K_e - br} = \frac{D_1}{K_e - g}$

$D_0 = 10$

Ke = 25%

g = 20%

$P_0 = \frac{10(1+0.20)}{0.25-0.20} = 240$

(b) Number of Shares = $\frac{24,00,000}{240}$ = 10,000 shares

(ii) (a) As per MM Hypothesis,

$P_o = \frac{P_1 + D_1}{1 + K_e}$

$240 = \frac{P_1 + 12}{1 + 0.25}$

$P_1 = 288$

(b) Calculation of Amount of earning required

V_f or nPo = $\frac{(n+\Delta n)P_1 - I + E}{(1+K_e)}$

Where,

V_f = Value of firm in the beginning of the period

n = Number of shares in the beginning of the period

Δn = Number of shares issued to raise the funds required

I = Amount required for investment

E = Total earnings during the period

$$24,00,000 = \frac{[(10,000 + 1,000) \times ₹\ 288] - ₹\ 5,00,000 + E}{(1 + 0.25)}$$

30,00,000 = 26,68,000 + E

Amount of earnings required (E) = ₹ 3,32,000

ICAI · P6 · Jan 2026 exam · question p. 11 · answer p. 14-15

597 **EXAM** **Sep 2025** Case scenario · Q2(b)(i) to Q2(b)(ii) 3 marks

CASE SCENARIO

Saraswati Ltd. has started its business a year back with a paid-up equity capital of ₹ 50,00,000. The other details are as under:

Earnings of company	₹ 5,00,000
Cost of Capital	₹ 159.09
Market price per share using Gordon's Model	8%
Internal rate of return on investment	12%
Number of shares	50,000

You are required to:

Q2(b)(i) 2 marks

Calculate the Dividend paid per share using Gordon's Model.

Q2(b)(ii) 1 mark

What will be Optimum dividend pay-out ratio according to Gordon's Model when r = K_e?

ICAI'S ANSWERS

Q2(b)(i)

$$P_0 = \frac{E(1-b)}{K_e - br}$$

$$159.09 = \frac{10(1-b)}{0.08 - 0.12b}$$

159.09(0.08–0.12b) = 10(1–b)

12.7272–10 = 19.0908b–10b

2.7272 = 9.0908b

B (retained earning ratio) = 0.30

Dividend payout ratio= (1-b) = 0.70

Dividend per share = EPS x (1–b) = 10×0.70 = ₹ 7

Q2(b)(ii)

When r = K_e, There is no optimum ratio

ICAI · P6 · Sep 2025 exam · question p. 13-14 · answer p. 15

598 EXAM May 2025 Case scenario · Q1(b)(i) to Q1(b)(iii) 5 marks

CASE SCENARIO

Y Ltd. produces energy drinks in different flavours. Due to high demand of its product, the rate of return on its earnings is 25%. Currently, the company retains 60% of its earnings and distributes the rest. The current P/E ratio is 8 and earnings per share is ₹ 10.

According to Gordon's Model:

Q1(b)(i) 2 marks

What will be retention ratio if the company wants to maintain its P/E ratio to 12 in current year, given that the expected rate of return for an investor is 20%?

Q1(b)(ii) 1 mark

What will be the expected price per share after one year if Y Ltd. achieves above-mentioned targeted P/E ratio?

Q1(b)(iii) 2 marks

Will there be any change in retention ratio if the company wants to maintain its P/E ratio to 10 in current year, given that the expected rate of return for an investor is 17.50%?

ICAI'S ANSWERS

ICAI's working for this case

The question can be solved with the help of Justified P/E Ratio linked to Gordon Growth Model.

Q1(b)(i)

Calculation of Retention Ratio

$$P/E = \frac{1-b}{K_e - br}$$

$$12 = \frac{1-b}{0.20 - 0.25b}$$

Or 2.4 – 3b =1-b

Or 2.4 – 1 = 3b – b

Or 2b = 1.4

Or b = 0.70 or 70%

So, Retention Ratio is 70%, if the company wants to maintain its P/E ratio to 12 in the current year.

Q1(b)(ii)

Calculation of Expected Price Per Share

$$P/E = \frac{\text{Market Price Share}}{\text{Earning Per Share}}$$

P/E Ratio is 12 and EPS is ₹ 10

So, Market Price Share = P/E Ratio x Earning Per Share

= 12 x 10 = ₹ 120

Retention Ratio is 70%, when P/E ratio is 12, Return on Earnings is 25%

So, New growth rate = 0.70 X 0.25 = 0.175 = 17.5%

Expected price after one year = ₹ 120 X (1+17.5%) = ₹ 141

Therefore, the expected price per share after one year is ₹ 141 if Y Ltd. achieves the targeted P/E ratio of 12.

Q1(b)(iii)

Calculation of change in Retention Ratio

$$P/E = \frac{1-b}{K_e - br}$$

$$10 = \frac{1-b}{0.175 - 0.25b}$$

1.75 – 2.5b = 1-b

1.75 – 1 = 2.5b – b

1.5 b = 0.75

$$b = \frac{0.75}{1.5} = 0.5 \text{ or } 50\%$$

So, Retention Ratio is 50%, if the company wants to maintain its P/E ratio to 10 in the current year.

ICAI · P6 · May 2025 exam · question p. 9 · answer p. 11-12, 12

599 EXAM May 2025 Q2(b) 2 marks

Following details are related to H Ltd.:

EPS	₹ 3.00
Return on investment	20%
Cost of equity	15%

As per Walter's Model, what would be the maximum and minimum price of share?

ICAI'S ANSWER

In this question, since r>Ke, as per the Walter's Model, the price will be maximum when the company retains all the earnings. Detailed calculations are as follows:

(i) Maximum Price when company retains all the earnings i.e. D = 0

$$Po = \frac{D + \frac{r}{Ke}(E-D)}{K_e}$$

$$= \frac{0 + \frac{0.20}{0.15}(3-0)}{0.15} = \frac{(1.333)\text{x } 3}{0.15} = \frac{4}{0.15}$$

= ₹ 26.67

(ii) Minimum Price is when company distributes all its earnings

$$Po = \frac{3 + \frac{0.20}{0.15}(3-3)}{0.15} = \frac{3+0}{0.15} = \frac{3}{0.15} = ₹\ 20$$

ICAI · P6 · May 2025 exam · question p. 15 · answer p. 17-18

600 EXAM Jan 2025 Case scenario · Q1(c)(i) to Q1(c)(ii) 5 marks

CASE SCENARIO

Following information have been provided by LP Ltd.:

Profit before Tax	₹ 40 Lakh
Tax Rate	30%
Equity Share Capital (₹ 10)	₹ 40 Lakh
Return on Investment	18%
Cost of Equity	15%
Dividend Payout Ratio	50%

You are required:

Q1(c)(i) 3 marks

to determine the price of Equity Share of the company as per Walter's Model;

Q1(c)(ii) 2 marks

to determine the Dividend Pay-out Ratio by applying Walter's Model assuming the price of equity share of the company is ₹ 48.

ICAI'S ANSWERS

Q1(c)(i)

Price per share as per Walter's Model

	₹ in lakhs
Profit before tax	40
Less: tax @ 30%	12
Earning for equity shareholders	28
Earning per share	28/4 = ₹ 7

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where,

P = Market Price of the share.

E = Earnings per share. = ₹ 7

D = Dividend per share. = ₹ 3.5

K_e = Cost of equity/ rate of capitalization/ discount rate. = 15%

r = Internal rate of return/ return on investment = 18%

Applying the above formula, price per share

$$P = \frac{3.5 + \frac{0.18}{0.15}(7 - 3.5)}{0.15}$$

Or, P = $\frac{3.5 + 4.2}{0.15}$ = ₹ 51.33

Q1(c)(ii)

Let, the dividend per share be D to get share price of ₹ 48

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$₹\ 48 = \frac{D + \frac{0.18}{0.15}(7 - D)}{0.15}$$

$$7.2 = \frac{0.15D + 1.26 - 0.18D}{0.15}$$

0.03D = 1.26-1.08

D = ₹ 6

$$\text{D/P ratio} = \frac{\text{DPS}}{\text{EPS}} \times 100 = \frac{6}{7} \times 100 = 85.714\%$$

ICAI · P6 · Jan 2025 exam · question p. 10 · answer p. 12-13, 13

601 EXAM **Sep 2024** Q4(b) 4 marks

List any four assumptions of Gorden's Model.

ICAI'S ANSWER

Gordon's model is based on the following assumptions:

- Firm is an all equity firm i.e. no debt.
- IRR will remain constant, because change in IRR will change the growth rate and consequently the value will be affected. Hence this assumption is necessary.
- K_e will remain constant, because change in discount rate will affect the present value.
- Retention ratio (b), once decide upon, is constant i.e. constant dividend payout ratio will be followed.
- Growth rate (g = br) is also constant, since retention ratio and IRR will remain unchanged and growth, which is the function of these two variable will remain unaffected.
- K_e> g, this assumption is necessary and based on the principles of series of sum of geometric progression for 'n' number of years.
- All investment proposals of the firm are to be financed through retained earnings only.

ICAI · P6 · Sep 2024 exam · question p. 22 · answer p. 23

602 EXAM **May 2024** Q3(b) 3 marks

Same question, different ICAI answer: #634 MTP May 2026 S-II

Vista Limited's retained earnings per share for the year ending 31.03.2023 being 40% is ₹ 3.60 per share. Company is foreseeing a growth rate of 10% per annum in the next two years. After that the growth rate is expected to stabilize at 8% per annum. Company will maintain its existing pay-out ratio. If the investor's required rate of return is 15%, Calculate the intrinsic value per share as of date using Dividend Discount model.

ICAI'S ANSWER

As per Dividend discount model, the price of share is calculated as follows:

Retained earning per share = ₹ 3.60

Dividend per share, D_0 = $\frac{₹3.60}{40\%}$ x 60% = ₹5.40

$$P = \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} + \frac{D_3}{(K_e - g)} \times \frac{1}{(1+K_e)^2}$$

Where,

P = Price per share

K_e = Required rate of return on equity

g = Growth rate

$$P = \frac{5.4x1.1}{(1+0.15)^1} + \frac{5.94x1.1}{(1+0.15)^2} + \frac{6.534x1.08}{(0.15-0.08)} \times \frac{1}{(1+0.15)^2}$$

P= 5.17 + 4.94 + 76.23 = ₹86.33

Intrinsic value of share is ₹ 86.33

ICAI · P6 · May 2024 exam · question p. 15 · answer p. 19

603 EXAM Old P8 Nov 2023 Q1(c)(i) 2 marks old scheme

EPS of a company is ₹ 60 and Dividend payout ratio is 60%. Multiplier is 5. Determine price per share as per Graham & Dodd model.

ICAI'S ANSWER

Price per share (P) = $m\left(D + \frac{E}{3}\right)$

Where,

m = Multiplier

D = Dividend

E = EPS

P = $5\left(60 \times 0.6 + \frac{60}{3}\right)$

P = 5(36 +20) = ₹ 280

ICAI · Old P8 · Nov 2023 exam · question p. 2 · answer p. 4-5

604 EXAM Old P8 Nov 2023 Q1(c)(ii) 3 marks old scheme

Last year's dividend is ₹ 6.34, adjustment factor is 45%, target payout ratio is 60% and current year's EPS is ₹ 12. Compute current year's dividend using Linter's model.

ICAI'S ANSWER

$D_1 = D_0 + [(EPS \times Target\ payout) - D_0] \times Adjustment\ factor$

$D_1 = 6.34 + [(12 \times 60\%) - 6.34] \times 0.45$

$D_1 = 6.34 + 0.387 = ₹ 6.727$

ICAI · Old P8 · Nov 2023 exam · question p. 2 · answer p. 5

605 EXAM Old P8 Nov 2023 Q5(b) 5 marks old scheme

INFO Ltd is a listed company having share capital of ₹ 2400 Crores of ₹ 5 each.

During the year 2022-23

Dividend distributed	1000%
Expected Annual growth rate in dividend	14%
Expected rate of return on its equity capital	18%

Required:

(a) Calculate price of share applying Gordon's growth Model.

(b) What will be the price of share if the Annual growth rate in dividend is only 10%?

(c) According to Gordon's growth Model, if Internal Rate of Return is 25%, then what should be the optimum dividend payout ratio in case of growing stage of company? Comment.

ICAI'S ANSWER

(a) In the present situation, the current MPS is as follows:

$$P = \frac{D_0(1+g)}{K_e - g}$$

Where

P = Market price per share

D0 = current year dividend

g = growth rate of dividends

Ke = cost of equity capital/ expected rate of return

$$P = \frac{50(1+0.14)}{0.18-0.14} = ₹\ 1425$$

(b) The impact of changes in growth rate to 10% on MPS will be as follows:

$$P = \frac{50(1+0.10)}{0.18-0.10} = ₹\ 687.5$$

(c) If Internal rate of return, r = 25% and Ke = 18%

As per Gordon's model, when r > Ke, optimum dividend payout ratio is 'Zero'. When IRR is greater than cost of capital, the price per share increases and dividend pay-out decreases.

ICAI · Old P8 · Nov 2023 exam · question p. 15 · answer p. 16-17

606 EXAM Old P8 May 2023 Q1(a) 5 marks old scheme

Following information are given for a company:

Earnings per share	₹ 10
P/E ratio	12.5
Rate of return on investment	12%
Market price per share as per Walter's Model	₹ 130

You are required to calculate:

(i) Dividend payout ratio.

(ii) Market price of share at optimum dividend payout ratio.

(iii) P/E ratio, at which the dividend policy will have no effect on the price of share.

(iv) Market price of share at this P/E ratio.

(v) Market price of share using Dividend growth model.

ICAI'S ANSWER

(i) The EPS of the firm is ₹ 10, r =12%. The P/E Ratio is given at 12.5 and the cost of capital (K_e) may be taken as the inverse of P/E ratio. Therefore, K_e is 8% (i.e., 1/12.5). The value of the share is ₹ 130 which may be equated with Walter Model as follows:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} \text{ or } P = \frac{D + \frac{12\%}{8\%}(10 - D)}{8\%}$$

or [D+1.5(10-D)]/0.08=130

or D+15-1.5D=10.4

or -0.5D=-4.6

So, D = ₹ 9.2

The firm has a dividend pay-out of 92% (i.e., 9.2/10).

(ii) Since the rate of return of the firm (r) is 12% and it is more than the Ke of 8%, therefore, by distributing 92% of earnings, the firm is not following an optimal dividend policy. The optimal dividend policy for the firm would be to pay zero dividend and in such a situation, the market price would be:

$$P = \frac{0 + \frac{12\%}{8\%}(10 - 0)}{8\%}$$

P = ₹ 187.5

So, theoretically the market price of the share can be increased by adopting a zero pay-out.

(iii) The P/E ratio at which the dividend policy will have no effect on the value of the share is such at which the K_e would be equal to the rate of return (r) of the firm. The K_e would be 12% (= r) at the P/E ratio of 1/12%=8.33. Therefore, at the P/E ratio of 8.33, the dividend policy would have no effect on the value of the share.

(iv) If the P/E is 8.33 instead of 12.5, then the K_e which is the inverse of P/E ratio, would be 12% and in such a situation k_e= r and the market price, as per Walter's model would be:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{9.2 + \frac{0.12}{0.12}(10 - 9.2)}{0.12} = ₹ 83.33$$

(v) Dividend Growth Model applying growth on dividend

K_e = 8%, r = 12%, D_0 = 9.2, b = 0.08

g = b.r

g = 0.08 x 0.12=0.96%

$D_1 = D_0$ (1+g) = 9.2 (1+0.0096) = ₹ 9.2883

P = $\frac{D1}{(Ke - g)}$ = 9.2883/(0.08 – 0.0096) = 9.2883/0.0704 = ₹ 131.936

Alternative

Alternatively, without applying growth on dividend

P = $\frac{E(1-b)}{Ke - br} = \frac{10(1-0.08)}{0.08 - (0.08 \times 0.12)}$ = ₹ 130.68

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607 EXAM **Old P8 May 2022** Q6(b) 4 marks old scheme

Briefly explain the assumptions of Walter's Model.

ICAI'S ANSWER

Assumptions of Walter's Model

- All investment proposals of the firm are to be financed **through retained earnings** only.
- **'r'** rate of return & 'K_e' cost of **capital are constant.**
- **Perfect capital markets**: The firm operates in a market in which all investors are rational and information is freely available to all.
- **No taxes or no tax discrimination** between dividend income and capital appreciation (capital gain). It means there is no difference in taxation of dividend income or capital gain. This assumption is necessary for the universal applicability of the theory, since, the tax rates may be different in different countries.
- **No floatation or transaction cost**: Similarly, these costs may differ country to country or market to market.
- The firm has **perpetual life.**

ICAI · Old P8 · May 2022 exam · question p. 16 · answer p. 16-17

608 EXAM **Old P8 Dec 2021** Q1(c) 5 marks old scheme

X Ltd. is a multinational company. Current market price per share is ₹ 2,185. During the F.Y. 2020-21, the company paid ₹ 140 as dividend per share. The company is expected to grow @ 12% p.a. for next four years, then 5% p.a. for an indefinite period. Expected rate of return of shareholders is 18% p.a.

(i) Find out intrinsic value per share.

(ii) State whether shares are overpriced or underpriced.

Year	1	2	3	4	5
Discounting Factor @ 18%	0.847	0.718	0.608	0.515	0.436

ICAI'S ANSWER

As per Dividend discount model, the price of share is calculated as follows:

$$P= \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} + \frac{D_3}{(1+K_e)^3} + \frac{D_4}{(1+K_e)^4} + \frac{D_4(1+g)}{(K_e - g)} \times \frac{1}{(1+K_e)^4}$$

Where,

P = Price per share

K_e = Required rate of return on equity

g = Growth rate

$$P= \frac{₹\ 140 \times 1.12}{(1+0.18)^1} + \frac{₹\ 156.80 \times 1.12}{(1+0.18)^2} + \frac{₹\ 175.62 \times 1.12}{(1+0.18)^3} + \frac{₹\ 196.69 \times 1.12}{(1+0.18)^4} + \frac{₹\ 220.29(1+0.05)}{(0.18-0.05)} \times \frac{1}{(1+0.18)^4}$$

P= 132.81 + 126.10 + 119.59 + 113.45 + 916.34 = ₹ 1,408.29

Intrinsic value of share is ₹ 1,408.29 as compared to latest market price of ₹2,185. **Market price of share is over-priced by ₹ 776.71.**

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609 EXAM **Old P8 Jul 2021** Q1(d) 5 marks old scheme

The following information relates to LMN Ltd.

Earning of the company	₹ 30,00,000
Dividend pay-out ratio	60%
No. of shares outstanding	5,00,000
Rate of return on investment	15%
Equity capitalized rate	13%

Required:

(i) Determine what would be the market value per share as per Walter's model.

(ii) Compute optimum dividend pay-out ratio according to Walter's model and the market value of company's share at that pay-out ratio.

ICAI'S ANSWER

(i) Calculation of market value per share as per Walter's model

$$P = \frac{D + \frac{r}{K_e}(E-D)}{K_e}$$

Where,

P = Market price per share.

E = Earnings per share = ₹ 30,00,000/5,00,000 = ₹ 6

D = Dividend per share = ₹ 6 x 0.60 = ₹ 3.6

r = Return earned on investment = 15%

K_e = Cost of equity capital = 13%

$$P = \frac{3.6 + \frac{0.15}{0.13}(6-3.6)}{0.13} = ₹\ 49$$

(ii) According to Walter's model, when the return on investment (r) is more than the cost of equity capital (K_e), the price per share increases as the dividend pay-out ratio decreases. Hence, the optimum dividend pay-out ratio in this case is nil.

So, at a pay-out ratio of zero, the market value of the company's share will be:

$$P = \frac{0 + \frac{0.15}{0.13}(6 - 0)}{0.13} = ₹\ 53.254$$

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610 **EXAM** **Old P8 Jan 2021** Q1(b) **5 marks** **old scheme**

The following information is taken from ABC Ltd.

Net Profit for the year	₹ 30,00,000
12% Preference share capital	₹ 1,00,00,000
Equity share capital (Share of ₹ 10 each)	₹ 60,00,000
Internal rate of return on investment	22%
Cost of Equity Capital	18%
Retention Ratio	75%

Calculate the market price of the share using:

(1) Gordon's Model

(2) Walter's Model

ICAI'S ANSWER

Market price per share by-

(1) Gordon's Model:

Present market price per share $(P_o)^* = \frac{D_o(1+g)}{K_e - g}$

OR

Present market price per share $(P_o) = \frac{D_1}{K_e - g}$

Where,

P_o = Present market price per share.

g = Growth rate (br) = 0.75 X 0.22 = 0.165

b = Retention ratio (i.e., % of earnings retained)

r = Internal rate of return (IRR)

D_0 = E x (1 – b) = 3 X (1 – 0.75) = 0.75

E = Earnings per share

$P_o = \frac{0.75\,(1 + 0.165)}{0.18 - 0.165} = \frac{0.874}{0.015}$ = **₹ 58.27 approx.**

*Alternatively, P_o can be calculated as $\frac{E(1-b)}{k - br}$ = ₹ 50.

(2) Walter's Model:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$= \frac{0.75 + \frac{0.22}{0.18}(3 - 0.75)}{0.18} = \textbf{₹ 19.44}$$

Workings:

1. Calculation of Earnings per share

Particulars	Amount (₹)
Net Profit for the year	30,00,000
Less: Preference dividend (12% of ₹ 1,00,00,000)	(12,00,000)
Earnings for equity shareholders	18,00,000
No. of equity shares (₹ 60,00,000/₹10)	6,00,000
Therefore, Earnings per share $\left(\frac{\text{Earning for equity shareholders}}{\text{No. of equity shares}}\right)$	₹ 18,00,000/6,00,000 = **₹ 3.00**

2. Calculation of Dividend per share

Particulars	
Earnings per share	₹ 3
Retention Ratio (b)	75%
Dividend pay-out ratio (1-b)	25%
Dividend per share (Earnings per share x Dividend pay-out ratio)	₹ 3 × 0.25 = **₹ 0.75**

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611 EXAM **Old P8 Nov 2020** Q1(c) 5 marks old scheme

The following figures are extracted from the annual report of RJ Ltd.:

Net Profit	₹ 50 Lakhs
Outstanding 13% preference shares	₹ 200 Lakhs
No. of Equity Shares	6 Lakhs
Return on Investment	25%
Cost of Capital (K_e)	15%

You are required to compute the approximate dividend pay-out ratio by keeping the share price at ₹ 40 by using Walter's Model.

ICAI'S ANSWER

Particulars	₹ in lakhs
Net Profit	50
Less: Preference dividend (₹ 200,00,000 × 13%)	26
Earning for equity shareholders	24
Therefore, earning per share = ₹ 24 lakh /6 lakh shares = **₹ 4**	

Let, the dividend per share be D to get share price of ₹ 40

$$P = \frac{D + \frac{r}{Ke}(E - D)}{K_e}$$

$$₹\ 40 = \frac{D + \frac{0.25}{0.15}(₹\ 4 - D)}{0.15}$$

$$6 = \frac{0.15D + 1 - 0.25D}{0.15}$$

$0.1D = 1 - 0.9$

$D = ₹\ 1$

$$\text{D/P ratio} = \frac{\text{DPS}}{\text{EPS}} \times 100 = \frac{₹\ 1}{₹\ 4} \times 100 = \mathbf{25\%}$$

So, the required dividend pay-out ratio will be = 25%

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612 EXAM **Old P8 Nov 2019** Q1(c) 5 marks old scheme

Following figures and information were extracted from the company A Ltd.

Earnings of the company	₹ 10,00,000
Dividend paid	₹ 6,00,000
No. of shares outstanding	2,00,000
Price Earnings Ratio	10
Rate of return on investment	20%

You are required to calculate:

(i) Current Market price of the share

(ii) Capitalisation rate of its risk class

(iii) What should be the optimum pay-out ratio?

(iv) What should be the market price per share at optimal pay-out ratio? (use Walter's Model)

ICAI'S ANSWER

(i) Current Market price of shares (applying Walter's Model)

- The EPS of the firm is ₹ 5 (i.e., Rs 10,00,000 / 2,00,000).

- Rate of return on Investment (r) = 20%.
- The Price Earnings (P/E) Ratio is given as 10, so capitalization rate (K_e), may be taken at the inverse of P/E Ratio. Therefore, K_e is 10% or .10 (i.e., 1/10).
- The firm is distributing total dividends of ₹ 6,00,000 among 2,00,000 shares, giving a dividend per share of ₹ 3.

The value of the share as per Walter's model may be found as follows:

Walter's model is given by-

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where,

P = Market price per share.

E = Earnings per share = ₹ 5

D = Dividend per share = ₹ 3

R = Return earned on investment = 20 %

K_e = Cost of equity capital = 10% or .10

$$P = \frac{3 + \frac{0.20}{0.10}(5 - 3)}{0.10} = ₹\ 70$$

Current Market Price of shares can also be calculated as follows:

$$\text{Price Earnings (P/E) Ratio} = \frac{\text{Market Price of Share}}{\text{Earnings per Shares}}$$

$$\text{Or, } 10 = \frac{\text{Market Price of Share}}{₹10{,}00{,}000/2{,}00{,}000}$$

$$\text{Or, } 10 = \frac{\text{Market Price of Share}}{₹5}$$

Market Price of Share = ₹ 50

(ii) Capitalization rate (K_e) of its risk class is 10% or .10 (i.e., 1/10).

(iii) Optimum dividend pay-out ratio

According to Walter's model when the return on investment is more than the cost of equity capital (10%), the price per share increases as the dividend pay-out ratio decreases. Hence, the optimum dividend pay-out ratio in this case is nil or 0 (zero).

(iv) Market price per share at optimum dividend pay-out ratio

At a pay-out ratio of zero, the market value of the company's share will be:

$$P = \frac{0 + \frac{0.20}{0.10}(5 - 0)}{0.10} = ₹\ 100$$

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613 RTP **Sep 2026** Q6

The stocks of Firms A and B are considered to be equally risky. Investors expect the share of Firm A, which does not intend to pay any dividend, to be worth ₹ 100 one year from now.

Similarly, investors expect a total payoff of ₹ 100 from the share of Firm B after one year, comprising ₹ 10 as dividend income and ₹ 90 as the expected share price.

Dividends are taxed at 25 percent, while capital gains are taxed at 12 percent.

Assuming that each stock is expected to provide a post-tax rate of return of 18 percent and that the radical position holds, determine the current market prices of the shares of Firms A and B.

ICAI'S ANSWER

	A	B
Next year's price	100	90
Dividend	0	10
Let the current price be	A	B
Post-tax capital appreciation	0.88(100-A)	0.88 (90-B)
Post-tax dividend income	0	0.75 × 10
Total return	$\frac{0.88(100\text{-}A)}{A}$ = 18%	$\frac{0.88(90\text{-}B)+7.5}{B}$ =18%
Current price (obtained by solving the preceding equation)	A = ₹ 83.02	B = ₹ 81.79

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614 RTP **May 2026** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

ABC Ltd, a newly incorporated company, commenced its operations in 2024 by issuing equity shares at face value. The company follows a pure equity capital structure and has 10,000 equity shares outstanding. During its first two years of operations, the company demonstrated stable financial performance. It earned a profit of ₹ 1,00,000 in 2024 and ₹ 1,04,000 in 2025, reflecting a consistent annual growth rate of 4%. The company's cost of equity (Ke) is estimated at 8%. Based on the Gordon Growth Model, the market price per share at the end of 2025 has been valued at ₹ 162.24.

Although the company has been following a traditional dividend payout policy, the management is now concerned about whether this policy maximizes shareholders' wealth. Specifically, the Board of Directors wants to assess whether the current dividend payout ratio is appropriate in light of the company's growth prospects and return on retained earnings.

To support this decision, the finance team has been instructed to evaluate the firm's dividend policy using both the Gordon Growth Model and Walter's Model. The objective is to determine the impact of the existing and alternative dividend payout strategies on the company's market price per share, growth rate, capital employed, and overall shareholder wealth maximization.

You are required to answer the following requirements:

Q1(i)

Based on the given information, what is the face value per share and the rate of return on retained earnings (r)?

A ₹ 10 and 8%

B ₹ 100 and 10%

C ₹ 100 and 8%

D ₹ 10 and 10%

Q1(ii)

What is the market price per share in 2025 under Walter's Model?

A ₹ 162.24

B ₹ 150.00

C ₹ 143.00

D ₹ 170.00

Q1(iii)

If the firm's return on retained earnings (10%) is greater than its cost of equity (8%), what is the optimum dividend payout ratio and corresponding market price per share?

A 60% payout; ₹ 143.00

B 40% payout; ₹ 150.00

C 0% payout; ₹ 162.50

D 100% payout; ₹ 130.00

Q1(iv)

What is the capital employed at the beginning of 2026, growth rate and estimated earnings for 2026 if company continue the existing (traditional) dividend payout ratio in 2025?

A ₹ 11,24,864; 4%; ₹ 1,12,486

B ₹ 11,89,760; 10%; ₹ 1,18,976

C ₹ 10,81,600; 4%; ₹ 1,08,160

D ₹ 11,24,864; 10%; ₹ 1,18,976

Q1(v)

What is the capital employed at the beginning of 2026, growth rate and estimated earnings for 2026 if company adopts optimum dividend payout ratio in 2025?

A ₹ 11,24,864; 4%; ₹ 1,12,486

B ₹ 11,89,760; 10%; ₹ 1,18,976

C ₹ 10,40,000; 8%; ₹ 1,04,000

D ₹ 11,44,000, 10%, ₹ 1,14,400

ICAI'S ANSWERS

Q1(i)

(B) ₹ 100 and 10%

Working Note 1: Earnings per Share (EPS) and Growth rate

$$EPS_{2025} = \frac{1,04,000}{10,000} = ₹\ 10.40$$

$$\text{Growth rate (g)} = \frac{1,04,000 - 1,00,000}{1,00,000} = 4\%$$

Working Note 2: Dividend Payout Ratio & Retention Ratio

Using Gordon Growth Model:

$P_0 = \frac{D_1}{Ke\text{-}g}$

$162.24 = \frac{D_1}{0.08\text{-}0.04} \Rightarrow D_1 = 6.4896$

$D_0 = \frac{6.4896}{1.04}$ = ₹ 6.24

Dividend Payout Ratio (DPR)

DPR = $\frac{6.24}{10.40}$ = 60%

Retention Ratio (b)

b = 1-0.60 = 40%

Working Note 3: Rate of Return on Retained Earnings (r)

g = b × r

0.04 = 0.40 × r ⇒ r = 10%

Earnings in 2024 = 1,00,000

Capital employed = $\frac{1,00,000}{10\%}$ = ₹ 10,00,000

Face value per share = $\frac{10,00,000}{10,000}$ = ₹ 100

Q1(ii)

(C) ₹ 143.00

Market Price per Share under Walter's Model (2025)

P = $\frac{D + \frac{r}{Ke}(E - D)}{Ke}$

Where:

E = 10.40

D = 6.24

r = 10%

Ke = 8%

$\frac{r}{Ke}$ = 1.25

E-D = 4.16

P = $\frac{6.24 + (1.25 \times 4.16)}{0.08}$

P = ₹ 143.00

Q1(iii)

(C) 0% payout; ₹ 162.50

Optimum Dividend Policy and Optimum Price (Walter's Model)

When r > Ke, optimum dividend payout ratio = 0%

D = 0

$$P = \frac{\frac{r}{Ke} \times E}{Ke}$$

$$P = \frac{1.25 \times 10.40}{0.08} = ₹\ 162.50$$

Q1(iv)

(C) ₹ 10,81,600; 4%; ₹ 1,08,160

Year	Capital Employed in Beginning (₹)	Retained Earning for the year (₹)
2024	₹ 10,00,000	1,00,000 × 40% = ₹ 40,000
2025	10,00,000 + 40,000 = ₹ 10,40,000	1,04,000 × 40% = ₹ 41,600
2026	10,40,000 + 41,600 = ₹ 10,81,600	-

Earnings 2026 = 10,81,600 x 10% = ₹ 1,08,160

Growth rate = 4%

Q1(v)

(D) ₹ 11,44,000, 10%, ₹ 1,14,400

Revised Growth rate under optimum dividend policy

G = b × r

G = 100 × 10% ⇒ g = 10%

Capital employed (Beginning 2026) = 10,40,000 + (1,04,00 × 100%) = ₹ 11,44,000

Earnings 2026 = 11,44,000 x 10% = ₹ 1,14,400

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615 RTP **Jan 2026** Q10

For the year ending 31.03.2025, ABC Industries Ltd declared 40% dividend. The current market price of the share is quoted at ₹ 57.50. If the retention ratio is reduced by 20 percent, the market price dropped by ₹ 2.50. Face value of the shares is ₹ 10 per share.

1. Find the return on investment and cost of equity using walter's model.
2. Comment on the dividend policy adopted by the company.
3. Find the limiting value of shares using walter's model.
4. Also find the intrinsic value of the shares under Gordon's Model.
5. Mr. X, the chief financial officer of the company, opines that the return on investment of the company is not having direct / indirect effect on the intrinsic value of the shares under Gordon's Model. Only cost of equity will be considered for the computation of the intrinsic value of the shares under Gordon's Model.

ICAI'S ANSWER

1. Price under Walter's Model (P) = $$\frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

	Situation 1	Situation 2
1. DP Ratio	40%	60%
2. Retention Ratio	60%	60% - 20% = 40%
3. Dividend per share = D	₹ 4	₹ 6
4. Retention per share = (E-D)	₹ 6	₹ 4
5. Market price per share = MPS	₹ 57.50	₹ 57.50 – ₹ 2.50 = ₹ 55
6. Equation = Let r/ke be y & Ke be x.	4 + 6y = 57.50x	6 + 4y = 55x
7. Equation for solving	57.50x – 6y = 4	55x – 4y = 6

Equation 1 * 4 = 230x – 24y = 16 .....(1)

Equation 2 * 6 = 330x – 24y = 36 .....(2)

By solving, we get x = 20% and y = 1.25.

y = $\frac{r}{K_e}$ = 1.25

$\frac{r}{20\%}$ = 1.25.

Hence, r = 25%

2. Since r 25% > ke 20%, the company should retain 100%. Since the company pays 40% Dividend, the company is not following optimum dividend policy.

3.

	Situation 1	Situation 2
1. DP Ratio	0%	100%
2. Retention Ratio	100%	0%
3. Dividend per share = D	₹ 0	₹ 10
4. Retention per share = (E-D)	₹ 10	₹ 0
5. Market price per share = MPS	$\frac{10 \text{ x } 1.25}{0.2}$ = 62.50	$\frac{10}{0.2}$ = 50

4. Price under Gordon's Model

Market price per share(P_0) = $\left[\frac{D_0(1+g)}{K_e - g}\right]$

	Situation 1	Situation 2
1. Dividend paid	₹ 4	₹ 6
2. Retention Ratio (b)	60%	40%
3. Return on investment (r)	25%	25%
4. Growth = g = br	15%	10%
5. Market price per share = MPS	$\frac{4(1+0.15)}{0.2-0.15}$ = ₹ 92	$\frac{6(1+0.10)}{0.2-0.1}$ = ₹ 66

5. Since, return on investment is used for computing growth rate, both cost of equity and return on investment are considered for computing price under Gordons model. Hence, the opinion of the chief financial officer is not valid.

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616 RTP **Sep 2025** Q8

In the month of April of the current Financial Year, shares of PQR Ltd. were sold for ₹ 1,570 per share. A long-term earnings growth rate of 8% is anticipated. PQR Ltd. paid dividend of ₹ 25 per share.

(i) CALCULATE rate of return an investor can expect to earn assuming that dividends are expected to grow along with earnings at 8% per year in perpetuity.

(ii) It is expected that PQR Ltd. will earn about 10% on retained earnings and shall retain 60% of earnings. In this case, STATE whether, there would be any change in growth rate and cost of Equity?

ICAI'S ANSWER

(i) According to Dividend Discount Model approach, the firm's expected or required return on equity is computed as follows:

$$K_e = \frac{D_1}{P_0} + g$$

$$K_e = \frac{25\ (1+0.08)}{1{,}570} + 0.08$$

= 0.0172 + 0.08 = 0.0972 or 9.72%

(ii) With rate of return on retained earnings (r) is 10% and retention ratio (b) is 60%, new growth rate will be as follows:

g = br = 0.10 × 0.60 = 0.06

Accordingly, dividend will also get changed and to calculate this, first we shall calculate previous retention ratio (b_1) and then EPS assuming that rate of return on retained earnings (r) is same.

With previous Growth Rate of 8% and r =10%, the retention ratio comes out to be:

0.08 = b_1 × 0.10

b_1 = 0.80. So, payout ratio = 0.20

With 0.20 payout ratio the EPS will be as follows:

$\frac{₹\ 25}{0.20}$ = ₹ 125

With new 0.40 (1 – 0.60) payout ratio, the new dividend will be

D_1 = ₹ 125 × 0.40 = ₹ 50

Accordingly, new K_e will be

$$K_e = \frac{50}{1{,}570} + 0.06$$

or, K_e = 9.18%

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617 RTP **May 2025** Q8

Mr. A had gathered the following information for his analysis –

(A) A Company pays regular dividend on quarterly basis and the last interim dividend declared for the quarter was ₹ 3 per share

(B) Owing to a wide market reach and presence, company's turnover has seen an annual compounded growth of 25% (CAGR) in the last 5 years and the turnover is expected to grow at the same rate in the future as well. The company expects the following Rate of Return (ROI) against the probabilities of likely achievement mentioned along with in different situations.

Scenario	ROI	Probability
I	20%	0.30
II	15%	0.60
III	12%	0.50

(C) The retention ratio over the last 5 years has been 40%, 65%, 50%, 45%, 30% respectively and company plans to retain based on the past average.

(D) The current interest rate on GOI Treasury bond is at 4.5% and the beta of the company is 1.3 and a market return of 12.5%

You are required to CALCULATE the theoretical market price of the company's share for Mr. A's decision-making using Gordon's model and Walter's model.

ICAI'S ANSWER

Calculation of the theoretical price (intrinsic price) denoted by 'Po' using Gordon's formula

$$P_0 = \frac{D_1}{K_e - g}$$

So we need to calculate 3 variables i.e g, D1 & Ke

(A) g = Retention Ratio x ROI

Retention ratio = $\frac{40 + 65 + 50 + 45 + 30}{5}$

= 0.46

Scenario	ROI	Probability	Expected ROI
I	20%	0.30	20 × 0.3 = 6
II	15%	0.60	15 × 0.6 = 9
III	12%	0.50	12 × 0.5 = 6
			Expected ROI = 21%

Therefore g = 0.46 x 21 = 9.66%

(B) D1 = Do + g

Do = ₹ 3 per quarter x 4

= ₹ 12 (Annually)

Therefore D1 = 12 + 9.66%

= ₹ 13.16

(C) Ke will be calculated using CAPM Model and as per CAPM

Ke = Rf + (Rm – Rf) x Beta

= 4.5 + (12.5 – 4.5) x 1.3

= 14.9%

$$P_0 = \frac{13.16}{0.149 - 0.0966}$$

= ₹ 251.15

Calculation of the theoretical price (intrinsic price) denoted by 'Po' using Walter's formula

As per Walter P = $\frac{D + \frac{r}{Ke}(E - D)}{K_e}$

Where 'D' is constant, so no growth would be added

EPS = Dividend / (1 – Retention ratio)

= 12 / (1 – 0.46)

= ₹ 22.22

$$P = \frac{12 + \frac{0.21}{0.149}(22.22 - 12)}{0.149}$$

Po = ₹ 177.21

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618 RTP **Jan 2025** Q2 MCQ

The cost of capital of a firm is 12% & its expected earning per share at the end of the year is ₹ 20. its existing payout ratio is 25%. the company is planning to increase its payout ratio to 50% what will be the effect of this change on the market price of equity share (MPS) of the company as per Gordon model, if the reinvestment rate of the company is 15%?

A It will increase by ₹ 444.45

B It will decrease by ₹ 444.45

C It will increase by ₹ 222.22

D It will decrease by ₹ 222.22

ICAI'S ANSWER

(B) It will decrease by ₹ 444.45

Current D1 = 20 x 25% = 5

Current g = 0.75 x 0.15 = 11.25%

Current MPS = 5/(0.12 - 0.1125) = 666.67

Proposed D1 = 20 x 50% = 10

proposed g = 0.5 x 0.15 = 0.075,

Proposed MPS = 10/(0.12 - 0.075) = 222.22

Change in MPS = 666.67 - 222.22 = ₹444.45

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619 **RTP** **Jan 2025** Q8

The following information is supplied to you:

Particulars	Amount (₹)
Total Earnings	4,50,000
No of Equity Shares (of ₹ 100 each)	25,000 shares
Retention ratio	40%
MPS	198

Applying Walter's Model:

(i) ANALYSE whether the company is following an optimal dividend policy.

(ii) COMPUTE P/E ratio at which the dividend policy will have no effect on the value of the share. Also calculate the MPS at such P/E ratio

(iii) Will your decision change if the P/E ratio is 4.5? ANALYSE.

ICAI'S ANSWER

(i) As per Walter,

If ROI > Ke, firm should retain everything and distribute nothing to maximize the share price. On the contrary, if ROI < Ke, firm should distribute everything and retain nothing to maximize the wealth of the equity owners.

ROI = Total Earnings / Equity Share capital

= 4,50,000 / 25,00,000

ROI = 18%

$$Ke = \frac{1}{PE}$$

P.E Ratio = MPS / EPS = 198 /18 = 11

Therefor Ke = 1/11 = 9.091%

Since ROI > Ke, optimal dividend policy of the firm should be to retain everything and distribute nothing. However, the firm has retained 40% and distributed 60%, hence it is not having an optimal dividend policy as per Walter's model.

(ii) When ROI = Ke, dividend policy of the company will have no effect on the value of the share as per Walter's model

Therefore, in that case, Ke should be equal to 18%

$$\text{P.E Ratio} = \frac{1}{\text{Ke}} = \frac{1}{0.18}$$

P.E Ratio = 5.56 times

MPS at the above P.E Ratio = 18 x 5.56 = ₹ 100.08

(iii) If P.E Ratio is 4.5,

$$\text{Ke} = \frac{1}{4.5} = 22.22\%$$

Since, ROI < Ke, optimal dividend policy of the firm should be to distribute everything and retain nothing, as the value of share would be maximum at that point thereby maximizing the wealth of the shareholder

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620 RTP **Sep 2024** Q9

The following information is taken from Gamma Ltd.

Net Profit for the year	₹ 30,00,000
12% Preference share capital	₹ 1,00,00,000
Equity share capital (Share of ₹ 10 each)	₹ 60,00,000
Internal rate of return on investment	22%
Cost of Equity Capital	18%
Retention Ratio	75%

CALCULATE the market price of the share using:

(1) Gordon's Model

(2) Walter's Model

ICAI'S ANSWER

Market price per share by-

(1) Gordon's Model:

$$\text{Present market price per share } (P_o)^* = \frac{D_o(1+g)}{K_e - g}$$

OR

Present market price per share (P_o) = $\frac{D_1}{K_e - g}$

Where,

P_o = Present market price per share.

g = Growth rate (br) = 0.75 X 0.22 = 0.165

b = Retention ratio (i.e., % of earnings retained)

r = Internal rate of return (IRR)

E = Earnings per share

$P_o = \frac{0.75(1 + 0.165)}{0.18 - 0.165} = \frac{0.874}{0.015}$ = **₹ 58.27 approx.**

(2) Walter's Model:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$= \frac{0.75 + \frac{0.22}{0.18}(3 - 0.75)}{0.18}$$ = **₹ 19.44**

Workings:

1. Calculation of Earnings per share

Particulars	Amount (₹)
Net Profit for the year	30,00,000
Less: Preference dividend (12% of ₹ 1,00,00,000)	(12,00,000)
Earnings for equity shareholders	18,00,000
No. of equity shares (₹ 60,00,000/₹ 10)	6,00,000
Therefore, Earnings per share $\left(\frac{\text{Earning for equity shareholders}}{\text{No. of equity shares}}\right)$	₹ 18,00,000/6,00,000 = **₹ 3.00**

2. Calculation of Dividend per share (D_0)

Particulars	
Earnings per share	₹ 3
Retention Ratio (b)	75%
Dividend pay-out ratio (1-b)	25%
Dividend per share (Earnings per share x Dividend pay-out ratio)	₹ 3 × 0.25 = **₹ 0.75**

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621 RTP **May 2024** Q6

MCO Ltd. has a paid-up share capital of ₹ 10,00,000, face value of ₹ 10 each. The current market price of the shares is ₹20 each. The Board of Directors of the company has an agenda of meeting to pay a

dividend of 25% to its shareholders. The company expects a net income of ₹ 5,20,000 at the end of the current financial year. Company also plans for a capital expenditure for the next financial year for a cost of ₹ 7,50,000, which can be financed through retained earnings and issue of new equity shares.

Company's desired rate of investment is 15%.

Required:

Following the Modigliani- Miller (MM) Hypothesis, DETERMINE value of the company when:

(i) It does not pay dividend and

(ii) It does pay dividend

ICAI'S ANSWER

As per MM Hypothesis, value of firm/ company is calculated as below:

Vf or $nP_0 = \frac{(n+\Delta n)P_1 - I + E}{(1+K_e)}$

Where,

V_f = Value of firm in the beginning of the period

n = number of shares in the beginning of the period

Δn = number of shares issued to raise the funds required

I = Amount required for investment

E = total earnings during the period

(i) **Value of the ZX Ltd. when dividends are not paid.**

$$nP_o = \frac{(n+\Delta n)P_1 - I + E}{1+K_e}$$

$$nP_0 = \frac{\left(1{,}00{,}000 + \frac{2{,}30{,}000}{23}\right) \times ₹23 - ₹7{,}50{,}000 + ₹5{,}20{,}000}{(1+0.15)}$$

$$= \frac{₹25{,}30{,}000 - ₹7{,}50{,}000 + ₹5{,}20{,}000}{(1+0.15)} = ₹20{,}00{,}000$$

Working notes:

1. Price of share at the end of the period (P_1)

$$P_o = \frac{P_1 + D_1}{1+K_e}$$

$20 = \frac{P_1 + 0}{1+0.15}$ or, P_1= ₹ 23

2. Calculation of funds required for investment

Earnings	₹ 5,20,000
Dividend distributed	Nil
Fund available for investment	₹5,20,000
Total Investment	₹7,50,000
Balance Funds required	₹2,30,000

3. Calculation of no. of shares required to be issued for balance fund

No. of shares (Δn) = $\frac{\text{Funds required}}{\text{Price at end}(P_1)} = \frac{2,30,000}{23}$ shares

= 10,000 shares

(ii) **Value of the ZX Ltd. when dividends are paid.**

$$nP_o = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

$$nP_0 = \frac{\left(1,00,000 + \frac{4,80,000}{20.5}\right) \times ₹\ 20.5 - ₹\ 7,50,000 + ₹\ 5,20,000}{(1 + 0.15)}$$

$= \frac{₹25,30,000 - ₹7,50,000 + ₹5,20,000}{(1 + 0.15)}$ = ₹20,00,000

Working notes:

4. Price of share at the end of the period (P_1)

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

$20 = \frac{P_1 + 2.5}{1 + 0.15}$ or, P_1 = ₹ 20.5

5. Calculation of funds required for investment

Earnings	₹ 5,20,000
Dividend distributed	₹ 2,50,000
Fund available for investment	₹ 2,70,000
Total Investment	₹ 7,50,000
Balance Funds required	₹ 4,80,000

6. Calculation of no. of shares required to be issued for balance fund

No. of shares (Δn) = $\frac{\text{Funds required}}{\text{Price at end}(P_1)} = \frac{4,80,000}{20.5}$

= 23,415 shares(approx.)

Note- As per MM-hypothesis of dividend irrelevance, value of firm remains same irrespective of dividend paid. In the solution, there may be variation in value, which is due to rounding off error.

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622 RTP **Old P8 Nov 2023** Q8 old scheme

HM Ltd. is listed on Bombay Stock Exchange which is currently been evaluated by Mr. A on certain parameters.

Mr. A collated following information:

(a) The company generally gives a quarterly interim dividend. ₹ 2.5 per share is the last dividend declared.

(b) The company's sales are growing by 20% on a 5-year Compounded Annual Growth Rate (CAGR) basis, however the company expects following retention amounts against probabilities mentioned as contention is dependent upon cash requirements for the company. Rate of return is 10% generated by the company.

Situation	Prob.	Retention Ratio
A	30%	50%
B	40%	60%
C	30%	50%

(c) The current risk-free rate is 3.75% and with a beta of 1.2 company is having a risk premium of 4.25%.

You are required to help Mr. A in calculating the current market price using Gordon's formula.

ICAI'S ANSWER

Market price using Gordon's formula

$$P_0 = \frac{D_0(1+g)}{k_e - g}$$

D_0 = 2.5×4 = 10 per share (annual)

g = br or retention ratio x rate of return

Calculation of expected retention ratio

Situation	Prob.	Retention Ratio	Expected Retention Ratio
A	30%	50%	0.15
B	40%	60%	0.24
C	30%	50%	0.15
Total			**0.54**

g = 0.54 × 0.10= 0.054 or 5.4%

$$P_0 = \frac{D_0(1+g)}{K_e - g}$$

$$P_0 = \frac{10(1+0.054)}{0.0885 - 0.054} = \frac{10.54}{0.0345} = 305.51$$

K_e = Risk free rate + (Beta x Risk Premium)

= 3.75% + (1.2 x 4.25%) = 8.85%

PassBook's note: Gordon's model and the CAPM cost of equity are in the January 2027 Study Material, so the item is in syllabus. As printed: "which is currently been evaluated", "as contention is dependent upon cash requirements", the first formula's lower-case k_e beside K_e in the second, and the cost of equity worked after the price that uses it.

ICAI · Old P8 · RTP Nov 2023 · question p. 4-5 · answer p. 11-12

623 RTP **Old P8 May 2023** Q7 old scheme

Rambo Limited Has 1,00,000 equity shares outstanding for the year 2022. The current market price of the shares is ₹ 100 each. Company is planning to pay dividend of ₹ 10 per share. Required rate of return is

15%. Based on Modigliani-Miller approach, calculate the market price of the share of the company when the recommended dividend is 1) declared and 2) not declared.

How many new shares are to be issued by the company at the end of the year on the assumption that net income for the year is ₹ 40 Lac and the investment budget is ₹ 50,00,000 when dividend is declared, or dividend is not declared.

PROOF that the market value of the company at the end of the accounting year will remain same whether dividends are distributed or not distributed.

ICAI'S ANSWER

CASE 1: Value of the firm when dividends are not paid.

Step 1: Calculate price at the end of the period

K_e = 15%, Po = ₹100, D1 = 0

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

$$₹100 = \frac{P_1 + 0}{1 + 0.15}$$

P1 = ₹115

Step 2: Calculation of funds required for investment

Earning	₹ 40,00,000
Dividend distributed	Nil
Fund available for investment	₹ 40,00,000
Total Investment	₹ 50,00,000
Balance Funds required	₹ 50,00,000 - ₹ 40,00,000 = ₹ 10,00,000

Step 3: Calculation of No. of shares required to be issued for balance funds

No. of shares = Funds required/P_1

Δn = ₹10,00,000/₹115

Step 4: Calculation of value of firm

nP_o = [(n+Δn)P1-I+E]/(1+Ke)

nPo = [(100000+1000000/₹115) ₹115 - ₹5000000 + ₹4000000]/(1.15)

= ₹1,00,00,000

CASE 2: Value of the firm when dividends are paid.

Step 1: Calculate price at the end of the period

K_e= 15%, Po= ₹100, D1= ₹10

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

$$₹100 = \frac{P_1 + 10}{1 + 0.15}$$

P1 = ₹105

Step 2: Calculation of funds required for investment

Earning	₹ 40,00,000
Dividend distributed	10,00,000
Fund available for investment	₹ 30,00,000
Total Investment	₹ 50,00,000
Balance Funds required	₹ 50,00,000 - ₹ 30,00,000 = ₹ 20,00,000

Step 3: Calculation of No. of shares required to be issued for balance fund

No. of shares = Funds Required/P1

Δn = ₹2000000/₹105

Step 4: Calculation of value of firm

$nP_o = [(n+\Delta n)P1 - I+E]/(1+Ke)$

nPo = [(100000 + 2000000/₹105) ₹105 – ₹5000000 + ₹4000000]/(1.15)= ₹1,00,00,000

Thus, it can be seen from the above calculations that the value of the firm remains the same in either case.

ICAI · Old P8 · RTP May 2023 · question p. 4 · answer p. 16-17

624 RTP **Old P8 Nov 2022** Q7 old scheme

Ordinary shares of a listed company are currently trading at ₹ 10 per share with two lakh shares outstanding. The company anticipates that its earnings for next year will be ₹ 5,00,000. Existing cost of capital for equity shares is 15%. The company has certain investment proposals under discussion which will cause an additional 26,089 ordinary shares to be issued if no dividend is paid or an additional 47,619 ordinary shares to be issued if dividend is paid.

Applying the MM hypothesis on dividend decisions, CALCULATE the amount of investment and dividend that is under consideration by the company.

ICAI'S ANSWER

P_0 = ₹ 10 n = 2,00,000, E = ₹ 5,00,000

K_e = 15%, Δn = 26,089, I = ?

$$P_0 = \frac{P_1}{1 + K_e}$$

$$10 = \frac{P_1}{1.15}$$

$\therefore P_1 = 11.5$

$$\Delta n = \frac{I - E + nD_1}{P_1}$$

$$26{,}089 = \frac{I - 5{,}00{,}000}{11.5}$$

I = 8,00,024

Now,

P_0 = ₹ 10, n = ₹ 2,00,000,

E = ₹ 5,00,000, I = 8,00,024,

K_e = 15%, Δn 47,619, D_1 = ?

$$P_0 = \frac{P_1 + D_1}{1 + K_e}$$

$$10 = \frac{P_1 + D_1}{1.15}$$

$P_1 + D_1 = 11.5$

$\therefore P_1 = 11.5 - D_1$ 1

$$\because \Delta n = \frac{I - E + nD_1}{P_1}$$

$$47{,}619 = \frac{8{,}00{,}024 - 5{,}00{,}000 + 2{,}00{,}000D_1}{P_1}$$

47,619 P_1 = 2,00,000 D_1 + 3,00,024

From 1,

47619 (11.5 – D_1) = 2,00,000 D_1 + 3,00,024

5,47,618.5 – 47,619D_1 = 2,00,000D_1 + 3,00,024

$\therefore$ 2,47,594.5 = 2,00,000D_1 + 47,619 D_1

$\therefore$ 2,47,594.5 = 2,47,619 D_1

$$\therefore D_1 = \frac{2{,}47{,}594.5}{2{,}47{,}619} = 0.99 \approx ₹\ 1$$

$\therefore P_1 = 11.5 - D_1$

$P_1 = 11.5 - 1$

$P_1 = 10.5$

$$\because n.P_0 = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

$$= \frac{(2{,}00{,}000 + 47{,}619)(10.5) - 8{,}00{,}024 + 5{,}00{,}000}{1.15}$$

n.P_0 = ₹19,99,979 ≈ ₹20,00,000

Using direct calculation,

n.P_0 = 2,00,000 ×10 = ₹ 20,00,000

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625 RTP **Old P8 May 2022** Q7 old scheme

The following figures have been collected from the annual report of ABC Ltd. for the current financial

year:

Net Profit	₹ 75 lakhs
Outstanding 12% preference shares	₹ 250 lakhs
No. of equity shares	7.50 lakhs
Return on Investment	20%
Cost of capital i.e. (K_e)	16%

(a) COMPUTE the approximate dividend pay-out ratio so as to keep the share price at ₹ 42 by using Walter's model?

(b) DETERMINE the optimum dividend pay-out ratio and the price of the share at such pay-out.

(c) PROVE that the dividend pay-out ratio as determined above in (b) is optimum by using random pay-out ratio.

ICAI'S ANSWER

(a)

	₹ in lakhs
Net Profit	75
Less: Preference dividend	30
Earning for equity shareholders	45
Earning per share	= 45/7.5 = ₹ 6.00

Let, the dividend per share be D to get share price of ₹ 42

$$P = \frac{D + \frac{r}{Ke}(E - D)}{K_e}$$

$$₹\ 42 = \frac{D + \frac{0.20}{0.16}(6 - D)}{0.16}$$

$$6.72 = \frac{0.16D + 1.2 - 0.20D}{0.16}$$

$0.04D = 1.2 - 1.0752$

$D = 3.12$

$$D/P \text{ ratio} = \frac{DPS}{EPS} \times 100 = \frac{3.12}{6} \times 100 = 52\%$$

So, the required dividend payout ratio will be = 52%

(b) Since r > K_e, the optimum dividend pay-out ratio would 'Zero' (i.e. D = 0),

Accordingly, value of a share:

$$P = \frac{D + \frac{r}{Ke}(E - D)}{K_e}$$

$$P = \frac{0 + \frac{0.20}{0.16}(6 - 0)}{0.16} = ₹ 46.875$$

(c) The optimality of the above pay-out ratio can be proved by using 25%, 50%, 75% and 100% as pay-out ratio:

At 25% pay-out ratio

$$P = \frac{1.5 + \frac{0.20}{0.16}(6 - 1.5)}{0.16} = ₹ 44.531$$

At 50% pay-out ratio

$$P = \frac{3 + \frac{0.20}{0.16}(6 - 3)}{0.16} = ₹ 42.188$$

At 75% pay-out ratio

$$P = \frac{4.5 + \frac{0.20}{0.16}(6 - 4.5)}{0.16} = ₹ 39.844$$

At 100% pay-out ratio

$$P = \frac{6 + \frac{0.20}{0.16}(6 - 6)}{0.16} = ₹ 37.50$$

From the above it can be seen that price of share is maximum when dividend pay-out ratio is 'zero' as determined in (b) above.

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626 RTP Old P8 Dec 2021 Q7 old scheme

Aakash Ltd. has 10 lakh equity shares outstanding at the start of the accounting year 2021. The existing market price per share is ₹ 150. Expected dividend is ₹ 8 per share. The rate of capitalization appropriate to the risk class to which the company belongs is 10%.

(i) CALCULATE the market price per share when expected dividends are: (a) declared, and (b) not declared, based on the Miller – Modigliani approach.

(ii) CALCULATE number of shares to be issued by the company at the end of the accounting year on the assumption that the net income for the year is ₹ 3 crore, investment budget is ₹ 6 crores, when (a) Dividends are declared, and (b) Dividends are not declared.

(iii) PROOF that the market value of the shares at the end of the accounting year will remain unchanged irrespective of whether (a) Dividends are declared, or (ii) Dividends are not declared.

ICAI'S ANSWER

(i) Calculation of market price per share

According to Miller – Modigliani (MM) Approach:

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

Where,

Existing market price (P_o) = ₹ 150

Expected dividend per share (D_1) = ₹ 8

Capitalization rate (k_e) = 0.10

Market price at year end (P_1) = to be determined

(a) If expected dividends are declared, then

$$₹\ 150 = \frac{P_1 + ₹\ 8}{1 + 0.10}$$

$\therefore P_1 = ₹\ 157$

(b) If expected dividends are not declared, then

$$₹\ 150 = \frac{P_1 + 0}{1 + 0.10}$$

$\therefore P_1 = ₹\ 165$

(ii) Calculation of number of shares to be issued

	(a)	(b)
	Dividends are declared (₹ lakh)	**Dividends are not Declared (₹ lakh)**
Net income	300	300
Total dividends	(80)	-
Retained earnings	220	300
Investment budget	600	600
Amount to be raised by new issues	380	300
Relevant market price (₹ per share)	157	165
No. of new shares to be issued (in lakh) (₹ 380 ÷ 157; ₹ 300 ÷ 165)	2.42	1.82

(iii) Calculation of market value of the shares

	(a)	(b)
	Dividends are declared	**Dividends are not Declared**
Existing shares (in lakhs)	10.00	10.00
New shares (in lakhs)	2.42	1.82
Total shares (in lakhs)	12.42	11.82
Market price per share (₹)	157	165
Total market value of shares at the end of the year (₹ in lakh)	12.42 × 157 **= 1,950 (approx.)**	11.82 × 165 **= 1,950 (approx.)**

Hence, it is proved that the total market value of shares remains unchanged irrespective of whether dividends are declared, or not declared.

ICAI · Old P8 · RTP Dec 2021 · question p. 4 · answer p. 13-14

627 RTP Old P8 Jul 2021 Q7 old scheme

Same question, different ICAI answer: #642 MTP May 2025 S-II

The following information is supplied to you:

	(₹)
Total Earnings	2,00,000
No. of equity shares (of ₹ 100 each)	20,000
Dividend paid	1,50,000
Price/ Earnings ratio	12.5

Applying Walter's Model:

(i) ANALYSE whether the company is following an optimal dividend policy.

(ii) COMPUTE P/E ratio at which the dividend policy will have no effect on the value of the share.

(iii) Will your decision change if the P/E ratio is 8 instead of 12.5? ANALYSE.

ICAI'S ANSWER

(i) The EPS of the firm is ₹ 10 (i.e., ₹ 2,00,000/ 20,000) and r = 2,00,000/ (20,000 shares × ₹100) = 10%. The P/E Ratio is given at 12.5 and the cost of capital, Ke, may be taken at the inverse of P/E ratio. Therefore, Ke is 8 (i.e., 1/12.5). The firm is distributing total dividends of ₹ 1,50,000 among 20,000 shares, giving a dividend per share of ₹ 7.50. the value of the share as per Walter's model may be found as follows:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{7.5 + \frac{0.1}{0.08}(10 - 7.5)}{0.08} = ₹\ 132.81$$

The firm has a dividend payout of 75% (i.e., ₹ 1,50,000) out of total earnings of ₹ 2,00,000. Since, the rate of return of the firm, r, is 10% and it is more than the Ke of 8%, therefore, by distributing 75% of earnings, the firm is not following an optimal dividend policy. The optimal dividend policy for the firm would be to pay zero dividend and in such a situation, the market price would be-

$$= \frac{0 + \frac{0.1}{0.08}(10 - 0)}{0.08} = ₹\ 156.25$$

So, theoretically the market price of the share can be increased by adopting a zero payout.

(ii) The P/E ratio at which the dividend policy will have no effect on the value of the share is such at which the Ke would be equal to the rate of return, r, of the firm. The Ke would be 10% (= r) at the P/E ratio of 10. Therefore, at the P/E ratio of 10, the dividend policy would have no effect on the value of the share.

(iii) If the P/E is 8 instead of 12.5, then the Ke which is the inverse of P/E ratio, would be 12.5 and in such a situation ke> r and the market price, as per Walter's model would be:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{7.5 + \frac{0.1}{0.125}(10 - 7.5)}{0.125} = ₹\ 76$$

ICAI · Old P8 · RTP Jul 2021 · question p. 5 · answer p. 16-17

628 RTP Old P8 Nov 2020 Q8 old scheme

The following information is given for QB Ltd.

Earnings per share	₹ 120
Dividend per share	₹ 36
Cost of capital	15%
Internal Rate of Return on investment	20%

CALCULATE the market price per share using

(a) Gordon's formula

(b) Walter's formula

ICAI'S ANSWER

(a) As per **Gordon's Model**, Price per share is computed using the formula:

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

Where,

P_0 = Price per share

E_1 = Earnings per share

b = Retention ratio; (1 - b = Pay-out ratio)

K_e = Cost of capital

r = IRR

br = Growth rate (g)

Applying the above formula, price per share

$$P_0 = \frac{120(1-0.7)}{0.15 - 0.70 \times 0.2} = \frac{36}{0.01} = ₹\ 3{,}600$$

(b) As per Walter's Model, Price per share is computed using the formula:

$$\text{Price (P)} = \frac{D + \frac{r}{Ke}(E - D)}{K_e}$$

Where,

P = Market Price of the share.

E = Earnings per share.

D = Dividend per share.

K_e = Cost of equity/ rate of capitalization/ discount rate.

r = Internal rate of return/ return on investment

Applying the above formula, price per share

$$\text{P} = \frac{36 + \frac{0.20}{0.15}(120 - 36)}{0.15}$$

$$\text{Or, P} = \frac{36 + 112}{0.15} = ₹\ 986.67$$

ICAI · Old P8 · RTP Nov 2020 · question p. 5 · answer p. 15

629 RTP **Old P8 May 2020** Q8 old scheme

Following information relating to Jee Ltd. is given:

Particulars	
Profit after tax	₹ 10,00,000
Dividend pay-out ratio	50%
Number of Equity Shares	50,000
Cost of Equity	10%
Rate of Return on Investment	12%

(i) CALCULATE market value per share as per Walter's Model?

(ii) What is the optimum dividend pay-out ratio according to Walter's Model and Market value of equity share at that pay-out ratio?

ICAI'S ANSWER

(i) Walter's model is given by –

$$P = \frac{D + (E - D)(r/K_e)}{K_e}$$

Where,

P = Market price per share,

E = Earnings per share = ₹ 10,00,000 ÷ 50,000 = ₹ 20

D = Dividend per share = 50% of 20 = ₹ 10

r = Return earned on investment = 12%

K_e = Cost of equity capital = 10%

$$\therefore P = \frac{10 + (20 - 10) \times \frac{0.12}{0.10}}{0.10} = \frac{22}{0.10} = ₹ 220$$

(ii) According to Walter's model when the return on investment is more than the cost of equity capital, the price per share increases as the dividend pay-out ratio decreases. Hence, the optimum dividend pay-out ratio in this case is Nil. So, at a pay-out ratio of zero, the market value of the company's share will be:

$$\frac{0 + (20 - 0) \times \frac{0.12}{0.10}}{0.10} = \frac{24}{0.10} = ₹ 240$$

ICAI · Old P8 · RTP May 2020 · question p. 5 · answer p. 12

630 RTP **Old P8 Nov 2019** Q7 old scheme

The following information pertains to SD Ltd.

Earnings of the Company	₹ 50,00,000
Dividend Payout ratio	60%
No. of shares outstanding	10,00,000
Equity capitalization rate	12%
Rate of return on investment	15%

(i) COMPUTE the market value per share as per Walter's model?

(ii) COMPUTE the optimum dividend payout ratio according to Walter's model and the market value of Company's share at that payout ratio?

ICAI'S ANSWER

(i) Walter's model is given by

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where

P = Market price per share.

E = Earnings per share = ₹ 5

D = Dividend per share = ₹ 3

R = Return earned on investment = 15%

Ke = Cost of equity capital = 12%

$$P = \frac{3 + \frac{0.15}{0.12}(5 - 3)}{0.12} = ₹\ 45.83$$

(ii) According to Walter's model when the return on investment is more than the cost of equity capital, the price per share increases as the dividend pay-out ratio decreases. Hence, the optimum dividend pay-out ratio in this case is nil.

So, at a pay-out ratio of zero, the market value of the company's share will be:

$$P = \frac{0 + \frac{0.15}{0.12}(5 - 0)}{0.12} = ₹52.08$$

ICAI · Old P8 · RTP Nov 2019 · question p. 4 · answer p. 11

631 MTP **Sep 2026 S-I** Q3(a) 7 marks

X Y Ltd. has a paid-up share capital of ₹ 1,00,00,000, face value of ₹ 100 each. The current market price of the shares is ₹ 100 each. The Board of Directors of the company has an agenda of meeting to pay a dividend of 50% to its shareholders. The company expects a net income of ₹ 75,00,000 at the end of the current financial year. Company also plans for a capital expenditure for the next financial year for a cost of ₹ 95,00,000, which can be financed through retained earnings and issue of new equity shares.

Company's desired rate of investment is 15%.

Required:

Following the Modigliani- Miller (MM) Hypothesis, DETERMINE value of the company when:

(i) It does not pay dividend and

(ii) It does pay dividend

ICAI'S ANSWER

As per MM Hypothesis, value of firm/ company is calculated as below:

V_f or nPo = $\frac{(n+\Delta n)P_1 - I + E}{(1+K_e)}$

Where,

V_f = Value of firm in the beginning of the period

n = number of shares in the beginning of the period

Δn = number of shares issued to raise the funds required

I = Amount required for investment

E = total earnings during the period

(i) Value of the ZX Ltd. when dividends are not paid.

$nP_o = \frac{(n+\Delta n)P_1 - I + E}{1+K_e}$

$nPo = \frac{\left(1{,}00{,}000 + \frac{20{,}00{,}000}{115}\right) \times ₹115 - ₹95{,}00{,}000 + ₹75{,}00{,}000}{(1+0.15)}$

$= \frac{₹1{,}35{,}00{,}000 - ₹95{,}00{,}000 + ₹75{,}00{,}000}{(1+0.15)} = ₹1{,}00{,}00{,}000$

Working notes:

1. Price of share at the end of the period (P_1)

$P_o = \frac{P_1 + D_1}{1+K_e}$

$100 = \frac{P_1 + 0}{1+0.15}$ or, $P_1 = 115$

2. Calculation of funds required for investment

Earnings	₹ 75,00,000
Dividend distributed	Nil
Fund available for investment	₹ 75,00,000
Total Investment	₹ 95,00,000
Balance Funds required	₹ 20,00,000

3. Calculation of no. of shares required to be issued for balance fund

No. of shares (Δn) = $\frac{\text{Funds required}}{\text{Price at end } (P_1)} = \frac{20{,}00{,}000}{115}$ shares

(ii) Value of the ZX Ltd. when dividends are paid.

$$nP_o = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

$$nPo = \frac{\left(1{,}00{,}000 + \frac{70{,}00{,}000}{65}\right) \times ₹65 - ₹95{,}00{,}000 + ₹75{,}00{,}000}{(1 + 0.15)}$$

$$= \frac{₹1{,}35{,}00{,}000 - ₹95{,}00{,}000 + ₹75{,}00{,}000}{(1 + 0.15)} = ₹1{,}00{,}00{,}000$$

Working notes:

4. Price of share at the end of the period (P_1)

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

$$100 = \frac{P_1 + 50}{1 + 0.15} \text{ or, } P_1 = ₹\ 65$$

5. Calculation of funds required for investment

Earnings	₹ 75,00,000
Dividend distributed	₹ 50,00,000
Fund available for investment	₹ 25,00,000
Total Investment	₹ 95,00,000
Balance Funds required	₹ 70,00,000

6. Calculation of no. of shares required to be issued for balance fund

$$\text{No. of shares } (\Delta n) = \frac{\text{Funds required}}{\text{Price at end } (P_1)} = \frac{70{,}00{,}000}{65} = 1{,}07{,}693 \text{ shares (approx.)}$$

Note- *As per MM-hypothesis of dividend irrelevance, value of firm remains same irrespective of dividend paid. In the solution, there may be variation in value, which is due to rounding off error.*

ICAI · P6 · MTP Sep 2026 · Series I · question p. 6 · answer p. 8-10

632 MTP **Sep 2026 S-II** Q1(b) 5 marks

The following information is supplied to you:

Particulars	Amount (₹)
Total Earnings	8,00,000
Equity Shares (of ₹ 100 each)	40,00,000
Dividend Paid	4,80,000
Price / Earnings (P/E) Ratio	10

Applying Walter's Model, answer the following:

(i) ANALYSE whether the company is following an optimal dividend policy.

(ii) COMPUTE the P/E ratio at which the dividend policy will have no effect on the value of the share.

(iii) Will your decision change if the P/E ratio is 4 instead of 10? ANALYSE.

ICAI'S ANSWER

Number of equity shares (₹ 40,00,000 ÷ ₹ 100)	40,000 shares
Earnings per share, EPS (₹ 8,00,000 ÷ 40,000)	₹ 20.00
Dividend per share, DPS (₹ 4,80,000 ÷ 40,000)	₹ 12.00
Rate of return, r (₹ 8,00,000 ÷ ₹ 40,00,000)	20%
Cost of equity, Ke (inverse of P/E ratio = 1 ÷ 10)	10%

(i) Is the Dividend Policy Optimal?

Walter's Model values a share as:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Substituting the values (D = ₹ 12, E = ₹ 20, r = 20%, Ke = 10%):

$$P = \frac{12 + \frac{0.20}{0.10}(20 - 12)}{0.10} = ₹\ 280.00$$

Since the firm's rate of return, r (20%), is greater than its cost of equity, Ke (10%), it is a growth firm. For a growth firm, Walter's Model prescribes a zero payout as the optimal policy. By retaining 100% of earnings, the value of the share would be:

$$P = \frac{0 + \frac{0.20}{0.10}(20 - 0)}{0.10} = ₹\ 400.00$$

Conclusion: Since r > Ke, the company is NOT following an optimal dividend policy. Retaining the entire earnings (zero payout) would raise the share value from ₹ 280 to ₹ 400.

(ii) P/E Ratio at which Dividend Policy Has No Effect

Dividend policy becomes irrelevant to value only when Ke equals r, i.e. Ke = 20%. The corresponding P/E ratio is:

$$P/E = \frac{1}{k_e} = \frac{1}{0.20} = 5$$

Conclusion: At a P/E ratio of 5 (Ke = r = 20%), dividend policy will have no effect on the value of the share.

(iii) Effect of a Lower P/E Ratio (P/E = 4)

If the P/E ratio is 4, the cost of equity becomes:

$$Ke = \frac{1}{4} = 25\%$$

Recomputing the share value with r = 20%, Ke = 25%, D = ₹ 12, E = ₹ 20

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$P = \frac{12 + \frac{0.20}{0.25}(20 - 12)}{0.25} = ₹\ 73.60$$

Here, r (20%) is now less than Ke (25%) the firm has become a declining firm. For a declining firm, Walter's Model prescribes a 100% payout as the optimal policy, the opposite of the conclusion in part

(i).

Conclusion: YES, the decision changes. With P/E = 4 (r < Ke), the optimal policy reverses to full (100%) payout, instead of zero payout.

ICAI · P6 · MTP Sep 2026 · Series II · question p. 5 · answer p. 4-5

633 MTP **May 2026 S-I** MCQ 7 2 marks MCQ

The following information is given for QB Ltd.

Earnings per share	₹ 180
Dividend per share	₹ 45
Cost of capital	17%
Internal Rate of Return on investment	20%

What is the market price per share using Walter's formula:

A ₹ 1,200

B ₹ 1,800

C ₹ 2,250

D ₹ 1,960

ICAI'S ANSWER

(A) ₹ 1,200

As per Walter's Model, Price per share is computed using the formula:

$$\text{Price (P)} = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where,

E = Earnings per share.

D = Dividend per share.

Ke = Cost of equity/ rate of capitalization/ discount rate.

r = Internal rate of return/ return on investment

$$\text{Price P} = \frac{45 + \frac{0.20}{0.17}(180 - 45)}{0.17}$$

$$\text{Or, P} = \frac{45 + 158.82}{0.17} = ₹\ 1{,}200 \text{ (approx.)}$$

ICAI · P6 · MTP May 2026 · Series I · question p. 4 · answer p. 3-4

634 MTP **May 2026 S-II** Q3(b) 3 marks

Same question, different ICAI answer: #602 Exam May 2024

Vista Limited's retained earnings per share for the year ending 31.03.2023 being 40% is ₹ 3.60 per share. Company is foreseeing a growth rate of 10% per annum in the next two years. After that the growth rate is expected to stabilize at 8% per annum. Company will maintain its existing pay-out ratio. If the

investor's required rate of return is 15%, CALCULATE the intrinsic value per share as of date using Dividend Discount model.

ICAI'S ANSWER

As per Dividend discount model, the price of share is calculated as follows:

Retained earning per share = ₹ 3.60

Dividend per share, D_0 = $\frac{₹\ 3.60}{40\%} \times 60\%$ = ₹ 5.40

$$P = \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} + \frac{D_3}{(K_e - g)} + \frac{1}{(1+K_e)^2}$$

Where,

P = Price per share

K_e = Required rate of return on equity

g = Growth rate

$$P = \frac{₹\ 5.4 \times 1.1}{(1+0.15)^1} + \frac{₹\ 5.94 \times 1.1}{(1+0.15)^2} + \frac{₹\ 6.534 \times 1.08}{(0.15-0.08)^1} \times \frac{1}{(1+0.15)^2}$$

P= 5.17 + 4.94 + 76.23 = ₹ 86.33

Intrinsic value of share is ₹ 86.33

ICAI · P6 · MTP May 2026 · Series II · question p. 8 · answer p. 9

635 MTP Jan 2026 S-I Q1(c) 5 marks

Perfact Ltd. has 8 lakhs equity shares outstanding at the beginning of the year. The current market price per share is ₹ 120. The Board of Directors of the company is contemplating ₹ 6.4 per share as dividend. The rate of capitalisation, appropriate to the risk-class to which the company belongs, is 9.6%:

(i) Based on M-M Approach, CALCULATE the market price of the share of the company, when the dividend is – (a) declared; and (b) not declared.

(ii) DETERMINE, how many new shares are to be issued by the company, if the company desires to fund an investment budget of ₹ 3.20 crores by the end of the year assuming net income for the year will be ₹ 1.60 crores?

ICAI'S ANSWER

Modigliani and Miller (M-M) – Dividend Irrelevancy Model:

$$P_0 = \frac{P_1 + D_1}{1 + K_e}$$

Where,

P_o = Existing market price per share i.e. ₹ 120

P_1 = Market price of share at the year-end (to be determined)

D_1 = Contemplated dividend per share i.e. ₹ 6.4

K_e = Capitalisation rate i.e. 9.6%.

(i) (a) Calculation of share price when dividend is declared:

$$P_0 = \frac{P_1 + D_1}{1 + K_e}$$

$$120 = \frac{P_1 + 6.4}{1 + 0.096}$$

$120 \times 1.096 = P_1 + 6.4$

$P_1 = 120 \times 1.096 - 6.4$

$= 125.12$

(b) Calculation of share price when dividend is not declared:

$$P_0 = \frac{P_1 + D_1}{1 + K_e}$$

$$120 = \frac{P_1 + 0}{1 + 0.096}$$

$120 \times 1.096 = P_1 + 0$

$P_1 = 131.52$

(ii) Calculation of No. of shares to be issued:

(₹ in lakhs)

Particulars	If dividend declared	If dividend not declared
Net Income	160	160
Less: Dividend paid	51.20	------
Retained earnings	108.80	160
Investment budget	320	320
Amount to be raised by issue of new shares (i)	211.20	160
Market price per share (ii)	125.12	131.52
No. of new shares to be issued (ii)	1,68,797.95	1,21,654.50
Or say	1,68,798	1,21,655

ICAI · P6 · MTP Jan 2026 · Series I · question p. 5 · answer p. 7-8

636 MTP **Jan 2026 S-I** Q4(c) 2 marks

STATE the Limitations of Walter's Model

ICAI'S ANSWER

Limitations of Walter's Model

1. The formula **does not consider all the factors** affecting dividend policy and share prices. Moreover, determination of market capitalisation rate is difficult.

2. Further, the formula **ignores such factors as taxation,** various legal and contractual obligations, management policy and attitude towards dividend policy and so on.

ICAI · P6 · MTP Jan 2026 · Series I · question p. 7 · answer p. 15

637 MTP **Jan 2026 S-II** MCQ 8 1 mark MCQ

ABC Ltd. has earnings of ₹ 50,00,000 and 10,00,000 equity shares. The company is considering two

dividend policies:

- Policy A: Pay 100% dividend
- Policy B: Retain all earnings

According to MM Proposition I (no taxes), the market value of the firm under both policies will be:

A ₹ 1,00,00,000 for Policy A; ₹ 1,20,00,000 for Policy B

B ₹ 1,00,00,000 for both policies

C ₹ 1,10,00,000 for Policy A; ₹ 1,00,00,000 for Policy B

D ₹ 90,00,000 for Policy A; ₹ 1,10,00,000 for Policy B

ICAI'S ANSWER

(B) ₹ 1,00,00,000 for both policies

Under MM (no taxes), dividend policy does not affect firm value.

ICAI · P6 · MTP Jan 2026 · Series II · question p. 4 · answer p. 4

638 MTP **Sep 2025 S-I** Q1(a) 5 marks

Profitable FMCG company PQR Ltd. is examining its dividend policy considering potential future investment opportunities. The company's objective is to maintain a good financial strategy by balancing market valuation, financing requirements, and shareholder expectations.

Following information has been provided:

Equity Share Capital: ₹ 20,00,000 (₹ 10 per share fully paid)

Reserves and Surplus: ₹ 30,00,000

Earnings After Tax (EAT): ₹ 10,00,000

Return on Investment (r): 15%

Cost of Equity (Ke): 12%

Market Price per Share: ₹80

Number of Equity Shares: 2,00,000

Current Debt-Equity Ratio: 0.5:1

Planned Investment Requirement: ₹12,00,000 (next financial year)

The Board of Directors of the company is considering three dividend payout options. Option A – 100% payout; Option B – 50% payout; and Option C – 0% payout.

Further, because of the market conditions, no fresh equity will be raised. Any financial shortage must be covered by debt, but not more than a 1:1 debt-to-equity ratio. The corporate tax rate is 30%. Interest on debt is 10% per annum.

You are required to:

(i) Calculate the market price per share under each dividend payout option using Walter's Model.

(ii) Evaluate the feasibility of internally funding the ₹ 12,00,000 investment under each payout scenario.

(iii) Determine the revised Debt-Equity Ratio assuming the shortfall is met through borrowing if the company adopts a 50% payout.

(iv) Recommend the most suitable dividend policy.

ICAI'S ANSWER

(i) Market Price Per Share using Walter's Model

Walter's Model Formula:

$$P = \frac{D + (r/Ke) \times (E - D)}{Ke}$$

Option A: 100% payout (D = ₹ 5)

$$P = \frac{[5 + (15\%/12\%) \times (5 - 5)]}{0.12} = ₹\ 41.67$$

Option B: 50% payout (D = ₹ 2.5)

$$P = \frac{[2.5 + (15\%/12\%) \times (5 - 2.5)]}{0.12} = ₹\ 72.92$$

Option C: 0% payout (D = ₹ 0)

$$P = \frac{[0 + (15\%/12\%) \times 5]}{0.12} = ₹\ 104.17$$

Since the company earns more on retained earnings (r > Ke), lower payout leads to higher market price per share.

(ii) Feasibility of Funding ₹ 12,00,000 Investment Internally

Investment Requirement = ₹ 12,00,000

EAT = ₹ 10,00,000

Retained earnings depend on payout ratio.

The present situation can be summarized in the following table:

Option	Retained Earnings (₹)	Shortfall (₹)	Feasibility
100% payout	0	12,00,000	Feasible
50% payout	5,00,000	7,00,000	Feasible
0% payout	10,00,000	2,00,000	Feasible

(iii) Revised Debt-Equity Ratio (if 50% payout is chosen)

Existing Equity = ₹ 50,00,000

Existing Debt = 0.5 × ₹ 50,00,000 = ₹ 25,00,000

New Equity (post-retention): ₹ 55,00,000

New Debt (after ₹ 7,00,000 borrowing): ₹ 32,00,000

Revised debt-equity Ratio = ₹ 32,00,000 / ₹55,00,000 = 0.582

It is well within the 1:1 cap.

(iv) Recommendation

0% payout is the best option as it has the highest market price which increases shareholders wealth.

ICAI · P6 · MTP Sep 2025 · Series I · question p. 4-5 · answer p. 4-5

639 MTP Sep 2025 S-II MCQ 8 1 mark MCQ

XYZ Ltd. has the following data:

Earnings per share (EPS): ₹10

Return on equity (ROE): 18%

Capitalisation rate (Ke): 15%

Dividend payout ratio: 60%

Based on Gordon's Model, what would be the likely effect on the market price per share if the company increases its dividend payout ratio to 80%, assuming all other factors remain constant?

A Market price per share will increase

B Market price per share will decrease

C Market price per share will remain unchanged

D Cannot be determined due to insufficient information

ICAI'S ANSWER

(B) Market price per share will decrease

According to Gordon's Model:

$P = [E \times (1 - b)] / (ke - b \times r)$

Where:

E = EPS = ₹10

b = Retention ratio = 1 – Dividend payout ratio

r = ROE = 18%

ke = Cost of equity = 15%

If payout ratio increases from 60% to 80%, then:

Retention ratio b falls from 40% to 20%

Hence, b × r i.e. growth rate becomes smaller (i.e., 0.2 × 18% = 3.6% instead of 7.2%)

Denominator (ke – b×r) becomes larger so price decreases.

So, increasing payout leads to a decrease in market price, since the firm is reducing investment in high-return (18%) projects while the cost of capital is lower (15%).

ICAI · P6 · MTP Sep 2025 · Series II · question p. 4 · answer p. 4

640 MTP Sep 2025 S-II Q1(a) 5 marks

ABC Ltd. is a listed company operating in the FMCG sector. The company is currently facing a dilemma regarding its dividend policy. The financial controller has been asked to prepare a report for the Board of Directors evaluating whether the company should retain more profits or distribute higher dividends.

The following financial data is available for the year ended 31st March 2025:

Earnings per share (EPS)	:	₹ 10
Dividend per share (proposed)	:	₹6
Number of equity shares	:	10,00,000
Return on investment (r)	:	15%
Cost of equity capital (Ke)	:	12%
Current market price per share	:	₹ 100
Retention ratio	:	40%
Tax rate	:	30%

Corporate bond yield: 8% (for comparison with cost of equity)

Further, the company has an investment proposal requiring ₹ 30,00,000 with an expected return of 14%. If the dividend is increased to ₹ 7/share, the share price is expected to rise to ₹ 104 and if the dividend is reduced to ₹ 4/share, the share price may fall to ₹ 95. The CFO believes that MM hypothesis should be followed, but the Board is inclined towards Walter's model of valuation.

You are required to:

(i) CALCULATE the value of the share using Walter's Model, and comment whether the current dividend policy is optimal.

(ii) Apply Gordon's Model to ESTIMATE the market value of the share.

(iii) HOW by way of MM's Dividend Irrelevance Theory, whether a change in dividend payout (from ₹ 6 to ₹ 7 or ₹4) has any impact on the firm's valuation, assuming no taxes and perfect capital markets.

ICAI'S ANSWER

(i) Calculation of value of the share using Walter's Model

$$P = \frac{\left[D + \frac{r}{ke}(E - D)\right]}{Ke}$$

$$= \frac{\left[6 + \frac{0.15}{0.12}(10 - 6)\right]}{0.12}$$

$$= \frac{[6 + 1.25 \text{ x } 4]}{0.12} = \frac{[6 + 5]}{0.12} = ₹ 91.67$$

Since the market price is ₹100, and Walter's model gives ₹91.67, the current dividend policy is not optimal.

(ii) Calculation of value of the share using Gordon's Model

$$P = \frac{[E(1 - b)]}{(Ke - br)}$$

g = br = 0.4 x 0.15 = 0.06

So, $P = \frac{[10 \text{ x } 0.6]}{(0.12 - 0.06)}$ = ₹100

This supports the current market price. Since the model gives the same value as the market, the current retention and dividend policy appears acceptable.

(iii) MM's Dividend Irrelevance Theory

$$P0 = \frac{[D1 + P1]}{(1 + Ke)}$$

Situation 1:

D = ₹6, P1 = ₹100, Ke = 12%

So, $P0 = \frac{[6 + 100]}{(1.12)}$ = ₹94.64

Situation 2:

D = ₹7, P1 = ₹104, Ke = 12%

$P0 = \frac{[7 + 104]}{(1.12)}$ = ₹99.11

Situation 3:

D = ₹4, P1 = ₹95, Ke = 12%

$P0 = \frac{[4 + 95]}{(1.12)}$ = ₹88.39

So, a change in dividend payout has indeed impacted the firm's valuation.

ICAI · P6 · MTP Sep 2025 · Series II · question p. 4-5 · answer p. 5-6

641 MTP **May 2025 S-I** Q1(a) 5 marks

Same question, different ICAI answer: #656 Old P8 MTP Dec 2021 S-I

M Ltd. belongs to a risk class for which the capitalization rate is 12%. It has 40,000 outstanding shares and the current market price is ₹ 200. It expects a net profit of ₹ 5,00,000 for the year and the Board is considering dividend of ₹ 10 per share.

M Ltd. requires to raise ₹ 10,00,000 for an approved investment expenditure. ILLUSTRATE, how the MM approach affects the value of M Ltd. if dividends are paid or not paid.

ICAI'S ANSWER

Given,

Cost of Equity (Ke)	12%
Number of shares in the beginning (n)	40,000
Current Market Price (P0)	₹200
Net Profit (E)	₹5,00,000
Expected Dividend (D1)	₹10 per share
Investment (I)	₹10,00,000

Situation 1 – When dividends are paid	Situation 2 – When dividends are not paid
(i) $P_0 = \frac{P1 + D1}{1 + Ke}$	(i) $P_0 = \frac{P1 + D1}{1 + Ke}$
$200 = \frac{P1 + 10}{1 + 0.12}$	$200 = \frac{P1 + 0}{1 + 0.12}$
P1 + 10 = 200 × 1.12	P1 + 0 = 200 × 1.12
P1 = 224 – 10 = 214	P1 = 224 – 0 = 224
(ii) Calculation of funds required	(ii) Calculation of funds required
= Total Investment - (Net profit - Dividend)	= Total Investment - (Net profit - Dividend)
= 10,00,000 - (5,00,000 – 4,00,000)	= 10,00,000 – (5,00,000 - 0)
= 9,00,000	= 5,00,000
(iii) No. of shares required to be issued for balance fund	(iii) No. of shares required to be issued for balance fund
No. of shares = $\frac{\text{Funds Required}}{\text{Price at end (P1)}}$	No. of shares = $\frac{\text{Funds Required}}{\text{Price at end (P1)}}$
$\Delta n = \frac{9,00,000}{214}$ = 4,205.61	$\Delta n = \frac{5,00,000}{224}$ = 2,232.14
(iv) Calculation of value of firm	(iv) Calculation of value of firm
$V_f = \frac{(n + \Delta n)P1 - I + E}{1 + Ke}$	$V_f = \frac{(n + \Delta n)P1 - I + E}{1 + Ke}$
$= \frac{(40,000 + \frac{9,00,000}{214})214 - 10,00,000 + 5,00,000}{1 + 0.12}$	$= \frac{(40,000 + \frac{5,00,000}{224})224 - 10,00,000 + 5,00,000}{1 + 0.12}$
$= \frac{94,60,000 - 5,00,000}{1.12}$ = 80,00,000	$= \frac{94,60,000 - 5,00,000}{1.12}$ = 80,00,000

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642 MTP **May 2025 S-II** Q2(a) 4 marks

Same question, different ICAI answer: #627 Old P8 RTP Jul 2021

The following information is supplied to you:

	(₹)
Total Earnings	2,00,000
No. of equity shares (of ₹ 100 each)	20,000
Dividend paid	1,50,000
Price/ Earnings ratio	12.5

Applying Walter's Model:

(i) ANALYSE whether the company is following an optimal dividend policy.

(ii) COMPUTE P/E ratio at which the dividend policy will have no effect on the value of the share.

(iii) Will your decision change if the P/E ratio is 8 instead of 12.5? ANALYSE.

ICAI'S ANSWER

(i) The EPS of the firm is ₹ 10 (i.e., ₹ 2,00,000/ 20,000) and r = 2,00,000/ (20,000 shares × ₹100) = 10%. The P/E Ratio is given at 12.5 and the cost of capital, Ke, may be taken at the inverse of P/E ratio. Therefore, Ke is 8 (i.e., 1/12.5). The firm is distributing total dividends of ₹ 1,50,000 among 20,000 shares, giving a dividend per share of ₹ 7.50. the value of the share as per Walter's model may be found as follows:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{7.5 + \frac{0.1}{0.08}(10 - 7.5)}{0.08} = ₹\ 132.81$$

The firm has a dividend payout of 75% (i.e., ₹ 1,50,000) out of total earnings of ₹ 2,00,000. Since, the rate of return of the firm, r, is 10% and it is more than the Ke of 8%, therefore, by distributing 75% of earnings, the firm is not following an optimal dividend policy. The optimal dividend policy for the firm would be to pay zero dividend and in such a situation, the market price would be-

$$= \frac{0 + \frac{0.1}{0.08}(10 - 0)}{0.08} = ₹\ 156.25$$

So, theoretically the market price of the share can be increased by adopting a zero payout.

(ii) The P/E ratio at which the dividend policy will have no effect on the value of the share is such at which the Ke would be equal to the rate of return, r, of the firm. The Ke would be 10% (= r) at the P/E ratio of 10. Therefore, at the P/E ratio of 10, the dividend policy would have no effect on the value of the share.

(iii) If the P/E is 8 instead of 12.5, then the Ke which is the inverse of P/E ratio, would be 12.5 and in such a situation ke> r and the market price, as per Walter's model would be:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{7.5 + \frac{0.1}{0.125}(10 - 7.5)}{0.125} = ₹\ 76$$

So, the market price will come down to ₹ 76.

ICAI · P6 · MTP May 2025 · Series II · question p. 7 · answer p. 10-11

643 MTP Jan 2025 S-I MCQ 8 1 mark MCQ

Under Modigliani and Miller's Dividend Irrelevance Theory, a company has ₹1,00,000 to distribute. If it chooses to retain the earnings instead of paying dividends, what happens to shareholder wealth?

A Increases due to reinvestment opportunities.

B Decreases due to lower immediate returns.

C Remains unchanged because value depends on earnings and investment policy.

D Depends on the dividend payout ratio

ICAI'S ANSWER

(C) Remains unchanged because value depends on earnings and investment policy.

(Explanation: M&M's theory suggests that dividend policy has no impact on shareholder wealth in a perfect market.)

ICAI · P6 · MTP Jan 2025 · Series I · question p. 3 · answer p. 3

644 MTP **Jan 2025 S-I Q2(b)** 4 marks

Paras TMT Ltd. is a TMT manufacturing company with a face value of ₹ 10 per share.

The following information is given about the company:

- The company is expected to grow @ 10% p.a. for next four years then 5% for an indefinite period.
- Rate of return expected by the shareholders on their share investments is 15%.
- Company paid ₹ 4 as dividend per share for the current Financial Year.

FIND out the intrinsic value per share

ICAI'S ANSWER

As per Dividend discount model, the price of share is calculated as follows:

$$P = \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} + \frac{D_3}{(1+K_e)^3} + \frac{D_4}{(1+K_e)^4} + \frac{D_5}{(K_e\text{-}g)} \times \frac{1}{(1+K_e)^4}$$

Where,

P = Price per share

K_e = Required rate of return on equity

g = Growth rate

Calculation PV of Dividends

Year	Dividend per share	PVF @ 15%	PV
1	4.4	0.870	3.828
2	4.84	0.756	3.660
3	5.324	0.658	3.503
4	5.856	0.572	3.350
Total			14.341

PV of Terminal Value = $\frac{₹5.856 \text{x} 1.05}{(0.15-0.05)^1} \times \frac{1}{(1+0.15)^4}$ = 61.488 x .572 = 35.171

Intrinsic value of share = PV of Dividends + PV of terminal value

= 14.341 + 35.171 = ₹ 49.512

ICAI · P6 · MTP Jan 2025 · Series I · question p. 5-6 · answer p. 7-8

645 MTP **Jan 2025 S-II Q2(b)** 4 marks

Return on Equity (ROE) is Satva Limited is 15% and the capitalization rate applicable to the company is at 20%. Satva Limited's Book Value per share (BVPS) is Rs 125. Calculate the intrinsic value of the share today using Gordon's model and Walter's model if the company's policy is to retain 65% of the earning.

ICAI'S ANSWER

EPS = ROE x BVPS (WN 1)

EPS = 0.15 x 125 = ₹ 18.75

Growth = ROE x Retention Ratio

= 0.15 x 0.65

= 9.75%

D1 = Do (1 + g)

= (18.75 x 35%)(1 + 0.0975)

= ₹ 7.20

Intrinsic Value of share today - Gordon's Formula

$$\text{Po} = \frac{D_1}{Ke - g}$$

$$= \frac{7.20}{0.20 - 0.0975}$$

Po = ₹ 70.24

Intrinsic Value of share today - Walter's Model

$$\text{Po} = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Here D = Do assuming it would remain constant through infinity

$$\text{Po} = \frac{6.5625 + \frac{0.15}{0.20}(18.75 - 6.5625)}{0.20}$$

Po = ₹ 78.51

WN 1 - Relationship between ROE-EPS-BVPS

$$\text{ROE} = \frac{\text{Earnings for Equity Shareholders}}{\text{Equity shareholders funds}}$$

If we divide the numerator and denominator with "**No of equity shares**"

$$\text{ROE} = \frac{\text{Earnings for Equity Shareholders / No of equity shares}}{\text{Equity shareholders funds / No of equity shares}}$$

Therefore, ROE = EPS / BVPS

ICAI · P6 · MTP Jan 2025 · Series II · question p. 6 · answer p. 6-7

646 MTP Sep 2024 S-I Q2(b) 4 marks

The following information is given for QB Ltd.

Earnings per share	₹ 180
Dividend per share	₹ 45
Cost of capital	17%
Internal Rate of Return on investment	20%

CALCULATE the market price per share using -

(a) Gordon's formula

(b) Walter's formula

ICAI'S ANSWER

(i) As per **Gordon's Model**, Price per share is computed using the formula:

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

Where,

P_0 = Price per share

E_1 = Earnings per share

Payout ratio = 45/180 = 25%

b = Retention ratio; (1 - b = Pay-out ratio) = 1-0.25 = 0.75

K_e = Cost of capital

r = IRR

br = Growth rate (g)

Applying the above formula, price per share

$$P_0 = \frac{180(1-0.75)}{0.17 - 0.75 \times 0.2} = \frac{45}{0.02} = ₹\ 2{,}250$$

(ii) As per Walter's Model, Price per share is computed using the formula:

$$\text{Price } \textbf{(P)} = \frac{D + \frac{r}{Ke}(E\text{-}D)}{K_e}$$

Where,

P = Market Price of the share.

E = Earnings per share.

D = Dividend per share.

K_e = Cost of equity/ rate of capitalization/ discount rate.

r = Internal rate of return/ return on investment

Applying the above formula, price per share

$$P = \frac{45 + \frac{0.20}{0.17}(180 - 45)}{0.17}$$

Or, $P = \frac{45 + 158.82}{0.17} = ₹\ 1{,}200$ (approx..)

ICAI · P6 · MTP Sep 2024 · Series I · question p. 5-6 · answer p. 7

647 **MTP** **Sep 2024 S-I** **Q3(b)** **2 marks**

DISCUSS the parameters of Lintner's Model.

ICAI'S ANSWER

Lintner's model has two parameters:

i. The target payout ratio,

ii. The spread at which current dividends adjust to the target.

ICAI · P6 · MTP Sep 2024 · Series I · question p. 6 · answer p. 9

648 MTP **May 2024 S-I** Q1(a) 5 marks

The following figures have been extracted from the annual report of Xee Ltd.:

Net Profit	₹ 54 lakhs
Outstanding 12% preference shares	₹ 200 lakhs
No. of equity shares	2 lakhs
Return on Investment	22%
Cost of capital i.e. (Ke)	15%

COMPUTE the approximate dividend pay-out ratio so as to keep the share price at ₹ 120 by using Walter's model?

(Decimal may be taken up to 2 units)

ICAI'S ANSWER

Particulars	(₹' in lakhs)
Net Profit	54
Less: Preference dividend	24
Earnings for equity shareholders	30
Earnings per share	30/2 = ₹ 15

Let, the dividend per share be D to get share price of ₹ 120.

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where,

P = Market price per share.

E = Earnings per share = ₹ 15

D = Dividend per share

R = Return earned on investment = 22%

Ke = Cost of equity capital = 15%

$$120 = \frac{D + \frac{0.22}{0.15}(15 - D)}{0.15}$$

$$18 = \frac{0.15D + 3.3 - 0.22D}{0.15}$$

$0.07D = 3.3 - 2.7$

$D = 8.57$

$$\text{D/P ratio} = \frac{DPS}{EPS} \times 100 = \frac{8.57}{15} \text{ x } 100 = 57.13\%$$

So, the required dividend pay-out ratio will be = 57.13%

ICAI · P6 · MTP May 2024 · Series I · question p. 3 · answer p. 2-3

649 MTP May 2024 S-II Q1(b) 5 marks

SOC Ltd has 10 lakh equity shares outstanding at the beginning of the accounting year 2024. The existing market price per share is Rs 600. Expected dividend is Rs 40 per share. The rate of capitalization appropriate to the risk class to which the company belongs is 20%.

(i) CALCULATE the market price per share by the end of the year when expected dividends are: (a) declared, and (b) not declared, based on the Miller – Modigliani approach.

(ii) CALCULATE the number of shares to be issued by the company at the end of the accounting year on the assumption that the net income for the year is Rs 15 crore; investment budget is Rs 20 crores, when (a) Dividends are declared, and (b) Dividends are not declared.

(iii) PROVE that the market value of the shares at the end of the accounting year will remain unchanged irrespective of whether (a) Dividends are declared, or (ii) Dividends are not declared.

ICAI'S ANSWER

(i) Calculation of market price per share

According to Miller – Modigliani (MM) Approach:

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

Where,

Existing market price (P_o) = ₹ 600

Expected dividend per share (D_1) = ₹ 40

Capitalization rate (k_e) = 0.20

Market price at year end (P_1) = ?

a. If expected dividends are declared, then

600=(P1+40)/(1+0.2)

600 x 1.2 = P1+40

P1 = 680

b. If expected dividends are not declared, then

600 = (P1+0)/(1 + 0.2)

600 x 1.2 = P1

P1= 720

(ii) Calculation of number of shares to be issued

	(a)	(b)
	Dividends are declared (₹ lakh)	**Dividends are not Declared (₹ lakh)**
Net income	1500	1500
Total dividends	(400)	-
Retained earnings	1100	1500

	(a)	(b)
Investment budget	2000	2000
Amount to be raised by new issues	900	500
Relevant market price (₹ per share)	680	720
No. of new shares to be issued (in lakh) (₹ 900 ÷ 680; ₹ 500 ÷ 720)	1.3235	0.6944

(iii) Calculation of market value of the shares

	(a)	(b)
Particulars	**Dividends are declared**	**Dividends are not Declared**
Existing shares (in lakhs)	10.00	10.00
New shares (in lakhs)	1.3235	0.6944
Total shares (in lakhs)	11.3235	10.6944
Market price per share (₹)	680	720
Total market value of shares at the end of the year (₹ in lakh)	11.3235 × 680 = 7,700 (approx.)	10.6944 × 720 = 7,700 (approx.)

Hence, it is proved that the total market value of shares remains unchanged irrespective of whether dividends are declared, or not declared.

ICAI · P6 · MTP May 2024 · Series II · question p. 4 · answer p. 4-5

650 MTP Old P8 Nov 2023 S-I Q2 10 marks old scheme

ZX Ltd. has a paid-up share capital of ₹ 2,00,00,000, face value of ₹ 100 each. The current market price of the shares is ₹ 100 each. The Board of Directors of the company has an agenda of meeting to pay a dividend of 50% to its shareholders. The company expects a net income of ₹ 1,50,00,000 at the end of the current financial year. Company also plans for a capital expenditure for the next financial year for a cost of ₹ 1,90,00,000, which can be financed through retained earnings and issue of new equity shares.

Company's desired rate of investment is 15%.

Required:

Following the Modigliani- Miller (MM) Hypothesis, DETERMINE value of the company when:

(i) It does not pay dividend and

(ii) It does pay dividend

ICAI'S ANSWER

As per MM Hypothesis, value of firm/ company is calculated as below:

$$V_f \text{ or } nP_0 = \frac{(n + \Delta n)P_1 - I + E}{(1 + K_e)}$$

Where,

V_f = Value of firm in the beginning of the period

n = number of shares in the beginning of the period

Δn = number of shares issued to raise the funds required

I = Amount required for investment

E = total earnings during the period

(i) Value of the ZX Ltd. when dividends are not paid.

$$nP_o = \frac{(n+\Delta n)P_1 - I + E}{1+K_e}$$

$$nP_0 = \frac{\left(2{,}00{,}000 + \frac{40{,}00{,}000}{115}\right) \times ₹115 - \text{Rs.}1{,}90{,}00{,}000 + ₹1{,}50{,}00{,}000}{(1+0.15)}$$

$$= \frac{₹2{,}70{,}00{,}000 - ₹1{,}90{,}00{,}000 + ₹1{,}50{,}00{,}000}{(1+0.15)} = ₹\ 2{,}00{,}00{,}000$$

Working notes:

1. Price of share at the end of the period (P_1)

$$P_o = \frac{P_1 + D_1}{1+K_e}$$

$$100 = \frac{P_1 + 0}{1+0.15} \text{ or, } P_1 = 115$$

2. Calculation of funds required for investment

Earnings	₹1,50,00,000
Dividend distributed	Nil
Fund available for investment	₹ 1,50,00,000
Total Investment	₹ 1,90,00,000
Balance Funds required	₹ 40,00,000

3. Calculation of no. of shares required to be issued for balance fund

$$\text{No. of shares } (\Delta n) = \frac{\text{Funds required}}{\text{Price at end}(P_1)} = \frac{40{,}00{,}000}{115} \text{ shares}$$

(ii) Value of the ZX Ltd. when dividends are paid.

$$nP_o = \frac{(n+\Delta n)P_1 - I + E}{1+K_e}$$

$$nP_0 = \frac{\left(2{,}00{,}000 + \frac{1{,}40{,}00{,}000}{65}\right) \times ₹65 - ₹1{,}90{,}00{,}000 + ₹1{,}50{,}00{,}000}{(1+0.15)}$$

$$= \frac{₹2{,}70{,}00{,}000 - ₹1{,}90{,}00{,}000 + ₹1{,}50{,}00{,}000}{(1+0.15)} = ₹\ 2{,}00{,}00{,}000$$

Working notes:

4. Price of share at the end of the period (P_1)

$P_o = \frac{P_1 + D_1}{1 + K_e}$

$100 = \frac{P_1 + 50}{1 + 0.15}$ or, P_1= ₹65

5. Calculation of funds required for investment

Earnings	₹ 1,50,00,000
Dividend distributed	₹ 1,00,00,000
Fund available for investment	₹ 50,00,000
Total Investment	₹ 1,90,00,000
Balance Funds required	₹ 1,40,00,000

6. Calculation of no. of shares required to be issued for balance fund

No. of shares (Δn) = $\frac{\text{Funds required}}{\text{Price at end}(P_1)} = \frac{1,\ 40,00,000}{65}$ = 2,15,385 shares(approx.)

Note- *As per MM-hypothesis of dividend irrelevance, value of firm remains same irrespective of dividend paid. In the solution, there may be variation in value, which is due to rounding off error.*

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 2 · answer p. 3-5

651 MTP **Old P8 Nov 2023 S-II** Q1(b) 5 marks old scheme

PQR Ltd. is a blue-chip company listed in NSE in India with a face value of ₹ 100 per share. The company is expected to grow @ 15% p.a. for next four years then 5% for an indefinite period. The shareholders expect 20% return on their share investments. Company paid ₹ 150 as dividend per share for the current Financial Year. The shares of the company traded at an average price of ₹ 2,052 on last day. FIND out the intrinsic value per share and state whether shares are overpriced or underpriced.

ICAI'S ANSWER

As per Dividend discount model, the price of share is calculated as follows:

$$P = \frac{D_1}{(1+K_e)^1} + \frac{D_2}{(1+K_e)^2} + \frac{D_3}{(1+K_e)^3} + \frac{D_4}{(1+K_e)^4} + \frac{D_5}{(K_e\text{-}g)} \times \frac{1}{(1+K_e)^4}$$

Where,

P = Price per share

K_e = Required rate of return on equity

g = Growth rate

$$P = \frac{₹150 \text{x} 1.15}{(1+0.2)^1} + \frac{₹172.5 \text{x} 1.15}{(1+0.2)^2} + \frac{₹198.38 \text{x} 1.15}{(1+0.2)^3} + \frac{₹228.13 \text{x} 1.15}{(1+0.2)^4} + \frac{₹262.35 \text{x} 1.05}{(0.2-0.05)^1} \times \frac{1}{(1+0.2)^4}$$

P = 143.75 + 137.76 + 132.02 + 126.52 + 885.63 = ₹ 1425.68

Intrinsic value of share is ₹ 1425.68 as compared to latest market price of ₹ 2052. Market price of a share is overpriced by ₹ 626.32.

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 1 · answer p. 1-2

652 MTP Old P8 May 2023 S-I Q1(b) 5 marks old scheme

Roma Nov Ltd. has a capital of ₹25,00,000 in equity shares of ₹100 each. The shares are currently quoted at ₹120. The company proposes to declare a dividend of ₹15 per share at the end of the current financial year. The capitalization rate for the risk class of which the company belongs is 15%. COMPUTE market price of the share at the end of the year, if

(i) Dividend is not declared.

(ii) Dividend is declared.

Assuming that the company pays the dividend and has net profits of ₹9,00,000 and makes new investments of ₹15,00,000 during the period, CALCULATE number of new shares to be issued? Use the MM model.

ICAI'S ANSWER

Given,

Cost of Equity (K_e)	15%
Number of shares in the beginning (n)	25,000
Current Market Price (P_0)	120
Net Profit (E)	9,00,000
Expected Dividend (D_1)	15
Investment (I)	15,00,000

Computation of market price per share, when:

(i) No dividend is declared:

$$\text{Po} = \frac{P_1 + D_1}{1 + k_e}$$

$$₹120 = \frac{P_1 + 0}{1 + 0.15}$$

P_1 = ₹138 – 0 = ₹ 138

(ii) Dividend is declared:

$$₹120 = \frac{P_1 + 15}{1 + 0.15}$$

P_1 = ₹138 – ₹15 = ₹ 123

Calculation of number of shares required for investment.

	₹
Earnings	9,00,000
Dividend distributed	3,75,000
Fund available for investment	5,25,000
Total Investment	15,00,000
Balance Funds required	15,00,000 – 5,25,000 = 9,75,000

No. of shares = $\frac{\text{Funds required}}{\text{Price at end}(P_1)}$

= $\frac{9{,}75{,}000}{123}$ = 7,927 Shares(approx.)

ICAI · Old P8 · MTP May 2023 · Series I · question p. 1 · answer p. 2

653 MTP **Old P8 May 2023 S-II** Q3 10 marks old scheme

Rex Ltd has 20 lakh equity shares outstanding at the start of the accounting year 2023. The existing market price per share is ₹ 300. Expected dividend is ₹ 20 per share. The rate of capitalization appropriate to the risk class to which the company belongs is 20%.

CALCULATE the market price per share when expected dividends are: (a) declared, and (b) not declared, based on the Miller – Modigliani approach.

CALCULATE number of shares to be issued by the company at the end of the accounting year on the assumption that the net income for the year is ₹ 5 crore; investment budget is ₹ 8 crores, when (a) Dividends are declared, and (b) Dividends are not declared.

PROVE that the market value of the shares at the end of the accounting year will remain unchanged irrespective of whether (a) Dividends are declared, or (ii) Dividends are not declared.

WHAT is the implied growth rate in dividends as per Gordon's model, if expected dividend payment is considered imminent?

ICAI'S ANSWER

(a) (i) Calculation of market price per share

According to Miller – Modigliani (MM) Approach:

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

Where,

Existing market price (P_o) = ₹ 300

Expected dividend per share (D_1) = ₹ 20

Capitalization rate (k_e) = 0.20 Market price at year end (P_1) = ?

a. If expected dividends are declared, then

300=(P1+20)/(1+0.2)

300 x 1.2 = P1+20

P1= 340

b. If expected dividends are not declared, then

300=(P1+0)/(1+0.2)

300 x 1.2 = P1

P1= 360

(ii) Calculation of number of shares to be issued

	(a)	(b)
	Dividends are declared. (₹ lakh)	**Dividends are not Declared (₹ lakh)**
Net income	500	500
Total dividends	(400)	-
Retained earnings	100	500
Investment budget	800	800
Amount to be raised by new issues	700	300
Relevant market price (₹ per share)	340	360
No. of new shares to be issued (in lakh) (₹ 700 ÷ 340; ₹ 300 ÷ 360)	2.0588	0.8333

(iii) Calculation of market value of the shares

	(a)	(b)
Particulars	**Dividends are declared**	**Dividends are not Declared**
Existing shares (in lakhs)	20.00	20.00
New shares (in lakhs)	2.0588	0.8333
Total shares (in lakhs)	22.0588	20.8333
Market price per share (₹)	340	360
Total market value of shares at the end of the year (₹ in lakh)	22.0588 × 340 = 7,500 (approx.)	20.8333 × 360 = 7,500 (approx.)

Hence, it is proved that the total market value of shares remains unchanged irrespective of whether dividends are declared, or not declared.

(iv) P0=D1/(Ke-g)

300 = 20/(0.2-g)

0.2-g = 20/300

0.2-g = 0.0667

G = 0.133333

g = 13.3333%

ICAI · Old P8 · MTP May 2023 · Series II · question p. 3 · answer p. 6-8

654 MTP Old P8 Nov 2022 S-I Q1(d) 5 marks old scheme

Following information is given for WN Ltd.:

Earnings	₹ 30 per share
Dividend	₹ 9 per share
Cost of capital	15%
Internal Rate of Return on investment	20%

You are required to CALCULATE the market price per share using-

(i) Gordon's formula

(ii) Walter's formula

ICAI'S ANSWER

(i) As per **Gordon's Model**, Price per share is computed using the formula:

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

Where,

P_0 = Price per share

E_1 = Earnings per share

b = Retention ratio; (1 - b = Pay-out ratio)

K_e = Cost of capital

r = IRR

br = Growth rate (g)

Applying the above formula, price per share

$$P_0 = \frac{30 \times 0.3*}{0.15 - 0.70 \times 0.2} = \frac{9}{0.01} = ₹\ 900$$

*Dividend pay-out ratio = $\frac{₹\ 9}{₹\ 30}$ = 0.3 or 30%

(ii) As per Walter's Model, Price per share is computed using the formula:

$$\text{Price (P)} = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where,

P = Market Price of the share

E = Earnings per share

D = Dividend per share

K_e = Cost of equity/ rate of capitalization/ discount rate

r = Internal rate of return/ return on investment

Applying the above formula, price per share

$$\text{P} = \frac{9 + \frac{0.20}{0.15}(30 - 9)}{0.15} = \frac{37}{0.15} = ₹\ 246.67$$

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 2 · answer p. 2-3

655 MTP **Old P8 May 2022 S-I** Q1(a) 5 marks old scheme

The following figures have been extracted from the annual report of Xee Ltd.:

Net Profit	₹ 75 lakhs
Outstanding 12% preference shares	₹ 250 lakhs
No. of equity shares	3 lakhs
Return on Investment	20%
Cost of capital i.e. (K_e)	16%

COMPUTE the approximate dividend pay-out ratio so as to keep the share price at ₹ 105 by using Walter's model?

ICAI'S ANSWER

Particulars	(₹' in lakhs)
Net Profit	75
Less: Preference dividend	30
Earnings for equity shareholders	45
Earnings per share	45/3 = ₹ 15

Let, the dividend per share be D to get share price of ₹ 105

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

$$105 = \frac{D + \frac{0.20}{0.16}(15 - D)}{0.16}$$

$$16.8 = \frac{0.16D + 3 - 0.20D}{0.16}$$

$0.04D = 3 - 2.688$

$D = 7.80$

$$\text{D/P ratio} = \frac{\text{DPS}}{\text{EPS}} \times 100 = \frac{7.80}{15} \times 100 = 52\%$$

So, the required dividend pay-out ratio will be = 52%

ICAI · Old P8 · MTP May 2022 · Series I · question p. 1 · answer p. 1

656 MTP **Old P8 Dec 2021 S-I** Q1(d) 5 marks old scheme

Same question, different ICAI answer: #641 MTP May 2025 S-I

M Ltd. belongs to a risk class for which the capitalization rate is 12%. It has 40,000 outstanding shares and the current market price is ₹ 200. It expects a net profit of ₹ 5,00,000 for the year and the Board is considering dividend of ₹ 10 per share.

M Ltd. requires to raise ₹ 10,00,000 for an approved investment expenditure. ILLUSTRATE, how the MM approach affects the value of M Ltd. if dividends are paid or not paid.

ICAI'S ANSWER

Given,

Cost of Equity (K_e)	12%
Number of shares in the beginning (n)	40,000
Current Market Price (P_0)	₹200
Net Profit (E)	₹5,00,000
Expected Dividend (D_1)	₹10 per share
Investment (I)	₹10,00,000

Situation 1 – When dividends are paid

(i) $P_0 = \frac{P1 + D1}{1 + Ke}$

$200 = \frac{P1 + 10}{1 + 0.12}$

P_1 + 10 = 200 x 1.12

P_1 = 224 – 10 = 214

(ii) Calculation of funds required

= Total Investment - (Net profit - Dividend)

= 10,00,000 - (5,00,000 – 4,00,000)

= 9,00,000

(iii) No. of shares required to be issued for balance fund

No. of shares = $\frac{\text{Funds Required}}{\text{Price at end (P1)}}$

$\Delta n = \frac{9,00,000}{214}$ = 4205.61

(iv) Calculation of value of firm

$V_f = \frac{(n + \Delta n)P1 - I + E}{1 + Ke}$

$= \frac{(40,000 + \frac{9,00,000}{214})214 - 10,00,000 + 5,00,000}{1 + 0.12}$

$= \frac{94,60,000 - 5,00,000}{1.12}$ = 80,00,000

Situation 2 – When dividends are not paid

(i) $P_0 = \frac{P1 + D1}{1 + Ke}$

$200 = \frac{P1 + 0}{1 + 0.12}$

P_1 + 0 = 200 x 1.12

P_1 = 224 – 0 = 224

(ii) Calculation of funds required

= Total Investment - (Net profit - Dividend)

= 10,00,000 – (5,00,000 - 0)

= 5,00,000

(iii) No. of shares required to be issued for balance fund

$$\text{No. of shares} = \frac{\text{Funds Required}}{\text{Price at end (P1)}}$$

$$\Delta n = \frac{5{,}00{,}000}{224} = 2232.14$$

(iv) Calculation of value of firm

$$V_f = \frac{(n + \Delta n)P1 - I + E}{1 + Ke}$$

$$= \frac{(40{,}000 + \frac{5{,}00{,}000}{224})224 - 10{,}00{,}000 + 5{,}00{,}000}{1 + 0.12}$$

$$= \frac{94{,}60{,}000 - 5{,}00{,}000}{1.12} = 80{,}00{,}000$$

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 2 · answer p. 3-4

657 MTP **Old P8 Dec 2021 S-II** Q1(d) 5 marks old scheme

The following information is supplied to you:

Particulars	₹
Total Earnings	5,00,000
Equity shares (of ₹ 100 each)	50,00,000
Dividend paid	3,75,000
Price/ Earnings ratio	12.5

Applying Walter's Model:

(i) ANALYSE whether the company is following an optimal dividend policy.

(ii) COMPUTE P/E ratio at which the dividend policy will have no effect on the value of the share.

(iii) Will your decision change, if the P/E ratio is 8 instead of 12.5? ANALYSE.

ICAI'S ANSWER

(i) The EPS of the firm is ₹ 10 (i.e. ₹ 5,00,000/ 50,000). r = 5,00,000/ 50,00,000 = 10%. The P/E Ratio is given at 12.5 and the cost of capital, K_e, may be taken at the inverse of P/E ratio. Therefore, K_e is 8 (i.e., 1/12.5). The firm is distributing total dividends of ₹ 3,75,000 among 50,000 shares, giving a dividend per share of ₹ 7.50. The value of the share as per Walter's model may be found as follows:

$$\text{P} = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{7.5 + \frac{0.1}{0.08}(10 - 7.5)}{0.08} = ₹\ 132.81$$

The firm has a dividend payout of 75% (i.e., ₹ 3,75,000) out of total earnings of ₹ 5,00,000. Since, the rate of return of the firm, r, is 10% and it is more than the K_e of 8%, therefore, by distributing 75% of earnings, the firm is not following an optimal dividend policy. The optimal dividend policy for the firm would be to pay zero dividend and in such a situation, the market price would be,

$$\frac{0+\frac{0.1}{0.08}(10-0)}{0.08} = ₹\ 156.25$$

So, theoretically, the market price of the share can be increased by adopting a zero payout.

(ii) The P/E ratio at which the dividend policy will have no effect on the value of the share is such at which the K_e would be equal to the rate of return, r, of the firm. The K_e would be 10% (= r) at the P/E ratio of 10. Therefore, at the P/E ratio of 10, the dividend policy would have no effect on the value of the share.

(iii) If the P/E is 8 instead of 12.5, then the Ke which is the inverse of P/E ratio, would be 12.5 and in such a situation k_e> r and the market price, as per Walter's model would be:

$$\text{P} = \frac{D+\frac{r}{K_e}(E-D)}{K_e} = \frac{7.5+\frac{0.1}{0.125}(10-7.5)}{0.125} = ₹\ 76$$

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 2 · answer p. 3

658 MTP **Old P8 Jul 2021 S-I** Q1(a) 5 marks old scheme

The following information is given:

Dividend per share (DPS)	Rs. 9
Cost of capital (K_e)	19%
Internal rate of return on investment	24%
Retention Ratio	25%

CALCULATE the market price per share by using:

(i) Walter's formula

(ii) Gordon's formula (Dividend Growth model)

ICAI'S ANSWER

Working:

Calculation of Earnings per share (EPS):

$$\text{EPS} = \frac{\text{DPS}}{\text{Dividend Payout Ratio}}$$

$$\text{EPS} = \frac{\text{Rs. } 9}{1\text{-}0.25} = \text{Rs.}12$$

Market price per share by

(i) Walter's model:

$$\text{P} = \frac{D+\frac{r}{Ke}(E-D)}{K_e}$$

$$= \frac{\text{Rs. } 9+\frac{0.24}{0.19}(\text{Rs. } 12-\text{Rs. } 9)}{0.19}$$

= Rs. 67.31

(ii) Gordon's model (Dividend Growth model):

$$P_o = \frac{D_o(1+g)}{K_e\text{-}g}$$

Where,

P_o = Present market price per share.

g = Growth rate (br) = 0.25 × 0.24 = 0.06

b = Retention ratio

k = Cost of Capital

r = Internal rate of return (IRR)

D_0 = Dividend per share

E = Earnings per share

$$= \frac{\text{Rs. } 9(1+0.06)}{0.19\text{-}0.06}$$

$$= \frac{\text{Rs. } 9.54}{0.13} = \text{Rs.}73.38$$

Alternatively,

$$P_o = \frac{E(1-b)}{k-br}$$

$$P_o = \frac{12\ (1\text{-}0.25)}{0.19-0.06} = \frac{9}{0.13} = \text{Rs. } 69.23$$

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 1 · answer p. 1

659 MTP **Old P8 Jul 2021 S-II** Q1(b) 5 marks old scheme

The following data is available in respect of N Ltd. for the year ended 31st March, 2021:

	Rs. (in Crore)
Share capital (@ Rs. 10 per share)	25.00
Reserves	15.00
Profit after tax (PAT)	3.70
Dividends paid	3.00
P/E ratio	26.70

Using Walter's Model:

(i) COMMENT on the firm's dividend policy;

(ii) DETERMINE the optimum payout ratio and

(iii) DETERMINE the P/E ratio at which dividend payout will have no effect on share price.

ICAI'S ANSWER

Workings:

1. Earnings per share (E) = $\frac{\text{PAT}}{\text{No. of shares}} = \frac{\text{Rs. } 3.7 \text{ crores}}{2.5 \text{ crore shares}} = \text{Rs. } 1.48$
2. Return on Investment (r) = $\frac{\text{PAT}}{\text{Net worth}} \text{ x } 100 = \frac{\text{Rs. } 3.7 \text{ crores}}{\text{Rs. } (25+15) \text{ crores}} \text{ x } 100 = 9.25\%$
3. Dividend per share (D) = $\frac{\text{Dividend paid}}{\text{No. of shares}} = \frac{\text{Rs. } 3 \text{ crores}}{2.5 \text{ crore shares}} = \text{Rs. } 1.2$

Dividend payout ratio = $\frac{\text{Dividend}}{\text{PAT}}$ x 100 = $\frac{\text{Rs. 3 crores}}{\text{Rs. 3.7 crores}}$ x 100 = 81.08%

4. Current Market Price (P_o) = P/E Ratio x E = 26.7 x Rs. 1.48 = Rs. 39.52
5. Growth rate (g) = b x r = (1 - 0.8108) x 0.0925 = 1.75%
6. Cost of Capital (K_e) = $\frac{D(1+g)}{P_o}$ + g = $\frac{\text{Rs. } 1.2\ (1+0.0175)}{\text{Rs. } 39.52}$ + 0.0175= 4.84%

(i) The value of the share as per Walter's model:

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e} = \frac{1.2 + \frac{0.0925}{0.0484}(1.48 - 1.2)}{0.0484} = \text{Rs. } 35.85$$

The firm has a dividend payout of 81.08% (i.e., Rs. 3 crores) out of Profit after tax of Rs. 3.7 crores with value of the share at Rs. 35.85. The rate of return on investment (r) is 9.25% and it is more than the K_e of 4.84%, therefore, by distributing 81.08% of earnings, the firm is not following an optimal dividend policy.

(ii) Under Walter's model, when return on investment is more than cost of capital (r > K_e), the market share price will be maximum if 100% retention policy is followed. So, the optimal payout ratio would be to pay zero dividend and in such a situation, the market price would be:

$$P = \frac{0 + \frac{0.0925}{0.0484}(1.48 - 0)}{0.0484} = \text{Rs. } 58.44$$

(iii) The P/E ratio at which dividend payout will have no effect on share price is at which the K_e would be equal to the rate of return (r) of the firm i.e. 9.25%.

So, K_e = $\frac{D\ (1+g)}{P_o}$ + g

0.0925 = $\frac{\text{Rs. } 1.2\ (1+0.0175)}{P_o}$ + 0.0175

$\therefore P_o$ = Rs. 16.28

If P_o is Rs. 16.28, then, P/E Ratio will be:

= $\frac{P_o}{E}$ = $\frac{\text{Rs. } 16.28}{\text{Rs. } 1.48}$ = 11 times

Therefore, at the P/E ratio of 11, the dividend payout will have no effect on share price.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 1 · answer p. 1-2

660 MTP **Old P8 Nov 2020 S-I** Q1(b) 5 marks old scheme

The annual report of XYZ Ltd. provides the following information for the Financial Year 2019-20:

Particulars	Amount (₹)
Net Profit	50 lakhs
Outstanding 15% preference shares	100 lakhs
No. of equity shares	5 lakhs
Return on Investment	20%
Cost of capital i.e. (K_e)	16%

CALCULATE price per share using Gordon's Model when dividend pay-out is-

(i) 25%;

(ii) 50%;

(iii) 100%.

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 1

661 MTP **Old P8 May 2020 S-I** Q2 10 marks old scheme

ZX Ltd. has a paid-up share capital of Rs.1,00,00,000, face value of Rs.100 each. The current market price of the shares is Rs.100 each. The Board of Directors of the company has an agenda of meeting to pay a dividend of 50% to its shareholders. The company expects a net income of Rs.75,00,000 at the end of the current financial year. Company also plans for a capital expenditure for the next financial year for a cost of Rs.95,00,000, which can be financed through retained earnings and issue of new equity shares.

Company's desired rate of investment is 15%.

Required:

Following the Modigliani- Miller (MM) Hypothesis, DETERMINE value of the company when:

(i) It does not pay dividend and

(ii) It does pay dividend

ICAI'S ANSWER

As per MM Hypothesis, value of firm/ company is calculated as below:

$$V_f \text{ or } nP_0 = \frac{(n + \Delta n)P_1 - I + E}{(1 + K_e)}$$

Where,

V_f	=	Value of firm in the beginning of the period
n	=	number of shares in the beginning of the period
Δn	=	number of shares issued to raise the funds required
I	=	Amount required for investment
E	=	total earnings during the period

(i) Value of the ZX Ltd. when dividends are not paid.

$$nP_o = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

$$nP_0 = \frac{\left(1{,}00{,}000 + \frac{20{,}00{,}000}{115}\right) \times \text{Rs.}115 - \text{Rs.}95{,}00{,}000 + \text{Rs.}75{,}00{,}000}{(1 + 0.15)}$$

$$= \frac{\text{Rs.}1{,}35{,}00{,}000 - \text{Rs.}95{,}00{,}000 + \text{Rs.}75{,}00{,}000}{(1 + 0.15)} = \text{Rs.}1{,}00{,}00{,}000$$

Working notes:

1. Price of share at the end of the period (P_1)

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

$$100 = \frac{P_1 + 0}{1 + 0.15} \text{ or, } P_1 = 115$$

2. Calculation of funds required for investment

Earnings	Rs.75,00,000
Dividend distributed	Nil
Fund available for investment	Rs.75,00,000
Total Investment	Rs.95,00,000
Balance Funds required	Rs.20,00,000

3. Calculation of no. of shares required to be issued for balance fund

$$\text{No. of shares } (\Delta n) = \frac{\text{Funds required}}{\text{Price at end}(P_1)} = \frac{20,00,000}{115} \text{ shares}$$

(ii) Value of the ZX Ltd. when dividends are paid.

$$nP_o = \frac{(n + \Delta n)P_1 - I + E}{1 + K_e}$$

$$nP_0 = \frac{\left(1,00,000 + \frac{70,00,000}{65}\right) \times \text{Rs.}65 - \text{Rs.}95,00,000 + \text{Rs.}75,00,000}{(1 + 0.15)}$$

$$= \frac{\text{Rs.}1,35,00,000 - \text{Rs.}95,00,000 + \text{Rs.}75,00,000}{(1 + 0.15)} = \text{Rs.}1,00,00,000$$

Working notes:

4. Price of share at the end of the period (P_1)

$$P_o = \frac{P_1 + D_1}{1 + K_e}$$

$$100 = \frac{P_1 + 50}{1 + 0.15} \text{ or, } P_1 = \text{Rs.}65$$

5. Calculation of funds required for investment

Earnings	Rs.75,00,000
Dividend distributed	Rs.50,00,000
Fund available for investment	Rs.25,00,000
Total Investment	Rs.95,00,000
Balance Funds required	Rs.70,00,000

6. Calculation of no. of shares required to be issued for balance fund

$$\text{No. of shares } (\Delta n) = \frac{\text{Funds required}}{\text{Price at end}(P_1)} = \frac{70,00,000}{65} = 1,07,693 \text{ shares(approx.)}$$

Note- *As per MM-hypothesis of dividend irrelevance, value of firm remains same irrespective of dividend paid. In the solution, there may be variation in value, which is due to rounding off error.*

ICAI · Old P8 · MTP May 2020 · Series I · question p. 2 · answer p. 4-6

662 MTP **Old P8 Nov 2019 S-I** Q1(a) 5 marks old scheme

The following figures are collected from the annual report of XYZ Ltd.:

Net Profit	Rs.60 lakhs
Outstanding 10% preference shares	Rs.100 lakhs
No. of equity shares	5 lakhs
Return on Investment	20%
Cost of capital i.e. (K_e)	14%

CALCULATE price per share using Gordon's Model when dividend pay-out is (i) 25%; (ii) 50% and (iii) 100%.

ICAI'S ANSWER

	Rs. in lakhs
Net Profit	60
Less: Preference dividend	10
Earning for equity shareholders	50
Therefore earning per share	50/5 = Rs.10.00

Price per share according to Gordon's Model is calculated as follows:

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

Here, E_1 = 10, K_e = 14%, r = 20%

(i) When dividend pay-out is 25%

$$P_0 = \frac{10 \times 0.25}{0.14 - (0.75 \times 0.2)} = \frac{2.5}{0.14 - 0.15} = -250$$

As per the Gordon's Dividend relevance model, the Cost of equity (K_e) should be greater than the growth rate i.e. br. In this case K_e is 14% and br = 15%, hence, the equity investors would prefer capital appreciation than dividend.

(ii) When dividend pay-out is 50%

$$P_0 = \frac{10 \times 0.5}{0.14 - (0.5 \times 0.2)} = \frac{5}{0.14 - 0.10} = 125$$

(iii) When dividend pay-out is 100%

$$P_0 = \frac{10 \times 1}{0.14 - (0 \times 0.2)} = \frac{10}{0.14} = 71.43$$

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 1 · answer p. 1

CHAPTER 9

Management of Working Capital

9.1 Working capital, the operating cycle and financing policy

24 questions · asked in 19 ICAI documents · 4 exam papers

663 EXAM Sep 2025 Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

RG Limited has estimated that number of operating cycle in year 2025-26 will be 3.

The amount of working capital required after taking an estimated 5% unforeseen contingency is ₹ 1,01,850. Other information is as under:

Opening stock of Finished goods is ₹ 31,100 and the closing stock will be lesser by ₹ 4,200 from the opening stock.

Opening stock of Raw Material is ₹ 24,450 and Closing stock of Raw material will be ₹ 28,200.

Opening stock of Work in Progress is ₹ 20,500 and Closing stock of Work in Progress will be more by ₹ 1,800 from the Opening stock.

Wages and Manufacturing expenses during the year will be ₹ 1,12,800.

Selling and Distribution expenses during the year will be ₹ 30,000.

Average Creditors for the year is ₹ 26,170.

All Purchases will be done on credit basis only.

Assume a 360 days year.

Using above information, you are required to answer the following questions 1 to 5:

MCQ 1 2 marks

What will be the Total Operating Expenses and Cost of Goods Sold for the year?

A ₹ 2,61,000 and ₹ 2,91,000

B ₹ 2,91,000 and ₹ 2,61,000

C ₹ 2,75,500 and ₹ 2,61,000

D ₹ 2,58,500 and ₹ 2,88,000

MCQ 2 2 marks

What will be the Cost of Production and Finished Goods storage period?

A ₹ 2,58,600 and 45 Days

B ₹ 2,58,600 and 40 Days

C ₹ 2,56,800 and 45 Days

D ₹ 2,56,800 and 40 Days

MCQ 3 2 marks

What will be the Raw Material Consumed and Credit purchases?

A ₹ 1,52,500 and ₹ 1,50,550

B ₹ 1,49,500 and ₹ 1,52,500

C ₹ 1,45,800 and ₹ 1,49,550

D ₹ 1,52,000 and ₹ 1,45,800

MCQ 4 2 marks

What will be the Raw material storage period and Work in Progress conversion period?

A 62 Days and 32 Days

B 65 Days and 30 Days

C 63 Days and 33 Days

D 63 Days and 45 Days

MCQ 5 2 marks

What will be the Creditors Payment period and Debtors Collection period?

A 60 Days and 50 Days

B 63 Days and 48 Days

C 62 Days and 45 Days

D 65 Days and 47 Days

ICAI'S ANSWERS

MCQ 1

Answer key: **(B)**

MCQ 2

Answer key: **(D)**

MCQ 3

Answer key: **(C)**

MCQ 4

Answer key: **(B)**

MCQ 5

Answer key: **(B)**

ICAI · P6 · Sep 2025 exam · question p. 1 · answer p. 7

664 EXAM Jan 2025 Case scenario · Q2(b)(i) to Q2(b)(iii) 3 marks

CASE SCENARIO

The following information pertain to MSD Limited for the year ending 31st March, 2024:

Particulars	Number of days
Raw material storage period	61 days
Work-in -progress conversion period	20 days
Finished goods storge period	30 days
Debt collection period	45 days
Creditors payment period	60 days

The annual operating cost (including depreciation of ₹ 4,80,000) was ₹ 60,00,000. Assume 360 days in a year.

You are required to calculate:

Q2(b)(i) 1 mark

Operating cycle period

Q2(b)(ii) 1 mark

Number of operating cycles in a year

Q2(b)(iii) 1 mark

Amount of working capital required for the company

ICAI'S ANSWERS

Q2(b)(i)

Net Operating Cycle Period

= R + W + F + D – C

Where,

R = Raw material storage period.

W = Work-in-progress holding period.

F = Finished goods storage period.

D = Debtors collection period.

C = Credit period availed.

= 61 + 20 + 30 + 45 – 60

= 96 days

Q2(b)(ii)

Number of Operating Cycles in the Year

$$= \frac{360}{\text{Operating Cycle Period}} = \frac{360}{96} = 3.75 \text{ times}$$

Q2(b)(iii)

Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{₹60,00,000}{3.75} = ₹\ 16,00,000$$

Alternative Solution on cash cost basis to Part (ii)

(iii) Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{₹60,00,000 - ₹4,80,000}{3.75} = ₹\ 14,72,000$$

ICAI · P6 · Jan 2025 exam · question p. 14 · answer p. 17, 17-18

665 EXAM **Old P8 May 2023** Q6(b) 4 marks old scheme

"Permanent working capital and fluctuating (temporary) working capital, both are necessary to facilitate production and sales through the operating cycle." - Describe.

ICAI'S ANSWER

Both kinds of working capital i.e. permanent and fluctuating (temporary) are necessary to facilitate production and sales through the operating cycle:

Permanent working capital refers to the base working capital, which is the minimum level of investment in the current assets that is carried by the entity at all times to carry its day to day activities. It generally stays invested in the business unless the operations are scaled up or down permanently which would also result in increase or decrease in permanent working capital. It is generally financed by long term sources of finance.

Temporary working capital refers to that part of total working capital, which is required by an entity in addition to the permanent working capital. It is also called variable or fluctuating working capital which is used to finance the short-term working capital requirements which arises due to fluctuation in sales volume. For instance, an organization would maintain increased levels of inventory to meet increased seasonal demand.

ICAI · Old P8 · May 2023 exam · question p. 20 · answer p. 21

666 EXAM **Old P8 Jan 2021** Q1(d) 5 marks old scheme

The following information is provided by MNP Ltd. for the year ending 31st March, 2020:

Raw Material Storage period	45 days
Work-in-Progress conversion period	20 days
Finished Goods storage period	25 days
Debt Collection period	30 days
Creditors payment period	60 days
Annual Operating Cost	₹ 25,00,000
(Including Depreciation of ₹ 2,50,000)	

Assume 360 days in a year.

You are required to calculate:

(i) Operating Cycle period

(ii) Number of Operating Cycle in a year.

(iii) Amount of working capital required for the company on a cost basis.

(iv) The company is a market leader in its product and it has no competitor in the market. Based on a market survey it is planning to discontinue sales on credit and deliver products based on pre-payments in order to reduce its working capital requirement substantially. You are required to compute the reduction in working capital requirement in such a scenario.

ICAI'S ANSWER

(i) Calculation of Operating Cycle Period:

Operating Cycle Period = R + W + F + D – C

= 45 + 20 + 25 + 30 – 60 = **60 days**

(ii) Number of Operating Cycle in a Year

$= \frac{360}{\text{Operating cycle period}} = \frac{360}{60} =$ **6**

(iii) Amount of Working Capital Required

$= \frac{\text{Annual operating cost}}{\text{Number of operating cycle}} = \frac{₹\ 25,00,000 - ₹\ 2,50,000}{6}$

$= \frac{₹\ 22,50,000}{6} =$ **₹ 3,75,000**

(iv) Reduction in Working Capital

Operating Cycle Period = R + W + F – C

= 45 + 20 + 25 – 60 = **30 days**

Amount of Working Capital Required = $\frac{₹\ 22,50,000}{360}$ x 30 = ₹ 1,87,500

Reduction in Working Capital = ₹ 3,75,000 – ₹ 1,87,500 = **₹ 1,87,500**

Note: If we use Total Cost basis, then amount of Working Capital required will be ₹ 4,16,666.67 (approx.) and Reduction in Working Capital will be ₹ 2,08,333.33 (approx.)

ICAI · Old P8 · Jan 2021 exam · question p. 2-3 · answer p. 5-6

667 **RTP** **Sep 2026** Q10(c)

Sunrise Healthcare Pvt. Ltd. operates a chain of pharmacies across India. The company has recently experienced rising raw material costs, fluctuations in exchange rates affecting imported medicines, and increasing inventory levels due to uncertain demand. Management is also reviewing its credit policy to improve cash flows and is considering various short-term financing options to meet working capital needs.

In the context of the above case, discuss the factors that should be considered by Sunrise Healthcare Pvt. Ltd. while planning its working capital requirements.

ICAI'S ANSWER

Some of the factors which need to be considered while planning for working capital requirement are:

1. **Need for Cash:** Identify the cash balance which allows for the business to **meet day-to-day expenses** but reduces cash holding costs (example - loss of interest on long term investment had the surplus cash invested therein).
2. **Desired level of Inventory: Identify the level of inventory** which allows for uninterrupted production but reduces the investment in raw materials and hence increases cash flow. The techniques like Just in Time (JIT) and Economic order quantity (EOQ) are used for this.
3. **Receivables:** Identify the **appropriate credit policy**, i.e., credit terms which will attract customers, such that any impact on cash flows and the cash conversion cycle will be offset by increased revenue and hence Return on Capital (or vice versa). The tools like Early Payment Discounts and allowances are used for this.
4. **Short-term Financing Options:** Inventory is ideally financed by credit granted by the supplier. However, depending on the cash conversion cycle, it may be necessary to utilize a bank loan (or overdraft), or to "convert debtors to cash" through "factoring" in order to finance working capital requirements.

5. **Nature of Business:** For e.g. in a business of restaurant, most of the sales are in Cash. Therefore, need for working capital is very less. On the other hand, there would be a higher inventory in case of a pharmacy or a bookstore.
6. **Market and Demand Conditions:** For e.g. if an item's demand far exceeds its production, the working capital requirement would be less as investment in finished goods inventory would be very less with continuous sales.
7. **Technology and Manufacturing Policies:** For e.g. in some businesses the **demand for goods is seasonal**, in that case a business may follow a policy for steady production throughout the whole year or rather may choose a policy of production only during the demand season.
8. **Operating Efficiency:** A company can reduce the working capital requirement by **eliminating waste, improving coordination, process improvements** etc.
9. **Price Level Changes & Exchange Rate Fluctuations:** For e.g. **rising prices necessitate the use of more funds** for maintaining an existing level of activity. For the same level of current assets, higher cash outlays are required. Therefore, the effect of rising prices is that a higher amount of working capital is required. Another example would be unfavorable exchange rate movement in case of imported raw materials would warrant additional cost of same.

ICAI · P6 · RTP Sep 2026 · question p. 9-10 · answer p. 26-27

668 RTP **May 2026** Q9

Following information is forecasted by the Puja Limited for the year ending 31st March, 2025:

	Balance as at 1st April, 20X6	Balance as at 31st March, 20X7
	(₹)	(₹)
Raw Material	45,000	65,356
Work-in-progress	35,000	51,300
Finished goods	60,181	70,175
Debtors	1,12,123	1,35,000
Creditors	50,079	70,469
Annual purchases of raw material (all credit)		4,00,000
Annual cost of production		7,50,000
Annual cost of goods sold		9,15,000
Annual operating cost		9,50,000
Annual sales (all credit)		11,00,000

You may take one year as equal to 365 days.

You are required to CALCULATE:

(i) Net operating cycle period.

(ii) Number of operating cycles in the year.

(iii) Amount of working capital requirement using operating cycles.

ICAI'S ANSWER

Working Notes:

1. Raw Material Storage Period (R)

$$= \frac{\text{Average Stock of Raw Material}}{\text{Annual Consumption of Raw Material}} \times 365$$

$$= \frac{\frac{₹45,000 + ₹65,356}{2}}{₹3,79,644} \times 365$$

= 53 days.

Annual Consumption of Raw Material

= Opening Stock +Purchases- Closing Stock

= ₹ 45,000 + ₹ 4,00,000 – ₹ 65,356

= ₹ 3,79,644

2. Work-in-Progress (WIP) Conversion Period (W)

$$\text{WIP Conversion Period} = \frac{\text{Average Stock of WIP}}{\text{Annual Cost of Production}} \times 365$$

$$= \frac{\frac{₹35,000 + ₹51,300}{2}}{₹7,50,000} \times 365$$

= 21 days

3. Finished Stock Storage Period (F)

$$= \frac{\text{Average Stock of Finished Goods}}{\text{Cost of Goods Sold}} \times 365$$

$$= \frac{₹65,178}{₹9,15,000} \times 365 = 26 \text{ days.}$$

$$\text{Average Stock} = \frac{₹60,181 + ₹70,175}{2}$$

= ₹ 65,178.

4. Debtors Collection Period (D)

$$= \frac{\text{Average Debtors}}{\text{Annual Credit Sales}} \times 365$$

$$= \frac{₹1,23,561.50}{₹11,00,000} \times 365$$

= 41 days

$$\text{Average debtors} = \frac{₹1,12,123 + ₹1,35,000}{2} = ₹1,23,561.50$$

5. Creditors Payment Period (C)

$$= \frac{\text{Average Creditors}}{\text{Annual Net Credit Purchases}} \times 365$$

$$= \frac{\left(\frac{₹50,079 + ₹70,469}{2}\right)}{₹4,00,000} \times 365$$

= 55 days

(i) Operating Cycle Period

= R + W + F+ D - C

= 53 + 21 + 26 + 41 - 55

= 86 days

(ii) Number of Operating Cycles in the Year

$$= \frac{365}{\text{Operating Cycle Period}} = \frac{365}{86} = 4.244$$

(iii) Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{₹9,50,000}{4.244} = ₹\ 2,\ 23,845.42$$

ICAI · P6 · RTP May 2026 · question p. 10 · answer p. 25-27

669 RTP **Sep 2025** Q10(c)

ABC Ltd. is a manufacturing company experiencing rapid sales growth. To meet the increased demand, the company has started maintaining higher inventory levels and offering more generous credit terms to customers. However, its cash reserves are depleting, and it is facing difficulties in paying suppliers on time. As a financial analyst, apply your understanding of the scope of working capital management to ANALYZE the trade-offs the company is making between different components of working capital (inventory, receivables, advances, payables, and cash).

ICAI'S ANSWER

The trade-off between the components of working capital can be summarised as follows:

Component of Working Capital	Advantages of higher side (Profitability)	Trade-off (between Profitability and Liquidity)	Advantages of lower side (Liquidity)
Inventory	Fewer stock-outs increase the profitability.	Use techniques like EOQ, JIT etc. to carry optimum level of inventory.	Lower inventory requires less capital but endangered stock-out and loss of goodwill.
Receivables	Higher Credit period attract customers and increase revenue (but can result in more bad debts)	Evaluate the credit policy; use the services of debt management (factoring) agencies.	Cash sales provide liquidity but fails to boost sales and revenue (due to lower credit period)
Pre-payment of expenses	Reduces uncertainty and profitable in inflationary environment.	Cost-benefit analysis required	Improves or maintains liquidity.
Cash and Cash equivalents	Payables are honoured in time, improves the goodwill and helpful in getting future discounts.	Cash budgets and other cash management techniques can be used	Cash can be invested in some other investment avenues
Payables and Expenses	Capital can be used in some other investment avenues	Evaluate the credit policy and related cost.	Payables are honoured in time, improves the goodwill and helpful in getting future discounts.

ICAI · P6 · RTP Sep 2025 · question p. 9 · answer p. 26-27

670 MTP **Sep 2026 S-I** Q2(b) 3 marks

Also asked in MTP Sep 2024 S-II Q2(b)

Determining the amount to be invested in current assets as working capital is a crucial policy decision for any entity. What FACTORS should a company consider when deciding the level of investment in working capital?

ICAI'S ANSWER

Level of investment depends on the various factors listed below:

(a) **Nature of Industry:** Construction companies, breweries etc. requires large investment in working capital due long gestation period.

(b) **Types of products:** Consumer durable has large inventory as compared to perishable products.

(c) **Manufacturing Vs Trading Vs Service:** A manufacturing entity has to maintain three levels of inventory i.e. raw material, work-in-process and finished goods whereas a trading and a service entity has to maintain inventory only in the form of trading stock and consumables respectively.

(d) **Volume of sales:** Where the sales are high, there is a possibility of high receivables as well.

(e) **Credit policy:** An entity whose credit policy is liberal has not only high level of receivables but may require more capital to fund raw material purchases as that will depend on credit period allowed by suppliers.

ICAI · P6 · MTP Sep 2026 · Series I · question p. 6 · answer p. 7-8

671 MTP Sep 2026 S-II Q4(b) 4 marks

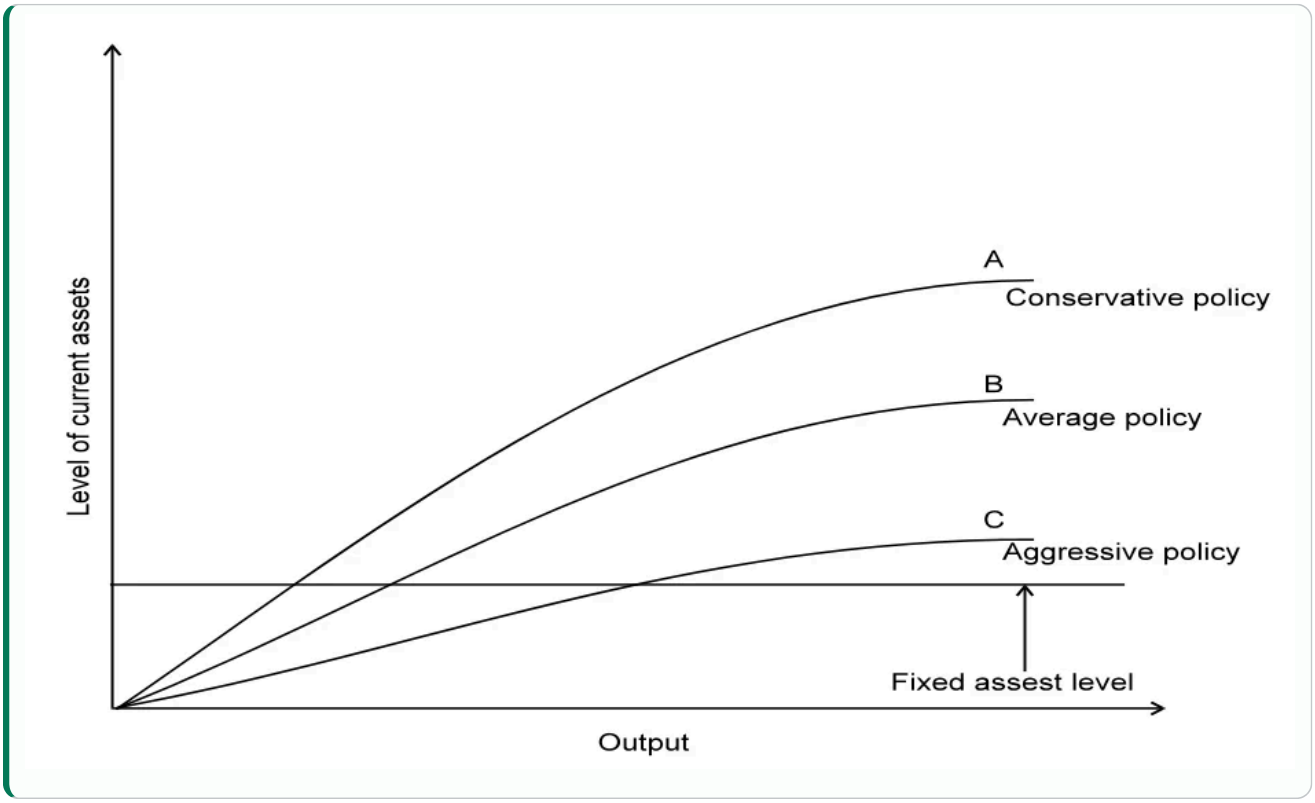
Also asked in MTP May 2024 S-II Q4(b)

DISCUSS the liquidity vs. profitability issue in management of working capital.

ICAI'S ANSWER

Liquidity versus Profitability Issue in Management of Working Capital: Working capital management entails the control and monitoring of all components of working capital i.e. cash, marketable securities, debtors, creditors etc. Finance manager has to pay particular attention to the levels of current assets and their financing. To decide the level of financing of current assets, the risk return trade off must be taken into account. The level of current assets can be measured by creating a relationship between current assets and fixed assets. A firm may follow a conservative, aggressive or moderate policy.

A conservative policy means lower return and risk while an aggressive policy produces higher return and risk. The two important aims of the working capital management are profitability and solvency. A liquid firm has less risk of insolvency i.e. it will hardly experience a cash shortage or a stock out situation. However, there is a cost associated with maintaining a sound liquidity position. So, to have a higher profitability the firm may have to sacrifice solvency and maintain a relatively low level of current assets.



ICAI · P6 · MTP Sep 2026 · Series II · question p. 8 · answer p. 11-12

672 MTP May 2026 S-II MCQ 8 1 mark MCQ

The following information is provided by the DVP Ltd. for the year ending 31st March, 2025.

Raw Material storage period	50 days
Work in progress conversion period	18 days
Finished Goods storage period	22 days
Debt Collection period	45 days
Creditors' payment period	55 days

Number of Operating Cycle in a Year (360 days) of DVP Ltd is:

A 4.50

B 3.10

C 4.25

D 2.20

ICAI'S ANSWER

(A) 4.50

Operating Cycle Period = R + W + F + D – C

= 50 + 18 + 22 + 45 – 55 = 80 days

Number of Operating Cycle in a Year

$$= \frac{360}{\text{Operating Cycle Period}} = \frac{360}{80} = 4.5 \text{ times}$$

673 MTP May 2026 S-II Q2(a) 8 marks

A company is in the process of finalising its financial strategy for the upcoming year, with a particular focus on balancing profitability and liquidity through its working capital policies. The management anticipates that the level of investment in current assets — especially inventories and receivables (book debts) will have a direct impact on sales volume and, consequently, on Earnings Before Interest and Tax (EBIT).

For the next financial year, the company has projected its investment in fixed assets at ₹ 2.60 crore, while current liabilities are expected to stand at ₹ 2.34 crore. With these baseline estimates in place, the company must decide how aggressively or conservatively it should invest in current assets.

The Financial Controller has been tasked with evaluating alternative working capital policies. Each policy represents a different level of investment in current assets, which in turn affects sales potential, operating efficiency, risk exposure, and profitability. A higher investment in inventories and receivables may support increased sales and improved customer service but could also lead to higher carrying costs and risk of obsolescence or bad debts. On the other hand, a more conservative approach may reduce costs and risks but could constrain sales growth and customer satisfaction.

In this context, the Financial Controller is analysing various scenarios of working capital investment (expressed in ₹ crore), comparing their impact on sales, EBIT, and overall financial performance. The objective is to identify an optimal policy that strikes a balance between risk and return while ensuring adequate liquidity for smooth business operations.

Working capital policy	Investment in current assets	Estimated sales	EBIT
Conservative	4.50	12.30	1.23
Moderate	3.90	11.50	1.15
Aggressive	2.60	10.00	1.00

After evaluating the working capital policy, the financial controller has advised the adoption of the moderate working capital policy. The company is now examining the use of long-term and short-term borrowings for financing its assets. The company will use ₹ 2.50 crore of the equity funds. The corporate tax rate is 35 per cent. The company is considering the following debt alternatives:

Financing Policy	Short-term debt	Long-term debt
Conservative	0.54	1.12
Moderate	1.00	0.66
Aggressive	1.50	0.16
Interest rate-Average (%)	12	16

Required:

(1) Working capital investment for each policy. (a) Net working capital position, (b) Rate of return, and (c) Current ratio

(2) Financing for each policy. (a) Net working capital position, (b) Rate of return on shareholder equity, and (c) Current ratio

ICAI'S ANSWER

Statement showing Working Capital for each policy

(1) Statement showing working capital investment under each policy

	Particulars	Working capital policy		
		Conservative	**Moderate**	**Aggressive**
(A)	Current assets	4.50	3.90	2.60
(B)	Fixed assets	2.60	2.60	2.60
(C)	Total assets (A) + (B)	7.10	6.50	5.20
(D)	Current liabilities	2.34	2.34	2.34
(E)	Net worth (C) – (D)	4.76	4.16	2.86
(F)	Estimated sales	12.30	11.50	10.00
(G)	EBIT	1.23	1.15	1.00
(i)	Net working capital position (A) – (D)	2.16	1.56	0.26
(ii)	Rate of return (G)/(C) (in per cent)	17.3	17.7	19.2
(ii)	Current ratio (A)/(D)	1.92	1.67	1.11

(2) Statement showing effect of financing under alternative financing policy

(₹ in crores)

	Particulars	Financing policy		
		Conservative	**Moderate**	**Aggressive**
(A)	Current assets	3.90	3.90	3.90
(B)	Fixed assets	2.60	2.60	2.60
(C)	Total assets (A) + (B)	6.50	6.50	6.50
(D)	Current liabilities	2.34	2.34	2.34
(E)	Short-term debt	0.54	1.00	1.50
(F)	Long-term debt	1.12	0.66	0.16
(G)	Equity capital	2.50	2.50	2.50
	Total liabilities (D) + (E) + (F) + (G)	6.50	6.50	6.50
	Estimated sales	11.50	11.50	11.50

	Particulars		Financing policy	
(H)	EBIT	1.15	1.15	1.15
(I)	*Less:* Interest on short-term debt [12% of (E)]	(0.06)	(0.12)	(0.18)
(J)	Interest on long-term debt [16% of (F)]	(0.18)	(0.11)	(0.03)
(K)	EBT [(H) – (I) – (J)]	0.91	0.92	0.94
(L)	*Less:* Taxes @ 35%	(0.32)	(0.32)	(0.33)
(M)	EAT [(K) – (L)]	0.59	0.60	0.61
(a)	Net working capital [(A) – (D) – (E)]	1.02	0.56	0.06
(b)	Rate of return on equity funds [(M)/(G)%]	23.6%	24%	24.4%
(c)	Current ratio [(A)/(D) + (E)	1.35	1.17	1.02

ICAI · P6 · MTP May 2026 · Series II · question p. 6-8 · answer p. 6-7

674 MTP **Jan 2026 S-I** MCQ 7 2 marks MCQ

Calculate number of operating cycles in a year from the information given below:

Raw material inventory consumed during the year	₹ 6,00,000
Average stock of raw material	₹ 41,667
Work-in-progress holding period	18 days
Finished Goods storage period	25 days
Average collection period from debtors	45 days
Average credit period availed	30 days
No. of days in a year	360 days

A 67

B 5.37

C 74

D 4.34

ICAI'S ANSWER

(D) 4.34

1. **Raw-material holding period**

$$\text{RM holding period} = \frac{\text{Average RM stock}}{\text{RM consumed}} \times 360 = \frac{41{,}667}{6{,}00{,}000} \times 360$$

$= 0.06945 \times 360 = 25$ days

2. **Operating Cycle (OC)**

OC = RM period + WIP + FG + Debtors − Creditors

= 25 + 18 + 25 + 45 - 30 = 83 days

3. **Number of Operating Cycles in a Year**

No. of cycles = $\frac{360}{83}$ = 4.34

ICAI · P6 · MTP Jan 2026 · Series I · question p. 4 · answer p. 4

675 MTP **Sep 2025 S-I** Q4(a) 4 marks

A finance executive of XYZ company claims that "Increasing the firm's current ratio by accumulating more inventory is a sound strategy for improving working capital efficiency."

As a financial analyst, do you agree with the statement? Justify your answer using relevant working capital concepts, and explain the potential risks associated with the strategy mentioned.

ICAI'S ANSWER

While accumulating more inventories may increase the current ratio (Current Assets / Current Liabilities), it does not necessarily improve working capital efficiency. The reasons are as follows:

1. **Inefficient Use of Funds:** Increasing inventory ties up cash in unsold goods, which may lead to higher holding costs (storage, insurance, obsolescence).
2. **False Sense of Liquidity:** Inventory is a less liquid current asset compared to cash or receivables. A high current ratio driven by inventory does not ensure actual liquidity.
3. **Decline in Inventory Turnover:** More inventories may reduce the inventory turnover ratio, indicating poor inventory management.
4. **Risk of Obsolescence:** Excess stock might become obsolete, especially in fast-moving industries (e.g., technology, fashion), resulting in losses.
5. **Working Capital Efficiency Metrics:** Efficient working capital management focuses on optimizing the cash conversion cycle (CCC) not just boosting ratios. Stockpiling inventory may increase CCC, reducing efficiency.
   Hence, it may be concluded that increasing inventory inflates the current ratio but harms working capital efficiency by locking funds in slow-moving assets. True efficiency comes from balancing liquidity with turnover, not simply increasing asset size.

ICAI · P6 · MTP Sep 2025 · Series I · question p. 9 · answer p. 16

676 MTP **Jan 2025 S-I** Q4(a) 4 marks

A company is evaluating two options for financing its current assets: using short-term loans or long-term loans. HOW should the company balance risk and return in making this decision, and WHAT factors should it consider to ensure optimal financing?

ICAI'S ANSWER

The financing of current assets involves a trade off between risk and return. A firm can choose from short or long term sources of finance. Short term financing is less expensive than long term financing but at the same time, short term financing involves greater risk than long term financing.

Depending on the mix of short term and long term financing, the approach followed by a company may be referred as matching approach, conservative approach and aggressive approach.

In matching approach, long-term finance is used to finance fixed assets and permanent current assets and short term financing to finance temporary or variable current assets. Under the conservative plan, the firm finances its permanent assets and also a part of temporary current assets with long term financing and hence less risk of facing the problem of shortage of funds.

An aggressive policy is said to be followed by the firm when it uses more short term financing than warranted by the matching plan and finances a part of its permanent current assets with short term financing.

ICAI · P6 · MTP Jan 2025 · Series I · question p. 6 · answer p. 11

677 MTP **Old P8 Nov 2023 S-II** Q5(b) 2 marks old scheme

The following information is provided by the Shrishti Ltd. for the year ending 31st March 2022.

Raw Material storage period	54 days
Work in progress conversion period	20 days
Finished Goods storage period	22 days
Debt Collection period	74 days
Creditors' payment period	25 days
Annual Operating Cost	45 crore

(Including depreciation of ₹42,00,000)

(1 year = 360 days)

You are required to CALCULATE Operating Cycle period and Number of Operating Cycles in a year.

ICAI'S ANSWER

Calculation of Operating Cycle Period and number of Operating Cycle in a Year

Operating Cycle Period = R + W + F + D – C

= 54 + 20 + 22 + 74 – 25 = 145 days

$$\text{Number of Operating Cycle in a Year} = \frac{360}{\text{Operating Cycle Period}}$$

= 360/145 = 2.48 times

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 3-4 · answer p. 8

678 MTP **Old P8 May 2022 S-I** Q3 10 marks old scheme

Following information is forecasted by Gween Limited for the year ending 31st March, 2022:

	Balance as at 31st March, 2022	Balance as at 31st March, 2021
	(₹ in lakh)	(₹ in lakh)
Raw Material	845	585
Work-in-progress	663	455

	Balance as at 31st March, 2022	Balance as at 31st March, 2021
Finished goods	910	780
Receivables	1,755	1,456
Payables	923	884
Annual purchases of raw material (all credit)	5,200	
Annual cost of production	5,850	
Annual cost of goods sold	6,825	
Annual operating cost	4,225	
Annual sales (all credit)	7,605	

Considering one year as equal to 365 days, CALCULATE:

(i) Net operating cycle period.

(ii) Number of operating cycles in the year.

(iii) Amount of working capital requirement.

ICAI'S ANSWER

Working Notes:

1. **Raw Material Storage Period (R)**

$$= \frac{\text{Average Stock of Raw Material}}{\text{Annual Consumption of Raw Material}} \times 365$$

$$= \frac{\frac{₹\ 585 + ₹\ 845}{2}}{₹\ 4{,}940} \times 365 = 53 \text{ days}$$

Annual Consumption of Raw Material = Opening Stock + Purchases - Closing Stock

= ₹ 585 + ₹ 5,200 – ₹ 845 = ₹ 4,940 lakh

2. **Work – in - Progress (WIP) Conversion Period (W)**

$$= \frac{\text{Average Stock of WIP}}{\text{Annual Cost of Production}} \times 365$$

$$= \frac{\frac{₹\ 455 + ₹\ 663}{2}}{₹\ 5{,}850} \times 365 = 35 \text{ days}$$

3. **Finished Stock Storage Period (F)**

$$= \frac{\text{Average Stock of Finished Goods}}{\text{Cost of Goods Sold}} \times 365$$

$$= \frac{\frac{₹\ 780 + ₹\ 910}{2}}{₹\ 6{,}825} \times 365 = 45 \text{ days.}$$

4. **Receivables (Debtors) Collection Period (D)**

$$= \frac{\text{Average Receivables}}{\text{Annual Credit Sales}} \times 365$$

$$= \frac{\frac{₹\ 1{,}456 + ₹\ 1{,}755}{2}}{₹\ 7{,}605} \times 365 = 77 \text{ days}$$

5. **Payables (Creditors) Payment Period (C)**

$$= \frac{\text{Average Payables for materials}}{\text{Annual Credit purchases}} \times 365$$

$$= \frac{\frac{₹\ 884 + ₹\ 923}{2}}{₹\ 5{,}200} \times 365 = 64 \text{ days}$$

(i) Net Operating Cycle Period

= R + W + F + D - C

= 53 + 35 + 45 + 77 – 64 = 146 days

(ii) Number of Operating Cycles in the Year

$$= \frac{365}{\text{Operating Cycle Period}} = \frac{365}{146} = 2.5 \text{ times}$$

(iii) Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{₹\ 4{,}225}{2.5} = ₹\ 1{,}690 \text{ lakh}$$

Note: Number of days may vary due to fraction.

ICAI · Old P8 · MTP May 2022 · Series I · question p. 2-3 · answer p. 4-5

679 MTP **Old P8 Nov 2020 S-I** Q6(c) 2 marks old scheme

STATE any four factors which need to be considered while planning for working capital requirement.

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 4

680 MTP **Old P8 May 2020 S-I** Q5(b) 2 marks old scheme

The following information is provided by the P Ltd. for the year ending 31st March, 2020.

Raw Material storage period	52 days
Work in progress conversion period	18 days
Finished Goods storage period	20 days
Debt Collection period	75 days
Creditors' payment period	25 days
Annual Operating Cost	45 crore

(Including depreciation of Rs.42,00,000)

(1 year = 360 days)

You are required to CALCULATE Operating Cycle period and Number of Operating Cycles in a year.

ICAI'S ANSWER

Calculation of Operating Cycle Period and number of Operating Cycle in a Year

Operating Cycle Period = R + W + F + D – C

= 52 + 18 + 20 + 75 – 25 = 140 days

$$\text{Number of Operating Cycle in a Year} = \frac{360}{\text{Operating Cycle Period}}$$

= 360/140 = 2.57 times

ICAI · Old P8 · MTP May 2020 · Series I · question p. 4 · answer p. 11

9.2 Estimating working capital requirement

49 questions · asked in 28 ICAI documents · 6 exam papers

681 EXAM Sep 2026 Q1(c) 5 marks

PQR Company Ltd. has applied for a short term loan to a commercial bank for financing its working capital requirement.

The following figures for the 12 months ending 31st March, 2026 are given to ascertain the requirement of working capital of the company on a cash cost basis.

Sales at 2 months credit	₹ 30,00,000
Materials consumed (suppliers credit is for 2 months)	₹ 7,50,000
Total wages (Paid at the beginning of next month)	₹ 6,00,000
Manufacturing expenses outstanding at the end of this year. These expenses are paid one month in arrears.	₹ 67,500
Total administration expenses	₹ 1,95,000
Sales promotion expenses paid quarterly and in advance	₹ 1,00,000
Rate of gross profit	20%
Cash is to be held to the extent of 50% of current liabilities	
Safety margin to be maintained	15%
These will be no work-in-progress	
Stock of Raw Materials and finished goods are kept at one month's requirement.	

You are required to prepare an estimate for the requirement of the working capital of the company on cash cost basis for the period ending 31st March, 2026.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 15-16

682 EXAM May 2025 Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

XYZ Ltd. wants to establish a shoe manufacturing unit. To setup this unit, it needs a loan. XYZ Ltd, approaches a commercial bank for working capital loan.

Bank has asked the company to present the proposal for such loan. To prepare such proposal, the company has appointed you and provided some information about the plan.

It wants to maintain margin of safety of 10% for contingencies. The company want to keep cash balance of ₹ 90,000.

The product will be sold at gross profit margin of 25% on COGS. Depreciation will be part of cost of production.

Stock of raw material will be held at 1.5 months of its consumption, while finished goods inventory will be maintained at one month's requirement. Finished goods will be valued at manufacturing cost.

The company will sell shoes in domestic as well as in foreign market. Total estimated annual sales will be ₹ 30,00,000. 20% of the sales are foreign sales and there will be a delay of 1/2 month to realise the sales proceed from foreign debtors. All domestic sales will be on credit and allowed a credit period of 2 months.

Raw material consumed will be 25% of estimated sales and the company will get credit period of 2 months of consumption from its supplier of raw material. Wage expenses will be 20% of sales and will be paid in 1/2 month's lag.

Outstanding cash manufacturing overhead expenses at the end of the year will be ₹ 60,000; it will be paid in one month arrear. Estimated total administrative cost (to be paid after one month) will be ₹ 1,80,000. Expected annual sales promotion expenses are ₹ 90,000, which will be paid quarterly in advance. The company want to submit the proposal to bank on cash cost basis.

You are required to answer the following question 1 to 5:

MCQ 1 2 marks

What will be the total manufacturing cost and total cost of sales on cash cost basis?

A ₹ 20,70,000, ₹ 23,40,000

B ₹ 24,00,000, ₹ 30,00,000

C ₹ 22,50,000, ₹ 23,40,000

D ₹ 16,80,000, ₹ 24,00,000

MCQ 2 2 marks

What will be the total estimated finished goods inventory and total debtors on cash cost basis?

A ₹ 2,00,000, ₹ 4,25,000

B ₹ 1,72,500, ₹ 3,40,000

C ₹ 1,72,500, ₹ 3,31,500

D ₹ 1,87,500, ₹ 3,31,500

MCQ 3 2 marks

What will be the total estimated current assets on cash cost basis?

A ₹ 8,31,250

B ₹ 7,18,750

C ₹ 7,25,250

D ₹ 7,10,250

MCQ 4 2 marks

What will be the total estimated current liabilities?

A ₹ 2,52,500

B ₹ 1,90,000

C ₹ 2,17,500

D ₹ 2,25,000

MCQ 5 2 marks

What will be the estimated working capital to be submitted by you in the proposal?

A ₹ 5,33,775

B ₹ 4,36,725

C ₹ 5,50,275

D ₹ 4,50,225

ICAI'S ANSWERS

MCQ 1

Answer key: **(A)**

MCQ 2

Answer key: **(C)**

MCQ 3

Answer key: **(D)**

MCQ 4

Answer key: **(D)**

MCQ 5

Answer key: **(A)**

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683 EXAM **Jan 2025** MCQ 8 MCQ

JKL Company Ltd. has current assets of ₹ 1,00,00,000 and current liabilities of ₹ 50,00,000. The Financial manager of the company desires to make a provision for contingencies @20% of working capital. Accordingly, what would be the amount of working capital requirement for the company?

A ₹ 55 Lakh

B ₹ 60 Lakh

C ₹ 65 Lakh

D ₹ 70 Lakh

ICAI'S ANSWER

Answer key: **(B)**

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684 EXAM Sep 2024 Q1(b) 5 marks

The following information is available for SK Limited for the year ended on 31st March,2024:

Particulars	₹
Cost of production	15,48,000
Cost of goods sold	14,61,000
Average stock of work-in-progress	94,600
Average stock of finished goods	2,43,500
Administration and Selling expenses	4,14,000
Receivables collection period	36 days
Raw Material Storage period	65 days
Creditors payment period	63 days

You are required to calculate the working capital requirement by operating cycle method. Assume a 360 days year.

ICAI'S ANSWER

Operating Cycle = R + W + F + D – C

Where,

R = Raw material storage period = 65 days

W = Work-in-progress inventory* holding period

F = Finished goods storage period

D = Receivables (Debtors) collection period = 36 days

C = Credit period allowed by suppliers (Creditors) = 63 days

Work-in-progress inventory holding period (W)

$$= \frac{\text{Average Work -in-progress inventory}}{\text{Average Cost of Production per day}}$$

$$= \frac{₹94{,}600}{₹15{,}48{,}000 \div 360 \text{ days}} = \mathbf{22 \text{ days}}$$

Finished Goods storage period (F)

$$= \frac{\text{Average stock of finished goods}}{\text{Average Cost of Goods Sold per day}}$$

$$= \frac{₹2{,}43{,}500}{₹14{,}61{,}000 \div 360 \text{ days}} = \mathbf{60 \text{ days}}$$

Net Operating Cycle = 65+22+60+36-63 = **120 days**

Number of Operating Cycles in a year

$$= \frac{\text{No. of days in a year}}{\text{Operating Cycle period}}$$

$$= \frac{360 \text{ days}}{120 \text{ days}} = \mathbf{3 \text{ times}}$$

Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{₹14,61,000 + ₹4,14,000}{3} = ₹\ 6,25,000$$

ICAI · P6 · Sep 2024 exam · question p. 10 · answer p. 12-13

685 EXAM Old P8 Nov 2023 Q1(d) 5 marks old scheme

X Ltd. has furnished following cost sheet of per unit cost;

Raw material cost	₹ 150
Direct labour cost	₹ 40
Overhead cost	₹ 60
Total Cost	₹ 250
Profit	₹ 50
Selling Price	₹ 300

The company keeps raw material in stock on an average for 2 months; work in progress on an average for 3 months and finished goods in stock on an average 1 month. The credit allowed by suppliers is 1.5 months and company allows 2 months credit to its debtors. The lag in payment of wages is 1 month and lag in payment of overhead expenses is 1.5 months. The company sells 25% of the output against cash and maintain cash in hand at bank put together at ₹ 1,50,000. Production is carried on evenly throughout the year and wages and overheads also similarly. Work in progress stock is 75% complete in all respects. Prepare statement showing estimate of working capital requirements to finance an activity level of 15,000 units of production.

ICAI'S ANSWER

Statement showing Estimate of Working Capital Needs

(Receivables (Debtors) are calculated based on Cost of goods sold)

		(₹)	(₹)
A.	**Current Assets**		
(i)	Inventories:		
	Raw material (2 months) (15,000 units × ₹ 150/12 months × 2 months)	3,75,000	
	WIP Inventory (3 months) (15,000 units × ₹ 250/12 months × 3 months) × 0.75	7,03,125	
	Finished goods inventory (1 months) (15,000 units × ₹ 250/12 months × 1months)	3,12,500	13,90,625
(ii)	Receivables (Debtors) (2 months) (15,000 units × ₹ 250/12 months × 2 months) x 0.75		4,68,750
(iii)	Cash and bank balance		1,50,000

		(₹)	(₹)
	Total Current Assets		20,09,375
B.	**Current Liabilities:**		
(i)	Payables (Creditors) for materials (1.5 months) (15,000 units × ₹ 150/12 months × 1.5 months)		2,81,250
(ii)	Outstanding wages (1 months) (15,000 units × ₹ 40/12 months × 1months)		50,000
(iii)	Outstanding overheads (1.5 months) (15,000 units × ₹ 60/12 months × 1.5 months)		1,12,500
	Total Current Liabilities		4,43,750
	Net Working Capital Needs (A – B)		15,65,625

Alternative Solution

Statement showing Estimate of Working Capital Needs

(Receivables (Debtors) are calculated based on Selling price)

		(₹)	(₹)
A.	**Current Assets**		
(i)	Inventories:		
	Raw material (2 months) (15,000 units × ₹ 150/12 months × 2 months)	3,75,000	
	WIP Inventory (3 months) (15,000 units × ₹ 250/12 months × 3 months) × 0.75	7,03,125	
	Finished goods inventory (1 months) (15,000 units × ₹ 250/12 months × 1months)	3,12,500	13,90,625
(ii)	Receivables (Debtors) (2 months) (15,000 units × ₹ 300/12 months × 2 months) x 0.75		5,62,500
(iii)	Cash and bank balance		1,50,000
	Total Current Assets		21,03,125
B.	**Current Liabilities:**		
(i)	Payables (Creditors) for materials (1.5 months) (15,000 units × ₹ 150/12 months × 1.5 months)		2,81,250
(ii)	Outstanding wages (1 months) (15,000 units × ₹ 40/12 months × 1months)		50,000
(iii)	Outstanding overheads (1.5 months) (15,000 units × ₹ 60/12 months × 1.5 months)		1,12,500

	(₹)	(₹)
Total Current Liabilities		4,43,750
Net Working Capital Needs (A – B)		16,59,375

ICAI · Old P8 · Nov 2023 exam · question p. 2 · answer p. 5-7

686 EXAM Old P8 Nov 2020 Q2 10 marks old scheme

PK Ltd., a manufacturing company, provides the following information:

	(₹)
Sales	1,08,00,000
Raw Material Consumed	27,00,000
Labour Paid	21,60,000
Manufacturing Overhead (Including Depreciation for the year ₹ 3,60,000)	32,40,000
Administrative & Selling Overhead	10,80,000

Additional Information:

(a) Receivables are allowed 3 months' credit.

(b) Raw Material Supplier extends 3 months' credit.

(c) Lag in payment of Labour is 1 month.

(d) Manufacturing Overhead are paid one month in arrear.

(e) Administrative & Selling Overhead is paid 1 month advance.

(f) Inventory holding period of Raw Material & Finished Goods are of 3 months.

(g) Work-in-Progress is Nil.

(h) PK Ltd. sells goods at Cost plus 33⅓%.

(i) Cash Balance ₹ 3,00,000.

(j) Safety Margin 10%.

You are required to compute the Working Capital Requirements of PK Ltd. on Cash Cost basis.

ICAI'S ANSWER

Statement showing the requirements of Working Capital (Cash Cost basis)

Particulars	(₹)	(₹)
A. Current Assets:		
Inventory:		
Stock of Raw material (₹ 27,00,000 × 3/12)	6,75,000	
Stock of Finished goods (₹ 77,40,000 × 3/12)	19,35,000	

Particulars	(₹)	(₹)
Receivables (₹ 88,20,000 × 3/12)	22,05,000	
Administrative and Selling Overhead (₹ 10,80,000 × 1/12)	90,000	
Cash in Hand	3,00,000	
Gross Working Capital	52,05,000	**52,05,000**
B. Current Liabilities:		
Payables for Raw materials* (₹ 27,00,000 × 3/12)	6,75,000	
Outstanding Expenses:		
Wages Expenses (₹ 21,60,000 × 1/12)	1,80,000	
Manufacturing Overhead (₹ 28,80,000 × 1/12)	2,40,000	
Total Current Liabilities	10,95,000	10,95,000
Net Working Capital (A-B)		**41,10,000**
Add: Safety margin @ 10%		4,11,000
Total Working Capital requirements		**45,21,000**

Working Notes:

(i)

(A)	Computation of Annual Cash Cost of Production	(₹)
	Raw Material consumed	27,00,000
	Wages (Labour paid)	21,60,000
	Manufacturing overhead (₹ 32,40,000 - ₹ 3,60,000)	28,80,000
	Total cash cost of production	77,40,000
(B)	**Computation of Annual Cash Cost of Sales**	**(₹)**
	Cash cost of production as in (A) above	77,40,000
	Administrative & Selling overhead	10,80,000
	Total cash cost of sales	88,20,000

*Purchase of Raw material can also be calculated by adjusting Closing Stock and Opening Stock (assumed nil). In that case Purchase will be Raw material consumed +Closing Stock-Opening Stock i.e ₹27,00,000 + ₹6,75,000 - Nil = ₹33,75,000. Accordingly, Total Working Capital requirements (₹ 43,35,375) can be calculated.

ICAI · Old P8 · Nov 2020 exam · question p. 6 · answer p. 6-7

687 RTP Jan 2026 Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

PPW Ltd is a mid-sized manufacturing company producing household appliances for the Indian market. With competition heating up, the company's management is becoming increasingly conscious about liquidity management and efficient use of working capital.

During the last financial year, the company recorded annual sales of ₹ 48,00,000, offering its customers a 3-month credit period to encourage bulk orders. While this has boosted sales, it has also tied up significant funds in debtors.

On the production side, the company incurs the following annual costs:

- **Materials consumed**: ₹ 12,00,000 (suppliers allow 2 months' credit)
- **Wages**: ₹ 9,60,000 (paid with a 1-month lag)
- **Cash manufacturing expenses**: ₹ 12,00,000 (settled 1 month in arrears)
- **Administrative expenses**: ₹ 3,60,000 (paid with a 1-month lag)
- **Sales promotion expenses**: ₹ 1,20,000 (paid monthly in advance to agencies)

The company sells its products at a gross profit margin of 20%. It also maintains 2 months' stock of raw materials and 2 months' stock of finished goods to ensure smooth production and timely delivery to distributors.

Further, depreciation is considered part of the cost of production, but since it does not require a cash outflow, management wants to ignore it while assessing the cash cost basis of working capital. A minimum cash balance of ₹ 1,00,000 is always retained for contingencies.

Now, the Board of Directors has asked the finance team to answer the following 5 questions:

Q1(i)

What will be the gross working capital of the company on cash cost basis?

A ₹ 16,00,000

B ₹ 17,20,000

C ₹ 18,30,000

D ₹ 19,60,000

Q1(ii)

PPW Ltd wants to compute its working capital requirement on a cash cost basis (ignoring WIP), assuming a 15% margin. Based on the given financial data, what will be the working capital requirement?

A ₹ 16,33,000

B ₹ 10,20,000

C ₹ 20,40,000

D ₹ 9,60,000

Q1(iii)

Excluding the 15% margin, what is the Current Ratio of PPW Ltd?

A 4.46 : 1

B 3.8 : 1

C 2.0 : 1

D 2.2 : 1

Q1(iv)

What is the Quick Ratio of PPW Ltd (excluding the 15% margin)?

A 2.92 : 1

B 1.05 : 1

C 2.59 : 1

D 1.85 : 1

Q1(v)

What is the Working Capital Turnover Ratio based on cash cost of sales?

A 4 times

B 3.38 times

C 6.2 times

D 2.7 times

ICAI'S ANSWERS

Q1(i)

(C) ₹18,30,000

Q1(ii)

(A) ₹16,33,000

Working Notes:

(i) Computation of Annual Cash Cost of Production	(₹)
Material consumed	12,00,000
Wages	9,60,000
Manufacturing expenses	12,00,000
Total cash cost of production	**33,60,000**
(ii) Computation of Annual Cash Cost of Sales:	**(₹)**
Total Cash cost of production as in (i) above	33,60,000
Administrative Expenses	3,60,000
Sales promotion expenses	1,20,000
Total cash cost of sales	**38,40,000**
Add: Gross Profit @ 20% on sales (25% on cost of sales)	9,60,000
Sales Value	**48,00,000**

Statement of Working Capital requirements (cash cost basis)

	(₹)	(₹)
A. Current Assets		
Inventory:		
- Raw materials (₹ 12,00,000/12months ×2 months)	2,00,000	
- Finished Goods (₹ 33,60,000/12months ×2 months)	5,60,000	
Receivables (Debtors) (₹ 38, 40,000/12months ×3 months)	9,60,000	
Sales Promotion expenses paid in advance (₹ 1,20,000/12 months ×1 month)	10,000	
Cash balance	1,00,000	18,30,000
Gross Working Capital		18,30,000
B. Current Liabilities:		
Payables:		
-Creditors for materials (₹ 12,00,000/12months ×2 months)	2,00,000	
Wages outstanding (₹ 9,60,000/12months ×1 month)	80,000	
Manufacturing expenses outstanding (₹ 12,00,000/12months ×1 month)	1,00,000	
Administrative expenses outstanding (₹ 3,60,000/12months ×1 month)	30,000	4,10,000
Net working capital (A - B)		14,20,000
Add: Safety margin @ 15%		2,13,000
Total Working Capital requirements		16,33,000

Q1(iii)

(A) 4.46 : 1

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$= \frac{18,30,000}{4,10,000} = 4.46$$

Q1(iv)

(C) 2.59 : 1

$$\text{Quick Ratio or Acid Test Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

$$\text{Quick Assets} = \text{Current Assets} - \text{Inventories} - \text{Prepaid expenses}$$

$$\text{Quick Ratio} = \frac{9,60,000 + 1,00,000}{4,10,000}$$

Or,

$$= \frac{18,30,000 - 2,00,000 - 5,60,000 - 10,000}{4,10,000}$$

$$= 2.59$$

Q1(v)

(D) 2.7 times

$$\text{Working Capital Turnover Ratio} = \frac{\text{Sales / Cost of Goods Sold}}{\text{Working Capital excluding Margin}}$$

$$= \frac{38,40,000}{14,20,000}$$

$$= 2.70$$

ICAI · P6 · RTP Jan 2026 · question p. 1-2 · answer p. 12, 12-13, 13, 14

688 RTP **May 2025** Q9

The management of Parshvam Limited is planning to expand its business at international level and consults you for preparing and estimation of working capital needs so that they can avail the finance from the bank. The estimated data of Parshvam Limited reveal the following information –

Particulars	Amount (₹)
Materials Used	
Domestic on 2 months credit	9,00,000
Imports on 3 months credit **	6,00,000
Lag in Payment of wages - 1 month	6,00,000
Lag in Payment of Manufacturing Overheads – ½ Month	26,40,000
Sales	
Domestic on 1.5 months credit	30,00,000
Export on 3 months credit (sale price 10% below domestic price)	24,80,000
Administrative expenses payable in advance for 2 months	3,60,000
Lag in payment of Selling & Distribution expenses – 1 month	3,00,000

Advance Income tax for ₹25,000 for the quarter falling in the next financial year is paid by the company. Manufacturing overheads is inclusive of depreciation on the new machine purchased for tailor-made export products. The purchase price for the new machine is ₹24,00,000 with a depreciation rate of 10%. Cash Gross profit is at 20% on domestic sales.

However, to promote exports, Export Promotion Council (EPC Board) provides a revenue subsidy of 2.5% for the new machine purchased. Furthermore, Parshvam Limited submits the letter of credit (LOC) to its bank and avails the all-Export Sales value within 1 month. Financial institution charges a fee of 5% for the same.

The company keeps one month stock of raw materials and finished goods each. Goods remain in process for half a month with 90% raw materials introduced in the process. The company believes in keeping cash and bank balance of ₹1,50,000. The management is of the opinion that the safety margin is to be kept at 15%.

**Raw materials imported will attract a custom duty at 20% to be paid up front with a duty drawback of 5% credited upfront. You are required to -

(A) PREPARE the estimated working capital statement for the next year.

(B) ADVISE whether Parshvam Limited should continue with the export business or not.

(Requisite assumptions and notes should form part of the solution).

ICAI'S ANSWER

(A) Statement for estimation of Working Capital using Cash Cost Basis

Parshvam Limited

Particulars	Amount (₹)	Amount (₹)
(A) Current Assets		
1. Raw materials 15,90,000 × 1/12	1,32,500	
2. WIP		
~ RM 15,90,000 × 0.5 /12 × 90%	59,625	
~ Wages 6,00,000 × 0.5 /12 × 50%	12,500	
~ Manufacturing OH 23,40,000 × 0.5 /12 × 50%	48,750	
~ Other OH 74,472 × 0.5 /12 × 50%	1,552	
3. FG (on COGS) 46,04,472 × 1/12	3,83,706	
4. Debtors		
~ Domestic 27,61,314 × 1.5/12	3,45,164	
~ Export 26,27,158 X 1/12 **WN - 5**	2,18,930	
5. Cash/bank balance (given)	1,50,000	
6. Prepaid admin exp 3,60,000 × 2/12	60,000	
7. Income tax paid in advance (given)	25,000	
Gross working capital		**14,37,727**
(B) Current Liabilities		
1. Creditors		
~ Domestic 9,00,000 × 2/12	1,50,000	
~ Import 6,00,000 × 3/12	1,50,000	
2. Lag in wages payment 6,00,000 × 1/12	50,000	
3. Lag in manufacturing OH 23,40,000 × 0.5/12	97,500	
Lag in other OH 74,472 × 0.5/12	3,103	
4. Lag in S&D exp 3,00,000 × 1/12	25,000	

Particulars	Amount (₹)	Amount (₹)
Excess of CA over CL		**9,62,124**
***Add:* 15% safety margin (9,87,124 × 15%)**		**1,44,319**
Net working capital		**11,06,443**

Notes –

(a) Working Capital is estimated on Cash Cost Basis

(b) Other Overheads are assumed to be the part of production.

(c) In absence of information on % completion for wages, manufacturing and other overheads, it is assumed to be 50% complete for the purpose of calculating WIP.

(d) Other Overheads are also assumed to be outstanding for a period of ½ month. In absence of specific information, it can also be assumed that nothing is outstanding or prepaid.

(B) If just the monetary aspects and factors are considered then, Parshvam limited should discontinue its operations at international level as the Cash Cost of sales for export at ₹ 26,27,158 is higher than the Export sales value which is just ₹ 24,80,000. In reality, non-monetary factors are also considered in decision making; exports will add a new customer base for the company. Furthermore, existence at international level brings on a high credibility and image to the company, etc.

WN 1 - Calculation of gross profit on Export Sales:

Let the domestic selling price be ₹100.

Therefore, Gross profit = ₹20, and cost per unit = ₹ 80

Now as given, Export price is 10% less than the domestic price = 100 – 10% = ₹ 90. However, the cost per unit to produce exported goods will remain at ₹ 80 only.

So gross profit on exports will be ₹ 90 - 80 = ₹ 10.

Therefore, Gross profit in % for Export Sales = 10 / 90 = 11.11%

	Domestic	Export	Total
Sales	30,00,000	24,80,000	54,80,000
Less: Gross Profit 20% for Domestic 11.11% for Export	(6,00,000)	(2,75,528)	(8,75,528)
COGS	24,00,000	22,04,472	46,04,472
Add: Admin Exp (To be Apportioned in the ratio of Sales)	1,97,080	1,62,920	3,60,000
Add: S&D Expense (To be Apportioned in the ratio of Sales)	1,64,234	1,35,766	3,00,000
Add: Bank Fees and charges for providing LOC services	-	1,24,000	1,24,000
Cash Cost of Sales	**27,61,314**	**26,27,158**	**53,88,472**

WN 2 - Preparation of Cost/Income Statement

Particulars	Amount (₹)
Raw Materials	
Domestic	9,00,000
Import **WN - 3**	6,90,000
Wages	6,00,000
Manufacturing Overheads (Cash) **WN - 4**	23,40,000
Other Overheads (Bal. Fig)	74,472
Cost of Production/Cost of Goods Sold	**46,04,472**
Add: Admin Exp	3,60,000
Add: S&D Exp	3,00,000
Add: Bank charges & Fees for L.O.C services	1,24,000
Cost of Sales	**53,88,472**

WN 3 – Calculation of Raw Materials Purchased - Imports

Purchase Price = ₹ 6,00,000

+ Custom Duty @ 20% = ₹ 1,20,000

(-) Upfront Duty Drawback @ 5% = ₹ (30,000)

Total Value of Raw materials = ₹ 6,90,000

WN 4 – Calculation of Cash Manufacturing Overheads

Manufacturing Overheads = ₹ 26,40,000

Less: Depreciation on Machinery

(24,00,000 x 10%) = ₹ (2,40,000)

Less: Revenue Subsidy from EPC Board ***

(24,00,000 x 2.5%) = ₹ (60,000)

Cash Manufacturing Overheads = ₹ 23,40,000

***Revenue subsidy shall not be capitalized but instead it will result in bringing down your manufacturing expenses which is revenue in nature. Had the it been the capital subsidy, then it would have reduced the purchase price of the machine and thereby changing the amount of depreciation.

WN 5 - Credit Period for Export customers

Since the company is availing benefit of Letter of Credit (L.O.C), the funds blocked in export customers would only be for 1 month and not 3 months; as the company would receive the entire Export Sales value in 1 month's time from the financial institution after paying the bank charges and fees.

ICAI · P6 · RTP May 2025 · question p. 8-10 · answer p. 26-29

689 RTP Jan 2025 Integrated Case Scenario · Q1(i) to Q1(v)

INTEGRATED CASE SCENARIO

Samvar Ltd, a leading FMCG company having its current presence in more than 150 Tier I and Tier II cities in India. The stores are operating in the brand name of GoMART competing with Reliance fresh, Walmart, BigBazaar and other chains. Owing to the increase in demand from Tier III cities and rural areas, it is planning for massive expansion and is contemplating to open up additional 50 stores which will have variety of FMCG products.

The CFO and his team estimate that the funds needed for massive expansion would be ₹ 200 lakhs per store. Such funds would be utilized for buying out a space and setting up a store, buying the other required fixed assets, etc. Central government will provide a revenue subsidy of 15% on Gross profit if the overall cost of capital doesn't exceed 10%

Apart from above, CFO and his team require an estimate on the additional capital needed based for the smooth running of fixed assets and its daily operations. Based on their market research, they have collected the other information for each store which is as follows-

Average Sales would be ₹ 120 lakhs p.a. with a GP margin of 18%. Customers pay through different digital modes and channels including POS systems (Debit and credit cards) which generally takes approx. 9 days for the funds to get credited in the bank account. 15% of the customers use debit and credit cards to make the payment. Installing a POS system comes with a fee of 2% of total sales through POS.

Being a FMCG outlet, inventories of multiple products need to be kept. Different products have different storage period. However primarily, products are classified into three broad categories, Durable, Semi Durable & Perishable. Perishable products comprise 60% of sales, whereas semi-durable is 25% and balance is for durable products. Inventory storage period for perishable, semi-durable & durable products are 10 days, 30 days & 60 days respectively. Suppliers of these products provide a credit period of average 30 days.

Each store will employ around 20 personnel of a different hierarchy and monthly average salaries to staff for each store is estimated at ₹ 4 lakhs per month. Company will pay employees' dues on the 1st of next month.

Samvar Ltd plans to keep optimum cash balance in hand as suggested by Baumol's model. Excess cash balance if any, will be invested in the marketable securities which will generate a return of 12% p.a. The total disbursement for the year is estimated at ₹ 1.50 lakhs per month with the transaction cost of ₹ 20 per transfer to the disbursement account.

The optimum capital structure with debt equity of 2:1 has been proven ideal for raising the finance and company wishes to follow the same pattern for the additional funds required for each store. Trade credit can also be utilized for financing the expansion needs.

The cost of raising debt and equity for each store is as per the slabs as under:

Project Cost *	Cost of Debt	Minimum rate expected by equity share holders
Upto 80 lakhs	10%	12.5%
Above 80 lakhs but upto 150 Lakhs	11.5%	13.5%
Above 150 lakhs & Upto 250 lakhs	12%	14%
Above 250 lakhs	13.5%	15%

*It means that upto 80 lakhs of project cost company can raise debt at 10% and equity at 12.5% and so on.

Tax rate applicable to the corporate is 25%

Based on the above details, calculate the following for each store:

Q1(i)

The optimum Cash balance is

A ₹ 7,071

B ₹ 26,500

C ₹ 7,150

D ₹ 24,495

Q1(ii)

The Gross and Net Working Capital for the next year would be

A ₹ 6.7730 L, (5.9396 L)

B ₹ 6.7730 L, 12.7125 L

C ₹ 200 L, (5.9396L)

D ₹ (5.9396 L), 6.7730 L

Q1(iii)

The amount of total funds needed to setup a store is

A ₹ 194.0605 L

B ₹ 200 L

C ₹ 6.7730 L

D ₹ 206.7730 L

Q1(iv)

The overall cost of capital for raising additional funds for setting up of each store is

A 10.01%

B 10.65%

C 9.90%

D 8.91%

Q1(v)

The amount of revenue subsidy granted by the central govt is

A ₹3L

B ₹ 3.24 L

C Nil

D ₹ 2.25 L

ICAI'S ANSWERS

Q1(i)

(D) ₹ 24,495

As per William J Baumol,

optimum cash balance = $\sqrt{\frac{2AT}{O}}$

A = Annual Cash disbursement

T = Cost per transfer

O = Opportunity cost

= $\sqrt{\frac{2 \text{ X}18{,}00{,}000 \text{ X } 20}{0.12}}$ **= ₹ 24,495/-**

Q1(ii)

(A) ₹6.7730 L, (5.9396 L)

Gross working capital is sum of total current assets and net working capital is Gross working capital less current liabilities.

Estimation of Working Capital Statement

	Amount (₹)	Amount (₹)
A) CURRENT ASSETS / GROSS W.C		
1. FG Inventory **WN - 1**	6,15,000	
2. Trade receivables **WN - 2**	37,800	
3. Cash/ bank balance **(Calculated in Solution 1)**	24,495	**6,77,295**
B) CURRENT LIABILTIES		
1. Trade payables **WN - 3**	8,71,250	
2. Outstanding salaries **WN - 4**	4,00,000	**12,71,250**
NET WORKING CAPITAL (A) - (B)		**(5,93,955)**

WN – 1 Calculation of FG Inventory

FG Inventory = COGS x $\frac{\text{FG STORAGE PERIOD (DAYS)}}{360}$

COGS = 120 Lakhs x 82% = 98.40 Lakhs

Perishable = 98.40 x 60% x 10 /360 = 1.64 Lakhs

Semi Durable = 98.40 X 25% X 30 /360 = 2.05 Lakhs

Durable = 98.40 X 15% X 60 /360 = 2.46 Lakhs

Total = **₹ 6.15 lakhs**

WN – 2 Calculation of Trade Receivables

Since, company is into FMCG industry, sales are always on cash basis as no credit is given to any of the custome₹ However, as mentioned in the case study, company will get the credit in the bank account only after 9 days for those customers that pay through POS (debit and credit cards). It means companies funds' get blocked for 9 days.

Company's trade receivable would only comprise of 15% of total sales as rest are through cash basis

Trade Receivables = Cost of Sales x Days Blocked / 360

= 15.12 L x 9 / 360

= **0.378 Lakhs**

Cost of Sales = COGS + POS Transaction fees

= (98.40 L x 0.15) + (120 L x 0.15 x 2%)

= 15.12 Lakhs

WN – 3 Calculation of Trade Payables

Trade Payables = Purchases x $\frac{\text{Average Credit period in days}}{360}$

= 104.55 x 30 /360

= **8.7125 Lakhs**

Purchases = COGS (+) Closing Stock (-) Opening Stock

Since, company is planning to open up new store, its opening stock would be NIL but there would be definitely a closing FG stock which is calculated in WN -1

Therefore, Purchases = 98.40 L + 6.15 L - 0 = **104.55 Lakhs**

WN – 4 Calculation of Outstanding salaries

Salaries are paid on 1st of next month, thereby meaning it has been outstanding for a period of 30 days assuming salaries accruing evenly throughout.

Outstanding salaries = 48,00,000 x 30 / 360

= **4,00,000**

Q1(iii)

(A) ₹ 194.0605 L

Total Capital needed = Total capital needs (Fixed assets) + Working Capital needs

= 200 L + (5,93,955)

= ₹ 194.0605 L

Q1(iv)

(C) 9.90%

Samvar Ltd would require financing of ₹ 194.0605 lakhs from debt and equity and not ₹ 200 lakhs as trade credit is also considered to be a source of finance as mentioned in the case study.

Furthermore, the overall cost of raising this additional fund for each store of ₹ 194.0605 needs to be calculated slab wise

Project Cost	Weights (W)	Cost (K)	W X K	Total cost (₹)
Upto 80 Lakhs	Debt = 0.67 Equity = 0.33	Kd = 10 (1 - 0.25) = 7.5 Ke = 12.5	Ko = 9.167%	= 80L x 9.167% **= 7.334 Lakhs**
Above 80 L upto 150 L	Debt = 0.67 Equity = 0.33	Kd = 11.5(1-0.25) = 8.625 Ke = 13.5	Ko = 10.25%	= 70L x 10.25% **= 7.175 Lakhs**
Above 150 L upto 250 L	Debt = 0.67 Equity = 0.33	K = 12 (1-0.25) = 9 Ke = 14	Ko = 10.667%	=44.0605L x 10.667% **= 4.7Lakhs**

Total Funds = 194.0605 L

Total Cost (₹) = 7.334 L + 7.175 L + 4.700 L = 19.209 L

Ko = Total Cost (₹) / Total Funds

= 19.209 / 194.0605

= 9.90%

Q1(v)

(B) ₹ 3.24 L

Since the Overall Cost of Capital is below 10%, Samvar Ltd is eligible for revenue subsidy

Revenue Subsidy = GP x 15%

= 21.6 L x 15%

= ₹ 3.24 Lakhs

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690 RTP **Sep 2024** Q10

TMT Limited is commencing a new project for manufacture of electric toys. The following cost information has been ascertained for annual production of 60,000 units at full capacity:

		Amount per unit (₹)
Raw materials		20
Direct labour		15
Manufacturing overheads:		
	₹	
Variable	15	
Fixed	10	25
Selling and Distribution overheads:		
	₹	
Variable	3	
Fixed	1	4

	Amount per unit (₹)
Total cost	64
Profit	16
Selling price	80

In the first year of operations expected production and sales are 40,000 units and 35,000 units respectively. To assess the need of working capital, the following additional information is available:

(i) Stock of Raw materials 3 months consumption.

(ii) Credit allowable for debtors 1½ months.

(iii) Credit allowable by creditors 4 months.

(iv) Lag in payment of wages 1 month.

(v) Lag in payment of overheads ½ month.

(vi) Cash in hand and Bank is expected to be ₹ 60,000.

(vii) Provision for contingencies is required @ 10% of working capital requirement including that provision.

You are required to PREPARE a projected statement of working capital requirement for the first year of operations. Debtors are taken at cost.

ICAI'S ANSWER

Statement Showing Cost and Sales for the First Year

Annual Production Capacity	60,000 units
Production	40,000 units
Sales	35,000 units

Particulars	₹
Sales Revenue (₹ 80 × 35,000)	28,00,000
Cost of Production:	
Materials @ ₹ 20 per unit	8,00,000
Direct Labour @ ₹ 15 per unit	6,00,000
Manufacturing Overheads	
Variable @ ₹ 15 per unit	6,00,000
Fixed (based on production capacity 60,000 units × ₹ 10)	6,00,000
Cost of Production	26,00,000
Less: Closing Stock (40,000 – 35,000 = 5,000 units)	

Particulars	₹
$\left(₹ \frac{26,00,000}{40,000} \times 5,000 \text{ units}\right)$	3,25,000
Cost of Goods Sold	22,75,000
Add: Selling & Distribution Overheads	
Variable @ ₹ 3 × 35,000 units = 1,05,000	
Fixed (Re. 1 × 60,000 units) = 60,000	1,65,000
Cost of Sales	24,40,000
Profit	3,60,000

Statement Showing Working Capital Requirement

		₹
A.	**Current Assets**	
	Stock of Raw Materials (₹ 8,00,000 × 3/12)	2,00,000
	Stock of Finished Goods	3,25,000
	Debtors at Cost (₹ 24,40,000 × 3/24)	3,05,000
	Cash and Bank	60,000
	Total (A)	8,90,000
B.	**Current Liabilities**	
	Creditors for Materials (₹ 10,00,000 × 4/12)	3,33,333
	Creditors for Expenses (₹ 13,65,000 × 1/24)	56,875
	Outstanding Wages (₹ 6,00,000 × 1/12)	50,000
	Total (B)	4,40,208
	Working Capital Requirement before Contingencies **(A – B)**	4,49,792
	Add: Provision for Contingencies (₹ 4,49,792 × 1/9)	49,977
	Estimated Working Capital Requirement	4,99,769

Workings Notes:

Purchase of Raw Material during the first year	₹
Raw Material consumed during the year	8,00,000
Add: Closing Stock of Raw Materials (3 months consumption)	2,00,000
	10,00,000
Less: Opening Stock of Raw Material	Nil
Purchases during the year	10,00,000

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691 RTP **May 2024** Division A: Case Scenario · Division A Q1(i) to Division A Q1(v)

DIVISION A: CASE SCENARIO

ArMore LLP is a newly established startup dealing in manufacture of a revolutionary product HDHMR which is a substitute to conventional wood and plywood. It is an economical substitute for manufacture of furniture and home furnishing. It has been asked by a venture capitalist for an estimated amount of funds required for setting up plant and also the amount of circulating capital required. A consultant hired by the entity has advised that the cost of setting up the plant would be ₹ 5 Crores and it will require 1 year to make the plant operational. The anticipated revenue and associated cost numbers are as follows:

Units to be sold = 3 lakh sq metres p.a.

Sale Price of each sq mtr = ₹ 1000

Raw Material cost = ₹ 200 per sq mtr

Labour cost = ₹ 50 per hour

Labour hours per sq mtr = 3 hours

Cash Manufacturing Overheads = ₹ 75 per machine hour

Machine hours per sq mtr = 2 hours

Selling and credit administration Overheads = ₹ 250 per sq mtr

Being a new product in the industry, the firm will have to give a longer credit period of 3 months to its customers. It will maintain a stock of raw material equal to 15% of annual consumption. Based on negotiation with the creditors, the payment period has been agreed to be 1 month from the date of purchase. The entity will hold finished goods equal to 2 months of units to be sold. All other expenses are to be paid one month in arrears. Cash and Bank balance to the tune of ₹ 25,00,000 is required to be maintained.

The entity is also considering reducing the working capital requirement by either of the two options: a) reducing the credit period to customers by a month which will lead to reduction in sales by 5%. b) Engaging with a factor for managing the receivables, who will charge a commission of 2% of invoice value and will also advance 65% of receivables @ 12% p.a. This will lead to savings in administration and bad debts cost to the extent of ₹ 20 lakhs and 16 lakhs respectively.

The entity is also considering funding a part of working capital by bank loan. For this purpose, bank has stipulated that it will grant 75% of net current assets as advance against working capital. The bank has quoted 16.5% rate of interest with a condition of opening a current account with it, which will require 10% of loan amount to be minimum average balance.

You being an finance manager, has been asked the following questions:

Division A Q1(i)

The anticipated profit before tax per annum after the plant is operational is .....

A 750 Lakhs

B 570 Lakhs

C 370 Lakhs

D 525 Lakhs

Division A Q1(ii)

The estimated current assets requirement in the first year of operation (debtors calculated at cost) is .....

A 9,42,50,000

B 2,17,08,333

C 7,25,41,667

D 67,08,333

Division A Q1(iii)

The net working capital requirement for the first year of operation is .....

A 9,42,50,000

B 2,17,08,333

C 7,25,41,667

D 67,08,333

Division A Q1(iv)

The annualised % cost of two options for reducing the working capital is .....

A 18.18% and 16.92%

B 18.33% and 16.92%

C 18.59% and 18.33%

D 16.92% and 19.05%

Division A Q1(v)

What will be the Maximum Permissible Bank Finance by the bank and annualised % cost of the same?

A 4,55,03,630 and 18.33%

B 5,44,06,250 and 18.33%

C 4,45,86,025 and 18.59%

D 3,45,89,020 and 19.85%

ICAI'S ANSWERS

Division A Q1(i)

(A) 750 Lakhs

	Units	Per unit (₹)	Amount (₹)
Raw Material consumption	3,50,000	200	7,00,00,000
labour cost	3,50,000	150	5,25,00,000
Production Overheads	3,50,000	150	5,25,00,000
Cost of Production	**3,50,000**	**500**	**17,50,00,000**
Less: Stock of FG	50,000	500	2,50,00,000
COGS	**3,00,000**	**500**	**15,00,00,000**
Selling and admin exp	3,00,000	250	7,50,00,000
Cost of Sales	**3,00,000**	**750**	**22,50,00,000**
Sales	3,00,000	1000	30,00,00,000
Profit	**3,00,000**	**250**	**7,50,00,000**

Stock of FG (sq. mtr.) = 30,00,000x2/12 = 50,000

Units sold = 3,00,000

Raw material consumed (sq. mtr.) = 3,50,000

Raw Material Purchases = Consumption + RM stock (15%)

= 7,00,00,000 + 1,05,00,000

= ₹ 8,05,00,000

Division A Q1(ii)

(A) 9,42,50,000

Stock of Raw Material (15% of 7,00,00,000) = 1,05,00,000

Stock of finished goods = 2,50,00,000

Debtors (22,50,00,000 x 3/12) = 5,62,50,000

Cash = 25,00,000

Total Current Assets = 9,42,50,000

Division A Q1(iii)

(C) 7,25,41,667

Working Capital Statement

	Amount (₹)
Stock of Raw Material (15% of 7,00,00,000)	1,05,00,000
Stock of finished goods	2,50,00,000

	Amount (₹)
Debtors (22,50,00,000 × 3/12)	5,62,50,000
Cash	25,00,000
Total Current Assets	**9,42,50,000**
Creditors (8,05,00,000 × 1/12)	67,08,333
O/s Exp (18,00,00,000 × 1/12)	1,50,00,000
Total Current Liabilities	**2,17,08,333**
Net Working Capital	**7,25,41,667**

Division A Q1(iv)

(A) 18.18% and 16.92%

Cost reducing debtors credit period

Debtors credit period = 2 months

Debtors balance = 21,37,50,000(2,85,000 units) x 2/12 = ₹3,56,25,000

Debtors credit period = 3 months

Debtors balance = 22,50,00,000 x 3/12

= ₹ 5,62,50,000

Amount released from debtors = ₹ 2,06,25,000

reduction in profit (15,000 units x ₹ 250) = ₹ 37,50,000

% p.a. cost (37,50,000/2,06,25,000) = **18.18%**

Costs of factoring

Commission (2% of 30 crores) = ₹ 60,00,000

Interest (30cr x 65% x 12% x 3/12) = ₹ 58,50,000

savings = ₹ 36,00,000

Net cost of factoring $\frac{82,50,000}{65\% \text{ of } 30\text{cr. i.e. } 19,50,00,000} \times \frac{12}{3}$

= ₹ 82,50,000

% p.a. cost = **16.92%**

Division A Q1(v)

(B) 5,44,06,250 and 18.33%

Maximum Permissible Bank Finance = 75% of 7,25,41,667

= ₹ 5,44,06,250

Annualised cost of bank loan = 16.5/90% = 18.33%

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692 RTP May 2024 Q7

PQ Ltd. has commenced new business segment in 2023-24. The following information has been ascertained for annual production of 25,000 units which is the full capacity.

	Cost per unit (₹)
Material	100
Labour and variable overhead expenses	50
Fixed manufacturing expenses	35
Depreciation	15
Selling expenses (80% variable)	10

In the first two years of operations, production and sales are expected to be as follows:

Year	Production (No. of units)	Sales (No. of units)
1	12,000	10,000
2	18,000	19,000

The selling price is expected to be **₹ 250** .

To assess the working capital requirements, the following additional information is available:

(a)	Stock of materials	2 months' average consumption
(b)	Debtors	1.5 month's average sales.
(c)	Cash balance	₹ 50,000
(d)	Creditors for supply of materials	1 month's average purchase during the year.
(e)	Expenses	All expenses will be paid 1 month in advance during the year.

Goods equal to 15% of the year's production (in terms of physical units) will be in process on the average requiring full materials but only 40% of the other expenses.

The management is also of the opinion to make 10% margin for contingencies on computed figure and value the closing stock at cost of production.

PREPARE, for the two years:

(i) A projected statement of Profit/Loss (Ignoring taxation); and

(ii) A projected statement of working capital requirements on a cash cost basis.

ICAI'S ANSWER

(i) **PQ Limited**

Projected Statement of Profit / Loss

(Ignoring Taxation)

	Year 1	Year 2
Production (Units)	12,000	18,000
Sales (Units)	10,000	19,000
	(₹)	**(₹)**
Sales revenue (A) (Sales unit × ₹ 250)	25,00,000	47,50,000
Cost of production:		
Materials cost (Units produced × ₹ 100)	12,00,000	18,00,000
Direct labour and variable expenses (Units produced × ₹ 50)	6,00,000	9,00,000
Fixed manufacturing expenses (Production Capacity: 25,000 units × ₹ 35)	8,75,000	8,75,000
Depreciation (Production Capacity: 25,000 units × ₹ 15)	3,75,000	3,75,000
Gross Factory Cost	**30,50,000**	**39,50,000**
Add: Opening W.I.P.	-	2,91,000
Less: Closing W.I.P.	2,91,000	3,99,000
Cost of goods produced	**27,59,000**	**38,42,000**
Add: Opening stock of finished goods (Year 1 : Nil; Year 2 : 2,000 units)	-	4,59,833
Cost of Goods available for sale (Year 1: 12,000 units; Year 2: 20,000 units)	27,59,000	43,01,833
Less: Closing stock of finished goods at average cost (year 1: 2000 units, year 2 : 1000 units) (Cost of Production × Closing stock/ units produced)	4,59,833	2,13,444
Cost of Goods Sold	**22,99,167**	**40,88,389**
Add: Selling expenses – Variable (Sales unit × ₹ 8)	80,000	1,52,000
Add: Selling expenses -Fixed (25,000 units × ₹ 2)	50,000	50,000
Cost of Sales : (B)	24,29,167	42,90,389
Profit (+) / Loss (-): (A - B)	**70,833**	**4,59,611**

Working Notes:

Calculation of Stock of Work-in-progress

Particulars	Year 1	Year 2
	(₹)	(₹)
Raw Material (material cost × 15%)	1,80,000	2,70,000
Labour & Mfg. Expenses (Labour & mfg. expenses × 15% × 40%)	88,500	1,06,500
Depreciation (Depreciation × 15% × 40%)	22,500	22,500
Total	2,91,000	3,99,000

1. Calculation of creditors for supply of materials:

	Year 1 (₹)	Year 2 (₹)
Materials consumed during the year	12,00,000	18,00,000
Add: Closing stock (2 month's average consumption)	2,00,000	3,00,000
	14,00,000	21,00,000
Less: Opening Stock	-	2,00,000
Purchases during the year	14,00,000	19,00,000
Average purchases per month (Creditors)	1,16,667	1,58,333

2. Prepayment for expenses:

	Year 1 (₹)	Year 2 (₹)
Direct labour and variable expenses	6,00,000	9,00,000
Fixed manufacturing expenses	8,75,000	8,75,000
Selling expenses (variable + fixed)	1,30,000	2,02,000
Total	16,05,000	19,77,000
Average per month	1,33,750	1,64,750

(ii) **Projected Statement of Working Capital Requirement**

(Cash Cost Basis)

	Year 1 (₹)	Year 2 (₹)
(A) Current Assets		
Inventories:		
- Stock of Raw Material (12,000 units ₹ 100 2/12); (18,000 units ₹ 100 2/12)	2,00,000	3,00,000
- Finished Goods (Refer working note 3)	4,01,083	1,92,611

	Year 1 (₹)	Year 2 (₹)
- Work In Process (Refer working note 5)	2,68,500	3,76,500
Receivables (Debtors) (Refer working note 4)	2,66,927	4,84,684
Prepayment for Expenses (Refer working note 2)	1,33,750	1,64,750
Minimum Cash balance	50,000	50,000
Total Current Assets/ Gross working capital (A)	13,20,260	15,68,545
(B) Current Liabilities		
Creditors for raw material (Refer working note 1)	1,16,667	1,58,333
Total Current Liabilities	1,16,667	1,58,333
Net Working Capital (A – B)	12,03,594	14,10,212
Add: 10% contingency margin	1,20,359	1,41,021
Total Working capital required	**13,23,953**	**15,51,233**

Working Note:

3. Cash Cost of Production:

	Year 1 (₹)	Year 2 (₹)
Gross Factory Cost as per projected Statement of P&L	30,50,000	39,50,000
Add: Opening W.I.P	-	2,68,500
Less: Closing W.I.P	2,68,500	3,76,500
Cost of goods produced	27,81,500	38,42,000
Less: Depreciation	(3,75,000)	(3,75,000)
Cash Cost of Production	24,06,500	34,67,000
Add: Opening Stock at Average Cost:	-	4,01,083
Cash Cost of Goods Available for sale	24,06,500	38,68,083
Less: Closing Stock at Avg. Cost	4,01,083	1,92,611
$\left(\frac{₹\ 24{,}06{,}500 \times 2{,}000}{12{,}000}\right)$ $\left(\frac{₹\ 34{,}67{,}000 \times 1{,}000}{18{,}000}\right)$		
Cash Cost of Goods Sold	20,05,417	36,75,472

4. Receivables (Debtors)

	Year 1 (₹)	Year 2 (₹)
Cash Cost of Goods Sold	20,05,417	36,75,472
Add: Selling expenses – Variable (Sales unit × ₹ 8)	80,000	1,52,000
Add: Selling expenses -Fixed (25,000 units × ₹ 2)	50,000	50,000
Cash Cost of Debtors	21,35,417	38,77,472
Average Debtors	2,66,927	4,84,684

Calculation of Stock of Work-in-progress (Cash Cost Basis)

Particulars		(₹)
Raw Material (material cost × 15%)	1,80,000	2,70,000
Labour & Mfg. Expenses (Labour & mfg. expenses × 15% × 40%)	88,500	1,06,500
Total	2,68,500	3,76,500

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693 RTP **Old P8 May 2023** Q9 old scheme

Kalyan limited has provided you the following information for the year 2021-22:

By working at 60% of its capacity the company was able to generate sales of ₹ 72,00,000. Direct labour cost per unit amounted to ₹ 20 per unit. Direct material cost per unit was 40% of the selling price per unit. Selling price was 3 times the direct labour cost per unit. Profit margin was 25% on the total cost.

For the year 2022-23, the company makes the following estimates:

Production and sales will increase to 90% of its capacity. Raw material per unit price will remain unchanged. Direct expense per unit will increase by 50%. Direct labour per unit will increase by 10%. Despite the fluctuations in the cost structure, the company wants to maintain the same profit margin on sales.

Raw materials will be in stock for one month whereas finished goods will remain in stock for two months. Production cycle is for 2 months. Credit period allowed by suppliers is 2 months. Sales are made to three zones:

Zone	Percentage of sale	Mode of Credit
A	50%	Credit period of 2 months
B	30%	Credit period of 3 months
C	20%	Cash Sales

There are no cash purchases and cash balance will be ₹ 1,11,000

The company plans to apply for a working capital financing from bank for the year 2022-23. ESTIMATE Net Working Capital of the Company receivables to be taken on sales and also COMPUTE the maximum

permissible bank finance for the company using 3 criteria of Tandon Committee Norms. (Assume stock of finished goods to be a core current asset)

ICAI'S ANSWER

Cost Structure

		2021-22			2022-23	
Particulars	**Calculations**	**P.U.**	**Amount (p.u. X units)**	**Calculations**	**P.U.**	**Amount (p.u. X units)**
Direct Material	40% of SP	₹24	₹28,80,000	Same as PY	₹24	₹43,20,000
Direct labour	Given	₹20	₹24,00,000	20*1.1	₹22	₹39,60,000
Direct Expenses	bal. fig.	₹4	₹4,80,000	4*1.5	₹6	₹10,80,000
Total Cost	SP - Profit	₹48	₹57,60,000		₹52	₹93,60,000
Profit	(SP/125×25)	₹12	₹14,40,000	52*25%	₹13	₹23,40,000
Sales	3 x Direct Labour p.u.	₹60	₹72,00,000		₹65	₹1,17,00,000
*units=		₹72,00,000 / ₹60 =1,20,000			1,20,000/60 x 90=1,80,000	

Operating Cycle

Raw material holding period	1 month
Finished Goods holding period	2 months
WIP conversion period	2 months
Creditor Payment Period	2 months
Receivables Collection Period	2/3 months

Estimation of Working Capital

Particulars	Calculation	Amount
Current Assets		
Stock of Raw Material	43,20,000 × 1/12	₹3,60,000
Stock of WIP		
RM cost	₹43,20,000	
Labour cost	₹19,80,000	

Particulars	Calculation	Amount
Direct Exp cost	₹5,40,000	
Total WIP Cost	₹68,40,000	
Stock of WIP	68,40,000 × 2/12	₹11,40,000
Stock of Finished Goods	93,60,000 × 2/12	₹15,60,000
Receivables (on sales)		
A	1,17,00,000 × 50% x 2/12	₹9,75,000
B	1,17,00,000 × 30% x 3/12	₹8,77,500
C	NIL	-
Cash Balance	Given	₹1,11,000
Total Current Assets		**₹ 50,23,500**
Current Liabilities		
Payables	*₹44,40,000 × 2/12	₹7,40,000
Net Working Capital		**₹ 42,83,500**

Opening RM stock = 28,80,000 x 1/12= ₹2,40,000

* RM purchased = RM consumed – Opening Stock + Closing Stock

= ₹43,20,000 – ₹2,40,000 + ₹3,60,000

= ₹44,40,000

Computation of Maximum Permissible Bank Finance

Method	Formula	Calculation	₹
I	75% x (Current Assets-Current Liabilities)	75% x (₹50,23,500 - ₹7,40,000)	₹32,12,625
II	75% x Current Assets-Current Liabilities	75% x ₹50,23,500 - ₹7,40,000	₹30,27,625
III	75% x (Current Assets-Core CA)-Current Liabilities	75% x (₹50,23,500- ₹15,60,000) - ₹7,40,000	₹18,57,625

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Trading and Profit and Loss Account of Beat Ltd. for the year ended 31st March, 2022 is given below:

Particulars	Amount (₹)	Amount (₹)	Particulars	Amount (₹)	Amount (₹)
To Opening Stock:			By Sales (Credit)		1,60,00,000

Particulars	Amount (₹)	Amount (₹)	Particulars	Amount (₹)	Amount (₹)
- Raw Materials	14,40,000		By Closing Stock:		
- Work-in- progress	4,80,000		- Raw Materials	16,00,000	
- Finished Goods	20,80,000	40,00,000	- Work-in-progress	8,00,000	
To Purchases (credit)		88,00,000	- Finished Goods	24,00,000	48,00,000
To Wages		24,00,000			
To Production Exp.		16,00,000			
To Gross Profit c/d		40,00,000			
		2,08,00,000			2,08,00,000
To Administration Exp.		14,00,000	By Gross Profit b/d		40,00,000
To Selling Exp.		6,00,000			
To Net Profit		20,00,000			
		40,00,000			40,00,000

The opening and closing payables for raw materials were ₹ 16,00,000 and ₹ 19,20,000 respectively whereas the opening and closing balances of receivables were ₹ 12,00,000 and ₹ 16,00,000 respectively.

You are required to ASCERTAIN the working capital requirement by operating cycle method.

ICAI'S ANSWER

Computation of Operating Cycle

(1) Raw Material Storage Period (R)

$$\text{Raw Material Storage Period (R)} = \frac{\text{Average Stock of Raw Material}}{\text{Daily Average Consumption of Raw material}}$$

$$= \frac{(14{,}40{,}000 + 16{,}00{,}000)/2}{86{,}40{,}000/365} = 64.21 \text{ Days}$$

Raw Material Consumed = Opening Stock + Purchases – Closing Stock

= ₹ 14,40,000+₹ 88,00,000–₹ 16,00,000 = ₹ 86,40,000

(2) Conversion/Work-in-Process Period (W)

$$\text{Conversion/Processing Period} = \frac{\text{Average Stock of WIP}}{\text{Daily Average Production cost}}$$

$$= \frac{(4{,}80{,}000 + 8{,}00{,}000)/2}{1{,}23{,}20{,}000/365} = 18.96 \text{ days}$$

Production Cost:	₹
Opening Stock of WIP	4,80,000
Add: Raw Material Consumed	86,40,000
Add: Wages	24,00,000
Add: Production Expenses	16,00,000
	1,31,20,000
Less: Closing Stock of WIP	8,00,000
Production Cost	1,23,20,000

(3) Finished Goods Storage Period (F)

$$\text{Finished Goods Storage Period} = \frac{\text{Average Stock of Finished Goods}}{\text{Daily Average Cost of Good Sold}}$$

$$= \frac{(20,80,000 + 24,00,000)/2}{1,20,00,000/365} = 68.13 \text{ Days}$$

Cost of Goods Sold	₹
Opening Stock of Finished Goods	20,80,000
Add: Production Cost	1,23,20,000
	1,44,00,000
Less: Closing Stock of Finished Goods	(24,00,000)
	1,20,00,000

(4) Receivables Collection Period (D)

$$\text{Receivables Collection Period} = \frac{\text{Average Receivables}}{\text{Daily average credit sales}}$$

$$= \frac{(12,00,000 + 16,00,000)/2}{1,60,00,000/365} = 31.94 \text{ Days}$$

(5) Payables Payment Period (C)

$$\text{Payables Payment Period} = \frac{\text{Average Payables}}{\text{Daily average credit purchase}}$$

$$= \frac{(16,00,000 + 19,20,000)/2}{88,00,000/365} = 73 \text{ Days}$$

(6) Duration of Operating Cycle (O)

O = R + W + F + D – C

= 64.21 + 18.96 + 68.13 + 31.94 – 73

= 110.24 days

Computation of Working Capital

(i) Number of Operating Cycles per Year

= 365/Duration Operating Cycle = 365/110.24 = 3.311

(ii) Total Operating Expenses

	₹
Total Cost of Goods sold	1,20,00,000
Add: Administration Expenses	14,00,000
Add: Selling Expenses	6,00,000
	1,40,00,000

(iii) Working Capital Required

$$\text{Working Capital Required} = \frac{\text{Total Operating Expenses}}{\text{Number of Operating Cycles per year}}$$

$$= \frac{1,40,00,000}{3.311} = ₹\ 42,28,329.81$$

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695 RTP **Old P8 May 2022** Q9 old scheme

PQR Ltd., a company newly commencing business in the year 2021-22, provides the following projected Profit and Loss Account:

	(₹)	(₹)
Sales		5,04,000
Cost of goods sold		3,67,200
Gross Profit		1,36,800
Administrative Expenses	33,600	
Selling Expenses	31,200	64,800
Profit before tax		72,000
Provision for taxation		24,000
Profit after tax		48,000
The cost of goods sold has been arrived at as under:		
Materials used	2,01,600	
Wages and manufacturing Expenses	1,50,000	
Depreciation	56,400	
	4,08,000	
Less: Stock of Finished goods (10% of goods produced not yet sold)	40,800	

	(₹)	(₹)
	3,67,200	

The figure given above relate only to finished goods and not to work-in-progress. Goods equal to 15% of the year's production (in terms of physical units) will be in process on the average requiring full materials but only 40% of the other expenses. The company believes in keeping materials equal to two months' consumption in stock.

All expenses will be paid one month in advance. Suppliers of materials will extend 1 -1/2 months credit. Sales will be 20% for cash and the rest at two months' credit. 70% of the Income tax will be paid in advance in quarterly instalments. The company wishes to keep ₹ 19,200 in cash. 10% must be added to the estimated figure for unforeseen contingencies.

PREPARE an estimate of working capital.

ICAI'S ANSWER

Statement showing the requirements of Working Capital

	Particulars	(₹)	(₹)
A.	**Current Assets:**		
	Inventory:		
	Stock of Raw material (₹ 2,31,840 × 2/12)	38,640	
	Stock of Work-in-progress (As per Working Note)	39,240	
	Stock of Finished goods (₹ 3,51,600 × 10/100)	35,160	
	Receivables (Debtors) (₹3,04,992 × 2/12)	50,832	
	Cash in Hand	19,200	
	Prepaid Expenses:		
	Wages & Mfg. Expenses (₹ 1,59,000 × 1/12)	13,250	
	Administrative expenses (₹ 33,600 × 1/12)	2,800	
	Selling & Distribution Expenses (₹ 31,200 × 1/12)	2,600	
	Advance taxes paid {(70% of ₹ 24,000) × 3/12}	4,200	
	Gross Working Capital	2,05,922	2,05,922
B.	**Current Liabilities:**		
	Payables for Raw materials (₹ 2,70,480 × 1.5/12)	33,810	
	Provision for Taxation (Net of Advance Tax) (₹ 24,000 × 30/100)	7,200	
	Total Current Liabilities	41,010	41,010
C.	**Excess of CA over CL**		1,64,912

	Particulars	(₹)	(₹)
	Add: 10% for unforeseen contingencies		16,491
	Net Working Capital requirements		1,81,403

Working Notes:

(i) Calculation of Stock of Work-in-progress

Particulars	(₹)
Raw Material (₹ 2,01,600 × 15%)	30,240
Wages & Mfg. Expenses (₹ 1,50,000 × 15% × 40%)	9,000
Total	39,240

(ii) Calculation of Stock of Finished Goods and Cost of Sales

Particulars	(₹)
Direct material Cost [₹ 2,01,600 + ₹ 30,240]	2,31,840
Wages & Mfg. Expenses [₹ 1,50,000 + ₹ 9,000]	1,59,000
Depreciation	0
Gross Factory Cost	3,90,840
Less: Closing W.I.P.	(39,240)
Cost of goods produced	3,51,600
Add: Administrative Expenses	33,600
	3,85,200
Less: Closing stock	(35,160)
Cost of Goods Sold	3,50,040
Add: Selling and Distribution Expenses	31,200
Total Cash Cost of Sales	3,81,240
Debtors (80% of cash cost of sales)	3,04,992

(iii) Calculation of Credit Purchase

Particulars	(₹)
Raw material consumed	2,31,840
Add: Closing Stock	38,640
Less: Opening Stock	-
Purchases	2,70,480

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696 RTP **Old P8 Dec 2021** Q9 old scheme

The management of Trux Company Ltd. is planning to expand its business and consults you to prepare an estimated working capital statement. The records of the company reveals the following annual information:

	(₹)
Sales – Domestic at one month's credit	18,00,000
Export at three month's credit (sales price 10% below domestic price)	8,10,000
Materials used (suppliers extend two months credit)	6,75,000
Lag in payment of wages – ½ month	5,40,000
Lag in payment of manufacturing expenses (cash) – 1 month	7,65,000
Lag in payment of Administration Expenses – 1 month	1,80,000
Selling expenses payable quarterly in advance	1,12,500
Income tax payable in four installments, of which one falls in the next financial year	1,68,000

Rate of gross profit is 20%. Ignore work-in-progress and depreciation.

The company keeps one month's stock of raw materials and finished goods (each) and believes in keeping ₹ 2,50,000 available to it including the overdraft limit of ₹ 75,000 not yet utilized by the company.

The management is also of the opinion to make 10% margin for contingencies on computed figure.

You are required to PREPARE the estimated working capital statement for the next year.

ICAI'S ANSWER

Preparation of Statement of Working Capital Requirement for Trux Company Ltd.

	(₹)	(₹)
A. Current Assets		
(i) Inventories:		

	(₹)	(₹)
Material (1 month) $\left(\frac{₹\ 6,75,000}{12\ \text{months}} \times 1\ \text{month}\right)$	56,250	
Finished goods (1 month) $\left(\frac{₹\ 21,60,000}{12\ \text{months}} \times 1\ \text{month}\right)$	1,80,000	2,36,250
(ii) Receivables (Debtors)		
For Domestic Sales $\left(\frac{₹\ 15,17,586}{12\ \text{months}} \times 1\ \text{month}\right)$	1,26,466	
For Export Sales $\left(\frac{₹\ 7,54,914}{12\ \text{months}} \times 3\ \text{months}\right)$	1,88,729	3,15,195
(iii) Prepayment of Selling expenses $\left(\frac{₹\ 1,12,500}{12\ \text{months}} \times 3\ \text{months}\right)$		28,125
(iii) Cash in hand & at bank		1,75,000
Total Current Assets		7,54,570
B. Current Liabilities:		
(i) Payables (Creditors) for materials (2 months) $\left(\frac{₹\ 6,75,000}{12\ \text{months}} \times 2\ \text{months}\right)$		1,12,500
(ii) Outstanding wages (0.5 months) $\left(\frac{₹\ 5,40,000}{12\ \text{months}} \times 0.5\ \text{month}\right)$		22,500
(iii) Outstanding manufacturing expenses $\left(\frac{₹\ 7,65,000}{12\ \text{months}} \times 1\ \text{month}\right)$		63,750
(iv) Outstanding administrative expenses $\left(\frac{₹\ 1,80,000}{12\ \text{months}} \times 1\ \text{month}\right)$		15,000
(v) Income tax payable		42,000
Total Current Liabilities		2,55,750
Net Working Capital (A – B)		4,98,820
Add: 10% contingency margin		49,882
Total Working Capital required		5,48,702

Working Notes:

1. **Calculation of Cost of Goods Sold and Cost of Sales**

	Domestic (₹)	Export (₹)	Total (₹)
Domestic Sales	18,00,000	8,10,000	26,10,000
Less: Gross profit @ 20% on domestic sales and 11.11% on export sales (Working note-2)	3,60,000	90,000	4,50,000
Cost of Goods Sold	14,40,000	7,20,000	21,60,000
Add: Selling expenses (Working note-3)	77,586	34,914	1,12,500
Cash Cost of Sales	15,17,586	7,54,914	22,72,500

2. **Calculation of gross profit on Export Sales**

Let domestic selling price is ₹ 100. Gross profit is ₹ 20, and then cost per unit is ₹ 80

Export price is 10% less than the domestic price i.e. ₹ 100 – (1- 0.1) = ₹ 90

Now, gross profit will be = ₹ 90 - ₹ 80 = ₹ 10

So, Gross profit ratio at export price will be = $\frac{₹\ 10}{₹\ 90} \times 100 = 11.11\%$

3. **Apportionment of Selling expenses between Domestic and Exports sales:**

Apportionment on the basis of sales value:

Domestic Sales = $\frac{₹\ 1,12,500}{₹\ 26,10,000} \times ₹\ 18,00,000 = ₹\ 77,586$

Exports Sales = $\frac{₹\ 1,12,500}{₹\ 26,10,000} \times ₹\ 8,10,000 = ₹\ 34,914$

4. **Assumptions**

(i) It is assumed that administrative expenses is related to production activities.

(ii) Value of opening and closing stocks are equal.

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MT Ltd. has been operating its manufacturing facilities till 31.3.202 1 on a single shift working with the following cost structure:

	Per unit (₹)
Cost of Materials	24
Wages (out of which 60% variable)	20
Overheads (out of which 20% variable)	20
	64
Profit	8
Selling Price	72

As at 31.3.2021 with the sales of ₹ 17,28,000, the company held:

	(₹)
Stock of raw materials (at cost)	1,44,000
Work-in-progress (valued at prime cost)	88,000
Finished goods (valued at total cost)	2,88,000
Sundry debtors	4,32,000

In view of increased market demand, it is proposed to double production by working an extra shift. It is expected that a 10% discount will be available from suppliers of raw materials in view of increased volume of business. Selling price will remain the same. The credit period allowed to customers will remain unaltered. Credit availed from suppliers will continue to remain at the present level i.e. 2 months. Lag in payment of wages and overheads will continue to remain at one month.

You are required to CALCULATE the additional working capital requirements, if the policy to increase output is implemented, to assess the impact of double shift for long term as a matter of production policy.

ICAI'S ANSWER

Workings:

(1) Statement of cost at single shift and double shift working

	24,000 units		48,000 Units	
	Per unit (₹)	**Total (₹)**	**Per unit (₹)**	**Total (₹)**
Raw materials	24	5,76,000	21.6	10,36,000
Wages:				
Variable	12	2,88,000	12	5,76,000
Fixed	8	1,92,000	4	1,92,000
Overheads:				
Variable	4	96,000	4	1,92,000
Fixed	16	3,84,000	8	3,84,000
Total cost	64	15,36,000	49.6	23,80,800
Profit	8	1,92,000	22.4	10,75,200
Sales	72	17,28,000	72	34,56,000

(2) Sales in units 2020-21 = $\frac{\text{Sales}}{\text{Unit selling price}} = \frac{₹\ 17,28,000}{₹\ 72} = 24,000$ units

(3) Stock of Raw Materials in units on 31.3.2021

$= \frac{\text{Value of stock}}{\text{Cost per unit}} = \frac{₹\ 1,44,000}{₹\ 24} = 6,000$ units

(4) Stock of work-in-progress in units on 31.3.2021

$$= \frac{\text{Value of work-in-progress}}{\text{Prime Cost per unit}} = \frac{₹\ 88{,}000}{₹\ (24+20)} = 2{,}000 \text{units}$$

(5) Stock of finished goods in units 2020-21

$$= \frac{\text{Value of stock}}{\text{Total Cost per unit}} = \frac{₹\ 2{,}88{,}000}{₹\ 64} = 4{,}500 \text{ units.}$$

Comparative Statement of Working Capital Requirement

	Single Shift (24,000 units)			Double Shift (48,000 units)		
	Units	**Rate (₹)**	**Amount (₹)**	**Units**	**Rate (₹)**	**Amount (₹)**
Current Assets						
Inventories:						
Raw Materials	6,000	24	1,44,000	12,000	21.6	2,59,200
Work-in-Progress	2,000	44	88,000	2,000	37.6	75,200
Finished Goods	4,500	64	2,88,000	9,000	49.6	4,46,400
Sundry Debtors	6,000	64	3,84,000	12,000	49.6	5,95,200
Total Current Assets (A)			9,04,000			13,76,000
Current Liabilities						
Creditors for Materials	4,000	24	96,000	8,000	21.6	1,72,800
Creditors for Wages	2,000	20	40,000	4,000	16	64,000
Creditors for Overheads	2,000	20	40,000	4,000	12	48,000
Total Current Liabilities (B)			1,76,000			2,84,800
Working Capital (A) – (B)			**7,28,000**			**10,91,200**

Analysis: Additional Working Capital requirement = ₹ 10,91,200 – ₹ 7,28,000 = ₹3,63,200, if the policy to increase output is implemented.

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698 RTP **Old P8 Jul 2021 Q9** old scheme

While applying for financing of working capital requirements to a commercial bank, TN Industries Ltd. projected the following information for the next year:

Cost Element	Per unit (₹)	Per unit (₹)
Raw materials		
X	30	
Y	7	
Z	6	43
Direct Labour		25
Manufacturing and administration overheads (excluding depreciation)		20
Depreciation		10
Selling overheads		15
		113

Additional Information:

(a) Raw Materials are purchased from different suppliers leading to different credit period allowed as follows:

X – 2 months; Y– 1 months; Z – ½ month

(b) Production cycle is of ½ month. Production process requires full unit of X and Y in the beginning of the production. Z is required only to the extent of half unit in the beginning and the remaining half unit is needed at a uniform rate during the production process.

(c) X is required to be stored for 2 months and other materials for 1 month.

(d) Finished goods are held for 1 month.

(e) 25% of the total sales is on cash basis and remaining on credit basis. The credit allowed by debtors is 2 months.

(f) Average time lag in payment of all overheads is 1 months and ½ months for direct labour.

(g) Minimum cash balance of ₹ 8,00,000 is to be maintained.

CALCULATE the estimated working capital required by the company on cash cost basis if the budgeted level of activity is 1,50,000 units for the next year. The company also intends to increase the estimated working capital requirement by 10% to meet the contingencies. (You may assume that production is carried on evenly throughout the year and direct labour and other overheads accrue similarly.)

ICAI'S ANSWER

Statement showing Working Capital Requirements of TN Industries Ltd. (on cash cost basis)

	Amount in (₹)	Amount in (₹)
A. Current Assets		
(i) Inventories:		
Raw material		
X (1,50,000units × ₹ 30/12 months × 2 months)	7,50,000	
Y (1,50,000units × ₹ 7/12 months × 1 month)	87,500	
Z (1,50,000units × ₹ 6/12 months × 1 month)	75,000	
WIP (1,50,000units × ₹ 64/12 months × 0.5 month)	4,00,000	
Finished goods (1,50,000units × ₹ 88/12 months × 1 month)	11,00,000	24,12,500
(ii) Receivables (Debtors) (1,50,000units × ₹ 103/12 months × 2 months) × 0.75		19,31,250
(iii) Cash and bank balance		8,00,000
Total Current Assets		**51,43,750**
B. Current Liabilities:		
(i) Payables (Creditors) for Raw materials		
X (1,50,000units × ₹ 30/12 months × 2 months)	7,50,000	
Y (1,50,000units × ₹ 7/12 months × 1 month)	87,500	
Z (1,50,000units × ₹ 6/12 months × 0.5 month)	37,500	8,75,000
(ii) Outstanding Direct Labour (1,50,000units × ₹ 25/12 months × 0.5 month)		1,56,250
(iii) Outstanding Manufacturing and administration overheads (1,50,000units × ₹ 20/12 months × 1 month)		2,50,000
(iv) Outstanding Selling overheads (1,50,000units × ₹ 15/12 months × 1 month)		1,87,500
Total Current Liabilities		**14,68,750**
Net Working Capital Needs (A – B)		**36,75,000**
Add: Provision for contingencies @ 10%		3,67,500
Working capital requirement		**40,42,500**

Workings:

1.

(i) Computation of Cash Cost of Production	Per unit (₹)
Raw Material consumed	43
Direct Labour	25
Manufacturing and administration overheads	20
Cash cost of production	88
(ii) Computation of Cash Cost of Sales	**Per unit (₹)**
Cash cost of production as in (i) above	88
Selling overheads	15
Cash cost of sales	103

2. Calculation of cost of WIP

Particulars	Per unit (₹)
Raw material (added at the beginning):	
X	30
Y	7
Z (₹ 6 × 50%)	3
Cost during the year:	
Z {(₹ 6 × 50%) x 50%}	1.5
Direct Labour (₹ 25 × 50%)	12.5
Manufacturing and administration overheads (₹ 20 × 50%)	10
	64

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699 RTP **Old P8 May 2020** Q9 old scheme

Day Ltd., a newly formed company has applied to the Private Bank for the first time for financing it's Working Capital Requirements. The following information is available about the projections for the current year:

Estimated Level of Activity	Completed Units of Production 31,200 plus unit of work in progress 12,000
Raw Material Cost	₹ 40 per unit
Direct Wages Cost	₹ 15 per unit
Overhead	₹ 40 per unit (inclusive of Depreciation ₹10 per unit)

Selling Price	₹ 130 per unit
Raw Material in Stock	Average 30 days consumption
Work in Progress Stock	Material 100% and Conversion Cost 50%
Finished Goods Stock	24,000 Units
Credit Allowed by the supplier	30 days
Credit Allowed to Purchasers	60 days
Direct Wages (Lag in payment)	15 days
Expected Cash Balance	₹ 2,00,000

Assume that production is carried on evenly throughout the year (360 days) and wages and overheads accrue similarly. All sales are on the credit basis. You are required to CALCULATE the Net Working Capital Requirement on Cash Cost Basis.

ICAI'S ANSWER

Calculation of Net Working Capital requirement:

	(₹)	(₹)
A. Current Assets:		
Inventories:		
Stock of Raw material (Refer to Working note (iii)	1,44,000	
Stock of Work in progress (Refer to Working note (ii)	7,50,000	
Stock of Finished goods (Refer to Working note (iv)	20,40,000	
Debtors for Sales(Refer to Working note (v)	1,02,000	
Cash	2,00,000	
Gross Working Capital	32,36,000	32,36,000
B. Current Liabilities:		
Creditors for Purchases (Refer to Working note (vi)	1,56,000	
Creditors for wages (Refer to Working note (vii)	23,250	
	1,79,250	1,79,250
Net Working Capital (A - B)		**30,56,750**

Working Notes:

(i) Annual cost of production

	(₹)
Raw material requirements {(31,200 × ₹ 40) + (12,000 x ₹ 40)}	17,28,000
Direct wages {(31,200 ×₹ 15) +(12,000 X ₹ 15 × 0.5)}	5,58,000
Overheads (exclusive of depreciation) {(31,200 × ₹ 30) + (12,000 x ₹ 30 × 0.5)}	11,16,000
Gross Factory Cost	34,02,000
Less: Closing W.I.P [12,000 (₹ 40 + ₹ 7.5 + ₹15)]	(7,50,000)
Cost of Goods Produced	26,52,000
Less: Closing Stock of Finished Goods (₹ 26,52,000 × 24,000/31,200)	(20,40,000)
Total Cash Cost of Sales*	6,12,000

[*Note: Alternatively, Total Cash Cost of Sales = (31,200 units – 24,000 units) x (₹ 40 + ₹ 15 + ₹ 30) = ₹ 6,12,000]

(ii) Work in progress stock

	(₹)
Raw material requirements (12,000 units × ₹40)	4,80,000
Direct wages (50% × 12,000 units × ₹ 15)	90,000
Overheads (50% × 12,000 units × ₹ 30)	1,80,000
	7,50,000

(iii) Raw material stock

It is given that raw material in stock is average 30 days consumption. Since, the company is newly formed; the raw material requirement for production and work in progress will be issued and consumed during the year. Hence, the raw material consumption for the year (360 days) is as follows:

	(₹)
For Finished goods (31,200 × ₹ 40)	12,48,000
For Work in progress (12,000 × ₹ 40)	4,80,000
	17,28,000

$$\text{Raw material stock} = \frac{₹17,28,000}{360 \text{ days}} \times 30 \text{ days} = ₹1,44,000$$

(iv) Finished goods stock:

24,000 units @ ₹ (40+15+30) per unit = ₹20,40,000

(v) Debtors for sale: ₹ 6,12,000 × $\frac{60 \text{ days}}{360 \text{ days}}$ = ₹ 1,02,000

(vi) Creditors for raw material Purchases [Working Note (iii)]:

Annual Material Consumed (₹12,48,000 + ₹4,80,000)	₹17,28,000
Add: Closing stock of raw material [(₹17,28,000 × 30 days) / 360 days]	₹ 1,44,000
	₹18,72,000

Credit allowed by suppliers = $\frac{₹18,72,000}{360 \text{ days}} \times 30 \text{ days} = ₹ 1,56,000$

(vii) Creditors for wages:

Outstanding wage payment = [(31,200 units x ₹ 15) + (12,000 units x ₹ 15 x .50)] x 15 days / 360 days

= $\frac{₹ 5,58,000}{360 \text{ days}} \times 15 \text{ days} = ₹ 23,250$

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700 RTP **Old P8 Nov 2019** Q8 old scheme

Following are cost information of KG Ltd., which has commenced a new project for an annual production of 24,000 units which is the full capacity:

	Costs per unit (₹)
Materials	80.00
Direct labour and variable expenses	40.00
Fixed manufacturing expenses	12.00
Depreciation	20.00
Fixed administration expenses	8.00
	160.00

The selling price per unit is expected to be ₹192 and the selling expenses ₹10 per unit, 80% of which is variable.

In the first two years of operations, production and sales are expected to be as follows:

Year	Production (No. of units)	Sales (No. of units)
1	12,000	10,000
2	18,000	17,000

To assess the working capital requirements, the following additional information is available:

(a)	Stock of materials	2 months' average consumption
(b)	Work-in-process	Nil
(c)	Debtors	2 month's average sales.
(d)	Cash balance	₹ 1,00,000
(e)	Creditors for supply of materials	1 month's average purchase during the year.
(f)	Creditors for expenses	1 month's average of all expenses during the year.

PREPARE, for the two years:

(i) A projected statement of Profit/Loss (Ignoring taxation); and

(ii) A projected statement of working capital requirements

ICAI'S ANSWER

(i) Projected Statement of Profit / Loss

(Ignoring Taxation)

	Year 1	Year 2
Production (Units)	12,000	18,000
Sales (Units)	10,000	17,000

	(₹)	(₹)
Sales revenue (A) (Sales unit × ₹192)	19,20,000	32,64,000
Cost of production:		
Materials cost (Units produced × ₹80)	9,60,000	14,40,000
Direct labour and variable expenses (Units produced × ₹40)	4,80,000	7,20,000
Fixed manufacturing expenses (Production Capacity: 24,000 units × ₹12)	2,88,000	2,88,000
Depreciation (Production Capacity : 24,000 units × ₹20)	4,80,000	4,80,000
Fixed administration expenses (Production Capacity : 24,000 units × ₹8)	1,92,000	1,92,000
Total Costs of Production	24,00,000	31,20,000
Add: Opening stock of finished goods (Year 1 : Nil; Year 2 : 2,000 units)	---	4,00,000
Cost of Goods available for sale (Year 1: 12,000 units; Year 2: 20,000 units)	24,00,000	35,20,000
Less: Closing stock of finished goods at average cost (year 1: 2000 units, year 2 : 3000 units) (Cost of Production × Closing stock/ units produced)	(4,00,000)	(5,28,000)
Cost of Goods Sold	20,00,000	29,92,000

	(₹)	(₹)
Add: Selling expenses – Variable (Sales unit × ₹8)	80,000	1,36,000
Add: Selling expenses -Fixed (24,000 units × ₹2)	48,000	48,000
Cost of Sales : (B)	21,28,000	31,76,000
Profit (+) / Loss (-): (A - B)	(-) 2,08,000	(+) 88,000

Working Notes:

1. Calculation of creditors for supply of materials:

	Year 1 (₹)	Year 2 (₹)
Materials consumed during the year	9,60,000	14,40,000
Add: Closing stock (2 month's average consumption)	1,60,000	2,40,000
	11,20,000	16,80,000
Less: Opening Stock	---	1,60,000
Purchases during the year	11,20,000	15,20,000
Average purchases per month (Creditors)	93,333	1,26,667

2. Creditors for expenses:

	Year 1 (₹)	Year 2 (₹)
Direct labour and variable expenses	4,80,000	7,20,000
Fixed manufacturing expenses	2,88,000	2,88,000
Fixed administration expenses	1,92,000	1,92,000
Selling expenses (variable + fixed)	1,28,000	1,84,000
Total	10,88,000	13,84,000
Average per month	90,667	1,15,333

(ii) Projected Statement of Working Capital requirements

	Year 1 (₹)	Year 2 (₹)
Current Assets:		
Inventories:		
-Stock of materials (2 month's average consumption)	1,60,000	2,40,000
-Finished goods	4,00,000	5,28,000

	Year 1 (₹)	Year 2 (₹)
Debtors (2 month's average sales) (including profit)	3,20,000	5,44,000
Cash	1,00,000	1,00,000
Total Current Assets/ Gross working capital (A)	9,80,000	14,12,000
Current Liabilities:		
Creditors for supply of materials (Refer to working note 1)	93,333	1,26,667
Creditors for expenses (Refer to working note 2)	90,667	1,15,333
Total Current Liabilities: (B)	1,84,000	2,42,000
Estimated Working Capital Requirements: (A-B)	7,96,000	11,70,000

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701 MTP **Jan 2026 S-I** Q2(a) 8 marks

A newly formed company has applied to the commercial bank for the first time for financing its working capital requirements. The following information is available about the projections for the current year:

Estimated level of activity: 1,04,000 completed units of production plus 4,000 units of work-in-progress. Based on the above activity, estimated cost per unit is:

Raw material	₹ 80 per unit
Direct wages	₹ 30 per unit
Overheads (exclusive of depreciation)	₹ 60 per unit
Total cost	₹ 170 per unit
Selling price	₹ 200 per unit

Raw materials in stock: Average 4 weeks consumption, work-in-progress (assume 50% completion stage in respect of conversion cost) (materials issued at the start of the processing).

Finished goods in stock	8,000 units
Credit allowed by suppliers	Average 4 weeks
Credit allowed to debtors/receivables	Average 8 weeks
Lag in payment of wages	Average $1\frac{1}{2}$ weeks

Cash at banks (for smooth operation) is expected to be ₹ 25,000

Assume that production is carried on evenly throughout the year (52 weeks) and wages and overheads accrue similarly. All sales are on credit basis only.

CALCULATE Net Working Capital.

ICAI'S ANSWER

Estimate of the Requirement of Working Capital

		(₹)	(₹)
A.	**Current Assets:**		
	Raw material stock	6,64,615	
	(Refer to Working note 3)		
	Work in progress stock	5,00,000	
	(Refer to Working note 2)		
	Finished goods stock	13,60,000	
	(Refer to Working note 4)		
	Debtors	25,10,769	
	(Refer to Working note 5)		
	Cash and Bank balance	25,000	50,60,384
B.	**Current Liabilities**:		
	Creditors for raw materials	7,15,740	
	(Refer to Working note 6)		
	Creditors for wages	91,731	8,07,471
	(Refer to Working note 7)		
	Net Working Capital (A - B)		42,52,913

Working Notes:

1. **Annual cost of production**

	₹
Raw material requirements (1,04,000 units × ₹ 80)	83,20,000
Direct wages (1,04,000 units × ₹ 30)	31,20,000
Overheads (exclusive of depreciation) (1,04,000 × ₹ 60)	62,40,000
	1,76,80,000

2. **Work in progress stock**

	₹
Raw material requirements (4,000 units × ₹ 80)	3,20,000
Direct wages (50% × 4,000 units × ₹ 30)	60,000
Overheads (50% × 4,000 units × ₹ 60)	1,20,000
	5,00,000

3. **Raw material stock**

It is given that raw material in stock is average 4 weeks' consumption. Since, the company is newly formed, the raw material requirement for production and work in progress will be issued and consumed during the year.

Hence, the raw material consumption for the year (52 weeks) is as follows:

	₹
For Finished goods	83,20,000
For Work in progress	3,20,000
	86,40,000

$$\text{Raw material stock} = \frac{₹\ 86,40,000}{52\ \text{weeks}} \times 4\ \text{weeks i.e. ₹ 6,64,615}$$

4. **Finished goods stock**

8,000 units @ ₹ 170 per unit = ₹13,60,000

5. **Debtors for sale**

Credit allowed to debtors	Average 8 weeks
Credit sales for year (52 weeks) i.e. (1,04,000 units - 8,000 units)	96,000 units
Cost per unit	₹ 170
Credit sales for the year (96,000 units × ₹170)	₹ 1,63,20,000
Debtors	₹1,63,20,000/52 weeks × 8 weeks i.e. ₹ 25,10,769

6. **Creditors for raw material:**

Credit allowed by suppliers	Average 4 weeks
Purchases during the year (52 weeks) i.e. (₹83,20,000 + ₹3,20,000 + ₹6,64,615) (Refer to Working notes 1,2 and 3 above)	₹ 93,04,615
Creditors	₹93,04,615/52 weeks × 4 weeks i.e. ₹ 7,15,740

7. **Creditors for wages**

Lag in payment of wages	Average $1\frac{1}{2}$ weeks
Direct wages for the year (52 weeks) i.e. (₹ 31,20,000 + ₹ 60,000) (Refer to Working notes 1 and 2 above)	₹ 31,80,000
Creditors	₹31,80,000/52 weeks × $1\frac{1}{2}$ weeks i.e. ₹ 91,731

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702 **MTP** **Jan 2026 S-II** Q2 **10 marks**

Sundaram Industries Ltd., a medium-sized manufacturing company based in Tamil Nadu, is engaged in producing a consumer product that involves significant working capital requirements. The company's finished goods are sold across state borders, and therefore all sales and purchases are subject to Integrated GST (IGST) at 18%.

The company spends a total of ₹ 90 lakhs annually in cash, incurred at a constant rate. Its selling price structure consists of the following elements:

- Raw Material – 60%
- Wages – 20%
- Direct Overheads – 10%
- Gross Profit – 10%

Thus, 90% of the selling price represents the cost of production.

During the current month, the company has processed 75% of its purchases into finished goods and sold them entirely. Wages and Overheads are paid 80% in the current month and 20% in the next month. The Cash-to-Cash Conversion Cycle (CCC) is 30 days shorter than the Gross Operating Cycle. The company's Cash Turnover is 9 times per year. At the end of the current month, the company has paid ₹ 45,000 towards IGST liability.

The company holds long-term investments which generate a monthly return of 1.5% net of 10% TDS. If needed, the company can borrow / invest funds at an interest rate equal to this pre-tax return. There is no WIP.

The company's supplier has offered a 2% cash discount if invoices are settled within 15 days, compared to the regular day credit period. Management must evaluate whether to accept this discount.

(i) PREPARE a Working Capital Statement as at the end of the month, clearly showing Current Assets and Current Liabilities, and compute the net Working Capital requirement.

(ii) Suppose company has offered credit terms from its major supplier of 2/15, net 60. CALCULATE the implied annual cost of foregoing the discount and advise whether the company should accept the offer given its current cost of capital.

(iii) Assuming invoice value is ₹ 10,000, COMPARE the net cost to the company if it accepts the discount versus if it refuses the discount and invests the funds between day 15 and day 60.

ICAI'S ANSWER

(i) Let the Purchase of Raw Material be X.

Hence, RM consumed = 0.75X.

Wages 1/3rd of RM = 0.25X

OH & GP 1/6th of RM each = 1/3rd of RM = 0.25X

Sales = 0.75X + 0.25X + 0.25X = 1.25X

GST paid = Output IGST - Input IGST

45,000 = 0.225X - 0.18X = 0.045X

Purchases(X) = 10,00,000

Sales = 1.25X = 12,50,000

Cash Turnover = 9 = 360/Cash cycle

Cash cycle = 360/9 = 40 days

Operating Cycle = 40 + 30 = 70 days

Closing Stock = 25% of Purchases = 2,50,000

RM Holding period = $\frac{2,50,000}{7,50,000/30 \text{ days}}$ = 10 days

Cash Cycle = Gross Operating Cycle – Credit Period by Supplier

40 days = 70 days - Credit Period by Supplier

Credit Period by Supplier = 30 days or 1 months

Gross Operating Cycle = Inventory Conversion + Debtors Credit period

Debtors Credit period = 70 - 10 = 60 days

Statement of Working Capital requirements

	(₹)	(₹)
A. Current Assets		
Inventory:		
-Raw materials (Refer WN)	2,50,000	
Receivables (Debtors) (12,50,000 × 2 months)	25,00,000	
Cash balance (90,00,000 / 9)	10,00,000	37,50,000
Gross Working Capital		37,50,000
B. Current Liabilities:		
Payables:		
-Creditors for materials (1 month)	10,00,000	
Wages and overhead outstanding (20% of Total wages & OH i.e 20% x 2,50,000 for 1 month)	50,000	10,50,000
Net working capital (A - B)		27,00,000

(ii) **Evaluating Supplier's Offer**

Gross Return for Additional borrowing/ investment = $\frac{1.5\%}{90\%}$ x 12 = 20%

If the company does not avail the cash discount and pays the amount after 45 days, the implied cost of interest per annum would be approximately:

$$\left(\frac{100}{100-d}\right)^{\frac{365}{t}} - 1$$

$$\left(\frac{100}{100-2}\right)^{\frac{360}{45}} - 1$$

= 17.54%

Advise: Company should not accept supplier's offer since the gross return is more than the supplier's offer.

(iii) **Comparison of Net Cost, if the invoice amount is 10,000**

Particulars	Refuse discount (₹)	Accept discount (₹)
Payment to supplier	10,000	9,800
Return from investing ₹ 9,800 between day 15 and day 60: 45/360 x ₹ 9,800 × 20%	(245)	–
Net Cost	**9,755**	**9,800**

Advise: Thus, it is better for the company to refuse the discount, as return on cash retained is more than the saving on account of discount.

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703 MTP **Sep 2025 S-I** Q1(c) 5 marks

Wint Limited is a mid-sized manufacturing company specializing in industrial equipment parts. It operates on a just-in-time procurement strategy but has recently faced delays in raw material deliveries due to supply chain disruptions in global markets. To maintain production continuity, the management has decided to maintain slightly higher inventories and build in a contingency reserve.

The following forecasted financial information is provided for the year ending 31st March, 2025:

Particulars	Balance as at 31st March, 2025 (₹ in lakh)	Balance as at 31st March, 2024 (₹ in lakh)
Raw Material	845	585
Work-in-progress	663	455
Finished goods	910	780
Receivables	1,755	1,456
Payables	923	884

Additional details for the year ending 31st March, 2025:

- Annual purchases of raw materials (on credit): ₹ 5,200 lakh
- Annual cost of production: ₹ 5,850 lakh
- Annual cost of goods sold (COGS): ₹ 6,825 lakh

- Annual operating cost (includes admin, manufacturing, distribution): ₹ 4,225 lakh
- Annual sales (on credit): ₹ 7,605 lakh

The finance team is currently evaluating the company's working capital requirement due to a planned expansion and potential volatility in receivables collection caused by credit extensions to new clients.

Management wants to set aside a 10% contingency reserve in the working capital to deal with unforeseen cash flow constraints (e.g., delayed collections, inventory build-up, or supplier hold-ups).

Assume one year = 365 days.

You are required to:

(i) Calculate the Net Operating Cycle Period (in days).

(ii) Determine the Number of Operating Cycles in the year.

(iii) Calculate the Working Capital Requirement including a 10% contingency reserve.

ICAI'S ANSWER

Working Notes:

1. Raw Material Storage Period (R)

$$\frac{\text{Average Stock of Raw Material}}{\text{Annual Consuption of Raw Material}} \text{ x } 365$$

$$\frac{\frac{585+845}{2}}{4{,}940} \text{ x } 365 = 52.83 \text{ or } 53 \text{ days}$$

Annual Consumption of Raw Material = Opening Stock + Purchases - Closing Stock

= ₹ 585 + ₹ 5,200 – ₹ 845 = ₹ 4,940 lakh

2. Work – in - Progress (WIP) Conversion Period (W)

$$= \frac{\text{Average Stock of WIP}}{\text{Annual Cost of Production}} \text{ x } 365$$

$$= \frac{\frac{455+663}{2}}{5{,}850} \text{ x } 365 = 34.87 \text{ or } 35 \text{ days}$$

3. Finished Stock Storage Period (F)

$$= \frac{\text{Average Stock of Finished Goods}}{\text{Cost of Goods Sold}} \text{ x } 365$$

$$= \frac{\frac{780+910}{2}}{6{,}825} \text{ x } 365 = 45.19 \text{ or } 45 \text{ days.}$$

4. Receivables (Debtors) Collection Period (D)

$$= \frac{\text{Average Receivables}}{\text{Annual Credit Sales}} \text{ x } 365$$

$$= \frac{\frac{1,456+1,755}{2}}{7{,}605} \text{ x } 365 = 77.06 \text{ or } 77 \text{ days}$$

5. Payables (Creditors) Payment Period (C)

$$= \frac{\text{Average Payables for Materials}}{\text{Annual Credit Purchases}} \text{ x } 365$$

$$= \frac{\frac{884+923}{2}}{5{,}200} \text{ x } 365 = 63.42 \text{ or } 63 \text{ days}$$

(i) Net Operating Cycle Period

=R+W+F+D-C

= 53 + 35 + 45 + 77 – 63 = 147 days

(ii) Number of Operating Cycles in the Year

$$= \frac{365}{\text{Operating Cycle Period}} = \frac{365}{147} = 2.48 \text{ times}$$

(iii) Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{4,225}{2.48} = ₹\ 1,703.63 \text{ lakh}$$

Add: 10% Contingency Reserve = ₹ 170.363 lakh

Total Working Capital Requirement (including contingency) = ₹ 1,874 lakh

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704 MTP **Jan 2025 S-II Q2(a)** 6 marks

ABC Ltd., a newly formed company has applied to the Private Bank for the first time for financing it's Working Capital Requirements. The following information is available about the projections for the current year: Estimated Level of Activity Completed Units of Production 31,200.

Raw Material Cost	Rs 40 per unit
Direct Wages Cost	Rs 25 per Unit
Overhead	Rs 40 per Unit (Incl Rs 10 of Depreciation)
Selling Price	Rs 150 per unit
GP Ratio (Cash Cost)	30%
Net Profit Ratio	25% (On Total cost)
Raw Material in Stock	Avg of 30 days consumption
Work in Progress Stock at 30% of FG Produced Units	****Valued at Prime Cost** Material – 90% into process Relevant Conversion Cost – 60% completed
Finished Goods Stock	2,500 units
Credit Allowed by the supplier	30 Days
Credit Allowed to Purchasers	45 Days
Direct Wages [Lag in payment]	15 Days
Expected Cash Balance	1,25,000

Safety margin is to be kept at 15% of the net working capital required inclusive of the margin amount. Assume that production is carried on evenly throughout the year (360 days) and wages and overheads accrue similarly. All sales are on the credit basis. You are required to CALCULATE the Net Working Capital Requirement.

ICAI'S ANSWER

Problem mentions that the company has applied to the Private Bank for financing its working capital needs. Ideally, banks would not finance for Depreciation cost being a non-cash cost and it would also not finance the profit for you. So, problem needs to be solved using Cash Cost Basis.

Estimation of working capital required (cash cost basis)

	Particulars		Amount
	A) Current Assets		
	A1) Stock of RM	15,84,960 × 30/360	1,32,080.00
	A2) Stock of WIP	(From Cost Statement)	4,77,360.00
	A3) Stock of FG	(From Cost Statement)	2,37,500.00
	A4) Debtors	32,74,686 × 45/360	4,09,335.75
	A5) Cash & Cash Equivalents	(Given)	1,25,000.00
	Gross Working Capital		**13,81,275.75**
Less:	**B) Current Liabilites**		
	B1) Creditors	17,17,040 × 30/360	1,43,086.67
	B2) Lag in Wages Payment	9,20,400 × 15/360	38,350.00
	Excess of Current Assets Over Current Liabilites	(A) - (B)	**11,99,839.08**
Add:	Safety Margin @ 15% Of Net Working Capital		2,11,736.31
	Net Working Capital		**14,11,575.39**

WN -1: Calculation of Profit

Profit = 25% of total cost i.e 20% of sales price

= {(31,200-2,500) x 150} x 20% = Rs. 8,61,000

WN – 2:

	Completed Units	WIP Units
	31,200	9,360
Raw Mat. Consumed	12,48,000	3,36,960
Direct Wages	7,80,000	1,40,400
Overheads	9,36,000	1,68,480
	29,64,000	6,45,840
Gross Factory Cost	**36,09,840**	

	Completed Units	WIP Units
Add: Op WIP	-	
Less: Cl. WIP **(At Prime Cost)**	4,77,360	
Cost of Production	**31,32,480**	
Add: Op FG Stock	-	
Less: Cl. FG Stock	2,37,500	
Cash Cost of Goods Sold	**28,94,980**	
Add: Selling & Distribution Expenses (Bal. Figure)	3,79,706	
Cost Of Sales	**32,74,686**	
Profit*	8,61,000	
Sales	**41,35,686**	

*It is assumed that profit is unchanged

WN 3 - Calculation of WIP stock (units) and WIP stock amount

WIP UNITS = 30% of FG produced units i.e 30% of 31,200 units

= 9,360 units

WIP amount (at prime cost)

Raw materials = 9,360 x 40 x 90% = 3,36,960

Direct wages = 9,360 x 25 x 60% = 1,40,400

WN 4 - Calculation of purchases from suppliers

Raw Materials Consumed = OP RM Stock + Purchases - Closing RM Stock

15,84,960 = 0 + Purchases – 1,32,080

Purchases = 17,17,040

WN 5 – Calculation of safety margin

Safety Margin = 15% Of Net Working Capital Needs

Excess Of CA Less CL	85	11,99,839.08
Safety Margin	15	2,11,736.31
Net Working Capital	100	1411575.388

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705 MTP May 2024 S-II Q1(a) 5 marks

The below information for Lever Ltd is provided on annual basis:

	₹
Sales at 3 months credit	48,00,000
Materials consumed (suppliers extend 2 months credit)	12,00,000
Wages paid (one month lag in payment)	9,60,000
Cash manufacturing expenses (paid on month in arrear)	12,00,000
Administrative expense (one month lag in payment)	3,60,000
Sales promotion expense (paid monthly in advance)	1,20,000

The Company sells its products at a gross profit of 20%.

The Company keeps two months stock of raw materials and two months stock of finished goods.

Depreciation is considered as a part of cost of production.

Cash balance is retained at ₹ 1,00,000,

Assuming a 15% margin, COMPUTE the working capital requirements of the Company on cash cost basis. Ignore work-in progress.

ICAI'S ANSWER

(i) Working Notes:

(i) Computation of Annual Cash Cost of Production	**(₹)**
Material consumed	12,00,000
Wages	9,60,000
Manufacturing expenses	12,00,000
Total cash cost of production	**33,60,000**
(ii) Computation of Annual Cash Cost of Sales:	**(₹)**
Total Cash cost of production as in (i) above	33,60,000
Administrative Expenses	3,60,000
Sales promotion expenses	1,20,000
Total cash cost of sales	**38,40,000**
Add: Gross Profit @ 20% on sales (25% on cost of sales)	9,60,000
Sales Value	**48,00,000**

Statement of Working Capital requirements (cash cost basis)

	(₹)	(₹)
A. Current Assets		
Inventory:		
- Raw materials $\left(\frac{₹\ 12{,}00{,}000}{12\text{months}} \times 2\ \text{months}\right)$	2,00,000	
- Finished Goods $\left(\frac{₹\ 33{,}60{,}000}{12\text{months}} \times 2\ \text{months}\right)$	5,60,000	
Receivables (Debtors) $\left(\frac{₹\ 38{,}40{,}000}{12\text{months}} \times 3\ \text{months}\right)$	9,60,000	
Sales Promotion expenses paid in advance $\left(\frac{₹1{,}20{,}000}{12\ \text{months}} \times 1\ \text{month}\right)$	10,000	
Cash balance	1,00,000	18,30,000
Gross Working Capital		18,30,000
B. Current Liabilities:		
Payables:		
- Creditors for materials $\left(\frac{₹\ 12{,}00{,}000}{12\text{months}} \times 2\ \text{months}\right)$	2,00,000	
Wages outstanding $\left(\frac{₹\ 9{,}60{,}000}{12\text{months}} \times 1\ \text{month}\right)$	80,000	
Manufacturing expenses outstanding $\left(\frac{₹\ 12{,}00{,}000}{12\text{months}} \times 1\ \text{month}\right)$	1,00,000	
Administrative expenses outstanding $\left(\frac{₹\ 3{,}60{,}000}{12\text{months}} \times 1\ \text{month}\right)$	30,000	4,10,000
Net working capital (A - B)		14,20,000
Add: Safety margin @ 15%		2,13,000
Total Working Capital requirements		16,33,000

ICAI · P6 · MTP May 2024 · Series II · question p. 3-4 · answer p. 3-4

706 MTP Old P8 Nov 2023 S-II Q5(a) 8 marks old scheme

Cost sheet of X&Y Ltd. provides the following particulars:

	Amount per unit (₹)
Raw materials cost	260.00
Direct labour cost	125.00
Overheads cost	200.00
Total cost	585.00
Profit	75.00
Selling Price	660.00

The Company keeps raw material in stock, on an average for four weeks; work-in-progress, on an average for one week; and finished goods in stock, on an average for two weeks.

The credit allowed by suppliers is three weeks and company allow four weeks credit to its debtors. The lag in payment of wages is one week and lag in payment of overhead expenses is two weeks.

The Company sells one-fifth of the output against cash and maintains cash-in-hand and at bank put together at ₹ 2,70,000.

Required:

PREPARE a statement showing estimate of Working Capital needed to finance an activity level of 2,40,000 units of production. Assume that production is carried on evenly throughout the year, and wages and overheads accrue similarly. Work-in-progress stock is 75% complete in all respects.

ICAI'S ANSWER

Statement showing Estimate of Working Capital Needs

	(Amount in ₹)	(Amount in ₹)
A. Current Assets		
(i) Inventories:		
Raw material (4 weeks) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹260}{52 \text{ weeks}} \times 4 \text{ weeks}\right)$	48,00,000	
WIP Inventory (1 week) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹585}{52 \text{ weeks}} \times 1 \text{ week}\right) \times 0.75$	20,25,000	
Finished goods inventory (2 weeks) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹585}{52 \text{ weeks}} \times 2 \text{ weeks}\right)$	54,00,000	**1,22,25,000**
(ii) Receivables (Debtors) (4 weeks) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹585}{52 \text{ weeks}} \times 4 \text{ weeks}\right) \times \frac{4}{5}$		86,40,000
(iii) Cash and bank balance		2,70,000
Total Current Assets		2,11,35,000

	(Amount in ₹)	(Amount in ₹)
B. Current Liabilities:		
(i) Payables (Creditors) for materials (3 weeks) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹260}{52 \text{ weeks}} \times 3 \text{ weeks}\right)$		36,00,000
(ii) Outstanding wages (1 week) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹125}{52 \text{ weeks}} \times 1 \text{ week}\right)$		5,76,923
(iii) Outstanding overheads (2 weeks) $\left(\frac{2{,}40{,}000 \text{ units} \times ₹200}{52 \text{ weeks}} \times 2 \text{ weeks}\right)$		18,46,154
Total Current Liabilities		60,23,077
Net Working Capital Needs (A – B)		1,51,11,923

ICAI · Old P8 · MTP Nov 2023 · Series II · question p. 3 · answer p. 7-8

707 MTP **Old P8 Nov 2022 S-II** Case scenario · Q1(a) to Q1(d) 20 marks old scheme

CASE SCENARIO

Answer the following:

Q1(a) 5 marks

PREPARE a working capital estimate to finance an activity level of 52,000 units a year (52 weeks) based on the following data:

Raw Materials - ₹ 400 per unit

Direct Wages - ₹ 150 per unit

Overheads (Manufacturing) - ₹200 per unit

Overheads (Selling & Distribution) - ₹100perunit

Selling Price - ₹ 1,000 per unit, Raw materials & Finished Goods remain in stock for 4 weeks, Work in process takes 4 weeks. Debtors are allowed 8 weeks for payment whereas creditors allow us 4 weeks.

Minimum cash balance expected is ₹50,000. Receivables are valued at Selling Price.

Q1(b) 5 marks

Axar Ltd. has a Sales of ₹ 68,00,000 with a Variable cost Ratio of 60%.

The company has fixed cost of ₹16,32,000. The capital of the company comprises of 12% long term debt, ₹1,00,000 Preference Shares of ₹ 10 each carrying dividend rate of 10% and 1,50,000 equity shares.

The tax rate applicable for the company is 30%.

At current sales level, DETERMINE the Interest, EPS and amount of debt for the firm if a 25% decline in Sales will wipe out all the EPS.

Q1(c) 5 marks

Avesh Pvt. Ltd. is considering relaxing its present credit policy for accounts receivable and is in the process of evaluating two proposed policies. Currently, the company has annual credit sales of ₹ 55 lakhs and accounts receivable turnover ratio of 5 times a year. The current level of loss due to bad debts is ₹ 2,00,000. The company is required to give a return of 15% on the investment in new accounts

receivable. The company's variable costs are 75% of the selling price. Given the following information, IDENTIFY which is the better policy?

(Amount in ₹)

Particulars	Present Policy	Proposed Policy 1	Proposed Policy 2
Annual credit sales	55,00,000	65,00,000	70,00,000
Accounts receivable turnover ratio	5 times	4 times	3 times
Bad debt losses	2,00,000	3,50,000	5,00,000

Q1(d) 5 marks

The annual report of XYZ Ltd. provides the following information for the Financial Year 2019-20:

Particulars	Amount (₹)
Net Profit	78 lakhs
Outstanding 15% preference shares	120 lakhs
No. of equity shares	6 lakhs
Return on Investment	20%
Cost of capital i.e. (K_e)	16%

CALCULATE price per share using Gordon's Model when dividend pay-out is-

(i) 30%;

(ii) 50%;

(iii) 100%.

ICAI'S ANSWERS

Q1(a)

Cost Structure for 52000 units	
Particulars	**Amount (₹)**
Raw Material @ ₹ 400	2,08,00,000
Direct Wages @ ₹ 150	78,00,000
Manufacturing Overheads@ ₹ 200	1,04,00,000
Selling and Distribution OH@ ₹ 100	52,00,000
Total Cost	4,42,00,000
Sales@₹1000	5,20,00,000

Particulars	Calculation	Amount (₹)
A. Current Assets:		

Particulars	Calculation	Amount (₹)
Raw Material Stock	2,08,00,000 × 4/52	16,00,000
Work in Progress (WIP) Stock	2,08,00,000 + (78,00,000 + 1,04,00,000)/2 × 4/52	23,00,000
Finished Goods Stock	4,42,00,000 × 4/52	34,00,000
Receivables	5,20,00,000 × 8/52	80,00,000
Cash		50,000
	Total Current Assets	1,53,50,000
B. Current Liabilities:		
Creditors	20800000 × 4/52	16,00,000
C. **Working Capital Estimates(A-B)**		1,37,50,000

Q1(b)

Break Even Sales = ₹ 6800000×0.75 = ₹ 51,00,000

Income Statement (Amount in ₹)

	Original	Calculation of Interest at BEP (backward calculation)	Now at present level
Sales	68,00,000	51,00,000	68,00,000
Less: Variable Cost	40,80,000	30,60,000	40,80,000
Contribution	27,20,000	20,40,000	27,20,000
Less: Fixed Cost	16,32,000	16,32,000	16,32,000
EBIT	10,88,000	4,08,000	10,88,000
Less: Interest (EBIT-PBT)	?	**3,93,714**	**3,93,714**
PBT	?	14,286(10,000/70%)	6,94,286
Less: Tax @ 30%(or PBT-PAT)	?	4,286	2,08,286
PAT	?	10,000(Nil+10,000)	4,86,000
Less: Preference Dividend	10,000	10,000	10,000
Earnings for Equity share holders	?	**Nil (at BEP)**	4,76,000
Number of Equity Shares	1,50,000	1,50,000	1,50,000

	Original	Calculation of Interest at BEP (backward calculation)	Now at present level
EPS	?	-	3.1733

So Interest=₹3,93,714, EPS=₹3.1733, Amount of debt=3,93,714/12%=₹ 32,80,950

Q1(c)

Statement showing the Evaluation of Accounts Receivable Policies

(Amount in ₹)

	Particulars	Present Policy	Proposed Policy 1	Proposed Policy 2
A	Expected Profit:			
	(a) Credit Sales	55,00,000	65,00,000	70,00,000
	(b) Total Cost other than Bad Debts:			
	(i) Variable Costs (75%)	41,25,000	48,75,000	52,50,000
	(c) Bad Debts	2,00,000	3,50,000	5,00,000
	(d) Expected Profit [(a) – (b) – (c)]	11,75,000	12,75,000	12,50,000
B	Opportunity Cost of Investments in Accounts Receivable (Working Note)	1,23,750	1,82,813	2,62,500
C	Net Benefits (A – B)	10,51,250	10,92,187	9,87,500

Recommendation: The Proposed Policy 1 should be adopted since the net benefits under this policy are higher as compared to other policies.

Working Note:

Calculation of Opportunity Cost of Average Investments

Opportunity Cost = Total Cost × Collection period/12 × Rate of Return/100

Present Policy = ₹ 41,25,000 × 2.4/12 × 15% = ₹1,23,750

Proposed Policy 1 = ₹ 48,75,000× 3/12 × 15% = ₹ 1,82,813

Proposed Policy 2 = ₹ 52,50,000× 4/12 × 15% = ₹ 2,62,500

Q1(d)

Price per share according to Gordon's Model is calculated as follows:

Particulars	Amount in ₹
Net Profit	78 lakhs
Less:Preference dividend(120 lakhs@15%)	18 lakhs
Earnings for equity shareholders	60 lakhs
Earnings Per Share	60 lakhs/6 lakhs = ₹ 10.00

Price per share according to Gordon's Model is calculated as follows:

$$P_0 = \frac{E_1(1-b)}{K_e - br}$$

Here, $E_1 = 10$, $K_e = 16\%$

(i) When dividend pay-out is 30%

$$P_0 = \frac{10 \times 0.30}{0.16 - (0.70 \times 0.2)} = \frac{3}{0.16 - 0.14} = ₹150$$

(ii) When dividend pay-out is 50%

$$P_0 = \frac{10 \times 0.5}{0.16 - (0.5 \times 0.2)} = \frac{5}{0.16 - 0.10} = ₹83.33$$

(iii) When dividend pay-out is 100%

$$P_0 = \frac{10 \times 1}{0.16 - (0 \times 0.2)} = \frac{10}{0.16} = ₹ 62.5$$

ICAI · Old P8 · MTP Nov 2022 · Series II · question p. 1 · answer p. 1, 1-2, 2, 2-3

708 MTP **Old P8 May 2022 S-II** Q3 10 marks old scheme

The following annual figures relate to manufacturing entity:

A.	Sales at one month credit	84,00,000
B.	Material consumption	60% of sales value
C.	Wages (paid in a lag of 15 days)	12,00,000
D.	Cash Manufacturing Expenses	3,00,000
E.	Administrative Expenses	2,40,000
F.	Creditors extend 3 months credit for payment.	
G.	Cash manufacturing and administrative expenses are paid 1 months in arrear.	

The company maintains stock of raw material equal to economic order quantity. The company incurs ₹ 100 as per ordering cost per order and opportunity cost of capital is 15% p.a. The optimum cash balance is determined using Baumol's model. The bank charges ₹ 10 for each cash withdrawal. Finished goods are held in stock for 1 month. The company maintains a bank balance of ₹12,00,000 on an average. Creditors are paid through net banking and all other expenses are incurred in cash which is withdrawn from bank.

Assuming a 20% safety margin, you are required to ESTIMATE the amount of working capital that needs to be invested by the Company.

ICAI'S ANSWER

Statement of working capital Requirement

Particular	(₹)	(₹)
A. Current Assets		
Stock of Raw Material (W.N. 2)	81,975	

Particular	(₹)	(₹)
Stock of finished Goods $\left(65{,}40{,}000 \times \frac{1}{12}\right)$	5,45,000	
Average Receivables (at Cost) $\left(67{,}80{,}000 \times \frac{1}{12}\right)$	5,65,000	
Bank Balance	12,00,000	
Cash Balance (W.N. 3)	15,232	
Gross Working Capital		24,07,207
B. Current Liabilities		
Average Creditor for materials $\left(50{,}40{,}000 \times \frac{3}{12}\right)$	12,60,000	
Outstanding Wages $\left(12{,}00{,}000 \times \frac{0.5}{12}\right)$	50,000	
Outstanding Cash Manufacturing Expenses $\left(3{,}00{,}000 \times \frac{1}{12}\right)$	25,000	
Outstanding administrative Expenses $\left(2{,}40{,}000 \times \frac{1}{12}\right)$	20,000	
		13,55,000
Net Working Capital (A-B)		10,52,207
Add: Safety Margin @ 20%		2,10,441
Total Working Capital Requirement		**12,62,648**

Working Notes:

1. Computation of annual cash Cost of Production & Sales

Material Consumed (84,00,000 × 60%)	50,40,000
Wages	12,00,000
Manufacturing expenses	3,00,000
Cash Cost of production	**65,40,000**
(+) Administrative Expenses	2,40,000
Cash Cost of Sales	**67,80,000**

2. Computation of stock of Raw Material

A = 50,40,000

B = 100

C = 0.15

$$\therefore \text{EOQ} = \sqrt{\frac{2AB}{C}} = \sqrt{\frac{2 \times 50{,}40{,}000 \times 100}{0.15}} = ₹\ 81{,}975$$

3. Calculation of Cash Balance

A = 12,00,000+3,00,000+2,40,000

A = 17,40,000

B = 10

C = 0.15

$$\text{Optimal Cash Balance} = \sqrt{\frac{2AB}{C}} = \sqrt{\frac{2 \times 17{,}40{,}000 \times 10}{0.15}} = ₹\ 15{,}232$$

ICAI · Old P8 · MTP May 2022 · Series II · question p. 2-3 · answer p. 7-8

709 MTP **Old P8 Dec 2021 S-II Q4** 10 marks old scheme

On 01st April, 2020, the Board of Director of ABC Ltd. wish to know the amount of working capital that will be required to meet the programme they have planned for the year. From the following information, PREPARE a working capital requirement forecast and a forecast profit and loss account and balance sheet:

Issued share capital	₹ 6,00,000
10% Debentures	₹ 1,00,000
Fixed Assets	₹ 4,50,000

Production during the previous year was 1,20,000 units; it is planned that this level of activity should be maintained during the present year.

The expected ratios of cost to selling price are: raw materials 60%, direct wages 10% overheads 20%

Raw materials are expected to remain in store for an average of two months before issue to production. Each unit of production is expected to be in process for one month. The time lag in wage payment is one month.

Finished goods will stay in the warehouse awaiting dispatch to customers for approximately three months.

Credit allowed by creditors is two months from the date of delivery of raw materials. Credit given to debtors is three months from the date of dispatch.

Selling price is ₹ 5 per unit.

There is a regular production and sales cycle and wages and overheads accrue evenly.

ICAI'S ANSWER

Forecast Profit and Loss Account for the period 01.04.2020 to 31.03.2021

Particulars	₹	Particulars	₹
Materials consumed 1,20,000 @ ₹ 3	3,60,000	By Sales 1,20,000 @ ₹ 5	6,00,000
Direct wages : 1,20,000 @ ₹ 0.50	60,000		

Particulars	₹	Particulars	₹
Overheads : 1,20,000 @ ₹ 1	1,20,000		
Gross profit c/d	60,000		
	6,00,000		**6,00,000**
Debenture interest (10% of 1,00,000)	10,000	By gross profit b/d	60,000
Net profit c/d	50,000		
	60,000		**60,000**

Working Capital Requirement Forecast for the year 01.04.2020 to 31.03.2021

Particulars	Period (Months)	Total (₹)	Current Assets (₹)				Current Liabilities (₹)
			Raw materials	Work-in-progress	Finished goods	Debtors	Creditors
1. Material							
In store	2		60,000				
In work-in-progress	1			30,000			
In finished goods	3				90,000		
Credit to debtors	3					90,000	
	9						
Less : Credit from creditors	2						60,000
Net block period	7	**2,10,000**					
2. Wages:							
In work-in-progress	1/2			2,500			
In finished goods	3				15,000		
Credit to debtors	3					15,000	
	6½						
Less : Time lag in payment	1						5,000

Particulars	Period (Months)	Total (₹)	Current Assets (₹)				Current Liabilities (₹)
Net block period	**5 ½**	**27,500**					
3. Overheads:							
In work-in-progress	½			5,000			
In finished goods	3				30,000		
Credit to debtors	3					30,000	
Net block period	**6½**	**65,000**					
4. Profit							
Credit to debtors	3					15,000	
Net block period	**3**	**15,000**					
Total (₹)		**3,17,500**	**60,000**	**37,500**	**1,35,000**	**1,50,000**	**65,000**

Forecast Balance Sheet as on 31.03.2021

	(₹)			(₹)
Issued share capital	6,00,000	Fixed Assets		4,50,000
Profit and Loss A/c	50,000	Current Assets:		
10% Debentures	1,00,000	Stock:		
Sundry creditors	65,000	Raw materials	60,000	
Bank overdraft-		Work-in-progress	37,500	
Balancing figure	17,500	Finished goods	1,35,000	2,32,500
		Debtors		1,50,000
	8,32,500			**8,32,500**

The Total amount of working capital, thus, stands as follows:

	₹
Requirement as per working capital	3,17,500
Less: Bank overdraft as per balance sheet	17,500
Net requirement	3,00,000

Notes:

1. Average monthly production: 1,20,000 ÷ 12 = 10,000 units

2. Average cost per month:

Raw Material	10,000 × (₹ 5 × 0.6) = ₹ 30,000
Direct wages	10,000 × (₹ 5 × 0.1) = ₹ 5,000
Overheads	10,000 × (₹ 5 × 0.2) = ₹ 10,000

3. Average profit per month: 10,000 × (₹ 5 × 0.1) = ₹ 5,000

4. Wages and overheads accrue evenly over the period and, hence, are assumed to be completely introduced for half the processing time.

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 4 · answer p. 5-8

710 MTP **Old P8 May 2020 S-I** Q5(a) 8 marks old scheme

Cost sheet of A&R Ltd. provides the following particulars:

	Amount per unit (Rs.)
Raw materials cost	200.00
Direct labour cost	75.00
Overheads cost	150.00
Total cost	425.00
Profit	75.00
Selling Price	500.00

The Company keeps raw material in stock, on an average for four weeks; work-in-progress, on an average for one week; and finished goods in stock, on an average for two weeks.

The credit allowed by suppliers is three weeks and company allows four weeks credit to its debtors. The lag in payment of wages is one week and lag in payment of overhead expenses is two weeks.

The Company sells one-fifth of the output against cash and maintains cash-in-hand and at bank put together at Rs.2,50,000.

Required:

PREPARE a statement showing estimate of Working Capital needed to finance an activity level of 2,60,000 units of production. Assume that production is carried on evenly throughout the year, and wages and overheads accrue similarly. Work-in-progress stock is 80% complete in all respects.

ICAI'S ANSWER

Statement showing Estimate of Working Capital Needs

	(Amount in Rs.)	(Amount in Rs.)
A. Current Assets		

	(Amount in Rs.)	(Amount in Rs.)
(i) Inventories:		
Raw material (4 weeks) (2,60,000 units × Rs.200/52 weeks × 4 weeks)	40,00,000	
WIP Inventory (1 week) (2,60,000 units × Rs.425/52 weeks × 1 week) × 0.8	17,00,000	
Finished goods inventory (2 weeks) (2,60,000 units × Rs.425/52 weeks × 2 weeks)	42,50,000	99,50,000
(ii) Receivables (Debtors) (4 weeks) (2,60,000 units × Rs.425/52 weeks × 4 weeks) × 4/5th		68,00,000
(iii) Cash and bank balance		2,50,000
Total Current Assets		1,70,00,000
B. Current Liabilities:		
(i) Payables (Creditors) for materials (3 weeks) (2,60,000 units × Rs.200/52 weeks × 3 weeks)		30,00,000
(ii) Outstanding wages (1 week) (2,60,000 units × Rs.75/52 weeks × 1 week)		3,75,000
(iii) Outstanding overheads (2 weeks) (2,60,000 units × Rs.150/52 weeks × 2 weeks)		15,00,000
Total Current Liabilities		48,75,000
Net Working Capital Needs (A – B)		1,21,25,000

ICAI · Old P8 · MTP May 2020 · Series I · question p. 3-4 · answer p. 10-11

9.3 Management of receivables, credit policy, factoring and forfaiting

40 questions · asked in 32 ICAI documents · 7 exam papers

711 EXAM May 2026 MCQ 6 2 marks MCQ

A company has credit sales of ₹ 20,00,000 with an average collection period of 35 days. It wants to liberalize its existing credit terms which are expected to increase sales to ₹ 25,00,000. However, average collection period is expected to decline to 10 days. Profit Volume ratio is 20% and opportunity cost (post-tax) is 6%. Tax rate is 40%. Assume 360 days in a year.

What is the incremental investment in debtors and its impact on post-tax profitability?

	Incremental Investment in debtors	Impact on Post -tax profitability
A	₹ (1,25,000), ₹ 7,500	
B	₹ (1,00,000), ₹ (6,000)	
C	₹ (1,25,000), ₹ (7,500)	

D ₹ (1,00,000), ₹ 6,000

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P6 · May 2026 exam · question p. 2-3 · answer p. 8

712 **EXAM** **Jan 2026** Q1(b) **5 marks**

X Ltd. has average debtors of ₹ 90,00,000. The company's average collection period is 60 days. At present, it incurs credit administration expenses of ₹ 4,00,000 per year and bad-debt losses amounting to 1% of credit sales.

A factoring company has proposed to purchase these receivables by charging a commission of 2% on credit sales. The factor will provide an advance of 85% of receivables, after keeping 15% as reserve, and will charge interest at 15% per annum for the collection period. Assume 360 days in a year.

Required:

(i) Calculate the effective cost of factoring.

(ii) As a financial consultant, advise whether X Ltd. should accept the factoring proposal.

ICAI'S ANSWER

(i) Working notes:

Credit Sales= $\frac{90,00,000 \times 360}{60}$ = ₹ 5,40,00,000

	Particulars	₹
A.	**Annual Savings (Benefit) on taking Factoring Service**	
	Cost of credit administration saved	₹ 4,00,000
	Bad debts avoided	₹ 5,40,000
	Total	₹ 9,40,000
B.	**Annual Cost of Factoring to the Firm:**	
	Factoring Commission [2% x 5,40,00,000]	₹10,80,000
	Interest Charged by Factor on advance (90,00,000 – 1,80,000 – 13,50,000) x 15%	₹ 11,20,500
	Total	₹ 22,00,500
C.	**Net Cost of Factoring to the Firm:**	**₹ 12,60,500**

Advance to be paid

= (90,00,000 – 1,80,000 – 13,50,000) – $\left(\frac{₹\ 74,70,000 \times 0.15 \times 60}{360 \times 100}\right)$

= ₹ 72,83,250

Rate of Effective Cost of Factoring to the Firm = $\frac{12,60,500}{72,83,250} \times 100$

= 17.307%

Rate of Effective Cost of Factoring to the Firm can also be also calculated for advance without taking Interest into consideration:

Rate of Effective Cost of Factoring to the Firm

$$= \frac{12,60,500}{74,70,000\ (90,00,000 - 1,80,000 - 13,50,000)} \times 100 = 16.874\%$$

(ii) Advice: Since the cost to the firm exceeds the saving to the firm on account of factoring, therefore, the proposal should not be acceptable.

ICAI · P6 · Jan 2026 exam · question p. 10-11 · answer p. 13-14

713 EXAM **Sep 2024** Q3(a) 5 marks

AB Enterprises deals in hardware materials having current turnover ₹ 30 Lakhs per annum. All sales are on credit and average collection period is 30 days with zero bad debts. The customers are requesting to increase the credit period. As a result of increase in credit period sales will also increase. Other information is as under:

Credit policy	Increase in collection period (days)	Increase in sales (₹)	Bad debts anticipated
A	15	3,00,000	1%
B	30	5,00,000	3.5%

The Selling price is ₹ 100/- per unit. Variable cost per unit is ₹ 50/- and fixed cost is ₹ 5,00,000. Required rate of return on additional investment is 20%. Creditors for variable cost are ready to give 15 days extra credit for the additional cost incurred. Assume a 360 days year.

You are required to analyse the present and proposed credit policies using the "Total Approach" method and recommend the credit policy to be adopted.

ICAI'S ANSWER

A. Statement showing the Evaluation of Credit Policies (Total Approach)

Particulars		Present Credit Policy	Proposed Credit Policy	
			A	**B**
	Credit Period (in days)	30	45	60
	Units sold	30,000	33,000	35,000
		₹	₹	₹
A	**Expected Profit:**			
	(a) Credit Sales @ ₹ 100 per unit	30,00,000	33,00,000	35,00,000
	(b) Total Cost other than Bad Debts			
	(i) Variable Costs @ ₹ 50 per unit	15,00,000	16,50,000	17,50,000
	(ii) Fixed Costs	5,00,000	5,00,000	5,00,000

Particulars		Present Credit Policy	Proposed Credit Policy	
		20,00,000	21,50,000	22,50,000
	(c) Bad Debts	-	33,000	1,22,500
	(d) Expected Profit [(a) – (b) – (c)]	**10,00,000**	**11,17,000**	**11,27,500**
B	**Opportunity Cost of Investments in Receivables (i)-(ii)**	**33,333**	**52,500**	**72,917**
C	**Net Benefits (A – B)**	**9,66,667**	**10,64,500**	**10,54,583**

Recommendation: The Proposed Policy A (i.e. increase in collection period by 15 days or total 45 days) should be adopted since the net benefits under this policy are higher as compared to other policies.

Working Notes:

(i) Calculation of Opportunity Cost of Average Investment in Receivables

Particulars		Present Credit Policy	Proposed Credit Policy	
			A	**B**
	Credit Period (in days)	30	45	60
		₹	₹	₹
(a)	Cost of Sales (Variable Cost + Fixed Cost)	20,00,000	21,50,000	22,50,000
(b)	Average Debtors = Cost of Sales × (Credit period) / 360	1,66,667	2,68,750	3,75,000
(c)	Average Creditors for extra variable cost [(Variable Cost) x 15/360]	-	6,250	10,417
(d)	**Average Investment in Receivables or Net Working Capital = (b) – (c)**	1,66,667	2,62,500	3,64,583
(e)	**Opportunity Cost @20% of average Investments in Receivables**	33,333	52,500	72,917

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714 EXAM May 2024 Q1(c) 5 marks

Following is the sales information in respect of Bright Ltd:

Annual Sales (90 % on credit)	₹ 7,50,00,000
Credit period	45 days
Average Collection period	70 days
Bad debts	0.75%
Credit administration cost (out of which 2/5th is avoidable)	₹ 18,60,000

A factor firm has offered to manage the company's debtors on a non-recourse basis at a service charge of 2%. Factor agrees to grant advance against debtors at in interest rate of 14% after withholding 20% as reserve. Payment period guaranteed by factor is 45 days. The cost of capital of the company is 12.5%. One time redundancy payment of ₹ 50,000 is required to be made to factor.

Calculate the effective cost of factoring to the company. (Assume 360 days in a year)

ICAI'S ANSWER

Evaluation of Factoring Proposal

	Particulars	₹	₹
A.	**Savings due to factoring**		
	Bad Debts saved	0.75% x 7.5 crores x 90%	₹ 5,06,250
	Administration cost saved	18.6 lakhs x 2/5	₹ 7,44,000
	Interest saved due to reduction in average collection period	7.5 crores x 90% x (70-45)/ 360 x 12.5%	₹ 5,85,937.5
	Total		**₹ 18,36,187.5**
B.	**Costs of factoring:**		
	Service charge	7.5 crores x 90% x 2%	₹ 13,50,000
	Interest cost	₹ 1,15,171.875 × 360/45	₹ 9,21,375
	Redundancy Payment		₹ 50,000
	Total		**₹ 23.21,375**
C.	**Net Annual cost to the Firm: (A-B)**		**₹ 4,85,187.5**
	Rate of effective cost of factoring	₹ 4,85,187.5/ ₹ 64,66,078.125 x 100	**7.504%**

Advice: Since the rate of effective cost of factoring is less than the existing cost of capital, therefore, the proposal is acceptable.

Credit Sales = ₹ 7.5 crores x 90%	= ₹ 6,75,00,000
Average level of receivables = ₹ 6.75 crores x 45/360	= ₹ 84,37,500
Service charge = 2% of ₹ 84,37,500	₹ 1,68,750
Reserve = 20% of ₹ 84,37,500	₹ 16,87,500
Total (i)	₹ 18,56,250
Thus, the amount available for advance is	
Average level of receivables	₹ 84,37,500
Less: Total (i) from above	₹ 18,56,250
(ii)	₹ 65,81,250
Less: Interest @ 14% p.a. for 45 days	₹ 1,15,171.875
Net Amount of Advance available.	₹ 64,66,078.125

Note: Alternatively, if redundancy cost is taken as irrelevant for decision making, then Net Annual cost to the Firm will be ₹ 4,35,187.5 and Rate of effective cost of factoring will be ₹ 4,35,187.5/₹ 64,66,078.125 x 100 = 6.730%

If average level of receivables is considered for 70 days then the calculation can be done in following way:

Evaluation of Factoring Proposal

Credit Sales = ₹ 7.5 crores X 90%	= ₹ 6,75,00,000
Average level of receivables = ₹ 6.75 crores x 70/360	= ₹ 1,31,25,000
Service charge = 2% of ₹ 1,31,25,000	₹ 2,62,500
Reserve = 20% of ₹ 1,31,25,000	₹ 26,25,000
Total (i)	₹ 28,87,500
Thus, the amount available for advance is	
Average level of receivables	₹ 1,31,25,000
Less: Total (i) from above	₹ 28,87,500
(ii)	₹ 1,02,37,500
Less: Interest @ 14% p.a. for 45 days	₹ 1,79,156.25
Net Amount of Advance available.	₹ 1,00,58,343.75

Note 1: Accordingly, interest cost will be ₹ 14,33,250 cost of factoring will be ₹ 28,33,250. Therefore, Rate of effective cost of factoring is 9.913%

Note 2: Alternatively, if redundancy cost is taken as irrelevant for decision making, then Net Annual cost to the Firm will be ₹ 9,47,062.5 and Rate of effective cost of factoring will be ₹ 9,47,062.5/ ₹ 1,00,58,343.75 x 100 = 9.416%.

Advice: Since the rate of effective cost of factoring is less than the existing cost of capital, therefore, the proposal is acceptable.

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715 EXAM **Old P8 May 2023** Q1(b) 5 marks old scheme

A company has current sale of ₹ 12 lakhs per year. The profit-volume ratio is 20% and post-tax cost of investment in receivables is 15%. The current credit terms are 1/10, net 50 days and average collection period is 40 days. 50% of customers in terms of sales revenue are availing cash discount and bad debt is 2% of sales.

In order to increase sales, the company want to liberalize its existing credit terms to 2/10, net 35 days. Due to which, expected sales will increase to ₹ 15 lakhs. Percentage of default in sales will remain same. Average collection period will decrease by 10 days. 80% of customers in terms of sales revenue are expected to avail cash discount under this proposed policy.

Tax rate is 30%.

ADVISE, should the company change its credit terms. (Assume 360 days in a year.)

ICAI'S ANSWER

(i) Calculation of Cash Discount

Cash Discount = Total credit sales × % of customers who take up discount × Rate

Present Policy = $\frac{12,00,000 \times 50 \times 0.01}{100}$ = ₹ 6,000

Proposed Policy = 15,00,000 × 0.80 × 0.02 = ₹ 24,000

(ii) Opportunity Cost of Investment in Receivables

Present Policy: Opportunity Cost = Total Cost × $\frac{\text{Collection period}}{360} \times \frac{\text{Rate of Return}}{100}$

= 9,60,000 × $\frac{40}{360} \times \frac{15}{100}$ = ₹ 16,000

Proposed Policy: = Total Cost × $\frac{\text{Collection period}}{360} \times \frac{\text{Rate of Return}}{100}$

= 12,00,000 × $\frac{30}{360} \times \frac{15}{100}$ = ₹ 15,000

Statement showing Evaluation of Credit Policies

Particulars	Present Policy	Proposed Policy
Credit Sales	12,00,000	15,00,000
Variable Cost @ 80%* of sales	9,60,000	12,00,000
Bad Debts @ 2%	24,000	30,000
Cash Discount	6,000	24,000
Profit before tax	**2,10,000**	**2,46,000**

Particulars	Present Policy	Proposed Policy
Tax @ 30%	63,000	73,800
Profit after Tax	**1,47,000**	**1,72,200**
Opportunity Cost of Investment in Receivables	16,000	15,000
Net Profit	**1,31,000**	**1,57,200**

*Only relevant or variable costs are considered for calculating the opportunity costs on the funds blocked in receivables. Since 20% is profit-volume ratio, hence the relevant costs are taken to be 80% of the respective sales.

Advise: Proposed policy should be adopted since the net benefit is increased by (₹ 1,57,200 - ₹ 1,31,000) = ₹ 26,200.

Alternative presentation using incremental approach

	₹
Incremental sales (15,00,000 – 12,00,000)	3,00,000
Less: Incremental variable cost (12,00,000 – 9,60,000)	2,40,000
Less: Incremental Bad debts (30,000 – 24,000)	6,000
Less: Incremental Cash discount (24,000 – 6,000)	18,000
Increase in Profit Before Tax	**36,000**
Less: Tax @ 30%	10,800
Increase in Profit After Tax	**25,200**
Add: **Savings in opportunity cost** (16,000 - 15,000)	**1,000**
Increase in Net Profit	**26,200**

Advise: Proposed policy should be adopted since the net benefit is increased by (₹ 1,57,200 - ₹ 1,31,000)

= ₹ 26,200.

ICAI · Old P8 · May 2023 exam · question p. 1 · answer p. 4-5

716 EXAM **Old P8 Dec 2021** Q1(a) 5 marks old scheme

A factoring firm has offered a company to buy its accounts receivables.

The relevant information is given below:

(i) The current average collection period for the company's debt is 80 days and ½% of debtors default. The factor has agreed to pay over money due to the company after 60 days and it will suffer all the losses of bad debts also.

(ii) Factor will charge commission @2%.

(iii) The company spends ₹ 1,00,000 p.a. on administration of debtor.

These are avoidable cost.

(iv) Annual credit sales are ₹ 90 lakhs. Total variable costs is 80% of sales. The company's cost of borrowing is 15% per annum. Assume 365 days in a year.

Should the company enter into agreement with factoring firm?

ICAI'S ANSWER

	Particulars	(₹)
A.	**Annual Savings (Benefit) on taking Factoring Service**	
	Cost of credit administration saved	1,00,000
	Bad debts avoided (₹ 90 lakh x ½%)	**45,000**
	Interest saved due to reduction in average collection period [₹ 90 lakh x 0.80 × 0.15 × (80 days – 60 days)/365 days]	**59,178**
	Total	**2,04,178**
B.	**Annual Cost of Factoring to the Firm:**	
	Factoring Commission [₹ 90 lakh × 2%]	1,80,000
	Total	**1,80,000**
C.	**Net Annual Benefit of Factoring to the Firm (A – B)**	**24,178**

Advice: Since savings to the firm exceeds the cost to the firm on account of factoring, therefore, the company should enter into agreement with the factoring firm.

ICAI · Old P8 · Dec 2021 exam · question p. 1 · answer p. 2-3

717 EXAM **Old P8 Jul 2021** Q1(a) 5 marks old scheme

Current annual sale of SKD Ltd. is ₹ 360 lakhs. It's directors are of the opinion that company's current expenditure on receivables management is too high and with a view to reduce the expenditure they are considering following two new alternate credit policies:

	Policy X	Policy Y
Average collection period	1.5 months	1 month
% of default	2%	1%
Annual collection expenditure	₹ 12 lakh	₹ 20 lakh

Selling price per unit of product is ₹ 150. Total cost per unit is ₹ 120.

Current credit terms are 2 months and percentage of default is 3%.

Current annual collection expenditure is ₹ 8 lakh. Required rate of return on investment of SKD Ltd. is 20%. Determine which credit policy SKD Ltd. should follow.

ICAI'S ANSWER

Statement showing the Evaluation of Credit policies (Total Approach)

Particulars		Present Policy (2 Months)	Proposed Policy X (1.5 Months)	Proposed Policy Y (1 Month)
		₹ in lakhs	₹ in lakhs	₹ in lakhs
A.	**Expected Profit:**			
	(a) Credit Sales*	360	360	360
	(b) Total Cost other than Bad Debts and collection expenditure (360/150 x 120)	288	288	288
	(c) Bad Debts	10.8 (360 × 0.03)	7.2 (360 × 0.02)	3.6 (360 × 0.01)
	(d) Collection expenditure	8	12	20
	(e) Expected Profit [(a) – (b) – (c) - (d)]	53.2	52.8	48.4
B.	**Opportunity Cost of Investments in Receivables (Working Note)**	9.6	7.2	4.8
C.	**Net Benefits (A – B)**	43.6	45.6	43.6

Recommendation: The Proposed Policy X should be followed since the net benefits under this policy are higher as compared to other policies.

***Note:** It is assumed that all sales are on credit.

Working Note:

Calculation of Opportunity Cost of Average Investments

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection period}}{12} \times \frac{\text{Rate of Return}}{100}$$

$$\text{Present Policy} = ₹\ 288 \text{ lakhs} \times \frac{2}{12} \times \frac{20}{100} = ₹\ 9.6 \text{ lakhs}$$

$$\text{Policy X} = ₹\ 288 \text{ lakhs} \times \frac{1.5}{12} \times \frac{20}{100} = ₹\ 7.2 \text{ lakhs}$$

$$\text{Policy Y} = ₹\ 288 \text{ lakhs} \times \frac{1}{12} \times \frac{20}{100} = ₹\ 4.8 \text{ lakhs}$$

Alternatively

Statement showing the Evaluation of Credit policies (Incremental Approach)

Particulars		Present Policy (2 Months)	Proposed Policy X (1.5 Months)	Proposed Policy Y (1 Month)
		₹ in lakhs	**₹ in lakhs**	**₹ in lakhs**
	(a) Credit Sales*	360	360	360
	(b) Cost of sales (360/150 x 120)	288	288	288
	(c) Receivables (Refer Working Note)	48	36	24
	(d) Reduction in receivables from present policy	-	12	24
(A)	**Savings in Opportunity Cost of Investment in Receivables (@ 20%)**	**-**	**2.4**	**4.8**
	(e) Bad Debts	10.8 (360 × 0.03)	7.2 (360 × 0.02)	3.6 (360 × 0.01)
(B)	**Reduction in bad debts from present policy**	**-**	**3.6**	**7.2**
	(f) Collection expenditure	8	12	20
(C)	**Increase in Collection expenditure from Present policy**	**-**	**4**	**12**
(D)	**Net Benefits (A +B-C)**		2	0

Recommendation: The Proposed Policy X should be followed since the net benefits under this policy are higher as compared to other policies.

***Note:** It is assumed that all sales are on credit.

Working Note:

Calculation of Investment in Receivables

$$= \text{Total Cost} \times \frac{\text{Collection period}}{12}$$

$$\text{Present Policy} = ₹\ 288 \text{ lakhs} \times \frac{2}{12} = ₹\ 48 \text{ lakhs}$$

$$\text{Policy X} = ₹\ 288 \text{ lakhs} \times \frac{1.5}{12} = ₹\ 36 \text{ lakhs}$$

$$\text{Policy Y} = ₹\ 288 \text{ lakhs} \times \frac{1}{12} = ₹\ 24 \text{ lakhs}$$

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718 EXAM **Old P8 Jul 2021** Q6(b) 4 marks old scheme

Describe the salient features of FORFAITING.

ICAI'S ANSWER

The Salient features of forfaiting are:

- It motivates exporters to **explore new geographies** as payment is assured.
- An overseas buyer (importer) can import goods and services on **deferred payment terms**.
- The exporter enjoys **reduced transaction costs and complexities** of international trade transactions.
- The exporter gets to **compete in the international market** and can continue to put his working capital to good use to scale up operations.
- While importers avail of forfaiting facility from international financial institutions in order to **finance their imports at competitive rates**.

ICAI · Old P8 · Jul 2021 exam · question p. 83 · answer p. 84

719 RTP **Sep 2026** Q8

A firm has total annual sales of ₹ 200 lakhs, of which 80% are made on credit. The company offers credit terms of 2/40, net 120. Out of the total credit customers, 50% avail themselves of the cash discount, while the remaining customers settle their accounts after 120 days.

Based on past experience, bad debt losses amount to approximately 1% of credit sales. In addition, the firm incurs ₹ 2,40,000 per annum in administering its credit sales. These costs can be avoided because a factor is willing to purchase the firm's receivables.

The factor will charge a 2% commission and provide advances against receivables at an interest rate of 18%, while retaining 10% of the receivables as a reserve.

Required:

(i) Determine the effective cost of factoring. Assume a 360-day year.

(ii) If bank finance for working capital is available at an interest rate of 14%, advise whether the firm should avail itself of the factoring service.

ICAI'S ANSWER

Particulars	(₹)
Total Sales	₹ 200 lakhs
Credit Sales (80%)	₹ 160 lakhs
Receivables for 40 days	₹ 80 lakhs
Receivables for 120 days	₹ 80 lakhs
Average collection period [(40 × 0.5) + (120 × 0.5)]	80 days
Average level of Receivables (₹ 1,60,00,000 × 80/360)	₹ 35,55,556
Factoring Commission (₹ 35,55,556 × 2/100)	₹ 71,111
Factoring Reserve (₹ 35,55,556 × 10/100)	₹ 3,55,556
Amount available for advance {₹ 35,55,556 - (3,55,556 + 71,111)}	₹ 31,28,889
• Factor will deduct his interest @ 18% : • Interest = $\frac{₹31,28,889 \times 18 \times 80}{100 \times 360}$	₹ 1,25,156
• Advance to be paid (₹ 31,28,889 – ₹ 1,25,156)	₹ 30,03,733

(i) **Statement Showing Evaluation of Factoring Proposal**

		₹
A.	**Annual Cost of Factoring to the Firm:**	
	Factoring commission (₹ 71,111 × 360/80)	3,20,000
	Interest charges (₹ 1,25,156 × 360/80)	5,63,200
	Total	8,83,200
B.	**Firm's Savings on taking Factoring Service:**	₹
	Cost of credit administration saved	2,40,000
	Bad Debts (₹ 160,00,000 × 1/100) avoided	1,60,000
	Total	4,00,000
C.	Net Cost to the firm (A – B) (₹ 8,83,200 – ₹ 4,00,000)	4,83,200

Effective cost of factoring = $\frac{₹\ 4,83,200}{₹30,03,733} \times 100 = 16.09*\%$

* If cost of factoring is calculated on the basis of total amount available for advance, then, it will be

$= \frac{₹4,83,200}{₹31,28,889} \times 100 = 15.44\%$

(ii) If Bank finance for working capital is available at 14%, firm will not avail factoring service as 14 % is less than 16.08% (or 15.44%)

ICAI · P6 · RTP Sep 2026 · question p. 8 · answer p. 20-21

720 RTP **Sep 2025** Q9

A firm has a total sale of ₹ 200 lakhs of which 80% is on credit. It is offering credit terms of 2/40, net 120. Of the total, 50% of customers avail of discount and the balance pay in 120 days. Past experience indicates that bad debt losses are around 1.5% of credit sales. The firm spends about ₹ 2,40,000 per annum to administer its credit sales. These are avoidable as a factor is prepared to buy the firm's receivables. He will charge 2% commission. He will pay advance against receivables to the firm at an interest rate of 18% after withholding 10% as reserve.

(i) DETERMINE the effective cost of factoring? Consider year as 360 days.

(ii) If bank finance for working capital is available at 12% interest, ADVISE, should the firm avail of factoring service

ICAI'S ANSWER

Particulars	(₹)
Total Sales	₹ 200 lakhs
Credit Sales (80%)	₹ 160 lakhs
Receivables for 40 days	₹ 80 lakhs
Receivables for 120 days	₹ 80 lakhs
Average collection period [(40 × 0.5) + (120 ×0.5)]	80 days
Average level of Receivables (₹ 1,60,00,000 × 80/360)	₹ 35,55,556
Factoring Commission (₹ 35,55,556 × 2/100)	₹ 71,111
Factoring Reserve (₹ 35,55,556 × 10/100)	₹ 3,55,556
Amount available for advance {₹ 35,55,556 - (3,55,556 + 71,111)}	₹ 31,28,889
• Factor will deduct his interest @ 18%:	
• Interest = $\frac{₹\ 31{,}28{,}889 \times 18 \times 80}{100 \times 360}$	₹ 1,25,156
• Advance to be paid (₹ 31,28,889 – ₹ 1,25,156)	₹ 30,03,733

(i) **Statement Showing Evaluation of Factoring Proposal**

		₹
A.	**Annual Cost of Factoring to the Firm:**	
	Factoring commission (₹ 71,111 × 360/80)	3,20,000
	Interest charges (₹ 1,25,156 × 360/80)	5,63,200

		₹
	Total	8,83,200
B.	**Firm's Savings on taking Factoring Service:**	₹
	Cost of credit administration saved	2,40,000
	Bad Debts (₹ 160,00,000 × 1.5/100) avoided	2,40,000
	Total	4,80,000
C.	Net Cost to the firm (A – B) (₹ 8,83,200 – ₹ 4,80,000)	4,03,200

Effective cost of factoring = $\frac{₹\ 4,03,200}{₹30,03,733} \times 100 = 13.42^{*}\ \%$

* If cost of factoring is calculated on the basis of total amount available for advance, then, it will be = $\frac{₹4,03,200}{₹31,28,889} \times 100 = 12.89\%$

(ii) If Bank finance for working capital is available at 12%, firm will not avail factoring service as 12% is less than 13.42% (or 12.89%)

ICAI · P6 · RTP Sep 2025 · question p. 8 · answer p. 24-25

721 RTP **Jan 2025** Q10

Nirmoh Limited wants to avail short-term loan from the bank. However, bank grants short term loan by keeping the collateral in the form of accounts receivable. A bank is analyzing the receivables of Nirmoh Limited to identify acceptable collateral for a short-term loan.

The current policy of the company is 3/10 net 40. Bank will lend only to the extent of 90% of acceptable receivables at an interest rate of 12% only if both the conditions mentioned below are fulfilled. Bank will keep a reserve of 5% for cash discount & returns

(a) Customers are not currently overdue for more than 5 days to the net period

(b) Average aging (payment period) of the customer should not exceed 15 days past the net period.

If any of the above conditions are not fulfilled, the bank will lend 65% of the receivables subject to a reserve of 15% and the interest rate will be charged at 15% on such accounts. The corporate tax rate applicable is 25%.

On the scrutiny of all the receivables, following are the acceptable receivables considered for lending-

Accounts	Amount (₹)	Outstanding in Days since invoiced	Average Aging (payment period) in Days
DR 01	50,000	37	40
DR 02	25,000	25	48
DR 03	1,20,000	47	49
DR 04	72,000	10	56
DR 05	45,000	30	30
DR 06	1,75,000	39	50

Accounts	Amount (₹)	Outstanding in Days since invoiced	Average Aging (payment period) in Days
DR 07	19,000	55	25
DR 08	54,000	44	54
DR 09	1,05,000	15	25
DR 10	37,000	22	75

You are required to CALCULATE:

(a) Total amount lend by the bank

(b) Effective Interest cost (%) to the company

ICAI'S ANSWER

(A) **Condition (a)** says that accounts shouldn't be overdue for more than 5 days to the net period. In other words, it means those accounts who are overdue by 45 days (40 days + 5 additional days), will not fulfill condition a) and thus will not be eligible for 90% lending.

Therefore, from the above, we can see that **Accounts DR 03 & DR 07** are overdue for more than 45 days and hence will not be eligible for 90% lending.

Condition (b) says that average receivables ageing (payment period) should not exceed 15 days to the net period i.e. it should not exceed 55 days (40 days + 15 days = 55 days). Therefore, from the above, we can see that **Accounts DR 04 & DR 10** has an ageing of more than 55 days. Hence, they would also not be eligible for 90% lending.

Amount of Bank Lending:

Accounts	Bank Lending at 90%	Bank Lending at 65%
DR 01	50,000	-
DR 02	25,000	-
DR 03	-	1,20,000
DR 04	-	72,000
DR 05	45,000	-
DR 06	1,75,000	-
DR 07	-	19,000
DR 08	54,000	-
DR 09	1,05,000	-
DR 10	-	37,000
Total	**4,54,000**	**2,48,000**
(-) Reserve	22,700 {4,54,000 × 5%}	37,200 {2,48,000 × 15%}

Accounts	Bank Lending at 90%	Bank Lending at 65%
Net	**4,31,300**	**2,10,800**
Loan	**3,88,170**	**1,37,020**

Total short-term loan granted by the bank = ₹ 5,25,190

(B) **Calculation of the Effective Interest Cost**

Interest at 12% (On 90% lending) = 3,88,170 x 0.12 = 46,580.4

Interest at 15% (On 65% lending) = 1,37,020 x 0.15 = 20,553

Total Interest = ₹ 67,133.4

Effective Interest Cost (%) = Interest (1-t) / Total Short-term Loan

= 67,133.4 (1-0.25) / 5,25,190

Effective Interest Cost (%) = 9.59%

ICAI · P6 · RTP Jan 2025 · question p. 9-10 · answer p. 27-28

722 RTP **Sep 2024** Q11

Banu Limited is considering relaxing its present credit policy and is in the process of evaluating two proposed policies. Currently, the firm has annual credit sales of ₹ 225 lakhs and accounts receivable turnover ratio of 5 times a year. The current level of loss due to bad debts is ₹ 7,50,000. The firm is required to give a return of 20% on the investment in new accounts receivables. Policy option II requires a manager to manage the receivables with salary of ₹ 50,000 per month. The company's variable costs are 60% of the selling price. Given the following information, which is a better option?

(Amount in lakhs)

	Present Policy	Policy Option I	Policy Option II
Annual credit sales (₹)	225	275	350
Accounts receivable turnover ratio	5	4	3
Bad debt losses (₹)	7.5	22.5	47.5

ICAI'S ANSWER

Statement showing Evaluation of Credit Policies (Amount in lakhs)

	Particulars	Present Policy (₹)	Proposed Policy I (₹)	Proposed Policy II (₹)
A	Expected Profit:			
	(a) Credit Sales	225.00	275.00	350.00
	(b) Total Cost other than Bad Debts:			
	Variable Costs other than manager salary	135.00	165.00	210.00
	(c) Salary of Manager	-	-	6

	Particulars	Present Policy (₹)	Proposed Policy I (₹)	Proposed Policy II (₹)
	(d) Bad Debts	7.50	22.50	47.50
	(e) Expected Profit [(a)-(b)-(c)-(d)]	82.50	87.50	86.50
B	Opportunity Cost of Investment in Receivables*	5.40	8.25	14.40
C	Net Benefits [A-B]	77.10	79.25	72.10

Recommendation: The Proposed Policy I should be adopted since the net benefits under this policy is higher than those under other policies.

Working Note:

***Calculation of Opportunity Cost of Average Investments**

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection Period}}{12} \times \frac{\text{Rate of Return}}{100}$$

Present Policy = ₹ 135 lakhs × 2.4/12 × 20% = ₹ 5.40 lakhs

Proposed Policy I = ₹ 165 lakhs × 3/12 × 20% = ₹ 8.25 lakhs

Proposed Policy II = ₹ 216 lakhs × 4/12 × 20% = ₹ 14.40 lakhs

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723 RTP Old P8 Nov 2023 Q6 old scheme

A regular customer of your company has approached to you for extension of credit facility for purchasing of goods. On analysis of past performance and on the basis of information supplied, the following pattern of payment schedule emerges:

Pattern of Payment Schedule

At the end of 30 days	20% of the bill
At the end of 60 days	30% of the bill.
At the end of 90 days	30% of the bill
At the end of 100 days	18% of the bill
Non-recovery	2% of the bill

The customer wants to enter into a firm commitment for purchase of goods of ₹ 40 lakhs in 2022, deliveries to be made in equal quantities on the first day of each quarter in the calendar year. The price per unit of commodity is ₹ 400 on which a profit of ₹ 20 per unit is expected to be made. It is anticipated that taking up of this contract would mean an extra recurring expenditure of ₹ 20,000 per annum. If the opportunity cost is 18% per annum, would you as the finance manager of the company RECOMMEND the grant of credit to the customer? Assume 1 year = 360 days.

ICAI'S ANSWER

Statement showing the Evaluation of credit Policies

Particulars	Proposed Policy ₹
A. Expected Profit:	
(a) Credit Sales	**40,00,000**
(b) Total Cost	
(i) Variable Costs (₹ 380 × 10000 units)	38,00,000
(ii) Recurring Costs	20,000
	38,20,000
(c) Bad Debts	**80,000**
(d) Expected Profit [(a) – (b) – (c)]	**1,00,000**
B. Opportunity Cost of Investments in Receivables	**1,31,790**
C. Net Benefits (A – B)	**(31,790)**

Recommendation: The Proposed Policy should not be adopted since the net benefits under this policy are negative.

Working Note: Calculation of Opportunity Cost of Average Investments

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection period}}{360} \text{ x } \frac{\text{Rate of Return}}{100}$$

Particulars	20%	30%	30%	18%	Total
A. Total Cost	7,64,000	11,46,000	11,46,000	6,87,600	**37,43,600**
B. Collection period	30/360	60/360	90/360	100/360	
C. Required Rate of Return	18%	18%	18%	18%	
D. Opportunity Cost (A × B × C)	11,460	34,380	51,570	34,380	**1,31,790**

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724 RTP **Old P8 May 2023** Q8 old scheme

River limited currently uses the credit terms of 1.5/15 net 45 days and average collection period was 30 days. The company presently having sales of ₹ 50,00,000 and 30% customers availing the discount. The chances of default are currently 5%. Variable cost constitutes 65% and total cost constitute 85% of sales. The company is planning liberalization of credit terms to 2/20 net 50 days. It is expected that sales are likely to increase by ₹ 5,00,000, the default chances are 10% and average collection period will decline to 25 days. There won't be any change in the fixed cost and 50% customers are expected to avail the discount. Tax rate is 35%.

EVALUATE this policy in comparison with the current policy and recommend whether the new policy should be implemented. Assume cost of capital to be 10% (post tax) and 360 days in a year.

ICAI'S ANSWER

Evaluation of Credit Policies

Particulars		1.5/15 net 45	2/20 net 50
A	Sales	₹50,00,000	₹55,00,000
B	Variable Cost (65%)	₹32,50,000	₹35,75,000
C	Fixed Cost (20% in 1st Case)	₹10,00,000	₹10,00,000
D	Bad Debts (5% and 10%)	₹2,50,000	₹5,50,000
E	Discounts		
	(₹5000000×30%x1.5%)	₹22,500	-
	(₹5500000×50%x2%)	-	₹55,000
F	PBT (A-B-C-D-E)	₹4,77,500	₹3,20,000
G	Tax @ 35%	₹1,67,125	₹1,12,000
H	PAT	₹3,10,375	₹2,08,000
I	Opportunity Cost		
	(₹3250000 + ₹1000000) x 30/360×10%	₹35,417	-
	(₹3575000 + ₹1000000) x 25/360 × 10%	-	₹31,771
J	**Net Benefit**	**₹2,74,958**	**₹1,76,229**

The new policy leads to lower net benefit for the company. Hence it should not be implemented.

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725 RTP **Old P8 Nov 2020** Q6 old scheme

A company wants to follow a more prudent policy to improve its sales for the region which is ₹ 9 lakhs per annum at present, having an average collection period of 45 days. After certain researches, the management consultant of the company reveals the following information:

Credit Policy	Increase in collection period	Increase in sales	Present default anticipated
W	15 days	₹ 60,000	1.5%
X	30 days	₹ 90,000	2%
Y	45 days	₹ 1,50,000	3%
Z	70 days	₹ 2,10,000	4%

The selling price per unit is ₹ 3. Average cost per unit is ₹ 2.25 and variable costs per unit are ₹ 2. The current bad debt loss is 1%. Required return on additional investment is 20%. (Assume 360 days year)

ANALYSE which of the above policies would you recommend for adoption?

ICAI'S ANSWER

A. Statement showing the Evaluation of Debtors Policies (Total Approach)

(Amount in ₹)

Particulars		Present Policy 45 days	Proposed Policy W 60 days	Proposed Policy X 75 days	Proposed Policy Y 90 days	Proposed Policy Z 115 days
I.	**Expected Profit:**					
	(a) Credit Sales	9,00,000	9,60,000	9,90,000	10,50,000	11,10,000
	(b) Total Cost other than Bad Debts					
	(i) Variable Costs [Sales × 2/ 3]	6,00,000	6,40,000	6,60,000	7,00,000	7,40,000
	(ii) Fixed Costs	75,000	75,000	75,000	75,000	75,000
		6,75,000	7,15,000	7,35,000	7,75,000	8,15,000
	(c) Bad Debts	9,000	14,400	19,800	31,500	44,400
	(d) Expected Profit [(a) – (b) – (c)]	2,16,000	2,30,600	2,35,200	2,43,500	2,50,600
II.	**Opportunity Cost of Investments in Receivables**	16,875	23,833	30,625	38,750	52,069
III.	**Net Benefits (I – II)**	1,99,125	2,06,767	2,04,575	2,04,750	1,98,531

Recommendation: The Proposed Policy W (i.e. increase in collection period by 15 days or total 60 days) should be adopted since the net benefits under this policy are higher as compared to other policies.

Working Notes:

(i) Calculation of Fixed Cost = [Average Cost per unit – Variable Cost per unit] × No. of Units sold

= [₹ 2.25 - ₹ 2.00] × (₹ 9,00,000/3)

= ₹ 0.25 × 3,00,000 = ₹ 75,000

(ii) Calculation of Opportunity Cost of Average Investments

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection period}}{360} \times \frac{\text{Rate of Return}}{100}$$

$$\text{Present Policy} = 6{,}75{,}000 \times \frac{45}{360} \times \frac{20}{100} = 16{,}875$$

$$\text{Policy W} = 7{,}15{,}000 \times \frac{60}{360} \times \frac{20}{100} = 23{,}833$$

$$\text{Policy X} = 7{,}35{,}000 \times \frac{75}{360} \times \frac{20}{100} = 30{,}625$$

$$\text{Policy Y} = 7{,}75{,}000 \times \frac{90}{360} \times \frac{20}{100} = 38{,}750$$

$$\text{Policy Z} = 8{,}15{,}000 \times \frac{115}{360} \times \frac{20}{100} = 52{,}069$$

B. Another method of solving the problem is **Incremental Approach**. Here we assume that sales are all credit sales.

(Amount in ₹)

Particulars		Present Policy 45 days	Proposed Policy W 60 days	Proposed Policy X 75 days	Proposed Policy Y 90 days	Proposed Policy Z 115 days
I.	**Incremental Expected Profit:**					
	(a) Incremental Credit Sales	0	60,000	90,000	1,50,000	2,10,000
	(b) Incremental Costs					
	(i) Variable Costs	6,00,000	40,000	60,000	1,00,000	1,40,000
	(ii) Fixed Costs	75,000	-	-	-	-
	(c) Incremental Bad Debt Losses	9,000	5,400	10,800	22,500	35,400
	(d) Incremental Expected Profit (a – b –c)]		14,600	19,200	27,500	34,600
II.	**Required Return on Incremental Investments:**					
	(a) Cost of Credit Sales	6,75,000	7,15,000	7,35,000	7,75,000	8,15,000
	(b) Collection period	45	60	75	90	115
	(c) Investment in Receivable (a × b/360)	84,375	1,19,167	1,53,125	1,93,750	2,60,347
	(d) Incremental Investment in Receivables	-	34,792	68,750	1,09,375	1,75,972
	(e) Required Rate of Return (in %)		20	20	20	20
	(f) Required Return on Incremental Investments (d × e)	**-**	**6,958**	**13,750**	**21,875**	**35,194**
III.	**Net Benefits (I – II)**	-	7,642	5,450	5,625	(594)

Recommendation: The Proposed Policy W should be adopted since the net benefits under this policy are higher than those under other policies.

C. Another method of solving the problem is by computing the **Expected Rate of Return.**

$$\text{Expected Rate of Return} = \frac{\text{Incremental Expected Profit}}{\text{Incremental Investment in Receivables}} \times 100$$

$$\text{For Policy W} = \frac{₹\ 14,600}{₹\ 34,792} \text{ x } 100 = 41.96\%$$

$$\text{For Policy X} = \frac{₹\ 19,200}{₹\ 68,750} \text{ x } 100 = 27.93\%$$

$$\text{For Policy Y} = \frac{₹\ 27,500}{₹\ 1,09,375} \text{ x } 100 = 25.14\%$$

$$\text{For Policy Z} = \frac{₹\ 34,600}{₹\ 1,75,972} \text{ x } 100 = 19.66\%$$

Recommendation: The Proposed Policy W should be adopted since the Expected Rate of Return (41.96%) is more than the Required Rate of Return (20%) and is highest among the given policies compared.

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726 RTP Old P8 May 2020 Q6 old scheme

TM Limited, a manufacturer of colour TV sets is considering the liberalization of existing credit terms to three of their large customers A, B and C. The credit period and likely quantity of TV sets that will be sold to the customers in addition to other sales are as follows:

Quantity sold (No. of TV Sets)

Credit Period (Days)	A	B	C
0	10,000	10,000	-
30	10,000	15,000	-
60	10,000	20,000	10,000
90	10,000	25,000	15,000

The selling price per TV set is ₹15,000. The expected contribution is 50% of the selling price. The cost of carrying receivable averages 20% per annum.

You are required to COMPUTE the credit period to be allowed to each customer.

(Assume 360 days in a year for calculation purposes).

ICAI'S ANSWER

In case of customer A, there is no increase in sales even if the credit is given. Hence comparative statement for B & C is given below:

Particulars	Customer B				Customer C			
1. Credit period (days)	0	30	60	90	0	30	60	90
2. Sales Units	10,000	15,000	20,000	25,000	-	-	10,000	15,000
	₹ in lakh				**₹in lakh**			
3. Sales Value	1,500	2,250	3,000	3,750	-	-	1,500	2,250
4. Contribution at 50% (A)	750	1,125	1,500	1,875	-	-	750	1,125
5. Receivables:- Credit Period × Sales/360	-	187.5	500	937.5	-	-	250	562.5
6. Debtors at cost	-	93.75	250	468.75	-	-	125	281.25

Particulars	Customer B				Customer C			
7. Cost of carrying debtors at 20% (B)	-	18.75	50	93.75	-	-	25	56.25
8. Excess of contributions over cost of carrying debtors (A – B)	750	1,106.25	1,406.25	1,781.25	-	-	725	1,068.75

The excess of contribution over cost of carrying Debtors is highest in case of credit period of 90 days in respect of both the customers B and C. Hence, credit period of 90 days should be allowed to B and C.

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727 RTP **Old P8 Nov 2019 Q9** old scheme

A regular customer of your company has approached to you for extension of credit facility for purchasing of goods. On analysis of past performance and on the basis of information supplied, the following pattern of payment schedule emerges:

Pattern of Payment Schedule

At the end of 30 days	20% of the bill
At the end of 60 days	30% of the bill.
At the end of 90 days	30% of the bill.
At the end of 100 days	18% of the bill.
Non-recovery	2% of the bill.

The customer wants to enter into a firm commitment for purchase of goods of ₹30 lakhs in 2019, deliveries to be made in equal quantities on the first day of each quarter in the calendar year. The price per unit of commodity is ₹300 on which a profit of ₹10 per unit is expected to be made. It is anticipated that taking up of this contract would mean an extra recurring expenditure of ₹10,000 per annum. If the opportunity cost is 18% per annum, would you as the finance manager of the company RECOMMEND the grant of credit to the customer? Assume 1 year = 360 days.

ICAI'S ANSWER

Statement showing the Evaluation of credit Policies

Particulars	Proposed Policy ₹
A. Expected Profit:	
(a) Credit Sales	30,00,000
(b) Total Cost	

Particulars	Proposed Policy ₹
(i) Variable Costs	29,00,000
(ii) Recurring Costs	10,000
	29,10,000
(c) Bad Debts	60,000
(d) Expected Profit [(a) – (b) – (c)]	30,000
B. Opportunity Cost of Investments in Receivables	1,00,395
C. Net Benefits (A – B)	(70,395)

Recommendation: The Proposed Policy should not be adopted since the net benefits under this policy are negative

Working Note: Calculation of Opportunity Cost of Average Investments

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection period}}{360} \times \frac{\text{Rate of Return}}{100}$$

Particulars	20%	30%	30%	18%	Total
A. Total Cost	5,82,000	8,73,000	8,73,000	5,23,800	28,51,800
B. Collection period	30/360	60/360	90/360	100/360	
C. Required Rate of Return	18%	18%	18%	18%	
D. Opportunity Cost (A × B × C)	8,730	26,190	39,285	26,190	1,00,395

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728 MTP **Sep 2026 S-I** Q2(a) 7 marks

Bright Vision Electronics Limited, a manufacturer of Colour Television sets, is considering the relaxation of its existing credit policy for three of its major customers, namely A, B, and C. The proposed credit periods and the estimated additional sales of Colour TV sets expected from these customers as a result of the revised credit terms are given below:

Quantity sold (No. of TV Sets)

Credit Period (Days)	A	B	C
0	1,000	1,000	-
30	1,000	1,500	-
60	1,000	2,000	1,000
90	1,000	2,500	1,500

The selling price per TV set is ₹ 9,000. The expected contribution is 20% of the selling price. The cost of carrying receivable averages 20% per annum.

You are required:

(a) Determine the credit period to be allowed to each customer.

(Assume 360 days in a year for calculation purposes).

(b) What other problems the company might face in allowing the credit period as determined in (a) above?

ICAI'S ANSWER

(a) In case of customer A, there is no increase in sales even if the credit is given. Hence comparative statement for B & C is given below:

Particulars	Customer B				Customer C			
1. Credit period (days)	0	30	60	90	0	30	60	90
2. Sales Units	1,000	1,500	2,000	2,500	-	-	1,000	1,500
				₹ in lakhs				**₹in lakhs**
3. Sales Value	90	135	180	225	-	-	90	135
4. Contribution at 20% (A)	18	27	36	45	-	-	18	27
5. Receivables: Credit Period × Sales/360	-	11.25	30	56.25	-	-	15	33.75
6. Debtors at cost i.e. 80% of 11.25	-	9	24	45	-	-	12	27
7. Cost of carrying debtors at 20% (B)	-	1.8	4.8	9	-	-	2.4	5.4
8. Excess of contributions over cost of carrying debtors (A – B)	18	25.2	31.2	36	-	-	15.6	21.6

The excess of contribution over cost of carrying Debtors is highest in case of credit period of 90 days in respect of both the customers B and C. Hence, credit period of 90 days should be allowed to B and C.

(b) **Problem**

(i) Customer A is taking 1000 TV sets whether credit is given or not. Customer C is taking 1000 TV sets at credit for 60 days. Hence A also may demand credit for 60 days compulsorily.

(ii) B will take 2500 TV sets at credit for 90 days whereas C would lift 1500 sets only. In such case B will demand further relaxation in credit period i.e. B may ask for 120 days credit.

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729 MTP Sep 2026 S-II Q1(a) 5 marks

The current credit sales of a firm is ₹ 20,00,000 and the firm still has an unutilized capacity. In order to boost its sales, the firm is willing to relax its credit policy. The firm proposes a new credit policy of 2/15 net 60 days as against the present policy of 1/15 net 45 days. The firm expects an increase in the sales by 15%. However, it is also expected that bad debts will go up to 2.5% of sales from 1.5%. The contribution to sales ratio of the firm is 25%. The firm's tax rate is 30% and the firm requires an after tax return of 12% on its investment.

Should the firm change the credit policy?

ICAI'S ANSWER

Evaluation of Credit Policies

Particulars	Present Policy (₹)	Proposed Policy (₹)
Credit Sales	20,00,000	23,00,000 (115% of 20,00,000)
Variable Cost (75%)	(15,00,000)	(17,25,000)
Contribution	**5,00,000**	**5,75,000**
Bad Debts	(30,000) (20,00,000 × 1.5%)	(57,500) (23,00,000 × 2.5%)
Profit Before Tax (PBT)	**4,70,000**	**5,17,500**
Tax @ 30%	(1,41,000)	(1,55,250)
Profit After Tax (PAT)	**3,29,000**	**3,62,250**
Opportunity Cost (Refer working note)	(22,500)	(34,500)
Net Profit	**3,06,500**	**3,27,750**

In the proposed scheme the net profit is more by ₹ 21,250 i.e. (₹ 3,27,750 – ₹ 3,06,500), hence, the company should change the credit policy.

Working Note:

Opportunity Cost on Investment in Debtors:

Present policy = Variable Cost × (Collection period / 360) × Required rate of return

$= 15,00,000 \times \frac{45}{360} \times 12\% = ₹\ 22,500$

Proposed policy $= 17,25,000 \times \frac{60}{360} \times 12\% = ₹\ 34,500$

Assumptions:

(i) Cash discount is not availed by the debtors.

(ii) Debtors are utilising the full credit period for payment.

(iii) Number of days in a year is 360 days.

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730 MTP May 2026 S-I Q3 10 marks

Being a financial controller of Shilpa Limited, they have provided you with the debt collection policy of the debtors' ageing analysis as on the date of analysis –

Name of Customers	Current	31 – 60 Days	61 – 90 Days	91 or More Days	Total
Customer A	3,12,500	1,05,000	-	-	4,17,500
Customer B	-	-	3,11,200	-	3,11,200
Customer C	6,81,600	-	-	-	6,81,600
Customer D	-	-	-	1,62,000	1,62,000
Customer E	-	3,01,500	2,10,000	38,000	5,49,500
	9,94,100	**4,06,500**	**5,21,200**	**2,00,000**	**21,21,800**

Collection Policy –

(a) Normal Credit terms are for 30 days

(b) Interest of 2% is charged on accounts overdue for 45 days

(c) Interest of 4% is charged on accounts overdue for more than 75 days

(d) Cash Discount of 2% is given to customers who make payment within 10 days of invoice getting due.

Additional Information –

(a) 40% of the invoice value after bad debt falling in the ageing criteria of 31-60 days' pay after 45th day but pay within the 60th day with 1.5% of the total being uncollectible

(b) 25% of the invoice value after bad debt falling in the ageing criteria of 61-90 days' pay after 75th day but pay within the 90th day with 3% of the total being uncollectible.

(c) 15% of the total invoice value overdue for 91 days or more are uncollectible.

(d) 50% of the customers (invoice value) avail the cash discount.

(e) Administration cost for managing the above customers would also be involved amounting to ₹ 50,000.

(i) CALCULATE the effective cost involved in the collection process assuming that the operating cost of sales ratio is 0.75.

(ii) EXPLAIN three aspects of management of receivables.

ICAI'S ANSWER

(i) (a) Total cost involved in the collection would include bad debts, cash discount & administration cost i.e. Cost of Carrying debtors

WN – 1 Calculation of Bad debts:

= 4,06,500 x 1.5% + 5,21,200 x 3% + 2,00,000 x 15%

= 6,098 + 15,636 + 30,000

= ₹ 51,734

WN – 2 Calculation of Cash Discount:

= (9,94,100 x 50%) x 2% = ₹ 9,941

WN – 3 Calculation of Interest Income charged on account of delayed payment:

	Amount
Total Invoice value for 31 – 60 days	4,06,500
Less: Bad debts @ 1.5%	(6,098)
	4,00,402
40% of above (payment after 45th day but within 60th day)	1,60,161
Interest @ 2% (160161 × 2%)	**3,203**

	Amount
Total Invoice value for 61 – 90 days	5,21,200
Less: Bad debts @ 3%	(15,636)
	5,05,564
25% of above (payment after 75th day but within 90th day)	1,26,391
Interest @ 4% (126391 × 4%)	**5,056**

∴ Total Interest Income = 3,203 + 5,056 = ₹ 8,259

Operating Cost of Sales = 21,21,800 x 0.75 = ₹ 15,91,350

Effective Cost of Debtors Policy

$$= \frac{\text{Bad debts} + \text{Cash discount} + \text{Administration Cost - Interest Charged Received}}{\text{Operating Cost of Sales}}$$

$$= \frac{51{,}734 + 9{,}941 + 50{,}000 - 8{,}259}{15{,}91{,}350}$$

∴ **Effective Cost of Debtors Policy = 6.50%**

Notes -

i. In absence of information on Opportunity Rate of return on Investment, Opportunity Cost of Investment in receivables is not calculated.

(ii) There are basically three aspects of management of receivables:

1. ***Credit Policy:*** A balanced credit policy should be determined for effective management of receivables. Decision of Credit standards, Credit terms and collection efforts is included in Credit policy. It involves a trade-off between the profits on additional sales that arise due to credit being extended on the one hand and the cost of carrying those debtors and bad debt losses on the other. This seeks to decide credit period, cash discount and other relevant matters.
2. ***Credit Analysis:*** This requires the finance manager to determine as to how risky it is to advance credit to a particular party. This involves due diligence or reputation check of the customers with respect to their credit worthiness.

3. ***Control of Receivable:*** This requires finance manager to follow up debtors and decide about a suitable credit collection policy. It involves both laying down of credit policies and execution of such policies.

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731 MTP **May 2026 S-II** Q2(b) 2 marks

EXPLAIN the 'Aging Schedule' in the context of monitoring of receivables.

ICAI'S ANSWER

Ageing Schedule: An important means to get an insight into collection pattern of debtors is the preparation of their 'Ageing Schedule'. Receivables are classified according to their age from the date of invoicing e.g. 0 – 30 days, 31 – 60 days, 61 – 90 days, 91 – 120 days and more. The ageing schedule can be compared with earlier month's figures or the corresponding month of the earlier year.

This classification helps the firm in its collection efforts and enables management to have a close control over the quality of individual accounts. The ageing schedule can be compared with other firms also.

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732 MTP **Sep 2025 S-I** MCQ (vii) 2 marks MCQ

A company has annual credit sales of ₹ 6,00,00,000. The average collection period is 90 days. A factor offers to buy the receivables with a 2% commission. The factor will advance on receivables to the company at an interest rate of 18% per annum after withholding 10% as reserve. Calculate the amount of advance to be paid assuming 360 days in a year.

A ₹ 1,32,00,000

B ₹ 1,28,92,500

C ₹ 1,40,38,500

D ₹ 1,26,06,000

ICAI'S ANSWER

Answer key: **(D)**

(d) ₹ 1,26,06,000

Average level of Receivables	₹ 6,00,00,000 × 90/360 = ₹1,50,00,000
Factoring Commission	₹ 1,50,00,000 × 2% = ₹ 3,00,000
Factoring Reserve	₹ 1,50,00,000 × 10% = ₹ 15,00,000
Amount available in advance	₹ 1,50,00,000 – (₹ 3,00,000 + ₹ 15,00,000) = ₹ 1,32,00,000
Factor will deduct interest @ 18%	
So, Interest	₹ 1,32,00,000 × 18/100 × 90/360 = 5,94,000
Hence, advance to be paid	₹ 1,32,00,000 - 5,94,000 = ₹ 1,26,06,000

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733 MTP Sep 2025 S-II Q3(a) 8 marks

Oggy Limited has a current credit sales of ₹ 7,20,000. It is considering revising its credit policy. The proposed terms of credit will be "2/10, net 30" against the present policy of "net 30".

As a result, Oggy Limited's credit sales are expected to increase by ₹ 20,000 and the average collection period will reduce from 30 days to 20 days. It is also expected that 50 percent of the customers will take the discounts and pay on the 10th day and rest of the customers will pay on the 30th day. Bad debt losses will remain at 2 percent of sales. The variable cost ratio is 70 percent.

Its corporate tax rate is 50 percent and opportunity cost of investment in receivables is 10 percent.

ADVISE whether Oggy Limited should change its credit period?

ICAI'S ANSWER

Advise to Oggy Limited regarding Change in Credit Policy

Particulars	(₹)
Current Credit Sales	7,20,000
Increase in Credit Sales	20,000
New Level of Credit Sales	7,40,000
Current Average Collection Period (Days)	30
New Average Collection Period (Days)	20
Current Level of Receivables $\left(₹\ 7{,}20{,}000 \times \frac{30}{360}\right)$	60,000
New Level of Receivables $\left(₹\ 7{,}40{,}000 \times \frac{20}{360}\right)$	41,111
Cash Discount	2%
Discount Period (Days)	10
Percentage of Customers Taking Discount	50%
Bad Debt Losses	2%
Variable Cost	70%
Corporate Tax Rate	50%
Opportunity Cost of Capital	10%
(A) Increased Credit Sales	20,000
(B) Contribution from Increased Credit Sales [A x (1-0.70)]	6,000
(C) Bad Debt Loss [A x 2%]	400
(D) Cost of Cash Discount [₹ 740,000 × 0.02 × 0.5]	7,400
(E) After-tax Profit [(B - C - D) x (1-0.5)]	(900)

Particulars	(₹)
(F) Decrease in Receivable Investment [₹ 41,111 - ₹ 60,000]	(18,889)
(G) Expected Return (E/F)	4.8%
(H) Net Gain % [10% -G]	5.2%

Therefore, Oggy Limited should change the credit policy because it results in net gain of 5.2%.

ICAI · P6 · MTP Sep 2025 · Series II · question p. 9 · answer p. 12

734 MTP **May 2025 S-I** Q2(b) 4 marks

Also asked in Old P8 MTP Nov 2020 S-I Q1(a)

ABC Pvt. Ltd. is considering relaxing its present credit policy for accounts receivable and is in the process of evaluating two proposed policies. Currently, the company has annual credit sales of ₹ 50 lakhs and accounts receivable turnover ratio of 4 times a year. The current level of loss due to bad debts is ₹ 1,50,000. The company is required to give a return of 20% on the investment in new accounts receivable. The company's variable costs are 70% of the selling price. Given the following information, IDENTIFY which is the better policy?

(Amount in ₹)

Particulars	Present Policy	Proposed Policy 1	Proposed Policy 2
Annual credit sales	50,00,000	60,00,000	67,50,000
Accounts receivable turnover ratio	4 times	3 times	2.4 times
Bad debt losses	1,50,000	3,00,000	4,50,000

ICAI'S ANSWER

Statement showing the Evaluation of Accounts Receivable Policies

(Amount in ₹)

	Particulars	Present Policy	Proposed Policy 1	Proposed Policy 2
A	**Expected Profit:**			
	(a) Credit Sales	50,00,000	60,00,000	67,50,000
	(b) Total Cost other than Bad Debts:			
	(i) Variable Costs (70%)	35,00,000	42,00,000	47,25,000
	(c) Bad Debts	1,50,000	3,00,000	4,50,000
	(d) Expected Profit [(a) – (b) – (c)]	13,50,000	15,00,000	15,75,000
B	**Opportunity Cost of Investments in Accounts Receivable** (Working Note)	1,75,000	2,80,000	3,93,750
C	**Net Benefits (A – B)**	11,75,000	12,20,000	11,81,250

Recommendation: The Proposed Policy 1 should be adopted since the net benefits under this policy are higher as compared to other policies.

Working Note:

Calculation of Opportunity Cost of Average Investments

Opportunity Cost = Total Cost × Collection period/12 × Rate of Return/100

Present Policy = ₹ 35,00,000 × 3/12 × 20% = ₹ 1,75,000

Proposed Policy 1 = ₹ 42,00,000 × 4/12 × 20% = ₹ 2,80,000

Proposed Policy 2 = ₹ 47,25,000 × 5/12 × 20% = ₹ 3,93,750

ICAI · P6 · MTP May 2025 · Series I · question p. 6 · answer p. 10-11

735 MTP **May 2025 S-II** Case Scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Ashrav Machines Private Limited (AMPL) was started as Joint venture company in 2017 by a leading group head quartered at Mumbai, engaged into the business of automatic washing machine product segment. The company received state-of-the-art technological know-how from a well-known Japanese company named Suzutsu Inc which was its JV partner. The company had acquired superior technology through JV, and its sales were expected to grow at 10 per cent per annum. The Company has two types of customer base, retail customers and dealership customers. The company experienced a significant growth in its sales during the first 5 years of its operations, and its actual growth rate averaged about 30 per cent per annum. The company had cost and technological advantages, and therefore, could keep its prices significantly lower as compared to its competitors. Over this period, the company established its name, and its products became well-known in the market.

As the competition intensified in the washing machine market, and the general market slow-down conditions engulfed the entire economy, AMPL started experiencing a decline in its sales from the beginning of FY 2022. The company identified poor inventory management on the retail front like piling up of inventory stocks, very high import of raw material, high selling price, etc. & poor receivables management on the dealership front like poor credit policy & collection process, bad debts, etc. as a major issue causing financial problems and decline. The company decided to work out a plan to reduce the inventory levels and rejig the receivables process and come out with feasible solutions. The concerned leadership teams in the company presented the following plans:

(1) Exploring selling the washing machines at discounted price. The company plans to offer a 5% average discount on its current price thereby resulting in a discounted price of ₹ 15,675 per machine. However, this didn't create any significant impact on sales as expected. The previous studies on market behavior to price responses in this category of product suggested less sensitivity to price reduction or discounts in the lower range. However, a price reduction above 10 per cent indicated a significant association with the increase in demand. Using these findings as the basis of their argument, the marketing team proposed a plan to provide discounts of up to 12 per cent. It provided an estimate of about 1,000 units to be sold every quarter during the year after implementing the suggestion of proposed discount.

(2) Owing to the recent trends observed in the exchange-sales scheme in the television and other electronic segments, the company wants to explore this scheme for the washing machine segment as well. Upon subsequent research, the marketing team argued that there was a good chance of an

exchange scheme becoming popular provided the customers were offered an attractive exchange scheme as most of the existing machines owned by the households did not have new features which the company's product offered. The marketing team gave an estimate of selling about 4,650 units in the next one year, provided the scheme offered, off-setting ₹ 3,000 to ₹ 4,000 of the prices to customers who purchase the company's product in exchange for their existing machines. The actual deduction, however, would depend on the condition of the machines and for that, the company estimated that an average offset of ₹ 3,500 per machine would be needed.

(3) The third proposal pertains to an all-round reduction in inventory levels. If the production takes place as per the sales plan, the total inventory holding period of the company works out about 3½ months. Of this, the finished goods holding period alone is 2 to 2½ months. The company, therefore, must hold finished goods for this period to sell its products in the market. Recently, due to a slow-down in the market, this period had increased to 4 – 4½ months. As a result, the working capital requirements of the company have increased significantly. The company recently explored the possibility of selling the goods to its customers under a scheme which would deliver the machine after 1½ months of placing the order. A pilot test was carried out with a select group of customers. Based on the results of this pilot study, the proposed scheme would offer the following package to a customer: on paying an advance of ₹ 3,000, the customer could order the product which would be delivered after 1½ months of time. As per the terms and conditions of the scheme, the customer would pay ₹ 9,500 at the time of delivery of the product. Through this scheme, the customer would get a benefit of ₹ 4,500 for this waiting period. The field tests had indicated that a section of customers would not mind waiting for 1 to 1½ months and taking this benefit. The marketing team gave an estimate that about 400 machines could be sold per month through this scheme.

(4) The current dealership sales of the company are ₹ 3,65,00,000 and it has a present policy of extending 60 days credit but most of its dealers in the recent period have stretched payments and the average collection period was about 120 days. Ninety per cent of the firm's sales are on a credit basis. Out of this, 15 per cent is sold to the customers whose financial position is not very sound and the entire bad debt losses amount to about 2 per cent of sales to these customers. The cost of Sales is 80% and the current collection expenses amount to ₹ 5,00,000 per year. After consulting the marketing and accounting staff and analyzing the status of the competition, the company has decided to change its credit policy for the dealership customers.

Policy A : To offer cash discount to motivate customers to pay early. The new credit terms would be "2/15, net 60". A quick study of sample customers indicated that about two-thirds of them might like to avail cash discounts. This change in policy would not change the expected sales, but the average collection period is likely to be reduced to 80 days for the remaining customers. There will be no changes in bad debts, but the collection cost will be reduced by 30% to the existing collection expense.

Policy B : To relax its credit standards by 20 Days to expand its sales. This is expected to increase sales by 10 per cent. The marginal customers, which would also include new customers, are not expected to take advantage of cash discounts and are likely to continue to take on an average of 150 days to pay. In the case of these customers, bad debt losses are expected to increase to 2.5 per cent on the sales to financially unsound customers, which will remain the same in proportion. The company would enforce collections with more vigour and diligence. It is expected that collection expenses will increase by ₹ 1,00,000 per annum.

The company generally requires a rate of return of 15 per cent from its investments and assume 360 days in a year.

Based on the above scenario, answer to the following MCQs:

MCQ 1 2 marks

Incremental benefit of adopting credit Policy A to the company is _____

- A ₹ 6,24,500
- B ₹ 58,500
- C ₹ 46,57,450
- D ₹ (-) 6,24,500

MCQ 2 2 marks

Incremental benefit of adopting credit Policy B to the company is _____

- A ₹ 27,294
- B ₹ 8,70,444
- C ₹ 46,84,744
- D ₹ 2,31,444

MCQ 3 2 marks

Evaluate all the proposed options of inventory management and select the best sequence as per their rankings based on benefit analysis to the company in the form of sales.

- A Proposal 1, 2 & 3
- B Proposal 2, 3 & 1
- C Proposal 2, 1 & 3
- D Proposal 3, 2 & 1

MCQ 4 2 marks

What are the bad debt expenses of policy A and policy B?

- A ₹ 98,550 and ₹ 98,550
- B ₹ 1,35,506 and ₹ 98,550
- C ₹ 98,550 and ₹ 1,35,506
- D ₹ 1,35,506 and ₹ 1,35,506

MCQ 5 2 marks

The optimum combination of inventory and credit policy comes out to be _____

- A Proposal 3 & Credit Policy 'A'
- B Proposal 3 & Credit Policy 'B'
- C Proposal 2 & Credit Policy 'B'
- D Proposal 2 & Credit Policy 'A'

ICAI'S ANSWERS

MCQ 1

(A) ₹ 6,24,500

Calculation of Incremental Benefit Statement (Current Policy v/s Policy A)

	Current Policy (₹)	Policy 'A' (₹)
Credit Sales (365 lakhs X 0.9)	3,28,50,000	3,28,50,000
(-) Cost Of Sales @ 0.8	2,62,80,000	2,62,80,000
Operating Profit	65,70,000	65,70,000
(-) Bad Debts Exp (Wn-1)	98,550	98,550
(-) Collection Exp	5,00,000	3,50,000 (30% Reduction)
(-) Cash Discount Exp (Wn-2)	-	4,38,000
(-) Opportunity Cost of Carrying Debtors (Wn – 3)	13,14,000	4,01,500
Net Benefit	**46,57,450**	**52,81,950**

Incremental Benefit in Policy 'A' = ₹ 6,24,500

WN – 1: Calculation of Bad debts expense

Of the total credit sales, 15% are sold to the customers who are not financially sound and 2% of such sales is bad debts.

Bad debts = ₹ 3,28,50,000 x 0.15 x 0.02 = ₹ 98,550 (same under Policy A as well)

WN – 2: Calculation of Cash Discount under Policy A

Credit Sales made to 2/3 of the customer's avail Cash discount @ rate of 2% as they would be paying their dues within maximum of 15 days.

Cash Discount = ₹ 3,28,50,000 x 2/3 x 2% = ₹ 4,38,000

WN – 3: Calculation of Opportunity Cost of Carrying Debtors assuming Debtors are valued at Cost of Sales

Current Policy -

a) Average Debtors = Cost of Sales x Avg Collection Period / 360

= (₹ 3,28,50,000 x 80%) x 120 / 360

= ₹ 87,60,000

b) Apply Rate of Return on above debtors calculated

= ₹ 87,60,000 X 0.15 = ₹ 13,14,000

Policy 'A'

Total Cost of Sales = ₹ 2,62,80,000

2/3 Customers

Cost of Sales = ₹ 1,75,20,000

Avg Collection Period = 15 days

Debtors = 1,75,20,000 x 15/360

= ₹ 7,30,000

Opportunity Cost = ₹ 7,30,000 x 0.15 = ₹ 1,09,500

1/3 Customers

Cost of Sales = ₹ 87,60,000

Avg Collection Period = 80 days

Debtors = 87,60,000 x 80/360 = ₹ 19,46,667

Opportunity Cost = ₹ 19,46,667 x 0.15 = ₹ 2,92,000

Total Opportunity Cost = ₹ 1,09,500 + ₹ 2,92,000 = ₹ 4,01,500

MCQ 2

(A) ₹ 27,294

Calculation of Incremental Benefit Statement (Current Policy v/s Policy B)

	Current Policy (₹)	Policy 'B' (₹)
Credit Sales (365 lakhs X 0.9)	3,28,50,000	3,61,35,000 (10% Increase)
(-) Cost Of Sales @ 0.8	2,62,80,000	2,89,08,000
Operating Profit	65,70,000	72,27,000
(-) Bad Debts Exp (Wn-1)	98,550	1,35,506
(-) Collection Exp	5,00,000	6,00,000
(-) Opportunity Cost of Carrying Debtors (Wn – 3)	13,14,000	18,06,750
Net Benefit	**46,57,450**	**46,84,744**

Incremental Benefit in Policy 'B' = ₹ 27,294

WN – 1: Calculation of Bad debts expense under Policy 'B'

Of the total credit sales, 15% are sold to the customers who are not financially sound (As proportion of those customer to remain same) and 2.5% of such sales is bad debts.

Bad debts = ₹ 3,61,35,000 x 0.15 x 0.025 = ₹ 1,35,506.25

WN – 3: Calculation of Opportunity Cost of Carrying Debtors assuming Debtors are valued at Cost of Sales under Policy 'B'.

Average Debtors = Cost of Sales x Avg Collection Period / 360

= ₹ 2,89,08,000 x 150 / 360

= ₹ 1,20,45,000

Therefore, Opportunity Cost of Carrying Debtors = ₹ 1,20,45,000 x 0.15 = ₹ 18,06,750

MCQ 3

(B) Proposal 2, 3 & 1

Evaluating Proposal, I - Selling washing machine at discount

Selling price (with 5% discount) = ₹ 15,675

Therefore, Selling Price (Without Discount) = ₹ 15,675 / 0.95 = ₹ 16,500 per unit.

To increase the demand, a minimum of 10% discount is necessary and the marketing team has proposed a 12% discount to get a competitive advantage.

Thus, the New Selling Price = ₹ 16,500 – 12% = ₹ 14,520 per unit.

Total Sales @ new Selling Price = 1,000 Units x 4 Quarters x ₹ 14,520

= ₹ 5,80,80,000

Evaluating Proposal II - Exchange-sales scheme

Total Sales (Units) = 4,650

Selling Price per unit = ₹ 16,500

(-) Exchange off set price = ₹ 3,500

Net Proceeds = ₹ 13,000

Total Sales = 4,650 x ₹ 13,000

= ₹ 6,04,50,000

Evaluating Proposal III - All-round reduction in Inventory levels

Selling Price = ₹ 3,000 advance + ₹ 9,500 on delivery = ₹ 12,500

Total Sales = 400 x 12 x ₹ 12,500

= ₹ 6,00,00,000

Rankings: Rank 1 – Proposal 2

Rank 2 – Proposal 3

Rank 3 – Proposal 1

NOTE: In absence of other info, we would be considering the benefits in the form sales and decide on the proposal to be selected

MCQ 4

(C) ₹ 98,550 and ₹ 1,35,506

MCQ 5

(D) The best combination of Inventory proposal and Credit policies comes to Proposal 2 & Credit Policy 'A'.

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736 MTP **Jan 2025 S-I** Q3 10 marks

Zomo Ltd. currently has a turnover of ₹ 120 lakhs, 75% of which is on credit. The variable cost ratio is 80%, and the credit terms offered are 2/10, net 30. On the current sales volume, the bad debts are 1%, and the company spends ₹ 1,20,000 annually on administering its credit sales, including staff salaries for credit checking and collection. These costs are avoidable.

In addition:

- 60% of customers avail of the 2% cash discount, and the remaining customers take 60 days on average to pay after the date of sale.
- The book debts are financed by a mix of bank borrowings and owned funds in a 1:1 ratio, with annual costs of 15% and 14%, respectively.

However, Zomo Ltd. is also considering dynamic discounting for its cash customers, which might incentivize more customers to pay earlier by increasing the discount rate. This could lead to a potential reduction in bad debts to 0.8% but may also increase the cost of the discount offered to 2.5%.

A factoring firm has proposed a deal with the following terms: (i) Factor reserve: 12% (ii) Guaranteed payment: 25 days (iii) Interest charges: 15% (iv) Commission: 4% of receivables.

In addition, the company also has the option to extend the credit period for its remaining customers (who do not avail of the discount) to 75 days, which might increase sales by 10% but could result in an increase in bad debts to 1.5%.

Given:

1. The cost of funds is expected to rise to 16% next year.
2. Zomo Ltd. plans to introduce late payment penalties (for customers who take more than 60 days) at 5% of outstanding receivables after 60 days.

Assume a 360-day year.

Required:

- SHOULD Zomo Ltd. opt for dynamic discounting or the factoring firm's offer?
- ANALYZE the impact of extending the credit period on the company's finances.

COMPARE all options and RECOMMEND whether to continue with in-house management, dynamic discounting, or accept the factoring firm's offer.

ICAI'S ANSWER

1. In-House Management of Receivables (With Dynamic Discounting)

Particulars:

1. **Cash Discount Cost:**
   - Revised discount rate: 2.5%
   - 60% of customers avail discount.
   - **Cost of Discount:** ₹ 90,00,000 × 60% × 2.5% = ₹ 1,35,000
2. **Bad Debts (Reduced to 0.8% due to dynamic discounting):**
   - ₹ 90,00,000 × 0.8% = ₹ 72,000
3. **Administration Cost:** ₹ 1,20,000
4. **Cost of Financing Receivables:**
   - **Working Note 1 (Average Collection Period):** (10 days × 60%) + (60 days × 40%) = 30 days
   - **Working Note 2 (Average Receivables):** ₹ 90,00,000 × (30/360) = ₹ 7,50,000
   - **Working Note 3 (Cost of Financing):**
     - Cost of Bank Funds: ₹ 7,50,000 × 1/2 × 15% = ₹ 56,250
     - Cost of Owned Funds: ₹ 7,50,000 × 1/2 × 14% = ₹ 52,500
     - **Total Cost of Financing Receivables:** ₹ 1,08,750

Total Cost with In-House Receivables Management and Dynamic Discounting:

Particulars	Amount (₹)
Cash Discount (₹ 90,00,000 × 60% × 2.5%)	1,35,000
Bad Debts (₹ 90,00,000 × 0.8%)	72,000
Admin Cost	1,20,000
Cost of Financing Receivables	1,08,750
Total Cost (In-House with Dynamic Discounting):	**4,35,750**

2. Factoring Firm's Offer:

Particulars:

1. **Factoring Commission:** ₹ 90,00,000 × 4% = ₹ 3,60,000
2. **Interest Charges on Receivables:** Factor Reserve: 12%, so financing on 88% of receivables. Interest for 25 days: (₹ 90,00,000-3,60,000) × 88% × 15% × (25/360) = ₹ 79,200
3. **Cost of Owned Funds (Receivables not factored):** ₹ 13,96,800 × 14% × (25/360) = ₹ 13580

Owned Funds: (₹ 90,00,000-3,60,000) × 12% + 3,60,000 = ₹ 13,96,800

Total Cost with Factoring Firm:

Particulars	Amount (₹)
Factoring Commission (₹ 90,00,000 × 4%)	3,60,000
Interest Charges on Receivables	79,200
Cost of Owned Funds	13,580
Total Cost with Factoring:	**4,52,780**

3. Impact of Extending Credit Period:

If Zomo Ltd. extends the credit period to 75 days:

- **Sales increase:** 10% of ₹ 120,00,000 = ₹ 12,00,000
  New total turnover = ₹ 120,00,000 + ₹ 12,00,000 = ₹ 1,32,00,000

  Credit Sales (75%) = ₹ 99,00,000
- **Increased Bad Debts (1.5%):** ₹ 99,00,000 × 1.5% = ₹ 1,48,500
- **Late Payment Penalty:** Customers delaying beyond 60 days (40%): ₹ 99,00,000 × 40% × 5% = ₹ 1,98,000

A. Cash Discount Cost:

- **Discount rate:** 2% (since there's no mention of dynamic discounting in this case)
- **Percentage of customers availing discount:** 60%
- **Calculation:** ₹ 99,00,000 × 60% × 2% = ₹ 1,18,800

B. Bad Debts (Increased to 1.5%):

- **Calculation:** ₹ 99,00,000 × 1.5% = ₹ 1,48,500

C. Administration Costs (Remains the same):

- The administration cost stays fixed at ₹ 1,20,000, as no change in admin structure is mentioned.

D. Cost of Financing Receivables (Based on the new extended credit period):

- **Working Note 1 (Average Collection Period):** Credit period has been extended to 75 days for customers who don't take the discount (40% of customers).
  - **Revised Average Collection Period:** (10 days × 60%) + (75 days × 40%) = 36 days
- **Working Note 2 (Average Receivables):** ₹ 99,00,000 × (36/360) = ₹ 9,90,000
- **Working Note 3 (Cost of Financing Receivables):**
  - **Cost of Bank Funds (15%):** ₹ 9,90,000× 1/2 × 15% = ₹ 74,250
  - **Cost of Owned Funds (14%):** ₹ 9,90,000 × 1/2 × 14% = ₹ 69,300
  - **Total Cost of Financing Receivables:** ₹ 74,250 + ₹ 69,300 = ₹ 1,43,550

Revised Bad Debts after Penalty:

- **Bad debts before penalty:** ₹ 1,48,500
- **Penalty earned:** ₹ 1,98,000
- **Net effect on bad debts:** ₹ 1,48,500 - ₹ 1,98,000 = (-₹ 49,500) (Zomo Ltd. would effectively earn ₹ 49,500 from penalties, reducing bad debt cost.)

4. Total Cost Calculation:

Now, summing up all the components:

Particulars	Amount (₹)
Cash Discount (₹ 99,00,000 × 60% × 2%)	1,18,800
Net Bad Debts after Penalty (–₹ 49,500)	-49,500
Administration Costs	1,20,000
Cost of Financing Receivables	1,43,550
Total Cost (In-House with Extended Credit Period)	**₹ 3,32,850**

5. Final Decision:

Option	Total Cost (₹)
In-House with Dynamic Discounting	4,35,750
Factoring Firm's Offer	4,52,780
In-House with Extended Credit Period	**3,32,850**

Recommendation: Zomo Ltd. should **extend the credit period** and continue in-house management. This option will not only reduce costs (due to lower bad debts offset by penalties) but also increase

sales by 10%. Factoring is the least beneficial due to its high commission charges, and dynamic discounting offers only marginal savings compared to the credit extension option.

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737 MTP **Jan 2025 S-II** Q4(a) 4 marks

EXPLAIN the difference between factoring and forfaiting

ICAI'S ANSWER

Particulars	Factoring	Forfaiting
A) Meaning	**Factoring** involves sales of receivables to the financial institution called factor in exchange for immediate cash payment	**Forfaiting** is a form of export financing where the exporter sells the rights to trade receivables to a forfaiter and receives instant cash
B) Recourse or non-recourse	May be on Recourse or Non-recourse basis	Always non-recourse
C) Amount paid	Firms are generally paid 80% to 90% upfront	100% on the value of exported goods is paid
D) Type of receivables	Receivables may either domestic or international	Receivables are international
E) Cost	Factoring cost in the form of factor commission or fees is to be borne by the seller	Overseas Buyer bears the forfaiting cost, if any
F) Secondary market	Factoring does not involve a secondary market for the receivables, meaning that the transaction is complete once the receivables are sold to the factor.	Forfaiting has a secondary market where the receivables can be traded, enhancing liquidity and providing additional opportunities for investors

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738 MTP **Sep 2024 S-I** Q2(a) 6 marks

The financial statements of Gurunath Ltd is furnished below –

Balance Sheet as at 31st March

Particulars as at 31st March		Note	₹
I	**EQUITY AND LIABILITIES:**		
(1)	Shareholders' Funds:		10,00,000
(2)	Non–Current Liabilities: 10% Debt		6,00,000
(3)	Current Liabilities		1,56,000
	Total		17,56,000
II	**ASSETS**		

Particulars as at 31st March		Note	₹
(1)	Non–Current Assets		16,56,000
(2)	Current Assets – Trade Receivables		1,00,000
	Total		17,56,000

Additional Information:

1. The existing credit terms are 1/10, net 45 days and average collection period is 30 days. The current bad debts loss is 1.5%. In order to accelerate the collection process further as also to increase sales, the company is contemplating liberalization of its existing credit terms to 2/10, net 45 days.

2. It is expected that sales are likely to increase by 1/3 of existing sales, bad debts increase to 2% of sales and average collection period to decline to 20 days.

3. Credit period allowed by the supplier is 60 days. Generally, operating expenses are paid 2 months in arrears. Total Variable expenses of the company constitute Purchases of stock in trade and operating expenses only.

4. Opportunity cost of investment in receivables is 15%. 50% and 80% of customers in terms of sales revenue are expected to avail cash discount under existing and liberalization scheme respectively. The tax rate is 30%.

5. The Company considers only the relevant or variable costs for calculating the opportunity costs on the funds blocked in receivables. Assume 360 days in a year and 30 days in a month.

Should the company change its credit terms?

ICAI'S ANSWER

Particulars	Result
Current liabilities	1,56,000
Total Variable expenses = Purchases & Operating Expenses	1,56,000 ÷ 60 × 360 = 9,36,000
Variable expenses % of Sales	9,36,000 ÷ 12,00,000 × 100 = 78%

Particulars	Present	Proposed
1. Sales	1 Lakh ÷ 30 × 360 = 12,00,000	12 Lakhs + 1/3rd = 16,00,000
2. Variable Cost at 78%	9,36,000	12,48,000
3. Cash Discount	12 Lakh × 50% × 1% = 6,000	16 Lakh × 80% × 2% = 25,600
4. Bad debts	12 Lakh × 1.5% = 18,000	16 Lakh × 2% = 32,000
5. Profit before Tax	2,40,000	2,94,400
6. Tax @ 30%	72,000	88,320
7. Profit after Tax	1,68,000	2,06,080

Particulars	Present	Proposed
8. Opportunity Cost of Invest. in Debtors	9,36,000 × 30/360 × 70% ×15% = 8,190	12,48,000 × 20/360 × 70% × 15% = 7,280
9. **Net Benefit**	**1,59,810**	**1,98,800**

Advise: Proposed policy should be adopted since the net benefit is increased by (₹ 1,98,800 - 1,59,810) = ₹ 38,990.

ICAI · P6 · MTP Sep 2024 · Series I · question p. 5 · answer p. 6-7

739 MTP **Sep 2024 S-II** Q2(a) 7 marks

Sukrut Limited has annual credit sales of ₹ 75,00,000/-. Actual credit terms are 30 days, but its management of receivables has been poor, and the average collection period is about 60 days. Bad debt is 1 per cent of total sales.

A factor has offered to take over the task of debt administration and credit checking, at an annual fee of 1.5 per cent of credit sales. Sukrut Limited estimates that it would save ₹ 45,000 per year in administration costs as a result. Due to the efficiency of the factor, the average collection period would come back to the original credit offered of 30 days and bad debts would come to 0.5% on recourse basis.

The factor would pay net advance of 80 percent to the company at an annual interest rate of 12 per cent after withholding a reserve of 10%.

Sukrut Limited is currently financing its receivables from an overdraft costing 10 per cent per year and will continue to finance the balance fund needed (which is not financed by factor) through the overdraft facility

If occurrence of credit sales is throughout the year, COMPUTE whether the factor's services should be accepted or rejected. Assume 360 days in a year.

ICAI'S ANSWER

Evaluation of Factoring Proposal -

	PARTICULARS	₹	₹
(A)	**Savings (Benefit) to the firm**		
	Administration Cost	45,000	45,000
	Bad Debts Cost (On Recourse basis) In House – 75 lakhs X 1% Factoring – 75 lakhs X 0.5% **Net Savings in bad debts cost**	(75 lakhs X 0.5%)	37,500
	Cost of Carrying Debtors Cost	**(WN – 1)**	1,06,750
	TOTAL		**1,89,250**
(B)	**Cost to the Firm:**		
	Factor Commission [Annual credit Sales × % of Commission]	75 lakhs X 1.5%	1,12,500
	Interest Cost on Net advances	**(See WN – 1)**	53,100

	PARTICULARS	₹	₹
	TOTAL		**1,65,600**
(C)	**Net Benefits to the Firm (A – B)**		**23,650**

Advice: Since the savings to the firm exceed the cost due to factoring, the proposal is acceptable.

WN-1 : Calculation of Savings in Interest Cost of Carrying Debtors

(I) In house Management:

Interest Cost = Credit Sales X Avg Collection Period / 360 X Interest (%) p.a

= 75,00,000 x 60/360 x 10%

= 1,25,000

(II) If Factoring services availed: If factoring services are availed, then Sukrut Limited must raise the funds blocked in receivables to the extent which is not funded by the factor (i.e amount of factor reserve **(+)** amount of factor commission for 30 days **(+)** 20% of net advances)

Calculation of Net Advances to the firm -

Debtors = 75 lakhs x 30/360 = 6,25,000

(-) Factor Reserve = 10% of above = (62,500)

(-) Factor Commission = 1.5% of Debtors = (9,375)

Net Advance = 5,53,125

Advance from Factor = 5,53,125 x 80% = 4,42,500

Int cost on Advance from Factor = 4,42,500 x 12% = 53,100

Now, the amount that is not funded by the factor (6,25,000 - 4,42,500) needs to be funded by Sukrut Limited from overdraft facility at 10%

Therefore, Int cost on Overdraft (Cost of carrying debtors) = **1,82,500 x 10% = 18,250**

Net Savings in Interest Cost of Carrying Debtors = 1,25,000 (-) 18,250 = 1,06,750

ICAI · P6 · MTP Sep 2024 · Series II · question p. 4 · answer p. 4-5

740 MTP **May 2024 S-I** Case scenario · MCQ 1(I) to MCQ 1(V) 10 marks

CASE SCENARIO

NV Industries Ltd. is a manufacturing industry which manages its accounts receivables internally by its sales and credit department. It supplies small articles to different industries. The total sales ledger of the company stands at ₹ 200 lakhs of which 80% is credit sales. The company has a credit policy of 2/40, net 120. Past experience of the company has been that on average out of the total, 50% of customers avail of discount and the balance of the receivables are collected on average in 120 days. The finance controller estimated, bad debt losses are around 1% of credit sales.

With escalating cost associated with the in-house management of the debtors coupled with the need to unburden the management with the task so as to focus on sales promotion, the CFO is examining the possibility of outsourcing its factoring service for managing its receivables. Currently, the firm spends about ₹ 2,40,000 per annum to administer its credit sales. These are avoidable as a factoring firm is prepared to buy the firm's receivables. The main elements of the proposal are : (i) It will charge

2% commission (ii) It will pay advance against receivables to the firm at an interest rate of 18% after withholding 10% as reserve.

Also, company has option to take long term loan at 15% interest or may take bank finance for working capital at 14% interest.

You were also present at the meeting; being a financial consultant, the CFO has asked you to be ready with the following questions:

Consider year as 360 days.

MCQ 1(I) 2 marks

What is average level of receivables of the company?

A ₹ 53,33,333

B ₹ 35,55,556

C ₹ 44,44,444

D ₹ 71,11,111

MCQ 1(II) 2 marks

How much advance factor will pay against receivables?

A ₹ 31,28,889

B ₹ 39,11,111

C ₹ 30,03,733

D ₹ 46,93,333

MCQ 1(III) 2 marks

What is the annual cost of factoring to the company?

A ₹ 8,83,200

B ₹ 4,26,667

C ₹ 5,51,823

D ₹ 4,00,000

MCQ 1(IV) 2 marks

What is the net cost to the company on taking factoring service?

A ₹ 4,00,000

B ₹ 4,26,667

C ₹ 3,50,000

D ₹ 4,83,200

MCQ 1(V) 2 marks

What is the effective cost of factoring on advance received?

A 16.09%

B 13.31%

C 12.78%

D 15.89%

ICAI'S ANSWERS

MCQ 1(I)

(B) ₹ 35,55,556

MCQ 1(II)

(C) ₹ 30,03,733

MCQ 1(III)

(A) ₹ 8,83,200

MCQ 1(IV)

(D) ₹ 4,83,200

MCQ 1(V)

(A) 16.09%

Working Note

Particulars	(₹)
Total Sales	₹ 200 lakhs
Credit Sales (80%)	₹ 160 lakhs
Receivables for 40 days	₹ 80 lakhs
Receivables for 120 days	₹ 80 lakhs
Average collection period [(40 × 0.5) + (120 ×0.5)]	80 days
Average level of Receivables (₹ 1,60,00,000 × 80/360)	₹ 35,55,556
Factoring Commission (₹ 35,55,556 × 2/100)	₹ 71,111
Factoring Reserve (₹ 35,55,556 × 10/100)	₹ 3,55,556
Amount available for advance {₹ 35,55,556 - (3,55,556 + 71,111)}	₹ 31,28,889
Factor will deduct his interest @ 18%: Interest = $\frac{₹31,28,889 \times 18 \times 80}{100 \times 360}$	₹ 1,25,156
Advance to be paid (₹ 31,28,889 – ₹ 1,25,156)	₹ 30,03,733

(i) Statement Showing Evaluation of Factoring Proposal

		₹
A.	**Annual Cost of Factoring to the Company:**	
	Factoring commission (₹ 71,111 × 360/80)	3,20,000
	Interest charges (₹ 1,25,156 × 360/80)	5,63,200
	Total	8,83,200
B.	**Company's Savings on taking Factoring Service:**	₹
	Cost of credit administration saved	2,40,000

		₹
	Bad Debts (₹ 160,00,000 × 1/100) avoided	1,60,000
	Total	4,00,000
C.	Net Cost to the company (A – B) (₹ 8,83,200 – ₹ 4,00,000)	4,83,200

Effective cost of factoring = $\frac{₹\ 4,83,200}{₹30,03,733} \times 100 = 16.09\%$

ICAI · P6 · MTP May 2024 · Series I · question p. 1-2 · answer p. 1, 1-2

741 **MTP** **Old P8 Nov 2022 S-I** Q3 **10 marks** old scheme

GT Ltd. is taking into account the revision of its credit policy with a view to increasing its sales and profit. Currently, all its sales are on one month credit. Other information is as follows:

Contribution	2/5th of Sales Revenue
Additional funds raising cost	20% per annum

The marketing manager of the company has given the following options along with estimates for considerations:

Particulars	Current Position	Option I	Option II	Option III
Sales Revenue (₹)	40,00,000	42,00,000	44,00,000	50,00,000
Credit period (in months)	1	1½	2	3
Bad debts (% of sales)	2	2½	3	5
Cost of Credit administration (₹)	24,000	26,000	30,000	60,000

You are required to ADVISE the company for the best option.

ICAI'S ANSWER

Statement Showing Evaluation of Credit Policies

(₹ in lakhs)

Particulars	Current position (1 month)	Option I (1.5 months)	Option II (2 months)	Option III (3 months)
Sales Revenue	40,00,000	42,00,000	44,00,000	50,00,000
Contribution @ 40%	16,00,000	16,80,000	17,60,000	20,00,000
Increase in contribution over current level (A)	–	80,000	1,60,000	4,00,000
Debtors = $\left(\frac{\text{Average Collection period x Credit Sale}}{12}\right)$	$\frac{1 \times 40,00,000}{12}$ = 3,33,333.33	$\frac{1.5 \times 42,00,000}{12}$ = 5,25,000	$\frac{2 \times 44,00,000}{12}$ = 7,33,333.33	$\frac{3 \times 50,00,000}{12}$ = 12,50,000
Increase in debtors over current level	–	1,91,666.67	4,00,000.00	9,16,666.67
Cost of funds for additional amount of debtors @ 20% (B)	–	38,333.33	80,000.00	1,83,333.33

Particulars	Current position (1 month)	Option I (1.5 months)	Option II (2 months)	Option III (3 months)
Credit administrative cost	24,000	26,000	30,000	60,000
Increase in credit administration cost over present level (C)	–	2,000	6,000	36,000
Bad debts	80,000	1,05,000	1,32,000	2,50,000
Increase in bad debts over current levels (D)	–	25,000	52,000	1,70,000
Net gain/loss A – (B + C + D)	–	14,666.67	22,000.00	10,666.67

Advise: It is suggested that the company GT Ltd. should implement Option II with a net gain of ₹ 22,000 which has a credit period of 2 months.

ICAI · Old P8 · MTP Nov 2022 · Series I · question p. 3 · answer p. 3-4

742 MTP **Old P8 Jul 2021 S-II** Q3 10 marks old scheme

WQ Limited is considering relaxing its present credit policy and is in the process of evaluating two proposed polices. Currently, the firm has annual credit sales of Rs. 180 lakh and Debtors turnover ratio of 4 times a year. The current level of loss due to bad debts is Rs. 6 lakh. The firm is required to give a return of 25% on the investment in new accounts receivables. The company's variable costs are 60% of the selling price. Given the following information, DETERMINE which is a better Policy?

(Amount in lakhs)

	Present Policy	Proposed Policy	
		Option I	**Option II**
Annual credit sales (Rs.)	180	220	280
Debtors turnover ratio	4	3.2	2.4
Bad debt losses (Rs.)	6	18	38

ICAI'S ANSWER

Statement showing evaluation of Credit Policies

(Amount in lakhs)

	Particulars	Present (Rs.)	Proposed Policy (Rs.)	
			Option I	**Option II**
A	Expected Profit:			
	(a) Credit Sales	180	220	280
	(b) Total Cost other than Bad Debts:			
	Variable Costs (60%)	108	132	168
	(c) Bad Debts	6	18	38

	Particulars	Present (Rs.)	Proposed Policy (Rs.)	
	(d) Expected Profit [(a)-(b)-(c)]	66	70	74
B	Opportunity Cost of Investment in Debtors (Refer workings)	6.75	10.31	17.5
C	Net Benefits [A - B]	59.25	59.69	56.5

Recommendation: The Proposed Policy I should be adopted since the net benefits under this policy is higher than those under other policies.

Workings:

Calculation of Opportunity Cost of Investment in Debtors

$$\text{Opportunity Cost} = \text{Total Cost} \times \frac{\text{Collection Period}^*}{12} \times \frac{\text{Rate of Return}}{100}$$

*Collection period (in months) = 12/Debtors turnover ratio

$$\text{Present Policy} = \text{Rs. } 108 \times \frac{12/4}{12} \times \frac{25}{100} = \text{Rs. } 6.75 \text{ lakhs}$$

$$\text{Proposed Policy I} = \text{Rs. } 132 \times \frac{12/3.2}{12} \times \frac{25}{100} = \text{Rs. } 10.31 \text{ lakhs}$$

$$\text{Proposed Policy II} = \text{Rs. } 168 \times \frac{12/2.4}{12} \times \frac{25}{100} = \text{Rs. } 17.5 \text{ lakhs}$$

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 2-3 · answer p. 4-5

9.4 Treasury and cash management

23 questions · asked in 18 ICAI documents · 8 exam papers

743 EXAM **Sep 2026** Q4(b) 4 marks

Explain the main features of Miller-Orr Cash Management Model for a business concern.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P6 · Sep 2026 exam · question p. 19

744 EXAM **May 2026** Case scenario · Q3(b)(i) to Q3(b)(ii) 6 marks

CASE SCENARIO

AR Ltd. has given following data for the year ended 31st March, 2026:

- Receivables are collected on an average after 60 days,
- Inventories are held for average period of 90 days before being sold.
- The company usually settles its payables with suppliers after 30 days.
- The total sales of the company amounts to ₹ 200 lakhs, which is earned evenly throughout the year. All sales are credit sales. Net profit margin is 10%. The company has not incurred any non-operating expenses during the year.
- Tax rate is 30%.
- Assume 360 days in a year.
- Opportunity cost of investment is 12%.

On the basis of the above data:

Q3(b)(i) 2 marks

Calculate the cash cycle and the minimum cash balance to be maintained to meet its obligations.

Q3(b)(ii) 4 marks

The company is planning to propose a scheme, whereby it will offer a cash discount of 2% to debtors to reduce the debtors' collection period by 30 days. It is expected that 60% of the debtors will avail the scheme.

(A) What will be reduction in the minimum cash to be maintained to meet obligation after the proposed scheme is adopted?

(B) What will be net saving due to such scheme?

(C) Should the company adopt the scheme?

ICAI'S ANSWERS

Q3(b)(i)

Cash cycle = 60 days + 90 days – 30 days =120 days (4 months)

Cash turnover = 12 months (360 days)/4months (120 days) = 3.

Total Cash Outlay = Sales – Net Profit

= 200 Lakhs–20 Lakhs = ₹ 180 Lakhs

Minimum operating cash = Total operating annual outlay/cash turnover

= ₹ 180 lakhs/3 = ₹ 60 lakhs.

Q3(b)(ii)

(A) Reduction in Minimum Cash Balance

First, we calculate the new average collection period and the new cash cycle.

New Average Collection Period: 30 days

New Cash Cycle: 30 + 90 –30 = 90 days

New Cash Turnover: 360/90 = 4 times

New Minimum Cash Balance: 180/4 = ₹ 45 Lakhs

Reduction in Cash Balance: 60 Lakhs–45 Lakhs = ₹ 15 Lakhs

(B) Net Saving Due to Scheme

Opportunity Cost Saving = 12% of reduction in cash balance (15 Lakhs) x (1-0.30) = ₹ 1.26Lakhs

Cost of Cash Discount = 2% of 60 % of Total Sales x (1-0.30)

= 0.02 × 0.60 × 200 Lakhs = ₹ 1.68 Lakhs

Net Saving =1.26 Lakhs–1.68 Lakhs = ₹ (0.42 Lakhs)

(C) Recommendation

No, the company should not adopt the scheme.

Solution for part (ii) can be done in following alternative way:

If we assume

- 60% of debtors will pay 30 days early (Collection period = 30 days).

- The remaining 40% of debtors will *not* avail the scheme and will continue to pay after the normal 60 days.

Therefore, the New Average Collection Period:

= (60%×30 days) + (40% × 60 days) = 18 days+24 days = 42 days

Cash Cycle & Cash Balance:

- New Cash Cycle = 90 (Inventory) + 42 (New Debtors) - 30 (Payables) = 102 days
- New Cash Turnover = 360 / 102 = 3.529 times
- New Minimum Cash Balance = ₹ 180 Lakhs / 3.529 = ₹ 51 Lakhs
- Reduction in Cash Balance = ₹ 60 Lakhs - ₹ 51 Lakhs = ₹ 9 Lakhs

(B) Net Savings:

- Opportunity Cost Saving: ₹ 9 Lakhs × 12% × (1 - 0.30) = ₹ 0.756 Lakhs
- Cost of Cash Discount: 2% × 60% × 200 Lakhs × (1 - 0.30) = ₹ 1.68 Lakhs
- Net Saving: ₹ 0.756 Lakhs - ₹ 1.68 Lakhs = (₹ 0.924 Lakhs) or (Net Loss of ₹ 92,400)

(C) No, the company should not adopt the scheme, because it results in a net financial loss of ₹ 0.924 Lakhs.

ICAI · P6 · May 2026 exam · question p. 17 · answer p. 19, 19-21

745 EXAM **Jan 2026** MCQ 7 MCQ

A firm maintains a separate account for cash disbursement. The administrative and transaction cost of transferring cash to the disbursement account is ₹ 25 per transfer. Marketable securities yield 10% per annum. The firm has determined its optimum cash balance at ₹ 30,000, according to the William J. Baumol Model.

What will be the Total Annual Cash Disbursements of the firm?

A ₹ 18,00,000

B ₹ 45,00,000

C ₹ 36,00,000

D ₹ 27,00,000

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P6 · Jan 2026 exam · question p. 4-5 · answer p. 9

746 EXAM **Jan 2026** Q4(b) 4 marks

Why the Virtual banking service is popular among the customers?

Discuss the advantages of Virtual Banking.

ICAI'S ANSWER

The popularity which virtual banking services have won among customers is due to the speed, convenience and round the clock access they offer.

The advantages of virtual banking services are as follows:

(i) Lower cost of handling a transaction.

(ii) The increased speed of response to customer requirements.

(iii) The lower cost of operating branch network along with reduced staff costs leads to cost efficiency.

(iv) Virtual banking allows the possibility of improved and a range of services being made available to the customer rapidly, accurately and at his convenience.

ICAI · P6 · Jan 2026 exam · question p. 22 · answer p. 23-24

747 EXAM Jan 2025 Q1(a) 5 marks

KP Ltd. has provided the following information:

(i) Estimated monthly sales:

Month	₹ in Lakh
April-2024	10
May-2024	12
June-2024	15
July-2024	10
August-2024	13
September-2024	14

(ii) Gross Profit Ratio is 20%.

(iii) Cost of Goods sold is paid in next month.

(iv) Sales are in credit and credit period is allowed for 2 months.

(v) Indirect Expenses are paid in the same month.

Monthly indirect expenses are as follows:

Month	₹ in Lakh
June-2024	1.0
July-2024	1.2
August-2024	1.0
September-2024	1.3

(vi) Dividend amounting September 2024, 3 Lakh will be paid in the month of September 2024

(vii) Cash Balance on 01/07/2024 was 1.5 Lakh.

(viii) The company has to maintain minimum cash balance of 1 Lakh. If there is cash balance deficit in any month, company would take a temporary short term loan and if cash balance exceed 2 Lakh, then company would invest for short term excess amount of 2 Lakh.

(ix) Ignore the interest on short term loans and short term investment.

You are required to prepare Cash Budget for three months starting from July 2024.

ICAI'S ANSWER

Cash Budget (₹ in lakh)

Particulars	July	August	September
Opening Balance	1.5	1	2
Add: Receipts			
Collection from debtors	12	15	10
Total cash available (A)	**13.5**	**16**	**12**
Less: Payments			
Payment of COGS	12	8	10.4
Payment of indirect exp	1.2	1	1.3
Payment of Dividend	-	-	3
Total payments (B)	**13.2**	**9**	**14.7**
Closing Balance before adjustment	0.3	7	-2.7
Minimum cash balance desired	1	1	1
Less: Temporary Investment	-	5	-
Add: Temporary Borrowing	0.7	-	3.7
Cash Closing Balance	1	2	1

W.N. 1

COGS payment	June	July	Aug	Sept
Sales	15	10	13	14
COGS (80%)	12	8	10.4	11.2
Paid in		12	8	10.4

ICAI · P6 · Jan 2025 exam · question p. 8-9 · answer p. 10-11

748 EXAM Jan 2025 Q4(c) 2 marks

State any two advantages of virtual banking.

ICAI'S ANSWER

Following are the advantages of virtual banking-

(i) Lower cost of handling a transaction.

(ii) The increased speed of response to customer requirement.

(iii) Lower cost of operating branch network along with reduced staff costs leads to cost efficiency.

(iv) It allows the possibility of improved and a range of services being made available to the customer rapidly, accurately and at his convenience.

ICAI · P6 · Jan 2025 exam · question p. 21 · answer p. 22

749 EXAM **Old P8 Nov 2022** Q1(a) 5 marks old scheme

K Ltd. has a Quarterly cash outflow of ₹ 9,00,000 arising uniformly during the Quarter. The company has an Investment portfolio of Marketable Securities. It plans to meet the demands for cash by periodically selling marketable securities. The marketable securities are generating a return of 12% p.a. Transaction cost of converting investments to cash is ₹ 60. The company uses Baumol model to find out the optimal transaction size for converting marketable securities into cash.

Consider 360 days in a year.

You are required to calculate

(i) Company's average cash balance,

(ii) Number of conversions each year and

(iii) Time interval between two conversions.

ICAI'S ANSWER

(i) Computation of Average Cash balance:

Annual cash outflow (U) = 9,00,000 x 4 = ₹ 36,00,000

Fixed cost per transaction (P) = ₹ 60

Opportunity cost of one rupee p.a. (S) = $\frac{12}{100} = 0.12$

Optimum cash balance (C) = $\sqrt{\frac{2UP}{S}} = \sqrt{\frac{2 \times 36,00,000 \times 60}{0.12}}$ = ₹ 60,000

∴ Average Cash balance = $\frac{(0 + 60,000)}{2}$ = ₹ 30,000

(ii) Number of conversions p.a.

Annual cash outflow = ₹ 36,00,000

Optimum cash balance = ₹ 60,000

∴ No. of conversions p.a. = $\frac{36,00,000}{60,000} = 60$

(iii) Time interval between two conversions

No. of days in a year = 360

No. of conversions p.a. = 60

∴ Time interval = $\frac{360}{60}$ = 6 days

ICAI · Old P8 · Nov 2022 exam · question p. 1 · answer p. 3

750 EXAM **Old P8 Nov 2022** Q6(b) 4 marks old scheme

Elucidate the fundamental tasks of treasury department of a firm.

ICAI'S ANSWER

Fundamental tasks of treasury department of a firm:

(i) Cash management: It involves efficient cash collection process and managing payment of cash both inside the organization and to third parties. Treasury will also manage surplus funds in an investment portfolio.

(ii) Currency management: The treasury department manages the foreign currency risk exposure of the company. In a large multi-national company, the first step will usually be to set off intra-group indebtedness. The use of matching receipts and payments in the same currency will save transaction costs and will save the organization from any unfavorable exchange movement.

(iii) Fund management: Treasury department is responsible for planning and sourcing the company's short, medium and long-term cash needs. It also facilitates temporary investment of surplus funds by mapping the time gap between funds inflow and outflow.

(iv) Banking: It is important that a company maintains a good relationship with its bankers. Treasury department carry out negotiations with bankers with respect to interest rates, foreign exchange rates etc. and act as the initial point of contact with them.

(v) Corporate finance: Treasury department is involved with both acquisition and divestment activities within the group. In addition, it will often have responsibility for investors' relations.

ICAI · Old P8 · Nov 2022 exam · question p. 15 · answer p. 15-16

751 EXAM Old P8 Dec 2021 Q1(d) 5 marks old scheme

A garment trader is preparing cash forecast for first three months of calendar year 2021. His estimated sales for the forecasted periods are as below:

	January (₹ '000)	February (₹ '000)	March (₹ '000)
Total sales	600	600	800

(i) The trader sells directly to public against cash payments and to other entities on credit. Credit sales are expected to be four times the value of direct sales to public. He expects 15% customers to pay in the month in which credit sales are made, 25% to pay in the next month and 58% to pay in the next to next month. The outstanding balance is expected to be written off.

(ii) Purchases of goods are made in the month prior to sales and it amounts to 90% of sales and are made on credit. Payments of these occur in the month after the purchase. No inventories of goods are held.

(iii) Cash balance as on 1st January, 2021 is ₹ 50,000.

(iv) Actual sales for the last two months of calendar year 2020 are as below:

	November (₹ '000)	December (₹ '000)
Total sales	640	880

You are required to prepare a monthly cash, budget for the three months from January to March, 2021.

ICAI'S ANSWER

Working Notes:

(1) Calculation of cash and credit sales (₹ in thousands)

	Nov.	Dec.	Jan.	Feb.	Mar.
Total Sales	640	880	600	600	800
Cash Sales (1/5th of total sales)	128	176	120	120	160
Credit Sales (4/5th of total sales)	512	704	480	480	640

(2) Calculation of Credit Sales Receipts (₹ in thousands)

Month	Nov.	Dec.	Jan.	Feb.	Mar.
Forecast Credit sales (Working note 1)	512.00	704.00	480.00	480.00	640.00
Receipts:					
15% in the month of sales			**72.00**	**72.00**	**96.00**
25% in next month			**176.00**	**120.00**	**120.00**
58% in next to next month			**296.96**	**408.32**	**278.40**
Total			544.96	600.32	494.40

Cash Budget (₹in thousands)

	Nov.	Dec.	Jan.	Feb.	Mar.
Opening Balance **(A)**			50.00	174.96	355.28
Sales	640.00	880.00	600.00	600.00	800.00
Receipts:					
Cash Collection (Working note 1)			120.00	120.00	160.00
Credit Collections (Working note 2)			544.96	600.32	494.40
Total **(B)**			664.96	720.32	654.40
Purchases (90% of sales in the month prior to sales)		540	540	720	
Payments:					
Payment for purchases (next month)			**540**	**540**	**720**
Total **(C)**			540	540	720
Closing balance**(D) = (A + B – C)**			174.96	355.28	**289.68**

ICAI · Old P8 · Dec 2021 exam · question p. 2 · answer p. 4-5

752 EXAM **Old P8 Jan 2021** Q6(b) 4 marks old scheme

Explain Electronic Cash Management System.

ICAI'S ANSWER

Electronic Cash Management System: Most of the cash management systems now-a-days are electronically based, since 'speed' is the essence of any cash management system. Electronically, transfer of data as well as funds play a key role in any cash management system. Various elements in the process of cash management are linked through a satellite. Various places that are interlinked may be the place where the instrument is collected, the place where cash is to be transferred in company's account, the place where the payment is to be transferred etc.

ICAI · Old P8 · Jan 2021 exam · question p. 13 · answer p. 14

753 EXAM Old P8 Nov 2019 Q3 10 marks old scheme

Slide Ltd. is preparing a cash flow forecast for the three months period from January to the end of March. The following sales volumes have been forecasted:

Months	December	January	February	March	April
Sales (units)	1,800	1,875	1,950	2,100	2,250

Selling price per unit is ₹ 600. Sales are all on one month credit. Production of goods for sale takes place one month before sales. Each unit produced requires two units of raw materials costing ₹ 150 per unit. No raw material inventory is held. Raw materials purchases are on one month credit. Variable overheads and wages equal to ₹ 100 per unit are incurred during production and paid in the month of production. The opening cash balance on 1st January is expected to be ₹ 35,000. A long term loan of ₹ 2,00,000 is expected to be received in the month of March. A machine costing ₹ 3,00,000 will be purchased in March.

(a) Prepare a cash budget for the months of January, February and March and calculate the cash balance at the end of each month in the three months period.

(b) Calculate the forecast current ratio at the end of the three months period.

ICAI'S ANSWER

Working Notes:

(1) Calculation of Collection from Trade Receivables:

Particulars	December	January	February	March
Sales (units)	1,800	1,875	1,950	2,100
Sales (@ ₹ 600 per unit) / Trade Receivables (Debtors) (₹)	10,80,000	11,25,000	11,70,000	12,60,000
Collection from Trade Receivables (Debtors) (₹)		10,80,000	11,25,000	11,70,000

(2) Calculation of Payment to Trade Payables:

Particulars	December	January	February	March
Output (units)	1,875	1,950	2,100	2,250
Raw Material (2 units per output) (units)	3,750	3,900	4,200	4,500
Raw Material (@ ₹ 150 per unit) / Trade Payables (Creditors) (₹)	5,62,500	5,85,000	6,30,000	6,75,000
Payment to Trade Payables (Creditors) (₹)		5,62,500	5,85,000	6,30,000

(3) Calculation of Variable Overheads and Wages:

Particulars	January	February	March
Output (units)	1,950	2,100	2,250
Payment in the same month @ ₹ 100 per unit (₹)	1,95,000	2,10,000	2,25,000

(a) Preparation of Cash Budget

Particulars	January (₹)	February (₹)	March (₹)
Opening Balance	35,000	3,57,500	6,87,500
Receipts:			
Collection from Trade Receivables (Debtors)	10,80,000	11,25,000	11,70,000
Receipt of Long-Term Loan			2,00,000
Total (A)	11,15,000	14,82,500	20,57,500
Payments:			
Trade Payables (Creditors) for Materials	5,62,500	5,85,000	6,30,000
Variable Overheads and Wages	1,95,000	2,10,000	2,25,000
Purchase of Machinery			3,00,000
Total (B)	7,57,500	7,95,000	11,55,000
Closing Balance (A – B)	**3,57,500**	**6,87,500**	**9,02,500**

(b) Calculation of Current Ratio

Particulars	March (₹)
Output Inventory (i.e. units produced in March) [(2,250 units x 2 units of raw material per unit of output x ₹ 150 per unit of raw material) + 2,250 units x ₹ 100 for variable overheads and wages] **or,** [6,75,000 + 2,25,000] from Working Notes 2 and 3	9,00,000

Particulars	March (₹)
Trade Receivables (Debtors)	12,60,000
Cash Balance	9,02,500
Current Assets	**30,62,500**
Trade Payables (Creditors)	6,75,000
Current Liabilities	**6,75,000**
Current Ratio (Current Assets / Current Liabilities)	**4.537 approx.**

ICAI · Old P8 · Nov 2019 exam · question p. 9 · answer p. 9-11

754 **RTP** **Sep 2026 Q9**

You have been asked to prepare a cash budget for the next quarter, January through March, for Jahanara Fashions. They have provided you with the following information:

a. Sales are expected to be: ₹ 400,000 in January, ₹ 400,000 in February, and ₹ 600,000 in March. All sales will be in cash.

b. The estimated purchases are: ₹380,000 in January, ₹ 360,000 in February, and ₹ 450,000 in March. Payments for purchases will be made after a lag of one month. Outstanding on account of purchases in December last are ₹ 350,000.

c. The rent per month is ₹ 10,000 and the partners' personal withdrawal per month is ₹25,000.

d. Salaries and other expenses, payable in cash, are expected to be: ₹ 25,000 in January, ₹ 20,000 in February, and ₹ 30,000 in March.

e. They plan to buy furniture worth ₹ 40,000 on cash payment in January.

f. The cash balance at present is ₹ 6,000. Their target cash balance, however, is ₹ 15,000. What will be surplus/ deficit of cash in relation to their target cash balance?

ICAI'S ANSWER

The projected cash inflows and outflows for the quarter, January through March, is shown below:

Month	December (₹)	January (₹)	February (₹)	March (₹)
Inflows:				
Sales collection		4,00,000	4,00,000	6,00,000
Outflows:				
Payment to sundry creditors		3,50,000	3,80,000	3,60,000
Rent		10,000	10,000	10,000
Drawings		25,000	25,000	25,000
Salaries & other expenses		25,000	20,000	30,000

Month	December (₹)	January (₹)	February (₹)	March (₹)
Purchase of furniture		40,000		
Total outflows 2 to 6)		**4,50,000**	**4,35,000**	**4,25,000**

Given an opening cash balance of ₹6,000 and a target cash balance of ₹ 15,000, the surplus/deficit in relation to the target cash balance is worked out below:

	January (₹)	February (₹)	March (₹)
1. Opening balance	6,000		
2. Inflows	400,000	400,000	600,000
3. Outflows	450,000	435,000	425,000
4. Net cash flow (2 - 3)	(50,000)	(35,000)	175,000
5. Cumulative net cash flow	(50,000)	(85,000)	90,000
6. Opening balance + Cumulative net cash flow	(44,000)	(79,000)	96,000
7. Minimum cash balance required	15,000	15,000	15,000
8. Surplus/(Deficit)	(59,000)	(94,000)	81,000

ICAI · P6 · RTP Sep 2026 · question p. 8-9 · answer p. 21-22

755 RTP **May 2026** Q10(b)

Same question, different ICAI answer: #758 MTP May 2026 S-II

"The treasury department of a business concern has evolved its importance over the years from being responsible for only cash handling issues to technical areas revolving around hedging forex risks and composition of capital structure." In the light of this statement, EXPLAIN briefly the fundamental tasks for which treasury department is responsible.

ICAI'S ANSWER

The fundamental tasks for which treasury department of any enterprise is responsible are:

1. **Cash Management:** It involves efficient cash collection process and managing payment of cash both inside the organisation and to third parties.

Treasury will also manage surplus funds in an investment portfolio. Investment policy will consider future needs for liquid funds and acceptable levels of risk as determined by company policy.

2. **Currency Management:** The treasury department manages the foreign currency risk exposure of the company. In a large multinational company (MNC) the first step will usually be to set off intra-group indebtedness. The use of matching receipts and payments in the same currency will save transaction costs and also will save the organization from any unfavorable exchange movements.

The treasury will manage any net exchange exposures in accordance with company policy. If risks are to be minimized then forward contracts can be used either to buy or sell currency forward.

3. **Fund Management:** Treasury department is responsible for planning and sourcing the company's short, medium and long-term cash needs. They also facilitate temporary investment of surplus funds by mapping the time gap between funds inflow and outflow.

4. **Banking:** It is important that a company maintains a good relationship with its bankers. Treasury department carry out negotiations with bankers with respect to interest rates, foreign exchange rates etc. and act as the initial point of contact with them.

5. **Corporate Finance:** Treasury department is involved with both acquisition and divestment activities within the group. In addition, it will often have responsibility for investor relations.

ICAI · P6 · RTP May 2026 · question p. 11 · answer p. 28-29

756 RTP **Old P8 Nov 2022 Q8** old scheme

A company was incorporated w.e.f. 1st April, 2021. Its authorised capital was ₹ 1,00,00,000 divided into 10 lakh equity shares of ₹ 10 each. It intends to raise capital by issuing equity shares of ₹ 50,00,000 (fully paid) on 1st April. Besides this, a loan of ₹ 6,50,000 @ 12% per annum will be obtained from a financial institution on 1st April and further borrowings will be made at same rate of interest on the first day of the month in which borrowing is required. All borrowings will be repaid along with interest on the expiry of one year. The company will make payment for the following assets in April.

Particulars	(₹)
Plant and Machinery	10,00,000
Land and Building	20,00,000
Furniture	5,00,000
Motor Vehicles	5,00,000
Stock of Raw Materials	5,00,000

The following further details are available:

(1) Projected Sales (April-September):

	(₹)
April	15,00,000
May	17,50,000
June	17,50,000
July	20,00,000
August	20,00,000
September	22,50,000

(2) Gross profit margin will be 25% on sales.

(3) The company will make credit sales only and these will be collected in the second month following sales.

(4) Creditors will be paid in the first month following credit purchases. There will be credit purchases only.

(5) The company will keep minimum stock of raw materials of ₹ 5,00,000.

(6) Depreciation will be charged @ 10% per annum on cost on all fixed assets.

(7) Payment of miscellaneous expenses of ₹ 50,000 will be made in April.

(8) Wages and salaries will be ₹ 1,00,000 each month and will be paid on the first day of the next month.

(9) Administrative expenses of ₹ 50,000 per month will be paid in the month of their incurrence.

(10) No minimum cash balance is required.

You are required to PREPARE the monthly cash budget (April-September), the projected Income Statement for the 6 months period and the projected Balance Sheet as on 30th September, 2021.

ICAI'S ANSWER

Monthly Cash Budget (April-September) (₹)

	April	May	June	July	August	September
Opening cash balance	-	10,50,000	-	1,37,500	5,25,000	7,25,000
A. Cash inflows						
Equity shares	50,00,000	-	-	-	-	-
Loans (Refer to working note 1)	6,50,000	1,25,000	-	-	-	-
Receipt from debtors	-	-	15,00,000	17,50,000	17,50,000	20,00,000
Total (A)	56,50,000	11,75,000	15,00,000	18,87,500	22,75,000	27,25,000
B. Cash Outflows						
Plant and Machinery	10,00,000	-	-	-	-	-
Land and Building	20,00,000	-	-	-	-	-
Furniture	5,00,000	-	-	-	-	-
Motor Vehicles	5,00,000	-	-	-	-	-
Stock of raw materials (Minimum stock)	5,00,000	-	-	-	-	-
Miscellaneous expenses	50,000	-	-	-	-	-
Payment to creditors for credit purchases (Refer to working note 2)	-	10,25,000	12,12,500	12,12,500	14,00,000	14,00,000
Wages and salaries	-	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000

	April	May	June	July	August	September
Admn. expenses	50,000	50,000	50,000	50,000	50,000	50,000
Total :(B)	46,00,000	11,75,000	13,62,500	13,62,500	15,50,000	15,50,000
Closing balance (A)-(B)	10,50,000	-	1,37,500	5,25,000	7,25,000	11,75,000

Budgeted Income Statement for six-month period ending 30th September

Particulars	(₹)	Particulars	(₹)
To Purchases	83,37,500	By Sales	1,12,50,000
To Wages and Salaries	6,00,000	By Closing stock	5,00,000
To Gross profit c/d	28,12,500		
	1,17,50,000		**1,17,50,000**
To Admn. expenses	3,00,000	By Gross profit b/d	28,12,500
To Depreciation (10% on ₹ 40 lakhs for six months)	2,00,000		
To Accrued interest on loan (Refer to working note 3)	45,250		
To Miscellaneous expenses	50,000		
To Net profit c/d	22,17,250		
	28,12,500		**28,12,500**

Projected Balance Sheet as on 30th September, 2021

Liabilities	Amount (₹)	Assets			Amount (₹)
Share Capital:		Fixed Assets:			
		Land and Building	20,00,000		
		Less: Depreciation	1,00,000	19,00,000	
Authorised capital 10,00,000 equity shares of ₹ 10 each	1,00,00,000	Plant and Machinery	10,00,000		
		Less: Depreciation	50,000	9,50,000	
		Furniture	5,00,000		

Liabilities		Amount (₹)	Assets			Amount (₹)
			Less: Depreciation	25,000	4,75,000	
Issued, subscribed and paid up capital 5,00,000 equity shares of ₹ 10 each		50,00,000	Motor Vehicles	5,00,000		
			Less: Depreciation	25,000	4,75,000	38,00,000
Reserve and Surplus:			Current Assets:			
			Stock		5,00,000	
			Sundry debtors		42,50,000	
Profit and Loss		22,17,250	Cash		11,75,000	59,25,000
Long-term loans		7,75,000				
Current liabilities and provisions:						
Sundry creditors	15,87,500					
Accrued interest	45,250					
Outstanding expenses	1,00,000	17,32,750				
		97,75,000				97,75,000

Working Notes:

Subsequent Borrowings Needed (₹)

	April	May	June	July	August	September
A. Cash Inflow						
Equity shares	50,00,000					
Loans	6,50,000					
Receipt from debtors	-	-	15,00,000	17,50,000	17,50,000	20,00,000
Total (A)	56,50,000	-	15,00,000	17,50,000	17,50,000	20,00,000
B. Cash Outflow						

	April	May	June	July	August	September
Purchase of fixed assets	40,00,000					
Stock	5,00,000					
Miscellaneous expenses	50,000					
Payment to creditors	-	10,25,000	12,12,500	12,12,500	14,00,000	14,00,000
Wages and salaries	-	1,00,000	1,00,000	1,00,000	1,00,000	1,00,000
Administrative expenses	50,000	50,000	50,000	50,000	50,000	50,000
Total	46,00,000	11,75,000	13,62,500	13,62,500	15,50,000	15,50,000
Surplus/ (Deficit)	10,50,000	(11,75,000)	1,37,500	3,87,500	2,00,000	4,50,000
Cumulative balance	10,50,000	(1,25,000)	12,500	4,00,000	6,00,000	10,50,000

1. There is shortage of cash in May of ₹ 1,25,000 which will be met by borrowings in May.
2. Payment to Creditors

Purchases = Cost of goods sold - Wages and salaries

Purchases for April = (75% of 15,00,000) - ₹ 1,00,000 = ₹ 10,25,000

(**Note:** Since gross margin is 25% of sales, cost of manufacture i.e. materials plus wages and salaries should be 75% of sales)

Hence, Purchases = Cost of manufacture minus wages and salaries of ₹ 1,00,000)

The creditors are paid in the first month following purchases.

Therefore, payment in May is ₹ 10,25,000

The same procedure will be followed for other months.

April	(75% of 15,00,000) - ₹ 1,00,000 = ₹ 10,25,000
May	(75% of 17,50,000) - ₹ 1,00,000 = ₹ 12,12,500
June	(75% of 17,50,000) - ₹ 1,00,000 = ₹ 12,12,500
July	(75% of 20,00,000) - ₹ 1,00,000 = ₹ 14,00,000
August	(75% of 20,00,000) - ₹ 1,00,000 = ₹ 14,00,000
September	(75% of 22,50,000) - ₹ 1,00,000 = ₹ 15,87,500
Minimum Stock	₹ 5,00,000
Total Purchases	₹ 83,37,500

3. Accrued Interest on Loan

12% interest on ₹ 6,50,000 for 6 months	39,000
Add: 12% interest on ₹ 1,25,000 for 5 months	6,250
	45,250

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757 RTP **Old P8 May 2022** Q8 old scheme

You are given below the Profit & Loss Accounts for two years for a company:

Profit and Loss Account

	Year 1	Year 2		Year 1	Year 2
	(₹)	**(₹)**		**(₹)**	**(₹)**
To Opening stock	32,00,000	40,00,000	By Sales	3,20,00,000	4,00,00,000
To Raw materials	1,20,00,000	1,60,00,000	By Closing stock	40,00,000	60,00,000
To Stores	38,40,000	48,00,000	By Misc. Income	4,00,000	4,00,000
To Manufacturing Expenses	51,20,000	64,00,000			
To Other Expenses	40,00,000	40,00,000			
To Depreciation	40,00,000	40,00,000			
To Net Profit	42,40,000	72,00,000		-	-
	3,64,00,000	**4,64,00,000**		**3,64,00,000**	**4,64,00,000**

Sales are expected to be ₹ 4,80,00,000 in year 3.

As a result, other expenses will increase by ₹ 20,00,000 besides other charges. Only raw materials are in stock. Assume sales and purchases are in cash terms and the closing stock is expected to go up by the same amount as between year 1 and 2. You may assume that no dividend is being paid. The Company can use 75% of the cash generated to service a loan. COMPUTE how much cash from operations will be available in year 3 for the purpose? Ignore income tax.

ICAI'S ANSWER

Projected Profit and Loss Account for the year 3

Particulars	Year 2 Actual (₹ in lakhs)	Year 3 Projected (₹ in lakhs)	Particulars	Year 2 Actual (₹ in lakhs)	Year 3 Projected (₹ in lakhs)
To Materials consumed	140.00	168.00	By Sales	400.00	480.00
To Stores	48.00	57.60	By Misc. Income	4.00	4.00
To Mfg. Expenses	64.00	76.80			
To Other expenses	40.00	60.00			
To Depreciation	40.00	40.00			
To Net profit	72.00	81.60			
	404.00	**484.00**		**484.00**	**484.00**

Cash Flow:

Particulars	(₹ in lakhs)
Profit	81.60
Add: Depreciation	40.00
	121.60
Less: Cash required for increase in stock	20.00
Net cash inflow	**101.60**

Available for servicing the loan: 75% of ₹ 1,01,60,000 or ₹ 76,20,000

Working Notes:

(i) Material consumed in year 1 = (32 + 120 – 40)/320 = 35%

Material consumed in year 2 = (40 + 160 – 60)/400 = 35%

Likely consumption in year 3 = $480 \times \frac{35}{100}$ = ₹ 168 (lakhs)

(ii) Stores are 12% of sales & Manufacturing expenses are 16% of sales for both the years.

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758 MTP **May 2026 S-II** Q4(b) 4 marks

Same question, different ICAI answer: #755 RTP May 2026

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ICAI'S ANSWER

The fundamental tasks for which treasury department of any enterprise is responsible are:

1. **Cash Management:** It involves efficient cash collection process and managing payment of cash both inside the organisation and to third parties.
   Treasury will also manage surplus funds in an investment portfolio. Investment policy will consider future needs for liquid funds and acceptable levels of risk as determined by company policy.
2. **Currency Management:** The treasury department manages the foreign currency risk exposure of the company. In a large multinational company (MNC) the first step will usually be to set off intra-group indebtedness. The use of matching receipts and payments in the same currency will save transaction costs and also will save the organization from any unfavorable exchange movements. The treasury will manage any net exchange exposures in accordance with company policy. If risks are to be minimized then forward contracts can be used either to buy or sell currency forward.
3. **Fund Management:** Treasury department is responsible for planning and sourcing the company's short, medium and long-term cash needs. They also facilitate temporary investment of surplus funds by mapping the time gap between funds inflow and outflow.
4. **Banking:** It is important that a company maintains a good relationship with its bankers. Treasury department carry out negotiations with bankers with respect to interest rates, foreign exchange rates etc. and act as the initial point of contact with them.
5. **Corporate Finance:** Treasury department is involved with both acquisition and divestment activities within the group. In addition, it will often have responsibility for investor relations.

* Any four points

ICAI · P6 · MTP May 2026 · Series II · question p. 9 · answer p. 10-11

759 MTP **Old P8 May 2023 S-I** Q3 10 marks old scheme

You are given the following information:

(i) Estimated monthly Sales are as follows:

	₹		₹
January	5,50,000	June	4,40,000
February	6,60,000	July	5,50,000
March	7,70,000	August	4,40,000
April	4,40,000	September	3,30,000
May	3,30,000	October	5,50,000

(ii) Wages and Salaries are estimated to be payable as follows:

	₹		₹
April	49,500	July	55,000
May	44,000	August	49,500
June	55,000	September	49,500

(iii) Of the sales, 75% is on credit and 25% for cash. 60% of the credit sales are collected within one month and the balance in two months. There are no bad debt losses.

(iv) Purchases amount to 75% of sales and are made and paid for in the month preceding the sales.

(v) The firm has taken a loan of ₹6,00,000. Interest @ 12% p.a. has to be paid quarterly in January, April and so on.

(vi) The firm is to make payment of tax of ₹26,000 in July 2023.

(vii) The firm had a cash balance of ₹35,000 on 1St April 2023 which is the minimum desired level of cash balance. Any cash surplus/deficit above/below this level is made up by temporary investments/liquidation of temporary investments or temporary borrowings at the end of each month (interest on these to be ignored).

Required:

PREPARE monthly cash budgets for six months beginning from April, 2023 on the basis of the above information.

ICAI'S ANSWER

Computation – Collections from Customers

Particulars	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
Total Sales	6,60,000	7,70,000	4,40,000	3,30,000	4,40,000	5,50,000	4,40,000	3,30,000
Credit Sales (75% of total Sales)	4,95,000	5,77,500	3,30,000	2,47,500	3,30,000	4,12,500	3,30,000	2,47,500
Collection (within one month)		2,97,000	3,46,500	1,98,000	1,48,500	1,98,000	2,47,500	1,98,000
Collection (within two months)			1,98,000	2,31,000	1,32,000	99,000	1,32,000	1,65,000
Total Collections			**5,44,500**	**4,29,000**	**2,80,500**	**2,97,000**	**3,79,500**	**3,63,000**

Monthly Cash Budget for Six Months: April to September 2023

Particulars	April	May	June	July	August	Sept.
	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)

Particulars	April	May	June	July	August	Sept.
Receipts:						
Opening Balance	35,000	35,000	35,000	35,000	35,000	35,000
Cash Sales	1,10,000	82,500	1,10,000	1,37,500	1,10,000	82,500
Collections from Debtors	5,44,500	4,29,000	2,80,500	2,97,000	3,79,500	3,63,000
Total Receipts (A)	**6,89,500**	**5,46,500**	**4,25,500**	**4,69,500**	**5,24,500**	**4,80,500**
Payments:						
Purchases	2,47,500	3,30,000	4,12,500	3,30,000	2,47,500	4,12,500
Wages and Salaries	49,500	44,000	55,000	55,000	49,500	49,500
Interest on Loan	18,000	-----	-----	18,000	-----	-----
Tax Payment	-----	-----	-----	26,000	-----	-----
Total Payment (B)	**3,15,000**	**3,74,000**	**4,67,500**	**4,29,000**	**2,97,000**	**4,62,000**
Minimum Cash Balance	35,000	35,000	35,000	35,000	35,000	35,000
Total Cash Required (C)	**3,50,000**	**4,09,000**	**5,02,500**	**4,64,000**	**3,32,000**	**4,97,000**
Surplus/ (Deficit) (A)-(C)	3,39,500	1,37,500	-77,000	5,500	1,92,500	-16,500
Investment/Financing:						
Total effect of (Invest)/ Financing (D)	-3,39,500	-1,37,500	77,000	-5,500	-1,92,500	16,500
Closing Cash Balance (A) + (D) - (B)	35,000	35,000	35,000	35,000	35,000	35,000

ICAI · Old P8 · MTP May 2023 · Series I · question p. 2-3 · answer p. 5-6

760 MTP **Old P8 Dec 2021 S-I** Q5(b) 2 marks old scheme

Write a short note on electronic fund transfer.

ICAI'S ANSWER

Electronic Fund Transfer: With the developments which took place in the information technology, the present banking system has switched over to the computerization of banks branches to offer efficient banking services and cash management services to their customers. The network will be linked to the different branches, banks. This helped the customers in the following ways:

(i) Instant updating of accounts.

(ii) Quick transfer of funds.

(iii) Instant information about foreign exchange rates.

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 4 · answer p. 10

761 MTP Old P8 Jul 2021 S-I Q3 10 marks old scheme

PREPARE monthly cash budget for the first six months of 2021 on the basis of the following information:

(i) Actual and estimated monthly sales are as follows:

Actual	(Rs.)	Estimated	(Rs.)
October 2020	2,00,000	January 2021	60,000
November 2020	2,20,000	February 2021	80,000
December 2020	2,40,000	March 2021	1,00,000
		April 2021	1,20,000
		May 2021	80,000
		June 2021	60,000
		July 2021	1,20,000

(ii) Operating Expenses (including salary & wages) are estimated to be payable as follows:

Month	(Rs.)	Month	(Rs.)
January 2021	22,000	April 2021	30,000
February 2021	25,000	May 2021	25,000
March 2021	30,000	June 2021	24,000

(iii) Of the sales, 75% is on credit and 25% for cash. 60% of the credit sales are collected after one month, 30% after two months and 10% after three months.

(iv) Purchases amount to 80% of sales and are made on credit and paid for in the month preceding the sales.

(v) The firm has 12% debentures of Rs.1,00,000. Interest on these has to be paid quarterly in January, April and so on.

(vi) The firm is to make an advance payment of tax of Rs. 5,000 in April.

(vii) The firm had a cash balance of Rs. 40,000 at 31st Dec. 2020, which is the minimum desired level of cash balance. Any cash surplus/deficit above/below this level is made up by temporary investments/liquidation of temporary investments or temporary borrowings at the end of each month (interest on these to be ignored).

ICAI'S ANSWER

Monthly Cash Budget for first six months of 2021

(Amount in Rs.)

Particulars	Jan.	Feb.	Mar.	April	May	June
Opening balance	40,000	40,000	40,000	40,000	40,000	40,000
Receipts:						

Particulars	Jan.	Feb.	Mar.	April	May	June
Cash sales	15,000	20,000	25,000	30,000	20,000	15,000
Collection from debtors	1,72,500	97,500	67,500	67,500	82,500	70,500
Total cash available (A)	2,27,500	1,57,500	1,32,500	1,37,500	1,42,500	1,25,500
Payments:						
Purchases	64,000	80,000	96,000	64,000	48,000	96,000
Operating Expenses	22,000	25,000	30,000	30,000	25,000	24,000
Interest on debentures	3,000	-	-	3,000	-	-
Tax payment	-	-	-	5,000	-	-
Total payments (B)	89,000	1,05,000	1,26,000	1,02,000	73,000	1,20,000
Minimum cash balance desired	40,000	40,000	40,000	40,000	40,000	40,000
Total cash needed (C)	1,29,000	1,45,000	1,66,000	1,42,000	1,13,000	1,60,000
Surplus/(deficit) (A - C)	98,500	12,500	(33,500)	(4,500)	29,500	(34,500)
Investment/financing						
Temporary Investments	(98,500)	(12,500)	-	-	(29,500)	-
Liquidation of temporary investments or temporary borrowings			33,500	4,500	-	34,500
Total effect of investment/financing(D)	(98,500)	(12,500)	33,500	4,500	(29,500)	34,500
Closing cash balance (A + D - B)	40,000	40,000	40,000	40,000	40,000	40,000

Workings:

1. **Collection from debtors:**

(Amount in Rs.)

	Year 2020			Year 2021					
	Oct.	**Nov.**	**Dec.**	**Jan.**	**Feb.**	**Mar.**	**April**	**May**	**June**
Total sales	2,00,000	2,20,000	2,40,000	60,000	80,000	1,00,000	1,20,000	80,000	60,000
Credit sales (75% of total sales)	1,50,000	1,65,000	1,80,000	45,000	60,000	75,000	90,000	60,000	45,000
Collections:									
One month		90,000	99,000	1,08,000	27,000	36,000	45,000	54,000	36,000
Two months			45,000	49,500	54,000	13,500	18,000	22,500	27,000
Three months				15,000	16,500	18,000	4,500	6,000	7,500
Total collections				**1,72,500**	**97,500**	**67,500**	**67,500**	**82,500**	**70,500**

2. **Payment to Creditors:**

(Amount in Rs.)

	Year 2021						
	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**
Total sales	60,000	80,000	1,00,000	1,20,000	80,000	60,000	1,20,000
Purchases (80% of total sales)	48,000	64,000	80,000	96,000	64,000	48,000	96,000
Payment:							
One month prior	**64,000**	**80,000**	**96,000**	**64,000**	**48,000**	**96,000**	

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 3 · answer p. 5-6

762 MTP **Old P8 Nov 2020 S-I** Q6(c) 2 marks old scheme

EXPLAIN Billing float and Mail float with reference to management of cash.

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 4

763 MTP Old P8 Nov 2019 S-I Q3 10 marks old scheme

You are given the following information:

(i) Estimated monthly Sales are as follows:

	Rs.		Rs.
January	1,00,000	June	80,000
February	1,20,000	July	1,00,000
March	1,40,000	August	80,000
April	80,000	September	60,000
May	60,000	October	1,00,000

(ii) Wages and Salaries are estimated to be payable as follows:

	Rs.		Rs.
April	9,000	July	10,000
May	8,000	August	9,000
June	10,000	September	9,000

(iii) Of the sales, 80% is on credit and 20% for cash. 75% of the credit sales are collected within one month and the balance in two months. There are no bad debt losses.

(iv) Purchases amount to 80% of sales and are made and paid for in the month preceding the sales.

(v) The firm has taken a loan of Rs.1,20,000. Interest @ 10% p.a. has to be paid quarterly in January, April and so on.

(vi) The firm is to make payment of tax of Rs. 5,000 in July, 2019.

(vii) The firm had a cash balance of Rs. 20,000 on 1st April, 2019 which is the minimum desired level of cash balance. Any cash surplus/deficit above/below this level is made up by temporary investments/liquidation of temporary investments or temporary borrowings at the end of each month (interest on these to be ignored).

Required

PREPARE monthly cash budgets for six months beginning from April, 2019 on the basis of the above information.

ICAI'S ANSWER

Computation – Collections from Debtors

Particulars	Feb (Rs.)	Mar (Rs.)	Apr (Rs.)	May (Rs.)	Jun (Rs.)	Jul (Rs.)	Aug (Rs.)	Sep (Rs.)
Total Sales	1,20,000	1,40,000	80,000	60,000	80,000	1,00,000	80,000	60,000
Credit Sales (80% of total Sales)	96,000	1,12,000	64,000	48,000	64,000	80,000	64,000	48,000
Collection (within one month)		72,000	84,000	48,000	36,000	48,000	60,000	48,000
Collection (within two months)			24,000	28,000	16,000	12,000	16,000	20,000
Total Collections			1,08,000	76,000	52,000	60,000	76,000	68,000

Monthly Cash Budget for Six Months: April to September, 2019

Particulars	April (Rs.)	May (Rs.)	June (Rs.)	July (Rs.)	August (Rs.)	Sept. (Rs.)
Receipts:						
Opening Balance	20,000	20,000	20,000	20,000	20,000	20,000
Cash Sales	16,000	12,000	16,000	20,000	16,000	12,000
Collections from Debtors	1,08,000	76,000	52,000	60,000	76,000	68,000
Total Receipts (A)	1,44,000	1,08,000	88,000	1,00,000	1,12,000	1,00,000
Payments:						
Purchases	48,000	64,000	80,000	64,000	48,000	80,000
Wages and Salaries	9,000	8,000	10,000	10,000	9,000	9,000
Interest on Loan	3,000	-----	-----	3,000	-----	-----
Tax Payment	-----	-----	-----	5,000	-----	-----
Total Payment (B)	60,000	72,000	90,000	82,000	57,000	89,000
Minimum Cash Balance	20,000	20,000	20,000	20,000	20,000	20,000
Total Cash Required (C)	80,000	92,000	1,10,000	1,02,000	77,000	1,09,000
Surplus/ (Deficit) (A)-(C)	64,000	16,000	(22,000)	(2,000)	35,000	(9,000)
Investment/Financing: Total effect of (Invest)/ Financing (D)	(64,000)	(16,000)	22,000	2,000	(35,000)	9,000

Particulars	April (Rs.)	May (Rs.)	June (Rs.)	July (Rs.)	August (Rs.)	Sept. (Rs.)
Closing Cash Balance (A) + (D) - (B)	20,000	20,000	20,000	20,000	20,000	20,000

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 2-3 · answer p. 4-5

764 MTP **Old P8 Nov 2019 S-I** Q5(b) 2 marks old scheme

A firm maintains a separate account for cash disbursement. Total disbursement are Rs.10,50,000 per month or Rs. 1,26,00,000 per year. Administrative and transaction cost of transferring cash to disbursement account is Rs.20 per transfer. Marketable securities yield is 8% per annum.

COMPUTE the optimum cash balance according to William J. Baumol model.

ICAI'S ANSWER

The optimum cash balance $C = \sqrt{\frac{2 \times \text{Rs.}1{,}26{,}00{,}000 \times \text{Rs.}20}{0.08}} = \text{Rs.}79{,}372.54$

ICAI · Old P8 · MTP Nov 2019 · Series I · question p. 4 · answer p. 7

9.5 Management of inventory

1 question · asked in 1 ICAI document · 1 exam paper

765 EXAM **Old P8 May 2022** Q1(c) 5 marks old scheme

A company requires 36,000 units of a product per year at cost of ₹ 100 per unit. Ordering cost per order is ₹ 250 and the carrying cost is 4.5% per year of the inventory cost. Normal lead time is 25 days and safety stock is NIL.

Assume 360 working days in a year.

(i) Calculate the Reorder Inventory Level.

(ii) Calculate the Economic Order Quantity (EOQ).

(iii) If the supplier offers 1% quantity discount for purchase in lots of 9,000 units or more, should the company accept the proposal?

ICAI'S ANSWER

Annual Consumption = 36,000 (A)

Ordering Cost = ₹ 250 per order (O)

Carrying Cost = $\frac{4.5}{100} \times 100$

= ₹ 4.5 (C)

Lead Time = 25 days

(i) Reorder Level = Lead Time × Daily Consumption

$= 25 \times \frac{36{,}000}{360}$

= **2,500 units**

(ii) Economic Order Quantity (EOQ) = $\sqrt{\frac{2AO}{C}}$

$$= \sqrt{\frac{2 \times 36{,}000 \times 250}{4.5}}$$

= **2,000 units**

(iii) Evaluation of Profitability of Quantity Discount Offer:

(a) When EOQ is ordered

		(₹)
Purchase Cost	(36,000 units × ₹ 100)	36,00,000
Ordering Cost	[(36,000 units/2,000 units) × ₹ 250]	4,500
Carrying Cost	(2,000 units × ½ × ₹ 4.5)	4,500
Total Cost		**36,09,000**

(b) When Quantity Discount is accepted

		(₹)
Purchase Cost	(36,000 units × ₹ 99*)	35,64,000
Ordering Cost	[(36,000 units/9,000 units) × ₹ 250]	1,000
Carrying Cost	(9,000 units × ½ × ₹ 99 × 4.5%)	20,048
Total Cost		**35,85,048**

*Unit Cost = ₹100

Less: Quantity Discount @ 1% = ₹ 1

Purchase Cost = ₹ 99

Advise – The total cost of inventory is lower if Quantity Discount is accepted. Hence, the company is advised to accept the proposal.

ICAI · Old P8 · May 2022 exam · question p. 2 · answer p. 5-6

9.6 Management of payables

2 questions · asked in 2 ICAI documents · 1 exam paper

766 EXAM **May 2026** Q2(b) 3 marks

B Ltd. has received an offer to get a discount of 2% on an invoice value of ₹75,000, if payment is made within one month instead of normal credit terms of 3 months. If the company makes early payment, it will need to finance the payment through a bank overdraft of 10% p.a. Tax rate is 30%.

Whether B Ltd. should accept the offer? Support your decision with calculations.

ICAI'S ANSWER

Discount on early payment = 2% x 75,000 = ₹1,500

Net Benefit (A) = ₹ 1,500 - Tax (30%) = ₹ 1,050

The company must borrow the discounted invoice amount via a bank overdraft for the 2-month period.

Amount to be Borrowed= 75,000 − 1,500 = ₹ 73,500

Bank overdraft interest for 2 months:

= P × R × T

= 73,500 × 10% × 2/12

= ₹ 1,225

After tax interest cost = 1,225 × (1−0.30)

= 1,225 × 0.70

= ₹ 857.50

Net Saving = Net Benefit – Net Interest Cost

= ₹ 1,050 – ₹ 857.50

= ₹ 192.50 (Net Saving)

Since the company gains a net benefit of ₹ 192.50 by making early payment, B Ltd. should accept the discount offer.

Alternatively, the question can be solved as below:

Annualized benefit of discount

$$= \frac{\text{Discount \%}}{\text{100-Discount \%}} \times \frac{\text{12 months}}{\text{Credit Period} - \text{Discount Period}}$$

= 2/98 x 12/2 x 100 =12.245 %

Post tax benefit = 12.245 x (1-0.3) =8.571

Post tax Bank OD rate = 10% x (1-0.3) =7%

Benefit of discount is more than cost of OD. Hence, discount should be accepted.

ICAI · P6 · May 2026 exam · question p. 14 · answer p. 16

767 MTP **Sep 2024 S-I** Q4(a) 4 marks

DISCUSS the Costs of Availing Trade Credit

ICAI'S ANSWER

Normally it is considered that the trade credit does not carry any cost. However, it carries the following costs:

(i) **Price:** There is often a discount on the price that the firm undergoes when it uses trade credit, since it can take advantage of the discount only if it pays immediately. This discount can translate into a high implicit cost.

(ii) **Loss of goodwill:** If the credit is overstepped, suppliers may discriminate against delinquent customers if supplies become short. As with the effect of any loss of goodwill, it depends very much on the relative market strengths of the parties involved.

(iii) **Cost of managing:** Management of creditors involves administrative and accounting costs that would otherwise be incurred.

(iv) **Conditions:** Sometimes most of the suppliers insist that for availing the credit facility the order should be of some minimum size or even on regular basis.

ICAI · P6 · MTP Sep 2024 · Series I · question p. 6 · answer p. 9-10

9.7 Financing of working capital and bank credit

6 questions · asked in 8 ICAI documents · 3 exam papers

768 EXAM **May 2024** Q4(b) 4 marks

ABC Ltd. is approaching the banks for financing its business activity. You are required to describe any four forms of bank credit for the consideration of the company.

ICAI'S ANSWER

Some of the forms of bank credit are:

(i) Cash Credit: This facility will be given by the banker to the customers by giving certain amount of credit facility on continuous basis. The borrower will not be allowed to exceed the limits sanctioned by the bank.

(ii) Bank Overdraft: It is a short-term borrowing facility made available to the companies in case of urgent need of funds. The banks will impose limits on the amount they can lend. When the borrowed funds are no longer required they can quickly and easily be repaid. The banks issue overdrafts with a right to call them in at short notice.

(iii) Bills Discounting: The Company which sells goods on credit will normally draw a bill on the buyer who will accept it and sends it to the seller of goods. The seller, in turn discounts the bill with his banker. The banker will generally earmark the discounting bill limit.

(iv) Bills Acceptance: To obtain finance under this type of arrangement a company draws a bill of exchange on bank. The bank accepts the bill thereby promising to pay out the amount of the bill at some specified future date.

(v) Line of Credit: Line of Credit is a commitment by a bank to lend a certain amount of funds on demand specifying the maximum amount.

(vi) Letter of Credit: It is an arrangement by which the issuing bank on the instructions of a customer or on its own behalf undertakes to pay or accept or negotiate or authorizes another bank to do so against stipulated documents subject to compliance with specified terms and conditions.

(vii) Bank Guarantees: Bank guarantee is one of the facilities that the commercial banks extend on behalf of their clients in favour of third parties who will be the beneficiaries of the guarantees.

(viii) Short Term Loans: In a loan account, the entire advance is disbursed at one time either in cash or by transfer to the current account of the borrower. It is a single advance and given against securities like shares, government securities, life insurance policies and fixed deposit receipts, etc.

(ix) Clean Overdrafts: Request for clean advances are entertained only from parties which are financially sound and reputed for their integrity. The bank has to rely upon the personal security of the borrowers.

(x) Advances against goods: Goods are charged to the bank either by way of pledge or by way of hypothecation. Goods include all forms of movables which are offered to the bank as security.

(xi) Usance bills maturing at a future date or sight are discounted by the banks for approved parties. The borrower is paid the present worth and the bank collects the full amount on maturity.

(xii) Advance against documents of title to goods: A document becomes a document of title to goods when its possession is recognised by law or business custom as possession of the goods like bill of lading, dock warehouse keeper's certificate, railway receipt, etc. An advance against the pledge of such documents is an advance against the pledge of goods themselves.

(xiii) Advance against supply of bills: Advances against bills for supply of goods to government or semi-government departments against firm orders after acceptance of tender fall under this category. It is this debt that is assigned to the bank by endorsement of supply bills and executing irrevocable power of attorney in favour of the banks for receiving the amount of supply bills from the Government departments.

ICAI · P6 · May 2024 exam · question p. 19 · answer p. 21-22

769 EXAM Old P8 Nov 2023 Q6(b) 4 marks old scheme

Also asked in RTP Jan 2026 Q11(b)

What do you understand by Spontaneous Sources of finance and explain its sources of finance?

ICAI'S ANSWER

Spontaneous sources of finance are those which naturally arise in the course of business operations. Trade credit, credit from employees, credit from suppliers of services, etc. are some of the examples which may be quoted in this respect.

Spontaneous Sources of Finance

(i) Trade Credit: Trade credit is a spontaneous source of finance which is normally extended to the purchaser organization by the sellers or services providers. It contributes to about one-third of the total short-term requirements.

(ii) Bills Payable: In the case of "Bills Payable" the purchaser will have to give a written promise to pay the amount of the bill/invoice either on demand or at a fixed future date to the seller or the bearer of the note.

(iii) Accrued Expenses: The accrued expenses refer to the services availed by the firm, but the payment for which has yet to be made. It is a built in and an automatic source of finance as most of the services like wages, salaries, taxes, duties etc., are paid at the end of the period.

ICAI · Old P8 · Nov 2023 exam · question p. 17 · answer p. 18

770 EXAM Old P8 May 2022 Q1(b) 5 marks old scheme

Balance sheet of X Ltd for the year ended 31st March,2022 is given below:

(₹ in lakhs)

Liabilities	Amount	Assets	Amount
Equity Shares ₹ 10 each	200	Fixed Assets	500
Retained earnings	200	Raw materials	150
11% Debentures	300	W.I.P	100
Public deposits (Short-Term)	100	Finished goods	50
Trade Creditors	80	Debtors	125
Bills Payable	100	Cash/Bank	55
	980		**980**

Calculate the amount of maximum permissible bank finance under three methods as per Tandon Committee lending norms.

The total core current assets are assumed to be ₹ 30 lakhs.

ICAI'S ANSWER

Current Assets = 150 + 100 + 50 + 125 + 55 = ₹ 480 Lakhs

Current Liabilities = 100 + 80 + 100 = ₹ 280 Lakhs

Maximum Permissible Banks Finance under Tandon Committee Norms:

Method I

Maximum Permissible Bank Finance = 75% of (Current Assets – Current Liabilities)

= 75% of (480 - 280)

= **₹ 150 Lakhs**

Method II

Maximum Permissible Bank Finance = 75% of Current Assets – Current Liabilities

= 75 % of 480 – 280

= **₹ 80 Lakhs**

Method III

Maximum Permissible Bank Finance = 75% of (Current Assets – Core Current Assets) – Current Liabilities

= 75 % of (480 - 30) – 280

= **₹ 57.5 Lakhs**

ICAI · Old P8 · May 2022 exam · question p. 1-2 · answer p. 4-5

771 RTP **Old P8 Dec 2021 Q8** old scheme

The Alliance Ltd., a Petrochemical sector company had just invested huge amount in its new expansion project. Due to huge capital investment, the company is in need of an additional ₹ 1,50,000 in working

capital immediately. The Finance Manger has determined the following three feasible sources of working capital funds:

(i) Bank loan: The Company's bank will lend ₹ 2,00,000 at 15%. A 10% compensating balance will be required, which otherwise would not be maintained by the company.

(ii) Trade credit: The company has been offered credit terms from its major supplier of 3/30, net 90 for purchasing raw materials worth ₹ 1,00,000 per month.

(iii) Factoring: A factoring firm will buy the company's receivables of ₹ 2,00,000 per month, which have a collection period of 60 days. The factor will advance up to 75% of the face value of the receivables at 12% on an annual basis. The factor will also charge commission of 2% on all receivables purchased. It has been estimated that the factor's services will save the company a credit department expense and bad debt expense of ₹ 1,250 and ₹ 1,750 per month respectively.

On the basis of annual percentage cost, ADVISE which alternative should the company select? Assume 360 days year.

ICAI'S ANSWER

(i) **Bank loan:** Since the compensating balance would not otherwise be maintained, the real annual cost of taking bank loan would be:

$= \frac{15}{90} \times 100 = 16.67\%$ p.a.

(ii) **Trade credit:** Amount upto ₹ 1,50,000 can be raised within 2 months or 60 days. The real annual cost of trade credit would be:

$= \frac{3}{97} \times \frac{360}{60} \times 100 = 18.56\%$ p.a.

(iii) **Factoring:**

Commission charges per year = 2% × (₹ 2,00,000 × 12) = ₹ 48,000

Total Savings per year = (₹ 1,250 + ₹ 1,750) × 12 = ₹ 36,000

Net factoring cost per year = ₹ 48,000 - ₹ 36,000 = ₹ 12,000

Annual Cost of Borrowing ₹ 1,50,000 receivables through factoring would be:

$= \frac{12\% \times 1{,}50{,}000 + ₹\ 12{,}000}{₹\ 1{,}50{,}000} \times 100$

$= \frac{₹\ 18{,}000 + ₹\ 12{,}000}{₹\ 1{,}50{,}000} \times 100$

= 20% p.a.

Advise: The company should select alternative of Bank Loan as it has the lowest annual cost i.e. 16.67% p.a.

ICAI · Old P8 · RTP Dec 2021 · question p. 4 · answer p. 14-15

772 MTP **Old P8 Nov 2023 S-I** Q6(c) 2 marks old scheme

Also asked in Old P8 MTP May 2020 S-I Q6(c)

EXPLAIN in short, the term Letter of Credit.

ICAI'S ANSWER

Letter of Credit: It is an arrangement by which the issuing bank on the instructions of a customer or on its own behalf undertakes to pay or accept or negotiate or authorizes another bank to do so

against stipulated documents subject to compliance with specified terms and conditions.

ICAI · Old P8 · MTP Nov 2023 · Series I · question p. 3 · answer p. 10

773 MTP **Old P8 May 2023 S-II** Q1(b) 5 marks old scheme

Sundaram limited a plastic manufacturing company had invested enormous amount of money in a new expansion project. Due to such a great amount of capital investment, Company needs an additional ₹ 2,00,00,000 in working capital immediately. The CFO has determined the following three feasible sources of working capital funds:

Bank Loan: The company's bank will lend ₹2,30,00,000 at 12% per annum. However, the bank will require 15% of the loan granted to be kept in a current account as the minimum average balance which otherwise would have been just ₹ 50,000.

Trade Credit: A major supplier with 2/20 net 80 credit terms has approached for supply of raw material worth ₹1,90,00,000 p.m.

Factoring: factoring firm will buy the companies receivables of ₹ 2,50,00,000 per month, which have a collection period of 60 days. factor will advance up to 75% of the face value of the receivables at 14 percent per annum. Factor Commission will amount to 2% on all receivables purchased. Factoring will save credit department expense and bad debts of ₹ 1,75,000 p.m. and ₹ 2,25,000 p.m.

Based on annual percentage cost, ADVISE which alternative should the company select. Assume 360 days a year

ICAI'S ANSWER

(i) Bank Loan: As the minimum average balance more than ₹ 50,000 need not be kept if loan is not undertaken, the incremental money made available by bank through bank loan is ₹ 2,30,00,000- (15% x ₹ 2,30,00,000-₹ 50,000) = ₹ 1,96,00,000. Real annual cost of bank loan = (₹ 2.3 crores x 12%) / ₹ 1.96 crores = 14.08%.

(ii) Trade Credit: The real annual cost of trade credit will be 2/98 x 360/60 x 100 = 12.24% .

(iii) Factoring:

Commission charges per year = 2% x 2.5 crores x 12 = ₹ 60,00,000

Savings per year = (1,75,000+2,25,000) x 12 = ₹ 48,00,000

Net Factoring cost per year = ₹ 60,00,000 – ₹ 48,00,000 = ₹12,00,000

Annual cost of borrowing ₹ 2.5 crores x 75% i.e. ₹ 1,87,50,000 will be

$$\frac{1,87,50,000 \text{ x } 14\% + ₹\ 12,00,000}{1,87,50,000} = 20.4\%$$

Conclusion: The company should select trade credit as a preferred mode of financing the working capital requirement as it results in lowest cost on an annual basis.

ICAI · Old P8 · MTP May 2023 · Series II · question p. 1-2 · answer p. 1-2

APPENDIX

Out of syllabus

Old-scheme questions on topics the January 2027 syllabus no longer has, and items ICAI itself marks out of syllabus. Skip them unless you are curious.

A.1 Near Chapter 7: Investment Decisions

37 questions ICAI marks out of syllabus, or whose topic has left the syllabus. For reference only.

774 EXAM Old P8 Nov 2023 Q5(a) 5 marks old scheme

BSB Ltd. is considering its new project with the following details:

Sr. No.	Particulars	Amount
1	Initial capital cost	5,00,00,000
2	Annual unit sales	6,00,000
3	Selling price per unit (in ₹)	120
4	Variable cost per unit (in ₹)	80
5	Fixed cost per year	36,00,000
6	Discount Rate	10%

Required:

a. To advise the company whether to invest in the new project or not based on the NPV concept.

b. Compute the impact on the project's NPV considering a 1% adverse variance in each variable. Which variable is having minimum effect?

Consider Life of the project as 3 years.

Year	1	2	3
PVF @ 10%	0.909	0.826	0.751
PVF @ 11%	0.901	0.812	0.731

ICAI'S ANSWER

1. Calculation of Net Cash Inflow per year

	Particulars	Amount (₹)
A	Selling price per unit	120
B	Variable cost per unit	80
C	Contribution per unit (A - B)	40
D	Number of units sold per year	6 lakhs
E	Total Contribution (C × D)	₹ 240 lakhs
F	Fixed cost per year	₹ 36 lakhs
G	Net cash inflow per year (E - F)	₹ 204 lakhs

Calculation of Net Present Value (NPV) of the Project

Year	Year Cash Flow (₹ in lakhs)	PV factor @ 10%	Present Value (PV) (₹ in lakhs)
0	(500.00)	1.000	(500.00)
1	204	0.909	185.44
2	204	0.826	168.50
3	204	0.751	153.20
Net Present Value			7.14

Since the NPV of the project is positive, the company should invest in the new project.

2. Sensitivity Analysis considering 1 % Adverse Variance in each variable

	Particulars	Base	Initial capital cost increased to ₹ 505 lakhs	Selling Price per Unit Reduced to ₹ 118.8	Variable Cost Per Unit increased to ₹ 80.80	Fixed Cost per year increased to ₹ 36.36 lakhs	Units sold per year reduced to 5.94 lakhs
		(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
A	Selling price per unit	120	120	118.8	120	120	120
B	Variable cost per unit	80	80	80	80.8	80	80
C	Contribution per unit (A - B)	40	40	38.8	39.2	40	40
		(₹ in lakhs)	(₹ in lakhs)	(₹ in lakhs)	(₹ in lakhs)	(₹ in lakhs)	(₹ in lakhs)
D	Number of units sold per year (units in lakhs)	6	6	6	6	6	5.94
E	Total Contribution (C × D)	240	240	232.8	235.2	240	237.6
F	Fixed cost per year	36	36	36	36	36.36	36
G	Net Cash Inflow per year (E - F)	204	204	196.8	199.2	203.64	201.6
H	PV of Net cash Inflow per year (G × 2.486)	507.14	507.14	489.24	495.21	506.25	501.18
I	Initial capital cost	500	505	500	500	500	500
J	NPV (H - I)	7.14	2.14	-10.76	-4.79	6.25	1.18
K	Percentage Change in NPV	-	-69.99%	-250.55%	-167.03%	-12.53%	-83.52%

The above table shows that by changing one variable at a time by 1% (adverse) while keeping the others constant, the impact in percentage terms on the NPV of the project can be calculated. Thus, it

can be seen that the change in fixed cost has the minimum effect on the NPV by 12.53%.

PassBook's note: Out of syllabus (old scheme): requirement b, the sensitivity analysis of the NPV to a 1% adverse change in each variable, is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter teaches payback, ARR, NPV, PI, IRR, MIRR, discounted payback, capital rationing and unequal lives, with no risk analysis). Requirement a (the NPV decision) is in syllabus: the tag names it. ICAI prints one (5 Marks) for both requirements, so the item is not split.

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775 EXAM **Old P8 May 2023** Q1(c) 5 marks old scheme

A company wants to invest in a project. This requires an initial investment of ₹ 4,50,000. Salvage value after estimated useful life of 5 years is ₹ 50,000. Other details of project are as follows:

	Worst case	Most likely	Best case
Contribution (₹)	3,30,000	5,40,000	6,31,250
Fixed cost (excluding depreciation) (₹)	75,000	1,50,000	2,00,000

Tax rate is 40%. Expected cost of capital of project is 12%. Ignore tax on capital gain.

(i) Calculate NPV in each scenario.

(ii) The company is certain about most likely result in first two years, but uncertain about remaining period. In such a situation, calculate NPV expecting worst case scenario during next two years and best case scenario in the remaining period.

Years	1	2	3	4	5
$PVIF_{0.12,t}$	0.893	0.797	0.712	0.636	0.567
$PVIFA_{0.12,t}$	0.893	1.690	2.402	3.038	3.605

ICAI'S ANSWER

(i) Initial Investment = ₹ 4,50,000

Salvage Value = ₹ 50,000

Useful Life = 5 years

Calculation of cash flow in each scenario

Particulars	Scenario		
	Worst case	**Most Likely**	**Best case**
Contribution	3,30,000	5,40,000	6,31,250
Less: Fixed Cost	75,000	1,50,000	2,00,000
Less: Depreciation	80,000	80,000	80,000
Profit before tax	1,75,000	3,10,000	3,51,250
Less: Taxes	70,000	1,24,000	1,40,500
Profit after tax	1,05,000	1,86,000	2,10,750

Particulars	Scenario		
Add: Depreciation	80,000	80,000	80,000
Cash Flow After Tax	**1,85,000**	**2,66,000**	**2,90,750**

The possible outcomes will be as follows:

Year	PVF @ 12%	Worst Case		Most likely		Best case	
		Cash Flow	**PV**	**Cash Flow**	**PV**	**Cash Flow**	**PV**
0	1	(4,50,000)	(4,50,000)	(4,50,000)	(4,50,000)	(4,50,000)	(4,50,000)
1-5	3.605	1,85,000	6,66,925	2,66,000	9,58,930	2,90,750	10,48,153.75
5	0.567	50,000	28,350	50,000	28,350	50,000	28,350
NPV			**2,45,275**		**5,37,280**		**6,26,503.75**

Alternative presentation

(i) Computation of NPV of different scenarios

Years	DF	Worst case		Most-likely case		Best case	
		CF	**DCF**	**CF**	**DCF**	**CF**	**DCF**
0	1.000	(450000)	(450000)	(450000)	(450000)	(450000)	(450000)
1	0.893	185000	165205	266000	237538	290750	259640
2	0.797	185000	147445	266000	212002	290750	231728
3	0.712	185000	131720	266000	189392	290750	207014
4	0.636	185000	117660	266000	169176	290750	184917
5	0.567	185000	104895	266000	150822	290750	164855
5	0.567	50000	28350	50000	28350	50000	28350
	NPV		**2,45,275**		**5,37,280**		**6,26,504**

(ii) If the company is certain about the most likely result in first two years but uncertain about the remaining period, then, NPV expecting worst case scenario during next two years and best-case scenario in remaining period will be as follows:

$$= -4,50,000 + \frac{₹2,66,000}{(1+0.12)} + \frac{₹2,66,000}{(1+0.12)^2} + \frac{₹1,85,000}{(1+0.12)^3} + \frac{₹1,85,000}{(1+0.12)^4} + \frac{₹2,90,750}{(1+0.12)^5} + \frac{₹50,000}{(1+0.12)^5}$$

= -4,50,000 + (2,66,000 × 0.893) + (2,66,000 × 0.797) + (1,85,000 × 0.712) + (1,85,000 × 0.636) + (2,90,750 × 0.567) + (50,000 × 0.567)

= -4,50,000 + 2,37,538 + 2,12,002 + 1,31,720 + 1,17,660 + 1,64,855 + 28,350

= ₹ 4,42,125

Alternative presentation

(ii) Computation of NPV on the basis of fixed scenarios

Years	Scenarios	DF	CF	DCF (DF*CF)
0	Initial Outflow	1.000	(450000)	**(450000)**
1	Most-likely case	0.893	266000	**237538**
2	Most-likely case	0.797	266000	**212002**
3	Worst case	0.712	185000	**131720**
4	Worst case	0.636	185000	**117660**
5	Best case	0.567	290750	**164855**
5	Salvage	0.567	50000	**28350**
				4,42,125

(All figures are in ₹)

PassBook's note: Lead ruling 2026-10-20: risk analysis in capital budgeting (scenario / best-worst-case NPVs, expected NPV, sensitivity, certainty equivalent, risk-adjusted discount rate) is not in the January 2027 Study Material, so this item is out of syllabus (oos) and keeps the nearest tag fm.capbud.npv.

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776 EXAM **Old P8 Nov 2022** Q1(d) 5 marks old scheme

Determine the Risk Adjusted Net Present Value of the following projects:

	A	B	C
Net cash outlays (₹)	70,000	1,20,000	2,20,000
Project life	5 years	5 years	5 years
Annual cash inflow (₹)	30,000	42,000	70,000
Coefficient of Variation	2.2	1.6	1.2

The company selects the risk-adjusted discount rate on the basis of the Coefficient of variation.

Coefficient of Variation	Applicable Risk adjusted discount rate (i)	PVIFA (i,5)
0	10%	3.791
0.4.	12%	3.605
0.8	14%	3.433
1.2	16%	3.274
1.6	18%	3.127
2	22%	2.864
>2.0	25%	2.689

Which project should be selected by the company based on Risk Adjusted NPV?

ICAI'S ANSWER

Selection of project on the basis of Risk Adjusted Net Present Value

Particulars	A	B	C
Co efficient of Variation	2.2	1.6	1.2
Applicable discount rate (%)	25	18	16
Annual cash inflow (₹)	30,000	42,000	70,000
Relevant PVIFA	2.689	3.127	3.274
PV of cash inflow (₹)	**80,670**	**1,31,334**	**2,29,180**
Less: Cash outflow (₹)	70,000	1,20,000	2,20,000
Risk adjusted NPV (₹)	**10,670**	**11,334**	**9,180**

Conclusion: Project B should be selected as its Risk adjusted NPV is high.

PassBook's note: Out of syllabus (old scheme): the risk-adjusted discount rate chosen by the coefficient of variation is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk analysis). The NPV arithmetic itself is in syllabus: the tag names it. ICAI's own text kept as printed: "0.4." in the discount-rate table and "Co efficient of Variation" in the answer.

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777 EXAM Old P8 May 2022 Q1(d) 5 marks old scheme

P Ltd. is considering a project with the following details:

Initial Project Cost	₹ 1,00,000			
Annual Cash Inflow (₹)	1	2	3	4
	30,000	40,000	50,000	60,000
Project Life (Years)	4			
Cost of Capital	10%			

(i) MEASURE the sensitivity of the project to change in initial project cost and Annual cash inflows (considering each factor at a time) such that NPV become zero.

(ii) IDENTIFY which of the two factors; the project is most sensitive to affect the acceptability of the project?

Year	1	2	3	4	5
$PVIF_{0.10,t}$	0.909	0.826	0.751	0.683	0.621

ICAI'S ANSWER

Computation of Net Present Value (NPV):

Year	PVF @ 10%	Original Cash Flows (₹)	PV (₹)	PV (₹)
0	1	(1,00,000)		(1,00,000)
1	0.909	30,000	27,270	
2	0.826	40,000	33,040	
3	0.751	50,000	37,550	
4	0.683	60,000	40,980	1,38,840
NPV				**38,840**

Determination of the most Sensitive factor:

(i) Sensitivity Analysis w.r.t. Initial Project cost (such that NPV becomes zero):

NPV of the project would be zero when the initial project cost is increased by ₹ 38,840.

$\therefore$ Percentage change in Initial project cost = $\frac{₹\ 38,840}{₹\ 1,00,000} \times 100$ = **38.84%**

(ii) Sensitivity Analysis w.r.t. Annual Cash inflows (such that NPV becomes zero):

NPV of the project would be zero when the Annual cash inflows is decreased by ₹ 38,840.

$\therefore$ Percentage change in the Annual cash inflows = $\frac{₹\ 38,840}{₹\ 1,38,840} \times 100$ = **27.97%**

Conclusion: Annual cash inflows factor is the most sensitive as only a change beyond 27.97% in savings makes the project unacceptable.

PassBook's note: Out of syllabus (old scheme): the item is sensitivity analysis of a project's NPV (the change in initial cost and in annual inflows that brings NPV to zero, and the most sensitive factor), which is Risk Analysis in Capital Budgeting. That topic is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter teaches payback, ARR, NPV, PI, IRR, MIRR, discounted payback, capital rationing and equivalent annual cost, with no risk analysis). The tag names the nearest sub-topic: the NPV computation the answer starts from.

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778 EXAM **Old P8 Dec 2021** Q6(b) 4 marks old scheme

Distinguish between Scenario Analysis & Sensitivity Analysis.

ICAI'S ANSWER

Scenario Analysis Vs Sensitivity Analysis

Sensitivity analysis calculates the impact of the change of a single input variable on the outcome of the project viz., NPV or IRR. The sensitivity analysis thus enables to identify that single critical variable which can impact the outcome in a huge way and the range of outcomes of the project given the change in the input variable.

Scenario analysis, on the other hand, is based on a scenario. The scenario may be recession or a boom wherein depending on the scenario, all input variables change. Scenario Analysis calculates the outcome of the project considering this scenario where the variables have changed simultaneously. Similarly, the outcome of the project would also be considered for the normal and recessionary situation. The variability in the outcome under the three different scenarios would help the management to assess the risk a project carries. Higher deviation in the outcome can be assessed as higher risk and lower to medium deviation can be assessed accordingly.

Scenario analysis is far more complex than sensitivity analysis because in scenario analysis all inputs are changed simultaneously, considering the situation in hand while in sensitivity analysis, only one input is changed, and others are kept constant.

PassBook's note: Out of syllabus (old scheme): sensitivity analysis and scenario analysis (risk analysis in capital budgeting) are not in the current Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · Dec 2021 exam · question p. 14 · answer p. 15

779 EXAM **Old P8 Dec 2021** Q6(c) 2 marks old scheme

Adjustment of risk is required in capital budgeting decision, give reasons for it.

ICAI'S ANSWER

Reasons for adjustment of Risk in Capital Budgeting decisions are as follows:

1. There is an opportunity cost involved while investing in a project for the level of risk. Adjustment of risk is necessary to help make the decision as to whether the returns out of the project are proportionate with the risks borne and whether it is worth investing in the project over the other investment options available.
2. Risk adjustment is required to know the real value of cash Inflows. Higher risk will lead to higher risk premium and also the expectation of higher return.

PassBook's note: Out of syllabus (old scheme): risk adjustment in capital budgeting (risk analysis) is not in the current Paper 6A syllabus or the January 2027 Study Material.

ICAI · Old P8 · Dec 2021 exam · question p. 14 · answer p. 16

780 EXAM Old P8 Jul 2021 Q1(c) 5 marks old scheme

K.P. Ltd. is investing ₹ 50 lakhs in a project. The life of the project is 4 years. Risk free rate of return is 6% and risk premium is 6%, other information is as under:

Sales of 1st year	₹ 50 lakhs
Sales of 2nd year	₹ 60 lakhs
Sales of 3rd year	₹ 70 lakhs
Sales of 4th year	₹ 80 lakhs
P/V Ratio (same in all the years)	50%
Fixed Cost (Excluding Depreciation) of 1st year	₹ 10 lakhs
Fixed Cost (Excluding Depreciation) of 2nd year	₹ 12 lakhs
Fixed Cost (Excluding Depreciation) of 3rd year	₹ 14 lakhs
Fixed Cost (Excluding Depreciation) of 4th year	₹ 16 lakhs

Ignoring interest and taxes,

You are required to calculate NPV of given project on the basis of Risk Adjusted Discount Rate.

Discount factor @ 6% and 12% are as under:

Year	1	2	3	4
Discount Factor @ 6%	0.943	0.890	0.840	0.792
Discount Factor@ 12%	0.893	0.797	0.712	0.636

ICAI'S ANSWER

Calculation of Cash Flow

Year	Sales (₹ in Lakhs) (A)	P/V ratio (B)	Contribution (₹ in Lakhs) (C) = (A x B)	Fixed Cost (₹ in Lakhs) (D)	Cash Flows (₹ in lakhs) (E) = (C – D)
1	50	50%	25	10	15
2	60	50%	30	12	18
3	70	50%	35	14	21
4	80	50%	40	16	24

When risk-free rate is 6% and the risk premium expected is 6%, then risk adjusted discount rate would be 6% + 6% =12%.

Calculation of NPV using Risk Adjusted Discount Rate (@ 12%)

Year	Cash flows (₹ in Lakhs)	Discounting Factor @ 12%	Present Value of Cash Flows (₹ in lakhs)
1	15	0.893	13.395
2	18	0.797	14.346
3	21	0.712	14.952

Year	Cash flows (₹ in Lakhs)	Discounting Factor @ 12%	Present Value of Cash Flows (₹ in lakhs)
4	24	0.636	15.264
Total of present value of Cash flow			57.957
Less: Initial Investment			50.000
Net Present value (NPV)			**7.957**

PassBook's note: Risk analysis in capital budgeting (the risk-adjusted discount rate method as such) is not in the new-scheme Intermediate syllabus or the January 2027 Study Material.

ICAI · Old P8 · Jul 2021 exam · question p. 65 · answer p. 68-69

781 EXAM **Old P8 Jan 2021** Q1(c) 5 marks old scheme

A project requires an initial outlay of ₹ 3,00,000.

The company uses certainty equivalent method approach to evaluate the project. The risk free rate is 7%.

Following information is available:

Year	CFAT (Cash Flow After Tax) ₹	CE (Certainty Equivalent Coefficient)
1.	1,00,000	0.90
2.	1,50,000	0.80
3.	1,15,000	0.60
4.	1,00,000	0.55
5.	50,000	0.50

PV Factor at 7%

Year	1	2	3	4	5
PV Factor	0.935	0.873	0.816	0.763	0.713

Evaluate the above. Is investment in the project beneficial?

ICAI'S ANSWER

Statement Showing the Net Present Value of Project

Year end	CFAT (₹)	C.E.	Adjusted Cash flow (₹)	Present value factor @ 7%	Total Present value (₹)
	(a)	**(b)**	**(c) = (a) × (b)**	**(d)**	**(e) = (c) × (d)**
1	1,00,000	0.90	90,000	0.935	84,150

Year end	CFAT (₹)	C.E.	Adjusted Cash flow (₹)	Present value factor @ 7%	Total Present value (₹)
2	1,50,000	0.80	1,20,000	0.873	1,04,760
3	1,15,000	0.60	69,000	0.816	56,304
4	1,00,000	0.55	55,000	0.763	41,965
5	50,000	0.50	25,000	0.713	17,825
PV of Cash Inflow					3,05,004
Less: Initial Investment					(3,00,000)
Net Present Value					**5,004**

Since the NPV of the project is positive, it is beneficial to invest in the project.

PassBook's note: Out of syllabus (old scheme): the item evaluates a project by the certainty equivalent method, which is Risk Analysis in Capital Budgeting. That topic is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter teaches payback, ARR, NPV, PI, IRR, MIRR, discounted payback, capital rationing and equivalent annual cost, with no risk analysis; The Study Material has no certainty equivalent). The tag names the nearest sub-topic: the NPV the answer computes.

ICAI · Old P8 · Jan 2021 exam · question p. 2 · answer p. 5

782 EXAM **Old P8 Nov 2019** Q1(b) 5 marks old scheme

Door Ltd. is considering an investment of ₹ 4,00,000. This investment is expected to generate substantial cash inflows over the next five years. Unfortunately, the annual cash flows from this investment is uncertain, and the following profitability distribution has been established.

Annual Cash Flow (₹)	Probability
50,000	0.3
1,00,000	0.3
1,50,000	0.4

At the end of its 5 years life, the investment is expected to have a residual value of ₹ 40,000.

The cost of capital is 5%

(i) Calculate NPV under the three different scenarios.

(ii) Calculate Expected Net Present Value.

(iii) Advise Door Ltd. on whether the investment is to be undertaken.

Year	1	2	3	4	5
DF @ 5%	0.952	0.907	0.864	0.823	0.784

ICAI'S ANSWER

(i) Calculation of NPV under three different scenarios (Amount in ₹)

Particulars	1st Scenario	2nd Scenario	3rd Scenario
Annual Cash Flow	50,000	1,00,000	1,50,000
PV of cash inflows (Annual Cash Flow × 4.33*)	2,16,500	4,33,000	6,49,500
PV of Residual Value (₹ 40,000 × 0.784)	31,360	31,360	31,360
Total PV of Cash Inflow	2,47,860	4,64,360	6,80,860
Initial investment	4,00,000	4,00,000	4,00,000
NPV	**(1,52,140)**	**64,360**	**2,80,860**

* .952 + .907 + .864 + .823 + .784 = 4.33

(ii) Calculation of Expected Net present Value under three different scenarios

Particulars	1st Scenario	2nd Scenario	3rd Scenario	Total (₹)
Annual Cash Flow	₹ 50,000	₹ 1,00,000	₹ 1,50,000	
Probability	0.3	0.3	0.4	
Expected Value	₹ 15,000	₹ 30,000	₹ 60,000	1,05,000
PV of Expected value (1,05,000 × 4.33)				4,54,650
PV of Residual Value (40,000 × 0.784)				31,360
Total PV of Cash Inflow				4,86,010
Initial investment				4,00,000
Expected Net Present Value				**86,010**

(iii) Since the expected net present value of the Investment is positive, the Investment should be undertaken.

PassBook's note: Old-scheme mapping (ours): the expected NPV from a probability distribution of cash flows belongs to risk analysis in capital budgeting, which the new-scheme Paper 6A syllabus and the January 2027 Study Material do not teach (its Investment Decisions chapter has no expected NPV or probability analysis), so the item is out of syllabus (oos, lead ruling 2026-10-20, consistent with the other old Paper 8 risk-analysis items) and keeps the nearest tag fm.capbud.npv.

ICAI · Old P8 · Nov 2019 exam · question p. 1-2 · answer p. 4

783 RTP **Old P8 Nov 2023** Q7 old scheme

An enterprise is investing ₹ 200 lakhs in a project. The risk-free rate of return is 7%. Risk premium expected by the Management is 7%. The life of the project is 5 years. Following are the cash flows that are estimated over the life of the project.

Year	Cash flows (₹ In lakhs)
1	50
2	120
3	150
4	160
5	130

CALCULATE Net Present Value of the project based on Risk free rate and also on the basis of Risks adjusted discount rate.

ICAI'S ANSWER

The Present Value of the Cash Flows for all the years by discounting the cash flow at 7% is calculated as below:

Year	Cash flows ₹In lakhs	Discounting Factor@7%	Present value of Cash Flows ₹ In Lakhs
1	50	0.935	46.75
2	120	0.873	104.76
3	150	0.816	122.40
4	160	0.763	122.08
5	130	0.713	92.69
Total of present value of Cash flow			488.68
Less: Initial investment			(200.00)
Net Present Value (NPV)			288.68

Now, the risk-free rate is 7 % and the risk premium expected by the Management is 7 %. So, the risk adjusted discount rate is 7 % + 7 % =14%.

Discounting the above cash flows using the Risk Adjusted Discount Rate would be as below:

Year	Cash flows ₹ in Lakhs	Discounting Factor@14%	Present Value of Cash Flows ₹ in lakhs
1	50	0.877	43.85
2	120	0.769	92.28
3	150	0.675	101.25
4	160	0.592	94.72
5	130	0.519	67.47

Year	Cash flows ₹ in Lakhs	Discounting Factor@14%	Present Value of Cash Flows ₹ in lakhs
Total of present value of Cash flow			399.57
Initial investment			(200.00)
Net present value (NPV)			199.79

PassBook's note: Out of syllabus (old scheme): the risk-adjusted discount rate is Risk Analysis in Capital Budgeting (ICAI prints the question under that chapter heading in this RTP), which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · RTP Nov 2023 · question p. 4 · answer p. 11

784 RTP **Old P8 May 2023** Q6 old scheme

Remi limited is a manufacturer of mobile phones in India. Currently the company is dependent on the foreign supplier for import of the battery. It is considering investment of ₹ 55,00,000 in a new machine for manufacturing battery of mobile phones. The expected life of machine is 5 years and has no scrap value. It is expected that 3 lakhs units will be produced and sold each year at a selling price of ₹ 20 per unit. The estimated variable costs and annual fixed costs will be₹ 12 per unit and ₹ 6,00,000 respectively. Consider 14% to be an appropriate cost of capital. Ignore the taxation and depreciation.

CALCULATE the expected net present value of the project.

You are also REQUIRED to measure the sensitivity of the projects NPV to a 10% decrease in the project variables sale price per unit and sales volume and 10% increase in Fixed Cost

ICAI'S ANSWER

Calculation of NPV and sensitivity for different scenarios

	Original Scenario	Sale Price reduced by 10%	Sales Volume reduced by 10%	Fixed Cost increased by 10%
Units	300000	300000	270000	300000
Sale Price p.u.	₹20	₹18	₹20	₹20
VC p.u.	₹12	₹12	₹12	₹12
Contribution p.u.	₹8	₹6	₹8	₹8
Contribution	₹24,00,000	₹18,00,000	₹21,60,000	₹24,00,000
Less: Fixed Cost	₹6,00,000	₹6,00,000	₹6,00,000	₹6,60,000
Profit/Cash Flows	₹18,00,000	₹12,00,000	₹15,60,000	₹17,40,000
PVAF (14%, 5 years)	3.4331	3.4331	3.4331	3.4331
PV of Cash Inflows	₹61,79,546	₹41,19,697	₹53,55,606	₹59,73,561

	Original Scenario	Sale Price reduced by 10%	Sales Volume reduced by 10%	Fixed Cost increased by 10%
Less: Initial Investment	₹ (55,00,000)	₹ (55,00,000)	₹ (55,00,000)	₹ (55,00,000)
NPV	₹6,79,546	₹ (13,80,303)	₹ (1,44,394)	₹4,73,561
% Change in NPV	NA	(303.12%)	(121.25%)	(30.31%)
Sensitivity	NA	30.3121 times	12.1249 times	3.0312 times

Sales Price is the most sensitive variable of all and needs to be given due attention during the life of the project.

PassBook's note: Out of syllabus (old scheme): ICAI places this question under its heading "Risk analysis in Capital Budgeting"; The expected NPV and the sensitivity of NPV to the sale price, volume and fixed cost are Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

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785 RTP **Old P8 Nov 2022** Q6 old scheme

Consider the below mentioned table for the risk premium and the coefficient of variation

Co-efficient of Variation	Risk Premium
0	0
0 to 0.25	2%
0.25 to 0.50	3%
0.50 to 0.75	4%
0.75 to 1	6%

A company is evaluating two projects with an initial investment of ₹ 1,50,000 for each project with cash inflows from them occurring at the end of 5th Year which depends on possible scenarios prevailing during the investment period. The details of the same are as follows:

Scenario	Project X		Project Y	
	Cash Flow (₹)	**Probability**	**Cash Flow (₹)**	**Probability**
Superb	5,00,000	0.20	4,00,000	0.30
Better	3,00,000	0.30	3,50,000	0.20
Moderate	1,50,000	0.15	2,50,000	0.20
Bad	50,000	0.20	75,000	0.20
Worse	10,000	0.15	5,000	0.10

If the ongoing government bond yield is 6%, identify WHICH project to be undertaken.

ICAI'S ANSWER

Calculation of Expected Cash Flow, Standard Deviation & Co-efficient of variation

(a) Project X

Probability (P)	Cash Flows (x)	P.x	P.x²
0.20	5,00,000	1,00,000	50,00,00,00,000
0.30	3,00,000	90,000	27,00,00,00,000
0.15	1,50,000	22,500	3,37,50,00,000
0.20	50,000	10,000	50,00,00,000
0.15	10,000	1,500	1,50,00,000
		2,24,000	80,89,00,00,000

Expected Cash flow = $\sum$ P.x = 2,24,000 = $\overline{X}$

Standard Deviation = $\sqrt{\sum P.x^2 - (\sum P.x)^2}$

= $\sqrt{80,89,00,00,000 - (2,24,000)^2}$

= $\sqrt{30,71,40,00,000}$

σ_x = 1,75,254

Co-efficient of variation = $\frac{\sigma_x}{\overline{X}}$

= $\frac{1,75,254}{2,24,000}$

COV_x = 0.7824

(b) Project Y

Probability (P)	Cash Flows (y)	P.y	P.y²
0.3	4,00,000	1,20,000	48,00,00,00,000
0.2	3,50,000	70,000	24,50,00,00,000
0.2	2,50,000	50,000	12,50,00,00,000
0.2	75,000	15,000	1,12,50,00,000
0.1	5,000	500	25,00,000
		2,55,500	86,12,75,00,000

Expected Cash flow = $\sum$ P.y = $\overline{y}$ = 2,55,500

Standard Deviation = $\sqrt{\sum P.y^2 - (\sum P.y)^2}$

= $\sqrt{86,12,75,00,000 - (2,55,500)^2}$

σ_y = 1,44,386

Co-efficient of variation = $\frac{\sigma_y}{\overline{Y}}$

$= \frac{1,44,386}{2,55,500}$

$COV_Y = 0.5651$

B. Calculation of Risk Adjusted Discount Rate

Project	COV	Risk Premium	RADR
X	0.7824	6%	6% + 6% = 12%
Y	0.5651	4%	6% + 4% = 10%

C. Calculation of NPV

Year	Project X			Project Y		
	Cash Flows	PVF @ 12%	PV	Cash Flows	PVF @ 10%	PV
0	(1,50,000)	1	(1,50,000)	(1,50,000)	1	(1,50,000)
5	2,24,000	0.5674	1,27,098	2,55,500	0.6209	1,58,640
NPV			(22,902)			8,640

∵ NPV of project Y is higher, Project Y should be selected.

PassBook's note: Out of syllabus (old scheme): expected cash flows, standard deviation and coefficient of variation of project cash flows and the risk-adjusted discount rate are Risk Analysis in Capital Budgeting (ICAI prints the question under that chapter heading in this RTP), which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

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786 RTP **Old P8 May 2022** Q6 old scheme

ASG Ltd. is considering a project "Z" with an initial outlay of ₹ 15,00,000 and life of 5 years. The estimates of project are as follows:

	Lower Estimates	Base	Upper Estimates
Sales (units)	9,000	10,000	11,000
	(₹)	**(₹)**	**(₹)**
Selling Price p.u.	175	200	225
Variable cost p.u.	100	125	150
Fixed Cost	1,00,000	1,50,000	2,00,000

Depreciation included in Fixed cost is ₹ 70,000 and corporate tax is 25%.

Assuming the cost of capital as 15%, DETERMINE NPV in three scenarios i.e worst, base and best case scenario.

PV factor for 5 years at 15% are as follows:

Years	1	2	3	4	5
P.V. factor	0.870	0.756	0.658	0.572	0.497

ICAI'S ANSWER

(i) Calculation of Yearly Cash Inflow

In worst case: High costs and Low price (Selling price) and volume (Sales units) are taken.

In best case: Low costs and High price (Selling price) and volume (Sales units) are taken.

	Worst Case	Base	Best Case
Sales (units) (A)	9,000	10,000	11,000
	(₹)	**(₹)**	**(₹)**
Selling Price p.u.	175	200	225
Less: Variable cost p.u.	150	125	100
Contribution p.u. (B)	25	75	125
Total Contribution (A x B)	2,25,000	7,50,000	13,75,000
Less: Fixed Cost	2,00,000	1,50,000	1,00,000
EBT	25,000	6,00,000	12,75,000
Less: Tax @ 25%	62,50	1,50,000	3,18,750
EAT	18,750	4,50,000	9,56,250
Add: Depreciation	70,000	70,000	70,000
Cash Inflow	88,750	5,20,000	10,26,250

(ii) Calculation of NPV in different scenarios

	Worst Case	Base	Best Case
Initial outlay (A) (₹)	15,00,000	15,00,000	15,00,000
Cash Inflow (c) (₹)	88,750	5,20,000	10,26,250
Cumulative PVF @ 15% for 5 years (d)	3.353	3.353	3.353
PV of Cash Inflow (B = c x d) (₹)	2,97,578.75	17,43,560.00	34,41,016.25
NPV (B - A) (₹)	(12,02,421.25)	2,43,560.00	19,41,016.25

PassBook's note: Out of syllabus (old scheme, ours; Lead ruling for old Paper 8): NPV in worst, base and best case scenarios is risk analysis in capital budgeting (ICAI's heading "Risk Analysis in Capital Budgeting"), which the new-scheme Paper 6A syllabus and the January 2027 Study Material leave out.

ICAI · Old P8 · RTP May 2022 · question p. 3-4 · answer p. 15-16

787 RTP Old P8 Dec 2021 Q6 old scheme

TIP Ltd. is considering two mutually exclusive projects M and N. You have been given below the Net Cash flow probability distribution of each project:

Project-M		Project-N	
Net Cash Flow (₹)	**Probability**	**Net Cash Flow (₹)**	**Probability**
62,500	0.30	1,62,500	0.20
75,000	0.30	1,37,500	0.30
87,500	0.40	1,12,500	0.50

(i) REQUIRED:

(a) Expected Net Cash Flow of each project.

(b) Variance of each project.

(c) Standard Deviation of each project.

(d) Coefficient of Variation of each project.

(ii) IDENTIFY which project would you recommend? Give reasons.

ICAI'S ANSWER

(i) (a) Expected Net Cash Flow (ENCF) of Projects

Project M			Project N		
Net Cash Flow (₹)	**Probability**	**Expected Net Cash Flow (₹)**	**Net Cash Flow (₹)**	**Probability**	**Expected Net Cash Flow (₹)**
62,500	0.3	18,750	1,62,500	0.2	32,500
75,000	0.3	22,500	1,37,500	0.3	41,250
87,500	0.4	35,000	1,12,500	0.5	56,250
ENCF		**76,250**			**1,30,000**

(b) Variance of Projects

Project M

$= (62,500 - 76,250)^2\ (0.3) + (75,000 - 76,250)^2\ (0.3) + (87,500 - 76,250)^2\ (0.4)$

$= 5,67,18,750 + 4,68,750 + 5,06,25,000$ **= 10,78,12,500**

Project N

$= (1,62,500 - 1,30,000)^2\ (0.2) + (1,37,500 - 1,30,000)^2\ (0.3) + (1,12,500 - 1,30,000)^2\ (0.5)$

$= 21,12,50,000 + 1,68,75,000 + 15,31,25,000$ **= 38,12,50,000**

(c) Standard Deviation of Projects

Project M

$= \sqrt{\text{Variance}} = \sqrt{10,78,12,500}$ **= 10,383.2798**

Project N

= $\sqrt{\text{Variance}} = \sqrt{38,12,50,000}$ = **19,525.6242**

(d) Coefficient of Variation of Projects

Project	Coefficient of variation $\left(\frac{\text{Standard Deviation}}{\text{Expected Net Cash Flow}}\right)$
M	$\frac{10,383.2798}{76,250}$ = **0.1362 or 13.62%**
N	$\frac{19,525.6242}{1,30,000}$ = **0.1502 or 15.02%**

(ii) Coefficient of Variation of Project M is 0.1362 and Project N is 0.1502. So, the risk per rupee of net cash flow is less of Project M, therefore, Project M is better than Project N.

PassBook's note: Out of syllabus (old scheme): risk analysis in capital budgeting (ICAI's heading "Risk Analysis in Capital Budgeting": the expected net cash flow, variance, standard deviation and coefficient of variation of each project) is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

ICAI · Old P8 · RTP Dec 2021 · question p. 3-4 · answer p. 12-13

788 RTP **Old P8 Jul 2021 Q6** old scheme

X Ltd is considering installation of new machine with the following details:

Sr. No.	Particulars	Figures
1	Initial Investment	₹ 1400 Crore
2	Annual unit sales	100 Crore
3	Selling price per unit	₹ 40
4	Variable cost per unit	₹ 20
5	Annual Fixed costs	₹ 500 Crore
6	Depreciation	₹ 200 Crore
7	Discount Rate	12%
8	Tax rate	30%

Consider Life of the project as 4 years with no salvage value.

Required:

(i) CALCULATE the expected NPV of the project.

(ii) COMPUTE the impact on the project's NPV if change in variables is as under and also compute which variable is having maximum impact on NPV.

Sr. No.	Variable	Figures
1	Unit sold per year	85 Crore
2	Selling price per unit	₹ 39
3	Variable cost per unit	₹ 22
4	Annual Fixed costs	₹ 575 Crore

PV factor at 12% are as follows:

Year	1	2	3	4
PV factor	0.893	0.797	0.712	0.636

ICAI'S ANSWER

(i) Calculation of Net Cash Inflow per year:

	Particulars	Amount (₹)
A	Selling Price Per Unit (A)	40
B	Variable Cost Per Unit (B)	20
C	Contribution Per Unit (C = A - B)	20
D	Number of Units Sold Per Year	100 Crore
E	Total Contribution (E = C × D)	2,000 Crore
F	Annual Fixed costs	500 Crore
G	Depreciation	200 Crore
H	Net Profit before taxes (H = E - F - G)	1,300 Crore
I	Tax @ 30% of H	390 Crore
J	Net Cash Inflow Per Year (J = H - I + G)	910 Crore

Calculation of expected Net Present Value (NPV) of the Project:

Year	Year Cash Flow (₹ in Crore)	PVAF @ 12%	Present Value (PV) (₹ in Crore)
0	(1400.00)	1.000	(1400.00)
1-4	910.00	3.038	2,764.58
		Net Present Value	1,364.58

(ii) Sensitivity Analysis under variable different value:

	Changes in variable	Base	Units sold per year is 85 Crore	Selling price per unit is ₹ 39	Variable cost per unit is ₹ 22	Annual Fixed costs is ₹ 575 Crore
	Particulars	**(₹)**	**(₹)**	**(₹)**	**(₹)**	**(₹)**

	Changes in variable	Base	Units sold per year is 85 Crore	Selling price per unit is ₹ 39	Variable cost per unit is ₹ 22	Annual Fixed costs is ₹ 575 Crore
A	Selling Price Per Unit	40	40	39	40	40
B	Variable Cost Per Unit	20	20	20	22	20
C	Contribution Per Unit (C = A - B)	20	20	19	18	20
D	Number of Units Sold Per Year *(in Crores)*	100	85	100	100	100
E	Total Contribution (E = C × D)	2,000	1,700	1,900	1,800	2,000
F	Annual Fixed Cost *(in Crores)*	500	500	500	500	575
G	Depreciation *(in Crores)*	200	200	200	200	200
H	Net Profit before taxes (H = E - F - G)	1,300	1,000	1,200	1,100	1,225
I	Tax @ 30% of H	390	300	360	330	367.50
J	Net Cash Inflow Per Year (J = H - I + G)	910	700	840	770	857.50
K	(J × 3.038)	2,764.58	2,126.60	2,551.92	2,339.26	2,605.09
L	Initial Cash Flow	1400	1400	1400	1400	1400
M	NPV (K - L)	1,364.58	726.60	1,151.92	939.26	1205.09
N	Change in NPV	-	(637.98)	(212.66)	(425.32)	(159.49)
O	Percentage Change in NPV	-	-46.75%	-15.58%	-31.17%	-11.69%

The above table shows that by varying one variable at a time while keeping the others constant, the impact in percentage terms on the NPV of the project. Thus, it can be seen that the change in units sold per year has the maximum effect on the NPV by 46.75%.

PassBook's note: Out of syllabus (old scheme): ICAI's own heading for this question is "Risk Analysis in Capital Budgeting"; It asks for the expected NPV and a sensitivity analysis of NPV to each variable, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

ICAI · Old P8 · RTP Jul 2021 · question p. 4-5 · answer p. 15-16

789 RTP **Old P8 Nov 2020 Q7** old scheme

A&R Ltd. is considering one of two mutually exclusive proposals, Projects- X and Y, which require cash outlays of ₹ 42,50,000 and ₹ 41,25,000 respectively. The certainty-equivalent (C.E) approach is used in incorporating risk in capital budgeting decisions. The current yield on government bonds is 5.5% and

this is used as the risk-free rate. The expected net cash flows and their certainty equivalents are as follows:

	Project X		Project Y	
Year-end	**Cash Flow (₹)**	**C.E.**	**Cash Flow (₹)**	**C.E.**
1	16,50,000	0.8	16,50,000	0.9
2	15,00,000	0.7	16,50,000	0.8
3	15,00,000	0.5	15,00,000	0.7
4	20,00,000	0.4	10,00,000	0.8
5	21,00,000	0.6	8,00,000	0.9

The Present value (PV) factor @ 5.5% is as follows:

Year	0	1	2	3	4	5
PV factor	1	0.947	0.898	0.851	0.807	0.765

Required:

DETERMINE the project that should be accepted?

ICAI'S ANSWER

Calculation of Net Present Value (NPV) of Project X

Year end	Cash Flow (₹) (a)	C.E. (b)	Adjusted Cash flow (₹) (c) = (a) × (b)	Present value factor at 5.5% (d)	Total Present value (₹) (e) = (c) × (d)
1	16,50,000	0.8	13,20,000	0.947	12,50,040
2	15,00,000	0.7	10,50,000	0.898	9,42,900
3	15,00,000	0.5	7,50,000	0.851	6,38,250
4	20,00,000	0.4	8,00,000	0.807	6,45,600
5	21,00,000	0.6	12,60,000	0.765	9,63,900
PV of total cash inflows					44,40,690
Less: Initial Investment					(42,50,000)
Net Present Value					**1,90,690**

Calculation of Net Present Value (NPV) of Project Y

Year end	Cash Flow (₹) (a)	C.E. (b)	Adjusted Cash flow (₹) (c) = (a) × (b)	Present value factor at 5.5% (d)	Total Present value (₹) (e) = (c) × (d)
1	16,50,000	0.9	14,85,000	0.947	14,06,295
2	16,50,000	0.8	13,20,000	0.898	11,85,360
3	15,00,000	0.7	10,50,000	0.851	8,93,550
4	10,00,000	0.8	8,00,000	0.807	6,45,600
5	8,00,000	0.9	7,20,000	0.765	5,50,800
PV of total cash inflows					46,81,605
Less: Initial Investment					(41,25,000)
Net Present Value					**5,56,605**

Project Y has NPV of ₹ 5,56,605/- which is higher than the NPV of Project X. Thus, A&R Ltd. should accept Project Y.

PassBook's note: Out of syllabus (old scheme): the certainty-equivalent approach to risk in capital budgeting is Risk Analysis in Capital Budgeting (ICAI prints the question under that chapter heading in this RTP), which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · RTP Nov 2020 · question p. 4 · answer p. 14

790 RTP **Old P8 May 2020** Q7 old scheme

G Ltd. using certainty-equivalent approach in the evaluation of risky proposals. The following information regarding a new project is as follows:

Year	Expected Cash flow	Certainty-equivalent quotient
0	(8,00,000)	1.0
1	6,40,000	0.8
2	5,60,000	0.7
3	5,20,000	0.6
4	4,80,000	0.4
5	3,20,000	0.3

Riskless rate of interest on the government securities is 6 per cent. DETERMINE whether the project should be accepted?

ICAI'S ANSWER

Determination of Net Present Value (NPV)

Year	Expected Cash flow (₹)	Certainty-equivalent (CE)	Adjusted Cash flow (Cash flow × CE) (₹)	PV factor (at 0.06)	Total PV (₹)
0	(8,00,000)	1.0	(8,00,000)	1.000	(8,00,000)
1	6,40,000	0.8	5,12,000	0.943	4,82,816
2	5,60,000	0.7	3,92,000	0.890	3,48,880
3	5,20,000	0.6	3,12,000	0.840	2,62,080
4	4,80,000	0.4	1,92,000	0.792	1,52,064
5	3,20,000	0.3	96,000	0.747	71,712
	NPV = (13,17,552 – 8,00,000)				5,17,552

As the Net Present Value is positive the project should be accepted.

PassBook's note: Out of syllabus (old scheme): the certainty-equivalent approach is Risk Analysis in Capital Budgeting (ICAI's heading for this question), which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

ICAI · Old P8 · RTP May 2020 · question p. 4-5 · answer p. 11-12

791 RTP **Old P8 Nov 2019** Q6 old scheme

SL Ltd. has invested ₹1,000 lakhs in a project. The risk-free rate of return is 5%. Risk premium expected by the Management is 10%. The life of the project is 5 years. Following are the cash flows that are estimated over the life of the project.

Year	Cash flows (₹ in lakhs)
1	125
2	300
3	375
4	400
5	325

CALCULATE Net Present Value of the project based on Risk free rate and also on the basis of Risks adjusted discount rate.

ICAI'S ANSWER

The Present Value of the Cash Flows for all the years by discounting the cash flow at 5% is calculated

as below:

Year	Cash flows ₹ in lakhs	Discounting Factor @5%	Present value of Cash Flows ₹ In Lakhs
1	125	0.952	119.00
2	300	0.907	272.10
3	375	0.863	323.62
4	400	0.822	328.80
5	325	0.783	254.47
	Total of present value of Cash flow		1,297.99
	Less: Initial investment		1,000.00
	Net Present Value (NPV)		297.99

Now when the risk-free rate is 5% and the risk premium expected by the Management is 10%. So the risk adjusted discount rate is 5% + 10% =15%.

Discounting the above cash flows using the Risk Adjusted Discount Rate would be as below:

Year	Cash flows ₹ in Lakhs	Discounting Factor@15%	Present Value of Cash Flows ₹in lakhs
1	125	0.869	108.62
2	300	0.756	226.80
3	375	0.657	246.37
4	400	0.571	228.40
5	325	0.497	161.52
	Total of present value of Cash flow		971.71
	Initial investment		1,000.00
	Net present value (NPV)		(28.29)

PassBook's note: Out of syllabus (old scheme, ours; It asks for the NPV at the risk-free rate and at the risk-adjusted discount rate, which is risk analysis in capital budgeting, not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

ICAI · Old P8 · RTP Nov 2019 · question p. 3-4 · answer p. 10-11

792 MTP **Old P8 Nov 2023 S-I** Q1(c) 5 marks old scheme

A new project "Ambar" requires an initial outlay of ₹ 4,50,000. The company uses certainty equivalent

method approach to evaluate the project. The risk-free rate is 7%. Following information is available:

Year	Cash Flow After Tax (₹)	Certainty Equivalent Coefficient
1	1,50,000	0.90
2	2,25,000	0.80
3	1,75,000	0.58
4	1,50,000	0.56
5	70,000	0.50

PV Factor at 7%

Year	1	2	3	4	5
PV Factor	0.935	0.873	0.816	0.763	0.713

Is investment in the project beneficial based on above information?

ICAI'S ANSWER

Calculation of Net Present Value of the Project

Year	Cash Inflows After Tax (in ₹)	C.E.	Adjusted Cash Inflows (in ₹)	Present Value Factor	Present Value (in ₹)
1	1,50,000	0.90	1,35,000	0.935	1,26,225
2	2,25,000	0.80	1,80,000	0.873	1,57,140
3	1,75,000	0.58	1,01,500	0.816	82,824
4	1,50,000	0.56	84,000	0.763	64,092
5	70,000	0.50	35,000	0.713	24,955
Total Present Value of Cash Inflows					**4,55,236**
Less: Initial Investment or Cash Outflow required for "Ambar"					(4,50,000)
Net Present Value					**5,236**

Conclusion: As the Net Present Value of the project after considering the Certainty Equivalent factors is still positive, it may be advised to invest in project "Ambar".

PassBook's note: Out of syllabus (old scheme): the item evaluates a project by the certainty equivalent method, which is Risk Analysis in Capital Budgeting. That topic is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter teaches payback, ARR, NPV, PI, IRR, MIRR, discounted payback, capital rationing and equivalent annual cost, with no risk analysis; The Study Material has no certainty equivalent). The tag names the nearest sub-topic: the NPV the answer computes.

793 MTP **Old P8 Nov 2023 S-II** Q4 10 marks old scheme

A&R Ltd. has undertaken a project which has an initial investment of ₹2,000 lakhs in plant & machinery and ₹ 600 lakhs for working capital. The plant & machinery would have a salvage value of ₹ 525 lakhs at the end of the fifth year. The plant & machinery would depreciate at the rate of 20% p.a. on the WDV method. The other details of the project for the five-year period are as follows:

Sales	10,00,000 units p.a.
Selling price per unit	₹500
Variable cost	50% of selling price
Fixed overheads (excluding depreciation)	₹400 lakh p.a.
Corporate tax rate	30%
Rate of interest on bank loan	12%
After tax required rate of return	15%

Required:

(i) CALCULATE net present value (NPV) of the project and DETERMINE the viability of the project.

(ii) DETERMINE the sensitivity of project's NPV under each of the following condition:

a. Decrease in selling price by 10%.

b. Increase in cost of plant & machinery by 10%.

PV factor	Year-1	Year-2	Year-3	Year-4	Year-5
12%	0.892	0.797	0.711	0.635	0.567
15%	0.869	0.756	0.657	0.571	0.497

ICAI'S ANSWER

(i) Calculation of Net Present Value (NPV):

	Year-1	Year-2	Year-3	Year-4	Year-5
Sales volume (Qty. in lakh)	10	10	10	10	10
Contribution per unit (₹) (Selling price – variable cost)	250	250	250	250	250
Total contribution (₹in lakh)	2,500	2,500	2,500	2,500	2,500
Less: Fixed overheads (₹ In lakh)	400	400	400	400	400
PBDT	2,100	2,100	2,100	2,100	2,100
Less: Depreciation (₹ in lakh) (Working note 1)	400	320	256	204.8	163.84

	Year-1	Year-2	Year-3	Year-4	Year-5
PBT	1,700	1,780	1,844	1,895.2	1936.16
Less: Tax @ 30%	510	534	553.2	568.56	580.85
PAT	1,190	1,246	1,290.8	1,326.64	1,355.31
Add: Depreciation	400	320	256	204.8	163.84
Add: Salvage value of plant & machinery	-	-	-	-	525
Add: Working capital	-	-	-	-	600
Net Cash inflow	1,590	1,566	1,546.8	1,531.44	2,644.15
P.V factor @15%	0.869	0.756	0.657	0.571	0.497
P.V of cash inflows	1,381.71	1,183.90	1,016.24	874.45	1,314.14

Net Present Value = P.V of cash inflows – P.V of cash outflows

= ₹ (1383.71+1183.9+1016.24+874.45+1314.14) – (₹2,000 + ₹ 600)

= 5772.44 – 2600 = ₹3172.44 lakh

The NPV of the project is positive, hence, the project is viable.

Working note 1:

	Year-1 (₹ in lakhs)	Year-2 (₹ in lakhs)	Year-3 (₹ in lakhs)	Year-4 (₹ in lakhs)	Year-5 (₹ in lakhs)
Opening balance	2,000	1,600	1,280	1024	819.20
Depreciation @20%	400	320	256	204.8	163.84
Closing WDV	1,600	1,280	1024	819.2	655.36

(ii) Determination of sensitivity of NPV w.r.t.

a. Decrease in selling price by 10%

	Year-1	Year-2	Year-3	Year-4	Year-5
Sales volume (Qty. in lakh)	10	10	10	10	10
New Selling price	450	450	450	450	450
Variable cost	250	250	250	250	250
Contribution per unit (₹) (Selling price – variable cost)	200	200	200	200	200
Total contribution (₹in lakh)	2,000	2,000	2,000	2,000	2,000
Less: Fixed overheads (₹ In lakh)	400	400	400	400	400

	Year-1	Year-2	Year-3	Year-4	Year-5
PBDT	1,600	1,600	1,600	1,600	1,600
Less: Depreciation (₹ in lakh) (Working note 1)	400	320	256	204.8	163.84
PBT	1,200	1,280	1,344	1,395.2	1,436.16
Less: Tax @ 30%	360	384	403.2	418.56	430.85
PAT	840	896	940.8	976.64	1,005.31
Add: Depreciation	400	320	256	204.8	163.84
Add: Salvage value of plant & machinery	-	-	-	-	525
Add: Working capital	-	-	-	-	600
Net Cash inflow	1,240	1,216	1,196.8	1,181.44	2,294.15
P.V factor @15%	0.869	0.756	0.657	0.571	0.497
P.V of cash inflows	1,077.56	919.30	786.30	674.60	1,140.19

NPV = ₹ (1,077.56+919.30+786.30+674.60+1,140.19) – (₹ 2,000 + ₹ 600)

= ₹ 4,597.95 – ₹ 2,600 = ₹1,997.95 lakh

10% reduction in selling price reduces the NPV by 37.02% (3172.44-1997.95/3172.44)

b. Increase in project cost by 10%

	Year-1	Year-2	Year-3	Year-4	Year-5
PBDT	2,100	2,100	2,100	2,100	2,100
Less: Depreciation (₹ in lakh) (Working note 2)	440	352	281.6	225.28	180.22
PBT	1660	1748	1818.4	1874.72	1919.78
Less: Tax @ 30%	498	524.40	545.52	562.42	575.93
PAT	1162	1223.6	1272.88	1312.30	1343.85
Add: Depreciation	440	352	281.60	225.28	180.22
Add: Salvage value of plant & machinery	-	-	-	-	525
Add: Working capital	-	-	-	-	600
Net Cash inflow	1,602	1,575.6	1,554.48	1,537.58	2,649.07
P.V factor @15%	0.869	0.756	0.657	0.571	0.497
P.V of cash inflows	1,392.14	1,191.15	1,021.29	877.96	1316.59

NPV = ₹ (1,392.14+1,191.15+1,021.29+877.96+1,316.59) – (₹ 2,200 + ₹ 600)

= ₹ 5,799.13 – ₹ 2,800 = ₹2,999.13 lakh

10% increase in project cost reduces the NPV only by 5.46% (3,172.44 - 2,999.13/3172.44)

Working note 2:

	Year-1	Year-2	Year-3	Year-4	Year-5
Opening balance	2,200	1,760	1408	1126.4	901.12
Depreciation @20%	440	352	281.6	225.28	180.22
Closing WDV	1,760	1408	1126.40	901.12	702.90

PassBook's note: Out of syllabus (old scheme): requirement (ii), the sensitivity of the NPV to a 10% fall in selling price and a 10% rise in the cost of plant, is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (as o-p8-pyq-nov2023 q5a). Requirement (i), the NPV decision, is in syllabus: the tags name it. ICAI prints one [10 Marks] for both requirements, so the item is not split. ICAI's own figures kept as printed: "1383.71" in the NPV sum where the table prints 1,381.71.

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794 MTP **Old P8 Nov 2023 S-II** Q6(b) 4 marks old scheme

EXPLAIN the concept of risk adjusted discount rate.

ICAI'S ANSWER

Risk Adjusted Discount Rate: The use of risk adjusted discount rate (RADR) is based on the concept that investors demands higher returns from the risky projects. The required rate of return on any investment should include compensation for delaying consumption plus compensation for inflation equal to risk free rate of return, plus compensation for any kind of risk taken. If the risk associated with any investment project is higher than risk involved in a similar kind of project, discount rate is adjusted upward in order to compensate this additional risk borne.

A risk adjusted discount rate is a **sum of risk free rate and risk premium.** The Risk Premium depends on the perception of risk by the investor of a particular investment and risk aversion of the Investor.

So Risks adjusted discount rate = Risk free rate+ Risk premium

Risk Free Rate: It is the rate of return on Investments that bear no risk. For e.g., Government securities yield a return of 6 % and bear no risk. In such case, 6 % is the risk-free rate.

Risk Premium: It is the rate of return over and above the risk-free rate, expected by the Investors as a reward for bearing extra risk. For high risk project, the risk premium will be high and for low risk projects, the risk premium would be lower.

PassBook's note: Out of syllabus (old scheme): the risk adjusted discount rate is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

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795 MTP **Old P8 May 2023 S-I** Q5 10 marks old scheme

BIG BLOCK Ltd. is considering two mutually exclusive projects X and Y. You have been given below the

Net Cash flow probability distribution of each project:

Project- X		Project – Y	
Net Cash Flow (₹)	**Probability**	**Net Cash Flow (₹)**	**Probability**
3,12,500	0.2	9,75,000	0.1
3,75,000	0.2	8,25,000	0.3
4,37,500	0.6	6,75,000	0.6

(i) REQUIRED:

(a) Expected Net Cash Flow of each project.

(b) Variance of each project.

(c) Standard Deviation of each project.

(d) Coefficient of Variation of each project.

(ii) IDENTIFY which project would you recommend? Give reasons.

ICAI'S ANSWER

(i) (a) Expected Net Cash Flow (ENCF) of Projects

Project X			Project Y		
Net Cash Flow (₹)	**Probability**	**Expected Net Cash Flow (₹)**	**Net Cash Flow (₹)**	**Probability**	**Expected Net Cash Flow (₹)**
3,12,500	0.2	62,500	9,75,000	0.1	97,500
3,75,000	0.2	75,000	8,25,000	0.3	2,47,500
4,37,500	0.6	2,62,500	6,75,000	0.6	4,05,000
		4,00,000			**7,50,000**

(b) Variance of Projects

Project X

$= (₹3,12,500 - ₹4,00,000)^2\ (0.2) + (₹3,75,000 - ₹4,00,000)^2\ (0.2) + (₹4,37,500 - ₹4,00,000)^2\ (0.6)$

$= ₹1,53,12,50,000 + ₹12,50,00,000 + ₹84,37,50,000$

$= ₹2,50,00,00,000$

Project Y

$= (₹9,75,000 - ₹7,50,000)^2\ (0.1) + ₹\ (8,25,000 - ₹7,50,000)^2\ (0.3) + (₹6,75,000 - ₹7,50,000)^2\ (0.6)$

$= ₹5,06,25,00,000 + ₹1,68,75,00,000 + ₹3,37,50,00,000$

$= ₹10,12,50,00,000$

(c) Standard Deviation of Projects

Project X

$= \sqrt{\text{Variance}}$

$= \sqrt{2,50,00,00,000} = ₹\ 50,000$

Project Y

$= \sqrt{\text{Variance}}$

$= \sqrt{10,12,50,00,000}$

= ₹10,0623.06

(d) Coefficient of Variation of Projects

Project	Coefficient of variation = $\frac{\text{Standard Deviation}}{\text{Expected Net Cash Flow}}$
X	$\frac{50,000}{4,00,000}$ **= 0.125 or 12.5%**
Y	$\frac{1,00,623.06}{7,50,000}$ **= 0.1342 or 13.42%**

(ii) Coefficient of Variation of Project X is 0.125 and Project Y is 0.1342. So, the risk per rupee of net cash flow is less of Project X, therefore, Project X is better than Project Y.

PassBook's note: Out of syllabus (old scheme): the expected value, variance, standard deviation and coefficient of variation of a project's cash flows (risk analysis in capital budgeting) are not in the current Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

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796 MTP **Old P8 May 2023 S-II** Q2 10 marks old scheme

Based on the details given below, you are required to UNDERTAKE NPV sensitivity analysis for the factors of Sales Price, Sales units, Variable cost, Fixed Cost and Initial Investment for a 10% adverse variation in each of the mentioned factors Assume straight line depreciation is allowed for tax purpose

Initial Investment	₹ 15,00,000
Tenure of Project (Years)	4
Sales Units	6000
Sales Price p.u.	₹ 250
Variable Cost p.u.	₹ 80
Fixed Cost	₹ 2,00,000
Tax Rate	30%
Salvage Value	₹ 1,50,000
Cost of Capital	14%

Use the below PV factors for calculation purpose

Year	0	1	2	3	4
PVF @ 14%	1.00	0.8772	0.7695	0.6750	0.5921

ICAI'S ANSWER

Computation of CFAT (year 1 to year 4)

Sr. No.	Particulars	Original Case	Sales Units reduced by 10%	SP reduced by 10%	Variable Cost increased by 10%
A	Sale Price p.u.	₹ 250	₹ 250	₹ 225	₹ 250
B	Sale units	6000	5400	6000	6000
C	Sales (A x B)	₹ 15,00,000	₹ 13,50,000	₹ 13,50,000	₹ 15,00,000
D	Variable Cost p.u.	₹ 80	₹ 80	₹ 80	₹ 88
E	Variable Cost (B x D)	₹ 4,80,000	₹ 4,32,000	₹ 4,80,000	₹ 5,28,000
F	Contribution (C - E)	₹ 10,20,000	₹ 9,18,000	₹ 8,70,000	₹ 9,72,000
G	Less: Fixed Cost	₹ -2,00,000	₹ -2,00,000	₹ -2,00,000	₹ -2,00,000
H	PBDT (F-G)	₹ 8,20,000	₹ 7,18,000	₹ 6,70,000	₹ 7,72,000
I	Less: Depreciation				
	(1500000-150000)/4	₹ -3,37,500	₹ -3,37,500	₹ -3,37,500	₹ -3,37,500
	(1600000-150000)/4	-	-	-	-
J	PBT	₹ 4,82,500	₹ 3,80,500	₹ 3,32,500	₹ 4,34,500
K	Less: Taxes @ 30%	₹ -1,44,750	₹ -1,14,150	₹ -99,750	₹ -1,30,350
L	PAT	₹ 3,37,750	₹ 2,66,350	₹ 2,32,750	₹ 3,04,150
M	Add: Depreciation	₹ 3,37,500	₹ 3,37,500	₹ 3,37,500	₹ 3,37,500
N	CFAT	₹ 6,75,250	₹ 6,03,850	₹ 5,70,250	₹ 6,41,650

Sr. No.	Particulars	Fixed Cost increased by 10%	Initial Investment increased by 10%
A	Sale Price p.u.	₹ 250	₹ 250
B	Sale units	6000	6000
C	Sales (A x B)	₹ 15,00,000	₹ 15,00,000
D	Variable Cost p.u.	₹ 80	₹ 80
E	Variable Cost (B x D)	₹ 4,80,000	₹ 4,80,000

Sr. No.	Particulars	Fixed Cost increased by 10%	Initial Investment increased by 10%
F	Contribution (C - E)	₹ 10,20,000	₹ 10,20,000
G	Less: Fixed Cost	₹ -2,20,000	₹ -2,00,000
H	PBDT (F-G)	₹ 8,00,000	₹ 8,20,000
I	Less: Depreciation		
	(1500000-150000)/4	₹ -3,37,500	-
	(1600000-150000)/4	-	₹ -3,62,500
J	PBT	₹ 4,62,500	₹ 4,57,500
K	Less: Taxes @ 30%	₹ -1,38,750	₹ -1,37,250
L	PAT	₹ 3,23,750	₹ 3,20,250
M	Add: Depreciation	₹ 3,37,500	₹ 3,62,500
N	CFAT	₹ 6,61,250	₹ 6,82,750

Calculation of NPV and sensitivity analysis

Sr. No.	Particulars	Original Case	Sales Units reduced by 10%	SP reduced by 10%	Variable Cost increased by 10%
I	CFAT	₹ 6,75,250	₹ 6,03,850	₹ 5,70,250	₹ 6,41,650
II	PVAF @ 14%	2.9138	2.9138	2.9138	2.9138
III	PV of CFATs (I x II)	₹ 19,67,543	₹ 17,59,498	₹ 16,61,594	₹ 18,69,640
IV	Salvage	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000
V	PVF @ 14%	0.5921	0.5921	0.5921	0.5921
VI	PV of Salvage (IV x V)	₹ 88,815	₹ 88,815	₹ 88,815	₹ 88,815
VII	Initial Investment	₹ 15,00,000	₹ 15,00,000	₹ 15,00,000	₹ 15,00,000
VIII	NPV (III+VI-VII)	₹ 5,56,358	₹ 3,48,313	₹ 2,50,409	₹ 4,58,455
IX	Sensitivity (Case NPV-Original NPV)/ Original NPV		-37.39%	-54.99%	-17.60%
				Most sensitive	

Sr. No.	Particulars	Fixed Cost increased by 10%	Initial Investment increased by 10%
I	CFAT	₹ 6,61,250	₹ 6,82,750
II	PVAF @ 14%	2.9138	2.9138
III	PV of CFATs (I x II)	₹ 19,26,750	₹ 19,89,397
IV	Salvage	₹ 1,50,000	₹ 1,50,000
V	PVF @ 14%	0.5921	0.5921
VI	PV of Salvage (IV x V)	₹ 88,815	₹ 88,815
VII	Initial Investment	₹ 15,00,000	₹ 16,50,000
VIII	NPV (III+VI-VII)	₹ 5,15,565	₹ 4,28,212
IX	Sensitivity Sensitivity (Case NPV-Original NPV)/ Original NPV	-7.33%	-23.03%
		Least sensitive	

PassBook's note: Out of syllabus (old scheme): NPV sensitivity analysis (risk analysis in capital budgeting) is not in the current Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

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797 MTP **Old P8 Nov 2022 S-I** Q4 10 marks old scheme

A Ltd. is considering investing in new project with the following details:

Initial capital cost	₹ 100 Crores
Annual unit sales	1.25 Crores
Selling price	₹ 100 per unit
Variable cost	₹ 50 per unit
Fixed cost	₹ 12.50 Crores per year
Discounting Rate	6%

Considering life of the project as 3 years, you are required to:

(a) CALCULATE the NPV of the project.

(b) COMPUTE the impact on the project's NPV considering a 5% adverse variance in following variables:

(i) Selling Price per Unit

(ii) Variable Cost Per Unit

(iii) Fixed Cost Per Unit

WHICH variable is having maximum effect?

(c) MEASURE the maximum sensitivity of the project to change in the variable (as found in part (b) above) such that NPV becomes zero.

PV factors @ 6% are:

Year	1	2	3	4	5
PV Factor	0.943	0.890	0.840	0.792	0.747

ICAI'S ANSWER

(a) Calculation of Net Cash Inflow per year

	Particulars	Amount (₹)
A	Selling price per unit	100
B	Variable cost per unit	50
C	Contribution per unit (A - B)	50
D	Number of units sold per year	1.25 Crores
E	Total Contribution (C × D)	₹ 62.50 Crores
F	Fixed cost per year	₹ 12.50 Crores
G	Net cash inflow per year (E - F)	₹ 50 Crores

Calculation of Net Present Value (NPV) of the Project

Year	Year Cash Flow (₹ in Cr.)	PV factor @ 6%	Present Value (PV) (₹ in Cr.)
0	(100.00)	1.000	(100.00)
1	50.00	0.943	47.15
2	50.00	0.890	44.50
3	50.00	0.840	42.00
Net Present Value			**33.65**

Here, NPV represent the most likely outcomes and not the actual outcomes. The actual outcome can be lower or higher than the expected outcome.

(b) Sensitivity Analysis considering 5 % Adverse Variance in following variable

	Particulars	Base	Selling Price per Unit Reduced to ₹ 95	Variable Cost Per Unit increased to ₹ 52.50	Fixed Cost increased to ₹ 13.125 crores per year
		(₹)	**(₹)**	**(₹)**	**(₹)**
A	Selling price per unit	100	95	100	100
B	Variable cost per unit	50	50	52.50	50
C	Contribution per unit (A - B)	50	45	47.50	50
		(₹ in Cr.)	**(₹ in Cr.)**	**(₹ in Cr.)**	**(₹ in Cr.)**

	Particulars	Base	Selling Price per Unit Reduced to ₹ 95	Variable Cost Per Unit increased to ₹ 52.50	Fixed Cost increased to ₹ 13.125 crores per year
D	Number of units sold per year (units in Crores)	1.25	1.25	1.25	1.25
E	Total Contribution (C × D)	62.50	56.25	59.375	62.50
F	Fixed cost per year	12.50	12.50	12.50	13.125
G	Net Cash Inflow per year (E - F)	50.00	43.75	46.875	49.375
H	PV of Net cash Inflow per year (G × 2.673)	133.65	116.94	125.30	131.98
I	Initial capital cost	100.00	100	100	100
J	NPV (H - I)	33.65	16.94	25.30	31.98
	Difference in NPV	-	(16.71)	(8.35)	(1.67)
	Percentage Change in NPV	**-**	**(49.66%)**	**(24.81%)**	**(4.96%)**

The above table shows that by changing one variable at a time by 5% (adverse) while keeping the others constant, the impact in percentage terms on the NPV of the project is maximum in selling price by 49.66%.

(c) For 49.66% change in NPV, Selling Price per Unit needs to be reduced by 5%

Thus, for 100% change in NPV (such that NPV becomes zero), **sensitivity to change in Selling price would be** = $\frac{5}{49.66} \times 100 = 10.07\%$

PassBook's note: Out of syllabus (old scheme): requirements (b) and (c), the impact on the NPV of a 5% adverse variance in selling price, variable cost and fixed cost, and the change in the most sensitive variable that makes the NPV zero, are sensitivity analysis (Risk Analysis in Capital Budgeting), which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section). Requirement (a), the NPV, is in syllabus: the tag names it. ICAI prints one [10 Marks] for the three requirements and answers them on the same data, so the item is not split. ICAI's answer reads "NPV represent the most likely outcomes".

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798 MTP **Old P8 Nov 2022 S-II** Q4 10 marks old scheme

The initial investment outlay for a capital investment project consists of ₹ 150 lacs for Plant & machinery and ₹ 60 lacs for Working Capital. Other details are summarized below:

Sales – 1.5 lakh units of output per year for years 1 to 5

Selling Price - 120 per unit of output

Variable cost - 60 per unit of output

Fixed overheads (excluding depreciation) - 22.5 lakhs per year for years 1 to 5

Salvage value of Plant & machinery Equal to the Written Down Value (WDV) at the end of year 5

Tax rate 40%

Time horizon 5 years

Post tax cut off rate 12%. PVAF (12%,5 Years) is 3.6048.PV @12% 5th year is 0.5674.

Rate of depreciation 20% on the Straight-Line Method (SLM)

Required:

(a) INDICATE the financial viability of the project by calculating the NPV

(b) DETERMINE the sensitivity of the project's NPV under each of the following conditions:

(i) Decrease in selling price by 10%

(ii) Increase in variable cost by 10%

(iii) Increase in cost of Plant & machinery by 10%

ICAI'S ANSWER

(a) Calculation of Cash Flow After Tax (CFAT) in original scenario

Sr. No.	Particulars	
1	Sales units	1,50,000
		(₹)
2	Sale Price p.u.	120
3	Sales	1,80,00,000
4	Variable Cost p.u.	60
5	Variable Cost	90,00,000
6	Contribution (3-4)	90,00,000
7	Fixed OH (Excluding Depreciation)	22,50,000
8	Depreciation	30,00,000
9	EBIT or PBT (6-7-8)	37,50,000
10	Tax@40%	15,00,000
11	Profit After Tax (PAT)	22,50,000
12	Add: Depreciation	30,00,000
13	CFAT	52,50,000

Calculation of NPV in original Scenario

Years	Particulars	Cash Flows	PVF	PV
0	Initial Investment	(1,50,00,000)	1	(1,50,00,000)
0	Initial WC introduced	(60,00,000)	1	(60,00,000)
1 to 5	CFAT	52,50,000	3.6048	189,25,075
5	WC released	60,00,000	0.5674	34,04,561
	NPV			13,29,636

Since the NPV of the project is Positive the project is viable.

(b) Sensitivity Analysis of the project under various conditions

Sr. No.	Particulars	(i)Selling Price reduced by 10%	(ii)Variable Cost increased by 10%	(iii)Plant &Machinery cost increased by 10%
1	Sales units	1,50,000	1,50,000	1,50,000
		₹	₹	₹
2	Sale Price p.u.	108	120	120
3	Sales	162,00,000	180,00,000	180,00,000
4	Variable Cost p.u.	60	66	60
5	Variable Cost	90,00,000	99,00,000	90,00,000
6	Contribution (3-4) [or Contribution per unit x1,50,000 units)	72,00,000	81,00,000	90,00,000
7	Fixed OH (Excluding Depreciation)	22,50,000	22,50,000	22,50,000
8	Depreciation	30,00,000	30,00,000	33,00,000
9	EBIT or PBT (6-7-8)	19,50,000	28,50,000	34,50,000
10	Tax	7,80,000	11,40,000	13,80,000
11	PAT	11,70,000	17,10,000	20,70,000
12	Add: Depreciation	30,00,000	30,00,000	33,00,000
13	CFAT	41,70,000	47,10,000	53,70,000
14	PV of CFAT (CFAT x3.6048)	150,31,917	169,78,496	193,57,648
15	WC released (60,00,000 x0.5674)	34,04,561	34,04,561	34,04,561

Sr. No.	Particulars	(i)Selling Price reduced by 10%	(ii)Variable Cost increased by 10%	(iii)Plant &Machinery cost increased by 10%
16	Initial Investment	(1,50,00,000)	(1,50,00,000)	(1,50,00,000)
17	Initial WC introduced	(60,00,000)	(60,00,000)	(60,00,000)
18	NPV (14+15-16-17)	(25,63,522)	(6,16,943)	2,62,209
19	% Change in NPV [Based on original NPV (13,29,636)]	-292.80%	-146.40%	-80.28%

From the above calculations it can be seen that change in selling price is most sensitive and has the maximum effect on the NPV.

PassBook's note: Out of syllabus (old scheme): requirement (b), the sensitivity of the NPV to a 10% fall in selling price, a 10% rise in variable cost and a 10% rise in the cost of plant, is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (as o-p8-mtp-nov2023-s2 q4 and o-p8-pyq-nov2023 q5a). Requirement (a), the NPV decision, is in syllabus: the tags name it. ICAI prints one [10 Marks] for both requirements, so the item is not split.

ICAI · Old P8 · MTP Nov 2022 · Series II · question p. 3 · answer p. 7-8

799 MTP **Old P8 May 2022 S-I** Q5(a) 8 marks old scheme

X Ltd. is considering two mutually exclusive projects A and B.

Net Cash flow probability distribution of each project has been given below:

Project-A		Project-B	
Net Cash Flow (₹)	**Probability**	**Net Cash Flow (₹)**	**Probability**
1,72,000	0.30	3,38,000	0.20
1,82,000	0.30	3,18,000	0.30
1,92,000	0.40	2,98,000	0.50

(i) COMPUTE the following:

(I) Expected Net Cash Flow of each project.

(II) Variance of each project.

(III) Standard Deviation of each project.

(IV) Coefficient of Variation of each project.

(ii) IDENTIFY which project do you recommend? Give reason.

ICAI'S ANSWER

(i) (I) Calculation of Expected Net Cash Flow (ENCF) of Project A and Project B

Project A			Project B		
Net Cash Flow (₹)	**Probability**	**Expected Net Cash Flow (₹)**	**Net Cash Flow (₹)**	**Probability**	**Expected Net Cash Flow (₹)**
1,72,000	0.30	51,600	3,38,000	0.20	67,600
1,82,000	0.30	54,600	3,18,000	0.30	95,400
1,92,000	0.40	76,800	2,98,000	0.50	1,49,000
ENCF		**1,83,000**			**3,12,000**

(II) Variance of Projects

Project A

Variance (σ^2) = $(1,72,000 - 1,83,000)^2 \times (0.3) + (1,82,000 - 1,83,000)^2 \times (0.3) + (1,92,000 - 1,83,000)^2 \times (0.4)$

= 3,63,00,000 + 3,00,000 + 3,24,00,000 **= 6,90,00,000**

Project B

Variance(σ^2) = $(3,38,000 - 3,12,000)^2 \times (0.2) + (3,18,000 - 3,12,000)^2 \times (0.3) + (2,98,000 - 3,12,000)^2 \times (0.5)$

= 13,52,00,000 + 1,08,00,000 + 9,80,00,000 **= 24,40,00,000**

(III) Standard Deviation of Projects

Project A

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{6,90,00,000}$ **= 8,306.624**

Project B

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{24,40,00,000}$ **= 15,620.499**

(IV) Coefficient of Variation of Projects

Projects	Coefficient of variation $\left(\frac{\text{Standard Deviation}}{\text{Expected Net Cash Flow}}\right)$	Risk	Expected Net Cash Flow
A	$\frac{8,306.624}{1,83,000}$ **= 0.045 or 4.5%**	Less	Less
B	$\frac{15,620.499}{3,12,000}$ **= 0.050 or 5.0%**	More	More

(ii) In project A risk per rupee of cash flow is 0.045 (approx.) while in project B it is 0.050 (approx.). Therefore, Project A is better than Project B.

PassBook's note: Out of syllabus (old scheme): the expected net cash flow, variance, standard deviation and coefficient of variation of a project's cash flows (probability distributions) are Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · MTP May 2022 · Series I · question p. 3 · answer p. 6

800 MTP **Old P8 May 2022 S-I** Q5(b) 2 marks old scheme

DISTINGUISH between Sensitivity Analysis & Scenario Analysis.

ICAI'S ANSWER

Sensitivity Analysis Vs. Scenario Analysis: Sensitivity analysis and Scenario analysis both help to understand the impact of the change in input variable on the outcome of the project. However, there are certain basic differences between the two.

Sensitivity analysis calculates the impact of the change of a **single** input variable on the outcome of the project viz., NPV or IRR. The sensitivity analysis thus enables to identify that single critical variable which can impact the outcome in a huge way and the range of outcomes of the project given the change in the input variable.

Scenario analysis, on the other hand, is based on a **scenario**. The scenario may be recession or a boom wherein depending on the scenario, all input variables change. Scenario Analysis calculates the outcome of the project considering this scenario where the variables have changed simultaneously. Similarly, the outcome of the project would also be considered for the normal and recessionary situation. The variability in the outcome under the three different scenarios would help the management to assess the risk a project carries. Higher deviation in the outcome can be assessed as higher risk and lower to medium deviation can be assessed accordingly.

Scenario analysis is far more **complex** than sensitivity analysis because in scenario analysis all inputs are changed simultaneously, considering the situation in hand while in sensitivity analysis, only one input is changed and others are kept constant.

PassBook's note: Out of syllabus (old scheme): sensitivity analysis and scenario analysis (risk analysis in capital budgeting) are not in the current Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · MTP May 2022 · Series I · question p. 3 · answer p. 7

801 MTP **Old P8 May 2022 S-II** Q1(c) 5 marks old scheme

An enterprise is investing ₹ 100 lakhs in a project. The risk-free rate of return is 7%. Risk premium expected by the Management is 7%. The life of the project is 5 years. Following are the cash flows that are estimated over the life of the project.

Year	Cash flows (₹ in lakhs)
1	25
2	60
3	75
4	80
5	65

CALCULATE Net Present Value of the project based on Risk free rate and also on the basis of Risks adjusted discount rate.

ICAI'S ANSWER

The Present Value of the Cash Flows for all the years by discounting the cash flow at 7% is calculated as below:

Year	Cash flows (₹ in lakhs)	Discounting Factor @7%	Present value of Cash Flows (₹ in Lakhs)
1	25	0.935	23.38

Year	Cash flows (₹ in lakhs)	Discounting Factor @7%	Present value of Cash Flows (₹ in Lakhs)
2	60	0.873	52.38
3	75	0.816	61.20
4	80	0.763	61.04
5	65	0.713	46.35
Total of present value of Cash flow			244.34
Less: Initial investment			100
Net Present Value (NPV)			**144.34**

Now, when the risk-free rate is 7 % and the risk premium expected by the Management is 7 %. So, the risk adjusted discount rate is 7 % + 7 % =14%.

Discounting the above cash flows using the Risk Adjusted Discount Rate would be as below:

Year	Cash flows (₹ in Lakhs)	Discounting Factor @14%	Present Value of Cash Flows (₹ in lakhs)
1	25	0.877	21.93
2	60	0.769	46.14
3	75	0.675	50.63
4	80	0.592	47.36
5	65	0.519	33.74
Total of present value of Cash flow			199.80
Initial investment			100
Net present value (NPV)			**99.80**

PassBook's note: Out of syllabus (old scheme): the risk-adjusted discount rate belongs to risk analysis in capital budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter covers NPV, IRR, PI and the other techniques but no risk analysis).

ICAI · Old P8 · MTP May 2022 · Series II · question p. 1-2 · answer p. 2-3

802 MTP **Old P8 Dec 2021 S-I** Q6(b) 4 marks old scheme

EXPLAIN sensitivity analysis and its various steps.

ICAI'S ANSWER

Sensitivity Analysis and its various steps: Sensitivity analysis put in simple terms is a modeling technique which is used in Capital Budgeting decisions which is used to study the **impact of changes in the variables on the outcome of the project**. In a project, several variables like weighted average cost of capital, consumer demand, price of the product, cost price per unit etc. operate simultaneously.

The changes in these variables impact the outcome of the project. It therefore becomes very difficult to assess change in which variable impacts the project outcome in a significant way. In Sensitivity Analysis, the project outcome is studied after taking into **change in only one variable.** The more sensitive is the NPV, the more critical is that variable. So, Sensitivity analysis is a way of finding impact in the project's NPV (or IRR) for a given change in one of the variables.

Sensitivity Analysis is conducted by following the steps as below:

(i) Finding variables, which have an influence on the NPV (or IRR) of the project.

(ii) Establishing mathematical relationship between the variables.

(iii) Analysis the effect of the change in each of the variables on the NPV (or IRR) of the project.

PassBook's note: Out of syllabus (old scheme): sensitivity analysis is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter teaches payback, ARR, NPV, PI, IRR, MIRR, discounted payback, capital rationing and equivalent annual cost, with no risk-analysis section).

ICAI · Old P8 · MTP Dec 2021 · Series I · question p. 4 · answer p. 10-11

803 MTP Old P8 Dec 2021 S-II Q5(a) 8 marks old scheme

PQR Ltd. has under its consideration a project with an initial investment of ₹ 2,25,00,000. Three probable cash inflow scenarios with their probabilities of occurrence have been estimated as below:

Annual cash inflow (₹)	50,00,000	75,00,000	1,00,00,000
Probability	0.2	0.7	0.1

The project life is 5 years and the desired rate of return is 12%. The estimated terminal values for the project assets under the three probability alternatives are ₹ 0, ₹ 50,00,000 and ₹ 75,00,000 respectively.

You are required to:

(i) CALCULATE the probable NPV;

(ii) CALCULATE the worst-case NPV and the best-case NPV; and

(iii) STATE the probability occurrence of the worst case, if the cash flows are perfectly positively correlated over time.

ICAI'S ANSWER

(i) Calculation of Net Present Value (NPV)

Year	Prob. = 0.2		Prob. = 0.7		Prob. = 0.1				
	Cash flow	Probable cash flow	Cash flow	Probable cash flow	Cash flow	Probable cash flow	Total Cash flow	PVF@ 12%	PV of Total cash flow
0							(2,25,00,000)	1.0000	(2,25,00,000)
1 to 5	50,00,000	10,00,000	75,00,000	52,50,000	1,00,00,000	10,00,000	72,50,000	3.6048	2,61,34,800
5	0	0	50,00,000	35,00,000	75,00,000	7,50,000	42,50,000	0.5674	24,11,450
Net Present Value (NPV)									**60,46,250**

(ii) Worst and Best case is the case where expected annual cash inflows are minimum and maximum respectively.

Calculation of Worst Case and Best Case NPV:

Year	PVF@ 12%	Worst case		Best Case	
		Cash flows	PV of Cash flows	Cash flows	PV of Cash flows
0	1.0000	(2,25,00,000)	(2,25,00,000)	(2,25,00,000)	(2,25,00,000)
1 to 5	3.6048	50,00,000	1,80,24,000	1,00,00,000	3,60,48,000
5	0.5674	0	0	75,00,000	42,55,500
NPV			**(44,76,000)**		**1,78,03,500**

Worst case NPV = ₹ (44,76,000)

Best Case NPV = ₹ 1,78,03,500

(iii) The cash flows are perfectly positively correlated over time means cash flow in first year will be cash flows in subsequent years. The cash flow of ₹ 50,00,000 is the worst case cash flow and its probability is 20%, thus, possibility of worst case is 20%.

PassBook's note: Out of syllabus (old scheme): the probable (expected) NPV from probability-weighted cash flows, worst- and best-case NPVs and the probability of the worst case under perfectly correlated cash flows are Risk Analysis in Capital Budgeting, which old Paper 8 taught; It is not in the new Paper 6A syllabus or the January 2027 Study Material (as o-p8-mtp-nov2022-s1 q4), so the item carries only its nearest chapter. ICAI's question asks for "the probability occurrence of the worst case".

ICAI · Old P8 · MTP Dec 2021 · Series II · question p. 4 · answer p. 8

804 MTP **Old P8 Jul 2021 S-I** Q4(a) 8 marks old scheme

N&B Ltd. is considering one of two mutually exclusive proposals, Projects A and B, which require cash outlays of Rs. 34,00,000 and Rs. 33,00,000 respectively. The certainty-equivalent (C.E) approach is used in

incorporating risk in capital budgeting decisions. The current yield on government bonds is 5% and this is used as the risk free rate. The expected net cash flows and their certainty equivalents are as follows:

Year-end	Project A		Project B	
	Cash Flow (Rs.)	**C.E.**	**Cash Flow (Rs.)**	**C.E.**
1	16,75,000	0.8	16,75,000	0.9
2	15,00,000	0.7	15,00,000	0.8
3	15,00,000	0.5	15,00,000	0.7
4	20,00,000	0.4	10,00,000	0.8
5	21,20,000	0.6	9,00,000	0.9

PV factor at 5% are as follows:

Year	1	2	3	4	5
PV factor	0.952	0.907	0.864	0.823	0.784

DETERMINE which project should be accepted.

ICAI'S ANSWER

Statement Showing the Net Present Value of Project A

Year end	Cash Flow (Rs.)	C.E.	Adjusted Cash flow (Rs.)	Present value factor at 5%	Total Present value (Rs.)
	(a)	**(b)**	**(c) = (a) × (b)**	**(d)**	**(e) = (c) ×(d)**
1	16,75,000	0.8	13,40,000	0.952	12,75,680
2	15,00,000	0.7	10,50,000	0.907	9,52,350
3	15,00,000	0.5	7,50,000	0.864	6,48,000
4	20,00,000	0.4	8,00,000	0.823	6,58,400
5	21,20,000	0.6	12,72,000	0.784	9,97,248
PV of total Cash Inflows					45,31,678
Less: Initial Investment					34,00,000
Net Present Value					**11,31,678**

Statement Showing the Net Present Value of Project B

Year end	Cash Flow (Rs.)	C.E.	Adjusted Cash flow (Rs.)	Present value factor at 5%	Total Present value (Rs.)
	(a)	**(b)**	**(c) = (a) × (b)**	**(d)**	**(e) = (c) ×(d)**

Year end	Cash Flow (Rs.)	C.E.	Adjusted Cash flow (Rs.)	Present value factor at 5%	Total Present value (Rs.)
1	16,75,000	0.9	15,07,500	0.952	14,35,140
2	15,00,000	0.8	12,00,000	0.907	10,88,400
3	15,00,000	0.7	10,50,000	0.864	9,07,200
4	10,00,000	0.8	8,00,000	0.823	6,58,400
5	9,00,000	0.9	8,10,000	0.784	6,35,040
PV of total Cash Inflows					47,24,180
Less: Initial Investment					33,00,000
Net Present Value					**14,24,180**

Project B has NPV of Rs. 14,24,180 which is higher than the NPV of Project A. Thus, N&B Ltd. should accept Project B.

PassBook's note: Out of syllabus (old scheme): the item evaluates two projects by the certainty equivalent approach, which is Risk Analysis in Capital Budgeting. That topic is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 3-4 · answer p. 6

805 MTP **Old P8 Jul 2021 S-I** Q4(b) 2 marks old scheme

DISCUSS the advantages of Certainty Equivalent Method.

ICAI'S ANSWER

Advantages of Certainty Equivalent Method:

1. The certainty equivalent method is simple and easy to understand and apply.
2. It can easily be calculated for different risk levels applicable to different cash flows. For example, if in a particular year, a higher risk is associated with the cash flow, it can be easily adjusted and the NPV can be recalculated accordingly.

PassBook's note: Out of syllabus (old scheme): the advantages of the certainty equivalent method belong to Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

ICAI · Old P8 · MTP Jul 2021 · Series I · question p. 4 · answer p. 7

806 MTP **Old P8 Jul 2021 S-II** Q2(a) 7 marks old scheme

SG Ltd. is considering a project "Z" with an initial outlay of Rs. 7,50,000 and life of 5 years. The estimates

of project are as follows:

	Lower Estimates	Base	Upper Estimates
Sales (units)	4,500	5,000	5,500
	(Rs.)	**(Rs.)**	**(Rs.)**
Selling Price p.u.	175	200	225
Variable cost p.u.	100	125	150
Fixed Cost	50,000	75,000	1,00,000

Depreciation included in Fixed cost is Rs. 35,000 and corporate tax is 25%.

Assuming the cost of capital as 15%, DETERMINE NPV in three scenarios i.e worst, base and best case scenario.

PV factor for 5 years at 15% are as follows:

Years	1	2	3	4	5
P.V. factor	0.870	0.756	0.658	0.572	0.497

ICAI'S ANSWER

(i) Calculation of Yearly Cash Inflow

In worst case: High costs and Low price (Selling price) and volume(Sales units) are taken.

In best case: Low costs and High price(Selling price) and volume(Sales units) are taken.

	Worst Case	Base	Best Case
Sales (units) (A)	4,500	5,000	5,500
	(Rs.)	**(Rs.)**	**(Rs.)**
Selling Price p.u.	175	200	225
Less: Variable cost p.u.	150	125	100
Contribution p.u. (B)	25	75	125
Total Contribution (A x B)	1,12,500	3,75,000	6,87,500
Less: Fixed Cost	1,00,000	75,000	50,000
EBT	12,500	3,00,000	6,37,500
Less: Tax @ 25%	3,125	75,000	1,59,375
EAT	9,375	2,25,000	4,78,125
Add: Depreciation	35,000	35,000	35,000
Cash Inflow	44,375	2,60,000	5,13,125

(ii) Calculation of NPV in different scenarios

	Worst Case	Base	Best Case
Initial outlay (A) (Rs.)	7,50,000	7,50,000	7,50,000
Cash Inflow (c) (Rs.)	44,375	2,60,000	5,13,125
Cumulative PVF @ 15% (d)	3.353	3.353	3.353
PV of Cash Inflow (B = c x d) (Rs.)	1,48,789.38	8,71,780	17,20,508.13
NPV (B - A) (Rs.)	(6,01,210.62)	1,21,780	9,70,508.13

PassBook's note: It is one table here. Lead ruling 2026-10-20: risk analysis in capital budgeting (scenario / best-worst-case NPVs, expected NPV, sensitivity, certainty equivalent, risk-adjusted discount rate) is not in the January 2027 Study Material, so this item is out of syllabus (oos) and keeps the nearest tag fm.capbud.npv.

ICAI · Old P8 · MTP Jul 2021 · Series II · question p. 2 · answer p. 3-4

807 MTP **Old P8 Nov 2020 S-I** Q1(c) 5 marks old scheme

ABC Ltd. is considering a project "X" with an initial outlay of ₹ 16,00,000 and the possible three cash inflow attached with the project is as follows:

(Amount in ₹ '000)

Particular	Year 1	Year 2	Year 3
Scenario 1	550	500	800
Scenario 2	650	550	900
Scenario 3	750	600	1000

Assuming the cost of capital as 9%.

(i) DETERMINE NPV in each scenario.

(ii) If ABC Ltd. is certain about the 1st and 2nd year's results in scenario 2 but uncertain about the third year's cash flow, DETERMINE NPV expecting scenario 1 in the third year.

Year	1	2	3
DF @ 9%	0.917	0.842	0.772

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

PassBook's note: Out of syllabus (old scheme): NPV under three cash-flow scenarios, and with the third year switched to another scenario, is scenario analysis (Risk Analysis in Capital Budgeting), which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 2

808 MTP **Old P8 Nov 2020 S-I** Q5(b) 2 marks old scheme

EXPLAIN certainty equivalents, one of the techniques of risk analysis.

ICAI'S ANSWER

ICAI's answers to this paper are no longer available online.

PassBook's note: Out of syllabus (old scheme): certainty equivalents are a technique of Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section).

ICAI · Old P8 · MTP Nov 2020 · Series I · question p. 4

809 MTP **Old P8 May 2020 S-I** Q3 10 marks old scheme

A&R Ltd. has undertaken a project which has an initial investment of Rs.2,000 lakhs in plant & machinery and Rs.800 lakhs for working capital. The plant & machinery would have a salvage value of Rs. 474.61 lakhs at the end of the fifth year. The plant & machinery would depreciate at the rate of 25% p.a. on WDV method. The other details of the project for the five year period are as follows:

Sales	10,00,000 units p.a.
Selling price per unit	Rs.500
Variable cost	50% of selling price
Fixed overheads (excluding depreciation)	Rs.300 lakh p.a.
Corporate tax rate	35%
Rate of interest on bank loan	12%
After tax required rate of return	15%

Required:

(i) CACULATE net present value (NPV) of the project and DETERMINE the viability of the project.

(ii) DETERMINE the sensitivity of project's NPV under each of the following condition:

a. Decrease in selling price by 10%;

b. Increase in cost of plant & machinery by 10%.

PV factor	Year-1	Year-2	Year-3	Year-4	Year-5
12%	0.892	0.797	0.711	0.635	0.567
15%	0.869	0.756	0.657	0.571	0.497

ICAI'S ANSWER

(i) Calculation of Net Present Value (NPV):

	Year-1	Year-2	Year-3	Year-4	Year-5
Sales volume (Qty. in lakh)	10	10	10	10	10
Contribution per unit (Rs.) (Selling price – variable cost)	250	250	250	250	250
Total contribution (Rs.in lakh)	2,500	2,500	2,500	2,500	2,500

	Year-1	Year-2	Year-3	Year-4	Year-5
Less: Fixed overheads (Rs. In lakh)	300	300	300	300	300
PBDT	2,200	2,200	2,200	2,200	2,200
Less: Depreciation (Rs. in lakh) (Working note-1)	500	375	281.25	210.94	158.20
PBT	1,700	1,825	1,918.75	1,989.06	2,041.80
Less: Tax @ 35%	595	638.75	671.56	696.17	714.63
PAT	1,105	1,186.25	1,247.19	1,292.89	1,327.17
Add: Depreciation	500	375	281.25	210.94	158.20
Add: Salvage value of plant & machinery	-	-	-	-	474.61
Add: Working capital	-	-	-	-	800
Net Cash inflow	1,605	1,561.25	1,528.44	1,503.83	2,759.98
P.V factor @15%	0.869	0.756	0.657	0.571	0.497
P.V of cash inflows	1,394.74	1,180.31	1,004.18	858.68	1,371.71

Net Present Value = P.V of cash inflows – P.V of cash outflows

= Rs. (1,394.74+1,180.31+1,004.18+858.68+1,371.71) – Rs.2,000

= Rs.3,809.62 lakh

The NPV of the project is positive, hence, the project is viable.

Working note-1:

	Year-1	Year-2	Year-3	Year-4	Year-5
Opening balance	2,000	1,500	1,125	843.75	632.81
Depreciation @25%	500	375	281.25	210.94	158.20
Closing WDV	1,500	1,125	843.75	632.81	474.61

(ii) Determination of sensitivity of NPV w.r.t.

a. Decrease in selling price by 10%

	Year-1	Year-2	Year-3	Year-4	Year-5
Sales volume (Qty. in lakh)	10	10	10	10	10
New Selling price	450	450	450	450	450
Variable cost	250	250	250	250	250

	Year-1	Year-2	Year-3	Year-4	Year-5
Contribution per unit (Rs.) (Selling price – variable cost)	200	200	200	200	200
Total contribution (Rs.in lakh)	2,000	2,000	2,000	2,000	2,000
Less: Fixed overheads (Rs. In lakh)	300	300	300	300	300
PBDT	1,700	1,700	1,700	1,700	1,700
Less: Depreciation (Rs. in lakh) (Working note-1)	500	375	281.25	210.94	158.20
PBT	1,200	1,325	1,418.75	1,489.06	1,541.80
Less: Tax @ 35%	420	463.75	496.56	521.17	539.63
PAT	780	861.25	922.19	967.89	1,002.17
Add: Depreciation	500	375	281.25	210.94	158.20
Add: Salvage value of plant & machinery	-	-	-	-	474.61
Add: Working capital	-	-	-	-	800
Net Cash inflow	1,280	1,236.25	1,203.44	1,178.83	2,434.98
P.V factor @15%	0.869	0.756	0.657	0.571	0.497
P.V of cash inflows	1,112.32	934.61	790.66	673.11	1,210.18

NPV = (1,112.32+934.61+790.66+673.11+1,210.18) – 2,000

= 4,720.88 – 2,000 = 2,720.88 lakh

10% reduction in selling price reduces the NPV by 28.58% (3,809.62-2,720.88/3809.62)

b. Increase in project cost by 10%

	Year-1	Year-2	Year-3	Year-4	Year-5
PBDT	2,200	2,200	2,200	2,200	2,200
Less: Depreciation (Rs. in lakh) (Working note-2)	550	412.5	309.37	232.03	174.03
PBT	1,650	1,787.50	1,890.63	1,967.97	2,025.97
Less: Tax @ 35%	577.50	625.63	661.72	688.79	709.09
PAT	1072.50	1,161.87	1,228.91	1,279.18	1,316.88
Add: Depreciation	550	412.5	309.37	232.03	174.03
Add: Salvage value of plant & machinery	-	-	-	-	474.61

	Year-1	Year-2	Year-3	Year-4	Year-5
Add: Working capital	-	-	-	-	800
Net Cash inflow	1,622.50	1,574.37	1,538.28	1,511.21	2,765.52
P.V factor @15%	0.869	0.756	0.657	0.571	0.497
P.V of cash inflows	1,409.95	1,190.22	1,010.65	862.90	1,374.46

NPV = (1,409.95+1,190.22+1,011.65+862.90+1,374.46) – 2,200

= 5,849.18 – 2,200 = 3,649.18 lakh

10% increase in project cost reduces the NPV only by 4.21% (3,809.62 - 3,649.18/3809.62)

Working note-2:

	Year-1	Year-2	Year-3	Year-4	Year-5
Opening balance	2,200	1,650	1,237.50	928.13	696.10
Depreciation @25%	550	412.5	309.37	232.03	174.03
Closing WDV	1,650	1,237.50	928.13	696.10	522.07

PassBook's note: Out of syllabus (old scheme): requirement (ii), the sensitivity of the NPV to a 10% fall in selling price and a 10% rise in the cost of plant & machinery, is Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material (its Investment Decisions chapter has no risk-analysis section); As o-p8-mtp-nov2023-s2 q4). Requirement (i), the NPV decision, is in syllabus. ICAI prints one [10 Marks] for both requirements, so the item is not split. ICAI's own wording and figures kept as printed: "CACULATE" in the question.

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810 MTP **Old P8 Nov 2019 S-I** Q5(a) 8 marks old scheme

CALCULATE Variance and Standard Deviation on the basis of following information:

Possible Event	Project A		Project B	
	Cash Flow (Rs.)	Probability	Cash Flow (Rs.)	Probability
A	80,000	0.10	2,40,000	0.10
B	1,00,000	0.20	2,00,000	0.15
C	1,20,000	0.40	1,60,000	0.50
D	1,40,000	0.20	1,20,000	0.15
E	1,60,000	0.10	80,000	0.10

ICAI'S ANSWER

Calculation of Expected Value for Project A and Project B

Possible Event	Project A			Project B		
	Net Cash Flow (Rs.)	Probability	Expected Value (Rs.)	Cash Flow (Rs.)	Probability	Expected Value (Rs.)
A	80,000	0.10	8,000	2,40,000	0.10	24,000
B	1,00,000	0.20	20,000	2,00,000	0.15	30,000
C	1,20,000	0.40	48,000	1,60,000	0.50	80,000
D	1,40,000	0.20	28,000	1,20,000	0.15	18,000
E	1,60,000	0.10	16,000	80,000	0.10	8,000
ENCF			1,20,000			1,60,000

Project A

Variance $(\sigma^2) = (80,000 - 1,20,000)^2 \times (0.1) + (1,00,000 - 1,20,000)^2 \times (0.2) + (1,20,000 - 1,20,000)^2 \times (0.4) + (1,40,000 - 1,20,000)^2 \times (0.2) + (1,60,000 - 1,20,000)^2 \times (0.1)$

$= 16,00,00,000 + 8,00,00,000 + 0 + 8,00,00,000 + 16,00,00,000$

$= 48,00,00,000$

Standard Deviation $(\sigma) = \sqrt{\text{Variance}\,(\sigma^2)} = \sqrt{48,00,00,000} = 21,908.90$

Project B:

Variance$(\sigma^2) = (2,40,000 - 1,60,000)^2 \times (0.1) + (2,00,000 - 1,60,000)^2 \times (0.15) + (1,60,000 - 1,60,000)^2 \times (0.5) + (1,20,000 - 1,60,000)^2 \times (0.15) + (80,000 - 1,60,000)^2 \times (0.1)$

$= 64,00,00,000 + 24,00,00,000 + 0 + 24,00,00,000 + 64,00,00,000$

$= 1,76,00,00,000$

Standard Deviation $(\sigma) = \sqrt{1,76,00,00,000} = 41,952.35$

PassBook's note: Out of syllabus (old scheme): the expected value, variance and standard deviation of a project's cash flows (probability distributions) are Risk Analysis in Capital Budgeting, which is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material.

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A.2 Topics no longer in the syllabus

3 questions ICAI marks out of syllabus, or whose topic has left the syllabus. For reference only.

811 EXAM Old P8 Nov 2020 Q4 10 marks old scheme

A Ltd. is considering two mutually exclusive projects X and Y.

You have been given below the Net Cash flow probability distribution of each project:

Project-X		Project-Y	
Net Cash Flow (₹)	**Probability**	**Net Cash Flow (₹)**	**Probability**
50,000	0.30	1,30,000	0.20
60,000	0.30	1,10,000	0.30
70,000	0.40	90,000	0.50

(i) Compute the following :

(a) Expected Net Cash Flow of each project.

(b) Variance of each project.

(c) Standard Deviation of each project.

(d) Coefficient of Variation of each project.

(ii) Identify which project do you recommend ? Give reason.

ICAI'S ANSWER

(i) (a) Calculation of Expected Net Cash Flow (ENCF) of Project X and Project Y

Project X			Project Y		
Net Cash Flow (₹)	**Probability**	**Expected Net Cash Flow (₹)**	**Net Cash Flow (₹)**	**Probability**	**Expected Net Cash Flow (₹)**
50,000	0.30	15,000	1,30,000	0.20	26,000
60,000	0.30	18,000	1,10,000	0.30	33,000
70,000	0.40	28,000	90,000	0.50	45,000
ENCF		**61,000**			**1,04,000**

(b) Variance of Projects

Project X

Variance $(\sigma^2) = (50,000 - 61,000)^2 \times (0.3) + (60,000 - 61,000)^2 \times (0.3) + (70,000 - 61,000)^2 \times (0.4)$

= 3,63,00,000 + 3,00,000 + 3,24,00,000 = **6,90,00,000**

Project Y

Variance$(\sigma^2) = (1,30,000 - 1,04,000)^2 \times (0.2) + (1,10,000 - 1,04,000)^2 \times (0.3) + (90,000 - 1,04,000)^2 \times (0.5)$

= 13,52,00,000 + 1,08,00,000 + 9,80,00,000 = **24,40,00,000**

(c) Standard Deviation of Projects

Project X

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{6,90,00,000}$ = **8,306.624**

Project Y

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{24,40,00,000}$ = **15,620.499**

(d) Coefficient of Variation of Projects

Projects	Coefficient of variation ($\frac{\text{Standard Deviation}}{\text{Expected Net Cash Flow}}$)	Risk	Expected Net Cash Flow
X	$\frac{8,306.24}{61,000}$ = **0.136 or 13.60%**	Less	Less
Y	$\frac{15,620.499}{1,04,000}$ = **0.150 or 15.00%**	More	More

(ii) In project X risk per rupee of cash flow is 0.136 (approx.) while in project Y it is 0.15 (approx.). Therefore, Project X is better than Project Y.

PassBook's note: Out of syllabus (old scheme): risk analysis in capital budgeting by statistical measures (expected net cash flow, variance, standard deviation and coefficient of variation of a project's cash flow distribution) was part of old Paper 8's Investment Decisions; Neither the new-scheme Paper 6A syllabus nor the January 2027 Study Material teaches it. No tag. ICAI's table (d) uses 8,306.24 for Project X's standard deviation, which (c) gives as 8,306.624.

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812 EXAM **Old P8 Nov 2020** Q6(c) 2 marks old scheme

What is Risk Adjusted Discount Rate ?

ICAI'S ANSWER

Risk Adjusted Discount Rate: A risk adjusted discount rate is a sum of risk-free rate and risk premium. The Risk Premium depends on the perception of risk by the investor of a particular investment and risk aversion of the Investor.

So, Risks adjusted discount rate = Risk free rate+ Risk premium.

PassBook's note: ICAI prints this as the OR alternative to Q6(c), with its own (2 Marks). Old scheme, out of syllabus (ours): the risk-adjusted discount rate, a technique of risk analysis in capital budgeting in old Paper 8's Investment Decisions, is not in the new-scheme Paper 6A syllabus or the January 2027 Study Material. No tag.

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813 EXAM **Old P8 Nov 2019** Q4 10 marks old scheme

Loft Ltd. is considering an investment in new technology that will reduce operating costs through increasing efficiency. The new technology will cost ₹ 5,00,000 and have a four year life at the end of which it will have a residual value of ₹ 50,000.

An annual license fee of ₹ 52,000 is payable to operate the machine. The purchase can be financed by 10% loan payable in equal installments at the end of each of four years. The depreciation is to be charged as per reducing balance method. The rate of depreciation is 25% per annum.

Alternatively, Loft Ltd. could lease the new technology. The Company would pay four annual lease rentals of ₹ 1,90,000 per year. The annual lease rentals include the cost of license fee. Tax rate is 30%. Compute the incremental cash flows under each option. What would be the appropriate rate at which these cash flows have to be discounted? Discount the incremental cash flows under each option and decide which option is better - buy or lease?

Year	1	2	3	4
DF @ 7%	0.935	0.873	0.816	0.763
DF @ 10%	0.909	0.826	0.751	0.683

ICAI'S ANSWER

(1) The buy or lease decision means computation of NPV arising from lease decision i.e. computation of valuation advantage of lease over buy. If the value is positive then we go for lease, otherwise we buy.

(2) The valuation process involves – a) finding incremental cash flow of lease over buy, and then, b) discounting the incremental cash flow by net of tax interest rate of equivalent loan (to purchase the asset in question).

If we go for lease, there would be cash outflow in the form of net of tax lease rent from year 1 to 4. Net of tax lease rent per annum = ₹ 1,90,000 x (1-.30) = ₹ 1,33,000.

Again, if the equipment had been purchases there would have been tax saving of depreciation = Depreciation x tax rate. Here, the tax saving or tax shield is available for 4 years. But under lease the benefit accrues to lessor. For lessee it is a negative cash flow as advantage is not available to him under lease arrangement as lessor is considered the legal owner of the asset for claiming depreciation under Income tax law. The depreciation schedule and tax shield on depreciation are given in table 1.

Table 1

Year	Cost/ opening balance (₹)	Depreciation @ 25% (₹)	Closing balance (₹)	Tax shield (₹)
1	5,00,000	1,25,000	3,75,000	37,500
2	3,75,000	93,750	2,81,250	28,125
3	2,81,250	70,312.50	2,10,937.50	21,093.75
4	2,10,937.50	52,734.38	1,58,203.12	15,820.31

(3) Further, if the equipment had been purchased there would have been tax saving of interest on loan = interest on loan x tax rate. Here, the tax saving or tax shield is available for 4 years. For lessee it is a negative cash flow as advantage is not available to him under lease arrangement.

The loan amount would have been repayable together with the interest at the rate of 10% in equal installment at the end of each year. The PVAF at the rate of 10% for 4 years is 3.169 (.909 + .826 + .751 + .683), the amount payable would have been-

Annual Installment = $\frac{₹5,00,000}{3.169}$ = ₹ 1,57,778 (approx.)

The interest expense schedule and tax shield on interest expense are given in table 2.

Table 2

Year	Total Installment (₹)	Interest (₹)	Principal (₹)	Principal amount outstanding (₹)	Tax shield (₹)
1	1,57,778	50,000	1,07,778	3,92,222	15,000
2	1,57,778	39,222	1,18,556	2,73,666	11,766.6
3	1,57,778	27,367	1,30,411	1,43,255	8,210.1
4	1,57,778	14,523 (bal. fig.)	1,43,255	------	4,356.9

(4) After 4 years the equipment is sold for ₹ 50,000 which is a cash outflow due to lease over buy

Loss on sale = ₹ (1,58,203.12 - 50,000)

= ₹ 1,08,203.12

Tax savings on loss = 30% of ₹ 1,08,203.12 = ₹ 32,460.94

This further tax shield has to be accounted for in the year 4.

(5) If the equipment is taken on lease, the cash outflow on a/c of lease rental, depreciation tax shield is given in table 3

Table 3

Year	Net of tax lease rental (₹)	Depreciation tax shield (₹) (from Table 1)	Interest tax shield (₹) (from Table 2)	Total (₹)
1	1,33,000	37,500	15,000	1,85,500
2	1,33,000	28,125	11,766.6	1,72,891.6
3	1,33,000	21,093.75	8,210.1	1,62,303.85
4	1,33,000	15,820.31	4,356.9	1,53,177.21

(6) Net of tax interest rate = 0.10X (1-.30) = 0.07

Calculation of NPV for lease over buy option (Amount in ₹)

Year	1	2	3	4
Loan Installment	1,57,778	1,57,778	1,57,778	1,57,778
License Fees (net of Taxes)	36,400	36,400	36,400	36,400
Amount from sale of Machine				(50,000)
Tax saving on loss on sale				(32,460.94)
Total Tax Shield on Lease Rent, Interest and Depreciation	(1,85,500)	(1,72,891.60)	(1,62,303.85)	(1,53,177.21)
Total Cash flow	8,678.00	21,286.40	31,874.15	(41,460.15)
Discounting Factor @7%	0.935	0.873	0.816	0.763
Discounted Cash Flow	8,113.93	18,583.03	26,009.31	(31,634.09)

Year	1	2	3	4
NPV [8,113.93 + 18,583.03 + 26,009.31 + (31,634.09)]				**21,072.17**

Since, NPV or value of the lease is positive, the equipment should be taken on lease.

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